

APPENDIX D

Box No. VIII (ii) DECLARATION: ENTITLEMENT TO APPLY FOR AND BE GRANTED A PATENT

The declaration must conform to the standardized wording provided for in Section 212; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (ii). If this Box is not used, this sheet should not be included in the request.

Declaration as to the applicant's entitlement, as at the international filing date, to apply for and be granted a patent (Rules 4.17(ii) and 51bis.1(a)(ii)), in a case where the declaration under Rule 4.17(iv) is not appropriate:

in relation to this international application,

trinamiX GmbH is/are entitled to apply for and be granted a patent by virtue of the following:

an Assignment from Schillen, Peter to trinamiX GmbH, dated 08 Sep 2021.
an Assignment from Guthier, Benjamin to trinamiX GmbH, dated 24 Sep 2021.
an Assignment from Schick, Friedrich to trinamiX GmbH, dated 21 Sep 2021.
an Assignment from Guenther, Manuel to trinamiX GmbH, dated 10 May 2021.
an Assignment from Diesselberg, Lars to trinamiX GmbH, dated 21 Apr 2021.
an Assignment from Lennartz, Christian to trinamiX GmbH, dated 20 Apr 2021.

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

Box No. VIII (iv) DECLARATION: INVENTORSHIP (only for the purposes of the designation of the United States of America)
The declaration must conform to the following standardized wording provided for in Section 214; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (iv). If this Box is not used, this sheet should not be included in the request.

**Declaration of inventorship (Rules 4.17(iv) and 51bis.1(a)(iv))
 for the purposes of the designation of the United States of America:**

I hereby declare that I believe I am the original inventor or an original joint inventor of a claimed invention in the application.

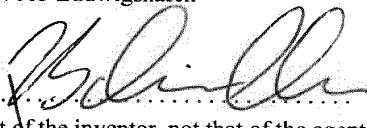
This declaration is directed to the international application of which it forms a part (if filing declaration with application).

This declaration is directed to international application No. (if furnishing declaration pursuant to Rule 26ter).

I hereby declare that the above-identified international application was made or authorized to be made by me.

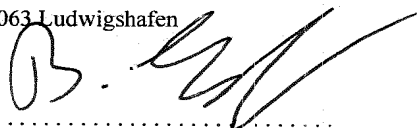
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Name: SCHILLEN, Peter
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

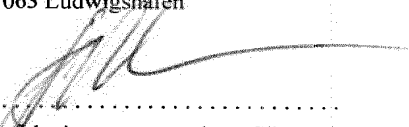
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Name: GUTHIER, Benjamin
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Inventor's Signature: 
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Date: 8.9.21

Name: SCHICK, Friedrich
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Inventor's Signature: 
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Date: 21.9.21

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: GUENTHER, Manuel
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8050 Zuerich

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Date: 10.05.2021

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Name: LENNARTZ, Christian
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Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Continuation of Box No. VIII (i) to (v) DECLARATION

If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.

Name: GUENTHER, Manuel
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Date:

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Residence: Germany
Mailing Address: Geibelstr. 22
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Inventor's Signature: 
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Date: 27.4.2021

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Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 27.4.21

PCT REQUEST

(Original in Electronic Form)

0	For receiving Office use only	
0-1	International Application No.	PCT/EP2022/053941
0-2	International Filing Date	17 FEB 2022 (17.02.2022)
0-3	Name of receiving Office and "PCT International Application"	RO/EP
0-4	Form PCT/RO/101 PCT Request	
0-4-1	Prepared Using	PCT Online Filing Version 3.51.000.273e MT/FOP 20141031/0.20.5.24
0-5	Petition The undersigned requests that the present international application be processed according to the Patent Cooperation Treaty	
0-6	Receiving Office (specified by the applicant)	European Patent Office (EPO) (RO/EP)
0-7	Applicant's or agent's file reference	B202595PC
I	Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK
II	Applicant	
II-1	This person is	Applicant only
II-2	Applicant for	All designated States
II-4	Name	trinamiX GmbH
II-5	Address	Industriestrasse 35 67063 Ludwigshafen am Rhein Germany
II-6	State of nationality	DE
II-7	State of residence	DE
III-1	Applicant and/or inventor	
III-1-1	This person is	Inventor only
III-1-3	Inventor for	All designated States
III-1-4	Name (LAST, First)	SCHILLEN, Peter
III-1-5	Address	Industriestraße 35 67063 Ludwigshafen am Rhein Germany
III-2	Applicant and/or inventor	
III-2-1	This person is	Inventor only
III-2-3	Inventor for	All designated States
III-2-4	Name (LAST, First)	GUTHIER, Benjamin
III-2-5	Address	Industriestraße 35 67063 Ludwigshafen am Rhein Germany

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(Original in Electronic Form)

III-3 III-3-1 III-3-3 III-3-4 III-3-5	Applicant and/or inventor This person is Inventor for Name (LAST, First) Address	Inventor only All designated States SCHICK, Friedrich Industriestraße 35 67063 Ludwigshafen am Rhein Germany
III-4 III-4-1 III-4-3 III-4-4 III-4-5	Applicant and/or inventor This person is Inventor for Name (LAST, First) Address	Inventor only All designated States GUENTHER, Manuel Andreastrasse 15 8050 Zuerich Switzerland
III-5 III-5-1 III-5-3 III-5-4 III-5-5	Applicant and/or inventor This person is Inventor for Name (LAST, First) Address	Inventor only All designated States DIESELBERG, Lars Geibelstr. 22 76185 Karlsruhe Germany
III-6 III-6-1 III-6-3 III-6-4 III-6-5	Applicant and/or inventor This person is Inventor for Name (LAST, First) Address	Inventor only All designated States LENNARTZ, Christian Industriestraße 35 67063 Ludwigshafen am Rhein Germany

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(Original in Electronic Form)

IV-1	Agent or common representative; or address for correspondence The person identified below is hereby/ has been appointed to act on behalf of the applicant(s) before the competent International Authorities as:	Agent
IV-1-1	Name	Altmann Stöbel Dick Patentanwälte PartG mbB
IV-1-2	Address	Dr. Matthias Stöbel Theodor-Heuss-Anlage 2 68165 Mannheim Germany
IV-1-3	Telephone No.	+49 (0) 621-391813400
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IV-1-5	e-mail	BASF-Post@asd-patent.com
IV-1-5(a))	E-mail authorization The receiving Office, the International Searching Authority, the International Bureau and the International Preliminary Examining Authority are authorized to use this e-mail address, if the Office or Authority so wishes, to send notifications issued in respect of this international application:	exclusively in electronic form (no paper notifications will be sent)
IV-1-6	Agent's registration No.	446
V	DESIGNATIONS	
V-1	The filing of this request constitutes under Rule 4.9(a), the designation of all Contracting States bound by the PCT on the international filing date, for the grant of every kind of protection available and, where applicable, for the grant of both regional and national patents.	
VI-1	Priority claim of earlier regional application	
VI-1-1	Filing date	18 February 2021 (18.02.2021)
VI-1-2	Number	21157800.0
VI-1-3	Regional Office	EP
VI-2	Priority document request The International Bureau is requested to obtain from a digital library a certified copy of the earlier application(s) identified above as item(s), using, where applicable, the access code(s) indicated:	VI-1 Access code: 8843
VI-3	Incorporation by reference : where an element of the international application referred to in Article 11(1)(iii)(d) or (e) or a part of the description, claims or drawings referred to in Rule 20.5(a), or an element or part of the description, claims or drawings referred to in Rule 20.5bis(a) is not otherwise contained in this international application but is completely contained in an earlier application whose priority is claimed on the date on which one or more elements referred to in Article 11(1)(iii) were first received by the receiving Office, that element or part is, subject to confirmation under Rule 20.6, incorporated by reference in this international application for the purposes of Rule 20.6.	
VII-1	International Searching Authority Chosen	European Patent Office (EPO) (ISA/EP)

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(Original in Electronic Form)

VII-2	Request to use results of earlier search; reference to that search		
VII-2-1	Filing date	18 February 2021 (18.02.2021)	
VII-2-2	Application Number	21157800.0	
VII-2-3	Country (or regional Office)	EP	
VII-2-5	Documents are available to the ISA in a form and a manner acceptable to it, and therefore do not need to be submitted by the applicant to the receiving Office, or to the ISA (Rules 12bis1.(c) and (d) and 12bis.2(b)):	A copy of the results of the earlier search A copy of the earlier application A copy of any document cited in the earlier search results	
VIII	Declarations	Number of declarations	
VIII-1	Declaration as to the identity of the inventor	–	
VIII-2	Declaration as to the applicant's entitlement, as at the international filing date, to apply for and be granted a patent	≠ 1*	*RO/EP
VIII-3	Declaration as to the applicant's entitlement, as at the international filing date, to claim the priority of the earlier application	–	
VIII-4	Declaration of inventorship (only for the purposes of the designation of the United States of America)	≠ 1*	*RO/EP
VIII-5	Declaration as to non-prejudicial disclosures or exceptions to lack of novelty	–	
IX	Check list	Number of sheets	Electronic file(s) attached
IX-1	Request (including declaration sheets)	[6–] 10**	✓ **RO/EP
IX-2	Description	60	✓
IX-3	Claims	3	✓
IX-4	Abstract	1	✓
IX-5	Drawings	2	✓
IX-6a	Sequence listing part of the description (also to be used for the purposes of international search)	–	–
IX-7	TOTAL	[72] 76**	**RO/EP

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	Accompanying Items	Paper document(s) attached	Electronic file(s) attached
IX-8	Fee calculation sheet	-	✓
IX-19	Other	Pre-conversion archive Statement (Section 706): This is a complete and accurate copy of the international application prior to its conversion into the electronic document format in which it is filed.	✓
IX-19	Other	[B202595PC-PCT-Decl--4-17-Inventor-signed---] ***	✓ ***RO/EP
IX-19	Other	[B202595PC-PCT-Decl--4-17-Applicant---] ***	✓ ***RO/EP
IX-20	Figure of the drawings which should accompany the abstract	3	
IX-21	Language of filing of the international application	English	
X-1	Signature of applicant, agent or common representative	/Dr. Matthias Stöbel/	
X-1-1	Name	Altmann Stöbel Dick Patentanwälte PartG mbB	
X-1-2	Name of signatory	Dr. Matthias Stöbel	
X-1-3	Capacity (if such capacity is not obvious from reading the request)	(Representative)	

FOR RECEIVING OFFICE USE ONLY

10-1	Date of actual receipt of the purported international application	17 FEB 2022 (17.02.2022)
10-2	Drawings:	
10-2-1	Received	received
10-2-2	Not received	
10-3	Corrected date of actual receipt due to later but timely received papers or drawings completing the purported international application	
10-4	Date of timely receipt of the required corrections under PCT Article 11(2)	
10-5	International Searching Authority	ISA/EP
10-6	Transmittal of search copy delayed until search fee is paid	

PCT REQUEST

(Original in Electronic Form)

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11-1	Date of receipt of the record copy by the International Bureau	
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FIG.3

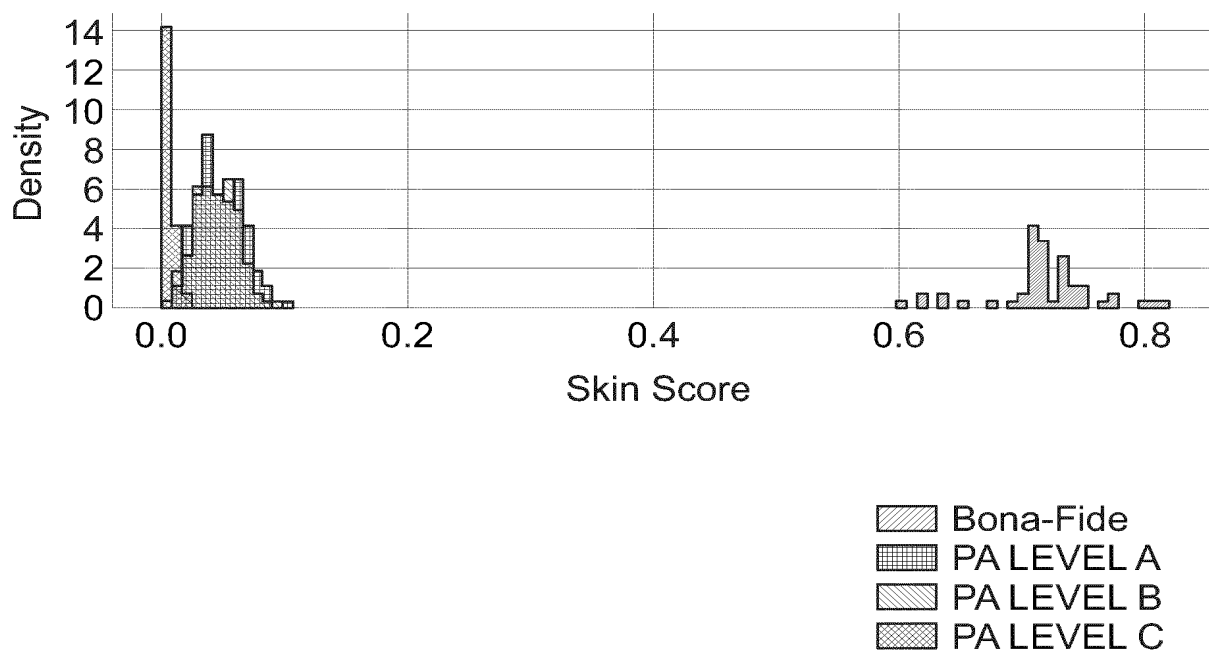


FIG.1

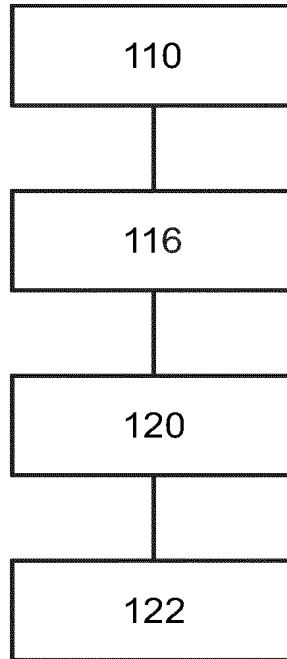
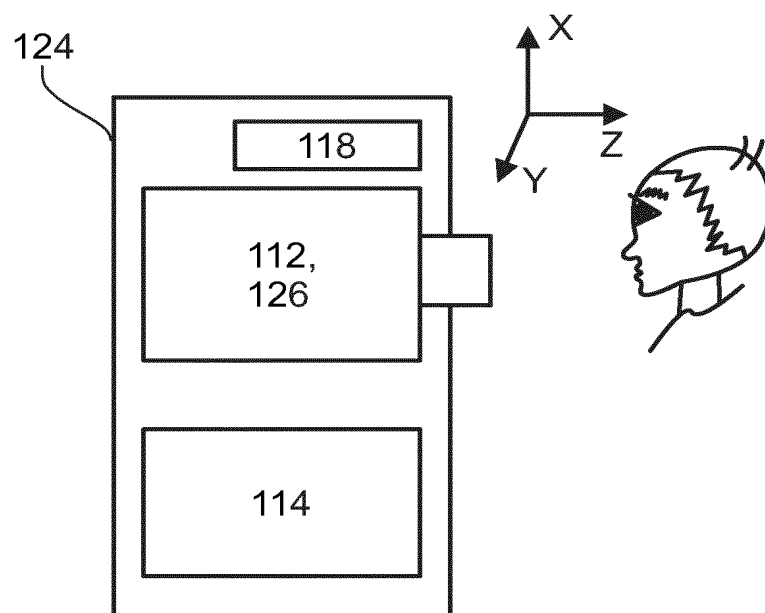


FIG.2



Optical Skin Detection for Face Unlock

Abstract

- 5 A method for face authentication is proposed. The method comprising the following steps:
- a) at least one face detection step (110) comprising determining at least one first image by using at least one camera (112);
 - b) at least one skin detection step (116) comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit (118), determining at least one second image using the at least one camera (112), determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile, and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit (114), wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
 - c) at least one 3D detection step (120) comprising determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit (114), wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
 - d) at least one authentication step (122) comprising authenticating the detected face by using at least one authentication unit if in step b) (116) the detected face is characterized as skin and in step c) (120) the detected face is characterized as 3D object.

30 (Figure 3)

13. A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding claims referring to a method.
- 5 14. A mobile device (124) comprising at least one camera (112), at least one illumination unit (118) and at least one processing unit (114), the mobile device (124) being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding claims referring to a method.
- 10 15. Use of the method according to any one of the preceding claims for biometric presentation attack detection.

- 5 3. The method according to the preceding claim, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.
- 10 4. The method according to any one of the preceding claims, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.
- 15 5. The method according to the preceding claim, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.
- 20 6. The method according to any one of the two preceding claims, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step (116) at least one region specific parametrized skin classification model is used.
- 25 7. The method according to any one of the preceding claims, wherein the illumination pattern comprises a periodic grid of laser spots.
- 30 8. The method according to any one of the preceding claims, wherein the illumination features have wavelengths in a near infrared (NIR) regime.
- 35 9. The method according to the preceding claim, wherein the illumination features have wavelengths of 940 nm.
- 35 10. The method according to any one of the preceding claims, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) (116) and/or for 3D detection in step c) (120).
- 40 11. The method according to any one of the preceding claims, wherein the camera (112) is or comprises at least one near infrared camera.
- 40 12. Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding claims, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding claims.

Claims

1. Method for face authentication comprising the following steps:

- 5 a) at least one face detection step (110), wherein the face detection step (110) comprises determining at least one first image by using at least one camera (112), wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step (110) comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit (114);
- 10 b) at least one skin detection step (116), wherein the skin detection step (116) comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit (118) and determining at least one second image using the at least one camera (112), wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit (114), wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- 15 c) at least one 3D detection step (120), wherein the 3D detection step (120) comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit (114), wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- 20 d) at least one authentication step (122), wherein the authentication step (122) comprises authenticating the detected face by using at least one authentication unit if in step b) (116) the detected face is characterized as skin and in step c) (120) the detected face is characterized as 3D object.
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- 30
- 35

2. The method according to the preceding claim, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.
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List of reference numbers

110	face detection step
112	camera
114	processing unit
116	skin detection step
118	illumination unit
120	3D detection step
122	authentication step
124	mobile device
126	optical sensor

Level A and Level B attacks are all based on 2D PAIs, whereas for Level C attacks, based on 3D masks. For Level C attacks, custom rigid masks constructed using a 3D printer were used. A test crew of 10 subjects has been used to obtain the reference distribution of skin-scores for bona fide presentations.

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Level	#PAI species	PAI species	#objects	#presentations	Results
A	6	4 printed, 2 digital replay (normal quality)	60	300 (60x5)	APCER=0%
B	5	4 printed, 1 digital replay (optimized for spoof)	50	250 (50x5)	APCER=0%
C	1	3D printed mask	10	50 (10x5)	APCER=0%
Bona Fide	-	-	10	50 (10x5)	BPCER=0%

The experiments with these PAIs show that the two classes of presentations (bona fide or PA) are clearly distinguishable based on the skin-score. Clear distinction between paper, 3D print and skin is possible using the method according to the present invention.

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the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

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The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides
10 depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.
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The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.
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No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the method may comprise sampling reflection features over several frames in order to reach a more stable classification.
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Figure 3 shows experimental results, in particular density as a function of skin score. On the x-axis the score is shown and on the y-axis the frequency. The score is a measure for the classification quality and has a value range between 0 and 1, wherein 1 indicates very high skin-similarity and 0 very low skin-similarity. A threshold for decision may be around 0.5. A reference distribution of skin-scores for bona fide presentations has been generated using 10 subjects. Skin-scores have also been recorded for presentation attacks (PA) of Level A, Level B and Level C (as defined in the relevant ISO standards). The experimental setup (target of evaluation, ToE) included a proprietary hardware device, e.g. as described in Figure 2, which includes the necessary sensors as well as the computational platform on which the PAD software is executed. The ToE was tested using six species of target of Level A PAIs (presentation attack instruments), five PAI species of Level B, and one PAI species of Level C. For each PAI species, 10 PAIs
30 have been used. The PAI species used in this study are listed in the table below. In the table, APCER is the Attack Presentation Classification Error Rate, which refers to the number of successful attacks/all attacks*100. In the table, BPCER is the Bona Fide Presentation Classification Error Rate, which refers to the number of rejected unlock-attempts/all unlock-attempts*100
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list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.

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3D features determined step c) 120 may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) 116 and c) 120 may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

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Preferably subsequent to steps a) 110, b) 116 and c) 120 the authentication step 122 may be performed. The authentication step 122 may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) 110 no face is detected and/or in step b) 116 the reflection features is determined not to be generated by skin and/or in step c) 120 the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) 116 the detected face is characterized as skin and in step c) 122 the detected face is characterized as 3D object.

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Steps a) to d) may be performed by using at least one device, for example the at least one mobile device¹²⁴ such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

25

The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. The enrolling may comprise a process of registering and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit 114, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order improve

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The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing unit 114 may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera 112 may comprise at least two optical sensors 126, wherein the light-sensitive areas of a first optical sensor 126 and of a second optical sensor 126 may be arranged such that the first optical sensor 126 is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor 126 is adapted to determine the second area of the beam profile of the reflection feature. The processing unit 114 may be adapted to integrate the first area and the second area. The processing unit 114 may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit 114 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

The processing unit 114 may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The processing unit 114 may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.

The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) 120 may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup

difference measure, application of a neural network, application of a machine learning algorithm. The processing unit 114 may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit 114 may filter the beam
5 profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit 114 may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit 114 may be configured
10 for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The processing unit 114 may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding
15 to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit 114 may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO
20 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

The longitudinal coordinate for the reflection feature may be a distance between the optical sensor 126 and the point of the scene remitting the corresponding illumination features. The analysis of the beam profile of one of the reflection features may comprise determining at least one
25 first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The processing unit 114 may be configured for integrating the first area and the second area. The processing unit 114 may be configured to derive a combined signal, in particular a quotient Q, by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of
30 the integrated first area and the integrated second area. The processing unit 114 may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit
35 114 may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit 114 may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit 114 may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The
40 processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit 114 may be configured for determining at least one first area and at least one second area of the beam profile. The area of the beam profile may be an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q.

The skin detection step 116 may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step 116, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit 114 may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates "human skin". The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit. The processing unit 114 is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit 114 may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

The 3D detection step 120 may be performed after the skin detection step 116 and/or the face detection step 110. However, other embodiments are feasible, in which the 3D detection step 120 is performed before the skin detection step 116 and/or the face detection step 110.

The 3D detection step 120 comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The second beam profile information may comprise a quotient Q of areas of the beam profile.

The analysis of the beam profile may comprise evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a

user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit 114 may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

While feature based approaches are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) 116 may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials.

Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step 116 at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The machine-learning may comprise a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The skin classification model may comprise a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The training may comprise a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

machine learning methods which may allow parametrization of a skin classification model. Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

5 The skin detection step 116 may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit 114. Specifically, the processing unit 114 is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. The at least one predetermined or predefined
10 criterion may be at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. Specifically, the processing unit 114 may be configured for skin detection, in particular for identifying if the detected face is human skin. The identification if the material is biological tissue, in particular human skin, may comprise to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in
20 particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or
25 non-living objects.

The processing unit 114 may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. The material property may be at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one
35 material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. The determining at least one material property may comprise assigning the material property to the detected face. The processing unit 114 may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined
40 and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a

The processing unit 114 may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit 114 may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit 114 may be configured for determining the beam profile of the respective reflection feature. The determining the beam profile may comprise identifying at least one reflection feature provided by the optical sensor 126 and/or selecting at least one reflection feature provided by the optical sensor 126 and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may be an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The processing unit 114 is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

The beam profile information may be or may comprise arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b) 116, beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. With respect to feature-based methods reference is made to the description above. The feature based methods may be used in combination with

least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

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The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor 126.

For example, the illumination unit 118 may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor 126, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor 126 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit 114 may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit 114 may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

nation unit 118 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points, for example the illumination unit 118 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit 118 is particularly preferred. The illumination unit 118 may be integrated into a housing of the camera 112 or may be separated from the camera 112.

The illumination pattern comprises at least one illumination feature adapted to illuminate at least one part of the scene. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points. The illumination unit 118 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit 118 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit 118.

The skin detection step 116 comprises determining the at least one second image, also denoted as reflection image, using the camera 112. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c). The reflection feature may be a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. The beam profile of the reflection feature may generally refer to at

a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

- 5 Specifically subsequent to the face detection step 110, the skin detection step 116 may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit 118. However, embodiments are feasible wherein the skin detection step 116 is performed before the face detection step 110.

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The illumination unit 118 may be configured for providing the illumination pattern for illumination of the scene. The illumination unit 118 may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit 118 may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit 118 may be configured for generating an illuminating light beam for illuminating the scene.

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- 20 The illumination unit 118 may comprise at least one light source. The illumination unit 118 may comprise a plurality of light sources. The illumination unit 118 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. The illumination unit 118 may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

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- The illumination unit 118 may be or may comprise at least one multiple beam light source. For example, the illumination unit 118 may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit 118 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumi-

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The face detection step 110 comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit 114. As an example, the at least one processing unit 114 may comprise a software code stored thereon comprising a number of computer commands. The processing unit 114 may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit 114. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit 114 may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit 114 and the camera 112 may fully or partially be integrated into a single device. Thus, generally, the processing unit 114 also may form part of the camera 112. Alternatively, the processing unit 114 and the camera 112 may fully or partially be embodied as separate devices.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The geometrical feature characteristic for faces may be at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit 114 may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying

The camera 112 further may comprise at least one transfer device (not shown here). The camera 112 may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The transfer device may be adapted to guide the light beam onto the optical sensor 126. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The camera 112 is configured for determining at least one image of the scene, in particular the first image. The scene may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. The first image is at least one two-dimensional image having information about transversal coordinates such as the dimensions of height and width.

or alternatively, the mobile device 124 may also refer to a tablet computer or another type of portable computer. An embodiment of a mobile device according to the present invention is shown in Figure 2.

5 Specifically, the camera 112 may be or may comprise at least one optical sensor 126 having at least one light-sensitive area. The optical sensor 126 specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor 126 may be sensitive in the infrared spectral range. The optical sensor 126 may comprise at
10 least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor 126 may be
15 or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such
20 as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor 126 may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an
25 InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge: Au photodiode, a Ge: Hg photodiode, a Ge: Cu photodiode, a Ge: Zn photodiode, a Si: Ga photodiode, a Si: As photodiode. Additionally or alternatively, the optical sensor 126 may comprise at least one photo-
30 conductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

Specifically, the optical sensor 126 may be sensitive in the near infrared region. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor 126,
35 specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor 126 may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any
40 other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit 114, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;

- c) (reference number 120) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit 114, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- d) (reference number 122) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The camera 112 may comprise at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera 112 may be a digital camera. As an example, the camera 112 may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera 112 may be or may comprise at least one near infrared camera. The image may relate to data recorded by using the camera 112, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera 112, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera 112 may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera 112 may also comprise one or more variable lenses which may be adjusted, automatically or manually.

The camera 112 may be a camera of a mobile device 124 such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera 112 may be part of the mobile device 124 which, besides the at least one camera 112, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The mobile device 124 may be a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart phone. Additionally

with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

5

Specifically, in the figures:

Figure 1 shows an embodiment of a method for face authentication according to the present invention;

10 Figure 2 shows an embodiment of a mobile device according to the present invention; and

Figure 3 shows experimental results.

15

Detailed description of the embodiments:

Figure 1 shows a flow chart of a method for face authentication according to the present invention. The face authentication may comprise verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

30

The method comprises the following steps:

- a) (reference number 110) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera 112, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit 114;
- 35
- b) (reference number 116) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit 118 and determining at least one second image using the at least one camera 112, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at
- 40

algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step at least one region specific parametrized skin classification model is used.

- 5 Embodiment 7 The method according to any one of the preceding embodiments, wherein the illumination pattern comprises a periodic grid of laser spots.

Embodiment 8 The method according to any one of the preceding embodiments, wherein the illumination features have wavelengths in a near infrared (NIR) regime.

10

Embodiment 9 The method according to the preceding embodiment, wherein the illumination features have wavelengths of 940 nm.

15

Embodiment 10 The method according to any one of the preceding embodiments, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) and/or for 3D detection in step c).

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Embodiment 11 The method according to any one of the preceding embodiments, wherein the camera is or comprises at least one near infrared camera.

25

Embodiment 12 Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding embodiments, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding embodiments.

30

Embodiment 13 A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding embodiments referring to a method.

35

Embodiment 14 A mobile device comprising at least one camera, at least one illumination unit and at least one processing unit, the mobile device being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding embodiments referring to a method.

40

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination

profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;

- c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

Embodiment 2 The method according to the preceding embodiment, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.

Embodiment 3 The method according to the preceding embodiment, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.

Embodiment 4 The method according to any one of the preceding embodiments, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.

Embodiment 5 The method according to the preceding embodiment, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.

Embodiment 6 The method according to any one of the two preceding embodiments, wherein the skin detection step comprises using at least one 2D face and facial landmark detection

- a computer program comprising program means for performing the method according to one of the embodiments described in this description while the computer program is being executed on a computer or on a computer network,
- a computer program comprising program means according to the preceding embodiment, wherein the program means are stored on a storage medium readable to a computer,
- a storage medium, wherein a data structure is stored on the storage medium and wherein the data structure is adapted to perform the methods according to one of the embodiments described in this description after having been loaded into a main and/or working storage of a computer or of a computer network, and
- a computer program product having program code means, wherein the program code means can be stored or are stored on a storage medium, for performing the method according to one of the embodiments described in this description, if the program code means are executed on a computer or on a computer network.

In a further aspect a mobile device comprising at least one camera, at least one illumination unit and at least one processing unit is disclosed. The mobile device is configured for performing at least steps a) to c), and optionally step d) of the method for face authentication according to the present invention. Step d) may be performed by using at least one authentication unit. The authentication unit may be a unit of the mobile device or may be an external authentication unit.

With respect to definitions and embodiments of the mobile device reference is made to definitions and embodiments described with respect to the method.

In a further aspect of the present invention, use of the method according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use for biometric presentation attack detection.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

- Embodiment 1 Method for face authentication comprising the following steps:
- a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
 - b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam

Thus, specifically, one, more than one or even all of method steps as indicated above may be performed by using a computer or a computer network, preferably by using a computer program.

5

In a further aspect a computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to the present invention.

10 Further disclosed and proposed herein is a data carrier having a data structure stored thereon, which, after loading into a computer or computer network, such as into a working memory or main memory of the computer or computer network, may execute the method according to one or more of the embodiments disclosed herein.

15 Further disclosed and proposed herein is a computer program product with program code means stored on a machine-readable carrier, in order to perform the method according to one or more of the embodiments disclosed herein, when the program is executed on a computer or computer network. As used herein, a computer program product refers to the program as a tradable product. The product may generally exist in an arbitrary format, such as in a paper format, or on a computer-readable data carrier and/or on a computer-readable storage medium. Specifically, the
20 computer program product may be distributed over a data network.

Finally, disclosed and proposed herein is a modulated data signal which contains instructions readable by a computer system or computer network, for performing the method according to one
25 or more of the embodiments disclosed herein.

Referring to the computer-implemented aspects of the invention, one or more of the method steps or even all of the method steps of the methods according to one or more of the embodiments disclosed herein may be performed by using a computer or computer network. Thus, generally,
30 any of the method steps including provision and/or manipulation of data may be performed by using a computer or computer network. Generally, these method steps may include any of the method steps, typically except for method steps requiring manual work.

Specifically, further disclosed herein are:

- 35
- a computer or computer network comprising at least one processor, wherein the processor is adapted to perform the method according to one of the embodiments described in this description,
 - a computer loadable data structure that is adapted to perform the method according to one of the embodiments described in this description while the data structure is being
40 executed on a computer,
 - a computer program, wherein the computer program is adapted to perform the method according to one of the embodiments described in this description while the program is being executed on a computer,

The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides
5 depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.
10

The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the
15 method may comprise sampling reflection features over several frames in order to reach a more stable classification.
20

Besides accuracy, also execution speed and power consumption are important requirements. Further restrictions on the availability of computational resources can be introduced by security
25 considerations. For example, steps a) to d) may have run in a secure zone of the processing unit to avoid any software-based manipulations during program execution. The compact nature of the above described material detection networks may solve this problem by showing excellent runtime behavior in the said secure zone, whereas traditional PAD solutions require the examination of several consecutive frames which results in large computational cost and longer
30 response times of the algorithm.

In a further aspect of the present invention a computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to the present invention, when executed on the computer or the computer network,
35 wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to the present invention. Specifically, the computer program may be stored on a computer-readable data carrier and/or on a computer-readable storage medium.

As used herein, the terms "computer-readable data carrier" and "computer-readable storage
40 medium" specifically may refer to non-transitory data storage means, such as a hard-ware storage medium having stored thereon computer-executable instructions. The computer-readable data carrier or storage medium specifically may be or may comprise a storage medium such as a random-access memory (RAM) and/or a read-only memory (ROM).

3D features determined step c) may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) and c) may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

Preferably subsequent to steps a) to c) the authentication step may be performed. The authentication step may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) no face is detected and/or in step b) the reflection features is determined not to be generated by skin and/or in step c) the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

Steps a) to d) may be performed by using at least one device, for example at least one mobile device such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. As used herein, the term “enrolling” may refer to a process of registering and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

As used herein, the term “longitudinal coordinate for the reflection feature” refers to a distance between the optical sensor and the point of the scene remitting the corresponding illumination features. The processing unit may be configured for using the at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

- 10 The processing unit may be configured for executing at least one depth-from-photon-ratio algorithm which computes distances for all reflection features with zero order and higher order.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

- 15 The processing unit may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. As used herein, the term “depth” or depth information may refer to a distance between the object and the optical sensor and may be given by the longitudinal coordinate. As used herein, the term “depth map” may refer to spatial distribution of depth. The processing unit may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.
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- 25

- The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.
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- 35
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from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring pixels; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The processing unit may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The quotient Q may be a signal which is generated by combining the center signal and the sum signal. Specifically, the determining may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the quotient Q may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

- determining at least one longitudinal coordinate z of the object by evaluating the quotient Q .

As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the display device, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. The term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the pixel having the highest sensor signal out of the plurality of sensor signals generated by the pixels of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the pixels of the matrix. The center signal may arise from a single pixel or from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of pixels may be added up, integrated or averaged, in order to determine the center signal. The group of pixels from which the center signal arises may be a group of neighboring pixels, such as pixels having less than a predetermined distance from the actual pixel having the highest sensor signal, or may be a group of pixels generating sensor signals being within a predetermined range from the highest sensor signal. The group of pixels from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The processing unit may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of pixels around the pixel having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the pixel in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from pixels in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of sensor signals

sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

- 5 In one embodiment, the illumination pattern may comprise at least point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 10 0.6 of the full radius.

The illumination pattern may comprise both at least one point pattern and at least one line pattern. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

- 15 The processing unit may be configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.

- 20 The processing unit may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The processing unit may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

25

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the beam profile, respectively, and E(x,y) denotes the beam profile.

- 30 Additionally or alternatively, the processing unit may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

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For example, in case of the optical sensor having a matrix of pixels, the processing unit may be configured for evaluating the beam profile, by

- determining the pixel having the highest sensor signal and forming at least one center signal;
- 40 - evaluating sensor signals of the matrix and forming at least one sum signal;
- determining the quotient Q by combining the center signal and the sum signal; and

and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as
5 first region.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional
10 beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area
15 of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The processing unit may be configured for determining an area integral of the beam profile. The processing unit may be configured for determining the edge information by integrating and/or summing of the first area. The processing unit may be configured for determining the center information by integrating and/or summing of the
20 second area. For example, the beam profile may be a trapezoid beam profile and the processing unit may be configured for determining an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and
25 deriving edge and center signals by geometric considerations.

In one embodiment, A1 may correspond to a full or complete area of a feature point on the optical sensor. A2 may be a central area of the feature point on the optical sensor. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius
30 from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In one embodiment, the illumination pattern may comprise at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on
35 the light sensitive area of the optical sensors. The line pattern on the optical sensor may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensor. The line width of the
40 central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical

may be configured for comparing at least two of the determined integrals. Specifically, the processing unit may be configured for determining at least one first area and at least one second area of the beam profile. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing unit may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera may comprise at least two optical sensors, wherein the light-sensitive areas of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the beam profile of the reflection feature. The processing unit may be adapted to integrate the first area and the second area. The processing unit may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined

step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The processing unit may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

The processing unit may be configured for determining at least one first area and at least one second area of the reflection beam profile of each of the reflection features and/or of the reflection features in at least one region of interest. The processing unit is configured for integrating the first area and the second area.

The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The processing unit may be configured for integrating the first area and the second area. The processing unit may be configured to derive a combined signal, in particular a quotient Q, by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The processing unit may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit

For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates "human skin". The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit. The processing unit is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

The 3D detection step may be performed after the skin detection step and/or the face detection step. However, other embodiments are feasible, in which the 3D detection step is performed before the skin detection step and/or the face detection step. The material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z in step d) such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} .

The 3D detection step comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The second beam profile information may comprise a quotient Q of areas of the beam profile.

As used herein, the term "analysis of the beam profile" may generally refer to evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis

While feature based approaches, as the approaches described above, are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The term “machine-learning” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The term “skin classification model” may refer to a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The term “training”, also denoted learning, as used herein, is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face.

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

- 5 The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

10

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

- For example, the material dependent image filter may be a threshold area filter. This material
 15 feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the
 function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein
 the function f is smaller than α times the maximum of f , but larger than a threshold value ε times
 the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the ar-
 20 eas may not simply correspond to an inner and an outer circle around the spot center. As an ex-
 ample, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material fea-
 ture may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

- 25 The processing unit may be configured for using at least one predetermined relationship be-
 tween the material feature φ_{2m} and the material property of the surface having generated the re-
 flection feature for determining the material property of the surface having generated the reflec-
 tion feature. The predetermined relationship may be one or more of an empirical relationship, a
 semi-empiric relationship and an analytically derived relationship. The processing unit may com-
 30 prise at least one data storage device for storing the predetermined relationship, such as a
 lookup list or a lookup table.

be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

5 The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1, g_2) = [p_{g_1, g_2}]$, whereas p_{g_1, g_2} is the occurrence rate of the grey combination $(g_1, g_2) = [f(x_1, y_1), f(x_2, y_2)]$, and the relation ρ defines the distance between (x_1, y_1) and (x_2, y_2) , which is $\rho(x, y) = (x+a, y+b)$ with a and b selected from 0,1.

The material feature of the grey-level-occurrence-based contrast filter may be given by

15

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

20 The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

25

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may

equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k .

The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term “not distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical. As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J - 1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1 - t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1 - t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or

by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter $\Phi_{2\text{other}}$ may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term "passes a hypothesis testing" refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term "beam profile image" refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k :

$(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values

tissue in case the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise determining a deviation, e.g. a sum of squared point to point distances, between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The processing unit may be adapted to compare the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit.

Additionally or alternatively, the first beam profile information may be determined by applying at least one image filter to the image of the area. As further used herein, the term “image” refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a material feature. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The processing unit may be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the surface of the scene having generated the reflection feature.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof

may refer to assigning the material property to respective reflection feature, in particular to the detected face. The processing unit may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

The reflection properties of skin may be characterized by the simultaneous occurrence of direct reflection at the surface (Lambertian-like) and subsurface scattering (volume scattering). This leads to a broadening of the laser spot on skin compared to the above-mentioned materials.

The first beam profile information may be a reflection profile. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering profile.

Specifically, the processing unit may be configured for comparing the reflection beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a device executing the method according to the present invention. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the processing unit or the device, e.g. by software, specifically by the app downloaded from an app store or the like. The reflection feature may be identified as to be generated by biological

application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. The feature-based methods may be explained in the following. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model. Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

The skin detection step may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit. Specifically, the processing unit is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. As used herein, the term “at least one predetermined or predefined criterion” refers to at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. As used herein, the term “indicate” refers to an arbitrary indication such as an electronic signal and/or at least one visual or acoustic indication. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. As used herein, the term “biological tissue” generally refers to biological material comprising living cells. Specifically, the processing unit may be configured for skin detection. The term “identification” of being generated by biological tissue, in particular human skin, may refer to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “determining at least one material property”

one reflection feature provided by the optical sensor and/or selecting at least one reflection feature provided by the optical sensor and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may be an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The reflection feature may cover or may extend over at least one pixel of the second image. For example, the reflection feature may cover or may extend over plurality of pixels. The processing unit may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The processing unit may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum_j r_{pixel}},$$

wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum_j l_{total}$ with j being the number of pixels j connected to and/or belonging to the reflection feature and l_{total} being the total intensity.

The processing unit is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

As used herein, the term “beam profile information” may refer to arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b), beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP

image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor.

For example, the illumination unit may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit may be configured for determining the beam profile of the respective reflection feature. As used herein, the term "determining the beam profile" refers to identifying at least

from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit may be adapted to generate and/or to project a cloud of points. The illumination unit may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit.

A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination unit may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination unit comprises at least one laser source and/or at least one laser diode which is designated for generating laser radiation. The illumination unit may comprise the at least one diffractive optical element (DOE). The illumination unit may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one periodic point pattern. As further used herein, the term “projecting at least one illumination pattern” may refer to providing the at least one illumination pattern for illuminating the at least one scene.

The skin detection step comprises determining the at least one second image, also denoted as reflection image, using the camera. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c).

The second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features. As used herein, the term “reflection feature” may refer to a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. As used herein, the term “beam profile” of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or

non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit may be adapted to generate and/or to project a cloud of points, for example the illumination unit may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit is particularly preferred. The illumination unit may be integrated into a housing of the camera or may be separated from the camera.

Further, the illumination unit may be configured for emitting modulated or non-modulated light. In case a plurality of illumination units is used, the different illumination units may have different modulation frequencies which later on may be used for distinguishing the light beams.

The light beam or light beams generated by the illumination unit generally may propagate parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The illumination unit may be configured such that the light beam or light beams propagate from the illumination unit towards the scene along an optical axis of the illumination unit and/or of the camera. For this purpose, the illumination unit and/or the camera may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the light beam or light beams, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the optical axis or off the optical axis. As an example, the light beam or light beams may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the scene. As used herein, the term "illumination feature" refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected

semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination unit may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used.

The illumination unit may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime.

The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination unit may be configured for emitting light at a single wavelength. In other embodiments, the illumination unit may be configured for emitting light with a plurality of wavelengths allowing additional measurements in other wavelengths channels.

As used herein, the term "ray" generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term "beam" generally refers to a collection of rays. In the following, the terms "ray" and "beam" will be used as synonyms. As further used herein, the term "light beam" generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however.

The illumination unit may be or may comprise at least one multiple beam light source. For example, the illumination unit may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively,

lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step, the skin detection step may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit. However, embodiments are feasible wherein the skin detection step is performed before the face detection step.

As used herein, the term "illumination unit", also denoted as illumination source, may generally refers to at least one arbitrary device configured for generating at least one illumination pattern. The illumination unit may be configured for providing the illumination pattern for illumination of the scene. The illumination unit may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit may comprise at least one light source. The illumination unit may comprise a plurality of light sources. The illumination unit may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one

The processing unit may be configured to one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the camera. The processing unit, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAPTTM), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The processing unit may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The processing unit may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The processing unit may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The processing unit may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The term "geometrical feature characteristic for faces" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a

the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to “pixels”, reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. The first image is at least one two-dimensional image. As used herein, the term “two dimensional image” may generally refer to an image having information about transversal coordinates such as the dimensions of height and width. The first image may be an RGB (red green blue) image. The term “determining at least one first image” may refer to capturing and/or recording the first image.

The face detection step comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit.

As further used herein, the term “processing unit” generally refers to an arbitrary data processing device adapted to perform the named operations such as by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one processing unit may comprise a software code stored thereon comprising a number of computer commands. The processing unit may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit and the camera may fully or partially be integrated into a single device. Thus, generally, the processing unit also may form part of the camera. Alternatively, the processing unit and the camera may fully or partially be embodied as separate devices.

The processing unit may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the processing unit may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the processing unit may comprise one or more data storage devices. Further, the processing unit may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the camera may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. In case of a plurality of optical sensors, the optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The camera is configured for determining at least one image of the scene, in particular the first image. As used herein, the term “scene” may refer to a spatial region. The scene may comprise

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable.

In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

The camera further may comprise at least one transfer device. The camera may further comprise one or more additional elements such as one or more additional optical elements. The camera may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensor. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a

ment selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor may be part of or constitute a pixelated optical device. For example, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the camera form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the camera specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. The optical sensor may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor each, independently, may be or may comprise at least one ele-

The term “camera” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a device having at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera may be a digital camera. As an example, the camera may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera may be or may comprise at least one near infrared camera.

As used herein, without limitation, the term “image” specifically may relate to data recorded by using a camera, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses.

As an example, the camera may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera may also comprise one or more variable lenses which may be adjusted, automatically or manually.

The camera may be a camera of a mobile device such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera may be part of a mobile device which, besides the at least one camera, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The term “mobile device” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart phone. Additionally or alternatively, the mobile device may also refer to a tablet computer or another type of portable computer.

Specifically, the camera may be or may comprise at least one optical sensor having at least one light-sensitive area. As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The camera may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal. In case the camera comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area

as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

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As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

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Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

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Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

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In a first aspect of the present invention a method for face authentication is disclosed. The face to be authenticated may be, specifically, a human face. The term “face authentication” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such

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clude displays as potential spoof objects, near infra-red (NIR) cameras can be used, since displays emit photons only in the visible regime of the electromagnetic spectrum. As a further counteraction, 3D cameras were introduced, which can clearly distinguish between planar photos or tablets with playback videos and a 3D face. Still, these systems can be attacked with high quality masks generated, for example, by 3D printing, careful 3D-arrangement of 2D photos, or hand-crafted silicone or latex masks to name a few approaches. Since masks are typically worn by humans, liveness detection based on small motions fails as well. However, these types of masks can be rejected by a PAD system that is able to classify human skin from other materials such as described in EP application No. 20159984.2 filed on February 28, 2020 and EP application No. 20 154 961.5190679 filed on January 31, 2020, the full content of which is included by reference.

A further problem might arise from differences in skin optical properties with respect ethnic origin. A reliable recognition and PAD technology needs to be fully agnostic with respect to these different origins.

Besides security considerations, also the speed of the unlock process and the required computational power are important to provide an acceptable user experience. High-speed face recognition can be used for several tasks after the successful unlock of a device, for example to check whether the user is still in front of the display or to launch further secure applications like banking apps by performing an on the fly check of the person in front of the display. Again, speed and computational resources have great impact on the user experience.

Current 3D algorithms are computationally very demanding and presentation attack detections require several video frames to be processed. Therefore, expensive hardware needs to be involved to deliver acceptable unlock performance. Furthermore, high power consumption is a consequence.

Summarizing, current methods for face unlock do not provide the ability to reliably detect spoof attacks by 3D masks and perform this task at a speed below human detection limit.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Although simple presentation attacks with photographs and videos of the legitimate face can be detected, an approach to reliably detect presentation attacks with 3D face masks is still missing. Specifically, another layer of security is needed to enable the replacement of passphrases/PIN codes by biometric features derived from faces to unlock digital devices. Moreover, a method is needed that operates fully agnostic with respect to different skin types originating from different ethnic origins.

Summary of the invention

Optical Skin Detection for Face Unlock

Description

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Field of the invention

The invention relates to a method for face authentication, a mobile device and various uses of the method. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

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Prior art

In today's digital world secure access to information technology is an essential requirement of any state-of-the-art system. Standard concepts like passphrases or PIN codes are currently extended, combined or even replaced by biometric methods like fingerprint or face recognition. Although passphrases can provide a high level of security if chosen carefully and at adequate length, this step requires attention and the need of memorizing several potentially long phrases depending on the IT landscape. Furthermore, it is never guaranteed that the person providing the passphrase is authorized by the owner of the passphrase or the digital device. In contrast biometric features like faces or fingerprints are unique and person-specific properties. Therefore, using features derived from these not only can be more convenient than passphrases/PIN codes but also more secure since they combine personal identity with an unlock process.

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Unfortunately, similarly to being able to spy passwords, fingerprints and faces can be also artificially created in order to impersonate a legitimate user. Automatic face recognition tools of the first generation use digital camera images or image streams and apply 2D image processing methods to extract characteristic features, while machine learning techniques are used to generate face templates to perform identity recognition based on those features. The second generation of face recognition algorithms uses deep convolutional neuronal networks instead of hand-crafted image features to generate the classification model.

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However, both approaches can be attacked, e.g., with a high-quality photograph that is nowadays freely available to be downloaded from the Internet. Therefore, the idea of presentation attack detection (PAD) gained significance. Early approaches were designed to protect against simple attacks like presenting the photograph of a legitimate user by recording a sequence of images and test for time dependent features, for example small but natural changes in the head position or eye blinking. These methods can again be tricked by playing a pre-recorded video of the user or by careful animations generated from publicly available photographs. In order to ex-

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Optical Skin Detection for Face Unlock

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For the President of the European Patent Office

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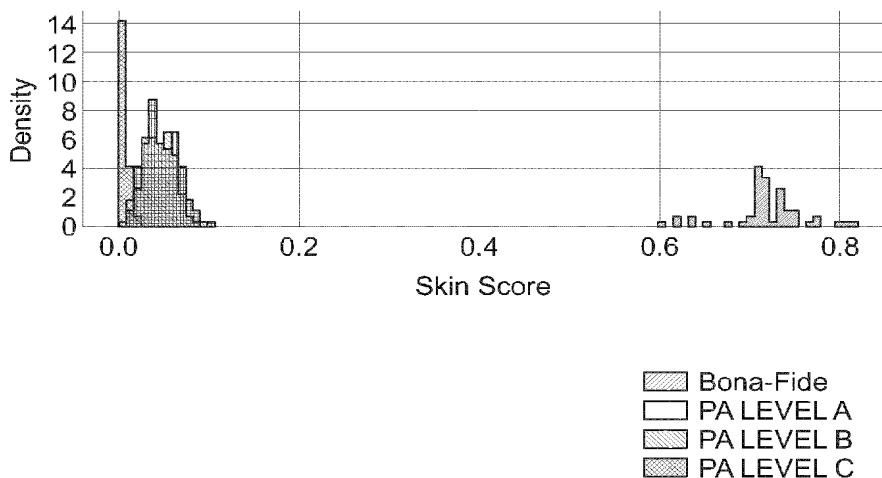
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(54) Title: OPTICAL SKIN DETECTION FOR FACE UNLOCK

FIG.3



(57) **Abstract:** A method for face authentication is proposed. The method comprising the following steps: a) at least one face detection step (110) comprising determining at least one first image by using at least one camera (112); b) at least one skin detection step (116) comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit (118), determining at least one second image using the at least one camera (112), determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile, and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit (114), wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin; c) at least one

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3D detection step (120) comprising determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit (114), wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects; d) at least one authentication step (122) comprising authenticating the detected face by using at least one authentication unit if in step b) (116) the detected face is characterized as skin and in step c) (120) the detected face is characterized as 3D object.

Optical Skin Detection for Face Unlock

Description

Field of the invention

The invention relates to a method for face authentication, a mobile device and various uses of the method. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

In today's digital world secure access to information technology is an essential requirement of any state-of-the-art system. Standard concepts like passphrases or PIN codes are currently extended, combined or even replaced by biometric methods like fingerprint or face recognition. Although passphrases can provide a high level of security if chosen carefully and at adequate length, this step requires attention and the need of memorizing several potentially long phrases depending on the IT landscape. Furthermore, it is never guaranteed that the person providing the passphrase is authorized by the owner of the passphrase or the digital device. In contrast biometric features like faces or fingerprints are unique and person-specific properties. Therefore, using features derived from these not only can be more convenient than passphrases/PIN codes but also more secure since they combine personal identity with an unlock process.

Unfortunately, similarly to being able to spy passwords, fingerprints and faces can be also artificially created in order to impersonate a legitimate user. Automatic face recognition tools of the first generation use digital camera images or image streams and apply 2D image processing methods to extract characteristic features, while machine learning techniques are used to generate face templates to perform identity recognition based on those features. The second generation of face recognition algorithms uses deep convolutional neuronal networks instead of hand-crafted image features to generate the classification model.

However, both approaches can be attacked, e.g., with a high-quality photograph that is nowadays freely available to be downloaded from the Internet. Therefore, the idea of presentation attack detection (PAD) gained significance. Early approaches were designed to protect against simple attacks like presenting the photograph of a legitimate user by recording a sequence of images and test for time dependent features, for example small but natural changes in the head position or eye blinking. These methods can again be tricked by playing a pre-recorded video of the user or by careful animations generated from publicly available photographs. In order to ex-

clude displays as potential spoof objects, near infra-red (NIR) cameras can be used, since displays emit photons only in the visible regime of the electromagnetic spectrum. As a further counteraction, 3D cameras were introduced, which can clearly distinguish between planar photos or tablets with playback videos and a 3D face. Still, these systems can be attacked with high quality masks generated, for example, by 3D printing, careful 3D-arrangement of 2D photos, or hand-crafted silicone or latex masks to name a few approaches. Since masks are typically worn by humans, liveness detection based on small motions fails as well. However, these types of masks can be rejected by a PAD system that is able to classify human skin from other materials such as described in EP application No. 20159984.2 filed on February 28, 2020 and EP application No. 20 154 961.5190679 filed on January 31, 2020, the full content of which is included by reference.

A further problem might arise from differences in skin optical properties with respect ethnic origin. A reliable recognition and PAD technology needs to be fully agnostic with respect to these different origins.

Besides security considerations, also the speed of the unlock process and the required computational power are important to provide an acceptable user experience. High-speed face recognition can be used for several tasks after the successful unlock of a device, for example to check whether the user is still in front of the display or to launch further secure applications like banking apps by performing an on the fly check of the person in front of the display. Again, speed and computational resources have great impact on the user experience.

Current 3D algorithms are computationally very demanding and presentation attack detections require several video frames to be processed. Therefore, expensive hardware needs to be involved to deliver acceptable unlock performance. Furthermore, high power consumption is a consequence.

Summarizing, current methods for face unlock do not provide the ability to reliably detect spoof attacks by 3D masks and perform this task at a speed below human detection limit.

US 2019/213309 A1 describes a system and method of authenticating a user's face with a ranging sensor. The ranging sensor includes a time of flight sensor and a reflectance sensor. The ranging sensor transmits a signal that is reflected off of a user and received back at the ranging sensor. The received signal can be used to determine distance between the user and the sensor, and the reflectance value of the user. With the distance or the reflectivity, a processor can activate a facial recognition process in response to the distance and the reflectivity.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Although simple presentation attacks with photographs and videos of the legitimate face can be detected, an approach

to reliably detect presentation attacks with 3D face masks is still missing. Specifically, another layer of security is needed to enable the replacement of passphrases/PIN codes by biometric features derived from faces to unlock digital devices. Moreover, a method is needed that operates fully agnostic with respect to different skin types originating from different ethnic origins.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a method for face authentication is disclosed. The face to be authenticated may be, specifically, a human face. The term “face authentication” as used

herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;

- d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

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The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

- 10 The term “camera” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a device having at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera may be a digital
15 camera. As an example, the camera may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera may be or may comprise at least one near infrared camera.

- As used herein, without limitation, the term “image” specifically may relate to data recorded by
20 using a camera, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera may also comprise one
25 or more variable lenses which may be adjusted, automatically or manually.

- The camera may be a camera of a mobile device such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera may be part of a mobile device which, besides the at least one camera, comprises one or more data
30 processing devices such as one or more data processors. Other cameras, however, are feasible. The term “mobile device” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart
35 phone. Additionally or alternatively, the mobile device may also refer to a tablet computer or another type of portable computer.

- Specifically, the camera may be or may comprise at least one optical sensor having at least one light-sensitive area. As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers
40 to an area of the optical sensor which may be illuminated externally, by the at least one light

beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The camera may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal. In case the camera comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the camera specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. The optical sensor may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor may be part of or constitute a pixelated optical device. For example, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the camera form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

The camera further may comprise at least one transfer device. The camera may further comprise one or more additional elements such as one or more additional optical elements. The camera may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical

sensor. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. As used herein, the term "focal length" of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a "focus" which may also be denoted as "focal point". Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. As used herein, the term "optical axis of the transfer device" generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the camera may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. In case of a plurality of optical sensors, the optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The camera is configured for determining at least one image of the scene, in particular the first image. As used herein, the term "scene" may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to "pixels", reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. The first image is at least one two-dimensional image. As used herein, the term "two dimensional image" may generally refer to an image having information about transversal coordinates such as the dimensions of height and width. The first image may be an RGB (red green blue) image. The term "determining at least one first image" may refer to capturing and/or recording the first image.

The face detection step comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit. Specifically, the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature which is characteristic for faces by using the at least one processing unit.

As further used herein, the term "processing unit" generally refers to an arbitrary data processing device adapted to perform the named operations such as by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one processing unit may comprise a software code stored thereon comprising a number of computer commands. The processing unit may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit and the camera may fully or partially be integrated into a single device. Thus, generally, the processing unit also may form

part of the camera. Alternatively, the processing unit and the camera may fully or partially be embodied as separate devices.

The processing unit may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the processing unit may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the processing unit may comprise one or more data storage devices. Further, the processing unit may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The processing unit may be configured to one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the camera. The processing unit, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The processing unit may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The processing unit may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The processing unit may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The processing unit may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial
5 connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The term "geometrical feature
10 characteristic for faces" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit may comprise at least
15 one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE,
20 2018, the full content of which is included by reference.

The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region
25 of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying
30 a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step, the skin detection step may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit. However, embodiments are feasible wherein the skin detection step is performed before the face detection step.

40

As used herein, the term “illumination unit”, also denoted as illumination source, may generally refers to at least one arbitrary device configured for generating at least one illumination pattern. The illumination unit may be configured for providing the illumination pattern for illumination of the scene. The illumination unit may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit may comprise at least one light source. The illumination unit may comprise a plurality of light sources. The illumination unit may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination unit may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used.

The illumination unit may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination unit may be configured for emitting light at a single wavelength. In other embodiments, the illumination unit may be configured for emitting light with a plurality of wavelengths allowing additional measurements in other wavelengths channels.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one

edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however.

5 The illumination unit may be or may comprise at least one multiple beam light source. For example, the illumination unit may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure la-
10 sers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively,
15 non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit may be adapted to generate and/or to project a cloud of points, for example the illumination unit may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least
20 one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit is particularly preferred. The illumination unit may be integrated into a housing of the camera or may be separated from the camera.

25 Further, the illumination unit may be configured for emitting modulated or non-modulated light. In case a plurality of illumination units is used, the different illumination units may have different modulation frequencies which later on may be used for distinguishing the light beams.

30 The light beam or light beams generated by the illumination unit generally may propagate parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The illumination unit may be configured such that the light beam or light beams propagates from the illumination unit towards the scene along an optical axis of the illumination unit and/or of the camera. For this purpose, the illumination unit and/or the camera may comprise at
35 least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the light beam or light beams, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the optical axis or off the optical axis. As an example, the light beam or light
40 beams may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

As used herein, the term “at least one illumination pattern” refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the scene. As used herein, the term “illumination feature” refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit may be adapted to generate and/or to project a cloud of points. The illumination unit may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit.

A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination unit may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination unit comprises at least one laser source and/or at least one laser diode which is designated for generating laser radiation. The illumination unit may comprise the at least one diffractive optical element (DOE). The illumination unit may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one periodic point pattern. As further used herein, the term “projecting at least one illumination pattern” may refer to providing the at least one illumination pattern for illuminating the at least one scene.

The skin detection step comprises determining the at least one second image, also denoted as reflection image, using the camera. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c).

The second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features. As used herein, the term “reflection feature” may refer to a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. As used herein, the term “beam profile” of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor.

For example, the illumination unit may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing

unit may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit may be configured for determining the beam profile of the respective reflection feature. As used herein, the term “determining the beam profile” refers to identifying at least one reflection feature provided by the optical sensor and/or selecting at least one reflection feature provided by the optical sensor and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may be an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The reflection feature may cover or may extend over at least one pixel of the second image. For example, the reflection feature may cover or may extend over plurality of pixels. The processing unit may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The processing unit may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum_j r_{pixel}},$$

wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum_j I_{total}$ with j being the number of pixels j connected to and/or belonging to the reflection feature and I_{total} being the total intensity.

The processing unit is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image

and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

5 As used herein, the term “beam profile information” may refer to arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b), beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto
10 object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP
15 application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. The feature-based methods may be explained in the following. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model. Alternatively
20 or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

Other methods for authenticating a user’s face are known, such as from US 2019/213309 A1. However, these methods use time of flight (ToF) sensors. A well-known working principle of a
25 ToF sensor is sending out light and measuring the time span until receiving the reflected light. In contrast, the proposed beam profile analysis uses a projected illumination pattern. For ToF sensor using such a projected pattern is not possible. Using illumination pattern may be advantageous e.g. in view of covering and therefore allowing to take into account different locations on the face. This may enhance reliability and security of the authentication of the user’s face.

30 The skin detection step may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit. Specifically, the processing unit is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one
35 predetermined or predefined criterion. As used herein, the term “at least one predetermined or predefined criterion” refers to at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated
40 as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. As used herein, the term “indicate” refers to an arbitrary indication such as an electronic signal and/or at least one visual or acoustic indication. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. As

used herein, the term “biological tissue” generally refers to biological material comprising living cells. Specifically, the processing unit may be configured for skin detection. The term “identification” of being generated by biological tissue, in particular human skin, may refer to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “determining at least one material property” may refer to assigning the material property to respective reflection feature, in particular to the detected face. The processing unit may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

The reflection properties of skin may be characterized by the simultaneous occurrence of direct reflection at the surface (Lambertian-like) and subsurface scattering (volume scattering). This leads to a broadening of the laser spot on skin compared to the above-mentioned materials.

The first beam profile information may be a reflection profile. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering

profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering profile.

Specifically, the processing unit may be configured for comparing the reflection beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a device executing the method according to the present invention. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the processing unit or the device, e.g. by software, specifically by the app downloaded from an app store or the like. The reflection feature may be identified as to be generated by biological tissue in case the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise determining a deviation, e.g. a sum of squared point to point distances, between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The processing unit may be adapted to compare the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit.

Additionally or alternatively, the first beam profile information may be determined by applying at least one image filter to the image of the area. As further used herein, the term “image” refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a material feature. Images may be

subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

- 5 The processing unit may be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the surface of the scene having generated the reflection feature.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

- 30 The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used
- 35 herein, the term “beam profile image” refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

- 40 wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k : $(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond

to a material classifier and a distance. The material classifier may be a label such as ‘Material A’, ‘Material B’, etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

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The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

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Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term “not distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical. As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis

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testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis

5 testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- 10
- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
 - $F(x) = 1 - CDF(x)$
 - $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete

15 Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can

20 be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However,

25 other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

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$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

- 10 For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

- 15 wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

- 20 For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

- 25 For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

- 30 For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse

scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1 g_2) = [p_{g_1, g_2}]$, whereas p_{g_1, g_2} is the occurrence rate of the grey combination $(g_1, g_2) = [f(x_1, y_1), f(x_2, y_2)]$, and the relation ρ defines the distance between (x_1, y_1) and (x_2, y_2) , which is $\rho(x, y) = (x+a, y+b)$ with a and b selected from 0,1.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i - j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

and

$$\begin{aligned} f_{k,E_5L_5}^*(x,y) &= \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T. \\ E &= \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy, \\ F &= \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy, \end{aligned}$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ϵ times

the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The processing unit may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the surface having generated the reflection feature for determining the material property of the surface having generated the reflection feature. The predetermined relationship may be one or more of an empirical relationship, a semi-empirical relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

While feature based approaches, as the approaches described above, are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The term "machine-learning" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The term "skin classification model" may refer to a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For

example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The term “training”, also denoted learning, as used herein, is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates “human skin”. The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit. The processing unit is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

The 3D detection step may be performed after the skin detection step and/or the face detection step. However, other embodiments are feasible, in which the 3D detection step is performed before the skin detection step and/or the face detection step. The material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z in step d) such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} .

The 3D detection step comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The 3D detection step may comprise determining a second beam profile information of at least four of the reflection features by analysis of their respective beam profile. The second beam profile information may comprise a quotient Q of areas of the beam profile.

As used herein, the term “analysis of the beam profile” may generally refer to evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The processing unit may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

The processing unit may be configured for determining at least one first area and at least one second area of the reflection beam profile of each of the reflection features and/or of the reflection features in at least one region of interest. The processing unit is configured for integrating the first area and the second area.

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The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The processing unit may be configured for integrating the first area and the second area. The processing unit may be configured to derive a combined signal, in particular a quotient Q , by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The processing unit may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit may be configured for determining at least one first area and at least one second area of the beam profile. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing unit may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera may comprise at least two optical sensors, wherein the light-sensitive areas of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the beam profile of the reflection feature. The processing unit may be adapted to integrate the first area and the second area. The processing unit may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

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The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The processing unit may be configured for determining an area integral of the beam profile. The processing unit may be configured for determining the edge information by integrating and/or summing of the first area. The processing unit may be configured for determining the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the processing unit may be configured for determining an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as

determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 5 In one embodiment, A1 may correspond to a full or complete area of a feature point on the optical sensor. A2 may be a central area of the feature point on the optical sensor. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.
- 10 In one embodiment, the illumination pattern may comprise at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensor may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the
- 15 line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensor. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width,
- 20 preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.
- 25 In one embodiment, the illumination pattern may comprise at least point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to
- 30 0.6 of the full radius.
- The illumination pattern may comprise both at least one point pattern and at least one line pattern. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.
- 35 The processing unit may be configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.
- 40 The processing unit may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The processing unit may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the beam profile, respectively, and E(x,y) denotes the beam profile.

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Additionally or alternatively, the processing unit may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

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For example, in case of the optical sensor having a matrix of pixels, the processing unit may be configured for evaluating the beam profile, by

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- determining the pixel having the highest sensor signal and forming at least one center signal;
- evaluating sensor signals of the matrix and forming at least one sum signal;
- determining the quotient Q by combining the center signal and the sum signal; and
- determining at least one longitudinal coordinate z of the object by evaluating the quotient

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Q.

As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the display device, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. The term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the pixel having the highest sensor signal out of the plurality of sensor signals generated by the pixels of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the pixels of the matrix. The center signal may arise from a single pixel or from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of pixels may be added up, integrated or averaged, in order to determine the center signal. The group of pixels from which the center signal arises may be a group of neighboring pixels, such as pixels having less than a predetermined distance from the actual pixel having the

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highest sensor signal, or may be a group of pixels generating sensor signals being within a predetermined range from the highest sensor signal. The group of pixels from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The processing unit may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of pixels around the pixel having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the pixel in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from pixels in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring pixels; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The processing unit may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of

the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 5 Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge
10 information in the center signal, the edge signal may further be subtracted from the center signal.

The quotient Q may be a signal which is generated by combining the center signal and the sum signal. Specifically, the determining may include one or more of: forming a quotient of the center
15 signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the quotient Q may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

20 As used herein, the term “longitudinal coordinate for the reflection feature” refers to a distance between the optical sensor and the point of the scene remitting the corresponding illumination features. The processing unit may be configured for using the at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal
25 coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

30 The processing unit may be configured for executing at least one depth-from-photon-ratio algorithm which computes distances for all reflection features with zero order and higher order.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

35 The processing unit may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. As used herein, the term “depth” or depth information may
40 refer to a distance between the object and the optical sensor and may be given by the longitudinal coordinate. As used herein, the term “depth map” may refer to spatial distribution of depth. The processing unit may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light,

beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.

- 5 The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature.
- 10 The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup list or
- 15 a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.
- 20 3D features determined step c) may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) and c) may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both
- 25 properties can be calculated from the same frame with low computational effort.

Preferably subsequent to steps a) to c) the authentication step may be performed. The authentication step may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) no face is detected and/or in step b) the reflection features is determined not to be generated by skin and/or in step c) the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

35 Steps a) to d) may be performed by using at least one device, for example at least one mobile device such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

40 The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. As used herein, the term "enrolling" may refer to a process of registering

and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device.

5 The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a
10 depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving
15 the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

20 The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides
25 depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.
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The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

35 No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the
40 method may comprise sampling reflection features over several frames in order to reach a more stable classification.

Besides accuracy, also execution speed and power consumption are important requirements. Further restrictions on the availability of computational resources can be introduced by security considerations. For example, steps a) to d) may have run in a secure zone of the processing unit to avoid any software-based manipulations during program execution. The compact nature of the above described material detection networks may solve this problem by showing excellent runtime behavior in the said secure zone, whereas traditional PAD solutions require the examination of several consecutive frames which results in large computational cost and longer response times of the algorithm.

In a further aspect of the present invention a computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to the present invention, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to the present invention. Specifically, the computer program may be stored on a computer-readable data carrier and/or on a computer-readable storage medium.

As used herein, the terms “computer-readable data carrier” and “computer-readable storage medium” specifically may refer to non-transitory data storage means, such as a hard-ware storage medium having stored thereon computer-executable instructions. The computer-readable data carrier or storage medium specifically may be or may comprise a storage medium such as a random-access memory (RAM) and/or a read-only memory (ROM).

Thus, specifically, one, more than one or even all of method steps as indicated above may be performed by using a computer or a computer network, preferably by using a computer program.

In a further aspect a computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to the present invention.

Further disclosed and proposed herein is a data carrier having a data structure stored thereon, which, after loading into a computer or computer network, such as into a working memory or main memory of the computer or computer network, may execute the method according to one or more of the embodiments disclosed herein.

Further disclosed and proposed herein is a computer program product with program code means stored on a machine-readable carrier, in order to perform the method according to one or more of the embodiments disclosed herein, when the program is executed on a computer or computer network. As used herein, a computer program product refers to the program as a tradable product.

The product may generally exist in an arbitrary format, such as in a paper format, or on a computer-readable data carrier and/or on a computer-readable storage medium. Specifically, the computer program product may be distributed over a data network.

Finally, disclosed and proposed herein is a modulated data signal which contains instructions readable by a computer system or computer network, for performing the method according to one or more of the embodiments disclosed herein.

5 Referring to the computer-implemented aspects of the invention, one or more of the method steps or even all of the method steps of the methods according to one or more of the embodiments disclosed herein may be performed by using a computer or computer network. Thus, generally, any of the method steps including provision and/or manipulation of data may be performed by using a computer or computer network. Generally, these method steps may include any of the
10 method steps, typically except for method steps requiring manual work.

Specifically, further disclosed herein are:

- 15 - a computer or computer network comprising at least one processor, wherein the processor is adapted to perform the method according to one of the embodiments described in this description,
- a computer loadable data structure that is adapted to perform the method according to one of the embodiments described in this description while the data structure is being executed on a computer,
- 20 - a computer program, wherein the computer program is adapted to perform the method according to one of the embodiments described in this description while the program is being executed on a computer,
- a computer program comprising program means for performing the method according to one of the embodiments described in this description while the computer program is being executed on a computer or on a computer network,
- 25 - a computer program comprising program means according to the preceding embodiment, wherein the program means are stored on a storage medium readable to a computer,
- a storage medium, wherein a data structure is stored on the storage medium and wherein the data structure is adapted to perform the methods according to one of the embodiments described in this description after having been loaded into a main and/or working storage
30 of a computer or of a computer network, and
- a computer program product having program code means, wherein the program code means can be stored or are stored on a storage medium, for performing the method according to one of the embodiments described in this description, if the program code means are executed on a computer or on a computer network.

35 In a further aspect a mobile device comprising at least one camera, at least one illumination unit and at least one processing unit is disclosed. The mobile device is configured for performing at least steps a) to c), and optionally step d) of the method for face authentication according to the present invention. Step d) may be performed by using at least one authentication unit. The authentication unit may be a unit of the mobile device or may be an external authentication unit.
40 With respect to definitions and embodiments of the mobile device reference is made to definitions and embodiments described with respect to the method.

In a further aspect of the present invention, use of the method according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use for biometric presentation attack detection.

- 5 Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1 Method for face authentication comprising the following steps:

- 10 a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- 15 b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- 20 c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- 25 d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.
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Embodiment 2 The method according to the preceding embodiment, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.

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Embodiment 3 The method according to the preceding embodiment, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.

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Embodiment 4 The method according to any one of the preceding embodiments, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.

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Embodiment 5 The method according to the preceding embodiment, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.

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Embodiment 6 The method according to any one of the two preceding embodiments, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step at least one region specific parametrized skin classification model is used.

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Embodiment 7 The method according to any one of the preceding embodiments, wherein the illumination pattern comprises a periodic grid of laser spots.

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Embodiment 8 The method according to any one of the preceding embodiments, wherein the illumination features have wavelengths in a near infrared (NIR) regime.

Embodiment 9 The method according to the preceding embodiment, wherein the illumination features have wavelengths of 940 nm.

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Embodiment 10 The method according to any one of the preceding embodiments, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) and/or for 3D detection in step c).

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Embodiment 11 The method according to any one of the preceding embodiments, wherein the camera is or comprises at least one near infrared camera.

Embodiment 12 Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding embodiments, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding embodiments.

Embodiment 13 A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding embodiments referring to a method.

Embodiment 14 A mobile device comprising at least one camera, at least one illumination unit and at least one processing unit, the mobile device being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding embodiments referring to a method.

Embodiment 15 Use of the method according to any one of the preceding claims for biometric presentation attack detection.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1 shows an embodiment of a method for face authentication according to the present invention;

Figure 2 shows an embodiment of a mobile device according to the present invention; and

Figure 3 shows experimental results.

Detailed description of the embodiments:

Figure 1 shows a flow chart of a method for face authentication according to the present invention. The face authentication may comprise verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may
5 comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one
10 identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- 15 a) (reference number 110) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera 112, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit 114;
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- b) (reference number 116) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit 118 and determining at least one second image using the at least one camera 112, wherein the second image
25 comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit 114, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
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- c) (reference number 120) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit 114, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
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- d) (reference number 122) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in

step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

5 The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

10 The camera 112 may comprise at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera 112 may be a digital camera. As an example, the camera 112 may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera 112 may be or may comprise at least one near infrared camera. The image may relate to data recorded by using the camera 112, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip.
15 The camera 112, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera 112 may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera 112 may also comprise one or more variable lenses which may be adjusted, automatically or manually.

20 The camera 112 may be a camera of a mobile device 124 such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera 112 may be part of the mobile device 124 which, besides the at least one camera 112, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The mobile device 124 may be a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart phone. Additionally or alternatively, the mobile device 124 may also refer to a tablet computer or another type of portable computer. An embodiment of a mobile device according to the present invention is shown in Figure 2.

30 Specifically, the camera 112 may be or may comprise at least one optical sensor 126 having at least one light-sensitive area. The optical sensor 126 specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor 126 may be sensitive in the infrared spectral range. The optical sensor 126 may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor 126 may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes

are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor 126 may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor 126 may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

Specifically, the optical sensor 126 may be sensitive in the near infrared region. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor 126, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor 126 may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The camera 112 further may comprise at least one transfer device (not shown here). The camera 112 may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The transfer device may be adapted to guide the light beam onto the optical sensor 126. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging

thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

- 10 The transfer device may have an optical axis. The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.
- 15
- 20 The camera 112 is configured for determining at least one image of the scene, in particular the first image. The scene may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. The first image is at least one two-dimensional image having information about transversal coordinates such as the dimensions of height and width.
- 25

The face detection step 110 comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit 114. As an example, the at least one processing unit 114 may comprise a software code stored thereon comprising a number of computer commands. The processing unit 114 may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit 114. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit 114 may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit 114 and the camera 112 may fully or partially be integrated into a single device. Thus, generally, the processing unit 114 also may form part of the camera 112. Alternatively, the processing unit 114 and the camera 112 may fully or partially be embodied as separate devices.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The geometrical feature characteristic for faces may be at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit 114 may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step 110, the skin detection step 116 may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit 118. However, embodiments are feasible wherein the skin detection step 116 is performed before the face detection step 110.

The illumination unit 118 may be configured for providing the illumination pattern for illumination of the scene. The illumination unit 118 may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit 118 may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit 118 may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit 118 may comprise at least one light source. The illumination unit 118 may comprise a plurality of light sources. The illumination unit 118 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp
5 and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. The illumination unit 118 may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark
10 and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

15 The illumination unit 118 may be or may comprise at least one multiple beam light source. For example, the illumination unit 118 may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit 118 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement hetero-
20 structure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or
25 alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit 118 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points, for example the illumination unit 118 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least
30 one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit 118 is particularly preferred. The illumination unit 118 may be integrated into a housing of the camera 112 or may be separated from the camera 112.

35 The illumination pattern comprises at least one illumination feature adapted to illuminate at least one part of the scene. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern;
40 at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern

may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points. The illumination unit 118 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit 118 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit 118.

The skin detection step 116 comprises determining the at least one second image, also denoted as reflection image, using the camera 112. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c). The reflection feature may be a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. The beam profile of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted

interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor 126.

For example, the illumination unit 118 may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor 126, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor 126 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit 114 may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit 114 may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit 114 may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit 114 may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit 114 may be configured for determining the beam profile of the respective reflection feature. The determining the beam profile may comprise identifying at least one reflection feature provided by the optical sensor 126 and/or selecting at least one reflection feature provided by the optical sensor 126 and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The processing unit 114 is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image
5 corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

The beam profile information may be or may comprise arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b) 116, beam profile analysis may be used. Specifically,
15 beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference.
20 Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. With respect to feature-based methods reference is made to the description above. The feature based methods may be used in combination with
25 machine learning methods which may allow parametrization of a skin classification model. Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

The skin detection step 116 may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit 114. Specifically,
30 the processing unit 114 is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. The at least one predetermined or predefined criterion may be at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be
35 or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. Specifically, the processing unit 114 may be configured for
40 skin detection, in particular for identifying if the detected face is human skin. The identification if the material is biological tissue, in particular human skin, may comprise to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in

particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit 114 may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. The material property may be at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. The determining at least one material property may comprise assigning the material property to the detected face. The processing unit 114 may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit 114 may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

While feature based approaches are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) 116 may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step 116 at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The machine-learning may comprise a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The skin classification model may comprise a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The training may comprise a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step 116 may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step 116, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit 114 may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion.

- 5 The reflection feature may be identified as to be generated by human skin in case the material property indicates "human skin". The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data
10 storage device of the processing unit. The processing unit 114 is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit 114 may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

- 15 The 3D detection step 120 may be performed after the skin detection step 116 and/or the face detection step 110. However, other embodiments are feasible, in which the 3D detection step 120 is performed before the skin detection step 116 and/or the face detection step 110.

- 20 The 3D detection step 120 comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The second beam profile information may comprise a quotient Q of areas of the beam profile.

- 25 The analysis of the beam profile may comprise evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit 114 may be configured for symmetrizing and/or for normalizing
30 and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit 114 may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit 114 may be
35 configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit 114 may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

- 40 The processing unit 114 may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit 114 may be configured

for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

5 The longitudinal coordinate for the reflection feature may be a distance between the optical sensor 126 and the point of the scene remitting the corresponding illumination features. The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile
10 may be an area A1 and the second area of the beam profile may be an area A2. The processing unit 114 may be configured for integrating the first area and the second area. The processing unit 114 may be configured to derive a combined signal, in particular a quotient Q, by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of
15 the integrated first area and the integrated second area. The processing unit 114 may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit 114 may be configured for determining a plurality of areas such as two, three, four, five, or up to
20 ten areas. The processing unit 114 may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit 114 may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit 114 may be configured for determining at least one first area and at
25 least one second area of the beam profile. The area of the beam profile may be an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q. The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the
30 beam profile may be not congruent in area. For example, the processing unit 114 may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the
35 camera 112 may comprise at least two optical sensors 126, wherein the light-sensitive areas of a first optical sensor 126 and of a second optical sensor 126 may be arranged such that the first optical sensor 126 is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor 126 is adapted to determine the second area of the beam profile of the reflection feature. The processing unit 114 may be adapted to integrate the
40 first area and the second area. The processing unit 114 may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The

processing unit 114 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

5 The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

10 The processing unit 114 may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The processing unit 114 may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be
15 crowded comprising a large amount of entries.

20 The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) 120 may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature
25 and/or a range for at least one curvature.
30

35 3D features determined step c) 120 may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) 116 and c) 120 may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

40 Preferably subsequent to steps a) 110, b) 116 and c) 120 the authentication step 122 may be performed. The authentication step 122 may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) 110 no face is detected and/or in step b) 116 the reflection features is determined not to be generated by skin and/or in step c) 120 the

depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) 116 the detected face is characterized as skin and in step c) 122 the detected face is characterized as 3D object.

5 Steps a) to d) may be performed by using at least one device, for example the at least one mobile device¹²⁴ such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

10

The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. The enrolling may comprise a process of registering and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may
15 comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit 114, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This
20 may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular
35 in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

40 Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.

The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

- 5 No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the method may comprise sampling reflection features over several frames in order to reach a more stable classification.

Figure 3 shows experimental results, in particular density as a function of skin score. On the x-axis the score is shown and on the y-axis the frequency. The score is a measure for the classification quality and has a value range between 0 and 1, wherein 1 indicates very high skin-similarity and 0 very low skin-similarity. A threshold for decision may be around 0.5. A reference distribution of skin-scores for bona fide presentations has been generated using 10 subjects. Skin-scores have also been recorded for presentation attacks (PA) of Level A, Level B and Level C (as defined in the relevant ISO standards). The experimental setup (target of evaluation, ToE) included a proprietary hardware device, e.g. as described in Figure 2, which includes the necessary sensors as well as the computational platform on which the PAD software is executed. The ToE was tested using six species of target of Level A PAIs (presentation attack instruments), five PAI species of Level B, and one PAI species of Level C. For each PAI species, 10 PAIs have been used. The PAI species used in this study are listed in the table below. In the table, APCER is the Attack Presentation Classification Error Rate, which refers to the number of successful attacks/all attacks*100. In the table, BPCER is the Bona Fide Presentation Classification Error Rate, which refers to the number of rejected unlock-attempts/all unlock-attempts*100. Level A and Level B attacks are all based on 2D PAIs, whereas for Level C attacks, based on 3D masks. For Level C attacks, custom rigid masks constructed using a 3D printer were used. A test crew of 10 subjects has been used to obtain the reference distribution of skin-scores for bona fide presentations.

Level	#PAI species	PAI species	#objects	#presentations	Results
A	6	4 printed, 2 digital replay (normal quality)	60	300 (60x5)	APCER=0%
B	5	4 printed, 1 digital replay (optimized for spoof)	50	250 (50x5)	APCER=0%

C	1	3D printed mask	10	50 (10x5)	APCER=0%
Bona Fide	-	-	10	50 (10x5)	BPCER=0%

The experiments with these PAIs show that the two classes of presentations (bona fide or PA) are clearly distinguishable based on the skin-score. Clear distinction between paper, 3D print and skin is possible using the method according to the present invention.

List of reference numbers

110	face detection step
112	camera
114	processing unit
116	skin detection step
118	illumination unit
120	3D detection step
122	authentication step
124	mobile device
126	optical sensor

Claims

1. Method for face authentication comprising the following steps:

- 5 a) at least one face detection step (110), wherein the face detection step (110) comprises determining at least one first image by using at least one camera (112), wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step (110) comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit (114);
- 10 b) at least one skin detection step (116), wherein the skin detection step (116) comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit (118) and determining at least one second image using the at least one camera (112), wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit (114), wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- 15 c) at least one 3D detection step (120), wherein the 3D detection step (120) comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit (114), wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- 20 d) at least one authentication step (122), wherein the authentication step (122) comprises authenticating the detected face by using at least one authentication unit if in step b) (116) the detected face is characterized as skin and in step c) (120) the detected face is characterized as 3D object.
- 25
- 30
- 35

2. The method according to the preceding claim, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.

40

3. The method according to the preceding claim, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.
5
4. The method according to any one of the preceding claims, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.
10
5. The method according to the preceding claim, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.
15
6. The method according to any one of the two preceding claims, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step (116) at least one region specific parametrized skin classification model is used.
20
7. The method according to any one of the preceding claims, wherein the illumination pattern comprises a periodic grid of laser spots.
25
8. The method according to any one of the preceding claims, wherein the illumination features have wavelengths in a near infrared (NIR) regime.
9. The method according to the preceding claim, wherein the illumination features have wavelengths of 940 nm.
30
10. The method according to any one of the preceding claims, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) (116) and/or for 3D detection in step c) (120).
35
11. The method according to any one of the preceding claims, wherein the camera (112) is or comprises at least one near infrared camera.
12. Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding claims, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding claims.
40

13. A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding claims referring to a method.
- 5 14. A non-transient computer-readable medium including instructions that, when executed by one or more processors, cause the one or more processors to perform at least steps a) to d) of the method according to any one of the preceding claims referring to a method.
- 10 15. A mobile device (124) comprising at least one camera (112), at least one illumination unit (118) and at least one processing unit (114), the mobile device (124) being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding claims referring to a method.
- 15 16. Use of the method according to any one of the preceding claims for biometric presentation attack detection.

FIG.1

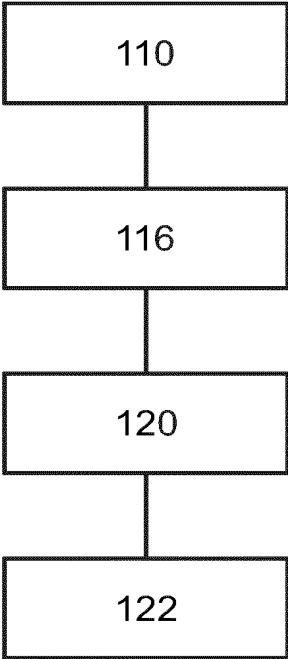
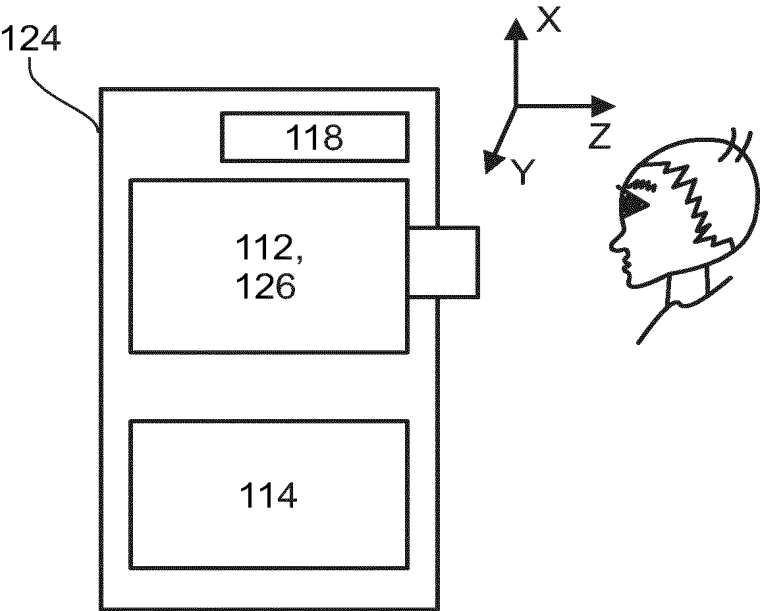
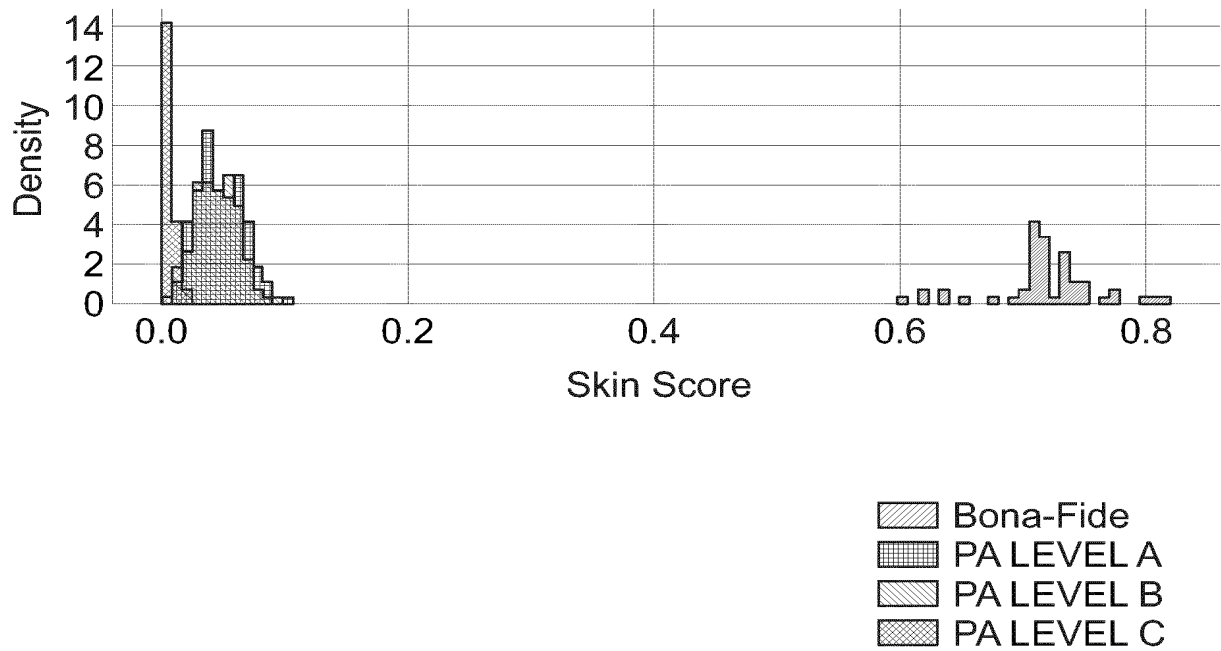


FIG.2



2/2

FIG.3





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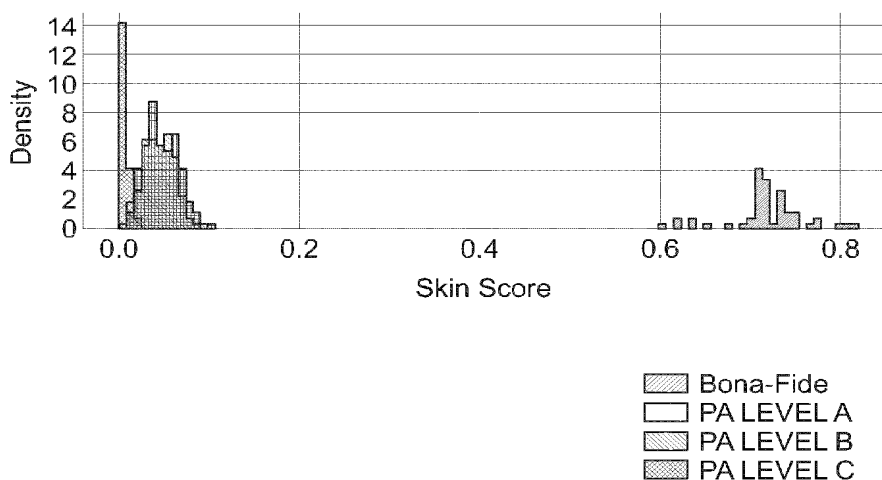
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(54) Title: OPTICAL SKIN DETECTION FOR FACE UNLOCK

FIG.3



(57) **Abstract:** A method for face authentication is proposed. The method comprising the following steps: a) at least one face detection step (110) comprising determining at least one first image by using at least one camera (112); b) at least one skin detection step (116) comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit (118), determining at least one second image using the at least one camera (112), determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile, and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit (114), wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin; c) at least one

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3D detection step (120) comprising determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit (114), wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects; d) at least one authentication step (122) comprising authenticating the detected face by using at least one authentication unit if in step b) (116) the detected face is characterized as skin and in step c) (120) the detected face is characterized as 3D object.

Optical Skin Detection for Face Unlock

Description

Field of the invention

The invention relates to a method for face authentication, a mobile device and various uses of the method. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

In today's digital world secure access to information technology is an essential requirement of any state-of-the-art system. Standard concepts like passphrases or PIN codes are currently extended, combined or even replaced by biometric methods like fingerprint or face recognition. Although passphrases can provide a high level of security if chosen carefully and at adequate length, this step requires attention and the need of memorizing several potentially long phrases depending on the IT landscape. Furthermore, it is never guaranteed that the person providing the passphrase is authorized by the owner of the passphrase or the digital device. In contrast biometric features like faces or fingerprints are unique and person-specific properties. Therefore, using features derived from these not only can be more convenient than passphrases/PIN codes but also more secure since they combine personal identity with an unlock process.

Unfortunately, similarly to being able to spy passwords, fingerprints and faces can be also artificially created in order to impersonate a legitimate user. Automatic face recognition tools of the first generation use digital camera images or image streams and apply 2D image processing methods to extract characteristic features, while machine learning techniques are used to generate face templates to perform identity recognition based on those features. The second generation of face recognition algorithms uses deep convolutional neuronal networks instead of hand-crafted image features to generate the classification model.

However, both approaches can be attacked, e.g., with a high-quality photograph that is nowadays freely available to be downloaded from the Internet. Therefore, the idea of presentation attack detection (PAD) gained significance. Early approaches were designed to protect against simple attacks like presenting the photograph of a legitimate user by recording a sequence of images and test for time dependent features, for example small but natural changes in the head position or eye blinking. These methods can again be tricked by playing a pre-recorded video of the user or by careful animations generated from publicly available photographs. In order to ex-

clude displays as potential spoof objects, near infra-red (NIR) cameras can be used, since displays emit photons only in the visible regime of the electromagnetic spectrum. As a further counteraction, 3D cameras were introduced, which can clearly distinguish between planar photos or tablets with playback videos and a 3D face. Still, these systems can be attacked with high quality masks generated, for example, by 3D printing, careful 3D-arrangement of 2D photos, or hand-crafted silicone or latex masks to name a few approaches. Since masks are typically worn by humans, liveness detection based on small motions fails as well. However, these types of masks can be rejected by a PAD system that is able to classify human skin from other materials such as described in EP application No. 20159984.2 filed on February 28, 2020 and EP application No. 20 154 961.5190679 filed on January 31, 2020, the full content of which is included by reference.

A further problem might arise from differences in skin optical properties with respect ethnic origin. A reliable recognition and PAD technology needs to be fully agnostic with respect to these different origins.

Besides security considerations, also the speed of the unlock process and the required computational power are important to provide an acceptable user experience. High-speed face recognition can be used for several tasks after the successful unlock of a device, for example to check whether the user is still in front of the display or to launch further secure applications like banking apps by performing an on the fly check of the person in front of the display. Again, speed and computational resources have great impact on the user experience.

Current 3D algorithms are computationally very demanding and presentation attack detections require several video frames to be processed. Therefore, expensive hardware needs to be involved to deliver acceptable unlock performance. Furthermore, high power consumption is a consequence.

Summarizing, current methods for face unlock do not provide the ability to reliably detect spoof attacks by 3D masks and perform this task at a speed below human detection limit.

US 2019/213309 A1 describes a system and method of authenticating a user's face with a ranging sensor. The ranging sensor includes a time of flight sensor and a reflectance sensor. The ranging sensor transmits a signal that is reflected off of a user and received back at the ranging sensor. The received signal can be used to determine distance between the user and the sensor, and the reflectance value of the user. With the distance or the reflectivity, a processor can activate a facial recognition process in response to the distance and the reflectivity.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Although simple presentation attacks with photographs and videos of the legitimate face can be detected, an approach

to reliably detect presentation attacks with 3D face masks is still missing. Specifically, another layer of security is needed to enable the replacement of passphrases/PIN codes by biometric features derived from faces to unlock digital devices. Moreover, a method is needed that operates fully agnostic with respect to different skin types originating from different ethnic origins.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, notwithstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a method for face authentication is disclosed. The face to be authenticated may be, specifically, a human face. The term “face authentication” as used

herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;

- d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

5

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

- 10 The term “camera” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a device having at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera may be a digital
15 camera. As an example, the camera may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera may be or may comprise at least one near infrared camera.

- As used herein, without limitation, the term “image” specifically may relate to data recorded by
20 using a camera, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera may also comprise one
25 or more variable lenses which may be adjusted, automatically or manually.

- The camera may be a camera of a mobile device such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera may be part of a mobile device which, besides the at least one camera, comprises one or more data
30 processing devices such as one or more data processors. Other cameras, however, are feasible. The term “mobile device” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart
35 phone. Additionally or alternatively, the mobile device may also refer to a tablet computer or another type of portable computer.

- Specifically, the camera may be or may comprise at least one optical sensor having at least one light-sensitive area. As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers
40 to an area of the optical sensor which may be illuminated externally, by the at least one light

beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The camera may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal. In case the camera comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the camera specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. The optical sensor may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor may be part of or constitute a pixelated optical device. For example, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the camera form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

The camera further may comprise at least one transfer device. The camera may further comprise one or more additional elements such as one or more additional optical elements. The camera may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical

sensor. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the camera may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. In case of a plurality of optical sensors, the optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The camera is configured for determining at least one image of the scene, in particular the first image. As used herein, the term “scene” may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to “pixels”, reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. The first image is at least one two-dimensional image. As used herein, the term “two dimensional image” may generally refer to an image having information about transversal coordinates such as the dimensions of height and width. The first image may be an RGB (red green blue) image. The term “determining at least one first image” may refer to capturing and/or recording the first image.

The face detection step comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit. Specifically, the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature which is characteristic for faces by using the at least one processing unit.

As further used herein, the term “processing unit” generally refers to an arbitrary data processing device adapted to perform the named operations such as by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one processing unit may comprise a software code stored thereon comprising a number of computer commands. The processing unit may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit and the camera may fully or partially be integrated into a single device. Thus, generally, the processing unit also may form

part of the camera. Alternatively, the processing unit and the camera may fully or partially be embodied as separate devices.

The processing unit may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the processing unit may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the processing unit may comprise one or more data storage devices. Further, the processing unit may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The processing unit may be configured to one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the camera. The processing unit, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The processing unit may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The processing unit may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The processing unit may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The processing unit may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial
5 connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The term "geometrical feature
10 characteristic for faces" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit may comprise at least
15 one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE,
20 2018, the full content of which is included by reference.

The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region
25 of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying
30 a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step, the skin detection step may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit. However, embodiments are feasible wherein the skin detection step is performed before the face detection step.

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As used herein, the term “illumination unit”, also denoted as illumination source, may generally refers to at least one arbitrary device configured for generating at least one illumination pattern. The illumination unit may be configured for providing the illumination pattern for illumination of the scene. The illumination unit may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit may comprise at least one light source. The illumination unit may comprise a plurality of light sources. The illumination unit may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination unit may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used.

The illumination unit may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination unit may be configured for emitting light at a single wavelength. In other embodiments, the illumination unit may be configured for emitting light with a plurality of wavelengths allowing additional measurements in other wavelengths channels.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one

edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however.

5 The illumination unit may be or may comprise at least one multiple beam light source. For example, the illumination unit may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure la-
10 sers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, ex-
15 tended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit may be adapted to generate and/or to project a cloud of points, for example the illumination unit may comprise one or more of at least one digital light
20 processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit is particularly preferred. The illumination unit may be integrated into a housing of the camera or may be sepa-
25 rated from the camera.

Further, the illumination unit may be configured for emitting modulated or non-modulated light. In case a plurality of illumination units is used, the different illumination units may have different modulation frequencies which later on may be used for distinguishing the light beams.

30 The light beam or light beams generated by the illumination unit generally may propagate parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The illumination unit may be configured such that the light beam or light beams propa-
35 gates from the illumination unit towards the scene along an optical axis of the illumination unit and/or of the camera. For this purpose, the illumination unit and/or the camera may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light
40 beam onto the optical axis. As an example, the light beam or light beams, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the optical axis or off the optical axis. As an example, the light beam or light
beams may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

As used herein, the term “at least one illumination pattern” refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the scene. As used herein, the term “illumination feature” refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit may be adapted to generate and/or to project a cloud of points. The illumination unit may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit.

A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination unit may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination unit comprises at least one laser source and/or at least one laser diode which is designated for generating laser radiation. The illumination unit may comprise the at least one diffractive optical element (DOE). The illumination unit may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one periodic point pattern. As further used herein, the term “projecting at least one illumination pattern” may refer to providing the at least one illumination pattern for illuminating the at least one scene.

The skin detection step comprises determining the at least one second image, also denoted as reflection image, using the camera. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c).

The second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features. As used herein, the term “reflection feature” may refer to a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. As used herein, the term “beam profile” of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor.

For example, the illumination unit may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing

unit may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit may be configured for determining the beam profile of the respective reflection feature. As used herein, the term “determining the beam profile” refers to identifying at least one reflection feature provided by the optical sensor and/or selecting at least one reflection feature provided by the optical sensor and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may be an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The reflection feature may cover or may extend over at least one pixel of the second image. For example, the reflection feature may cover or may extend over plurality of pixels. The processing unit may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The processing unit may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum_j r_{pixel}},$$

wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum_j I_{total}$ with j being the number of pixels j connected to and/or belonging to the reflection feature and I_{total} being the total intensity.

The processing unit is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image

and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

5 As used herein, the term “beam profile information” may refer to arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b), beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto
10 object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP
15 application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. The feature-based methods may be explained in the following. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model. Alternatively
20 or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

Other methods for authenticating a user’s face are known, such as from US 2019/213309 A1. However, these methods use time of flight (ToF) sensors. A well-known working principle of a
25 ToF sensor is sending out light and measuring the time span until receiving the reflected light. In contrast, the proposed beam profile analysis uses a projected illumination pattern. For ToF sensor using such a projected pattern is not possible. Using illumination pattern may be advantageous e.g. in view of covering and therefore allowing to take into account different locations on the face. This may enhance reliability and security of the authentication of the user’s face.

30 The skin detection step may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit. Specifically, the processing unit is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one
35 predetermined or predefined criterion. As used herein, the term “at least one predetermined or predefined criterion” refers to at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated
40 as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. As used herein, the term “indicate” refers to an arbitrary indication such as an electronic signal and/or at least one visual or acoustic indication. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. As

used herein, the term “biological tissue” generally refers to biological material comprising living cells. Specifically, the processing unit may be configured for skin detection. The term “identification” of being generated by biological tissue, in particular human skin, may refer to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “determining at least one material property” may refer to assigning the material property to respective reflection feature, in particular to the detected face. The processing unit may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

The reflection properties of skin may be characterized by the simultaneous occurrence of direct reflection at the surface (Lambertian-like) and subsurface scattering (volume scattering). This leads to a broadening of the laser spot on skin compared to the above-mentioned materials.

The first beam profile information may be a reflection profile. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering

profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering profile.

Specifically, the processing unit may be configured for comparing the reflection beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a device executing the method according to the present invention. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the processing unit or the device, e.g. by software, specifically by the app downloaded from an app store or the like. The reflection feature may be identified as to be generated by biological tissue in case the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise determining a deviation, e.g. a sum of squared point to point distances, between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The processing unit may be adapted to compare the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit.

Additionally or alternatively, the first beam profile information may be determined by applying at least one image filter to the image of the area. As further used herein, the term “image” refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a material feature. Images may be

subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

- 5 The processing unit may be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the surface of the scene having generated the reflection feature.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

- 30 The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used
- 35 herein, the term “beam profile image” refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

- 40 wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k : $(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond

to a material classifier and a distance. The material classifier may be a label such as ‘Material A’, ‘Material B’, etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

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The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

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Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term “not distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical. As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis

testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis

5 testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- 10
- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
 - $F(x) = 1 - CDF(x)$
 - $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete

- 15 Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can
- 20 be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

- 25 For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by
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$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

- 10 For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

- 15 wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

- 20 For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

- 25 For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

- 30 For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse

scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image

5 filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

20 Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1 g_2) = [p_{g_1, g_2}]$, whereas p_{g_1, g_2} is the occurrence rate of the grey combination $(g_1, g_2) = [f(x_1, y_1), f(x_2, y_2)]$, and the relation ρ defines the distance between (x_1, y_1) and (x_2, y_2) , which is $\rho(x, y) = (x+a, y+b)$ with a and b selected from

25 0,1.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij} (i - j)^2.$$

For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based energy filter may be given by

5

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i - j|}.$$

10 For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

15 For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

and

20

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

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For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ϵ times

the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The processing unit may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the surface having generated the reflection feature for determining the material property of the surface having generated the reflection feature. The predetermined relationship may be one or more of an empirical relationship, a semi-empirical relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

While feature based approaches, as the approaches described above, are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The term "machine-learning" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The term "skin classification model" may refer to a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For

example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The term “training”, also denoted learning, as used herein, is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates “human skin”. The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit. The processing unit is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

The 3D detection step may be performed after the skin detection step and/or the face detection step. However, other embodiments are feasible, in which the 3D detection step is performed before the skin detection step and/or the face detection step. The material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z in step d) such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} .

The 3D detection step comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The 3D detection step may comprise determining a second beam profile information of at least four of the reflection features by analysis of their respective beam profile. The second beam profile information may comprise a quotient Q of areas of the beam profile.

As used herein, the term “analysis of the beam profile” may generally refer to evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The processing unit may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

The processing unit may be configured for determining at least one first area and at least one second area of the reflection beam profile of each of the reflection features and/or of the reflection features in at least one region of interest. The processing unit is configured for integrating the first area and the second area.

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The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The processing unit may be configured for integrating the first area and the second area. The processing unit may be configured to derive a combined signal, in particular a quotient Q , by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The processing unit may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit may be configured for determining at least one first area and at least one second area of the beam profile. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing unit may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera may comprise at least two optical sensors, wherein the light-sensitive areas of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the beam profile of the reflection feature. The processing unit may be adapted to integrate the first area and the second area. The processing unit may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

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The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The processing unit may be configured for determining an area integral of the beam profile. The processing unit may be configured for determining the edge information by integrating and/or summing of the first area. The processing unit may be configured for determining the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the processing unit may be configured for determining an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as

determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 5 In one embodiment, A1 may correspond to a full or complete area of a feature point on the optical sensor. A2 may be a central area of the feature point on the optical sensor. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.
- 10 In one embodiment, the illumination pattern may comprise at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensor may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the
- 15 line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensor. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width,
- 20 preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.
- 25 In one embodiment, the illumination pattern may comprise at least point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to
- 30 0.6 of the full radius.
- The illumination pattern may comprise both at least one point pattern and at least one line pattern. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.
- 35 The processing unit may be configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.
- 40 The processing unit may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The processing unit may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the beam profile, respectively, and E(x,y) denotes the beam profile.

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Additionally or alternatively, the processing unit may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

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For example, in case of the optical sensor having a matrix of pixels, the processing unit may be configured for evaluating the beam profile, by

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- determining the pixel having the highest sensor signal and forming at least one center signal;
- evaluating sensor signals of the matrix and forming at least one sum signal;
- determining the quotient Q by combining the center signal and the sum signal; and
- determining at least one longitudinal coordinate z of the object by evaluating the quotient

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Q.

As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the display device, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. The term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the pixel having the highest sensor signal out of the plurality of sensor signals generated by the pixels of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the pixels of the matrix. The center signal may arise from a single pixel or from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of pixels may be added up, integrated or averaged, in order to determine the center signal. The group of pixels from which the center signal arises may be a group of neighboring pixels, such as pixels having less than a predetermined distance from the actual pixel having the

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highest sensor signal, or may be a group of pixels generating sensor signals being within a predetermined range from the highest sensor signal. The group of pixels from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The processing unit may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of pixels around the pixel having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the pixel in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from pixels in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring pixels; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The processing unit may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of

the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 5 Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge
10 information in the center signal, the edge signal may further be subtracted from the center signal.

The quotient Q may be a signal which is generated by combining the center signal and the sum signal. Specifically, the determining may include one or more of: forming a quotient of the center
15 signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the quotient Q may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

20 As used herein, the term “longitudinal coordinate for the reflection feature” refers to a distance between the optical sensor and the point of the scene remitting the corresponding illumination features. The processing unit may be configured for using the at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal
25 coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

30 The processing unit may be configured for executing at least one depth-from-photon-ratio algorithm which computes distances for all reflection features with zero order and higher order.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

35 The processing unit may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. As used herein, the term “depth” or depth information may
40 refer to a distance between the object and the optical sensor and may be given by the longitudinal coordinate. As used herein, the term “depth map” may refer to spatial distribution of depth. The processing unit may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light,

beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.

- 5 The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature.
- 10 The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup list or
- 15 a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.
- 20 3D features determined step c) may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) and c) may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both
- 25 properties can be calculated from the same frame with low computational effort.

Preferably subsequent to steps a) to c) the authentication step may be performed. The authentication step may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) no face is detected and/or in step b) the reflection features is determined not to be generated by skin and/or in step c) the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

35 Steps a) to d) may be performed by using at least one device, for example at least one mobile device such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

40 The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. As used herein, the term "enrolling" may refer to a process of registering

and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device.

5 The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images.

10 The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving
15 the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

20 The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides
25 depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.
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The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

35 No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the
40 method may comprise sampling reflection features over several frames in order to reach a more stable classification.

Besides accuracy, also execution speed and power consumption are important requirements. Further restrictions on the availability of computational resources can be introduced by security considerations. For example, steps a) to d) may have run in a secure zone of the processing unit to avoid any software-based manipulations during program execution. The compact nature of the above described material detection networks may solve this problem by showing excellent runtime behavior in the said secure zone, whereas traditional PAD solutions require the examination of several consecutive frames which results in large computational cost and longer response times of the algorithm.

In a further aspect of the present invention a computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to the present invention, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to the present invention. Specifically, the computer program may be stored on a computer-readable data carrier and/or on a computer-readable storage medium.

As used herein, the terms “computer-readable data carrier” and “computer-readable storage medium” specifically may refer to non-transitory data storage means, such as a hard-ware storage medium having stored thereon computer-executable instructions. The computer-readable data carrier or storage medium specifically may be or may comprise a storage medium such as a random-access memory (RAM) and/or a read-only memory (ROM).

Thus, specifically, one, more than one or even all of method steps as indicated above may be performed by using a computer or a computer network, preferably by using a computer program.

In a further aspect a computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to the present invention.

Further disclosed and proposed herein is a data carrier having a data structure stored thereon, which, after loading into a computer or computer network, such as into a working memory or main memory of the computer or computer network, may execute the method according to one or more of the embodiments disclosed herein.

Further disclosed and proposed herein is a computer program product with program code means stored on a machine-readable carrier, in order to perform the method according to one or more of the embodiments disclosed herein, when the program is executed on a computer or computer network. As used herein, a computer program product refers to the program as a tradable product.

The product may generally exist in an arbitrary format, such as in a paper format, or on a computer-readable data carrier and/or on a computer-readable storage medium. Specifically, the computer program product may be distributed over a data network.

Finally, disclosed and proposed herein is a modulated data signal which contains instructions readable by a computer system or computer network, for performing the method according to one or more of the embodiments disclosed herein.

5 Referring to the computer-implemented aspects of the invention, one or more of the method steps or even all of the method steps of the methods according to one or more of the embodiments disclosed herein may be performed by using a computer or computer network. Thus, generally, any of the method steps including provision and/or manipulation of data may be performed by using a computer or computer network. Generally, these method steps may include any of the
10 method steps, typically except for method steps requiring manual work.

Specifically, further disclosed herein are:

- 15 - a computer or computer network comprising at least one processor, wherein the processor is adapted to perform the method according to one of the embodiments described in this description,
- a computer loadable data structure that is adapted to perform the method according to one of the embodiments described in this description while the data structure is being executed on a computer,
- 20 - a computer program, wherein the computer program is adapted to perform the method according to one of the embodiments described in this description while the program is being executed on a computer,
- a computer program comprising program means for performing the method according to one of the embodiments described in this description while the computer program is being executed on a computer or on a computer network,
- 25 - a computer program comprising program means according to the preceding embodiment, wherein the program means are stored on a storage medium readable to a computer,
- a storage medium, wherein a data structure is stored on the storage medium and wherein the data structure is adapted to perform the methods according to one of the embodiments described in this description after having been loaded into a main and/or working storage
30 of a computer or of a computer network, and
- a computer program product having program code means, wherein the program code means can be stored or are stored on a storage medium, for performing the method according to one of the embodiments described in this description, if the program code means are executed on a computer or on a computer network.

35 In a further aspect a mobile device comprising at least one camera, at least one illumination unit and at least one processing unit is disclosed. The mobile device is configured for performing at least steps a) to c), and optionally step d) of the method for face authentication according to the present invention. Step d) may be performed by using at least one authentication unit. The authentication unit may be a unit of the mobile device or may be an external authentication unit.
40 With respect to definitions and embodiments of the mobile device reference is made to definitions and embodiments described with respect to the method.

In a further aspect of the present invention, use of the method according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use for biometric presentation attack detection.

- 5 Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1 Method for face authentication comprising the following steps:

- 10 a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- 15 b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- 20 c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- 25 d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.
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Embodiment 2 The method according to the preceding embodiment, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.

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Embodiment 3 The method according to the preceding embodiment, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.

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Embodiment 4 The method according to any one of the preceding embodiments, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.

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Embodiment 5 The method according to the preceding embodiment, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.

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Embodiment 6 The method according to any one of the two preceding embodiments, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step at least one region specific parametrized skin classification model is used.

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Embodiment 7 The method according to any one of the preceding embodiments, wherein the illumination pattern comprises a periodic grid of laser spots.

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Embodiment 8 The method according to any one of the preceding embodiments, wherein the illumination features have wavelengths in a near infrared (NIR) regime.

Embodiment 9 The method according to the preceding embodiment, wherein the illumination features have wavelengths of 940 nm.

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Embodiment 10 The method according to any one of the preceding embodiments, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) and/or for 3D detection in step c).

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Embodiment 11 The method according to any one of the preceding embodiments, wherein the camera is or comprises at least one near infrared camera.

Embodiment 12 Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding embodiments, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding embodiments.

Embodiment 13 A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding embodiments referring to a method.

Embodiment 14 A mobile device comprising at least one camera, at least one illumination unit and at least one processing unit, the mobile device being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding embodiments referring to a method.

Embodiment 15 Use of the method according to any one of the preceding claims for biometric presentation attack detection.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1 shows an embodiment of a method for face authentication according to the present invention;

Figure 2 shows an embodiment of a mobile device according to the present invention; and

Figure 3 shows experimental results.

Detailed description of the embodiments:

Figure 1 shows a flow chart of a method for face authentication according to the present invention. The face authentication may comprise verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) (reference number 110) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera 112, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit 114;
- b) (reference number 116) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit 118 and determining at least one second image using the at least one camera 112, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit 114, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- c) (reference number 120) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit 114, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- d) (reference number 122) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in

step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

5 The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

10 The camera 112 may comprise at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera 112 may be a digital camera. As an example, the camera 112 may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera 112 may be or may comprise at least one near infrared camera. The image may relate to data recorded by using the camera 112, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip.
15 The camera 112, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera 112 may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera 112 may also comprise one or more variable lenses which may be adjusted, automatically or manually.

20 The camera 112 may be a camera of a mobile device 124 such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera 112 may be part of the mobile device 124 which, besides the at least one camera 112, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The mobile device 124 may be a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart phone. Additionally or alternatively, the mobile device 124 may also refer to a tablet computer or another type of portable computer. An embodiment of a mobile device according to the present invention is shown in Figure 2.

30 Specifically, the camera 112 may be or may comprise at least one optical sensor 126 having at least one light-sensitive area. The optical sensor 126 specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor 126 may be sensitive in the infrared spectral range. The optical sensor 126 may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor 126 may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes

are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor 126 may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor 126 may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

Specifically, the optical sensor 126 may be sensitive in the near infrared region. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor 126, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor 126 may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The camera 112 further may comprise at least one transfer device (not shown here). The camera 112 may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The transfer device may be adapted to guide the light beam onto the optical sensor 126. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging

thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

- 10 The transfer device may have an optical axis. The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.
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- 20 The camera 112 is configured for determining at least one image of the scene, in particular the first image. The scene may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. The first image is at least one two-dimensional image having information about transversal coordinates such as the dimensions of height and width.
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- The face detection step 110 comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit 114. As an example, the at least one processing unit 114 may comprise a software code stored thereon comprising a number of computer commands. The processing unit 114 may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit 114. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit 114 may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit 114 and the camera 112 may fully or partially be integrated into a single device. Thus, generally, the processing unit 114 also may form part of the camera 112. Alternatively, the processing unit 114 and the camera 112 may fully or partially be embodied as separate devices.
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The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The geometrical feature characteristic for faces may be at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit 114 may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step 110, the skin detection step 116 may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit 118. However, embodiments are feasible wherein the skin detection step 116 is performed before the face detection step 110.

The illumination unit 118 may be configured for providing the illumination pattern for illumination of the scene. The illumination unit 118 may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit 118 may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit 118 may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit 118 may comprise at least one light source. The illumination unit 118 may comprise a plurality of light sources. The illumination unit 118 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp
5 and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. The illumination unit 118 may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark
10 and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

15 The illumination unit 118 may be or may comprise at least one multiple beam light source. For example, the illumination unit 118 may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit 118 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement hetero-
20 structure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or
25 alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit 118 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points, for example the illumination unit 118 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least
30 one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit 118 is particularly preferred. The illumination unit 118 may be integrated into a housing of the camera 112 or may be separated from the camera 112.

35 The illumination pattern comprises at least one illumination feature adapted to illuminate at least one part of the scene. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern
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may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points. The illumination unit 118 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit 118 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit 118.

The skin detection step 116 comprises determining the at least one second image, also denoted as reflection image, using the camera 112. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c). The reflection feature may be a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. The beam profile of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted

interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor 126.

For example, the illumination unit 118 may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor 126, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor 126 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit 114 may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit 114 may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit 114 may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit 114 may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit 114 may be configured for determining the beam profile of the respective reflection feature. The determining the beam profile may comprise identifying at least one reflection feature provided by the optical sensor 126 and/or selecting at least one reflection feature provided by the optical sensor 126 and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The processing unit 114 is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

The beam profile information may be or may comprise arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b) 116, beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. With respect to feature-based methods reference is made to the description above. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model. Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

The skin detection step 116 may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit 114. Specifically, the processing unit 114 is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. The at least one predetermined or predefined criterion may be at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. Specifically, the processing unit 114 may be configured for skin detection, in particular for identifying if the detected face is human skin. The identification if the material is biological tissue, in particular human skin, may comprise to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in

particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit 114 may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. The material property may be at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. The determining at least one material property may comprise assigning the material property to the detected face. The processing unit 114 may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit 114 may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

While feature based approaches are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) 116 may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step 116 at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The machine-learning may comprise a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The skin classification model may comprise a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The training may comprise a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step 116 may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step 116, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit 114 may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion.

- 5 The reflection feature may be identified as to be generated by human skin in case the material property indicates "human skin". The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data
10 storage device of the processing unit. The processing unit 114 is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit 114 may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

- 15 The 3D detection step 120 may be performed after the skin detection step 116 and/or the face detection step 110. However, other embodiments are feasible, in which the 3D detection step 120 is performed before the skin detection step 116 and/or the face detection step 110.

- 20 The 3D detection step 120 comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The second beam profile information may comprise a quotient Q of areas of the beam profile.

- 25 The analysis of the beam profile may comprise evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit 114 may be configured for symmetrizing and/or for normalizing
30 and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit 114 may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit 114 may be
35 configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit 114 may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

- 40 The processing unit 114 may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit 114 may be configured

for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

5 The longitudinal coordinate for the reflection feature may be a distance between the optical sensor 126 and the point of the scene remitting the corresponding illumination features. The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile
10 may be an area A1 and the second area of the beam profile may be an area A2. The processing unit 114 may be configured for integrating the first area and the second area. The processing unit 114 may be configured to derive a combined signal, in particular a quotient Q, by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of
15 the integrated first area and the integrated second area. The processing unit 114 may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit 114 may be configured for determining a plurality of areas such as two, three, four, five, or up to
20 ten areas. The processing unit 114 may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit 114 may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit 114 may be configured for determining at least one first area and at
25 least one second area of the beam profile. The area of the beam profile may be an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q. The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the
30 beam profile may be not congruent in area. For example, the processing unit 114 may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the
35 camera 112 may comprise at least two optical sensors 126, wherein the light-sensitive areas of a first optical sensor 126 and of a second optical sensor 126 may be arranged such that the first optical sensor 126 is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor 126 is adapted to determine the second area of the beam profile of the reflection feature. The processing unit 114 may be adapted to integrate the
40 first area and the second area. The processing unit 114 may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The

processing unit 114 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

5 The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

10 The processing unit 114 may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The processing unit 114 may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be
15 crowded comprising a large amount of entries.

20 The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) 120 may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature
25 and/or a range for at least one curvature.
30

35 3D features determined step c) 120 may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) 116 and c) 120 may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

40 Preferably subsequent to steps a) 110, b) 116 and c) 120 the authentication step 122 may be performed. The authentication step 122 may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) 110 no face is detected and/or in step b) 116 the reflection features is determined not to be generated by skin and/or in step c) 120 the

depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) 116 the detected face is characterized as skin and in step c) 122 the detected face is characterized as 3D object.

5 Steps a) to d) may be performed by using at least one device, for example the at least one mobile device¹²⁴ such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

10

The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. The enrolling may comprise a process of registering and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit 114, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

40 Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.

The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

- 5 No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the method may comprise sampling reflection features over several frames in order to reach a more stable classification.

Figure 3 shows experimental results, in particular density as a function of skin score. On the x-axis the score is shown and on the y-axis the frequency. The score is a measure for the classification quality and has a value range between 0 and 1, wherein 1 indicates very high skin-similarity and 0 very low skin-similarity. A threshold for decision may be around 0.5. A reference distribution of skin-scores for bona fide presentations has been generated using 10 subjects. Skin-scores have also been recorded for presentation attacks (PA) of Level A, Level B and Level C (as defined in the relevant ISO standards). The experimental setup (target of evaluation, ToE) included a proprietary hardware device, e.g. as described in Figure 2, which includes the necessary sensors as well as the computational platform on which the PAD software is executed. The ToE was tested using six species of target of Level A PAIs (presentation attack instruments), five PAI species of Level B, and one PAI species of Level C. For each PAI species, 10 PAIs have been used. The PAI species used in this study are listed in the table below. In the table, APCER is the Attack Presentation Classification Error Rate, which refers to the number of successful attacks/all attacks*100. In the table, BPCER is the Bona Fide Presentation Classification Error Rate, which refers to the number of rejected unlock-attempts/all unlock-attempts*100. Level A and Level B attacks are all based on 2D PAIs, whereas for Level C attacks, based on 3D masks. For Level C attacks, custom rigid masks constructed using a 3D printer were used. A test crew of 10 subjects has been used to obtain the reference distribution of skin-scores for bona fide presentations.

Level	#PAI species	PAI species	#objects	#presentations	Results
A	6	4 printed, 2 digital replay (normal quality)	60	300 (60x5)	APCER=0%
B	5	4 printed, 1 digital replay (optimized for spoof)	50	250 (50x5)	APCER=0%

C	1	3D printed mask	10	50 (10x5)	APCER=0%
Bona Fide	-	-	10	50 (10x5)	BPCER=0%

The experiments with these PAIs show that the two classes of presentations (bona fide or PA) are clearly distinguishable based on the skin-score. Clear distinction between paper, 3D print and skin is possible using the method according to the present invention.

List of reference numbers

110	face detection step
112	camera
114	processing unit
116	skin detection step
118	illumination unit
120	3D detection step
122	authentication step
124	mobile device
126	optical sensor

Claims

1. Method for face authentication comprising the following steps:

- 5 a) at least one face detection step (110), wherein the face detection step (110) comprises determining at least one first image by using at least one camera (112), wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step (110) comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit (114);
- 10 b) at least one skin detection step (116), wherein the skin detection step (116) comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit (118) and determining at least one second image using the at least one camera (112), wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit (114), wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- 15 c) at least one 3D detection step (120), wherein the 3D detection step (120) comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit (114), wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- 20 d) at least one authentication step (122), wherein the authentication step (122) comprises authenticating the detected face by using at least one authentication unit if in step b) (116) the detected face is characterized as skin and in step c) (120) the detected face is characterized as 3D object.
- 25
- 30
- 35

2. The method according to the preceding claim, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.

40

3. The method according to the preceding claim, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.
5
4. The method according to any one of the preceding claims, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.
10
5. The method according to the preceding claim, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.
15
6. The method according to any one of the two preceding claims, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step (116) at least one region specific parametrized skin classification model is used.
20
7. The method according to any one of the preceding claims, wherein the illumination pattern comprises a periodic grid of laser spots.
25
8. The method according to any one of the preceding claims, wherein the illumination features have wavelengths in a near infrared (NIR) regime.
9. The method according to the preceding claim, wherein the illumination features have wavelengths of 940 nm.
30
10. The method according to any one of the preceding claims, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) (116) and/or for 3D detection in step c) (120).
35
11. The method according to any one of the preceding claims, wherein the camera (112) is or comprises at least one near infrared camera.
12. Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding claims, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding claims.
40

13. A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding claims referring to a method.
- 5 14. A non-transient computer-readable medium including instructions that, when executed by one or more processors, cause the one or more processors to perform at least steps a) to d) of the method according to any one of the preceding claims referring to a method.
- 10 15. A mobile device (124) comprising at least one camera (112), at least one illumination unit (118) and at least one processing unit (114), the mobile device (124) being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding claims referring to a method.
- 15 16. Use of the method according to any one of the preceding claims for biometric presentation attack detection.

FIG.1

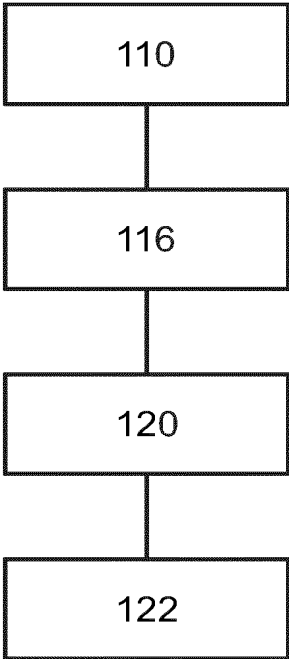


FIG.2

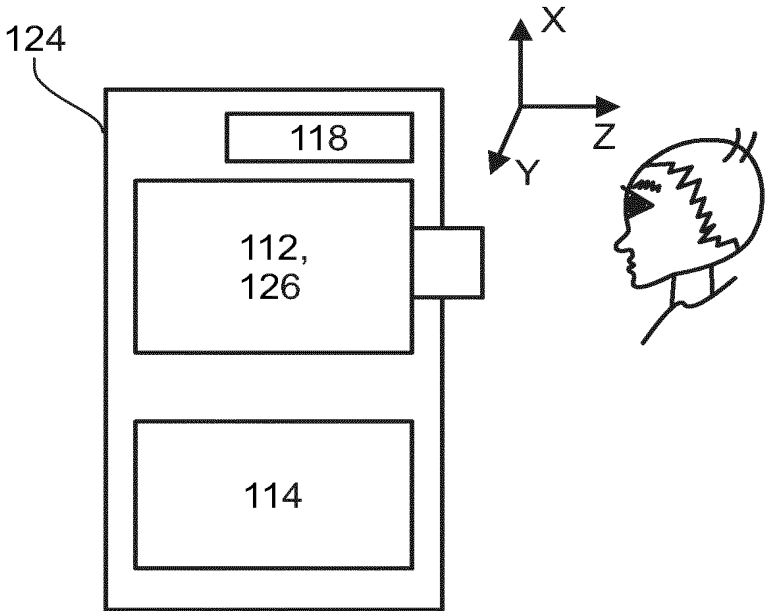
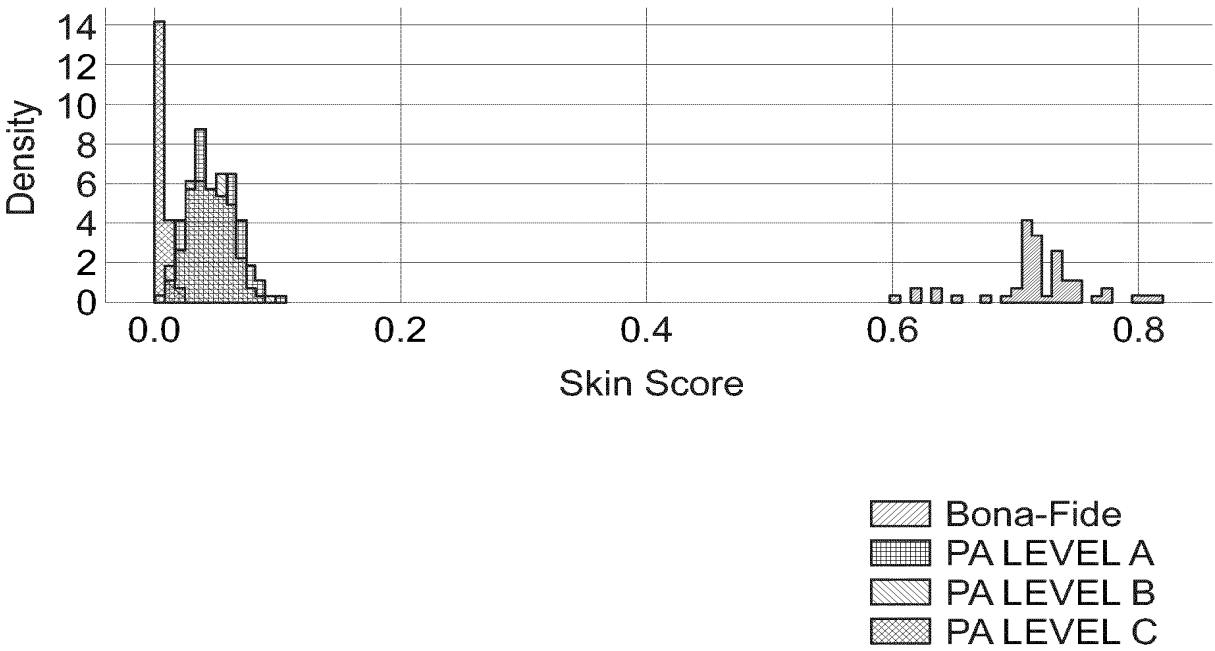


FIG.3



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Application Number	
Filing Date	August 4, 2023
First Named Inventor	Peter SCHILLEN
Title	OPTICAL SKIN DETECTION FOR FACE UNLOCK
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2262 202595US01

SIGNATURE of Applicant or Patent Practitioner

Signature	/Christopher M. Goff/	Date (Optional)	2023-08-04
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2020/06/09

Name

Dr. Hans Graf

Title

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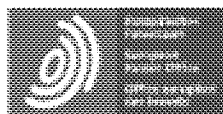
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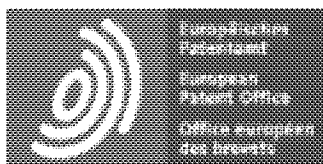
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Anti-spoof real face sensing for enhancing the security of facial recognition

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- **cooperative:** G06V10/145 (CN); G06V10/147 (CN);
G06V40/166 (CN); G06V40/45 (CN)**Application number:** CN20198004265 20191231 Global Dossier**Priority number(s):** US201962788749P 20190104 ; WO2019CN130509 20191231**Abstract of CN111164610 (A)**

The technology of devices, systems, technologies, and processes is used to provide anti-spoofing features for facial recognition through the use of optical sensing and other sensing mechanisms. The technology has enhanced security against facial spoofing devices or technologies to explore some of the unique characteristics of human faces that are lacking or difficult to replicate in most fraudulent devices made of artificial materials. The technology includes optical sensing based on the unique optical absorption or reflection characteristics of the biological parts of the human face.



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DESCRIPTION CN111164610A

Anti-spoofing true face sensing for enhanced facial recognition security

[0001]

Cross References to Related Applications

[0002]

This patent document requires the applicant to be Shenzhen Goodix Technology Co., Ltd., with the application number 62/788,749 filed on January 4, 2019, and the title of the invention is "Anti-spoofing real face sensing for enhanced facial recognition security" The benefit and priority of the U.S. Provisional Patent Application of , which is incorporated by reference as part of the disclosure of this patent document.

[0003]

technical field

[0004]

This patent document relates to the detection of facial features of a person as part of user authentication for authorizing access to a secure electronic device, secure digital system, secure facility or location.

[0005]

Background technique

[0006]

Various sensors can be implemented in electronic devices or systems to detect unique identifying characteristics from a person or person and provide user authentication to grant or deny access to devices including portable or mobile computing devices (e.g., laptops, tablets, smart Mobile phones), game systems, various databases, information systems or security devices or systems for large computer control systems.

Various authentication mechanisms can be used to protect sensitive or confidential data and prevent unauthorized access.

User authentication on electronic devices may be performed by one or more forms of biometric identifiers, which may be used alone or in addition to conventional password authentication methods.

A popular form of biometric identifier is a person's fingerprint pattern.

A fingerprint sensor may be built into the electronic device to read the user's fingerprint pattern so that the device can only be unlocked by an authorized user of the device by authenticating the authorized user's fingerprint pattern.

Another example is an imaging device that detects a person's face to extract the person's unique facial features as a biometric identifier for user authentication. Some of Apple's latest iPhone models are examples of devices that use facial imaging to provide facial recognition to authenticate a person as an authorized user.

[0007]

However, a person's biometric identifiers, such as facial features, can be copied or fabricated through the use of spoofing tools or devices to evade facial recognition-based user authentication mechanisms.

[0008]

Contents of the invention

[0009]

This patent document discloses devices, systems, techniques and processes for providing anti-spoofing features for facial recognition with enhanced security against facial spoofing devices or techniques by using optical sensing and other sensing mechanisms.

The face of a real person has certain unique features that are absent or difficult to replicate in most spoofing devices made of artificial materials.

The technology disclosed in this patent application uses this unique feature to provide anti-spoofing features for facial recognition, including optical sensing based on the unique optical absorption or reflection characteristics of biological parts of a person's face.

[0010]

In one aspect, the disclosed technology can be implemented to provide a system capable of facial recognition to determine whether a person is an authorized user.

The system may include, for example, a device structure on which is located an optical imaging system including an optical sensor array of optical detectors for capturing one or more facial images of a person's face for facial recognition and one or more probe light sources on said device structure for generating two or more probe lights of different optical illumination wavelengths to illuminate said person's face.

The one or more probe light sources are configured such that the two or more different optical illumination wavelengths of the probe light cause different levels of optical absorption by one or more portions of the real person's face to produce the respective two or more different light signals of different optical illumination wavelengths that are measurably different at the optical detection module to indicate whether the detected face is from a real person, rather than for evasion. A spoofing device for facial recognition. The system includes: an optical detection light detection module located on the device structure, separated from the optical sensor array of the optical imaging system, for detecting the two or more light caused by illumination of probe light of different optical illumination wavelengths to generate a probe detection signal; and a device control module coupled to receive information in the probe detection signal from the optical probe light detection module and collected information from the probe detection signal. The one or more facial images of the optical imaging system, the device control module is configured to process the collected one or more facial images to extract the facial features of the person, and the extracted facial signature is compared to the authorized user's known and existing digital data of facial features to determine if there is a match, as part of facial recognition. The device control module is configured and operable to process the detected light on the face of the person caused by the illumination of the probe light to measure two of the two or more different optical illumination wavelengths. The difference between the detected light wavelengths is optically illuminated, and the measured difference is used to determine whether the detected face is from a real person, as an additional part of facial recognition.

[0011]

In another aspect, the disclosed technology can be implemented to provide a method for performing facial recognition of a person to determine whether the person is an authorized user.

An example of the method includes: operating an optical imaging system to acquire one or more facial images of a person's face, the optical imaging system including an optical sensor array of optical detectors; processing the acquired one or more facial images to extract the facial features of said person, compare said extracted facial features with the authorized user's known and existing digital data of facial features to determine if there is a match, as part of facial recognition; guide the two probing light of one or more different optical illumination wavelengths to illuminate the face of the person; operating an optical probing light detection module separate from the optical sensor array of the optical imaging system to detect the face of the person the light caused by the illumination of the probe light of the two or more different optical illumination wavelengths on the face; the light caused by the illumination of the probe light on the detected face of the person is processed to measuring a difference in said light detected between two of said two or more different optical illumination wavelengths; and using said measured difference to determine whether said detected face is from a real person, as a face additional part of the identification.

[0012]

In some implementations, the method above can include selecting the two or more different optical illumination wavelengths of the probe light such that optical reflections from the pupils of the human's eyes are measurably different in signal strength, To allow the detected difference in the measurement of the optical reflections from the pupils of the person's eyes to be used as an indicator for whether the detected face is from a real person.

In other implementations, the above method may include: selecting the two or more different optical illumination wavelengths of the probe light to cause different levels of optical absorption by the facial skin of a real person such that the two or more The beam spot size of the probe light that is penetrating into the facial skin and scattered by the facial skin of two different optical illumination wavelengths is measurably different; the detection module is operated to collect the beam at the optical sensor array an image of the spot; processing the acquired images of the beam spot at the two or more different optical illumination wavelengths to measure a difference in the beam spot; and using the measured difference to determine the detected face as an additional part of facial recognition.

[0013]

In some implementations, the above method may further include placing two or more detection light sources adjacent to the optical sensor array, so that the detection lights of the two or more different optical illumination wavelengths are guided to illuminate the the person's face to cause a near retroreflection from the person's eyes to the probe light detection module when the person looks at the probe light detection module.

[0014]

In some implementations, the above method may further include: causing the probe light directed towards the person to be polarized light with a first polarization; detect light in a second polarization when the reflected probe light is reflected; and detect the power level of said reflected probe light at a particular polarization as whether said reflected probe light is from the eye of a person depolarizing light when reflected or indicators produced by reflections from man-made objects.

[0015]

In yet another aspect, the disclosed technology can be implemented to provide an apparatus for assisting facial recognition of a person to determine whether the person is a real person.

The device includes: an optical detection detection module, an optical detection sensor array including an optical detection detector, for detecting light from a person's face or capturing an image of a person's face to assist in facial recognition; and one or more detection A light source, located near the optical detection sensor array, is used to generate two or more detection lights of different optical illumination wavelengths to illuminate the face of the person.

The one or more probe light sources are configured such that the two or more different optical illumination wavelengths of the probe light cause different levels of optical absorption by one or more portions of the real person's face to produce the respective Two or more different light signals of different optical illumination wavelengths that are measurably different at the

optical detection module to indicate whether the detected face is from a real person, rather than for evasion A spoofing device for facial recognition.

In some implementations, the device may include: one or more optical polarizers, respectively placed in the optical paths of the one or more detection light sources, for generating the detection light directed toward the person, as polarized light of a first polarization; and one or more detection optical polarizers positioned to select only light of incident light to said array of optical detection sensors at a second polarization to reach said array of optical detection sensors for detection A power level of said light in said second polarization is received as an indicator of whether said light detected is a reflection from a human eye that depolarizes light when the light is reflected, or from an artificial object.

[0016]

In some implementations of the above apparatus, the one or more probe light sources may be configured to set (1) one of the two or more different optical illumination wavelengths of the probe light to be at or near 942 nm , at which wavelength the vitreous humor in the pupil of the real person or facial skin exhibits strong optical absorption and will (2) the other of the two or more different optical illumination wavelengths of the probe light Set to be positioned away from the 942 nm spectrum at selected optical illumination wavelengths at which the vitreous humor in the pupils of the real person or facial skin exhibits measurably less optical absorption.

[0017]

In some implementations of the above-mentioned device, the device may further include: one or more optical polarizers respectively placed in the optical path of the one or more detection light sources for generating the detection light directed to the person light being polarized at a first polarization; and one or more detection optical polarizers positioned to select only light of incident light to said array of optical detection sensors at a second polarization to reach said optical detection sensor an array to detect the power level of said light received at said second polarization as an indicator of whether said light detected is from a reflection from a human eye depolarizing light when the light is reflected, or from an artificial object .

[0018]

In some implementations of the above systems, methods or devices, the triggering mechanism or module may be implemented to trigger, turn on, activate or start face ID detection and processing when it is determined that the user intends to access the device.

One triggering technique is to use one or more pressure sensors on the mobile device to sense whether the user is holding or gripping the phone.

Another example of this trigger mechanism is the use of an acceleration sensor.

Yet another example of such a trigger mechanism is through an optical trigger that monitors the user's eyes.

[0019]

The following figures, description and claims provide a more detailed description of the above and other aspects of the disclosed technology, its implementations and features.

[0020]

Description of drawings

[0021]

1 is a block diagram of an example of a security device or system with face recognition for user authentication, the security device or system includes an optical imaging module, an optical authenticity detection module, and a user authentication processor, the optical imaging module is used to collect faces The image of the part is used for face recognition, the optical authenticity detection module is used to determine whether the object from which the face image is collected is a real person's face, and the user authentication processor is based on the information from the two modules used for face recognition. Determines whether to grant or deny access.

[0022]

FIG. 2 shows an example of the front of a mobile phone performing optical real-face detection in FIG. 1 .

[0023]

Figure 3 shows an example of an optical true face detection module package with an optical detection array and one or more detection light sources for generating detection light at two or more different optical illumination wavelengths , to induce different levels of optical absorption by one or more parts of a real person's face to produce a distinct optical signal that can be measured as different to indicate whether a detected face is from a real person, rather than a spoof used to evade facial recognition equipment.

In Figure 3, the following legend is used:

[0024]

29 - connector;

[0025]

31 - circuit board;

[0026]

33-miniature camera driver board;

[0027]

35 - miniature camera;

[0028]

35a-filter layer;

[0029]

35b - polarizer;

[0030]

35c - photodiode array;

[0031]

No. 37-1 light source;

[0032]

Optical transmitter No. 37a-1;

[0033]

Optical Processor 37b-1;

[0034]

No. 39-2 light source;

[0035]

Optical transmitter No. 39a-2;

[0036]

Optical Processor No. 39b-2;

[0037]

41 - camera lens assembly;

[0038]

41a - lens;

[0039]

No. 43-1 window material;

[0040]

45 - Proximity to the light source;

[0041]

45a - proximity light emitter;

[0042]

45b - proximity light processor; and

[0043]

47-2 window material.

[0044]

Figure 4 shows an example of the light absorption properties of water and other substances and two reflection images from a real person's eye at 942nm and 850nm respectively.

[0045]

Figure 5 shows a graphical representation of the beam expansion of infrared probe light entering the skin and being scattered by skin tissue at 942nm and 850nm.

[0046]

Figure 6 illustrates the use of light polarization in conjunction with the depolarizing properties of the human retina when reflecting polarized light.

[0047]

7A and 7B illustrate the optical detection of the face orientation of a face based on the image positions of certain face parts on a person's face.

[0048]

FIG. 8 shows an example of a real face detection algorithm based on the optical real face detection disclosed in this patent document.

[0049]

Detailed ways

[0050]

The facial recognition technology disclosed in this patent document provides optical sensing of unique features of a person's face when interacting with light to provide an anti-spoofing feature for facial recognition.

An electronic device or system can be equipped with the disclosed facial recognition mechanism to improve the security of accessing the device or system.

Such electronic devices or systems can include portable or mobile computing devices, such as smartphones, tablets, wrist-worn devices, and other wearable or portable devices, as well as larger electronic devices or systems, such as portable or desktop personal computers, Automatic Teller Machine (Automatic Teller Machine, ATM), various terminals for commercial or government purposes to various electronic systems, databases or information systems, and motor transportation systems including automobiles, ships, trains, airplanes and others.

[0051]

1 is a block diagram of an example of a security device or system with facial recognition for user authentication, including (1) an optical imaging module 110, (2) an optical authenticity detection module 120, and (3) a user authentication processor 130. The optical imaging module 110 is used to collect the image of the face for face recognition, and the optical authenticity detection module 120 is used to identify one or more objects from the facial image collected by the optical imaging module 110. The optical features are optically sensed and the user authentication processor 130 determines whether to grant or deny access to the security device or system 101 based on the information from the two modules 110 and 120 for facial recognition.

The user authentication processor 130 is in communication with the two optical sensing modules 110 and 120 to receive their output signals or data and to process such received signals or data to based on the face image from the optical imaging module 110 and from the optical authenticity. The authenticity detection of the detection module 120 is used to determine whether the object from which the face image is captured is a face of a real person.

In this example, the optical imaging module 110 for face recognition can be implemented in various configurations, such as the optical face ID system used in some of Apple's latest iPhone models.

This optical imaging module 110 for face recognition includes one or more high-resolution cameras for capturing detailed images of the face to extract facial features for face ID.

In some implementations, the optical imaging module 110 for facial recognition may include imaging hardware for acquiring 2D images in some designs, and may also include imaging hardware for acquiring 3D images in other designs.

The image of the face from the optical imaging module 110 is sent in the form of digital imaging data to the user authentication processor 130 for processing.

The user authentication processor 130 is programmed to perform facial recognition operations, including extracting facial features from images obtained at the optical imaging module 110 and performing computational operations, comparing the extracted facial features of a person with known and existing Facial features are compared to determine if a match exists.

Known and existing facial features of authorized users are typically stored in local memory on the device or system, and may also be stored on a connected server in the network or cloud.

In some facial recognition systems, access is granted when there is a match.

Under the techniques disclosed in this patent document, additional sensing is performed through the operation of the optical authenticity detection module 120 and through further processing performed in the user authentication processor 130 to determine the identity of a subject exhibiting matching facial features. Detect if an object is a real face.

[0052]

The optical authenticity detection module 120 is configured to optically detect a putative face or object presented to the device by optically detecting or measuring one or more optical parameters or characteristics of the putative face or object to determine whether the putative face or object is real face.

In some implementations, this can be achieved by directing probe light at two or more different optical illumination wavelengths to illuminate the face, and using designated optical probes separate from one or more optical sensor arrays of optical imaging module 110 . The light detection module is implemented to detect the light on the face of the person caused by the illumination of the probe light of two or more different optical illumination wavelengths.

This probe detection can be designed into the hardware of the optical truth detection module 120 to measure certain optical properties of the light reflected or scattered by the face due to the probe illumination, where certain optical properties are associated with unique characteristics. The unique characteristics are represented by a real person's face and lack artificial materials or structures commonly used in spoofing devices such as fake masks.

In some implementations, the optical authenticity detection module 120 may include at least one optical sensor array, which can capture images of faces, process the captured images for real face detection or face direction detection, and Auxiliary face ID authentication.

Thus, in this example, the optical authenticity detection module 120 is an additional feature added to the face ID mechanism for enhancing the security of the face ID authentication.

In practice, the disclosed optical true face detection can be used in different face ID authentication systems.

[0053]

In operation, optical measurements for real faces obtained by optical authenticity detection module 120 are sent to user authentication processor 130, which is also programmed to process detected Light caused by illumination of light to measure the difference in detected light between two optical illumination wavelengths of two or more different optical illumination wavelengths, and use the measured difference to determine whether a detected face is from a real person, as a face additional part of the identification.

Two specific examples of optical characteristics associated with human face skin and human eyes are provided below to illustrate the operation and function of the optical authenticity detection module 120 .

[0054]

In practice, as shown in FIG. 1 , the above facial recognition system with modules 110 , 120 and 130 may be coupled to device or system 101 to secure and restrict access to only one or more authorized users.

Devices or systems 101 that can benefit from such facial recognition systems can be a wide variety of different devices or systems including, for example, smartphones, tablet computers, wrist-worn devices and other wearable or portable devices, larger Electronic equipment or systems, portable or desktop personal computers, ATMs, various terminals for commercial or government use to various electronic systems, databases or information systems, and motorized transportation systems including automobiles, ships, trains, airplanes and others .

In a car equipped with such a facial recognition system, two optical modules 110 and 120 may be installed near an entrance to the car, such as each door of the driver or passenger (in some vehicles, including the entrance to the rear of the vehicle, For example, the rear door or hatch of a sports utility vehicle (Sport Utility Vehicle, SUV).

[0055]

FIG. 2 shows an example of the front of the mobile phone 11 performing optical real-face detection in FIG. 1 .

The mobile phone 11 includes a front camera and a face ID imaging module marked "21" on the upper right side of the front.

The module 21 includes the optical imaging module 110 in FIG. 1 .

The front optical real face sensor 27 on the upper left side of the front is the optical truth detection module 120 in FIG. 1 .

Other sensors are shown in FIG. 2 , including a pressure or force sensor 19 shown on the side of the mobile phone 11 .

In some implementations, one or more of such pressure or force sensors can be used to detect whether the user is gripping or holding the phone, and such sensors are used as triggers for turning on or activating the face shown in FIG. ID operation.

Pressure or force sensors are specific examples of such Face ID triggers or activation mechanisms, and in some designs other examples include proximity sensors or motion sensors.

[0056]

Figure 3 shows an example of an optical true face detection module package with an optical detection array and one or more detection light sources for generating detection light at two or more different optical illumination wavelengths , to induce different levels of optical absorption by one or more parts of a real person's face to produce a distinct optical signal that can be measured as different to indicate whether a detected face is from a real person, rather than a spoof used to evade facial recognition equipment.

This is an example implementation of optical true face detection module 120 in FIG. 1 .

In this example, a miniature infrared (IR) camera 35 and probe light sources 37 and 39 are assembled on a circuit board 31 .

The miniature IR camera 35 includes an optical sensor array of IR optical sensors for optical detection, detection light sources 37 and 39 for generating different illumination detection lights of different optical wavelengths, which are specially selected for detecting whether an object is a real person face.

Connector 29 connects the whole package with the mobile phone platform.

This design allows the optical true face detection module package to be integrated with different mobile phone platforms or different devices or systems using optical face ID authentication.

[0057]

Specifically, in the example shown in FIG. 3, the miniature IR camera 35 includes a lens or lens assembly 41a, an optical filter layer 35a, an optical polarizer 35b, and an optical sensor array 35c, which may be a photodiode of an IR sensing diode. array.

The probe light sources 37 , 39 and the proximity light source 45 are designed to cooperate with the miniature IR camera 35 .

In the light source, optical processors 37b, 39b, and 45b are provided so as to process beam divergence, polarization characteristics, and the like for functions.

The probe light can be linearly polarized or rotated, eg circularly polarized.

Light processors may include linear polarizers, wave plates, and liquid crystal or optically birefringent crystal materials to control the light polarization properties.

The light processor may also include a diffractive optical element (Diffraction Optical Element, DOE) for forming a light spot at a proper position.

[0058]

The example in Figure 3 also shows the application of a micro-driver board 33 to operate a micro-camera and detect a light source.

In this example, the probe light sources 37 and 39 are mounted close to the lens axis of the miniature camera to get a near retro reflection from a face looking at the moving screen.

Proximity light source 45 is mounted at a desired distance from the lens axis of the miniature camera and is used to provide optical sensing of object proximity, such as eye-to-camera distance.

[0059]

Figure 4 shows an example of the light absorption properties of water and other substances and two reflection images from a real person's eye at 942nm and 850nm respectively.

When a searchlight source such as 37 or 39 is placed close to the lens axis of a miniature camera, retro reflection may occur when the camera captures images from the human eye if the eye looks directly at the camera.

The reflection of light from the eye has a unique optical absorption feature at or near 942nm, which can be used for authenticity detection.

[0060]

Figure 4 further shows that the optical absorption in water is strong at the 942nm window and relatively weak at other wavelengths, such as the 850nm window which is still within the IP spectral range.

Nearly 98-99% of the vitreous humor of the human eye is water, therefore, light at or near 942 nm is strongly absorbed and hardly reaches the retina of the human eye behind the vitreous humor.

Based on this characteristic, one or more probe light sources emitting light at or near 942nm can be used as one of the retroreflecting light sources, and other IR probe light sources at 850nm or other IR wavelengths selected as the other retroreflecting light emitting IR probe light source, the IR probe light is not as strongly absorbed as the 942nm probe light.

[0061]

Due to the strong absorption at the 942nm window in the spectral range, the retroreflection of the human eye is weak when the 942nm light source is turned on.

However, at other IR wavelengths away from 942nm, such as 850nm, this IR light is not strongly absorbed by the vitreous humor of the human eye, so that it can reach the retina, which reflects this light, to create a strong optical reflection. This reflection, once picked up by the IR camera, produces a strongly retroreflected image. Using optical reflections at 942nm and another IR wavelength (eg, 850nm), the detected optical reflections at two different IR wavelengths are different in signal intensity based on the water-induced spectral signature. Measurements of this optical reflection can be evaluated to determine whether the retrograde signal is from a real eye, ie as an indicator of whether the object is the eye of a real person.

[0062]

Figure 5 shows a graphical representation of the beam expansion of infrared probe light entering the skin and being scattered by skin tissue at 942nm and 850nm.

When the light spot is projected onto the real face, part of the light is scattered on the surface, and part of the light is refracted into the surface. The refractive component is strongly scattered, expanding the spot profile to larger sizes and shapes. Since facial skin tissue (due to water) has strong absorption at or near 942nm, once the probe light at or around 942nm enters the skin, the probe light is strongly attenuated, and thus the beam spot shows weak expansion. Conversely, due to the weak absorption of facial skin tissue at 850nm or another IR wavelength of probe light away from 942nm, the 850nm (or other suitable wavelength band) probe light entering the facial skin will produce a strong beam profile Extended spot. A camera in the optical real-face module can be used to capture such images, and the captured images can be used to analyze facial realism.

[0063]

6 illustrates an example of the use of light polarization in conjunction with the depolarization properties of the human retina when reflecting polarized light in some implementations of the disclosed technology.

This aspect of the disclosed technology is based on the realization that the light reflection of the human eye can be simulated with a lens and a reflector in its focal plane, in particular the retina of the human eye strongly depolarizes incident light and is therefore the only reflector. This unique property of the retina can be used to detect whether an object is a real eye.

[0064]

Specifically, when polarized light is incident on a real live human eye, the retroreflection contains a large proportion (-80% or higher) of the depolarized component.

Even with a rough reflector at the focal plane, the lens hardly produces similar depolarization characteristics. Therefore, a polarizer, such as a linear polarizer, may be installed on each detection light source in the optical real face detection module 110 in FIG. 1 to ensure that the detection light from each detection light source has a desired polarization. Based on this, a detection optical polarizer was installed in the optical path of the miniature camera to evaluate the depolarization of the returning light from the reflective object. Reverse light reflected from a real living eye is almost completely depolarized. The retrograde light reflected from the fake eye is either partially polarized or not depolarized. The polarization direction of the detection optical polarizer at the camera may be oriented perpendicular to the polarization of the detection optical polarizer at the detection light source. This can be used to detect the authenticity of objects.

[0065]

In addition, the depolarization characteristics of the reflected light spot images on the face can be analyzed in combination with the absorption characteristics discussed in Figures 4 to 5 to further enhance the trueness detection, because it is usually difficult to find light wavelength windows with strong absorption in the 942nm window and 850nm window. A fake face material with the same depolarization characteristics.

By analyzing the difference in spot expansion and depolarization characteristics between the 942nm wavelength band and other reference wavelength bands, we can obtain the criteria for judging whether the face is real.

[0066]

The processing of Face ID is a computationally intensive process and tends to drain the battery.

When it is determined that a user intends to access a device such as a mobile phone, various trigger mechanisms can be used to turn on, activate or begin Face ID detection and processing

[0067]

One triggering technique is to use one or more pressure sensors on the mobile device to sense whether the user is holding or gripping the phone.

Referring to Fig. 2, one or more grip force sensors or pressure sensors 19 may be integrated in the frame of the mobile platform. This sensor can be visible as indicated at 19, or it can be completely hidden in the frame so that it cannot be seen.

[0068]

Another example of this trigger mechanism is a G-sensor (or acceleration sensor) in a mobile device or phone to sense the movement of the platform, such as a smartphone.

Almost all smartphones have a G sensor integrated in the system, and the sensor output can be used to trigger Face ID detection and processing.

[0069]

Yet another example of such a trigger mechanism is through an optical trigger that monitors the user's eyes.

Security sensors such as face ID sensors or fingerprint sensors are activated when optical sensors detect that the user's eyes are looking at a platform such as a smartphone. In this regard, the probe light detection module, which may be part of the optical truth detection module 120 in FIG. The authentication processor 130 processes the image locations of key feature points on the recognized person's face to determine whether the person is looking towards the probe light detection module.

[0070]

7A and 7B illustrate an example of optical detection of the face orientation of a face based on the image positions of certain face parts on a person's face.

When a person is gazing at the face ID camera, the images of the eyes and the image positions of other face parts (eg, nose tip, jaw, etc.) are generally symmetrical in the image, as shown in the upper left panel of FIG. 7A. When the person is facing other directions, the image positions of the eyes and other face parts (eg, the tip of the nose) are asymmetric in the image captured by

the face ID camera, as shown in the upper right figure of FIG. 7 . Based on image analysis, the Face ID sensor can tell if the user is looking at the platform.

[0071]

FIG. 8 shows an example of a real face detection algorithm based on the optical real face detection disclosed in this patent document.

The process first detects user state based on one or more trigger signals, such as the user's hand movement and grip relative to the orientation of the user's face relative to the phone screen. If this detection generates a trigger, the Face ID hardware is turned on for optical Face ID sensing and processing for Face ID authentication. Access is granted as long as facial feature matching and true face detection are successful. This combined use of facial feature matching and true face detection can reduce the success of face spoofing attempts, thereby enhancing the security level of Face ID authentication.

[0072]

While this patent document contains many specifics, these should not be construed as limitations on the scope of any invention or of what may be claimed, but rather as descriptions of features specific to particular embodiments of particular inventions.

Certain features that are described in this patent document, in the context of separate embodiments, can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Furthermore, although the features described above work in some combinations and in originally claimed combinations, one or more features from a claimed combination can in some cases be departed from the combination, and the claimed combination Morphs that can be directed to partial assemblies or partial assemblies.

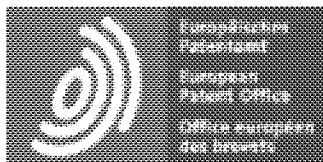
[0073]

Similarly, while operations are depicted in the figures in a particular order, this should not be construed as requiring that those operations be performed in the particular order shown, or in sequential order, or that all illustrated operations be performed, to achieve desirable results.

Moreover, the isolation of various system components in the embodiments described in this patent document should not be understood as requiring such separation in all embodiments.

[0074]

Only a few implementations and examples are described, and other implementations, enhancements and changes can be made based on what is described and illustrated in this patent document.



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CLAIMS CN111164610A

1.

A method for performing facial recognition of a person to determine whether the person is an authorized user, comprising:

operating an optical imaging system to acquire one or more facial images of a person's face, the optical imaging system including an optical sensor array of optical detectors;

processing said one or more facial images captured to extract facial features of said person, comparing said extracted facial features with digital data of known and existing facial features of authorized users to determine whether there is matching, as part of facial recognition;

directing probe light of two or more different optical illumination wavelengths to illuminate the face of the person;

operating an optical probe light detection module separate from the optical sensor array of the optical imaging system to detect illumination on the face of the person caused by the probe light of the two or more different optical illumination wavelengths the light;

processing said light on the detected face of said person caused by illumination of said probe light to measure said light detected between two of said two or more different optical illumination wavelengths difference in light; and

The difference measured is used to determine whether the detected face is from a real person as an additional part of facial recognition.

2.

The method of claim 1, wherein:

The two or more different optical illumination wavelengths of the probe light are selected such that the optical reflections from the pupils of the eyes of the person are measurably different in signal strength to allow the detected light from the person to The measured difference in the optical reflection of the pupils of the eyes serves as an indicator for whether the face being detected is from a real person.

3.

The method of claim 1, wherein:

The two or more different optical illumination wavelengths of the probe light are selected to induce different levels of optical absorption by the facial skin of a real person such that positive penetration of the two or more different optical illumination wavelengths into the facial skin from which the beam spot size of the probe light scattered by the facial skin is measurably different;

operating the probing detection module to acquire an image of a beam spot at the optical sensor array;

processing the acquired images of the beam spot at the two or more different optical illumination wavelengths to measure a difference in the beam spot; and

The difference measured is used to determine whether the detected face is from a real person as an additional part of facial recognition.

4.

A method according to claim 1, 2 or 3, wherein:

one of said two or more different optical illumination wavelengths of said probe light is at or near 942 nm, at which wavelength the vitreous humor in the pupil of said real person or facial skin exhibits strong optical absorption; and

The other of the two or more different optical illumination wavelengths of the probe light is positioned away from the 942 nm spectrum at a selected optical illumination wavelength at which vitreous humor in the pupils of the real person or facial skin exhibits measurably less optical absorption.

5.

The method of claim 4, wherein:

The selected optical illumination wavelength of another of the two or more different optical illumination wavelengths of the probe light spectrally located away from 942 nm is near 850 nm.

6.

The method according to claim 1, further comprising:

Two or more detection light sources are positioned adjacent to the optical sensor array such that the detection light of the two or more different optical illumination wavelengths is directed to illuminate the face of the person to A near retroreflection from the eye of the person to the probe light detection module is caused when directed toward the probe light detection module.

7.

The method according to claim 1, further comprising:

causing the probe light directed toward the person to be polarized at a first polarization;

operating the probe light detection module to detect light at a second polarization upon detection of reflected probe light from a pupil of an eye of the person; and

The detected power level of the reflected probe light at a particular polarization is taken as an indicator of whether the reflected probe light is a reflection from a human eye that depolarizes light when the light is reflected, or from an artificial object.

8.

The method of claim 7, wherein the second polarization is orthogonal to the first polarization.

9.

The method of claim 8, wherein the first and second polarizations are linear polarizations that are orthogonal to each other.

10.

The method of claim 8, wherein the first and second polarizations are circular polarizations that are orthogonal to each other.

11.

The method of claim 1, wherein:

The detection module includes an infrared optical sensor array of infrared optical detectors configured to detect light in the infrared spectral range.

12.

The method of claim 11, wherein:

The detection module includes:

a first infrared optical sensor array of the infrared optical detector configured to detect light at a first optical illumination wavelength; and

The separate second infrared optical sensor array of the infrared optical detector is configured to detect light at a different second optical illumination wavelength.

13.

The method according to claim 1, comprising:

Before starting said facial recognition,

operating the probe light detection module to capture one or more human face images to identify image locations of key feature points on the person's face,

processing the identified image locations of key feature points on the person's face to determine whether the person is looking at the probe light detection module, and

After determining that the person is looking at the probe light detection module, the facial recognition is initiated to determine whether there is a match between the extracted facial features and known and existing facial features of authorized users, and based on the detection A difference in the measurements of the optical reflectance of the two or more different optical illumination wavelengths determines whether the detected face is from a real person.

14.

The method of claim 13, wherein:

The image positions of the key feature points on the face of the person include the image positions of the eyes and the image positions of the nose or mouth,

The processing of the identified image locations of key feature points on the person's face includes evaluating the relative image locations of the eyes relative to the nose or mouth.

15.

The method according to claim 1, comprising:

Before starting said facial recognition,

operating one or more pressure sensors placed on the handheld device to measure one or more forces exerted on the one or more pressure sensors as an indicator of whether the person is holding the handheld device, the The optical imaging system and the detection module are located on the handheld device,

processing the measured one or more forces to generate an affirmative signal indicating whether the person is holding the hand-held device, and

After determining that the person is holding the handheld device, the facial recognition is initiated to determine whether there is a match between the extracted facial features and known and existing facial features of authorized users, and based on the detected The difference in the measurements of the optical reflectance of two or more different optical illumination wavelengths determines whether said detected face is from a real person.

16.

The method according to claim 1, comprising:

Before starting said facial recognition,

operating an accelerometer placed on a handheld device on which the optical imaging system and the optical detection module are located to measure a state of motion of the handheld device, and

After the measured state from the accelerometer indicates that the person is moving the hand-held device, the facial recognition is initiated to determine the difference between the extracted facial features and known and existing facial features of an authorized user, and determining whether the detected face is from a real person based on the detected difference in measured optical reflectance of the two or more different optical illumination wavelengths.

17.

An apparatus for assisting facial recognition of a person to determine whether said person is a real person, comprising:

an optical detection module comprising an optical detection sensor array of an optical detection detector for detecting light from a person's face or capturing an image of a person's face to assist in facial recognition; and

one or more detection light sources, located adjacent to the optical detection sensor array, for generating two or more detection lights of different optical illumination wavelengths to illuminate the face of the person, wherein the one or more The probe light source is configured such that the two or more different optical illumination wavelengths of the probe light cause different levels of optical absorption by one or more portions of the real person's face to respectively produce the two or more Different optical signals of different optical illumination wavelengths, which are measurably different at the optical detection module to indicate whether the detected face is from a real person and not a spoofing device used to evade the facial recognition .

18.

The apparatus of claim 17, wherein:

The one or more probe light sources are configured to set (1) one of the two or more different optical illumination wavelengths of the probe light at or near 942 nm at which the human Vitreous humor in the pupil of the or facial skin exhibits strong optical absorption and (2) the other of said two or more different optical illumination wavelengths of said probe light is set at a selected optical illumination wavelength Locate away from the 942nm spectrum, at which wavelength the vitreous humor in the pupils of the real person or the facial skin exhibit measurably less optical absorption.

19.

The apparatus of claim 17, comprising:

one or more optical polarizers respectively positioned in the optical path of said one or more probe light sources for generating said probe light directed towards said person as polarized light with a first polarization; and

one or more detection optical polarizers positioned to select only light in a second polarization of incident light to the optical detection sensor array to reach the optical detection sensor array to detect received light in the second polarization The power level of the light as an indicator of whether the detected light is from a human eye that depolarizes light when it is reflected, or from a reflection from an artificial object.

20.

The apparatus of claim 19, wherein the second polarization is orthogonal to the first polarization.

21.

A system capable of facial recognition to determine whether a person is an authorized user, comprising:

a device structure on which is located an optical imaging system comprising an optical sensor array of optical detectors for capturing one or more facial images of a person's face for facial recognition;

One or more probe light sources on the device structure for generating probe light of two or more different optical illumination wavelengths to illuminate the face of the person, wherein the one or more probe light sources are configured such that the two or more different optical illumination wavelengths of the probe light cause different levels of optical absorption by one or more portions of the real person's face to respectively produce the two or more different optical illumination different wavelengths of optical signals that are measurably different at the optical detection module to indicate whether the detected face is from a real person, rather than a spoofing device used to evade said facial recognition;

an optical detection light detection module located on the device structure, separate from the optical sensor array of the optical imaging system, for detecting light induced by illumination of the probe light at a wavelength to produce a probe detection signal; and

a device control module, coupled to receive information in the probe detection signal from the optical probe light detection module and the one or more facial images collected from the optical imaging system, the device control module configured to process said one or more facial images captured to extract facial features of said person, to compare said extracted facial features to digital data of known and existing facial features of an authorized user, to determining whether there is a match, as part of facial recognition, the device control module being configured and operable to process the detected light on the face of the person caused by the illumination of the probe light to measure the a difference in said light detected between two of two or more different optical illumination wavelengths, and using said difference measured to determine whether said detected face is from a real person, as an additional feature of facial recognition part.

22.

The system of claim 21, wherein:

The device structure is a part of the mobile device and is configured such that when the user looks at the display screen of the mobile device, the optical imaging system and the optical detection module are placed facing the user.

23.

The system of claim 22, comprising:

One or more pressure sensors on the device structure of the mobile phone can be used to determine whether the person is holding the mobile device by measuring one or more forces exerted on the one or more pressure sensors. indicator of the ; and

The device control module is operable, prior to initiating the facial recognition, to process the one or more measured forces to generate an affirmative signal indicative of whether the person is holding the mobile device, and upon determining that the person While the mobile device is being held, initiating the facial recognition to determine whether there is a match between the extracted facial features and known and existing facial features of an authorized user, and based on the two or more differences Said difference in said light detected between two of the optical illumination wavelengths determines whether said detected face is from a real person.

24.

The system of claim 23, wherein a pressure sensor is located on an edge of the mobile device.

25.

The system of claim 22, wherein:

the mobile device includes an accelerometer for measuring a state of movement of the device; and

The device control module is operable, after the measured state from the accelerometer indicates that the person is moving the handheld device, to perform the facial recognition to determine that the extracted facial features are consistent with those of an authorized user. whether there is a match between known and existing facial features, and based on said difference in said light detected between two of said two or more different optical illumination wavelengths, determining detection Whether the received face is from a real person.

26.

The system of claim 21, wherein:

The device structure is part of a laptop computer and is configured such that the optical imaging system and the optical detection module are positioned facing the user when the user looks at the display screen of the laptop computer.

27.

The system of claim 21, wherein:

The device structure is part of the entrance to the secure place and is configured such that the optical imaging system and the optical detection module are placed facing the user when the user looks at the display screen.

28.

The system of claim 21, wherein:

The device structure is a part of the vehicle and is configured such that the optical imaging system and the optical detection module are placed facing the user when the user looks at the display screen.

29.

The system of claim 21, wherein:

The one or more probe light sources are configured to generate the two or more different optical beams of the probe light in such a way that the optical reflection from the pupil of the human eye is measurably different in signal strength. Illumination wavelengths to allow the detected difference in measurements of optical reflections from the pupils of the person's eyes as an indicator of whether the detected face is from a real person.

30.

The system of claim 21, wherein:

The one or more probe light sources are configured to generate the two or more different optical illumination wavelengths of the probe light in a manner that causes different levels of light absorption by the facial skin of a real person such that the two or The beam spot sizes of the probe light being diffused by the facial skin penetrating into the facial skin for a plurality of different optical illumination wavelengths are measurably different;

the optical detection module is operable to acquire an image of the beam spot; and

The device control module is operable to process the acquired images of the beam spot at the two or more different optical illumination wavelengths to measure a difference in the beam spot; and use the measured difference to determine Detect if said face is from a real person, as an additional part of facial recognition.

31.

The system of claim 21, wherein:

one of said two or more different optical illumination wavelengths of said probe light is at or near 942 nm, at which wavelength the vitreous humor in the pupil of said real person or facial skin exhibits strong optical absorption; and

The other of the two or more different optical illumination wavelengths of the probe light is positioned away from the 942 nm spectrum at a selected optical illumination wavelength at which vitreous humor in the pupils of the real person or facial skin exhibits measurably less optical absorption.

32.

The system of claim 21, wherein:

The selected optical illumination wavelength of another of the two or more different optical illumination wavelengths of the probe light spectrally located away from 942 nm is near 850 nm.

33.

The system of claim 21, wherein the optical detection module comprises:

a first infrared optical sensor array of infrared optical detectors configured to detect light at a first optical illumination wavelength;
and

A separate second infrared optical sensor array of infrared optical detectors is configured to detect light at a different second optical illumination wavelength.



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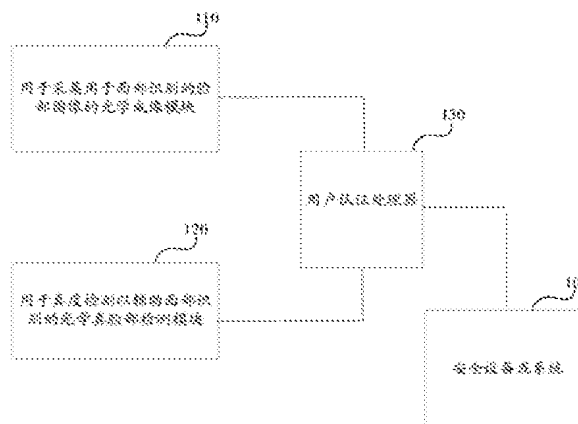
权利要求书5页 说明书9页 附图9页

(54)发明名称

用于增强面部识别安全性的反欺骗真脸部
感测

(57)摘要

设备、系统、技术和过程的技术用于通过使用光学感测和其他感测机制来提供用于面部识别的反欺骗特征,其针对面部欺骗设备或技术具有增强的安全性,以探究真人的脸部的在大多数由人工材料制成的欺骗设备中缺乏的或难以复制的某些独特的特征,包括基于人的脸部的生物部分的独特光学吸收或反射特征的光学感测。



1. 一种用于进行人的面部识别以确定所述人是否为授权用户的方法,包括:

操作光学成像系统以采集人的脸部的一个或多个脸部图像,所述光学成像系统包括光学检测器的光学传感器阵列;

处理采集的所述一个或多个脸部图像以提取所述人的面部特征,将提取的所述面部特征与授权用户的已知和现有的面部特征的数字数据进行比较,以确定是否存在匹配,作为面部识别的一部分;

引导两个或更多个不同光学照明波长的探测光以照亮所述人的脸部;

操作与所述光学成像系统的所述光学传感器阵列分离的光学探测光检测模块,以检测所述人的脸部上的由所述两个或更多个不同光学照明波长的探测光的照明引起的光;

处理检测的所述人的脸部上的由所述探测光的照明引起的所述光,以测量所述两个或更多个不同光学照明波长中两个光学照明波长之间检测的所述光的差异;以及

使用测量的所述差异来确定检测的所述脸部是否来自真人,作为面部识别的附加部分。

2. 根据权利要求1所述的方法,其中:

选择所述探测光的所述两个或更多个不同光学照明波长,使得来自所述人的眼睛的瞳孔的光学反射在信号强度上可测量到不同,以允许将检测的来自所述人的眼睛的瞳孔的光学反射的测量的差异作为用于检测的所述脸部是否来自真人的指示符。

3. 根据权利要求1所述的方法,其中:

选择所述探测光的所述两个或更多个不同光学照明波长,以引起真人的面部皮肤不同水平的光学吸收,使得所述两个或更多个不同光学照明波长的正渗透入所述面部皮肤而被所述面部皮肤散射的所述探测光的束斑尺寸可测量到不同;

操作所述探测检测模块以采集所述光学传感器阵列处束斑的图像;

处理采集的所述两个或更多个不同光学照明波长的所述束斑的图像,以测量所述束斑的差异;以及

使用测量的所述差异来确定检测的所述脸部是否来自真人,作为面部识别的附加部分。

4. 根据权利要求1、2或3所述的方法,其中:

所述探测光的所述两个或更多个不同光学照明波长之一为942nm或接近942nm,在该波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出强光学吸收;并且

所述探测光的所述两个或更多个不同光学照明波长中的另一个在选定的光学照明波长处远离942nm光谱定位,在该波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出可测量较少的光学吸收。

5. 根据权利要求4所述的方法,其中:

远离942nm光谱定位的所述探测光的所述两个或更多个不同光学照明波长中的另一个的所述选定的光学照明波长接近850nm。

6. 根据权利要求1所述的方法,还包括:

两个或更多个探测光源与所述光学传感器阵列相邻放置,从而所述两个或更多个不同光学照明波长的探测光被引导照射所述人的脸部,以当所述人看向所述探测光检测模块时引起从所述人的眼睛向所述探测光检测模块的近回射。

7. 根据权利要求1所述的方法,还包括:

使引导向所述人的所述探测光为以第一偏振的偏振光;

操作所述探测光检测模块以在检测来自所述人的眼睛的瞳孔的反射探测光时检测处于第二偏振的光;以及

将检测的以特定偏振的所述反射探测光的功率水平作为所述反射探测光是由来自当反射光时消偏振光的人的眼睛还是来自人造物体的反射产生的指示符。

8. 根据权利要求7所述的方法,其中,所述第二偏振与所述第一偏振正交。

9. 根据权利要求8所述的方法,其中,所述第一偏振和第二偏振是彼此正交的线性偏振。

10. 根据权利要求8所述的方法,其中,所述第一偏振和第二偏振是彼此正交的圆偏振。

11. 根据权利要求1所述的方法,其中:

所述探测检测模块包括红外光学检测器的红外光学传感器阵列,所述红外光学感测器阵列被构造为检测红外光谱范围内的光。

12. 根据权利要求11所述的方法,其中:

所述探测检测模块包括:

所述红外光学检测器的第一红外光学传感器阵列,被构造为检测第一光学照明波长的光;以及

所述红外光学检测器的分离的第二红外光学传感器阵列,被构造为检测不同的第二光学照明波长的光。

13. 根据权利要求1所述的方法,包括:

在开始所述面部识别之前,

操作所述探测光检测模块以采集一个或多个人脸图像,以识别所述人的脸部上关键特征点的图像位置,

处理识别的所述人的脸部上关键特征点的图像位置,以确定所述人是否正看向所述探测光检测模块,以及

在确定所述人正看向所述探测光检测模块后,开始所述面部识别以确定提取的所述面部特征与授权用户的已知和现有的面部特征之间是否存在匹配,并基于检测的所述两个或更多个不同光学照明波长的光学反射的测量的差异,确定检测的所述脸部是否来自真人。

14. 根据权利要求13所述的方法,其中:

所述人的脸部上关键特征点的图像位置包括眼睛的图像位置和鼻子或嘴巴的图像位置,

所述处理识别的所述人的脸部上关键特征点的图像位置包括评估相对于鼻子或嘴巴的眼睛的相对图像位置。

15. 根据权利要求1所述的方法,包括:

在开始所述面部识别之前,

操作放置在手持设备上的一个或多个压力传感器,以测量施加在所述一个或多个压力传感器上的一个或多个力,作为所述人是否正手持所述手持设备的指示符,所述光学成像系统和所述探测检测模块位于所述手持设备上,

处理测量的所述一个或多个力以产生指示所述人是否正手持所述手持设备的肯定信

号,以及

在确定所述人正手持所述手持设备后,开始所述面部识别以确定提取的所述面部特征与授权用户的已知和现有的面部特征之间是否存在匹配,并基于检测的所述两个或更多个不同光学照明波长的光学反射的测量的差异,确定检测的所述脸部是否来自真人。

16. 根据权利要求1所述的方法,包括:

在开始所述面部识别之前,

操作放置在手持设备上的加速计,以测量所述手持设备的运动的状态,所述光学成像系统和所述光学探测检测模块位于所述手持设备上,以及

在测量的来自所述加速计的所述状态指示所述人正移动所述手持设备后,开始所述面部识别以确定提取的所述面部特征与授权用户的已知和现有的面部特征之间是否存在匹配,并基于检测的所述两个或更多个不同光学照明波长的光学反射的测量的差异,确定检测的所述脸部是否来自真人。

17. 一种用于辅助人的面部识别以确定所述人是否为真人的设备,包括:

光学探测检测模块,包括光学探测检测器的光学探测传感器阵列,用于检测来自人的脸部的光或采集人的脸部的图像,以辅助面部识别;以及

一个或多个探测光源,位于所述光学探测传感器阵列附近,用于产生两个或更多个不同光学照明波长的探测光以照亮所述人的脸部,其中,所述一个或多个探测光源被构造使得所述探测光的所述两个或更多个不同光学照明波长引起真人的脸部的一个或多个部分不同水平的光学吸收,以分别产生所述两个或更多个不同光学照明波长的不同的光信号,所述光信号在所述光学探测检测模块处可测量到不同以指示检测的所述脸部是否来自真人,而不是用于逃避所述面部识别的欺骗设备。

18. 根据权利要求17所述的设备,其中:

所述一个或多个探测光源被构造为将(1)所述探测光的所述两个或更多个不同光学照明波长之一设置为942nm或接近942nm,在所述波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出强光学吸收以及将(2)所述探测光的所述两个或更多个不同光学照明波长中的另一个设置为在选定的光学照明波长处远离942nm光谱定位,在所述波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出可测量的较少的光学吸收。

19. 根据权利要求17所述的设备,包括:

一个或多个光学偏振器,分别放置在所述一个或多个探测光源的光路中,用于产生引导向所述人的所述探测光,成为以第一偏振的偏振光;以及

一个或多个检测光学偏振器,被定位成仅选择处于第二偏振的至所述光学探测传感器阵列的入射光中的光到达所述光学探测传感器阵列,以检测接收的处于所述第二偏振的所述光的功率水平,作为检测的所述光是由来自当反射光时消偏振光的人的眼睛还是来自人造物体的反射产生的指示符。

20. 根据权利要求19所述的设备,其中,所述第二偏振与所述第一偏振正交。

21. 一种能够进行面部识别以确定人是否为授权用户的系统,包括:

光学成像系统位于其上的设备结构,所述光学成像系统包括光学检测器的光学传感器阵列,用于采集人的脸部的一个或多个脸部图像用于面部识别;

所述设备结构上的一个或多个探测光源,用于产生两个或更多个不同光学照明波长的

探测光以照亮所述人的脸部,其中,所述一个或多个探测光源被构造使得所述探测光的所述两个或更多个不同光学照明波长引起真人的脸部的一个或多个部分不同水平的光学吸收,以分别产生所述两个或更多个不同光学照明波长的不同的光信号,所述光信号在光学探测检测模块处可测量到不同以指示检测的所述脸部是否来自真人,而不是用于逃避所述面部识别的欺骗设备;

位于所述设备结构上的光学探测光检测模块,与所述光学成像系统的所述光学传感器阵列分离,用于检测所述人的脸部上的由所述两个或更多个不同光学照明波长的探测光的照明引起的光以产生探测检测信号;以及

设备控制模块,被耦合以接收来自所述光学探测光检测模块的所述探测检测信号中的信息以及采集的来自所述光学成像系统的所述一个或多个脸部图像,所述设备控制模块被构造为处理采集的所述一个或多个脸部图像以提取所述人的面部特征,将提取的所述面部特征与授权用户的已知和现有的面部特征的数字数据进行比较,以确定是否存在匹配,作为面部识别的一部分,所述设备控制模块被构造且可操作为处理检测的所述人的脸部上的由所述探测光的照明引起的所述光,以测量所述两个或更多个不同光学照明波长中两个光学照明波长之间检测的所述光的差异,并且应用测量的所述差异来确定检测的所述脸部是否来自真人,作为面部识别的附加部分。

22. 根据权利要求21所述的系统,其中:

所述设备结构是移动设备的一部分,并且被构造使得当用户看所述移动设备的显示屏时,所述光学成像系统和所述光学探测检测模块被放置成面向用户。

23. 根据权利要求22所述的系统,包括:

所述移动电话的所述设备结构上的一个或多个压力传感器,可通过测量施加在所述一个或多个压力传感器上的一个或多个力,作为所述人是否正手持所述移动设备的指示符;以及

所述设备控制模块可操作为,在开始所述面部识别之前,处理测量的所述一个或多个力以产生指示所述人是否正手持所述移动设备的肯定信号,以及在确定所述人正手持所述移动设备后,开始所述面部识别以确定提取的所述面部特征与授权用户的已知和现有的面部特征之间是否存在匹配,并基于所述两个或更多个不同光学照明波长中两个光学照明波长之间检测的所述光的所述差异,确定检测的所述脸部是否来自真人。

24. 根据权利要求23所述的系统,其中,一个压力传感器位于所述移动设备的边缘上。

25. 根据权利要求22所述的系统,其中:

所述移动设备包括加速计,用于测量所述设备的移动的状态;以及

所述设备控制模块可操作为,在测量的来自所述加速计的所述状态指示所述人正移动所述手持设备之后,进行所述面部识别以确定提取的所述面部特征与授权用户的已知的和现有的面部特征之间是否存在匹配,并基于所述两个或更多个不同光学照明波长中的两个光学照明波长之间检测的所述光的所述差异,确定检测到的脸部是否来自真人。

26. 根据权利要求21所述的系统,其中:

所述设备结构是膝上型计算机的一部分,并且被构造使得当用户看所述膝上型计算机的显示屏时,所述光学成像系统和所述光学探测检测模块被放置成面向用户。

27. 根据权利要求21所述的系统,其中:

所述设备结构是向安全场所的入口的一部分,并且被构造使得当用户看显示屏时,所述光学成像系统和所述光学探测检测模块被放置成面向用户。

28. 根据权利要求21所述的系统,其中:

所述设备结构是车辆的一部分,并且被构造使得当用户看显示屏时,所述光学成像系统和所述光学探测检测模块被放置成面向用户。

29. 根据权利要求21所述的系统,其中:

所述一个或多个探测光源被构造为以来自所述人的眼睛的瞳孔的光学反射在信号强度上可测量到不同的方式来生成所述探测光的所述两个或更多个不同光学照明波长,以允许将检测的来自所述人的眼睛的瞳孔的光学反射的测量的差异作为检测的所述脸部是否来自真人的指示符。

30. 根据权利要求21所述的系统,其中:

所述一个或多个探测光源被构造为以引起真人的面部皮肤不同水平的光吸收的方式来生成所述探测光的所述两个或更多个不同光学照明波长,使得所述两个或更多个不同光学照明波长的正渗透入所述面部皮肤而被所述面部皮肤散射的所述探测光的束斑尺寸为可测量到不同;

所述光学探测检测模块可操作为采集所述束斑的图像;以及

所述设备控制模块可操作性为处理采集的所述两个或更多个不同光学照明波长的所述束斑的图像,以测量所述束斑的差异;以及使用测量的所述差异来确定检测的所述脸部是否来自真人,作为面部识别的附加部分。

31. 根据权利要求21所述的系统,其中:

所述探测光的所述两个或更多个不同光学照明波长之一为942nm或接近942nm,在该波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出强光学吸收;并且

所述探测光的所述两个或更多个不同光学照明波长中的另一个在选定的光学照明波长处远离942nm光谱定位,在该波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出可测量较少的光学吸收。

32. 根据权利要求21所述的系统,其中:

远离942nm光谱定位的所述探测光的所述两个或更多个不同光学照明波长中的另一个的所述选定的光学照明波长接近850nm。

33. 根据权利要求21所述的系统,其中,所述光学探测检测模块包括:

红外光学检测器的第一红外光学传感器阵列,被构造为检测第一光学照明波长的光;以及

红外光学检测器的分离的第二红外光学传感器阵列,被构造为检测不同的第二光学照明波长的光。

用于增强面部识别安全性的反欺骗真脸部感测

[0001] 相关申请的交叉引用

[0002] 本专利文件要求申请人为深圳市汇顶科技股份有限公司于2019年1月4日提交的申请号为62/788,749、发明名称为“用于增强面部识别安全性的反欺骗真脸部感测”的美国临时专利申请的权益和优先权,该申请通过引用并入作为本专利文件公开内容的一部分。

技术领域

[0003] 本专利文件涉及人的面部特征的检测,作为用于授权访问安全电子设备、安全数字系统、安全设施或场所的用户认证的一部分。

背景技术

[0004] 各种传感器可以在电子设备或系统中实现,用于检测来自人或人的独特的识别特征并提供用户认证,以准许或拒绝访问包括便携式或移动计算设备(例如,笔记本电脑、平板电脑、智能手机)、游戏系统、各种数据库、信息系统或大型计算机控制系统的安全设备或系统。可以使用各种认证机制来保护敏感或机密数据并阻止未经授权的访问。电子设备上的用户认证可以通过一种或多种形式的生物特征标识符执行,其可以单独使用或除常规密码认证方法之外使用。生物特征标识符的一种流行形式就是人的指纹图案。指纹传感器可以内置在该电子设备中,以读取用户的指纹图案,使得该设备只能由该设备的授权用户通过认证该授权用户的指纹图案来解锁。另一个示例是成像设备,该成像设备检测人的脸部以提取人的独特面部特征作为用于用户认证的生物特征标识符。苹果公司的一些最新iPhone型号是使用面部成像来提供脸部识别以认证人是否为授权用户的设备的示例。

[0005] 然而,人的生物特征标识符,例如面部特征,可以通过使用欺骗工具或设备来复制或制造,以逃避基于面部识别的用户认证机制。

发明内容

[0006] 本专利文件公开了通过使用光学感测和其他感测机制来提供用于面部识别的反欺骗特征的设备、系统、技术和过程,其针对面部欺骗设备或技术具有增强的安全性。真人的脸部具有大多数由人工材料制成的欺骗设备中缺乏的或难以复制的某些独特的特征。本专利申请中公开的技术使用这种独特的特征来提供用于面部识别的反欺骗特征,包括基于人的脸部的生物部分的独特光学吸收或反射特征的光学感测。

[0007] 一方面,所公开的技术可以被实现为提供一种能够进行面部识别以确定人是否为授权用户的系统。该系统可以包括,例如:光学成像系统位于其上的设备结构,所述光学成像系统包括光学检测器的光学传感器阵列,用于采集人的脸部的一个或多个脸部图像用于面部识别;以及所述设备结构上的一个或多个探测光源,用于产生两个或更多个不同光学照明波长的探测光以照亮所述人的脸部。所述一个或多个探测光源被构造使得所述探测光的所述两个或更多个不同光学照明波长引起真人的脸部的一个或多个部分不同水平的光学吸收,以分别产生所述两个或更多个不同光学照明波长的不同的光信号,所述光信号在

所述光学探测检测模块处可测量到不同以指示检测的所述脸部是否来自真人,而不是用于逃避所述面部识别的欺骗设备。该系统包括:位于所述设备结构上的光学探测光检测模块,与所述光学成像系统的所述光学传感器阵列分离,用于检测所述人的脸部上的由所述两个或更多个不同光学照明波长的探测光的照明引起的光以产生探测检测信号;以及设备控制模块,被耦合以接收来自所述光学探测光检测模块的所述探测检测信号中的信息以及采集的来自所述光学成像系统的所述一个或多个脸部图像,所述设备控制模块被构造为处理采集的所述一个或多个脸部图像以提取所述人的面部特征,将提取的所述面部特征与授权用户的已知和现有的面部特征的数字数据进行比较,以确定是否存在匹配,作为面部识别的一部分。设备控制模块被构造且可操作为处理检测的所述人的脸部上的由所述探测光的照明引起的所述光,以测量所述两个或更多个不同光学照明波长中两个光学照明波长之间检测的所述光的差异,并且应用测量的所述差异来确定检测的所述脸部是否来自真人,作为面部识别的附加部分。

[0008] 另一方面,所公开的技术可以被实现为提供一种用于进行人的面部识别以确定所述人是否为授权用户的方法。该方法的一个示例包括:操作光学成像系统以采集人的脸部的一个或多个脸部图像,所述光学成像系统包括光学检测器的光学传感器阵列;处理采集的所述一个或多个脸部图像以提取所述人的面部特征,将提取的所述面部特征与授权用户的已知和现有的面部特征的数字数据进行比较,以确定是否存在匹配,作为面部识别的一部分;引导两个或更多个不同光学照明波长的探测光以照亮所述人的脸部;操作与所述光学成像系统的所述光学传感器阵列分离的光学探测光检测模块,以检测所述人的脸部上的由所述两个或更多个不同光学照明波长的探测光的照明引起的光;处理检测的所述人的脸部上的由所述探测光的照明引起的所述光,以测量所述两个或更多个不同光学照明波长中两个光学照明波长之间检测的所述光的差异;以及使用测量的所述差异来确定检测的所述脸部是否来自真人,作为面部识别的附加部分。

[0009] 在一些实现方式中,上述方法可以包括选择所述探测光的所述两个或更多个不同光学照明波长,使得来自所述人的眼睛的瞳孔的光学反射在信号强度上可测量到不同,以允许将检测的来自所述人的眼睛的瞳孔的光学反射的测量的差异作为用于检测的所述脸部是否来自真人的指示符。在其他实现方式中,上述方法可以包括:选择所述探测光的所述两个或更多个不同光学照明波长,以引起真人的面部皮肤不同水平的光学吸收,使得所述两个或更多个不同光学照明波长的正渗透入所述面部皮肤而被所述面部皮肤散射的所述探测光的束斑尺寸为可测量到不同;操作所述探测检测模块以采集所述光学传感器阵列处束斑的图像;处理采集的所述两个或更多个不同光学照明波长的所述束斑的图像,以测量所述束斑的差异;以及使用测量的所述差异来确定检测的所述脸部是否来自真人,作为面部识别的附加部分。

[0010] 在一些实现方式中,上述方法还可以包括两个或更多个探测光源与所述光学传感器阵列相邻放置,从而将所述两个或更多个不同光学照明波长的探测光被引导照射所述人的脸部,以当所述人看向所述探测光检测模块时引起从所述人的眼睛向所述探测光检测模块的近回射。

[0011] 在一些实现方式中,上述方法还可以包括:使引导向所述人的所述探测光为以第一偏振的偏振光;操作所述探测光检测模块以在检测来自所述人的眼睛的瞳孔的反射探测

光时检测处于第二偏振的光;以及将检测的以特定偏振的所述反射探测光的功率水平作为所述反射探测光是由来自当反射光时消偏振光的人的眼睛还是来自人造物体的反射产生的指示符。

[0012] 又一方面,所公开的技术可以被实现为提供一种用于辅助人的面部识别以确定所述人是否为真人的设备。该设备包括:光学探测检测模块,包括光学探测检测器的光学探测传感器阵列,用于检测来自人的脸部的光或采集人的脸部的图像,以辅助面部识别;以及一个或多个探测光源,位于所述光学探测传感器阵列附近,用于产生两个或更多个不同光学照明波长的探测光以照亮所述人的脸部。所述一个或多个探测光源被构造使得所述探测光的所述两个或更多个不同光学照明波长引起真人的脸部的一个或多个部分不同水平的光学吸收,以分别产生所述两个或更多个不同光学照明波长的不同的光信号,所述光信号在所述光学探测检测模块处可测量到不同以指示检测的所述脸部是否来自真人,而不是用于逃避所述面部识别的欺骗设备。在一些实现方式中,该设备可以包括:一个或多个光学偏振器,分别放置在所述一个或多个探测光源的光路中,用于产生引导向所述人的所述探测光,成为以第一偏振的偏振光;以及一个或多个检测光学偏振器,被定位成仅选择处于第二偏振的向所述光学探测传感器阵列的入射光中的光到达所述光学探测传感器阵列,以检测接收的处于所述第二偏振的所述光的功率水平,作为检测的所述光是由来自当反射光时消偏振光的人的眼睛还是来自人造物体的反射产生的指示符。

[0013] 在上述设备的一些实现方式中,所述一个或多个探测光源可以被构造为将(1)所述探测光的所述两个或更多个不同光学照明波长之一设置为942nm或接近942nm,在所述波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出强光学吸收以及将(2)所述探测光的所述两个或更多个不同光学照明波长中的另一个设置为在选定的光学照明波长处远离942nm光谱定位,在所述波长处,所述真人的瞳孔中的玻璃体液或面部皮肤表现出可测量的较少的光学吸收。

[0014] 在上述设备的一些实现方式中,该设备还可以包括:一个或多个光学偏振器,分别放置在所述一个或多个探测光源的光路中,用于产生引导向所述人的所述探测光,成为以第一偏振的偏振光;以及一个或多个检测光学偏振器,被定位成仅选择处于第二偏振的向所述光学探测传感器阵列的入射光中的光到达所述光学探测传感器阵列,以检测接收的处于所述第二偏振的所述光的功率水平,作为检测的所述光是由来自当反射光时消偏振光的人的眼睛还是来自人造物体的反射产生的指示符。

[0015] 在上述系统、方法或设备的一些实现方式中,触发机制或模块可以被实现为,当确定用户打算访问设备时,触发、开启、激活或开始脸部ID检测和处理。一种触发技术是使用移动设备上的一个或多个压力传感器来感测用户是正手持电话还是正抓着电话。该触发机制的另一个示例是使用加速度传感器。这种触发机制的又一示例是通过监测用户的眼睛的光学触发器。

[0016] 以下附图、说明书和权利要求对所公开技术的上述和其它方面、其实现方式及特征提供了更详细的描述。

附图说明

[0017] 图1是具有用于用户认证的脸部识别的安全设备或系统的示例的框图,该安全设

备或系统包括光学成像模块、光学真度检测模块和用户认证处理器,光学成像模块用于采集脸部的图像用于脸部识别,光学真度检测模块用于确定从其采集脸部图像的物体是否为真人的脸部,用户认证处理器基于用于脸部识别的来自两个模块的信息来确定是准许还是拒绝访问。

[0018] 图2示出了执行图1中的光学真脸部检测的移动电话的正面的示例。

[0019] 图3示出了具有光学探测检测阵列和一个或多个探测光源的光学真脸部检测模块封装的示例,一个或多个探测光源用于生成两个或更多个不同光学照明波长的探测光,以引起真人的脸部的一个或多个部分不同水平的光学吸收,以产生可测量为不同的不同的光信号,以指示检测的脸部是否来自真人,而不是用于逃避面部识别的欺骗设备。图3中,使用以下图例:

[0020] 29-连接器;

[0021] 31-电路板;

[0022] 33-微型相机驱动板;

[0023] 35-微型相机;

[0024] 35a-滤光层;

[0025] 35b-偏振器;

[0026] 35c-光电二极管阵列;

[0027] 37-1号光源;

[0028] 37a-1号光发射器;

[0029] 37b-1号光处理器;

[0030] 39-2号光源;

[0031] 39a-2号光发射器;

[0032] 39b-2号光处理器;

[0033] 41-相机镜头组件;

[0034] 41a-透镜;

[0035] 43-1号窗口材料;

[0036] 45-邻近光源;

[0037] 45a-邻近光发射器;

[0038] 45b-邻近光处理器;和

[0039] 47-2号窗口材料。

[0040] 图4示出了水和其他物质的光吸收特性的示例以及分别在942nm和850nm处的来自真人的眼睛的两个反射图像。

[0041] 图5示出了942nm和850nm的进入皮肤并被皮肤组织散射的红外探测光的光束扩展的图示。

[0042] 图6示出了在反射偏振光时结合人的视网膜的消偏振特性的光偏振的使用。

[0043] 图7A和图7B示出了基于人的脸部上的某些面部部分的图像位置的脸部的脸部方向的光学检测。

[0044] 图8示出了基于本专利文件中所公开的光学真脸部检测的真脸部检测算法的示例。

具体实施方式

[0045] 该专利文件中公开的脸部识别技术提供了在与光相互作用时的人的脸部的独特的特征的光学感测,以提供用于面部识别的反欺骗特征。电子设备或系统可以配备所公开的脸部识别机制,以改进访问设备或系统的安全性。这种电子设备或系统可以包括便携式或移动计算设备,例如智能手机、平板计算机、腕戴式设备和其他可穿戴或便携式设备,还包括更大的电子设备或系统,例如便携式或台式的个人计算机、自动柜员机(Automatic Teller Machine, ATM)、用于商业或政府用途的各种终端到各种电子系统、数据库或信息系统、以及包括汽车、船、火车、飞机和其他的机动交通系统。

[0046] 图1是具有用于用户认证的脸部识别的安全设备或系统的示例的框图,该安全设备或系统包括(1)光学成像模块110、(2)光学真度检测模块120和(3)用户认证处理器130,光学成像模块110用于采集脸部的图像用于脸部识别,光学真度检测模块120用于对从其由光学成像模块110采集的脸部图像的物体的一个或多个光学特征进行光学感测,用户认证处理器130基于用于脸部识别的来自两个模块110和120的信息来确定是准许还是拒绝访问安全设备或系统101。用户认证处理器130与两个光学感测模块110和120通信,以接收其输出信号或数据并处理接收的这种信号或数据,以基于来自光学成像模块110的脸部图像和来自光学真度检测模块120的真度检测来确定从其采集脸部图像的物体是否为真人的脸部。在该示例中,用于脸部识别的光学成像模块110可以以各种配置来实现,例如,在苹果公司的一些最新iPhone型号中使用的光学脸部ID系统。这种用于脸部识别的光学成像模块110包括一个或多个高分辨率相机,用于采集脸部的详细图像以提取用于脸部ID的面部特征。在一些实现方式中,用于脸部识别的光学成像模块110在一些设计中可以包括用于采集2D图像的成像硬件,并且在其他设计中还可以包括用于采集3D图像的成像硬件。来自光学成像模块110的脸部的图像以数字成像数据的形式被发送至用户认证处理器130以进行处理。用户认证处理器130被编程为进行脸部识别操作,包括从在光学成像模块110处获得的图像中提取面部特征并进行计算操作,将提取的人的面部特征与授权用户的已知和现有面部特征进行比较,以确定是否存在匹配。授权用户的已知和现有面部特征通常存储在设备或系统中的本地存储器中,也可以存储在网络或云中的连接服务器中。在一些面部识别系统中,当存在匹配时会准许访问。在本专利文件中所公开的技术下,通过光学真度检测模块120的操作以及通过在用户认证处理器130中执行的进一步处理来进行附加的感测,以确定表现出匹配的面部特征的被检测物体是否为真人的脸部。

[0047] 光学真度检测模块120用于通过光学检测或测量假定脸部或物体的一个或多个光学参数或特征来光学探测呈现给设备的假定脸部或物体,以确定假定脸部或物体是否为真人的脸部。在一些实现方式中,这可以通过引导两个或更多个不同光学照明波长的探测光以照亮脸部,并使用与光学成像模块110的一个或多个光学传感器阵列分离的指定的光学探测光检测模块来检测人的脸部上的由两个或更多个不同光学照明波长的探测光的照明引起的光来实现。这种探测检测可以设计在光学真度检测模块120的硬件中,以测量由于探测照明而由脸部反射或散射的光的某些光学特性,其中,某些光学特性与独特的特性相关联,独特的特性由真人的脸部表现并且缺少在诸如假面罩的欺骗设备中通常使用的人造材料或结构。在一些实现方式中,光学真度检测模块120可以包括至少一个光学传感器阵列,该光学传感器阵列能够采集脸部的图像,处理采集的图像以进行真脸部检测或进行脸部方

向的检测,以辅助脸部ID认证。这样,在该示例中,该光学真度检测模块120是添加到脸部ID机制的附加特征,用于增强脸部ID认证的安全性。实际应用中,所公开的光学真脸部检测可以用于不同的脸部ID认证系统。

[0048] 操作中,由光学真度检测模块120获得的用于真脸的光学测量结果被发送至用户认证处理器130,该用户认证处理器130还被编程为处理检测的人的脸部上的由探测光的照明引起的光,以测量两个或更多个不同光学照明波长中两个光学照明波长之间检测的光的差异,并且使用测量的差异来确定检测的脸部是否来自真人,作为面部识别的附加部分。下面提供了与人脸皮肤和人眼相关联的光学特性的两个具体示例,以说明光学真度检测模块120的操作和功能。

[0049] 实际应用中,如图1所示,具有模块110、120和130的上述脸部识别系统可以耦合到设备或系统101,以保护访问并将访问限制为仅对一个或多个授权用户进行访问。可以从这种脸部识别系统中受益的设备或系统101可以是各种各样的不同设备或系统,包括例如智能手机、平板计算机、腕戴式设备和其他可穿戴或便携式设备、更大的电子设备或系统、便携式或台式的个人计算机、ATM、用于商业或政府用途的各种终端到各种电子系统、数据库或信息系统、以及包括汽车、船、火车、飞机和其他的机动交通系统。在配备有这种脸部识别系统的汽车中,两个光学模块110和120可以安装在汽车的入口附近,例如驾驶员或乘客的每个车门(在一些车辆中,包括车辆后部的入口,例如运动型实用汽车(Sport Utility Vehicle, SUV)的后门或舱门)。

[0050] 图2示出了执行图1中的光学真脸部检测的移动电话11的正面的示例。该移动电话11包括前置相机和在正面右上侧标记为“21”的脸部ID成像模块。该模块21包括图1中的光学成像模块110。正面左上侧的正面光学真脸部传感器27是图1中的光学真度检测模块120。图2中示出了其他传感器,包括被示为位于移动电话11一侧的压力或力传感器19。在一些实现方式中,一个或多个这种压力或力传感器可以用于检测用户是正抓着电话还是正手持电话,并且这种传感器被用作触发器用于开启或激活图1所示的脸部ID操作。压力或力传感器是这种脸部ID触发器或激活机制的具体示例,在一些设计中,其他示例包括接近传感器或运动传感器。

[0051] 图3示出了具有光学探测检测阵列和一个或多个探测光源的光学真脸部检测模块封装的示例,一个或多个探测光源用于生成两个或更多个不同光学照明波长的探测光,以引起真人的脸部的一个或多个部分不同水平的光学吸收,以产生可测量为不同的不同的光信号,以指示检测的脸部是否来自真人,而不是用于逃避面部识别的欺骗设备。这是图1中的光学真脸部检测模块120的示例实现方式。在该示例中,将微型红外(IR)相机35以及探测光源37和39组装到电路板31上。微型IR相机35包括IR光学传感器的光学传感器阵列,用于光学检测,探测光源37和39用于生成不同光学波长的不同照明探测光,该不同照明探测光被特别选择用于检测物体是否为真人的脸部。连接器29将整个封装与移动电话平台连接。这种设计允许光学真脸部检测模块封装与使用光学脸部ID认证的不同移动电话平台或不同设备或系统集成在一起。

[0052] 具体地,在图3中所示的示例中,微型IR相机35包括透镜或透镜组件41a、光学滤光层35a、光学偏振器35b以及光学传感器阵列35c,其可以是IR感测二极管的光电二极管阵列。探测光源37,39和接近光源45被设计为与微型IR相机35配合。光源中,设置光处理器

37b, 39b和45b,使得针对功能处理光束立散角和偏振特性等。探测光可以是线性偏振或旋转偏振,例如圆偏振。光处理器可以包括线性偏振器、波片以及液晶或光学双折射晶体材料,从而控制光偏振特性。光处理器还可以包括衍射光学元件(Diffraction Optical Element, DOE)用于在适当位置形成光斑。

[0053] 图3中的示例还示出了应用微型驱动板33以运行微型相机和探测光源。在该示例中,探测光源37和39被安装在靠近微型相机的镜头轴的位置,以得到来自看着移动屏幕的脸部的近回射(nearly retro reflection)。邻近光源45安装在距微型相机的镜头轴所需距离的位置,并用于提供物体接近的光学感应,例如眼睛到相机的距离。

[0054] 图4示出了水和其他物质的光吸收特性的示例以及分别在942nm和850nm处的来自真人的眼睛的两个反射图像。当诸如37或39的探照光源靠近微型相机的镜头轴放置时,如果眼睛直视相机,则当相机采集来自人眼的图像时可能会出现回射(retro reflection)现象。来自眼睛的光的反射在942nm或942nm附近具有独特的光学吸收特征,此特性可以用于真度检测。

[0055] 图4进一步示出了水中的光学吸收在942nm窗口处较强,而在其他波长处相对较弱,例如仍在IP光谱范围内的850nm窗口。人眼球的玻璃体液的近98~99%是水,因此,942nm或942nm附近的光被强烈吸收并且几乎无法到达玻璃体液后的人眼视网膜。基于此特性,发射942nm或942nm附近的光的一个或多个探测光源可以被用作回射光源之一,并选择850nm或其他IR波长的其他IR探测光源作为发射IR探测光的另一个回射光源,该IR探测光不像942nm的探测光被强烈吸收。

[0056] 由于在光谱范围内的942nm窗口处具有强吸收,因此,当打开942nm光源时,人眼的回射弱。然而,在远离942nm的其他IR波长下,例如850nm,这种IR光不会被人眼球的玻璃体液强烈吸收,从而可以到达反射该光的视网膜,以产生强光学反射。这种反射一旦被IR相机接收,就会产生强回射图像。利用942nm和另一IR波长(例如850nm)的光学反射,基于水引起的光谱特征,检测到的两个不同IR波长的光学反射在信号强度上是不同的。可以对这种光学反射的测量结果进行评估,以确定逆向信号是否来自真眼球,即作为物体是否为真人的眼睛的指示符。

[0057] 图5示出了942nm和850nm的进入皮肤并被皮肤组织散射的红外探测光的光束扩展的图示。当光斑投射到真脸部上时,部分光散射在表面上,而部分光折射入表面。折射分量被强烈散射,使得光斑轮廓扩展至更大的尺寸和形状。由于在942nm或942nm附近脸部皮肤组织(由于水)具有强吸收,所以,一旦942nm或942nm附近的探测光进入皮肤,探测光被强烈衰减,因此,束斑表现出弱扩展。相反,由于在850nm或远离942nm的另一IR波长的探测光下脸部皮肤组织具有弱吸收,所以850nm(或其他合适的波长带)的探测光进入脸部皮肤后会产生表现出强光束轮廓扩展的光斑。光学真脸部模块中的相机可以用于采集这种图像,并且采集的图像可以用于分析脸部真度。

[0058] 图6示出了所公开技术的一些实现方式中的在反射偏振光时结合人的视网膜的消偏振特性的光偏振的使用的示例。所公开技术的这一方面基于以下认识:可以用透镜和在其焦平面中的反射器模拟人眼球的光反射,尤其人眼视网膜强烈地消偏振入射光,因此是唯一的反射器。视网膜的这种独特的特性可以用于检测物体是否为真人的眼睛。

[0059] 具体地,当偏振光入射到真实的活人眼中时,回射包含较大比例(~80%或更高)

的消偏振分量。即使在焦平面上有粗糙的反射器,透镜也几乎不会产生类似的消偏振特征。因此,可以将偏振器,例如线性偏振器,安装在图1中的光学真脸部检测模块110中的每个探测光源上,以确保来自每个探测光源的探测光具有期望的偏振。基于此,将检测光学偏振器安装在微型相机的光路中,以评估来自反射物体的返回光的消偏振。从真实的活眼睛反射的逆向光几乎被完全消偏振。从假眼睛反射的逆向光被部分偏振或未被消偏振。相机处的检测光学偏振器的偏振方向可以被定向为垂直于探测光源处的光学偏振器的偏振。这可用于检测物体的真度。

[0060] 另外,可以结合图4至5讨论的吸收特性来分析脸部反射光斑图像的消偏振特性,以进一步增强真度检测,因为通常很难找到在强吸收942nm窗口和850nm窗口的光波长窗口下具有相同消偏振特性的假脸材料。通过分析942nm波带与其他参考波长带之间的光斑扩展差异和消偏振特性,我们可以得出判别脸部是否为真的标准。

[0061] 脸部ID的处理是计算密集型过程,并且往往会消耗电池电量。当确定用户打算访问诸如移动电话的设备时,可以使用各种触发机制来开启、激活或开始脸部ID检测和处理

[0062] 一种触发技术是使用移动设备上的一个或多个压力传感器来感测用户是正手持电话还是正抓着电话。参考图2,一个或多个握持力传感器或压力传感器19可以集成在移动平台的框架中。这种传感器可以如19所标识的那样能够被看到,也可以完全隐藏在框架中使其不可见。

[0063] 此触发机制的另一个示例是在移动设备或电话中的G传感器(或加速度传感器)来感测平台的移动,例如智能手机。几乎所有智能手机都具有在系统中集成的G传感器,并且传感器输出可以用于触发脸部ID检测和处理。

[0064] 这种触发机制的又一示例是通过监测用户的眼睛的光学触发器。当光学传感器检测到用户的眼睛正看着平台例如智能手机时,安全传感器,例如脸部ID传感器或指纹传感器被激活。这点而言,可以是图1中光学真度检测模块120的一部分的探测光检测模块用于采集一个或多个脸部图像,以识别人脸部的关键特征点的图像位置,并且用户认证处理器130处理识别的人脸部的关键特征点的图像位置,以确定该人是否正看向探测光检测模块。

[0065] 图7A和图7B示出了基于人的脸部上的某些面部部分的图像位置的脸部的脸部方向的光学检测的示例。当人正凝视脸部ID相机时,图像中,眼睛的图像和其他脸部部分(例如,鼻尖,下颌等)的图像位置通常是对称的,如图7A左上图所示。当人正面对其他方向时,图像中,眼睛的图像和其他脸部部分(例如,鼻尖)的图像位置在由脸部ID相机采集的图像中不对称,如图7右上图所示。基于图像分析,脸部ID传感器可以判别用户是否正看着平台。

[0066] 图8示出了基于本专利文件中所公开的光学真脸部检测的真脸部检测算法的示例。该过程首先基于一个或多个触发信号来检测用户状态,例如,相对于电话屏幕的用户脸部方向,用户手部的移动和握持力。如果此检测生成触发器,则开启脸部ID硬件以进行光学脸部ID感测和处理以进行脸部ID认证。只要面部特征匹配和真脸部检测均成功,系统就会准许访问。面部特征匹配和真脸部检测的这种结合使用可以减少脸部欺骗尝试的成功,从而增强脸部ID认证的安全级别。

[0067] 虽然本专利文件包含许多细节,但是不应将这些细节解释为对任何发明或可要求保护的范围的限制,而应解释为对特定发明的特定实施例的特有的特征的描述。本专利文

件中描述的某些特征在单独实施例的上下文中还可以在单个实施例中组合实现。相反,在单个实施例的上下文中描述的各种特征还可以在多个实施例中单独实现或以任何合适的子组合实现。此外,虽然上文描述的特征在某些组合和原始要求保护的组合中起作用,但是来自要求保护的组合的一个或多个特征在某些情况下可以从组合中脱离,并且要求保护的组合可以导向到部分组合或部分组合的变形。

[0068] 类似地,尽管在附图中以特定顺序描述了操作,但这不应理解为要求这些操作以所示的特定顺序或按相继顺序次序执行,或者执行所有所示的操作以实现期望的结果。而且,在本专利文件中描述的实施例中的单独的各种系统部件不应理解为在所有实施例中需要这种分离。

[0069] 仅描述了几个实现方式和示例,并且可以基于本专利文件中描述和示出的内容来做出其他实现方式、增强和变化。

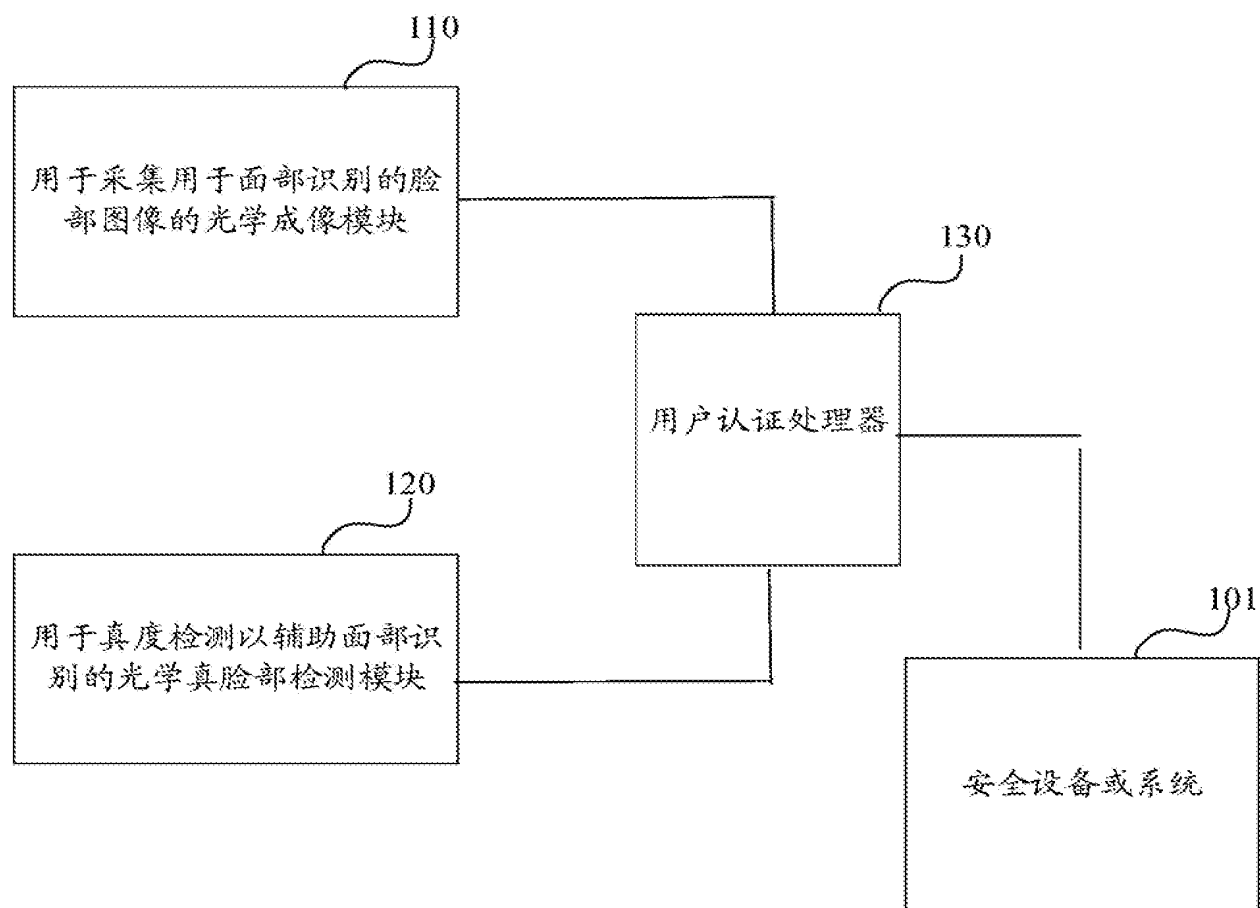


图1

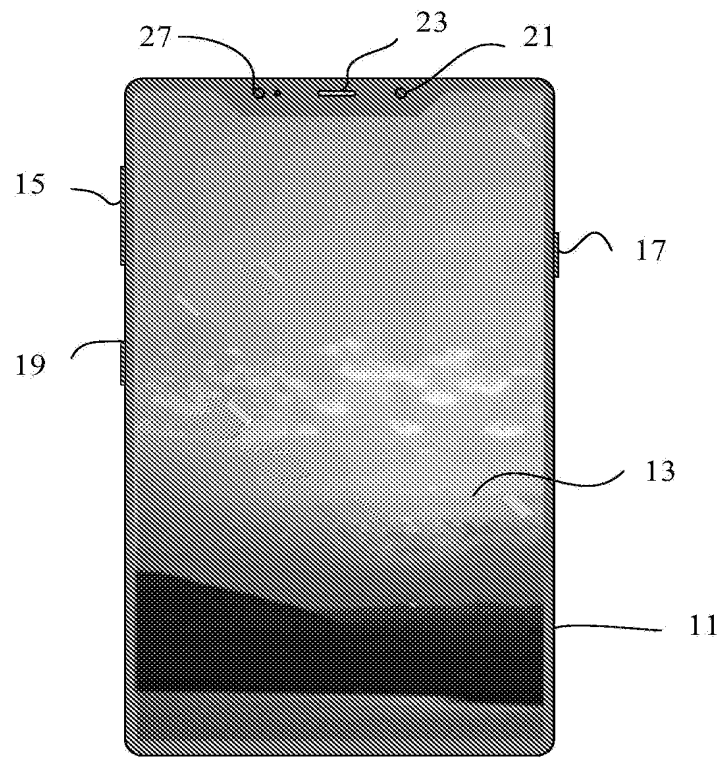


图2

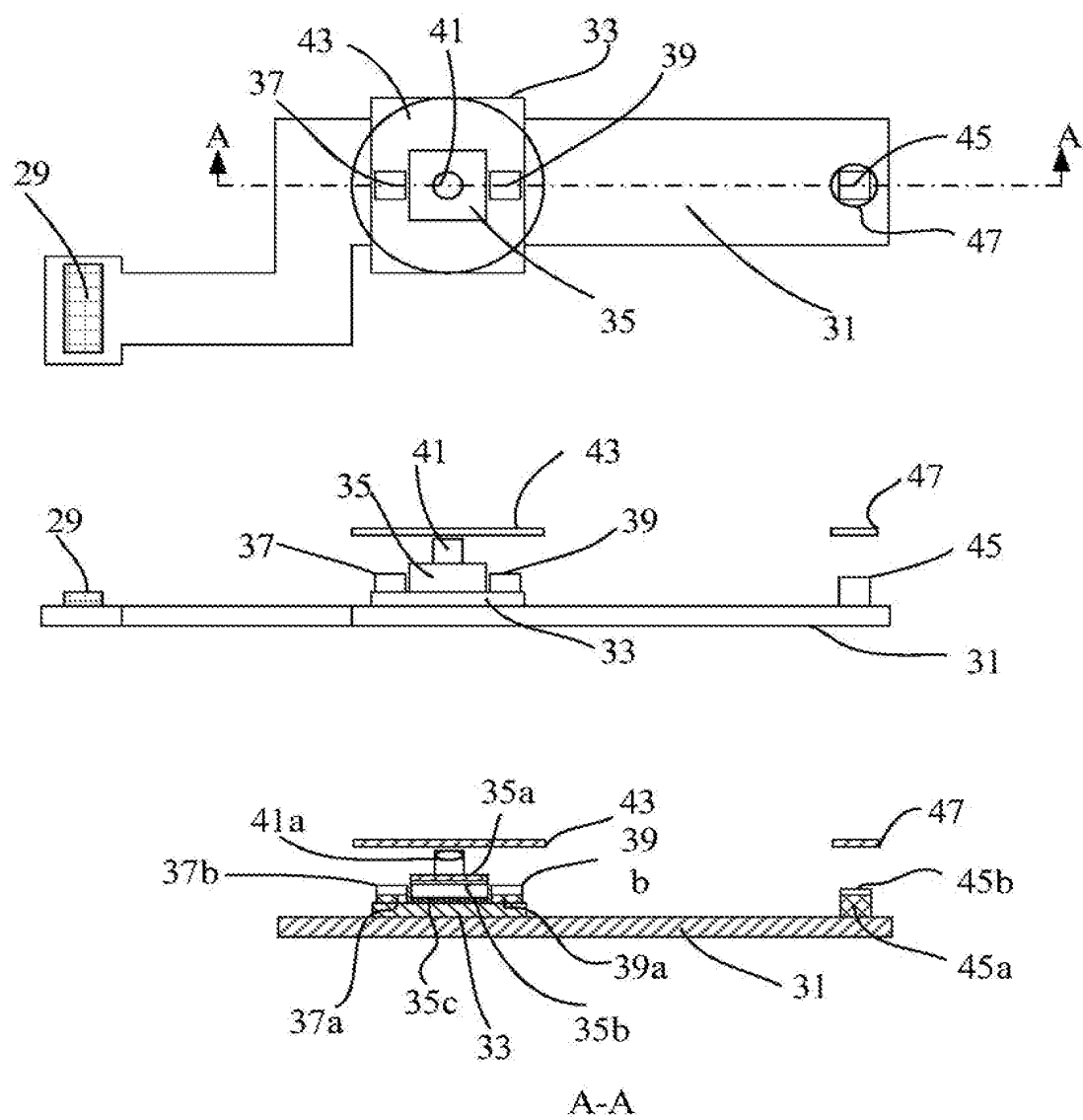
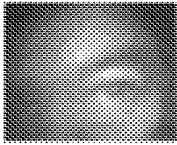
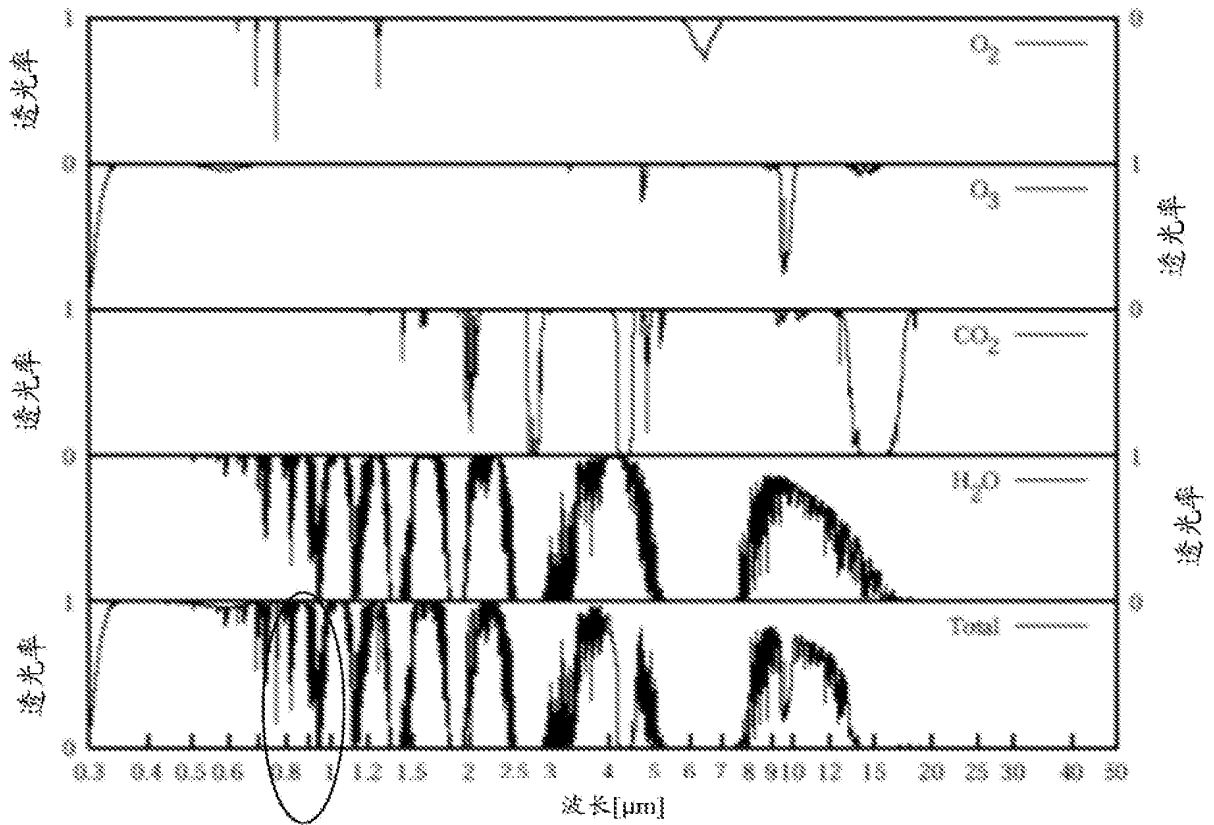
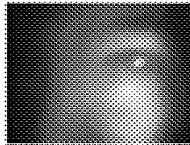


图3



942 nm照明下的弱反射



850 nm照明下的强反射

图4

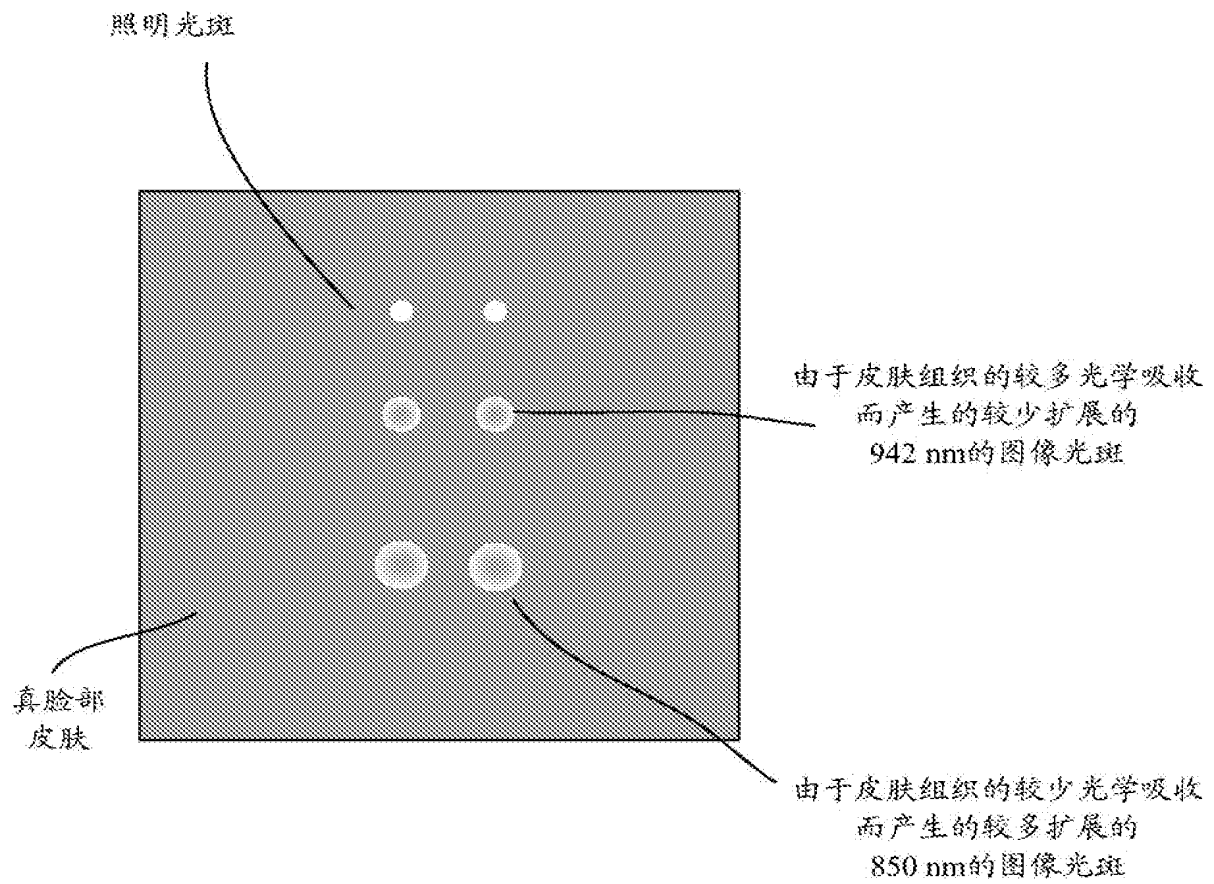
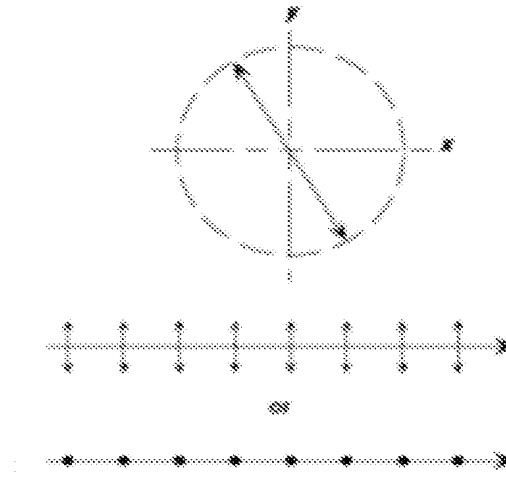
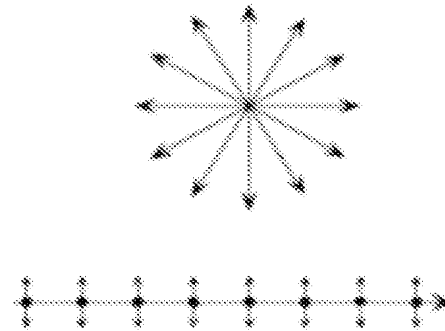


图5

偏振光



完全消偏振光



部分消偏振光

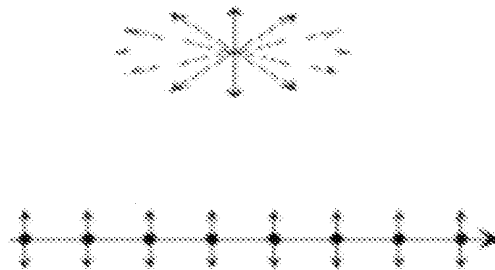


图6

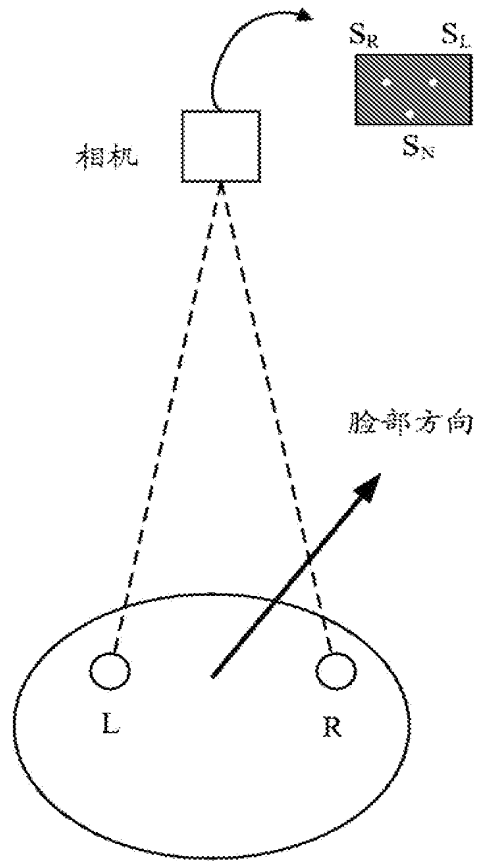


图7A

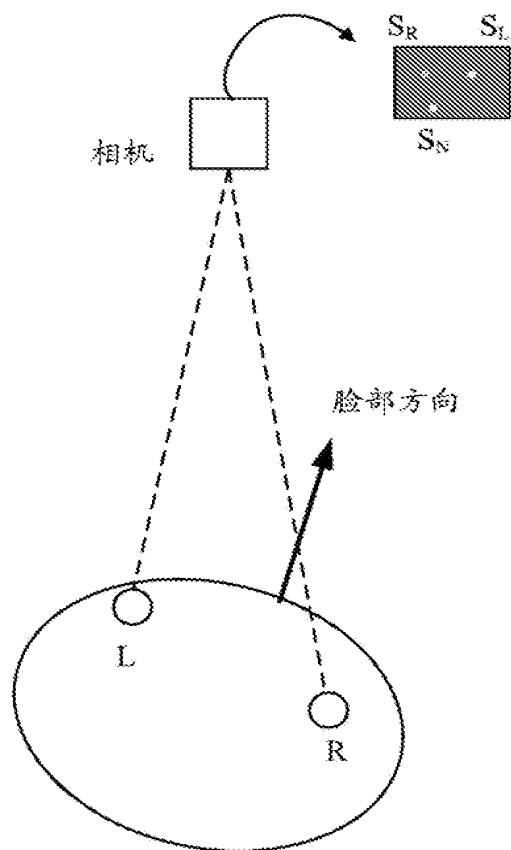


图7B

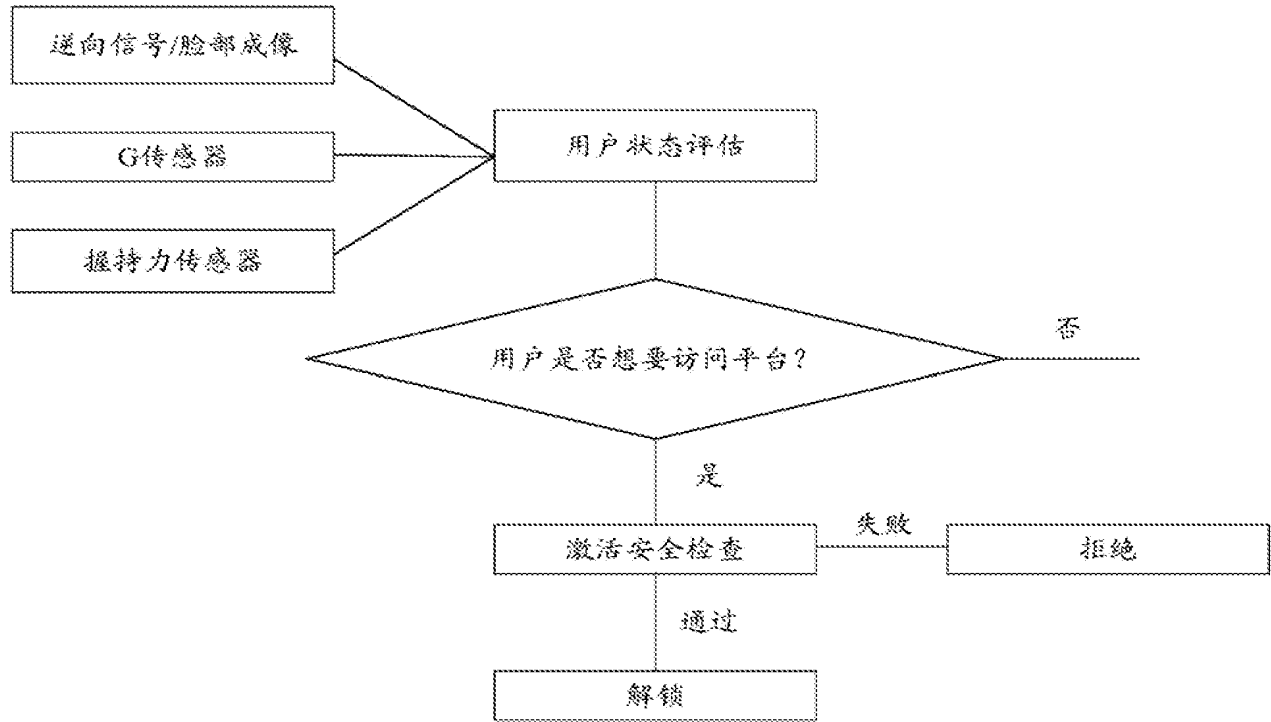


图8



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Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE Utility - U.S. National Stage under 35 USC 371

PATENT # -

CONFIRMATION # 2486

FILED BY Lauren Ferro

PATENT CENTER # 62560500

FILING DATE -

CUSTOMER # 76656

FIRST NAMED INVENTOR Peter SCHILLEN

INTL. APPLICATION # PCT/EP 2022/053941

INTL. FILING DATE 02/17/2022

CORRESPONDENCE ADDRESS -

AUTHORIZED BY Christopher Goff

Documents

TOTAL DOCUMENTS: 16

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-08-04_ADS-27843-2262.pdf	10	Application Data Sheet	2561 KB
2023-08-04_PCT_EP_22_53941.pdf	67	-	4263 KB
2023-08-04_PCT_EP_22_53941-ABST.pdf	(1-2) 2	Abstract	99 KB
2023-08-04_PCT_EP_22_53941-SPEC.pdf	(3-62) 60	Specification	3972 KB

2023-08-04_PCT_EP_22_53941-CLM.pdf	(63-65)	3	Claims	136 KB
2023-08-04_PCT_EP_22_53941-371P.pdf	(66-67)	2	Documents submitted with 371 (National Stage) Applications	55 KB
2023-08-04_PrelimAmendment_27843-2262.pdf		8	-	160 KB
2023-08-04_PrelimAmendment_27843-2262-A.PE.pdf	(1-1)	1	Preliminary Amendment	100 KB
2023-08-04_PrelimAmendment_27843-2262-SPEC.pdf	(2-2)	1	Specification	86 KB
2023-08-04_PrelimAmendment_27843-2262-ABST.pdf	(3-3)	1	Abstract	104 KB
2023-08-04_PrelimAmendment_27843-2262-CLM.pdf	(4-7)	4	Claims	150 KB
2023-08-04_PrelimAmendment_27843-2262-REM.pdf	(8-8)	1	Applicant Arguments/Remarks Made in an Amendment	86 KB
2023-08-04_POA_27843-2262.pdf		2	Power of Attorney	2289 KB
2023-08-04_371_Transmittal_27843-2262.pdf		4	Transmittal of New Application	1957 KB
2023-08-		3	Oath or Declaration filed	143 KB

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2023-08-04_IDS_27843-2262.pdf	4	Information Disclosure Statement (IDS) Form (SB08)	1382 KB
FOR_CN111164610.pdf	44	Foreign Reference	2188 KB
NPL_ISR_WOSA_27843-2262.pdf	12	Non Patent Literature	448 KB

Digest

DOCUMENT

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Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	2486	FILED BY	Lauren Ferro
PATENT CENTER #	62560500	AUTHORIZED BY	Christopher Goff
CUSTOMER #	76656	FILING DATE	-
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CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Peter SCHILLEN

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PRE-AUTHORIZED ACCOUNT 012384	PRE-AUTHORIZED CATEGORY 37 CFR 1.17 (Patent application and reexamination processing fees); 37 CFR 1.19 (Document supply fees); 37 CFR 1.21 (Miscellaneous fees and charges); 37 CFR 1.492 (National application filing, search, and examination fees)	

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1631	BASIC NATIONAL STAGE FEE	320.00	1	320.00
1633	NATIONAL STAGE EXAMINATION FEE - ALL OTHER SITUATIONS	800.00	1	800.00
1642	NATIONAL STAGE SEARCH FEE - SEARCH REPORT PREPARED AND PROVIDED TO USPTO	540.00	1	540.00
TOTAL				\$1,660.00

AMOUNT:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Peter		SCHILLEN		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen am Rhein	Country of Residence ⁱ		DE	
Mailing Address of Inventor:					
Address 1		Industriestraße 35			
Address 2					
City	Ludwigshafen am Rhein	State/Province			
Postal Code	67063	Country ⁱ	DE		
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Benjamin		GUTHIER		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen am Rhein	Country of Residence ⁱ		DE	
Mailing Address of Inventor:					
Address 1		Industriestraße 35			
Address 2					
City	Ludwigshafen am Rhein	State/Province			
Postal Code	67063	Country ⁱ	DE		
Inventor 3					Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Friedrich		SCHICK	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Ludwigshafen am Rhein	Country of Residence ⁱ	DE	

Mailing Address of Inventor:				
Address 1		Industriestraße 35		
Address 2				
City	Ludwigshafen am Rhein	State/Province		
Postal Code	67063	Country ⁱ	DE	
Inventor 4				Remove

Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Manuel		GUENTHER	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Zuerich	Country of Residence ⁱ	CH	

Mailing Address of Inventor:				
Address 1		Andreastrasse 15		
Address 2				
City	Zuerich	State/Province		
Postal Code	8050	Country ⁱ	CH	
Inventor 5				Remove

Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Lars		DIESELBERG	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Karlsruhe	Country of Residence ⁱ	DE	

Mailing Address of Inventor:				
Address 1		Geibelstr 22		
Address 2				
City	Karlsruhe	State/Province		
Postal Code	76185	Country ⁱ	DE	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

Inventor 6 Remove				
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Christian		LENNARTZ	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Ludwigshafen am Rhein	Country of Residence ⁱ	DE	
Mailing Address of Inventor:				
Address 1	Industriestraße 35			
Address 2				
City	Ludwigshafen am Rhein	State/Province		
Postal Code	67063	Country ⁱ	DE	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	76656		
Email Address	uspatents@atllp.com	Add Email	Remove Email

Application Information:

Title of the Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		
Attorney Docket Number	27843-2262 202595US01	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	2	Suggested Figure for Publication (if any)	3

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	76656		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	a 371 of international	PCT/EP22/53941	2022-02-17

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

[Add](#)

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
21157800.0	EP	2021-02-18	

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear	
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117
☐ Person to whom the inventor is obligated to assign.	☐ Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:	
Name of the Deceased or Legally Incapacitated Inventor:	
If the Applicant is an Organization check here. ☒	
Organization Name	trinamiX GmbH
Mailing Address Information For Applicant:	
Address 1	Industriestrasse 35
Address 2	
City	Ludwigshafen am Rhein
State/Province	
Country	DE
Postal Code	67063
Phone Number	
Fax Number	
Email Address	
Additional Applicant Data may be generated within this form by selecting the Add button. Add	

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

[Remove](#)

If the Assignee or Non-Applicant Assignee is an Organization check here. ☐

Prefix	Given Name	Middle Name	Family Name	Suffix

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1			
Address 2			
City		State/Province	
Country ⁱ		Postal Code	
Phone Number		Fax Number	
Email Address			

Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

[Add](#)**Signature:**[Remove](#)

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Christopher M. Goff/			Date (YYYY-MM-DD)	2023-08-04
First Name	Christopher M.	Last Name	Goff	Registration Number	41785

Additional Signature may be generated within this form by selecting the Add button.

[Add](#)

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Box No. VIII (iv) DECLARATION: INVENTORSHIP (only for the purposes of the designation of the United States of America)
The declaration must conform to the following standardized wording provided for in Section 214; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (iv). If this Box is not used, this sheet should not be included in the request.

**Declaration of inventorship (Rules 4.17(iv) and 51bis.1(a)(iv))
 for the purposes of the designation of the United States of America:**

I hereby declare that I believe I am the original inventor or an original joint inventor of a claimed invention in the application.

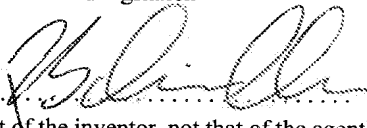
This declaration is directed to the international application of which it forms a part (if filing declaration with application).

This declaration is directed to international application No. (if furnishing declaration pursuant to Rule 26ter).

I hereby declare that the above-identified international application was made or authorized to be made by me.

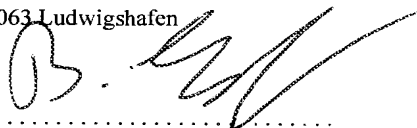
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Name: SCHILLEN, Peter
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

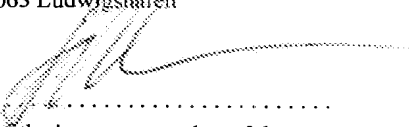
Date: 08.09.21

Name: GUTHIER, Benjamin
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

Date: 8.9.21

Name: SCHICK, Friedrich
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

Date: 21.9.21

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: GUENTHER, Manuel
Residence: Switzerland
Mailing Address: Leutschenbachstrasse 46
8050 Zuerich

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date: 10.05.2021

Name: DIESELBERG, Lars
Residence: Germany
Mailing Address: Geibelstr. 22
76185 Karlsruhe

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: LENNARTZ, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Continuation of Box No. VIII (i) to (v) DECLARATION

If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.

Name: GUENTHER, Manuel
Residence: Switzerland
Mailing Address: Leutschenbachstrasse 46
8050 Zuerich

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: DIESSELBERG, Lars
Residence: Germany
Mailing Address: Geibelstr. 22
76185 Karlsruhe

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 27.4.2021

Name: LENNARTZ, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 27.4.21

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Peter Schillen

Serial No.: TBD

Filed: August 4, 2023

Confirmation No.: TBD

For: OPTICAL SKIN DETECTION FOR FACE UNLOCK

August 4, 2023

PRELIMINARY AMENDMENT

TO THE COMMISSIONER FOR PATENTS:

Please enter the following amendments prior to calculation of fees due in the present application.

Amendments to the Specification are on page 2.

Amendments to the Abstract are on page 3.

Amendments to the Claims begin on page 4.

Remarks are on page 8.

AMENDMENTS TO THE SPECIFICATION

Please enter the following amendments to the specification.

After the TITLE, please add the following heading and paragraph:

-- CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase Application of International Patent Application No. PCT/EP2022/053941, filed February 17, 2022, which claims priority to EP Patent Application No. 21157800.0, filed February 18, 2021, each of which is hereby incorporated by reference herein.

AMENDMENTS TO THE ABSTRACT

Please amend the Abstract as follows:

Disclosed herein is ~~A~~ **a** method for face authentication ~~is proposed~~. The method ~~comprising~~ **includes** the following steps:

- a) at least one face detection step ~~[(110)]~~ **comprising including** determining at least one first image by using at least one camera ~~[(112)]~~;
- b) at least one skin detection step ~~[(116)]~~ **comprising including** projecting at least one illumination pattern ~~comprising including~~ a plurality of illumination features on the scene by using at least one illumination unit ~~[(118)]~~, determining at least one second image using the at least one camera ~~[(112)]~~, **and** determining a first beam profile information ~~of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile, and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit (114), wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;~~
- c) at least one 3D detection step ~~[(120)]~~ **comprising including** determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image ~~comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit (114), wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects; and~~
- d) at least one authentication step ~~[(122)]~~ **comprising including** authenticating the detected face by using at least one authentication unit ~~if in step b) (116) the detected face is characterized as skin and in step c) [(120)] the detected face is characterized as 3D object.~~

(Figure 3)

AMENDMENTS TO THE CLAIMS

1. (Currently Amended) ~~Method~~ **A method** for face authentication comprising the following steps:

- a) at least one face detection step **[(110)]**, wherein the face detection step **[(110)]** comprises determining at least one first image by using at least one camera **[(112)]**, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step **[(110)]** comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit **[(114)]**;
- b) at least one skin detection step **[(116)]**, wherein the skin detection step **[(116)]** comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit **[(118)]** and determining at least one second image using the at least one camera **[(112)]**, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit **[(114)]**, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- c) at least one 3D detection step **[(120)]**, wherein the 3D detection step **[(120)]** comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing

- unit **[(114)]**, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects; **and**
- d) at least one authentication step **[(122)]**, wherein the authentication step **[(122)]** comprises authenticating the detected face by using at least one authentication unit if in step b) **[(116)]** the detected face is characterized as skin and in step c) **[(120)]** the detected face is characterized as 3D object.
2. (Currently Amended) The method according to ~~the preceding~~ claim **1**, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.
 3. (Currently Amended) The method according to ~~the preceding~~ claim **2**, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.
 4. (Currently Amended) The method according to ~~any one of the preceding claims~~ **claim 1**, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.
 5. (Currently Amended) The method according to ~~the preceding~~ claim **4**, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.
 6. (Currently Amended) The method according to ~~any one of the two preceding claims~~ **claim 4**, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of

characteristic points of a human face, wherein in the skin detection step ~~[(116)]~~ at least one region specific parametrized skin classification model is used.

7. (Currently Amended) The method according to ~~any one of the preceding claims~~ claim 1, wherein the illumination pattern comprises a periodic grid of laser spots.
8. (Currently Amended) The method according to ~~any one of the preceding claims~~ claim 1, wherein the illumination features have wavelengths in a near infrared (NIR) regime.
9. (Currently Amended) The method according to ~~the preceding~~ claim 8, wherein the illumination features have wavelengths of 940 nm.
10. (Currently Amended) The method according to ~~any one of the preceding claims~~ claim 1, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) ~~[(116)]~~ and/or for 3D detection in step c) ~~[(120)]~~.
11. (Currently Amended) The method according to ~~any one of the preceding claims~~ claim 1, wherein the camera ~~[(112)]~~ ~~is or~~ comprises at least one near infrared camera.
12. (Currently Amended) ~~Computer~~ A computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to ~~any one of the preceding claims~~ claim 1, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) ~~of the method according to any one of the preceding claims~~.
13. (Currently Amended) A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to ~~any one of the preceding claims~~ claim 1 ~~referring to a method~~.

14. (Currently Amended) A non-transient computer-readable medium **including comprising** instructions that, when executed by one or more processors, cause the one or more processors to perform at least steps a) to d) of the method according to ~~any one of the preceding claims~~ **claim 1 referring to a method.**
15. (Currently Amended) A mobile device **[[(124)]]** comprising at least one camera **[[(112)]]**, at least one illumination unit **[[(118)]]**, and at least one processing unit **[[(114)]]**, the mobile device **[[(124)]]** being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to ~~any one of the preceding claims~~ **claim 1 referring to a method.**
16. (Currently Amended) ~~Use of the~~ **A method of using the method for face authentication method** according to ~~any one of the preceding claims~~ **claim 1, wherein the method comprising using the method for face authentication** for biometric presentation attack detection.

REMARKS

The Specification has been amended to include the priority paragraph.

The Abstract has been amended to remove legal language.

Claims 1-16 will be pending after entry of this Preliminary Amendment. Claims 1-16 are amended herein. No new matter has been added in this amendment.

The Commissioner is hereby authorized to charge any fees under 37 CFR 1.16 and 1.17 which may be required except fees for excess claims or multiple dependent claims during the entire pendency of this application to Account No. 012384.

Respectfully submitted,

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CMG/EMF/ZSW/mbn
VIA EFS

TRANSMITTAL LETTER TO THE UNITED STATES DESIGNATED/ELECTED OFFICE (DO/EO/US) CONCERNING A SUBMISSION UNDER 35 U.S.C. 371		Attorney Docket No. 27843-2262 202595US01
		U.S. Application No. (if known, see 37 CFR 1.5)
International Application No. PCT/EP22/53941	International Filing Date 2022-02-17	Priority Date Claimed 2021-02-18
Title of Invention OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK		
First Named Inventor Peter SCHILLEN		
Applicant herewith submits to the United States Designated/Elected Office (DO/EO/US) the following items and other information. 1. ☒ This is an express request to begin national examination procedures (35 U.S.C. 371(f)). NOTE: The express request under 35 U.S.C. 371(f) will not be effective unless the requirements under 35 U.S.C. 371(c)(1), (2), and (4) for payment of the basic national fee, copy of the International Application and English translation thereof (if required), and the oath or declaration of the inventor(s) have been received. 2. ☒ A copy of the International Application (35 U.S.C. 371(c)(2)) is attached hereto (not required if the International Application was previously communicated by the International Bureau or was filed in the United States Receiving Office (RO/US)). 3. An English language translation of the International Application (35 U.S.C. 371(c)(2)) a. ☒ is attached hereto. b. ☐ has been previously submitted under 35 U.S.C. 154(d)(4). 4. An oath or declaration of the inventor(s) (35 U.S.C. 371(c)(4)) a. ☒ is attached. b. ☐ was previously filed in the international phase under PCT Rule 4.17(iv). Items 5 to 8 below concern amendments made in the international phase. <u>PCT Article 19 and 34 amendments</u> 5. ☐ Amendments to the claims under PCT Article 19 are attached (not required if communicated by the International Bureau) (35 U.S.C. 371(c)(3)). 6. ☐ English translation of the PCT Article 19 amendment is attached (35 U.S.C. 371(c)(3)). 7. ☐ English translation of annexes (Article 19 and/or 34 amendments only) of the International Preliminary Examination Report is attached (35 U.S.C. 371(c)(5)). <u>Cancellation of amendments made in the international phase</u> 8a. ☐ Do not enter the amendment made in the international phase under PCT Article 19. 8b. ☐ Do not enter the amendment made in the international phase under PCT Article 34. NOTE: A proper amendment made in English under Article 19 or 34 will be entered in the U.S. national phase application absent a clear instruction from applicant not to enter the amendment(s). The following items 9 to 17 concern a document(s) or information included. 9. ☒ An Information Disclosure Statement under 37 CFR 1.97 and 1.98. 10. ☒ A preliminary amendment. 11. ☒ An Application Data Sheet under 37 CFR 1.76. 12. ☐ A substitute specification. NOTE: A substitute specification cannot include claims. See 37 CFR 1.125(b). 13. ☒ A power of attorney and/or change of address letter. 14. ☐ A computer-readable form of the sequence listing in accordance with PCT Rule 13ter.3 and 37 CFR 1.821-1.825 (not required if sequence listing in text format was indicated on the PCT Request as part of the International Application and the sequence listing was published as part of the international application). 15. ☐ Assignment papers (cover sheet and document(s)). Name of Assignee: _____ 16. ☐ 37 CFR 3.73(c) Statement (when there is an Assignee). _____		

This collection of information is required by 37 CFR 1.414 and 1.491-1.492. The information is required to obtain or retain a benefit by the public, which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 15 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Mail Stop PCT, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

U.S. APPLN. No. (if known – see 37 CFR 1.5)	INTERNATIONAL APPLICATION No. PCT/EP22/53941	ATTORNEY DOCKET No. 27843-2262 202595US01
17. ☒ Other items or information: International Search Report and Written Opinion for Corresponding PCT/EP22/53941, mailed June 9, 2022, 12 pages		
The following fees have been submitted.		CALCULATIONS
18. ☒ Basic national fee (37 CFR 1.492(a)) \$320		\$ 320.00
19. ☒ Examination fee (37 CFR 1.492(c)) • If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)..... \$0 • All other situations \$800		\$ 800.00
20. ☒ Search fee (37 CFR 1.492(b)) • If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)..... \$0 • Search fee (37 CFR 1.445(a)(2)) has been paid on the international application to the USPTO as an International Searching Authority \$140 • International Search Report prepared by an ISA other than the US and provided to the Office or previously communicated to the US by the IB..... \$540 • All other situations \$700		\$ 540.00
TOTAL OF 18, 19, and 20 =		\$ 1660.00
☐ Additional fee for specification and drawings filed in paper over 100 sheets (excluding sequence listing in compliance with 37 CFR 1.821(c) or (e) in an electronic medium or computer program listing in an electronic medium) (37 CFR 1.492(j)). Fee for each additional 50 sheets of paper or fraction thereof \$420		\$
Total Sheets	Extra Sheets	RATE
- 100 =	/ 50 =	x \$420
Surcharge for furnishing any of the search fee, examination fee, or the oath or declaration after the date of commencement of the national stage (37 CFR 1.492(h))..... \$160		\$
CLAIMS	NUMBER FILED	NUMBER EXTRA
Total claims	16 - 20 =	x \$100
Independent claims	1 - 3 =	x \$480
MULTIPLE DEPENDENT CLAIM(S) (if applicable)		+ \$860
Fee for submission of Sequence Listing text file of 300 MB to 800 MB (37 CFR 1.21(o)(1))..... \$1,060		\$
Fee for submission of Sequence Listing text file of more than 800 MB (37 CFR 1.21(o)(2))..... \$10,500		\$
Processing fee for furnishing the English translation later than 30 months from the earliest claimed priority date (37 CFR 1.492(i))..... \$140 +		\$
TOTAL OF ABOVE CALCULATIONS =		\$ 1660.00
☐ Applicant asserts small entity status. See 37 CFR 1.27. Fees above are reduced by 60%.		
☐ Applicant certifies micro entity status. See 37 CFR 1.29. Fees above are reduced by 80%. Applicant must attach form PTO/SB/15A or B or equivalent.		
TOTAL NATIONAL FEE =		\$ 1660.00
Fee for recording the enclosed assignment (37 CFR 1.21(h)). The assignment must be accompanied by an appropriate cover sheet (37 CFR 3.28, 3.31)..... \$50.00 per property +		\$
TOTAL FEES ENCLOSED =		\$ 1660.00

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

- a. ☐ A check in the amount of \$ _____ to cover the above fees is enclosed.
- b. ☒ Please charge my Deposit Account No. 012384 in the amount of \$ 1660.00 to cover the above fees.
- c. ☒ The Director is hereby authorized to charge additional fees which may be required, or credit any overpayment, to Deposit Account No. 012384 as follows:
- i. ☒ any required fee.
- ii. ☐ any required fee except for excess claims fees required under 37 CFR 1.492(d) and (e) and multiple dependent claim fee required under 37 CFR 1.492(f).
- d. ☐ Fees are to be charged to a credit card. **WARNING:** Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. The PTO-2038 should only be mailed or faxed to the USPTO. However, when paying the basic national fee, the PTO-2038 may NOT be faxed to the USPTO.

ADVISORY: If filing by EFS-Web, do NOT attach the PTO-2038 form as a PDF along with your EFS-Web submission. Please be advised that this is not recommended and by doing so your credit card information may be displayed via PAIR. To protect your information, it is recommended to pay fees online by using the electronic payment method.

NOTE: Where an appropriate time limit under 37 CFR 1.495 has not been met, a petition to revive (37 CFR 1.137(a) or (b)) must be filed and granted to restore the International Application to pending status.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

- ☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013, and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE 1: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

NOTE 2: A U.S. national stage application may not claim priority to the international application of which it is the national phase. The filing date of a U.S. national stage application is the international filing date. See 35 U.S.C. 363.

Correspondence Address

- ☒ The address associated with Customer Number: 76656 OR ☐ Correspondence address below

Name					
Address					
City		State		Zip Code	
Country				Telephone	
Email					

Signature	/Christopher M. Goff/	Date	2023-08-04
Name (Print/Type)	Christopher M. Goff	Registration No. (Attorney/Agent)	41785

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference B202595PC	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/EP2022/053941	International filing date (<i>day/month/year</i>) 17 February 2022 (17-02-2022)	(Earliest) Priority Date (<i>day/month/year</i>) 18 February 2021 (18-02-2021)
Applicant TRINAMIX GMBH		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 4 sheets.

☒ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the **language**, the international search was carried out on the basis of:

- ☒ the international application in the language in which it was filed
- ☐ a translation of the international application into _____, which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b))

b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6bis(a)).

c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (See Box No. II)

3. ☐ **Unity of invention is lacking** (see Box No III)

4. With regard to the **title**,

- ☒ the text is approved as submitted by the applicant
- ☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

- ☒ the text is approved as submitted by the applicant
- ☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority

6. With regard to the **drawings**,

- a. the figure of the **drawings** to be published with the abstract is Figure No. 3
- ☒ as suggested by the applicant
- ☐ as selected by this Authority, because the applicant failed to suggest a figure
- ☐ as selected by this Authority, because this figure better characterizes the invention
- b. ☐ none of the figures is to be published with the abstract

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2022/053941

A. CLASSIFICATION OF SUBJECT MATTER

INV. **G06F21/32** **G06V10/141** **G06V10/60** **G06V20/64** **G06V40/16**
 G06V40/40

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G06F **G06V**

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2019/213309 A1 (MORESTIN FREDERIC [US] ET AL) 11 July 2019 (2019-07-11) abstract figure 3 paragraph [0025] paragraph [0036] - paragraph [0039] paragraph [0050] paragraph [0053] paragraph [0076] paragraph [0079] paragraph [0081] paragraph [0085] - paragraph [0087] ----- -/--	1-16

☒ Further documents are listed in the continuation of Box C.

☒ See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

31 May 2022

Date of mailing of the international search report

09/06/2022

Name and mailing address of the ISA/
European Patent Office, P.B. 5818 Patentlaan 2
NL - 2280 HV Rijswijk
Tel. (+31-70) 340-2040,
Fax: (+31-70) 340-3016

Authorized officer

Martinez, Francis

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2022/053941

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	ZITONG YU ET AL: "Face Anti-Spoofing with Human Material Perception", ARXIV.ORG, CORNELL UNIVERSITY LIBRARY, 201 OLIN LIBRARY CORNELL UNIVERSITY ITHACA, NY 14853, 4 July 2020 (2020-07-04), XP081716228, the whole document -----	1-16
A	CN 111 164 610 A (SHENZHEN GOODIX TECH CO LTD) 15 May 2020 (2020-05-15) the whole document -----	1-16
A	US 2020/082155 A1 (KALSCHEUR MICAH P [US] ET AL) 12 March 2020 (2020-03-12) the whole document -----	1-16

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2022/053941

Patent document cited in search report		Publication date	Patent family member(s)	Publication date
US 2019213309	A1	11-07-2019	NONE	

CN 111164610	A	15-05-2020	NONE	

US 2020082155	A1	12-03-2020	CN 110895689 A	20-03-2020
			US 2020082155 A1	12-03-2020

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To:

see form PCT/ISA/220

PCT

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY (PCT Rule 43bis.1)

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/EP2022/053941

International filing date (day/month/year)
17.02.2022

Priority date (day/month/year)
18.02.2021

International Patent Classification (IPC) or both national classification and IPC
INV. G06F21/32 G06V10/141 G06V10/60 G06V20/64 G06V40/16 G06V40/40

Applicant
TRINAMIX GMBH

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☐ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. **FURTHER ACTION**

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA:



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D-80298 Munich
Tel. +49 89 2399 - 0
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Date of completion of
this opinion

see form
PCT/ISA/210

Authorized Officer

Martinez, Francis

Telephone No. +49 89 2399-0



Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43bis.1(a))
3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:
 - a. ☐ forming part of the international application as filed:
 - ☐ in the form of an Annex C/ST.25 text file.
 - ☐ on paper or in the form of an image file.
 - b. ☐ furnished together with the international application under PCT Rule 13ter.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.
 - c. ☐ furnished subsequent to the international filing date for the purposes of international search only:
 - ☐ in the form of an Annex C/ST.25 text file (Rule 13ter.1(a)).
 - ☐ on paper or in the form of an image file (Rule 13ter.1(b) and Administrative Instructions, Section 713).
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

Box No. V Reasoned statement under Rule 43*bis*.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Yes: Claims	<u>1-16</u>
	No: Claims	
Inventive step (IS)	Yes: Claims	
	No: Claims	<u>1-16</u>
Industrial applicability (IA)	Yes: Claims	<u>1-16</u>
	No: Claims	

2. Citations and explanations

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

Re Item VIII

Certain observations on the international application

1 The present application does not meet the criteria of Article 6 PCT for the following reasons.

1.1 Claim 1

1.1.1 The claim is unclear because it refers to "identifying [...] geometrical feature characteristic" while later it refers to "the identified geometrical feature".

In addition, it is not clear how geometrical feature characteristic are identified.

1.1.2 The claim is unclear because "each of the reflection features comprises at least one beam profile" is not clear. It is not clear how a beam profile is defined and thus it is not clear how beam profile information of reflection feature is determined. In addition, it is not clear how a reflection feature is defined to have multiple beam profiles.

1.2 Claim 12

The claim is unclear because it is not clear how the computer or computer network partially performs the the method according to any one of the preceding claims. What is effectively performed ?

1.3 Claims 13, 14

The claims are unclear because it recites "the method according to any one of the preceding claims referring to a method", while the preceding claims also encompass a computer program claim or a storage medium claim, which are all referring to a method. The following formulation is suggested : "the method according to any one of claims 1-11".

1.4 Claim 15

The claim is unclear because it recites "the method according to any one of the preceding claims", while the preceding claims also encompass a computer program claim, a storage medium claim and a device claim. The following formulation is suggested : "the method according to any one of claims 1-11".

Re Item V

Reasoned statement with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

2 Reference is made to the following documents:

D1 US 2019/213309 A1 (MORESTIN FREDERIC [US] ET AL) 11 July 2019 (2019-07-11)

- D2 ZITONG YU ET AL: "Face Anti-Spoofing with Human Material Perception",
ARXIV.ORG, CORNELL UNIVERSITY LIBRARY, 201 OLIN LIBRARY CORNELL
UNIVERSITY ITHACA, NY 14853, 4 July 2020 (2020-07-04), XP081716228,
- D3 CN 111 164 610 A (SHENZHEN GOODIX TECH CO LTD) 15 May 2020 (2020-05-15)
- D4 US 2020/082155 A1 (KALSCHEUR MICAH P [US] ET AL) 12 March 2020 (2020-03-12)

3 The present application does not meet the criteria of Article 33(1) PCT because the subject-matter of **claim 1** does not involve an inventive step within the meaning of Article 33(3) PCT.

3.1 Document D1 discloses (the references in bold apply to D1):

Method for face authentication (**abstract**) comprising the following steps:

a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;

[0085] "After the RGB camera 240 is woken up at block 340, the method 300 proceeds to block 350. At block 350, the method 300 includes detecting a face of the user using the RGB camera 240. The RGB sensor 240 detects the user's face by detecting the ROI of a human face, detecting the facial characteristics of the user (e.g., the location of the lips, eyes, nose, eyebrows, ears, the uniqueness of the shape of the lips, eyes, nose, eyebrows, etc.), and may create a face map based on the sensed information."

b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;

[0036] "For example, objects including metal may have a high reflectance compared to the reflectance of a human's skin. Generally, a surface reflectance value of a human being is different from that of other objects such as metal, plastic or the like. The reflectance sensor 104 in conjunction with a processor in the device 100 may have a pre-stored range of the surface reflectance value of a person, and may easily detect whether the reflectance value substantially matches those of a human compared to a non-human object."

[0039] "The skin reflectance value is a value that differs based on surface characteristics of objects. Objects such as human, mannequin, plastics, or metals all have different surface characteristics, which cause the reflectance value to differ. In one embodiment, surface reflectance or skin reflectance (if the reflected object is a human) may be measurements of albedo, the ratio of reflected light over incident light. In other embodiments, the surface reflectance or skin reflectance may be measured based on other parameters or methods. For example, amplitudes of the reflected signal, wavelength of the reflected signal, ratio of light reflectance and absorption, etc., may be used. The skin reflectance, which indicates the surface reflectance of a human, may be within a certain range almost common to most or all human beings. The skin reflectance value may be between a first reflectance threshold and a second reflectance threshold. The first reflectance threshold is a minimum value of skin reflectance that can be measured from a human skin (e.g., lips, nose, eyes, arms, legs, etc.) and the second reflectance threshold is a maximum value of skin reflectance that can be measured from a human skin. That is, if a light is reflected on a human skin, the detected reflectance value will be between the first reflectance threshold and the second reflectance threshold. A surface reflectance value that is not within this range may indicate that the object is not of human skin, but rather, for example, a mannequin."

[0050] "The time of flight sensor can detect both distance from an object and the skin reflectance value of the object."

[0053] "The time of flight sensor 225 may also detect a surface reflectance value along with the distance measurements. When a signal emitted from the sensor 225 is reflected, the reflected signal has a different amplitude from the original amplitude depending on which kind of object reflected the emitted signal. The range of the signal amplitudes detected based on different objects vary, and the processor 255 is capable of determining whether the object 204 that reflected the emitted signal is a human or a non-human object based on the change in signal amplitude. Accordingly, the time of flight sensor 225 is capable of measuring distance, as well as surface reflectance. Due to these features of the time of flight

sensor 225, in some embodiments, the ranging sensor 102 and the reflectance sensor 104 in FIG. 1 may be incorporated into one time of flight sensor 225. This reduces the size of the camera module incorporated in the electronic device 200."

[0086] "If the skin reflectance value is within a first reflectance and a second reflectance threshold"

c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of ~~at least four~~ of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;

[0050] "The time of flight sensor can detect both distance from an object and the skin reflectance value of the object."

[0037] "The reflectance sensor 104 also assists in preventing any attempts of face authentication fraud by using a picture of an authorized user's face for unlocking the device 100. The surface reflectance characteristics of a real human skin and a photo will differ significantly, and the reflectance sensor 104 will stop recognition if there are any attempts at spoofing the sensor. That is, the reflectance sensor 104 will analyze the distances to determine that the detected face is indeed a three-dimensional face, as opposed to a photograph or print of a human face."

d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

[0086] "If the skin reflectance value is within a first reflectance and a second reflectance threshold, and the distance detected is within a face recognizable distance as defined as being between a first presence detection threshold and a second presence detection threshold, this may activate the facial authentication process by proceeding the method 300 to block 340 or 350. That is, a facial authentication process may activate in response to the face recognizable distance and the skin reflectance value"

The subject-matter of claim 1 differs from this in D1 in that at least four of the reflection features are used to determine the second beam profile information. In D1, distances, i.e. at least two reflection features, are analyzed to determine whether the detected face is a 3D face (see [0037] of D1).

The problem to be solved can thus be regarded as how to adapt the method of D1 to improve determination of whether the detected face is a 3D face.

Starting from D1, in particular [0037] of D1, and trying to solve the aforesaid technical problem, it would be an obvious choice to the skilled person to use more distance measurements in order to more accurately determine whether a detected face is a 3D face.

Claim 1 does not therefore involve an inventive step (Art. 33(3) PCT).

4 The additional features of **claims 2 and 3** are also known from D1.

4.1 **Claim 2:** Fig.3, [0081] and [0087] "unlock the device".

4.2 **Claim 3:** [0087] "At block 360, the method 300 includes identifying the user's face with the pre-stored face of the authorized user. The processor 255 may retrieve the face information of the authorized user from the memory 265 and compare the face with the information sensed using the RGB camera 240. If the processor 255 determines at block 370 that the matching level exceeds a minimum threshold for ensuring that the faces are substantially the same face of the authorized user, the method 300 may proceed to block 376 and the processor 255 may unlock the device 200."

4.3 Claims 2 and 3 do not therefore involve an inventive step (Art. 33(3) PCT).

5 Dependent **claims 4-11** do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the PCT with respect to inventive step (Art. 33(3) PCT).

5.1 **Claims 4-5:** Applying a machine learning-based parametric classification model instead of thresholds to detect skin would be obvious to the skilled person in order to improve the accuracy of skin detection.

5.2 **Claims 6-9 and 11:** In view of the ToF sensor in D1, the technical features proposed in the claims are mere design choices which the skilled person would select, depending on the circumstances, without exercising inventive skill.

5.3 **Claim 10:** Using additional images in order to improve classification results is a well-known practice to the skilled person.

6 The reasoning for claim 1 applies similarly to **claims 12-15** (see also [0025], [0076] and [0079] of D1). Claims 12-14 do not therefore involve an inventive step (Art. 33(3) PCT).

7 It is well-known that a face authentication method such as the one disclosed in D1 can be used in the context of detecting presentation attacks (see e.g. abstract of D2). **Claim 16** does not therefore involve an inventive step (Art. 33(3) PCT).

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		
	First Named Inventor	Peter SCHILLEN	
	Art Unit		
	Examiner Name	TBD	
	Attorney Docket Number	27843-2262 202595US01	

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	1	20190213309	A1	2019-07-11	MORESTIN et al.	
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**INFORMATION DISCLOSURE
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(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		
First Named Inventor	Peter SCHILLEN	
Art Unit		
Examiner Name	TBD	
Attorney Docket Number	27843-2262 202595US01	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report and Written Opinion for corresponding PCT/EP2022/053941, mailed June 9, 2022, 12 pages	☐

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	
Filing Date	
First Named Inventor	Peter SCHILLEN
Art Unit	
Examiner Name	TBD
Attorney Docket Number	27843-2262 202595US01

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2023-08-04
Name/Print	Christopher M. Goff	Registration Number	41785

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/264,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

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Postal Code	67063	Country ⁱ	DE		
Inventor 3					Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/264,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		

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Inventor 4				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Manuel		GUENTHER	
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Inventor 5				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
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Mailing Address of Inventor:

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Address 2				
City	Karlsruhe	State/Province		
Postal Code	76185	Country ⁱ	DE	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/254,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Inventor 6					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Christian		LENNARTZ		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
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Address 1	Industriestraße 35				
Address 2					
City	Ludwigshafen am Rhein	State/Province			
Postal Code	67063	Country ⁱ	DE		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.					

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	76656		
Email Address	uspatents@atllp.com	Add Email	Remove Email

Application Information:

Title of the Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		
Attorney Docket Number	27843-2262 202595US01	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	2	Suggested Figure for Publication (if any)	3

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/264,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under

☐ 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not be** the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	76656		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	a 371 of international	PCT/EP22/53941	2022-02-17
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/264,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
21157800.0	EP	2021-02-18	

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/264,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/284,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

☒ Assignee
 ☐ Legal Representative under 35 U.S.C. 117
 ☐ Joint Inventor

☐ Person to whom the inventor is obligated to assign.
 ☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here. ☒

Organization Name

Mailing Address Information For Applicant:

Address 1		Industriestrasse 35	
Address 2			
City	Ludwigshafen am Rhein	State/Province	
Country	DE	Postal Code	67063
Phone Number		Fax Number	
Email Address			

Additional Applicant Data may be generated within this form by selecting the Add button.

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/264,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City			State/Province	
Country ⁱ			Postal Code	
Phone Number			Fax Number	
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.				

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Christopher M. Goff/			Date (YYYY-MM-DD)	2023-08-04 2023-08-08
First Name	Christopher M.	Last Name	Goff	Registration Number	41785
Additional Signature may be generated within this form by selecting the Add button.					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2262 202595US01
		Application Number	18/264,326
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK OPTICAL SKIN DETECTION FOR FACE UNLOCK		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/264,326

RECEIPT DATE / TIME
08/08/2023 04:53:19 PM ET

ATTORNEY DOCKET #
27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE Utility - U.S. National Stage under 35 USC 371

PATENT # -

CONFIRMATION # 2486

FILED BY Lauren Ferro

PATENT CENTER # 62579616

FILING DATE -

CUSTOMER # 76656

FIRST NAMED INVENTOR Peter SCHILLEN

INTL. APPLICATION # -

INTL. FILING DATE -

CORRESPONDENCE ADDRESS

AUTHORIZED BY Christopher Goff

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-08-08_Corrected_ADS-27843-2262.pdf	10	Application Data Sheet	3296 KB

Warning: This is not a USPTO supplied ADS fillable form. Data in the form cannot be automatically loaded to other USPTO systems.

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DOCUMENT	MESSAGE DIGEST(SHA-512)
2023-08-08 Corrected ADS-	20B7B12CEE0B77C5DBAC48CBE41AE90C3A80DC4449131357

27843-2262.pdf

BF9F194682CF3A8EA17FE299CEC11BCF59B428586E5BB3CC
805CA3D2988AD39D65D34814E27BC16F

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/264,326

RECEIPT DATE / TIME
09/13/2023 10:23:11 AM ET

ATTORNEY DOCKET #
27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE Utility - U.S. National Stage under 35 USC 371

PATENT # -

CONFIRMATION # 2486

FILED BY Barbara McGregor

PATENT CENTER # 62786660

FILING DATE -

CUSTOMER # 76656

FIRST NAMED INVENTOR Peter SCHILLEN

INTL. APPLICATION # -

INTL. FILING DATE -

CORRESPONDENCE ADDRESS -

AUTHORIZED BY Christopher Goff

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-09-13 SupplIDS 27843-2262.pdf	4	Information Disclosure Statement (IDS) Form (SB08)	1371 KB
NPL_Yu.pdf	20	Non Patent Literature	9546 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2023-09-13 SupplIDS 27843-2262.pdf	7D5FCA146F666BDE9F2E157A7D02AD39CF3DEAA7F229F4477616271B8470A3310841727B1E384630989C1B8E3459702716F

5EA37C8C272104C5664755DD8BDB9

NPL_Yu.pdf

913ED3B33490BCB7947440BBB8F9A9FCE8140DF40E94625BC
535737D96443F821D40319545D29C683BF12068B49DE312CAB
0021B1682EA9063D61C32CA69256A

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		18264326
	Filing Date		2023-08-04
	First Named Inventor	Peter SCHILLEN	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	27843-2262 202595US01	

U.S.PATENTS							Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	
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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵	

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	18264326
Filing Date	2023-08-04
First Named Inventor	Peter SCHILLEN
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2262 202595US01

	1	YU et al., "Face Anti-Spoofing with Human Material Perception", Arxiv.org, Cornell University Library, 20 pages (2020)	☐
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If you wish to add additional non-patent literature document citation information please click the Add button

Add

EXAMINER SIGNATURE

Examiner Signature		Date Considered	
--------------------	--	-----------------	--

*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	18264326
Filing Date	2023-08-04
First Named Inventor	Peter SCHILLEN
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2262 202595US01

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2023-09-13
Name/Print	Christopher M. Goff	Registration Number	41785

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

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The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		18264326	
	Filing Date		2023-08-04	
	First Named Inventor	Peter Schillen		
	Art Unit			
	Examiner Name			
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4	2018091640	WO	A2	2018-05-24	TRINAMIX GMBH		☐
5	2021170791	WO	A1	2021-09-02	TRINAMIX GMBH		☐
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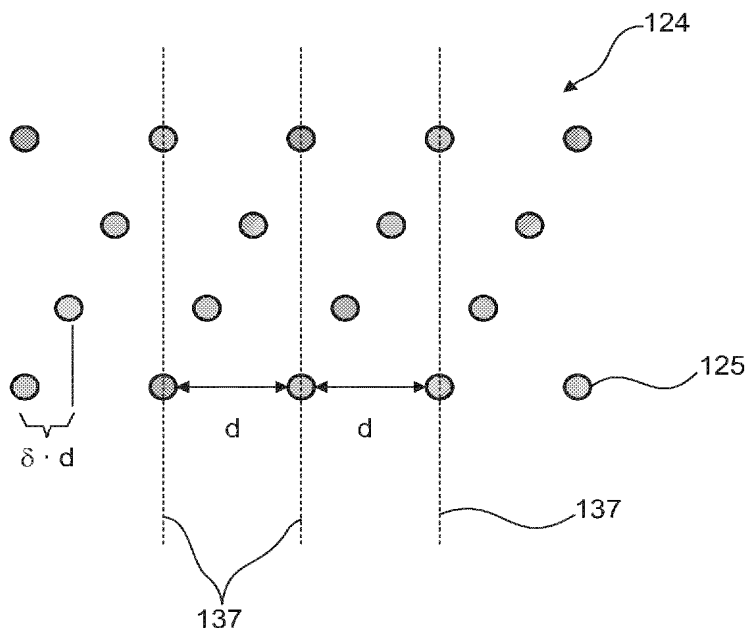
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(54) Title: ILLUMINATION PATTERN FOR OBJECT DEPTH MEASUREMENT

FIG.2A



(57) Abstract: A detector (110) for determining a position of at least one object (112) is disclosed. The detector (110) comprises: - at least one projector (122) for illuminating the object (112) with at least one illumination pattern (124), wherein the illumination pattern (124) comprises a plurality of illumination features (125), wherein the illumination features (125) are spatially modulated; - at least one sensor element (114) having a matrix (116) of optical sensors (118), the optical sensors (118) each having a light-sensitive area (120), wherein each optical sensor (118) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (120) by a reflection light beam propagating from the object to the detector (110), wherein the sensor element (114) is configured to determine at least one reflection image (142); - at least one evaluation device (144), wherein the evaluation device (144) is configured to select at least one reflection feature of the reflection image (142), wherein the evaluation device (144) is configured

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for determining at least one longitudinal coordinate z of the selected reflection feature of the reflection image (142) by using a depth-from-photon-ratio technique by evaluating a combined signal Q from the sensor signals, wherein the illumination features (125) are patterned illumination features (125), wherein each of the patterned illumination features (125) comprises a plurality of sub-features (141), and/or wherein the illumination features (125) are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features (125) have an offset, wherein the offset of neighboring rows differ.

ILLUMINATION PATTERN FOR OBJECT DEPTH MEASUREMENT

Description

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Field of the invention

10 The invention relates to a detector and a method for determining a position of at least one object. The invention further relates to various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

Prior art

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The depth-from-photon-ratio (DPR) technique allows reliable distance determination even in case of environment causing multiple reflections, with biasing light sources, or reflective measurement objects with reduced computational demand, in particular with reduced processing power. For example, WO 2018/091640 describes a detector for determining a position of at least one object. The detector comprises: at least one transfer device, wherein the transfer device has at least one focal length in response to at least one incident light beam propagating from the object to the detector; at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam, at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q from the sensor signals. The detector is adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from the object size in an object plane.

35 However, despite the achievements of the DPR technique, technical challenges remain.

For distance determination using the DPR technique a system comprising a mono-camera may be used in combination with a projector which projects a Laser points pattern or grid on a scene. Each point is evaluated and a distance is determined therefrom. The distance determined with DPR technique may provide a distance estimation per Laser point and can be refined by triangulation methods for known positions of camera and projector. In order to calculate the triangulation, the so-called correspondence problem needs to be solved. The correspondence problem describes assignment of a detected laser point and the position of said point in

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the projected pattern. If this assignment is known, the distance can be determined via triangulation. However, DPR evaluation may be noisy such that the correspondence problem cannot be solved unambiguously for high point densities of the laser point pattern. Specifically, solvability of the correspondence problem may depend on distance of laser points along an epipolar line.

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A further challenging problem with DPR technology is that the reflected beam spot area, shape and 2D intensity profile change confusingly if the reflective objects in the imaged scene are unknown. DPR analyzes the response of the reflecting objects when exposed to individual laser beams. This makes the properties of the reflected laser spots, such as area, shape and 2D intensity profile, mainly depend on the reflector. Analyzing the information provided by these properties of a laser spot, helps in identifying the type of material that reflected this spot, and consequently makes it possible to achieve better accuracy when estimating the distance between the reflective object and the image sensor or the laser beams projector. However, various materials positioned at various distances from the image sensor or the laser beams projector might generate almost identical combinations of laser spot properties, and this consequently makes it difficult to identify the material of the reflecting object and to estimate its distance to the projector or camera.

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Another challenging problem with the present DPR technique is to estimate the orientation of the reflecting surface.

A further such a challenging problem is the difficulty of analyzing a laser spot that is reflected on a sharp edge, when a part of the laser spot is reflected on one side of this sharp edge and another part of it is reflected on the other side of the edge.

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Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods using the DPR technique which may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost, with enhanced precision.

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Summary of the invention

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This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

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In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed.

As used herein, the term “object” refers to an arbitrary object, in particular a surface or region, which is configured to reflect at least partially at least one light beam impinging on the object. The light beam may originate from the projector illuminating the object, wherein the light beam is reflected or scattered by the object.

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As used herein, the term “position” refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

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20 The detector comprises:

- at least one projector for illuminating the object with at least one periodic illumination pattern, wherein the illumination pattern comprises a plurality of illumination features, wherein the illumination features are spatially modulated;
- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is configured to determine at least one reflection image;
- at least one evaluation device, wherein the evaluation device is configured to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal coordinate z of the selected reflection feature of the reflection image by using a depth-from-photon-ratio technique by evaluating a combined signal Q from the sensor signals,

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wherein the illumination features are patterned illumination features, wherein each of the patterned illumination features comprises a plurality of sub-features, and/or wherein the illumination features are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features have an offset, wherein the offset of neighboring rows differ.

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As used herein, the term “projector”, also denoted as light projector, refers to an optical device configured to project at least one illumination pattern onto the object, specifically onto a surface of the object. The projector may comprise at least one light source, also denoted illumination device or illumination source, configured for generating at least one light beam. The projector may be configured for generating at least one pattern and for projecting the pattern towards at

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least one surface or scene comprising the object. The projector may be configured such that the illumination patterns propagate from the projector, in particular from at least one opening of a housing of the projector, towards the object. The projector may be configured for generate and/or to project a cloud of points, for example the projector may comprise at least one digital light processing (DLP) projector, at least one LCoS projector, at least one laser source, at least one array of laser sources; at least one light emitting diode; at least one array of light emitting diodes. Additionally, an additional illumination pattern may be generated by at least one ambient light source.

As used herein, the term “pattern” refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term “illumination pattern” refers to a pattern which illuminates the object. As used herein, the term “illumination feature” refers to at least one arbitrary shaped feature of the illumination pattern. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the projector may be configured for generate and/or to project a cloud of points. For example, the projector may be configured for generate a cloud of points such that the illumination pattern may comprise a plurality of point features.

The projector may be configured to generate a plurality of illumination patterns each comprising a plurality of illumination features. The projector may be configured to project two, three, four, five or more illumination patterns each comprising a plurality of illumination features. The illumination patterns may differ, specifically in one or more of number of illumination features, arrangement of illumination features, shape of illumination features, wavelength of illumination features, intensity of illumination features, opening angle and the like.

The projector may comprise at least one transfer device, in particular at least one diffractive optical element, configured for generate the illumination pattern from at least one light beam generated by the laser source. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are configured for modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element,

preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

5 The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path,
10 with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g.
15 by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter
20 case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

25 The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate z . Any direction perpendicular to the z -axis may be considered a transversal direction, and
30 the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam”
35 generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light pulse may comprise at least one beam profile. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam
40 profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam may be a Gaussian light beam or a linear combination of Gaussian light beams. As used herein, the term “beam profile” relates to a spatial distribution, in particular in at least one plane perpendicular to the propagation of the light beam, of an intensity

of the light beam. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Other embodiments are feasible, however.

- 5 The projector may comprise the at least one transfer device which may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

- 10 The illumination features are spatially modulated. As used herein, the term “spatially modulated” may refer to specific and additional spatial arrangement of the illumination features or parts of the illumination feature in comparison to known, regular pattern. The illumination pattern, in particular the spatial arrangement of illumination features, may be designed with respect to a field of view of the sensor element. Specifically, the illumination features are patterned illumination features, wherein each of the patterned illumination features comprises a plurality of sub-features, and/or the illumination features are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features have an offset, wherein the offset of neighboring rows differ.

- 20 The illumination features are patterned illumination features. Each of the patterned illumination features comprises a plurality of sub-features. As used herein, the term “pattern illumination feature” may refer to an illumination feature comprising a plurality of elements, denoted sub-features. The sub-features belonging to the same illumination feature may be shaped identical. For example, the illumination feature may comprise a plurality of circles each having a center and a radius. The sub-features belonging to the same illumination feature may be arranged at different spatial positions in the illumination pattern. Specifically, the centers of the sub-features are arranged at different spatial positions in the illumination pattern. The extension of the sub-features may be selected such that they are clearly distinguishable. For example, the patterned illumination feature may be or may comprise a patterned light spot comprising a number of smaller light spots, or a cluster of few smaller light spots, packed densely forming a certain pattern. As used herein, a “light spot” generally refers to a visible or detectable round or non-round illumination of the object by a light beam. Rotated versions such as rotated by 45, 90 or 180 degrees of these patterned illumination features can be used as well. The chosen patterned illumination feature may be replicated such as 1000 to 2000 times to form the illumination pattern. In other words, the projected illumination pattern may comprise e.g. 1000 to 2000 copies of the chosen patterned illumination feature.

- 40 For example, the projector may comprise a single light source, in particular a single laser source, configured for generating at least one light beam, also denoted laser beam. The projector may comprise at least one transfer device for diffracting and for replicating the laser beam generated by the single laser source for generating the illumination pattern comprising the patterned illumination features. In particular, the projector comprises the at least one diffractive optical element for diffracting and replicating the light beam. The diffractive optical element may be configured for beam shaping and/or beam splitting. As used herein, the term “replicating” may

refer to generating a plurality of light beams from one light beam, in particular multiplying the light beam.

5 Additionally or alternatively, for example, the projector may comprise at least one array of densely packed light sources, in particular laser sources, according to a certain pattern configured for generating a cluster of light beams. As used herein, the term “densely packed” light sources may refer to a plurality of light sources arranged in a cluster. The density of the light sources may depend on extension of a housing of the individual light sources and distinguishability of the light beams. The projector may comprise the at least one transfer device for diffracting and replicating the cluster of light beams for generating the illumination pattern comprising
10 patterned illumination features.

15 Additionally or alternatively, the illumination features are arranged in a periodic pattern equidistant in rows. Each of the rows of illumination features has an offset, wherein the offset of neighboring rows differ. As used herein, the term “offset” may refer to a spatial distance between neighboring rows. The sensor element and the projector may be positioned such that the rows run parallel to epipolar lines. The illumination pattern may be selected such that two neighboring illumination features have on an epipolar line a suitable distance. The distance between two illumination features may be such that it is possible to assign unambiguously two points on the epipolar line via depth-from-photon-ratio technique. The suitable distance may depend on distance
20 error of the depth-from-photon-ratio technique and/or from a basis line of the sensor element and the projector.

25 The illumination features may be arranged as follows. The illumination pattern may comprise a number of rows on which the illumination features are arranged in equidistant positions with distance d . The rows are orthogonal with respect to the epipolar lines. A distance between the rows may be constant. A different offset may be applied to each of the rows in the same direction. The offset may result in that the illumination features of a row are shifted. The offset δ may be $\delta = a/b$, wherein a and b are positive integer numbers such that the illumination pattern is a periodic pattern. For example, δ may be $1/3$ or $2/5$. The offset and density of illumination features may enhance robustness for solving the correspondence problem.

35 As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. As further used herein, the term “matrix” generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix
40 having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also

feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible. The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. Thus, the term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one CCD and/or CMOS device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the setup. Other embodiments, however, are feasible. Thus, as an example, an optical device comprising two, three, four or more than four light-sensitive areas may be used which is regarded as two, three, four or more than four optical sensors in the context of the present invention. As outlined above, the sensor element comprises a matrix of optical sensors. Thus, as an example, the optical sensors may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be

identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

Specifically, the optical sensors may be or may comprise inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Cu photodiode, a Ge:Hg photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The matrix may be composed of independent optical sensors. Thus, a matrix may be composed of inorganic photodiodes. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1, 4:3, 16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2, n=2$ or $m=3, n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term “essentially perpendicular”, reference may be made to the definition given above. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 50

rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other
5 embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Preferably, the sensor element may be oriented essentially perpendicular to an optical axis of
10 the detector. Again, with respect to the term “essentially perpendicular”, reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter
15 cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The reflected light beams may propagate from the object towards the detector. The projector may illuminate the object with the illumination pattern and the light is reflected or scattered by
20 the object and, thereby, is at least partially directed as reflection light beam towards the detector.

The reflection light beam specifically may fully illuminate the sensor element such that the sensor element is fully located within the light beam with a width of the light beam being larger than the matrix. Contrarily, preferably, the reflection light beam specifically may create a light spot on
25 the entire matrix which is smaller than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may
35 comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be configured for process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

40 The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one

interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

As used herein, the term “reflection image” refers to an image determined by the optical sensor comprising at least one reflection feature. As used herein, the term “reflection feature” refers to a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. The reflection image may comprise the at least one reflection pattern comprising the at least one reflection feature. As used herein, the term “determining at least one reflection image” refers to one or more of imaging, recording and generating of the reflection image.

The sensor element may be configured for determine at least one reflection pattern. As used herein, the term “reflection pattern” refers to a response pattern generated by reflection or scattering of light at the surface of the object, in particular generated by the object in response to illumination by the illumination pattern. The reflection pattern may comprise at least one feature corresponding to at least one feature of the illumination pattern. The reflection pattern may com-

prise, in comparison to the illumination pattern, at least one distorted pattern, wherein the distortion depends on the distance of the object, such as surface properties of the object. The evaluation device may be configured for select at least one feature of the reflection pattern and to determine the longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals, as described above and in more detail below.

As further used herein, the term “evaluation device” generally refers to an arbitrary device configured for perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the relationships, as will be outlined below, may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As used herein, the term “select at least one reflection feature” refers to one or more of identifying, determining and choosing at least one reflection feature of the reflection image. The detector may be configured for determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection image from the combined signal. Thus, the detector may be configured for pre-classify the at least one reflection feature of the reflection image and/or to provide a distance estimate for the reflection feature. Specifically, the detector may be configured for determine at least one more accurate distance information of the object by using triangulation and/or structured light techniques considering the pre-classification and/or distance estimate.

The evaluation device may be configured for perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals

at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device is configured for determining at least one longitudinal coordinate z of the selected reflection feature of the reflection image by using a depth-from-photon-ratio technique by evaluating a combined signal Q from the sensor signals. The depth-from-photon-ratio technique is generally known to the skilled person such as from WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference. The technique of determining distance as described in WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1 by using photon ratio is denoted herein "depth-from-photon-ratio" or "beam profile analysis".

As used herein, the term "combined signal Q " refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region.

For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A_1 and A_2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A_1 and area A_2 may differ. In particular, A_1 and A_2 are not congruent. Thus, A_1 and A_2 may differ in one or more of the shape or content.

Generally the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$, $E(x, y; z_o) = L \cdot S$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determi-

nation of the distance z_0 independent from the object size. Thus, the combined signal allows determination of the distance z_0 independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

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Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q. The light-sensitive areas may be arranged such that a first sensor signal
10 comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

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The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the

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beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e.

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proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term

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“essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information

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of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile
may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

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Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first

area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

5 The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be configured for determining an area integral of the beam profile. The evaluation device may be configured for determine the edge information by integrating and/or summing of the first area. The evaluation device may be configured for determine the center information by integrating and/or summing of
10 the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be configured for determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and
15 deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be configured for determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot
20 may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

In one embodiment, the light beam propagating from the object to the detector may illuminate
25 the sensor element with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on
30 the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the
35 full radius.

The evaluation device may be configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the
40 center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

For example, the evaluation device may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - 5 c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining the at least one longitudinal coordinate z of the selected feature by evaluating the combined signal.
- 10 As explained, e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot may be characterized by two measurement variables: a measurement signal
- 15 measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their re-
- 20 spective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of
- 25 the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming the combined signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the quotient signal for deriving the longitudinal coordinate.
- 30 The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus,
- 35 may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.
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The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or
5 alternatively, as will be outlined in further detail below, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

Consequently, in accordance with the present invention, the term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. For example, the center signal may be the signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the ma-
10 trix. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from
15 which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from which the center signal arises may be chosen as large as possible in order to allow maximum
20 dynamic range. The evaluation device may be configured for determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be configured for determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal
25 may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from
30 the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical

sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be configured for determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be configured for determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

Additionally or alternatively, the evaluation device may be configured for determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center

signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

The evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the object by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

Raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations, however, are feasible.

As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be performed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center

signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the combined signal Q . This determining may be a one-step process, such

as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined signal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

The evaluation device may be configured for using at least one predetermined relationship between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The combined signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider. The divider may fully or partially be integrated into the sensor element answers or may fully or partially be embodied independent from the sensor element.

Depth measurements using depth-from-photon-ratio technique allow reliable distance determination even in case of environment causing multiple reflections, with biasing light sources, or reflective measurement objects with reduced computational demand, in particular with reduced processing power. The present invention allows identifying material of the reflecting object and to estimate its distance to the projector or sensor element even in the presence of various materials positioned at various distances from the sensor element or if the projector generates almost identical combinations of laser spot properties. Moreover, the present invention allows estimating the orientation of the reflecting surface and edge detection. The illumination pattern comprising spatially modulated illumination features may make at least one property or parameter of the resulted modulated illumination feature more robust against the deformation caused by the physical properties of the materials of the reflecting target objects. This at-least-one property or parameter may depend only on the distance between the reflecting object and the projector or the sensor element. The other properties or parameters of the illumination features may be allowed to be changed and deformed to provide information about the type and the physical properties of the reflecting material and its distance to the projector or the sensor element. As outlined above, the illumination features may be patterned illumination features, wherein each of the patterned illumination features comprises a plurality of sub-features. The illumination feature impinging on the object may illuminate the object. The object may generate, e.g. by reflection, in response to the impinging illumination feature the corresponding reflection feature. The illumination feature is patterned thus, the reflection feature is patterned, too. The evaluation device may be configured for determining information about a distance between the

object and the projector and/or the sensor element by analyzing the patterned reflection features of the reflection image. The evaluation device may be configured for finding and segmenting each patterned reflection feature. As used herein, the term "finding" may refer to identifying in the reflection image the reflection feature being generated in response to the corresponding illumination feature. The identifying may comprise performing at least one object recognition algorithm for identifying features in the reflection image. The identifying may further comprise assigning the identified reflection features to the corresponding illumination features. As used herein, the term "segmenting" may refer to a process of partitioning the reflection feature into multiple segments. The segments may correspond to the sub-features of the reflection image being generated in response to the corresponding sub-feature of the illumination feature. The evaluation device may be configured for determining a center of each of the sub-features. The evaluation device may be configured for determining distances between the centers of the sub-features. The evaluation device may be configured for determining the information about the distance between the object and the projector and/or the sensor element from the distances between the centers of the sub-features by using a pre-determined relationship. The evaluation device may be configured for determining a corrected longitudinal coordinate z considering the determined distance between the object and the projector and/or the sensor element. The correct longitudinal coordinate may be a longitudinal coordinate with enhanced accuracy.

The evaluation device may be configured for identifying a material property of the object by speckle contrast estimation within each sub-feature. The evaluation device may be configured for determining a corrected longitudinal coordinate z considering the material property. Specifically, laser spot profile analysis of the depth-from-defocus technique may be combined with speckle contrast estimation or computation within each laser spot to identify the material of the reflecting object and to further estimate its distance to the projector or sensor element, more accurately. Distance determination by using the depth-from-photon-ratio (DPR) technique as described herein is a different technique for determining distances compared to depth-from-defocus. However, the information determined by depth-from-photon-ratio and depth-from-defocus may be combined for even further enhanced distance determination.

The evaluation device may be configured for edge detection. The evaluation device may be configured for determining if the patterned reflection feature was generated by an edge. The illumination feature may comprise at least three sub-features. If said illumination feature impinges on the object, it may generate a reflection feature comprising at least three sub-features corresponding to the sub-features of the illumination feature having generated the reflection feature. The evaluation device may be configured for determining if the sub-features of the reflection feature were generated by a flat reflecting surface of the object or by an edge of the object. The evaluation device may be configured for determining for each of the sub-features a longitudinal coordinate of the object by evaluating a combined signal Q from the sensor signals of the respective sub-feature. The evaluation device may be configured for comparing the longitudinal coordinates of the sub-features. In case the longitudinal coordinates of the sub-features are identical within tolerances, the reflecting surface having generated said sub-features of the reflection feature may be considered as flat. Otherwise the object may be considered to be an

edge or to comprise an edge. The evaluation device may be configured for determining normal and local orientation of a reflecting surface of the object from the longitudinal coordinates determined from the sub-features. The evaluation device may be configured for determining a plane spanned by the sub-features. The evaluation device may be configured for determining a surface normal of the plane and/or an orientation of the plane in space. For example, the orientation may be defined by an angle between z-axis, in particular optical axis, and surface normal. Only one patterned illumination feature may be enough to be able to estimate the normal and the local orientation of the reflecting surface, and to know if this patterned laser spot is reflected on an edge or not.

In addition, in the case of problematic reflecting surfaces, the probability to get a usable laser spot reflection is higher when projecting such dense clusters of spots, compared to single spots that are relatively sparsely distributed.

The present invention may allow enhancing performance of the distance and depth measurement using depth-from-photon-ratio technology. This can be done by a simple modification on the hardware side and a simple computationally efficient addition to the currently used algorithm. The resulting detector may get much better performance and can be used for numerous additional applications.

The distance determined with DPR technique may provide a distance estimation per illumination feature and can be refined by triangulation methods for known positions of sensor element and projector. In order to calculate a refined longitudinal coordinate using triangulation, the so-called correspondence problem needs to be solved. In case of using regular patterns, such as a rectangle patterns, it may happen that two neighboring points may be directly neighbors on the epipolar line. For a robust correspondence problem, the physical distance of the features of the pattern may be large such that the point density of the pattern may be low. As outlined above, the present application proposes that the illumination features are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features have an offset, wherein the offset of neighboring rows differ. The offset may allow the reach at a feature density having a large distance on the epipolar line and enhanced robustness of the correspondence problem. The selection of difference offsets may be such that the structure of the illumination pattern can be designed with respect to a field of view of the sensor element.

The evaluation device may be configured to determine a longitudinal region of the selected reflection feature by evaluating the combined signal Q , wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \epsilon$. As used herein, the term "longitudinal region" refers to at least one uncertainty interval defined by the longitudinal coordinate z and the measurement uncertainty $\pm \epsilon$ of the determination of the longitudinal coordinate from the combined signal Q . Error ϵ may depend on measurement uncertainty of the optical sensor. The measurement uncertainty of the optical sensors may be pre-determined and/or estimated and/or may be deposited in at least one data storage unit of the evaluation device. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$.

The evaluation device may be configured to determine at least one displacement region in at least one reference image corresponding to the longitudinal region. As used herein, the term “reference image” refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be configured for determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. As outlined above, the evaluation device may be configured for perform an image analysis and to identify features of the reflection image. The evaluation device may be configured for identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term “essentially identical” refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“, Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The evaluation device may be configured for determine an epipolar line in the reference image. Relative position of the reference image and reflection image may be known. For example, relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be configured for determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teach-in of the epipolar geometry of the specific camera.

As used herein, the term “displacement region” refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the

reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term “reference feature” refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be configured for determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance measurement using the combined signal Q may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extent in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device may determine a displacement region around the image position of the reflection feature. The evaluation device may be configured for determine the longitudinal coordinate z for the reflection feature and an error interval $\pm\epsilon$ from the combined signal Q to determine the displacement region along the epipolar line corresponding to $z \pm \epsilon$. The evaluation device may be configured for match the selected reflection feature with at least one reference feature within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be configured for match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be configured for determine the epipolar line closest to and/or within the displacement region. The evaluation device may be configured for determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be configured for determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be configured for assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be configured for determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be configured to perform the following steps:

- Determining the displacement region for the image position of each reflection feature;
- 5 - Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or
10 within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

15 Additionally or alternatively, the evaluation device may be configured for decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ϵ -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or reference feature in shorter distance and/or ϵ -weighted distance to the reference feature and/or reflection feature.
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Preferably, the detector may be configured for pre-classify the selected reflection feature using the combined signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference image may have a relative distance to each
25 other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line.

The evaluation device may be configured for determine a displacement of a matched reference feature and the selected reflection feature. As used herein, the term "displacement" refers to difference between a position in the reference image to a position in the reflection image. The evaluation device may be configured for determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term "longitudinal information" refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a distance value. The
30 evaluation device may be configured for determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be configured for select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at
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least one storage device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship. The evaluation device may be configured for store parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector. The evaluation device may be configured for generate the parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector such as by performing a Tsai camera calibration. The evaluation device may be configured for compute and/or estimate parameters such as the focal length of the transfer device, the radial lens distortion coefficient, the coordinates of the center of radial lens distortion, scale factors to account for any uncertainty due to imperfections in hardware timing for scanning and digitization, rotation angles for the transformation between the world and camera coordinates, translation components for the transformation between the world and camera coordinates, aperture angles, image sensor format, principal point, skew coefficients, camera center, camera heading, baseline, rotation or translation parameters between camera and/or illumination source, apertures, focal distance, or the like.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce the possible number of solutions along the epipolar line significantly. The number of possible solutions may even be reduced to one. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may enhance accuracy and/or speed and/or may lower computational demand in comparison to conventional triangulation systems. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be reduced to increase the light intensity in each illumination point such as to compete with ambient light while complying to eye safety regulations. A reduced number of illumination features in a conventional triangulation system might increase the difficulty to match reflection features and reference features. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be increased, such as to increase the resolution of the distance measurement, such as to increase the resolution of the obtained depth map without increasing the processing power of the evaluation device such as in a mobile application.

For example, the reference image may be an image of the illumination pattern at an image plane at a position of the projector. The evaluation device may be configured for determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be configured for match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The projector and the sensor element may be separated by a fixed distance.

For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be configured for determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be configured for determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be configured for select at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The evaluation device may be configured for select the at least one reflection feature in the reflection pattern and to determine the longitudinal region of the selected feature by evaluating the combined signal Q from the sensor signals. The evaluation device may be configured for determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be configured for match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*,

Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be configured for combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used. The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method comprises the following method steps:

- Illuminating the object with at least one illumination pattern generated by the at least one projector of the detector, wherein the illumination pattern comprises a plurality of illumination features, wherein the illumination features are spatially modulated;
- generating for each reflection light beam impinging on the light-sensitive areas of the optical sensors of the sensor element having the matrix of optical sensors at least one sensor signal in response to an illumination;
- determining at least one reflection image by using the sensor element;
- selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal coordinate z of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to

the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application. With respect to further uses of the detector and devices of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

As used herein, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, notwithstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

- 5 Embodiment 1: A detector for determining a position of at least one object, the detector comprising:
- at least one projector for illuminating the object with at least one illumination pattern, wherein the illumination pattern comprises a plurality of illumination features, wherein the illumination features are spatially modulated;
 - 10 - at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is configured to determine at least one reflection image;
 - 15 - at least one evaluation device, wherein the evaluation device is configured to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal coordinate z of the selected reflection feature of the reflection image by using a depth-from-photon-ratio technique by evaluating a combined signal Q from the sensor signals,
 - 20 wherein the illumination features are patterned illumination features, wherein each of the patterned illumination features comprises a plurality of sub-features, and/or wherein the illumination features are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features have an offset, wherein the offset of neighboring rows differ.
- 25 Embodiment 2: The detector according to the preceding embodiment, wherein the projector comprises at least one light source configured for generating at least one light beam, wherein the projector comprises at least one transfer device configured for diffracting and for replicating the light beam for generating the illumination pattern comprising patterned illumination features.
- 30 Embodiment 3: The detector according to any one of the preceding embodiments, wherein the projector comprises at least one array of densely packed light sources according to a certain pattern configured for generating a cluster of light beams, wherein the projector comprises at least one transfer device for diffracting and replicating the cluster of light beams for generating the illumination pattern comprising patterned illumination features.
- 35 Embodiment 4: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining information about a distance between the object and the projector and/or the sensor element by analyzing the patterned reflection features of the reflection image.
- 40 Embodiment 5: The detector according to the preceding embodiment, wherein the evaluation device is configured for finding and segmenting each patterned illumination feature, wherein the evaluation device is configured for determining a center of each of the sub-features, wherein the

evaluation device is configured for determining distances between the centers of the sub-features, wherein the evaluation device is configured for determining the information about the distance between the object and the projector and/or the sensor element from the distances between the centers of the sub-features by using a pre-determined relationship.

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Embodiment 6: The detector according to any one of the two preceding embodiments, wherein the evaluation device is configured for determining a corrected longitudinal coordinate z considering the determined distance between the object and the projector and/or the sensor element.

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Embodiment 7: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for identifying a material property of the object by speckle contrast estimation within each sub-feature, wherein the evaluation device is configured for determining a corrected longitudinal coordinate z considering the material property.

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Embodiment 8: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for edge detection, wherein the illumination feature comprises at least three sub-features, wherein the evaluation device is configured for determining for each of the sub-features a longitudinal coordinate of the object by evaluating a combined signal Q from the sensor signals of the respective sub-feature, wherein the evaluation device is configured for determining normal and local orientation of a reflecting surface of the object from the longitudinal coordinates determined from the sub-features.

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Embodiment 9: The detector according to any one of the preceding embodiments, wherein the sensor element and the projector are positioned such that the rows run parallel to epipolar lines.

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Embodiment 10: The detector according to any one of the 9 preceding embodiments, wherein the offset δ is $\delta = \frac{a}{b}$, wherein a and b are integer numbers.

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Embodiment 11: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate z for determining the longitudinal coordinate.

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Embodiment 12: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured to determine a longitudinal region of the selected reflection feature by evaluating the combined signal Q , wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \epsilon$, wherein the evaluation device is configured to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the evaluation device is configured to match the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured to determine a displacement of the matched reference feature and the

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selected reflection feature, wherein the evaluation device is configured to determine a longitudinal information of the matched reference feature using a predetermined relationship between the longitudinal information and the displacement.

5 Embodiment 13: A method for determining a position of at least one object by using at least one detector according to any one of the preceding embodiments, the method comprising the following steps:

- 10 - Illuminating the object with at least one illumination pattern generated by the at least one projector of the detector, wherein the illumination pattern comprises a plurality of illumination features, wherein the illumination features are spatially modulated, wherein the illumination features are patterned illumination features, wherein each of the patterned illumination features comprises a plurality of sub-features, and/or wherein the illumination features are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features have an offset, wherein the offset of neighboring rows differ;
- 15 - generating for each reflection light beam impinging on the light-sensitive areas of the optical sensors of the sensor element having the matrix of optical sensors at least one sensor signal in response to an illumination;
- determining at least one reflection image by using the sensor element;
- 20 - selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal coordinate z of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals.

25 Embodiment 14: A use of the detector according to any one of the preceding embodiments referring to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a
30 quality control application; a manufacturing application.

Brief description of the figures

35 Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements
40 which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1 shows an embodiment of a detector according to the present invention;

Figures 2A and 2B show embodiments of illumination patterns; and

5 Figures 3A and 3B show embodiments of illumination features and a further embodiment of an illumination pattern.

Detailed description of the embodiments:

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Fig. 1 shows in a highly schematic fashion an embodiment of a detector 110 for determining a position of at least one object 112 according to the present invention. The detector 110 comprises at least one sensor element 114 having a matrix 116 of optical sensors 118. The optical sensors 118 each have a light-sensitive area 120.

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The sensor element 114 may be formed as a unitary, single device or as a combination of several devices. The matrix 116 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix 116 may be a single row of pixels. Other arrangements are feasible.

20

The optical sensors 118 of the matrix 116 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 120 of all optical sensors 118 of the matrix 116 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 120 may specifically be located on a surface of the respective optical sensor 118. Other embodiments, however, are feasible.

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The optical sensors 118 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 118 may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 120. Preferably the detector is configured such that the optical sensors 118 are exposed simultaneously within a certain time period, denoted as frame or imaging frame. For example, the optical sensors 118 may be part of or constitute at least one global shutter CMOS.

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The optical sensors 118 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 118 may be sensitive in the infrared spectral range. All of the optical sensors 118 of the matrix 116 or at least a group of the optical

- sensors 118 of the matrix 116 specifically may be identical. Groups of identical optical sensors 118 of the matrix 116 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 118 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix
- 5 116 may be composed of independent optical sensors 118. Thus, a matrix 116 may be composed of inorganic photodiodes. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.
- 10 The optical sensors 118 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 118, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or
- 15 higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.
- 20 The matrix 116 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 116 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably
- 25 at least 100 columns. The matrix 116 may comprise at least 50 optical sensors 118, preferably at least 100 optical sensors 118, more preferably at least 500 optical sensors 118. The matrix 116 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.
- 30 The detector 110 further comprises a projector 122 for illuminating the object 112 with at least one illumination pattern 124. The projector 122 may comprise at least one laser source 126, in particular for generating at least one light beam. The projector 122 comprises the at least one diffractive optical element 128, in particular for generating and/or forming the illumination pattern 124 from the light beam of the laser source 126. The projector 122 may be configured such that
- 35 the illumination pattern 124 propagates from the projector 122, in particular from at least one opening 130 of a housing of the projector 122, towards the object 112. The projector 122 may be configured for generate and/or to project a cloud of points, for example the projector 122 may comprise at least one digital light processing (DLP) projector, at least one LCoS projector, at least one laser source, at least one array of laser sources; at least one light emitting diode; at
- 40 least one array of light emitting diodes. The laser source 126 may comprise focusing optics 134. The projector 122 may comprise a plurality of laser sources 126. Additionally, an additional illumination pattern may be generated by at least one ambient light source.

The projector 122 may comprise at least one control unit 136. The control unit 136 may be configured for controlling the laser source 126. The control unit 136 may comprise at least one processing device, in particular at least one processor and/or at least one application-specific integrated circuit (ASIC). The control unit 136 may comprise one or more programmable devices
5 such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the controlling of the laser source 126. The control unit 136 may comprise at least one processing device having a software code stored thereon comprising a number of computer commands. The control unit 136 may provide one or more hardware elements for performing controlling the laser source 126 and/or may provide one or more processors with software running
10 thereon for performing controlling of the laser source. The control unit 136 may be configured to issue and/or to generate at least one electronic signal for controlling the laser source. The control unit 136 may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for controlling the laser source 126. The control unit 136 and the laser
15 source may be interconnected by one or more connectors and/or by one or more interfaces.

The illumination pattern 124 comprises a plurality of illumination features 125. The illumination pattern 124 may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi
20 random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the
25 projector 122 may be configured for generate and/or to project a cloud of points. For example, the projector 122 may be configured for generate a cloud of points such that the illumination pattern may comprise a plurality of point features.

The illumination features 125 are spatially modulated. The illumination pattern, in particular the
30 spatial arrangement of illumination features 125, may be designed with respect to a field of view of the sensor element 114. Specifically, the illumination features 125 are patterned illumination features 125, wherein each of the patterned illumination features 125 comprises a plurality of sub-features, and/or the illumination features 125 are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features 125 have an offset, wherein the offset
35 of neighboring rows differ.

As shown in Figure 2A, the illumination features 125 may be arranged in a periodic pattern equidistant in rows. The distance between neighboring illumination features on a row may be d . Each of the rows of illumination features 125 may have an offset δ , wherein the offset of neighboring rows differ. The offset δ may be a spatial distance between neighboring rows. The sensor element 114 and the projector 122 of Figure 1 may be positioned such that the rows run
40 parallel to epipolar lines 137. The illumination pattern 124 may be selected such that two neighboring illumination features 125 have on an epipolar line 124 a suitable distance. The distance

between two illumination features 125 may be such that it is possible to assign unambiguously two points on the epipolar line 124 via depth-from-photon-ratio technique. The suitable distance may depend on distance error of the depth-from-photon-ratio technique and/or from a basis line of the sensor element 114 and the projector 122.

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The illumination features 125 may be arranged as follows. The illumination pattern 124 may comprise a number of rows on which the illumination features 125 are arranged in equidistant positions with distance d . The rows are orthogonal with respect to the epipolar lines 137. A distance between the rows may be constant. A different offset may be applied to each of the rows in the same direction. The offset may result in that the illumination features of a row are shifted. The offset δ may be $\delta = a/b$, wherein a and b are positive integer numbers such that the illumination pattern is a periodic pattern. For example, δ may be $1/3$ or $2/5$. The so constructed illumination pattern 124 reveals a shifted grid in comparison to the initial regular rectangular pattern. The distance between features on the epipolar lines 137 for this grid arrangement is three times larger compared to the initial regular rectangular pattern. The offset and density of illumination features 125 may enhance robustness for solving the correspondence problem. Figure 2B shows the illumination pattern 124 in the field of view 139 of the sensor element 114. By using the offset, the illumination features 125 can be arranged such that the illumination pattern 124 matches with the field of view 139 of the sensor element 114.

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The illumination features 125 may be patterned illumination features. Each of the patterned illumination features may comprise a plurality of sub-features 141. The sub-features 141 belonging to the same illumination feature 125 may be shaped identical. For example, the illumination feature 125 may comprise a plurality of circles each having a center and a radius. The sub-features 141 belonging to the same illumination feature 125 may be arranged at different spatial positions in the illumination pattern 124. Specifically, the centers of the sub-features 141 are arranged at different spatial positions in the illumination pattern 124. The extension of the sub-features 141 may be selected such that they are clearly distinguishable. For example, the patterned illumination feature 125 may be or may comprise a patterned light spot comprising a number of smaller light spots, or a cluster of few smaller light spots, packed densely forming a certain pattern. Rotated versions such as rotated by 45, 90 or 180 degrees of these patterned illumination features can be used as well. The chosen patterned illumination feature 125 may be replicated such as 1000 to 2000 times to form the illumination pattern 124. In other words, the projected illumination pattern 124 may comprise e.g. 1000 to 2000 copies of the chosen patterned illumination feature 125.

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For example, the projector 122 of Figure 1 may comprise a single light source, in particular a single laser source 126, configured for generating at least one light beam, also denoted laser beam. The projector 122 may comprise the at least one transfer device, in particular the DOE 128, for diffracting and for replicating the laser beam generated by the single laser source for generating the illumination pattern 124 comprising the patterned illumination features. The diffractive optical element 128 may be configured for beam shaping and/or beam splitting.

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For example, the projector 122 may comprise at least one array of densely packed light sources, in particular laser sources 126, according to a certain pattern configured for generating a cluster of light beams. The density of the laser sources 126 may depend on extension of a housing of the individual light sources and distinguishability of the light beams. The projector
5 122 may comprise the at least one transfer device, in particular the DOE 128, for diffracting and replicating the cluster of light beams for generating the illumination pattern 124 comprising patterned illumination features.

Figure 3A shows an example of patterned illumination features that can be used to form the illumination pattern 124 that is projected onto the object 112. The illumination features 125 of the
10 upper row comprise four sub-features 141 in different spatial arrangements. In the lower row from left to right, an illumination feature 125 with two sub-features 141, an illumination feature 125 with three sub-features 141 and an illumination feature 125 with seven sub-features 141 are depicted. Fig. 3B shows a projected illumination pattern 124 comprising 26 patterned illumination features 125 with sub-features 142.
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In Figure 1, each optical sensor 118 is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 120 by a reflection light beam propagating from the object 112 to the detector 110. Furthermore, the sensor element 114 is
20 configured to determine at least one reflection image 142 comprising at least one reflection pattern 138. The reflection image 142 may comprise points as reflection features. These points result from reflection light beams originating from the object 112. The sensor element 114 may be configured to determine the reflection pattern 138. The reflection pattern 138 may comprise at least one feature corresponding to at least one illumination feature 125 of the illumination pattern 124. The reflection pattern 138 may comprise, in comparison to the illumination pattern
25 124, at least one distorted pattern, wherein the distortion depends on the distance of the object 112, such as surface properties of the object 112.

The detector 110 may comprise the at least one transfer device 140 comprising one or more of:
30 at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In
35 particular, the transfer device 140 may comprise at least one collimating lens configured to focus at least one object point in an image plane.

The detector 110 comprises at least one evaluation device 144. The evaluation device 144 is configured to select at least one reflection feature of the reflection image 142. The evaluation
40 device 144 may be configured to select at least one feature of the reflection pattern 138 and to determine the longitudinal coordinate of the selected feature of the reflection pattern by evaluating a combined signal Q from the sensor signals, as described above. Thus, the detector 110 may be configured to pre-classify the at least one reflection feature of the reflection image 142.

The evaluation device 144 may be configured to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors 118.

The evaluation device 144 is configured for determining at least one longitudinal coordinate z of the selected reflection feature of the reflection image 142 by evaluating a combined signal Q from the sensor signals. The evaluation device 144 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device 144 may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region. For example, the evaluation device 144 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A_1 and area A_2 may differ. In particular, A_1 and A_2 are not congruent. Thus, A_1 and A_2 may differ in one or more of the shape or content. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$, $E(x, y; z_o) = L \cdot S$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal

allows determination of the distance z_0 independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

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Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. The light-sensitive areas 120 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

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The evaluation device 144 may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably, the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1. Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

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The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 144 may be configured

for determining an area integral of the beam profile. The evaluation device 144 may be configured to determine the edge information by integrating and/or summing of the first area. The evaluation device 144 may be configured to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be configured to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device 144 may be configured to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

For example, the evaluation device 144 may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor 118 having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors 118 of the matrix 116 and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the at least one longitudinal coordinate z of the selected feature by evaluating the combined signal.

For example, the center signal may be the signal of the at least one optical sensor 116 having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors 118 of the entire matrix 116 or of a region of interest within the matrix 116, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix 116. The center signal may arise from a single optical sensor 118 or from a group of optical sensors 118, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors 118 may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors 118 from which the center signal arises may be a group of neighboring optical sensors, such as optical sensors 118 having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors 118 from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device 144 may be configured to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal.

The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. The detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be performed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device 144 may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor 118 having the highest sensor signal and a predetermined group of neighboring optical sensors 118; a sum of sensor signals from a group of optical sensors 118 containing the optical sensor 118 having the highest sensor signal and a predetermined group of neighboring optical sensors 118; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors 118 containing the optical sensor 118 having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix 116 or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors 118 of the matrix 116. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors 118 from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device 144 may be configured to determine the sum signal by integrating signals of the entire matrix 116, or of the region of interest within the matrix 116. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

The sum signal may be derived from all sensor signals of the matrix 116, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors 118 contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix 116; a sum of all sensor signals of the matrix 116; an integral of all sensor signals of the matrix 116; an average over all sensor signals of the matrix 116 except for sensor signals from those optical sensors 118 contributing to the center signal; a sum of all sensor signals of the matrix 116 except for sensor signals from those optical sensors 118 contributing to the center signal; an integral of all sensor signals of the matrix 116 except for sensor signals from those optical sensors 118 contributing to the center signal; a sum of sensor signals of optical sensors 118 within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor 118 having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor 118 having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors 118 being located within a predetermined range from the optical sensor 118 having the highest sensor signal. Other options, however, exist. The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device 144 may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal. The evaluation device 144 may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device 144 may be configured to determine the at least one coordinate z of the by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

The comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q, derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device 144 may be configured for forming the one or more quotient signals. The evaluation device 144 may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient signal.

The evaluation device 144 specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}} / s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the

center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

The evaluation device 144 may be configured for using at least one predetermined relationship between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 144 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

Depth measurements using depth-from-photon-ratio technique allow reliable distance determination even in case of environment causing multiple reflections, with biasing light sources, or reflective measurement objects with reduced computational demand, in particular with reduced processing power. The present invention allows identifying material of the reflecting object and to estimate its distance to the projector 122 or sensor element 114 even in the presence of various materials positioned at various distances from the sensor element 114 or if the projector 122 generates almost identical combinations of laser spot properties. Moreover, the present invention allows estimating the orientation of the reflecting surface and edge detection. The illumination pattern 124 comprising spatially modulated illumination features 125 may make at least one property or parameter of the resulted modulated illumination feature more robust against the deformation caused by the physical properties of the materials of the reflecting target objects. This at-least-one property or parameter may depend only on the distance between the reflecting object 112 and the projector 122 or the sensor element 114. The other properties or parameters of the illumination features 125 may be allowed to be changed and deformed to provide information about the type and the physical properties of the reflecting material and its distance to the projector 122 or the sensor element 114. As outlined above, the illumination features 125 may be patterned illumination features, wherein each of the patterned illumination features 125 comprises a plurality of sub-features 141. The evaluation device 144 may be configured for determining information about a distance between the object 112 and the projector 122 and/or the sensor element 114 by analyzing the patterned reflection features of the reflection image 142. The evaluation device 144 may be configured for finding and segmenting each patterned illumination feature. The evaluation device 144 may be configured for determining a center of each of the sub-features. The evaluation device 144 may be configured for determining distances between the centers of the sub-features 141. The evaluation device 144 may be configured for determining the information about the distance between the object 112 and the projector 122 and/or the sensor element 114 from the distances between the centers of the sub-features 141 by using a pre-determined relationship. The evaluation device 144 may be configured for determining a corrected longitudinal coordinate z considering the determined distance between the object 112 and the projector 122 and/or the sensor element 114. The correct longitudinal coordinate may be a longitudinal coordinate with enhanced accuracy.

The evaluation device 144 may be configured for edge detection. The illumination feature 125 may comprise at least three sub-features 141. The evaluation device 144 may be configured for

determining for each of the sub-features 141 a longitudinal coordinate of the object 112 by evaluating a combined signal Q from the sensor signals of the respective sub-feature 141. The evaluation device 144 may be configured for determining normal and local orientation of a reflecting surface of the object 112 from the longitudinal coordinates determined from the sub-features 141. Only one patterned illumination feature may be enough to be able to estimate the normal and the local orientation of the reflecting surface, and to know if this patterned laser spot is reflected on an edge or not.

In addition, in the case of problematic reflecting surfaces, the probability to get a usable laser spot reflection is higher when projecting such dense clusters of spots, compared to single spots that are relatively sparsely distributed.

The present invention may allow enhancing performance of the distance and depth measurement using depth-from-photon-ratio technology. This can be done by a simple modification on the hardware side and a simple computationally efficient addition to the currently used algorithm. The resulting detector 110 may get much better performance and can be used for numerous additional applications.

The distance determined with DPR technique may provide a distance estimation per illumination feature and can be refined by triangulation methods for known positions of sensor element 114 and projector 122. In order to calculate a refined longitudinal coordinate using triangulation, the so-called correspondence problem needs to be solved. In case of using regular patterns, such as a rectangle patterns, it may happen that two neighboring points may be directly neighbors on the epipolar line. For a robust correspondence problem, the physical distance of the features of the pattern may be large such that the point density of the pattern may be low. As outlined above, the present application proposes that the illumination features 125 are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features have an offset, wherein the offset of neighboring rows differ. The offset may allow the reach at a feature density having a large distance on the epipolar line and enhanced robustness of the correspondence problem. The selection of difference offsets may be such that the structure of the illumination pattern 124 can be designed with respect to a field of view of the sensor element 114.

The evaluation device 144 may be configured to determine a longitudinal region of the selected reflection feature by evaluating the combined signal Q, wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \epsilon$. Error ϵ may depend on measurement uncertainty of the optical sensor 118. The measurement uncertainty of the optical sensors 118 may be pre-determined and/or estimated and/or may be deposited in at least one data storage unit of the evaluation device 144. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$.

The evaluation device 144 may be configured to determine at least one displacement region in at least one reference image corresponding to the longitudinal region. The reference image may

be an image different from the reflection image which is determined at a different spatial position compared to the reflection image 142. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device 144 may be configured to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device 144 may be configured to perform an image analysis and to identify features of the reflection image 142. The evaluation device 144 may be configured to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference image and the reflection image 142 may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The reference image and the reflection image 142 may be images of the object 112 determined at different spatial positions having a fixed distance. The evaluation device 144 may be configured to determine an epipolar line 137 in the reference image. Relative position of the reference image and reflection image may be known. For example, relative position of the reference image and reflection image 142 may be stored within at least one storage unit of the evaluation device 144. The evaluation device 144 may be configured to determine a straight line extending from the selected reflection feature of the reflection image 142. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line 137, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line 137. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines 137 may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector 110, whereas the calibration is comparable to a teach-in of the epipolar geometry of the specific camera.

Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object 112, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar

line 137 or a section of an epipolar line 137. The displacement region may comprise more than one epipolar line 137 or more sections of more than one epipolar line 137. The displacement region may extend along the epipolar line 137, orthogonal to an epipolar line 137, or both. The evaluation device 144 may be configured to determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line 137 corresponding to the error interval $\pm\epsilon$ or orthogonal to an epipolar line 137. The measurement uncertainty of the distance measurement using the combined signal Q may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line 137 or epipolar lines 137 may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line 137 or epipolar lines 137. The displacement region may comprise an extent in an orthogonal direction with respect to the epipolar line 137 or epipolar lines 137. The evaluation device may determine a displacement region around the image position of the reflection feature. The evaluation device 144 may be configured to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm\epsilon$ from the combined signal Q to determine the displacement region along the epipolar line 137 corresponding to $z\pm\epsilon$. The evaluation device may be configured to match the selected reflection feature with at least one reference feature within the displacement region. The matching may comprise determining and/or evaluating corresponding reference and reflection features. The evaluation device 144 may be configured to match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. The evaluation device 144 may be configured to determine the epipolar line 137 closest to and/or within the displacement region. The evaluation device 144 may be configured to determine the epipolar line 137 closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line 137 may be larger than the extent of the displacement region orthogonal to the epipolar line 137. The evaluation device 144 may be configured to determine an epipolar line 137 before determining a corresponding reference feature. The evaluation device 144 may determine a displacement region around the image position of each reflection feature. The evaluation device 144 may be configured to assign an epipolar line 137 to each displacement region of each image position of the reflection features, such as by assigning the epipolar line 137 closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line 137. The evaluation device 144 may be configured to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line 137 and/or within the assigned displacement region along the assigned epipolar line 137.

Additionally or alternatively, the evaluation device 144 may be configured to perform the following steps:

- Determining the displacement region for the image position of each reflection feature;

- Assigning an epipolar line 137 to the displacement region of each reflection feature such as by assigning the epipolar line 137 closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line 137;

- 5 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line 137 and/or within the assigned displacement region along the assigned epipolar line 137.

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Additionally or alternatively, the evaluation device 144 may be configured to decide between more than one epipolar line 137 and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines 137 within the reference image and/or by comparing error weighted distances, such as ϵ -weighted distances of reflection features and/or epipolar lines 137 within the reference image and assigning the epipolar line 137 and/or reference feature in shorter distance and/or ϵ -weighted distance to the reference feature and/or reflection feature.

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Preferably, the detector 110 may be configured to pre-classify the selected reflection feature using the combined signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features 125 of the illumination pattern 124 may be arranged such that corresponding reference features of the reference image may have a relative distance to each other as large as possible on the epipolar line.

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25 The evaluation device 144 may be configured to determine a displacement of the matched reference feature and the selected reflection feature. The evaluation device 144 may be configured to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The evaluation device 144 may be configured to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device 144 may be configured to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device 144. The evaluation device 144 may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predeter-

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mined relationship. The evaluation device 144 may be configured to store parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector 110. The evaluation device 144 may be configured to generate the parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector such as by performing a Tsai camera calibration. The evaluation device 144 may be configured to compute and/or estimate parameters such as the focal length of the transfer device, the radial lens distortion coefficient, the coordinates of the center of radial lens distortion, scale factors to account for any uncertainty due to imperfections in hardware timing for scanning and digitization, rotation angles for the transformation between the world and camera coordinates, translation components for the transformation between the world and camera coordinates, aperture angles, image sensor format, principal point, skew coefficients, camera center, camera heading, baseline, rotation or translation parameters between camera and/or illumination source, apertures, focal distance, or the like.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce the possible number of solutions along the epipolar line significantly. The number of possible solutions may even be reduced to one. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may enhance accuracy and/or speed and/or may lower computational demand in comparison to conventional triangulation systems. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be reduced to increase the light intensity in each illumination point such as to compete with ambient light while complying to eye safety regulations. A reduced number of illumination features in a conventional triangulation system might increase the difficulty to match reflection features and reference features. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be increased, such as to increase the resolution of the distance measurement, such as to increase the resolution of the obtained depth map without increasing the processing power of the evaluation device 144 such as in a mobile application.

List of reference numbers

110	detector
112	object
114	sensor element
116	matrix
118	optical sensor
120	light-sensitive area
122	projector
124	illumination pattern
125	illumination feature
126	laser source
128	DOE
130	opening
132	housing
134	optics
136	control unit
137	epipolar line
138	reflection pattern
139	field of view
140	transfer device
141	sub-feature
142	reflection image
144	evaluation device

Claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) comprising:
- at least one projector (122) for illuminating the object (112) with at least one illumination pattern (124), wherein the illumination pattern (124) comprises a plurality of illumination features (125), wherein the illumination features (125) are spatially modulated such that the illumination features (125) are patterned illumination features;
 - at least one sensor element (114) having a matrix (116) of optical sensors (118), the optical sensors (118) each having a light-sensitive area (120), wherein each optical sensor (118) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (120) by a reflection light beam propagating from the object to the detector (110), wherein the sensor element (114) is configured to determine at least one reflection image (142);
 - at least one evaluation device (144), wherein the evaluation device (144) is configured to select at least one reflection feature of the reflection image (142), wherein the reflection feature is a feature in an image plane generated by the object (112) in response to illumination by at least one illumination feature (125), wherein the evaluation device (144) is configured for determining at least one longitudinal coordinate z of the selected reflection feature of the reflection image (142) by using a depth-from-photon-ratio technique by evaluating a combined signal Q from the sensor signals, wherein the combined signal Q is a signal which is generated by combining the sensor signals, wherein the evaluation device (144) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device (144) is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region, wherein each of the patterned illumination features (125) comprises a plurality of sub-features (141), and wherein the illumination features (125) are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features (125) have an offset, wherein the offset of neighboring rows differ.
2. The detector (110) according to the preceding claim, wherein the projector (122) comprises at least one light source configured for generating at least one light beam, wherein the projector (122) comprises at least one transfer device configured for diffracting and for replicating the light beam for generating the illumination pattern (124) comprising patterned illumination features (125).
3. The detector (110) according to any one of the preceding claims, wherein the projector (122) comprises at least one array of densely packed light sources according to a certain pattern configured for generating a cluster of light beams, wherein the projector (122) comprises at

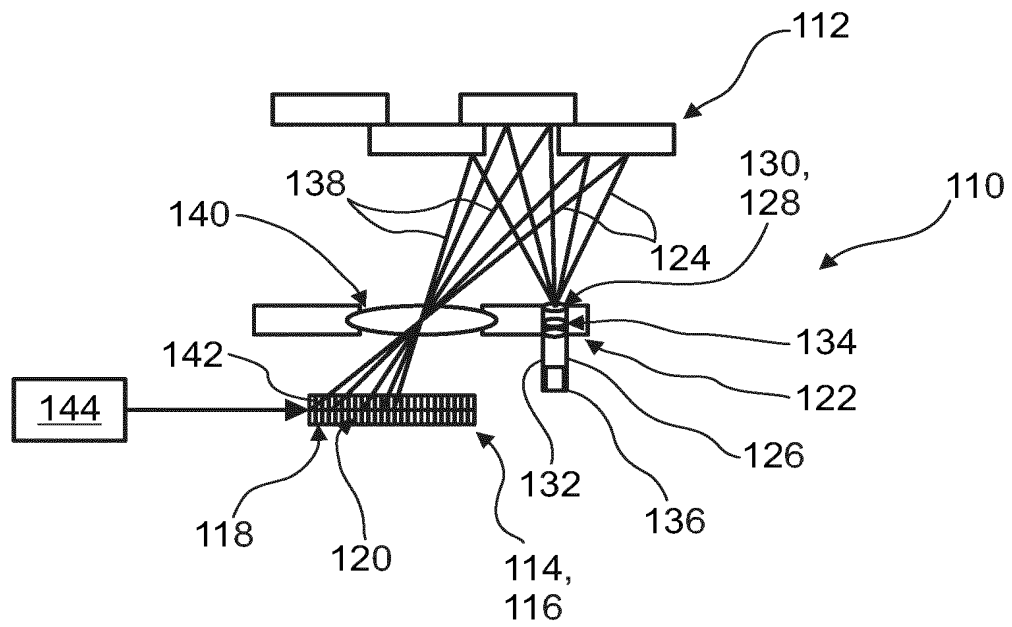
least one transfer device for diffracting and replicating the cluster of light beams for generating the illumination pattern (124) comprising patterned illumination features (125).

4. The detector (110) according to any one of the preceding claims, wherein the evaluation device (144) is configured for determining information about a distance between the object (112) and the projector (122) by analyzing the patterned reflection features of the reflection image (142) wherein the evaluation device (144) is configured for finding and segmenting each patterned reflection feature, wherein the evaluation device (144) is configured for determining a center of each of the sub-features (141), wherein the evaluation device (144) is configured for determining distances between the centers of the sub-features (141), wherein the evaluation device (144) is configured for determining the information about the distance between the object (112) and the projector (122) from the distances between the centers of the sub-features (141) by using a pre-determined relationship.
5. The detector (110) according to the preceding claim, wherein the evaluation device (144) is configured for determining a corrected longitudinal coordinate z considering the determined distance between the object (112) and the projector (122).
6. The detector (110) according to any one of the preceding claims, wherein the sensor element (114) and the projector (122) are positioned such that the rows run parallel to epipolar lines (137).
7. The detector (110) according to any one of the preceding claims, wherein the offset δ is $\delta = \frac{a}{b}$, wherein a and b are integer numbers, wherein the offset is an offset δ .
8. The detector (110) according to any one of the preceding claims, wherein the evaluation device (144) is configured to determine a longitudinal region of the selected reflection feature by evaluating the combined signal Q , wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \epsilon$, wherein the evaluation device (144) is configured to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the reference image is an image different from the reflection image which is determined at a different spatial position compared to the reflection image (142), wherein the evaluation device (144) is configured to match the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device (144) is configured to determine a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device (144) is configured to determine a longitudinal information of the matched reference feature using a predetermined relationship between the longitudinal information and the displacement.
9. A method for determining a position of at least one object (112) by using at least one detector (110) according to any one of the preceding claims, the method comprising the following steps:

- Illuminating the object (112) with at least one illumination pattern generated by the at least one projector (122) of the detector (110), wherein the illumination pattern (124) comprises a plurality of illumination features (125), wherein the illumination features (125) are spatially modulated, wherein the illumination features (125) are patterned illumination features (125), wherein each of the patterned illumination features (125) comprises a plurality of sub-features (141), and/or wherein the illumination features (125) are arranged in a periodic pattern equidistant in rows, wherein each of the rows of illumination features (125) have an offset, wherein the offset of neighboring rows differ;
- generating for each reflection light beam impinging on the light-sensitive areas (120) of the optical sensors (118) of the sensor element (114) having the matrix (116) of optical sensors (118) at least one sensor signal in response to an illumination;
- determining at least one reflection image (142) by using the sensor element (114);
- selecting at least one reflection feature of the reflection image (142) and evaluating the sensor signals by using at least one evaluation device (144), thereby, determining at least one longitudinal coordinate z of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals.

10. A use of the detector (110) according to any one of the preceding claims referring to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

FIG.1



2/3

FIG.2A

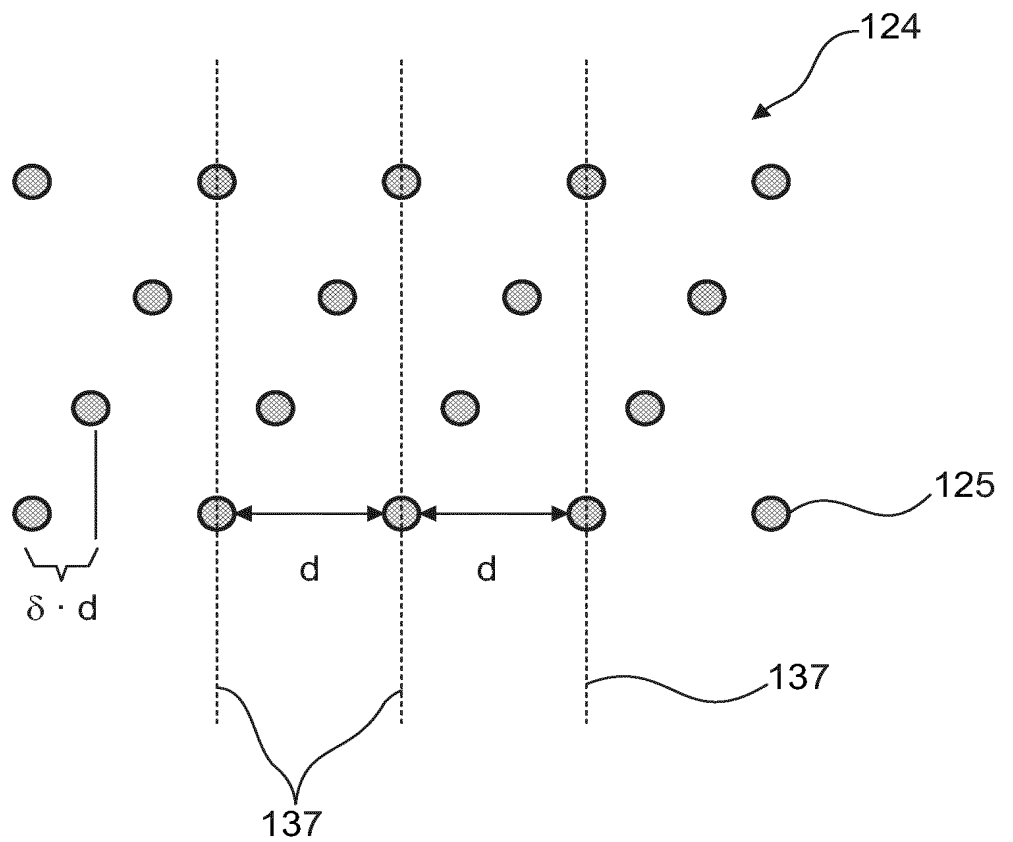


FIG.2B

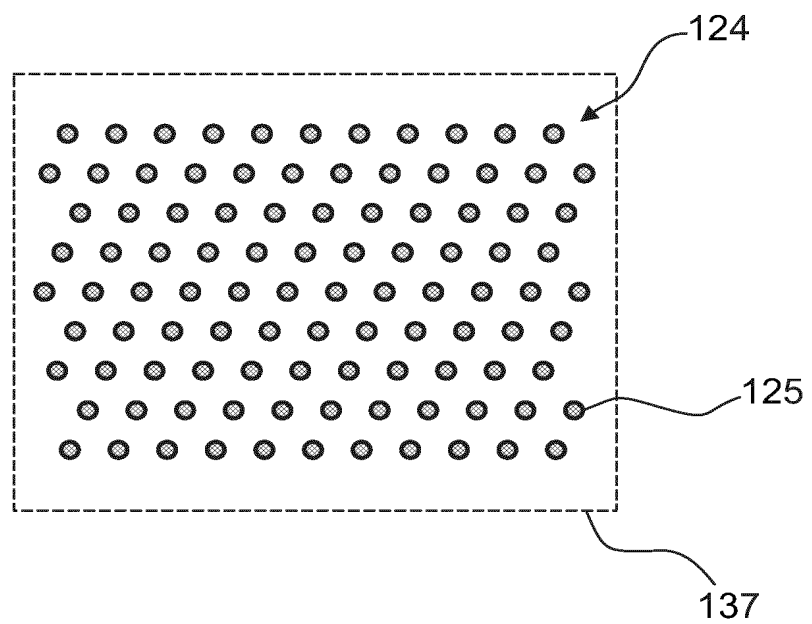


FIG.3A

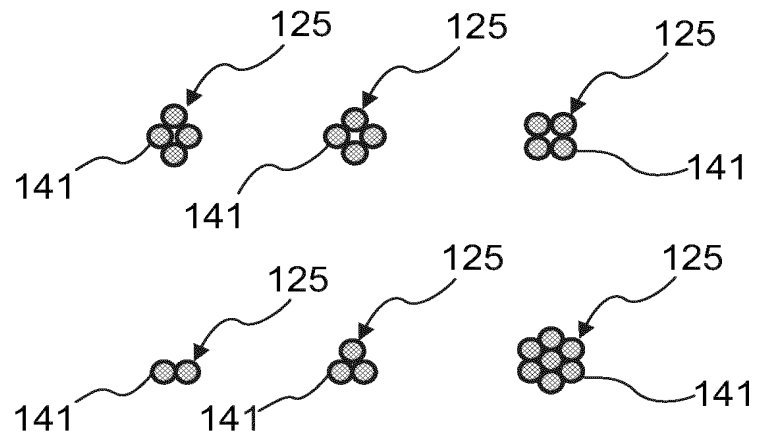
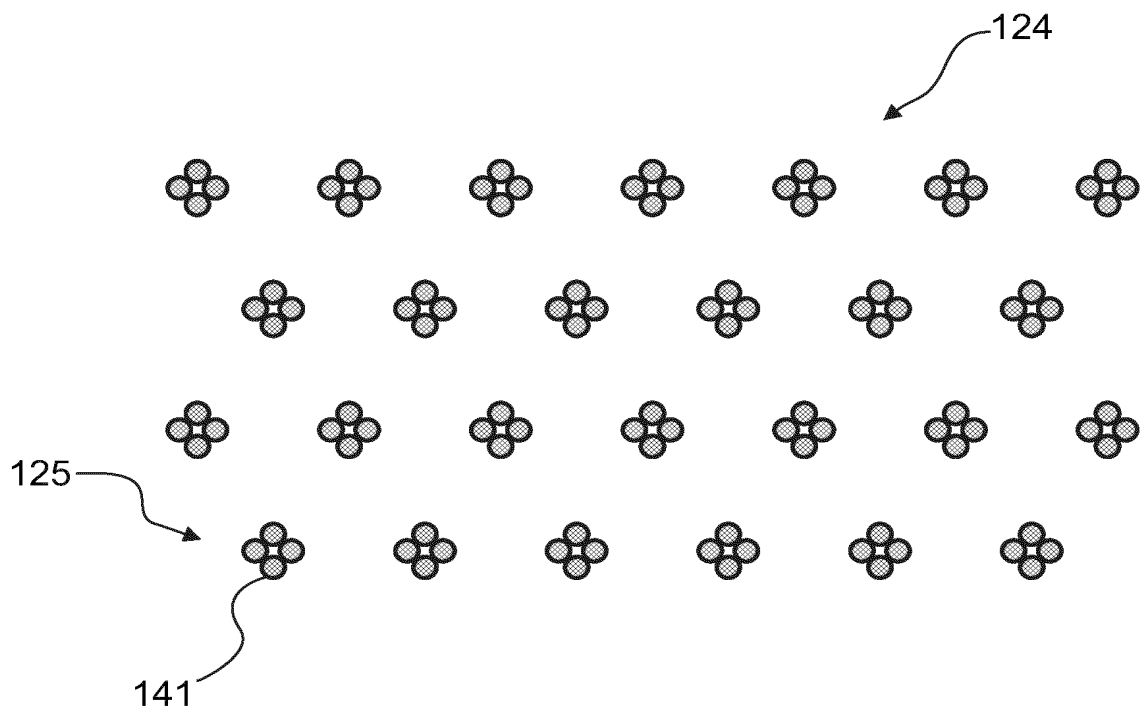


FIG.3B



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2021/060347

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S7/481 G01B11/02 G01S17/48
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S G01B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	WO 2018/091640 A2 (TRINAMIX GMBH [DE]) 24 May 2018 (2018-05-24) cited in the application the whole document -----	1-10



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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Date of the actual completion of the international search

21 June 2021

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Damp, Stephan

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2021/060347

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
WO 2018091640 A2	24-05-2018	CN 109964144 A	02-07-2019
		CN 109964148 A	02-07-2019
		CN 110178045 A	27-08-2019
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		EP 3571522 A1	27-11-2019
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		WO 2018091640 A2	24-05-2018
		WO 2018091649 A1	24-05-2018



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Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE Utility - U.S. National Stage under 35 USC 371

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Documents

TOTAL DOCUMENTS: 11

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-09-25 Supp-IDS 27843-2262.pdf	5	Information Disclosure Statement (IDS) Form (SB08)	619 KB
FOR_WO2020187719A1.pdf	142	Foreign Reference	8652 KB
FOR_WO2018091649A1.pdf	109	Foreign Reference	7051 KB
FOR_WO2018091638A1.pdf	106	Foreign Reference	6755 KB
FOR_WO2018091640A2.pdf	258	Foreign Reference	15282 KB
FOR_WO2021170791A1.pdf	66	Foreign Reference	4338 KB

FOR_WO2021152070A1.pdf	69	Foreign Reference	4400 KB
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FOR_WO2018091649A1.pdf	3011BF7828E40BC0EBFDBDD4250B263FCD636D68B509576DF54C7519BB4A48670E58ED09101803FCD36C7945B1DCFB84D5E2767C7F21CCA83BA5BEBE0F180D08
FOR_WO2018091638A1.pdf	64A4BEB25ACB971D3EDE7DE404859AABBEBE97F2338C788E162AC4BF4033A4FB4FDC9090AE06A174643AA874CD70DCD3B76CCECF10B3F7D0A812209841961CAB
FOR_WO2018091640A2.pdf	7C79057C90FFE4C1CC2FD101352D2460418944CD2B39195C1794AED519B2691B4B9E3AEC8EFEC4E6A05ED98A9A37A1C7E4FF84B232D29E5449CF3F14040EA580
FOR_WO2021170791A1.pdf	C80FB90FE145748ADB9CB8A63995F9356C19A57EF2D9A1D6FF732206BCF48DC3A1842F41FB73244E5FDB851D8F0CA893AA313AF200220AEC6792AFFB12629C1D

FOR_WO2021152070A1.pdf	835F94D414FDB5F027BC59ADC6664A74795C5EE078FD07195 6770182709F6206FA02DBB1BEB6D295AF07783CC8BC3C1E00 E358CA8EDE9224D16A49B096027989
FOR_WO2021214123A1.pdf	5A70B73C176E38E53CF4F3281128AB9C35A8C32513DB4D52C AAB9D3EBEBB0A904F6F74CE9BAA49F45B3B793A7E28A0CDD 7A616A97241278592882CEAFF066DB2
NPL_Yu.pdf	056C578BA8D5BB6FEB4B7EFBA93B4565A3546BCB9A6796078 527B042AE6A2F08BC1512FFE66AA5678A47DEBBCC06FC157 B3BDE09E17B8056F9C8B007B118EDD9
NPL_Masi.pdf	CBE3D5CAF6C640964E0FFB0405A61C910C8171643845C2922 1CEA9EEC647443F6AE0D89707E32C45A6E5AF5764C61B0DC BA3FF9BE831BD1DD2679491FE5315D9
NPL_J Eichler_Lasertechnik.pdf	9DBC3A75AE9EE7861A4C02E4B98A18306B3FE0CE60D6299D B1D84FC08D17FD6BE0EBC673F0194AFFFDAB33CB7288EC11 4E8090874EE5BF30DAF1B678653BD0B6

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

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New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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G06K 9/20 (2006.01) G06K 9/62 (2006.01)
G06K 9/34 (2006.01) G06K 9/22 (2006.01)

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(71) Applicant: **TRINAMIX GMBH** [DE/DE]; Industries-
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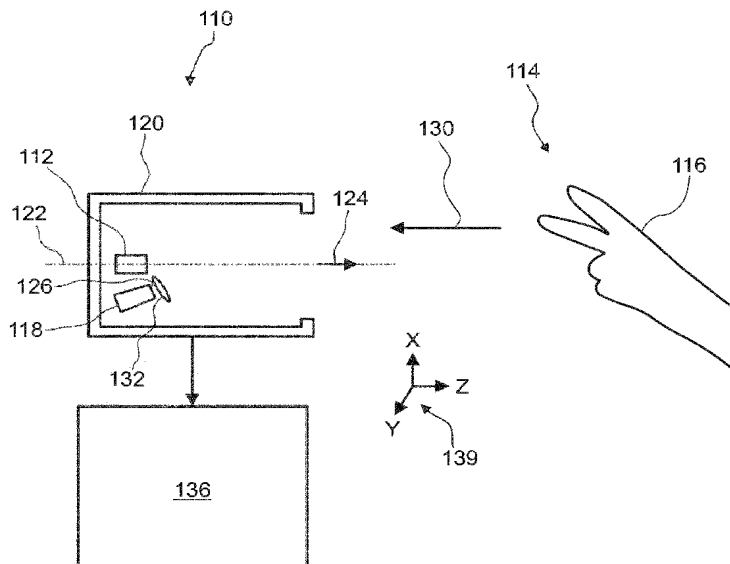
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DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN,
HR, HU, ID, IL, IN, IR, IS, IT, JO, JP, KE, KG, KH, KN,
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(54) Title: GESTURE RECOGNITION

FIG.1



(57) Abstract: Detector (110) for gesture detection comprising - at least one illumination source (112) configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area (114) comprising at least one object (116), wherein the object (116) comprises at least partially at least one human hand; - at least one optical sensor (118) having at least one light-sensitive area (126), wherein the optical sensor (118) is configured for determining at least one image (128) of the area, wherein the image (128) comprises a plurality of reflection features (130) generated by the area (114) in response to illumination by the illumination features; - at least one evaluation device (136), wherein the evaluation device (136) is configured for determining at least one depth map of the area by determining at least one depth information for each of the reflection features (130), wherein the evaluation device (136) is configured for finding the object (116) by identifying the reflection features (130) which were generated by

NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW,
SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN,
TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

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Declarations under Rule 4.17:

- *as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))*
- *of inventorship (Rule 4.17(iv))*

Published:

- *with international search report (Art. 21(3))*

illuminating biological tissue, wherein the evaluation device (136) is configured for determining at least one reflection beam profile of each of the reflection features (130), wherein the evaluation device (136) is configured for identifying a reflection feature (130) as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion, wherein the evaluation device (136) is configured for identifying the reflection feature (130) as to be background otherwise, wherein the evaluation device (136) is configured for segmenting the image (128) of the area by using at least one segmentation algorithm, wherein the reflection features (130) identified as to be generated by illuminating biological tissue are used as seed points and the reflection features (130) identified as background are used as background seed points for the segmentation algorithm, wherein the evaluation device (136) is configured for determining position and/or orientation of the object (116) in space considering the segmented image and the depth map.

Gesture Recognition

5 Description

Field of the invention

10 The invention relates to a detector for gesture detection and a method for gesture detection and various uses of the detector. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the
15 sciences. However, other applications are also possible.

Prior art

20 The trend towards automated vehicles is not only pushing towards autonomous driving but is also affecting assistance functions in the cabin. Driver monitoring, in-cabin surveillance and gesture tracking are basic concepts for next generation cars. First systems are already available and rely on 2D images, 3D depth maps, or a combination of both.

25 From a technical point of view, detection of human postures and gestures for in-cabin scenarios can be highly challenging. Existing methods based on 2D image processing rely on color features, e.g. as described in A. R. Várkonyi-Kóczy, B. Tusor, "Human-Computer Interaction for Smart Environment Applications Using Fuzzy Hand Posture and Gesture Models", IEEE Trans. Instrumentation and Measurement, vol. 60, no. 5, pp. 1505-1514, 2011 or C. C. Hsieh, D. H. Liou, "Novel Haar features for real-time hand gesture recognition", J. Real Time Image Pro-
30 cessing, vol. 10, pp. 357370, 2015, and shape features, see e.g. E. Ong, R. Bowden, "A boosted classifier tree for hand shape detection," Sixth IEEE International Conference on Automatic Face and Gesture Recognition, 2004. Proceedings., Seoul, South Korea, 2004, pp. 889-894. However, due to the high variation in appearance of hands, shadows and a broad range of light conditions, the robustness of these methods is rather limited, see M. Bergh, L. Gool, "Com-
35 bining RGB and ToF cameras for real-time 3D hand gesture interaction", In: Workshop on applications of computer vision (WACV), IEEE, pp. 66-72, 2011.

Motion analysis, e.g. using the optical flow, can fill this gap under certain conditions, see e.g. R. Cutler, M. Turk. "View-based Interpretation of Realtime Optical Flow for Gesture Recognition",
40 In Proc. IEEE Intl. Conference on Automatic Face and Gesture Recognition, pp. 416-421, 1998, but still lacks for static scenes or insufficient contrast. Depth maps, e.g. from time of flight cameras or stereo approaches, provide complementary data that can considerably improve detection accuracy. The combination of 2D cameras with 3D sensors provides more reliable detection

accuracy but demands comprehensive fusion concepts, careful calibration and causes significantly higher costs. Moreover, separating multiple objects positioned in a similar depth is still challenging.

- 5 In a conventional 2D or 3D gesture detection setup, the hand has to be recognized within a 2D picture or 3D point cloud. The hand recognition taken alone is a challenging part, as contrast to the background can be low in the 2D picture, hand color can vary strongly dependent on skin type and illumination conditions, and especially, the hand has to be recognized as a hand, for which model data such as shape, color, etc need to match. Shape however varies strongly, since the shape transmits part of the information in the gesture. Fingers might even be missing due to permanent injuries or may be just obscured by other objects.

- 10 Moreover, WO 2018/091640 and WO 2019/042956 describe accurately detecting distances using a single CMOS sensor combined with a laser grid projector for 2D pictures and depth measurement. All data is coming from a single CMOS sensor allowing for an easy and quick integration. Thus, system complexity is significantly reduced compared to the combination of separate image and depth sensors.

- 20 US 2016/253821 A1 describes an object which is identified or tracked within a volume by projecting a light beam encoded with one or more predefined properties to have a predefined optical structure into the volume. A detector captures light from the predefined optical structure reflected from the volume. By analyzing one or more characteristics of light from the predefined optical structure reflected from the object, the object is segmented from the volume.

25 Problem addressed by the invention

- It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which allow reliable gesture detection with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

- 35 This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

- 40 As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are

present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a device for gesture detection is disclosed. As used herein, the term “gesture” may refer to at least one state and/or at least one action of at least one part of a human’s body. The gesture may comprise body gesture such as of at least one part of the body, in particular hand gesture. The gesture may be static or dynamic. The gesture may comprise movement of at least one part of a hand, such as of a finger, movement of one or both hands, face, or other parts of the body. As used herein, the term “gesture detection” may refer to determining presence or absence of a gesture and/or gesture recognition. As used herein, the term “gesture recognition” may refer to interpreting of human gestures via mathematical algorithms. As used herein, the term “detector for gesture detection” may refer to at least one arbitrary device configured for detection of at least one gesture.

The detector for gesture detection comprises

- at least one illumination source configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area comprising at least one object, wherein the object comprises at least partially at least one human hand;
- at least one optical sensor having at least one light-sensitive area, wherein the optical sensor is configured for determining at least one image of the area, wherein the image comprises a plurality of reflection features generated by the area in response to illumination by the illumination features;

- at least one evaluation device, wherein the evaluation device is configured for determining at least one depth map of the area by determining at least one depth information for each of the reflection features,

wherein the evaluation device is configured for finding the object by identifying the reflection

features which were generated by illuminating biological tissue, wherein the evaluation device is configured for determining at least one reflection beam profile of each of the reflection features, wherein the evaluation device is configured for identifying a reflection feature as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion, wherein the evaluation device is configured for identifying the reflection feature as to be background otherwise,

wherein the evaluation device is configured for segmenting the image of the area by using at least one segmentation algorithm, wherein the reflection features identified as to be generated by illuminating biological tissue are used as seed points and the reflection features identified as background are used as background seed points for the segmentation algorithm,

wherein the evaluation device is configured for determining position and/or orientation of the object in space considering the segmented image and the depth map.

As used herein, the term “illumination source” may generally refers to at least one arbitrary device configured for generating at least one illumination pattern. The illumination source may be configured for providing the illumination pattern for illumination of the area. The illumination source may be adapted to directly or indirectly illuminating the area, wherein the illumination pattern is reflected or scattered by surfaces of the area and, thereby, is at least partially directed towards the optical sensor. The illumination source may be configured for illuminating the area, for example, by directing a light beam towards the area, which reflects the light beam. The illumination source may be configured for generating an illuminating light beam for illuminating the area.

As used herein, the term “area” may refer to at least one arbitrary object or spatial region. The area may comprise the at least one object and a surrounding environment. As used herein, the term “object” refers to a point or region emitting at least one light beam, in particular the at least one reflection pattern. For example, the object may be at least one object selected from the group consisting of: a scene, a human such as a human, wood, carpet, foam, an animal such as a cow, a plant, a piece of tissue, a metal, a toy, a metallic object, a beverage, a food such as a fruit, meat, fish, a dish, a cosmetics product, an applied cosmetics product, cloth, fur, hair, a maintenance product, a plant, a body, a part of a body, organic material, inorganic material, a reflective material, a screen, a display, a wall, a sheet of paper, such as photograph. The object may comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector. The object may, specifically, be a human body or at least one part of a human body such as at least one arm, at least one hand, at least one finger or a face. The object comprises at least partially at least one human hand. As used herein, the term “at least partially at least one human hand” refers to embodiments wherein the object is the human hand and/or embodiments wherein the object is at least one finger and/or at least one part of a palm.

The illumination source may comprise at least one light source. The illumination source may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. The illumination source may be configured for generating the at least one illumination pattern in the infrared region. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however.

The illumination source may be configured for emitting light at a single wavelength. Specifically, the wavelength may be in the near infrared region. In other embodiments, the illumination source may be configured for emitting light with a plurality of wavelengths allowing additional measurements in other wavelengths channels.

The illumination source may be or may comprise at least one multiple beam light source. For example, the illumination source may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination

source may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source may be integrated into a housing of the detector.

Further, the illumination source may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which later on may be used for distinguishing the light beams.

The light beam or light beams generated by the illumination source generally may propagate parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The detector may be configured such that the light beam or light beams propagate from the detector towards the area along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the light beam or light beams, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the optical axis or off the optical axis. As an example, the light beam or light beams may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the area. As used herein, the term "illumination feature" refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected

from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source.

A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination source may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination source comprises at least one laser source and/or at least one laser diode which is designated for generating laser radiation. The illumination source may comprise the at least one diffractive optical element (DOE). The detector may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one periodic point pattern.

As further used herein, the term “projecting at least one illumination pattern” refers to providing the at least one illumination pattern for illuminating the at least one area.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The detector may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal. In case the detector comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically

simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

5 Preferably, the light sensitive area may be oriented essentially perpendicular to an optical axis of the detector. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

10 The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. The optical sensor may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one inorganic photodiode
15 which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the
25 brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively,
30 the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Ga photodiode, a Ge:As photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.
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The optical sensor may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically,
40 cally, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers.

For example, the optical sensor each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor may be part of or constitute a pixelated optical device. For example, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the opti-

cal sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

5 The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

30 The detector further may comprise at least one transfer device. The detector may further comprise one or more additional elements such as one or more additional optical elements. The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensor. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term

“focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging
5 elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered
10 as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular
15 the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The
20 optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector may comprise the transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one
25 or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not
30 counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. In case of a plurality of optical
35 sensors, the optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The
40 coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a lon-

gitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

- 5 The detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.
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- 15 Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.
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The optical sensor is configured for determining at least one image of the area. When the area is illuminated by the illumination features the area, i.e. the parts of the area which are illuminated by the illumination features, reflects and/or remits the impinging light beams of the illumination features towards the detector, which can thus image these reflected and/or remitted illumination features. Thereby, the area may generate the reflection features. The image comprises a plurality of reflection features generated by the area in response to illumination by the illumination features. As used herein, without limitation, the term "image" specifically may relate to data recorded by using the optical sensor, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element. The image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to "pixels", reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. The image may be at least one two-dimensional image. As used herein, the term "two dimensional image" may generally refer to an image having information about transversal coordinates such as the dimensions of height and width. The image may be an RGB (red green blue) image. As used herein, the term "reflection feature" may refer to a feature in an image plane generated by the area in response to illumination, specifically with at least one illumination feature.

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The detector comprises the at least one evaluation device. The evaluation device is configured for evaluating the image. As further used herein, the term "evaluation device" generally refers to

an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images. Specifically the determining a beam profile and indication of the surface, may be performed by the at least one evaluation device. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of

a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

The evaluation device is configured for determining at least one depth map of the area by determining at least one depth information for each of the reflection features. The term “each of the reflection features” may refer to each of the reflection features of the image or of a region of interest of the image. As used herein, the term “depth” may refer to a distance between the object and the optical sensor and may be given by the longitudinal coordinate. As used herein, the term “depth map” may refer to spatial distribution of depth. The evaluation device may be configured for determining the depth information for each of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The evaluation device may be configured for considering the depth information for segmenting the object, in particular a hand region in the image, from the background in terms of the depth map.

In the following beam-profile analysis is described, in particular the so called depth-from-photon-ratio (DPR) technique. For further details, reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A the full content of which is included by reference,

which describe the beam-profile analysis in more detail, in particular determining of a combined signal and determining depth therefrom.

For example, the evaluation device may be configured for determining the depth information for each of the reflection features by using depth-from-photon-ratio technique. Each of the reflection features may comprise at least one beam profile, also denoted reflection beam profile. As used herein, the term “beam profile” of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device may be configured for determining depth information for each of the reflection features by analysis of their beam profiles.

The evaluation device may be configured for determining at least one longitudinal coordinate z_{DPR} for each of the reflection features by analysis of their beam profiles. As used herein, the term “analysis of the beam profile” may generally refer to evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least one symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The evaluation device may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The evaluation device may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The evaluation device may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The evaluation device may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The reflection feature may cover or may extend over at least one pixel of the image. For example, the reflection feature may cover or may extend over plurality of pixels. The evaluation device may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The evaluation device may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum_j r_{pixel}},$$

wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum_j I_{total}$ with j being the number of pixels j connected to and/or belonging to the reflection feature and I_{total} being the total intensity.

The evaluation device may be configured for determining the longitudinal coordinate z_{DPR} for each of the reflection features by using depth-from-photon-ratio technique. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

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The evaluation device may be configured for determining the beam profile of each of the reflection features. As used herein, the term “determining the beam profile” refers to identifying at least one reflection feature provided by the optical sensor and/or selecting at least one reflection feature provided by the optical sensor and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

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The depth-from-photon-ratio technique may comprise determining at least one first area and at least one second area of the reflection beam profile of each of the reflection features and/or of the reflection features in at least one region of interest. The depth-from-photon-ratio technique may further comprise integrating the first area and the second area. The depth-from-photon-ratio technique may further comprise deriving a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.

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The evaluation device may be configured for determining at least one first area and at least one second area of the reflection beam profile of each of the reflection features and/or of the reflection features in at least one region of interest. The evaluation device is configured for integrating the first area and the second area.

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The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area $A1$ and the second area of the beam profile may be an area $A2$. The evaluation device may be configured for integrating the first area and the second area. The evaluation device may be configured to derive a combined signal, in particular a quotient Q , by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The evaluation device may configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the

areas may be possible as long as the areas are not congruent. For example, the evaluation device may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The evaluation device may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The evaluation device may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The evaluation device may be configured for comparing at least two of the determined integrals. Specifically, the evaluation device may be configured for determining at least one first area and at least one second area of the beam profile. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the evaluation device may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the evaluation device may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the detector may comprise at least two optical sensors, wherein the light-sensitive areas of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the beam profile of the reflection feature. The evaluation device may be adapted to integrate the first area and the second area. The evaluation device may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the

center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be configured for determining an area integral of the beam profile. The evaluation device may be configured for determining the edge information by integrating and/or summing of the first area. The evaluation device may be configured for determining the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be configured for determining an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

In one embodiment, A1 may correspond to a full or complete area of a feature point on the optical sensor. A2 may be a central area of the feature point on the optical sensor. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In one embodiment, the illumination pattern may comprise at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensor may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the

line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensor. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

In one embodiment, the illumination pattern may comprise at least point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The illumination pattern may comprise both at least one point pattern and at least one line pattern. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

The evaluation device may be configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.

The evaluation device may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The evaluation device may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the beam profile, respectively, and E(x,y) denotes the beam profile.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

For example, in case of the optical sensor having a matrix of pixels, the evaluation device may be configured for evaluating the beam profile, by

- determining the pixel having the highest sensor signal and forming at least one center signal;
- 5 - evaluating sensor signals of the matrix and forming at least one sum signal;
- determining the quotient Q by combining the center signal and the sum signal; and
- determining at least one longitudinal coordinate z of the object by evaluating the quotient Q .

10 As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More
15 specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the display device, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. The term “center signal” generally refers to the at least one sensor signal comprising essentially
20 center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the pixel having the highest sensor signal out of the plurality of sensor signals generated by the pixels of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the pixels of the matrix. The center signal may arise from a single pixel or from a
25 group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of pixels may be added up, integrated or averaged, in order to determine the center signal. The group of pixels from which the center signal arises may be a group of neighboring pixels, such as pixels having less than a predetermined distance from the actual pixel having the
30 highest sensor signal, or may be a group of pixels generating sensor signals being within a predetermined range from the highest sensor signal. The group of pixels from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of pixels around the pixel having the highest sensor
35 signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

40 As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the pixel in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from pixels in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or

more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring pixels; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The quotient Q may be a signal which is generated by combining the center signal and the sum signal. Specifically, the determining may include one or more of: forming a quotient of the center

signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the quotient Q may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

As used herein, the term “longitudinal coordinate of the object” refers to a distance between the optical sensor and the object. The evaluation device may be configured for using the at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The evaluation device may be configured for executing at least one depth-from-photon-ratio algorithm which computes distances for all reflection features with zero order and higher order.

The evaluation of the image comprises identifying the reflection features of the image. The evaluation device may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a

Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor.

5 For example, the illumination source may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more
10 pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device may
15 be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The evaluation device is configured for finding the object by identifying the reflection features
20 which were generated by illuminating biological tissue. The evaluation device is configured for determining the at least one reflection beam profile of each of the reflection features. The evaluation device is configured for identifying a reflection feature as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion. The evaluation device is configured for identifying the reflection feature as to be
25 background otherwise.

As used herein, the term "biological tissue" generally refers to biological material comprising living cells. The detector may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The term "identification" of being generated by biological tissue may refer to determining and/or validating whether a surface to be examined or under test
30 is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the
35 biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector may be adapted to distinguish animal tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector may be adapted to distinguish plant tissue or parts
40 thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector may be adapted to distinguish different types of food such as a

fruit, meat, and fish. The detector may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from human skin. The detector may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector may be adapted to distinguish human skin from cloth. The detector may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector may be adapted to distinguish organic material from inorganic material. The detector may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector may be used, in particular, for non-therapeutic and non-diagnostic applications.

As used herein, the term “at least one predetermined or predefined criterion” refers to at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills at least one predetermined or predefined criterion. As used herein, the term “indicate” refers to an arbitrary indication such as an electronic signal and/or at least one visual or acoustic indication.

The evaluation device may be configured for determining at least one material property of the object by evaluating the beam profile of the reflection feature. As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “identifying at least one material property” refers to one or more of determining and assigning the material property to the object. The detector may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by the user of the detector. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The detector may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering profile.

Specifically, the evaluation device may be configured for comparing the reflection beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a mobile device comprising the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the mobile device, e.g. by software, specifically by the app downloaded from an app store or the like. The reflection feature may be identified as to be generated by biological tissue in case the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise determining a deviation, e.g. a sum of squared point to point distances, between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The evaluation device may be adapted to compare the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the detector.

Additionally or alternatively, for identification if the reflection feature was generated by biological tissue, the evaluation device may be configured for applying at least one image filter to the image of the area. As further used herein, the term “image” refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a

feature, in particular a material feature. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

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The evaluation device may be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the surface of the area having generated the reflection feature.

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The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

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The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term “beam profile image” refers to a sum of N_B Gaussian radial basis functions,

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$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

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wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k :

$(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

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Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J, $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

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Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k .

The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is

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the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m': \mu_m \neq \mu_{m'}$. As used herein, the term "not distinguish between material classifiers" refers to that the expectation values of the material classifiers are identical. As used

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herein, the term "distinguishes material classifiers" refers to that at least two expectation values of the material classifiers differ. As used herein "distinguishes at least two material classifiers" is

used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number

5 of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

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$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

15 Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For exam-

20 ple, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

25 In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

30 For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int f(x) dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1 g_2) = [p_{g_1, g_2}]$, whereas p_{g_1, g_2} is the occurrence rate of the grey combination $(g_1, g_2) = [f(x_1, y_1), f(x_2, y_2)]$, and the relation ρ defines the distance between (x_1, y_1) and (x_2, y_2) , which is $\rho(x, y) = (x+a, y+b)$ with a and b selected from $0, 1$.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

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The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

10 The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

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$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

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$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i,y+j) L_5(E_5)^T$$

and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i,y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

25 Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ε times the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

10 wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The evaluation device may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the surface having generated the reflection feature for determining the material property of the surface having generated the reflection feature. The predetermined relationship may be one or more of an empirical relationship, a semi-empirical relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

20 The evaluation device is configured for identifying a reflection feature as to be generated by illuminating biological tissue in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by biological tissue in case the material property indicates "biological tissue". The reflection feature may be identified as to be generated by biological tissue in case the material property is below or equal at least one threshold or range, wherein in case the determined deviation is below and/or equal the threshold the reflection feature is identified as to be generated by biological tissue and/or the detection of biological tissue is confirmed. At least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the detector. The evaluation device is configured for identifying the reflection feature as to be background otherwise.

30 Thus, the evaluation device may be configured for assigning each projected spot with a depth information and a material property, e.g. skin yes or no.

The material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} .

The evaluation device is configured for segmenting the image of the area by using at least one segmentation algorithm. As used herein, the term "segment" may refer to a part of the image and/or a sub-part of the image and/or a region of the image. The segment may comprise a set of pixels. As used herein, the term "segmenting" may refer to a process of partitioning the image

into multiple segments. The segmentation may comprise assigning at least one label to every pixel of the image such that pixels with the same label share at least one characteristic. The labels may be assigned under a pre-defined target. The segmentation may be a binary segmentation. The binary segmentation may comprise labeling the pixels of the image as “skin-pixels” and “background pixel”. All non-skin pixels may be considered as background. Skin-pixels may be considered as foreground seeds, also denoted seed points, serving as input for an image-based segmentation algorithm. With respect to image segmenting reference is made to https://en.wikipedia.org/wiki/Image_segmentation. The segmentation algorithm may comprise at least one region growing image segmentation method.

The segmentation algorithm may be configured for fusing and/or considering a-priori knowledge. Specifically, the segmentation algorithm may be configured for fusing and/or considering the material information determined from the beam profile analysis. This may allow to obtain or filter position data for motions of visible skin only. In contrast, in conventional setups, a recognition and separation of e.g. what is a finger/arm and what is not, needs to be done first. In addition to using the material information, the evaluation device may be configured for considering the depth information for segmenting the object, in particular a hand region in the image, from the background in terms of the depth map.

The segmentation algorithm may be based on energy or cost functions such as one or more of graph cut, level-set, fast marching and Markov random field approaches. Segmentation of the image may be driven by color homogeneity and edge indicators, wherein the seed points constitute edge- and color homogeneity criteria. As used herein, the term “seed point” may refer to starting points for the segmentation algorithm. The reflection features identified as to be generated by illuminating biological tissue are used as seed points and the reflection features identified as background are used as background seed points for the segmentation algorithm. These seed points may constitute, edge- and color homogeneity criteria, and thus provide an appropriate initialization of the segmentation of the target: “Which color and/or reflection and/or appearance has the skin in the image?”.

For example, graph cut segmentation may be used. Graph cut segmentation may be configured for fusing the a-priori knowledge as it additionally provides real-time capability when combining it with pre-clustering algorithms such as superpixels, see R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, “Slic superpixels compared to state-of-the-art superpixel methods,” IEEE TPAMI, vol. 34, no. 11, 2012. For example, a modified lazy snapping graph cut may be used. With respect to modified lazy snapping graph cut reference is made to Li, Yin and Sun, Jian and Tang, Chi-Keung and Shum, Heung-Yeung, “Lazy Snapping”, ACM Trans. Graph., vol. 23, no. 3, 2004. The Gibbs energy term may comprise an additional term to incorporate depth information for the graph cut segmentation routine. Instead of watershed-based pre-segmentation as proposed in Li, Yin and Sun, Jian and Tang, Chi-Keung and Shum, Heung-Yeung, “Lazy Snapping”, ACM Trans, a superpixel clustering may be used to ensure real-time capability.

The evaluation device is configured for determining position and/or orientation of the object in space considering the segmented image and the depth map. As used herein, the term “segmented image” may refer to the image after the segmentation process and/or an outcome of the segmentation algorithm. As used herein, the term “position” refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The evaluation device may be configured for determining at least one hand pose or gesture from the position and/or the orientation of the object in space. The evaluation device may be configured for recognition of gestures from the segmented image, in particular from the parts of the segmented image labeled as object. The evaluation device may be configured for identifying image coordinates of palm and finger in the segmented image. Palm and finger identification may be obtained from the segmentation. The evaluation device may be configured for extracting features such as color and/or brightness and/or gradient values from the parts of the segmented image labeled as object. For example, palm and finger tips may be detected using standard OpenCV routines.

The evaluation device may be configured for determining at least one three-dimensional finger vector considering image coordinates of palm and finger tip and the depth map. As used herein, the term “three-dimensional finger vector” may refer to a 3D vector spanned by a point of a finger tip and a point of the palm. The 3D finger vector may specify the position and orientation of the finger in space. The evaluation device may be configured for determining the 3D finger vector from the image and depth map. The 3D finger vector may provide the basis for hand pose estimation and scene interpretation. For the recognition of hand gestures, the evaluation device may be configured for using at least one classifier such as a hidden Markov model (HMM), a support vector machine (SVM), a conditional random field (CRF) and the like. The classifier may be configured for discriminating hand gestures.

The use of the beam profile analysis sensor fusion concept according to the present invention have the advantages for providing robust detection results, no need for complex model assumptions, a single sensor concept, easy integration and real-time capability.

In a further aspect, the present invention discloses a method for gesture detection, wherein a detector according to the present invention is used. The method comprises the following steps:

- a) projecting at least one illumination pattern comprising a plurality of illumination features on at least one area comprising at least one object, wherein the object comprises at least partially at least one human hand;
- b) determining at least one image of the area using at least one optical sensor having at least one light-sensitive area, wherein the image comprises a plurality of reflection features generated by the area in response to illumination by the illumination features;
- c) determining at least one depth map of the area by determining at least one depth information for each of the reflection features by using at least one evaluation device;
- d) finding the object by using the evaluation device by identifying the reflection features which were generated by illuminating biological tissue, wherein at least one reflection beam profile of each of the reflection features is determined, wherein a reflection feature is identified as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion, wherein the reflection feature otherwise is identified as background;
- e) segmenting the image of the area by using the evaluation device by using at least one segmentation algorithm, wherein the reflection features identified as to be generated by illuminating biological tissue are used as seed points and the reflection features identified as background are used as background seed points for the segmentation algorithm;
- f) determining position and/or orientation of the object in space considering the segmented image and the depth map by using the evaluation device.

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

The at least one evaluation device may be configured for performing at least one computer program, such as at least one computer program configured for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which may detect the gesture.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine

vision application; a robotics application; a quality control application; a manufacturing application.

With respect to further uses of the detector of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: Detector for gesture detection comprising

- at least one illumination source configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area comprising at least one object, wherein the object comprises at least partially at least one human hand;
- at least one optical sensor having at least one light-sensitive area, wherein the optical sensor is configured for determining at least one image of the area, wherein the image comprises a plurality of reflection features generated by the area in response to illumination by the illumination features;
- at least one evaluation device, wherein the evaluation device is configured for determining at least one depth map of the area by determining at least one depth information for each of the reflection features, wherein the evaluation device is configured for finding the object by identifying the reflection features which were generated by illuminating biological tissue, wherein the evaluation device is configured for determining at least one reflection beam profile of each of the reflection features, wherein the evaluation device is configured for identifying a reflection feature as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion, wherein the evaluation device is configured for identifying the reflection feature as to be background otherwise, wherein the evaluation device is configured for segmenting the image of the area by using at least one segmentation algorithm, wherein the reflection features identified as to be generated by illuminating biological tissue are used as seed points and the reflection features identified as background are used as background seed points for the segmentation algorithm, wherein the evaluation device is configured for determining position and/or orientation of the object in space considering the segmented image and the depth map.

Embodiment 2: The detector according to the preceding embodiment, wherein the evaluation device is configured for identifying image coordinates of palm and finger in the segmented image, wherein the evaluation device is configured for determining at least one three-dimensional finger vector considering image coordinates of palm and finger and the depth map.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining at least one hand pose or gesture from the position and/or the orientation of the object in space.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the segmentation algorithm is based on energy or cost functions such as one or more of graph cut, level-set, fast marching and Markov random field approaches.

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Embodiment 5: The detector according to any one of the preceding embodiments, wherein segmentation of the image is driven by color homogeneity and edge indicators, wherein the seed points constitute edge- and color homogeneity criterions.

10 Embodiment 6: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for comparing the reflection beam profile of each of the reflection features with at least one predetermined and/or prerecorded and/or predefined beam profile.

15 Embodiment 7: The detector according to the preceding embodiment, wherein the comparison comprises overlaying the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile such that their centers of intensity match, wherein the comparison comprises determining a deviation between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile, wherein the evaluation device is configured for comparing the determined deviation with at least one threshold, wherein in case the
20 determined deviation is below and/or equal the threshold the reflection feature is indicated as biological tissue.

25 Embodiment 8: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining the depth information for each of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors.

30 Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining the depth information for each of the reflection features by using depth-from-photon-ratio technique, wherein the evaluation device is configured for determining at least one first area and at least one second area of a beam profile of at least one of the reflection features, wherein the evaluation device is configured for integrating
35 the first area and the second area, wherein the evaluation device is configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.

40 Embodiment 10: The detector according to the preceding embodiment, wherein the first area of the reflection beam profile comprises essentially edge information of the reflection beam profile and the second area of the reflection beam profile comprises essentially center information of the reflection beam profile, and/or wherein the first area of the reflection beam profile comprises

essentially information about a left part of the reflection beam profile and the second area of the reflection beam profile comprises essentially information about a right part of the reflection beam profile.

- 5 Embodiment 11: The detector according to any one of the two preceding embodiments, wherein the evaluation device is configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the reflection beam profile, respectively, and E(x,y) denotes the reflection beam profile.

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Embodiment 12: The detector according to any one of the preceding embodiments, wherein the illumination source is configured for generating the at least one illumination pattern in the near infrared region (NIR).

- 15 Embodiment 13: The detector according to any one of the preceding embodiments, wherein the optical sensor comprises at least one CMOS sensor.

Embodiment 14: Method for gesture detection, wherein at least one detector according to the preceding embodiments is used, wherein the method comprises the following steps:

- 20 a) projecting at least one illumination pattern comprising a plurality of illumination features on at least one area comprising at least one object, wherein the object comprises at least partially at least one human hand;
- b) determining at least one image of the area using at least one optical sensor having at least one light-sensitive area, wherein the image comprises a plurality of reflection features generated by the area in response to illumination by the illumination features;
- 25 c) determining at least one depth map of the area by determining at least one depth information for each of the reflection features by using at least one evaluation device;
- d) finding the object by using the evaluation device by identifying the reflection features which were generated by illuminating biological tissue, wherein at least one reflection beam profile of each of the reflection features is determined, wherein a reflection feature is identified as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion, wherein the reflection feature otherwise is identified as background;
- 30 e) segmenting the image of the area by using the evaluation device by using at least one segmentation algorithm, wherein the reflection features identified as to be generated by illuminating biological tissue are used as seed points and the reflection features identified as background are used as background seed points for the segmentation algorithm;
- 35 f) determining position and/or orientation of the object in space considering the segmented image and the depth map by using the evaluation device.

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Embodiment 15: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: driver monitoring; in-cabin surveillance; gesture tracking; a security application; a safety application; a human-machine interface application; an information technology application; an agriculture application; a crop protection application; a medical application; a maintenance application; a cosmetics application.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1 shows an embodiment of a detector according to the present invention; and

Figures 2A to 2C show pose detection according to the present invention; and

Detailed description of the embodiments:

Fig. 1 shows in a highly schematic fashion an embodiment of a detector 110 for gesture detection. The gesture may comprise body gesture such as of at least one part of the body, in particular hand gesture. The gesture may be static or dynamic. The gesture may comprise movement of at least one part of a hand, such as of a finger, movement of one or both hands, face, or other parts of the body. The gesture detection may comprise determining presence or absence of a gesture and/or gesture recognition. The gesture recognition may comprise interpreting of human gestures via mathematical algorithms.

The detector 110 comprises at least one illumination source 112 configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area 114 comprising at least one object 116. The object 116 comprises at least partially at least one human hand.

The illumination source 112 may be configured for providing the illumination pattern for illumination of the area 114. The illumination source 112 may be adapted to directly or indirectly illuminating the area 114, wherein the illumination pattern is reflected or scattered by surfaces of the area 114 and, thereby, is at least partially directed towards at least one optical sensor 118. The illumination source 112 may be configured for illuminating the area 114, for example, by directing a light beam towards the area 114, which reflects the light beam. The illumination source 112 may be configured for generating an illuminating light beam for illuminating the area 114.

The area may comprise the at least one object 116 and a surrounding environment. For example, the object may be at least one object selected from the group consisting of: a scene, a human such as a human, wood, carpet, foam, an animal such as a cow, a plant, a piece of tissue, a metal, a toy, a metallic object, a beverage, a food such as a fruit, meat, fish, a dish, a cosmetics product, an applied cosmetics product, cloth, fur, hair, a maintenance product, a plant, a body, a part of a body, organic material, inorganic material, a reflective material, a screen, a display, a wall, a sheet of paper, such as photograph. The object 116 may comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector 110. The object 116 may, specifically, be a human body or at least one part of a human body such as at least one arm, at least one hand, at least one finger or a face. The object 116 comprises at least partially at least one human hand. The object 116 may be a human hand and/or at least one finger and/or at least one part of a palm.

The illumination source 112 may comprise at least one light source. The illumination source 112 may comprise a plurality of light sources. The illumination source 112 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. The illumination source may be configured for generating the at least one illumination pattern in the infrared region. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination source 112 may be configured for emitting light at a single wavelength. Specifically, the wavelength may be in the near infrared region. In other embodiments, the illumination source 112 may be configured for emitting light with a plurality of wavelengths allowing additional measurements in other wavelengths channels.

The illumination source 112 may be or may comprise at least one multiple beam light source. For example, the illumination source 112 may comprise at least one laser source and one or

more diffractive optical elements (DOEs). Specifically, the illumination source 112 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grat-
5 ing lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumi-
10 nation source 112 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source 112 may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at
15 least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source 112 may be integrated into a housing 120 of the detector 110.

20 The light beam or light beams generated by the illumination source 112 generally may propagate parallel to an optical axis 122 or tilted with respect to the optical axis 112, e.g. including an angle with the optical axis. The detector 110 may be configured such that the light beam or light beams propagates from the detector 110 towards the area 114 along an optical axis of the de-
25 tector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis 122. As an example, the light beam or light beams, such as the laser light beam, and the optical axis 122 may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the opti-
30 cal axis or off the optical axis. As an example, the light beam or light beams may be parallel to the optical axis 122 having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

35 The illumination pattern may comprise at least one illumination feature adapted to illuminate at least one part of the area 114. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern com-
40 prising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one

point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one
5 quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source 112 may be adapted to generate
10 and/or to project a cloud of points. The illumination source 112 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source 112 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source 112.

15 A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination source 112 may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination source comprises at least one laser
20 source and/or at least one laser diode which is designated for generating laser radiation. The illumination source 112 may comprise the at least one diffractive optical element (DOE). The detector 110 may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one periodic point pattern.

25 The illumination source 112 may illuminate the at least one object 116 with the illumination pattern. The illumination pattern may comprise a plurality of points. These points are illustrated as light beam 124 emerging from the illumination source 112.

30 The detector 110 comprises the at least one optical sensor 118 having at least one light-sensitive area 126. The optical sensor 118 is configured for determining at least one image 128 of the area 114. An embodiment of an image is shown in Figure 2A. The image 128 comprises a plurality of reflection features 130 generated by the area 114 in response to illumination by the illumination features. The detector 110 may comprise a plurality of optical sensors 118 each having a light sensitive area 126. Preferably, the light sensitive area 126 may be oriented essentially perpendicular to the optical axis 122 of the detector 110.
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The optical sensor 118 specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor 118 may be sensitive in the infrared spectral range. The optical sensor 118 may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be
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provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor 118 may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor 118 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor 118 may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor 118 may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Cu photodiode, a Ge:Hg photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor 118 may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor 118 may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor 118 may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensor 118 may be sensitive in the near infrared region. Specifically, the optical sensor 118 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor 118, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor 118 each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor 118 may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor 118 may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor 118 may be part of or constitute a pixelated optical device. For example, the optical sensor 118 may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor 118 may be part of or constitute at least one

CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in

which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

5 Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of
10 optical sensors.

The detector 110 further may comprise at least one transfer device 132. The detector 110 may further comprise one or more additional elements such as one or more additional optical elements. The detector 110 may comprise at least one optical element selected from the group
15 consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The transfer device 132 may be adapted to guide the light beam onto the optical sensor. The transfer device 132 specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel
20 lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The focal length constitutes a measure of an ability of the transfer device 132 to converge an impinging light beam. Thus, the transfer device 132 may comprise one or more imaging elements
25 which can have the effect of a converging lens. By way of example, the transfer device 132 can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as
30 the transfer device may be focused into a single spot. Additionally, the transfer device 132 can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device 132 can, in principle, be realized individually or in any desired combination.

The transfer device 132 may have an optical axis. In particular, the detector 110 and the transfer device 132 have a common optical axis 122. The optical axis 122 may be a line of symmetry
40 of the optical setup of the detector 110. The transfer device 132 may constitute a coordinate system 134, wherein a longitudinal coordinate is a coordinate along the optical axis 122 and wherein d is a spatial offset from the optical axis 122. The coordinate system 134 may be a polar coordinate system in which the optical axis of the transfer device 132 forms a z-axis and in

which a distance from the z-axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The optical sensor 118 is configured for determining the at least one image 128 of the area 114. The image 128 comprises a plurality of reflection features 130 generated by the area in response to illumination by the illumination features. The image 128 may be at least one two-dimensional image. The image 128 may be an RGB (red green blue) image.

The detector comprises the at least one evaluation device 136. The evaluation device 136 is configured for evaluating the image 128. The evaluation device 136 may comprise at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device 136 may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device 136 may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images. Specifically the determining a beam profile and indication of the surface, may be performed by the at least one evaluation device 136. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the evaluation device 136 may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device 136 may also fully or partially be embodied by hardware.

The evaluation device 136 and the detector 110 may fully or partially be integrated into a single device. Thus, generally, the evaluation device 136 also may form part of the detector 110. Alternatively, the evaluation device 136 and the detector 110 may fully or partially be embodied as separate devices. The detector 110 may comprise further components.

The evaluation device 136 is configured for determining at least one depth map of the area 114 by determining at least one depth information for each of the reflection features 130. The evaluation device 136 may be configured for determining the depth information for each of the reflection features 130 by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The evaluation device 136 may be configured for considering the depth information for segmenting the object 116, in particular a hand region in the image 128, from the background in terms of the depth map.

For example, the evaluation device 136 may be configured for determining the depth information for each of the reflection features 130 by using depth-from-photon-ratio technique. Each of the reflection features 130 may comprise at least one beam profile, also denoted reflection beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device 136 may be configured for determining depth information for each of the reflection features 130 by analysis of their beam profiles.

The evaluation device 136 may be configured for determining at least one longitudinal coordinate Z_{DPR} for each of the reflection features 130 by analysis of their beam profiles. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The evaluation device 136 may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The evaluation device 136 may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The evaluation device 136 may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The evaluation device 136 may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The evaluation device 136 may be configured for determining the longitudinal coordinate Z_{DPR} for each of the reflection features by using depth-from-photon-ratio technique. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference. The evaluation device 136 may be configured for executing at least one depth-from-photon-ratio algorithm which computes distances for all reflection features 130 with zero order and higher order.

The evaluation of the image 128 comprises identifying the reflection features 130 of the image 128. The evaluation device 136 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features 130. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a

maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image 128 generated by the optical sensor 118.

For example, the illumination source 112 may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor 118, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor 118 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device 136 may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device 136 may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device 136 may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device 136 may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The evaluation device 136 is configured for finding the object 116 by identifying the reflection features 130 which were generated by illuminating biological tissue. The evaluation device 136 is configured for determining the at least one reflection beam profile of each of the reflection features 130. The evaluation device 136 is configured for identifying a reflection feature 130 as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion. The evaluation device 136 is configured for identifying the reflection feature 130 as to be background otherwise.

The detector 110 may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The identification of being generated by biological tissue may comprise determining and/or validating whether a surface to be examined or under test is or

comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector 110 may be adapted to distinguish animal tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector 110 may be adapted to distinguish plant tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector 110 may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector 110 may be adapted to distinguish different types of food such as a fruit, meat, and fish. The detector 110 may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from human skin. The detector 110 may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector 110 may be adapted to distinguish human skin from cloth. The detector 110 may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector 110 may be adapted to distinguish organic material from inorganic material. The detector 110 may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector 110 may be used, in particular, for non-therapeutic and non-diagnostic applications.

The predetermined or predefined criterion may be or may comprise at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature 130 may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills at least one predetermined or predefined criterion. The indication may be an arbitrary indication such as an electronic signal and/or at least one visual or acoustic indication.

The evaluation device 136 may be configured for determining at least one material property of the object by evaluating the beam profile of the reflection feature. The material property may be or may comprise at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. The identifying at least one material property may comprise one or more of determining

and assigning the material property to the object. The detector 110 may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by the user of the detector 110. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The detector 110 may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering profile.

Specifically, the evaluation device 136 may be configured for comparing the reflection beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a mobile device comprising the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the mobile device, e.g. by software, specifically by the app downloaded from an app store or the like. The reflection feature 130 may be identified as to be generated by biological tissue in case the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise determining a deviation, e.g. a sum of squared point to point distances, between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The evaluation device 136 may be adapted to compare the determined deviation with at least

one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the detector

Additionally or alternatively, for identification if the reflection feature 130 was generated by biological tissue, the evaluation device 136 may be configured for applying at least one image filter to the image 128 of the area 114. The evaluation device 136 may be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the image. The material feature may be or may comprise at least one information about the at least one material property of the surface of the area 114 having generated the reflection feature 130.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

The evaluation device 136 may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the surface having generated the reflection feature 130 for determining the material property of the surface having generated the reflection feature 130. The predetermined relationship may be one or more of an empirical relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 136 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The evaluation device 136 is configured for identifying a reflection feature 130 as to be generated by illuminating biological tissue in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature 130 may be identified as to be generated by biological tissue in case the material property indicates "biological tissue".

The reflection feature 130 may be identified as to be generated by biological tissue in case the material property is below or equal at least one threshold or range, wherein in case the determined deviation is below and/or equal the threshold the reflection feature is identified as to be generated by biological tissue and/or the detection of biological tissue is confirmed. At least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the detector. The evaluation device 136 is configured for identifying the reflection feature as to be background otherwise. Thus, the evaluation device 136 may be configured for assigning each projected spot with a depth information and a material property, e.g. skin yes or no.

The material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} .

The evaluation device 136 is configured for segmenting the image 128 of the area 114 by using at least one segmentation algorithm. The segment may comprise a set of pixels. The segmenting may comprise a process of partitioning the image 128 into multiple segments. The segmentation may comprise assigning at least one label to every pixel of the image 128 such that pixels with the same label share at least one characteristic. The labels may be assigned under a pre-defined target. The segmentation may be a binary segmentation. The binary segmentation may comprise labeling the pixels of the image 128 as "skin-pixels" 138 and "background pixel" 140. All non-skin pixels may be considered as background. Skin-pixels may be considered as foreground seeds, also denoted seed points, serving as input for an image-based segmentation algorithm. With respect to image segmenting reference is made to https://en.wikipedia.org/wiki/Image_segmentation. The segmentation algorithm may comprise at least one region growing image segmentation method.

The segmentation algorithm may be configured for fusioning and/or considering a-priori knowledge. Specifically, the segmentation algorithm may be configured for fusioning and/or considering the material information determined from the beam profile analysis. This may allow to obtain or filter position data for motions of visible skin only. In contrast, in conventional set-ups, a recognition and separation of e.g. what is a finger/arm and what is not, needs to be done first. In addition to using the material information, the evaluation device 136 may be configured for considering the depth information for segmenting the object, in particular a hand region in the image, from the background in terms of the depth map.

The segmentation algorithm may be based on energy or cost functions such as one or more of graph cut, level-set, fast marching and Markov random field approaches. Segmentation of the image may be driven by color homogeneity and edge indicators, wherein the seed points constitute edge- and color homogeneity criterions. The reflection features 130 identified as to be generated by illuminating biological tissue are used as seed points and the reflection features identified as background are used as background seed points for the segmentation algorithm. These

seed points may constitute, edge- and color homogeneity criterions, and thus provide an appropriate initialization of the segmentation of the target: "Which color and/or reflection and/or appearance has the skin in the image?".

5 For example, graph cut segmentation may be used. Graph cut segmentation may be configured for fusing the a-priory knowledge as it additionally provides real-time capability when combining it with pre-clustering algorithms such as superpixels, see R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "Slic superpixels compared to state-of-the-art superpixel methods," IEEE TPAMI, vol. 34, no. 11, 2012. For example, a modified lazy snapping graph cut
10 may be used. With respect to modified lazy snapping graph cut reference is made to Li, Yin and Sun, Jian and Tang, Chi-Keung and Shum, Heung-Yeung, "Lazy Snapping", ACM Trans. Graph., vol. 23, no. 3, 2004. The Gibbs energy term may comprise an additional term to incorporate depth information for the graph cut segmentation routine. Instead of watershed-based pre-segmentation as proposed in Li, Yin and Sun, Jian and Tang, Chi-Keung and Shum,
15 Heung-Yeung, "Lazy Snapping", ACM Trans, a superpixel clustering may be used to ensure real-time capability.

The image 128 before the segmentation is shown in Figure 2A. Each projected spot is assigned with a depth and a material property (skin yes/no). To obtain the binary segmentation illustrated
20 in Figure 2B, all non-skin points are considered as background, skin-points are considered as foreground seeds serving as input for an image-based segmentation routine. In this example, a modified lazy snapping graph cut was used. The Gibbs energy term includes an additional term to incorporate depth information for the graph cut segmentation routine. Furthermore, a superpixel clustering was used to ensure real-time capability.

25 The evaluation device 136 is configured for determining position and/or orientation of the object 116 in space considering the segmented image and the depth map. The evaluation device 136 may be configured for determining at least one hand pose or gesture from the position and/or the orientation of the object 116 in space. The evaluation device 136 may be configured for
30 recognition of gestures from the segmented image, in particular from the parts of the segmented image labeled as object 116. The evaluation device 136 may be configured for identifying image coordinates of palm and finger in the segmented image. Palm and finger identification may be obtained from the segmentation. The evaluation device 136 may be configured for extracting features such as color and/or brightness and/or gradient values from the parts of the segmented
35 image labeled as object. For example, palm and finger tips may be detected using standard OpenCV routines.

The evaluation device 136 may be configured for determining at least one three-dimensional finger vector 141 considering image coordinates of palm 142, such as of a center point 144 of the
40 palm, and finger tip 146 and the depth map. Figure 2C shows an embodiment of the 3D finger vector 140. The 3D finger vector 140 may specify the position and orientation of the finger in space. The evaluation device 136 may be configured for determining the 3D finger vector 140 from the image 128 and depth map. The 3D finger vector 140 may provide the basis for hand

pose estimation and scene interpretation. For the recognition of hand gestures, the evaluation device 136 may be configured for using at least one classifier such as a hidden Markov model (HMM), a support vector machine (SVM), a conditional random field (CRF) and the like. The classifier may be configured for discriminating hand gestures.

5

The use of the beam profile analysis sensor fusion concept according to the present invention have the advantages for providing robust detection results, no need for complex model assumptions, a single sensor concept, easy integration and real-time capability.

List of reference numbers

110	detector
112	illumination source
114	area
116	object
118	optical sensor
120	housing
122	optical axis
124	Light beam
126	light-sensitive area
128	image
130	Reflection feature
132	transfer device
134	coordinate system
136	evaluation device
138	skin-pixels
140	background pixel
141	three-dimensional finger vector
142	palm
144	center point
146	finger tip

Claims

1. Detector (110) for gesture detection comprising
- 5 - at least one illumination source (112) configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area (114) comprising at least one object (116), wherein the object (116) comprises at least partially at least one human hand;
- 10 - at least one optical sensor (118) having at least one light-sensitive area (126), wherein the optical sensor (118) is configured for determining at least one image (128) of the area, wherein the image (128) comprises a plurality of reflection features (130) generated by the area (114) in response to illumination by the illumination features;
- 15 - at least one evaluation device (136), wherein the evaluation device (136) is configured for determining at least one depth map of the area by determining at least one depth information for each of the reflection features (130), wherein the evaluation device (136) is configured for finding the object (116) by identifying the reflection features (130) which were generated by illuminating biological tissue, wherein the evaluation device (136) is configured for determining at least one reflection beam profile of each of the reflection features (130), wherein the evaluation device (136) is configured for identifying a reflection feature (130) as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one predetermined or predefined criterion, wherein the predetermined or predefined criterion is or comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property, wherein the evaluation device (136) is configured for identifying the reflection feature (130) as to be background otherwise,
- 20 wherein the evaluation device (136) is configured for segmenting the image (128) of the area by using at least one segmentation algorithm, wherein the reflection features (130) identified as to be generated by illuminating biological tissue are used as seed points and the reflection features (130) identified as background are used as background seed points for the segmentation algorithm,
- 25 wherein the evaluation device (136) is configured for determining position and/or orientation of the object (116) in space considering the segmented image and the depth map.
- 30
- 35
2. The detector (110) according to the preceding claim, wherein the evaluation device (136) is configured for identifying image coordinates of palm and finger in the segmented image, wherein the evaluation device (136) is configured for determining at least one three-dimensional finger vector (141) considering image coordinates of palm and finger and the depth map.
- 40

3. The detector (110) according to any one of the preceding claims, wherein the evaluation device (136) is configured for determining at least one hand pose or gesture from the position and/or the orientation of the object (116) in space.
- 5 4. The detector (110) according to any one of the preceding claims, wherein the segmentation algorithm is based on energy or cost functions such as one or more of graph cut, level-set, fast marching and Markov random field approaches.
- 10 5. The detector (110) according to any one of the preceding claims, wherein segmentation of the image (128) is driven by color homogeneity and edge indicators, wherein the seed points constitute edge- and color homogeneity criterions.
- 15 6. The detector (110) according to any one of the preceding claims, wherein the evaluation device (136) is configured for comparing the reflection beam profile of each of the reflection features (130) with at least one predetermined and/or prerecorded and/or predefined beam profile.
- 20 7. The detector (110) according to the preceding claim, wherein the comparison comprises overlaying the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile such that their centers of intensity match, wherein the comparison comprises determining a deviation between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile, wherein the evaluation device (136) is configured for comparing the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the reflection feature (130) is indicated as biological tissue.
- 25 8. The detector (110) according to any one of the preceding claims, wherein the evaluation device (136) is configured for determining the depth information for each of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors.
- 30 9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (136) is configured for determining the depth information for each of the reflection features (130) by using depth-from-photon-ratio technique, wherein the evaluation device is configured for determining at least one first area and at least one second area of a beam profile of at least one of the reflection features, wherein the evaluation device (136) is configured for integrating the first area and the second area,
- 35 40 wherein the evaluation device (136) is configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.

10. The detector (110) according to the preceding claim, wherein the first area of the reflection beam profile comprises essentially edge information of the reflection beam profile and the second area of the reflection beam profile comprises essentially center information of the reflection beam profile, and/or wherein the first area of the reflection beam profile comprises essentially information about a left part of the reflection beam profile and the second area of the reflection beam profile comprises essentially information about a right part of the reflection beam profile.

11. The detector (110) according to any one of the two preceding claims, wherein the evaluation device (136) is configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the reflection beam profile, respectively, and E(x,y) denotes the reflection beam profile.

12. The detector (110) according to any one of the preceding claims, wherein the illumination source (112) is configured for generating the at least one illumination pattern in the near infrared region (NIR).

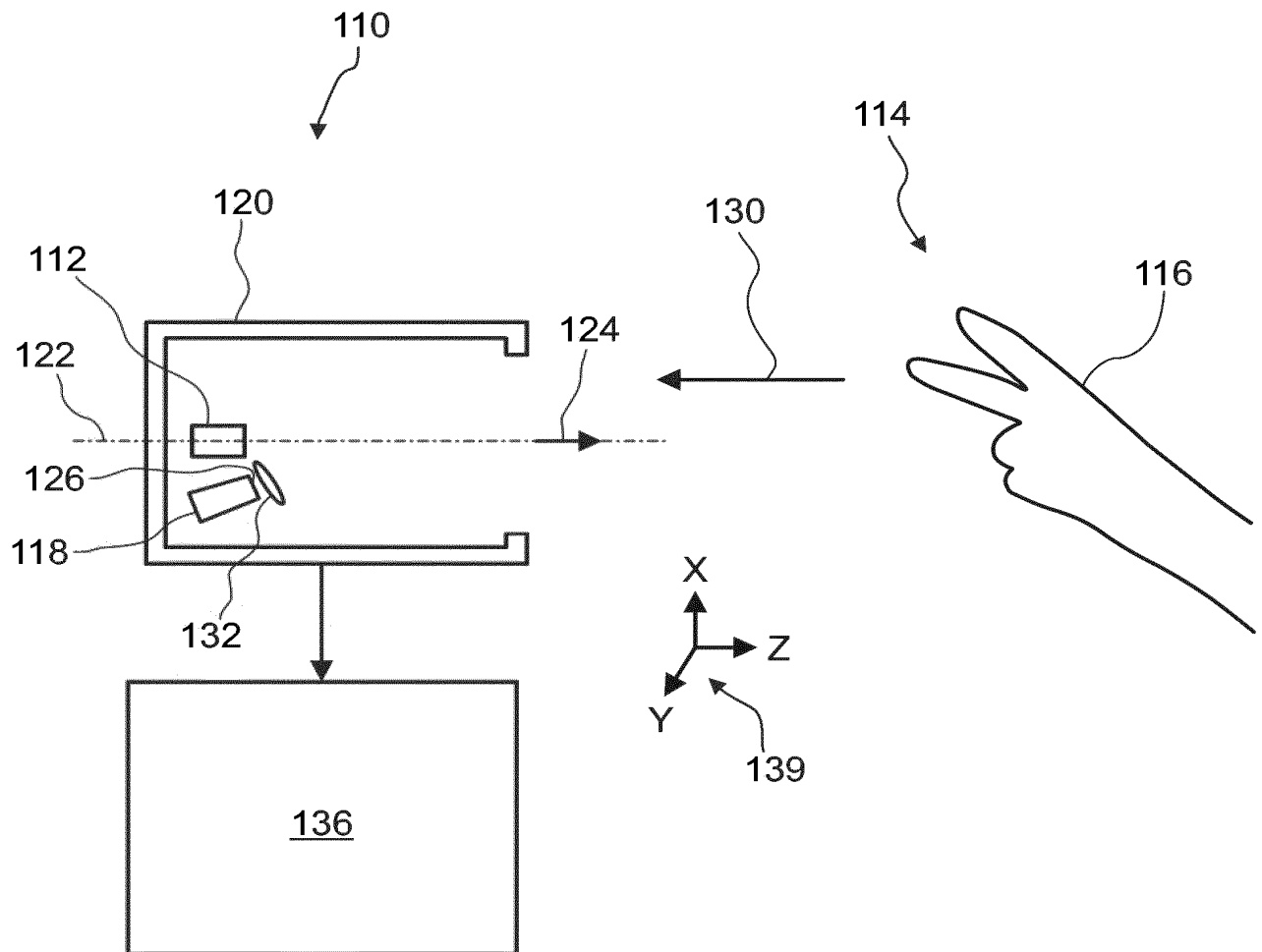
13. The detector (110) according to any one of the preceding claims, wherein the optical sensor (118) comprises at least one CMOS sensor.

14. Method for gesture detection, wherein at least one detector (110) according to the preceding claims is used, wherein the method comprises the following steps:

- a) projecting at least one illumination pattern comprising a plurality of illumination features on at least one area (114) comprising at least one object (116), wherein the object (116) comprises at least partially at least one human hand;
- b) determining at least one image (128) of the area (114) using at least one optical sensor (118) having at least one light-sensitive area (126), wherein the image (128) comprises a plurality of reflection features (130) generated by the area (114) in response to illumination by the illumination features;
- c) determining at least one depth map of the area (114) by determining at least one depth information for each of the reflection features (130) by using at least one evaluation device (136);
- d) finding the object (116) by using the evaluation device (136) by identifying the reflection features (130) which were generated by illuminating biological tissue, wherein at least one reflection beam profile of each of the reflection features (130) is determined, wherein a reflection feature (130) is identified as to be generated by illuminating biological tissue in case its reflection beam profile fulfills at least one

- predetermined or predefined criterion, wherein the predetermined or predefined criterion is or comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property, wherein the reflection feature (130) otherwise is identified as background;
- 5 e) segmenting the image (128) of the area (114) by using the evaluation device (136) by using at least one segmentation algorithm, wherein the reflection features (130) identified as to be generated by illuminating biological tissue are used as seed points and the reflection features (130) identified as background are used as background seed points for the segmentation algorithm;
- 10 f) determining position and/or orientation of the object (116) in space considering the segmented image and the depth map by using the evaluation device (136).
15. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: driver monitoring;
- 15 in-cabin surveillance; gesture tracking; a security application; a safety application; a human-machine interface application; an information technology application; an agriculture application; a crop protection application; a medical application; a maintenance application; a cosmetics application.

FIG.1



2/2

FIG.2A

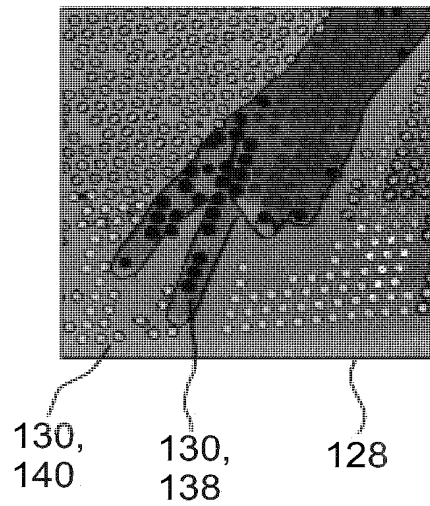


FIG.2B

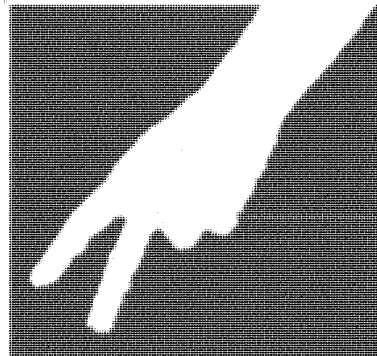
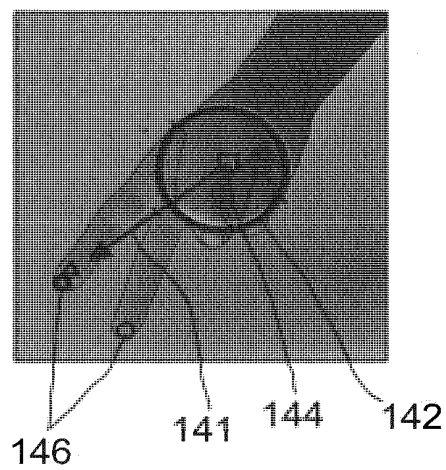


FIG.2C



INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2021/054811

A. CLASSIFICATION OF SUBJECT MATTER

INV. G06K9/00 G06K9/20 G06K9/34 G06K9/46 G06K9/62
G06K9/22

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G06K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EP0-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

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Further documents are listed in the continuation of Box C.



See patent family annex.

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"O" document referring to an oral disclosure, use, exhibition or other means

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"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

14 May 2021

Date of mailing of the international search report

26/05/2021

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2
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Atmatzidis, Lazaros

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2021/054811

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

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		WO 2018091640 A2	24-05-2018
		WO 2018091649 A1	24-05-2018



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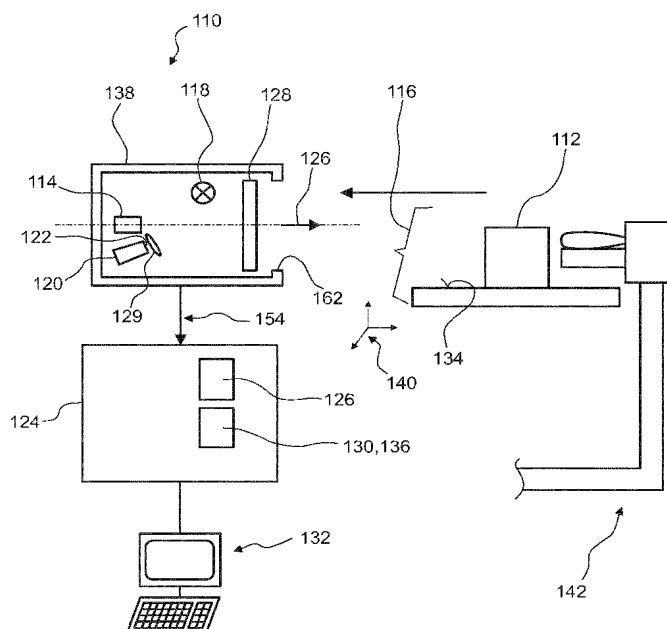
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FIG.1



(57) Abstract: A detector (110) for object recognition comprising - at least one illumination source (114) configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area (116) comprising at least one object (110); - an optical sensor (120) having at least one light sensitive area (122), wherein the optical sensor (120) is configured for determining at least one first image comprising at least one two dimensional image of the area, wherein the optical sensor (120) is configured for determining at least one second image comprising a plurality of reflection features generated by the area (116) in response to illumination by the illumination features; - at least one evaluation device (124), wherein the evaluation device (124) is configured for evaluating the first image and the second image, wherein each of the reflections features comprises at least one beam profile, wherein the evaluation device (124) is configured for determining beam profile information for each of the reflection features

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by analysis of their beam profiles, wherein the evaluation device (124) is configured for determining at least one three-dimensional image using the determined beam profile information, wherein the evaluation of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature, wherein the evaluation device (124) is configured for identifying the reflection features which are located inside an image region the geometrical feature and/or for identifying the reflection features which are located outside the image region of the geometrical feature, wherein the evaluation device (124) is configured for determining at least one depth level from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature, wherein the evaluation device (124) is configured for determining at least one material property of the object from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature, wherein the evaluation device (124) is configured for determining at least one position and/or orientation of the object by considering the depth level and/or the material property and pre-determined or predefined information about shape and/or size of the object.

Detector for object recognition

Description

5 Field of the invention

The invention relates to a detector and a method for object recognition and various uses of the detector. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for
10 arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

15 Prior art

Automatic object recognition of metal object is challenging. Depending on angle metal object do not or only sparsely reflect light from an illumination source such that a reliable three dimensional image generation is not possible. In order to allow automatic object recognition of metal
20 objects it is known to combine 3D and 2D image information even if the 3D measurement cannot provide utilizable data. The 2D image may comprise image information which cannot be recorded via a 3D measurement. However, for 3D image sensors such as 3D time-of-Flight (ToF)-cameras no 2D image information or only 2D image information with very limit resolution is available. Moreover, it is necessary to determine during image capturing whether a 2D or 3D
25 image was recorded in order to ensure correct analysis by an image analysis software. This is not possible by using purely the software driver controller of the camera due to missing real time behavior. In known methods high resolution 2D image data is recorded in addition to the 3D data by using a further camera. But position and angle of view of the further camera needs to be calibration in addition to the 3D camera. This calibration adds further uncertainty due to external
30 influences such as decalibration due to temperature changes or mechanical stress. Moreover, it is very complex to synchronize both camera systems in order to record 2D and 3D images.

Other techniques such as structured light in principle have the possibility to generate 2D image data with high resolution. But 2D imaging is not performed since the Laser needs to be shut
35 down for recording of the 2D image and in case of using band pass filter in the infrared wavelength range additional illumination may be necessary for image recording.

US 2016/0238377 A1 describes a modeling arrangement for modeling the topography of a three dimensional surface. The arrangement includes a light source arranged to produce substantially
40 monochromatic and coherent electromagnetic radiation; a camera arranged to photograph the surface to be modeled at wavelengths emitted by the light source as well as wavelengths detected by the human eye; and a grating provided in connection with the first light source. The light source and the grating provided in connection with the light source are arranged jointly to produce a diffraction pattern of a known geometry on the surface to be modeled.

Chen Guo-Hua et al. "Transparent object detection and location based on RGB-D camera", JOURNAL OF PHYSICS: CONFERENCE SERIES, vol. 1183, 1 March 2019, page 012011, XP055707266, GB ISSN: 1742-6588, DOI: 10.1088/1742-6596/1183/1/012011 describes a method for transparent object detection and location that utilize depth image, RGB image and IR image. In detection process, an active depth sensor (Re-alSense) is firstly employed to retrieve the transparent candidates from the depth image and the corresponding candidates in the RGB image and IR image are then extracted separately. A transparent candidate classification algorithm is subsequently presented that uses SIFT features to recognize the transparent ones from the candidates. In location process, a group of RGB images and IR images was obtained by adjusting camera orientation to make its optical axis perpendicular to the normal direction of the plane on which the object is placed. The object contours in RGB image and IR image are then extracted, respectively. The three-dimensional object is finally reconstructed by means of stereo matching of the two contours, and the current pose information of the object is calculated in the end.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which allow reliable object recognition with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases,

when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

5 Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

15 In a first aspect of the present invention a detector for object recognition is disclosed.

As used herein, the term “detector” may generally refer to an arbitrary sensor device configured for determining and/or detecting and/or sensing the at least one object. The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

25 As used herein, the term “object” may generally refer to an arbitrary physical body whose orientation and/or position is to be determined. The object may be at least one article. For example, the object may be at least one object selected from the group consisting of: a box, a bottle, a plate, a sheet of paper, a bag, a screw, a washer, a machined metal piece, a rubber seal, plastic pieces, wrapping, packing material. As used herein, the term “object recognition” may generally refer to identifying the object and determining at least one information about the position and/or orientation of the object. As used herein, the term “position” may refer to at least one item of information regarding a location of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. As used herein, the term “orientation” refers to angular position of the object in space. The orientation may be given by three spatial angles.

The detector comprises:

- at least one illumination source configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area comprising at least one object;
 - an optical sensor having at least one light sensitive area, wherein the optical sensor is configured for determining at least one first image comprising at least one two dimensional image of the area, wherein the optical sensor is configured for determining at least one second image comprising a plurality of reflection features generated by the area in response to illumination by the illumination features;
 - at least one evaluation device, wherein the evaluation device is configured for evaluating the first image and the second image, wherein each of the reflections features comprises at least one beam profile, wherein the evaluation device is configured for determining beam profile information for each of the reflection features by analysis of their beam profiles, wherein the evaluation device is configured for determining at least one three-dimensional image using the determined beam profile information, wherein the evaluation of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature, wherein the evaluation device is configured for identifying the reflection features which are located inside an image region the geometrical feature and/or for identifying the reflection features which are located outside the image region of the geometrical feature, wherein the evaluation device is configured for determining at least one depth level from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature, wherein the evaluation device is configured for determining at least one material property of the object from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature, wherein the evaluation device is configured for determining at least one position and/or orientation of the object by considering the depth level and/or the material property and pre-determined or predefined information about shape and/or size of the object.
- The object may be located within a scene and/or may have a surrounding environment. Specifically, the object may be located in at least one area. As used herein, the term “area” in this context may generally refer to at least one surface and/or region. As used herein, the term “area comprising the object” may generally refer to at least one surface on which the object is located and/or at least one region in which the object is located. The area may comprise additional elements such as the surrounding environment.

The illumination source is configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area comprising at least one object. As used herein, the term “illumination source” may generally refers to at least one arbitrary device adapted to provide the at least one illumination light beam for illumination of the object. The illumination source may be adapted to directly or indirectly illuminating the object, wherein the illumination pattern is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source may be adapted to illuminate the object, for

example, by directing a light beam towards the object, which reflects the light beam. The illumination source may be configured for generating an illuminating light beam for illuminating the object.

- 5 The illumination source may comprise at least one light source. The illumination source may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. The illumination source may be configured for generating the at least one illumination pattern in the infrared region. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The illumination source may be configured for emitting light at a single wavelength. Specifically, the wavelength may be in the near infrared region. Using near infrared light may be advantageous since in the near infrared region human skin exhibits a unique reflection beam profile and significant absorption and diffusion, as will be outlined below. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels

The illumination source may be or may comprise at least one multiple beam light source. For example, the illumination source may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure

ture lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source may be integrated into a housing of the detector.

The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination device may be integrated in a mobile device or attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

Further, the illumination source may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams.

The light beam or light beams generated by the illumination source generally may propagate parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The detector may be configured such that the light beam or light beams propagate from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the light beam or light beams, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the optical axis or off the optical axis. As an example, the light beam or light beams may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the object. As used herein, the term "illumination feature" refers to at least one at least partially extended

feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source.

A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination source may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination source comprises at least one laser source and/or at least one laser diode which is designated for generating laser radiation. The illumination source may comprise the at least one diffractive optical element (DOE). The detector may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one point pattern.

As further used herein, the term “projecting at least one illumination pattern” refers to providing the at least one illumination pattern for illuminating the at least one object.

For example, the projected illumination pattern may be a periodic point pattern. The projected illumination pattern may have a low point density. For example, the illumination pattern may comprise at least one periodic point pattern having a low point density, wherein the illumination pattern has ≤ 2500 points per field of view. In comparison with structured light having typically a point density of 10k - 30k in a field of view of $55 \times 38^\circ$ the illumination pattern according to the present invention may be less dense. This may allow more power per point such that the proposed technique is less dependent on ambient light compared to structured light.

The detector may comprise at least one further illumination source. The further illumination source may comprise one or more of at least one further light source such as at least one light emitting diode (LED) or at least one vertical-cavity surface-emitting laser (VCSEL) array. The further illumination source may comprise at least one optical element such as at least one diffuser or at least one lens. The further illumination source may be configured for providing additional illumination for imaging of the first image. For example, the further illumination source may be used in situations in which it is not possible or difficult for recording the reflection pattern, e.g. in case of highly reflective metallic surfaces, in order to ensure a good illumination and, thus, contrasts for two-dimensional images such that a two-dimensional image recognition is possible.

The detector may comprise a single camera comprising the optical sensor. The detector may comprise a plurality of cameras each comprising an optical sensor or a plurality of optical sensors.

The optical sensor has at least one light sensitive area. As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The detector may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal. In case the detector comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

Preferably, the light sensitive area may be oriented essentially perpendicular to an optical axis of the detector. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge: Au photodiode, a Ge: Hg photodiode, a Ge: Cu photodiode, a Ge: Zn photodiode, a Si: Ga photodiode, a Si: As photodiode. Additionally or alternatively, the optical sensor may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor may be part of or constitute a pixelated optical device. For example, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example,

tolerances of less than 20°, specifically less than 10° or even less than 5°, may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

The optical sensor is configured for determining at least one first image comprising at least one two dimensional image of the area.

As used herein, without limitation, the term “image” specifically may relate to data recorded by using the optical sensor, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element. The image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to “pixels”, reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly.

As used herein, the term “two dimensional image” may generally refer to an image having information about transversal coordinates such as the dimensions of height and width only. As used herein, the term “three dimensional image” may generally refer to an image having information about transversal coordinates and additionally about the longitudinal coordinate such as the dimensions of height, width and depth.

The optical sensor is configured for determining at least one second image comprising a plurality of reflection features generated by the area in response to illumination by the illumination features. As used herein, the term “reflection feature” may refer to a feature in an image plane generated by the object in response to illumination, specifically with at least one illumination feature.

The first image and the second image may be determined, in particular recorded, at different time points. Recording of the first image and the second time limit may be performed with a temporal shift. Specifically, a single camera comprising the optical sensor may record with a

temporal shift a two-dimensional image and an image of a projected pattern. Recording the first and the second image at different time points may ensure that the evaluation device can distinguish between the first and the second image and can apply the appropriate evaluation routine. Moreover, it is possible to adapt the illumination situation for the first image if necessary and in particular independent from the illumination for the second image. The detector may comprise at least one control unit. As used herein, the term “control unit” may refer to an arbitrary device configured for controlling operation of one or more components or elements of the detector. The control unit may be designed as hardware component of the detector. In particular, the control unit may comprise at least one microcontroller. The control unit may be configured for controlling the optical sensor and/or the illumination source. The control unit may be configured for triggering projecting of the illumination pattern and/or imaging of the second image. Specifically, the control unit may be configured for controlling the optical sensor, in particular frame rate and/or illumination time, via trigger signals. The control unit may be configured for adapting and/or adjusting the illumination time from frame to frame. As used herein, the term “frame” may refer to a time range for determining one image. This may allow adapting and/or adjusting illumination time for the first image, e.g. in order to have contrasts at the edges, and at the same time adapting and/or adjusting illumination time for the second image to maintain contrast of the reflection features. Additionally, the control unit may, at the same time and independently, control the elements of the illumination source and/or the further illumination source.

Specifically, the control unit may be configured for adapting exposure time for projection of the illumination pattern. The second image may be recorded with different illumination times. Dark regions of the area may require more light in comparison to lighter regions, which may result to run into saturation for the lighter regions. Therefore, the detector may be configured for recording a plurality of images of the reflection pattern, wherein the images may be recorded with different illumination times. The detector may be configured for generating and/or composing the second image from said images. The evaluation device may be configured for performing at least one algorithm on said images which were recorded with different illumination times.

As outlined above, the detector may comprise the further illumination source configured for illuminating the area for determining the first image. The control unit may be configured for controlling the further illumination source. The control unit may be configured for triggering illumination of the area by light generated by the further illumination source and imaging of the first image. The control unit may be configured for adapting exposure time for projection of the illumination pattern and illumination by light generated by the further illumination source.

The detector may comprise at least one first filter element. As used herein, the term “filter element” may refer to at least one arbitrary optical element configured for selectively blocking and transmitting light depending on wavelength. The first filter element may be configured for transmitting light in the infrared spectral range and for at least partially blocking light of other spectral ranges. The first filter element may be a monochromatic bandpass filter configured for transmitting light in a small spectral range. For example, the spectral range or bandwidth may be ± 100

nm, preferably ± 50 nm, most preferably ± 35 nm or even less. For example, the first filter element may be configured for transmitting light having a central wavelength of 808 nm, 830 nm, 850 nm, 905 nm or 940 nm. For example, the first filter element may be configured for transmitting light having a central wavelength of 850 nm with a bandwidth of 70 nm or less. The first
5 filter element may have a minimal angle dependency such that the spectral range can be small. This may result in a low dependency on ambient light, wherein at the same time an enhanced vignetting effect can be prevented. For example, the detector may comprise the single camera having the optical sensor and, in addition, the first filter element. The first filter element may ensure that even in presence of ambient light recording of the reflection pattern is possible and at
10 the same time to maintain laser output power low such that eye safety operation in laser class 1 is ensured.

Additionally or alternatively to the first filter element, the detector may comprise at least one polarization filter. The polarization filter may be positioned 90° rotated with respect to the polari-
15 zation of the illumination source such as of the laser. This may allow weaken direct back-reflections from e.g. metallic materials, and/or may allow detection of points from the projected pattern which can be evaluated using the depth-from-photon-ratio technique even if it is directly reflected and not diffusely scattered. The illumination source for determining the two-dimensional image may be non-polarized such that the 2D imaging may not be influenced by
20 the polarization filter, or only by that brightness is reduced.

Additionally or alternatively, the detector may comprise at least one second filter element. The second filter element may be a band-pass filter. The second filter element may be configured for transmitting light in the visual spectral range and for at least partially blocking light of other
25 spectral ranges.

The spectrum of the illumination source and/or of the further illumination source may be selected depending on the used filter elements. For example, in case of the first filter element having a central wavelength of 850 nm, the illumination source may comprise at least one light source
30 generating a wavelength of 850 nm such as at least one infrared (IR)-LED.

The evaluation device is configured for evaluating the first image and the second image. As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and,
35 more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors

with software running thereon for performing one or more of the named operations. Operations, including evaluating the images. Specifically the determining the reflection beam profile and indication of the surface, may be performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned instructions may be implemented in software and/or hardware. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by

buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

The evaluation of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature. As used herein, the term “geometrical feature” refers to at least one characteristic element of the object. The geometrical feature may be at least one characteristic element of the object selected from the group consisting of: a shape, a relative position of at least one edge, at least one borehole, at least one reflection point, at least one line, at least one surface, at least one circle, at least one disk, the full object, a part of the object and the like. The evaluation device may comprise at least one data storage device. The data storage device may comprise at least one table and/or at least one lookup table of geometrical features and/or pre-determined or predefined information about shape and/or size of the object. Additionally or alternatively, the detector may comprise at least one user interface via which a user can enter the at least one geometrical feature.

The evaluation device may be configured for evaluating in a first step the second image. The evaluation of the second image may provide, as will be outlined in more detail below, 3D information of the reflection features. The evaluation device may be configured for estimating a location of the geometrical feature in the first image by considering the 3D information of the reflection features. This may reduce effort of search for geometrical feature in the first image significantly.

The evaluation device may be configured for identifying the geometrical feature by using at least one image processing process. The image processing process may comprise one or more of at least one template matching algorithm; at least one Hough-transformation; applying a Canny

edge filter; applying a Sobel filter; applying a combination of filters. The evaluation device may be configured for performing at least one plausibility check. The plausibility check may comprise comparing the identified geometrical feature compared to at least one known geometrical feature of the object. For example, a user may enter a known geometrical feature via the user interface for the plausibility check.

The evaluation device is configured for evaluating of the second image. The evaluation of the second image may comprise generating a three-dimensional image.

Each of the reflections features comprises at least one beam profile. As used herein, the term "beam profile" of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device is configured for determining beam profile information for each of the reflection features by analysis of their beam profiles.

The evaluation device may be configured for determining the beam profile of each of the reflection features. As used herein, the term "determining the beam profile" refers to identifying at least one reflection feature provided by the optical sensor and/or selecting at least one reflection feature provided by the optical sensor and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The evaluation device may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest

point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within the image generated by the optical sensor.

For example, the illumination source may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device may be adapted to remove influences from background light from the reflection beam profile, for example, by an imaging without further illumination.

As used herein, the term "analysis of the beam profile" may generally refer to evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The evaluation device may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The evaluation device may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The evaluation device may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The evaluation device may be con-

figured for removing influences from background light from the reflection beam profile, for example, by an imaging without illumination.

The reflection feature may cover or may extend over at least one pixel of the image. For example, the reflection feature may cover or may extend over plurality of pixels. The evaluation device may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The evaluation device may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum_j r_{pixel}},$$

wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum_j I_{total}$ with j being the number of pixels j connected to and/or belonging to the reflection feature and I_{total} being the total intensity.

The evaluation device is configured for determining the beam profile information for each of the reflection features by analysis of their beam profiles. As used herein, the term “beam profile information” may generally refer to information about an intensity distribution of a light spot on the light sensitive area of the optical sensor. The beam profile information may comprise information about the longitudinal coordinate of the surface point or region having reflected the illumination feature. Additionally, the beam profile information may comprise information about a material property of said surface point or region having reflected the illumination feature.

The beam profile information may be the longitudinal coordinate of the surface point or region having reflected the illumination feature. The evaluation device may be configured for determining the beam profile information for each of the reflection features by using depth-from-photon-ratio technique. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The evaluation device may be configured for integrating the first area and the second area. The evaluation device may be configured to derive a combined signal, in particular a quotient Q, by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The evaluation device may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the evaluation device may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The evaluation device may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The evaluation device may be configured for deter-

mining for at least two of the areas an integral of the beam profile over the respective area. The evaluation device may be configured for comparing at least two of the determined integrals. Specifically, the evaluation device may be configured for determining at least one first area and at least one second area of the reflection beam profile. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the reflection beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the evaluation device may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the evaluation device may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the detector may comprise at least two optical sensors, wherein the light-sensitive areas of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the reflection beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the reflection beam profile of the reflection feature. The evaluation device may be adapted to integrate the first area and the second area. The evaluation device may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge

information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be configured for determining an area integral of the beam profile. The evaluation device may be configured for determining the edge information by integrating and/or summing of the first area. The evaluation device may be configured for determining the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be configured for determining an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

In one embodiment, A1 may correspond to a full or complete area of a feature point on the optical sensor. A2 may be a central area of the feature point on the optical sensor. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In one embodiment, the illumination pattern may comprise at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensor may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensor. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line

width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

In one embodiment, the illumination pattern may comprise at least point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The illumination pattern may comprise both at least one point pattern and at least one line pattern. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

The evaluation device may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The evaluation device may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the beam profile, respectively, and E(x,y) denotes the beam profile.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

For example, in case of the optical sensor having a matrix of pixels, the evaluation device may be configured for evaluating the beam profile, by

- determining the pixel having the highest sensor signal and forming at least one center signal;
- evaluating sensor signals of the matrix and forming at least one sum signal;
- determining the quotient Q by combining the center signal and the sum signal; and
- determining at least one longitudinal coordinate z of the object by evaluating the quotient

Q.

As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the pixel having the highest sensor signal out of the plurality of sensor signals generated by the pixels of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the pixels of the matrix. The center signal may arise from a single pixel or from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of pixels may be added up, integrated or averaged, in order to determine the center signal. The group of pixels from which the center signal arises may be a group of neighboring pixels, such as pixels having less than a predetermined distance from the actual pixel having the highest sensor signal, or may be a group of pixels generating sensor signals being within a predetermined range from the highest sensor signal. The group of pixels from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of pixels around the pixel having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the pixel in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from pixels in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor sig-

nals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring pixels; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term “sum signal” generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The quotient Q may be a signal which is generated by combining the center signal and the sum signal. Specifically, the determining may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the quotient Q may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

As used herein, the term “longitudinal coordinate of the object” refers to a distance between the optical sensor and the object. The evaluation device may be configured for using the at least one predetermined relationship between the quotient Q and the longitudinal coordinate for de-

termining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

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The evaluation device may be configured for determining at least one three-dimensional image and/or 3D-data using the determined beam profile information. The image or images recorded by the camera comprising the reflection pattern may be a two-dimensional image or two-dimensional images. As outlined above, the evaluation device may be configured for determining for each of the reflection features a longitudinal coordinate. The evaluation device may be configured for generating 3D-data and/or the three-dimensional image by merging the two-dimensional image or images of the reflection pattern with the determined longitudinal coordinate of the respective reflection feature.

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The evaluation device may be configured for merging and/or fusing the determined 3D-data and/or the three-dimensional image and the information determined from the first image, i.e. the at least one geometrical feature and its location, in order to identify the object in a scene, in particular in the area.

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The evaluation device is configured for identifying the reflection features which are located inside an image region the geometrical feature and/or for identifying the reflection features which are located outside the image region of the geometrical feature. The evaluation device may be configured for determining an image position of the identified geometrical feature in the first image. The image position may be defined by pixel coordinates, e.g. x and y coordinates, of pixels of the geometrical feature. The evaluation device may be configured for determining and/or assigning and/or selecting at least one border and/or limit of the geometrical feature in the first image. The border and/or limit may be given by at least one edge or at least one contours of the geometrical feature. The evaluation device may be configured for determining the pixels of the first image inside the border and/or limit and their image position in the first image. The evaluation device may be configured for determining at least one image region of the second image corresponding to the geometrical feature in the first image by identifying the pixels of the second image corresponding to the pixels of the first image inside the border and/or limit of the geometrical feature. As used herein, the term "image region" may refer to an area of an image, e.g. given by an amount of pixels and/or pixel coordinates. As used herein, the term "located inside" an image region refers to pixels of the image region and/or belonging to the image region. As used herein, the term "located outside" the image region may refer to pixels at a different location or region of the image as the pixels inside the image region.

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The evaluation device is configured for determining the at least one depth level from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature. The area comprising the object may comprise a plurality of elements at different depth levels. As used herein, the term "depth level" may refer to a bin or step of a depth map of the pixels of the second image. As outlined above, the evaluation device may be

configured for determining for each of the reflection features a longitudinal coordinate from their beam profiles. The evaluation device may be configured for determining the depth levels from the longitudinal coordinates of the reflection features located inside and/or outside of the image region of the geometrical feature. Metallic objects often cannot be identified in the second image correctly. However, levels can be correctly identified, which may be defined by the ground or cover of said metallic objects since these often are made of cardboard. The evaluation device may be configured for determining the depth level on which the object is located from the depth level of the reflection features located inside and/or outside of the image region of the geometrical feature.

The evaluation device is configured for determining the position and/or the orientation of the object by considering the depth level and pre-determined or predefined information about shape and/or size of the object. For example, the information about shape and/or size may be entered by a user via the user interface of the detector. For example, the information about shape and size may be measured in an additional measurement. As outlined above, the evaluation device is configured for determining the depth level on which the object is located. If in addition, the shape and/or size of the object are known the evaluation device can determine the position and orientation of the object.

For example, in case a task may be to detect and measure with the detector at least one object such as bottles in a box. The detector, in particular the optical sensor, may be installed on a robot arm such that the detector can move to different positions with respect to the objects in the box. The task may be that the robot should move to the objects and take it out of the box. Additionally, the user knows the object, in this example the bottles, in detail, such that the size, form and shape may be also known and may be programmed into the evaluation device.

The optical sensor may determine the two dimensional image and a resulting 3d depth map. The depth map may estimate the position of the detector and the objects. The depth map can also be distorted by different effects like to shiny objects, e.g. metal, and/or the 3d depth map may be to sparse. The present invention propose to get additional information by a 2d image that corresponds to the 3d depth map. In the example with the bottles, the task is to detect bottles in a box. In addition, it may be known that the bottles are rotationally symmetric. Certain features of the botte can helps for object detection, e.g. round bottle caps. This may lead to search for circles or ellipsoids in the 2d image for the object detection with image processing algorithms. A rough estimation of the size of the ellipsoids may be computed by the 3d depth information. For a detailed object detection, the detected ellipsoids in the 2d image and the known relation of the projection between detector and the real world can be used to determine the size and position of the circles in the real word. A relationship between the projection between detector and the real world lcan be used to determine size, position and orientation by using at least one system of equations.

The evaluation device is configured for determining the at least one material property of the object from the beam profile information of the reflection features located inside and/or outside of

the image region of the geometrical feature. With respect to details of determining the material property reference is made to European patent application 19 163 250.4 filed on March 15, 2019, the full content of which is included by reference.

- 5 The beam profile information may comprise information about a material property of the surface point or region having reflected the illumination feature. The object may comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector.
- 10 As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.
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- 20 The evaluation device may be configured for determining the material property of the surface point having reflected the illumination feature. As used herein, the term “determining the material property” refers to assigning the material property to the object. The detector may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties
- 25 may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by the user of the detector. The material property may additionally be assigned to a material classifier such as one or more of a
- 30 material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The database may comprise a list and/or table comprising the material properties and associated material name and/or material group.
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- Specifically, the detector may be configured for detection of biological tissue, in particular human skin. As used herein, the term “biological tissue” generally refers to biological material comprising living cells. The detector may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The term “detection of biological tissue”
- 40 refers to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. mus-

cle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector may be adapted to distinguish animal tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector may be adapted to distinguish plant tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector may be adapted to distinguish different types of food such as a fruit, meat, and fish. The detector may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from human skin. The detector may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector may be adapted to distinguish human skin from cloth. The detector may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector may be adapted to distinguish organic material from inorganic material. The detector may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector may be used, in particular, for non-therapeutic and non-diagnostic applications.

For example, the material property may be information whether the object is or comprises biological tissue. Without wishing to be bound by this theory, human skin may have a reflection beam profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering beam profile. The surface may be determined as biological tissue in case the reflection beam profile fulfills at least one predetermined or predefined criterion. The at least one predetermined or predefined criterion may be at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. Specifically, the evaluation device may be adapted for comparing the beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a mobile device comprising the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the mobile device, e.g. by software, specifically by the app downloaded from an app

store or the like. The surface may be indicated as biological tissue in case the beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise
5 determining a deviation, e.g. a sum of squared point to point distances, between the beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The evaluation device may be adapted to compare the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value
10 may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the detector.

Additionally or alternatively, the evaluation device may be configured for comparing the quotient Q with at least one predetermined or predefined quotient threshold value, wherein in case the
15 quotient Q is below and/or equal the quotient threshold the surface is indicated as biological tissue. The quotient threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the detector. For example, in case A_1 and A_2 were chosen to comprise essentially edge and essentially center information, respectively, the surface reflection may contribute mostly to the
20 center signal, whereas the diffuse reflection from skin penetration may contribute mostly to the edge integral.

As a further criterion for distinguishing, for example, human skin and non-skin objects, peak intensity of the beam profile may be used in combination with a distance between the detector
25 and the object. The peak intensity of the beam profile may be distance dependent. For example, the peak intensity of the beam profile multiplied by the square of the distance between object and detector may be used as a criterion. In case of using this criterion, the output light intensity of the illumination source may be monitored and the criterion may be corrected for deviations, such as by using a corrected peak intensity of the beam profile which is the peak intensity of the
30 reflection beam profile divided by the output light intensity of the illumination source. The distance between the object and the detector may be obtained by using depth-from-photon-ratio, as outlined above.

Additionally or alternatively, the evaluation device may be configured for determining the material property m by evaluation of the respective beam profiles of the reflection features. As further
35 used herein, the term "evaluating the beam profile" may refer to applying at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. The evaluation device may be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection feature. As further
40 used herein, the term "image" refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term "image filter" refers to

at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device may be configured for determining the material property m by evaluating the material feature φ_{2m} . As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the object.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter $\Phi_{2\text{other}}$ may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term “beam profile image” refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k :

$(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-10})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-11})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-12})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term "not distinguish between material classifiers" refers to that the expectation values of the material classifiers are identical. As used herein, the term "distinguishes material classifiers" refers to that at least two expectation values

of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

10 The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- 15 ○ $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq$

20 0.075, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

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In the following, image filters are described assuming that the reflection feature comprises a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

30 For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using depth-from-photon ratio technique, wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int f(x) dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1 g_2) = [p_{g_1, g_2}]$, whereas p_{g_1, g_2} is the occurrence rate of the grey combination $(g_1, g_2) = [f(x_1, y_1), f(x_2, y_2)]$, and the relation ρ defines the distance between (x_1, y_1) and (x_2, y_2) , which is $\rho(x, y) = (x+a, y+b)$ with a and b selected from $0, 1$.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

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The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

10 The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

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$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

20

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

25 Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ε times the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The material property m may be determined by using a predetermined relationship between φ_{2m} and m and/or the longitudinal coordinate z of the reflection feature. The evaluation device may be configured for determining the material property m by evaluating the feature φ_{2m} . The evaluation device may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the object for determining the material property of the object. The predetermined relationship may be one or more of an empirical relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. For example, the material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} . Specifically, the material property m by a function $m(z, \varphi_{2m})$. The function may be predefined and/or predetermined. For example, the function may be a linear function.

As outlined above, the detector may be configured for classify the material of the elements of the area comprising the object. In contrast to structured light the detector according to the present invention may be configured for evaluating each of the reflection features of the second image such that for each reflection feature it may be possible to determine information about its material property.

The evaluation device is configured for determining at least one position and/or orientation of the object by considering the material property and the pre-determined or predefined information about shape and/or size of the object. Generally, identification of the object may be possible using only of the 2d image information or the 3D depth map. However, quality can be enhanced by fusion of 2d and 3d information. Reflecting surfaces are generally problematic for optical 3D measurements. In case of reflecting surfaces using 2d image information only may be possible. In case of objects, which are highly reflective, 3d measurements may relate to erroneous depth map. For identification of such object the 2d information may be essential.

The detector may fully or partially be integrated into at least one housing.

The detector may further comprise one or more additional elements such as one or more additional optical elements. The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g.

by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path
5 may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

10 The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate.
15 Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z -axis and in which, additionally, an x -axis and a y -axis may be provided which are perpendicular to the z -axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system,
20 a direction parallel or antiparallel to the z -axis may be regarded as a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x - and/or y -coordinate may be considered a transversal coordinate.

25 Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the
30 z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As outlined above, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole
40 object or of one or more parts thereof. For example, the detector may be configured for determining the longitudinal coordinate of the object by using the depth-from-photon-ratio technique as outlined above. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, spe-

cifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. At least one of the optical sensors may be determined from which a center signal arises. This may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured for determining at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, a center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object. The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

In a further aspect, the present invention discloses a method for object recognition, wherein a detector according to the present invention is used. The method comprises the following steps:

- a) projecting at least one illumination pattern comprising a plurality of illumination features on at least one area of comprising at least one object;
- 5 b) determining at least one first image comprising at least one two dimensional image of the area using an optical sensor, wherein the optical sensor has at least one light sensitive area;
- c) determining at least one second image comprising a plurality of reflection features comprising a plurality of reflection features generated by the area in response to il-
- 10 lumination by the illumination features by using the optical sensor;
- d) evaluating the first image by using at least one evaluation device, wherein the evaluating of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature;
- e) evaluating the second image by using the evaluation device, wherein each of the
- 15 reflections features comprises at least one beam profile, wherein the evaluation of the second image comprises determining beam profile information for each of the reflection features by analysis of their beam profiles and determining at least one three-dimensional image using the determined beam profile information;
- f) identifying the reflection features which are located inside the geometrical feature
- 20 and/or for identifying the reflection features which are located outside of the geometrical feature by using the evaluation device;
- g) determining at least one depth level from the beam profile information of the reflection features located inside and/or outside of the geometrical feature by using the evaluation device;
- 25 h) determining at least one material property of the object from the beam profile information of the re-flection features located inside and/or outside of the image region of the geometrical feature by using the evaluation device;
- i) determining at least one position and/or orientation of the object by considering the
- 30 depth level and/or the material property and pre-determined or predefined information about shape and/or size of the object by using the evaluation device.

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details,

35 options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

40 The at least one evaluation device may be configured for performing at least one computer program, such as at least one computer program configured for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an

example, one or more algorithms may be implemented which may determine the position of the object.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

With respect to further uses of the detector and devices of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

Specifically, the present invention may be applied in the field of machine control such as for robotic application. For example, the present invention may be applied for controlling at least one gripper of a robot arm. As outlined above, the detector may be configured for determining position of objects, in particular metallic objects, which can be used for control of a gripper or vacuum gripper.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: Detector for object recognition comprising

- at least one illumination source configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area comprising at least one object;
- an optical sensor having at least one light sensitive area, wherein the optical sensor is configured for determining at least one first image comprising at least one two dimensional image of the area, wherein the optical sensor is configured for determining at least one second image comprising a plurality of reflection features generated by the area in response to illumination by the illumination features;
- at least one evaluation device, wherein the evaluation device is configured for evaluating the first image and the second image, wherein each of the reflections features comprises at least one beam profile, wherein the evaluation device is configured for determining beam profile information for each of the reflection features by analysis of their beam profiles, wherein the evaluation device is configured for determining at least one three-dimensional image using the determined beam profile information, wherein the evaluation of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature, wherein the evaluation device is configured for identifying the reflection features which are located inside an image region

the geometrical feature and/or for identifying the reflection features which are located outside the image region of the geometrical feature,

wherein the evaluation device is configured for determining at least one depth level from the beam profile information of the reflection features located inside and/or outside of the image

5 region of the geometrical feature,

wherein the evaluation device is configured for determining at least one material property of the object from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature,

10 wherein the evaluation device is configured for determining at least one position and/or orientation of the object by considering the depth level and/or the material property and pre-determined or predefined information about shape and/or size of the object.

Embodiment 2: The detector according to the preceding embodiment, wherein the first image and the second image are determined at different time points.

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Embodiment 3: The detector according to any one of the preceding embodiments, wherein the geometrical feature is at least one characteristic element of the object selected from the group consisting of: a shape, a relative position of at least one edge, at least one borehole, at least one reflection point, at least one line, at least one surface, at least one circle, at least one disk, 20 the full object, a part of the object and the like.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the evaluation device comprises at least one data storage device, wherein the data storage device comprises at least one table and/or at least one lookup table of geometrical features and/or pre- 25 determined or predefined information about shape and/or size of the object.

Embodiment 5: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for identifying the geometrical feature by using at least one image processing process, wherein the image processing process comprises one or more of at 30 least one template matching algorithm; at least one Hough-transformation; applying a Canny edge filter; applying a Sobel filter; applying a combination of filters.

Embodiment 6: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for performing at least one plausibility check, wherein the identified geometrical feature are compared to at least one known geometrical features of the object. 35

Embodiment 7: The detector according to any one of the preceding embodiments, wherein the illumination source is configured for generating the at least one illumination pattern in the infra-red region.

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Embodiment 8: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one first filter element, wherein the first filter element is configured

for transmitting light in the infrared spectral range and for at least partially blocking light of other spectral ranges.

5 Embodiment 9: The detector according to the preceding embodiment, wherein the first filter element is a monochromatic bandpass filter configured for transmitting light in a small spectral range, wherein the spectral range is ± 100 nm, preferably ± 50 nm, most preferably ± 35 nm.

10 Embodiment 10: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one second filter element, wherein the second filter element is configured for transmitting light in the visual spectral range and for at least partially blocking light of other spectral ranges.

15 Embodiment 11: The detector according to any one of the preceding embodiments, wherein the illumination pattern comprises at least one periodic point pattern having a low point density, wherein the illumination pattern has ≤ 2500 points per field of view.

20 Embodiment 12: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one control unit, wherein the control unit is configured for controlling the optical sensor and/or the illumination source.

Embodiment 13: The detector according to the preceding embodiment, wherein the control unit is configured for triggering projecting of the illumination pattern and/or imaging of the second image.

25 Embodiment 14: The detector according to any one of the two preceding embodiments, wherein the control unit is configured for adapting exposure time for projection of the illumination pattern.

30 Embodiment 15: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least one further illumination source configured for illuminating the area for determining the first image, wherein control unit is configured for controlling the further illumination source, wherein the control unit is configured for triggering illumination of the area by light generated by the further illumination source and imaging of the first image.

35 Embodiment 16: The detector according to the preceding embodiment, wherein the further illumination source comprises one or more of at least one light source such as at least one light emitting diode (LED) or at least one VCSEL array, wherein the further illumination source comprises at least one optical element such as at least one diffusor or at least one lens.

40 Embodiment 17: The detector according to any one of the two preceding embodiments, wherein the control unit is configured for adapting exposure time for projection of the illumination pattern and illumination by light generated by the further illumination source.

Embodiment 18: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining the beam profile information for each of the re-reflection features by using depth-from-photon-ratio technique.

- 5 Embodiment 19: The detector according to any one of the preceding embodiments, wherein the optical sensor comprises at least one CMOS sensor.

Embodiment 20: Method for object recognition, wherein at least one detector according to the preceding embodiments is used, wherein the method comprises the following steps:

- 10 a) projecting at least one illumination pattern comprising a plurality of illumination features on at least one area of comprising at least one object;
- b) determining at least one first image comprising at least one two dimensional image of the area using an optical sensor, wherein the optical sensor has at least one light sensitive area;
- 15 c) determining at least one second image comprising a plurality of reflection features comprising a plurality of reflection features generated by the area in response to illumination by the illumination features by using the optical sensor;
- d) evaluating the first image by using at least one evaluation device, wherein the evaluating of the first image comprises identifying at least one pre-defined or pre-
- 20 determined geometrical feature;
- e) evaluating the second image by using the evaluation device, wherein each of the reflections features comprises at least one beam profile, wherein the evaluation of the second image comprises determining beam profile information for each of the reflection features by analysis of their beam profiles and determining at least one three-
- 25 dimensional image using the determined beam profile information;
- f) identifying the reflection features which are located inside the geometrical feature and/or for identifying the reflection features which are located outside of the geometrical feature by using the evaluation device;
- g) determining at least one depth level from the beam profile information of the reflection
- 30 features located inside and/or outside of the geometrical feature by using the evaluation device;
- h) determining at least one material property of the object from the beam profile information of the re-reflection features located inside and/or outside of the image region of the geometrical feature by using the evaluation device;
- 35 i) determining at least one position and/or orientation of the object by considering the depth level and/or the material property and pre-determined or predefined information about shape and/or size of the object by using the evaluation device.

- 40 Embodiment 21: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping

application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

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Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

Figure 1 shows an embodiment of a detector according to the present invention.

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Detailed description of the embodiments:

Fig. 1 shows in a highly schematic fashion an embodiment of a detector for object recognition. The object 112 may generally refer to an arbitrary physical body whose orientation and/or position is to be determined. The object 112 may be at least one article. For example, the object may be at least one object selected from the group consisting of: a box, a bottle, a plate, a sheet of paper, a bag, a screw, a washer, a machined metal piece, a rubber seal, plastic pieces, wrapping, packing material.

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The detector 110 comprises at least one illumination source 114 configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area 116 comprising at least one object 112. The object 112 may be located within a scene and/or may have a surrounding environment. Specifically, the object 112 may be located in the at least one area 116. The area 116 may be at least one surface and/or region. The area 116 may comprise additional elements such as the surrounding environment.

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The illumination source 114 may be adapted to directly or indirectly illuminating the object 112, wherein the illumination pattern is reflected or scattered by the object 112 and, thereby, is at least partially directed towards the detector 110. The illumination source 114 may be adapted to illuminate the object 112, for example, by directing a light beam towards the object 112, which reflects the light beam. The illumination source 114 may be configured for generating an illuminating light beam for illuminating the object 112.

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The illumination source 114 may comprise at least one light source. The illumination source 114 may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 114 may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used.

The illumination source 114 may be configured for generating the at least one illumination pattern in the infrared region. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination source 114 may be or may comprise at least one multiple beam light source. For example, the illumination source 114 may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination source 114 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 114 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source 114 may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source 114 may be integrated into a housing of the detector 110.

The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one

point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one
5 quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source 114 may be adapted to generate
10 and/or to project a cloud of points. The illumination source 114 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source 114 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source 114.

15 Specifically, the illumination source 114 comprises at least one laser source and/or at least one laser diode which is designated for generating laser radiation. The illumination source 114 may comprise the at least one diffractive optical element (DOE). The detector 110 may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project
20 at least one point pattern.

For example, the projected illumination pattern may be a periodic point pattern. The projected illumination pattern may have a low point density. For example, the illumination pattern may
25 comprise at least one periodic point pattern having a low point density, wherein the illumination pattern has ≤ 2500 points per field of view. In comparison with structured light having typically a point density of 10k - 30k in a field of view of $55 \times 38^\circ$ the illumination pattern according to the present invention may be less dense. This may allow more power per point such that the proposed technique is less dependent on ambient light compared to structured light.

30 The detector 110 may comprise at least one further illumination source 118. The further illumination source 118 may comprise one or more of at least one further light source such as at least one light emitting diode (LED) or at least one vertical-cavity surface-emitting laser (VCSEL) array. The further illumination source 118 may comprise at least one optical element such as at least one diffusor or at least one lens. The further illumination source 118 may be configured for
35 providing additional illumination for imaging of the first image. For example, the further illumination source 118 may be used in situations in which it is not possible or difficult for recording the reflection pattern, e.g. in case of highly reflective metallic surfaces, in order to ensure a good illumination and, thus, contrasts for two-dimensional images such that a two-dimensional image recognition is possible.

40 The detector 110 comprises an optical sensor 120 having at least one light sensitive area 122. The optical sensor 120 is configured for determining at least one first image comprising at least one two dimensional image of the area 116. The optical sensor 120 is configured for determin-

ing at least one second image comprising a plurality of reflection features generated by the area 116 in response to illumination by the illumination features. The detector 110 may comprise a single camera comprising the optical sensor 120. The detector 110 may comprise a plurality of cameras each comprising an optical sensor 120 or a plurality of optical sensors 120.

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The optical sensor 120 specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor 120 may be sensitive in the infrared spectral range. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor 120 may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor 120 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor 120 may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor 120 may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge: Au photodiode, a Ge: Hg photodiode, a Ge: Cu photodiode, a Ge: Zn photodiode, a Si: Ga photodiode, a Si: As photodiode. Additionally or alternatively, the optical sensor 120 may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor 120 may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor 120 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor 120, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor 120 may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransis-

tor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor 120 may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor 120 may be part of or constitute a pixelated optical device. For example, the optical sensor 120 may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor 120 may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may be formed as a unitary, single device or as a combination of several devices. The matrix specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

The pixels of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other embodiments, however, are feasible. The optical sensor 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensor 120 may be part of or constitute a pixelated optical device. As an example, the optical sensor 120 may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

The optical sensor 120 is configured for determining at least one first image comprising at least one two dimensional image of the area 116. The image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. The optical sensor 120 is configured for determining at least one second image comprising a plurality of reflection features generated by the area 116 in response to illumination by the illumination features.

The first image and the second image may be determined, in particular recorded, at different time points. Recording of the first image and the second time limit may be performed with a temporal shift. Specifically, a single camera comprising the optical sensor 120 may record with a temporal shift a two-dimensional image and an image of a projected pattern. Recording the first and the second image at different time points may ensure that an evaluation device 124 can distinguish between the first and the second image and can apply the appropriate evaluation routine. Moreover, it is possible to adapt the illumination situation for the first image if nec-

essary and in particular independent from the illumination for the second image. The detector 110 may comprise at least one control unit 126. The control unit 126 may be designed as hardware component of the detector 110. In particular, the control unit 126 may comprise at least one microcontroller. The control unit 126 may be configured for controlling the optical sensor 120 and/or the illumination source 114. The control unit 126 may be configured for triggering projecting of the illumination pattern and/or imaging of the second image. Specifically, the control unit 126 may be configured for controlling the optical sensor 120, in particular frame rate and/or illumination time, via trigger signals. The control unit 126 may be configured for adapting and/or adjusting the illumination time from frame to frame. This may allow adapting and/or adjusting illumination time for the first image, e.g. in order to have contrasts at the edges, and at the same time adapting and/or adjusting illumination time for the second image to maintain contrast of the reflection features. Additionally, the control unit 126 may, at the same time and independently, control the elements of the illumination source 114 and/or the further illumination source 118.

Specifically, the control unit 126 may be configured for adapting exposure time for projection of the illumination pattern. The second image may be recorded with different illumination times. Dark regions of the area 116 may require more light in comparison to lighter regions, which may result to run into saturation for the lighter regions. Therefore, the detector 110 may be configured for recording a plurality of images of the reflection pattern, wherein the images may be recorded with different illumination times. The detector 110 may be configured for generating and/or composing the second image from said images. The evaluation device 124 may be configured for performing at least one algorithm on said images which were recorded with different illumination times.

The control unit 126 may be configured for controlling the further illumination source 118. The control unit 126 may be configured for triggering illumination of the area by light generated by the further illumination source 118 and imaging of the first image. The control unit 126 may be configured for adapting exposure time for projection of the illumination pattern and illumination by light generated by the further illumination source 118.

The detector 110 may comprise at least one first filter element 128. The first filter element 128 may be configured for transmitting light in the infrared spectral range and for at least partially blocking light of other spectral ranges. The first filter element 128 may be a monochromatic bandpass filter configured for transmitting light in a small spectral range. For example, the spectral range or bandwidth may be ± 100 nm, preferably ± 50 nm, most preferably ± 35 nm or even less. For example, the first filter element 128 may be configured for transmitting light having a central wavelength of 808 nm, 830 nm, 850 nm, 905 nm or 940 nm. For example, the first filter element 128 may be configured for transmitting light having a central wavelength of 850 nm with a bandwidth of 70 nm or less. The first filter element 128 may have a minimal angle dependency such that the spectral range can be small. This may result in a low dependency on ambient light, wherein at the same time an enhanced vignetting effect can be prevented. For example,

the detector 110 may comprise the single camera having the optical sensor 120 and, in addition, the first filter element 128. The first filter element 128 may ensure that even in presence of ambient light recording of the reflection pattern is possible and at the same time to maintain laser output power low such that eye safety operation in laser class 1 is ensured.

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Additionally or alternatively, the detector 110 may comprise at least one second filter element, not shown here. The second filter element may be a band-pass filter. For example, the first filter element may be a long pass filter configured for blocking visual light and for let pass light above a wavelength of 780 nm. The band pass filter may be positioned between the light-sensitive area 122, for example of a CMOS chip, and transfer device 129.

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The spectrum of the illumination source 114 and/or of the further illumination source 118 may be selected depending on the used filter elements. For example, in case of the first filter element 128 having a central wavelength of 850 nm, the illumination source 114 may comprise at least one light source generating a wavelength of 850 nm such as at least one infrared (IR)-LED.

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The detector 110 may comprise at least one transfer device 129 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 129 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

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The evaluation device 124 is configured for evaluating the first image and the second image.

The evaluation of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature. The geometrical feature may be at least one characteristic element of the object 112 selected from the group consisting of: a shape, a relative position of at least one edge, at least one borehole, at least one reflection point, at least one line, at least one surface, at least one circle, at least one disk, the full object, a part of the object and the like. The evaluation device 124 may comprise at least one data storage device 130. The data storage device 130 may comprise at least one table and/or at least one lookup table of geometrical features and/or pre-determined or predefined information about shape and/or size of the object 112. Additionally or alternatively, the detector 110 may comprise at least one user interface 132 via which a user can enter the at least one geometrical feature.

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The evaluation device 124 may be configured for evaluating in a first step the second image. The evaluation of the second image may provide, as will be outlined in more detail below, 3D information of the reflection features. The evaluation device 124 may be configured for estimating a location of the geometrical feature in the first image by considering the 3D information of

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the reflection features. This may reduce effort of search for geometrical feature in the first image significantly.

5 The evaluation device 124 may be configured for identifying the geometrical feature by using at least one image processing process. The image processing process may comprise one or more of at least one template matching algorithm; at least one Hough-transformation; applying a Canny edge filter; applying a Sobel filter; applying a combination of filters. The evaluation device may be configured for performing at least one plausibility check. The plausibility check may comprise comparing the identified geometrical feature compared to at least one known geomet-
10 rical feature of the object. For example, a user may enter a known geometrical feature via the user interface for the plausibility check.

The evaluation device 124 is configured for evaluating of the second image. The evaluation of the second image may comprise generating a three-dimensional image.

15 Each of the reflections features comprises at least one beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device 124 is configured for determining beam profile information for each of the reflection features by analy-
20 sis of their beam profiles.

The evaluation device 124 may be configured for determining the beam profile of each of the reflection features. The determining the beam profile may comprise identifying at least one re-
25 flection feature provided by the optical sensor 120 and/or selecting at least one reflection feature provided by the optical sensor 120 and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for deter-
mining the intensity distribution, such as a three-dimensional intensity distribution or a two-
dimensional intensity distribution, such as along an axis or line through the matrix. As an exam-
ple, a center of illumination by the light beam may be determined, such as by determining the at
30 least one pixel having the highest illumination, and a cross-sectional axis may be chosen
through the center of illumination. The intensity distribution may an intensity distribution as a
function of a coordinate along this cross-sectional axis through the center of illumination. Other
evaluation algorithms are feasible.

35 The evaluation device 124 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor
40 signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a de-
composition into hue; saturation; and brightness channels; a frequency decomposition; a singu-

- lar value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within the image generated by the optical sensor 120.
- For example, the illumination source 114 may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device 124 may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object 112. For example, the evaluation device 124 may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.
- The evaluation device 124 may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device 124 may be adapted to remove influences from background light from the reflection beam profile, for example, by an imaging without further illumination.
- The analysis of the beam profile may comprise evaluating of the beam profile. The analysis of the beam profile may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The evaluation device 124 may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The evaluation device 124 may filter the beam profile by removing high spatial frequencies such as by spatial frequency analy-

sis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The evaluation device 124 may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The evaluation device 5 124 may be configured for removing influences from background light from the reflection beam profile, for example, by an imaging without illumination.

The reflection feature may cover or may extend over at least one pixel of the image. For example, the reflection feature may cover or may extend over plurality of pixels. The evaluation device 10 124 may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The evaluation device 124 may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum_j r_{pixel}},$$

wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum_j I_{total}$ with j 15 being the number of pixels j connected to and/or belonging to the reflection feature and I_{total} being the total intensity.

The evaluation device 124 is configured for determining the beam profile information for each of the reflection features by analysis of their beam profiles. The beam profile information may 20 comprise information about the longitudinal coordinate of the surface point or region having reflected the illumination feature. Additionally, the beam profile information may comprise information about a material property of said surface point or region having reflected the illumination feature.

25 The beam profile information may be the longitudinal coordinate of the surface point or region having reflected the illumination feature. The evaluation device 124 may be configured for determining the beam profile information for each of the reflection features by using depth-from-photon-ratio technique. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content 30 of which is included by reference.

The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The 35 evaluation device 124 may be configured for integrating the first area and the second area. The evaluation device 124 may be configured to derive a combined signal, in particular a quotient Q , by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The evaluation device 124 may config- 40 ured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the evaluation device 124 may be configured for determining a plurality of areas such as two, three, four, five, or

up to ten areas. The evaluation device 124 may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The evaluation device 124 may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The evaluation device 124 may be configured for comparing at least two of the determined integrals. Specifically, the evaluation device 124 may be configured for determining at least one first area and at least one second area of the reflection beam profile. The first area of the beam profile and the second area of the reflection beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the evaluation device 124 may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the evaluation device may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the detector 110 may comprise at least two optical sensors 120, wherein the light-sensitive areas 122 of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the reflection beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the reflection beam profile of the reflection feature. The evaluation device 124 may be adapted to integrate the first area and the second area. The evaluation device 124 may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 124 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the fall-

ing edges of the cross section. For example, the whole area of the cross section may be determined as first region.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

The evaluation device 124 may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The evaluation device 124 may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the beam profile, respectively, and E(x,y) denotes the beam profile.

The evaluation device 124 may be configured for determining at least one three-dimensional image and/or 3D-data using the determined beam profile information. The image or images recorded by the camera comprising the reflection pattern may be a two-dimensional image or two-dimensional images. As outlined above, the evaluation device 124 may be configured for determining for each of the reflection features a longitudinal coordinate. The evaluation device 124 may be configured for generating 3D-data and/or the three-dimensional image by merging the two-dimensional image or images of the reflection pattern with the determined longitudinal coordinate of the respective reflection feature.

The evaluation device 124 may be configured for merging and/or fusing the determined 3D-data and/or the three-dimensional image and the information determined from the first image, i.e. the at least one geometrical feature and its location, in order to identify the object in a scene, in particular in the area.

The evaluation device 124 is configured for identifying the reflection features which are located inside an image region the geometrical feature and/or for identifying the reflection features which are located outside the image region of the geometrical feature. The evaluation device 124 may be configured for determining an image position of the identified geometrical feature in the first image. The image position may be defined by pixel coordinates, e.g. x and y coordinates, of pixels of the geometrical feature. The evaluation device 124 may be configured for determining and/or assigning and/or selecting at least one border and/or limit of the geometrical feature in the first image. The border and/or limit may be given by at least one edge or at least one contours of the geometrical feature. The evaluation device 124 may be configured for de-

termining the pixels of the first image inside the border and/or limit and their image position in the first image. The evaluation device 124 may be configured for determining at least one image region of the second image corresponding to the geometrical feature in the first image by identifying the pixels of the second image corresponding to the pixels of the first image inside the border and/or limit of the geometrical feature.

The evaluation device 124 is configured for determining the at least one depth level from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature. The area comprising the object may comprise a plurality of elements at different depth levels. The depth level may be a bin or step of a depth map of the pixels of the second image. As outlined above, the evaluation device 124 may be configured for determining for each of the reflection features a longitudinal coordinate from their beam profiles. The evaluation device 124 may be configured for determining the depth levels from the longitudinal coordinates of the reflection features located inside and/or outside of the image region of the geometrical feature. Metallic objects often cannot be identified in the second image correctly. However, levels can be correctly identified, which may be defined by the ground or cover of said metallic objects since these often are made of cardboard. Figure 1 shows an example, wherein the area 116 comprises a surface 134 on which the object 112 is located. The evaluation device 124 may be configured for determining the depth level on which the object 112 is located from the depth level of the reflection features located inside and/or outside of the image region of the geometrical feature.

The evaluation device 124 is configured for determining the position and/or the orientation of the object by considering the depth level and pre-determined or predefined information about shape and/or size of the object 112. For example, the information about shape and/or size may be entered by a user via the user interface 132. For example, the information about shape and size may be measured in an additional measurement. As outlined above, the evaluation device 124 is configured for determining the depth level on which the object 112 is located. If in addition, the shape and/or size of the object 112 are known the evaluation device 124 can determine the position and orientation of the object.

For example, in case a task may be to detect and measure with the detector 110 at least one object 112 such as bottles in a box. The detector 110, in particular the optical sensor 120, may be installed on a robot arm 142 such that the detector 110 can move to different positions with respect to the objects in the box. The task may be that the robot should move to the objects 112 and take it out of the box. Additionally, the user knows the object 112, in this example the bottles, in detail, such that the size, form and shape may be also known and may be programmed into the evaluation device 124.

The optical sensor 120 may determine the two dimensional image and a resulting 3d depth map. The depth map may estimate the position of the detector 110 and the objects 112. The depth map can also be distorted by different effects like to shiny objects, e.g. metal, and/or the 3d depth map may be to sparse. The present invention propose to get additional information by

a 2d image that corresponds to the 3d depth map. In the example with the bottles, the task is to detect bottles in a box. In addition, it may be known that the bottles are rotationally symmetric. Certain features of the bottle can help for object detection, e.g. round bottle caps. This may lead to search for circles or ellipsoids in the 2d image for the object detection with image processing algorithms. A rough estimation of the size of the ellipsoids may be computed by the 3d depth information. For a detailed object detection, the detected ellipsoids in the 2d image and the known relation of the projection between detector 110 and the real world can be used to determine the size and position of the circles in the real world. A relationship between the projection between detector 110 and the real world can be used to determine size, position and orientation by using at least one system of equations.

The evaluation device 124 is configured for determining at least one material property of the object from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature. The beam profile information may comprise information about a material property of the surface point or region having reflected the illumination feature. The object 112 may comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector 110. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

The evaluation device 124 may be configured for determining the material property of the surface point having reflected the illumination feature. The detector 110 may comprise at least one database 136 comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector 110 according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by the user of the detector 110. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The database 136 may comprise a list and/or table comprising the material properties and associated material name and/or material group.

The evaluation device 124 may be configured for determining the material property m by evaluation of the respective beam profiles of the reflection features. The evaluation device 124 may

be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection feature. The image may be a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. The image filter may be at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation. The evaluation device 124 may be configured for determining the material property m by evaluating the material feature φ_{2m} . The material feature may be or may comprise at least one information about the at least one material property of the object 112.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

As outlined above, the detector 110 may be configured for classify the material of the elements of the area 116 comprising the object 112. In contrast to structured light the detector 110 according to the present invention may be configured for evaluating each of the reflection features of the second image such that for each reflection feature it may be possible to determine information about its material property.

The evaluation device 124 is configured for determining at least one position and/or orientation of the object by considering the material property and the pre-determined or predefined information about shape and/or size of the object. Generally, identification of the object 112 may be

possible using only of the 2d image information or the 3D depth map. However, quality can be enhanced by fusion of 2d and 3d information. Reflecting surfaces are generally problematic for optical 3D measurements. In case of reflecting surfaces using 2d image information only may be possible. In case of objects, which are highly reflective, 3d measurements may relate to erroneous depth map. For identification of such object the 2d information may be essential.

The detector 110 may fully or partially be integrated into at least one housing 138.

With regard to the coordinate system for determining the position of the object 112, which may be a coordinate system of the detector 110, the detector may constitute a coordinate system 140 in which an optical axis of the detector 110 forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector 110 and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

The present invention may be applied in the field of machine control such as for robotic application. For example, as shown in figure 1, the present invention may be applied for controlling at least one gripper of a robot arm 142. As outlined above, the detector 110 may be configured for determining position of objects, in particular metallic objects, which can be used for control of the robot arm 142. For example, the object 112 may be at least one article. For example, the object 112 may be at least one object selected from the group consisting of: a box, a bottle, a plate, a sheet of paper, a bag, a screw, a washer, a machined metal piece, a rubber seal, plastic pieces, wrapping, packing material

List of reference numbers

110	detector
112	object
114	illumination source
116	area
118	further illumination source
120	optical sensor
122	light-sensitive area
124	evaluation device
126	control unit
128	first filter element
129	transfer device
130	data storage device
132	user interface
134	surface
136	database
138	detector system
140	coordinate system
142	robot arm

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Chen Guo-Hua et al. “Transparent object detection and location based on RGB-D camera”, JOURNAL OF PHYSICS: CONFERENCE SERIES, vol. 1183, 1 March 2019, page 012011, XP055707266, GB ISSN: 1742-6588, DOI: 10.1088/1742-6596/1183/1/012011

Claims

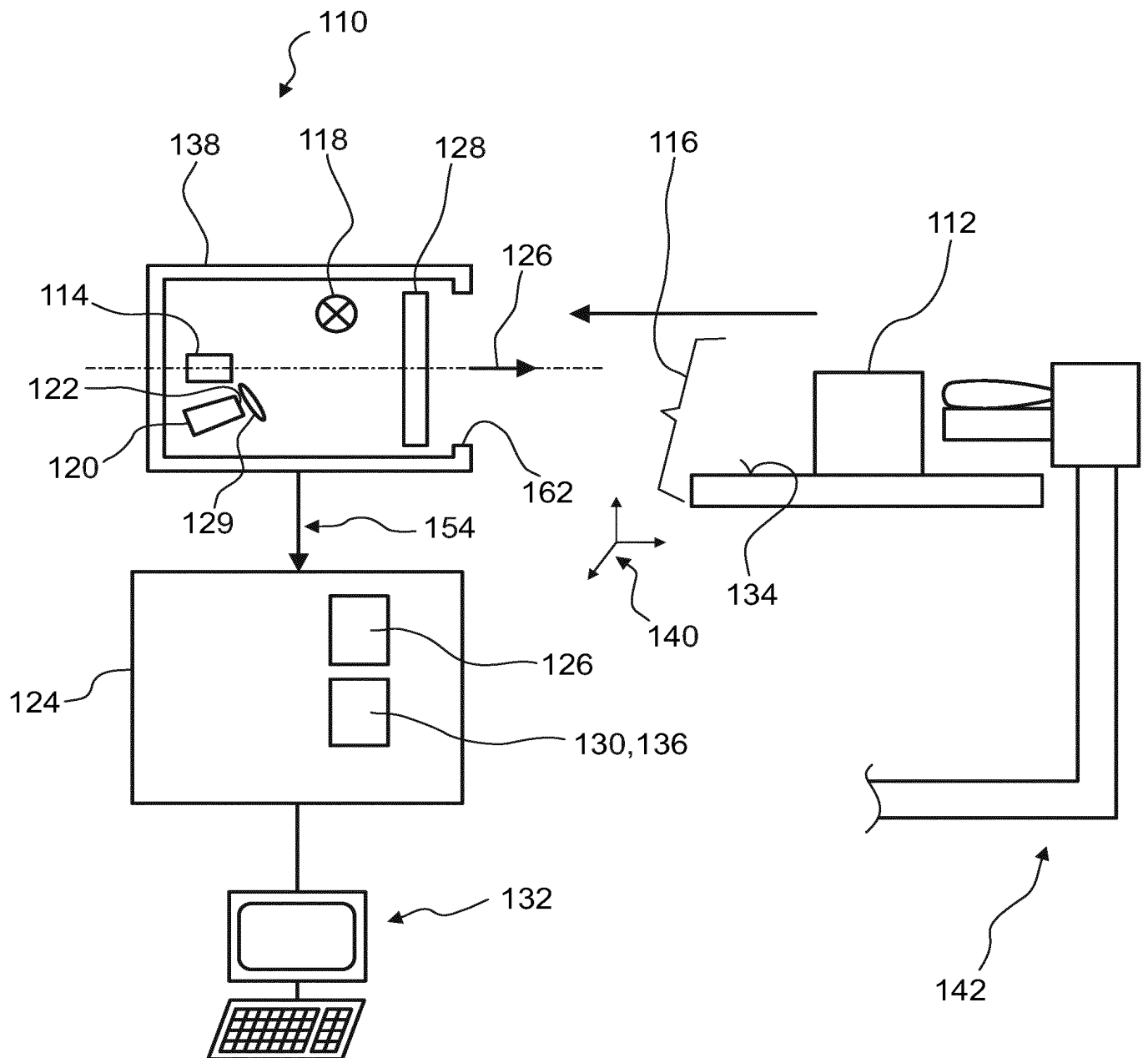
1. Detector (110) for object recognition comprising
 - at least one illumination source (114) configured for projecting at least one illumination pattern comprising a plurality of illumination features on at least one area (116) comprising at least one object (110);
 - an optical sensor (120) having at least one light sensitive area (122), wherein the optical sensor (120) is configured for determining at least one first image comprising at least one two dimensional image of the area, wherein the optical sensor (120) is configured for determining at least one second image comprising a plurality of reflection features generated by the area (116) in response to illumination by the illumination features;
 - at least one evaluation device (124), wherein the evaluation device (124) is configured for evaluating the first image and the second image, wherein each of the reflection features comprises at least one beam profile, wherein the evaluation device (124) is configured for determining beam profile information for each of the reflection features by analysis of their beam profiles, wherein the beam profile information is information about an intensity distribution of a light spot on the light sensitive area (122) of the optical sensor (120), wherein the evaluation device (124) is configured for determining at least one three-dimensional image using the determined beam profile information, wherein the evaluation of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature, wherein the evaluation device (124) is configured for identifying the reflection features which are located inside an image region the geometrical feature and/or for identifying the reflection features which are located outside the image region of the geometrical feature, wherein the evaluation device (124) is configured for determining at least one depth level from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature, wherein the evaluation device (124) is configured for determining at least one material property of the object from the beam profile information of the reflection features located inside and/or outside of the image region of the geometrical feature, wherein the evaluation device (124) is configured for determining at least one position and/or orientation of the object by considering the depth level and/or the material property and pre-determined or predefined information about shape and/or size of the object.
2. The detector (110) according to the preceding claim, wherein the first image and the second image are determined at different time points.
3. The detector (110) according to any one of the preceding claims, wherein the geometrical feature is at least one characteristic element of the object (112) selected from the

group consisting of: a shape, a relative position of at least one edge, at least one borehole, at least one reflection point, at least one line, at least one surface, at least one circle, at least one disk, the full object (112), a part of the object (112).

- 5 4. The detector (110) according to any one of the preceding claims, wherein the evaluation device (124) comprises at least one data storage device (130), wherein the data storage device (130) comprises at least one table and/or at least one lookup table of geometrical features and/or pre-determined or predefined information about shape and/or size of the object (112).
- 10
5. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one first filter element (128), wherein the first filter element (128) is configured for transmitting light in the infrared spectral range and for at least partially blocking light of other spectral ranges.
- 15
6. The detector (110) according to any one of the preceding claims, wherein the illumination pattern comprises at least one periodic point pattern having a low point density, wherein the illumination pattern has ≤ 2500 points per field of view.
- 20
7. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one control unit (126), wherein the control unit (126) is configured for controlling the optical sensor (120) and/or the illumination source (114), wherein the control unit (126) is configured for triggering projecting of the illumination pattern and/or imaging of the second image.
- 25
8. The detector (110) according to the preceding claim, wherein the control unit is configured for adapting exposure time for projection of the illumination pattern.
- 30
9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (124) is configured for determining the beam profile information for each of the reflection features by using depth-from-photon-ratio technique.
- 35
10. The detector (110) according to any one of the preceding claims, wherein the optical sensor (120) comprises at least one CMOS sensor.
- 40
11. Method for object recognition, wherein at least one detector (110) according to the preceding claims is used, wherein the method comprises the following steps:
- a) projecting at least one illumination pattern comprising a plurality of illumination features on at least one area (116) of comprising at least one object (112);
- b) determining at least one first image comprising at least one two dimensional image of the area (116) using an optical sensor (116), wherein the optical sensor (120) has at least one light sensitive area (122);

- 5 c) determining at least one second image comprising a plurality of reflection features comprising a plurality of reflection features generated by the area (116) in response to illumination by the illumination features by using the optical sensor (120);
- 10 d) evaluating the first image by using at least one evaluation device (124), wherein the evaluating of the first image comprises identifying at least one pre-defined or pre-determined geometrical feature;
- 15 e) evaluating the second image by using the evaluation device (124), wherein each of the reflection features comprises at least one beam profile, wherein the evaluation of the second image comprises determining beam profile information for each of the reflection features by analysis of their beam profiles and determining at least one three-dimensional image using the determined beam profile information;
- 20 f) identifying the reflection features which are located inside the geometrical feature and/or for identifying the reflection features which are located outside of the geometrical feature by using the evaluation device (124);
- 25 g) determining at least one depth level from the beam profile information of the reflection features located inside and/or outside of the geometrical feature by using the evaluation device (124);
- h) determining at least one material property of the object (112) from the beam profile information of the re-reflection features located inside and/or outside of the image region of the geometrical feature by using the evaluation device (124);
- i) determining at least one position and/or orientation of the object (112) by considering the depth level and/or the material property and pre-determined or predefined information about shape and/or size of the object (112) by using the evaluation device (124).
- 30 12. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application;
- 35 a quality control application; a manufacturing application.

FIG.1



INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2021/052075

A. CLASSIFICATION OF SUBJECT MATTER

INV. G06K9/20 G06K9/46 G06K9/00 G06T7/521 G06T7/60
G06T7/70 G01N21/00 G01S17/46

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G06K G06T G01N G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	CHEN GUO-HUA ET AL: "Transparent object detection and location based on RGB-D camera", JOURNAL OF PHYSICS: CONFERENCE SERIES, vol. 1183, 1 March 2019 (2019-03-01), page 012011, XP055707266, GB ISSN: 1742-6588, DOI: 10.1088/1742-6596/1183/1/012011 abstract figures 5, 9 section 3 section 4 ----- -/-	1-12



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

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Authorized officer

Martinez, Francis

INTERNATIONAL SEARCH REPORT

International application No

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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	KIM JUNGJUN ET AL: "RGBD Camera Based Material Recognition via Surface Roughness Estimation", 2018 IEEE WINTER CONFERENCE ON APPLICATIONS OF COMPUTER VISION (WACV), IEEE, 12 March 2018 (2018-03-12), pages 1963-1971, XP033337822, DOI: 10.1109/WACV.2018.00217 section 2.1 -----	1-12
A	SEUNGKYU LEE: "Joint Detection of Translucency and Depth Using a Time-of-Flight Camera", JOURNAL OF IMAGING SCIENCE AND TECHNOLOGY., vol. 61, no. 3, 1 May 2017 (2017-05-01), pages 305081-305086, XP055707508, US ISSN: 1062-3701, DOI: 10.2352/J.ImagingSci.Technol.2017.61.3.030 508 the whole document -----	1-12



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G01S 5/16 (2006.01) *G01S 7/48* (2006.01)

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 17195398.7 09 October 2017 (09.10.2017) EP

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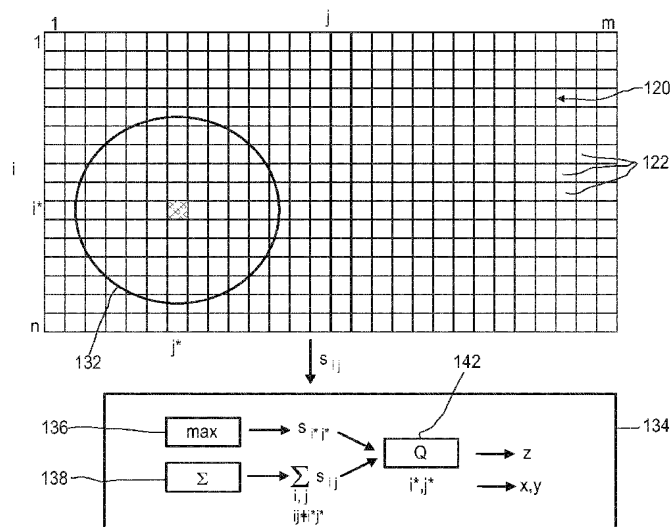
English

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 16199397.7 17 November 2016 (17.11.2016) EP
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(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.3



(57) **Abstract:** A detector (110) for determining a position of at least one object (112) is disclosed. The detector (110) comprises: - at least one sensor element (118) having a matrix (120) of optical sensors (122), the optical sensors (122) each having a light-sensitive area (124), wherein each optical sensor (122) is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area (124) by at least one light beam (116) propagating from the object (112) to the detector (110); - at least one evaluation device (134) configured for evaluating the sensor signals, by a) determining at least one optical sensor (122) having the highest sensor signal and forming at least one center signal; b) evaluating the sensor signals of the optical sensors (122) of the matrix (120) and forming at least one sum signal; c) determining at least one combined signal by combining the center signal and the sum signal; and d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.

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Declarations under Rule 4.17:

- *as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))*
- *of inventorship (Rule 4.17(iv))*

Published:

- *with international search report (Art. 21(3))*

Detector for optically detecting at least one object

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, systems, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

20 Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultraviolet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and for reasons of large-area processing, sensors comprising at least one organic sensor material are being used, as described for example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO 2009/013282 A1.

A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for opti-

cally detecting objects are triangulation systems, by means of which distance measurements can likewise be carried out. Kurt Konolige et al., A Low-Cost Laser Distance Sensor, 2008 IEEE International Conference on Robotics and Automation, Pasadena, CA, USA, May 19-23, 2008, discuss competing technologies using triangulation for planar laser distance sensor (LDS).

5 Structured line devices use a light stripe laser and offset camera to determine range to a set of points. Because the laser energy is spread over a line, it is difficult to achieve accurate range, especially in the presence of ambient light, or with darker objects. Point scan devices for 3D scanning of small objects typically use a scanning mirror to direct a point laser beam and redirect the laser return to an optical receiver. Such devices cannot be miniaturized, and their cost and mechanical fragility will remain high. Centroid point modules typically use position-sensitive devices (PSD). These devices measure the centroid of all light impinging on their surface. Although modulation techniques can be used to offset some of the effects of ambient light, PSDs do not perform well unless the laser spot has a very strong reflection, limiting their use to ranges of a meter or less. Pixel-based point modules search the pixel with maximum signal intensity to determine the position of the light spot on the sensor. Typically CMOS line arrays are used for detection. Konolige et al. introduce a low-cost version of a Pixel-based point module.

US 5 235 377 A describes a measuring apparatus for a camera, wherein light is projected to an object from a projection section. Light reflected on the object is received at a first light receiving section and a second light receiving section arranged to be adjacent to each other. In the first light receiving section, a first and second signals relating to a light receiving position and a quantity of received light are output, and in the second light receiving section, a third signal relating to a quantity of light of the reflected light is output. In a discriminating and calculating section, if it is discriminated that the reflected light is positioned at a place, which is nearer than a predetermined distance, a measuring distance can be calculated from an output ratio of the first signal to the second signal by a measuring calculation circuit. On the other hand, if it is discriminated that the reflected light is positioned at a place, which is farther than the predetermined distance, a measuring distance can be calculated from the output ratio of the sum of the first and second signals to the third signal by the measuring calculation circuit.

US 5 512 997 A describes a distance measuring device having an infrared light-emitting diode by which an infrared light is emitted towards a subject, and first and second linear line sensors on which an image corresponding to light reflected from the subject is formed. The first and second linear line sensors are disposed adjacently and in parallel to each other, and offset by an amount corresponding to half of the predetermined A/D converting step. Analog distance data outputted from the linear line sensors are converted to digital distance data using the predetermined A/D converting step, and then, the digital distance data are added together to obtain a subject distance with a high accuracy.

DE 42 11 875 A1 describes an optical rangefinder with electronic correction for spot eccentricity. The optical rangefinder has an evaluation circuit, which determines if an output distance measurement value is valid or erroneous due to e.g. inclination of object surface. The distance signal is determined by evaluation a position a light spot on a first detector. A second control

signal, which characterizes inclination of the object surface, is determined by evaluating eccentricity of a light spot on a second detector, arranged in a different plane.

WO 2016/051323 A1 describes a detector and a method for optically determining a position of at least one object. The detector comprises at least one optical sensor for determining a position of at least one light beam and at least one evaluation device for generating at least one item of information on a transversal position of the object and at least one item of information on a longitudinal position of the object. The sensor has at least a first electrode and a second electrode. At least one photovoltaic material is embedded in between the first electrode and the second electrode. The first electrode or the second electrode is a split electrode having at least three partial electrodes. The detector and the method can determine three-dimensional coordinates of an object in a fast and efficient way.

In WO 2012/110924 A1, the content of which is herewith included by reference, a detector for optically detecting at least one object is proposed. The detector comprises at least one optical sensor. The optical sensor has at least one sensor region. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area. The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

WO 2014/097181 A1, the full content of which is herewith included by reference, discloses a method and a detector for determining a position of at least one object, by using at least one transversal optical sensor and at least one optical sensor. Specifically, the use of sensor stacks is disclosed, in order to determine a longitudinal position of the object with a high degree of accuracy and without ambiguity.

WO 2015/024871 A1 and US 2015/286340 A1, the full content of which is herewith included by reference, discloses an optical detector, comprising:

- at least one spatial light modulator being adapted to modify at least one property of a light beam in a spatially resolved fashion, having a matrix of pixels, each pixel being controllable to individually modify the at least one optical property of a portion of the light beam passing the pixel;
- at least one optical sensor adapted to detect the light beam after passing the matrix of pixels of the spatial light modulator and to generate at least one sensor signal;
- at least one modulator device adapted for periodically controlling at least two of the pixels with different modulation frequencies; and
- at least one evaluation device adapted for performing a frequency analysis in order to determine signal components of the sensor signal for the modulation frequencies.

US 4,767,211 discloses an apparatus for and a method of measuring a boundary surface of a sample in which a ratio of the light quantity of a part of reflected light from a sample which travels in the vicinity of the optical axis of the reflected light, to the light quantity of another part of the reflected light which is directed to a position deviating from the optical axis by a predetermined distance is used to accurately measure a boundary surface of a sample. Since the accuracy of measurement is increased by using the above ratio, light capable of passing through the sample can be used as incident light. Thus, a deep hole in the surface of the sample and a void such as an air bubble in a living being sample, which cannot be measured by the prior art, can be measured very accurately.

WO 2014/198629 A1, the full content of which is herewith included by reference, discloses a detector for determining a position of at least one object, comprising:

- at least one optical sensor, the optical sensor being adapted to detect a light beam propagating from the object towards the detector, the optical sensor having at least one matrix of pixels; and
- at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

Further, generally, for various other detector concepts, reference may be made to WO 2014/198626 A1, WO 2014/198629 A1 and WO 2014/198625 A1, the full content of which is herewith included by reference. Further, referring to potential materials and optical sensors which may also be employed in the context of the present invention, reference may be made to European patent applications EP 15 153 215.7, filed on January 30, 2015, EP 15 157 363.1, filed on March 3, 2015, EP 15 164 653.6, filed on April 22, 2015, EP 15177275.3, filed on July 17, 2015, EP 15180354.1 and EP 15180353.3, both filed on August 10, 2015, and EP 15 185 005.4, filed on September 14, 2015, EP 15 196 238.8 and EP 15 196 239.6, both filed on November 25, 2015, EP 15 197 744.4, filed on December 3, 2015, the full content of all of which is herewith also included by reference.

US 2006/044546 A1 describes a ranging apparatus capable of ranging simultaneously to a three dimensional scene. An illumination means illuminates a scene with a two dimensional array of spots. A detector is located near to the illumination means and arranged to look toward the scene. A processor responds to the output from the detector and, from the location of a spot in the image of the scene, determines the range to that spot.

US 2008/13005 A1 describes driving schemes in which rows are selected one at a time and column data voltages are inverted to provide inversion schemes for display devices comprising pixels arranged in rows and columns. The order in which rows are selected is such that a first group of first polarity rows is selected in a first order, a first group of second polarity rows is selected in a second order, a second group of first polarity rows is selected in the second order,

and a second group of second polarity rows is selected in the first order, the first order being one of ascending or descending row number order, and the second order being the other of ascending or descending row number order.

- 5 Despite the advantages implied by the above-mentioned devices and detectors, several technical challenges remain. Thus, generally, a need exists for detectors for detecting a position of an object in space which is both reliable and may be manufactured at low cost. Specifically, a need exists for 3D-sensing concepts. Various known concepts are at least partially based on using so-called FiP sensors, such as several of the above-mentioned concepts. Therein, as an
- 10 example, large area sensors may be used, in which the individual sensor pixels are significantly larger than the light spot and which are fixed to a specific size. Still, large area sensors in many cases are inherently limited in the use of the FiP measurement principle, specifically in case more than one light spot is to be investigated simultaneously.
- 15 A further challenge using FiP detectors is the detector area or active area. Typically, for distance measurements, a large active area of the detector is used or even is required. This area, however, may cause noise problems, specifically when the tetralateral conductivity concept is employed to build a PSD. This often results in poor signal-to-noise-ratios and slow detector response times due to the large capacitance in conjunction with the series resistance of the detector. A further challenge using FiP detectors is cost of manufacturing. Thus, in many cases, typical FiP sensors are expensive, as compared to e.g. conventional Si-based photodiodes. Further, the evaluation of the measurement results of measurements using FiP-sensors remains an
- 20 issue, specifically in case the total power of a light beam is unknown. In case a plurality of FiP-sensors is used, located at different positions along an axis of propagation of the light beam, a range of measurement typically is limited to the range in between the two positions of the sensors. Further, many FiP-detectors show a luminance dependency which renders the evaluation of the measurement result more difficult, and, additionally, FiP-measurements in many cases are dependent on a target spot size.

30 Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a

35 position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

- 40 This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed. As used herein, the term “position” refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;
- at least one evaluation device configured for evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices.

As further used herein, the term “matrix” generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

As further used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane. As will further be outlined in detail below, each optical sensor, preferably, may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-

sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, preferably, each optical sensor is a single area optical sensor rather than a pixelated optical sensor like e.g. a sensor matrix. Other embodiments, however, are feasible. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photosensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the setup, one with a small light-sensitive area and one with a larger light-sensitive area, as will be explained in further detail below.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. The sensor signal may comprise at least one information of at least one beam profile of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The present invention may implement a general idea for determining the at least one longitudinal coordinate of the object, which might be characterized as follows. Thus, as explained in more detail in one or more of the above-mentioned prior art documents, e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot, may

be characterized by two measurement variables: a measurement signal measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their respective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming a combined signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the combined signal for deriving the longitudinal coordinate.

The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus, may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.

The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, as will be outlined in further detail below, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

Consequently, in accordance with the present invention, the term "center signal" generally refers to the at least one sensor signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. As used herein, the term "highest sensor signal" refers to one or both of a local maximum or a maximum in a

region of interest. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range.

The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

Similarly, the term "sum signal" generally refers to a signal derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be adapted to determine the sum signal by integrating of signals of the entire matrix, of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of sum and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of the center signal or sum signal from at least one slice or cut of the light spot. Slices or cuts may be realized for example by using one or more specific rows and/or columns. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

Similarly, the term "combined signal", as used herein, generally refers to a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an

arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

5 The light beam propagating from the object to the detector specifically may fully illuminate the at least one optical sensor from which the center signal is generated, such that the at least one optical sensor from which the center signal arises is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the at least one optical sensor from which the sensor signal arises. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the entire matrix which is smaller than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a "light spot" generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

The light beam may propagate from the object towards the detector. As will be outlined in further detail below, the light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source, as an example, may be or may comprise one or more of an external illumination source, and illumination source integrated into the detector or an illumination source integrated into a beacon device being one or more of attached to the object, integrated into the object or held by the object.

As used herein, the term "ray" generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term "beam" generally refers to a collection of rays. In the following, the terms "ray" and "beam" will be used as synonyms. As further used herein, the term "light beam" generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. As used herein, the term "beam profile" generally refers to a transverse intensity profile of the light beam. In particular, the term "beam profile" relates to a spatial distribution, in particular in at least one plane perpendicular to the propagation of the light beam, of an intensity of the light beam. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The detector may comprise at least one transfer device which may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The optical sensors of the sensor element may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers, and/or in the visible spectral range, specifically, in the range of 380 nm to 780 nm. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm.

Specifically, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the combined signal and/or any secondary signal derived thereof and the longitudinal coordinate. The combined signal, specifically, as will be outlined in further detail below, may be a quotient signal.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the opti-

cal sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold;
5 an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

10 As outlined above, raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by
15 using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image
20 created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition.; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a
25 Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or maybe determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations,
30 however, are feasible.
40

As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be per-

formed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more combined signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa;

forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one combined signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sum signal and the center signal, dividing multiples of the sum signal and the center

signal, dividing linear combinations of the sum signal and the center signal. For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 5 wherein x and y are transversal coordinates, A1 and A2 are areas of at least one beam profile at the position of the sensor element, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The center signal may be a sensor signal comprising essentially center information of the beam profile. The sum signal may be a
- 10 signal comprising essentially edge information of the beam profile. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.
- 15 Each of the center signal and the sum signal may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the position of the sensor element used for determining the combined signal Q. The light-sensitive regions may be arranged such that the sum signal signal comprises information of a first area of the beam profile and the
- 20 center signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.
- 25 The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the
- 30 beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e.
- 35 proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information com-

pared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, the center signal may be a signal corresponding to a rectangular stripe comprising the center of the light spot. The evaluation device may be adapted to form a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. In particular, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile. The first and the second area, as an example, may be adjacent, with a minimum separation from one another. However, the separation between adjacent areas may also be varied such as enlarged to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent areas may be increased by not evaluating the area between the first and the second area. This may reduce the light on one of the areas relative and/or absolute as compared to the adjacent area which will may increase a quotient of the signal of the two areas. Further, one or both of the areas may consist of separated subareas which may be adjacent to each other and/or which may be separated by areas that are not evaluated and/or which may be evaluated as part of a different quotient. Further, the first and the second area may consist of a linear combination of subareas, whereas the signal contributed by each subarea may be weighted differently when forming the center signal and/or the sum signal. This may further be beneficial for increasing the dynamic range of the quotient system.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determi-

nation of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by one or more line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The detector may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors. For example, the detector may comprise a plurality of optical sensors arranged in a matrix. The detector may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, a defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector may be adapted to determine the depth information using the radiance ratio. This principle is called Distance by Photon Ratio (DPR).

The DPR principle can be applied, for example, to many sub-regions within a segmented image profile, such as a segmented image of at least one feature generated by the at least one light beam on the matrix of optical sensors, which is demonstrated hereafter. The quotient $Q_A(z)$ in the two-dimensional case can be written as

$$Q_A(z) = \frac{\iint_{-r_{out}}^{r_{out}} P(x,y,z) dx dy}{\iint_{-r_{in}}^{r_{in}} P(x,y,z) dx dy},$$

where $P(x, y, z)$ is a two-dimensional beam profile and r_{in} and r_{out} are inner and outer circle radii, respectively. For line-segmented quotients $Q_y(z)$ along the y-dimension, this can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{out}}^{r_{out}} P(x,z) dx}{\int_{-r_{in}}^{r_{in}} P(x,z) dx}.$$

Without wishing to be bound by this theory, the overall defocused beam profile $P(x, z)$ can be regarded as a superposition of defocused image point profiles $p(x, z)$ along the image width $b(x, z)$. This relationship can be modelled as a convolution given by

$$P(x, z) = b(x, z) * p(x, z),$$

wherein $p(x, z)$ represents the Point-Spread-Function (PSF) of a lens in defocus, which in the field of paraxial optics is also known as Circle of Confusion (CoC). By inserting $P(x, z)$ into $Q_y(z)$ the above described quotient $Q_y(z)$ can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{out}}^{r_{out}} b(x, z) * p(x, z) dx}{\int_{-r_{in}}^{r_{in}} b(x, z) * p(x, z) dx} = \frac{\int_{-r_{out}}^{r_{out}} b(x, z) dx \cdot \int_{-r_{out}}^{r_{out}} p(x, z) dx}{\int_{-r_{in}}^{r_{in}} b(x, z) dx \cdot \int_{-r_{in}}^{r_{in}} p(x, z) dx}.$$

Furthermore, with the matrix of optical sensors a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates. Accordingly, the parameterization of a single line may be given by

$$\omega = x \cos(\theta) + y \sin(\theta).$$

The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega, \theta)}\{f(x, y)\} = \iint_{-\infty}^{\infty} f(x, y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy,$$

where δ denotes the Dirac delta function and $f(x, y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is given by

$$R = \frac{\Re_{(\omega, \theta)}\{f'(x, y)\}}{\Re_{(\omega, \theta)}\{f(x, y)\}},$$

with $f'(x, y)$ as the inner region. The variation of θ may yield different ratios R for skewed object surfaces at a certain distance. It is sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

In one embodiment, the light beam propagating from the object to the detector may illuminate the sensor element with at least one pattern comprising at least one feature point. As used herein, the term “feature point” refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central

area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

For example, the light beam propagating from the object to the detector may illuminate the sensor element with at least one point pattern. The point pattern may be generated by the object, for example in response to an illumination by the at least one light source with an illumination pattern comprising the at least one point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The light beam propagating from the object to the detector may illuminate the sensor element with a reflection pattern comprising both point patterns and line patterns. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the combined signal such as the combined signal Q. This determining may be a one-step process, such as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined signal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

The evaluation device may be configured for using at least one predetermined relationship between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the combined signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider. The divider may fully or partially be integrated into the

sensor element answers or may fully or partially be embodied independent from the sensor element.

As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

Specifically, the optical sensors may be or may comprise inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Cu photodiode, a Ge:Hg photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The matrix may be composed of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1, 4:3, 16:9$ or similar. As an example, the array may be a square array,

having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

As further outlined above, the matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term “essentially perpendicular”, reference may be made to the definition given above. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, as outlined above, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

As further outlined above, preferably, the sensor element may be oriented essentially perpendicular to an optical axis of the detector. Again, with respect to the term “essentially perpendicular”, reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The matrix may comprise two or four separate light sensitive areas and/or may be sub-divided into two or more segments, in particular two or four active areas. The light sensitive areas and/or segments may be separated by a dividing line. The dividing line, for example, of two Si-diodes, can be arranged parallel to the baseline of the detector, in particular parallel to a spot movement on the optical sensors. However, other arrangements are feasible. For example, the quadrant diode may comprise at least two dividing lines. The at least two dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the combined signal for near field and far field applications independently from each other. In addition to determining the combined signal of sum signal and center signal, the evaluation device may be adapted to determine a second quotient using other signals of the matrix. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotient signals allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient signal may be determined by dividing the sensor signals of two top quadrants, also called top segment, by the sensor signals of two bottom quadrants, also called bottom segment. Using the quotient signal of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient

without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between the top and bottom segment is parallel to the baseline, the quotient signal determined from top segment divided by bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient signal by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals of the quadrants such that distance measurement is possible over a wide range with a large resolution.

In view of the technical challenges involved in the prior art documents discussed above, specifically in view of the technical effort which is required for generating the FiP effect, it has to be noted that the present invention specifically may be realized by using non-FiP optical sensors. In fact, since optical sensors having the FiP characteristic typically exhibit a strong peak in the respective sensor signals at a focal point, the range of measurement of a detector according to the present invention using FiP sensors as optical sensors may be limited to a range in between the positions and which the optical sensors are in focus of the light beam. When using linear optical sensors, however, i.e. optical sensors not exhibiting the FiP effect, this problem, with the setup of the present invention, generally may be avoided. Consequently, the optical sensors each may have, at least within a range of measurement, a linear signal characteristic such that the respective sensor signals are dependent on the total power of illumination of the respective optical sensor.

The detector may further comprise an illumination source for illuminating the object. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

As outlined above, the illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source, as outlined above, specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at

least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

5 The detector may comprise the at least one illumination source. The illumination source may be adapted to illuminate the object with at least one illumination light beam. The illumination light beam may be fully or partially reflected by the object and may travel back towards the detector. The illumination source may be arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis. As used herein, the term “essentially parallel to the optical axis” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. The illumination source may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis do not intersect in space. In particular, the illumination may be on-axis. In order to provide such an on-axis illumination, as an example, one or more reflective elements may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms. The reflected light beam may be essentially parallel to the optical axis. Thus, according to the present invention, the object distance is not determined from an angle of the light beam, i.e. a position of the light spot on a sensor, as done in triangulation methods. Instead, the object distance may be determined using light beams essential parallel to the optical axis. Determination of the object distance by a non-triangulation method but by combining the at least two sensor signals, allows arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis.

25 The light beam may be a monochromatic light beam. The light beam may comprise a narrow band of wavelengths, preferably the light beam may comprise a single wavelength. The at least one light source may be adapted to generate at least one monochromatic light beam and/or the detector may comprise at least one filter element adapted to filter a narrow band of wavelength, such as a monochromator.

30 The illumination source may be adapted to generate at least one illumination pattern for illuminating the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term “pattern” refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term “illumination pattern” refers to a pat-

tern which illuminates the object. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines; at least one fringe pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that a hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source may be adapted to generate pulsed illumination. The illumination source may be adapted to generate at least one light pulse. As used herein, the term "light pulse" or "pulsed illumination" refers to a light beam limited in time. The light pulse may have a pre-defined length or time duration, for example in the nanoseconds range. For example, the illumination source may be adapted to generate pulses with a pulse length of less than a nanosecond, such as a tenth of a nanosecond, up to a tenth of a second. The illumination source may

be adapted to periodically generate the light pulse. For example, the illumination source may be adapted to generate the light pulse with a frequency of 10 Hz to 10 GHz.

5 The illumination source may be adapted to generate a pulsed light beam. For example, the illumination source may be adapted to generate a continuous illumination light beam and the detector may comprise at least one interruption device adapted to interrupt the illumination, in particular periodically. The interruption device may comprise at least one shutter and/or a beam chopper or some other type of mechanical or electronical periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates
10 at constant speed and which can thus periodically interrupt the illumination. By way of example, the at least one interruption device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

15 Preferably, the illumination source may be a movable and/or mobile illumination source. For example, the illumination source may be moved during the measurement such as to illuminate the object from different positions and/or angles. However, embodiments are feasible in which the illumination source may be positioned in at least one fixed position, for example during a complete measurement time. The evaluation device may be adapted to illumination sources with unclear position such as due to high manufacturing tolerances and/or user interaction
20 and/or user assembly or the like.

The illumination source may illuminate the object with a convergent and/or divergent and/or collimated light beam.

25 The illumination source and the sensor element may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the optical sensors may be arranged in a twisted arrangement.

30 The illumination source and the optical axis may be separated by a small baseline. As used herein, the term "baseline", also denoted as basis line, refers to a distance, for example in a xy - plane, between the optical axis and the illumination source, in particular a distance between the optical axis and a z-component of the illumination light beam. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical
35 sensors and transfer device. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. For example, the detector may be a compact device without further optical elements, wherein the illumination source may be placed as close as possible to an edge of the transfer device. Thus, the baseline
40 may be close to half a diameter of the transfer device, in particular the lens diameter and housings of lens and light source. For example, the detector may be an even more compact device, wherein a mirror, in particular a small mirror, may be positioned in front of the transfer device, in particular in a center, for example a geometrical center, or close to the center of the transfer

device, in order to couple the illumination beam into the beam path. Thus, the baseline may be less than half the diameter of the transfer device. The illumination source may be arranged such that the baseline is as small as possible. By arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis and that the illumination source and the optical axis are separated by the small baseline, very compact devices are possible. For example, a distance from the center of the transfer device to the illumination source, in particular along a connecting line from the center of the transfer device to the illumination source, may be preferably less than 2.5 times a distance from the center of the transfer device to an edge if the transfer device, more preferably less than 1.5 times the distance center to edge of the transfer device, and most preferably less than 1 times the distance center to edge of the transfer device. The transfer device may have an arbitrary shape, in particular a non-circular shapes are possible. At small distances an aperture of the illumination source may be small and the baseline may be small. At large distances an aperture of the illumination source may be large and the baseline may be small. This is contrarily as in triangulation methods, wherein at large distances a large baseline is necessary. Further, triangulation based systems have a minimum detection range significantly greater than zero, for example such as 20 cm from the detector system, due to necessary spatial extent of the baseline. Such a large baseline may result in that the illuminated light scattered from the object may not reach the light sensitive area of the optical sensor behind the transfer device. In addition, in triangulation based systems, using a small baseline would reduce the minimum detection range, however, at the same time would reduce a maximum detection range. Further, triangulation based systems require a plurality of light sensitive areas and sensor signals, for example sensor signals of at least one detector row. According to the invention, determination of the longitudinal coordinate z is possible with a reduced number of sensor signals, in particular with less than 20, preferably less than 10 and more preferably less than 5 sensor signals.

The detector may further comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

The optical sensor may be positioned off focus. As used herein, the term "focus" generally refers to one or both of a minimum extent of a circle of confusion of the light beam, in particular of at least one light beam emitted from one point of the object, caused by the transfer device or a focal length of the transfer device. As used herein, the term "circle of confusion" refers to a light spot caused by a cone of light rays of the light beam focused by the transfer device. The circle of confusion may depend on a focal length f of the transfer device, a longitudinal distance from the object to the transfer device, a diameter of an exit pupil of the transfer device, a longitudinal distance from the transfer device to the light sensitive area, a distance from the transfer device

to an image of the object. For example, for Gaussian beams, a diameter of the circle of confusion may be a width of the Gaussian beam. In particular, for a point like object situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from the object into a focus point at the focal length of the transfer device. For non-point like objects situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from at least one point of the object into a focus plane at the focal length of the transfer device. For point like objects not situated or placed at infinite distance from the detector, the circle of confusion may have a minimum extent at least at one longitudinal coordinate. For non-point like objects not situated or placed at infinite distance from the detector, the circle of confusion of the light beam from at least one point of the object may have a minimum extent at least at one longitudinal coordinate. As used herein, the term "positioned off focus" generally refers to a position other than the minimum extent of a circle of confusion of the light beam caused by the transfer device or a focal length of the transfer device. In particular, the focal point or minimum extent of the circle of confusion may be at a longitudinal coordinate l_{focus} , whereas the position of the optical sensor may have a longitudinal coordinate l_{sensor} different from l_{focus} . For example, the longitudinal coordinate l_{sensor} may be, in a longitudinal direction, arranged closer to the position of the transfer device than the longitudinal coordinate l_{focus} or may be arranged further away from the position of the transfer device than the longitudinal coordinate l_{focus} . Thus, the longitudinal coordinate l_{sensor} and the longitudinal coordinate l_{focus} may be situated at different distances from the transfer device. For example, the optical sensors may be spaced apart from the minimum extent of the circle of confusion in longitudinal direction by $\pm 2\%$ focal length, preferably by $\pm 10\%$ focal length, most preferably $\pm 20\%$ focal length. For example, at a focal length of the transfer device may be 20mm and the longitudinal coordinate l_{sensor} may be 19,5 mm, i.e. the optical sensor may be positioned at 97,5% focal length, such that l_{sensor} is spaced apart from the focus by 2,5% of focal length. The extent of the circle of confusion may be larger than the extent of the optical sensor. For example, the optical sensor may be positioned as such that the circle of confusion extends beyond the optical sensor. The optical sensor may thus only partially evaluate the beam profile of the light beam. The evaluation device may be adapted to extrapolate a beam profile such as due to symmetry considerations or due to comparing the partial beam profile to prerecorded beam profiles or the like. Further, the evaluation device may be adapted to evaluate a partial sum signal and a partial center signal of a partial beam profile and convert it into a sum signal and a center signal of an extrapolated and/or fitted and/or prerecorded beam profile or the like.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the

light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

The illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or object recognition method.

As outlined above, the detector may comprise the at least one illumination source. A distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors may be small. The distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. The illumination source and the optical axis may be separated by a small baseline. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and of the at least one optional transfer device, which will be described in detail below. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device.

At least one of the optical sensors may be adapted to generate at least one sensor signal dependent on a time-of-flight (TOF) the illumination light beam has traveled from the illumination source to the object and the reflection light beam has traveled from the object to the light sensitive area of the optical sensor. The evaluation device may be configured for determining at least one TOF-longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. The optical sensor adapted to generate the TOF sensor signal may be designed as time-of-flight detector. The time-of-flight detector may be selected from the group consisting of: at least one pulsed time-of-flight detector; at least one phase modulated time-of-flight detector; at least one direct time-of-flight detector; at least one indirect time-of-flight detector. For example, the pulsed time-of-flight detector may be at least one range gated imager and/or at least one direct time-of-flight imager. For example the phase modulated time-of-flight detector may be at least one RF-modulated light source with at least one phase detector. The optical sensor may be adapted to

determine a time delay between emission of the illumination light beam by the illumination source and receipt of the reflection light beam.

For example, the optical sensor adapted to generate the TOF sensor signal may be designed as pulsed time-of-flight detector. The detector may comprise the at least one interruption device, such as at least one shutter element, adapted to generate a pulsed light beam. The optical sensor may be adapted to store the TOF sensor signal dependent on the receiving time of the reflection light beam in a plurality of time windows, in particular subsequent time windows. The optical sensor may be adapted to store dependent on receiving time of the reflection light beam the generated TOF sensor signal in at least one first time window and/or in at least one second time window. The first and second time windows may be correlated with the opening and closure of the interruption device. Duration of first and second time windows may be pre-defined. For example, the TOF sensor signal may be stored in the first time window during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may be stored in the second time window. Other durations of time windows are thinkable. The first and the second time window may comprise information about background, signal height and signal shift.

For example, the optical sensor adapted to generate the TOF sensor signal may be designed as direct time-of-flight imager. The direct time-of-flight imager may comprise the at least one illumination source adapted to generate at least one single laser pulse. The single laser pulse may be reflected back from the object onto the optical sensor. The optical sensor may comprise at least one photodiode, for example at least one Avalanche Photo Diode (APD), such as at least one Si APD, or such as at least one InGaAs APD, or at least one PIN photo detector array, or at least one Single-Photon Avalanche Photodiode (SPAD), adapted to image the reflection light beam. The direct time-of flight imager may be adapted to image at least one image comprising spatial and temporal data.

For example, the optical sensor adapted to generate the TOF sensor signal may be designed as phase modulated time-of-flight modulator. The phase modulated time-of-flight modulator may be adapted to measure a difference in phase, in particular a phase shift, by determining a correlated signal, for example by multiplying a received signal, i.e. of the reflection light beam, with the emitted signal, i.e. the illumination light beam. A DC component of the correlated signal may comprise an information about the difference in phase. The evaluation device may be adapted to determine the second longitudinal coordinate of the object from the phase difference. For example, the illumination source and the optical sensor adapted to generate the TOF sensor signal may be designed as RF-modulated light source with at least one phase detector. The illumination source may comprise, for example, at least one LED and/or at least one laser. The illumination source may comprise at least one modulation device adapted to modulate the light beam having a pre-defined phase shift. For example, the modulation device may comprise at least one radio frequency module. The radio frequency module may be adapted to modulate the illumination beam with an RF carrier. The optical sensor may be adapted to determine a phase shift of the reflective light beam impinging on the optical sensor.

The optical sensor may be designed as and/or may comprise at least one time-of-flight pixel. Preferably, the detector may comprise at least two optical sensors, wherein each optical sensor is designed as and/or comprises at least one TOF pixel. For example, the detector, in particular the optical sensor may comprise a quadrant diode adapted to generate the TOF sensor signal.

5 For example, the detector, in particular the optical sensor, may comprise at least one pixelated TOF-imager.

The evaluation device may be configured for determining the at least one TOF longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. As used herein, the term “TOF longitudinal coordinate” refers to a longitudinal coordinate derived from the TOF sensor signal. As described above, the illumination light source may be adapted to periodically generate at least one light pulse. The detector may be adapted to generate the first longitudinal sensor signal for each period. The evaluation device may be adapted to determine in which pulse period the TOF sensor signal was generated using the second longitudinal coordinate. The detector
10 may be adapted to uniquely assign to which period the ToF-signal refers to by using the combined sensor signal. Both, the TOF sensor signal and the combined sensor signal, may be non-monotonous functions of the longitudinal coordinate z_{real} . Thus, the longitudinal coordinate may not be uniquely determined from one of the TOF sensor signal or the combined sensor signal alone and a measurement range may not be restricted to a longitudinal range in which the signals are unique functions of z_{real} . The term “measurement range” generally refers to a range
20 from the object to the detector in which determination of the longitudinal coordinate is possible. The term “longitudinal range” generally refers to a range from the object to the detector in which unique determination of the longitudinal coordinate is possible. Below and/or above certain distances from the object to the detector a determination of the longitudinal coordinate may not be possible. For example, time-of-flight measurements are not possible below a certain distance
25 from the object to the detector, possibly due to the minimum measurement time of the internal clock. Furthermore, in time-of-flight measurements the sensor signal may be unique within a longitudinal period, but the sensor signal may be the same in case integer multiples of the longitudinal period are added such that the determined longitudinal coordinate may be non-unique. Thus, the same TOF sensor signal will be obtained for a distance z_1 and a distance $z_1 + n \cdot z_{1p}$,
30 where n is an integer denoting the longitudinal period and z_{1p} is the longitudinal period of the TOF sensor signal, wherein the distances z_1 and $z_1 + n \cdot z_{1p}$ are within the measurement range. As used herein, the term “longitudinal period” refers to partitions of a period, in particular a distance range, in which the longitudinal coordinate can be unambiguously determined from the
35 TOF sensor signal. Non-unique longitudinal coordinates may be denoted as relative longitudinal coordinates and the unique longitudinal coordinates may be denoted as absolute longitudinal coordinates.

If both the TOF sensor signal $F1$ and the combined sensor signals $F2$ are available, it may be
40 possible to uniquely determine the longitudinal coordinate and to extend the longitudinal range, as long as each signal pair ($F1, F2$) corresponds to a unique distance and vice versa. In particular, if a unique signal pair ($F1, F2$) exists for each longitudinal coordinate and vice versa, the evaluation device may be adapted to determine the unique combined longitudinal coordinate by

(1) selecting at least one first selected signal such as the TOF sensor signal and/or the combined sensor signal and determining non-unique first longitudinal coordinates;

(2) selecting a second selected signal such as the combined signal Q and/or the TOF sensor signal, which was not selected in step (1), and determining non-unique second longitudinal coordinates;

(3) determining whether one of the non-unique first longitudinal coordinates and the non-unique second longitudinal coordinates match up to a predetermined tolerance threshold;

(4) setting a combined unique longitudinal coordinate to be a matching longitudinal coordinate.

In step (1) and step (2), signals may be selected in the given order or may be performed in a different order. For example, in step (1) the TOF sensor signal may be selected and in step (2) the combined signal Q may be selected. In another example, in step (1) the combined sensor signal may be selected, and the non-unique first longitudinal sensor signal may be determined therefrom. In step (2), the TOF sensor signal may be selected.

Additionally or alternatively to step (4), the evaluation device may be adapted to output an error signal in case no matching coordinates are found and/or to output an error signal in case more than one matching coordinates are found. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be stored in a look-up table. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be approximated or described by an analytical function which is evaluated to find the longitudinal coordinate corresponding to a given signal pair.

The evaluation device may comprise at least two memory elements. As used herein, the term “memory element” refers to a device adapted to store information. The evaluation device may be adapted to receive and to store information provided by the optical sensors, for example the at least one first sensor signal. Such information may comprise raw sensor data and/or processed sensor data. For example, the memory element may be adapted to store information for further evaluation by the evaluation device. The memory element may be a volatile or non-volatile memory element.

As described above, the optical sensor may be designed as and/or may comprise at least one ToF pixel. The detector may comprise at least two switches. Each of the switches may be connected to the optical sensor adapted to generate the first sensor signal, for example by at least one connector. In particular, each of the switches may be connected to the ToF pixel. The switches are adapted to provide the TOF sensor signal to one of the memory elements. In particular, the switches may be adapted to let, dependent on receiving time of the reflection light beam, the generated TOF sensor signal pass through one of the switches. For example, the TOF sensor signal may pass one of the switches during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may pass the other switch. Each of the switches may be controlled by a control signal having a pulse length identical to a pulse length of a light pulse generated by the illumination source. The control signal of one of the switches may be delayed. For example, the delay may correspond to the pulse

length of the light pulse. The evaluation device may be adapted to sample and/or store depending on the delay a first part, or fraction, of the TOF sensor signal through a first switch in a first memory element and the other, second part, or fraction, of the TOF sensor signal through a second switch in a second memory element. The evaluation device may be adapted to determine the first longitudinal coordinate by evaluating the first part and second part of the TOF sensor signal. The evaluation device may be adapted to determine the first longitudinal coordinate z_1 by

$$z_1 = \frac{1}{2} \cdot c \cdot t_0 \cdot \frac{S_{12}}{S_{11} + S_{12}} + z_0;$$

wherein c is the speed of light, t_0 is the pulse length of the illumination light beam, z_0 is a distance offset often determined by the resolution of the time measurement, and S_{11} and S_{12} are the first part and second part of the TOF sensor signal, respectively.

The detector may be adapted to determine at least one distance information of the object by using triangulation and/or structured light techniques. In known 3D sensing devices, such as devices using triangulation or structured light techniques, due to correspondence problems regular, constant or periodic pattern are not suitable since each measured point has to be assigned to one reference point of a reference pattern. The at least one light beam propagating from the object to the detector may be adapted to generate at least one reflection pattern on the matrix of optical sensors. The term "reflection pattern" refers to at least one image determined by the optical sensors in response to illumination of the light sensitive areas by the light beam propagating from the object to the detector which was generated in response to illumination by the illumination pattern. The reflection pattern may comprise at least one reflection feature depending on the corresponding illumination feature of the illumination pattern. The detector may be adapted to determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection pattern from the quotient signal. Thus, the detector may be adapted to pre-classify the at least one reflection feature of the reflection pattern. This allows using illumination patterns comprising regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal patterns may be preferred.

The evaluation device may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator;

applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device may be adapted to determine at least one reference feature in at least one reference pattern corresponding to the at least one reflection feature. The term "reference pattern" refers to an image of comparison at a specific spatial position comprising at least one reference feature. As outlined above, the evaluation device may be adapted to perform an image analysis and to identify features of the reflection pattern. The evaluation device may be adapted to identify at least one reference feature in the reference pattern having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference pattern and the reflection pattern may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The distance may be a relative distance, also denoted as baseline. The evaluation device may be adapted to determine an epipolar line in the reference pattern. The relative position of the reference pattern and reflection pattern may be known. For example, the relative position of the reference pattern and the reflection pattern may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature to the real world feature from which it originates. Thus, the straight line may comprise possible object features corresponding to the selected reflection feature. The straight line and the baseline span an epipolar plane. As the reference pattern is determined at a different relative constellation from the reflection pattern, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference pattern. The epipolar line may be the intersection of the epipolar plane and the reference pattern. Thus, a feature of the reference pattern corresponding to the selected feature of the reflection pattern lies on the epipolar line.

Depending on the distance to the object, the reference feature corresponding to the image position of the reflection feature may be displaced within the reference pattern compared to the image position in the reflection pattern. The reference pattern may comprise at least one displacement region in which the reference feature corresponding to the selected reflection feature may be imaged. The displacement region may comprise only one reference feature. The displacement region may extend along the epipolar line. The evaluation device may be adapted to determine the reference feature along the epipolar line. The evaluation device may be adapted to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm \varepsilon$ from the quotient signal Q to determine a displacement region along the epipolar line corresponding to $z \pm \varepsilon$. The evaluation device may be adapted to match the selected reflection feature with at

least one reference feature within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating the corresponding reference and reflection features. The evaluation device may be adapted to match the selected feature of the reflection pattern with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. Preferably, the detector may be adapted to pre-classify the selected reflection feature using the quotient signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference pattern may have a relative distance to each other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line. For example, the illumination pattern may comprise at least one hexagonal pattern. Preferably, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Preferably, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other.

The evaluation device may be adapted to determine a displacement of the reference feature and the reflection feature. The evaluation device may be adapted to determine the displacement of the matched reference feature and the selected reflection feature. As used herein, the term “displacement” refers to the difference between an image position in the reference image to an image position in the reflection image. The evaluation device may be adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate z_{triang} . For example, the longitudinal information may be a distance value. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case the position of the selected reflection feature in the reflection pattern and the position of the matched reference feature and/or the relative displacement of the selected reflection feature and the matched reference feature are known, the longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. The displacement and the corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one

data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

5 The detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, at least one optical sensors of the matrix of optical sensors may be adapted to generate a so called FiP signal. For example, each of the optical sensors of the matrix may be adapted to generate the at least one FiP signal. For example, the matrix of optical sensors may be a pixelated FiP-sensor. Additionally or alternatively, the FiP signal may be determined, such as extracted and/or simulated, from
10 the sensor signals of the matrix and the quotient as outlined above may be determined from the sensor signals. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. According to the FiP-effect, the longitudinal sensor signal, given the same total power, may exhibit at least one pronounced maximum for one or a plurality of focusings and/or for one or a plurality of specific sizes of the light spot on the sensor region or within the sensor region. For purposes of comparison, an observation of a maximum of the longitudinal sensor signal in a condition in which the corresponding material is impinged by a light beam with the smallest possible cross-section, such as when the material may be located at or near a focal point as affected by an optical lens, may be denominated as a "positive FiP-effect". The negative FiP effect describes an observation of a minimum of the longitudinal sensor signal under a condition in which the corresponding material is impinged by a light beam with the smallest available beam cross-section, in particular, when the material may be located at or near a focal point as effected by an optical lens. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

As outlined above, the detector may further comprise one or more additional elements such as
30 one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing. The detector specifically may comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensor. The transfer device may comprise one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at
35 least one multi-lens system. Therein, generally, the term "transfer device" may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical
40 sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of , preferably at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection

element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

5 In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

30 In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

35 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source.

Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined.

Alternatively, the illumination source of the scanning system may emit two or more individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning device. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as

mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with a several corners and surfaces. One or more flexibly movable laser sources may be attached to the scanning system. The one or more laser sources may be moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system may be measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information may be stored in a mobile phone application. The laser sources may be moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and/or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning device may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning device may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of

optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning device is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning device.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The optional transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The optional

transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. Most preferably, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the sensor element can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the sensor element. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses an inertial measurement unit for use in an electronic device. The electronic device may be a mobile electronic device. The electronic device may be a camera. The electronic device may be a mobile phone. The inertial measurement unit is adapted to receive data determined by at least one inertial measurement unit comprising at least one detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. As used herein, the term "data determined by the at least one detector" refers to at least one information on the at least one longitudinal coordinate z . The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. As used herein, the term "data determined by the at least one further sensor" refers to at least one information selected from the group consisting of: angle information; speed information; information on turn rate; information on inclination. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed. The inertial measurement unit may comprise at least one processor. The processor may be adapted to evaluate data recorded by the further sensor. In particular, the processor may be adapted to determine and/or calculate one or more of spatial position, spatial orientation, movement and velocity. The inertial measurement unit may comprise a plurality of further sensors. The inertial measurement unit may be adapted to fuse information determined from at least two of the further sensors. The inertial measurement unit may be adapted to fuse information of at least two further sensors by using at least one Kalman filter. As outlined

above, the detector may be adapted to provide an absolute measurement of the longitudinal coordinate z . The processor, e.g. the evaluation device as described above, may be adapted to fuse the information of the at least two further sensors considering the longitudinal coordinate z . The various sensor signals may be used within a Kalman filter or a linear quadratic estimated to take into account that each sensor signal is subject to measurement errors and inaccuracies. Fusion of these sensor signals within a Kalman filter may yield an improved estimate such as for the measurement of the longitudinal coordinate.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used.

The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps are as follows:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;
- evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Specifically, the evaluation of the first and second sensor signal may comprise deriving a combined signal Q by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Further, the determination of the longitudinal coordinate may comprise evaluating the combined signal Q . The evaluation of the combined signal specifically may comprise using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the longitudinal coordinate.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

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In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application;.

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The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

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As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

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Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially land-

mark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for house-

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hold applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

5 The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with
10 mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus,
15 specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or
20 other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance,
25 scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head-mounted displays.
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Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention
40 may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries,

obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in

combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

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Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

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Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems, or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present

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invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.

5 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. 10 Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, 20 such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions. 30

Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used 40 for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endo-

scopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric surgery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment.

Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe. In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

Further, the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a

surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

5 Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation
10 etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

15 Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

20 The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

25 The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

30 Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to
35 avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the
40 devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the

same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D
5 imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially
10 useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection,
15 such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections, inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed
20 and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a
25 guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to
30 active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications.

35 Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft when landing or starting. Therein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to

detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or accelerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

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Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racquet, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light-emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the

present invention may be used in drones or multicopters to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concertized movements of several drones or recharging or refueling in the air or the like.

Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as cat litter robots, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, window washing robots, toy robots, telepresence robots, social robots providing company to less mobile people, or robots translating and speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate inse-

cure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

Further, devices according to the present invention may be used for navigation purposes, where
5 Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that are can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a
10 heavy device.

Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition device, a heating device, a television, an audio
15 device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.

20 Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.

25 The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may
30 be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or greenhouse or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.

35 Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific
40 groups of animals such as bees.

Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or virus-

es or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location. Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location-specific information, such as the current position on a map, or information on goods sold, or the like.

Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.

Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices according to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.

As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, the devices according to the present invention may be used in logistics applications. Thus, the

5 devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

20 Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

35 Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes

the position. The detector generally may be a device adapted for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device.

- 5 Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

- 10 The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

- 15 The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside a camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based algorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

- 30 The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

- 40 The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared.

5 The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

15 The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

20 The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

40 The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The

evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and/or processing 3D information.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device

and/or the data processing device may be especially suitable for such integration into automotive design.

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

As outlined above, the human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon

device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device

may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller

might include one or more data processing devices, such as one or more computers and/or microcontrollers.

- 5 As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may
- 10 fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.
- 15 The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group
- 20 of data or data pair comprising at least one position information and at least one time information.
- The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device
- 25 and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.
- 30 The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual
- 35 position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
- 40 As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the

article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The proposed devices and methods provide a large number of advantages over known detectors of this kind. Thus, the detector generally may avoid the shortcomings of the known prior art systems disclosed above. Specifically, the detector may avoid the use of FiP sensors, thereby allowing for e.g. using simple and cheap and commercially available semiconductor sensors such as silicon photodiodes. These photodiodes generally do not show a luminance dependency, and the method disclosed above is generally independent from the brightness of the scenery and/or the brightness of the light spot on the light beam. Consequently, a range of measurement in terms of luminance or total power of the light beam entering the detector is generally larger in the present invention as compared to many of the devices disclosed above. Further, the measurement by using the detector according to the present invention is generally independent from the target spot size, i.e. from either the size of the object, the size of a light spot projected on to the object or the size of a beacon device being one or more of attached to the object, integrated into the object or held by the object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional option of measuring one or more transversal coordinates, thereby integrating the functionality of a PSD.

The detector specifically may use linear photodetectors, such as the near silicon photodiodes. The use of a beam splitter is optional but is not generally necessary, which allows for using a simple optical setup.

The result of the measurement, i.e. the position determined by the detector such as the at least one longitudinal coordinate, may be rendered widely independent from the brightness of the scenery and/or the brightness of the object, the brightness of the at least one beacon device, the brightness of the at least one illumination source or the total power of the light beam propagating from the object to the detector. Further, due to this independence and wide range of measurement in terms of brightness, reflective objects or non-reflective objects may be used.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention, such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the combined signal, as mentioned above, contrarily, provides a wide range of

a continuously and monotonously decreasing or increasing functions which may be used to determine the longitudinal coordinate from the combined signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the combined signal from the total power of the light beam, at least as long as no saturation of the optical sensors is reached, the present invention also provides a very wide range of measurement in terms of brightness, i.e. in terms of total power of the light beam.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector having

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;

- at least one evaluation device configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;

- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;

- c) determining at least one combined signal by combining the center signal and the sum signal; and

- d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 2: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the detector, specifically the evaluation device, is configured for transforming the sensor signals of

the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals.

5 Embodiment 4: The detector according to the preceding embodiment, wherein the transforming of the sensor signals comprises at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a
10 decomposition into color channels; a decomposition into hue, saturation, and brightness channels, a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image.

20 Embodiment 5: The detector according to any one of the preceding embodiments, wherein the evaluation device comprises at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal.

25 Embodiment 6: The detector according to the preceding embodiment, wherein the center detector is fully or partially integrated into the sensor element.

Embodiment 7: The detector according to any one of the two preceding embodiments, wherein the center detector is fully or partially embodied in one or both of software or hardware.

30 Embodiment 8: The detector according to any one of the preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all
35 sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor
40 signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.

45 Embodiment 9: The detector according to the preceding embodiment, wherein the evaluation device comprises at least one summing device for forming the sum signal.

Embodiment 10: The detector according to the preceding embodiment, wherein the summing device is fully or partially integrated into the sensor element.

- 5 Embodiment 11: The detector according to any one of the two preceding embodiments, wherein the summing device is fully or partially embodied in one or both of software or hardware.

Embodiment 12: The detector according to any one of the preceding embodiments, wherein the combined signal is a combined signal Q, derived by one or more of: forming a quotient of the
 10 center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

- 15 Embodiment 13: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 20 wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o.

Embodiment 14: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for using at least one predetermined relationship between the
 25 combined signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 15: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an
 30 analytically derived relationship.

Embodiment 16: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

- 35 Embodiment 17: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the combined signal.

Embodiment 18: The detector according to the preceding embodiment, wherein the divider is
 40 fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 19: The detector according to any one of the two preceding embodiments, wherein the divider is fully or partially integrated into the sensor element.

5 Embodiment 20: The detector according to any one of the preceding embodiments, wherein the optical sensors are photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors.

10 Embodiment 21: The detector according to any one of the preceding embodiments, wherein the optical sensors are sensitive in the infrared spectral range.

Embodiment 22: The detector according to any one of the preceding embodiments, wherein the matrix comprises one or both of a CCD detector, preferably a CCD detector chip, or a CMOS detector, preferably a CMOS detector chip.

15 Embodiment 23: The detector according to any one of the preceding embodiments, wherein the matrix is a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns.

20 Embodiment 24: The detector according to the preceding embodiment, wherein the rows and columns are oriented essentially perpendicular.

Embodiment 25: The detector according to any one of the two preceding embodiments, wherein the matrix has at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows.

25 Embodiment 26: The detector according to any one of the three preceding embodiments, wherein the matrix has at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns.

30 Embodiment 27: The detector according to any one of the preceding embodiments, wherein the matrix comprises at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors.

35 Embodiment 28: The detector according to any one of the four preceding embodiments, wherein the ratio of the number of rows and the number of columns is close to 1 preferable larger than 1:3, more preferably larger than 1:2.

Embodiment 29: The detector according to any one of the preceding embodiments, wherein the sensor element is oriented essentially perpendicular to an optical axis of the detector.

40 Embodiment 30: The detector according to any one of the preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the respective sensor signals are dependent on the total power of illumination of the respective optical sensor.

45 Embodiment 31: The detector according to any one of the preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

Embodiment 32: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

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Embodiment 33: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

10 Embodiment 34: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at
15 least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings.

20 Embodiment 35: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the combined signal Q .

25 Embodiment 36: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

30 Embodiment 37: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one
35 beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

Embodiment 38: The detector according to any one of the preceding embodiments, wherein the evaluation device is further configured for determining at least one transversal coordinate of the
40 object by evaluating a transversal position of the at least one optical sensor having the highest sensor signal.

Embodiment 39: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for distinguishing different light beams having different modulations.
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Embodiment 40: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is adapted to illuminate the object with at least one illumination light beam, wherein the illumination source is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis.

Embodiment 41: The detector according to any one of the preceding embodiments, wherein the detector comprises the at least one illumination source, wherein the illumination source and the optical axis are separated by a small baseline.

Embodiment 42: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

Embodiment 43: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 44: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 45: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 46: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 47: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 48: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of the preceding embodiment referring to a detector, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

Embodiment 49: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;
- evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 50: The method according to the preceding embodiment, wherein the combined signal is a combined signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

Embodiment 51: The method according to the preceding embodiment, wherein the determining the longitudinal coordinate comprises evaluating the combined signal Q , wherein the evaluating the combined signal comprises using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 52: The method according to any one of the preceding method embodiments, wherein the method comprises using the detector according to any one of the preceding claims referring to a detector.

Embodiment 53: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figures 1 and 2 show different embodiments of a detector according to the present invention;

Figure 3 shows an example of a light spot on a matrix of optical sensors and the evaluation thereof;

Figure 4 shows an exemplary embodiment of a detector according to the present invention; a detector system, a human machine interface, an entertainment device, a tracking system, a scanning system and a camera;

Figure 5 shows an exemplary embodiment of a combined signal Q as a function of a longitudinal coordinate z , for various illumination intensities; and

Figures 6A and 6B show a modification of the embodiment of Figure 2, with an off-axis illumination light beam;

Figures 6C and 6D show a comparison of two experimental setups using a detector setup according to Figure 6A with a Bi-cell as optical sensors and experimental results;

Figures 7A and 7B show a further exemplary embodiment of a detector according to the present invention with a small baseline;

Figure 8 shows an embodiment of determining a longitudinal coordinate z using a line pattern;

Figures 9A and B show a defocused beam profile subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates and Radon transformation different image regions on a matrix of optical sensors;

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Figures 10A to C show three embodiments of a hexagonal illumination pattern; and

Figure 11 shows an embodiment of a scanning device.

10 Detailed description of the embodiments

In Figure 1, a schematic view of an embodiment of a detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises a beacon device 114, from which a light beam 116 propagates towards a sensor element 118. The sensor element 118 comprises a matrix 120 of optical sensors 122, each optical sensor 122 having a light-sensitive area 124 facing the object 112. The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110, further, may comprise at least one transfer device 128, such as at least one lens and/or at least one lens system, specifically for beam shaping. Consequently, the light beam 116 may be focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of the above-mentioned prior art documents, such as to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Figure 1, the light beam 116 generates a light spot 132 on the matrix 120. In Figure 3, an exemplary view of the light spot 132 on the matrix 120 is shown. As can be seen, in this exemplary embodiment, the matrix 120 specifically may be a rectangular matrix, with rows numbered by "i", from 1 to n , and with columns, denoted by "j", from 1 to m , with n , m being integers. The center of the light spot 132, in this exemplary embodiment, is located in the sensor element denoted by i^* , j^* . The optical sensors 122 may provide sensor signals s_{ij} to an evaluation device 134 which, out of the sensor signals, determines at least one center signal, denoted symbolically by $s_{i^*j^*}$. As outlined in further detail above, for generating the center signal, the evaluation device 134 may comprise at least one center detector 136. As an example, the center detector simply may determine the maximum sensor signal out of the plurality of sensor signals generated by the optical sensors 122. Alternative methods are feasible. Thus, as an example, instead of determining a single maximum optical sensor signal, a plurality of sensor signals may be used for generating the center signal. Thus, as an example, neighboring optical sensors which are adjacent to the optical sensor i^* , j^* may contribute to the center signal, such as optical sensors with the coordinates $i^*-1, \dots, i^*+1$ and $j^*-1, \dots, j^*+1$. These coordinates, in the simple exemplary embodiment, may form a square around the optical sensor i^* , j^* . Instead of a square having a side length of 3, as in this embodiment, other environments around the optical sensor

having the highest sensor signal may be used. Further, additionally or alternatively, the center signal may be generated by adding up and/or averaging over sensor signals within a certain range from the maximum sensor signal which may for example be beneficial to the measurement precision concerning noise such as pixel noise. Further, additionally or alternatively, for the determination of the center signal or sum signal, image processing techniques such as sub-pixel processing, interpolation, normalization or the like may be employed. Other alternatives are feasible. For example, the beam profile may be a trapezoid beam profile and the evaluation device 134 may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid. Further, when trapezoid beam profiles may be assumed, the evaluation device 134 may be adopted to determine the sum and center signals by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Further, the evaluation device 134 is configured for determining at least one sum signal out of the sensor signals of the matrix 120. For this purpose, the evaluation device 134 may comprise at least one summing device 138. The summing device 138 may be configured for adding up, integrating or averaging over the sensor signals of the entire matrix 120, of a region of interest within the matrix 120, each option with or without the optical sensors from which the center signal is generated. Thus, in the exemplary embodiment shown in Figure 3, the summing device is simply configured for summing over the sensor signals s_{ij} of the entire matrix 120, except for the center optical detector with the coordinates i^* , j^* . Other options, however, are feasible.

The evaluation device 134 further is configured for forming at least one combined signal out of the center signal and the sum signal. For this purpose, the evaluation device 134, as an example, may comprise at least one combining device 140, such as at least one divider 142. As a very simple embodiment, a quotient Q may be formed, by dividing the center signal by the sum signal or vice versa. Other options are feasible and are given above.

Finally, the evaluation device 134 is configured for determining at least one longitudinal coordinate z of the object by evaluating the combined signal. For this purpose, the evaluation device may comprise at least one further component, such as at least one evaluation component, which is not depicted in the figures. It shall be noted that the components of the evaluation device 134 shown in Figure 3 may fully or partially be embodied in hardware and/or software. Further, the components may fully or partially be embodied as independent or separate components, and/or may fully or partially be embodied as components which are integrated into the sensor element 118.

The embodiment of Figure 3 further shows that, in addition to the longitudinal coordinate z , at least one item of information on a transversal coordinate of the object 112 and/or the beacon device 114 may be generated. Thus, the coordinates i^* and j^* provide additional items of information on a transversal position of the object 112 and/or the beacon device 114. In the setup of Figure 1, the beacon device 114, for the sake of simplicity, is positioned in the center, i.e. on the

optical axis 126. In this case, the light spot 132 is likely to be centered in the middle of the matrix 120. In the embodiment shown in Figure 3, however, as can easily be detected, the light spot 132 is off-centered. This off-centering is characterized by the coordinates i^* , j^* . By using known optical relationships between this off-centering and a transversal position of the object 112 and/or the beacon device 114, such as by using the lens equation, at least one transversal coordinate of the object 112 and/or the beacon device 114 may be generated. This option is also shown in the exemplary embodiment of Figure 3.

In Figure 2, a modification of the embodiment of Figure 1 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Figure 1. Instead of using an active beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 itself comprises at least one illumination source 144. The illumination source 144, as an example, may comprise at least one laser, whereas, in Figure 1, as an example, the beacon device 114 may comprise a light-emitting diode (LED). Other embodiments, however, are feasible. The illumination source 144 may be configured for generating at least one illumination light beam 146 for fully or partially illuminating the object 112. The illumination light beam 146 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116.

As shown in Figure 2, as an example, the illumination light beam 146 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too, as will be shown in the context of Fig. 6A and 6B below. In order to provide an on-axis illumination, as shown in Figure 2, as an example, one or more reflective elements 148 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Figure 2 corresponds to the setup in Figure 1. Thus, again, an evaluation device 134 may be used, having e.g. at least one divider 142 for forming the combined signal Q, and, as an example, at least one position evaluation device 143, for deriving the at least one longitudinal coordinate z from the combined signal Q and/or another type of combined signal. It shall be noted that the evaluation device 134, again, may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 136, 138, 140, 142, 143 may fully or partially be embodied by appropriate software components and/or may fully or partially be embodied by hardware components.

The optical sensors 122 of the matrix 120, as an example, may be pixels of a pixelated optical sensor, such as a CCD and/or a CMOS sensor chip. Thus, as an example, the optical sensors 122 may have a side length and/or an equivalent diameter in the range of a few micrometers to several hundred micrometers. It shall be noted, however, that larger pixels or optical sensors 122 may be used. Further, instead of using an integrated sensor element 118 such as a CCD and/or CMOS sensor chip, non-integrated matrices may be used.

Figure 4 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments in Figures 1 or 2. The detector 110, specifically, may be embodied as a camera 149 and/or may be part of a camera 149. The camera 149 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Figure 4 further shows an embodiment of a detector system 150, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached to and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Figure 4 further shows an exemplary embodiment of a human-machine interface 151, which comprises the at least one detector system 150 and, further, an entertainment device 152, which comprises the human-machine interface 151. The figure further shows an embodiment of a tracking system 153 for tracking a position of the object 112, which comprises the detector system 150. The components of the devices and systems shall be explained in further detail below.

Figure 4 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the at least one object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110 and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136, which is not shown. The illumination source 136, generally, may be configured to emit the at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. It shall be noted, however, that an active beacon device, as e.g. shown in the setup of Figure 1, may also be used, and, thus, that setups with no integrated illumination source 136 are also feasible. The scanning system 134 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112 and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Figure 4 is shown in Figures 1 and 2. Thus, the detector 110, besides the sensor element 118, comprises the at least one evaluation device 134, having, e.g., the at least one center detector 136, the at least one summing device 138, the at least one combining device 140, the at least one divider 142, the at least one position evaluation device 143 and/or combinations thereof. These components, which may optionally be present, are symbolically depicted in Figure 4. The components of the evaluation device 134 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the components of the evaluation device 134 and one or more of the components of the sensor element 118 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Figure 4. Further, the one or more connectors 156 may

comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 134 may fully or partially be integrated into one or both of the sensor element 118 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 134 may
5 fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object
10 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the sensor element 118. The sensor
15 118, wherein one or more of the sensor elements 118 may be provided, may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 may be comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard
20 to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. A coordinate system 168 may be defined, in which a direction parallel or antiparallel to the optical axis 126, such as parallel or antiparallel to the direction of view 166, may be defined as a transversal direction. In the coordinate system 168, symbolically depicted in Figure 4,
25 a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the sensor element 118 as well as, optionally, one or more further optical sensors. A non-branched beam path may be used or, alternatively, a branched
30 beam path may be possible, with, e.g., additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining the transversal coordinates of the object 112 and/or parts thereof. As outlined above, however, in the context of Figure 3, the at least one transversal coordinate may also be determined by the sensor element 118 itself, such as by determining
35 the transversal coordinates of the center of the light spot 132.

One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as outlined above in the context
40 of Figures 1 to 3, the evaluation device 134 may be configured to evaluate the sensor signals provided by the optical sensors 122 of the matrix 120 of the sensor element 118. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 122 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110,

specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may comprise active beacon devices with an integrated illumination source such as light-emitting diodes. In case the illumination source 136 is used, the beacon devices do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the sensor element 118. For details of the evaluation, reference may be made to Figures 1 to 3 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 151, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Figure 4, the machine 170 may be a computer and/or may comprise a computer.

Other embodiments are feasible. The evaluation device 134 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Figure 4 also depicts an example of a tracking system 153, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 153 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may fully or partially be integrated into the machine 170, specifically the computer, as indicated in Figure 4, and/or into the evaluation device 134.

Similarly, as outlined above, the human-machine interface 151 may form part of an entertainment device 152. The machine 170, specifically the computer, may also form part of the entertainment device 152. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Figure 5, experimental data are shown which demonstrate an exemplary embodiment of the present invention in terms of measurement data. The figure shows a combined signal Q as a function of a longitudinal coordinate z, given in millimeters, for various illumination intensities.

In order to gain the experimental data shown in Figure 5, an experimental setup was used with a sensor element 118 formed by a Basler AC 1920-40GC camera, with a transfer device 128 formed by a Nikkor 50 mm lens.

As a beacon device 114, a light-emitting diode (LED) was used, having a central nominal wavelength of 532 nm. A diffusor was used in front of the LED and a diaphragm, in order to provide a

well-defined light-emitting area having a diameter of 5 mm. The intensity of the LED was varied, by varying a drive current of the LED between 5 mA and 150 mA.

In the experiments, the distance z between the LED and the lens was varied from 300 mm to 1700 mm. The signal of the Basler AC 1920-40GC camera was evaluated by the following procedure. As a center signal, an accumulated signal of an inner circle having a radius of 15 pixels around the optical axis was determined, with the light spot centered at the optical axis. As a sum signal, the sum of the signals of all pixels of the camera within the light spot was generated. A combined signal was formed, by dividing the sum signal by the center signal.

In Figure 5, the solid curves, overlapping, show the combined signal curves for LED currents of 150 mA, 125 mA, 100 mA and 50 mA. As can be seen, there are basically no differences between these curves, within the whole measurement range. This experiment clearly shows that the combined signal is widely independent on the total power of the light beam. Only at lower intensities, the curves may be distinguished. Thus, the dotted curve shows measurements at an LED current of 25 mA, the dashed-dotted line at an LED current of 10 mA, and the dashed line at an LED current of 5 mA. Still, even at these low intensities, the measurement curves are very close to the solid curves, which shows the high tolerances of the measurement.

In Fig. 6A and 6B, an alternative embodiment of the detector 110 is shown which is a modification of the setup shown in Fig. 2. Thus, for most elements and optional details as well as further elements not shown in the schematic Figures 6A and 6B, reference may be made to the description of Fig. 2 above.

In Figure 2, the illumination light beam 146, as discussed above, preferably travels along the optical axis 126, i.e. parallel to the optical axis 126 or even on the optical axis 126. In the setup, the position of the center of the light spot 132 typically does not depend on the z -coordinate of the object 112, such as on a distance between the object 112 and the detector 110. In other words, the diameter or equivalent diameter of the light spot 132 changes with the distance between the object 112 and the detector 110 whereas, typically, the position of the light spot 132 on the matrix 120 does not.

Contrarily, in Fig. 6A and 6B, a setup of the detector 110 is shown in which an illumination light beam 146 travels off-axis, i.e. one or both of at an angle other than 0° with the optical axis 126 or parallel to the optical axis 126 but shifted from the optical axis 126. This embodiment, as will be discussed in further detail below, demonstrates that the method according to the present invention can be further enhanced by increasing the z -dependency of a combined sensor signal. Thus, in Fig. 6A, a side view is shown with two different positions of the object 112, i.e. a first position at z_1 , drawn in solid lines, and a second position at z_2 , drawn in dashed lines. As can be seen, the illumination light beam 146 which, as an example, propagates at an angle of 5° to 30° , e.g. 10° to 20° , with the optical axis 126, hits the object 112 in both cases at different positions. From these points of the object 112 illuminated by the illumination light beam 146, light beams 116 propagate towards the detector 110, wherein, again, the light beam 116 for the

object 112 being located at position z_1 is drawn in solid lines, wherein the light beam 116 for the object 112 being located at position z_2 is drawn in dashed lines.

In Fig. 6B, the matrix 120 is shown in an enlarged fashion. As can be seen in this setup, the position of the light spot 132 moves with the longitudinal position z of the object 112. Thus, not only is the size of the light spot 132 affected by the longitudinal position z but also is the position on the matrix 120 of the light spot 132 changed. In Fig. 6B, this movement of the light spot 132 is denoted by arrow z .

Consequently, by this movement of the light spot 132, the z -dependency of a combined signal taking into account at least two sensor signals of the optical sensor 122 may be increased.

In the situation shown in Fig. 2, the position of the light spot 132 does not depend on z . With a change in z , depending on the optical situation, the spot will become larger or smaller, such as by becoming more diffuse or more focused. In case the spot size increases and the spot becomes more diffuse such that the combined signal $Q = s_{\text{center}}/s_{\text{sum}}$ decreases.

Contrarily, in the situation of Fig. 6A, both the size and the position of the light spot 132 are dependent on the z -coordinate. Thus, the tendency of the z -dependency of the combined signal such as the combined signal Q will be increased. In the situation of Figure 2, depending on the z -coordinate, the sensor signal of at least one optical sensor of the matrix will increase and simultaneously the sensor signal of at least one different sensor will decrease, resulting in the z -dependent combined signal Q . In the situation of Fig. 6A, the position dependency of the light spot 132 can result in three different situations depending on the relative position of light source, optical axis, and sensor, especially in case the center signal and sum signal are determined from one or more optical sensors, whereas the optical sensors used for determining the center signal and sum signal are kept fixed, independent of the z -coordinate and/or position of the light spot: Firstly, the position dependency of the light spot may result in a further decrease of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a further increase of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Secondly, the position dependency of the light spot may result in a reduced decrease or even increase of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a reduced increase or even decrease of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Thirdly, the position dependency of the light spot may be as such that the z -dependence of the sensor signals is largely unchanged compared to the situation in Figure 2. However, according to the present invention, object distance is not determined from the position of the light spot on a sensor as done in triangulation methods. Instead, movement of the light spot 132 on the matrix 120 may be used to enhance dynamic of the sensor signals and or the resulting combined signal Q which may result in an enhanced dynamic of the z -dependency.

Figure 6C shows a comparison of two experimental setups using a detector setup according to Figure 6A with a Bi-cell as optical sensors 120 with two light sensitive areas. In a first experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 132 may move in parallel to the linear boundary of the two optical sensors 120 of the Bi-cell along a direction of movement 174 in dependence of the object distance. Since the direction of movement 174 of the light spot 132 is in parallel to the linear boundary of the two light sensitive areas in dependence of the object distance, the resulting sensor signals are identical to a situation with no movement of the light spot 132 depending on object distance as shown in Figure 2. In a second experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 132 may move as such that the distance of the center of the light spot 132 to the boundary of the two optical sensors 120 of the Bi-cell changes in dependence of the object distance such as a movement orthogonal to the boundary of the two optical sensors 120 such as a movement along a direction of movement 176 in dependence of the object distance. The detector setup allowing movement of the light spot 132 may be a modification of the setup shown in Figure 6A. Thus, for most elements and optional details as well as further elements, reference may be made to the description of Fig. 6A above. In Figure 6C, the optical sensors 120 may be 1x2 matrix.

Figure 6D shows experimental results of the comparison of the two experimental setups using a detector setup according to Figure 6A, allowing movement of the light spot 132 according to Figure 6C with movement of the light spot depending on the object distance along directions of movement 174 and 176. Curve 178 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup allowing movement of the light spot 132 along a direction of movement 174 as shown in Figure 6C which is in parallel to the boundary of the optical sensors of the Bi-Cell and, which is a situation equivalent to Figure 2 without a movement of the light spot depending on the object distance. Curve 180 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup according to Figure 6A and using a detector setup allowing movement of the light spot 186 with movement of the light spot 132 according to Figure 6C with movement of the light spot depending on the object distance along a direction of movement 176. The experimental setup was as follows: The optical sensors 122 may be a 1x2 matrix, in particular a Si-Bi-Cell. The illumination source 144 may be a 950 nm laser with a spot size of 4 mm. The transfer device 128 may have a focal length of 20 mm, e.g. a lens available as Thorlabs Asphere, $f = 20$ mm. The distance of the object 112 was varied from 0 to 3000 mm. Determination of the longitudinal coordinate z may be possible without allowing movement of the light spot 132. In particular, according to the present invention, movement of the light spot may not be essential for determination of the longitudinal coordinate z . With the detector setup allowing movement of the light spot 132 along a direction 174 or without any movement determination of object distance is possible at very small distance, whereas with movement along a direction 176 determination of object distance is possible for object distance such as distances greater than 500mm.

Figures 7A and 7B show an exemplary embodiment of a detector 110 with a small baseline 182. Figure 7A shows a side view of the detector 110. The detector 110 may comprise the at least one illumination source 144. The illumination source 144 may be adapted to illuminate the object 112 with at least one illumination light beam 146. The illumination light beam 146 may be fully or partially reflected by the object and may travel back towards the optical sensor 122. The illumination source 144 may be arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis 126. The illumination source 144 may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis 126 do not intersect in space. The illumination source 144 and the optical axis 126 may be separated by a small baseline 182. The illumination source 144 may be spaced apart from the optical axis 126 by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and transfer device. Figure 7B shows a front view on the illumination light beam 144 and the transfer device 128. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the detector 110 may be a compact device without further optical elements, wherein the illumination source 144 may be placed as close as possible to an edge of the transfer device 128. Thus, the baseline 182 may be close to half a diameter of the transfer device 128, in particular the lens diameter and housings of lens and light source. The illumination source 144 may be arranged such that the baseline 182 is as small as possible. For example, a distance from the center of the transfer device 128 to the illumination source 144, in particular along a connecting line from the center of the transfer device 128 to the illumination source 144, may be preferably less than 2.5 times a distance from the center of the transfer device 128 to an edge if the transfer device 128, more preferably less than 1.5 times the distance center to edge of the transfer device 128, and most preferably less 1 times the distance center to edge of the transfer device 128.

Referring to Figure 8, the detector 110 such as the detector as described with respect to Figures 1 to 7 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 122. For example, the detector 110 may comprise a plurality of optical sensors arranged in the matrix 120. The detector 110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector 110 may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. As outlined above, this principle is called Distance by Photon Ratio (DPR). In one embodiment, the light beam 116 may illuminate the sensor element with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 134 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are areas of the beam profile at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . A_1 may correspond to a full or complete area of a feature point on the optical sensors. A_2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 8, the light beam 116 propagating from the object 112 to the detector 110 may illuminate the matrix 120 with at least one line pattern 184. The line pattern 184 may be generated by the object 112 for example in response to an illumination by the at least one illumination source 144 with an illumination pattern comprising at least one illumination line pattern. A_1 may correspond to an area with a full line width of the line pattern 184 in the matrix 120. The line pattern 184 in the matrix 120 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a line width in the matrix 120 is increased. In particular, the line width of the line pattern 184 in the matrix 120 may change from one column to another column. A_2 may be a central area of the line pattern 184 in the matrix 120. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern 184 may be segmented in the matrix 120. Each of the columns may comprise center information of intensity in the central area of the line pattern 184 and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern 184.

Referring to Figures 9A and 9B, as outlined above, the detector 110 may comprise the at least one matrix 120 of optical sensors 122. With the aid of such a pixelated imaging device, a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates as shown in Figure 9A. Accordingly, the parameterization of a single line would be given by $\omega = x \cos(\theta) + y \sin(\theta)$. The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega, \theta)}\{f(x, y)\} = \iint_{-\infty}^{\infty} f(x, y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy$$

where δ denotes the Dirac delta function and $f(x, y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is then given by

$$R = \frac{\Re_{(\omega, \theta)}\{f'(x, y)\}}{\Re_{(\omega, \theta)}\{f(x, y)\}}$$

with $f'(x, y)$ as the overshined image region highlighted in Fig. 9. The variation of θ may yield different ratios R for skewed object surfaces at a certain distance. It is sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

- 5 Figures 10A to 10C show three embodiments of a hexagonal illumination pattern. The illumination source 144 may be adapted to generate at least one illumination pattern for illuminating the object 112. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. Specifically, the illumination source 144 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor
- 10 lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination pattern may comprise at least one feature such as a point or symbol. The illumination pattern may comprise a plurality of features. The illumination pattern may comprise an arrangement of periodic or non-periodic features. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one
- 15 illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasi-periodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines; at least one fringe pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern
- 20 may comprise as much as possible features per area such that hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.
- 25
- 30 The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on an epipolar line. As shown in Figure 10A, the illumination pattern may comprise at least one hexagonal pattern, wherein the individual points are positioned on epipolar lines 186. As shown in Figure 10B, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Such a positioning
- 35 of the illumination features allows enhancing distance between the individual points on each epipolar line. For example as shown in Figure 10C, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the
- 40 distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other. Such a positioning allows to enhance the number of possible features per area.

Figure 11 shows an embodiment of a scanning system 154. The scanning system 154 may be adapted as a line scanning device. In particular, the scanning system 154 may comprise at least one sensor line or row of optical sensors 113. Furthermore, the scanning system may comprise the at least one transfer device 128 and the at least one illumination source 144. Triangulation systems require a sufficient baseline, however due to the baseline in the near field no detection may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. Further, these near field and far field problems can be overcome by using the scanning system 154 of Figure 11. The scanning system 154 may be adapted to detect a plurality of light beams 116 propagating from the object 112 to the scanning device 154 on the CMOS line. The light beams 116 may be generated at different position on the object 112 or by movement of the object 112. The scanning system 154 may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the combined signal Q as described above.

List of reference numbers

110	detector
112	object
114	beacon device
116	light beam
118	sensor element
120	matrix
122	optical sensor
124	light-sensitive area
126	optical axis
128	transfer device
130	focal point
132	light spot
134	evaluation device
136	center detector
138	summing device
140	combining device
142	divider
143	position evaluation device
144	illumination source
146	illumination light beam
148	reflective element
149	camera
150	detector system
151	human-machine interface
152	entertainment device
153	tracking system
154	scanning system
156	connector
158	housing
160	control device
162	user
164	opening
166	direction of view
168	coordinate system
170	machine
172	track controller
174	direction of movement
176	direction of movement
178	curve
180	curve
182	baseline

- 184 Line pattern
- 186 Epipolar line

Patent claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) having
- 5 - at least one sensor element (118) having a matrix (120) of optical sensors (122), the optical sensors (122) each having a light-sensitive area (124), wherein each optical sensor (122) is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area (124) by at least one light beam (116) propagating from the object (112) to the detector (110);
- 10 - at least one evaluation device (134) configured for evaluating the sensor signals, by
- a) determining at least one optical sensor (122) having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors (122) of the matrix (120) and forming at least one sum signal;
- 15 c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.
- 20 2. The detector (110) according to the preceding claim, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors (122) containing the optical sensor (122) having the highest sensor signal and a predetermined group of neighboring optical sensors (122); a
- 25 sum of sensor signals from a group of optical sensors (122) containing the optical sensor (122) having the highest sensor signal and a predetermined group of neighboring optical sensors (122); a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an inte-
- 30 gral of sensor signals from a group of optical sensors (122) containing the optical sensor (122) having the highest sensor signal and a predetermined group of neighboring optical sensors (122); an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a pre-
- 35 determined threshold.
3. The detector (110) according to any one of the preceding claims, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix (120); a sum of all sensor signals of the matrix (120); an integral of all sensor signals of the matrix (120); an average over all sensor signals of the matrix (120) except for sensor signals from those
- 40 optical sensors (122) contributing to the center signal; a sum of all sensor signals of the matrix (120) except for sensor signals from those optical sensors (122) contributing to the center signal; an integral of all sensor signals of the matrix (120) except for sensor signals from those optical sensors (122) contributing to the center signal; a sum of sensor signals of optical sensors (122) within a predetermined range from the optical sensor (122) having the highest sensor

signal; an integral of sensor signals of optical sensors (122) within a predetermined range from the optical sensor (122) having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors (122) being located within a predetermined range from the optical sensor (122) having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors (122) being located within a predetermined range from the optical sensor (122) having the highest sensor signal.

4. The detector (110) according to any one of the preceding claims, wherein the combined signal is a combined signal Q, derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

5. The detector (110) according to the preceding claim, wherein the evaluation device (134) is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

6. The detector (110) according to any one of the two preceding claims, wherein the evaluation device (134) comprises at least one divider (142), wherein the divider (142) is configured for deriving the combined signal.

7. The detector (110) according to any one of the preceding claims, wherein the evaluation device (134) is further configured for determining at least one transversal coordinate of the object (112) by evaluating a transversal position of the at least one optical sensor (122) having the highest sensor signal.

8. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (144), wherein the illumination source (144) is adapted to illuminate the object (112) with at least one illumination light beam, wherein the illumination source (144) is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis (126).

9. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises the at least one illumination source (144), wherein the illumination source (144) and the optical axis (126) are separated by a small baseline (182).

10. A detector system (150) for determining a position of at least one object (112), the detector system (150) comprising at least one detector (110) according to any one of the preceding claims, the detector system (150) further comprising at least one beacon device (114) adapted to direct at least one light beam (116) towards the detector (110), wherein the beacon device (114) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).

11. A human-machine interface (151) for exchanging at least one item of information between a user (162) and a machine (170), wherein the human-machine interface (151) comprises at least one detector system (150) according to the preceding claim, wherein the at least one beacon device (114) is adapted to be at least one of directly or indirectly attached to the user (162) and held by the user (162), wherein the human-machine interface (151) is designed to determine at least one position of the user (162) by means of the detector system (150), wherein the human-machine interface (151) is designed to assign to the position at least one item of information.
12. An entertainment device (152) for carrying out at least one entertainment function, wherein the entertainment device (152) comprises at least one human-machine interface (151) according to the preceding claim, wherein the entertainment device (152) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (151), wherein the entertainment device (152) is designed to vary the entertainment function in accordance with the information.
13. A tracking system (153) for tracking a position of at least one movable object (112), the tracking system (153) comprising at least one detector system (150) according to any one of the preceding claims referring to a detector system (150), the tracking system (153) further comprising at least one track controller (172), wherein the track controller (172) is adapted to track a series of positions of the object (112) at specific points in time.
14. A scanning system (154) for determining a depth profile of a scenery, the scanning system (154) comprising at least one detector (110) according to any of the preceding claims referring to a detector (110), the scanning system (154) further comprising at least one illumination source (144) adapted to scan the scenery with at least one light beam (146, 116).
15. A camera (149) for imaging at least one object (112), the camera (149) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
16. A method for determining a position of at least one object (112) by using at least one detector (110), the method comprising the following steps:
- illuminating at least one sensor element (118) of the detector (110) with at least one light beam (116) propagating from the object (112) to the detector (110), the detector (110) having a matrix (120) of optical sensors (122), the optical sensors (122) each having a light-sensitive area (124), wherein each optical sensor (122) generates at least one sensor signal in response to the illumination;
 - evaluating the sensor signals, by
 - a) determining at least one optical sensor (122) having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors (122) of the matrix (120) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and

d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.

- 5 17. The method according to the preceding claim, wherein the combined signal is a combined signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.
- 10 18. A use of the detector (110) according to any one of the preceding claims relating to a detector (110), for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface (151) application; a tracking application; a photography application; a logistics application; a machine vision application; a
- 15 robotics application; a quality control application; a manufacturing application.

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FIG.1

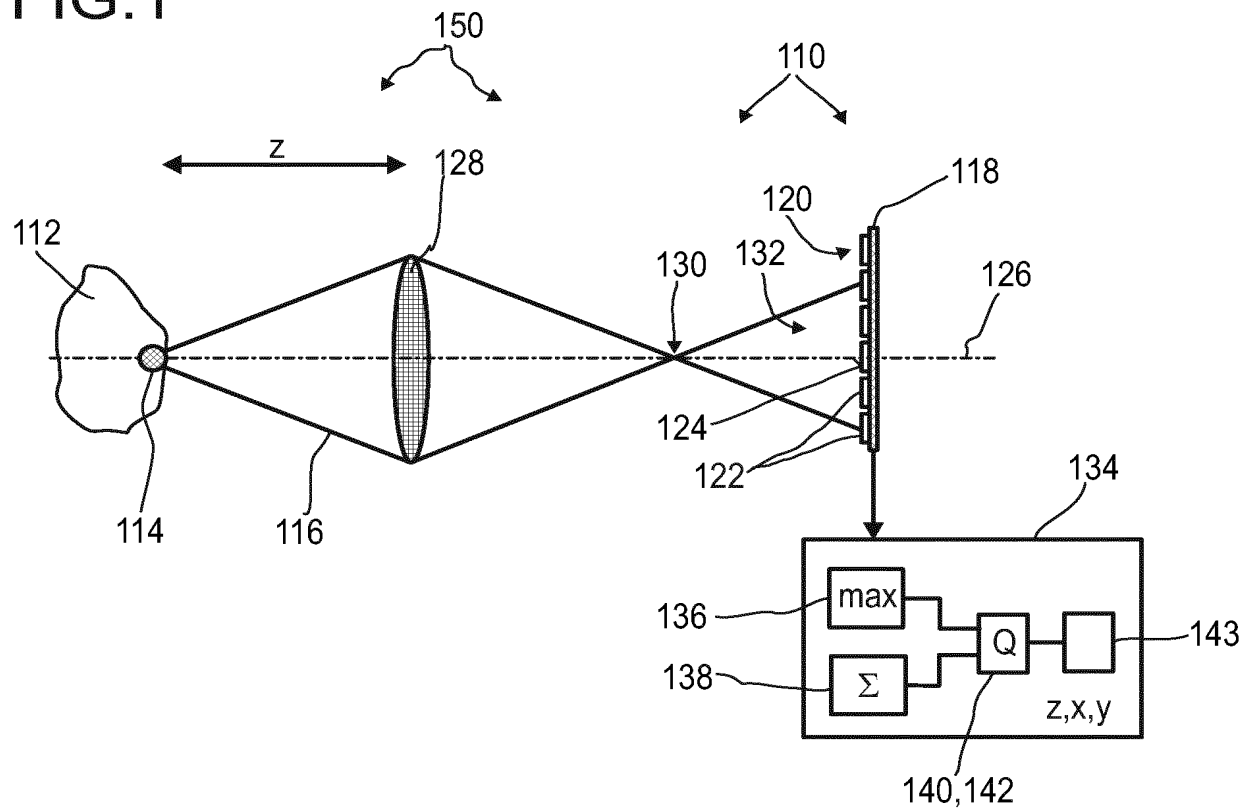


FIG.2

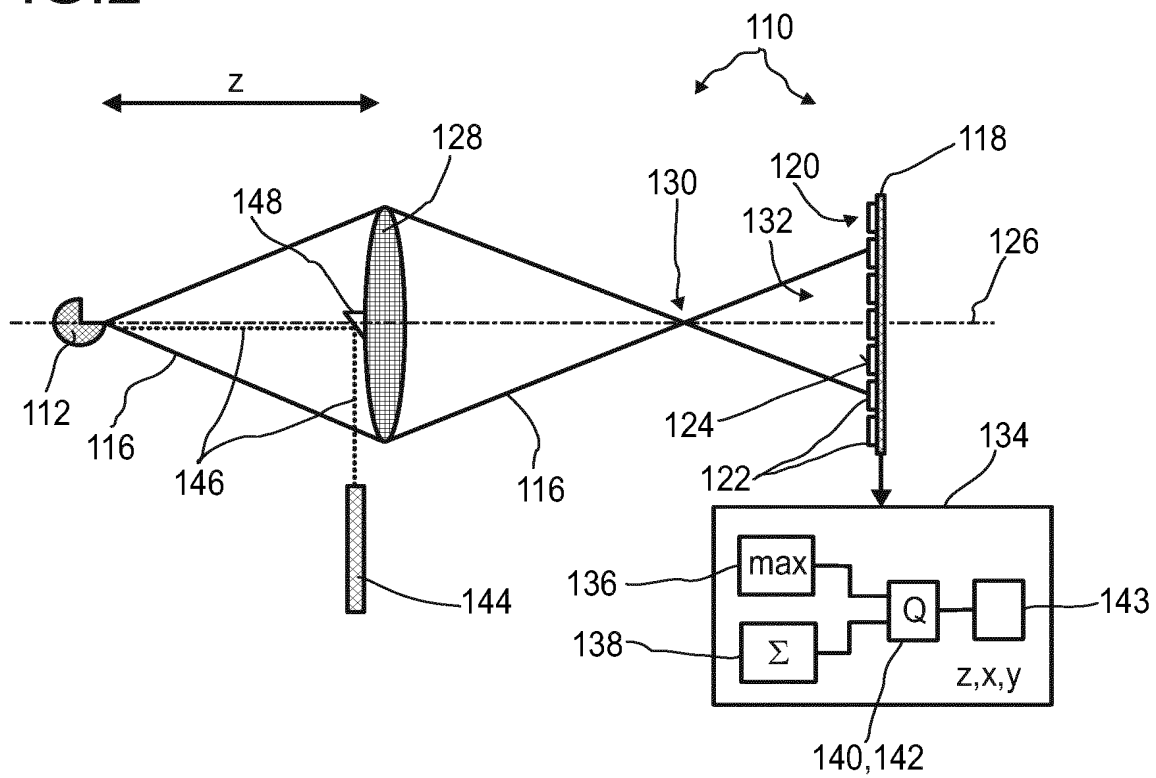


FIG.3

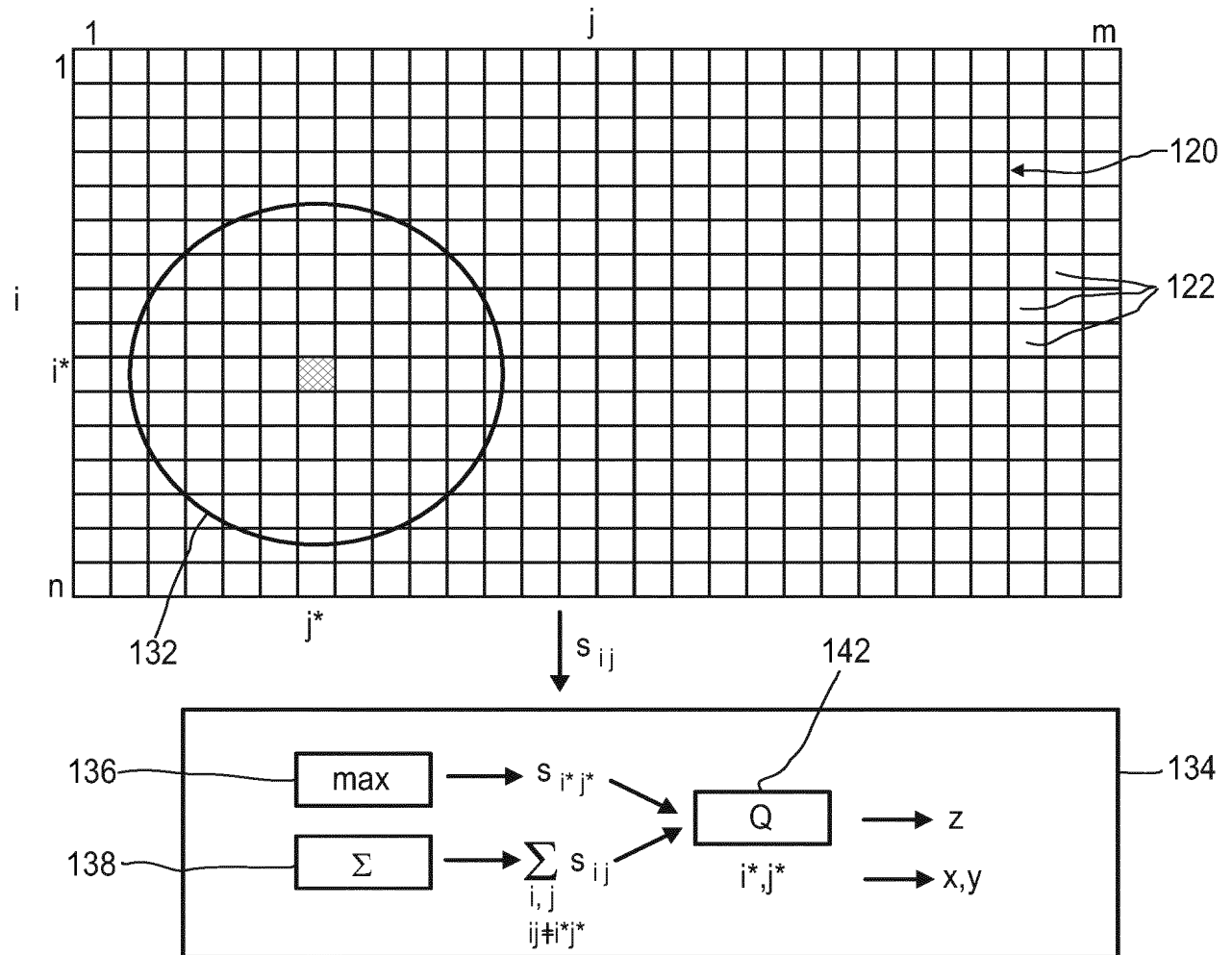
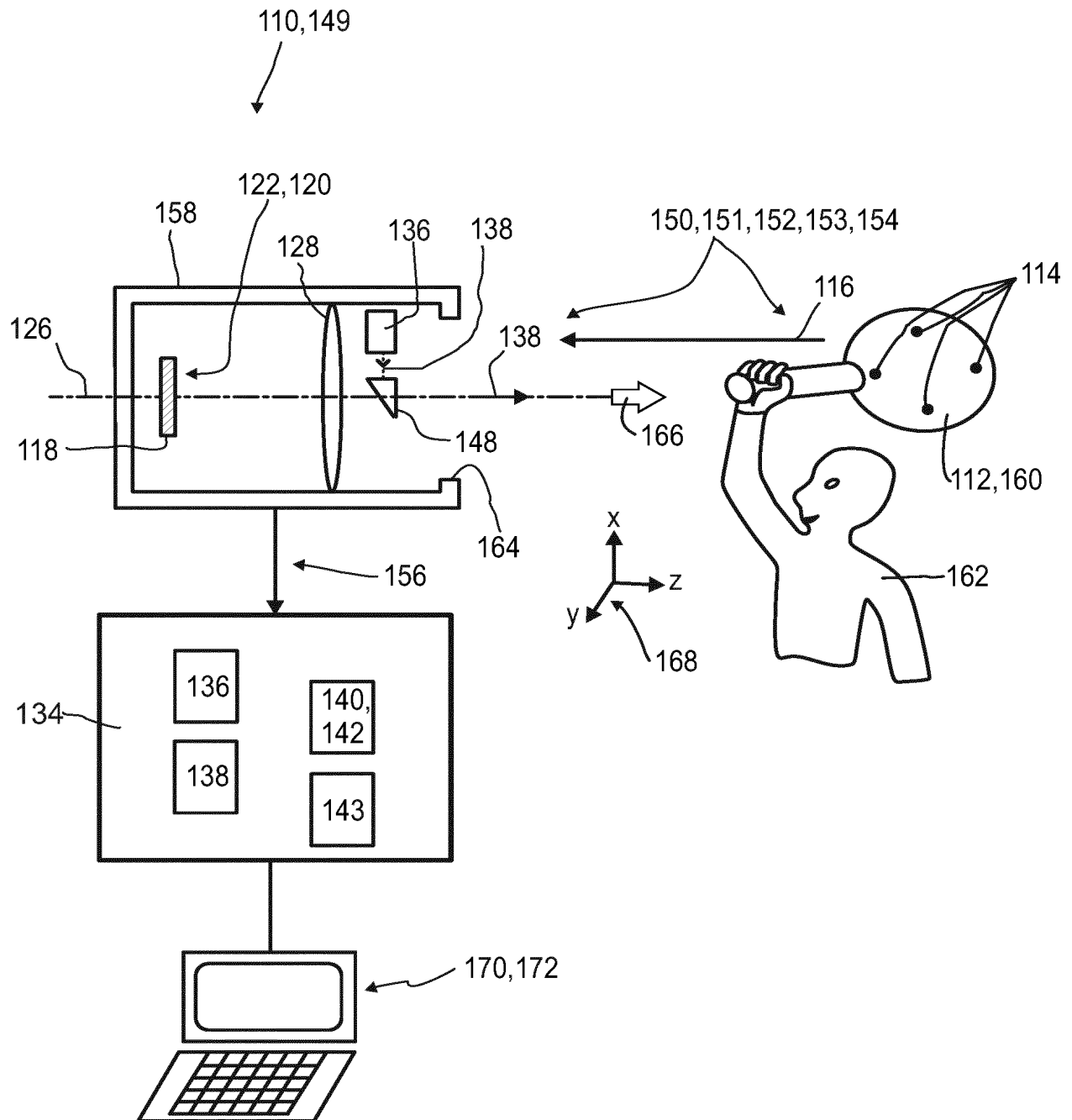


FIG.4



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FIG.5

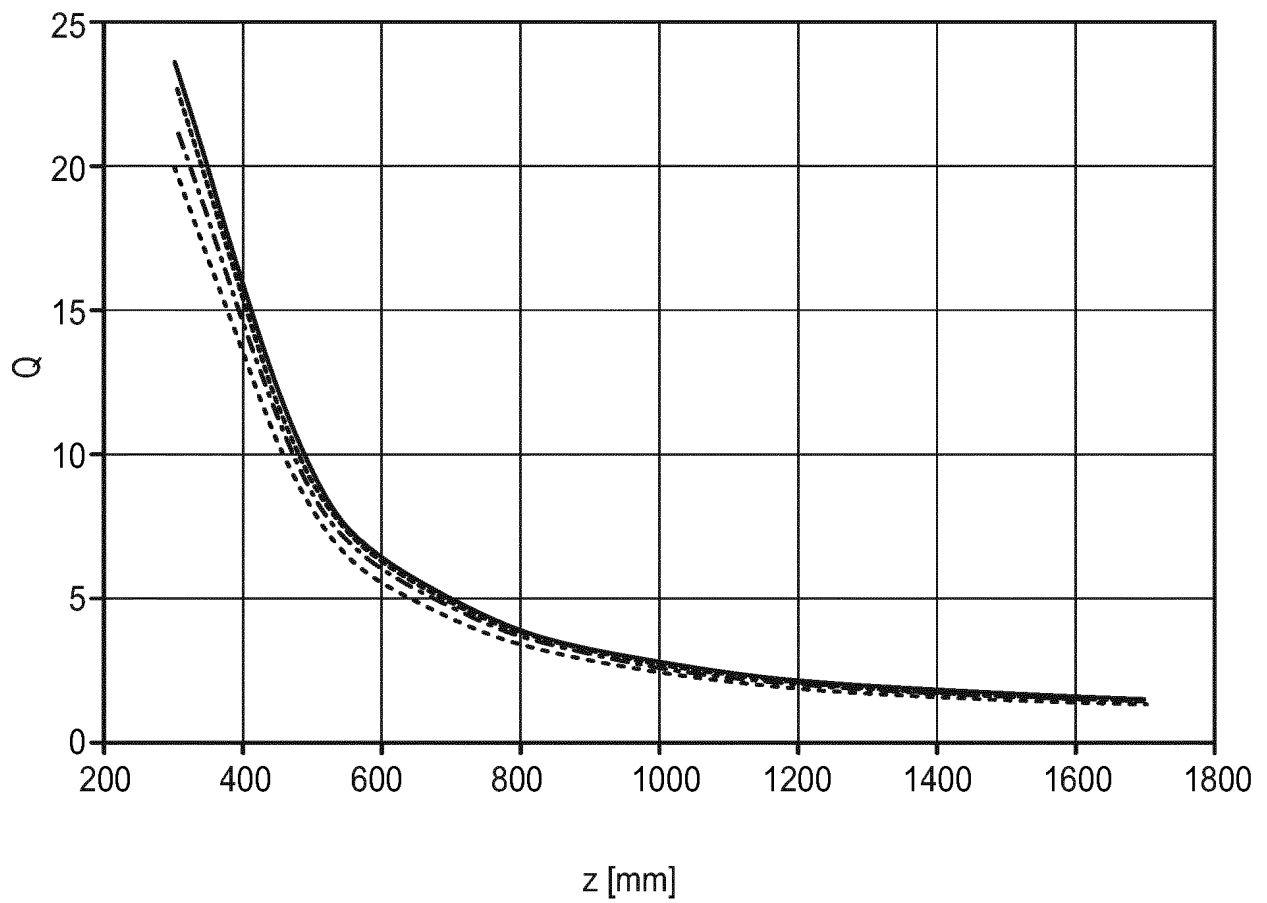


FIG.6A

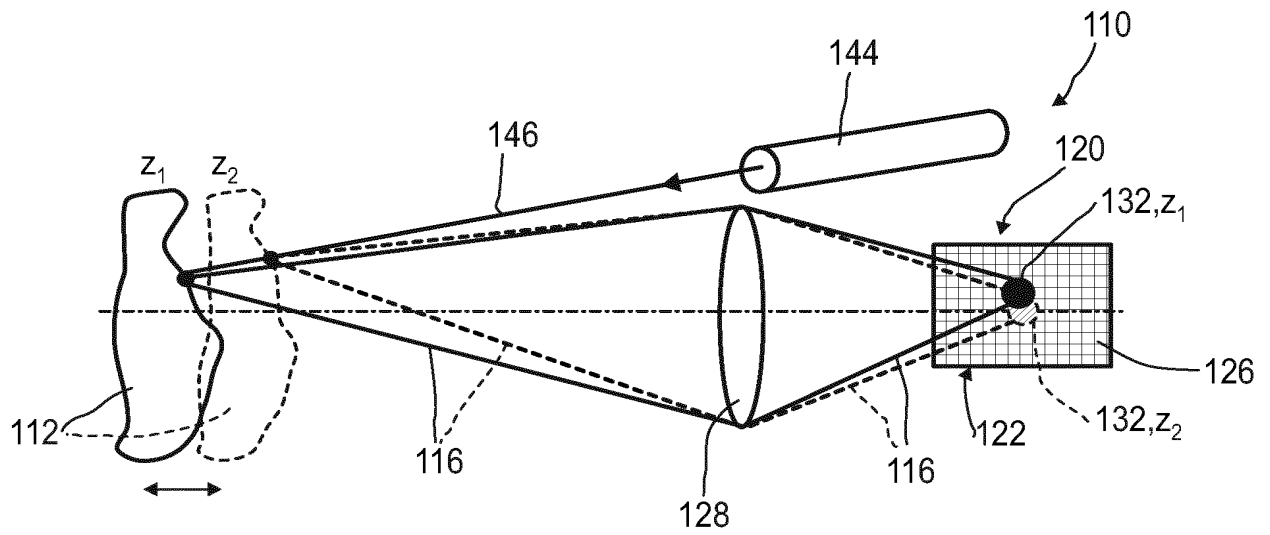


FIG.6B

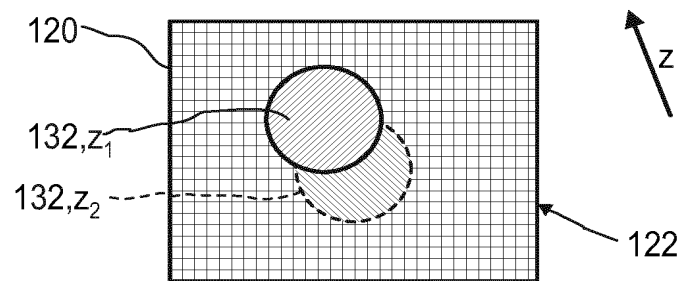


FIG.6C

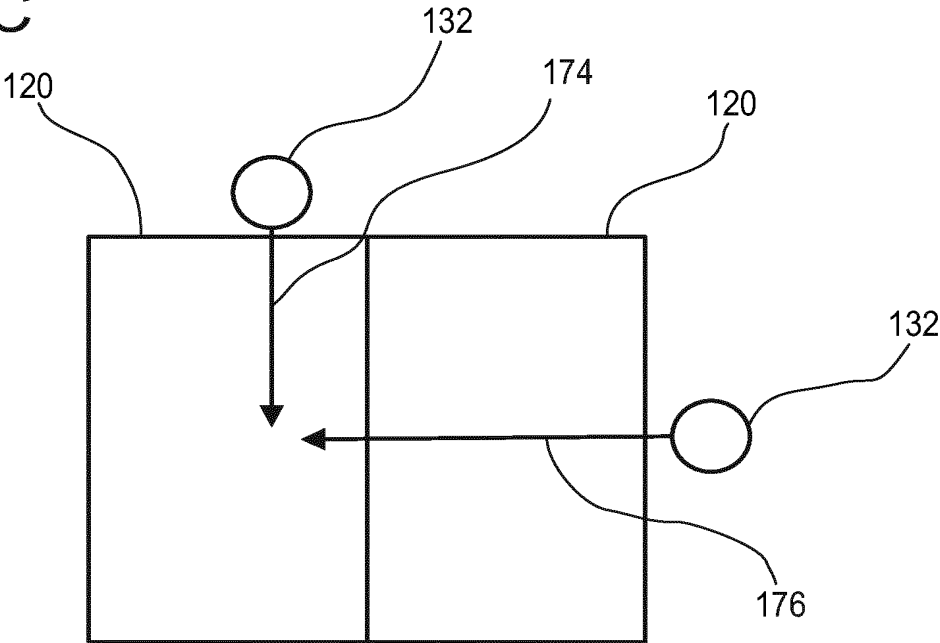
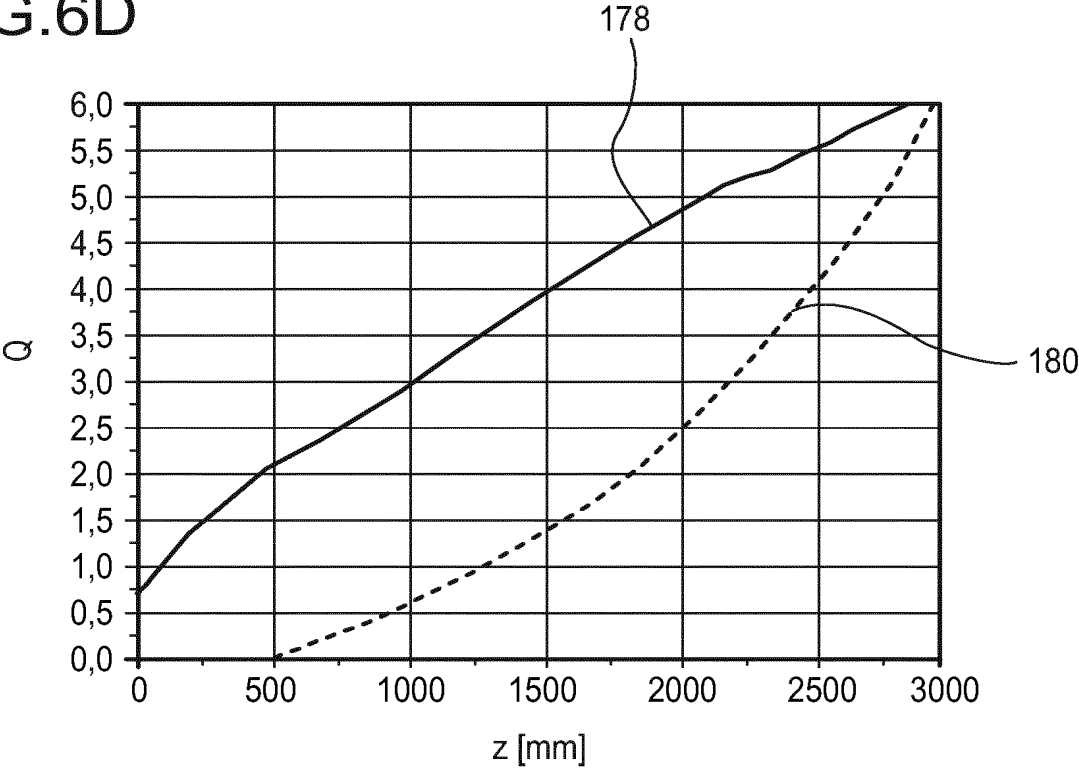


FIG.6D



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FIG.7A

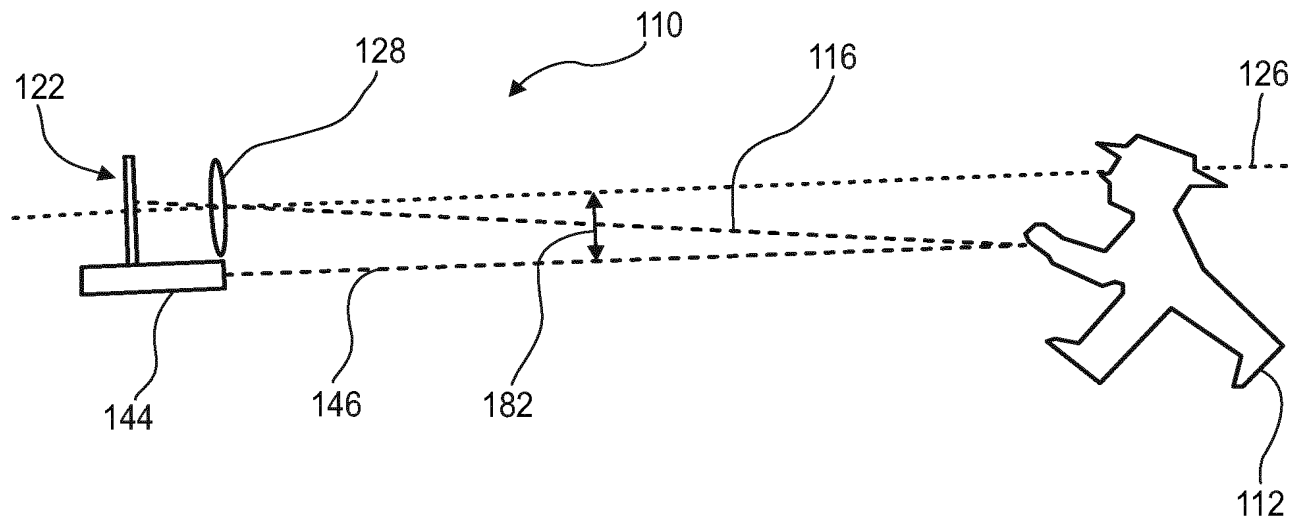


FIG.7B

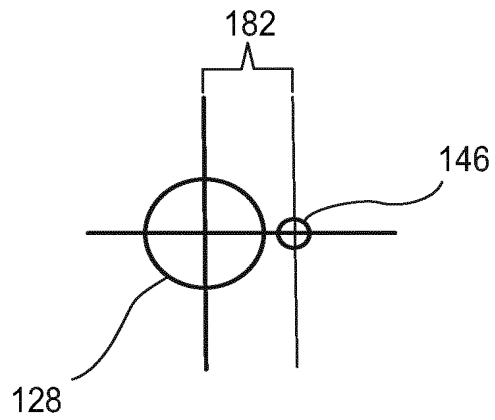


FIG.8

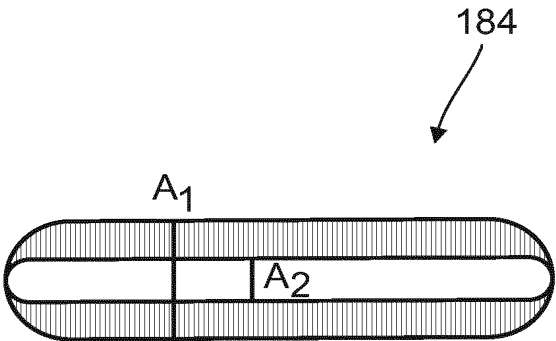


FIG.9A

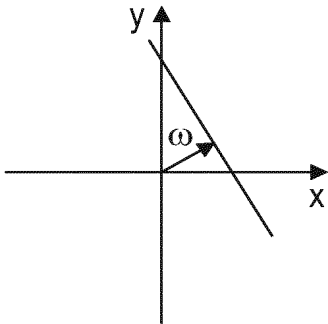


FIG.9B

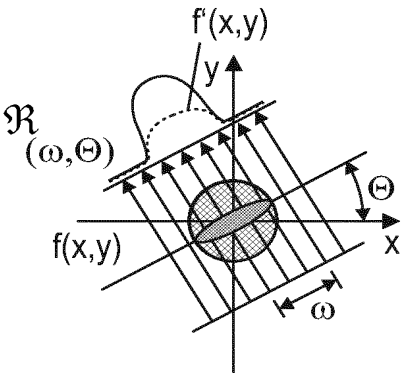


FIG.10A

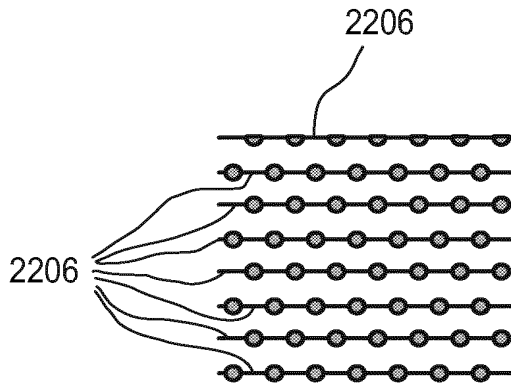


FIG.10B

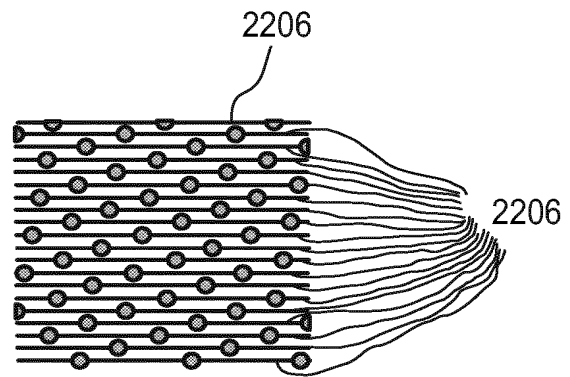


FIG.10C

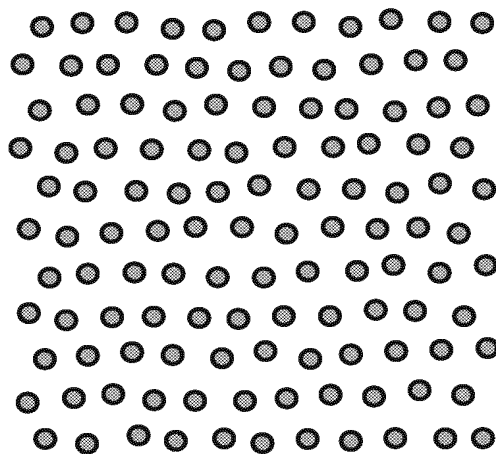
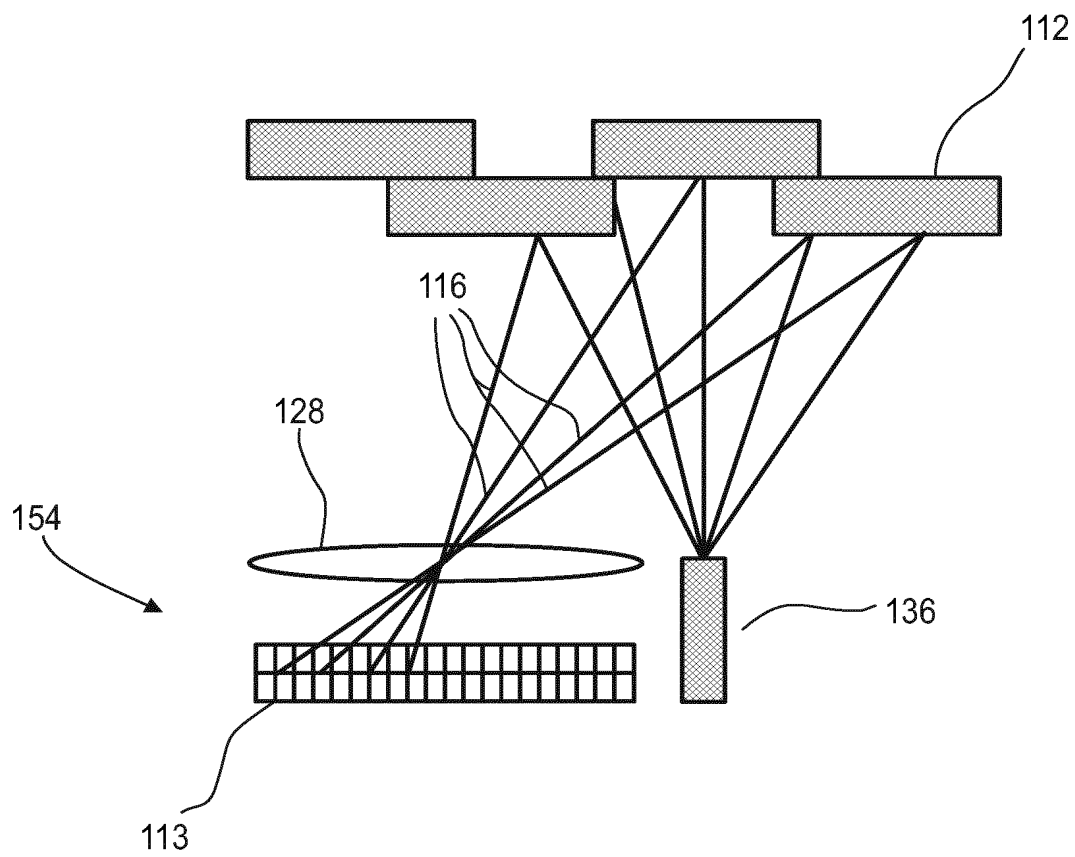


FIG. 11



INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2017/079577

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S17/46 G01S7/481 G01S5/16 G01S7/491 G01S11/12
G01S7/48

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data, INSPEC

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2006/044546 A1 (LEWIN ANDREW C [GB] ET AL) 2 March 2006 (2006-03-02)	1-18
Y	paragraphs [0001], [0030], [0093], [0110], [0118], [0119], [0121] - paragraphs [0123], [0129]; figures 2a, 2b	10
X	US 2008/130005 A1 (WASLOWSKI KAI [DE] ET AL) 5 June 2008 (2008-06-05)	1-3, 16
	paragraphs [0011], [0014] - paragraphs [0016], [0030], [0037], [0041], [0049], [0051], [0054], [0063], [0086], [0091]; figure 1	
Y	US 2015/286340 A1 (SEND ROBERT [DE] ET AL) 8 October 2015 (2015-10-08)	10
	cited in the application paragraph [0519]	



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

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"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

2 February 2018

Date of mailing of the international search report

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Beer, Mark

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2017/079577

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		US 2017082486 A1 23-03-2017	
		WO 2015024871 A1 26-02-2015	



(51) International Patent Classification:

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G01S 17/46 (2006.01) *G01S* 7/48 (2006.01)
G01S 3/783 (2006.01) *G01S* 11/12 (2006.01)

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17195427.4 09 October 2017 (09.10.2017) EP

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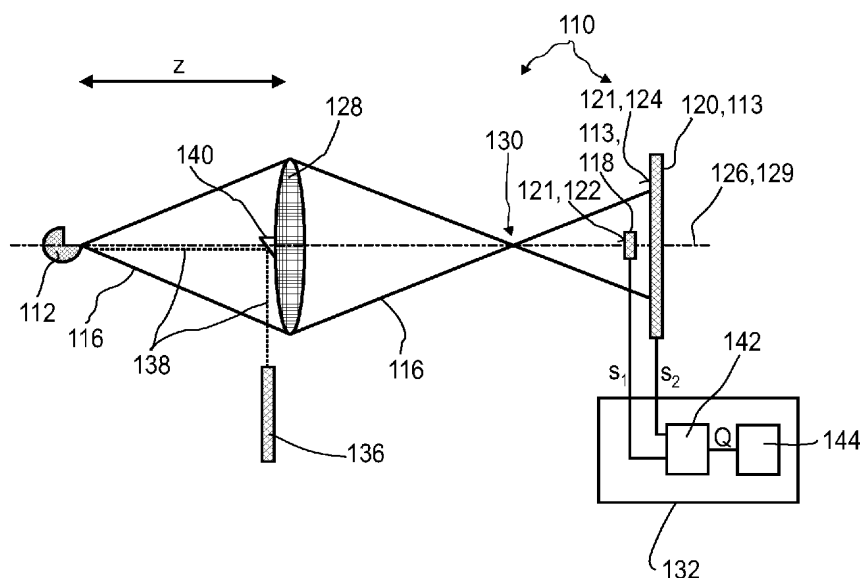
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(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.2



(57) Abstract: A detector (110, 1110, 2110) for determining a position of at least one object (112) is proposed. The detector (110, 1110, 2110) comprises: at least one transfer device (128, 1128), wherein the transfer device (128, 1128) has at least one focal length in response to at least one incident light beam (116, 1116) propagating from the object (112, 1112) to the detector (110, 1110, 2110); at least two optical sensors (113, 1118, 1120), wherein each optical sensor (113, 1118, 1120) has at least one light sensitive area (121, 1122, 1124), wherein each optical sensor (113, 1118, 1120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam (116, 1116), at least one evaluation device (132, 1132) being configured for determining at least one longitudinal coordinate z of the object (112, 1112) by evaluating a quotient signal Q from the sensor signals. The detector is adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from

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(81) **Designated States** (*unless otherwise indicated, for every kind of national protection available*): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

(84) **Designated States** (*unless otherwise indicated, for every kind of regional protection available*): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:

- *without international search report and to be republished upon receipt of that report (Rule 48.2(g))*

Detector for optically detecting at least one object

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, systems, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

20 Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultra-violet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1, US 2007/0176165 A1, WO 2009/013282 A1, WO 2012/110924 A1, WO 2014/097181 A1, WO 2015/024871 A1, US 4,767,211, WO 2014/198629 A1, WO 2014/198626 A1, WO 2014/198629 A1 and WO 2014/198625 A1, the full content of which is herewith included by reference. Further, referring to potential materials and optical sensors which may also be employed in the context of the present invention, reference may be made to European patent applications EP 15 153 215.7, filed on January 30, 2015, EP 15 157 363.1, filed on March 3, 2015, EP 15 164 653.6, filed on April 22, 2015, EP 15177275.3, filed on July 17, 2015, EP 15180354.1 and EP 15180353.3, both filed on August 10, 2015, and EP 15 185 005.4, filed on September 14, 2015, EP 15 196 238.8 and EP 15 196 239.6, both filed on November 25, 2015, EP 15 197 744.4, filed on December 3, 2015, the full content of all of which is herewith also included by reference.

Further, reference may be made to detector concepts comparing signals of at least two different sources for determining a position of an object. Thus, as an example, reference may be made to EP 16155834.1, EP 16155835.8 or EP 16155845.7, all filed on February 16, 2016, the full

disclosure of which is herewith included by reference. Further, P. Bartu, R. Koeppe, N. Arnold, A. Neulinger, L. Fallon, and S. Bauer, Conformable large-area position-sensitive photodetectors based on luminescence collecting silicone waveguides, J. Appl. Phys. 107, 123101 (2010), describe a kind of position sensitive detector (PSD) device which might be suitable for large areas and on curved surfaces.

For example, US 5,323,222 A describes a distance measuring device contains a light-projecting section for projecting luminous flux onto a subject. A light-receiving section is placed the base length away from the light-projecting section, receives the reflected rays of light from the subject, and produces a photoelectric current according to the amount of light received. The light-receiving section is divided into at least four light-receiving elements by a first dividing line that crosses the base length at right angles and by a second dividing line that crosses the first dividing line obliquely. A first ratio calculating section calculates the ratio of the sum of the photoelectric currents generated by the light-receiving elements on one side of the first dividing line to the sum of the photoelectric currents produced by the light-receiving elements on the other side. A second ratio calculating section calculates the ratio of the sum of the photoelectric currents generated by the light-receiving elements on one side of the second dividing line to the sum of the photoelectric currents produced by the light-receiving elements on the other side. A distance-determining section calculates the sum of the outputs of the first and second ratio calculating sections, and based on the resulting value, determines the distance from the device to the subject.

US 4,675,517 A describes a photosensitive element having first and second sensitive regions arranged to receive a light spot in such a way that as the light spot lies astride the boundary there between, the ratio of the areas of the two parts of the spot light which are received by the first and second sensitive regions, respectively, varies depending on the received position of the light spot, whereby the received position of the light spot is detected from the variation of the ratio of the outputs of the first and second sensitive regions, wherein an error of variation of the ratio of the outputs of the first and second sensitive regions for the received position of the light spot, which takes place when the diameter of the received spot light is so large as to overrun the first and second sensitive regions, is corrected by providing a supplementary sensitive region surrounding the first or second sensitive region.

WO 2015/024871 A1 describes an optical detector. The optical detector comprising: - at least one spatial light modulator being adapted to modify at least one property of a light beam in a spatially resolved fashion, having a matrix of pixels, each pixel being controllable to individually modify the at least one optical property of a portion of the light beam passing the pixel; - at least one optical sensor adapted to detect the light beam after passing the matrix of pixels of the spatial light modulator and to generate at least one sensor signal; - at least one modulator device adapted for periodically controlling at least two of the pixels with different modulation frequencies; and - at least one evaluation device adapted for performing a frequency analysis in order to determine signal components of the sensor signal for the modulation frequencies.

However, determination of distance to an object using these known methods and devices depends on the object size. In particular, the optical sensors may show a luminance dependency which renders the evaluation of the measurement result more difficult, and, additionally, may dependent on a target spot size. Dependency on object size may result in susceptibility of distance determination on manufacturing tolerances and environmental conditions. Furthermore, these methods and devices require a large baseline, and thus, large dimensions in order to reach a reliable distance value.

10 Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

20 This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

25 As used in the following, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

35 Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with op-

tional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for determining a position of at least one object is disclosed. As used herein, the term "object" refers to a point or region emitting at least one light beam. The light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least one transfer device, wherein the transfer device has at least one focal length in response to at least one incident light beam propagating from the object to the detector;
- at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam,
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q from the sensor signals.

The detector may be adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from the object size in an object plane.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term “at least two optical sensors each having at least one light sensitive area” refers to configurations with two single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having at least two light sensitive areas. Thus, the term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one CCD and/or CMOS device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible. Thus, as an example, an optical device comprising two, three, four or more than four light-sensitive areas may be used which is regarded as two, three, four or more than four optical sensors in the context of the present invention. As an example, the optical device may comprise a matrix of light-sensitive areas. Thus, as an example, the optical sensors may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces

of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to an optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The light beam may propagate from the object towards the detector. As will be outlined in further detail below, the light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source, as an example, may be or may comprise one or more of an external illumination source, an illumination source integrated into the detector or an illumination source integrated into a beacon device being one or more of attached to the object, integrated into the object or held by the object. Thus, the detector may be used in active and/or passive illumination scenarios. For example, the illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. Additionally or alternatively, the object may be adapted to generate and/or to emit the at least one light beam. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. As used herein, the term “beam profile” generally refers to a transverse intensity profile of the light beam. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The light beam may be a monochromatic light beam. The light beam may comprise a narrow band of wavelengths, preferably the light beam may comprise a single wavelength. The at least one light source may be adapted to generate at least one monochromatic light beam and/or the detector may comprise at least one filter element adapted to filter a narrow band of wavelength, such as a monochromator.

The illumination source may be adapted to generate at least one illumination pattern for illuminating the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "illumination pattern" refers to a pattern which illuminates the object. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines; at least one fringe pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that a hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may com-

prise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing
5 lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination
10 source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source may be adapted to generate pulsed illumination. The illumination source
15 may be adapted to generate at least one light pulse. As used herein, the term "light pulse" or "pulsed illumination" refers to a light beam limited in time. The light pulse may have a pre-defined length or time duration, for example in the nanoseconds range. For example, the illumination source may be adapted to generate pulses with a pulse length of less than a nanosecond, such as a tenth of a nanosecond, up to a tenth of a second. The illumination source may
20 be adapted to periodically generate the light pulse. For example, the illumination source may be adapted to generate the light pulse with a frequency of 10 Hz to 10 GHz.

The illumination source may be adapted to generate a pulsed light beam. For example, the illumination source may be adapted to generate a continuous illumination light beam and the de-
25 tector may comprise at least one interruption device adapted to interrupt the illumination, in particular periodically. The interruption device may comprise at least one shutter and/or a beam chopper or some other type of mechanical or electronical periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates at constant speed and which can thus periodically interrupt the illumination. By way of example,
30 the at least one interruption device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range
35 from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 microme-
40 ters. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a

CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes. As further used herein, the term "photosensitive element" generally refers to an element which is sensitive against illumination in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the photosensitive element may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used.

In view of the technical challenges involved in the prior art documents discussed above, specifically in view of the technical effort which is required for generating the so called FiP effect as described in WO 2015/024871 it has to be noted that the present invention specifically may be realized by using non-FiP optical sensors. In fact, since optical sensors having the FiP characteristic typically exhibit a strong peak in the respective sensor signals at a focal point, the range of measurement of a detector according to the present invention using FiP sensors as optical sensors may be limited to a range in between the two positions and which the first and second optical sensors are in focus of the light beam. When using linear optical sensors, however, i.e. optical sensors not exhibiting the FiP effect, this problem, with the setup of the present invention, generally may be avoided. Consequently, the first and second optical sensor may each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals may be dependent on the total power of illumination of the respective optical sensor and may be independent from a diameter of a light spot of the illumination. It shall be noted, however, that other embodiments are feasible, too.

The first and second optical sensors each specifically may be semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes. Thus, as opposed to complex and expensive FiP sensors, the present invention simply may be realized by using commercially available inorganic photodiodes, i.e. one small photodiode and one large area photodiode. Thus, the setup of the present invention may be realized in a cheap and inexpensive fashion.

However, embodiments are feasible in which the detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871, as will be outlined in detail below.

The detector may further comprise at least one illumination source for illuminating the object. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector. For this

purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source, as outlined above, specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the optical sensors may be arranged in a twisted arrangement.

The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. As an example, the illuminating light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illuminating light beam may be on the optical axis or off the optical axis. As an example, the illuminating light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

The detector comprises at least one transfer device. The term "transfer device", also denoted as "transfer system", may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one aspheric lens; at least one diffractive optical

element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

- 5 As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of
- 10 example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimat-
- 15 ed light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a pre-defined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the
- 20 transfer device can, in principle, be realized individually or in any desired combination.

- The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The
- 25 optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or
- 30 tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the
- 35 detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis and the condition noted above generally may refer to each beam path independently.. The optical sensors may be located in one
- 40 and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths. In case the optical sensors are distributed over different partial beam paths, the above-mentioned condition may be described such that at least one first optical sensor is located in at least one first partial beam path, being offset from the

optical axis of the first partial beam path by a first spatial offset, and at least one second optical sensor is located in at least one second partial beam path, being offset from the optical axis of the second partial beam path by at least one second spatial offset, wherein the first spatial offset and the second spatial offset are different.

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The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

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The optical sensors may be positioned off focus. As used herein, the term “focus” generally refers to one or both of a minimum extend of a circle of confusion of the light beam, in particular of at least one light beam emitted from one point of the object, caused by the transfer device or a focal length of the transfer device. As used herein, the term “circle of confusion” refers to a light spot caused by a cone of light rays of the light beam focused by the transfer device. The circle of confusion may depend on a focal length f of the transfer device, a longitudinal distance from the object to the transfer device, a diameter of an exit pupil of the transfer device, a longitudinal distance from the transfer device to the light sensitive area, a distance from the transfer device to an image of the object. For example, for Gaussian beams, a diameter of the circle of confusion may be a width of the Gaussian beam. In particular, for a point like object situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from the object into a focus point at the focal length of the transfer device. For non-point like objects situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from at least one point of the object into a focus plane at the focal length of the transfer device. For point like objects not situated or placed at infinite distance from the detector, the circle of confusion may have a minimum extend at least at one longitudinal coordinate. For non-point like objects not situated or placed at infinite distance from the detector, the circle of confusion of the light beam from at least one point of the object may have a minimum extend at least at one longitudinal coordinate. As used herein, the term “positioned off focus” generally refers to a position other than the minimum extent of a circle of confusion of the light beam caused by the transfer device or a focal length of the transfer device. In particular, the focal point or minimum extend of the circle of confusion may be at a longitudinal coordinate l_{focus} , whereas the position of each of the optical sensors may have a longitudinal coordinate l_{sensor} different from l_{focus} . For example, the longitudinal coordinate l_{sensor} may be, in a longitudinal direction, arranged closer to the position of the transfer device than the longitudinal coordinate l_{focus} or may be arranged further away from the position of the transfer device than the longitudinal coordinate l_{focus} . Thus, the longitudinal coordinate l_{sensor} and the longitudinal coordinate l_{focus} may be situated at different distances from the transfer device. For example, the optical sensors may be spaced apart from the minimum extend of the circle of confusion in longitu-

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dinal direction by $\pm 2\%$ of focal length, preferably by $\pm 10\%$ of focal length, most preferably $\pm 20\%$ of focal length. For example, at a focal length of the transfer device may be 20mm and the longitudinal coordinate l_{sensor} may be 19,5 mm, i.e. the sensors may be positioned at 97,5% focal length, such that l_{sensor} is spaced apart from the focus by 2,5% of focal length.

- 5 The optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.

Each light-sensitive area may have a geometrical center. As used herein, the term “geometrical center” of an area generally may refer to a center of gravity of the area. As an example, if an
10 arbitrary point inside or outside the area is chosen, and if an integral is formed over the vectors interconnecting this arbitrary point with each and every point of the area, the integral is a function of the position of the arbitrary point. When the arbitrary point is located in the geometrical center of the area, the integral of the absolute value of the integral is minimized. Thus, in other
15 words, the geometrical center may be a point inside or outside the area with a minimum overall or sum distance from all points of the area.

For example, each geometrical center of each light-sensitive area may be arranged at a longitudinal coordinate $l_{\text{center},i}$, wherein i denotes the number of the respective optical sensor. In the case of the detector comprising precisely two optical sensors and in the case of the detector
20 comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor, wherein the first optical sensor, in particular the geometrical center, being arranged at a first longitudinal coordinate $l_{\text{center},1}$, and at least one second optical sensor, wherein the second optical sensor, in particular the geometrical center, being at a second longitudinal coordinate $l_{\text{center},2}$, wherein the first longitudinal coordinate and the second longitudinal coordi-
25 nate differ. For example, the first optical sensor and the second optical sensor may be located in different planes which are offset in a direction of the optical axis. The first optical sensor may be arranged in front of the second optical sensor. Thus, as an example, the first optical sensor may simply be placed on the surface of the second optical sensor. Additionally or alternatively, the first optical sensor may be spaced apart from the second optical sensor, for example, by no
30 more than five times the square root of a surface area of the first light-sensitive area. Additionally or alternatively, the first optical sensor may be arranged in front of the second optical sensor and may be spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm. Relative distance of the first optical sensor and second optical sensor may depend, for example, on focal length or object distance.

35 The longitudinal coordinates of the optical sensors may also be identical, as long as one of the above-mentioned conditions is fulfilled. For example, the longitudinal coordinates of the optical sensors may be identical, but the light-sensitive areas may be spaced apart from the optical axis and/or the surface areas differ.

40 Each geometrical center of each light-sensitive area may be spaced apart from the optical axis of the transfer device, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located. The distance, in particular in transversal direc-

tion, between the geometrical center and the optical axis is denoted by the term “spatial offset”
In the case of the detector comprising precisely two optical sensors and in the case of the de-
tector comprising more than two optical sensors, the optical sensors may comprise at least one
first optical sensor being spaced apart from the optical axis by a first spatial offset and at least
5 one second optical sensor being spaced apart from the optical axis by a second spatial offset,
wherein the first spatial offset and the second spatial offset differ. The first and second spatial
offsets, as an example, may differ by at least a factor of 1.2, more preferably by at least a factor
of 1.5, more preferably by at least a factor of two. The spatial offsets may also be zero or may
assume negative values, as long as one of the above-mentioned conditions is fulfilled.

10 As used herein, the term “surface area” generally refers to both of a shape and a content of at
least one light-sensitive area. In the case of the detector comprising precisely two optical sen-
sors and in the case of the detector comprising more than two optical sensors, the optical sen-
sors may comprise at least one first optical sensor having a first surface area and at least one
15 second optical sensor having a second surface area. In the case of the detector comprising
more than two optical sensors, e.g. a sensor element comprising a matrix of optical sensors, a
first group of optical sensors or at least one of the optical sensors of the matrix may form a first
surface area, wherein a second group of optical sensors or at least one other optical sensor of
the matrix may form a second surface area. The first surface area and the second surface area
20 may differ. In particular, the first surface area and the second surface area are not congruent.
Thus, the surface area of the first optical sensor and the second optical sensor may differ in one
or more of the shape or content. For example, the first surface area may be smaller than the
second surface area. As an example, both the first surface area and the second surface area
may have the shape of a square or of a rectangle, wherein side lengths of the square or rectan-
25 gle of the first surface area are smaller than corresponding side lengths of the square or rectan-
gle of the second surface area. Alternatively, as an example, both the first surface area and the
second surface area may have the shape of a circle, wherein a diameter of the first surface area
is smaller than a diameter of the second surface area. Again, alternatively, as an example, the
first surface area may have a first equivalent diameter, and the second surface area may have a
30 second equivalent diameter, wherein the first equivalent diameter is smaller than the second
equivalent diameter. The surface areas may be congruent, as long as one of the above-
mentioned conditions is fulfilled.

35 The optical sensors, in particular the light-sensitive areas, may overlap or may be arranged
such that no overlap between the optical sensors is given.

As further used herein, the term “evaluation device” generally refers to an arbitrary device
adapted to perform the named operations, preferably by using at least one data processing de-
vice and, more preferably, by using at least one processor and/or at least one application-
40 specific integrated circuit. Thus, as an example, the at least one evaluation device may com-
prise at least one data processing device having a software code stored thereon comprising a
number of computer commands. The evaluation device may provide one or more hardware el-

ements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

5 The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Field Programmable Gate Arrays (FPGAs), or digital
10 signal processors (DSPs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

15 The evaluation device is configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q of the sensor signals A as used herein, the term "quotient signal Q " refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. In the following, the quotient signal is also denoted as combined signal or combined sensor signal.

20 The object size may be a priori unknown. An object image size, i.e. the size of the object in the image plane, may be distance dependent. However, different materials may have different scattering properties such that deviations from Lambertian distribution may occur. In particular, a luminance dependence may deviate significantly from a $1/z^2$. In addition, due type of scattering
25 and/or due to reflective properties and/or scattering properties of the object, for example depending on a material of the object, the object image size, in particular diameter of the spot, may vary. Thus, methods and devices using the spot size, for example by counting pixels, or absolute photon numbers for distance determination may be not suitable and may require calibrations depending on the material of the object. Methods and devices adapted for determination
30 of the distance independently from the object size are necessary in order to allow robust distance determination for any materials with one calibration, only. Further, the object size may be altered by the light source itself. For example, the properties of the optics may be altered by one or more of dirt, raindrops, scratches or manufacturing precision of the light source and/or its optics. Thus, the light beam emitted by the light source may be changed in one or more of its
35 properties such as diameter, beam profile, divergence or the like. Thus, methods and devices adapted for determination of the distance independently from the object size are necessary in order to allow robust distance determination for higher manufacturing tolerances or harsher environmental conditions or the like with one calibration, only. Further, object size may be altered depending on temperature as the optical properties of the detector and or the light source may
40 change such as due to a temperature dependent change of the distance between emitter and optics of the light source or due to a temperature dependent change of the distance between sensor and optics of the detector or due to a temperature dependent change of the refracting properties of the lens or the like. Object size independence is, thus, important for measure-

ments with little or no a priori knowledge on one or more of the measured object, the measurement environment, the temperature of the detector and its environment, the manufacturing quality of the sensor and/or the light source. Further, object size independence is, thus, important for measurements that require high flexibility concerning one or more of the measured object, the measurement environment, the temperature of the detector and its environment, the manufacturing quality of the sensor and/or the light source. Further, object size independence is important, when diverging or converging illumination light beams shall be used, such as when diverging or converging laser light beams are used where the size of the illuminated spot changes with distance. Further, object size independence is, thus, important for measurements that require a high robustness in one or more of measurement quality independent of the measured object, the measurement environment, the temperature of the detector and its environment, the manufacturing quality of the sensor and/or the light source.

Known 3D sensing methods are dependent on the object size. For example US 4,675,517 and US 5,323,222 describe devices and methods exhibiting dependence on object size. Specifically, US 4,675,517 describes with respect to Figures 3A to 3C, column 2, line 16 that the distance information changes with spot size. In particular, it is described that in actual practice, with the use of such a system, as the diameter of the image of the reflection of the projected image spot P changes at random, or the optical position deviates, the defined object distance boundaries are caused to shift largely. In particular, problems due to too large spot sizes are addressed. In case the spot size is too large, the sensor is outshined and the resulting ratio is wrong. US 4,675,517 describes to correct this spot size dependence by enlarging the sensor area. However, the spot size dependence is not eliminated. In particular, problems due to too small spot sizes are not addressed. Furthermore, specifically, US 5,323,222 describes that the ratio used for determining the distance depends on a chip size of the light source. In particular, in column 2, line 11 it is outlined that solid lines I1 and I2 in Fig 14A show the relationship between the reciprocal of the distance a and the calculation output $I1/(I1+I2)$ for the light-projecting chip sizes t1 and t2 ($t1 < t2$), respectively. Furthermore, it is described that if the distance between the projection lens and light emitting element is fT, the base length L, and the chip size of the light emitting element t, then the distance measuring range S will be: $S = \infty$ through $fT \cdot L/t$. However, US 5,323,222 does not further address eliminating impact of object size dependence.

As used herein, the term “object size”, generally refers to a size of the object in an object plane, in particular to the size of a spot and/or region and/or the entire object emitting the at least one light beam. As further used herein, the terms “independent from the object size” and “object size independence” refer to the fact that variations of the object size do not influence the determination of the longitudinal coordinate z. As will be outlined in detail below, the quotient signal Q may be used for determining the longitudinal coordinate. Thus, object size independence furthermore refers to the fact that the quotient signal is essentially independent from the object size such that, in case of identical object distance, a first quotient signal determined at a first object size and a second quotient signal determined at a different object size are identical, with a tolerance of $\pm 20\%$ or less, preferably a tolerance of $\pm 10\%$ or less, more preferably a tolerance of $\pm 5\%$ or less. Requirement or condition of object size independence can be stated as follows:

the circle of confusion of at least one light beam emitted from at least one point of the object is greater than an image size of the object, i.e. a size of the object in an image plane. The detector may be adapted to determine the longitudinal coordinate z of the object independent from the object size if the circle of confusion of the at least one light beam emitted from the at least one point of the object is greater than an image size of the object. This condition may be calculated by $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$, wherein z_0 is a longitudinal distance from the object to the transfer device; E_x is a diameter of an exit pupil of the transfer device; z_s is a longitudinal distance from the transfer device to the light sensitive area, z_i is a distance from the transfer device to an image of the object; O_{size} is an object size of the object in the object plane. As used herein, the term “diameter of an exit pupil” refers to an aperture of the transfer device. Given a measurement range the condition can be fulfilled by for example varying the sensor position, the diameter of the exit pupil or the image of the object, for example by varying the focal length. The detector may be adapted to determine the longitudinal coordinate z of the object independent from the object size if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for at least one distance z_0 . For example, the detector may be adapted to determine the longitudinal coordinate z at the at least one distance z_0 if the object size varies and/or changes within a large range, e.g. by more than 100% of the object size. The detector may be adapted to determine the longitudinal coordinate z at distances, at which the above-mentioned condition $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ is fulfilled only weakly, if the object size varies and/or changes within a smaller range, in particular if the object size varies and/or changes by $\pm 20\%$ or less, preferably by $\pm 10\%$ or less, more preferably by $\pm 5\%$ or less. The detector may be adapted to determine the longitudinal coordinate z over the whole measurement range if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for the at least one distance z_0 and if the object size and if the object size varies and/or changes by $\pm 20\%$ or less, preferably by $\pm 10\%$ or less, more preferably by $\pm 5\%$ or less. As used herein, the term “holds true for at least one distance” refers to that at the at least one distance the longitudinal coordinate z may be determined independent from the object size and that the detector is adapted to determine the longitudinal coordinate z at other distances, in particular within the whole or entire measurement range, if the object size varies and/or changes by $\pm 20\%$ or less, preferably by $\pm 10\%$ or less, more preferably by $\pm 5\%$ or less.

As used herein, the term “measurement range” generally refers to an arbitrary range in which the determination of the longitudinal coordinate z is performed. The measurement range may be adjustable by adjusting one or more parameters selected from the group consisting of: the longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; the diameter of the exit pupil of the transfer device E_x ; the longitudinal distance from the transfer device to the light sensitive area z_s , the distance from the transfer device to an image of the object z_i ; the object size O_{size} of the object in the object plane. For example, if the object size does not exceed an object-size limit, a unique relation of the quotient signal Q to the longitudinal distance from the object to the transfer device exists. The object-size limit may depend on the longitudinal distance from the transfer device to the light sensitive area, the longitudinal distance from the object to the transfer device, and an F-number of the transfer device $F\#$, i.e. a ratio of the focal length of the transfer device and the diameter of an exit pupil of the transfer device. For example, the measurement range may be adapted by selecting and/or choosing one or more of the longitudinal distance from the transfer device to the light sensitive area z_s , the dis-

- tance from the transfer device to an image of the object z_i , the longitudinal distance from the transfer device to the light sensitive area, the focal length and the F-number of the transfer device. For example, the focal length and/or F-number may be adjusted by using a zoom objective. In particular, the focal length may be between 10 and 200 mm, preferably between 20 and 150 mm. The F-number may be between 1 and 10, preferably between 1,5 and 6. The longitudinal distance from the transfer device may be as short as possible. The longitudinal distance from the transfer device to the light sensitive area may be between 0 and 200 mm, preferably between 20 and 50 mm. For a given system setup (f , $F\#$, z_s) a unique value of the object-size limit can be calculated. For example, in case the focal length is 3,5 mm and the F-number is 2,0, the object size may be smaller than 1,75 mm. Preferably a lower object size limit may be 0.5 μm or more, more preferably 1 μm or more and most preferably 10 μm or more. The lower object size limit refers to a minimum object size using an active measurement system, in particular to a laser spot size.
- The evaluation device may be configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate. For example, the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- wherein x and y are transversal coordinates, A_1 and A_2 are areas of the beam profile at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A_1 and area A_2 may differ. In particular, A_1 and A_2 are not congruent. Thus, A_1 and A_2 may differ in one or more of the shape or content. As used herein, the term “beam profile” relates to a spatial distribution, in particular in at least one plane perpendicular to the propagation of the light beam, of an intensity of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$, $E(x, y; z_o) = L \cdot S$. Thus, by deriving the quotient signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the quotient signal allows determination of the distance z_o independent from the object size. Thus, the quotient signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. Further-

more, even at very large spot sizes, for example at an outshining, reliable determination of the longitudinal coordinate of the object may be possible.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the quotient signal Q. The light-sensitive regions may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term "essentially edge information" generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, the center signal may be a signal corresponding to a rectangular stripe comprising the center of the light spot. The evaluation device may be adapted to form a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. In par-

5 ticular, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile. The first and the second area, as an example, may be adjacent, with a minimum separation from one another. However, the separation between adjacent areas may also be varied such as enlarged to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent areas may be increased by not evaluating the area between the first and the second area. This may reduce the light on one of the areas relative and/or absolute as compared to the adjacent area which will may increase a quotient of the signal of the two
10 areas. Further, one or both of the areas may consist of separated subareas which may be adjacent to each other and/or which may be separated by areas that are not evaluated and/or which may be evaluated as part of a different quotient. Further, the first and the second area may consist of a linear combination of subareas, whereas the signal contributed by each subarea may be weighted differently when forming the center signal and/or the sum signal. This may further
15 be beneficial for increasing the dynamic range of the quotient system.

The extent of the circle of confusion may be larger than the extent of the optical sensor. For example, the optical sensor may be positioned as such that the circle of confusion extends beyond the optical sensor. The optical sensor may thus only partially evaluate the beam profile of
20 the light beam. The evaluation device may be adapted to extrapolate a beam profile such as due to symmetry considerations or due to comparing the partial beam profile to prerecorded beam profiles or the like. Further, the evaluation device may be adapted to evaluate a partial sum signal and a partial center signal of a partial beam profile and convert it into a sum signal and a center signal of an extrapolated and/or fitted and/or prerecorded beam profile or the like.

25 The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the
30 second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by
35 equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

40 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot

may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The detector may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors. For example, the detector may comprise a plurality of optical sensors arranged in a matrix. The detector may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, a defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. This principle is called Distance by Photon Ratio (DPR).

The DPR principle can be applied, for example, to many sub-regions within a segmented image profile, such as a segmented image of at least one feature generated by the at least one light beam on the matrix of optical sensors, which is demonstrated hereafter. The quotient $Q_A(z)$ in the two-dimensional case can be written as

$$Q_A(z) = \frac{\iint_{-r_{out}}^{r_{out}} P(x,y,z) dx dy}{\iint_{-r_{in}}^{r_{in}} P(x,y,z) dx dy},$$

where $P(x, y, z)$ is a two-dimensional beam profile and r_{in} and r_{out} are inner and outer circle radii, respectively. For line-segmented quotients $Q_y(z)$ along the y -dimension, this can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{out}}^{r_{out}} P(x,z) dx}{\int_{-r_{in}}^{r_{in}} P(x,z) dx}.$$

Without wishing to be bound by this theory, the overall defocused beam profile $P(x, z)$ can be regarded as a superposition of defocused image point profiles $p(x, z)$ along the image width $b(x, z)$. This relationship can be modelled as a convolution given by

$$P(x, z) = b(x, z) * p(x, z),$$

wherein $p(x, z)$ represents the Point-Spread-Function (PSF) of a lens in defocus, which in the field of paraxial optics is also known as Circle of Confusion (CoC). By inserting $P(x, z)$ into $Q_y(z)$ the above described quotient $Q_y(z)$ can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{in}}^{r_{out}} b(x,z) * p(x,z) dx}{\int_{-r_{in}}^{r_{in}} b(x,z) * p(x,z) dx} = \frac{\int_{-r_{out}}^{r_{out}} b(x,z) dx \cdot \int_{-r_{out}}^{r_{out}} p(x,z) dx}{\int_{-r_{in}}^{r_{in}} b(x,z) dx \cdot \int_{-r_{in}}^{r_{in}} p(x,z) dx}.$$

The CoC radius may be r_c and r_o may be an image spot radius of the feature on the matrix of optical sensors, then $r_{out} = r_c + r_o \Rightarrow r_{out} \geq r_c$ and $r_{out} \geq r_o$, which yields to

$$Q_y(z) = \frac{\int_{-r_o}^{r_o} b(x,z)dx \cdot \int_{-r_c}^{r_c} p(x,z)dx}{\int_{-r_{in}}^{r_{in}} b(x,z)dx \cdot \int_{-r_{in}}^{r_{in}} p(x,z)dx}.$$

In case of $r_o \leq r_{in} < r_c$,

$$Q_y(z) = \frac{\int_{-r_o}^{r_o} b(x,z)dx \cdot \int_{-r_c}^{r_c} p(x,z)dx}{\int_{-r_o}^{r_o} b(x,z)dx \cdot \int_{-r_{in}}^{r_{in}} p(x,z)dx} = \frac{\int_{-r_c}^{r_c} p(x,z)dx}{\int_{-r_{in}}^{r_{in}} p(x,z)dx},$$

which indicates that the quotient is independent of the object size on condition that $r_o < r_c$.

The Object Size Independence (OSI) in DPR applies as long as the image width i remains below the Circle of Confusion (CoC) width denoted as c . In the one-dimensional case, this is represented by diameters $d_o < d_c$. Using paraxial optics, these diameters can be substituted, such that

$$\frac{i}{a} o_s < \frac{D_x}{i} |d - i|,$$

with a as the object distance, i as the image distance, o_s as the object size, D_x as the lens exit pupil diameter and d as the sensor position with respect to image-side principal plane. After re-arranging, this can be written as

$$o_s < \frac{a D_x}{i^2} |d - i|.$$

When focusing a single lens to infinity, this simplifies to

$$o_s < \frac{f}{F\#} \left| 1 - \frac{f}{a} \right|,$$

where f is the focal length and $F\#$ the F-number of the lens.

Furthermore, with a matrix of optical sensors a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates. Accordingly, the parameterization of a single line may be given by

$$\omega = x \cos(\theta) + y \sin(\theta).$$

The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega,\theta)}\{f(x,y)\} = \iint_{-\infty}^{\infty} f(x,y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy,$$

where δ denotes the Dirac delta function and $f(x,y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is given by

$$R = \frac{\Re_{(\omega,\theta)}\{f'(x,y)\}}{\Re_{(\omega,\theta)}\{f(x,y)\}},$$

with $f'(x,y)$ as the inner region. The variation of θ may yield different ratios R for skewed object surfaces at a certain distance. It is sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

In one embodiment, the light beam propagating from the object to the detector may illuminate the sensor element with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at

least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

For example, the light beam propagating from the object to the detector may illuminate the sensor element with at least one point pattern. The point pattern may be generated by the object, for example in response to an illumination by the at least one light source with an illumination pattern comprising the at least one point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The light beam propagating from the object to the detector may illuminate the sensor element with a reflection pattern comprising both point patterns and line patterns. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

For example, the optical sensors may comprise a first optical sensor having a first light sensitive area and a second optical sensor having a second light sensitive area, wherein the first and the second light sensitive areas are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$

is satisfied, wherein “a” is a ratio of photons hitting both an inner region of a plane perpendicular to the optical axis intersecting the optical axis at a distance equal to half of a focal length of the transfer device and the first light sensitive area, “b” is a ratio of photons hitting both the inner region of the plane and the second light sensitive area, “c” is a ratio of photons hitting both an outer region of the plane and the first light-sensitive area, “d” is a ratio of the photons hitting both the outer region of the plane and the second light sensitive area. The first light sensitive area and the second light sensitive areas may be arranged such that the inner region may have an area with a geometrical center point on the optical axis and an extension such that half of the photons hit the plane within the inner region and the other half hit the plane outside the inner region. The first light sensitive area and the second light sensitive areas may be arranged such that the inner region may be designed as a circle with a center point on the optical axis and a radius r which is chosen such that half of the photons hit the plane within the circle and the other half hit the plane outside the circle.

As outlined above, the detector may comprise at least one illumination source. A distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors may be small. The distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. The illumination source and the optical axis may be separated by a small baseline. As used herein, the term “baseline”, also denoted as basis line, refers to a distance, for example in a xy -plane, between the optical axis and the illumination source, in particular a distance between the optical axis and a z-component of the illumination light beam. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and of the at least one optional transfer device, which will be described in detail below. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device.

At least one of the optical sensors may be adapted to generate at least one sensor signal dependent on a time-of-flight (TOF) the illumination light beam has traveled from the illumination source to the object and the reflection light beam has traveled from the object to the light sensitive area of the optical sensor. The evaluation device may be configured for determining at least one TOF-longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. The optical sensor adapted to generate the TOF sensor signal may be designed as time-of-flight detector. The time-of-flight detector may be selected from the group consisting of: at least one pulsed time-of-flight detector; at least one phase modulated time-of-flight detector; at least one direct time-of-flight detector; at least one indirect time-of-flight detector. For example, the pulsed time-of-flight detector may be at least one range gated imager and/or at least one direct time-of-flight imager. For example the phase modulated time-of-flight detector may be at least one RF-

modulated light source with at least one phase detector. The optical sensor may be adapted to determine a time delay between emission of the illumination light beam by the illumination source and receipt of the reflection light beam.

- 5 For example, the optical sensor adapted to generate the TOF sensor signal may be designed as pulsed time-of-flight detector. The detector may comprise the at least one interruption device, such as at least one shutter element, adapted to generate a pulsed light beam. The optical sensor may be adapted to store the TOF sensor signal dependent on the receiving time of the reflection light beam in a plurality of time windows, in particular subsequent time windows. The
- 10 optical sensor may be adapted to store dependent on receiving time of the reflection light beam the generated TOF sensor signal in at least one first time window and/or in at least one second time window. The first and second time windows may be correlated with the opening and closure of the interruption device. Duration of first and second time windows may be pre-defined. For example, the TOF sensor signal may be stored in the first time window during opening of
- 15 the interruption device, whereas during closure of the interruption device the TOF sensor signal may be stored in the second time window. Other durations of time windows are thinkable. The first and the second time window may comprise information about background, signal height and signal shift.
- 20 For example, the optical sensor adapted to generate the TOF sensor signal may be designed as direct time-of-flight imager. The direct time-of-flight imager may comprise the at least one illumination source adapted to generate at least one single laser pulse. The single laser pulse may be reflected back from the object onto the optical sensor. The optical sensor may comprise at least one photodiode, for example at least one Avalanche Photo Diode (APD), such as at
- 25 least one Si APD, or such as at least one InGaAs APD, or at least one PIN photo detector array, or at least one Single-Photon Avalanche Photodiode (SPAD), adapted to image the reflection light beam. The direct time-of flight imager may be adapted to image at least one image comprising spatial and temporal data.
- 30 For example, the optical sensor adapted to generate the TOF sensor signal may be designed as phase modulated time-of-flight modulator. The phase modulated time-of-flight modulator may be adapted to measure a difference in phase, in particular a phase shift, by determining a correlated signal, for example by multiplying a received signal, i.e. of the reflection light beam, with the emitted signal, i.e. the illumination light beam. A DC component of the correlated signal may
- 35 comprise an information about the difference in phase. The evaluation device may be adapted to determine the second longitudinal coordinate of the object from the phase difference. For example, the illumination source and the optical sensor adapted to generate the TOF sensor signal may be designed as RF-modulated light source with at least one phase detector. The illumination source may comprise, for example, at least one LED and/or at least one laser. The
- 40 illumination source may comprise at least one modulation device adapted to modulate the light beam having a pre-defined phase shift. For example, the modulation device may comprise at least one radio frequency module. The radio frequency module may be adapted to modulate the

illumination beam with an RF carrier. The optical sensor may be adapted to determine a phase shift of the reflective light beam impinging on the optical sensor.

The optical sensor may be designed as and/or may comprise at least one time-of-flight pixel.

- 5 Preferably, the detector may comprise at least two optical sensors, wherein each optical sensor is designed as and/or comprises at least one TOF pixel. For example, the detector, in particular the optical sensor may comprise a quadrant diode adapted to generate the TOF sensor signal. For example, the detector, in particular the optical sensor, may comprise at least one pixelated TOF-imager.

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The evaluation device may be configured for determining the at least one TOF longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. As used herein, the term "TOF longitudinal coordinate" refers to a longitudinal coordinate derived from the TOF sensor signal.

- 15 As described above, the illumination light source may be adapted to periodically generate at least one light pulse. The detector may be adapted to generate the first longitudinal sensor signal for each period. The evaluation device may be adapted to determine in which pulse period the TOF sensor signal was generated using the second longitudinal coordinate. The detector may be adapted to uniquely assign to which period the ToF-signal refers to by using the combined sensor signal. Both, the TOF sensor signal and the combined sensor signal, may be non-monotonous functions of the longitudinal coordinate z_{real} . Thus, the longitudinal coordinate may not be uniquely determined from one of the TOF sensor signal or the combined sensor signal alone and a measurement range may not be restricted to a longitudinal range in which the signals are unique functions of z_{real} . The term "measurement range" generally refers to a range from the object to the detector in which determination of the longitudinal coordinate is possible.
- 20 The term "longitudinal range" generally refers to a range from the object to the detector in which unique determination of the longitudinal coordinate is possible. Below and/or above certain distances from the object to the detector a determination of the longitudinal coordinate may not be possible. For example, time-of-flight measurements are not possible below a certain distance from the object to the detector, possibly due to the minimum measurement time of the internal clock. Furthermore, in time-of-flight measurements the sensor signal may be unique within a longitudinal period, but the sensor signal may be the same in case integer multiples of the longitudinal period are added such that the determined longitudinal coordinate may be non-unique. Thus, the same TOF sensor signal will be obtained for a distance z_1 and a distance $z_1 + n \cdot z_{1p}$, where n is an integer denoting the longitudinal period and z_{1p} is the longitudinal period of the
- 25 TOF sensor signal, wherein the distances z_1 and $z_1 + n \cdot z_{1p}$ are within the measurement range. As used herein, the term "longitudinal period" refers to partitions of a period, in particular a distance range, in which the longitudinal coordinate can be unambiguously determined from the TOF sensor signal. Non-unique longitudinal coordinates may be denoted as relative longitudinal coordinates and the unique longitudinal coordinates may be denoted as absolute longitudinal
- 30 coordinates.
- 35
- 40

If both the TOF sensor signal $F1$ and the combined sensor signals $F2$ are available, it may be possible to uniquely determine the longitudinal coordinate and to extend the longitudinal range,

as long as each signal pair (F1,F2) corresponds to a unique distance and vice versa. In particular, if a unique signal pair (F1,F2) exists for each longitudinal coordinate and vice versa, the evaluation device may be adapted to determine the unique combined longitudinal coordinate by

- (1) selecting at least one first selected signal such as the TOF sensor signal and/or the combined sensor signal and determining non-unique first longitudinal coordinates;
- (2) selecting a second selected signal such as the combined signal Q and/or the TOF sensor signal, which was not selected in step (1), and determining non-unique second longitudinal coordinates;
- (3) determining whether one of the non-unique first longitudinal coordinates and the non-unique second longitudinal coordinates match up to a predetermined tolerance threshold;
- (4) setting a combined unique longitudinal coordinate to be a matching longitudinal coordinate.

In step (1) and step (2), signals may be selected in the given order or may be performed in a different order. For example, in step (1) the TOF sensor signal may be selected and in step (2) the combined signal Q may be selected. In another example, in step (1) the combined sensor signal may be selected, and the non-unique first longitudinal sensor signal may be determined therefrom. In step (2), the TOF sensor signal may be selected.

Additionally or alternatively to step (4), the evaluation device may be adapted to output an error signal in case no matching coordinates are found and/or to output an error signal in case more than one matching coordinates are found. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be stored in a look-up table. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be approximated or described by an analytical function which is evaluated to find the longitudinal coordinate corresponding to a given signal pair.

The evaluation device may comprise at least two memory elements. As used herein, the term “memory element” refers to a device adapted to store information. The evaluation device may be adapted to receive and to store information provided by the optical sensors, for example the at least one first sensor signal. Such information may comprise raw sensor data and/or processed sensor data. For example, the memory element may be adapted to store information for further evaluation by the evaluation device. The memory element may be a volatile or non-volatile memory element.

As described above, the optical sensor may be designed as and/or may comprise at least one ToF pixel. The detector may comprise at least two switches. Each of the switches may be connected to the optical sensor adapted to generate the first sensor signal, for example by at least one connector. In particular, each of the switches may be connected to the ToF pixel. The switches are adapted to provide the TOF sensor signal to one of the memory elements. In particular, the switches may be adapted to let, dependent on receiving time of the reflection light beam, the generated TOF sensor signal pass through one of the switches. For example, the TOF sensor signal may pass one of the switches during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may pass the other

switch. Each of the switches may be controlled by a control signal having a pulse length identical to a pulse length of a light pulse generated by the illumination source. The control signal of one of the switches may be delayed. For example, the delay may correspond to the pulse length of the light pulse. The evaluation device may be adapted to sample and/or store depending on the delay a first part, or fraction, of the TOF sensor signal through a first switch in a first memory element and the other, second part, or fraction, of the TOF sensor signal through a second switch in a second memory element. The evaluation device may be adapted to determine the first longitudinal coordinate by evaluating the first part and second part of the TOF sensor signal. The evaluation device may be adapted to determine the first longitudinal coordinate z_1 by

$$z_1 = \frac{1}{2} \cdot c \cdot t_0 \cdot \frac{S_{12}}{S_{11} + S_{12}} + z_0;$$

wherein c is the speed of light, t_0 is the pulse length of the illumination light beam, z_0 is a distance offset often determined by the resolution of the time measurement, and S_{11} and S_{12} are the first part and second part of the TOF sensor signal, respectively.

As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

In a first preferred embodiment of the present invention, the detector may comprise:

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor may be configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
- at least one second optical sensor having a second light-sensitive area, wherein the second optical sensor may be configured to generate at least one second sensor signal in response to an illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area may be smaller than the second light-sensitive area; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

In this first preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their longitudinal coordinate and/or their surface areas.

As will further be outlined in detail below, each optical sensor, preferably, may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, preferably, each optical sensor is a single area optical sensor rather than a pixelated optical sensor like e.g. a sensor matrix. Other embodiments, however, are feasible. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and effi-

cient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the setup, one with a small light-sensitive area and one with a larger light-sensitive area, as will be explained in further detail below.

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The first and second light-sensitive areas specifically may be oriented towards the object.

The light beam propagating from the object to the detector specifically may fully illuminate the first light-sensitive area, such that the first light-sensitive area is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the first optical sensor. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the second light-sensitive area, which is smaller than the second light-sensitive area, such that the light spot is fully located within the second light-sensitive area. Within the light spot on the second light-sensitive area, a shadow created by the first optical sensor may be located. Thus, generally, the first optical sensor, having the smaller first light-sensitive area, may be located in front of the second optical sensor, as seen from the object, with the first light-sensitive area being fully located within the light beam and with the light beam generating a light spot on the second light-sensitive area being smaller than the second light-sensitive area, and with further a shadow created by the first optical sensor within the light spot. The situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a light spot generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

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As outlined above, the first light-sensitive area is smaller than the second light-sensitive area. As used therein, the term "is smaller than" refers to the fact that the surface area of the first light-sensitive area is smaller than the surface area of the second light-sensitive area, such as by at least a factor of 0.9, e.g. at least a factor of 0.7 or even by at least a factor of 0.5. As an example, both the first light-sensitive area and the second light-sensitive area may have the shape of a square or of a rectangle, wherein side lengths of the square or rectangle of the first light-sensitive area are smaller than corresponding side lengths of the square or rectangle of the second light-sensitive area. Alternatively, as an example, both the first light-sensitive area and the second light-sensitive area may have the shape of a circle, wherein a diameter of the first light-sensitive area is smaller than a diameter of the second light-sensitive area. Again, alternatively, as an example, the first light-sensitive area may have a first equivalent diameter, and the second light-sensitive area may have a second equivalent diameter, wherein the first equivalent diameter is smaller than the second equivalent diameter.

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The first light-sensitive area specifically may overlap with the second light-sensitive area in a direction of propagation of the light beam. The light beam may illuminate both the first light-sensitive area and, fully or partially, the second light-sensitive area. Thus, as an example, as seen from an object located on an optical axis of the detector, the first light-sensitive area may

be located in front of the second light-sensitive area, such that the first light-sensitive area, as seen from the object, is fully located within the second light-sensitive area. When the light beam from this object propagates towards the first and second light-sensitive areas, as outlined above, the light beam may fully illuminate the first light-sensitive area and may create a light spot on the second light-sensitive area, wherein a shadow created by the first optical sensor is located within the light spot. It shall be noted, however, that other embodiments are feasible.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between first and second sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the object by using at least one known, determinable or predetermined relationship between the quotient signal derived from the first and second sensor signals and the longitudinal coordinate.

Thus, the evaluation device specifically may be configured for deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example, Q may simply be determined as

$$Q = s_1/s_2$$

or

$$Q = s_2/s_1,$$

with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as

$$Q = a \cdot s_1 / b \cdot s_2$$

or

$$Q = b \cdot s_2 / a \cdot s_1,$$

with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as

$$Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2),$$

with a , b , c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as

$$Q = s_1 / (s_1 + s_2).$$

Other quotient signals are feasible.

Typically, in the setup described above, Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q=s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the first signal s_1 and the second signal s_2 decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the first signal s_1 decreases more rapidly than the second signal s_2 , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the first and second sensor signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

The evaluation device is configured for determining the longitudinal coordinate by evaluating the quotient signal Q . The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The quotient signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

The first and second optical sensors specifically may be arranged linearly in one and the same beam path of the detector. As used herein, the term “linearly” generally refers to that the sensors are arranged along one axis. Thus, as an example, the first and second optical sensors both may be located on an optical axis of the detector. Specifically, the first and second optical sensors may be arranged concentrically with respect to an optical axis of the detector.

The first optical sensor may be arranged in front of the second optical sensor. Thus, as an example, the first optical sensor may simply be placed on the surface of the second optical sensor. Additionally or alternatively, the first optical sensor may be spaced apart from the second optical sensor by no more than five times the square root of a surface area of the first light-sensitive area. Additionally or alternatively, the first optical sensor may be arranged in front of the second optical sensor and may be spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm.

As outlined above, the second light-sensitive area is larger than the first light-sensitive area. Thus, as an example, the second light-sensitive area may be larger by at least a factor of two, more preferably by at least a factor of three and most preferably by at least a factor of five and the first light-sensitive area.

The first light-sensitive area specifically may be a small light-sensitive area, such that, preferably, the light beam fully illuminates this light-sensitive area. Thus, as an example which may be applicable to typical optical configurations, the first light-sensitive area may have a surface area of 1 mm² to 150 mm², more preferably a surface area of 10 mm² to 100 mm².

The second light-sensitive area specifically may be a large area. Thus, preferably, within a measurement range of the detector, light spots generated by a light beam propagating from the object of the detector may fully be located within the second light-sensitive area, such that the light spot is fully located within the borders of the second light-sensitive area. As an example, which is applicable e.g. in typical optical setups, the second light-sensitive area may have a surface area of 160 mm² to 1000 mm², more preferably a surface area of 200 mm² to 600 mm².

Infrared optical sensors which may be used for the first optical sensor, for the second optical sensor or for both the first and second optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany. Thus, as an example, the first optical sensor, the second optical sensor or both the first and the second optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the first optical sensor, the second optical sensor or both the first and the second optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor pho-

todiode selected from the group consisting of: a Ge:Ga photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the first optical sensor, the second optical sensor or both the first and the second optical sensor may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The first and second optical sensors each, independently, may be opaque, transparent or semi-transparent. For the sake of simplicity, however, opaque sensors which are not transparent for the light beam, may be used, since these opaque sensors generally are widely commercially available.

The first and second optical sensors each specifically may be uniform sensors having a single light-sensitive area each. Thus, the first and second optical sensors specifically may be non-pixelated optical sensors.

As outlined above, by evaluating the first and second sensor signals, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more additional transversal sensors may be used for determining at least one transversal coordinate of the object. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object.

In a further embodiment, the detector may comprise:

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor may be configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;

- at least one second optical sensor having a second light-sensitive area, wherein the second optical sensor may be configured to generate at least one second sensor signal in response to an illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area and the second light sensitive area may be arranged such that the light-sensitive areas differ in their longitudinal coordinate; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

With respect to definitions and embodiments reference can be made to the description of the first preferred embodiment. In particular, the first light-sensitive area may be smaller than the second light-sensitive area.

In a further preferred embodiment of the present invention, the detector may comprise:

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor may be configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;
- at least one evaluation device may be configured for evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

In this further preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in spatial offset and/or surface areas.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. As further used herein, the term "matrix" generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane.

As explained in more detail in e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot, may be characterized by two measurement variables: a measurement signal measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their respective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming the quotient signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the quotient signal for deriving the longitudinal coordinate.

The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus, may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.

The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or

alternatively, as will be outlined in further detail below, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

5 Consequently, in accordance with the present invention, the term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals
10 generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from
15 which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from
20 which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the
25 trapezoid, in particular of a plateau of the trapezoid.

Similarly, the term “sum signal” generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the
30 entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation
35 device may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of
40 the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

Similarly, the term “combined signal”, as used herein, generally refers to a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

The light beam propagating from the object to the detector specifically may fully illuminate the at least one optical sensor from which the center signal is generated, such that the at least one optical sensor from which the center signal arises is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the at least one optical sensor from which the sensor signal arises. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the entire matrix which is smaller than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a “light spot” generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal

may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

As outlined above, raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition.; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or maybe determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by

the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations, however, are feasible.

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As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be performed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

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As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

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The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

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As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the quotient signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined sig-

nal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the quotient signal Q such as the combined signal. This determining may be a one-step process, such as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined signal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

The matrix may be composed of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

As further outlined above, the matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term “essentially perpendicular”, reference may be made to the definition given above. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10

rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, as outlined above, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

As further outlined above, preferably, the sensor element may be oriented essentially perpendicular to an optical axis of the detector. Again, with respect to the term "essentially perpendicular", reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The detector may be adapted to determine at least one distance information of the object by using triangulation and/or structured light techniques. In known 3D sensing devices, such as devices using triangulation or structured light techniques, due to correspondence problems regular, constant or periodic pattern are not suitable since each measured point has to be assigned to one reference point of a reference pattern. The at least one light beam propagating from the object to the detector may be adapted to generate at least one reflection pattern on the matrix of optical sensors. The term "reflection pattern" refers to at least one image determined by the optical sensors in response to illumination of the light sensitive areas by the light beam propagating from the object to the detector which was generated in response to illumination by the illumination pattern. The reflection pattern may comprise at least one reflection feature depending on the corresponding illumination feature of the illumination pattern. The detector may be adapted to determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection pattern from the quotient signal. Thus, the detector may be adapted to pre-classify the at least one reflection feature of the reflection pattern. This allows using illumination patterns comprising regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal patterns may be preferred.

The evaluation device may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor

signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device may be adapted to determine at least one reference feature in at least one reference pattern corresponding to the at least one reflection feature. The term "reference pattern" refers to an image of comparison at a specific spatial position comprising at least one reference feature. As outlined above, the evaluation device may be adapted to perform an image analysis and to identify features of the reflection pattern. The evaluation device may be adapted to identify at least one reference feature in the reference pattern having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference pattern and the reflection pattern may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The distance may be a relative distance, also denoted as baseline. The evaluation device may be adapted to determine an epipolar line in the reference pattern. The relative position of the reference pattern and reflection pattern may be known. For example, the relative position of the reference pattern and the reflection pattern may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature to the real world feature from which it originates. Thus, the straight line may comprise possible object features corresponding to the selected reflection feature. The straight line and the baseline span an epipolar plane. As the reference pattern is determined at a different relative constellation from the reflection pattern, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference pattern. The epipolar line may be the intersection of the epipolar plane and the reference pattern. Thus, a feature of the reference pattern corresponding to the selected feature of the reflection pattern lies on the epipolar line.

Depending on the distance to the object, the reference feature corresponding to the image position of the reflection feature may be displaced within the reference pattern compared to the image position in the reflection pattern. The reference pattern may comprise at least one displacement region in which the reference feature corresponding to the selected reflection feature may be imaged. The displacement region may comprise only one reference feature. The dis-

placement region may extend along the epipolar line. The evaluation device may be adapted to determine the reference feature along the epipolar line. The evaluation device may be adapted to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm \varepsilon$ from the quotient signal Q to determine a displacement region along the epipolar line corresponding to $z \pm \varepsilon$. The evaluation device may be adapted to match the selected reflection feature with at least one reference feature within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating the corresponding reference and reflection features. The evaluation device may be adapted to match the selected feature of the reflection pattern with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. Preferably, the detector may be adapted to pre-classify the selected reflection feature using the quotient signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference pattern may have a relative distance to each other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line. For example, the illumination pattern may comprise at least one hexagonal pattern. Preferably, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Preferably, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other.

The evaluation device may be adapted to determine a displacement of the reference feature and the reflection feature. The evaluation device may be adapted to determine the displacement of the matched reference feature and the selected reflection feature. As used herein, the term “displacement” refers to the difference between an image position in the reference image to an image position in the reflection image. The evaluation device may be adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate z_{triang} . For example, the longitudinal information may be a distance value. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case the position of the selected reflection feature in the reflection pattern and the position of the matched reference feature and/or the relative displacement of the selected reflection feature and the matched reference feature are known, the longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. The displacement and the corresponding distance value may be stored in at least one storage

device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

The detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, at least one optical sensors of the matrix of optical sensors may be adapted to generate a so called FiP signal. For example, each of the optical sensors of the matrix may be adapted to generate the at least one FiP signal. For example, the matrix of optical sensors may be a pixelated FiP-sensor. Additionally or alternatively, the FiP signal may be determined, such as extracted and/or simulated, from the sensor signals of the matrix and the quotient as outlined above may be determined from the sensor signals.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image

may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

The illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or object recognition method.

Preferably, the illumination source may be a movable and/or mobile illumination source. For example, the illumination source may be moved during the measurement such as to illuminate the object from different positions and/or angles. However, embodiments are feasible in which the illumination source may be positioned in at least one fixed position, for example during a complete measurement time. The evaluation device may be adapted to illumination sources with unclear position such as due to high manufacturing tolerances and/or user interaction and/or user assembly or the like.

The illumination source may illuminate the object with a convergent and/or divergent and/or collimated light beam.

In a further preferred embodiment the detector may comprise:

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

10 In this further preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas

The light-sensitive areas of the optical sensors may overlap, as visible from the object, or may not overlap, i.e. may be placed next to each other without overlap. The light-sensitive areas may be spaced apart from each other or may directly be adjacent.

The detector may comprise more than two optical sensors. In any case, i.e. in the case of the detector comprising precisely two optical sensors and in the case of the detector comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor being spaced apart from the optical axis by a first spatial offset and at least one second optical sensor being spaced apart from the optical axis by a second spatial offset, wherein the first spatial offset and the second spatial offset differ. In case further optical sensors are provided, besides the first and second optical sensors, these additional optical sensors may also fulfill the condition or, alternatively, may be spaced apart from the optical axis by the first spatial offset, by the second spatial offset or by a different spatial offset. The first and second spatial offsets, as an example, may differ by at least a factor of 1.2, more preferably by at least a factor of 1.5, more preferably by at least a factor of two. The spatial offsets may also be zero or may assume negative values, as long as the above-mentioned conditions are fulfilled.

30 As outlined above, each light-sensitive area has a geometrical center. Each geometrical center of each light-sensitive area may be spaced apart from the optical axis of the detector, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located.

35 As outlined above, the optical sensors specifically may be located in one and the same plane, which, preferably, is a plane perpendicular to the optical axis. Other configurations, however, are possible. Thus, two or more of the optical sensors may also be spaced apart in a direction parallel to the optical axis.

40 For example, the optical sensors may be partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector. As used herein, the term "partial diode" may comprise several diodes that are connected in series or in parallel. This example is rather simple and cost-efficiently realizable. Thus, as an example, bi-cell diodes

or quadrant diodes are widely commercially available at low cost, and driving schemes for these bi-cell diodes or quadrant diodes are generally known. As used herein, the term "bi-cell diode" generally refers to a diode having two partial diodes in one packaging. Bi-cell and quadrant diodes may have two or four separate light sensitive areas, in particular two or four active areas.

5 As an example, the bi-cell diodes may each form independent diodes having the full functionality of a diode. As an example, each of the bi-cell diodes may have a square or rectangular shape, and the two diodes may be placed in one plane such that the two partial diodes, in total, form a 1 x 2 or 2 x 1 matrix having a rectangular shape. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes and quadrant diode is pro-
10 posed, as will be outlined in further detail below. Generally, however, the optical sensors specifically may be partial diodes of a quadrant diode, with a center of the quadrant diode being off-centered from the optical axis of the detector. As used herein, the term "quadrant diode" generally refers to a diode having four partial diodes in one packaging. As an example, the four partial diodes may each form independent diodes having the full functionality of a diode. As an exam-
15 ple, the four partial diodes may each have a square or rectangular shape, and the four partial diodes may be placed in one plane such that the four partial diodes, in total, form a 2 x 2 matrix having a rectangular or square shape. In a further example, the four partial diodes, in total, may form a 2 x 2 matrix having a circular or elliptical shape. The partial diodes, as an example, may be adjacent, with a minimum separation from one another. However, the separation between
20 adjacent diodes may also be varied to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent diodes may be increased by using an opaque mask. This may reduce the light on one of the diodes relative and/or absolute as compared to the adjacent diode which will may increase a quotient of the signal of the two diodes.

25 In case a quadrant diode is used, having a 2x2 matrix of partial diodes, the center of the quadrant diode specifically may be off-centered or offset from the optical axis. Thus, as an example, the center of the quadrant diodes, which may be an intersection of the geometrical centers of the optical sensors of the quadrant diode, may be off-centered from the optical axis by at least 0.2 mm, more preferably by at least 0.5 mm, more preferably by at least 1.0 mm or even 2.0
30 mm. Similarly, when using other types of optical sensors setups having a plurality of optical sensors, an overall center of the optical sensors may be offset from the optical axis by the same distance.

As outlined above, bi-cell and quadrant diodes may have two or four separate light sensitive
35 areas, in particular two or four active areas. The light sensitive areas may be separated by a dividing line. The dividing line, for example, of two Si-diodes, can be arranged parallel to the baseline of the detector, in particular parallel to a spot movement on the optical sensors. However, other arrangements are feasible. For example, the quadrant diode may comprise two di-
40 viding lines. The dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other. In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device may be adapted to determine a second quotient using at least three or all four sensor signals of the

quadrant diode. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotient signals allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient signal may be determined by dividing the sensor signals of two top quadrants, also called top segment, by the sensor signals of two bottom quadrants, also called bottom segment. Using the quotient signal of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between the top and bottom segment is parallel to the baseline, the quotient signal determined from top segment divided by bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient signal by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals of the quadrants such that distance measurement is possible over a wide range with a large resolution.

Generally, the light-sensitive areas of the optical sensors may have an arbitrary surface area or size. Preferably, however, specifically in view of a simplified evaluation of the sensor signals, the light-sensitive areas of the optical sensors are substantially equal, such as within a tolerance of less than 10%, preferably less than 5% or even less than 1%. This, specifically, is the case in typical commercially available quadrant diodes.

The detector may comprise at least one quadrant diode comprising four quadrants. For example, an upper left quadrant of the quadrant diode may be denoted "ol", a lower left quadrant "ul", an upper right quadrant may be denoted "or" and a lower right quadrant may be denoted "ur".

The light spot may move from the left in a small distance range to the left in a far distance range. The detector may comprise at least one neutral density filter. The quadrants of the small distance range may be covered by the neutral density filter. In the small distance range the quotient ol/ul may be used to determine the longitudinal coordinate z . In the far distance range a quotient or/ur and/or a quotient from a combination of all quadrants may be used. The neutral density filter allows that objects in the small distance range are adapted to illuminate the optical sensors brighter by constant amplification without saturating the amplifier. At the same time it is possible to obtain in the far distance range more of the signal. Thus, the neutral density filter allows to enhance dynamic of the detector.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between sensor signals and/or any secondary signal derived thereof and the longitudinal coordinate. Thus, the evaluation device may be config-

ured for determining at least one combined sensor signal out of the at least two sensor signals, i.e. of the at least one sensor signal of at least one first optical sensor and out of the at least one sensor signal of at least one second optical sensor.

5 The detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the bi-cell or quadrant diode may be adapted to generate a so called FiP signal. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. According to the FiP-effect, the longitudinal sensor signal, given the same total power, may exhibit at least one pronounced maximum for one or a plurality of focusings and/or for one or a plurality of specific sizes of the light spot on the sensor region or within the sensor region. For purposes of comparison, an observation of a maximum of the longitudinal sensor signal in a condition in which the corresponding material is impinged by a light beam with the smallest possible cross-section, such as when the material may be located at or near a focal point as affected by an optical lens, may be denominated as a "positive FiP-effect". The negative FiP effect describes an observation of a minimum of the longitudinal sensor signal under a condition in which the corresponding material is impinged by a light beam with the smallest available beam cross-section, in particular, when the material may be located at or near a focal point as effected by an optical lens. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

25 As outlined above, the detector may comprise the at least one illumination source. A distance perpendicular to an optical axis of the detector between the illumination source and the bi-cell or quadrant diode may be small. The distance perpendicular to the optical axis of the detector between the illumination source and the bi-cell or quadrant diode may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. The illumination source and the optical axis may be separated by a small baseline. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the bi-cell or quadrant diode and of the at least one optional transfer device. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device.

40 As generally used herein, the term "combine" generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least

one comparison signal or comparison result. As will be outlined in further detail below, the combined sensor signal or secondary signal may be or may comprise at least one quotient signal.

In case more than two sensor signals are provided, as an example, more than one quotient signal may be generated, such as by forming quotient signals of more than one pair of sensor signals.

Thus, as an example, in case more than two optical sensors are provided, the above-mentioned quotient formation may take place between two of the sensor signals generated by these optical sensors or may take place between more than two of the sensor signals. Thus, instead of using the first one of the sensor signals and the second one of the sensor signals in the formulae given above, combined sensor signals may be used for quotient formation.

The quotient signal Q generally is an example for an asymmetry parameter denoting an asymmetry or an asymmetric distribution of the light spot generated by the light beam on the light-sensitive areas. The quotient of the two or more optical sensors, such as the two or more photodiodes, may provide a combined signal which typically is monotonously dependent on the distance between the detector and the object from which the light beam travels towards the detector, as will be shown by experimental data below. In addition or as an alternative to the quotient signal, other types of combined functions implementing the sensor signals of two or more sensors in the setup of the present invention may be used, which also may show a dependency on the distance between the object and the detector. The asymmetry or the asymmetry parameter of the light spot, as an example, may be an indication of the width of a light beam. If this asymmetry parameter depends on the distance only, the measurement can be used to determine the distance.

In typical setups, commercially available quadrant diodes such as quadrant photodiodes are used for positioning, i.e. for adjusting and/or measuring a transversal coordinate of a light spot in the plane of the quadrant photodiode. Thus, as an example, laser beam positioning by using quadrant photodiodes is well known. According to a typical prejudice, however, quadrant photodiodes are used for xy-positioning, only. According to this assumption, quadrant photodiodes are not suitable for measuring distances. The above-mentioned findings, however, using an off-centered quadrant photodiode with regard to an optical axis of the detector, show otherwise, as will be shown in further measurements below. Thus, as indicated above, in quadrant photodiodes, the asymmetry of the spot can be measured by shifting the quadrant diode slightly off-axis, such as by the above-mentioned offset. Thereby, a monotonously z-dependent function may be generated, such as by forming the quotient signal Q of two or more of the sensor signals of two or more partial photodiodes, i.e. quadrants, of the quadrant photodiode. Therein, in principle, only two photodiodes are necessary for the measurement. The other two diodes may be used for noise cancellation or to obtain a more precise measurement.

In addition or as an alternative to using a quadrant diode or quadrant photodiode, other types of optical sensors may be used. Thus, as will be shown in further detail below, staggered optical sensors may be used.

- 5 The use of quadrant diodes provides a large number of advantages over known optical detectors. Thus, quadrant diodes are used in a large number of applications in combination with LEDs or active targets and are widely commercially available at very low price, with various optical properties such as spectral sensitivities and in various sizes. No specific manufacturing process has to be established, since commercially available products may be implemented into
10 the detector according to the present invention.

- The detector according to the present invention specifically may be used in multilayer optical storage discs, such as disclosed by international patent application number PCT/IB2015/052233, filed on March 26, 2015. Measurements performed by using the detector
15 according to the present invention specifically may be used in order to optimize the focus position in optical storage discs.

- As will be outlined in further detail below, the distance measurement by using the detector according to the present invention may be enhanced by implementing one or more additional distance measurement means into the detector and/or by combining the detector with other types
20 of distance measurement means. Thus, as an example, the detector may comprise or may be combined with at least one triangulation distance measurement device. Thus, the distance measurement can be enhanced by making use of a combination of the measurement principle discussed above and a triangulation type distance measurement. Further, means for measuring
25 one or more other coordinates, such as x- and/or y-coordinates, may be provided.

- In case a quadrant diode is used, the quadrant diode may also be used for additional purposes. Thus, the quadrant diode may also be used for conventional x-y-measurements of a light spot, as generally known in the art of optoelectronics and laser physics. Thus, as an example, the
30 lens or detector position can be adjusted using the conventional xy-position information of the quadrant diode to optimize the position of the spot for the distance measurement. As a practical example, the light spot, initially, may be located right in the center of the quadrant diode, which typically does not allow for the above-mentioned distance measurement using the quotient function Q. Thus, firstly, conventional quadrant photodiode techniques may be used for off-centering
35 a position of the light spot on the quadrant photodiode, such that, e.g., the spot position on the quadrant diode is optimal for the measurement. Thus, as an example, the different off-centering of the optical sensors of the detector may simply be a starting point for movement of the optical sensors relative to the optical axis such that the light spot is off-centered with respect to the optical axis and with respect to a geometrical center of the array of the optical sensors.

- 40 Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned quadrant diode. The case $m=2$, $n=2$ is the case of

the quadrant diode or quadrant optical sensor, which, for practical reasons, is one of the preferred cases, since quadrant photodiodes are widely available.

As a starting point, a geometrical center of the optical sensors within the array may be off-centered from the optical axis, such as by the above-mentioned offset. The sensor array specifically may be movable relative to the optical axis, for example along a gradient, preferably automatically, such as by moving the sensor array, e.g. in a plane perpendicular to the optical axis, and/or by moving the optical axis itself, e.g. shifting the optical axis in a parallel shift and/or tilting the optical axis. Thus, the sensor array may be shifted in order to adjust a position of a light spot generated by the light beam in the plane of the sensor array. Additionally or alternatively, the optical axis may be shifted and/or tilted by using appropriate elements, such as by using one or more deflection elements and/or one or more lenses. The movement, as an example, may take place by using one or more appropriate actuators, such as one or more piezo actuators and/or one or more electromagnetic actuators and/or one or more pneumatic or mechanical actuators, which, e.g., move and/or shift the array and/or move and/or shift and/or tilt one or more optical elements in the beam path in order to move the optical axis, such as parallel shifting the optical axis and/or tilting the optical axis. The evaluation device specifically may be adjusted to control a relative position of the sensor array to the optical axis, e.g. in the plane perpendicular to the optical axis. An adjustment procedure may take place in that the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly, moving the array relative to the optical axis, such as by moving the array and/or the optical axis, e.g. by moving the array in the plane to the optical axis until the light spot is off-centered and/or by tilting a lens until the light spot is off-centered. As used therein, a transversal position may be a position in a plane perpendicular to the optical axis, which may also be referred to as the x-y-plane. For the measurement of the transversal coordinate, as an example, the sensor signals of the optical sensors may be compared. As an example, in case the sensor signals are found to be equal and, thus, in case it is determined that the light spot is located symmetrically with respect to the optical sensors, such as in the center of the quadrant diodes, a shifting of the array and/or a tilting of a lens may take place, in order to off-center the light spot in the array. Thus, as outlined above, the off-centering of the array from the optical axis, such as by off-centering the center of the quadrant photodiode from the optical axis, may simply be a starting point in order to avoid the situation which is typical, in which the light spot is located on the optical axis and, thus, is centered. By off-centering the array relative to the optical axis, thus, the light spot should be off-centered. In case this is found not to be true, such that the light spot, incidentally, is located in the center of the array and equally illuminates all optical sensors, the above-mentioned shifting of the array relative to the optical axis may take place, preferably automatically, in order to off-center the light spot on the array. Thereby, a reliable distance measurement may take place.

Further, in a scanning system with a movable light source, the position of the light spot on the quadrant diode may not be fixed. This is still possible, but may necessitate that different calibrations are used, dependent on the xy-position of the spot in the diode.

Further, the use of the above-mentioned quotient signal Q is a very reliable method for distance measurements. Typically, Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q=s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, the sensor signals, such as the above-mentioned first sensor signal s_1 and the above-mentioned second sensor signal s_2 , decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, due to the off-centering, the one of the sensor signals decreases more rapidly than the other, since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. By spreading the light spot, however, the portion of the light illuminating the one or more optical sensors outside the center of the light spot increases, as compared to a situation of a very small light spot. Thus, the quotient of the sensor signals continuously changes, i.e. increases or decreases, with increasing diameter of the light beam or diameter of the light spot. The quotient, further, may further be rendered mainly independent from the total power of the light beam, since the total power of the light beam forms a factor in all sensor signals. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the sensor signals and the size or diameter of the light beam.

As outlined above, specifically, quadrant photodiodes may be used. As an example, commercially available quadrant photodiodes may be integrated in order to provide four optical sensors, such as one or more quadrant photodiodes available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany, such as quadrant Si PIN photodiodes of the type S4349, which are sensitive in the UV spectral range to the near IR spectral range. In case an array of optical sensors is used, the array may be a naked chip or may be an encapsulated array, such as encapsulated in a TO-5 metal package. Additionally or alternatively, a surface mounted device may be used, such as TT Electronics OPR5911 available from TT Electronics plc, Fourth Floor, St Andrews House, West Street Woking Surrey, GU21 6EB, England. It shall be noted that other optical sensors may also be used.

Further, it shall be noted that, besides the option of using precisely one quadrant photodiode, two or more quadrant photodiodes may also be used. Thus, as an example, a first quadrant photodiode may be used for the distance measurement, as described above, providing the two or more optical sensors. Another quadrant photodiode may be used, e.g. in a second partial beam path split off from the beam path of the first quadrant photodiode, for a transversal position measurement, such as for using at least one transversal coordinate x and/or y . The second quadrant photodiode, as an example, may be located on-axis with respect to the optical axis.

Further, it shall be noted that, besides the option of using one or more quadrant photodiodes, one or more quadrant photodiodes or further photodiode arrays may also be replaced or mim-

icked by separated photodiodes that are arranged or assembled close to each other, preferably in a symmetric shape such as a rectangular matrix, such as a 2 x 2 matrix. However further arrangements are feasible. In such an arrangement or assembly, the photodiodes may be arranged or assembled in a housing or mount, such as all photodiodes in a single housing or mount or groups of photodiodes in one housing or mount, or each of the photodiodes in a separate housing or mount. Further, the photodiodes may also be assembled directly on a circuit board. In such arrangements or assemblies, photodiodes may be arranged as such that the separation between the active area of the photodiodes, has a distinct value less than one centimeter, preferably less than one millimeter, more preferably as small as possible. Further, to avoid optical reflexes, distortions, or the like that may deteriorate the measurement, the space between the active areas may be either empty or filled with a material, preferably with a light absorbing material such as a black polymer, such as black silicon, black polyoxymethylene, or the like, more preferably optically absorbing and electrically insulating material, such as black ceramics or insulating black polymers such as black silicon, or the like. Further, the distinct value of the photodiode separation may also be realized by adding a distinct building block between the photodiodes such as a plastic separator. Further embodiments are feasible. The replacement of quadrant photodiodes by single diodes arranged in a similar setup such as in a 2 x 2 rectangular matrix with minimal distance between the active areas may further minimize the costs for the optical detector. Further, two or more diodes from a quadrant diode may be connected in parallel or in series to form a single light sensitive area.

The optical sensors each specifically may be uniform sensors having a single light-sensitive area each. Thus, the optical sensors specifically may be non-pixelated optical sensors.

In a further preferred embodiment the detector may comprise

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
- at least one second optical sensor having:
 - o at least one fluorescent waveguiding sheet forming a second light-sensitive area, wherein the fluorescent waveguiding sheet is oriented towards the object such that at least one light beam propagating from the object towards the detector generates at least one light spot in the second light-sensitive area, wherein the fluorescent waveguiding sheet contains at least one fluorescent material, wherein the fluorescent material is adapted to generate fluorescence light in response to the illumination by the light beam; and
 - o at least one photosensitive element located at at least one edge of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive element by the fluorescent waveguiding sheet and capable of generating at least one second sensor signal in response to the illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and

- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

In this further preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas. The first and second light-sensitive areas specifically may be oriented towards the object

The light beam propagating from the object to the detector specifically may fully illuminate the first light-sensitive area, such that the first light-sensitive area is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the first optical sensor. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the second light-sensitive area, which is smaller than the second light-sensitive area, such that the light spot is fully located within the second light-sensitive area. Within the light spot on the second light-sensitive area, a shadow created by the first optical sensor may be located. Thus, generally, the first optical sensor, having the smaller first light-sensitive area, may be located in front of the second optical sensor, as seen from the object, with the first light-sensitive area being fully located within the light beam and with the light beam generating a light spot on the second light-sensitive area being smaller than the second light-sensitive area, and with further a shadow created by the first optical sensor within the light spot. The situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a light spot generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

The first and second optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. The first and second optical sensors may be sensitive in one and the same spectral range or may have differing spectral sensitivities.

As further used herein, a fluorescent waveguiding sheet generally refers to an element with both waveguiding properties and fluorescent properties. Therein, waveguiding generally refers to the property or a plurality of an element to guide the light in one or more of the ultraviolet, visible or the infrared spectral range by internal reflection, specifically by internal total reflection. The term "fluorescence" generally refers to the property of an element or a material to emit secondary light, also referred to as fluorescence light, in one or more of the ultraviolet, visible or infrared spectral range, in response to excitation by electromagnetic radiation, also referred to as primary radiation or excitation radiation, such as primary light or excitation light. In most cases, the emitted light, fluorescence light or secondary light has a longer wavelength and a lower energy than the primary radiation. The primary radiation typically induces the presence of excited states within the fluorescent material, such as so-called excitons. Typically, excited state decay times for photon emissions with energies from the UV to near infrared are within the range of 0.5 to 20 nanoseconds. Other examples, however, are feasible within the present invention. Similarly, as used herein, the term "fluorescent material" generally refers to a material having fluorescence

properties. The term “fluorescence light” generally refers to the secondary light generated during the above-mentioned fluorescence process.

For potential embodiments of the fluorescent waveguiding sheet, generally, reference may be made to the so-called “Q-Foil Technology”, as disclosed e.g. in the above-mentioned article by P. Bartu, R. Koeppel, N. Arnold, A. Neuling, L. Fallon, and S. Bauer, Conformable large-area position-sensitive photodetectors based on luminescence collecting silicone waveguides, J. Appl. Phys. 107, 123101 (2010). Fluorescent waveguiding sheets and/or second optical sensors as disclosed therein may also be used in the context of the present invention, within the second optical sensor.

The fluorescent waveguiding sheet, as will be outlined in further detail below, specifically may be an element or may comprise an element which has a sheet-like shape or which is a sheet. As used herein, a “sheet” generally refers to an element which has a lateral extension, such as a diameter or an equivalent diameter, which significantly exceeds a thickness of the element, such as by at least a factor of 5, more preferably by at least a factor of 10 or even more preferably by at least a factor of 20, a factor of 50 or even a factor of 100. The sheet specifically may be flexible, deformable or rigid.

As will be outlined in further detail below, the fluorescent waveguiding sheet specifically may be or may comprise a transparent material, specifically a transparent sheet. The transparency may be a transparency of at least 50 to 70% in the visible spectral range or in a part thereof, such as in a range of 500 nm to 700 nm. Other embodiments are feasible.

As further used herein, the term “sensitive area” generally refers to a two-dimensional or three-dimensional region of an element, specifically of the fluorescent waveguiding sheet or the optical sensor, which is sensitive to external influences and, e.g., produces at least one reaction in response to an external stimulus. In this case, specifically, the sensitive area may be sensitive to an optical excitation. The sensitive area specifically may be a part of a surface or the volume of the fluorescent waveguiding sheet, such as the whole surface of the fluorescent waveguiding sheet or a part thereof.

As further used herein, the term “fluorescent waveguiding sheet is oriented towards the object” generally refers to the situation that the surface of the fluorescent waveguiding sheet or a part of this surface, specifically the sensitive area, is fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the sensitive area may form an angle with a surface element of the sensitive area or of the fluorescent waveguiding sheet which is different from 0°, such as an angle in the range of 20°-90°. Other embodiments, however, are feasible.

Most preferably, however, the at least one fluorescent waveguiding sheet, the sensitive area or a part thereof is oriented essentially perpendicular to an optical axis of the optical sensor and/or of the detector.

As further used herein, the term “edge” of the fluorescent waveguiding sheet generally refers to a boundary of the at least one fluorescent waveguiding sheet, such as a side boundary or side edge or a front or back face of the fluorescent waveguiding sheet. Thus, as the skilled person will recognize, the term “edge” of the fluorescent waveguiding sheet may generally refer to an interface or boundary between the fluorescent waveguiding sheet and a surrounding atmosphere, such as air. Specifically, the edge may be a border of a second light-sensitive area formed by the fluorescent waveguiding sheet. The term “located at” generally refers to the fact that the photosensitive elements are either located directly on the edge or in close proximity to the edge. As an example, at least 50% of the light emerging from the edge into the direction of the photosensitive element may be collected by the respective photosensitive element, without being lost by scattering, beam widening or other losses. As an example, the photosensitive element may be located at a position spaced apart from the edge of no more than 10 mm, more preferably of no more than 5 mm. It shall be noted, however, that other embodiments are feasible for connecting the fluorescence light. Most preferably, all photosensitive elements are located relative to their respective edges of the fluorescent waveguiding sheet in the same way, in order to provide similar measurement conditions for all photosensitive elements.

The at least one photosensitive element located at the at least one edge of the fluorescent waveguiding sheet may, as an example, be fully or partially located in the same plane as the fluorescent waveguiding sheet and/or may fully or partially be located in a different plane. In the latter case, as will be outlined in further detail below, as an example, an optical coupling between the edge of the fluorescent waveguiding sheet and the at least one photosensitive element may take place, by using at least one optical coupling element. Further, in case a plurality of photosensitive elements is provided in the second optical sensor, at least one of the photosensitive elements may be located in the same plane as the fluorescent waveguiding sheet and at least one of the photosensitive elements may be located outside the plane of the fluorescent waveguiding sheet. Further, a direction of view of the at least one photosensitive element may be parallel to the plane of the fluorescent waveguiding sheet or may be directed otherwise, such as perpendicular to the plane. Therein, when talking about a “plane” of the fluorescent waveguiding sheet, this term does not necessarily imply that the fluorescent waveguiding sheet is fully planar. Thus, as an example, the fluorescent waveguiding sheet may also be curved or bent, and the plane of the fluorescent waveguiding sheet at the location of the at least one photosensitive element may be a local tangential plane.

The term “edge” of the fluorescent waveguiding sheet, as used herein, may refer to a straight line or straight border area of the fluorescent waveguiding sheet, in the following also referred to as a “straight edge”, or may also refer to a non-straight line or non-straight border area of the fluorescent waveguiding sheet, such as a corner of the fluorescent waveguiding sheet. Thus, the at least one photosensitive element or, in case a plurality of photosensitive elements is provided, at least one of the photosensitive elements, may also be located at at least one corner of the fluorescent waveguiding sheet, such as at at least one corner of the second light-sensitive area formed by the fluorescent waveguiding sheet. Thus, specifically, the edge may comprise a

rim or a portion of a rim of the fluorescent waveguiding sheet, such as a corner and/or a straight rim portion. Additionally or alternatively, however, the edge may also comprise a flat surface of the fluorescent waveguiding sheet, such as a front side or back side.

- 5 As already outlined above, in order to improve feeding fluorescence light guided from the light spot towards the at least one photosensitive element into the at least one respective photosensitive element, at least one optical coupling may take place, by using at least one optical coupling element in between the fluorescent waveguiding sheet and the respective photosensitive element. Thus, the at least one photosensitive element, or, in case a plurality of photosensitive
- 10 elements is provided, at least one of the photosensitive elements may be optically coupled to the fluorescent waveguiding sheet by at least one optical coupling element configured for at least partially coupling the fluorescence light guided by the fluorescent waveguiding sheet out of the fluorescent waveguiding sheet and, preferably, at least partially into the photosensitive element. As used herein, the term "optical coupling element" generally refers to an arbitrary element which is configured for one or more of disturbing, diminishing or interrupting an internal
- 15 total reflection within the fluorescent waveguiding sheet which takes place during waveguiding within the fluorescent waveguiding sheet. Thus, as an example, the optical coupling element may be an arbitrary transparent element having an index of refraction in between an index of refraction of the fluorescent waveguiding sheet and the photosensitive element and/or the ambient atmosphere, such as air. Thus, as an example, in case an index of refraction of the fluorescent waveguiding sheet is denoted by n_1 , and an index of refraction of the photosensitive element is denoted by n_2 , an index of refraction n_3 of the optical coupling element may be
- 20 $n_1 < n_3 < n_2$ or $n_1 > n_3 > n_2$.
- 25 The optical coupling element may be in direct contact with the fluorescent waveguiding sheet, such as with at least one surface, such as a surface facing the object and/or a surface facing away from the object, of the fluorescent waveguiding sheet. Further, the optical coupling element may also be in direct contact with the at least one photosensitive element. Further, for each photosensitive element, an independent optical coupling element may be provided, or alternatively, a plurality of photosensitive elements may share a common optical coupling element, or, alternatively, a plurality of optical coupling elements may be coupled to one photosensitive element.
- 30

- Various ways of optical coupling are generally known to the skilled person and may also be
- 35 used for coupling fluorescence light from the fluorescent waveguiding sheet into the photosensitive element. Thus, as an example, the at least one optical coupling element may comprise at least one element selected from the group consisting of: a portion of transparent adhesive attaching the photosensitive element to the fluorescent waveguiding sheet; an etched portion within the fluorescent waveguiding sheet, such as within a surface of the fluorescent waveguiding sheet, such as a surface facing the object and/or facing away from the object; a scratch in the fluorescent waveguiding sheet, such as a scratch in the surface of the fluorescent waveguiding sheet, such as a surface facing the object and/or facing away from the object; a prism. Additionally or alternatively, other optical coupling elements are generally known and may also
- 40

be used in the present invention. In a most simple case, the at least one photosensitive element may simply be adhered or glued to a surface of the fluorescent waveguiding sheet, such as by at least one transparent glue or adhesive, e.g. a transparent epoxy. Other ways of optical coupling are feasible.

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The first optical sensor and the second optical sensor specifically may have an essentially identical electrical capacitance. When referring to a capacitance of an optical sensor, generally, the capacity C as may be determined in an oscillating circuit is referred to, as the person skilled in the art of electronics generally will recognize. Thus, when implementing the first optical sensor or, alternatively, the second optical sensor into an oscillating circuit, the oscillating circuit, as an example, may have essentially the same resonance frequency. As used herein, the term “essentially identical” generally refers to the fact that, as an example, the capacitance of the first optical sensor and the capacitance of the second optical sensor differ by no more than a factor of 10, more preferably a by no more than a factor of 5 or even by no more than a factor of 2.

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Further, when referring to the capacitance of the second optical sensor, reference is made to the capacitance of the at least one photosensitive element of the second optical sensor, since the photosensitive element determines the electrical behavior and, specifically, the high-frequency behavior of the second optical sensor. In case a plurality of photosensitive elements are present in the second optical sensor, the capacitance of at least one of these photosensitive elements is meant.

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As outlined above, for example with respect to the first preferred embodiment, for evaluating the first and second sensor signals at least one predetermined or determinable relationship between the first and second sensor signals and the longitudinal coordinate may be used. With respect to evaluation of the first and second sensor signals reference can be made to the description of the first preferred embodiment. Thus, the evaluation device specifically may be configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. The evaluation device may be configured for determining the longitudinal coordinate by evaluating the quotient signal Q .

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Typically, in the setup described above, Q may be a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. This is due to the fact that, in the preferred setup described above, the first light-sensitive area is fully illuminated and preferably is located in the center of the light spot, such as on the optical axis. Consequently, with increasing diameter of the light spot, the peak intensity of the illumination, which mainly may be located in the first light-sensitive area, decreases and, thus, the first sensor signal decreases. The second sensor signal, however, may be more or less independent from the size of the light spot, as long as the light spot, except for the shadow created by the first optical sensor, is located on the second light-sensitive area. Consequently, the second sensor signal may remain more or less constant. The quotient of the first and second sensor

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signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the quotient
5 Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship
10 between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal
15 coordinate of the object, or both.

The first optical sensor and the second optical sensor may be positioned in various ways. Thus, in a first example, the first optical sensor is positioned in front of the second optical sensor, such that the light beam, firstly, passes the first optical sensor and, secondly, thereafter, reaches the
20 second optical sensor. Alternatively, however, a reverse order is possible. Thus, as an alternative, the second optical sensor may be placed in front of the first optical sensor, such that the light beam, firstly, reaches the second optical sensor, fully or partially passes through the second optical sensor and, secondly, thereafter, reaches the first optical sensor. The latter alternative is mainly possible due to the fact that the fluorescent waveguiding sheet of the second optical
25 sensor may, in this embodiment or other embodiments, be embodied fully or partially transparent, specifically in the spectral range of the light beam. Thus, the smaller first optical sensor may be placed on a reverse side of the second optical sensor, from the perspective of the object.

The second optical sensor and, more specifically, the fluorescent waveguiding sheet of the second optical sensor, may be placed in various positions in one or more beam paths. Thus, as an example, the second optical sensor and/or the fluorescent waveguiding sheet of the second optical sensor may be placed close to a focal point within a beam path. Alternatively, however, specifically in order to improve the function of summing, the second optical sensor may also be
30 placed away from the focal point, such as by one, two or more than two Rayleigh wavelengths.

As outlined above, the second optical sensor comprises at least one photosensitive element, from which the at least one second sensor signal arises. Therein, precisely one photosensitive
40 element may be used, such as for collecting at least a part of the fluorescence light and for creating a second sensor signal in response. Alternatively, however, the second optical sensor may comprise two, three, four or more photosensitive elements which, preferably, are located at at least two edges of the fluorescent waveguiding sheet, such as at least two opposing portion

of a rim of the fluorescent waveguiding sheet. The photosensitive elements may be capable of detecting fluorescence light guided from the light spot towards the photosensitive elements by the fluorescent waveguiding sheet and capable of generating at least one sensor signal each. Thus, a plurality of second sensor signals may exist.

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In case at least two photosensitive elements are provided, being located at at least two edges of the fluorescent waveguiding sheet, a plurality of possibilities exists. Thus, as an example, the at least two photosensitive elements may be located at one or more of: at least two straight edges of the fluorescent waveguiding sheet, such as at least two opposing edges, such as opposing rim portions; at least two corners of the fluorescent waveguiding sheet, such as at least two opposing corners; at least one corner of the fluorescent waveguiding sheet and at least one straight edge, such as at least one straight rim portion, of the fluorescent waveguiding sheet. Other possibilities are generally given.

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The detector and, preferably, the evaluation device may specifically be embodied to combine at least two of these second sensor signals to form at least one common second sensor signal. This at least one common second sensor signal, specifically, may be used, in conjunction with the first second sensor signal, for determining the longitudinal coordinate. Additionally, at least one common second sensor signal may be used for acquiring additional information, such as at least one item of information on a transversal position of the object or a part thereof, as will be outlined in further detail below.

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Thus, for combining the at least two second sensor signals, the evaluation device may comprise at least one summing device configured for summing up the sensor signals of the at least two photosensitive elements, thereby forming a sum signal S. As an example, the sum signal may be derived by using the formula

$$S = \sum_i c_i s_i, \quad (1)$$

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with s_i denoting the second sensor signals, with $i = 1 \dots N$ with N being a positive integer indicating the number of the photosensitive elements or a smaller positive integer, and with c_i being respective calibration coefficients.

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As outlined above, the evaluation device specifically may be configured to determine the at least one longitudinal coordinate z of the object by using at least one predetermined relationship between the first sensor signal, the at least one second sensor signal and the longitudinal coordinate z. In case a plurality of second sensor signals is given, as outlined above, at least one combined second sensor signal may be used in this relationship. Thus, specifically, the sum signal S may be used. Thus, as an example, the evaluation device may be configured to determine the at least one longitudinal coordinate z by using at least one predetermined relationship between the first sensor signal, the sum signal S of the second sensor signals and the longitudinal coordinate z.

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In addition to the at least one longitudinal coordinate, the detector may further be configured for determining at least one transversal coordinate of the object and/or of a part thereof. For this purpose, the detector may comprise at least one transversal optical sensor. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or WO 2014/198629 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Thus, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Preferably, however, the at least one second optical sensor itself may be used as a transversal optical sensor, for determining at least one transversal coordinate of the object. This setup is specifically advantageous in case a plurality of photosensitive elements is used in the second optical sensor, such as in different positions and located at different edges of the fluorescent waveguiding sheet. The evaluation device may be configured to determine at least one transversal coordinate x , y of the object by evaluating the sensor signals of the photosensitive elements. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

Thus, generally, the evaluation device may further be configured to determine at least one transversal coordinate x , y of the object by evaluating the second sensor signals of the photosensitive elements. For the purpose of determining the at least one transversal coordinate in one or more directions, the sensor signals of the photosensitive elements may be compared. Thus, as is evident to the skilled person, the sensor signal of a respective photosensitive element, which represents the fluorescence light guided to the photosensitive elements by the fluorescent waveguiding sheet from the light spot and, thus, from the location of generation of the fluorescence light, depends on a distance between the light spot and the respective photosensitive element. Generally, with an increasing distance between the light spot and the photosensitive element, the sensor signal of the respective photosensitive element will decrease, such as due to losses during waveguiding and/or due to spreading of the fluorescence light. By comparing the sensor signals of the photosensitive elements located at different, known positions, the lateral or transversal position of the light spot on the fluorescent waveguiding sheet may be determined and, therefrom, by using e.g. a known or determinable relationship between the transversal position of the light spot and the transversal coordinate of the object, the transversal coordinate of the object. Again, empirical relationships and/or semiempirical relationships and/or analytical relationships may be used, such as the lens equation which is generally known to the skilled person.

Thus, the evaluation device specifically may be configured for evaluating at least two second sensor signals of at least two different photosensitive elements. As an example, these at least two second sensor signals may be compared in order to determine a transversal position of the light spot on the fluorescent waveguiding sheet and/or for directly deriving a transversal position of the object. Thus, as an example, a difference between at least two second sensor signals may be determined in order to derive at least one item of information on a transversal position of the object. Consequently, the evaluation device may comprise at least one subtracting device configured to form at least one difference signal D between second sensor signals generated by at least two of the photosensitive elements. The second sensor signals may comprise at least one second sensor signal s_{21} and at least one second sensor signal s_{22} , wherein the at least one difference signal D is proportional to $a \cdot s_{21} - b \cdot s_{22}$:

$$D \sim a \cdot s_{21} - b \cdot s_{22}, \quad (2)$$

with a, b being real number coefficients, preferably with $a=1$ and $b=1$. Additionally or alternatively, the at least one difference signal D may be derived according to the formula:

$$D = (a \cdot s_{21} - b \cdot s_{22}) / (a \cdot s_{21} + b \cdot s_{22}). \quad (3)$$

The subtracting device specifically may be configured to form at least one first difference signal D_x from which at least one first transversal coordinate x of the object is derived. Additionally or alternatively, the subtracting device may be further configured to form at least one second difference signal D_y from which at least one second transversal coordinate y of the object is derived. Thus, the first difference signal D_x may be generated from at least two second sensor signals s_{x1} , s_{x2} of at least two photosensitive elements located at opposing edges of the waveguiding sheet in a first dimension. The second difference signal D_y may be generated from at least two second sensor signals s_{y1} , s_{y2} of at least two photosensitive elements located at opposing edges of the waveguiding sheet in a second dimension. The at least one first difference signal D_x may be derived according to the formula

$$D_x = (a \cdot s_{x1} - b \cdot s_{x2}) / (a \cdot s_{x1} + b \cdot s_{x2}), \quad (4)$$

and the at least one second difference signal D_y may be derived according to the formula

$$D_y = (c \cdot s_{y1} - d \cdot s_{y2}) / (c \cdot s_{y1} + d \cdot s_{y2}), \quad (5)$$

with a, b, c, d being real number coefficients, preferably with $a=1$, $b=1$, $c=1$ and $d=1$.

As outlined above, in case the second optical sensor comprises a plurality of photosensitive elements, the second optical sensor specifically may comprise at least two photosensitive elements located at opposing edges of the fluorescent waveguiding sheet. The photosensitive elements may comprise at least one first pair of photosensitive elements located at opposing edges of the fluorescent waveguiding sheet in a first dimension of a coordinate system, and the

photosensitive elements may further comprise at least one second pair of photosensitive elements located at opposing edges of the fluorescent waveguiding sheet in a second dimension of the coordinate system.

- 5 Further optional details may refer to the at least one fluorescent waveguiding sheet and/or to the second light-sensitive area. Thus, as an example, the second light-sensitive area specifically may be a homogeneous light-sensitive area. Thus, the second light-sensitive area may not be subdivided physically into partial areas, such as pixels. Contrarily, the sensitive area may be one homogeneous area which forms a uniform fluorescence.

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The second light-sensitive area specifically may be a large sensitive area. Thus, as an example, the second light-sensitive area may have a surface of at least 5 mm², preferably of at least 10 mm², more preferably of at least 100 mm², more preferably of at least 400 mm². As an example, the sensitive area may have a surface of 5 mm² to 10,000 mm², such as 100 mm² to 2500 mm².

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The large-area design of the sensitive area is advantageous in many ways. Thus, specifically, by increasing the surface of the second light-sensitive area, a resolution of the determination of the transversal coordinates may be increased. Further, the field of view of the detector, e.g. the viewing angle, may be widened by using a large second light-sensitive area.

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The fluorescent waveguiding sheet specifically may comprise at least one planar sheet. Therein, however, slight curvatures still may be tolerated. In other embodiments, however, the fluorescent waveguiding sheet may also be embodied as a curved fluorescent waveguiding sheet, such as in order to provoke specific optical effects which might be desirable in certain applications. Thus, one of the advantages of the present detector may reside in the fact that the fluorescent waveguiding sheet specifically may be curved, flexible or having a specific geometry.

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The fluorescent waveguiding sheet may have a thickness of 10 µm to 3 mm, preferably a thickness of 100 µm to 1 mm, e.g. a thickness of 50 µm to 2 mm. The thickness of the waveguiding sheet specifically may be dimensioned of the waveguiding sheet along an optical axis of the detector. The thickness may be adapted to improve or optimize waveguiding properties of the fluorescence light.

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The fluorescent waveguiding sheet may fully or partially be rigid or, alternatively, may fully or partially be embodied flexible or deformable.

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The fluorescent waveguiding sheet may comprise at least one matrix material. As used herein, the term "matrix material" generally refers to a material which forms the main part of the fluorescent waveguiding sheet and which defines the main body of the fluorescent waveguiding sheet. As an example, the matrix material may be a material which is capable of receiving one or more additional materials that therein, such as by intermixing, chemical bonding, dispersion or solving. Thus, the at least one fluorescent material may be one or more of mixed into the matrix material, dispersed into the matrix material, chemically bound to the matrix material or dissolved in the matrix material.

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The matrix material specifically may be or may comprise at least one plastic material. The plastic material specifically may be or may comprise at least one polymer material. The plastic material, as an example, may be or may comprise at least one material selected from the group consisting of a polycarbonate, a poly(methyl methacrylate), a polystyrene, a polyurethane, a polypropylene, a polyethylene terephthalate, a polyvinylchloride. Other materials, however, are feasible.

The fluorescent material specifically may comprise an arbitrary fluorophore. Specifically, the at least one fluorescent material may comprise at least one fluorescent dye. A wide variety of fluorescent dyes is generally known to the skilled person. Thereof, as outlined above, most of these fluorescent dyes exhibit the above-mentioned saturation effects, thereby rendering the fluorescence being a nonlinear function of the excitation. As an example, the fluorescent dye may be capable of being saturated by the light beam, such that a total power of the fluorescence light generated by the fluorescent dye is a nonlinear function of an intensity of the light beam. Specifically, the total power of the fluorescence light may be sub-proportional to the intensity of the light beam.

The fluorescent dye specifically may comprise at least one organic fluorescent dye. Inorganic dyes, however, may be used additionally or alternatively. The fluorescent dye specifically may be selected from the group consisting of: a xanthene derivative, preferably one or more of fluorescein, rhodamine, oregon green, eosin, texas red, or a derivative of any component thereof; a cyanine derivative, preferably one or more of cyanine, indocarbocyanine, oxacarbocyanine, thiocarbocyanine, merocyanine, or a derivative of any component thereof; a squaraine derivative or a ring-substituted squaraine, preferably one or more of Seta, SeTau, and Square dyes, or a derivative of any component thereof; a naphthalene derivative, preferably one or more of a dansyl or a prodan derivative thereof; a coumarin derivative; a oxadiazole derivative, preferably one or more of pyridyloxazole, nitrobenzoxadiazole, benzoxadiazole, or a derivative of any component thereof; an anthracene derivative, preferably one or more of an anthraquinone, DRAQ5, DRAQ7, CyTRAK orange, or a derivative of any component thereof; a pyrene derivative, preferably cascade blue; an oxazine derivative, preferably one or more of nile red, nile blue, cresyl violet, oxazine 170, or a derivative of any component thereof; an acridine derivative, preferably one or more of proflavin, acridine orange, acridine yellow, or a derivative of any component thereof; an arylmethine derivative, preferably one or more of auramine, crystal violet, malachite green, or a derivative of any component thereof; a tetrapyrrole derivative, preferably one or more of porphin, phthalocyanine, bilirubin; a rylene dye or any derivative thereof, such as a perylene dye; a naphthalene imid or perylene imide; a naphthoilene benzimidazole dye such as published in WO 2012/168395 A1; or a derivative of any component of the listed substances. It shall be noted, however, that other dyes may be used additionally or alternatively.

Further optional details of the detector referred to the at least one photosensitive element or to the plurality of photosensitive elements. Thus, as an example, the at least one photosensitive element may comprise at least one photodiode, preferably at least one inorganic photodiode, such as a silicon photodiode. The at least one photosensitive element may comprise at least

one elongated photosensitive element extending along at least one segment of an edge or rim portion of the waveguiding sheet.

As outlined above, the fluorescent waveguiding sheet specifically may be a rectangular fluorescent waveguiding sheet, preferably a square waveguiding sheet. In the setup, as an example, the photosensitive elements may be located at each of the four edges, such as at each of the four straight rim sections or rim portions, of the waveguiding sheet. Other embodiments are feasible.

10 The first and second optical sensors each, independently, may be opaque, transparent or semi-transparent. For the sake of simplicity, however, opaque sensors which are not transparent for the light beam, may be used, since these opaque sensors generally are widely commercially available. The fluorescent waveguiding sheet, however, may fully or partially be transparent.

15 The first optical sensor specifically may be a uniform sensor having a uniform light-sensitive area. Thus, specifically, the first optical sensor specifically may be a non-pixelated optical sensor. Alternatively, however, the at least one first optical sensor may also comprise one or more pixelated sensors, each having two or more pixels. Other embodiments are feasible.

20 Similarly, as discussed in further detail below, the at least one second optical sensor may comprise precisely one second optical sensor or a plurality of optical sensors. Further, the at least one fluorescent waveguiding sheet of the second optical sensor may be precisely one coherent fluorescent waveguiding sheet. Alternatively, however, the at least one second optical sensor also may comprise a plurality of second optical sensors. The latter alternative, as an example,
25 may be realized by providing a plurality of fluorescent waveguiding sheets. Thus, a plurality of fluorescent waveguiding sheets may be provided, each having one or more photosensitive elements. Other embodiments are feasible.

The detector, such as of the second optical sensor, may further comprise at least one optical
30 filter element, preferably at least one optical short-pass filter. Thus, as an example, only photons which are capable of exciting fluorescence light may pass through to the fluorescent waveguiding sheet. Other embodiments, however, are feasible.

The first and second optical sensors specifically may be arranged linearly in one and the same
35 beam path of the detector. Thus, as an example, the first and second optical sensors may be arranged concentrically with respect to an optical axis of the detector, specifically with the first optical sensor being placed in front of the second optical sensor. It shall be noted, however, that other embodiments are feasible, such as embodiments with the first optical sensor and the second optical sensor being placed in separate and different beam paths, wherein the beam path of
40 the detector is split by one or more beam-splitting elements. The linear setup with the first and second optical sensors being placed in one and the same beam path, however, is preferred, due to the simple optical setup and the low volume required for the detector. Thus, the present invention specifically allows for a simple, efficient and inexpensive optical setup.

As outlined above, the first optical sensor specifically may be a semiconductor sensor, preferably an inorganic semiconductor sensor, more preferably a photodiode and most preferably a silicon photodiode. As an example, the first optical sensor may be a uniform sensor having a single light-sensitive area. As an example, a commercially available silicon photodiode may be used.

In a further preferred embodiment the detector may comprise

- at least one angle dependent optical element adapted to generate at least one light beam having at least one beam profile depending on an angle of incidence of an incident light beam propagating from the object towards the detector and illuminating the angle dependent optical element;
- at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element;
- the at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the quotient signal Q from the sensor signals.

The angle dependent optical element may comprise at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element, at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one prism; at least one lens; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter; at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.

At least one of the light-sensitive areas may be oriented towards the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. Additionally or alternatively, at least one of the light-sensitive areas may be oriented differing from an orientation towards the object. For example, at least one of the optical sensors may be oriented perpendicular or under an arbitrary angle to the optical axis and with respect to the object. The angle dependent optical element may be adapted to generate the light beam such that the light beam impinges on the light-sensitive areas. For example, in case at least one of the light-sensitive areas is oriented under the arbitrary angle with respect to the optical axis,

the angle dependent optical element may be adapted to guide the light beam onto the light-sensitive area.

As used herein, the term “angle dependent optical element” refers to an optical element adapted to generate the at least one light beam having at least one beam profile depending on the angle of incidence of the incident light beam propagating from the object towards the detector and illuminating the angle dependent optical element. In particular, the angle dependent optical element may be adapted to influence and/or change and/or adjust the beam profile of the incident light beam. For example, the angle dependent optical element may have one or more of angle dependent transmission properties, angle dependent reflection properties or angle dependent absorption properties. The light beam generated by the angle dependent optical element may comprise at least one transmission light beam and/or at least one reflection light beam. The angle of incidence may be measured with respect to an optical axis of the angle dependent optical element.

An electromagnetic wave impinging on a first side, for example a surface and/or an entrance, of the angle dependent optical element may be partly, depending on the properties of the angle dependent optical element, absorbed and/or reflected and/or transmitted. The term “absorption” refers to a reduction of power and/or intensity of the incident light beam by the angle dependent optical element. For example, the power and/or intensity of the incident light beam may be transformed by the angle dependent optical element to heat or another type of energy. As used herein, the term “transmission” refers to a part of the electromagnetic wave which is measurable outside the angle dependent optical element in a half-space with angles from 90° and higher with respect to the optical axis. For example, transmission may be a remaining part of the electromagnetic wave impinging on the first side of the angle dependent optical element, penetrating the angle dependent optical element and leaving the angle dependent optical element at a second side, for example an opposite side and/or an exit. The term “reflection” refers to a part of the electromagnetic wave which is measurable outside the angle dependent optical element in a half-space with angles below 90° with respect to the optical axis. For example, reflection may be a change in direction of a wavefront of the incident light beam due to interaction with the angle dependent optical element.

The total power of the electromagnetic wave impinging on the angle dependent optical element may be distributed by the angle dependent optical element in at least three components, i.e. an absorption component, a reflection component and a transmission component. A degree of transmission may be defined as power of the transmission component normalized by the total power of the electromagnetic wave impinging on the angle dependent optical element. A degree of absorption may be defined as power of the absorption component normalized by the total power of the electromagnetic wave impinging on the angle dependent optical element. A degree of reflection may be defined as power of the reflection component normalized by the total power of the electromagnetic wave impinging on the angle dependent optical element.

As used herein, "angle dependent transmission" refers to the fact that the degree of transmission depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element. As outlined above, the angle of incident may be measured with respect to an optical axis of the angle dependent optical element. The angle dependent optical element may be arranged in the direction of propagation behind at least one transfer device. The transfer device may, for example, comprise at least one collimating lens. The angle dependent optical element may be designed to weaken rays impinging with larger angles compared to rays impinging with a smaller angle. For example, the degree of transmission may be highest for light rays parallel to the optical axis, i.e. at 0°, and may decrease for higher angles. In particular, at at least one cut-off angle the degree of transmission may steeply fall to zero. Thus, light rays having a large angle of incidence may be cut-off.

As used herein, the term "angle dependent absorption" refers to the fact that the degree of absorption depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element. As used herein, the term "angle dependent absorption" refers to the fact that a degree of absorption depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element. For example, photon energy and/or intensity of the light beam propagating from the object to the detector may be reduced depending on the angle of incidence.

As used herein, the term "angle dependent reflection" refers to the fact that the degree of reflection depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element.

For example, the angle dependent optical element comprises at least one optical fiber. The optical fiber may be designed such that the degree of transmission may be highest for incoming light rays parallel, i.e. at an angle of 0°, to the optical fiber, neglecting reflection effects. The optical fiber may be designed such that for higher angles, for example angles from 1° to 10°, the degree of transmission may decrease smoothly to around 80% of the degree of transmission for parallel light rays and may remain at this level constantly up to an acceptance angle of the optical fiber. As used herein, the term "acceptance angle" may refer to an angle above which total reflection within the optical fiber is not possible such that the light rays are reflected out of the optical fiber. The optical fiber may be designed that at the acceptance angle, the degree of transmission may steeply fall to zero. Light rays having a large angle of incidence may be cut-off.

The optical fiber may be adapted to transmit at least parts of the incident light beam which are not absorbed and/or reflected, between two ends of the optical fiber. The optical fiber may have a length and may be adapted to permit transmission over a distance. The optical fiber may comprise at least one material selected from the group consisting of: silica, aluminosilicate glass, germane silicate glass, fluorozirconate, rare earth doped glass, fluoride glass, chalco-

genide glasses, sapphire, doped variants, especially for silica glass, phosphate glass, PMMA, polystyrene, fluoropolymers such as poly(perfluoro-butenylvinyl ether), or the like. The optical fiber may be a single or multi-mode fiber. The optical fiber may be a step index fiber, a polarizing fiber, a polarization maintaining fiber, a plastic optical fiber or the like. The optical fiber may
5 comprise at least one fiber core which is surrounded by at least one fiber cladding having a lower index of refraction as the fiber core. The fiber cladding may also be a double or multiple cladding. The fiber cladding may comprise a so-called outer jacket. The fiber cladding may be coated by a so-called buffer adapted to protect the optical fiber from damages and moisture. The buffer may comprise at least one UV-cured urethane acrylate composite and/or at least one
10 polyimide material. In one embodiment, a refractive index of the fiber core may be higher than the refractive index of the fiber cladding material and the optical fiber may be adapted to guide the incoming light beam by total internal reflection below the angle of acceptance. In one embodiment, the optical fiber may comprise at least one hollow core fiber, also called photonic bandgap fiber. The hollow-core fiber may be adapted to guide the incoming light beam essentially
15 within a so-called hollow region, wherein a minor portion of the light beam is lost due to propagation into the fiber cladding material.

The optical fiber may comprise one or more fiber connectors at the end of the fiber. The optical fiber may comprise end caps such as coreless end caps. The optical fiber may comprise one or
20 more of a fiber coupler, a fiber Bragg grating, a fiber polarizer, a fiber amplifier, a fiber coupled diode laser, a fiber collimator, a fiber joint, a fiber splicing, a fiber connector, a mechanical splicing, a fusion splicing, or the like. The optical fiber may comprise a polymer coating.

The optical fiber may comprise at least two or more fibers. The optical fiber may be at least one
25 multifurcated optical fiber, in particular at least one bifurcated optical fiber. For example, the bifurcated optical fiber may comprise two fibers, in particular at least one first fiber and at least one second fiber. The first fiber and the second fiber may be arranged close to each other at an entrance end of the bifurcated optical fiber and may split into two legs separated by a distance at an exit end of the bifurcated optical fiber. The first and second fiber may be designed as fi-
30 bers having identical properties or may be fibers of different type. The first fiber may be adapted to generate at least one first transmission light beam and the second fiber may be adapted to generate at least one second transmission light beam. The bifurcated optical fiber may be arranged such that the incident light beam may impinge at a first angle of incidence into the first fiber and at a second angle of incidence, different from the first angle, into the second fiber,
35 such that the degree of transmission is different for the first transmission light beam and the second transmission light beam. One of the optical sensors may be arranged at an exit end of the first fiber and the other optical sensor may be arranged at an exit end of the second fiber. The optical fiber may comprise more than two fibers, for example three, four or more fibers. For example, the multifurcated may comprise multiple fibers wherein each fiber may comprise at
40 least one of a core, a cladding, a buffer, a jacket, and one or more fibers may partially or entirely be bundled by a further jacket such as a polymer hose to ensure that the fibers stay close to each other such as at one end of the fiber.

The detector may comprise a plurality of optical fibers, for example a plurality of single optical fibers or a plurality of multifurcated optical fibers. For example, the detector may comprise a plurality of single optical fibers, for example optical fibers having identical properties. The optical fibers, i.e. the single optical fibers or multifurcated optical fibers, may be arranged such that the incident light beam may impinge at different angles of incidence into each of the optical fibers such that the degree of transmission is different for each of the optical fibers. At the exit ends of each optical fiber at least one optical sensor may be arranged. Alternatively, at least two or more of the optical fibers may use the same optical sensor.

The illumination source may be adapted to illuminate the object through the angle dependent optical element. The optical fiber may comprise at least one illumination fiber adapted to transmit at least one incoming light beam generated by the illumination source such that it illuminates the object. The illumination source may be adapted to couple the at least one light beam generated by the illumination source into the illumination fiber. The detector may comprise at least one coupling element, for example at least one in-coupling element and/or at least one out-coupling element, which may be arranged in front of and/or behind the optical fiber. The coupling element may be or may comprise at least one transfer device.

The detector may comprise a further coupling element, in particular a further in-coupling element, adapted to couple the light beam travelling from the object to the detector into the illumination fiber. The further coupling element may be arranged in direction of propagation of the light beam travelling from the object to the detector in front of the optical fiber. The further coupling element may be or may comprise at least one transfer device.

In case the detector comprises a plurality of optical fibers, the detector may comprise one illumination source or a plurality of identical illumination sources and/or a plurality of different illumination sources. For example, the plurality of illumination sources may comprise at least two illumination sources generating light with different properties such as color or modulation frequencies. In one embodiment, at least two of the optical fibers of the plurality of the optical fibers may be illuminated using the same illumination source and/or two identical illumination sources. At the exit ends of each optical fiber illuminated by the same illumination source and/or by identical illumination sources at least one optical sensor may be arranged. Alternatively, at least two or more of the optical fibers illuminated by the same illumination source may use the same optical sensor. Such a detector, in which for a plurality of optical fibers the same illumination source and same optical sensors are used, may be used as proximity sensor. In one embodiment, at least two of the optical fibers of the plurality of optical fibers may be illuminated by using at least two different illumination sources. At the exit ends of each optical fiber at least one optical sensor may be arranged. Alternatively, at least two or more of the optical fibers may use the same optical sensor.

As outlined above, the distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors may be small. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may

be 21 mm. In particular in case the angle dependent optical element is an optical fiber, the baseline may even be smaller than a radius of the transfer device.

For example, the angle dependent optical element may be an optical fiber. The illumination
5 source may be guided through an optical fiber wherein the illumination beam may exit the optical fiber behind the transfer device within a baseline that is smaller than the radius of the transfer device. The illumination source may be adapted to generate at least one illumination light beam, wherein the optical fiber guiding the illumination beam may be attached to the transfer device such as by a polymer or glue or the like, to reduce reflections at interfaces with larger
10 differences in refractive index. For example, the detector may be a compact device without further optical elements, wherein the illumination source may be placed as close as possible to an edge of the transfer device. Thus, the baseline may be close to half a diameter of the transfer device, in particular the lens diameter and housings of lens and light source. For example, the detector may be an even more compact device, wherein a mirror, in particular a small mirror,
15 may be positioned in front of the transfer device, in particular in a center, for example a geometrical center, or close to the center of the transfer device, in order to couple the illumination beam into the beam path. Thus, the baseline may be less than half the diameter of the transfer device. The illumination source may be arranged such that the baseline is as small as possible. By arranging the illumination source such that the direction of propagation of the illumination light
20 beam is essentially parallel to the optical axis and that the illumination source and the optical axis are separated by the small baseline, very compact devices are possible. For example, a distance from the center of the transfer device to the illumination source, in particular along a connecting line from the center of the transfer device to the illumination source, may be preferably less than 2.5 times a distance from the center of the transfer device to an edge if the transfer
25 device, more preferably less than 1.5 times the distance center to edge of the transfer device, and most preferably less 1 times the distance center to edge of the transfer device. The transfer device may have an arbitrary shape, in particular a non-circular shapes are possible. At small distances an aperture of the illumination source may be small and the baseline may be small. At large distances an aperture of the illumination source may be large and the baseline may be
30 small. This is contrarily as in triangulation methods, wherein at large distances a large baseline is necessary. Further, triangulation based systems have a minimum detection range significantly greater than zero, for example such as 20 cm from the detector system, due to necessary spatial extend of the baseline. Such a large baseline may result in that the illuminated light scattered from the object may not reach the light sensitive area of the optical sensor behind the
35 transfer device. In addition, in triangulation based systems, using a small baseline would reduce the minimum detection range, however, at the same time would reduce a maximum detection range. Further, triangulation based systems require a plurality of light sensitive areas and sensor signals, for example sensor signals of at least one detector row. According to the invention, determination of the longitudinal coordinate z is possible with a reduced number of sensor signals, in particular with less than 20, preferably less than 10 and more preferably less than 5
40 sensor signals. The illumination source and the angle dependent optical element may be arranged in the direction of propagation of the light beam traveling from the object to the detector behind the transfer device, which will be described in detail below. The distance perpendicular

to the optical axis of the detector between the illumination source and the optical sensors may be less than the radius of the transfer device.

5 The illumination source and the optical sensors and/or the illumination source and the angle dependent optical element and/or the optical sensors may be arranged with a relative spatial offset from the optical axis of the detector. Such an arrangement may allow enhancing the tendency of the quotient, and thus, accuracy of the distance measurement. In particular, with increasing spatial offset a slope in a Q vs distance diagram increases and thus allows distinguishing similar distances more accurately. For example, one of the illumination source and the optical sensors may be arranged on the optical axis and the other one may be arranged spaced apart from the optical axis. For example, both of illumination source and the optical sensors may be arranged spaced apart from the optical axis by at least one different distance, in particular perpendicular to the optical axis. For example, the at least two fibers of a multifurcated optical fiber may be arranged at different distances from the optical axis. The angle dependent optical element is adapted to mimic a larger distance compared to the real distance perpendicular to an optical axis between the illumination source and the optical sensors without moving the illumination source and/or optical sensors.

20 By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

30 Alternatively to the linear arrangement of the two optical sensors, the optical sensors may be arranged in different beam paths of the detector. The angle dependent optical element may be adapted to generate the first light beam and the second light beam. The first light beam and the second light beam may be generated with different degree of transmission. The first optical sensor may be configured to generate the first sensor signal in response to the illumination of the first light-sensitive area by the first light beam generated by the angle dependent optical element. The second optical sensor may be configured to generate the second sensor signal in response to the illumination of the second light-sensitive area by the second light beam generated by the angle dependent optical element. For example, as outlined above, the angle dependent optical element may comprise at least one multifurcated optical fiber which may be arranged such that the incident light beam may impinge at the first angle of incidence into the first fiber and at the second angle of incidence, different from the first angle, into the second fiber, such that the degree of transmission is different for the first light beam, in this case a first

transmission light beam, and the second light beam, in this case a second transmission light beam. One of the first and second optical sensors may be arranged at the exit end of the first fiber and the other optical sensor may be arranged at the exit end of the second fiber.

- 5 At least one sensor element may have a matrix of optical sensors, wherein the optical sensors each having a light-sensitive area, wherein each optical sensor may be configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by the light beam generated by the angle dependent optical element.
- 10 For example, the detector may comprise two sensor elements, in particular at least one first sensor element and at least one second sensor element, arranged in different beam paths of the detector. The angle dependent optical element may be adapted to generate the first light beam and the second light beam. The first light beam and the second light beam may be generated with different degree of transmission. The first sensor element may be configured to
- 15 generate the first sensor signal in response to the illumination by the first light beam generated by the angle dependent optical element. The second sensor element may be configured to generate the second sensor signal in response to an illumination by the second light beam generated by the angle dependent optical element. For example, as outlined above, the angle dependent optical element may comprise at least one multifurcated optical fiber arranged such that the
- 20 incident light beam may impinge at the first angle of incidence into the first fiber and at the second angle of incidence, different from the first angle, into the second fiber, such that the degree of transmission is different for the first transmission light beam and the second transmission light beam. One of the first and second sensors elements may be arranged at the exit end of the first fiber and the other sensor element may be arranged at the exit end of the second fiber.
- 25 The detector may comprise at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in
- 30 response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element.

- The evaluation device may be configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals. In this further embodiment the optical
- 35 sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas

- The detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. For example, in case the angle
- 40 dependent optical element comprises at least one multifurcated optical fiber, each of the fibers of the multifurcated optical fiber may constitute an independent beam path. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis, and the condi-

tion noted above generally may refer to each beam path independently. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths. In case the optical sensors are distributed over different partial beam paths, the above-mentioned condition may be described such that at least one first optical sensor is located in at least one first partial beam path, being offset from the optical axis of the first partial beam path by a first spatial offset, and at least one second optical sensor is located in at least one second partial beam path, being offset from the optical axis of the second partial beam path by at least one second spatial offset, wherein the first spatial offset and the second spatial offset are different.

As outlined above, each light-sensitive area has a geometrical center. Each geometrical center of each light-sensitive area may be spaced apart from the optical axis of the detector, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located.

As outlined above, the optical sensors specifically may be located in one and the same plane, which, preferably, is a plane perpendicular to the optical axis. Other configurations, however, are possible. Thus, two or more of the optical sensors may also be spaced apart in a direction parallel to the optical axis.

For example, the optical sensors may be partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector. As an example, bi-cell diodes or quadrant diodes are widely commercially available at low cost, and driving schemes for these bi-cell diodes or quadrant diodes are generally known. As an example, the bi-cell diodes may each form independent diodes having the full functionality of a diode. As an example, each of the bi-cell diodes may have a square or rectangular shape, and the two diodes may be placed in one plane such that the two partial diodes, in total, form a 1 x 2 or 2 x 1 matrix having a rectangular shape. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes and quadrant diode is proposed, as will be outlined in further detail below. Generally, however, the optical sensors specifically may be partial diodes of a quadrant diode, with a center of the quadrant diode being off-centered from the optical axis of the detector. As an example, the four partial diodes may each form independent diodes having the full functionality of a diode. As an example, the four partial diodes may each have a square or rectangular shape, and the four partial diodes may be placed in one plane such that the four partial diodes, in total, form a 2 x 2 matrix having a rectangular or square shape. In a further example, the four partial diodes, in total, may form a 2 x 2 matrix having a circular or elliptical shape. The partial diodes, as an example, may be adjacent, with a minimum separation from one another.

In case a quadrant diode is used, having a 2x2 matrix of partial diodes, the center of the quadrant diode specifically may be off-centered or offset from the optical axis. Thus, as an example, the center of the quadrant diodes, which may be an intersection of the geometrical centers of the optical sensors of the quadrant diode, may be off-centered from the optical axis by at least

0.2 mm, more preferably by at least 0.5 mm, more preferably by at least 1.0 mm or even 2.0 mm. Similarly, when using other types of optical sensors setups having a plurality of optical sensors, an overall center of the optical sensors may be offset from the optical axis by the same distance.

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In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector
10 system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more
15 illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the
20 one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein,
25 elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

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In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein,
35 the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

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In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is config-

ured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment function in accordance with the information.

- 5 In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

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- In a further aspect of the present invention, a readout device for optical storage media is proposed. The readout device comprises at least one detector according to any one of the preceding embodiments referring to a detector. As used therein, a readout device for optical storage media generally refers to a device which is capable of optically retrieving information stored in optical storage media such as optical storage discs, e.g. CCD, DVD or Blu-ray discs. Thus, the above-described measurement principle of the detector according to the present invention may be used for detecting data modules within an optical storage medium such as in optical storage discs. As an example, in case a reflective data module is present and reflects the illuminating light beam, the detector will not only detect the reflected light beam according to the above-mentioned measurement principle but will also detect a distance between the detector and the reflective data module, i.e. a depth of the reflective data module within the optical storage medium. Thus, as an example, the detector may be used for detecting different layers of information modules or data modules within the optical storage medium. Thereby, as an example, two layer discs or three layer discs or even discs having more than three layers may be generated and read out.

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- In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination

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light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning.

Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning.

The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-

related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object.

- 5 For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.
- 10 Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission
- 15 in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning device. In particular, the scanning system may, thus, use at
- 20 least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be
- 25 used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a
- 30 scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination
- 35 source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated
- 40 out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with a several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning device may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning device may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning

system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning device is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or drill at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning device.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon

photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses an inertial measurement unit for use in an electronic device. The electronic device may be a mobile electronic device. The electronic device may be a camera. The electronic device may be a mobile phone. The inertial measurement unit is adapted to receive data determined by at least one inertial measurement unit comprising at least one detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. As used herein, the term "data determined by the at least one detector" refers to at least one information on the at least one longitudinal coordinate z. The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. As used herein, the term "data determined by the at least one further sensor" refers to at least one information selected from the group consisting of: angle information; speed information; information on turn rate; information on inclination. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed. The inertial measurement unit may comprise at least one processor. The processor may be adapted to evaluate data recorded by the further sensor. In particular, the processor may be adapted to determine and/or calculate one or more of spatial position, spatial orientation, movement and velocity. The inertial measurement unit may comprise a plurality of further sensors. The inertial measurement unit may be adapted to fuse information determined from at least two of the further sensors. The inertial measurement unit may be adapted to fuse information of at least two further sensors by using at least one Kalman filter. As outlined

above, the detector may be adapted to provide an absolute measurement of the longitudinal coordinate z . The processor, e.g. the evaluation device as described above, may be adapted to fuse the information of the at least two further sensors considering the longitudinal coordinate z . The various sensor signals may be used within a Kalman filter or a linear quadratic estimated to take into account that each sensor signal is subject to measurement errors and inaccuracies. Fusion of these sensor signals within a Kalman filter may yield an improved estimate such as for the measurement of the longitudinal coordinate.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used. The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method comprises the following method steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by the light beam, wherein the detector has at least one transfer device, wherein the transfer device has at least one focal length in response to the light beam propagating from the object to the detector;
- illuminating each of light-sensitive areas of at least two optical sensors of the detector with a light beam propagating from the object to the detector, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving a quotient signal Q of the sensor signals.

The longitudinal coordinate z of the object may be determined in at least one measurement range independent from the object size in an object plane.

The deriving of the quotient signal Q may comprise one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

The method may comprise adjusting the measurement range by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the object in the object plane.

The transfer device may have an optical axis. With respect to embodiments of the transfer device reference is made to the embodiments of the transfer device described with respect to a first aspect above.

5 For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

10 In a first preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

15 The method steps may be as follows:

- illuminating at least one first light-sensitive area of at least one first optical sensor of the detector with a light beam propagating from the object to the detector and, thereby, generating at least one first sensor signal;
- illuminating at least one second light-sensitive area of at least one second optical
20 sensor of the detector with the light beam and, thereby, generating at least one second sensor signal, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
- evaluating the first and second sensor signals, thereby determining at least one longitudinal coordinate z of the object.

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Specifically, evaluating the first and second sensor signal may comprise deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. Further, the determining the longitudinal coordinate may comprise evaluating the quotient signal

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Q .

In a second preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed.

35 Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps may be as follows:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical
40 sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;
- evaluating the sensor signals, by

- determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - a) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - b) determining at least one combined signal by combining the center signal and the sum signal; and
 - c) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Specifically, the evaluation of the first and second sensor signal may comprise deriving the quotient signal Q by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Further, the determination of the longitudinal coordinate may comprise evaluating the quotient signal Q . The evaluation of the quotient signal specifically may comprise using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate, in order to determine the longitudinal coordinate.

In a third preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps may be as follows:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light;
- illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

Specifically, evaluating the first and second sensor signal may comprise deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q . Further, the determining of the longitudinal coordinate may comprise evaluating the quotient signal Q .

In a further preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps may be as follows:

- illuminating at least one first light-sensitive area of at least one first optical sensor with a light beam propagating from the object to the detector and generating at least one first sensor signal by the first optical sensor, in response to the illumination of the first light-sensitive area;
 - illuminating at least one second light-sensitive area of at least one second optical sensor with the light beam and generating at least one second sensor signal by the second optical sensor, in response to the illumination of the second light-sensitive area, the second optical sensor having:
 - o at least one fluorescent waveguiding sheet forming a second light-sensitive area, wherein the fluorescent waveguiding sheet is oriented towards the object such that at least one light beam propagating from the object towards the detector generates at least one light spot in the second light-sensitive area, wherein the fluorescent waveguiding sheet contains at least one fluorescent material, wherein the fluorescent material is adapted to generate fluorescence light in response to the illumination by the light beam; and
 - o at least one photosensitive element located at at least one edge of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive element by the fluorescent waveguiding sheet and capable of generating the at least one second sensor signal in response to the illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
 - determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.
- Specifically, as discussed above, the evaluating the first and second sensor signals may comprise deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. The determining the at least one longitudinal coordinate of the object may comprise evaluating the quotient signal Q . as discussed above, at least one predetermined relationship between the quotient signal and the longitudinal coordinate may be used for determining the longitudinal coordinate.

As also discussed above, the second optical sensor specifically may comprise at least two photosensitive elements. The method may further comprise determining at least one transversal coordinate x , y of the object by evaluating the sensor signals of the photosensitive elements.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to

the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further preferred embodiment, the method comprises the following steps:

- 5 - providing at least one angle dependent optical element and generating at least one light beam having at least one beam profile depending on an angle of incidence;
- providing at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light
- 10 beam generated by the angle dependent optical element;
- illuminating each of the light-sensitive areas of the at least two optical sensors of the detector with the light beam generated by the angle dependent optical element, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- 15 evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving the quotient signal Q of the sensor signals.

Specifically, evaluating the first and second sensor signal may comprise deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals.

20 Further, the determining the longitudinal coordinate may comprise evaluating the quotient signal Q .

As also discussed above, the second optical sensor specifically may comprise at least two photosensitive elements. The method may further comprise determining at least one transversal

25 coordinate x , y of the object by evaluating the sensor signals of the photosensitive elements.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or

30 given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position

35 measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; a machine vision application; a robotics application;

40 a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector sys-

tem. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

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As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

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Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for household applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

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The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus, specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling

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other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

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Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head-mounted displays.

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Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries, obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

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Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

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Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further

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be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general need for voluminous beam-splitters. Further, the devices according to the present invention

generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

- 5 Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems, or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.
- 30 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the

present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions.

Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endoscopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric surgery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment. Further, the devices according to the present invention may be used in wearable robots such as in exoskeletons or prosthesis or the like.

Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe.

In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

Further the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection, such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections,

inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, air-
5 planes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed
10 and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a
15 guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to
20 active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for air-
25 craft when landing or starting. Wherein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or ac-
30 celerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use
35 with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or
40 facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light

sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racket, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the present invention may be used in drones or multicopters to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concertized movements of several drones or recharging or refueling in the air or the like.

Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network)

to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as dog or cat litter robots, charging robot for electrical vehicles, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, robotic shopping carts, luggage carrying robots, line following robots, laundry robots, ironing robots, window washing robots, toy robots, , patient monitoring robots, baby monitoring robots, elderly monitoring robots, children monitoring robots, transport robots, telepresence robots, professional service robots, programmable toy robots, pathfinder robots, social robots providing company to less mobile people, following robots, smart card following robots, psychotherapy robots, or robots translating and speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the present invention may be used in humanoid robots, especially in the context of using humanoid hands to pick up or hold or place objects. Further, the devices according to the present invention may be used in combination with audio interfaces especially in combination with household robots which may serve as a digital assistant with interfaces to online or offline computer applications. Further, the devices according to the present invention may be used in robots that can control switches and buttons in industrial and household purposes. Further, the devices according to the present invention may be used in smart home robots such as Mayfield's Kuri. Further the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate insecure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

Further, devices according to the present invention may be used for the inspection of adhesive beads, sealing beads, or the like, such as to recognize disruptions, slubs, contractions, asy-

metries, local defects, or the like. Further, devices according to the present invention may be used to count objects such as dry fruits on a conveyer belt, such as in difficult situations, such as when fruit of similar color and shape may be in direct contact with each other. Further, devices according to the present invention may be used in quality control of die cast or injection molded parts such as to ensure flawless casting or molding, recognize surface damages, worn out toolings or the like. Further, devices according to the present invention may be used for laserscribing such as for quality control and positioning of the laser. Further, devices according to the present invention may be used for sorting systems, such as to detect position, rotation, and shape of an object, compare it to a database of objects, and classify the object. Further, devices according to the present invention may be used for stamping part inspection, packaging inspection, such as food and pharma packaging inspection, filament inspection, or the like.

Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that are can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.

Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition devices, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.

Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.

The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or green-

house or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.

Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific groups of animals such as bees.

Further, devices according to the present invention may be used to guide users during a shaving, hair cutting, or cosmetics procedure, or the like. Further, devices according to the present invention may be used to record or monitor what is played on an instrument, such as a violin. Further, devices according to the present invention may be used in smart household appliances such as a smart refrigerator, such as to monitor the contents of the refrigerator and transmit notifications depending on the contents. Further, devices according to the present invention may be used for monitoring or tracking populations of humans, animals, or plants, such as deer or tree populations in forests. Further, devices according to the present invention may be used in harvesting machines, such as for harvesting crops, flowers or fruits, such as grapes, corn, hops, apples, grains, rice, strawberries, asparagus, tulips, roses, soy beans, or the like. Further, devices according to the present invention may be used to monitor the growth of plants, animals, algae, fish, or the like, such as in breeding, food production, agriculture or research applications, to control irrigation, fertilization, humidity, temperature, use of herbicides, insecticides, fungicides, rodenticides, or the like. Further, devices according to the present invention may be used in feeding machines for animals or pets, such as for cows, pigs, cats, dogs, birds, fish, or the like. Further, devices according to the present invention may be used in animal product production processes, such as for collecting milk, eggs, fur, meat, or the like, such as in automated milking or butchering processes. Further, devices according to the present invention may be used for automated seeding machines, or sowing machines, or planting machines such as for planting corn, garlic, trees, salad or the like. Further, devices according to the present invention may be used to assess or monitor weather phenomena, such as clouds, fog, or the like, or to warn from danger of avalanches, tsunamis, gales, earthquakes, thunder storms, or the like. Further, devices according to the present invention may be used to measure motions, shocks, concussions, or the like such as to monitor earthquake risk. Further, devices according to the present invention may be used in traffic technology to monitor dangerous crossings, to control traffic lights depending on traffic, to monitor public spaces, to monitor roads, gyms, stadiums, ski resorts, public events, or the like. Further, devices according to the present invention may be used in medical applications such as to monitor or analyze tissues, medical or biological assays, changes in tissues such as in moles or melanoma or the like, to count bacteria, blood cells, cells, algae, or the like, for retina scans, breath or pulse measurements, gastroscopy, patient surveillance, or the like. Further, devices according to the present invention may be used to monitor the shape, size, or circumference of drops, streams, jets, or the like or to analyze, assess, or monitor profiles or gas or liquid currents such as in a wind channel, or the like. Further,

devices according to the present invention may be used to warn drivers such as car or train drivers when they are getting sick or tired or the like. Further, devices according to the present invention may be used in material testing to recognize strains or tensions or fissures, or the like. Further, devices according to the present invention may be used in sailing to monitor and optimize sail positions such as automatically. Further, devices according to the present invention may be used for fuel level gauges.

Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or viruses or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location. Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location specific information, such as the current position on a map, or information on goods sold, or the like.

Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.

Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices ac-

According to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.

5

As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate

along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

- 5 The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device adapted for detecting light beams, such as
10 the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The
15 detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the de-
20 tector are feasible.

The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as
25 simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based al-
30 gorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

35 The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying
40 distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor,

or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

5

The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared. Further, the devices according to the present invention may be used in the context of gesture recognition. In this context, gesture
10 recognition in combination with devices according to the present invention may, in particular, be used as a human-machine interface for transmitting information via motion of a body, of body parts or of objects to a machine. Herein, the information may, preferably, be transmitted via a motion of hands or hand parts, such as fingers, in particular, by pointing at objects, applying sign language, such as for deaf people, making signs for numbers, approval, disapproval, or the
15 like, by waving the hand, such as when asking someone to approach, to leave, or to greet a person, to press an object, to take an object, or, in the field of sports or music, in a hand or finger exercise, such as a warm-up exercise. Further, the information may be transmitted by motion of arms or legs, such as rotating, kicking, grabbing, twisting, rotating, scrolling, browsing, pushing, bending, punching, shaking, arms, legs, both arms, or both legs, or a combination of
20 arms and legs, such as for a purpose of sports or music, such as for entertainment, exercise, or training function of a machine. Further, the information may be transmitted by motion of the whole body or major parts thereof, such as jumping, rotating, or making complex signs, such as sign language used at airports or by traffic police in order to transmit information, such as "turn right", "turn left", "proceed", "slow down", "stop", or "stop engines", or by pretending to swim, to
25 dive, to run, to shoot, or the like, or by making complex motions or body positions such as in yoga, pilates, judo, karate, dancing, or ballet. Further, the information may be transmitted by using a real or mock-up device for controlling a virtual device corresponding to the mock-up device, such as using a mock-up guitar for controlling a virtual guitar function in a computer program, using a real guitar for controlling a virtual guitar function in a computer program, using
30 a real or a mock-up book for reading an e-book or moving pages or browsing through in a virtual document, using a real or mock-up pen for drawing in a computer program, or the like. Further, the transmission of the information may be coupled to a feedback to the user, such as a sound, a vibration, or a motion.

35 In the context of music and/or instruments, devices according to the present invention in combination with gesture recognition may be used for exercising purposes, control of instruments, recording of instruments, playing or recording of music via use of a mock-up instrument or by only pretending to have a instrument present such as playing air guitar, such as to avoid noise or make recordings, or, for conducting of a virtual orchestra, ensemble, band, big band, choir, or
40 the like, for practicing, exercising, recording or entertainment purposes or the like.

Further, in the context of safety and surveillance, devices according to the present invention in combination with gesture recognition may be used to recognize motion profiles of persons, such

as recognizing a person by the way of walking or moving the body, or to use hand signs or movements or signs or movements of body parts or the whole body as access or identification control such as a personal identification sign or a personal identification movement.

- 5 Further, in the context of smart home applications or internet of things, devices according to the present invention in combination with gesture recognition may be used for central or non-central control of household devices which may be part of an interconnecting network of home appliances and/or household devices, such as refrigerators, central heating, air condition, microwave ovens, ice cube makers, or water boilers, or entertainment devices, such as television sets,
10 smart phones, game consoles, video recorders, DVD players, personal computers, laptops, tablets, or combinations thereof, or a combination of household devices and entertainment devices.

- 15 Further, in the context of virtual reality or of augmented reality, devices according to the present invention in combination with gesture recognition may be used to control movements or function of the virtual reality application or of the augmented reality application, such as playing or controlling a game using signs, gestures, body movements or body part movements or the like, moving through a virtual world, manipulating virtual objects, practicing, exercising or playing sports, arts, crafts, music or games using virtual objects such as a ball, chess figures, go
20 stones, instruments, tools, brushes.

- Further, in the context of medicine, devices according to the present invention in combination with gesture recognition may be used to support rehabilitation training, remote diagnostics, or to monitor or survey surgery or treatment, to overlay and display medical images with positions of
25 medical devices, or to overlay display prerecorded medical images such as from magnetic resonance tomography or x-ray or the like with images from endoscopes or ultra sound or the like that are recorded during an surgery or treatment.

- 30 Further, in the context of manufacturing and process automation, devices according to the present invention in combination with gesture recognition may be used to control, teach, or program robots, drones, unmanned autonomous vehicles, service robots, movable objects, or the like, such as for programming, controlling, manufacturing, manipulating, repairing, or teaching purposes, or for remote manipulating of objects or areas, such as for safety reasons, or for maintenance purposes.

- 35 Further, in the context of business intelligence metrics, devices according to the present invention in combination with gesture recognition may be used for people counting, surveying customer movements, areas where customers spend time, objects, customers test, take, probe, or the like.

- 40 Further, devices according to the present invention may be used in the context of do-it-yourself or professional tools, especially electric or motor driven tools or power tools, such as drilling machines, saws, chisels, hammers, wrenches, staple guns, disc cutters, metals shears and nib-

blers, angle grinders, die grinders, drills, hammer drills, heat guns, wrenches, sanders, engrav-
ers, nailers, jig saws, biscuit joiners, wood routers, planers, polishers, tile cutters, washers,
rollers, wall chasers, lathes, impact drivers, jointers, paint rollers, spray guns, morticers, or
welders, in particular, to support precision in manufacturing, keeping a minimum or maximum
5 distance, or for safety measures.

Further, the devices according to the present invention may be used to aid visually impaired
persons. Further, devices according to the present invention may be used in touch screen such
as to avoid direct contact such as for hygienic reasons, which may be used in retail environ-
10 ments, in medical applications, in production environments, or the like. Further, devices accord-
ing to the present invention may be used in agricultural production environments such as in sta-
ble cleaning robots, egg collecting machines, milking machines, harvesting machines, farm ma-
chinery, harvesters, forwarders, combine harvesters, tractors, cultivators, ploughs, destoners,
harrows, strip tills, broadcast seeders, planters such as potato planters, manure spreaders,
15 sprayers, sprinkler systems, swathers, balers, loaders, forklifts, mowers, or the like.

Further, devices according to the present invention may be used for selection and/or adaption of
clothing, shoes, glasses, hats, prosthesis, dental braces, for persons or animals with limited
communication skills or possibilities, such as children or impaired persons, or the like. Further,
20 devices according to the present invention may be used in the context of warehouses, logistics,
distribution, shipping, loading, unloading, smart manufacturing, industry 4.0, or the like. Further,
in a manufacturing context, devices according to the present invention may be used in the con-
text of processing, dispensing, bending, material handling, or the like.

25 The evaluation device may be or may comprise one or more integrated circuits, such as one or
more application-specific integrated circuits (ASICs), and/or one or more data processing devic-
es, such as one or more computers, preferably one or more microcomputers and/or microcon-
trollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may
30 be comprised, such as one or more preprocessing devices and/or data acquisition devices,
such as one or more devices for receiving and/or preprocessing of the sensor signals, such as
one or more AD-converters and/or one or more filters. Further, the evaluation device may com-
prise one or more measurement devices, such as one or more measurement devices for meas-
uring electrical currents and/or electrical voltages. Further, the evaluation device may comprise
35 one or more data storage devices. Further, the evaluation device may comprise one or more
interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be adapted to perform at least one computer program,
such as at least one computer program adapted for performing or supporting one or more or
40 even all of the method steps of the method according to the present invention. As an example,
one or more algorithms may be implemented which, by using the sensor signals as input varia-
bles, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which

will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

As outlined above, the human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device.

Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element,

the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a

computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

5 The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one
10 item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements
15 may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

20 The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which
25 might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

30 As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may
35 fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

40 The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the

object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.

- 5 The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly at-
10 tachable to or integratable into the object to be tracked.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other
15 types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detec-
20 tor.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial
25 applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication
30 systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional
35 option of measuring one or more transversal coordinates, thereby integrating the functionality of a PSD.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention,
40 such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the quotient signal, as mentioned above, contrarily, provides a wide range of a

continuously and monotonously decreasing or increasing functions which may be used to determine the longitudinal coordinate from the quotient signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the quotient signal from the total power of the light beam, at least as long as no saturation of one or both of the optical sensors is reached, also provides a very wide range of measurement in terms of brightness, i.e. in terms of total power of the light beam.

The light beam, within the detector, generally may propagate along an optical axis of the detector. The first and second optical sensors may be placed on the optical axis. The light beam, however, may also propagate in other ways than along the optical axis. As an example, an illumination light beam may be generated which propagates along the optical axis or which propagates parallel to the optical axis or at an angle to the optical axis which is different from 0° , such as an angle of 1° to 20° . Other embodiments are feasible.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector comprising:

- at least one transfer device, wherein the transfer device has at least one focal length in response to at least one incident light beam propagating from the object to the detector;
- at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam,
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q from the sensor signals.

Embodiment 2: The detector according to the preceding embodiment, wherein the detector is adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from the object size in an object plane.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the measurement range is adjustable by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the object in the object plane.

Embodiment 4: The detector according to the preceding embodiment, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object

size if a circle of confusion of at least one light beam emitted from at least one point of the object is greater than an image size of the object.

Embodiment 5: The detector according to the preceding embodiment, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object size if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for at least one distance.

Embodiment 6: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 7: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 8: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are areas of the beam profile at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o .

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the optical sensors are positioned off focus.

Embodiment 10: The detector according to any one of the preceding embodiments, wherein the transfer device has an optical axis, wherein the transfer device constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis, wherein the optical sensors are arranged such that the light-sensitive areas of the optical sensors differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.

Embodiment 11: The detector according to any one of the preceding embodiments, wherein the light beam may be a monochromatic light beam.

Embodiment 12: The detector according to any one of the preceding embodiments, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of at least one beam profile of the light beam.

Embodiment 13: The detector according to the preceding embodiment, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

5

Embodiment 14: The detector according to any one of the two preceding embodiments, wherein the light-sensitive regions are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

10

Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile.

15

Embodiment 16: The detector according to any one of the four preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.

20

Embodiment 17: The detector according to any one of the preceding embodiments, wherein the edge information comprises an information relating to a number of photons in the first area of the beam profile and the center information comprises an information relating to a number of photons in the second area of the beam profile.

25

Embodiment 18: The detector according to any one of the two the preceding embodiments, wherein the evaluation device is configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.

30

Embodiment 19: The detector according to any one of the three preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensors with at least one line pattern, wherein A1 corresponds to an area with a full line width of the line pattern on the optical sensors, wherein A2 is a central area of the line pattern on the optical sensors.

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Embodiment 20: The detector according to any one of the four preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensor with at least one point pattern, wherein A1 corresponds to an area with a full radius of a point of the point pattern on the optical sensors, wherein A2 is a central area of the point in the point pattern on the optical sensors.

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Embodiment 21: The detector according to any one of the preceding embodiments, wherein the optical sensors comprise a first optical sensor having a first light sensitive area and a second optical sensor having a second light sensitive area, wherein the first and the second light sensitive areas are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$

- 5 is satisfied, wherein "a" is a ratio of photons hitting both an inner region of a plane perpendicular to the optical axis intersecting the optical axis at a distance equal to half of a focal length of the transfer device and the first light sensitive area, "b" is a ratio of photons hitting both the inner region of the plane and the second light sensitive area, "c" is a ratio of photons hitting both an outer region of the plane and the first light-sensitive area, "d" is a ratio of the photons hitting
10 both the outer region of the plane and the second light sensitive area.

Embodiment 22: The detector according to the preceding embodiment, wherein the first light sensitive area and the second light sensitive area are arranged such that the inner region has an area with a geometrical center point on the optical axis and an extension such that half of the
15 photons hit the plane within the inner region and the other half hit the plane outside the inner region.

Embodiment 23: The detector according to the preceding embodiment, wherein the first light sensitive area and the second light sensitive area are arranged such that the inner region is
20 designed as a circle with a center point on the optical axis and a radius r which is chosen such that half of the photons hit the plane within the circle and the other half hit the plane outside the circle.

- Embodiment 24: A detector for determining a position of at least one object, the detector having
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- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
 - at least one second optical sensor having a second light-sensitive area, wherein the
30 second optical sensor is configured to generate at least one second sensor signal in response to an illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
 - at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.
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Embodiment 25: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device is configured for determining the longitudinal coordinate by evaluating the quotient signal Q.
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Embodiment 26: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.

- 5 Embodiment 27: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

- 10 Embodiment 28: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

- 15 Embodiment 29: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

Embodiment 30: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

- 20 Embodiment 31: The detector according to any one of the seven preceding embodiments, wherein the first and second optical sensors are arranged linearly in one and the same beam path of the detector.

- 25 Embodiment 32: The detector according to the preceding embodiment, wherein the first and second optical sensors are arranged concentrically with respect to an optical axis of the detector.

- 30 Embodiment 33: The detector according to any one of the eight preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than five times the square root of a surface area of the first light-sensitive area.

- 35 Embodiment 34: The detector according to any one of the nine preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm.

- 40 Embodiment 35: The detector according to any one of the ten preceding embodiments, wherein the second light-sensitive area is larger by at least a factor of two, more preferably by at least a factor of three and most preferably by at least a factor of five as compared to the first light-sensitive area.

Embodiment 36: The detector according to any one of the eleven preceding embodiments, wherein the first light-sensitive area has a surface area of 1 mm² to 150 mm², more preferably a surface area of 10 mm² to 100 mm².

- 5 Embodiment 37: The detector according to any one of the twelve preceding embodiments, specifically according to the preceding embodiment, wherein the second light-sensitive area has a surface area of 160 mm² to 1000 mm², more preferably a surface area of 200 mm² to 600 mm².

- 10 Embodiment 38: The detector according to any one of the thirteen preceding embodiments, wherein the first and second optical sensor each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

- 15 Embodiment 39: The detector according to any one of the fourteen preceding embodiments, wherein the first and second optical sensors each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

- 20 Embodiment 40: The detector according to any one of the fifteen preceding embodiments, wherein the first and second optical sensors each are uniform sensors having a single light-sensitive area each.

- 25 Embodiment 41: The detector according to any one of the sixteen preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

- 30 Embodiment 42: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

- 35 Embodiment 43: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

- 40 Embodiment 44: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pat-

tern comprising convex uniform tilings; at least one fringe pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

5 Embodiment 45: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the quotient signal Q .

10 Embodiment 46: The detector according to any one of the twenty-one preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

15 Embodiment 47: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

20 Embodiment 48: The detector according to any one of the twenty-three preceding embodiments, wherein the first light-sensitive area, in a direction of propagation of the light beam, overlaps with the second light-sensitive area.

25 Embodiment 49: A detector for determining a position of at least one object, the detector having - at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;

30 - at least one evaluation device configured for evaluating the sensor signals, by
a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one
35 sum signal;
c) determining at least one combined signal by combining the center signal and the sum signal; and
d) determining at least one longitudinal coordinate z of the object by evaluating the combined
40 signal.

Embodiment 50: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal;

an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Embodiment 51: The detector according to any one of the two preceding embodiments, wherein the detector, specifically the evaluation device, is configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals.

Embodiment 52: The detector according to the preceding embodiment, wherein the transforming of the sensor signals comprises at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and brightness channels, a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image.

Embodiment 53: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal.

Embodiment 54: The detector according to the preceding embodiment, wherein the center detector is fully or partially integrated into the sensor element.

Embodiment 55: The detector according to any one of the two preceding embodiments, wherein the center detector is fully or partially embodied in one or both of software or hardware.

Embodiment 56: The detector according to any one of the seven preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.

Embodiment 57: The detector according to the preceding embodiment, wherein the evaluation device comprises at least one summing device for forming the sum signal.

Embodiment 58: The detector according to the preceding embodiment, wherein the summing device is fully or partially integrated into the sensor element.

Embodiment 59: The detector according to any one of the two preceding embodiments, wherein the summing device is fully or partially embodied in one or both of software or hardware.

Embodiment 60: The detector according to any one of the eleven preceding embodiments, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

Embodiment 61: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 62: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

Embodiment 63: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

Embodiment 64: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

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Embodiment 65: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 66: The detector according to any one of the two preceding embodiments, wherein the divider is fully or partially integrated into the sensor element.

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Embodiment 67: The detector according to any one of the eighteen preceding embodiments, wherein the optical sensors are photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors.

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Embodiment 68: The detector according to any one of the nineteen preceding embodiments, wherein the optical sensors are sensitive in the infrared spectral range.

Embodiment 69: The detector according to any one of the twenty preceding embodiments, wherein the matrix comprises one or both of a CCD detector, preferably a CCD detector chip, or a CMOS detector, preferably a CMOS detector chip.

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Embodiment 70: The detector according to any one of the twenty-one preceding embodiments, wherein the matrix is a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns.

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Embodiment 71: The detector according to the preceding embodiment, wherein the rows and columns are oriented essentially perpendicular.

Embodiment 72: The detector according to any one of the two preceding embodiments, wherein the matrix has at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows.

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Embodiment 73: The detector according to any one of the three preceding embodiments, wherein the matrix has at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns.

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Embodiment 74: The detector according to any one of the four preceding embodiments, wherein the ratio of the number of rows and the number of columns is close to 1 preferable larger than 1:3, more preferably larger than 1:2.

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Embodiment 75: The detector according to any one of the twenty-six preceding embodiments, wherein the matrix comprises at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors.

Embodiment 76: The detector according to any one of the twenty-seven preceding embodiments, wherein the sensor element is oriented essentially perpendicular to an optical axis of the detector.

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Embodiment 77: The detector according to any one of the twenty-eight preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the respective sensor signals are dependent on the total power of illumination of the respective optical sensor.

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Embodiment 78: The detector according to any one of the twenty-nine preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

Embodiment 79: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

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Embodiment 80: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

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Embodiment 81: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one fringe pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

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Embodiment 82: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the quotient signal Q .

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Embodiment 83: The detector according to any one of the thirty-four preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

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Embodiment 84: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

Embodiment 85: The detector according to any one of the thirty-six preceding embodiments, wherein the evaluation device is further configured for determining at least one transversal coordinate of the object by evaluating a transversal position of the at least one optical sensor having the highest sensor signal.

Embodiment 86: The detector according to any one of the thirty-eight preceding embodiments, wherein the evaluation device is configured for distinguishing different light beams having different modulations.

Embodiment 87: A detector for determining a position of at least one object, the detector having

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

Embodiment 88: The detector according to the preceding embodiment, wherein the optical sensors are located in the same plane.

Embodiment 89: The detector according to any one of the two preceding embodiments, wherein the optical sensors are partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector.

Embodiment 90: The detector according to any one of the three preceding embodiments, wherein the optical sensors are part of a sensor array, such as an array comprising 2-4 optical sensors such as a quadrant photodiode.

Embodiment 91: The detector according to the preceding embodiment, wherein a geometrical center of the sensor array is offset from the optical axis.

Embodiment 92: The detector according to any one of the two preceding embodiments, wherein the sensor array is movable, preferably automatically, relative to the optical axis.

Embodiment 93: The detector according to the preceding embodiment, wherein the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly,
5 moving the sensor array relative to the optical axis until the light spot is off-centered.

Embodiment 94: The detector according to any one of the seven preceding embodiments, wherein the optical sensors are partial diodes of a quadrant diode, with a geometrical center of the quadrant diode being off-centered from the optical axis of the detector.
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Embodiment 95: The detector according to any one of the eight preceding embodiments, wherein the light-sensitive areas of the optical sensors are equal.

Embodiment 96: The detector according to any one of the nine preceding embodiments, wherein the detector comprises at least one lens, wherein the optical axis of the detector is an axis of symmetry of the lens.
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Embodiment 97: The detector according to any one of the ten preceding embodiments, wherein the evaluation device is configured for deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the evaluation device is further configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
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Embodiment 98: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.
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Embodiment 99: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.
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Embodiment 100: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.
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Embodiment 101: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

Embodiment 102: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.
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Embodiment 103: The detector according to any one of the sixteen preceding embodiments, wherein the optical sensors are arranged in one and the same beam path of the detector.

Embodiment 104: The detector according to any one of the seventeen preceding embodiments, wherein the optical sensors each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

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Embodiment 105: The detector according to any one of the eighteen preceding embodiments, wherein the optical sensors each are uniform sensors having a single light-sensitive area each.

Embodiment 106: The detector according to any one of the nineteen preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

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Embodiment 107: The detector according to any one of the twenty preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object, e.g. an illumination source comprising at least one laser, such as an illumination source providing at least one illuminating light beam being on the optical axis or being not on the optical axis.

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Embodiment 108: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

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Embodiment 109: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

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Embodiment 110: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one fringe pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

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Embodiment 111: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the quotient signal

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Q.

Embodiment 112: The detector according to any one of the twenty-five preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

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Embodiment 113: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example, at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one diffractive optical element; at least one Fresnel lens; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

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Embodiment 114: A detector for determining a position of at least one object, the detector having

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
- at least one second optical sensor having:
 - o at least one fluorescent waveguiding sheet forming a second light-sensitive area, wherein the fluorescent waveguiding sheet is oriented towards the object such that at least one light beam propagating from the object towards the detector generates at least one light spot in the second light-sensitive area, wherein the fluorescent waveguiding sheet contains at least one fluorescent material, wherein the fluorescent material is adapted to generate fluorescence light in response to the illumination by the light beam; and
 - o at least one photosensitive element located at at least one edge of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive element by the fluorescent waveguiding sheet and capable of generating at least one second sensor signal in response to the illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

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Embodiment 115: The detector according to the preceding embodiment, wherein the at least one photosensitive element is located at at least one of: a corner of the fluorescent waveguiding sheet; a straight edge, e.g. a straight rim portion, of the fluorescent waveguiding sheet.

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Embodiment 116: The detector according to any one of the two preceding embodiments, wherein the at least one photosensitive element is optically coupled to the fluorescent waveguiding sheet by at least one optical coupling element configured for at least partially coupling the fluo-

rescence light guided by the fluorescent waveguiding sheet out of the fluorescent waveguiding sheet.

5 Embodiment 117: The detector according to the preceding embodiment, wherein the optical coupling element is configured for at least partially coupling the fluorescence light coupled out of the fluorescent waveguiding sheet into the photosensitive element.

10 Embodiment 118: The detector according to any one of the two preceding embodiments, wherein the optical coupling element is selected from the group consisting of: a portion of transparent adhesive attaching the photosensitive element to the fluorescent waveguiding sheet; an etched portion within the fluorescent waveguiding sheet; a scratch in the fluorescent waveguiding sheet; a prism.

15 Embodiment 119: The detector according to any one of the five preceding embodiments, wherein the at least one first optical sensor and the at least one photosensitive element of the at least one second optical sensor have essentially identical electrical capacitance.

20 Embodiment 120: The detector according to any one of the six preceding embodiments, wherein the evaluation device is configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device is configured for determining the longitudinal coordinate by evaluating the quotient signal Q .

25 Embodiment 121: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.

30 Embodiment 122: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

35 Embodiment 123: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

Embodiment 124: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

40 Embodiment 125: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 126: The detector according to any one of the twelve preceding embodiments, wherein the first optical sensor is located in front of the second optical sensor, preferably such

that the light beam, in the given order, illuminates the first optical sensor and, afterwards, the second optical sensor.

Embodiment 127: The detector according to any one of the thirteen preceding embodiments, wherein the second optical sensor comprises at least two, preferably at least three and most preferably at least four photosensitive elements located at at least two edges of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive elements by the fluorescent waveguiding sheet and capable of generating at least one sensor signal each.

Embodiment 128: The detector according to the preceding embodiment, wherein the at least two photosensitive elements are located at one or more of: at least two straight edges, e.g. straight rim portions, of the fluorescent waveguiding sheet, at least two corners of the fluorescent waveguiding sheet, at least one corner of the fluorescent waveguiding sheet and at least one straight edge of the fluorescent waveguiding sheet.

Embodiment 129: The detector according to the preceding embodiment, wherein the evaluation device comprises at least one summing device configured for summing up the sensor signals of the at least two photosensitive elements, thereby forming a sum signal S.

Embodiment 130: The detector according to the preceding embodiment, wherein the sum signal is derived by using the formula $S = \sum_i c_i s_i$, with s_i being the sensor signals, with $i = 1 \dots N$ with N being a positive integer indicating the number of the photosensitive elements or a smaller positive integer, and with c_i being respective calibration coefficients.

Embodiment 131: The detector according to any one of the two preceding embodiments, wherein the evaluation device is configured to determine the at least one longitudinal coordinate z of the object by using at least one predetermined relationship between the first sensor signal, the sum signal S of the second sensor signals and the longitudinal coordinate z.

Embodiment 132: The detector according to any one of the four preceding embodiments, wherein the evaluation device is further configured to determine at least one transversal coordinate x, y of the object by evaluating the sensor signals of the photosensitive elements.

Embodiment 133: The detector according to the preceding embodiment, wherein the evaluation device comprises at least one subtracting device configured to form at least one difference signal D between second sensor signals generated by at least two of the photosensitive elements.

Embodiment 134: The detector according to the preceding embodiment, wherein the second sensor signals comprise at least one second sensor signal s_{21} and at least one second sensor signal s_{22} , wherein the at least one difference signal D is proportional to $a \cdot s_{21} - b \cdot s_{22}$, with a, b being real number coefficients, preferably with $a=1$ and $b=1$.

Embodiment 135: The detector according to the preceding embodiment, wherein the at least one difference signal D is derived according to the formula $D = (a \cdot s_{21} - b \cdot s_{22}) / (a \cdot s_{21} + b \cdot s_{22})$.

Embodiment 136: The detector according to any one of the three preceding embodiments,

5 wherein the subtracting device is configured to form at least one first difference signal D_x from which at least one first transversal coordinate x of the object is derived, wherein the subtracting device is further configured to form at least one second difference signal D_y from which at least one second transversal coordinate y of the object is derived.

10 Embodiment 137: The detector according to the preceding embodiment, wherein the first difference signal D_x is generated from at least two second sensor signals s_{x1} , s_{x2} of at least two photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the waveguiding sheet in a first dimension, and wherein the second difference signal D_y is generated from at least two second sensor signals s_{y1} , s_{y2} of at least two photosensitive elements
15 located at opposing edges, e.g. opposing rim portions and/or corners, of the waveguiding sheet in a second dimension.

Embodiment 138: The detector according to the preceding embodiment, wherein the at least one first difference signal D_x is derived according to the formula $D_x = (a \cdot s_{x1} - b \cdot s_{x2}) / (a \cdot s_{x1} + b \cdot s_{x2})$, and wherein the at least one second difference signal D_y is derived according to the formula $D_y = (c \cdot s_{y1} - d \cdot s_{y2}) / (c \cdot s_{y1} + d \cdot s_{y2})$, with a , b , c , d being real number coefficients, preferably with $a=1$, $b=1$, $c=1$ and $d=1$.
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Embodiment 139: The detector according to any one of the 11 preceding embodiments, wherein
25 the photosensitive elements comprise at least two photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the fluorescent waveguiding sheet.

Embodiment 140: The detector according to any one of the 12 preceding embodiments, wherein the photosensitive elements comprise at least one first pair of photosensitive elements located
30 at opposing edges, e.g. opposing rim portions and/or corners, of the fluorescent waveguiding sheet in a first dimension of a coordinate system, wherein the photosensitive elements further comprise at least one second pair of photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the fluorescent waveguiding sheet in a second dimension of the coordinate system.
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Embodiment 141: The detector according to any one of the 27 preceding embodiments, wherein the second sensitive area is a homogeneous sensitive area.

Embodiment 142: The detector according to any one of the 28 preceding embodiments, wherein
40 the second sensitive area has a surface area of at least 5 mm^2 , preferably of at least 10 mm^2 , more preferably of at least 100 mm^2 , more preferably of at least 400 mm^2 .

Embodiment 143: The detector according to any one of the 29 preceding embodiments, wherein the fluorescent waveguiding sheet comprises at least one planar sheet.

Embodiment 144: The detector according to any one of the 30 preceding embodiments, wherein the fluorescent waveguiding sheet has a thickness of 10 μm to 3 mm, preferably a thickness of 100 μm to 1 mm, such as a thickness of 50 μm to 2 mm.

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Embodiment 145: The detector according to any one of the 31 preceding embodiments, wherein the fluorescent waveguiding sheet is flexible or deformable.

Embodiment 146: The detector according to any one of the 32 preceding embodiments, wherein the fluorescent waveguiding sheet comprises at least one matrix material, wherein the at least one fluorescent material is one or more of mixed into the matrix material, dispersed into the matrix material, chemically bound to the matrix material or dissolved in the matrix material.

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Embodiment 147: The detector according to the preceding embodiment, wherein the matrix material comprises at least one plastic material.

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Embodiment 148: The detector according to the preceding embodiment, wherein the plastic material comprises at least one polymer material.

Embodiment 149: The detector according to any one of the two preceding embodiments, wherein the plastic material comprises at least one material selected from the group consisting of: a polycarbonate, a poly(methyl methacrylate), a polystyrene, a polyurethane, a polypropylene, a polyethylene terephthalate, a polyvinylchloride.

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Embodiment 150: The detector according to any one of the 36 preceding embodiments, wherein the fluorescent material comprises at least one fluorescent dye.

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Embodiment 151: The detector according to the preceding embodiment, wherein the fluorescent dye comprises at least one organic fluorescent dye.

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Embodiment 152: The detector according to any one of the two preceding embodiments, wherein the fluorescent dye is selected from the group consisting of: a xanthene derivative, preferably one or more of fluorescein, rhodamine, oregon green, eosin, texas red, or a derivative of any component thereof; a cyanine derivative, preferably one or more of cyanine, indocarbocyanine, oxacarbocyanine, thiocarbocyanine, merocyanine, or a derivative of any component thereof; a squaraine derivative or a ring-substituted squaraine, preferably one or more of Seta, SeTau, and Square dyes, or a derivative of any component thereof; a naphthalene derivative, preferably one or more of a dansyl or a prodan derivative thereof; a coumarin derivative; a oxadiazole derivative, preferably one or more of pyridyloxazole, nitrobenzoxadiazole, benzoxadiazole, or a derivative of any component thereof; an anthracene derivative, preferably one or more of an anthraquinone, DRAQ5, DRAQ7, CyTRAK orange, or a derivative of any component thereof; a pyrene derivative, preferably cascade blue; an oxazine derivative, preferably one or more of nile red, nile blue, cresyl violet, oxazine 170, or a derivative of any component thereof; an acridine derivative, preferably one or more of proflavin, acridine orange, acridine yellow, or a derivative

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of any component thereof; an arylmethine derivative, preferably one or more of auramine, crystal violet, malachite green, or a derivative of any component thereof; a tetrapyrrole derivative, preferably one or more of porphyrin, phthalocyanine, bilirubin; a rylene dye or any derivative thereof, such as a perylene dye; a naphthalene imide or perylene imide; a naphthoiline benzimidazole dye such as published in WO 2012/168395 A1; or a derivative of any component thereof.

Embodiment 153: The detector according to any one of the 39 preceding embodiments, wherein the at least one photosensitive element comprises at least one elongated photosensitive element extending along at least one segment of an edge of the fluorescent waveguiding sheet.

Embodiment 154: The detector according to any one of the 40 preceding embodiments, wherein the fluorescent waveguiding sheet is a rectangular fluorescent waveguiding sheet, preferably a square fluorescent waveguiding sheet, wherein photosensitive elements are located at each of the four edges, e.g. each of the four rim portions and/or corners, of the fluorescent waveguiding sheet.

Embodiment 155: The detector according to any one of the 41 preceding embodiments, wherein the detector further comprises at least one optical filter element, preferably at least one optical short-pass filter.

Embodiment 156: The detector according to any one of the 42 preceding embodiments, wherein the first and second optical sensors are arranged linearly in one and the same beam path of the detector.

Embodiment 157: The detector according to the preceding embodiment, wherein the first and second optical sensors are arranged concentrically with respect to an optical axis of the detector.

Embodiment 158: The detector according to any one of the 44 preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than five times the square root of a surface area of the first light-sensitive area.

Embodiment 159: The detector according to any one of the 45 preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm.

Embodiment 160: The detector according to any one of the 46 preceding embodiments, wherein the second light-sensitive area is larger by at least a factor of two, more preferably by at least a factor of three and most preferably by at least a factor of five as compared to the first light-sensitive area.

Embodiment 161: The detector according to any one of the 47 preceding embodiments, wherein the first light-sensitive area has a surface area of 1 mm² to 150 mm², more preferably a surface area of 10 mm² to 100 mm².

- 5 Embodiment 162: The detector according to any one of the 48 preceding embodiments, specifically according to the preceding embodiment, wherein the second light-sensitive area has a surface area of 160 mm² to 1000 mm², more preferably a surface area of 200 mm² to 600 mm².

- 10 Embodiment 163: The detector according to any one of the 49 preceding embodiments, wherein the first and second optical sensor each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

- 15 Embodiment 164: The detector according to any one of the 50 preceding embodiments, wherein the first optical sensor is a semiconductor sensor, preferably an inorganic semiconductor sensor, more preferably a photodiode and most preferably a silicon photodiode.

- 20 Embodiment 165: The detector according to any one of the 51 preceding embodiments, wherein the first optical sensor is a uniform sensor having a single light-sensitive area.

Embodiment 166: The detector according to any one of the 52 preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

- 25 Embodiment 167: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

- 30 Embodiment 168: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

- 35 Embodiment 169: The detector according to any one of the 55 preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

- 40 Embodiment 170: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, at least one focus-tunable lens, at least one spheric lens, at least one aspheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

Embodiment 171: The detector according to any one of the 57 preceding embodiments, wherein the first light-sensitive area, in a direction of propagation of the light beam, overlaps with the second light-sensitive area.

- 5 Embodiment 172: A detector for determining a position of at least one object, the detector comprising:
- at least one angle dependent optical element adapted to generate at least one transmission light beam having at least one beam profile depending on an angle of incidence of an incident light beam propagating from the object towards the detector and illuminating the angle dependent optical element;
 - 10 - at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element;
 - 15 - at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q from the sensor signals.

Embodiment 173: The detector according to the preceding embodiment, wherein the angle dependent optical element comprises at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element; at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter;

20 at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.

30 Embodiment 174: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least one transfer device, wherein the transfer device has at least one focal length in response to at least one incident light beam propagating from the object to the detector.

35 Embodiment 175: The detector according to any one of the three preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

Embodiment 176: The detector according to the preceding embodiment, wherein the illumination source is adapted to illuminate the object through the angle dependent optical element.

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Embodiment 177: The detector according to any one of the two preceding embodiments, wherein the illumination source and the angle dependent optical element are arranged in the direction

of propagation of the light beam traveling from the object to the detector behind the transfer device.

5 Embodiment 178: The detector according to any one of the three preceding embodiments, wherein a distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors is small.

10 Embodiment 179: The detector according to the preceding embodiment, wherein the distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors is less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m.

15 Embodiment 180: The detector according to any one of the two preceding embodiments, wherein the angle dependent optical element is an optical fiber, wherein the illumination source is guided through the optical fiber, wherein the illumination source is adapted to generate at least one illumination beam, wherein the illumination beam exits the optical fiber behind the transfer device within a baseline that is smaller than a radius of the transfer device, wherein the optical fiber guiding the illumination beam may be attached to the transfer device to reduce reflections at interfaces with larger differences in refractive index.

20 Embodiment 181: The detector according to the preceding embodiment, wherein the optical fiber is attached to the transfer device by one or more of a polymer, a glue or other means for attachment.

25 Embodiment 182: The detector according to any one of the four preceding embodiments, wherein the distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors is less than the radius of the transfer device.

30 Embodiment 183: The detector according to any one of the eleven preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

35 Embodiment 184: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 185: The detector according to any one of the thirteen preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are areas of the beam profile at a sensor position of the optical sensors, and $E(x,y,z_0)$ denotes the beam profile given at the object distance z_0 .

- 5 Embodiment 186: The detector according to any one of the fourteen preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

- 10 Embodiment 187: The detector according to any one of the fifteen preceding embodiments, wherein the optical sensors are positioned off focus.

- 15 Embodiment 188: The detector according to any one of the sixteen preceding embodiments, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the beam profile of the light beam generated by the angle dependent optical element.

- 20 Embodiment 189: The detector according to the preceding embodiment, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

- 25 Embodiment 190: The detector according to any one of the two preceding embodiments, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- 30 Embodiment 191: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile.

- 35 Embodiment 192: The detector according to any one of the four preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.

- 40 Embodiment 193: The detector according to any one of the twenty preceding embodiments, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

- Embodiment 194: The detector according to any one of the two preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensors with at least one line pattern, wherein $A1$ corresponds to an area with a full line width of the line pat-

tern on the optical sensors, wherein A2 is a central area of the line pattern on the optical sensors.

Embodiment 195: The detector according to any one of the four preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensors with at least one point pattern, wherein A1 corresponds to an area with a full radius of a point of the point pattern on the optical sensors, wherein A2 is a central area of the point in the point pattern on the optical sensors.

Embodiment 196: The detector according to any one of the two the preceding embodiments, wherein the evaluation device is configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center in-formation.

Embodiment 197: The detector according to any one of the 23 preceding embodiments, wherein the detector has at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by the light beam generated by the angle dependent optical element.

Embodiment 198: The detector according to the preceding embodiment, wherein the evaluation device is configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 199: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor

signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

- 5 Embodiment 200: The detector according to any one of the two preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for
10 sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a
15 sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.
- 20 Embodiment 201: The detector according to any one of the three preceding embodiments, wherein the combined signal is the quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a
25 quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

Embodiment 202: The detector according to any one of the 27 preceding embodiments, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets,
30 wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element.

35 Embodiment 204: The detector according to the preceding embodiment, wherein the optical sensors are part of a sensor array, wherein a geometrical center of the sensor array is offset from the optical axis.

Embodiment 205: The detector according to the preceding embodiment, wherein the sensor
40 array is movable relative to the optical axis.

Embodiment 206: The detector according to the preceding embodiment, wherein the evaluation device is configured for, firstly, determining the transversal position of a light spot generated by

the light beam on the sensor array by using the sensor signals and for, secondly, moving the sensor array relative to the optical axis until the light spot is off-centered.

5 Embodiment 207: The detector according to any one of the four preceding embodiments, wherein the optical sensors are partial diodes of a bi-cell or quadrant diode, with a geometrical center of the quadrant diode being off-centered from the optical axis of the detector.

10 Embodiment 208: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

15 Embodiment 209: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to
20 assign to the position at least one item of information.

Embodiment 210: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one
25 item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

30 Embodiment 211: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

35 Embodiment 212: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

40 Embodiment 213: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 214: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of the preceding embodiment referring to a detector, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

Embodiment 215: A readout device for optical storage media, the readout device comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 216: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by the light beam, wherein the detector has at least one transfer device, wherein the transfer device has at least one focal length in response to the light beam propagating from the object to the detector;
- illuminating each of the light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving a quotient signal Q of the sensor signals.

Embodiment 217: The method according to the preceding embodiment, wherein the longitudinal coordinate z of the object is determined in at least one measurement range independent from the object size in an object plane.

Embodiment 218: The method according to the preceding embodiment, wherein the deriving of the quotient signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 219: The method according to any one of the preceding embodiments referring to a method, wherein the method comprises adjusting the measurement range by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive

area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the object in the object plane.

Embodiment 220: The method according to any one of the preceding embodiments referring to a method, wherein the transfer device has an optical axis, wherein the transfer device constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis, wherein the method comprises arranging the optical sensors such that the light-sensitive areas of the optical sensors differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.

Embodiment 221: The method according to any one of the preceding embodiments referring to a method, wherein each of the sensor signals comprises at least one information of at least one beam profile of the light beam, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles, wherein the light-sensitive regions are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

Embodiment 222: The method according to the preceding embodiment, wherein the evaluating comprises determining the first area of the beam profile and the second area of the beam profile.

Embodiment 223: The method according to any one of the two preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises an information relating to a number of photons in the first area of the beam profile and the center information comprises an information relating to a number of photons in the second area of the beam profile, wherein the evaluating comprises deriving the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.

Embodiment 224: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- illuminating at least one first light-sensitive area of at least one first optical sensor of the detector with a light beam propagating from the object to the detector and, thereby, generating at least one first sensor signal;
- illuminating at least one second light-sensitive area of at least one second optical sensor of the detector with the light beam and, thereby, generating at least one sec-

ond sensor signal, wherein the first light-sensitive area is smaller than the second light-sensitive area; and

- evaluating the first and second sensor signals, thereby determining at least one longitudinal coordinate z of the object.

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Embodiment 225: The method according to the preceding embodiment, wherein the evaluating the first and second sensor signal comprises deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, and wherein the determining the longitudinal coordinate comprises evaluating the quotient signal Q .

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Embodiment 226: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;

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- evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;

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- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;

- c) determining at least one combined signal by combining the center signal and the sum signal; and

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- d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 227: The method according to the preceding embodiment, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

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Embodiment 228: The method according to the preceding embodiment, wherein determining the longitudinal coordinate comprises evaluating the quotient signal Q , wherein evaluating the quotient signal comprises using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

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Embodiment 229: The method according to any one of the three preceding method embodiments, wherein the method comprises using the detector according to any one of the preceding embodiments referring to a detector.

Embodiment 230: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam;
- illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

Embodiment 231: The method according to the preceding embodiment, wherein evaluating of the sensor signals comprises deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q .

Embodiment 232: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least one angle dependent optical element and generating at least one light beam having at least one beam profile depending on an angle of incidence;
- providing at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element;
- illuminating each of the light-sensitive areas of the at least two optical sensors of the detector with the light beam generated by the angle dependent optical element, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving a combined signal Q of the sensor signals.

Embodiment 233: The method according to the preceding embodiment, wherein the deriving of the combined signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 234: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage appli-

cation; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use in combination with optical data storage and readout.

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Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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| 20 | Figures 1 and 2 | show different embodiments of a detector according to the present invention; |
| | Figure 3 | shows a quotient signal Q as a function of a longitudinal coordinate z of an object, for various intensities of a light source; |
| 25 | Figure 4 | shows a quotient signal Q as a function of a longitudinal coordinate z of an object for various target sizes; |
| 30 | Figure 5 | shows an exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera. |
| | Figures 6 and 7 | show further different embodiments of a detector according to the present invention; |
| 35 | Figure 8 | shows an example of a light spot on a matrix of optical sensors and the evaluation thereof; |
| 40 | Figure 9 | shows a further exemplary embodiment of a detector according to the present invention; a detector system, a human machine interface, an entertainment device, a tracking system, a scanning system and a camera; and |
| | Figure 10 | shows an exemplary embodiment of a quotient signal Q as a function of a longitudinal coordinate z , for various illumination intensities; |

- Figures 11 and 12 show further different embodiments of a detector according to the present invention;
- 5 Figure 13 shows a further exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera;
- 10 Figure 14 shows an off-centered position of the optical axis of the detector and of a light spot on a quadrant diode;
- Figure 15 shows an alternative embodiment of a detector with staggered optical sensors;
- 15 Figures 16 to 18 show a quotient signal Q of two sensor signals of two photodiodes of a quadrant photodiode as a function of a longitudinal coordinate z of an object under various measurement conditions;
- 20 Figures 19A and 19B show a modification of the embodiment of Figure 12, with an off-axis illumination light beam;
- Figures 19C to 19E shows a comparison of two experimental setups using a detector setup according to Figure 19A with a Bi-cell as optical sensors and experimental results;
- 25 Figures 20 and 21 show further different embodiments of a detector according to the present invention;
- 30 Figures 22A and 22B show different views of an exemplary embodiment of a second optical sensor which may be used in the present invention;
- Figure 23 shows a top view onto the light-sensitive area of the second optical sensor shown in Figures 22A and 22B, with a light spot generated by a light beam;
- 35 Figure 24 shows a further exemplary schematic setup of an evaluation device;
- Figure 25 shows a further exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera;
- 40 Figures 26A and 26B show an alternative embodiment of a second optical sensor which may be used in the present invention;

	Figures 27A and 27 B	show further exemplary embodiments of a detector according to the present invention;
5	Figure 28	shows experimental results of a distance determination with a detector according to the present invention;
	Figure 29	shows a further exemplary embodiment of a detector according to the present invention;
10	Figure 30	shows an exemplary cloud of points impinging on a sensor element;
	Figures 31 A to O	show further exemplary configurations of optical sensors according to the present invention;
15	Figure 32	shows experimental results of a determination of a longitudinal coordinate z for different object sizes;
20	Figures 33A and B	show an exemplary beam profile and determination of first area and second area of the beam profile;
	Figure 34	shows a further exemplary embodiment of the detector;
25	Figure 35	shows an exemplary embodiment of a detector according to the present invention;
	Figure 36	shows an embodiment with a multifurcated optical fiber;
30	Figure 37	shows a cut through the optical fiber of Figure 36;
	Figures 38	visualizes angle dependent transmission of an angle dependent optical element;
35	Figure 39	shows angle dependent transmission power of the optical fiber at constant irradiated power;
	Figures 40A and B	show experimental results of distance measurement;
40	Figure 41	shows an exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera;

- Figure 42 shows an embodiment of determining a longitudinal coordinate z using a line pattern;
- 5 Figure 43 shows a simulation testing of object size independence for a one-dimensional case using computational ray tracing;
- 10 Figures 44A and B show a defocused beam profile subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates and Radon transformation different image regions on a matrix of optical sensors;
- Figures 45A and B show further embodiments of the detector according to the present invention comprising at least one bi-cell;
- 15 Figure 46 shows spot diameter independence of the combined sensor signal;
- Figures 47A to C show three embodiments of a hexagonal illumination pattern;
- 20 Figure 48 shows an embodiment of a scanning device;

Detailed description of the embodiments:

- 25 In Fig. 1, a schematic view of a first embodiment of a detector 110 for determining a position of at least one object 112 is depicted. The detector 110 comprises at least two optical sensors 113, for example a first optical sensor 118 and a second optical sensor 120, each having at least one light-sensitive area 121. In this case, the object 112 comprises a beacon device 114, from which a light beam 116 propagates towards the first optical sensor 118 and the second
- 30 optical sensor 120. The first optical sensor 118 may comprise a first light-sensitive area 122, and the second optical sensor 120 may comprise a second light-sensitive area 124. The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110, further, comprises at least one transfer device 128, such as at least one lens or a lens system, specifically for beam shaping.
- 35 The transfer device 128 has at least one focal length in response to the incident light beam 116 propagating from the object 112 to the detector 110. The transfer device 128 has an optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. The transfer device 128 constitutes a coordinate system. A direction parallel or anti-parallel to the optical axis 126, 129 may be defined as a longitudinal direction, whereas direc-
- 40 tions perpendicular to the optical axis 126, 129 may be defined as transversal directions, wherein a longitudinal coordinate l is a coordinate along the optical axis 126, 129 and wherein d is a spatial offset from the optical axis 126, 129. Consequently, the light beam 116 is focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a

longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. The optical sensors 118, 120 may be positioned off focus. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

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In this first preferred embodiment the optical sensors 118, 120 may be arranged such that the light-sensitive areas 122, 124 differ in their longitudinal coordinate and/or their surface areas and/or their surface shapes. As can be seen in Fig. 1, the first optical sensor 118 is a small optical sensor, whereas the second optical sensor 120 is a large optical sensor. Thus, the width of the light beam 116 fully covers the first light-sensitive area 122, whereas, on the second light-sensitive area 124, a light spot is generated which is smaller than the second light-sensitive area 124, such that the light spot is fully located within the second light-sensitive area 124. As an example, the first light-sensitive area 122 may have a surface area of 1 mm^2 to 100 mm^2 , whereas the second light-sensitive area 124 may have a surface area of 50 to 600 mm^2 . Other embodiments, however, are feasible.

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The first optical sensor 118, in response to the illumination by the light beam 116, may generate a first sensor signal s_1 , whereas the second optical sensor 120 may generate a second sensor signal s_2 . Preferably, the optical sensors 118, 120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 116 or of the portion of the light beam 116 illuminating their respective light-sensitive areas 122, 124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination. In other words, preferably, the optical sensors 118, 120 do not exhibit the above-described FiP effect.

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The sensor signals s_1 and s_2 are provided to an evaluation device 132 of the detector 110. The evaluation device 132, as symbolically shown in Fig. 1, is embodied to derive a quotient signal Q , as explained above. From the quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 112 and/or the beacon device 114, from which the light beam 116 propagates towards the detector 110. For further details of this evaluation, reference is made to Fig. 3 and 4 below.

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The detector 110, in combination with the at least one beacon device 114, may be referred to as a detector system 134, as will be explained in further detail below with reference to Fig. 5.

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In Fig. 2, a modification of the embodiment of Fig. 1 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Fig. 1. Instead of using an active light source, i.e. a beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 comprises at least one illumination source 136. The illumination source 136, as an example, may comprise a laser, whereas, in Fig. 1, as an example, the beacon device 114 may comprise a light-emitting diode (LED). The illumination source 136 may be configured for generating at least one illumi-

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nation light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116.

- 5 As shown in Fig. 2, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too. In order to provide an on-axis illumination, as shown in Fig. 2, as an example, one or more reflective elements 140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

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Apart from these modifications, the setup of the embodiment in Fig. 2 corresponds to the setup in Fig. 1. Thus, again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q , and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the quotient signal Q . It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

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It shall further be noted that the embodiments shown in Fig. 1 and 2 simply provide embodiments for determining the longitudinal coordinate z of the object 112. It is also feasible, however, to modify the setups of Fig. 1 and 2 to provide additional information on a transversal coordinate of the object 112 and/or of parts thereof. As an example, e.g. in between the transfer device 128 and the optical sensors 118, 120, one or more parts of the light beam 116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 112 and/or of parts thereof. The transversal coordinate may be used to verify and/or enhance the quality of the distance information. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

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In Fig. 3 and 4 typical quotient signals Q are depicted, as a function of the longitudinal coordinate z of an object 112 in a test setup. Therein, a simple quotient s_1/s_2 is shown, for an exemplary setup of the detector 110. Fig. 3 and 4, each, show a bundle of experiments which are not resolved in these figures. Thus, in Fig. 3, various curves are given for the setup shown in Fig. 1, with an active beacon device 114 having an LED. The current of the LED target of the beacon device 114, in this experiment, is changed from 1000 mA to 25 mA. Basically, no difference in the quotient signal, as a function of the longitudinal coordinate z (given in mm) can be detected over the spatial measurement range of 250 mm to 2,250 mm. The experiment clearly shows that the setup of the detector 110 according to the present invention is independent from the total power of the light beam 116. Thus, no additional information on the total power of the light beam, and, thus, no additional illumination on the luminance is required in order to derive the

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longitudinal coordinate. Thus, as shown in Fig. 3, as an example, a unique relationship between a quotient signal Q^* as measured in an experiment and a longitudinal coordinate z^* exists.

Thus, the curves as shown in Fig. 3, as an example, may be used as calibration curves for indicating a unique and predetermined or determinable relationship between the quotient signal Q and the longitudinal coordinate. The curves as shown in Fig. 3, as an example, may be stored in a data storage and/or in a lookup table. The calibration curves Q may simply be determined by calibration experiments. It is also feasible, however, to derive these curves by one or more of modelling, analytically, semi-empirically and empirically.

The experiment shown in Fig. 3 clearly demonstrates that the setup of the detector 110 according to the present invention provides a large range of measurement, both in terms of space (e.g. a measurement range from 270 to 2,250 mm) and in terms of brightness or total power of the light beam 116. In Fig. 4, an additional experiment is shown which demonstrates that the setup is widely independent from the target size, i.e. the lateral diameter of the beacon device 114. For this experiment, again, an LED beacon device 114 was used, similar to the setup shown in Fig. 1, wherein the size of the target, i.e. the visible part of the LED, was changed by using a diffuser and an adjustable aperture. Thereby, the aperture or the size of the target was varied from 1 mm to 25 mm in diameter. Without resolving the curves shown in Fig. 4 in detail, it is clearly visible that the quotient signal Q , again, is widely independent from the target size, in between a target size of 1 mm to 25 mm. Thus, again, a unique relationship between the quotient signal Q and the longitudinal coordinate z can be derived, for various target sizes, which may be used for evaluation.

The results shown in Fig. 3 and 4 were derived experimentally, by varying the named parameters and by measuring appropriate signals. The results, however, may also be derived analytically, semi-analytically or by modelling. Comparable results were obtained.

Fig. 5 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments shown in Fig. 1 or 2. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 5 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 5 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 148. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises the detector system 134. The components of the devices and systems shall be explained in further detail below.

Fig. 5 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136. The illumination source 136, generally, is configured to emit at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 5 is shown in Fig. 1 and 2. Thus, the detector 110, besides the optical sensors 118, 120, comprises at least one evaluation device 132, having e.g. the at least one divider 142 and/or the at least one position evaluation device 144, as symbolically depicted in Fig. 5. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 118, 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Fig. 5. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 118, 120 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the optical sensors 118, 120. The optical sensors 118, 120 may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. A coordinate system 168 may be defined, in which a direction parallel or anti-parallel to the optical axis 126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 126 may be defined as transversal directions. In the coordinate system 128, symbolically depicted in Fig. 5, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the optical sensors 118, 120 as well as, optionally, further optical sensors. The optical sensors 118, 120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 118 covers a portion of the second optical sensor 120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 112 and/or of parts thereof. Alternatively, however, the optical sensors 118, 120 may be located at the same longitudinal coordinate.

One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as explained above in the context of Fig. 1 to 4, the evaluation device 132 is configured to evaluate sensor signals provided by the optical sensors 118, 120. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 118, 120 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 136 is used, the beacon devices 114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflected beacon devices 114 having at least one reflective surface such as a mirror, retro reflector, reflective film, or the like. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the light-sensitive areas 122, 124 of the optical sensors 118, 120. For details of the evaluation, reference may be made to Fig. 1 to 4 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Fig. 5, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Fig. 5 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may be fully or partially integrated into the machine 170, specifically the computer, as indicated in Fig. 5 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Figure 6, a schematic view of a further embodiment of the detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises the at least one beacon device 114, from which the light beam 116 propagates towards at least one sensor element 115. The sensor element 115 comprises a matrix 117 of optical sensors 113, each optical sensor 113 having at least one light-sensitive area 121 facing the object 112. In this second preferred embodiment the optical sensors 118, 120 may be arranged such that the light-sensitive areas of the optical sensors 113 differ in spatial offset and/or surface areas. The light beam 116, as an example, may propagate along the optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110 comprises the at least one transfer device 128, such as at least one lens and/or at least one lens system, specifically for beam shaping. Consequently, the light beam 116 may be focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on the longitudinal coordinate z of the object 112, such as on the distance between the detector 110 and the beacon device 114 and/or the object 112. The transfer device 128 constitutes the optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. Consequently, the light beam 116 is focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. The optical sensors 118, 120 are positioned off focus. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Figure 6, the light beam 116 generates a light spot 131 on the matrix 117. In Figure 8, an exemplary view of the light spot 131 on the matrix 117 is shown. As can be seen, in this exemplary embodiment, the matrix 117 specifically may be a rectangular matrix, with rows numbered by "i", from 1 to n , and with columns, denoted by "j", from 1 to m , with n , m being integers. The center of the light spot 131, in this exemplary embodiment, is located in the sensor

element denoted by i^* , j^* . The optical sensors 113 may provide sensor signals s_{ij} to an evaluation device 132 which, out of the sensor signals, determines at least one center signal, denoted symbolically by $s_{i^*j^*}$. As outlined in further detail above, for generating the center signal, the evaluation device 132 may comprise at least one center detector 133. As an example, the center detector 133 simply may determine the maximum sensor signal out of the plurality of sensor signals generated by the optical sensors 113. Alternative methods are feasible. Thus, as an example, instead of determining a single maximum optical sensor signal, a plurality of sensor signals may be used for generating the center signal. Thus, as an example, neighboring optical sensors which are adjacent to the optical sensor i^* , j^* may contribute to the center signal, such as optical sensors with the coordinates $i^*-1, \dots, i^*+1$ and $j^*-1, \dots, j^*+1$. These coordinates, in the simple exemplary embodiment, may form a square around the optical sensor i^* , j^* . Instead of a square having a side length of 3, as in this embodiment, other environments around the optical sensor having the highest sensor signal may be used such as to optimize the signal to noise ratio of the detector signal and or of the distance information. Further, additionally or alternatively, the center signal may be generated by adding up and/or averaging over sensor signals within a certain range from the maximum sensor signal which may for example be beneficial to the measurement precision concerning noise such as pixel noise. Further, additionally or alternatively, for the determination of the center signal or sum signal, image processing techniques such as sub-pixel processing, interpolation, normalization or the like may be employed. Other alternatives are feasible. The evaluation device 132 may be adapted to determine the center signal by integrating of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device 132 may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid. Further, when trapezoid beam profiles may be assumed, the evaluation device 132 may be adopted to determine the edge and center signals by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations. Additionally or alternatively, the evaluation device 132 may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

Further, the evaluation device 132 is configured for determining at least one sum signal out of the sensor signals of the matrix 117. For this purpose, the evaluation device 132 may comprise at least one summing device 135. The summing device 135 may be configured for adding up, integrating or averaging over the sensor signals of the entire matrix 117, of a region of interest within the matrix 117, each option with or without the optical sensors from which the center signal is generated. Thus, in the exemplary embodiment shown in Figure 8, the summing device 135 is simply configured for summing over the sensor signals s_{ij} of the entire matrix 117, except for the center optical detector with the coordinates i^* , j^* . Other options, however, are feasible.

The evaluation device 132 may be adapted to determine the sum signal by integrating of signals of the entire matrix 117, of the region of interest within the matrix 117. For example, the beam profile may be a trapezoid beam profile and the evaluation device 132 may be adapted to determine an integral of the entire trapezoid.

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The evaluation device 132 may be adapted to determine at least one region of interest within the matrix, such as one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform at least one filtering, for example at least one object recognition method.

10 The region of interest may be determined manually by a user or maybe determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device 132 further is configured for forming at least one combined signal out of the center signal and the sum signal. For this purpose, the evaluation device 132, as an example, may comprise at least one combining device 137, such as at least one divider 142. As a very simple embodiment, a quotient Q may be formed, by dividing the center signal by the sum signal or vice versa. Other options are feasible and are given above.

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Finally, the evaluation device 132 is configured for determining at least one longitudinal coordinate z of the object by evaluating the combined signal. For this purpose, the evaluation device may comprise at least one further component, such as at least one evaluation component, for example a position evaluation device 144. It shall be noted that the components of the evaluation device 132 shown in Figure 8 may fully or partially be embodied in hardware and/or software. Further, the components may fully or partially be embodied as independent or separate components, and/or may fully or partially be embodied as components which are integrated into the sensor element 115.

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The embodiment of Figure 8 further shows that, in addition to the longitudinal coordinate z , at least one item of information on a transversal coordinate of the object 112 and/or the beacon device 114 may be generated. Thus, the coordinates i^* and j^* provide additional items of information on a transversal position of the object 112 and/or the beacon device 114. In the setup of Figure 6, the beacon device 114, for the sake of simplicity, is positioned in the center, i.e. on the optical axis 126, 129. In this case, the light spot 131 is likely to be centered in the middle of the matrix 117. In the embodiment shown in Figure 8, however, as can easily be detected, the light spot 131 is off-centered. This off-centering is characterized by the coordinates i^* , j^* . By using known optical relationships between this off-centering and a transversal position of the object 112 and/or the beacon device 114, such as by using the lens equation, at least one transversal coordinate of the object 112 and/or the beacon device 114 may be generated. This option is also shown in the exemplary embodiment of Figure 8.

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In Figure 7, a modification of the embodiment of Figure 6 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Figure 6. Instead of using an active beacon device 114 with light-emitting

properties for generating the light beam 116, however, the detector 110 itself comprises at least one illumination source 136. The illumination source 136, as an example, may comprise at least one laser, whereas, in Figure 6, as an example, the beacon device 114 may comprise a light-emitting diode (LED). Other embodiments, however, are feasible. The illumination source 136
5 may be configured for generating at least one illumination light beam 138 for fully or partially illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116.

As shown in Figure 7, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too. In order to provide an on-axis illumination, as shown in Figure 7, as an example, one or more reflective elements 140 may be used, such as one or more
10 prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Figure 7 corresponds to the setup in Figure 6. Thus, again, an evaluation device 132 may be used, having e.g. at least one divider 142 for forming the quotient signal Q, and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the quotient signal Q and/or another type of combined signal. It shall be noted that the evaluation device 132, again,
20 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 133, 135, 137, 142, 144 may fully or partially be embodied by appropriate software components and/or may fully or partially be embodied by hardware components.

The optical sensors 113 of the matrix 117, as an example, may be pixels of a pixelated optical sensor, such as a CCD and/or a CMOS sensor chip. Thus, as an example, the optical sensors
25 113 may have a side length and/or an equivalent diameter in the range of a few micrometers to several hundred micrometers. It shall be noted, however, that larger pixels or optical sensors 113 may be used. Further, instead of using an integrated sensor element 115 such as a CCD and/or CMOS sensor chip, non-integrated matrices may be used.

Figure 9 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments in Figures 6 or 7. The detector 110, specifically, may be embodied as the camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or
30 image sequences such as digital video clips. Other embodiments are feasible.

Figure 9 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached to and/or integrated into an object 112, the position of which shall be detected by using
40 the detector 110. Figure 9 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 151. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises

the detector system 134. The components of the devices and systems shall be explained in further detail below.

Figure 9 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the at least one object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110 and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136, which is not shown. The illumination source 136, generally, may be configured to emit the at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. It shall be noted, however, that an active beacon device, as e.g. shown in the setup of Figure 6, may also be used, and, thus, that setups with no integrated illumination source 136 are also feasible. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112 and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Figure 9 is shown in Figures 6 and 7. Thus, the detector 110, besides the sensor element 115, comprises the at least one evaluation device 132, having, e.g., the at least one center detector 133, the at least one summing device 135, the at least one combining device 140, the at least one divider 142, the at least one position evaluation device 144 and/or combinations thereof. These components, which may optionally be present, are symbolically depicted in Figure 9. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the components of the evaluation device 132 and one or more of the components of the sensor element 115 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Figure 9. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the sensor element 115 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the sensor element 115. The sensor 115, wherein one or more of the sensor elements 115 may be provided, may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

5 An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. In the coordinate system 168, symbolically depicted in Figure 9, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other
10 types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the sensor element 115 as well as, optionally, one or more further optical sensors. A non-branched beam path may be used or, alternatively, a branched beam path may be possible, with, e.g., additional optical sensors in one or more additional
15 beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining the transversal coordinates of the object 112 and/or parts thereof. As outlined above, however, in the context of Figure 8, the at least one transversal coordinate may also be determined by the sensor element 115 itself, such as by determining the transversal coordinates of the center of the light spot 131.

20 One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as outlined above in the context of Figures 6 to 8, the evaluation device 132 may be configured to evaluate the sensor signals
25 provided by the optical sensors 113 of the matrix 117 of the sensor element 115. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 113 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may comprise
30 active beacon devices with an integrated illumination source such as light-emitting diodes. In case the illumination source 136 is used, the beacon devices do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by
35 one or more lenses, illuminates the sensor element 118. For details of the evaluation, reference may be made to Figures 6 to 8 above.

40 As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Figure 9, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Figure 9 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may fully or partially be integrated into the machine 170, specifically the computer, as indicated in Figure 9, and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Figure 10, experimental data are shown which demonstrate an exemplary embodiment of the present invention in terms of measurement data. The figure shows a quotient signal Q as a function of a longitudinal coordinate z , given in millimeters, for various illumination intensities.

In order to gain the experimental data shown in Figure 10, an experimental setup was used with a sensor element 118 formed by a Basler AC 1920-40GC camera, with a transfer device 128 formed by a Nikkor 50 mm lens.

As a beacon device 114, a light-emitting diode (LED) was used, having a central nominal wavelength of 532 nm. A diffusor made from Teflon film was used in front of the LED and a diaphragm, in order to provide a well-defined light-emitting area having a diameter of 5 mm. The intensity of the LED was varied, by varying a drive current of the LED between 5 mA and 150 mA.

In the experiments, the distance z between the LED and the lens was varied from 300 mm to 1700 mm. The signal of the Basler AC 1920-40GC camera was evaluated by the following procedure. As a center signal, an accumulated signal of an inner circle having a radius of 15 pixels around the optical axis was determined, with the light spot centered at the optical axis. As a sum signal, the sum of the signals of all pixels of the camera within the light spot was generated. A quotient signal was formed, by dividing the sum signal by the center signal.

In Figure 10, the solid curves, overlapping, show the quotient signal curves for LED currents of 150 mA, 125 mA, 100 mA and 50 mA. As can be seen, there are basically no differences between these curves, within the whole measurement range. This experiment clearly shows that the quotient signal is widely independent on the total power of the light beam. Only at lower intensities, the curves may be distinguished. Thus, the dotted curve shows measurements at an

LED current of 25 mA, the dashed-dotted line at an LED current of 10 mA, and the dashed line at an LED current of 5 mA. Still, even at these low intensities, the measurement curves are very close to the solid curves, which shows the high tolerances of the measurement.

- 5 In Fig. 11, a schematic view of a further embodiment of a detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises the beacon device 114, from which the light beam 116 propagates towards the first optical sensor 118 and the second optical sensor 120. The first optical sensor 118 may comprise a first light-sensitive area 122, and the second optical sensor 120 comprises the second light-sensitive area 124.
- 10 The optical sensors 118, 120, as shown e.g. in Fig. 14, may be part of an array 174 of optical sensors 176, such as the first optical sensor 118 being the optical sensor 176 in the upper left corner of the array 174 and the second optical sensor 120 being the optical sensor 176 in the lower right corner of the array 174. Other choices are feasible. The array 174, as an example, may be a quadrant photodiode 178, and the optical sensors 176 may be partial diodes of the
- 15 quadrant photodiode 178.

The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110 comprises the at least one transfer device 128, such as at least one lens and/or at least one lens system, specifically

20 for beam shaping. Consequently, the light beam 116 may be focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on the longitudinal coordinate z of the object 112, such as on the distance between the detector 110 and the beacon device 114 and/or the object 112. The optical sensors 118, 120 are positioned off focus. In this

25 third preferred embodiment the optical sensors 118, 120 may be arranged such that the light-sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Fig. 14, the setup of the detector 110 is off-centered in various ways. Thus, a

30 geometrical center 180 of the array 174 may be off-centered from the optical axis 126, 129 by offset d_0 . Further, a geometrical center 182 of the first optical sensor 118 is off-centered from the optical axis 126 by offset d_1 , and a geometrical center 184 of the second optical sensor 120 is off-centered from the optical axis 126 by offset d_2 , wherein $d_1 \neq d_2$. In other words, a light spot 186 is formed, which is unequally distributed over the light-sensitive areas 122, 124.

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As will be shown in further detail below, the detector 110 may be configured for automatically establishing the off-centered position of the light spot 186 on the array 174. For this purpose, firstly, the detector 110 may be configured for determining whether the sensor signals generated by the optical sensors 176 of the array 174 are equal. If this should be the case, the detector

40 110 may be configured to determine that the light spot 186 is centered in the array 174 and, consequently, may shift the light spot 186 out of the geometrical center 180 of the array 174, such as by shifting the whole array 174 in a plane perpendicular to the optical axis 126, 129. For

this purpose, as will be shown in further detail below with respect to Fig. 13, one or more actuators may be provided in the detector 110.

Turning back to the setup of Fig. 11, the first optical sensor 118, in response to the illumination by the light beam 116, generates a first sensor signal s_1 , whereas the second optical sensor 120 generates a second sensor signal s_2 . Preferably, the optical sensors 118, 120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 116 or of the portion of the light beam 116 illuminating their respective light-sensitive areas 122, 124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination. In other words, preferably, the optical sensors 118, 120 do not exhibit the above-described FiP effect.

The sensor signals s_1 and s_2 are provided to an evaluation device 132 of the detector 110. The evaluation device 110, as symbolically shown in Fig. 1, may specifically be embodied to derive a quotient signal Q , as explained above. The quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 112 and/or the beacon device 114, from which the light beam 116 propagates towards the detector 110, as will be explained in further detail with reference to the corrosion signals shown in Fig. 16 to 18 below.

The detector 110, in combination with the at least one beacon device 114, may be referred to as the detector system 134, as will be explained in further detail below with reference to Fig. 13.

In Fig. 12, a modification of the embodiment of Fig. 11 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Fig. 11. Instead of using an active light source, i.e. a beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 comprises the at least one illumination source 136. The illumination source 136, as an example, may comprise a laser, whereas, in Fig. 11, as an example, the beacon device 114 may comprise a light-emitting diode (LED). The illumination source 136 may be configured for generating at least one illumination light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The illumination source 136, as an example, may comprise one or more diaphragms 190, such as an adjustable diaphragm 190, e.g. an adjustable iris diaphragm and/or a pin hole.

The setup shown in Fig. 12, as an example, may also be used in or as a readout device 192 for optical storage media. Thus, as an example, the object 112 may be an optical storage medium such as in optical storage discs, e.g. a CCD, DVD or Blu-ray disc. By measuring the presence or non-presence of data storage modules and the depth of the same within the object 112, by using the above-mentioned measurement principle, a data readout may take place.

The light beam 116, specifically, may travel along the optical axis 126 of the detector 110. As shown in Fig. 12, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too, as will be shown in the context of Fig. 19A and 19B below. In order to provide an on-axis illumination, as shown in Fig. 12, as an example, one or more reflective elements 140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Fig. 12 corresponds to the setup in Fig. 11. Thus, again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q, and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the at least one quotient signal Q. It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

It shall further be noted that the embodiments shown in Fig. 11 and 12 simply provide embodiments for determining the longitudinal coordinate z of the object 112. It is also feasible, however, to modify the setups of Fig. 11 and 12 to provide additional information on a transversal coordinate of the object 112 and/or of parts thereof. As an example, e.g. in between the transfer device 128 and the optical sensors 118, 120, one or more parts of the light beam 116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or additional quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 112 and/or of parts thereof. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

Fig. 13 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments shown in Fig. 11 or 12. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 13 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 13 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 148. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises the detector system 134. The components of the devices and systems shall be explained in further detail below.

Fig. 13 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136. The illumination source 136, generally, is configured to emit at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 13 is shown in Fig. 11 and 12 or will be shown, as an alternative embodiment, in Fig. 15 below. Thus, the detector 110, besides the optical sensors 118, 120, comprises at least one evaluation device 132, having e.g. the at least one divider 142 and/or the at least one position evaluation device 144, as symbolically depicted in Fig. 13. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 118, 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Fig. 15. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 118, 120 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In Fig. 13, as an example, one or more reflective elements 140 may be used, for example partially transparent, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

The detector 110 as symbolically shown in the exemplary embodiment of Fig. 13 may also comprise at least one actuator 188 for moving the array 174 of the optical sensors 176 relative to the optical axis 126. As outlined above, for providing this movement, the optical axis 126 may be moved in relation to the array 174 by moving the optical axis 126, by moving the array 174 or both. Thus, as an example, the optical axis may be moved by using one or more of deflecting elements and/or by using the transfer device 128. As a simple example, a lens of the transfer device 128 may be tilted, such as by using one or more actuators 188 (not depicted). Additionally or alternatively, the array 174 may be shifted by the one or more actuators 188, preferably in

a plane perpendicular to the optical axis 126. As an example, one or more electromechanical actuators may be used, such as one electromechanical actuator for an x-direction and another electromechanical actuator for a y-direction. Other embodiments are feasible. Thereby, the above-mentioned off-centering procedure may be implemented for establishing an off-centered situation as shown e.g. in Fig. 14.

In the exemplary embodiment shown in Fig. 13, further, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the optical sensors 176, including at least the first optical sensor 118 and the second optical sensor 120. The optical sensors 176 may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. In the coordinate system 128, symbolically depicted in Fig. 15, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the optical sensors 118, 120 as well as, optionally, further optical sensors. The optical sensors 118, 120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 118 covers a portion of the second optical sensor 120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 112 and/or of parts thereof.

One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as explained above in the context of Fig. 11, 12 and 14, the evaluation device 132 is configured to evaluate sensor signals provided by the optical sensors 118, 120. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 118, 120 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may be or may comprise active beacon devices with an inte-

grated illumination source such as a light-emitting diode. In case the illumination source 136 is used, the beacon devices 114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the light-sensitive areas 122, 124 of the optical sensors 118, 120. For details of the evaluation, reference may be made to Fig. 11, 12 and 14 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Fig. 13, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Fig. 13 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may be fully or partially integrated into the machine 170, specifically the computer, as indicated in Fig. 13 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In the setup of the detectors 110 as shown in Fig. 11, 12 and 13, the optical sensors 176 are part of an array 174, and all optical sensors 176 may be located in one and the same plane oriented essentially perpendicular to the optical axis 126. As noted in this context, when mentioning “perpendicular” or “essentially perpendicular”, preferably, a 90° orientation is given. However, tolerances may be present, such as angular tolerances of no more than 20°, preferably of no more than 10° or more preferably of no more than 5°. The optical sensors 176, however, not necessarily have to be located in an array 174 and not necessarily have to be located in one and the same plane, as is shown in an alternative setup of the detector 110 shown in Fig. 15. In this figure, only the optical components are shown. For other components, reference may be made to Fig. 11, 12 and 13 above.

As can be seen, in this alternative setup, two or more optical sensors 176 are present, comprising at least one first optical sensor 118 and at least one second optical sensor 120 located in different planes which are offset in a direction of the optical axis 126, also referred to as the z-direction. Thus, further, as can also be seen, the optical sensors 118, 120 may overlap, whereas in the previous embodiments, preferably, no overlap between the optical sensors 176 is given. Apart from these modifications, the functionality and the evaluation of the sensor signals generally corresponds to the embodiment of Fig. 11, 12 and 13 above.

As discussed above, for evaluating the at least two sensor signals of the at least two optical sensors 176 and for deriving an information on the longitudinal position of the object 112 thereof, such as a distance between the detector 110 and the object 112 and/or a z-coordinate of the object 112, preferably, at least one combined sensor signal is generated by the evaluation device 132. The combined sensor signal, as long as this combined sensor signal provides, at least over a measurement range, a unique function of the distance, may be used for deriving the longitudinal coordinate. As an example, the combined sensor signal may be or may comprise at least one quotient signal Q. In Fig. 16 to 18, quotient signals Q of two sensor signals of two optical sensors 176 are shown under various measurement conditions. In each case, the quotient signal Q is denoted on the vertical axis, as a function of the longitudinal coordinate z of the object 112 on the horizontal axis, the latter given in centimeters.

In all experiments, a setup as shown in Fig. 12 was used. As an illumination source 136, in the experiments of Fig. 16 and 17, a 980 nm Picotronic laser source was used, in conjunction with a lens having a focal length of 100 mm. In the experiment of Fig. 18, a Laser Components laser light source having a wavelength of 850 nm was used, in conjunction with a lens having a focal length of 79 mm. In all experiments, the laser beam was aligned on the optical axis 126 via a small prism in front of the lens 128, forming a reflective element 140. A diaphragm 190 in front of the laser source was used to vary the spot size. The quadrant diode 178 was used to measure the reflection of the laser source on different materials. In all experiments, the distance dependency is given by the quotient Q of two adjacent quadrant currents.

In Fig. 16, the laser power was varied during the experiment, from 8 nA laser current, denoted by the dotted line, to 106 nA, denoted by the solid line. Therein, since the laser current typically does not provide a measure for the laser intensity, the laser current indicated therein is a current of a silicon photodetector in a measurement setup in which the laser illuminates a white sheet of paper at a distance of 330 mm from the lens. As is clearly visible, the curves are nearly identical and, at least within this range of variation of the laser power, do not significantly depend on the laser power. This experiment shows that the quotient signal provides a reliable and monotonous function of the longitudinal coordinate, independent from the influence of the brightness of the illumination source.

In Fig. 17, a spot size of the illumination source 136 was varied, by varying the open diameter of the diaphragm 190 in front of the laser. The spot size was varied from 1.5 mm, denoted by the dotted line, to 3.5 mm, denoted by the solid line, in steps of 0.5 mm. As can be seen, up to a

distance of approximately 200 cm, the quotient signal Q does not depend on the spot size and, thus, again, is not negatively affected by this variation.

In Fig. 18, a material of the object 112 illuminated by the laser beam was varied. Therein, the dotted line denotes white paper, the dashed line with the smallest dashes denotes black paper, the dashed line with the medium dashes denotes wood, and the dashed line with the largest dashes denotes an aluminum plate. As can be seen, at least up to a measurement range of approximately 250 cm, the experiment does not strongly depend on the type of material used for the object 112.

The experiments shown in Fig. 16 to 18, thus, clearly demonstrate that the quotient signal Q provides a reliable function of the distance. At least within a range of measurement, the function monotonously rises with the distance. The function is not strongly influenced by the most significant variations which may occur in real life measurements, such as the brightness of the illumination source, the spot size of the illumination source or the material of the object 112. Thus, by evaluating the quotient signal Q of two or more optical sensors 176, reliable distance information may be generated. Thus, as an example, the curves shown in Fig. 16 to 18 directly may be used as calibration curves for the purpose of the evaluation device 132. Other evaluation methods, however, are feasible.

In Fig. 19A and 19B, an alternative embodiment of the detector 110 is shown which is a modification of the setup shown in Fig. 2. Thus, for most elements and optional details as well as further elements not shown in the schematic Figures 19A and 19B, reference may be made to the description of Fig. 12 above.

In Figure 12, the illumination light beam 138, as discussed above, preferably travels along the optical axis 126, i.e. parallel to the optical axis 126 or even on the optical axis 126. In the setup, the position of the center of the light spot 186 typically does not depend on the z-coordinate of the object 112, such as on a distance between the object 112 and the detector 110. In other words, the diameter or equivalent diameter of the light spot 186 changes with the distance between the object 112 and the detector 110 whereas, typically, the position of the light spot 186 on the array 174 does not.

Contrarily, in Fig. 19A and 19B, a setup of the detector 110 is shown in which an illumination light beam 138 travels off-axis, i.e. one or both of at an angle other than 0° with the optical axis 126 or parallel to the optical axis 126 but shifted from the optical axis 126. This embodiment, as will be discussed in further detail below, demonstrates that the method according to the present invention can be further enhanced by increasing the z-dependency of a combined sensor signal. Thus, in Fig. 19A, a side view is shown with two different positions of the object 112, i.e. a first position at z_1 , drawn in solid lines, and a second position at z_2 , drawn in dashed lines. As can be seen, the illumination light beam 138 which, as an example, propagates at an angle of 5° to 30° , e.g. 10° to 20° , with the optical axis 126, hits the object 112 in both cases at different positions. From these points of the object 112 illuminated by the illumination light beam 138,

light beams 116 propagate towards the detector 110, wherein, again, the light beam 116 for the object 112 being located at position z_1 is drawn in solid lines, wherein the light beam 116 for the object 112 being located at position z_2 is drawn in dashed lines.

5 In Fig. 19B, the array 174, e.g. a quadrant photodiode, is shown in an enlarged fashion. As can be seen in this setup, the position of the light spot 186 moves with the longitudinal position z of the object 112. Thus, not only is the size of the light spot 186 affected by the longitudinal position z but also is the position on the array 174 of the light spot 186 changed. In Fig. 19B, this movement of the light spot 186 is denoted by arrow z .

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Consequently, by this movement of the light spot 186, the z -dependency of a combined sensor signal taking into account at least two sensor signals of the optical sensors 176 may be increased. As an example, the four diodes of the array 174, in Fig. 19B, are denoted by D1-D4.

15 The quotient signal Q , as an example, may be formed as $Q = i(D1)/i(D4)$, with $i(D1)$ being the sensor signal of photodiode D1, and $i(D4)$ being the sensor signal of photodiode D4. As shown in Figure 19B, the quadrant diode may comprise two dividing lines. The dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other.

20 In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device 132 may be adapted to determine a second quotient using at least three or all four sensor signals of the quadrant diode. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotients allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient may be determined

25 by $Q = i(D1+D2)/i(D3+D4)$, wherein the sensor signals of the two top quadrants, also called top segment, are divided by the sensor signals of the two bottom quadrants, also called bottom segment. Using the quotient of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing

30 line between top and bottom segment is parallel to the baseline, the quotient signal determined from the top segment divided by the bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom

35 segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments, i.e. $Q = i(D1+D3)/i(D2+D4)$, may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient by dividing sensor

40 signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals $i(D1)$, $i(D2)$, $i(D3)$ and $i(D4)$ of the quadrants such that distance measurement is possible over a wide range with a large resolution.

In the situation shown in Fig. 12, the position of the light spot 186 does not depend on z . With a change in z , depending on the optical situation, the spot will become larger or smaller, such as by becoming more diffuse or more focused. In case the spot size increases and the spot becomes more diffuse, $i(D4)$ will increase more rapidly than $i(D1)$, such that the quotient signal Q decreases.

Contrarily, in the situation of Fig. 19A, both the size and the position of the light spot 186 are dependent on the z -coordinate. Thus, the tendency of the z -dependency of the combined sensor signal such as the quotient signal Q will be increased. In the situation of Figure 12, depending on the z -coordinate, the sensor signal of at least one sensor will increase and simultaneously the sensor signal of at least one different sensor will decrease, resulting in the z -dependent quotient signal Q . In the situation of Fig. 19A, the position dependency of the light spot 186 can result in three different situations depending on the relative position of light source, optical axis, and sensor: Firstly, the position dependency of the light spot 186 may result in a further decrease of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot 186 may result in a further increase of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 12. Secondly, the position dependency of the light spot 186 may result in a reduced decrease or even increase of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot 186 may result in a reduced increase or even decrease of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 12. Thirdly, the position dependency of the light spot 186 may be as such that the z -dependence of the sensor signals is largely unchanged compared to the situation in Figure 12. However, according to the present invention, object distance is not determined from the position of the light spot 186 on a sensor as done in triangulation methods. Instead, movement of the light spot 186 on the array 174 may be used to enhance dynamic of the sensor signals and or the resulting quotient signal Q which may result in an enhanced dynamic of the z -dependency. Furthermore, movement of the light spot 186 on the array 174 during measurement may be used to establish and/or to enhance object size independence for the whole measurement range by suitable relative positioning of the optical sensor 176 and the illumination source 136. Thus, movement of the light spot 186 may not be used for the purpose of triangulation but for the purpose of object size independence.

Additionally, as known from the prior art, the sensor signals $i(D1)$, $i(D2)$, $i(D3)$, $i(D4)$ may also be used for determining a transversal position x , y of the object 112. Further, the sensor signals may also be used for verifying the z -coordinate determined by the present invention.

Figure 19C shows a comparison of two experimental setups using a detector setup according to Figure 19A with a Bi-cell as optical sensors 176 with two light sensitive areas. In a first experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 186 may move in parallel to the linear boundary of the two optical sensors 176 of the Bi-cell along a direction of movement 210 in dependence of the object distance. Since the direction of movement 210 of the light spot 186 is in parallel to the linear

boundary of the two light sensitive areas in dependence of the object distance, the resulting sensor signals are identical to a situation with no movement of the light spot 186 depending on object distance as shown in Figure 12. In a second experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 186 may move as such that the distance of the center of the light spot 186 to the boundary of the two optical sensors 176 of the Bi-cell changes in dependence of the object distance such as a movement orthogonal to the boundary of the two optical sensors 176 such as a movement along a direction of movement 208 in dependence of the object distance. The detector setup allowing movement of the light spot 186 may be a modification of the setup shown in Figure 19A. Thus, for most elements and optional details as well as further elements, reference may be made to the description of Fig. 19A above. In Figure 19C, the optical sensors 176 may be a bi-cell diode.

Figure 19D shows experimental results of the comparison of the two experimental setups using a detector setup according to Figure 19A, allowing movement of the light spot 186 according to Figure 19C with movement of the light spot depending on the object distance along directions of movement 210 and 208. Curve 212 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup allowing movement of the light spot 186 along a direction of movement 210 as shown in Figure 19C which is in parallel to the boundary of the optical sensors of the Bi-Cell and, which is a situation equivalent to Figure 12 without a movement of the light spot depending on the object distance. Curve 214 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup according to Figure 19A and using a detector setup allowing movement of the light spot 186 with movement of the light spot 186 according to Figure 19C with movement of the light spot depending on the object distance along a direction of movement 208. The experimental setup was as follows: The optical sensors 176 may be a bi-cell diode, in particular a Si-Bi-Cell. The illumination source 136 may be a 950 nm laser with a spot size of 4 mm. The transfer device 128 may have a focal length of 20 mm, e.g. a lens available as Thorlabs Asphere, $f = 20$ mm. The distance of the object 112 was varied from 0 to 3000 mm. Determination of the longitudinal coordinate z may be possible without allowing movement of the light spot 186. In particular, according to the present invention, movement of the light spot may not be essential for determination of the longitudinal coordinate z. With the detector setup allowing movement of the light spot 186 along a direction 210 or without any movement determination of object distance is possible at very small distance, whereas with movement along a direction 208 determination of object distance is possible for object distance such as distances greater than 500mm.

Figure 19E shows object independence of the two experimental setups using a detector setup according to Figure 19A, allowing movement of the light spot 186 according to Figure 19C with movement of the light spot depending on the object distance along directions of movement 208 and 210. In addition, for both experimental setups, the object size was varied from 1mm (dashed line), 2mm (dotted line), 6mm (solid line) and 12mm (loosely dotted line) by varying the aperture of the laser illumination source. Set of curves 216 shows dependency of quotient Q on the longitudinal coordinate z for the experimental setup allowing movement of the light spot 186

along a direction 208. Set of curves 218 shows dependency of quotient Q on the longitudinal coordinate z for the experimental setup allowing movement of the light spot 186 along a direction 210 or without any movement. Set of curves 216 show only small deviations, in particular less than 5%, whereas set of curves 218 show larger deviations, in particular with increasing distance z . Thus, movement of the light spot 186 on the array 174 during measurement may be used to establish and/or to enhance object size independence for the whole measurement range by suitable relative positioning of the optical sensor 176 and the illumination source 136.

In Fig. 20, a schematic view of a further embodiment of a detector 1110 for determining a position of at least one object 1112 is depicted. In this case, the object 1112 may comprise a beacon device 1114, from which a light beam 1116 propagates towards a first optical sensor 1118 and a second optical sensor 1120. The first optical sensor 1118 comprises a first light-sensitive area 1122, and the second optical sensor 1120 comprises a second light-sensitive area 1124. Details of the second optical sensor 1120 and the second light-sensitive area 124 will be explained in further detail below, with reference to Fig. 22A, 22B and 23.

It shall be noted therein, that, in the embodiment shown in Fig. 20, the first optical sensor 1118 is positioned in front of the second optical sensor 1120, such that the light beam 1116 reaches the first optical sensor 1118 before the second optical sensor 1120. As discussed above, however, another order is feasible. Thus, as an example, the second optical sensor 1120 may be positioned in front of the first optical sensor 1118. The latter option, which is not depicted herein, is specifically possible in case the second light-sensitive area 1124 is fully or partially transparent, such as by providing a transparent fluorescent waveguiding sheet 1174, as will be outlined in further detail below.

The light beam 1116, as an example, may propagate along an optical axis 1126 of the detector 1110. Other embodiments, however, are feasible.

The detector 1110, further, may comprise at least one transfer device 1128, such as at least one lens or a lens system, specifically for beam shaping. Consequently, the light beam 1116 may be focused, such as in one or more focal points 1130, and a beam width of the light beam 1116 may depend on a longitudinal coordinate z of the object 1112, such as on a distance between the detector 1110 and the beacon device 1114 and/or the object 1112. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Fig. 20, the first optical sensor 1118 is a small optical sensor, whereas the second optical sensor 1120 is a large optical sensor. Thus, the width of the light beam 1116 fully may cover the first light-sensitive area 1122, whereas, on the second light-sensitive area 1124, a light spot is generated which is smaller than the light-sensitive area 1124, such that the light spot is fully located within the second light-sensitive area 1124. Possible embodiments will be explained below with reference to Fig. 23. Thus, as an example, the first light-sensitive area 1122 may have a surface area of 10 mm^2 to 100 mm^2 , whereas the second light-sensitive area

1124 may have a surface area of more than 100 mm², such as 200 mm² or more, e.g. 200 to 600 mm² or 500 mm² or more. Other embodiments, however, are feasible.

5 The first optical sensor 1118, in response to the illumination by the light beam 1116, may generate a first sensor signal s_1 , and the second optical sensor 1120 may generate at least one second sensor signal s_2 . As an example, the first optical sensor 1118 may be a linear optical sensor, i.e. the sensor signal s_1 is dependent on the total power of the light beam 1116 or on the portion of the light beam 1116 illuminating the first light-sensitive area 1122, whereas the sensor signal s_1 is independent from the actual size of the light spot of illumination. In other words, the
10 first optical sensor 1118, preferably, does not exhibit the above-described FiP effect.

The sensor signals s_1 and s_2 may be provided to an evaluation device 1132 of the detector 1110. The evaluation device 1110, as symbolically depicted in Fig. 20, may specifically be embodied to derive a quotient signal Q , as explained above. The quotient signal Q , derived by dividing e.g. the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used
15 for deriving at least one item of information on a longitudinal coordinate z of the object 1112 and/or the beacon device 1114, from which the light beam 1116 propagates towards the detector 1110. Thus, as an example, a unique evaluation curve may exist, in which, for each quotient signal Q , a longitudinal coordinate z is assigned.

20 The detector 1110, in combination with the at least one beacon device 1114, may be referred to as a detector system 1134, as will be explained in further detail below, with reference to Fig. 25.

In Fig. 21, a modification of the embodiment of Fig. 20 is shown, which forms an alternative detector 1110. The alternative embodiment of the detector 1110 widely corresponds to the embodiment shown in Fig. 20. Instead of using an active light source, i.e. a beacon device 1114 with light-emitting properties for generating the light beam 1116, however, the detector 1110 may comprise at least one illumination source 1136. The illumination source 1136, as an example, may comprise a laser, whereas, in Fig. 20, as an example, the beacon device 1114 may comprise a light-emitting diode (LED). The illumination source 1136 may be configured for generating at least one illumination light beam 1138 for illuminating the object 1112. The illumination light beam 1138 may fully or partially be reflected by the object 1112 and may travel back towards the detector 1110, thereby forming the light beam 1116.

35 As shown in Fig. 20, as an example, the illumination light beam 1138 may be parallel to the optical axis 1126 of the detector 1110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too. In order to provide an on-axis illumination, as shown in Fig. 21, as an example, one or more reflective elements 1140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

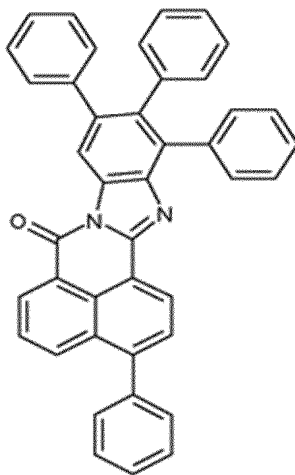
40 Apart from these modifications, the setup of the embodiment in Fig. 21 corresponds to the setup in Fig. 20. Thus, again, an evaluation device 1132 may be used, having, e.g., at least one divider 1142 for forming the quotient signal Q , and, as an example, at least one position evaluation

device 1144, for deriving the at least one longitudinal coordinate z from the quotient signal Q . It shall be noted that the evaluation device 1132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 1142, 1144 may be embodied by appropriate software components.

It shall be further noted that the embodiments shown in Fig. 20 and 21 simply provide embodiments for determining the longitudinal coordinate of the object 1112. As will be outlined in further detail below with reference to Fig. 22A and 22B as well as to Fig. 23, the detector 1110 may also be used for providing additional information on at least one transversal coordinate of the object 1112 and/or of parts thereof.

In Fig. 22A and 22B, a top view (Fig. 22A) and a cross-sectional view of the second optical sensor 1120, which may be used in the setups e.g. of Fig. 20 and/or 21, is shown. The second optical sensor 1120 may comprise a fluorescent waveguiding sheet 1174 which forms the second light-sensitive area 1124 facing towards the object 1112. The fluorescent waveguiding sheet 1174, in this exemplary embodiment, may be designed as a flat waveguiding sheet, in which, as symbolically depicted by the arrow 1176 in Fig. 22B, waveguiding by internal reflection may take place, specifically by internal total reflection, specifically a waveguiding of fluorescence light generated within the fluorescent waveguiding sheet 1174. The fluorescent waveguiding sheet 1174, as an example, may have a lateral extension of at least 25 mm^2 , such as at least 100 mm^2 , more preferably of at least 400 mm^2 . As an example, a $10 \text{ mm} \times 10 \text{ mm}$ square sheet, a $20 \text{ mm} \times 20 \text{ mm}$ square sheet, a $50 \text{ mm} \times 50 \text{ mm}$ square sheet or another dimension may be used. It shall be noted, however, that non-square geometries or even non-rectangular geometries may be used, such as circular or oval geometries or polygonal geometries.

The fluorescent waveguiding sheet 1174, as an example, may comprise a matrix material 1178 and at least one fluorescent material 1180 disposed therein, such as at least one fluorophore, e.g. a fluorescent dye. For exemplary embodiments, reference may be made to the above-mentioned materials, such as one or more of the materials listed in WO 2012/168395 A1. As an example, the following fluorescent material may be used:



This fluorescent material is disclosed as substance 34.2 in WO 2012/168395 A1, including potential synthesis methods. The material may be immersed in polystyrene, such as at a concentration of 0.001 - 0.5 wt.%. The fluorescent material 1180 may be designed to generate fluorescence light in response to an illumination by the light beam 1116.

The fluorescent material 1180 and/or the concentration of the fluorescent material 1180 within the matrix material 1178, specifically may be chosen to show linear properties, at least within a range of measurement, i.e. within a range of intensities, such that the total power of the fluorescence light generated in response to an excitation is a linear function of the intensity of the illumination by the excitation light, i.e. by the light beam 1116. As an example, the materials and/or intensities may be chosen such that saturation effects are avoided.

The second optical sensor 1120 further, in this embodiment, may comprise a plurality of photosensitive elements 1182, 1184, 1186, 1188, in Fig. 22A and 22B referred to as PD1-PD4, located at respective edges 190, 192, 194, 196 of the fluorescent waveguiding sheet 174, e.g. rim portions of the fluorescent waveguiding sheet 1174. In this exemplary embodiment, the fluorescent waveguiding sheet 1174 may have a rectangular shape, such that pairs of edges are opposing each other, such as the pair of edges 1190, 1192 and the pair of edges 1194, 1196. The sides of the rectangular shape of the fluorescent waveguiding sheet 174 may define a Cartesian coordinate system, with an x-dimension defined by an interconnection between edges 1190 and 192, and a y-dimension defined by an interconnection between edges 1196, 1194, as indicated in Fig. 22A. It shall be noted, however, that other coordinate systems are feasible.

The photosensitive elements 1182, 1184, 1186, 1188, as an example, may comprise photodiodes. Specifically, these photosensitive elements 1182, 1184, 1186, 1188 may have, each, a comparable, preferably an identical, electrical capacity as the first optical sensor 1118. It shall be noted, however, that other embodiments are feasible. The photosensitive elements 1182, 1184, 1186, 1188, as an example, may be or may comprise strip-shaped photodiodes covering, preferably, the full length of the respective edges 1190, 1192, 1194, 1196, or, preferably, covering at least 50 % or more preferably at least 70 % of the length of these respective edges 1190, 1192, 1194, 1196. Other embodiments, however, are feasible, such as embodiments in which more than one photosensitive element is located at a respective edge.

The photosensitive elements 1182, 1184, 1186, 1188 each produce at least one sensor signal, in response to the light, specifically the fluorescence light, detected by these photosensitive elements 1182, 1184, 1186, 1188. All of these sensor signals are referred to as second sensor signals, wherein, in the following, PD1 creates sensor signal $s_{2,1}$, PD2 creates sensor signal $s_{2,2}$, PD3 creates sensor signal $s_{2,3}$, and PD4 creates sensor signal $s_{2,4}$, with the first index 2 denoting the fact that these sensor signals are second sensor signals, and with the second index, from 1 to 4, indicating the respective photosensitive element 1182, 1184, 1186, 1188 from which the respective sensor signal originates.

As outlined above in Fig. 20 and 21, the at least one first sensor signal s_1 and the second sensor signals $s_{2,j}$ (with $j = 1, \dots, 4$) are provided to the evaluation device 1132 of the detector 1110, the function of which will be explained in further detail below, specifically with reference to Fig. 24.

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The evaluation device 1132 is configured to determine at least one longitudinal coordinate z of the object 1112, which is not depicted in these figures, and from which the light beam 1116 propagates towards the detector 1110, by evaluating the first and second sensor signals. Additionally as will be outlined in further detail below, at least one transversal coordinate x and/or y may be determined, as will also be outlined in further detail below, with reference to Fig. 3 and 24.

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The second optical sensor 1120, as depicted in Fig. 22B, may further optionally comprise at least one optical filter element 1198. The optical filter element 1198 may be placed in front of an optional reference photosensitive element 1200, which may further, with or without the optical filter element 1198, be present in the detector 1110. As an example, the reference photosensitive element 1200 may comprise a large area photodiode. Other setups, however, are feasible. Thus, it shall be noted, that the reference photosensitive element 1200 may also be left out in this embodiment, since the first optical sensor 1118 may also take over the functionality of the reference photosensitive element 1200. Specifically, in case a transparent fluorescent waveguiding sheet 1174 is used and in case the first optical sensor 1118 is placed behind the second optical sensor 1120, the first optical sensor 1118 may also take over the functionality of the reference photosensitive element 1200. It shall further be noted that one or both of the first optical sensor 1118 and the second optical sensor 1120 may be a uniform optical sensor, having a single light-sensitive area 1122, 1124, each, or that one or both of these optical sensors 1118, 1120 may be pixelated.

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As an example, the at least one optical filter element 1198 may be designed to prevent fluorescence light from entering the reference photosensitive element 1200 or, at least, may attenuate fluorescence light by at least 70 %, or, preferably, by at least 80 %.

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In Fig. 23, an illumination of the second light-sensitive area 1124 by the light beam 1116 is shown. Therein, two different situations are depicted, representing different distances between the object 1112 and from which the light beam 1116 propagates towards the detector 1110, and the detector 1110 itself, resulting in two different spot sizes of light spots generated by the light beam in the fluorescent waveguiding sheet 1174. Firstly, a small light spot 1202 and, secondly, a large light spot 1204. In both cases, the overall power of the light beam remains the same over light spots 1202, 1204. Further, a shadow 1206 is depicted, which is generated by the first optical sensor 1118 being placed in front of the second optical sensor 1120. In the following, it is assumed that the first optical sensor 1118 is still fully illuminated by the light beam 1116.

The illumination by the light beam 1116 induces fluorescence which, as depicted in Fig. 22B above, is fully or partially transported by waveguiding towards the photosensitive elements

1182, 1184, 1186, 1188. As indicated above, corresponding second sensor signals are generated by these photosensitive elements, and are provided to the evaluation device 1132, in conjunction with the first sensor signal and, optionally, further in conjunction with at least one reference sensor signal generated by the at least one reference photosensitive element 1200.

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The evaluation device 1132, as symbolically depicted in Fig. 24, is designed to evaluate the sensor signals which, therein, are represented as outlined above. The sensor signals may be evaluated by the evaluation device in various ways, in order to determine a location information and/or a geometrical information of the object 1112, such as at least one longitudinal coordinate z of the object 1112 and, optionally, one or more transversal coordinates of the object 1112.

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Firstly, the evaluation device 1132 may comprise at least one summing device 1208 configured to form a sum signal S of the sensor signals PD1 to PD4, such as according to formula (1) above, for the second sensor signals $s_{2,i}$, with $i = 1, \dots, 4$ (the first index, for the sake of simplicity, is left out in the above-mentioned formula (1)). This sum signal S may replace the second sensor signal s_2 in general and/or, for a part of the further evaluation, may be used as "the" second sensor signal of the second optical sensor 1120. This sum signal S may represent the total power of the fluorescence light generated by the light beam 1116. Even though, some losses may occur, since, generally, not all of the fluorescence light will actually reach the photosensitive elements 1182, 1184, 1186, 1188. Thus, as an example, losses in waveguiding may occur, or some of the fluorescence light may actually be emitted from the edges 1190, 1192, 1194, 1196, in a direction which is not covered by the photosensitive elements 1182, 1184, 1186, 1188. Still, the sum signal S provides a fairly good measure for the total power of the fluorescence generated within the fluorescent waveguiding sheet 1174.

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The evaluation device 1132 may comprise at least one divider 1142 which, as symbolically depicted in Fig. 24, may be part of a position evaluation device 1144 and which may be configured for forming at least one quotient signal out of the first and second sensor signals s_1 , s_2 , with s_2 , as an example, being the sum signal S of the respective second sensor signals, as outlined above. Thus, as an example, the divider 1142 may be configured for one or more of dividing the first and second sensor signals, dividing multiples of the first and second sensor signals or dividing linear combinations of the first and second sensor signals. The position evaluation device 1144 further may be configured for determining the at least one longitudinal coordinate z by evaluating the quotient signal Q , such as by using at least one predetermined or determinable relationship between the quotient signal Q and the longitudinal coordinate. As an example, calibration curves may be used. The divider 1142 and/or the position evaluation device 1144 may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the quotient signal, at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

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As outlined above, additional information may be derived from the second sensor signals $s_{2,1}$, $s_{2,2}$, $s_{2,3}$ and $s_{2,4}$, besides the at least one longitudinal coordinate z of the object. Thus, additionally, at least one transversal coordinate x , y may be derived. This is mainly due to the fact that the distances between a center of the light spots 1202, 1204 and the photosensitive elements 1182, 1184, 1186, 1188 are non-equal. Thus, the center of the light spot 1202, 1204 has a distance from the photosensitive element 1182 of l_1 , a distance from the photosensitive element 1184 of l_2 , from the photosensitive element 1186 of l_3 and from the photosensitive element 1188 of l_4 . Due to the differences in these distances between the location of the generation of the fluorescence light and the photosensitive elements detecting said fluorescence light, the sensor signals will differ. This is due to various effects. Firstly, again, internal losses will occur during waveguiding, since each internal total reflection implies a certain loss, such that the fluorescence light will be attenuated on its way, depending on the length of the path. The longer the distance of travel, the higher the attenuation and the higher the losses. Further, absorption effects will occur. Thirdly, a spreading of the light will have to be considered. The longer the distance between the light spot 1202, 1204 to the respective photosensitive element 1182, 1184, 1186, 1188, the higher the probability that a photon will be directed into a direction other than the photosensitive element. Consequently, by comparing the sensor signals of the photosensitive elements 1182, 1184, 1186, 1188, at least one item of information on a transversal coordinate of the light spot 1202, 1204 and, thus, of the object 1112 may be generated.

The comparison of the sensor signals may take place in various ways. Thus, generally, the evaluation device 1132 may be designed to compare the sensor signals in order to derive the at least one transversal coordinate of the object 1112 and/or of the light spot 1202, 1204. As an example, the evaluation device 1132 may comprise at least one subtracting device 1210 and/or any other device which provides a function which is dependent on at least one transversal coordinate, such as on the coordinates x , y , of the object 1112. For exemplary embodiments, the subtracting device 1210 and/or any other device may provide a function which is dependent on at least one transversal coordinate, such as on the coordinates x , y . For exemplary embodiments, the subtracting device 1210 may be designed to generate at least one difference signal, such as a signal according to formula (4) and/or (5) above, for one or each of dimensions x , y in Fig. 23. As an example, a simple difference between PD1 and PD2, such as $PD1 - PD2 / (PD1 + PD2)$ may be used, as a measure for the x -coordinate, and a difference between PD3 and PD4, such as $(PD3 - PD4) / (PD3 + PD4)$, may be used as measure for the y -coordinate. A transformation of the transversal coordinates of the light spot 1202, 1204 in the plane of the second light-sensitive area 1124, as an example, into transversal coordinates of the object from which the light beam 1116 propagates to the detector 1110, may simply be made by using the well-known lens equation. For further details, as an example, reference may be made to WO 2014/097181 A1.

It shall be noted, however, that other transformations or other algorithms for processing the sensor signals by evaluating device 1140 are feasible. Thus, besides subtractions or the linear combinations with positive or negative coefficients, non-linear transformations are generally feasible. As an example, for transforming the sensor signals into z -coordinates and/or x , y -

coordinates, one or more known or determinable relationships may be used, which, as an example, may be derived empirically, such as by calibrating experiments with the object placed at various distances from the detector 1110 and/or by calibrating experiments with the object placed at various transversal positions or three-dimensional positions, and by recording the respective sensor signals.

Fig. 25 shows, in a highly schematic illustration, an exemplary embodiment of a detector 1110, e.g. according to the embodiments shown in Fig. 20 or 21. The detector 1110 specifically may be embodied as a camera 1146 and/or may be part of a camera 1146. The camera 1146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 25 further shows an embodiment of a detector system 1134, which, besides the at least one detector 1110, comprises one or more beacon devices 1114, which, in this example, may be attached and/or integrated into an object 1112, the position of which shall be detected by using the detector 1110. Fig. 25 further shows an exemplary embodiment of a human-machine interface 1148, which comprises the at least one detector system 1134 and, further, an entertainment device 1150, which comprises the human-machine interface 1148. The figure further shows an embodiment of a tracking system 1152 for tracking a position of the object 1112, which comprises the detector system 1134. The components of the devices and systems shall be explained in further detail below.

Fig. 25 further shows an exemplary embodiment of a scanning system 1154 for scanning a scenery comprising the object 1112, such as for scanning the object 1112 and/or for determining at least one position of the at least one object 1112. The scanning system 1154 comprises the at least one detector 1110, and, further, optionally, the at least one illumination source 1136 as well as, optionally, at least one further illumination source 1136. The illumination source 1136, generally, is configured to emit at least one illumination light beam 1138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 1114 and/or on a surface of the object 1112. The scanning system 1154 may be designed to generate a profile of the scenery including the object 1112 and/or a profile of the object 1112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 1154, specifically the detector 1110, by using the at least one detector 1110. In Fig. 13, as an example, one or more reflective elements 1140 may be used, for example partially transparent, such as one or more prisms and

As outlined above, an exemplary embodiment of the detector 1110 which may be used in the setup of Fig. 25 is shown in Fig. 20 and 21. Thus, the detector 1110, besides the optical sensors 1118, 1120, comprises at least one evaluation device 1132, having e.g. the at least one divider 1142 and/or the at least one position evaluation device 1144, as symbolically depicted in Fig. 25. The components of the evaluation device 1132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 1110. Besides the possibility of fully or partially combining two or more components, one or

more of the optical sensors 1118, 1120 and one or more of the components of the evaluation device 1132 may be interconnected by one or more connectors 1156 and/or by one or more interfaces, as symbolically depicted in Fig. 25. Further, the one or more connectors 1156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 1156, the evaluation device 1132 may fully or partially be integrated into one or both of the optical sensors 1118, 1120 and/or into a housing 1158 of the detector 1110. Additionally or alternatively, the evaluation device 1132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 1112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 1160, the position of which may be manipulated by a user 1162. As an example, the object 1112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 1112 are possible. Further, the user 1162 himself or herself may be considered as the object 1112, the position of which shall be detected.

As outlined above, the detector 1110 comprises at least the optical sensors 1118, 1120. The optical sensors 1118, 1120 may be located inside the housing 1158 of the detector 1110. Further, the at least one transfer device 1128 may be comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 1164 inside the housing 1158, which, preferably, is located concentrically with regard to the optical axis 1126 of the detector 1110, preferably defines a direction of view 1166 of the detector 1110. A coordinate system 1168 may be defined, in which a direction parallel or anti-parallel to the optical axis 1126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 1126 may be defined as transversal directions. In the coordinate system 1128, symbolically depicted in Fig. 25, a longitudinal direction is denoted by z , and transversal directions are denoted by x and y , respectively. Other types of coordinate systems 1168 are feasible, such as non-Cartesian coordinate systems.

The detector 1110 may comprise the optical sensors 1118, 1120 as well as, optionally, further optical sensors. The optical sensors 1118, 1120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 1118 covers a portion of the second optical sensor 1120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 1112 and/or of parts thereof.

One or more light beams 1116 are propagating from the object 1112 and/or from one or more of the beacon devices 1114, towards the detector 1110. The detector 1110 is configured for determining a position of the at least one object 1112. For this purpose, as explained above in the context of Fig. 20 to 23, the evaluation device 1132 is configured to evaluate sensor signals

provided by the optical sensors 1118, 1120. The detector 1110 is adapted to determine a position of the object 1112, and the optical sensors 1118, 1120 are adapted to detect the light beam 1116 propagating from the object 1112 towards the detector 1110, specifically from one or more of the beacon devices 1114. In case no illumination source 1136 is used, the beacon devices 1114 and/or at least one of these beacon devices 1114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 1136 is used, the beacon devices 1114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 1112 may be used, such as integrated reflected beacon devices 1114 having at least one reflective surface. The light beam 1116, directly and/or after being modified by the transfer device 1128, such as being focused by one or more lenses, illuminates the light-sensitive areas 1122, 1124 of the optical sensors 1118, 1120. For details of the evaluation, reference may be made to Fig. 20 to 23 above.

As outlined above, the determination of the position of the object 1112 and/or a part thereof by using the detector 1110 may be used for providing a human-machine interface 1148, in order to provide at least one item of information to a machine 1170. In the embodiments schematically depicted in Fig. 25, the machine 1170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 1132 may even be fully or partially integrated into the machine 1170, such as into the computer.

As outlined above, Fig. 25 also depicts an example of a tracking system 1152, configured for tracking the position of the at least one object 1112 and/or of parts thereof. The tracking system 1152 comprises the detector 1110 and at least one track controller 1172. The track controller 1172 may be adapted to track a series of positions of the object 1112 at specific points in time. The track controller 1172 may be an independent device and/or may be fully or partially integrated into the machine 1170, specifically the computer, as indicated in Fig. 25 and/or into the evaluation device 1132.

Similarly, as outlined above, the human-machine interface 1148 may form part of an entertainment device 1150. The machine 1170, specifically the computer, may also form part of the entertainment device 1150. Thus, by means of the user 1162 functioning as the object 1112 and/or by means of the user 1162 handling a control device 1160 functioning as the object 1112, the user 1162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Fig. 26A and 26B, an alternative embodiment of the second optical sensor 1120 is shown, in a top view (Fig. 26A) and in a cross-sectional view (Fig. 26B). For most of the details of the second optical sensor 1120, reference may be made to Fig. 22A and 22B above. The embodiment, however, shows various variations from the embodiment of Fig. 22A and 22B, which may be realized in an isolated fashion or in combination.

Thus, firstly, the embodiment shows variations of the placement of the photosensitive elements. Besides the photosensitive elements 1182, 1184, 1186, 1188 located at opposing edges 1190, 1192, 1194, 1196, which, in this embodiment, are straight edges, additional photosensitive elements 1212 are located at corners 1214 of the fluorescent waveguiding sheet 1174. The edges 1190, 1192, 1194, 1196 in combination may form a rim of the fluorescent waveguiding sheet 1174, such as a rectangular rim. The rim itself may be roughened or even blackened in order to avoid back reflections from the rim. The corners 1214 also are part of the edges of the fluorescent waveguiding sheet 1174. The photosensitive elements 1212 located at the corners 1214 may provide additional second sensor signals which may be evaluated in a similar fashion as shown e.g. in Fig. 24. They may provide an increased accuracy of the determination of the z-coordinate and/or of the x, y-coordinate. Thus, as an example, these additional sensor signals may be included in the sum signal, such as formed by using formula (1) above. Additionally or alternatively, these additional sensor signals may be implemented into the formation of difference signals, such as according to formulae (2) and/or (3) above. As an example, difference signals between two photosensitive elements 1212 located at opposing corners 1214 may be formed and/or difference signals between one photosensitive element 1212 located at a corner 1214 and one photosensitive element located at a straight edge, e.g. a straight rim portion, may be formed. The difference signal D, in each case, may denote a location of the light spot on an axis interconnecting the two photosensitive elements.

Further, the embodiment of Fig. 26A and 26B shows a variation of the placement of the photosensitive elements 1182, 1184, 1186, 1188, 1212 with respect to the fluorescent waveguiding sheet 1174. Thus, in the embodiment of Fig. 22A and 22B, the photosensitive elements 1182, 1184, 1186, 1188 may be located within the plane of the fluorescent waveguiding sheet 1174. Additionally or alternatively, as shown in the embodiment of Fig. 26A and 26B, some or even all of the photosensitive elements 1182, 1184, 1186, 1188, 1212 may be located outside the plane of the fluorescent waveguiding sheet 1174. Specifically, as shown in the cross-sectional view of figure 26B, as an example, the photosensitive elements 1182, 1184, 1186, 1188, 1212 may be optically coupled to the fluorescent waveguiding sheet 1174 by optical coupling elements 1216. As an example, the photosensitive elements 1182, 1184, 1186, 1188, 1212 simply may be glued to the fluorescent waveguiding sheet 1174 by using one or more transparent adhesives, such as an epoxy adhesive.

Further, the embodiment of Fig. 26A and 26B shows a variation of the size and shape of the photosensitive elements 1182, 1184, 1186, 1188, 1212. Thus, the photosensitive elements 1182, 1184, 1186, 1188, 1212 do not necessarily have to be strip-shaped photosensitive elements. As an example, very small photodiodes may be used, such as rectangular photodiodes or even point-like or spot-like photodiodes. As outlined above, a small size of the photodiodes generally may lead to a lower electrical capacitance and, thus, may lead to a faster response of the second optical sensor 1120.

Further, the embodiment of Fig. 26A and 26B shows that no reference photosensitive element 1200 is necessary. Thus, as discussed above, the sum signal itself may replace the function of

the reference photosensitive element 1200. Thus, the second optical sensor 1120 as shown in the embodiment of Fig. 26A and 26B provides a fully functional and, optionally, transparent PSD. No further PSDs are required.

5 Figures 27A and 27 B show a schematic view of a further exemplary embodiment of a detector 110 according to the present invention. In Figure 27 A, the detector 110 comprises at least two optical sensors 113, for example a first optical sensor 118 and a second optical sensor 120, each having at least one light-sensitive area 121. The optical detector 110, further, comprises at least one transfer device 128, such as at least one lens or a lens system, specifically for beam
10 shaping. The transfer device 128 has an optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. The detector 110 may comprise at least one illumination source 136. The illumination source 136, as an example, may comprise a laser source. The illumination source 136 may be arranged such that the illumination light beam 138 is one or both of non-parallel to the optical axis 126, but off-axis, or shifted
15 from the optical axis 126. The illumination source 136 may be configured for generating at least one illumination light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The light beam 116 propagates from the object 112 towards the first optical sensor 118 and the second optical sensor 120. The first optical sensor 118 may com-
20 prise a first light-sensitive area 122, and the second optical sensor 120 may comprise a second light-sensitive area 124. In this embodiment the optical sensors 118, 120 may be arranged such that the light-sensitive areas 122, 124 have identical surface areas. For example, the optical sensors 118, 120 may be identical. The detector 110 may further comprise the reflective element 140, such as at least one beam splitter, which is adapted to lead the light beam 116 from
25 the transfer device 128 to both of the optical sensors 118, 120. The first optical sensor 118 may have a distance db_1 from the beam splitter and the second optical sensor 120 may have a distance db_2 from the beam splitter, wherein $db_1 \neq db_2$. Again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q, and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordi-
30 nate z from the quotient signal Q. It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

In Figure 27 B, the detector 110 comprises at least two optical sensors 113, for example a first
35 optical sensor 118 and a second optical sensor 120, each having at least one light-sensitive area 121. The optical detector 110, further, may comprise at least one transfer device 128, such as at least one lens or a lens system. The transfer device 128 has an optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. The detector 110 may comprise at least one illumination source 136. The illumination source
40 136, as an example, may comprise a laser source, for example with a 1550nm laser source. The illumination source 136 may be arranged such that the illumination light beam 138 is one or both of non-parallel to the optical axis 126, but off-axis, or shifted from the optical axis 126. The illumination source 136 may be configured for generating at least one illumination light beam

138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The light beam 116 propagates from the object 112 towards the first optical sensor 118 and the second optical sensor 120. The first optical sensor 118 may comprise the first light-sensitive area 122, and the second optical sensor 120 may comprise the second light-sensitive area 124. As can be seen in Fig. 27B, the first optical sensor 118 is a small optical sensor, whereas the second optical sensor 120 is a large optical sensor. The optical sensors 118, 120 may be Ge-sensors. The first optical sensor 118 may have a first distance from the transfer device 128 and the second optical sensor 120 may have a second distance from the transfer device 128. In Figure 27B, the first optical sensor 118 may be close to the transfer device 128, whereas the second optical sensor 120 may be arranged further away in direction to the focus. The first optical sensor 118 may be arranged such that, independent from a distance from the object, a sensor signal of the first optical sensor 118 may be proportional to the total power of the light beam passing the transfer device 128. Again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q , and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the quotient signal Q . It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

In Figure 28 experimental results of a distance determination with the detector 110 is shown. In this experimental setup, the transfer device 128 was a plano-convex lens having a focal length of 150 mm, a diameter of 75 mm and coated with an anti-reflective coating for a range of 1050-1700 nm, available as Thorlabs LA1002-C. The object 112, in this case a piece of carpet, was illuminated by a laser diode with 30mW CW-power output at a wavelength of 1550 nm, available as Schäfter+Kirchhoff 55cm-1550-30-Q04-T12-C-6. The illumination source 136 was placed laterally next to the transfer device and was operated at 367Hz with a 50:50 rectangle modulation. A second optical sensor 120, in this experimental setup, a Ge photodiode with dimensions of 10mm x 10mm, available as Thorlabs FDG1010, was arranged directly on the transfer device, and a first optical sensor 118 having a diameter of 5mm, available as Thorlabs FDG05 was placed with a distance of 0,85m from the transfer device 128. Figure 28 shows a dependency of distance d in m, corresponding to the longitudinal coordinate z of the object, of the determined quotient signal Q .

In Figure 29, a further exemplary embodiment of the detector 110 is depicted. For details of the optical sensor 113 reference is made to Figure 6 above. As in Figures 27A and B, the illumination source 136 may be positioned off-axis. The illumination source 136 may be adapted to generate and/or to project a cloud of points, for example the illumination source 136 may comprise one optical element 194, in particular one or more optical elements selected from the group consisting of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The sensor element 115 may comprise a matrix 117 of optical sensors 113, each optical sensor 113 having at least one light-

sensitive area 121 facing the object 112. The sensor element 115 may comprise at least one CMOS sensor. In Figure 30, schematically the cloud of points impinging on the sensor element 115 is depicted. Additionally, disturbances may be present on the matrix 117 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device 132 may be adapted to determine at least one region of interest 196, for example one or more pixels illuminated by the light beam 116 which are used for determination of the longitudinal coordinate of the object 112. In Figure 30, regions of interest 196 are shown exemplary as circular areas with dashed lines. For example, the evaluation device 132 may be adapted to perform a filtering method, for example, a blob-analysis and/or object recognition method.

Figure 31 A to O show further exemplary configurations of optical sensors according to the present invention, in particular top view in direction of propagation of the light beam 116. In Figure 31A, a top view of two rectangular optical sensors 113 is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120. The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset, in particular in a transversal direction y , from the optical axis 126. In Figures 31B and 31C, top view of a large rectangular optical sensor 120 is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120 having a triangle shaped (Figure 31B) or star-shaped (Figure 31C) light-sensitive area 121. In Figures 31 M to O, a top view of two rectangular optical sensors 113 is shown, wherein the first optical sensor 118 and the second optical sensor 120 are rectangular sensors with the same size. In Figures 31 M to O a mask 119 is arranged in front of the first and second optical sensors 118, 120. The mask 119 may be arranged with a different offset from the optical axis 126. The mask 119 may have an arbitrary size and shape, for example, the mask may be rectangular shaped (Figure 31M), triangle shaped (Figure 31N) or star-shaped (Figure 31O). However, other sizes and shapes are feasible. Mask 119 may be adapted to prevent light impinging on the light sensitive areas of the first and second optical sensors 118, 120. If used in a situation comparable to the situation illustrated in Figure 19A, the mask may result in a further z -dependent decrease of a decreasing sensor signal, resulting in an increased z -dependency of the resulting quotient signal Q .

The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset from the optical axis 126. Figure 31K shows two circular shaped optical sensors 113, wherein the first optical sensor 118 is a small optical sensor in front of the larger second optical sensor 120. In Figures 31D, the light sensitive area of the first optical sensor 118 is square-shaped, and the light sensitive area of the second optical sensor 120 is rectangular, such that the surface areas in x and y differ. In addition, a center of the first optical sensor 118 and a center of second optical sensor 120 may have different x coordinates such that the optical sensors 118, 120 may have different spatial offset in one or more of x and y direction from the optical axis. In Figure 31H, both the first optical sensor 118 and the second optical sensor 120 may be rectangular. The first optical sensor 118 and the second optical sensor 120 may be arranged such that the center of the first optical sensor 118 and the center of second optical sensor 120 may have different x coordinates and that the surface areas in x and y differ. The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset from the

optical axis 126. In Figure 31L, the first optical sensor 118 may have a deviating shape from the shape of the second optical sensor 120 such as a circular or semicircular shape. Figures 31 E, F, G, I, J show sensor element 115 having the matrix of pixels 117. In Figures 31 E, F, G the sensor element 115 has a rectangular shape, whereas in Figures 31 I and J the sensor element 115 has a circular shape. Rows and columns may be arranged equidistant or non-equidistant. In case of equidistant rows and/or columns the sensor element 115 may be arranged with a spatial offset to the optical axis 126.

Figure 32 shows experimental results of a determination of a longitudinal coordinate z for different object sizes. The experimental setup was comparable to the setup shown in Figure 19A. In the measurement setup the object 112, a paper target, was illuminated by laser 136 with a wavelength of 905 nm, 1,6 mW and modulated with 23 Hz. Light reflected from the object 112 was led to a quadrant diode 178, available as OSI Optoelectronics, OSI Spot-4D. Between the object 112 and quadrant diode 178 a lens 128 having an aspherical effective focal length of 20,0 mm, a diameter of 25,0 mm was placed, available as Thorlabs AL2520M-B. A distance from quadrant diode 178 to lens 128 was 19,7 mm and the quadrant diode 178 had an offset from the optical axis in $y = 0,5$ mm. Further, different from the situation in Figure 19A and not shown in Fig. 19A, in the situation of Figure 32, an iris diaphragm or a further lens was placed in front of the laser 136 between the laser 136 and the object 112, to modify the illumination light beam 138. The iris diaphragm was used to modify the width of the illumination light beam 138. The further lens was used to obtain a diverging illumination light beam 138 with a beam width decreasing with the distance from the laser 136. Figure 32 shows the quotient Q of two adjacent quadrant currents as a function of the distance, i.e. longitudinal coordinate of the object 112, z in mm. In a first experiment, a diameter of a illumination light beam 138 was varied by an iris diaphragm from 1 mm, solid line, to 3,5 mm, loosely dashed line, and to 5 mm, dash-dot line. In a second experiment, the diameter of the illumination light beam 138 was varied by the further lens such that the beam width of the illumination light beam 138 diverges with increasing distance from the further lens. To characterize the diverging illumination light beam 138, the beam width at 1m, 2m, and 3m from the lens 128 is given. The dashed line shows the quotient Q , wherein the beam width was 10 mm at 1 m distance, 16 mm at 2 m distance and 22 mm at 3 m distance from the lens 128. The dotted line shows the quotient Q , wherein the beam width was 15 mm at 1 m distance, 32 mm at 2 m distance and 49 mm at 3 m distance from the lens 128. Below $z = 2300$ mm all curves show the same dependency of Q from z and deviations below ± 5 % and thus independence from the beam width. In the situation of Figure 32 the beam width at the object 112 corresponds to the object size that is measured. The independence of the quotient Q from the beam width and thus from the object size clearly demonstrates the property of object size independence. In an application, the influence of the further lens leading to a diverging illumination light beam may be caused by a liquid drop, or rain, or dirt or the like such as on the laser module. Thus, object size independence is an important property for robust measurements.

Figures 33A and B show an exemplary beam profile and determination of first area 198 and second area 200 of the beam profile. In Figure 33A normalized intensity I_{norm} as a function of the

transversal coordinate x in mm is depicted. The object size was 20mm and the distance object to sensor was 1200mm. The first area 198 of the beam profile may comprise essentially edge information of the beam profile and the second area 200 of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, a maximum value of the beam profile and/or a center point of a plateau of the beam profile. In Figure 33A the center of the plateau may be at 500mm. The beam profile may further comprise falling edges extending from the plateau. The second area 200 may comprise inner regions of the cross section and the first area 198 may comprise outer regions of the cross section. At least one area of the beam profile may be determined and/or selected as first area 198 of the beam profile if it comprises at least parts of the falling edges of the cross section. In Figure 33A, the first area 198 at both sides from the center is depicted in dark grey. At least one area of the beam profile may be determined and/or selected as second area 200 of the beam profile if it is close or around the center and comprises essentially center information. In Figure 33A, the second area 200 is depicted in light grey. Figure 33B shows the corresponding light spot of the intensity distribution as shown in Figure 33A and the corresponding first area 198, and second area 200.

Figure 34 shows a further exemplary embodiment of the detector 110. The optical sensors 113 may comprise the first optical sensor 118 having the first light sensitive area 122 and the second optical sensor 120 having the second light sensitive area 124. The first light sensitive area 122 and the second light sensitive area 124 are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$

is satisfied. "a" is a ratio of photons hitting both an inner region 202 of a plane 204 perpendicular to the optical axis 126 intersecting the optical axis 126 at a distance equal to half of a focal length f of the transfer device 128 and the first light sensitive area 122. "b" is a ratio of photons hitting both the inner region 202 of the plane 204 and the second light sensitive area 124. "c" is a ratio of photons hitting both an outer region 206 of the plane 204 and the first light-sensitive area 122. "d" is a ratio of the photons hitting both the outer region 206 of the plane 204 and the second light sensitive area 124. The inner region 202 may have an area with a geometrical center point on the optical axis 126 and an extension such that half of the photons hit the plane 204 within the inner region 202 and the other half hit the plane outside the inner region 202. The inner region 202 may be designed as a circle with a center point on the optical axis 126 and a radius r which is chosen such that half of the photons hit the plane 204 within the circle and the other half hit the plane outside the circle.

In Fig. 25, a schematic view of an exemplary embodiment of a detector 2110 for determining a position of at least one object 2112 is depicted. In Figure 35, the object 2112 is depicted for two different object distances. The detector 2110 comprises at least two optical sensors 2113, for example a first optical sensor 2118 and a second optical sensor 2120, each having at least one light-sensitive area 2121. The object 2112 may comprise at least one beacon device 2114, from which a light beam 2116, also denoted as incident light beam, propagates towards the detector 2110. Additionally or alternatively, the detector may comprise at least one illumination source 2115 for illuminating the object 2112. As an example, the illumination source 2115 may be con-

figured for generating an illuminating light beam for illuminating the object 2112. Specifically, the illumination source 2115 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 2115 may
5 comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 2115 may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the
10 range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. Further, the illumination source 2115 may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources 2115 is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below,
15 later on may be used for distinguishing the light beams.

The first optical sensor 2118 may comprise a first light-sensitive area 2122, and the second optical sensor 2120 may comprise a second light-sensitive area 2124. The light beam 2116, as an example, may propagate along an optical axis 2126 of the detector 2110. Other embodiments,
20 however, are feasible. The first light-sensitive area 2122 and the second light-sensitive area may be oriented towards the object 2112. The optical detector 2110, further, may comprise at least one transfer device 2128, such as at least one lens or a lens system, specifically for beam shaping. The transfer device 2128 may have at least one focal length in response to the incident light beam 2116 propagating from the object 2112 to the detector 2110. The transfer device 2128 may have an optical axis 2129, wherein the transfer device 2128 and the optical detector preferably may have a common optical axis. The transfer device 2128 may constitute a coordinate system. A direction parallel or anti-parallel to the optical axis 2126, 2129 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 2126, 2129 may be defined as transversal directions, wherein a longitudinal coordinate l is a coordinate along the optical axis 2126, 2129 and wherein d is a spatial offset from the optical axis 2126, 2129. Consequently, the light beam 2116 is focused, such as in one or more focal points, and a beam width of the light beam 2116 may depend on a longitudinal coordinate z of the object 2112, such as on a distance between the detector 2110 and the beacon device 2114 and/or the object 2112. The optical sensors 2118, 2120 may be positioned off focus. For details of this
30 beam width dependency on the longitudinal coordinate, reference may be made to one or more of the WO 2012/110924 A1 and/or WO 2014/097181 A1.
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The detector comprises at least one angle dependent optical element 2130 adapted to generate at least one light beam 2131 having at least one beam profile depending on an angle of incidence of an incident light beam propagating from the object 2112 towards the detector 2110 and illuminating the angle dependent optical element 2130. The angle dependent optical element 2130 may have angle dependent transmission properties such that an electromagnetic wave impinging on a first side 2132, for example a surface and/or an entrance, of the angle de-
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pendent optical element 2130 may be partly, depending on the properties of the angle dependent optical element, absorbed and/or reflected and/or transmitted. A degree of transmission may be defined as quotient of transmitted power of the electromagnetic wave, i.e. the power behind the angle dependent optical element 2130, and the incident power of the electromagnetic wave, i.e. the power before impinging on the angle dependent optical element 2130. The angle dependent optical element 2130 may be designed such that the degree of transmission depends on an angle of incidence at which the incident light beam propagating from the object towards the detector 2116 impinges on the angle dependent optical element 2130. The angle of incident may be measured with respect to an optical axis of the angle dependent optical element 2130.

The angle dependent optical element 2130 may be arranged in the direction of propagation behind the transfer device 2128. The transfer device may, for example, comprise at least one collimating lens. The angle dependent optical element 2130 may be designed to weaken rays impinging with larger angles compared to rays impinging with a smaller angle. For example, the degree of transmission may be highest for light rays parallel to the optical axis, i.e. at 0° , and may decrease for higher angles. In particular, at at least one cut-off angle the degree of transmission may steeply fall to zero. Thus, light rays having a large angle of incidence may be cut-off.

The angle dependent optical element 2130 may comprise at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element; at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one prism; at least one lens; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter; at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.

The first optical sensor 2118, in response to the illumination by the light beam 2131, may generate a first sensor signal s_1 , whereas the second optical sensor 2120 may generate a second sensor signal s_2 . Preferably, the optical sensors 2118, 2120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 131 or of the portion of the light beam 2131 illuminating their respective light-sensitive areas 2122, 2124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination.

The sensor signals s_1 and s_2 are provided to an evaluation device 2133 of the detector 2110. The evaluation device 2133 is embodied to derive a quotient signal Q , as explained above.

From the quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 2112 and/or the beacon device 2114, from which the light beam 2116 propagates towards the detector 2110. The evaluation device 2133 may have at least one di-

vider 2134 for forming the combined signal Q, and, as an example, at least one position evaluation device 2136, for deriving the at least one longitudinal coordinate z from the combined signal Q. It shall be noted that the evaluation device 2133 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 2134, 2136 may be embodied by appropriate software components.

In Fig. 36, a modification of the embodiment of Fig. 35 is shown, which forms an alternative detector 2110. The alternative embodiment of the detector 2110 widely corresponds to the embodiment shown in Fig. 35. In Figure 36, the angle dependent optical element 2130 may comprise at least one optical fiber 2138. The optical fiber 2138 may be adapted to transmit at least parts of incident light beam 2116 which are not absorbed and/or reflected, between two ends of the optical fiber. The optical fiber 2138 may have a length and may be adapted to permit transmission over a distance. The optical fiber 2138 may comprise at least one fiber core which is surrounded by at least one fiber cladding having a lower index of refraction as the fiber core. Below the angle of acceptance, the optical fiber 2138 may be adapted to guide the incoming light beam by total internal reflection.

The optical fiber 2138 may be designed such that the degree of transmission may be highest for incoming light rays parallel, i.e. at an angle of 0° , to the optical fiber, neglecting reflection effects. The optical fiber 2130 may be designed such that for higher angles, for example angles from 1° to 10° , the degree of transmission may decrease smoothly to around 80% of the degree of transmission for parallel light rays and may remain at this level constantly up to an acceptance angle of the optical fiber 2138. The optical fiber 2138 may be designed such that above the acceptance angle total reflection within the optical fiber 2138 is not possible such that the light rays are reflected out of the optical fiber 2138. The optical fiber 2138 may be designed that at the acceptance angle, the degree of transmission may steeply fall to zero. Light rays having a large angle of incidence may be cut-off.

As shown in Figure 36, the illumination source 2115 may be adapted to illuminate the object 2112 through the angle dependent optical element 2130. The optical fiber 2138 may comprise at least one illumination fiber 2140 adapted to transmit the light beam 2142 generated by the illumination source 2115 such that it illuminates the object 2112. The illumination source 2115 may be adapted to couple the at least one light beam 2142 generated by the illumination source 2115 into the illumination fiber 2140.

The optical fiber 2138 may comprise at least two or more fibers. The optical fiber 2138 may be at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber. In the embodiment of Figure 36, and as shown in the cut through in Figure 37, the optical fiber 2138 may comprise four fibers. In particular the optical fiber may comprise the illumination fiber 2138 and at least two fibers each for generating at least one light beam 2131, denoted as first fiber 2144 and second fiber 2146. As shown in Figure 37, the first fiber 2144 and the second fiber 2146 may be arranged close to each other at an entrance end 2148 of the optical fiber 2138 and may split into legs separated by a distance at an exit end 2150 of the optical fiber 2138.

The first fiber 2144 and second fiber 2146 may be designed as fibers having identical properties or may be fibers of different type. The first fiber 2144 may be adapted to generate at least one first light beam 2152 and the second fiber 2146 may be adapted to generate at least one second light beam 2154. The optical fiber 138 may be arranged such that the incident light beam 2116 may impinge at a first angle of incidence into the first fiber 2144 and at a second angle of incidence, different from the first angle, into the second fiber 2146, such that the degree of transmission is different for the first light beam 2152 and the second light beam 2154. One of the optical sensors 2113 may be arranged at an exit end of the first fiber 2144 and the other optical sensor 2113 may be arranged at an exit end of the second fiber 2146. The optical fiber may comprise more than three fibers, for example four fibers as depicted in Figure 37.

It shall further be noted that the embodiments shown in Fig. 35 and 36 simply provide embodiments for determining the longitudinal coordinate z of the object 2112. It is also feasible, however, to modify the setups of Fig. 35 and 36 to provide additional information on a transversal coordinate of the object 2112 and/or of parts thereof. As an example, e.g. in between the transfer device 2128 and the optical sensors 2118, 2120, one or more parts of the light beam 2116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 2112 and/or of parts thereof. The transversal coordinate may be used to verify and/or enhance the quality of the distance information. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

Figures 38 visualizes angle dependent transmission of an angle dependent optical element 2130. The angle dependent optical element 2130 may be designed such that the degree of transmission depends on an angle of incidence at which the incident light beam propagating from the object towards the detector 2116 impinges on the angle dependent optical element 2130. The angle dependent optical element 2130 may be designed to weaken rays impinging with larger angles compared to rays impinging with a smaller angle. In particular, at the cutoff angle the degree of transmission may steeply fall to zero and the light rays having a large angle of incidence may be cut-off. As shown in Figure 38 regions of the incident light beam 2116 are cut-off by the angle dependent optical element 2130 in the generated light beam 2131.

Figure 39 shows a dependency of the transmission power P in W of the optical fiber at constant irradiated power as a function of angle of incidence A in degree. The acceptance angle is shown as vertical line. The degree of transmission may be highest for incoming light rays parallel, i.e. at an angle of 0° , to the optical fiber, neglecting reflection effects. For higher angles, for example angles from 1° to 10° , the degree of transmission may decrease smoothly to around 80% of the degree of transmission for parallel light rays and may remain at this level constantly up to an acceptance angle of the optical fiber 2138. At the acceptance angle, the degree of transmission may steeply fall to zero. Light rays having a large angle of incidence may be cut-off.

Figures 40A and 40B show experimental results of distance measurements. In The determined distance z_{meas} in mm is shown as a function of the object distance z_{obj} in mm. As illumination source 115 a Laser was used having a wavelength of 980 nm and average power of 2,4 mW available under Flexpoint® Laser components module. Two Si-photodetectors were used as optical sensors 113. As optical fiber 2138 and transfer device 2128 available under Thorlabs Fixed Focus Collimation package F220SMA-980 was used. In Figure 40A, the solid line indicates where $z_{\text{meas}} = z_{\text{obj}}$. For the measurement the object distance was varied and two different types of object were used, in particular a black paper object, curve 2156 (dotted line), and a white paper object, curve 2158 (dashed line). The determined object distance is in agreement with the real distance within 2% for small and medium distances and within 10% for large distances. In Figure 40B, the combined signal Q determined by dividing the signals of the two photodetectors as a function of the distance z_{obj} in mm is shown for the black paper object (dotted line) and the white paper object (dashed line). The determined quotient for both object types is in agreement within 2% for small and medium distances and within 10% for large distances.

Figure 41 shows, in a highly schematic illustration, an exemplary embodiment of a detector 2110, for example according to the embodiments shown in Figure 35 or 36. The detector 2110 specifically may be embodied as a camera 2156 and/or may be part of a camera 2156. The camera 156 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 41 further shows an embodiment of a detector system 2158, which, besides the at least one detector 2110, comprises one or more beacon devices 2114, which, in this example, may be attached and/or integrated into an object 2112, the position of which shall be detected by using the detector 2110. Fig. 41 further shows an exemplary embodiment of a human-machine interface 2160, which comprises the at least one detector system 2158 and, further, an entertainment device 2162, which comprises the human-machine interface 2160. The figure further shows an embodiment of a tracking system 2164 for tracking a position of the object 2112, which comprises the detector system 2158. The components of the devices and systems shall be explained in further detail below.

Fig. 41 further shows an exemplary embodiment of a scanning system 2166 for scanning a scenery comprising the object 2112, such as for scanning the object 2112 and/or for determining at least one position of the at least one object 2112. The scanning system 2166 comprises the at least one detector 2110, and, further, optionally, the at least one illumination source 2115 as well as, optionally, at least one further illumination source 2115. The illumination source 2115, generally, is configured to emit at least one illumination light beam 2142, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 2114 and/or on a surface of the object 2112. The scanning system 2166 may be designed to generate a profile of the scenery including the object 2112 and/or a profile of the object 2112, and/or may be designed to generate at least one item of information about the dis-

tance between the at least one dot and the scanning system 2166, specifically the detector 2110, by using the at least one detector 2110.

As outlined above, an exemplary embodiment of the detector 2110 which may be used in the setup of Fig. 41 is shown in Fig. 35 and 36. Thus, the detector 2110, besides the optical sensors 2118, 2120, comprises at least one evaluation device 2133, having e.g. the at least one divider 2134 and/or the at least one position evaluation device 2136, as symbolically depicted in Fig. 41. The components of the evaluation device 2133 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 2110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 2118, 2120 and one or more of the components of the evaluation device 2133 may be interconnected by one or more connectors 2168 and/or by one or more interfaces, as symbolically depicted in Fig. 41. Further, the one or more connectors 2168 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 2168, the evaluation device 2133 may fully or partially be integrated into one or both of the optical sensors 2118, 2120 and/or into a housing 2170 of the detector 2110. Additionally or alternatively, the evaluation device 2133 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 2112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 2172, the position of which may be manipulated by a user 2174. As an example, the object 2112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 2112 are possible. Further, the user 2174 himself or herself may be considered as the object 2112, the position of which shall be detected.

As outlined above, the detector 2110 comprises at least the optical sensors 2118, 2120. The optical sensors 2118, 2120 may be located inside the housing 2170 of the detector 2110. Further, the at least one transfer device 2128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 2176 inside the housing 2170, which, preferably, is located concentrically with regard to the optical axis 2126 of the detector 2110, preferably defines a direction of view 2178 of the detector 2110. A coordinate system 2180 may be defined, in which a direction parallel or anti-parallel to the optical axis 2126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 126 may be defined as transversal directions. In the coordinate system 2180, symbolically depicted in Fig. 41, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 2180 are feasible, such as non-Cartesian coordinate systems.

The detector 2110 may comprise the optical sensors 2118, 2120 as well as, optionally, further optical sensors. The optical sensors 2118, 2120 may be located in one and the same beam

path, for example one behind the other, such that the first optical sensor 2118 covers a portion of the second optical sensor 2120. Alternatively, however, a branched beam path may be possible, for example using a multifurcated optical fiber. The branched beam path may comprise additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 2112 and/or of parts thereof. Alternatively, however, the optical sensors 2118, 2120 may be located at the same longitudinal coordinate.

One or more light beams 2116 are propagating from the object 2112 and/or from one or more of the beacon devices 2114, towards the detector 2110. The detector 2110 is configured for determining a position of the at least one object 2112. For this purpose, as explained above in the context of Fig. 35 to 40, the evaluation device 2133 is configured to evaluate sensor signals provided by the optical sensors 2118, 2120. The detector 2110 is adapted to determine a position of the object 2112, and the optical sensors 2118, 2120 are adapted to detect the light beam 2131. In case no illumination source 2115 is used, the beacon devices 2114 and/or at least one of these beacon devices 2114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 2115 is used, the beacon devices 2114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 2112 may be used, such as integrated reflected beacon devices 2114 having at least one reflective surface such as a mirror, retro reflector, reflective film, or the like. The light beam 2116, directly and/or after being modified by the transfer device 2128, such as being focused by one or more lenses, impinges on the angle dependent element 2130 which generates the at least one light beam which illuminates the light-sensitive areas 2122, 2124 of the optical sensors 2118, 2120. For details of the evaluation, reference may be made to Fig. 35 to 40 above.

As outlined above, the determination of the position of the object 2112 and/or a part thereof by using the detector 2110 may be used for providing a human-machine interface 2160, in order to provide at least one item of information to a machine 2182. In the embodiments schematically depicted in Fig. 41, the machine 2182 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 2133 may even be fully or partially integrated into the machine 2182, such as into the computer.

As outlined above, Fig. 41 also depicts an example of a tracking system 2164, configured for tracking the position of the at least one object 2112 and/or of parts thereof. The tracking system 2164 comprises the detector 2110 and at least one track controller 2184. The track controller 2184 may be adapted to track a series of positions of the object 2112 at specific points in time. The track controller 2184 may be an independent device and/or may be fully or partially integrated into the machine 2182, specifically the computer, as indicated in Fig. 41 and/or into the evaluation device 2133.

Similarly, as outlined above, the human-machine interface 2160 may form part of an entertainment device 2162. The machine 2182, specifically the computer, may also form part of the en-

tertainment device 2162. Thus, by means of the user 2174 functioning as the object 2112 and/or by means of the user 2174 handling a control device 2172 functioning as the object 2112, the user 2174 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Referring to Figure 42, the detector 110, 1110, 2110, such as the detector as described with respect to Figures 1 to 41 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 113, 1118, 1120, 2113. For example, the detector 110, 1110, 2110 may comprise a plurality of optical sensors arranged in the matrix 117. The detector 110, 1110, 2110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector 110, 1110, 2110 may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. As outlined above, this principle is called Distance by Photon Ratio (DPR). In one embodiment, the light beam 116, 1116 may illuminate the sensor element with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 132 may be configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 42, the light beam 116, 1116 propagating from the object 112, 1112 to the detector 110, 1110, 2110 may illuminate the matrix 117 with at least one line pattern 2186. The line pattern 2186 may be generated by the object 112, 1112, for example in response to an illumination by the at least one illumination source 136 with an illumination pattern comprising at least one illumination line pattern. A1 may correspond to an area with a full line width of the line pattern 2186 in the matrix 117. The line pattern 2186 in the matrix 117 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a

line width in the matrix 117 is increased. In particular, the line width of the line pattern 2186 in the matrix 117 may change from one column to another column. A2 may be a central area of the line pattern 2186 in the matrix 117. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern 2186 may be segmented in the matrix 117. Each of the columns may comprise center information of intensity in the central area of the line pattern 2186 and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern 2186.

Figure 43 shows a simulation testing of object size independence for a one-dimensional case using computational ray tracing. In a simulation testing, an aspheric lens with $f=10$ mm, 10 mm pupil diameter in a distance range of 100 mm to 600 mm was used. Using this specification, object size independence of about 10 mm was obtained, such that the spot size was varied from 1 mm to 25 mm. In Fig. 43, the quotient $Q(z)$ over the longitudinal coordinate z is shown for 1 mm variation (curve 2188), 2 mm variation (curve 2190), 5 mm variation (curve 2192), 15 mm variation (curve 2194), 20 mm variation (curve 2196) and 25 mm variation (curve 2198). It can be seen that the quotient for object sizes above 10 mm deviates, whereas that of object sizes smaller than 10 mm yield identical ratios. This feature reduces calibration efforts for targets of varying size and is inherent to DPR analysis.

Referring to Figures 44A and B, as outlined above, the detector 110, 1110, 2110 may comprise the at least one matrix 117 of optical sensors 113, 1118, 1120, 2113. With the aid of such a pixelated imaging device, a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates as shown in Figure 44A. Accordingly, the parameterization of a single line would be given by $\omega = x \cos(\theta) + y \sin(\theta)$. The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega,\theta)}\{f(x,y)\} = \iint_{-\infty}^{\infty} f(x,y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy$$

where δ denotes the Dirac delta function and $f(x,y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is then given by

$$R = \frac{\Re_{(\omega,\theta)}\{f'(x,y)\}}{\Re_{(\omega,\theta)}\{f(x,y)\}}$$

with $f'(x,y)$ as the overshined image region highlighted in Fig. 44B. It is expected that the variation of θ yields different ratios R for skewed object surfaces. It may be sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

Figures 45A and B show further embodiments of the detector 110 according to the present invention comprising at least one bi-cell. The illumination source 136, such as a laser source, may generate the light beam 138 illuminating object 112. The reflected light beam 116 may propagate from the object 112 to the transfer device 128 and may impinge on the bi-cell of optical

sensors 176. In Figure 45A a side view is shown and in Figure 45B a front view is shown. The detector 110 may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the bi-cell in Figures 45 A and B may be adapted to generate a so called FiP signal. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

Figure 46 shows experimental results, in particular spot diameter independence and luminance independence of the combined sensor signal, determined using the detector setup shown in Figure 46. In particular, the bi-cell was a PbS-Bi-cell and a 1550 nm laser was used with a laser spot size of 4 mm. The baseline was 12.5 mm. The transfer device was a Thorlabs Asphere lens with focal length of $f = 20$ mm and diameter of $D = 25$ mm. Figure 46 shows quotient Q over the longitudinal coordinate z for different luminance and spot diameter, in particular for luminance of 2.6 mW and spot diameter of 12 mm (curve 2200), 2.4 mW and 6 mm (curve 2202) and 1.2 mW and spot diameter of 3 mm (curve 2204). All curves show identical curve shape and thus, spot diameter independence.

Figures 47A to C show three embodiments of a hexagonal illumination pattern. The illumination source 136 may be adapted to generate at least one illumination pattern for illuminating the object 112. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. Specifically, the illumination source 136 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination pattern may comprise at least one feature such as a point or symbol. The illumination pattern may comprise a plurality of features. The illumination pattern may comprise an arrangement of periodic or non-periodic features. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasi-periodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as

possible features per area such that hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

- 5 The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on an epipolar line. As shown in Figure 47A, the illumination pattern may comprise at least one hexagonal pattern, wherein the individual points are positioned on epipolar lines 2206. As shown in Figure 47B, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Such a positioning
- 10 of the illumination features allows enhancing distance between the individual points on each epipolar line. For example as shown in Figure 47C, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the
- 15 distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other. Such a positioning allows to enhance the number of possible features per area.
- 20 Figure 48 shows an embodiment of a scanning device 154. The scanning device may be adapted as a line scanning device. In particular, the scanning device 154 may comprise at least one sensor line or row of optical sensors 113. Furthermore, the scanning device may comprise the at least one transfer device 128 and the at least one illumination source 136. Triangulation systems require a sufficient baseline, however due to the baseline in the near field no detection
- 25 may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. Further, these near field and far field
- 30 problems can be overcome by using the scanning device 154 of Figure 48. The scanning device 154 may be adapted to detect a plurality of light beams 116 propagating from the object 112 to the scanning device 154 on the CMOS line. The light beams 116 may be generated at different position on the object 112 or by movement of the object 112. The scanning device 154 may be
- 35 adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above.

List of reference numbers

110	detector
112	object
113	optical sensors
114	beacon device
115	sensor element
116	light beam
117	matrix
118	first optical sensor
119	mask
120	second optical sensor
121	light-sensitive area
122	first light-sensitive area
124	second light-sensitive area
126	optical axis of the detector
128	transfer device
129	optical axis of the transfer device
130	focal point
131	light spot
132	evaluation device
133	center detector
134	detector system
135	summing device
136	illumination source
137	combining device
138	illumination light beam
140	reflective element
142	divider
144	position evaluation device
146	camera
148	human-machine interface
150	entertainment device
152	tracking system
154	scanning system
156	connector
158	housing

160 control device
162 user
164 opening
166 direction of view
168 coordinate system
170 machine
172 track controller
174 array
176 optical sensor
178 quadrant photodiode
180 geometrical center of every
182 geometrical center of first optical sensor
184 geometrical center of second optical sensor
186 light spot
188 actuator
190 diaphragm
192 readout device for optical storage media
194 optical element
196 region of interest
198 first area
200 second area
202 inner region
204 plane
206 outer region
208 direction of movement
210 direction of movement
212 curve
214 curve
216 set of curves
218 set of curves
1110 detector
1112 object
1114 beacon device
1116 light beam
1118 first optical sensor
1120 second optical sensor
1122 first light-sensitive area

1124 second light-sensitive area
1126 optical axis
1128 transfer device
1130 focal point
1132 evaluation device
1134 detector system
1136 illumination source
1138 illumination light beam
1140 reflective element
1142 divider
1144 position evaluation device
1146 camera
1148 human-machine interface
1150 entertainment device
1152 tracking system
1154 scanning system
1156 connector
1158 housing
1160 control device
1162 user
1164 opening
1166 direction of view
1168 coordinate system
1170 machine
1172 track controller
1174 fluorescent waveguiding sheet
1176 waveguiding
1178 matrix material
1180 fluorescent material
1182 photosensitive element
1184 photosensitive element
1186 photosensitive element
1188 photosensitive element
1190 edge
1192 edge
1194 edge
1196 edge

1198 optical filter element
1200 reference photosensitive element
1202 small light spot
1204 large light spot
1206 shadow
1208 summing device
1210 subtracting device
1212 photosensitive element
1214 corner
1216 optical coupling element
2110 detector
2112 object
2113 optical sensors
2114 beacon device
2115 Illumination source
2116 light beam
2118 first optical sensor
2120 second optical sensor
2121 light-sensitive area
2122 first light-sensitive area
2124 second light-sensitive area
2126 optical axis of the detector
2128 transfer device
2129 optical axis of the transfer device
2130 angle dependent optical element
2131 light beam
2132 first side
2133 evaluation device
2134 divider
2136 position evaluation device
2138 Optical fiber
2140 Illumination fiber
2142 Light beam
2144 First fiber
2146 Second fiber
2148 entrance end
2150 exit end

2152 first light beam
2154 Second light beam
2156 camera
2158 Detector system
2160 Human-machine interface
2162 Entertainment device
2164 Tracking system
2166 Scanning system
2168 connector
2170 housing
2172 Control device
2174 user
2176 opening
2178 Direction of view
2180 Coordinate system
2182 machine
2184 Track controller
2186 Line pattern
2188 curve
2190 curve
2192 curve
2194 curve
2196 curve
2198 curve
2200 curve
2202 curve
2204 curve
2206 Epipolar line

Patent claims

1. A detector (110, 1110, 2110) for determining a position of at least one object (112, 1112), the detector (110, 1110, 2110) comprising:
 - 5 - at least one transfer device (128, 1128, 2128), wherein the transfer device (128, 1128, 2128) has at least one focal length in response to at least one incident light beam (116, 1116) propagating from the object (112, 1112, 2112) to the detector (110, 1110, 2110);
 - 10 - at least two optical sensors (113, 1118, 1120, 2113), wherein each optical sensor (113, 1118, 1120) has at least one light sensitive area (121, 1122, 1124, 2121), wherein each optical sensor (113, 1118, 1120, 2113) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam (116, 1116, 2131),
 - 15 - at least one evaluation device (132, 1132, 2133) being configured for determining at least one longitudinal coordinate z of the object (112, 1112, 2112) by evaluating a quotient signal Q from the sensor signals;
 wherein the detector is adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from the object size in an object plane.
- 20 2. The detector (110, 1110, 2110) according to the preceding claim, wherein the measurement range is adjustable by adjusting one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to
 25 an image of the object z_i ; an object size O_{size} of the object in the object plane.
3. The detector (110, 1110, 2110) according to the preceding claim, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object size if a circle of confusion of at least one light beam emitted from at least one point of the
 30 object is greater than an image size of the object.
4. The detector (110, 1110, 2110) according to the preceding claim, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object size if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for at least one distance.
- 35 5. The detector (110, 1110, 2110) according to the preceding claim, wherein a lower object size limit is preferably 0.5 μm or more, more preferably 1 μm or more and most preferably 10 μm or more.
- 40 6. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for deriving the quotient signal Q by one or more of di-

viding the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

7. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.
8. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o.

9. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the optical sensors (113) are positioned off focus.
10. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the transfer device (128) has an optical axis (129), wherein the transfer device (128) constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis (129) and wherein d is a spatial offset from the optical axis (129), wherein the optical sensors (113) are arranged such that the light-sensitive areas (121) of the optical sensors (113) differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.
11. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of at least one beam profile of the light beam.
12. The detector (110, 1110, 2110) according to the preceding claim, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.
13. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the light-sensitive regions are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a

second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

14. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured to determine the first area of the beam profile and the second area of the beam profile.
15. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.
16. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
17. The detector (110, 1110, 2110) according to any one of the two the preceding claims, wherein the evaluation device (132) is configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.
18. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the optical sensors (113) comprise a first optical sensor (118) having a first light sensitive area (122) and a second optical sensor (120) having a second light sensitive area (124), wherein the first and the second light sensitive areas (122, 124) are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$
 is satisfied, wherein a is a ratio of photons hitting both an inner region of a plane perpendicular to the optical axis (129) intersecting the optical axis (129) at a distance equal to half of a focal length of the transfer device (129) and the first light sensitive area (122), b is a ratio of photons hitting both the inner region of the plane and the second light sensitive area (124), c is a ratio of photons hitting both an outer region of the plane and the first light-sensitive area (122), d is a ratio of the photons hitting both the outer region of the plane and the second light sensitive area (124).
19. The detector (110, 1110, 2110) according to the preceding claim, wherein the first light sensitive area (118) and the second light sensitive area (122) are arranged such that the inner region has an area with a geometrical center point on the optical axis (129) and an ex-

tension such that half of the photons hit the plane within the inner region and the other half hit the plane outside the inner region.

20. The detector (110, 1110, 2110) according to the preceding claim, wherein the first light sensitive area (118) and the second light sensitive area (122) are arranged such that the inner region is designed as a circle with a center point on the optical axis and a radius r which is chosen such that half of the photons hit the plane within the circle and the other half hit the plane outside the circle.
21. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the optical sensors (113) comprise a first optical sensor (118) having a first light sensitive area (122) and a second optical sensor (120) having a second light sensitive area (124),
- wherein the first optical sensor (118) is configured to generate at least one first sensor signal in response to the illumination of the first light-sensitive area (122) by the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110);
 - wherein the second optical sensor (120) is configured to generate at least one second sensor signal in response to the illumination of the second light-sensitive area (124) by the light beam (116), wherein the first light-sensitive area (122) is smaller than the second light-sensitive area (124).
22. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured for deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device (132) is configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
23. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the first and second optical sensors (118, 120) are arranged linearly in one and the same beam path of the detector (110, 1110, 2110).
24. The detector (110, 1110, 2110) according to the preceding claim, wherein the first and second optical sensors (118, 120) are arranged concentrically with respect to an optical axis (126) of the detector (110, 1110, 2110).
25. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the first and second optical sensors (118, 120) each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

26. The detector (110, 1110, 2110) according to any one of the five preceding claims, wherein the first and second optical sensors (118, 120) each are uniform sensors having a single light-sensitive area each.
- 5 27. The detector (110, 1110, 2110) according to any one of the six preceding claims, wherein the detector (110, 1110, 2110) further comprises an illumination source (136) for illuminating the object (112).
- 10 28. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the detector (110, 1110, 2110) has
- at least one sensor element (115) having a matrix (117) of optical sensors (113), the optical sensors (113) each having a light-sensitive area (121), wherein each optical sensor (113) is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area (121) by at least one light beam (116) propagating from the object (112) to the detector (110, 1110, 2110);
 - 15 - the at least one evaluation device (132) configured for evaluating the sensor signals, by
 - a) determining at least one optical sensor (113) having the highest sensor signal and forming at least one center signal;
 - 20 b) evaluating the sensor signals of the optical sensors (113) of the matrix (117) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - 25 d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.
29. The detector (110, 1110, 2110) according to the preceding claim, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors (113) containing the optical sensor (113) having the highest sensor signal and a predetermined group of neighboring optical sensors (113); a sum of sensor signals from a group of optical sensors (113) containing the optical sensor (113) having the highest sensor signal and a predetermined group of neighboring optical sensors (113); a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors (113) containing the optical sensor (113) having the highest sensor signal and a predetermined group of neighboring optical sensors (113); an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.
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30. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix (117); a sum of all sensor signals of the matrix (117); an integral of all sensor signals of the matrix (117); an average over all sensor signals of the matrix (117) except for sensor signals from those optical sensors (113) contributing to the center signal; a sum of all sensor signals of the matrix (117) except for sensor signals from those optical sensors (113) contributing to the center signal; an integral of all sensor signals of the matrix (117) except for sensor signals from those optical sensors (113) contributing to the center signal; a sum of sensor signals of optical sensors (113) within a predetermined range from the optical sensor (113) having the highest sensor signal; an integral of sensor signals of optical sensors (113) within a predetermined range from the optical sensor (113) having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors (113) being located within a predetermined range from the optical sensor (113) having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors (113) being located within a predetermined range from the optical sensor (113) having the highest sensor signal.
31. The detector (110, 1110, 2110) according to any one of the three preceding claims, wherein the combined signal is the quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.
32. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.
33. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the evaluation device (132) comprises at least one divider (142), wherein the divider (142) is configured for deriving the quotient signal.
34. The detector (110, 1110, 2110) according to any one of the six preceding claims, wherein the evaluation device (132) is further configured for determining at least one transversal coordinate of the object (112) by evaluating a transversal position of the at least one optical sensor (113) having the highest sensor signal.
35. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical

axis (126) of the detector (110, 1110, 2110) by different spatial offsets, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light beam (116) propagating from the object (112) to the detector (110, 1110, 2110).

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36. The detector (110, 1110, 2110) according to the preceding claim, wherein the optical sensors (118, 120, 176) are part of a sensor array (174), wherein a geometrical center (180) of the sensor array (174) is offset from the optical axis (126).

10 37. The detector (110, 1110, 2110) according to the preceding claim, wherein the sensor array (174) is movable relative to the optical axis (126).

38. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured for, firstly, determining the transversal position of a light spot (186) generated by the light beam (116) on the sensor array (174) by using the sensor signals and for, secondly, moving the sensor array (174) relative to the optical axis (126) until the light spot (186) is off-centered.

15 39. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the optical sensors (118, 120, 176) are partial diodes of a quadrant diode (178), with a geometrical center (180) of the quadrant diode (178) being off-centered from the optical axis (126) of the detector (110, 1110, 2110).

20 40. The detector (110, 1110, 2110) according to any one of the preceding claims, the detector (110, 1110, 2110) having

- at least one first optical sensor (1118) having a first light-sensitive area (1122), wherein the first optical sensor (1118) is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area (122) by a light beam (1116) propagating from the object (1112) to the detector (1110);
- 30 - at least one second optical sensor (1120) having:
 - at least one fluorescent waveguiding sheet (1174) forming a second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) is oriented towards the object (1112) such that at least one light beam (1116) propagating from the object (1112) towards the detector (1110) generates at least one light spot (1202, 1204) in the second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) contains at least one fluorescent material (1180), wherein the fluorescent material (1180) is adapted to generate fluorescence light in response to the illumination by the light beam (1116); and
 - 35 ○ at least one photosensitive element (1182, 1184, 1186, 1188, 1212) located at at least one edge (1190, 1192, 1194, 1196) of the fluorescent waveguiding sheet (1174), capable of detecting fluorescence light guided from the light spot (1202, 1204) towards the photosensitive element (1182, 1184, 1186, 1188, 1212) by the fluorescent waveguiding sheet (1174) and capable of generating at least one second sensor signal in response to the illumination of the second light-sensitive
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area (1124) by the light beam (1116), wherein the first light-sensitive area (1122) is smaller than the second light-sensitive area (1124); and

- at least one evaluation device (1132) being configured for determining at least one longitudinal coordinate z of the object (1112) by evaluating the first and second sensor signals.

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41. The detector (110, 1110, 2110) according to the preceding claim, wherein the at least one photosensitive element (1182, 1184, 1186, 1188, 1212) is optically coupled to the fluorescent waveguiding sheet (1174) by at least one optical coupling element (1216) configured for at least partially coupling the fluorescence light guided by the fluorescent waveguiding sheet (1174) out of the fluorescent waveguiding sheet (1174).
42. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the evaluation device (1132) is configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device (1132) is configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
43. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (1132) is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.
44. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the first optical sensor (1118) is located in front of the second optical sensor (1120), preferably such that the light beam (1116), in the given order, illuminates the first optical sensor (1118) and, afterwards, the second optical sensor (1120).
45. The detector (110, 1110, 2110) according to any one of the five preceding claims, wherein the second optical sensor (1120) comprises at least two photosensitive elements (1182, 1184, 1186, 1188, 1212) located at at least two edges (1190, 1192, 1194, 1196) of the fluorescent waveguiding sheet (1174), capable of detecting fluorescence light guided from the light spot (1202, 1204) towards the photosensitive elements (1182, 1184, 1186, 1188, 1212) by the fluorescent waveguiding sheet (1174) and capable of generating at least one sensor signal each.
46. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (1132) comprises at least one summing device (1208) configured for summing up the sensor signals of the at least two photosensitive elements (1182, 1184, 1186, 1188, 1212), thereby forming a sum signal S , wherein the evaluation device (1132) is configured to determine the at least one longitudinal coordinate z of the object (1112) by using at least one predetermined relationship between the first sensor signal, the sum signal S of the second sensor signals and the longitudinal coordinate z .

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47. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the evaluation device (1132) is further configured to determine at least one transversal coordinate x, y of the object (1112) by evaluating the sensor signals of the photosensitive elements (1182, 1184, 1186, 1188, 1212).
- 5 48. The detector (110, 1110, 2110) according to any one of the eight preceding claims, wherein the second light-sensitive area (1124) is larger by at least a factor of two than the first light-sensitive area (1122).
- 10 49. The detector (110, 1110, 2110) according to any one of the nine preceding claims, wherein the detector (110, 1110, 2110) further comprises an illumination source (1136) for illuminating the object (1112).
- 15 50. The detector (110, 1110, 2110) according to any one of the preceding claims, the detector (2110) comprising:
- at least one angle dependent optical element (2130) adapted to generate at least one light beam (2131) having at least one beam profile depending on an angle of incidence of an incident light beam (2116) propagating from the object (2112) towards the detector (2110) and illuminating the angle dependent optical element (2130);
 - 20 - at least two optical sensors (2113), wherein each optical sensor (2113) has at least one light sensitive area (2121), wherein each optical sensor (2113) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam (2131) generated by the angle dependent optical element (130);
 - 25 - at least one evaluation device (2133) being configured for determining at least one longitudinal coordinate z of the object (2112) by evaluating the quotient signal Q from the sensor signals.
- 30 51. The detector (2110) according to the preceding claim, wherein the angle dependent optical element (2130) comprises at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element; at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one prism; at least one lens; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter; at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.
- 35 52. A detector system (134) for determining a position of at least one object (112), the detector system (134) comprising at least one detector (110, 1110, 2110) according to any one of the preceding claims, the detector system (134) further comprising at least one beacon device (114) adapted to direct at least one light beam (116) towards the detector (110, 1110, 2110),
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wherein the beacon device (114) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).

53. A human-machine interface (148) for exchanging at least one item of information between a user (162) and a machine (170), wherein the human-machine interface (148) comprises at least one detector system (134) according to the preceding claim, wherein the at least one beacon device (114) is adapted to be at least one of directly or indirectly attached to the user (162) and held by the user (162), wherein the human-machine interface (148) is designed to determine at least one position of the user (162) by means of the detector system (134), wherein the human-machine interface (148) is designed to assign to the position at least one item of information.
54. An entertainment device (150) for carrying out at least one entertainment function, wherein the entertainment device (150) comprises at least one human-machine interface (148) according to the preceding claim, wherein the entertainment device (150) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (148), wherein the entertainment device (150) is designed to vary the entertainment function in accordance with the information.
55. A tracking system (152) for tracking a position of at least one movable object (112), the tracking system (152) comprising at least one detector system (134) according to any one of the preceding claims referring to a detector system (134), the tracking system (152) further comprising at least one track controller (172), wherein the track controller (172) is adapted to track a series of positions of the object (112) at specific points in time.
56. A scanning system (154) for determining a depth profile of a scenery, the scanning system (154) comprising at least one detector (110, 1110, 2110) according to any of the preceding claims referring to a detector (110, 1110, 2110), the scanning system (154) further comprising at least one illumination source (136) adapted to scan the scenery with at least one light beam (116).
57. A camera (146) for imaging at least one object (112), the camera (146) comprising at least one detector (110, 1110, 2110) according to any one of the preceding claims referring to a detector (110, 1110, 2110).
58. A readout device (192) for optical storage media, the readout device (192) comprising at least one detector (110, 1110, 2110) according to any one of the preceding claims referring to a detector (110, 1110, 2110).
59. A method for determining a position of at least one object (112) by using at least one detector (110, 1110, 2110), the method comprising the following steps:
- providing at least two optical sensors (113), each optical sensor (113) having a light-sensitive area (121), wherein each optical sensor (113) is configured to generate a sen-

- sor signal in response to an illumination of its respective light-sensitive area by a light beam (116), wherein the detector has at least one transfer device (128), wherein the transfer device (128) has at least one focal length in response to the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110);
- 5 - illuminating each of the light-sensitive areas (121) of the at least two optical sensors (113) of the detector with the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110), wherein, thereby, each of the light-sensitive areas (121) generates at least one sensor signal; and
- 10 - evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object (112), wherein the evaluating comprises deriving a quotient signal Q of the sensor signals,
- wherein the longitudinal coordinate z of the object is determined in at least one measurement range independent from the object size in an object plane.
- 15 60. The method according to the preceding claim, wherein the deriving of the quotient signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.
- 20 61. The method according to any one of the preceding claims referring to a method, wherein the method comprises adjusting the measurement range by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the
- 25 object in the object plane.
- 30 62. The method according to any one of the preceding claims referring to a method, wherein the transfer device (128) has an optical axis (129), wherein the transfer device (128) constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis (129) and wherein d is a spatial offset from the optical axis (129), wherein the method comprises arranging the optical sensors (113) such that the light-sensitive areas (121) of the optical sensors (113) differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.
- 35 63. The method according to any one of the preceding claims referring to a method, wherein each of the sensor signals comprises at least one information of at least one beam profile of the light beam, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles, wherein the light-sensitive regions are arranged such that a first
- 40 sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

64. The method according to the preceding claim, wherein the evaluating comprises determining the first area of the beam profile and the second area of the beam profile.
- 5 65. The method according to any one of the two preceding claims, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises an information relating to a number of photons in the first area of the beam profile and the center information comprises an information relating to a number of photons in the second area of the beam profile, wherein the evaluating comprises deriving the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.
- 10
- 15 66. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- illuminating at least one first light-sensitive area (122) of at least one first optical sensor (118) of the detector (110, 1110, 2110) with the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110) and, thereby, generating at least one first sensor signal;
 - illuminating at least one second light-sensitive area (124) of at least one second optical sensor (120) of the detector (110, 1110, 2110) with the light beam (116) and, thereby, generating at least one second sensor signal, wherein the first light-sensitive area (122) is smaller than the second light-sensitive area (124).
- 20
- 25
67. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- illuminating at least one sensor element (115) of the detector (110, 1110, 2110) with the at least one light beam (116) propagating from the object (112) to the detector (110, 1110, 2110), the detector (110, 1110, 2110) having a matrix (117) of optical sensors (113), the optical sensors (113) each having a light-sensitive area (121), wherein each optical sensor (113) generates at least one sensor signal in response to the illumination;
 - evaluating the sensor signals, by
 - a) determining at least one optical sensor (113) having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors (113) of the matrix (117) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.
- 30
- 35
- 40

68. The method according to the preceding claim, wherein the combined signal is the quotient signal Q, derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.
69. The method according to any one of the preceding claims referring to a method, wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110, 1110, 2110) by different separations.
70. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- illuminating at least one first light-sensitive area (1122) of at least one first optical sensor (1118) with a light beam (1116) propagating from the object (1112) to the detector (110, 1110, 2110) and generating at least one first sensor signal by the first optical sensor (1118), in response to the illumination of the first light-sensitive area (1122);
 - illuminating at least one second light-sensitive area (1124) of at least one second optical sensor (1120) with the light beam (1116) and generating at least one second sensor signal by the second optical sensor (1120), in response to the illumination of the second light-sensitive area (1124), the second optical sensor (1120) having:
 - o at least one fluorescent waveguiding sheet (1174) forming a second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) is oriented towards the object (1112) such that at least one light beam (1116) propagating from the object (1112) towards the detector (110, 1110, 2110) generates at least one light spot (1202, 1204) in the second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) contains at least one fluorescent material (1180), wherein the fluorescent material (1180) is adapted to generate fluorescence light in response to the illumination by the light beam (1116); and
 - o at least one photosensitive element (1182, 1184, 1186, 1188, 1212) located at at least one edge (1190, 1192, 1194, 1196) of the fluorescent waveguiding sheet (1174), capable of detecting fluorescence light guided from the light spot (1202, 1204) towards the photosensitive element (1182, 1184, 1186, 1188, 1212) by the fluorescent waveguiding sheet (1174) and capable of generating the at least one second sensor signal in response to the illumination of the second light-sensitive area (1124) by the light beam (1116), wherein the first light-sensitive area (1122) is smaller than the second light-sensitive area (1124); and
 - determining at least one longitudinal coordinate z of the object (1112) by evaluating the first and second sensor signals.

71. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- providing at least one angle dependent optical element (130) and generating at least one light beam (131) having at least one beam profile depending on an angle of incidence;
 - providing at least two optical sensors (113), wherein each optical sensor (113) has at least one light sensitive area (121), wherein each optical sensor (113) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (121) by the light beam (131) generated by the angle dependent optical element (130);
 - illuminating each of the light-sensitive areas (131) of the at least two optical sensors (113) of the detector (110) with the light beam (131) generated by the angle dependent optical element (130), wherein, thereby, each of the light-sensitive areas (121) generates at least one sensor signal; and
 - evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object (112), wherein the evaluating comprises deriving the quotient signal Q of the sensor signals.
72. A use of the detector (110, 1110, 2110) according to any one of the preceding claims relating to a detector (110, 1110, 2110), for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface (148) application; a logistics application; a tracking application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use in combination with optical data storage and readout.

FIG.1

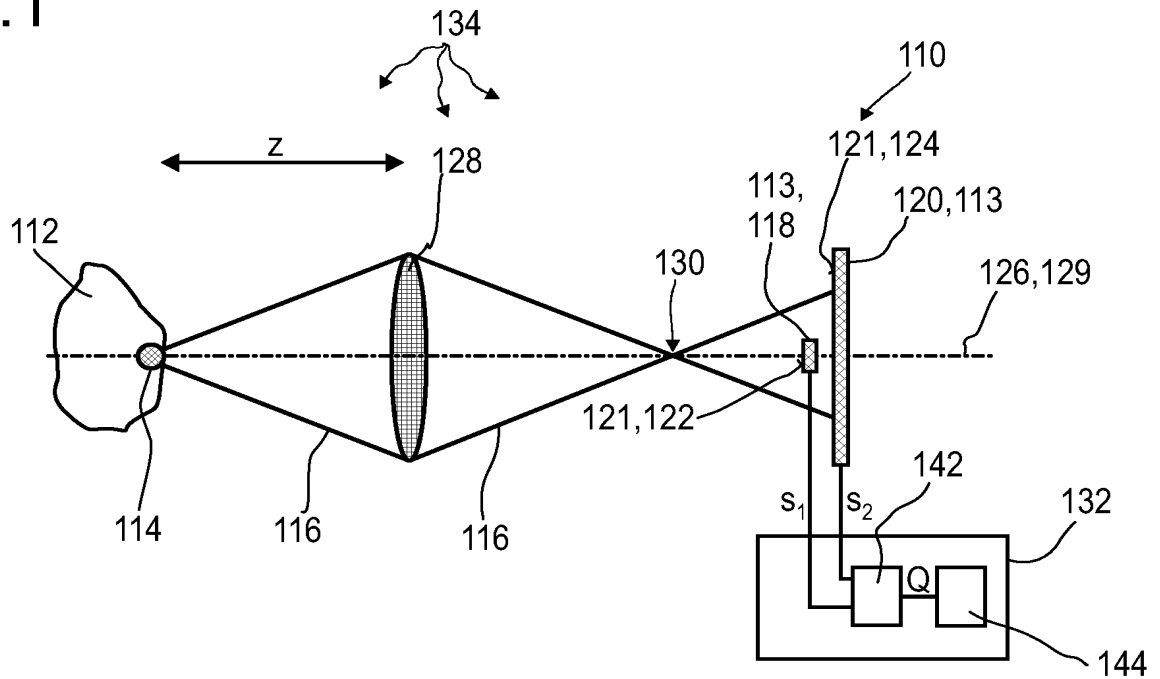


FIG.2

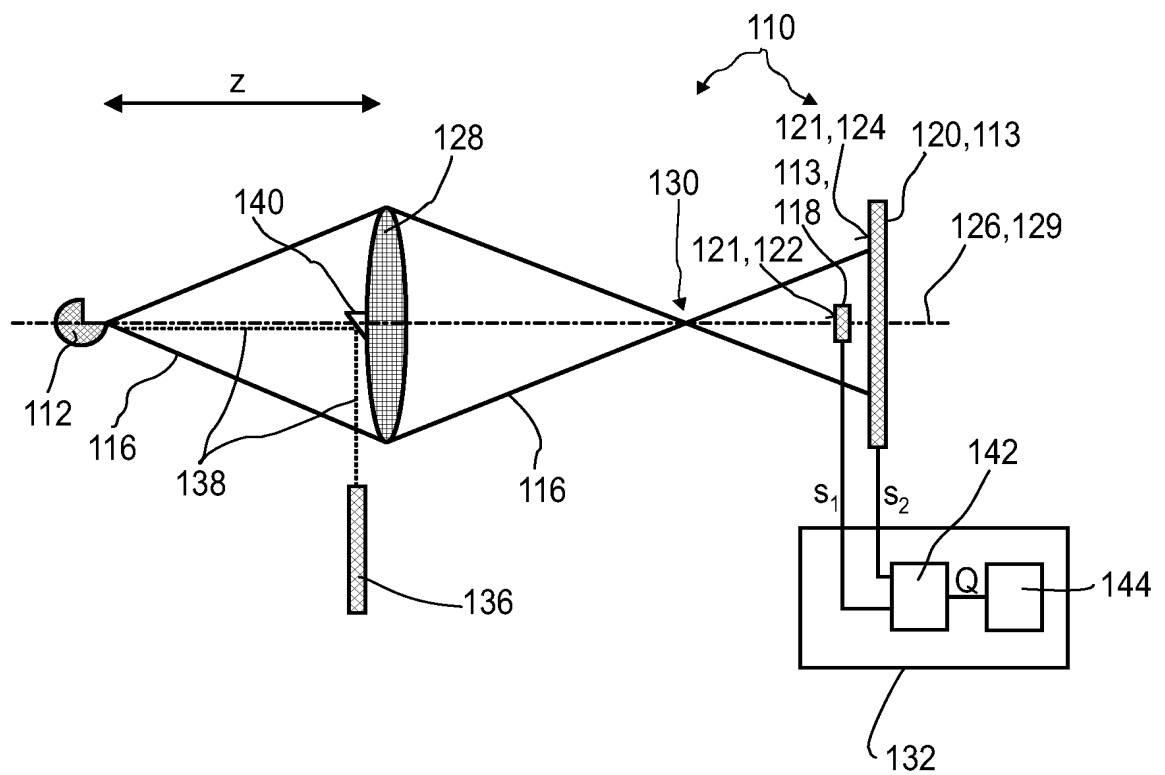


FIG.3

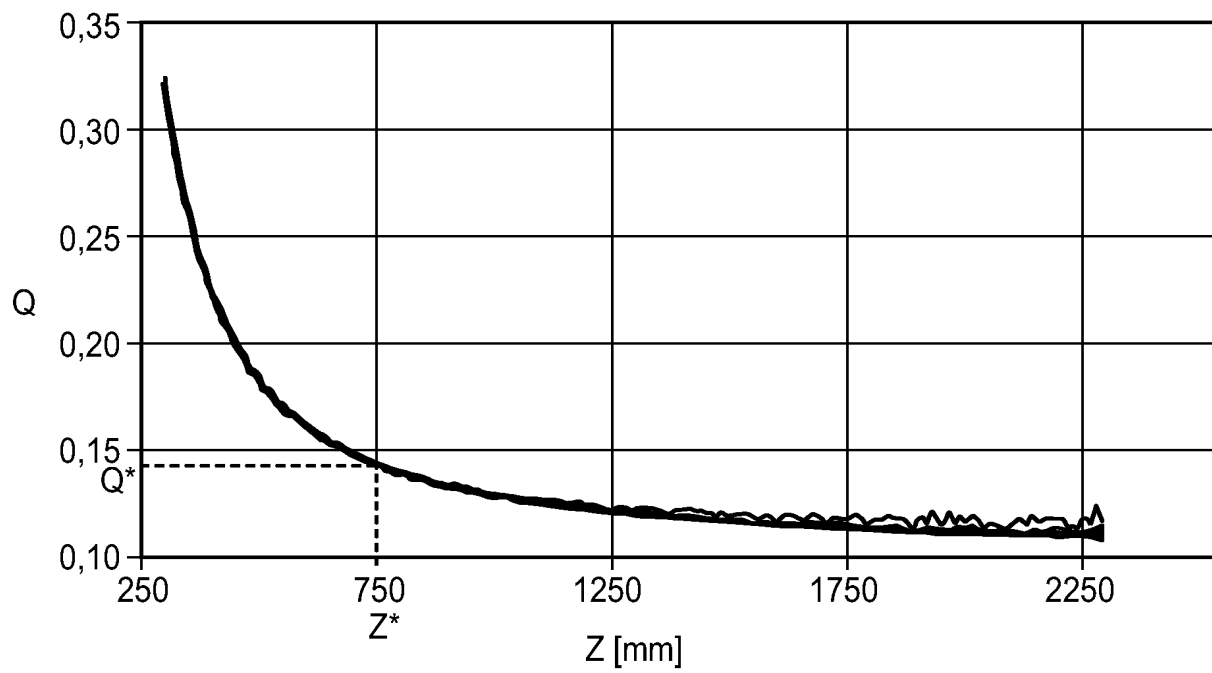


FIG.4

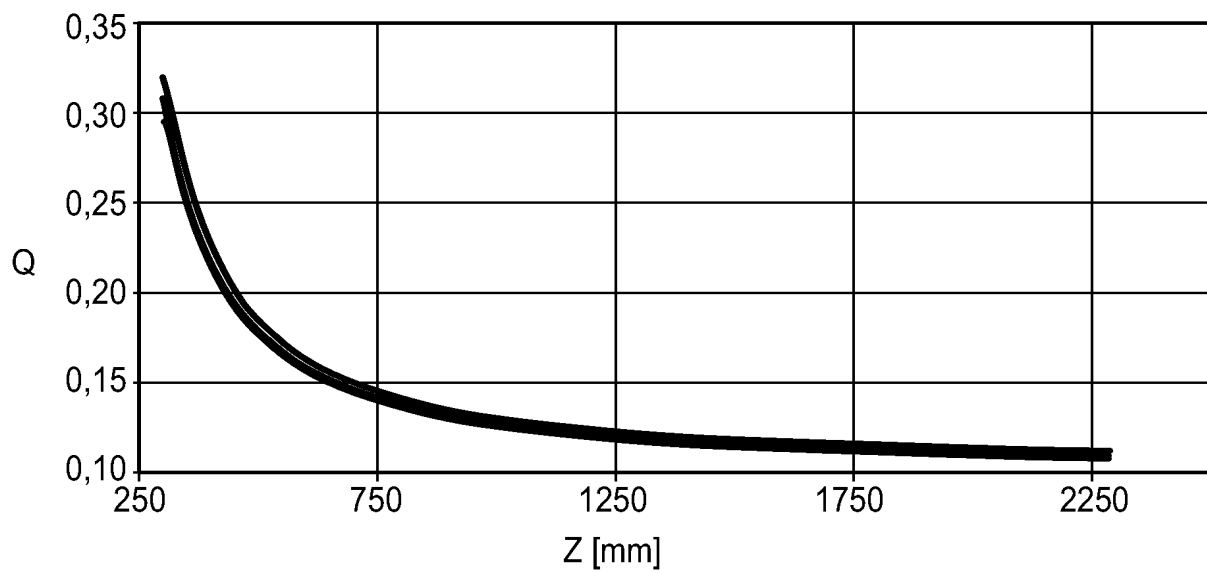


FIG.5

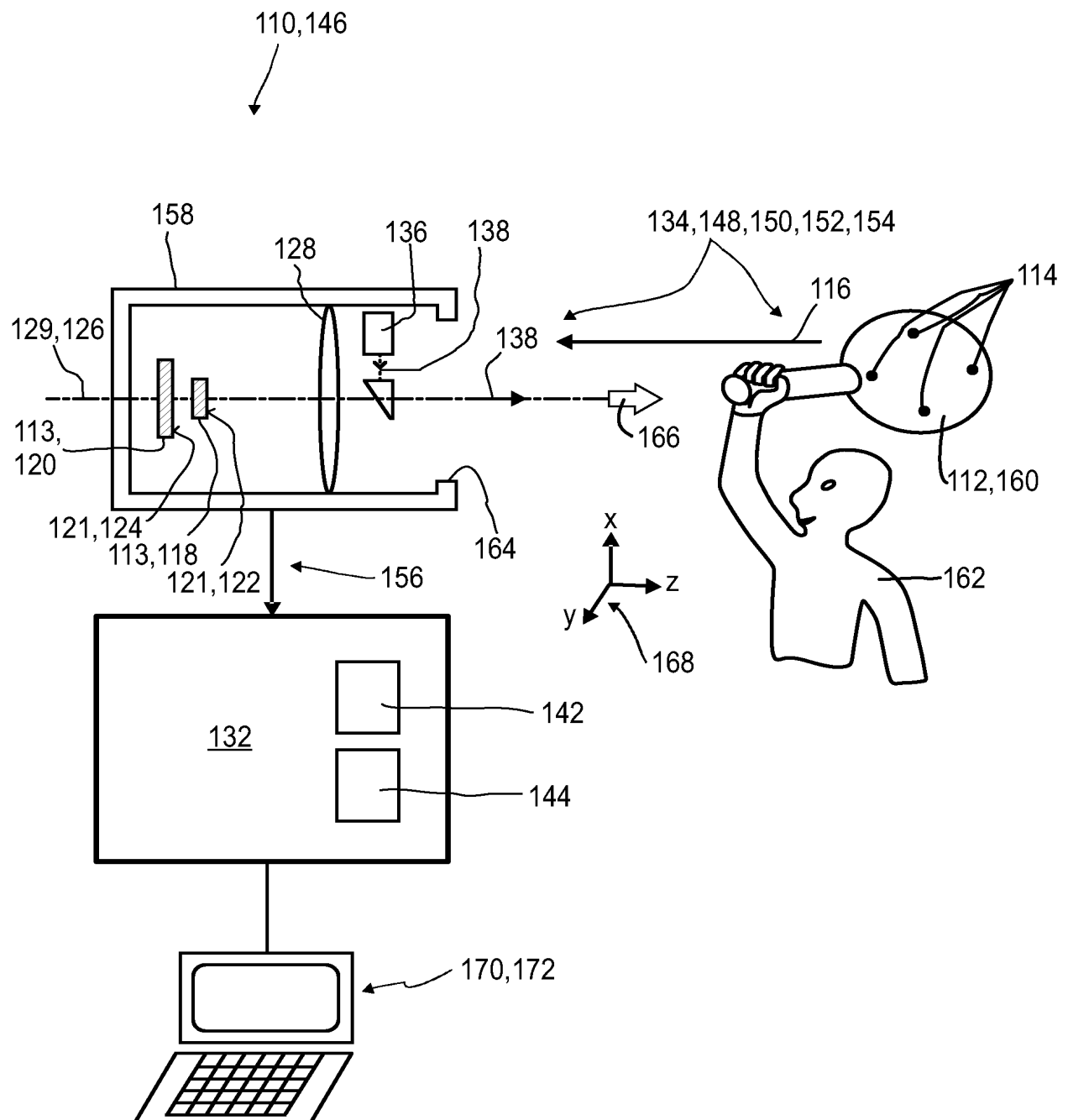


FIG.6

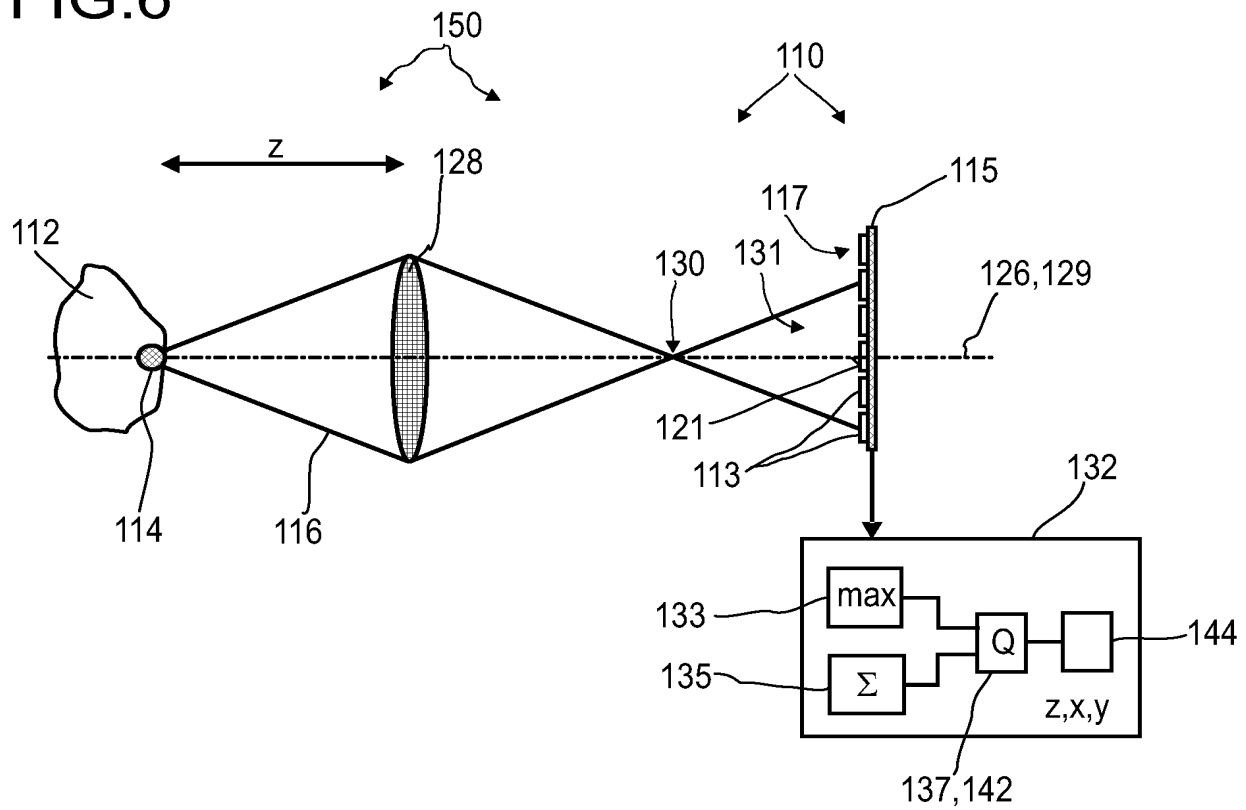


FIG.7

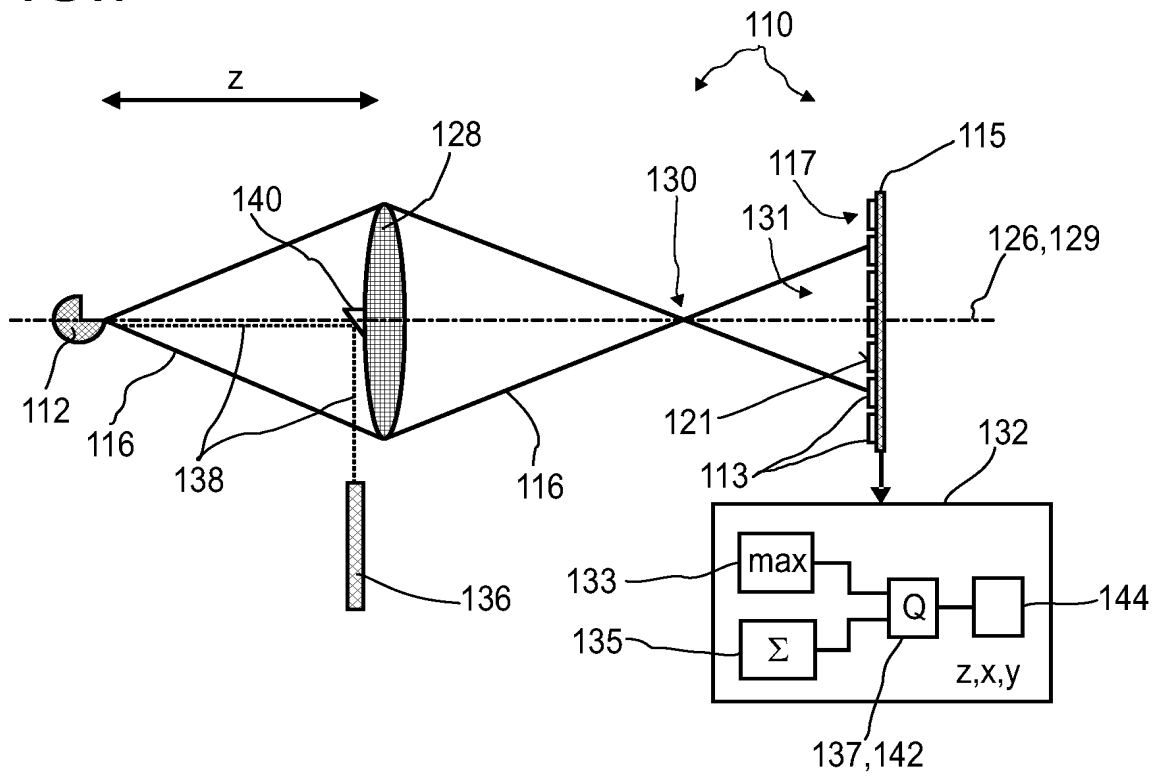


FIG.8

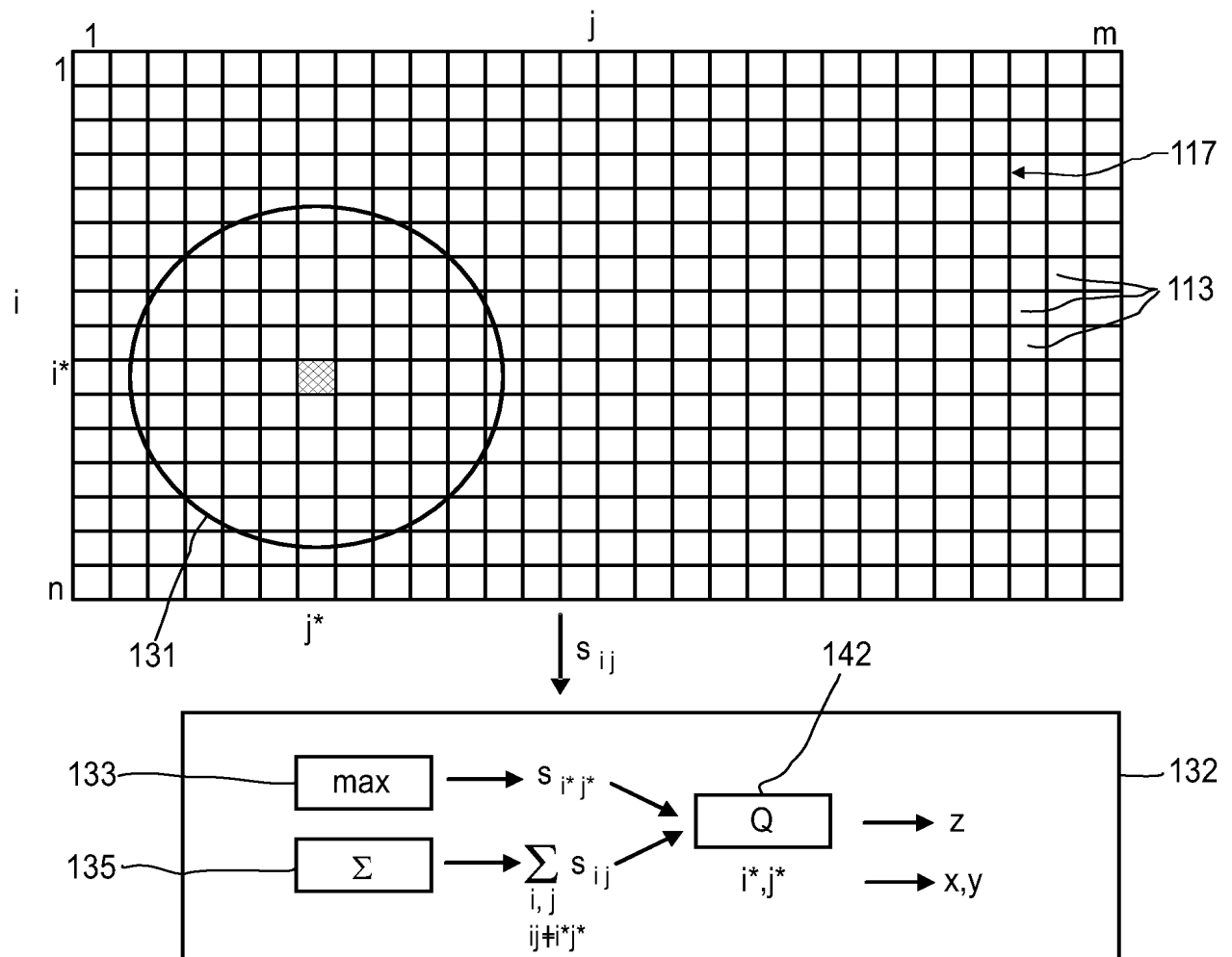


FIG.9

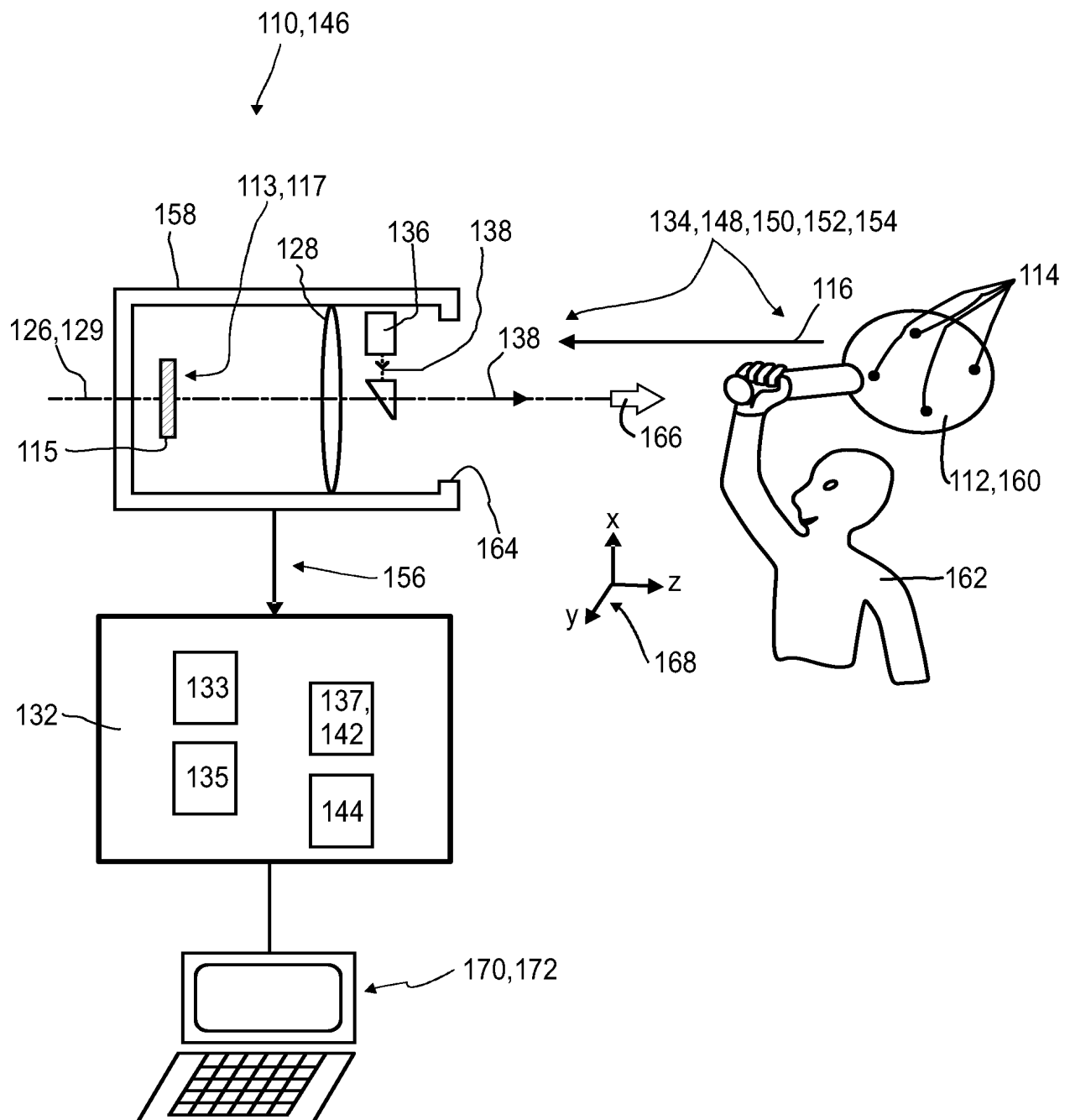


FIG.10

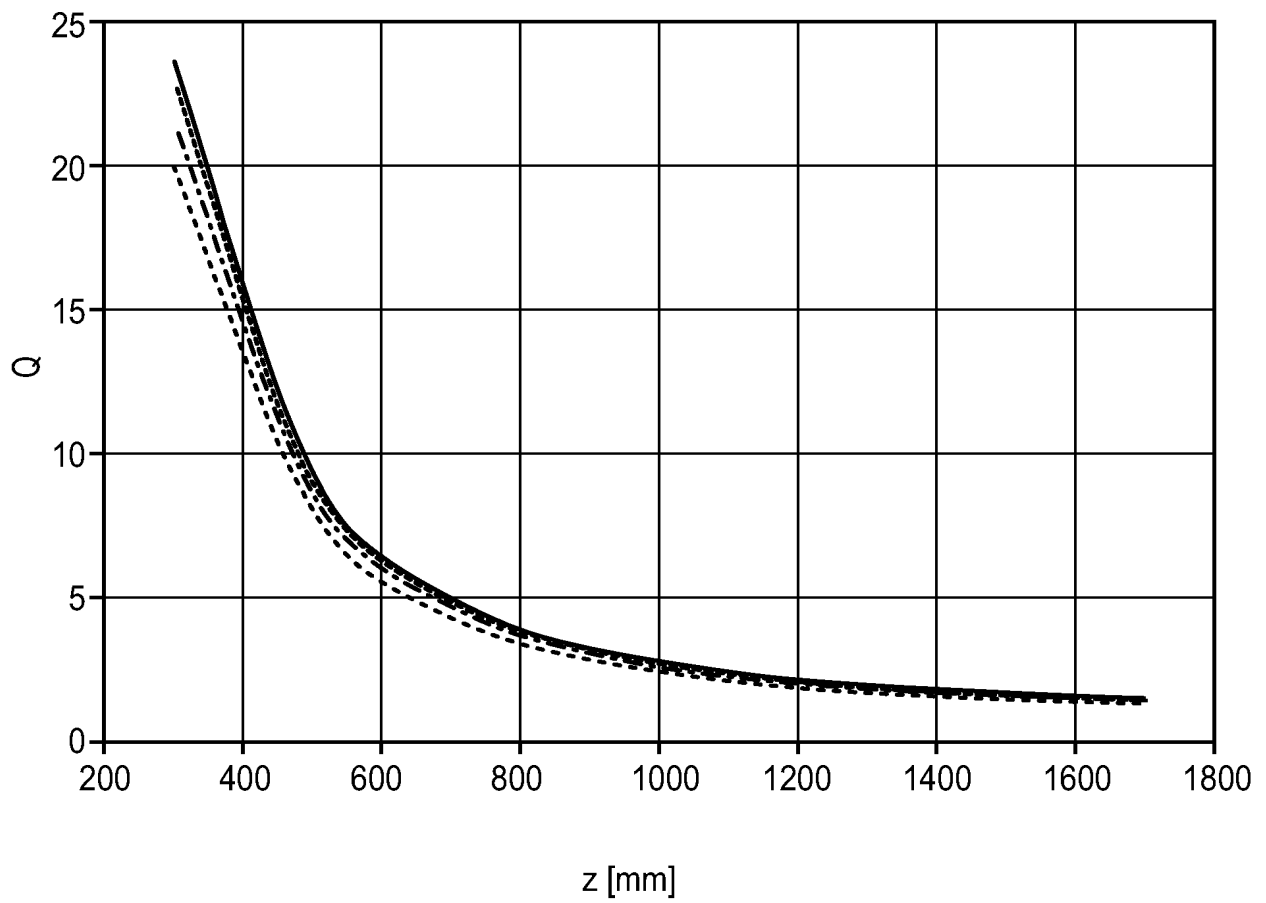


FIG.11

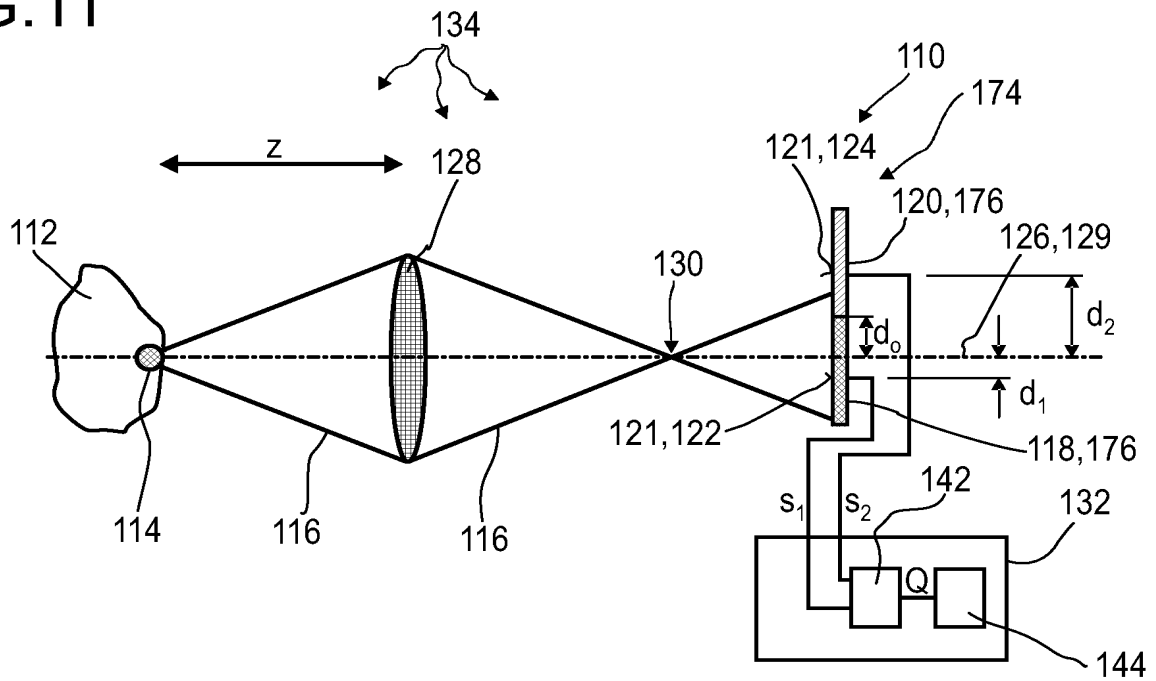


FIG.12

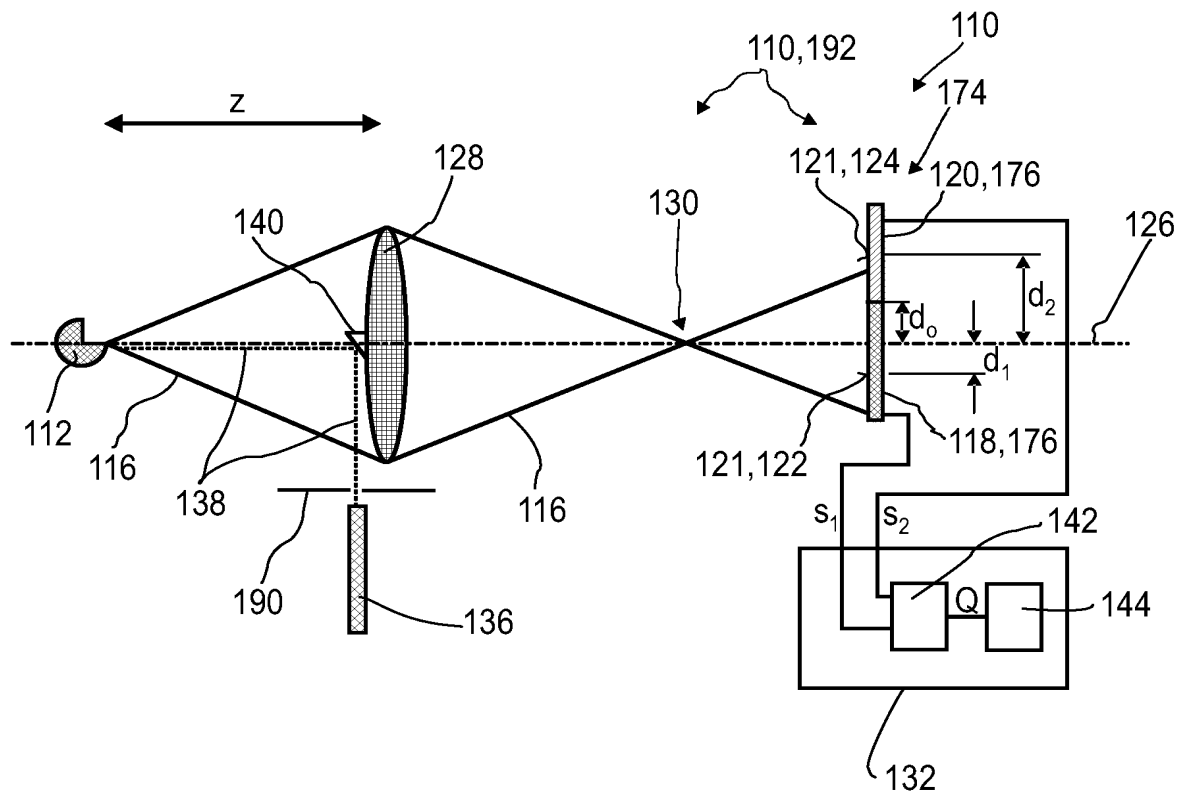


FIG.13

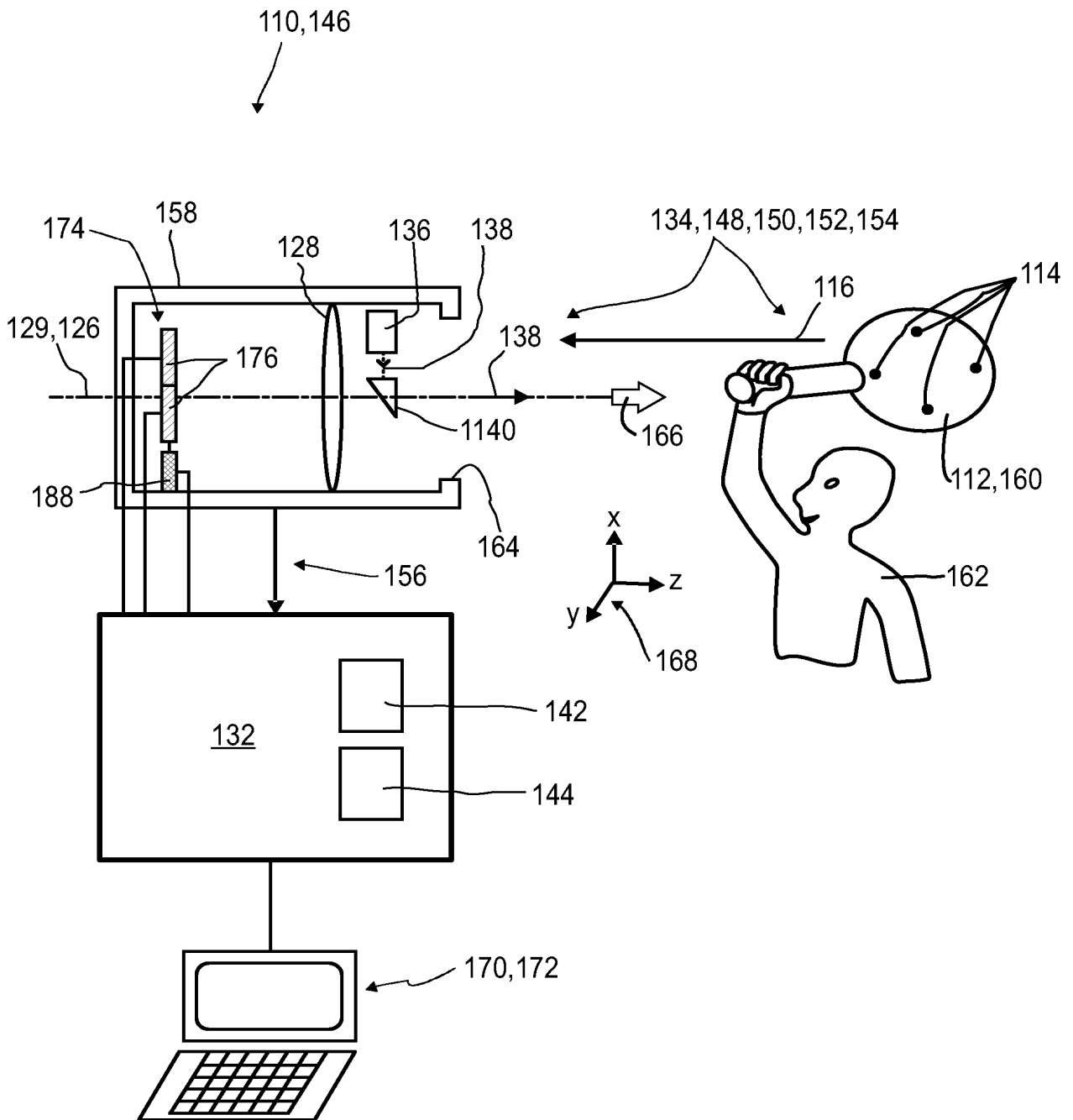


FIG.14

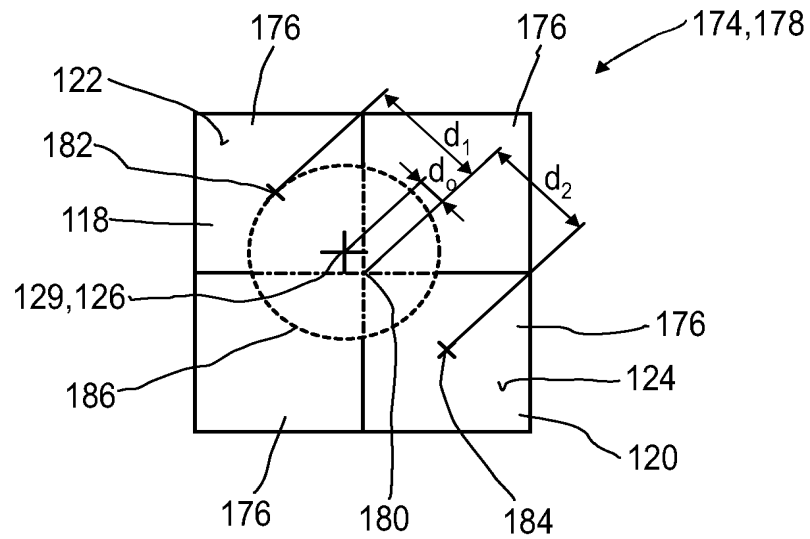


FIG.15

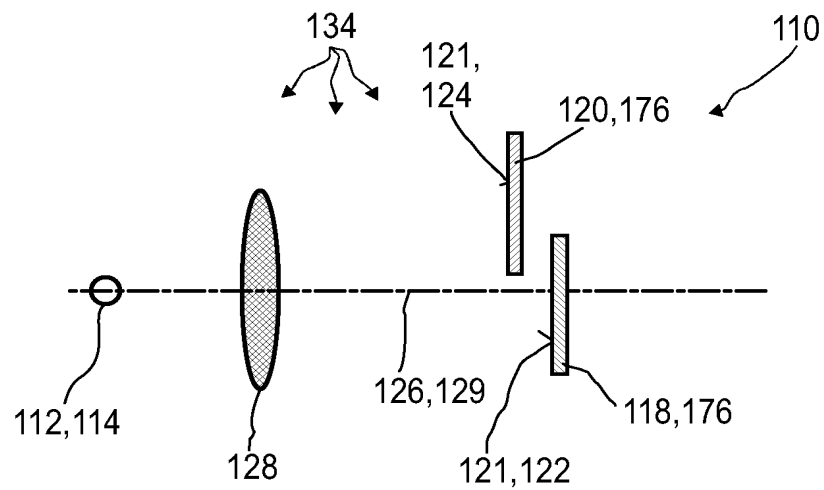


FIG.16

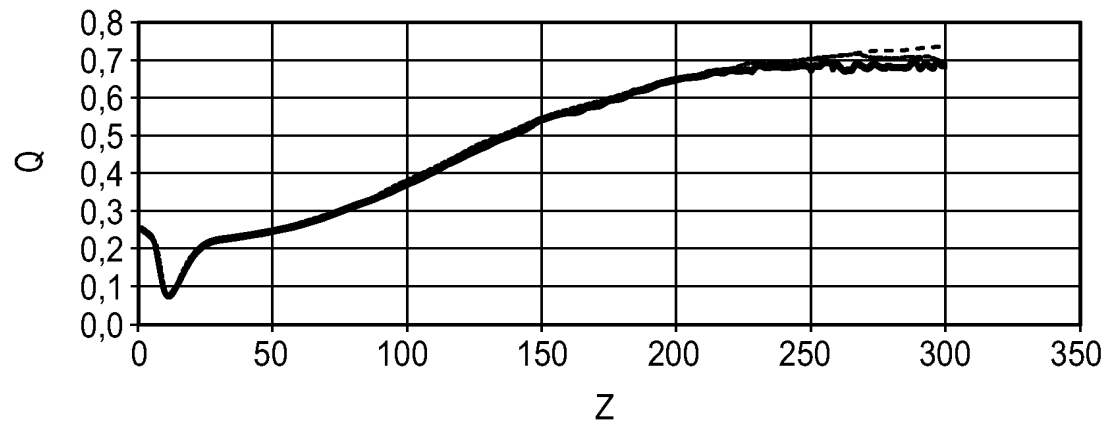


FIG.17

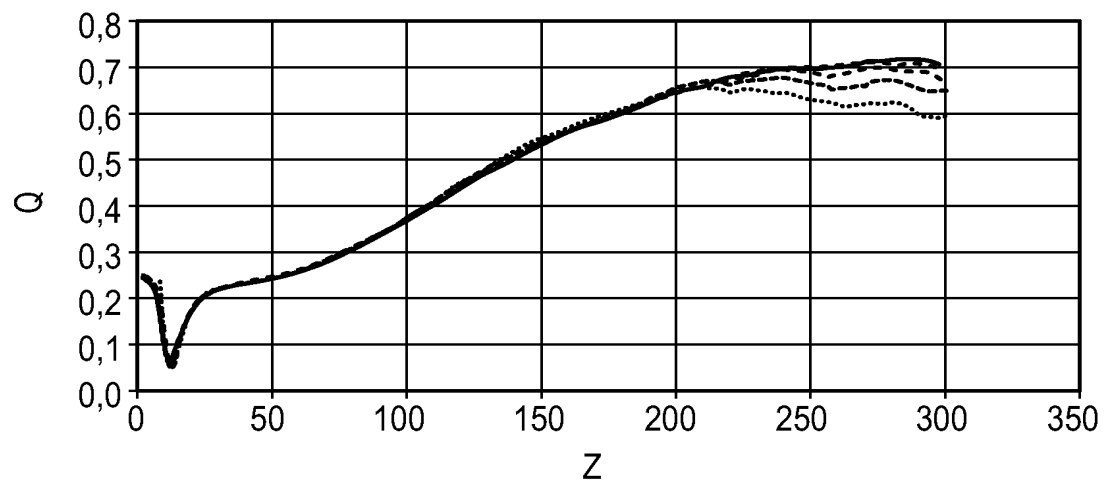


FIG.18

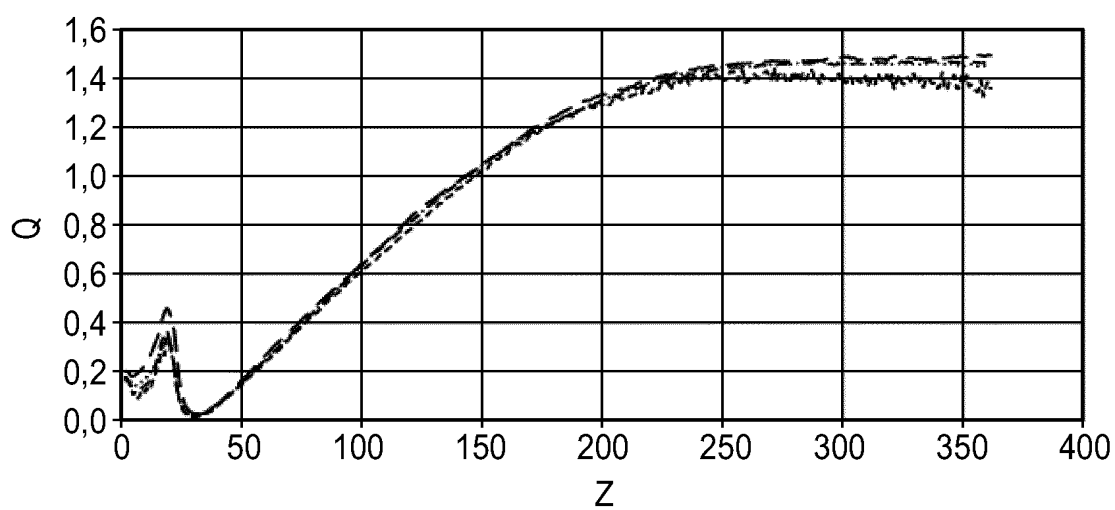


FIG.19A

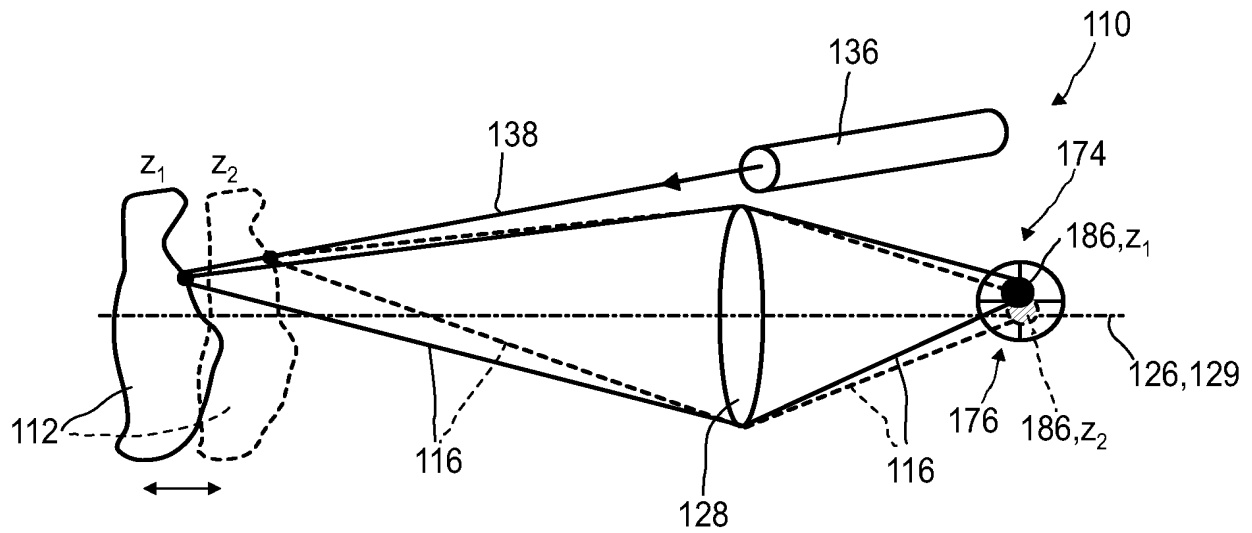


FIG.19B

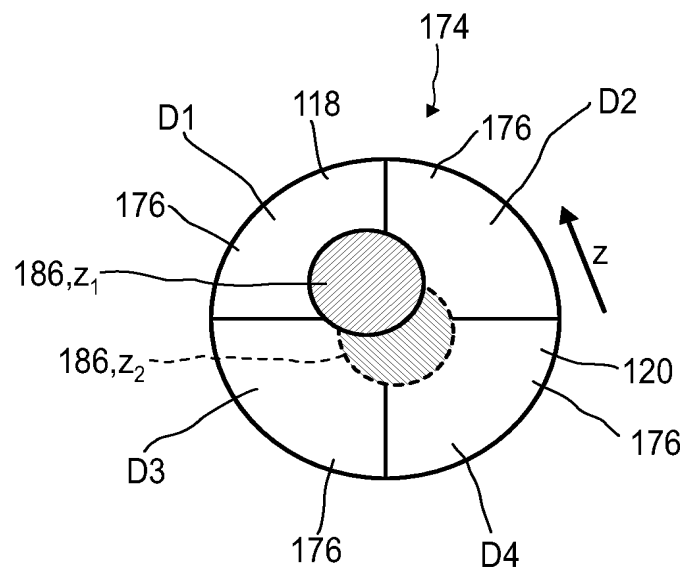


FIG.19C

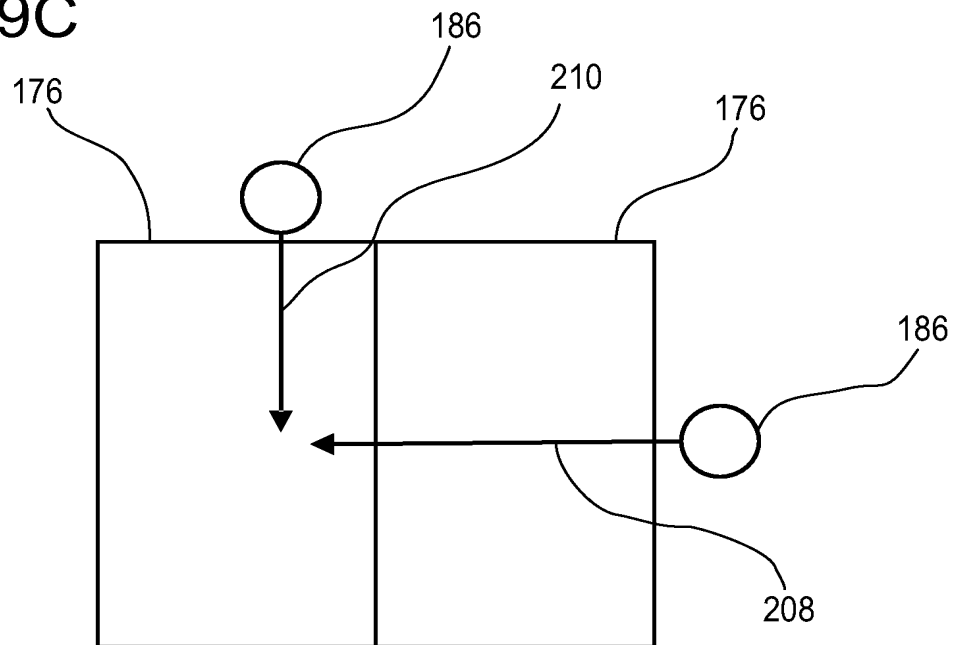


FIG.19D

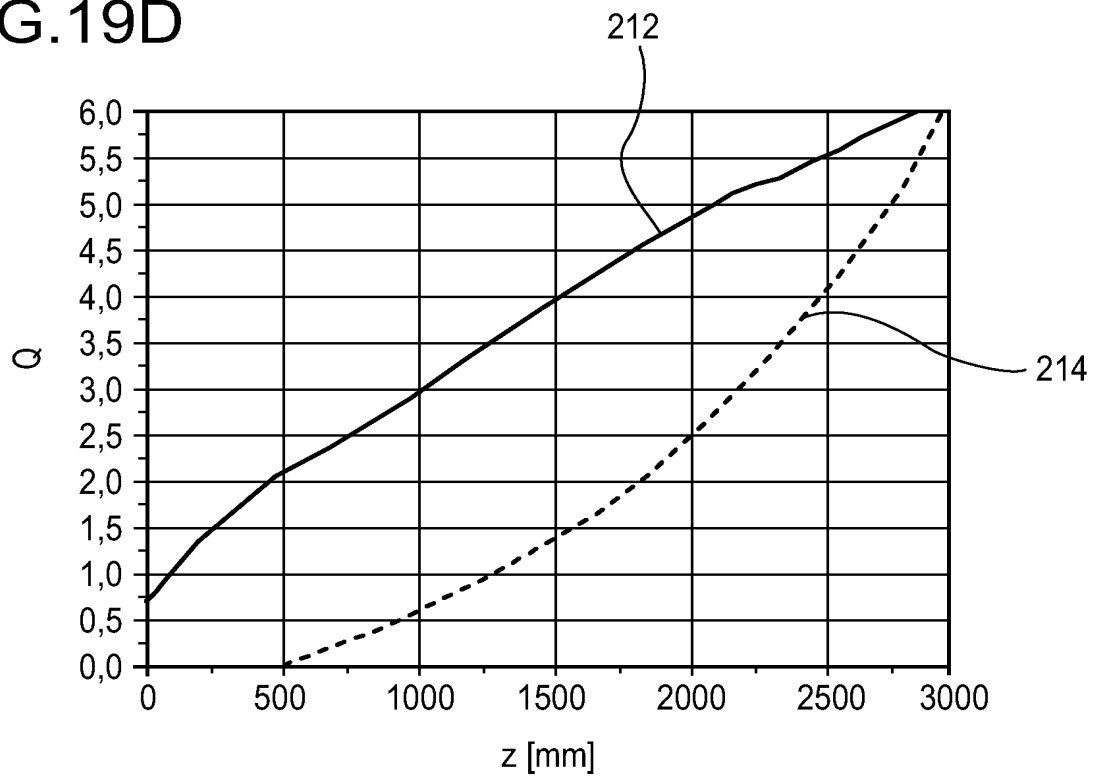


FIG.19E

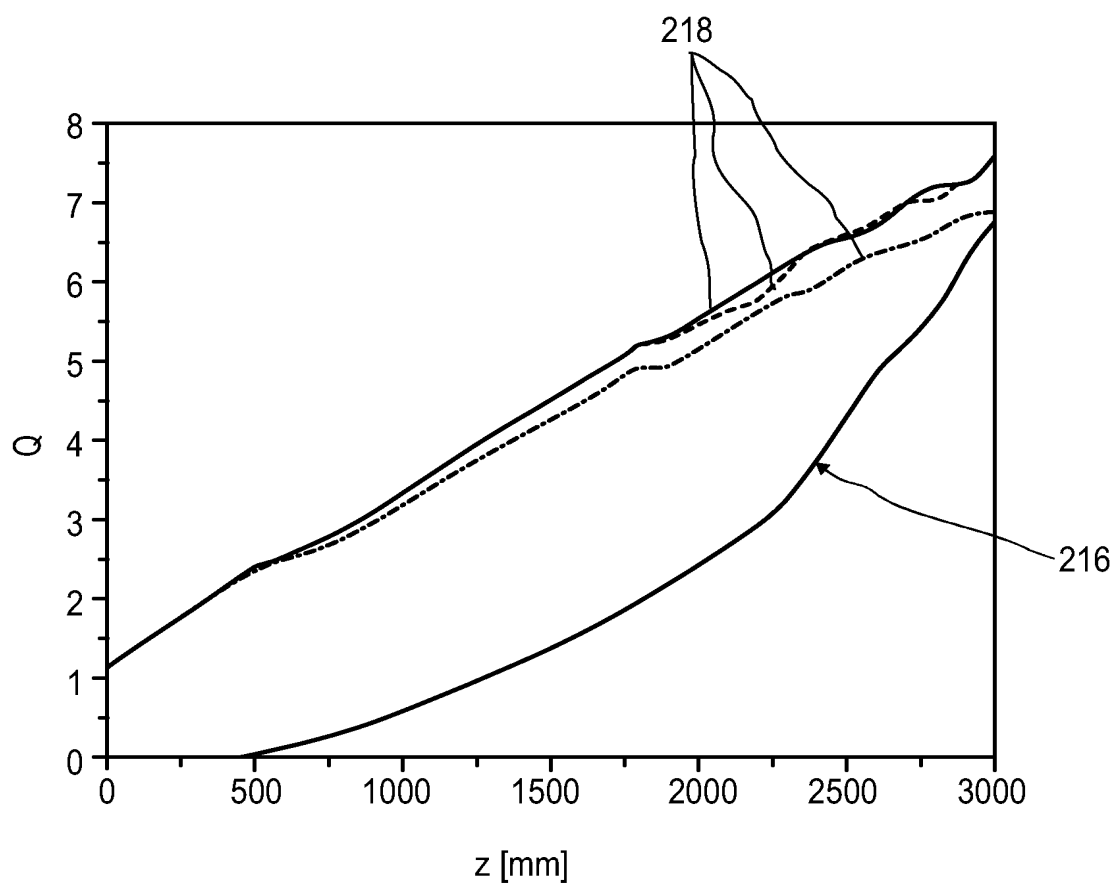


FIG.20

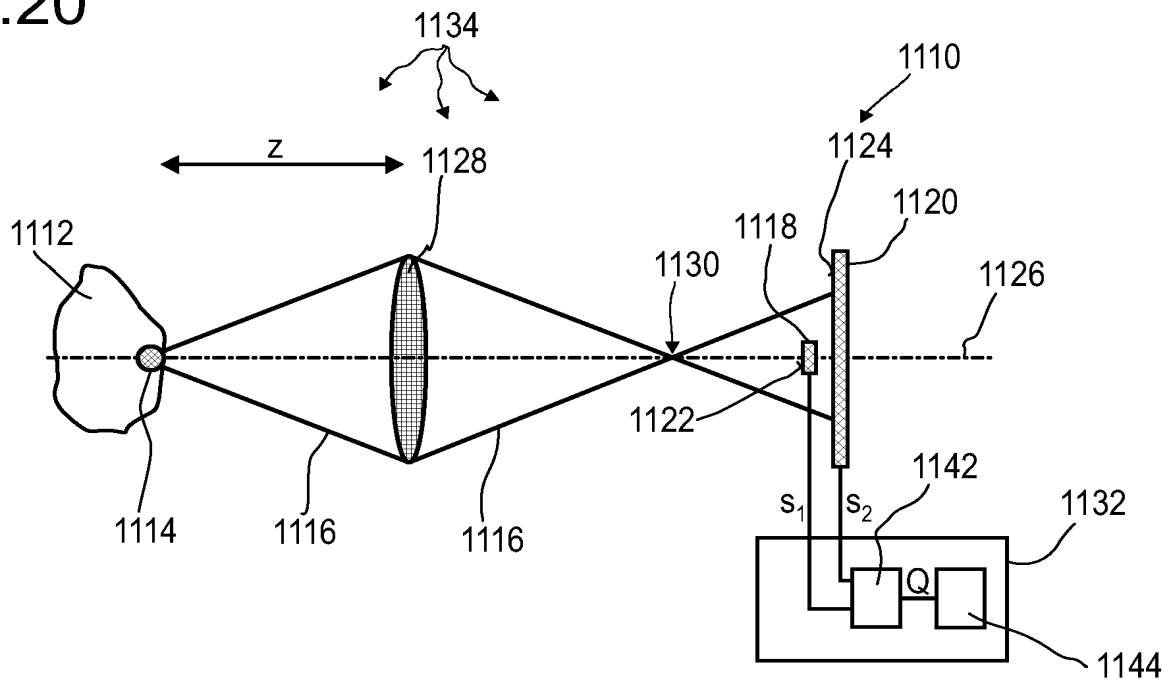


FIG.21

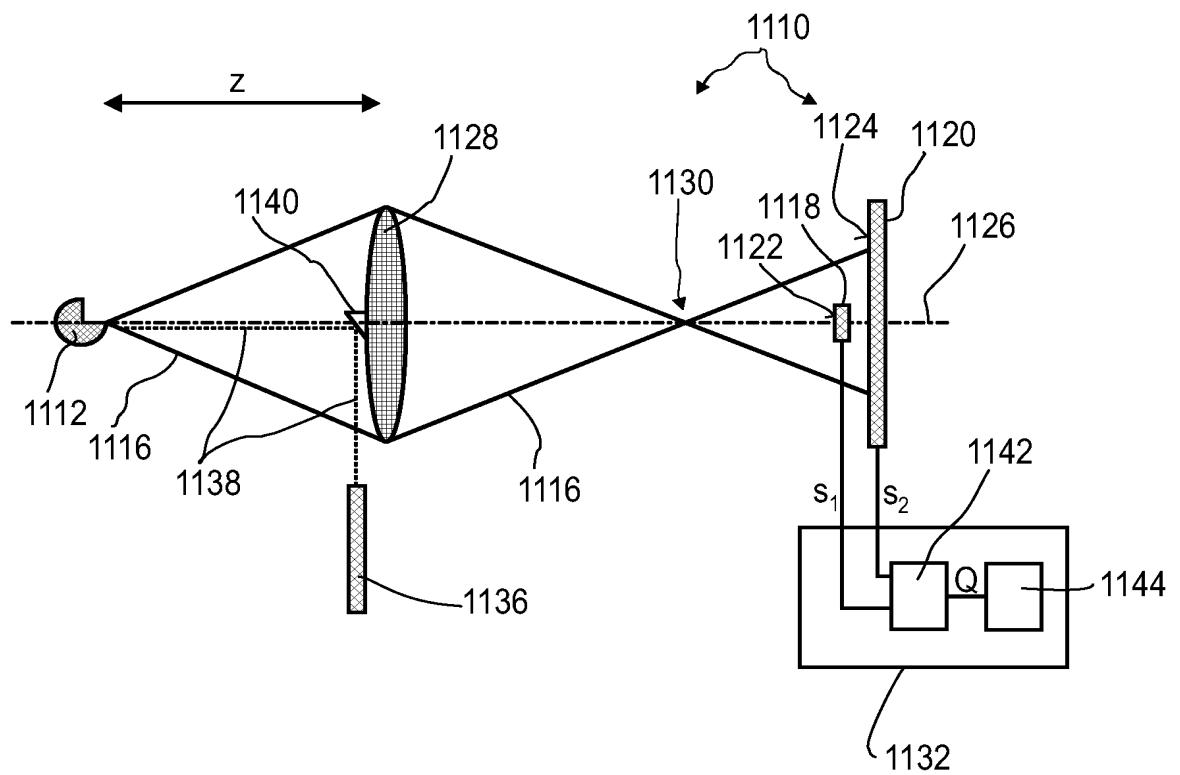


FIG.22A

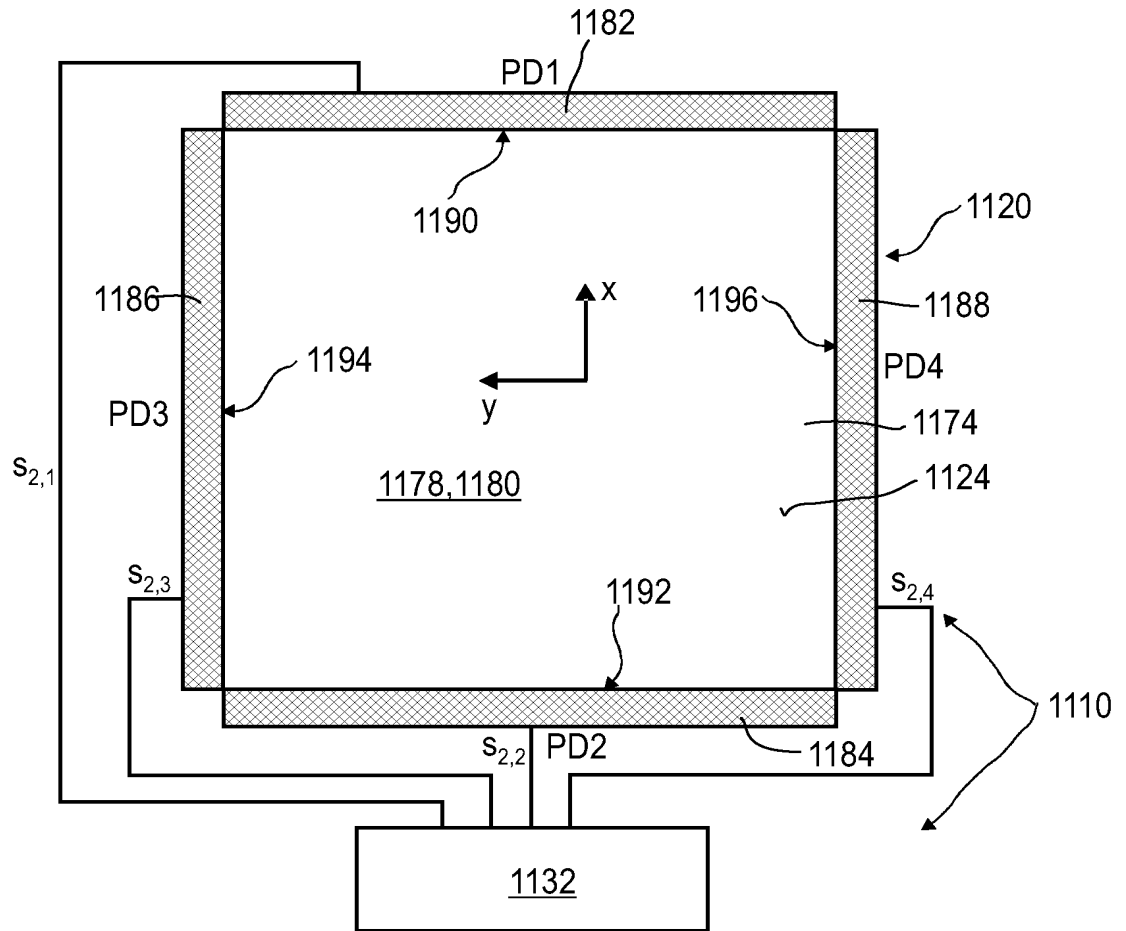


FIG.22B

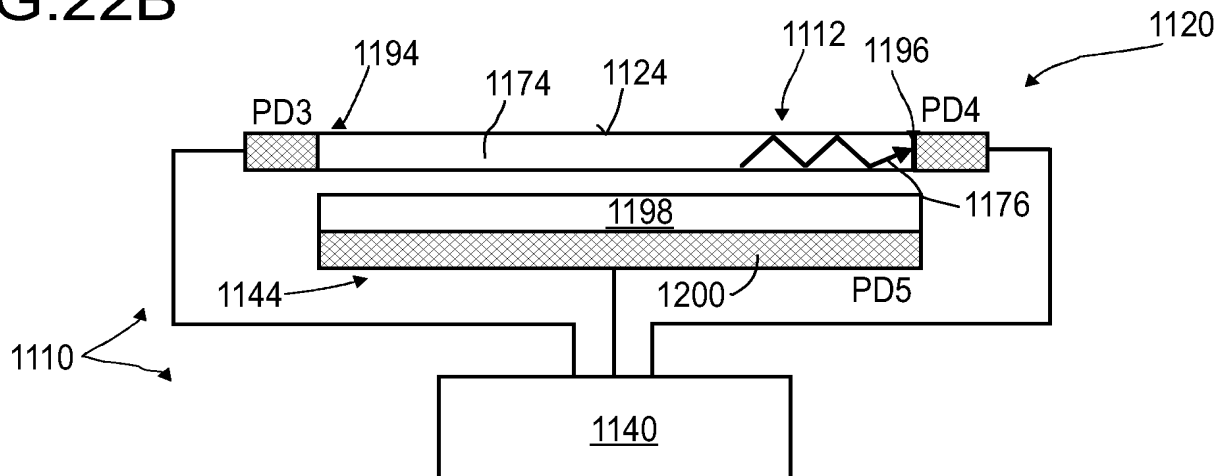


FIG.23

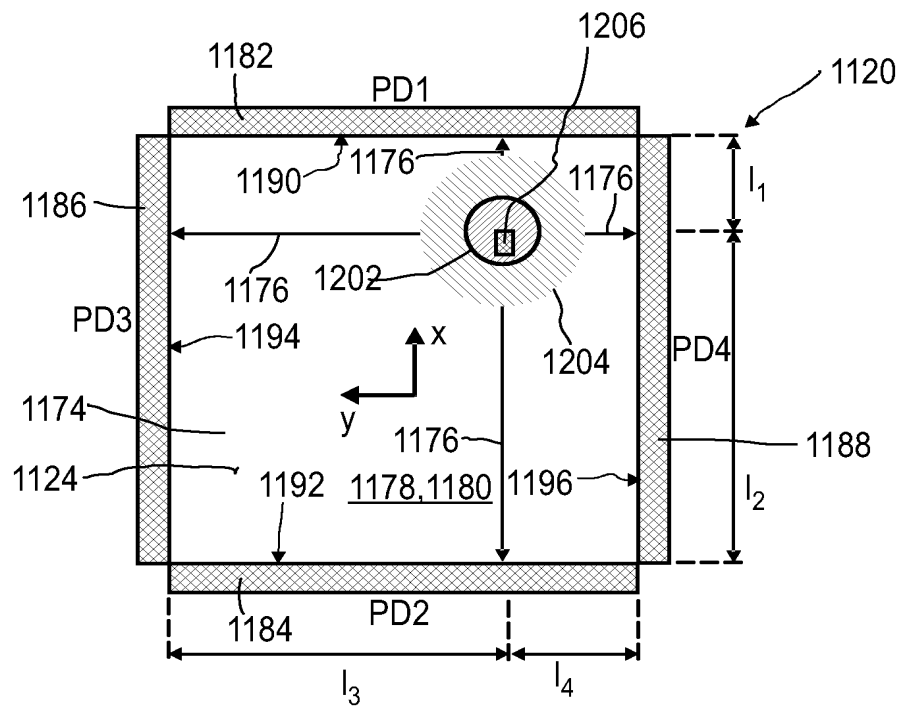


FIG.24

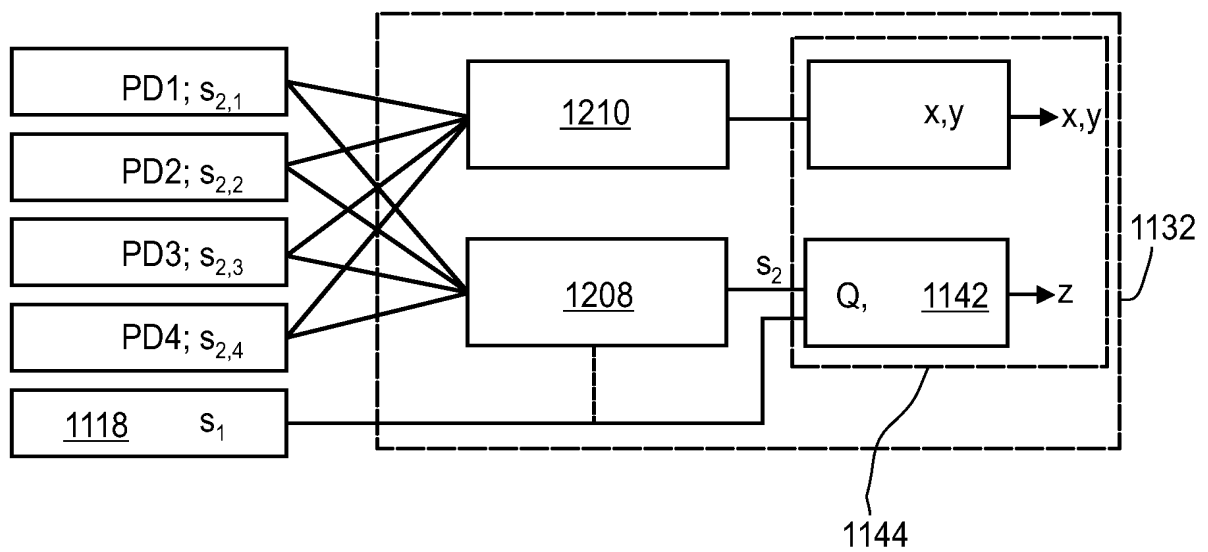


FIG.25

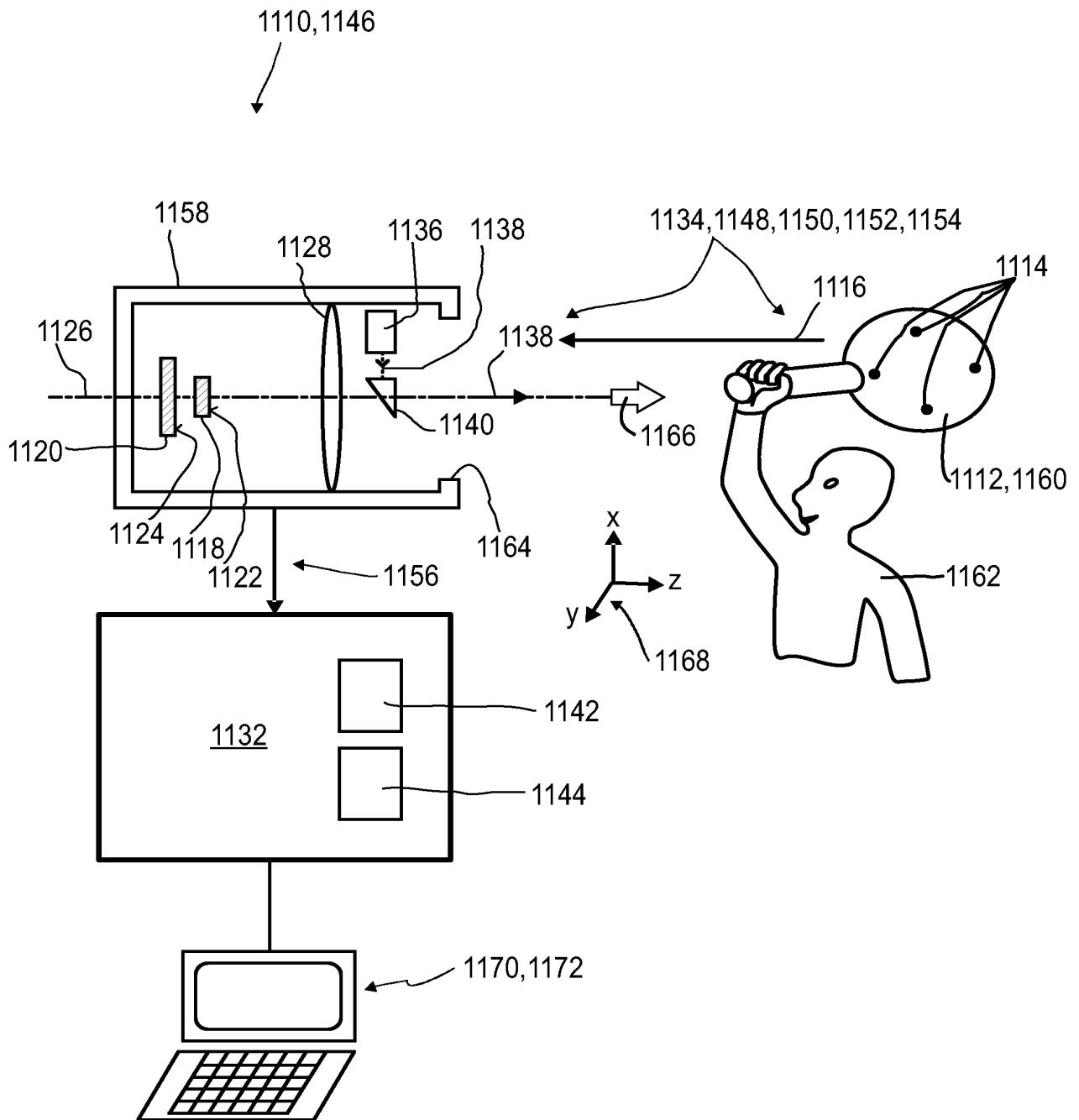


FIG.26A

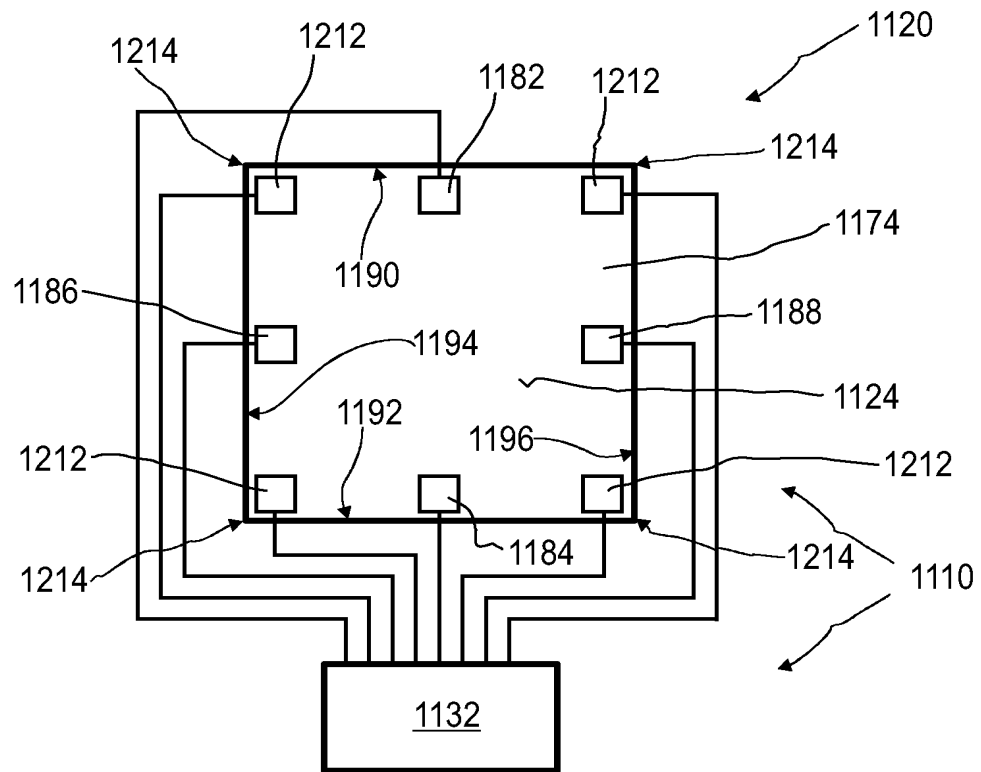


FIG.26B

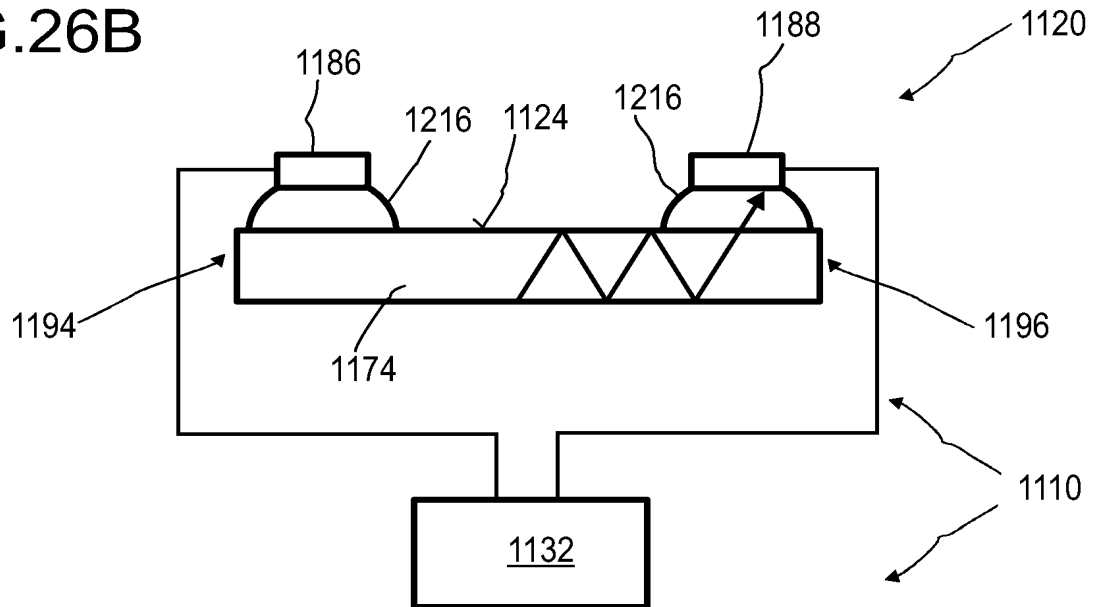


FIG.27A

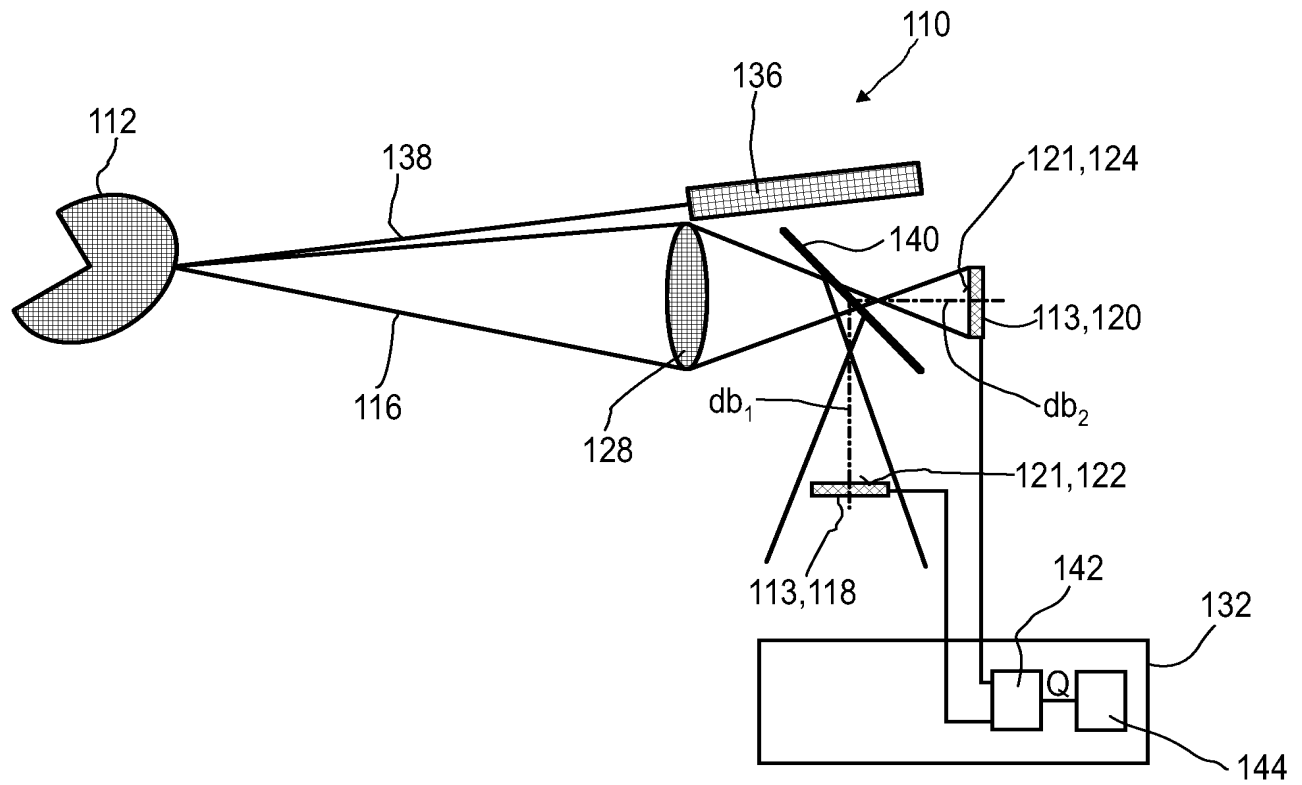


FIG.27B

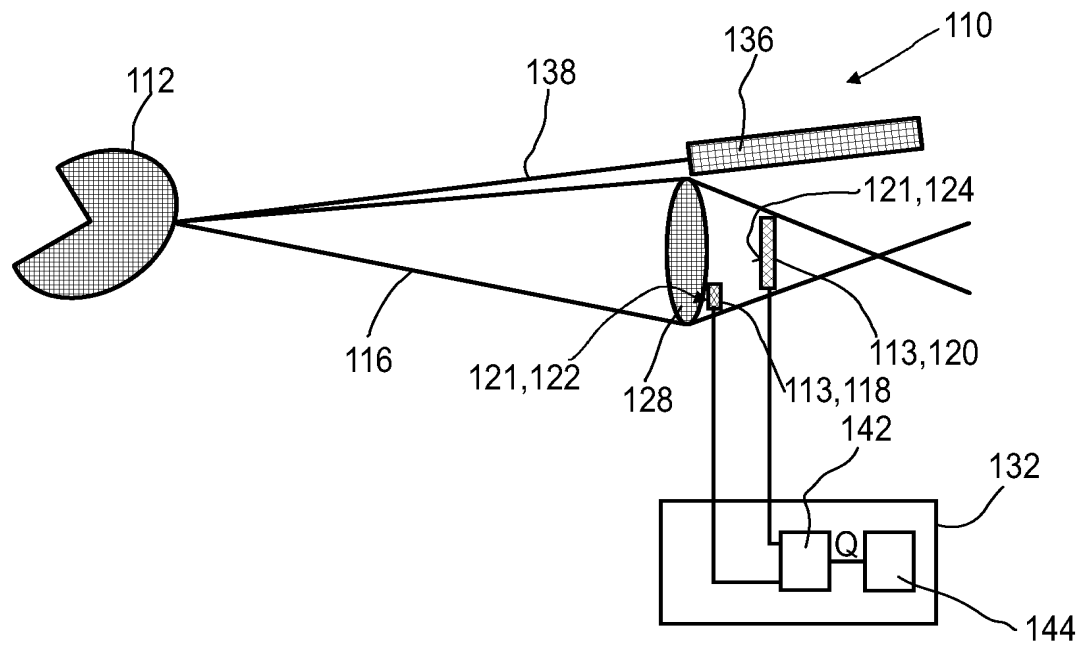


FIG.28

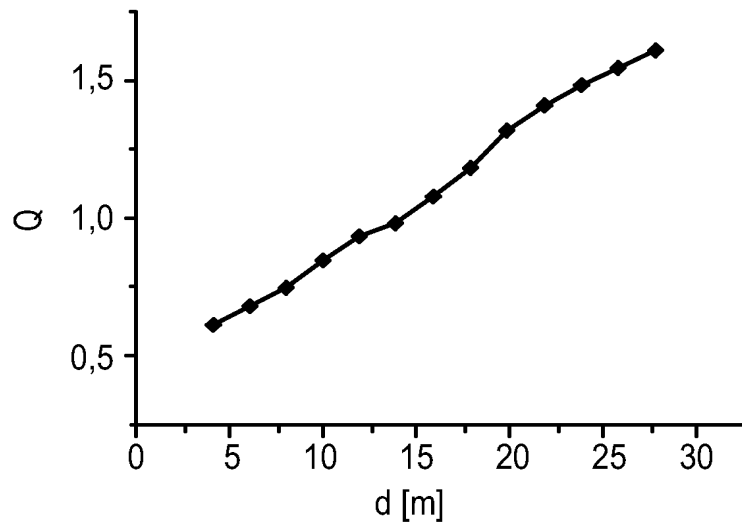


FIG.29

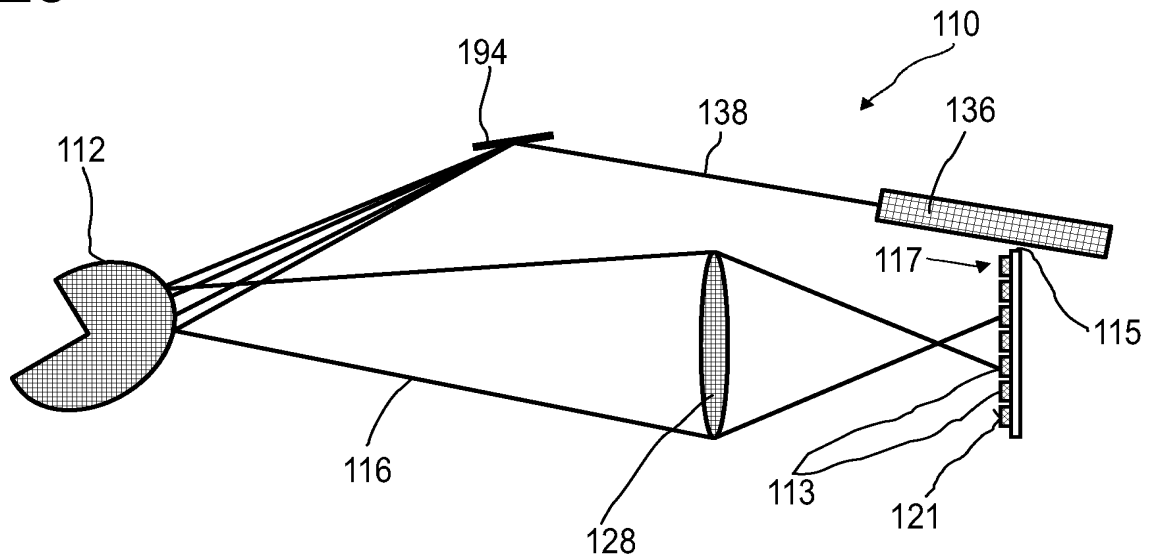


FIG.30

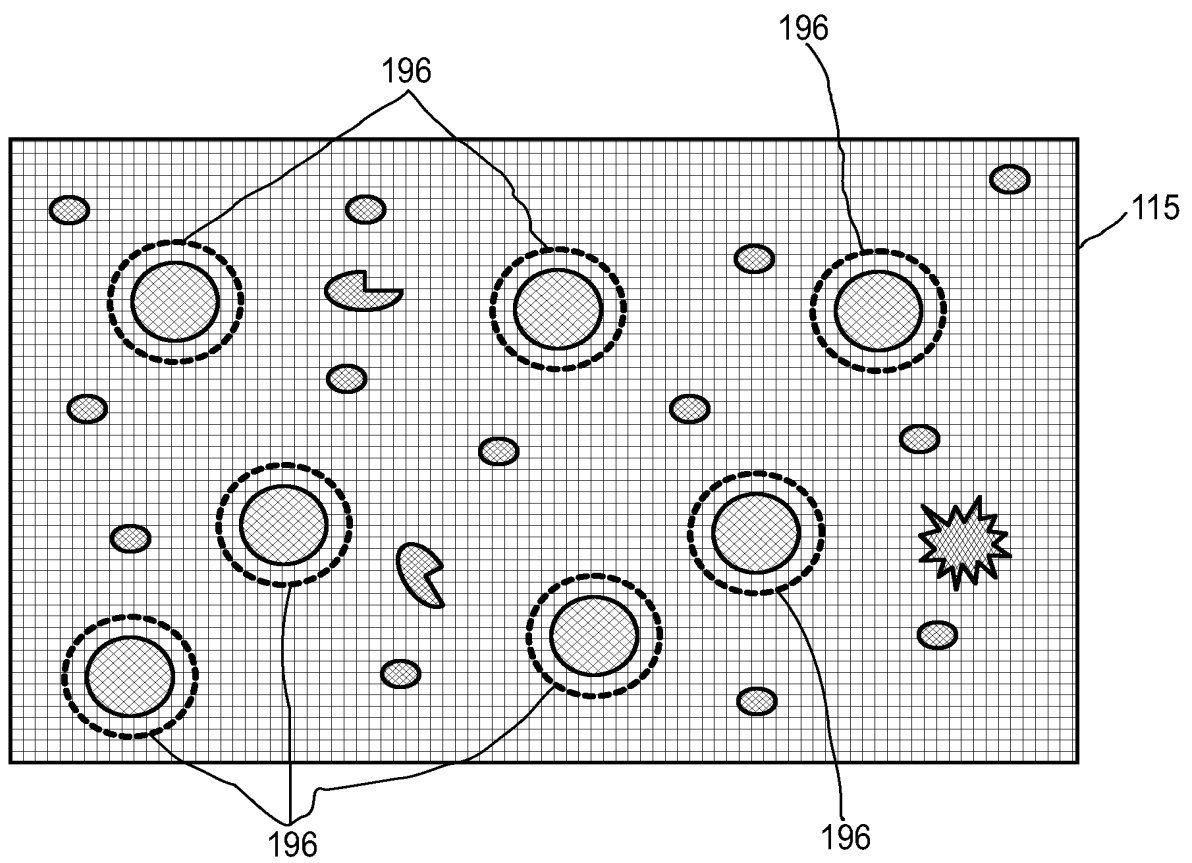


FIG.31

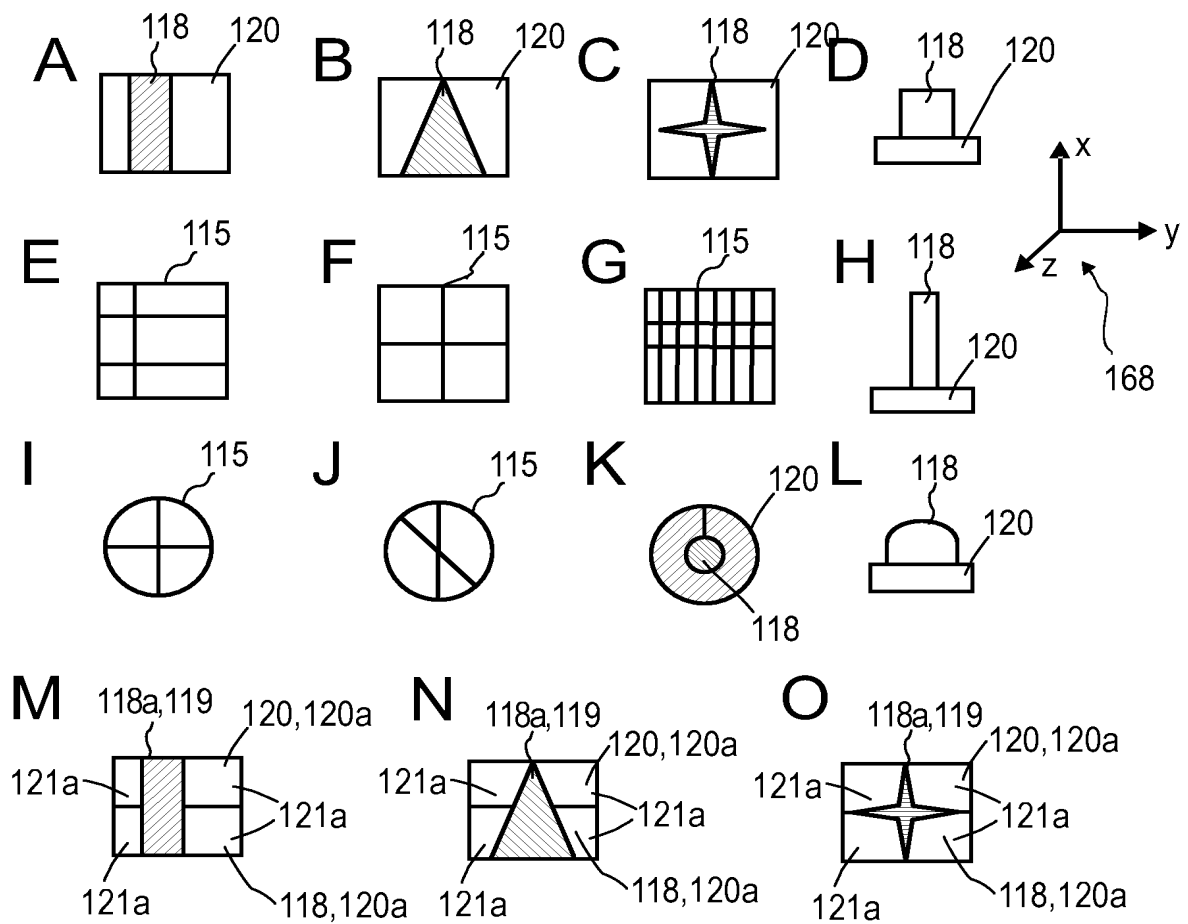


FIG.32

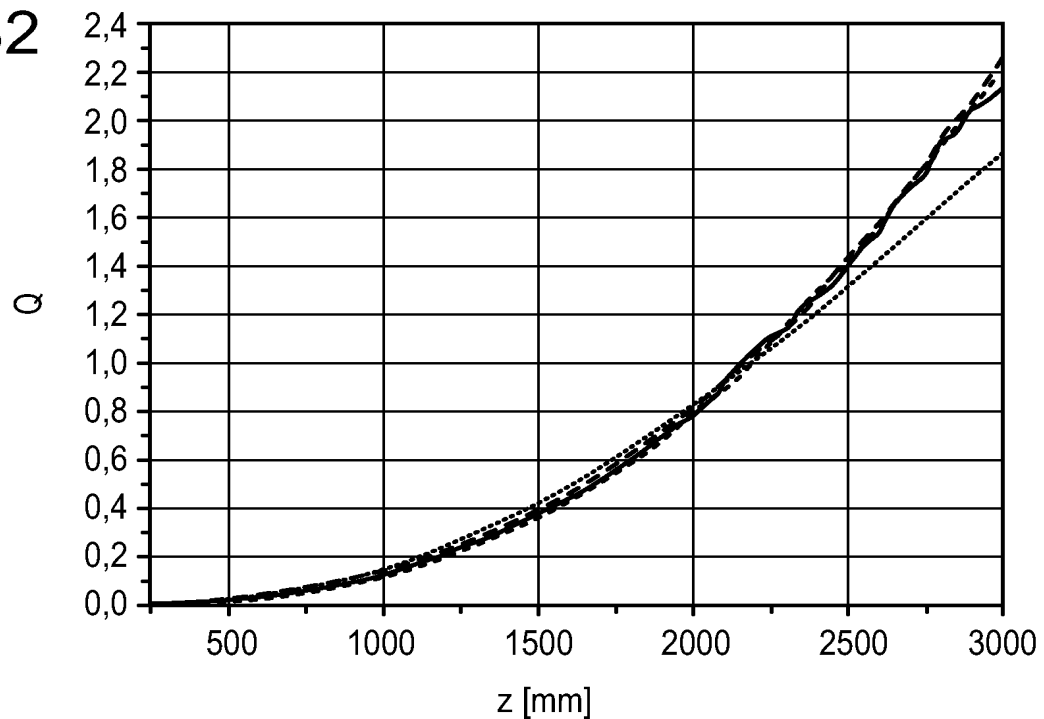


FIG.33A

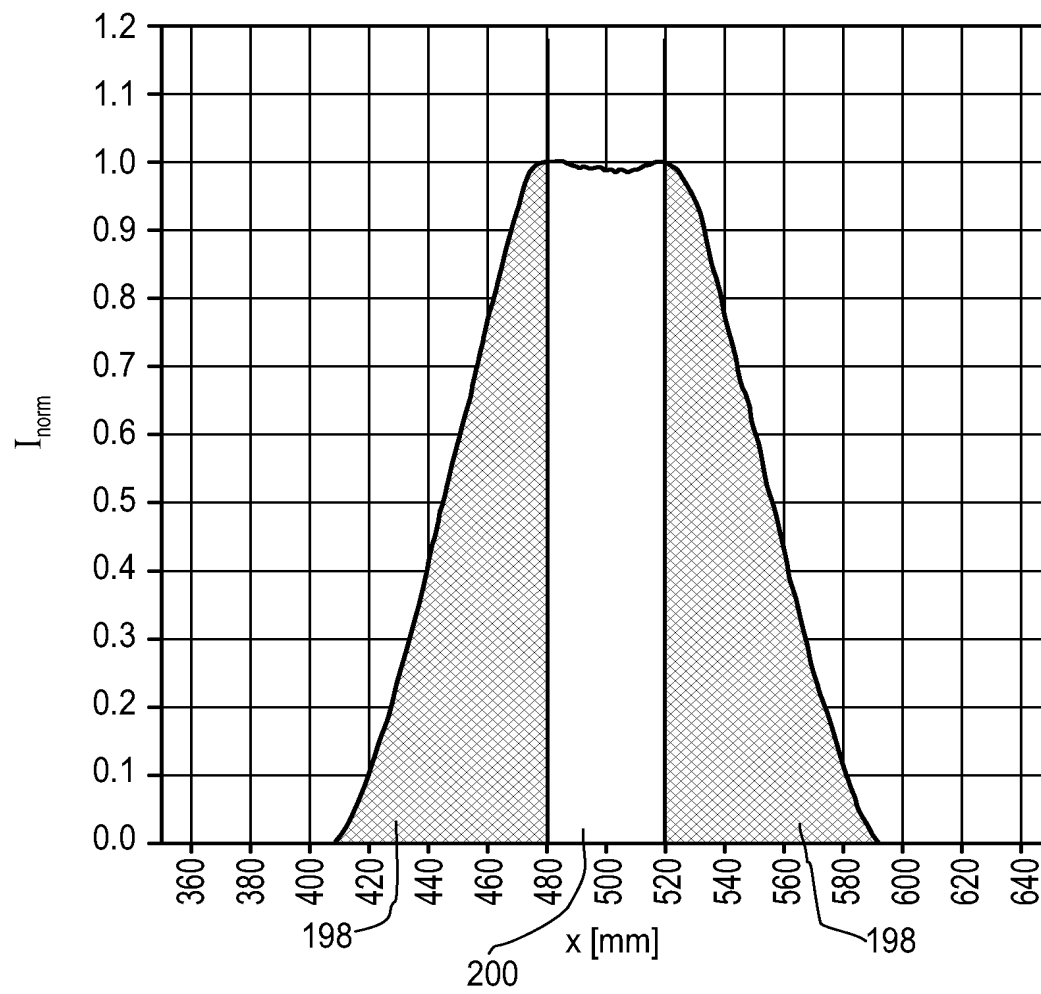


FIG.33B

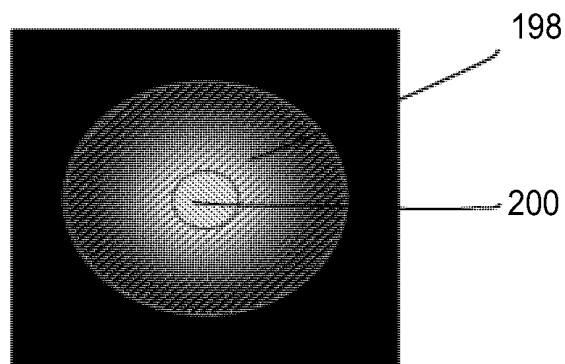


FIG.34

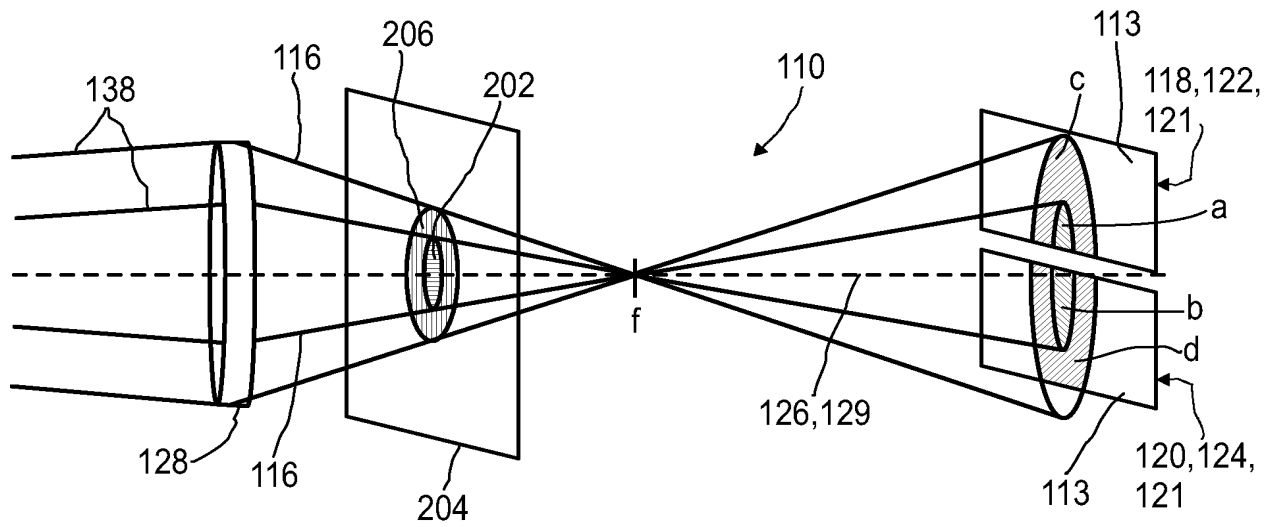


FIG.35

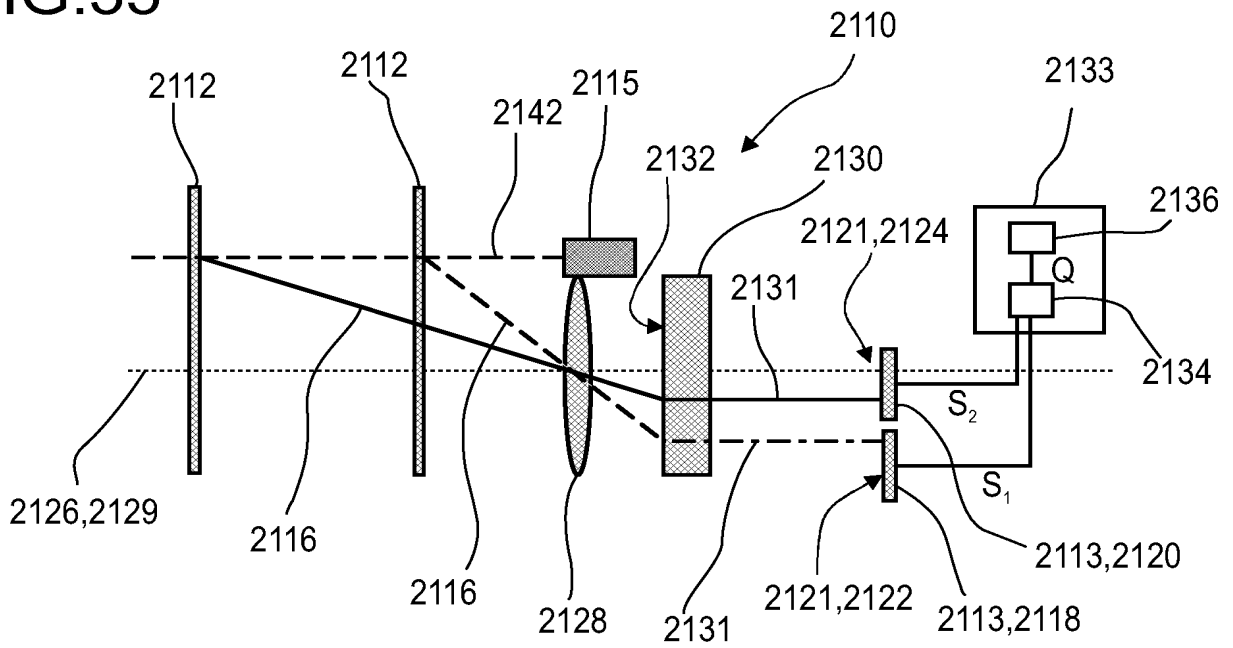


FIG.36

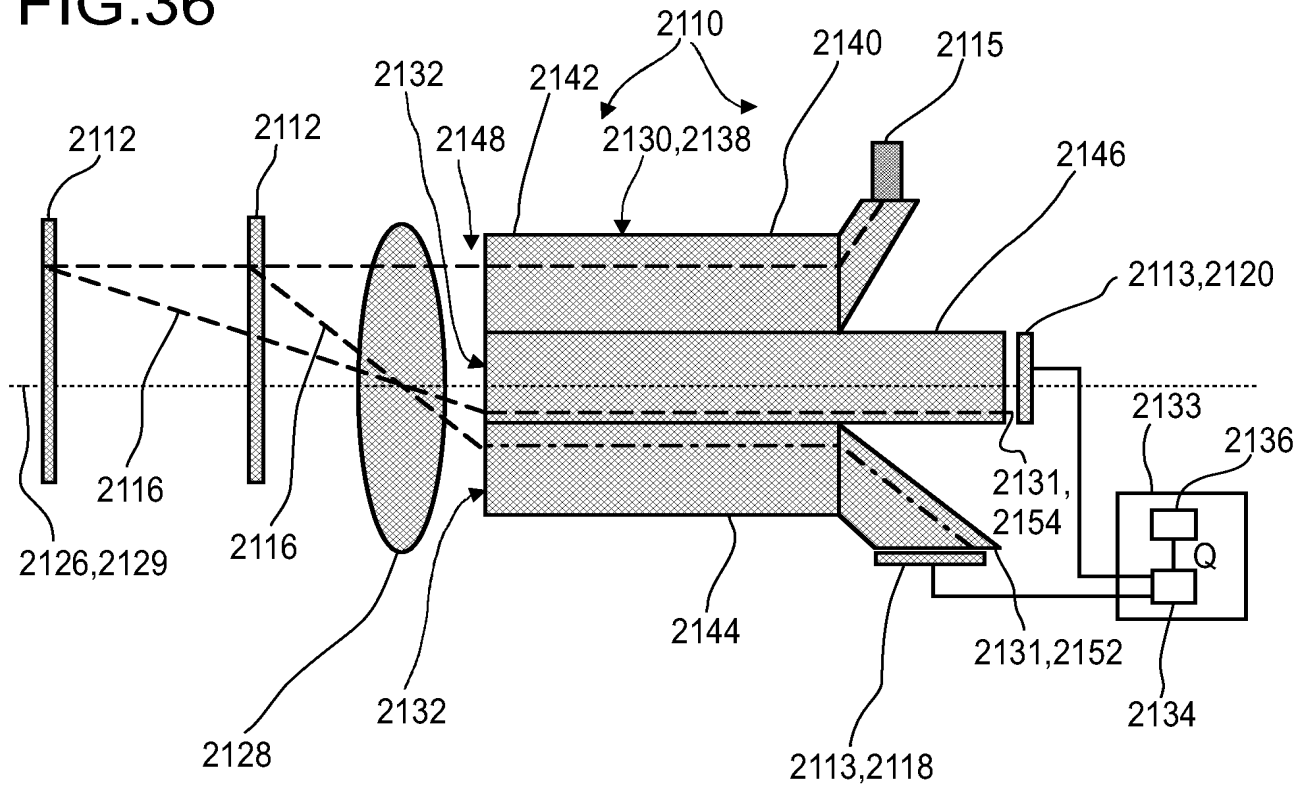


FIG.37

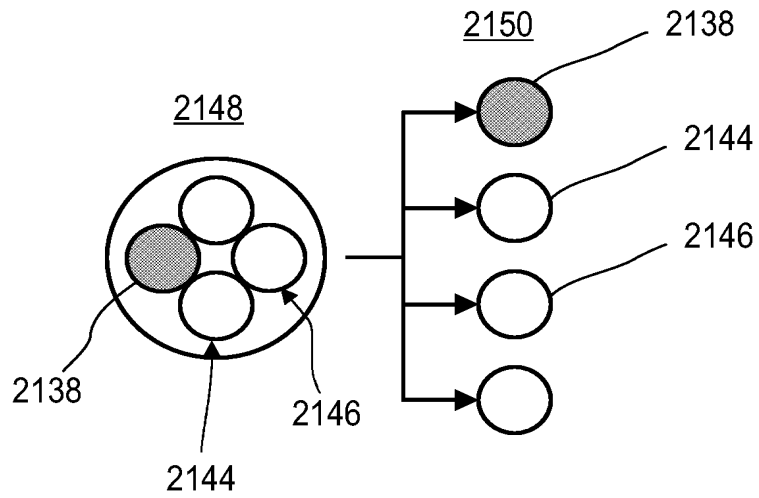


FIG.38

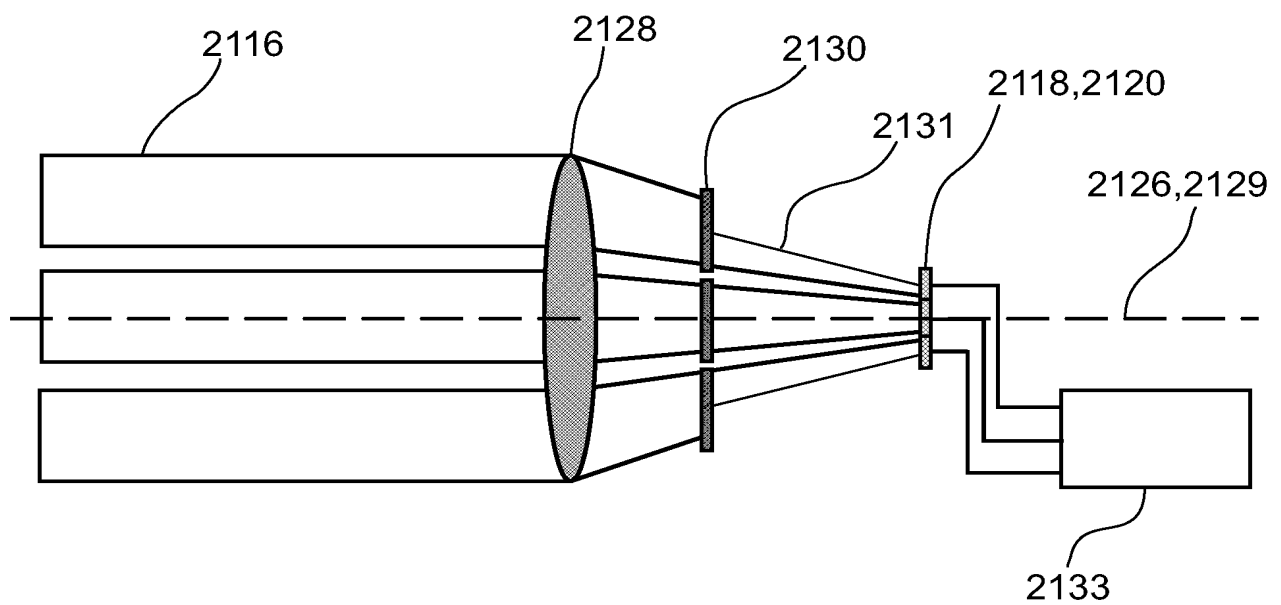


FIG.39

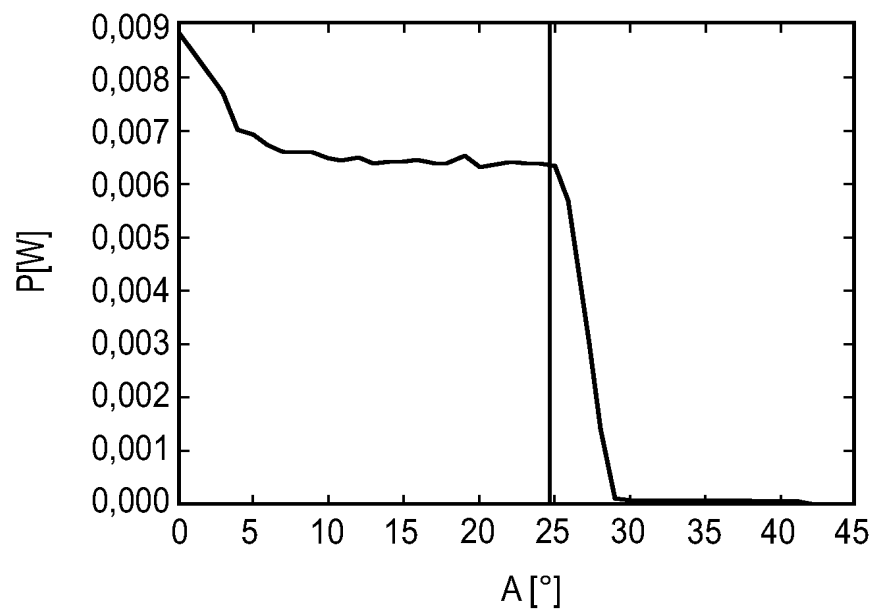


FIG.40A

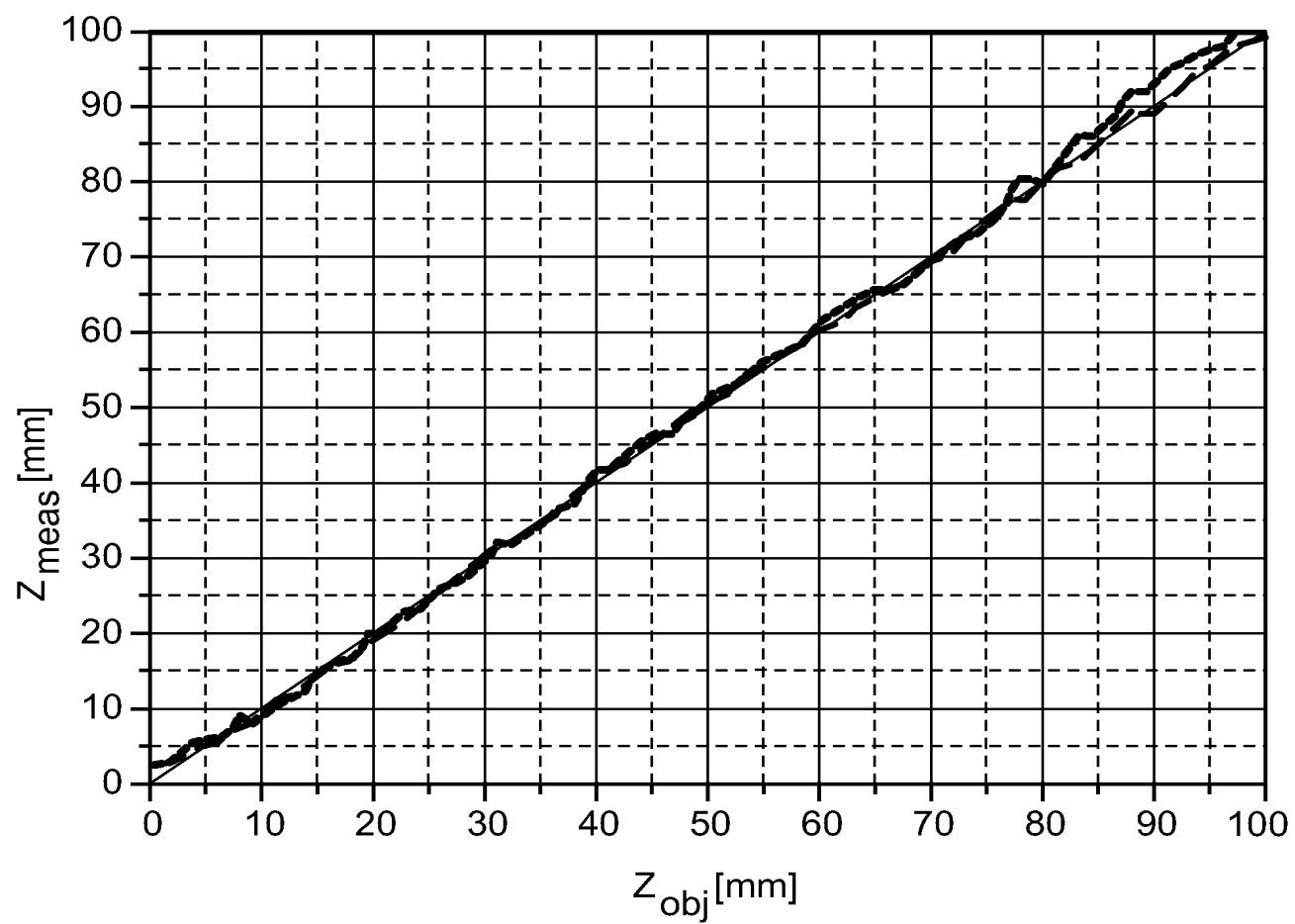


FIG.40B

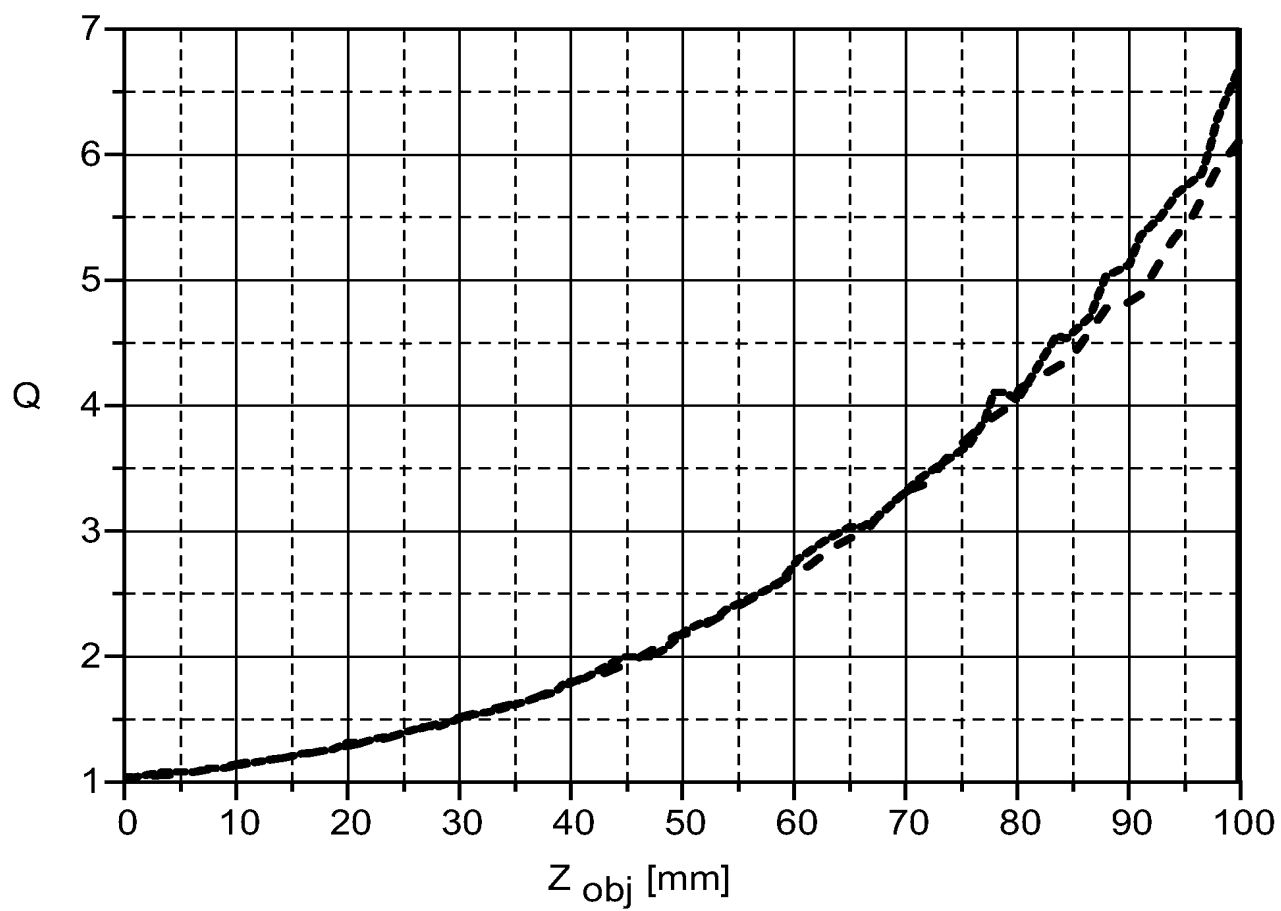


FIG.41

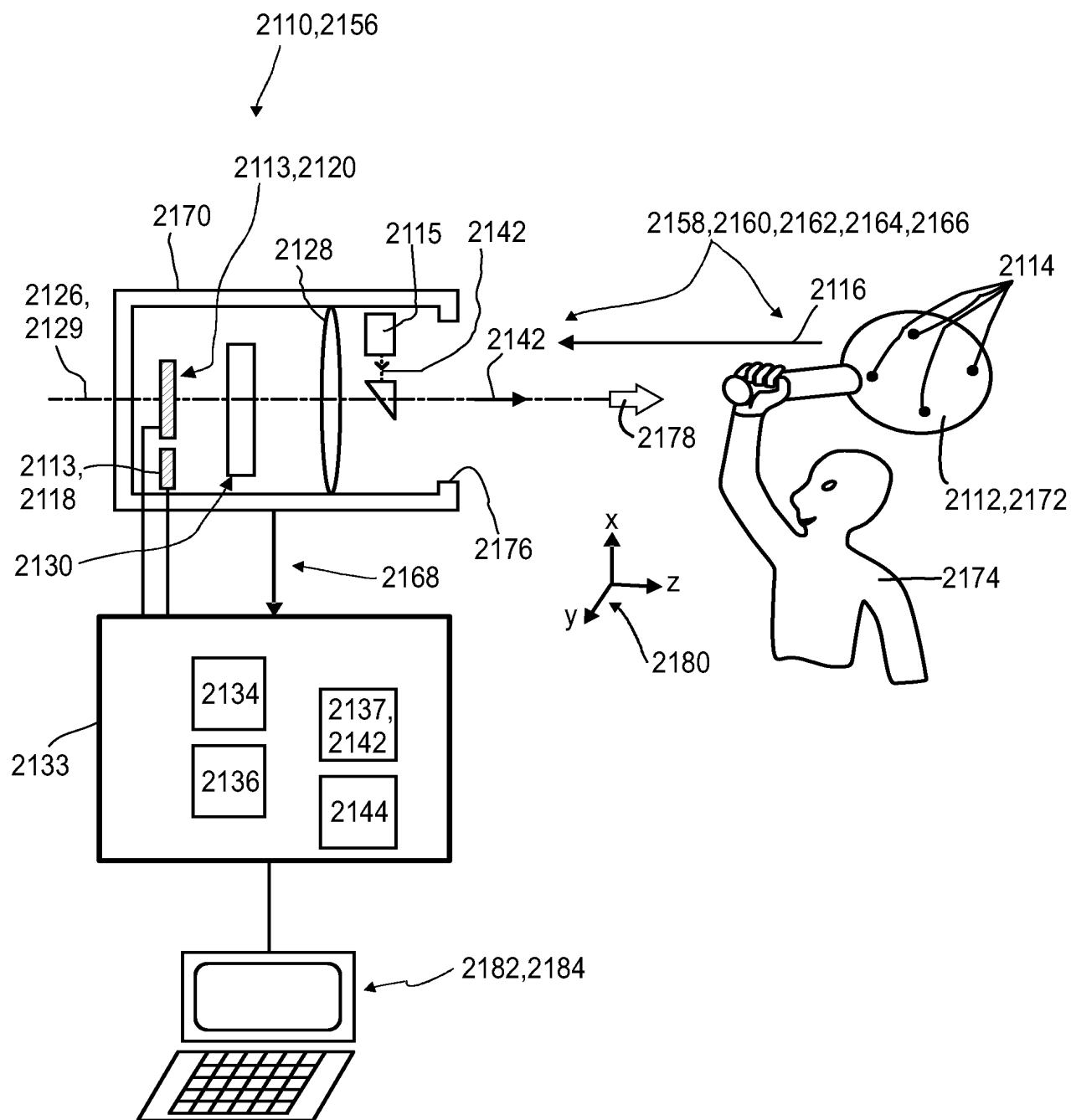


FIG.42

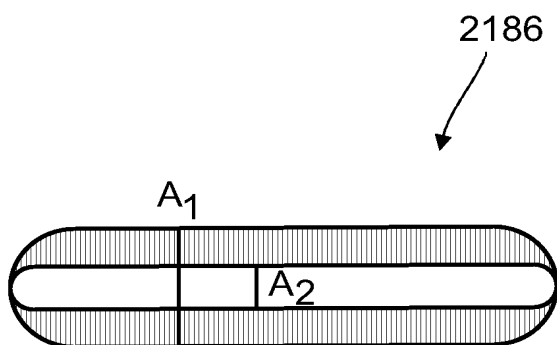


FIG.43

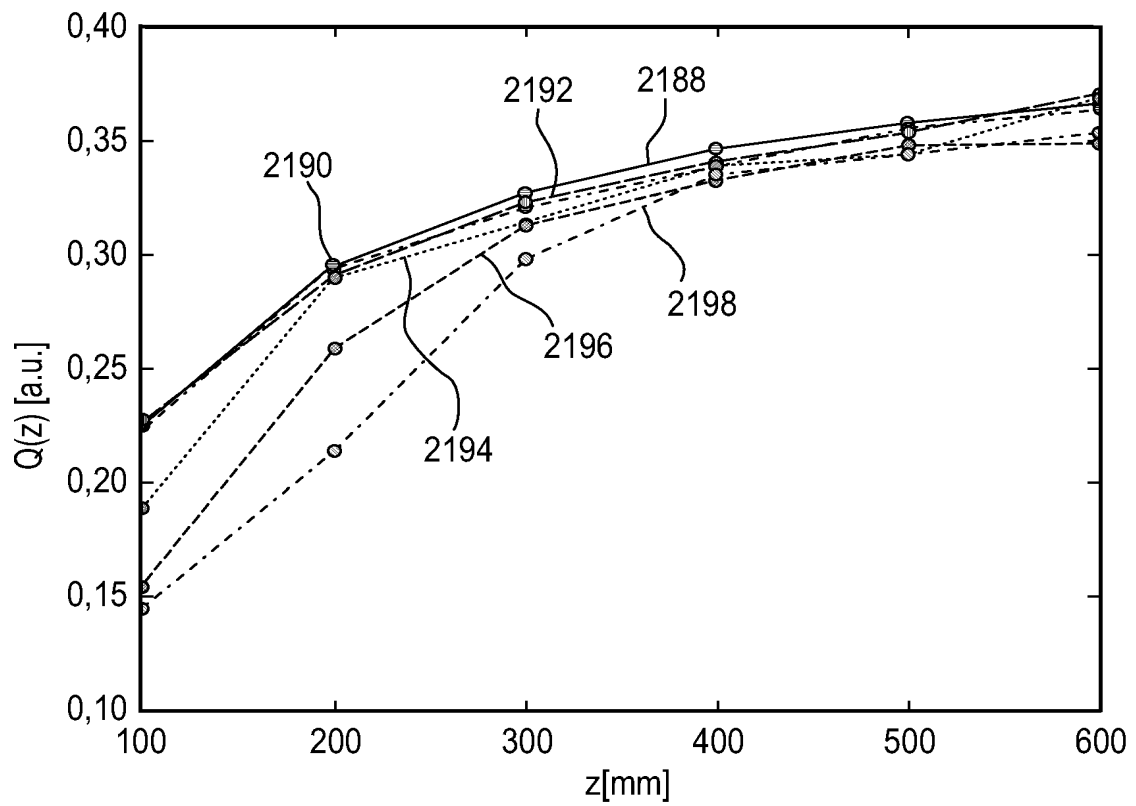


FIG.44A

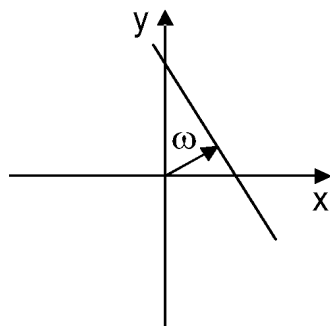


FIG.44B

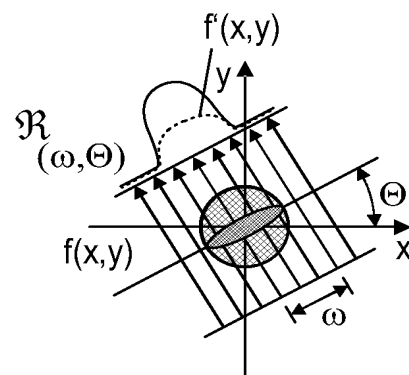


FIG.45A

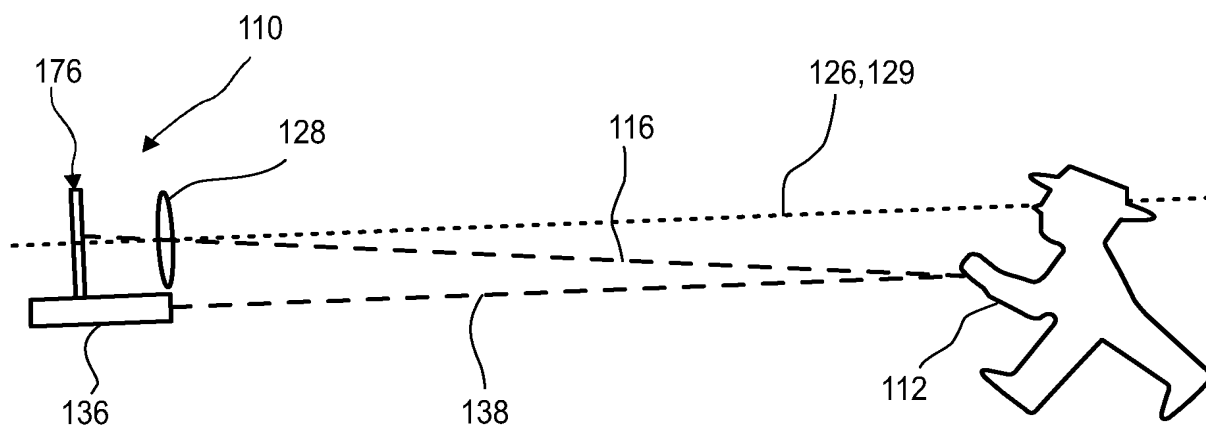


FIG.45B

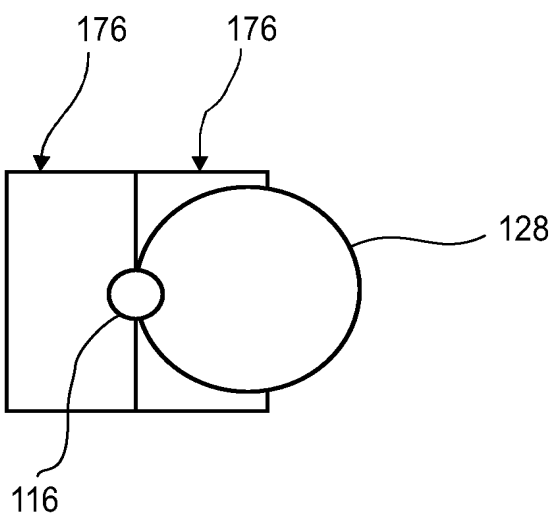


FIG.46

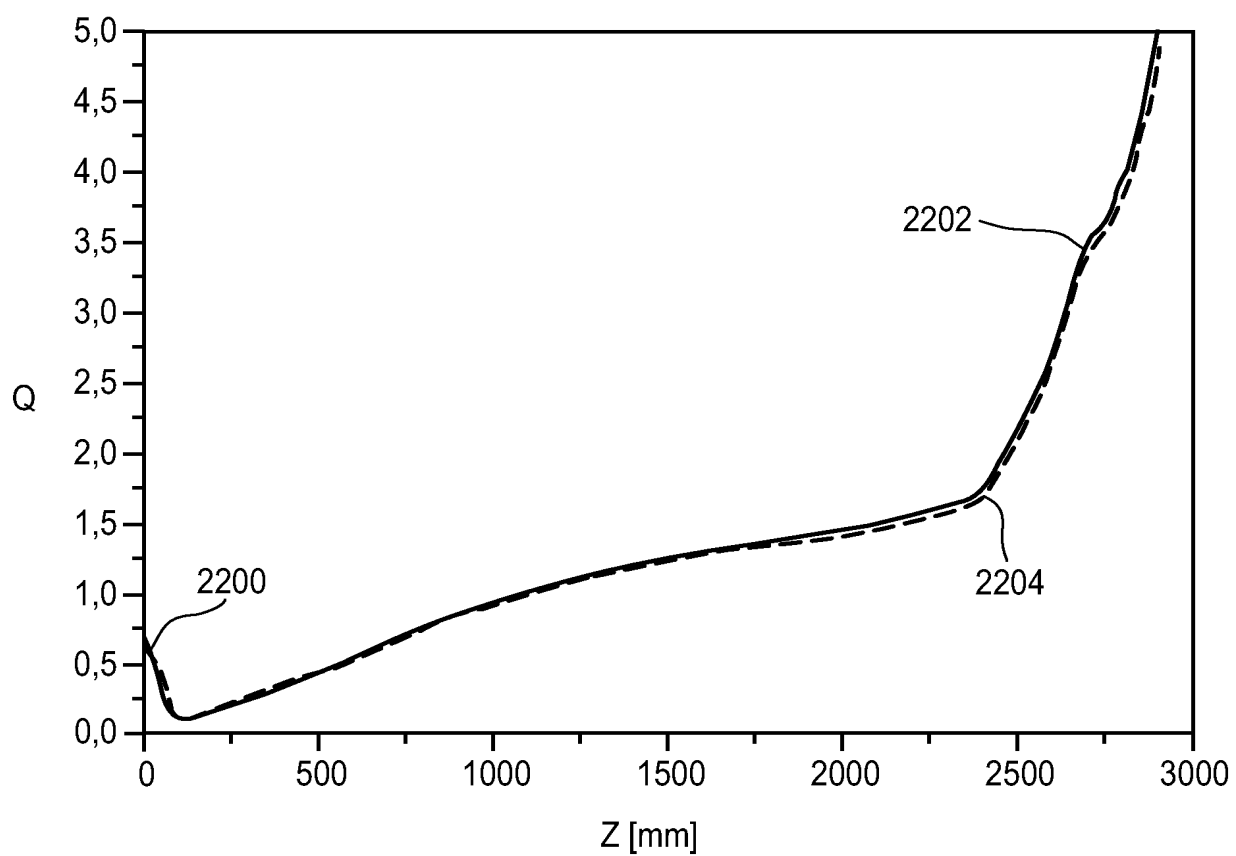


FIG.47A

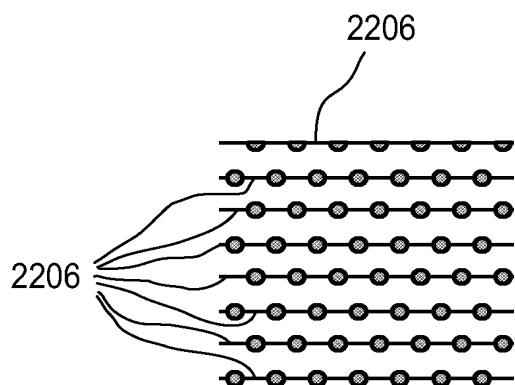


FIG.47B

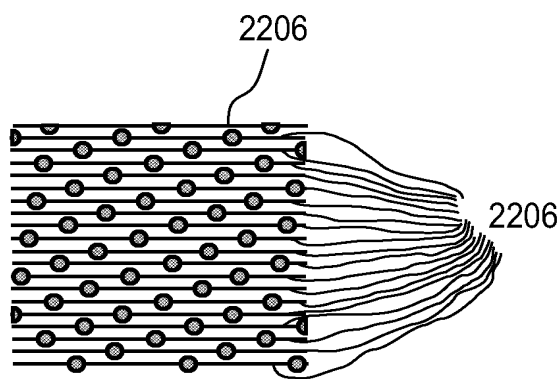


FIG.47C

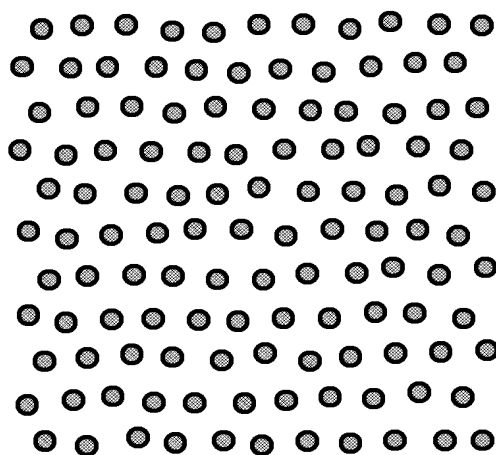
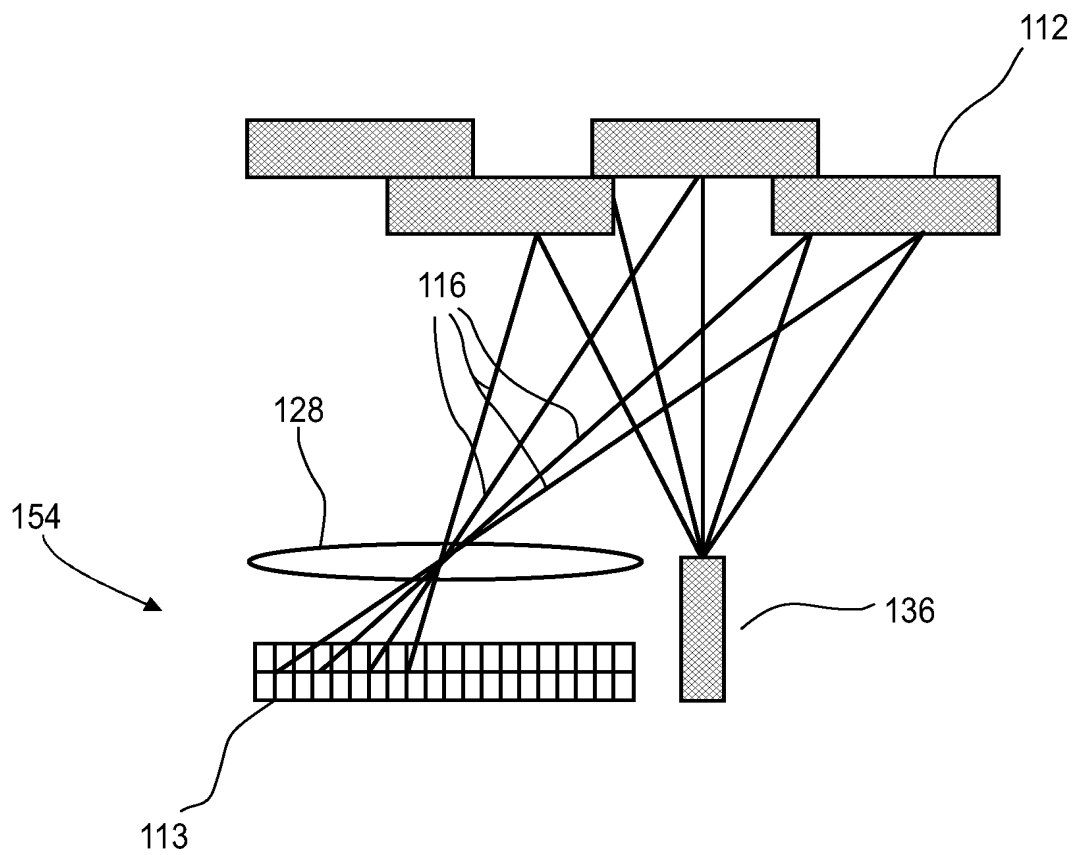


FIG.48





(51) International Patent Classification:

G01S 7/48 (2006.01) *G06K 9/00* (2006.01)
G01N 21/47 (2006.01) *G06T 7/00* (2017.01)
G01S 17/46 (2006.01) *G06T 7/521* (2017.01)

(21) International Application Number:

PCT/EP2020/056759

(22) International Filing Date:

13 March 2020 (13.03.2020)

(25) Filing Language:

English

(26) Publication Language:

English

(30) Priority Data:

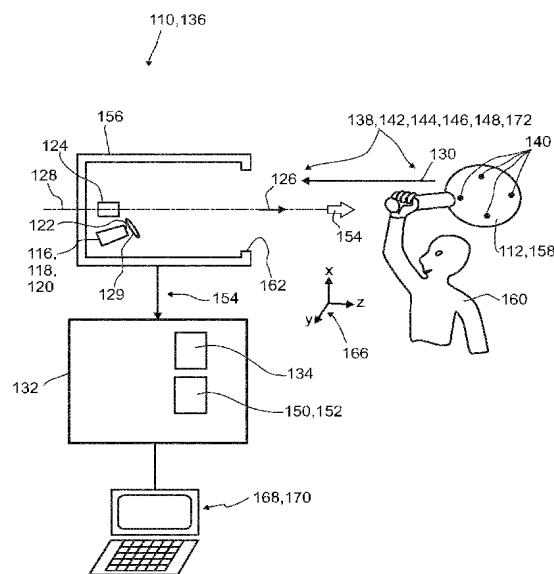
19163250.4 15 March 2019 (15.03.2019) EP

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(54) Title: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

FIG. 1



(57) **Abstract:** A detector (110) for identifying at least one material property m is proposed. The detector (110) comprises at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122). The sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112). The detector (110) comprises at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image. The evaluation device (132) is configured for determining at least one distance feature ϕ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The evaluation device (132) is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a

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grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

Detector for identifying at least one material property

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining at least one material property of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

- 20 A large number of methods for material classification and identification are known from the prior art. For example, material identification can be used in machine vision applications, medical applications and security applications, for image classification purposes or in gesture recognition algorithms.
- 25 For example, US 2016/0206216 A1 describes a device, system and method for skin detection. The device comprises a thermal sensor input for obtaining thermal sensor data of a scene, a light sensor input for obtaining light sensor data of the scene, and an evaluation unit for analyzing the obtained thermal sensor data and the obtained light sensor data and for detecting skin areas within the scene based on said analysis. US 2016/155006 A1 describes a device and a
- 30 corresponding method for skin detection. The device comprises an illumination unit configured to project a predetermined illumination pattern onto a scene, an imaging unit configured to acquire an image of the scene, and an evaluation unit configured to evaluate the acquired image by analyzing the imaged illumination pattern as reproduced in the image and to detect skin areas within the image and distinguish them from non-skin areas within the image based on said
- 35 analysis.

Material classification and identification is usually done in through beam mode, namely shining light through a sample and analyzing extinction. Further, material classification and identification can be done in reflective mode in controlled environments. However, in reflective mode usually wavelengths larger than 1000 nm are necessary in order to yield reliable results. Thus, despite the advantages of known methods for material identification, new concepts for reliable identification of materials are desired. Specifically, identifying a material from its reflection beam profile would be highly desirable. However, most image filters used for beam profile analysis (BPA) yield features that depend on both, distance and material such that reliable identification and

classification of the material is not possible. A typical example is the width of a beam profile reflected from a translucent material.

DE 198 46 619 A1 describes structured surface appearance quality determining equipment, which evaluates electrical measurement signal from photosensor array to derive structure code characterizing structure-dependent characteristic of measurement surface.

CN 108 363 482 A describes a method for controlling smart televisions via three-dimensional gestures on basis of binocular structured light. The method comprises the following steps of: synchronously acquiring images through a binocular camera; reconstructing a three-dimensional image according to an obtained left-right view; carrying out preprocessing; recognizing a gesture action after three-dimensional gestures are segmented; and converting the recognized gesture action into an operation instruction for a smart television, and executing the operation instruction.

US 2018/033146 A1 describes systems and methods for determining a depth map and a reflectivity map from a structured light image. The depth map can be determined by capturing the structured light image and then using a triangulation method to determine a depth map based on the dots in the captured structured light image. The reflectivity map can be determined based on the depth map and based on performing additional analysis of the dots in the captured structured light image.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may identify at least one material property of an object, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are

present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for identifying at least one material property m is disclosed.

As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “identifying at least one material property” refers to one or more of determining and assigning the material property to the object. The detector may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by

the user of the detector. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The detector may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

Specifically, the detector may be configured for detection of biological tissue, in particular human skin. As used herein, the term "biological tissue" generally refers to biological material comprising living cells. The detector may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The term "detection of biological tissue" refers to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector may be adapted to distinguish animal tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector may be adapted to distinguish plant tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector may be adapted to distinguish different types of food such as a fruit, meat, and fish. The detector may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from human skin. The detector may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector may be adapted to distinguish human skin from cloth. The detector may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector may be adapted to distinguish organic material from inorganic material. The detector may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector may be used, in particular, for non-therapeutic and non-diagnostic applications.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter, or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented

essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The detector may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured for generating at least one output signal. In case the detector comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge: Au photodiode, a Ge: Hg photodiode, a Ge: Cu photodiode, a Ge: Zn photodiode, a Si: Ga photodiode, a Si: As photodiode. Additionally or alternatively, the optical sensors may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly,

as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

Preferably, the light sensitive area may be oriented essentially perpendicular to an optical axis of the detector. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The sensor element is configured for recording at least one reflection image of a light beam originating from at least one object. As used herein, the term “reflection image” refers to an image determined by the sensor element comprising at least one reflection feature. As furthermore used herein, without limitation, the term “reflection image” specifically may relate to data recorded by using the sensor element, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element. The reflection image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to “pixels”, reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. As used herein, the term “reflection feature” refers to a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. As used herein, the term “determining at least one reflection feature” refers to imaging and/or recording at least one light beam generated by the object in response to illumination with a light beam, in particular with the at least one illumination feature. In particular, the sensor element may be configured for determining and/or imaging and/or recording at least reflection image. The reflection image may comprise at least one reflection pattern comprising the at least one reflection feature.

The detector comprises at least one illumination source. The detector may comprise at least one illumination source configured for illuminating the object with at least one illumination beam.

The illumination source may be configured for projecting at least one illumination pattern comprising the at least one illumination feature on at least one surface of the object. As used herein, the term “at least one illumination source” refers to at least one arbitrary device configured for providing at least one illumination light beam, specifically the at least one illumination pattern, for illumination of the object. The illumination source may be adapted to directly or indirectly illuminating the object, wherein the illumination light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam.

The illumination source may comprise at least one light source. The illumination source may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of

handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source may be integrated into a housing of the detector.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The detector may be configured such that the illumination light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the illumination light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illumination light beam may be on the optical axis or off the optical axis. As an example, the illumination light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

Specifically, the illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.

The illumination source may be configured for generating at least one illumination pattern for illumination of the object. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one
5 quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. As used herein, the term "pattern" refers to an arbitrary known or pre-
10 determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the object.
15 As used herein, the term "illumination feature" refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. For example, the illumination pattern may comprise at least one line pattern. For example, the illumination pattern may comprise at least one stripe pattern. For example, the illumination pattern may comprise at least
20 one checkerboard pattern. For example, the illumination pattern may comprise at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing
25 lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. For example, the illumination source may be adapted to generate and/or to project a cloud of points. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. The illumination source may comprise the at least one light
30 source configured for generating the at least one illumination pattern. Specifically, for generating and projecting the illumination pattern, the illumination source may comprise at least one laser source and at least one diffractive optical element (DOE). The detector may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at
35 least one point pattern. As further used herein, the term "projecting at least one illumination pattern" refers to providing the at least one illumination pattern for illuminating the at least one object. The projected illumination pattern may be scarce, as only a single illumination feature, such as a single point, may be present. For increasing reliability, the illumination pattern may comprise several illumination features such as several points. If the pattern is scarce, a single image
40 can be used for detection of biological tissue and face identification simultaneously.

For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The

illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern
5 may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination
10 source may be integrated in a mobile device or attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers
15 to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may
20 be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining
25 and determining the beam profile, in particular a shape of the beam profile.

As used herein, the term “object” refers to a point or region emitting at least one light beam, in particular the at least one reflection pattern. For example, the object may be at least one object selected from the group consisting of: a scene, a human such as a human, wood, carpet, foam,
30 an animal such as a cow, a plant, a piece of tissue, a metal, a toy, a metallic object, a beverage, a food such as a fruit, meat, fish, a dish, a cosmetics product, an applied cosmetics product, cloth, fur, hair, a maintenance product, a cream, an oil, a powder, a carpet, a juice, a suspension, a paint, a plant, a body, a part of a body, organic material, inorganic material, a reflective material, a screen, a display, a wall, a sheet of paper, such as photograph. The object may
35 comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrat-
40 ing the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase

with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These material properties may be used to distinguish skin from other materials, specifically by analyzing the back scattering profile.

The least one light beam may propagate from the object towards the detector. The light beam may originate from the object or may originate from an illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. Additionally or alternatively to the at least one illumination source, the detector may use radiation already present in the scene such as from at least one ambient light source.

The sensor element may be configured for recording a beam profile of at least one reflection feature of the reflection image. The evaluation device may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element. The evaluation device may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-

transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

For example, the illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device may be adapted to remove influences from background light from the reflection beam profile, for example, by an imaging without further illumination.

The evaluation device may be configured for determining the material property m by evaluating the beam profile of the reflection image. As used herein, the term "beam profile of the reflection image" refers to at least one intensity distribution of at least one of the reflection features of the reflection image, such as of a light spot on the sensor element, as a function of the pixel. The beam profile of the reflection image, also denoted reflection beam profile, may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. As used herein, the term "evaluating the beam profile" refers to applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. As used herein, the term "image" refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term "image filter" refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. As used herein, the term “distance” refers to the distance of the object, specifically between the object and the detector.

As used herein, the term “distance dependent” image filter refers to an image having a distance dependent output. The output of the distance dependent image filter is denoted herein “distance feature φ_{1z} ” or “distance dependent feature φ_{1z} ”. The distance feature may be or may comprise at least one information about the distance of the object such as at least one measure for the distance of the object, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter $\Phi_{1\text{other}}$ may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{\text{cov}((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j))}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one cur-

rent signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. As generally used herein, the term “combine” generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least one comparison signal or comparison result. As used herein, the term “combined signal Q” refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. In particular, the combined signal may be a quotient signal. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature ϕ_{1z} for determining the distance feature ϕ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy} ,$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing

precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature ϕ_{1z} may be a function of the combined signal Q , $\phi_{1z} = \phi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q . Further, as an example, the object distance z_0 may be a function of the distance dependent feature ϕ_{1z} , $z_0 = z_0(\phi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in ϕ_{1z} . Thus, the object distance z_0 may be a function of the combined signal Q , $z_0 = z_0(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q .

The light-sensitive areas of at least two optical sensors may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q .

The evaluation device may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for de-

termining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example, Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1 / b \cdot s_2$ or $Q = b \cdot s_2 / a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a , b , c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

Typically, the combined signal Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the setup described above, both the first signal s_1 and the second signal s_2 decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the first signal s_1 decreases more rapidly than the second signal s_2 , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus,

is spread over a larger area. The quotient of the first and second sensor signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the incident light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

With respect to further details and embodiments of evaluation of the combined signal Q reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device may be configured for determining from the image the distance feature φ_{1z} of the object by optimizing at least one blurring function f_a . The determined distance feature φ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature φ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. As used herein, the term “blurring function f_a ”, also referred to as blur kernel or point spread function, refers to a response function of the detector to the illumination from the object. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature φ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \| (i'_b * f_a(\sigma(\varphi_{1z})) - i_b) \|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. As used herein, the term “sharp” or “sharp image” refers to a blurred image having a maximum contrast. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature ϕ_{1z} .

The evaluation device may be configured for determining at least one combined distance information z considering the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield advantageous distance measurement results with enhanced precision.

The evaluation device may be configured for determining at least one combined distance information using at least one recursive filter. The recursive filter may be at least one Kalman filter or at least one Extended Kalman filter (EKF). The combined distance information z may be obtained using a real function $z=f(z_{DPR}, z_{DFD})$, wherein z_{DPR} is the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and z_{DFD} is the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter, such as the arithmetic or geometric mean, a polynomial, preferably a polynomial up to the eighth order in z_{DPR} and z_{DFD} . The function f may be or may be based on a look-up table of prerecorded values. For example, the evaluation device may comprise at least one data storage device configured for storing the prerecorded values and/or one or more look-up tables. The function f may be based on a look-up table in combination with an interpolation scheme for interpolation between the values in the look-up table. The interpolation scheme may be a linear interpolation, a splines interpolation, or the like. The z_{DFD} and z_{DPR} may be used as input variables within the recursive filter, in combination with a model or model function such as the function f , concerning the relationship between z , z_{DFD} , and z_{DPR} . The model or model function may comprise statistical assumptions and/or statistical models concerning the distances z , z_{DFD} , and z_{DPR} such as a distribution, such as a Gaussian distribution of a measured distance z , z_{DFD} , and/or z_{DPR} around a real distance z_{real} .

The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters. The further parameters may comprise further information from the sensor element, for example the CMOS sensor, such as information about quality and/or noise of recorded data and/or information about overexposure and/or information about underexposure or the like. The detector may comprise at least one further sensor configured for determining the further sensor data. The recursive filter may be configured for

determining the combined distance information considering the further sensor data. The further sensor may be at least one sensor selected from the group consisting of: a temperature sensor, an illumination sensor such as a control sensor for determining illumination information, an inertial measurement unit; a gyroscope. The model and/or the Kalman filter may comprise further input parameters such as the further sensor data, e.g. temperature and/or detector motion from a gyroscope and/or information from an inertial measurement unit, and/or information from an illumination sensor) and/or further parameters such as the quality/noise of recorded data, overexposure, underexposure, or the like. The further sensor data may be provided by the sensor element, in particular by at least one CMOS sensor, and/or by further image analysis. The model and/or the Kalman filter may comprise the at least one material feature ϕ_{2m} as input variable.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and for selecting at least one image determined by the other one of the first sensor element or the second sensor element as reference image. As used herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature ϕ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm \epsilon$.

The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Hei-

delberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. As used herein, the term “displacement region” refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term “reference feature” refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \epsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evalua-

tion device may determine the displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\varphi_{1z} \pm \varepsilon$.

5 The evaluation device may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at
10 least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the
15 extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the
20 epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region
25 along the assigned epipolar line.

The evaluation device may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device may be configured
30 for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. As used herein, the term “displacement” refers to a difference between a position in the reference image to a position in the reflection image. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a
35 distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device may be adapted to determine the predetermined relationship by using triangulation methods. In case position of the selected
40 reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent

and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device.

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Additionally or alternatively, the evaluation device may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ε -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or reference feature in shorter distance and/or ε -weighted distance to the reference feature and/or reflection feature.

The evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. As used herein, the term "material dependent" image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein "material feature φ_{2m} " or "material dependent feature φ_{2m} ". The material feature may be or may comprise at least one information about the at least one material property of the object.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof

by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter $\Phi_{2\text{other}}$ may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term "passes a hypothesis testing" refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term "beam profile image" refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k :

$(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J, $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values

from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k .

The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term "not distinguish between material classifiers" refers to that the expectation values of the material classifiers are identical. As used herein, the term "distinguishes material classifiers" refers to that at least two expectation values of the material classifiers differ. As used herein "distinguishes at least two material classifiers" is used synonymous to "suitable material classifier". The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq$

0.075, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

- 5 For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1g_2) = [p_{g_1,g_2}]$, whereas p_{g_1,g_2} is the occurrence rate of the grey combination $(g_1,g_2)=[f(x_1,y_1),f(x_2,y_2)]$, and the relation ρ defines the distance between (x_1,y_1) and (x_2,y_2) , which is $\rho(x,y)=(x+a,y+b)$ with a and b selected from 0,1.

10

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

- For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

15

The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

- For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

20

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1+|i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

- 25 The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

5 and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ε times the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The material property m and/or the longitudinal coordinate z may be determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m . The evaluation device may be configured for determining the material property m and/or the longitudinal coordinate z by evaluating the features φ_{2m} , φ_{1z} . For example, the evaluation device may be configured for using at least one predetermined relationship between the distance feature φ_{1z} and the longitudinal coordinate of the object for determining the longitudinal coordinate z of the object. The evaluation device may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the object for determining the material property of the object. The predetermined relationship may be one or more of an empirical relationship, a semi-empirical relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. For example, the material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z such that the information about the

longitudinal coordinate z can be considered for evaluating of ϕ_{2m} . Specifically, the material property m and/or the longitudinal coordinate z may be determined by a function $z(\phi_{1z}, \phi_{2m})$ and/or by a function $m(\phi_{1z}, \phi_{2m})$. The function may be predefined and/or predetermined. For example, the function may be a linear function.

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In the ideal case, an image filter would yield features that are only dependent on distance or material properties. However, image filters used in beam profile analysis may yield features that depend on distance and material properties, such as translucency. At least one of the material dependent image filter or the distance dependent image filter may be a function of the feature ϕ_{1z} or ϕ_{2m} of the at least one other image filter. At least of the material dependent image filter or the distance dependent image filter may be a function of the at least one other image filter. The evaluation device may be configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature ϕ_{1z} or ϕ_{1m} of the other image filter and/or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter. Specifically, the evaluation device may be configured for determining a correlation coefficient of the material dependent image filter and the distance dependent image filter. In case the correlation coefficient of the material dependent image filter with the distance dependent image filter is close to 1 or -1, the distance may be projected out, by projecting the material feature on the principal axis with the lowest variance. As an example, the material feature may be projected onto an axis orthogonal to the correlating main component. In other words, the material feature may be projected onto the second main component. This may be done using a principal component analysis as known to the person skilled in the art. The evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image. In particular, the evaluation device may be configured for simultaneously determining features that are strongly correlated with distance and weakly correlated with material properties and features that are weakly correlated with distance and strongly correlated with material properties. Alternatively, the evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image. The evaluation device may be configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of ϕ_{1z} and ϕ_{2m} to the reflection image.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. The above-mentioned operations, including analysis of the beam profile, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more

programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured for performing the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

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The detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

10 The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a
15 direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element,
20 preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more
25 imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging
30 thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on
35 the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

40 The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having

at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As outlined above, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. For example, the detector may be configured for determining the longitudinal coordinate of the object by using at least one distance dependent filter using the depth-from-photon-ratio technique and/or the depth-from-defocus technique as outlined above. The detector may be configured for determining a position of the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. At least one of the optical sensors may
5 determined from which a center signal arises. This may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal
10 sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also
15 be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one
20 transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created.
25 Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured for determining at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

30 The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, a center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. The use of the matrix of optical
35 sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object. The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at
40 least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p, is smaller or equal than a pre-defined level of significance, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

In a further aspect of the present invention, a detector system is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect

one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light
5 beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least
10 one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least
15 one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-
20 machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured for enabling at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured for varying the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one
30 track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.
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In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention,
40 such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As

used herein, the term “scenery” generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term “scan” generally refers to a consecutive measurement in different regions.

Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term “dot” refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a

case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using

the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

- 5 As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mir-
10 rors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on
15 the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors
20 comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or
25 more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices
30 as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety
35 curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenu-
40 ate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated

out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

5 As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system
10 and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3D model by connecting adjacent points with planar surfaces. The 3D model may be stored and
15 processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser
20 sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters
25 or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

30 As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection
35 in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system
40 may be adapted to determine at least one longitudinal coordinate for each of the light points as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. This feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor.

5 The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least
10 one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination
15 source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively
20 or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic
25 radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence
30 and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be
35 regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the
40 visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of

3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the

grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
 - b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking

application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

With respect to further uses of the detector and devices of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

Specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device configured for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements configured for generating electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device configured for performing 3D photography. The camera generally may be configured for acquiring a single image, such as a single 3D image, or may be configured for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera configured for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be configured for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression “position” generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device configured for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be configured for performing at least one computer program, such as at least one computer program configured for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

The human-machine interface may comprise a plurality of beacon devices which are configured for being at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or

by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device configured for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements configured for reflecting a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be configured in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the

user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

- 5 The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source configured for generating at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam proper-
- 10 ties. Additionally or alternatively, the beacon device may comprise at least one reflector configured for reflecting light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

- The object, which may form part of the detector system, may generally have an arbitrary shape.
- 15 Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be
- 20 part of the human-machine interface and/or of the entertainment device.

- As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably
- 25 computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

- The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one
- 35 item of information preferably may comprise at least one command configured for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements
- 40 may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is configured for gathering information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be configured for providing information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is configured for tracking a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.

The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.

The tracking system may be configured for initiating one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be configured for initiating at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards

the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

In a further aspect of the present invention, an inertial measurement unit for use in an electronic device disclosed. The inertial measurement unit is adapted to receive data determined by at least one detector according to the present invention. The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

The inertial measurement unit may comprise the detector according to the present invention and/or may be connected to the detector, via at least one data connection. The evaluation device and/or at least one processing device of the inertial measurement unit may be configured for determining the at least one combined distance information, in particular using the at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters such as further sensor data from the further sensor of the inertial measurement unit. With respect to definitions and embodiments of the inertial measurement unit reference is made to the description of the detector.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;

- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 2: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

Embodiment 3: The detector according to the preceding embodiment, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

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Embodiment 5: The detector according to any one of the preceding embodiments, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.80$.

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Embodiment 6: The detector according to any one of the preceding embodiments, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}},\Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}},\Phi_m}| \geq 0.80$.

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Embodiment 7: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

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Embodiment 8: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

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Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

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Embodiment 10: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

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Embodiment 11: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and/or φ_{2m} to the reflection image.

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Embodiment 12: The detector according to any one of the preceding embodiments, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .

Embodiment 13: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

5 wherein x and y are transversal coordinates, A1 and A2 are different areas of the beam profile and E(x,y,z_o) denotes the beam profile given at the object distance z_o, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

10 Embodiment 14: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

15 Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a
20 number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

25 Embodiment 16: The detector according to any one of the preceding embodiments, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

30 Embodiment 17: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

35
$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 18: The detector according to any one of the two preceding embodiments, wherein the at least one blurring function f_a is a function or composite function composed from at least

one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

- 5 Embodiment 19: The detector according to any one of the preceding embodiments, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 21: The detector according to the preceding embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 23: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 24: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \epsilon$.

Embodiment 25: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
 - at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
- wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 26: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.

Embodiment 27: The detector according to any one of embodiments 25 and 26, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 28: The detector according to any one of embodiments 25 to 27, wherein the at least one material property is a property selected from the group consisting of: a scattering coef-

ficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

Embodiment 29: The detector according to any one of embodiments 25 to 28, wherein the further distance dependent image filter $\Phi_{10\text{ther}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{10\text{ther}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{10\text{ther}}, \Phi_z}| \geq 0.80$.

Embodiment 30: The detector according to any one of embodiments 25 to 29, wherein the further material dependent image filter $\Phi_{20\text{ther}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.80$.

Embodiment 31: The detector according to any one of embodiments 25 to 30, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{20\text{ther}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

Embodiment 32: The detector according to any one of embodiments 25 to 31, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

Embodiment 33: The detector according to any one of embodiments 25 to 32, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

Embodiment 34: The detector according to any one of embodiments 25 to 33, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

Embodiment 35: The detector according to any one of embodiments 25 to 34, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the

at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

- 5 Embodiment 36: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

- 10 Embodiment 37: The detector according to any one of embodiments 25 to 36, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .
- 15

Embodiment 38: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 20 wherein x and y are transversal coordinates, A_1 and A_2 are different areas of the beam profile and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

- 25 Embodiment 39: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- 30 Embodiment 40: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
- 35

Embodiment 41: The detector according to any one of embodiments 25 to 40, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

Embodiment 42: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 43: The detector according to any one of embodiments 25 to 42, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

Embodiment 44: The detector according to any one of embodiments 25 to 43, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 45: The detector according to any one of embodiments 25 to 44, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 46: The detector according to the preceding embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 47: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflec-

tion image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 48: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 49: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 50: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,

wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

5 Embodiment 51: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a
10 pre-defined level of significance.

Embodiment 52: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.
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Embodiment 53: The detector according to any one of embodiments 50 and 52, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 54: The detector according to any one of embodiments 50 to 53, wherein the at
20 least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

Embodiment 55: The detector according to any one of embodiments 50 to 54, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.
25

Embodiment 56: The detector according to any one of embodiments 50 to 55, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.
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Embodiment 57: The detector according to any one of embodiments 50 to 56, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness
35 filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof, or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by
40 $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared

norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

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Embodiment 58: The detector according to any one of embodiments 50 to 57, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

10 Embodiment 59: The detector according to any one of embodiments 50 to 58, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

15 Embodiment 60: The detector according to any one of embodiments 50 to 59, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

20 Embodiment 61: The detector according to any one of embodiments 50 to 60, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

25 Embodiment 62: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

30 Embodiment 63: The detector according to any one of embodiments 50 to 62, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .

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Embodiment 64: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are different areas of the beam profile and $E(x,y,z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

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Embodiment 65: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

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Embodiment 66: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

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Embodiment 67: The detector according to any one of embodiments 50 to 66, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

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Embodiment 68: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

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$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 69: The detector according to any one of embodiments 50 to 68, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

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Embodiment 70: The detector according to any one of embodiments 50 to 69, wherein the sensor element comprises at least one CMOS sensor.

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Embodiment 71: The detector according to any one of embodiments 50 to 70, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 72: The detector according to the embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 73: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 74: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 75: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 76: A detector system, the detector system comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

Embodiment 77: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 78: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 79: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 80: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of embodiments 1 to 24, 25 to 49, or 50 to 75, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 81: A camera for imaging at least one object, the camera comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75.

Embodiment 82: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data

from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

5 Embodiment 83: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 1 to 24, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,
 - b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 84: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 25 to 49, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,
 - b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 85: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 50 to 75, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

- b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
 b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

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Embodiment 86: A use of the detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

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Brief description of the figures

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Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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Figure 1 shows an embodiment of a detector according to the present invention;

Figure 2 shows the statistics of the resulting material features of the spot shape material dependent image filter applied to the data set; and

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Figure 3 shows a beam profile image.

Detailed description of the embodiments:

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Fig. 1 shows in a highly schematic fashion an embodiment of a detector 110 for identifying at least one material property m of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.

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The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the ele-

ments are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

5 The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other
10 embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors 120 may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

15 The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical
20 sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photo-
25 diodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

30 The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As
35 an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

40 The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more prefer-

ably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

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The detector 110 may further comprise an illumination source 124. As an example, the illumination source 124 may be configured for generating an illumination light beam 126 for illuminating the object 112. The detector 110 may be configured such that the illumination light beam 126 propagates from the detector 110 towards the object 112 along an optical axis 128 of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis 128.

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The illumination source 124 may comprise at least one light source. The illumination source 124 may comprise a plurality of light sources. The illumination source 124 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 124 may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source 124 may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

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Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 124 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points, for example the illumination source 124 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least

one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source 124 may be integrated into a housing of the detector 110.

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The illumination light beam 126 may propagate from the object 112 towards the detector 110. The detector 110 may be used in active and/or passive illumination scenarios. For example, the at least one illumination source 124 may be adapted to illuminate the object 112, for example, by directing a light beam towards the object 112, which reflects the light beam. Additionally or

10 alternatively to the at least one illumination source 124, the detector 110 may use radiation already present in the scene such as from at least one ambient light source.

The illumination source 124 may illuminate the at least one object 112 with at least one illumination pattern. The illumination pattern may comprise a plurality of points as image features.

15 These points are illustrated as light beam 126 emerging from the illumination source 124.

The detector 110 may comprise at least one transfer device 129 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel

20 lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 129 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

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Each optical sensor 120 may be designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam 130 propagating from the object 112 to the detector 110. The sensor element 116 may be configured for determining at least one reflection image of at least one reflection feature generated by the reflection light beam 130 on the optical sensors 120. The reflection image may be data recorded by

30 using the sensor element 116, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element 116. The reflection image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix 118 of the sensor element 116. The matrix 118 may comprise the reflection image 126. For example, in case of illumination with a point pattern, the reflection image may comprise points as reflection features. These points may

35 result from reflection light beams 130 originating from the at least one object 112.

The sensor element 116 may be configured for recording a beam profile of at least one reflection feature of the reflection image. The detector 110 comprises at least one evaluation device

40 132. The evaluation device 132 may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element 116. The evaluation device 132 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image anal-

ysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element 116. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

The evaluation device 132 may be configured for determining the material property m by evaluating the beam profile of the reflection image. The beam profile of the reflection image may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device 132 may be configured for applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device 132 is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance feature may be or may comprise at least one information about the distance of the object 112 such as at least one measure for the distance of the object 112, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter Φ_{1other} may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{cov\left((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j)\right)}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device 132, as an example, may comprise at least one divider 134, wherein the divider 134 is configured for deriving the quotient signal. The divider 134 may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear com-

binations of the sensor signals. The evaluation device 132 may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance L(z_o) and beam shape S(x,y;z_o). Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object 112 and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature φ_{1z} may be a function of the combined signal Q, $\varphi_{1z} = \varphi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q. Further, as an example, the object distance z_o may be a function of the distance dependent feature φ_{1z} , $z_o = z_o(\varphi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in φ_{1z} . Thus, the object distance z_o may be a function of the combined signal Q, $z_o = z_o(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q.

The light-sensitive areas 122 of at least two optical sensors 120 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

The evaluation device 132 may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the

center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 132 may be adapted for determining an area integral of the beam profile. The evaluation device 132 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 132 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device 132 specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example,

Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1/b \cdot s_2$ or $Q = b \cdot s_2/a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a, b, c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

With respect to further details and embodiments of evaluation of the combined signal Q and determination of the longitudinal coordinate z reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device 132 may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device 132 may be configured for determining from the image the distance feature ϕ_{1z} of the object by optimizing at least one blurring function f_a . The determined distance feature ϕ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature ϕ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature ϕ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature ϕ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature ϕ_{1z} .

The evaluation device 132 may be configured for determining at least one combined distance information z considering the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ

z_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature φ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature φ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield to advantageous distance measurement results with enhanced precision.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. The detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device 132 may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device 132 may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm \epsilon$.

The evaluation device 132 may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device 132 may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device 132 may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The evaluation device 132 may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device 132 may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the

reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device 132 may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object 112, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device 132 may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \varepsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device 132 may determine the displacement region around the image position of the reflection feature. The evaluation device 132 may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\varphi_{1z} \pm \varepsilon$.

The evaluation device 132 may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The evaluation device 132 may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device 132 may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device 132 may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to

determine an epipolar line before determining a corresponding reference feature. The evaluation device 132 may determine a displacement region around the image position of each reflection feature. The evaluation device 132 may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device 132 may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device 132 may be configured for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 132 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device 132 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device 132 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device 132.

Additionally or alternatively, the evaluation device 132 may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material feature may be or may comprise at least one information about the at least one material property of the object 112.

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The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$. With respect to the description of exemplary material dependent image filter, reference is made to the description of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter such as a Gaussian filter or median filter, the grey-level-occurrence-based contrast filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, and the threshold area filter given above.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. The beam profile image may be given by a sum of N_B Gaussian radial basis functions,

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$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all beam profile images. The center-positions, x_{lk} , y_{lk} , are identical for all images $f_k: (x_0, x_1, \dots, x_{N_B}), (y_0, y_1, \dots, y_{N_B})$. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

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Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed. Figure 3 shows an embodiment of a beam profile image defined by an exemplary data set.

An exemplary data set is presented in the following. The following parameter tables list the center positions, $x_{lk} = x_l, y_{lk} = y_l$, of all Gaussian functions g_{lk} for all images f_k :

l	0	1	2	3	4	5	6	7
x_l	14	11	10	19	15	20	12	17
y_l	17	11	17	19	20	20	12	20
l	8	9	10	11	12	13	14	15
x_l	13	18	10	14	19	15	16	12
y_l	17	19	8	18	14	11	22	17
l	16	17	18	19	20	21	22	23
x_l	25	13	18	8	9	10	14	19
y_l	15	20	20	15	14	13	24	24
l	24	25	26	27	28	29	30	31
x_l	10	9	14	12	19	20	17	15
y_l	24	16	15	11	13	19	13	14
l	32	33	34	35	36	37	38	39
x_l	16	13	12	20	18	17	21	13
y_l	19	10	22	14	10	18	15	23
l	40	41	42	43	44	45	46	47
x_l	8	10	8	14	19	11	15	14
y_l	12	14	18	21	23	15	16	8
l	48	49	50	51	52	53	54	55
x_l	16	20	11	17	16	15	21	13
y_l	13	16	16	8	16	13	21	13

l	56	57	58	59	60	61	62	63
x_l	22	18	23	14	19	11	15	16
y_l	23	15	12	22	18	10	23	10

The following parameter tables list the material classifiers, in particular for a white Teflon target (denoted control), a fabric, a dark skin (denoted as dark_skin), a pale skin (denoted pale_skin), and a highly translucent skin (denoted trans), and the distance z for each image f_k , referenced

5 by the image index k :

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
0	control	0.4	40	dark_skin	0.4	80	control	0.6
1	control	0.4	41	dark_skin	0.4	81	control	0.6
2	control	0.4	42	dark_skin	0.4	82	control	0.6
3	control	0.4	43	dark_skin	0.4	83	control	0.6
4	control	0.4	44	dark_skin	0.4	84	control	0.6
5	control	0.4	45	dark_skin	0.4	85	control	0.6
6	control	0.4	46	dark_skin	0.4	86	control	0.6
7	control	0.4	47	dark_skin	0.4	87	control	0.6
8	control	0.4	48	pale_skin	0.4	88	control	0.6
9	control	0.4	49	pale_skin	0.4	89	control	0.6
10	control	0.4	50	pale_skin	0.4	90	control	0.6
11	control	0.4	51	pale_skin	0.4	91	control	0.6
12	control	0.4	52	pale_skin	0.4	92	control	0.6
13	control	0.4	53	pale_skin	0.4	93	control	0.6
14	control	0.4	54	pale_skin	0.4	94	control	0.6
15	control	0.4	55	pale_skin	0.4	95	control	0.6
16	fabric	0.4	56	pale_skin	0.4	96	fabric	0.6
17	fabric	0.4	57	pale_skin	0.4	97	fabric	0.6
18	fabric	0.4	58	pale_skin	0.4	98	fabric	0.6
19	fabric	0.4	59	pale_skin	0.4	99	fabric	0.6
20	fabric	0.4	60	pale_skin	0.4	100	fabric	0.6
21	fabric	0.4	61	pale_skin	0.4	101	fabric	0.6
22	fabric	0.4	62	pale_skin	0.4	102	fabric	0.6
23	fabric	0.4	63	pale_skin	0.4	103	fabric	0.6
24	fabric	0.4	64	trans	0.4	104	fabric	0.6
25	fabric	0.4	65	trans	0.4	105	fabric	0.6
26	fabric	0.4	66	trans	0.4	106	fabric	0.6
27	fabric	0.4	67	trans	0.4	107	fabric	0.6
28	fabric	0.4	68	trans	0.4	108	fabric	0.6
29	fabric	0.4	69	trans	0.4	109	fabric	0.6
30	fabric	0.4	70	trans	0.4	110	fabric	0.6
31	fabric	0.4	71	trans	0.4	111	fabric	0.6
32	dark_skin	0.4	72	trans	0.4	112	dark_skin	0.6
33	dark_skin	0.4	73	trans	0.4	113	dark_skin	0.6
34	dark_skin	0.4	74	trans	0.4	114	dark_skin	0.6
35	dark_skin	0.4	75	trans	0.4	115	dark_skin	0.6
36	dark_skin	0.4	76	trans	0.4	116	dark_skin	0.6
37	dark_skin	0.4	77	trans	0.4	117	dark_skin	0.6
38	dark_skin	0.4	78	trans	0.4	118	dark_skin	0.6
39	dark_skin	0.4	79	trans	0.4	119	dark_skin	0.6

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
120	dark_skin	0.6	160	control	0.8	200	dark_skin	0.8
121	dark_skin	0.6	161	control	0.8	201	dark_skin	0.8
122	dark_skin	0.6	162	control	0.8	202	dark_skin	0.8
123	dark_skin	0.6	163	control	0.8	203	dark_skin	0.8
124	dark_skin	0.6	164	control	0.8	204	dark_skin	0.8
125	dark_skin	0.6	165	control	0.8	205	dark_skin	0.8
126	dark_skin	0.6	166	control	0.8	206	dark_skin	0.8
127	dark_skin	0.6	167	control	0.8	207	dark_skin	0.8
128	pale_skin	0.6	168	control	0.8	208	pale_skin	0.8
129	pale_skin	0.6	169	control	0.8	209	pale_skin	0.8
130	pale_skin	0.6	170	control	0.8	210	pale_skin	0.8
131	pale_skin	0.6	171	control	0.8	211	pale_skin	0.8
132	pale_skin	0.6	172	control	0.8	212	pale_skin	0.8
133	pale_skin	0.6	173	control	0.8	213	pale_skin	0.8
134	pale_skin	0.6	174	control	0.8	214	pale_skin	0.8
135	pale_skin	0.6	175	control	0.8	215	pale_skin	0.8
136	pale_skin	0.6	176	fabric	0.8	216	pale_skin	0.8
137	pale_skin	0.6	177	fabric	0.8	217	pale_skin	0.8
138	pale_skin	0.6	178	fabric	0.8	218	pale_skin	0.8
139	pale_skin	0.6	179	fabric	0.8	219	pale_skin	0.8
140	pale_skin	0.6	180	fabric	0.8	220	pale_skin	0.8
141	pale_skin	0.6	181	fabric	0.8	221	pale_skin	0.8
142	pale_skin	0.6	182	fabric	0.8	222	pale_skin	0.8
143	pale_skin	0.6	183	fabric	0.8	223	pale_skin	0.8
144	trans	0.6	184	fabric	0.8	224	trans	0.8
145	trans	0.6	185	fabric	0.8	225	trans	0.8
146	trans	0.6	186	fabric	0.8	226	trans	0.8
147	trans	0.6	187	fabric	0.8	227	trans	0.8
148	trans	0.6	188	fabric	0.8	228	trans	0.8
149	trans	0.6	189	fabric	0.8	229	trans	0.8
150	trans	0.6	190	fabric	0.8	230	trans	0.8
151	trans	0.6	191	fabric	0.8	231	trans	0.8
152	trans	0.6	192	dark_skin	0.8	232	trans	0.8
153	trans	0.6	193	dark_skin	0.8	233	trans	0.8
154	trans	0.6	194	dark_skin	0.8	234	trans	0.8
155	trans	0.6	195	dark_skin	0.8	235	trans	0.8
156	trans	0.6	196	dark_skin	0.8	236	trans	0.8
157	trans	0.6	197	dark_skin	0.8	237	trans	0.8
158	trans	0.6	198	dark_skin	0.8	238	trans	0.8
159	trans	0.6	199	dark_skin	0.8	239	trans	0.8

The following parameter tables list the prefactors a_{lk} for the Gaussian functions g_{lk} in each image f_k referenced by the image index k and the Gaussian index l :

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
0	291.94	-7.20	-14.72	-22.75	32.57	16.92	-45.50	-23.31
1	300.91	-7.58	-15.22	-14.41	32.18	15.96	-44.41	-23.92
2	297.04	-6.42	-13.94	-10.30	31.19	16.44	-43.68	-22.27
3	289.98	-7.67	-14.20	-13.60	31.68	17.16	-44.04	-21.41
4	295.94	-7.72	-12.93	-16.81	32.77	15.92	-45.15	-22.30
5	295.37	-6.87	-13.30	-15.51	33.10	16.70	-45.18	-22.29
6	298.03	-6.79	-14.89	-12.68	32.43	14.46	-44.47	-22.71

7	296.95	-7.09	-14.98	-13.54	31.56	15.69	-45.09	-20.03
8	306.69	-6.91	-14.94	-14.27	33.28	16.02	-43.93	-20.21
9	291.40	-7.73	-13.91	-14.95	30.51	16.18	-45.86	-21.61
10	301.21	-7.21	-14.05	-17.46	32.02	17.26	-45.02	-22.89
11	309.78	-6.93	-13.47	-19.23	33.01	14.94	-43.61	-20.86
12	298.62	-7.09	-14.75	-18.43	32.59	18.97	-44.11	-22.73
13	296.09	-7.18	-14.25	-15.83	31.94	14.62	-43.72	-19.77
14	301.13	-7.24	-14.44	-19.11	33.54	17.01	-44.08	-21.39
15	304.28	-7.08	-14.10	-17.26	32.54	16.81	-44.40	-21.14
16	246.71	-3.77	-13.62	1.43	19.38	-2.03	-52.50	-19.62
17	258.85	-3.11	-12.88	-3.91	19.02	-1.05	-49.64	-21.50
18	263.06	-2.91	-13.72	1.65	19.03	-1.26	-51.02	-21.58
19	251.12	-4.15	-14.47	-6.19	19.00	-1.26	-50.65	-22.40
20	250.05	-2.91	-13.84	0.30	19.48	-1.79	-49.89	-20.92
21	249.35	-2.80	-13.64	-0.25	18.90	0.42	-51.29	-19.21
22	260.34	-2.57	-14.30	1.83	17.72	-1.18	-51.27	-24.07
23	246.64	-3.68	-13.92	-2.29	19.79	-0.20	-50.33	-23.09
24	245.58	-2.91	-15.25	-6.59	18.34	-1.75	-50.89	-24.38
25	249.09	-3.09	-14.13	3.85	19.58	-1.58	-50.05	-22.87
26	255.98	-2.91	-13.67	6.77	18.09	-2.15	-48.40	-21.74
27	249.87	-3.43	-13.89	1.80	17.14	-2.32	-50.95	-25.65
28	247.42	-3.34	-13.32	2.32	17.69	-1.29	-50.40	-23.82
29	253.00	-3.32	-12.60	4.98	17.09	-1.16	-49.54	-25.89
30	254.40	-3.44	-14.69	1.64	20.18	-1.15	-49.66	-22.73
31	256.40	-3.75	-13.62	-8.47	19.57	-1.08	-52.04	-21.62
32	24.07	-2.06	2.98	-22.09	14.88	8.52	3.28	-4.27
33	24.00	-1.30	3.15	-22.28	14.65	9.22	3.94	-4.75
34	20.76	-1.69	3.24	-22.34	14.81	8.95	3.76	-4.69
35	22.23	-2.51	3.10	-22.85	14.01	9.71	3.29	-4.61
36	20.33	-2.23	3.04	-21.40	15.61	8.99	3.00	-4.12
37	21.12	-2.09	3.25	-26.46	14.68	8.75	4.67	-5.05
38	21.89	-1.92	3.05	-22.32	14.65	8.25	3.77	-4.15
39	14.81	-2.07	2.82	-20.92	14.96	9.05	3.73	-5.40

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
0	-95.90	-116.79	0.49	-92.61	-76.01	36.70	0.46	78.40
1	-99.07	-117.11	0.57	-89.09	-76.31	37.12	-0.60	86.28
2	-92.44	-115.78	0.59	-93.89	-76.95	35.15	0.42	83.74
3	-96.50	-115.93	0.49	-92.28	-75.29	36.48	-0.17	70.45
4	-100.83	-120.58	0.57	-95.39	-73.94	37.36	-0.26	79.80
5	-94.89	-117.93	0.59	-93.45	-75.67	36.88	0.22	74.29
6	-94.27	-119.97	0.51	-91.14	-74.61	35.76	-0.72	73.46
7	-94.78	-114.15	0.55	-91.40	-71.49	36.55	-0.61	73.29
8	-97.53	-112.61	0.54	-91.30	-70.94	36.62	0.39	75.30
9	-97.85	-108.91	0.57	-94.07	-72.52	36.34	-0.32	76.34
10	-97.23	-115.13	0.64	-95.38	-74.62	36.22	-0.33	74.11
11	-96.18	-115.91	0.45	-93.13	-77.72	37.20	-0.07	81.82
12	-100.79	-113.16	0.52	-90.33	-72.38	36.46	-1.02	74.17
13	-98.70	-124.58	0.53	-93.48	-74.52	36.90	-0.78	74.38
14	-93.07	-116.00	0.52	-92.98	-75.28	36.84	-0.77	76.49
15	-106.83	-117.44	0.52	-94.22	-75.28	36.42	-0.51	79.82
16	-88.58	-87.41	0.63	-70.05	-37.06	27.50	3.58	84.26
17	-87.71	-82.32	0.54	-68.18	-40.98	28.10	3.20	88.34
18	-76.19	-80.81	0.53	-66.95	-35.36	27.23	3.61	87.52
19	-80.42	-84.95	0.55	-67.91	-38.39	27.70	3.39	90.71

20	-87.87	-76.07	0.74	-66.04	-39.03	27.55	2.84	83.83
21	-93.60	-80.04	0.45	-69.48	-38.83	27.01	3.71	83.32
22	-88.04	-78.17	0.62	-72.00	-39.30	27.24	3.08	89.23
23	-88.92	-76.16	0.62	-66.79	-39.81	27.64	3.01	84.46
24	-90.47	-77.18	0.58	-66.41	-35.84	27.85	3.13	94.66
25	-89.84	-83.96	0.70	-64.36	-36.65	26.40	3.38	86.08
26	-81.82	-72.08	0.59	-66.39	-38.13	27.45	3.47	83.27
27	-80.57	-82.11	0.65	-66.61	-35.78	28.23	3.99	86.32
28	-93.40	-81.19	0.66	-68.45	-39.71	26.68	3.19	88.66
29	-92.92	-79.14	0.68	-64.90	-38.61	28.11	3.79	83.05
30	-91.82	-79.09	0.57	-66.39	-37.74	27.24	4.20	82.26
31	-90.60	-82.20	0.51	-67.61	-35.58	28.10	3.02	89.40
32	-0.54	-16.54	-0.31	-3.67	-33.80	4.91	1.87	11.78
33	-0.17	-16.06	-0.31	-2.39	-32.09	4.41	1.86	11.66
34	-3.04	-15.03	-0.34	-5.36	-31.98	4.17	1.76	11.91
35	-5.65	-15.79	-0.35	-2.71	-32.21	4.77	1.90	10.71
36	-6.54	-15.97	-0.29	-3.29	-33.09	4.26	1.69	14.99
37	-1.22	-16.20	-0.29	-2.79	-32.90	4.42	1.93	9.92
38	-3.85	-13.17	-0.22	-2.98	-33.98	4.68	1.84	11.96
39	-5.99	-12.87	-0.34	-2.53	-34.59	3.97	1.80	11.77

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
0	3.86	18.38	66.68	70.89	46.30	51.83	63.61	6.33
1	-1.41	18.18	66.65	-116.27	47.93	53.00	-19.71	6.14
2	5.49	17.64	67.40	86.57	47.99	51.08	-20.20	6.06
3	1.41	18.29	68.83	-73.86	49.40	52.06	-9.49	6.14
4	0.58	17.86	70.11	73.41	47.45	52.04	19.20	6.06
5	-0.96	18.19	71.29	12.08	48.40	52.10	36.63	6.18
6	-0.98	18.97	69.33	107.17	48.12	51.68	-30.34	6.21
7	1.44	17.76	67.10	-119.11	46.64	51.64	-19.85	6.11
8	2.83	18.10	66.62	42.19	47.28	52.29	-24.88	6.17
9	-0.87	17.50	68.71	-129.91	45.89	51.63	-43.44	6.23
10	1.24	18.26	66.88	-12.32	47.49	52.45	45.87	6.14
11	0.39	18.50	68.03	-103.43	47.42	52.40	-25.17	6.01
12	1.60	18.13	67.42	-159.30	48.41	52.43	-8.06	6.08
13	2.92	17.75	68.36	139.40	45.89	52.74	-16.56	6.19
14	1.69	17.80	68.48	-16.76	46.43	52.86	-15.34	6.20
15	0.66	17.51	68.30	-16.84	46.61	52.23	-12.07	6.00
16	1.53	10.12	45.53	-105.48	51.47	53.43	29.32	3.91
17	-0.83	10.67	47.22	-65.54	49.74	55.32	-3.14	3.90
18	-0.05	11.56	46.07	-38.72	50.65	54.17	-71.99	3.87
19	1.11	10.54	47.56	-22.99	50.63	54.47	19.05	3.81
20	-2.77	10.45	46.29	102.12	50.39	54.05	-21.67	3.88
21	1.51	11.03	46.20	-61.41	52.28	55.89	51.76	4.01
22	-2.08	11.82	45.96	121.36	49.18	54.28	-44.05	3.89
23	-0.13	10.89	46.21	34.94	52.82	54.93	26.48	3.82
24	-3.52	10.89	49.08	97.01	50.10	55.70	4.07	3.65
25	2.07	9.39	46.08	-53.59	51.65	53.65	4.94	3.97
26	0.44	11.04	47.23	-188.70	49.01	55.21	30.06	4.08
27	-4.62	9.33	47.59	141.42	51.09	54.20	-1.25	3.95
28	-1.78	10.70	45.70	-47.29	49.84	54.81	40.79	3.71
29	-2.45	10.65	47.01	-32.51	51.18	53.56	11.77	3.93
30	0.03	10.49	43.45	-149.80	50.93	53.95	14.36	3.83
31	3.37	10.57	46.51	20.87	49.67	54.60	22.79	3.96
32	0.86	7.55	19.63	-6.16	7.11	9.03	-20.02	1.08

33	-1.54	7.76	20.22	38.99	7.26	8.99	-10.41	1.12
34	-0.19	7.43	18.78	21.86	7.75	8.86	-16.09	1.15
35	-2.22	7.79	19.15	4.23	7.44	8.75	19.94	1.06
36	-1.11	7.83	20.15	20.92	6.25	8.80	-31.48	1.06
37	-0.31	7.90	19.64	8.33	7.22	9.07	18.83	1.12
38	-1.59	7.27	19.37	-18.56	6.82	8.51	21.93	1.10
39	-1.18	7.41	19.11	85.48	7.72	8.69	-17.48	1.01

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
0	-2.45	11.03	80.28	37.44	26.08	-32.86	-12.63	179.34
1	4.01	11.52	78.78	37.44	24.32	-34.91	-12.16	180.51
2	-3.35	10.90	81.50	37.26	24.80	-33.69	-10.77	180.89
3	1.73	10.84	80.38	37.51	25.47	-36.57	-15.88	183.66
4	1.27	9.42	81.14	38.14	25.13	-35.66	-13.87	180.51
5	-1.05	10.89	78.62	35.99	24.44	-34.31	-10.65	181.52
6	-0.46	10.30	81.92	35.89	24.74	-34.63	-15.73	180.07
7	2.40	10.57	78.99	37.01	23.94	-34.87	-12.35	183.67
8	1.32	10.56	80.87	37.45	24.70	-35.58	-16.41	180.05
9	1.05	9.87	79.26	37.63	24.52	-33.58	-12.25	178.55
10	-0.78	9.63	81.05	38.38	25.20	-33.14	-13.41	176.95
11	3.17	10.63	81.55	37.98	24.25	-32.39	-8.97	178.12
12	-0.12	10.57	81.11	38.18	25.06	-32.51	-14.47	180.98
13	-3.92	11.23	81.59	37.07	24.28	-35.06	-13.10	179.20
14	-1.14	11.04	81.63	36.75	25.25	-34.70	-11.65	180.54
15	1.46	11.05	80.55	36.57	23.89	-33.64	-12.62	181.91
16	-5.70	11.70	54.00	34.44	12.44	-14.39	-12.01	171.54
17	-0.70	12.09	55.21	34.71	9.92	-17.13	-11.70	168.02
18	-3.57	11.69	53.52	34.81	11.95	-13.55	-12.37	172.64
19	-3.32	12.04	54.84	35.03	11.31	-16.56	-8.97	169.62
20	2.59	12.21	55.48	35.95	11.23	-16.53	-9.56	169.59
21	2.54	12.19	54.25	35.20	10.76	-16.90	-12.28	169.58
22	-3.44	11.69	55.10	35.89	12.10	-18.59	-8.83	171.87
23	0.56	12.26	53.50	35.39	10.43	-17.85	-6.90	169.62
24	1.18	11.85	54.90	34.61	11.57	-14.31	-8.59	170.29
25	4.59	13.16	55.42	35.65	11.43	-20.08	-11.70	170.01
26	-0.12	11.06	54.09	34.94	11.37	-15.16	-13.76	172.27
27	1.10	11.81	55.45	34.85	10.99	-17.00	-9.50	170.91
28	2.78	11.51	56.06	34.44	10.57	-13.57	-8.79	172.45
29	-1.28	11.48	55.08	35.02	10.89	-15.74	-9.63	169.75
30	2.04	12.16	55.11	35.67	10.86	-17.71	-6.55	170.86
31	0.32	12.82	53.12	34.92	10.54	-18.42	-8.44	169.18
32	0.20	0.74	50.97	-0.31	16.61	-0.13	-0.95	4.08
33	-2.15	1.02	49.63	-0.32	16.22	0.23	0.22	1.95
34	1.20	0.57	50.53	-0.69	16.60	0.05	-0.14	2.81
35	2.07	0.84	50.16	-0.49	16.35	0.75	-0.83	2.72
36	0.80	0.74	49.51	0.39	16.42	-0.53	-1.24	3.37
37	-1.45	1.05	52.96	-0.54	16.22	0.77	-0.23	3.04
38	2.48	0.88	49.48	0.67	16.29	-0.46	-1.51	3.83
39	-0.81	0.65	50.20	-0.45	16.38	-0.34	-0.14	3.97

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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0	-19.43	-15.41	-0.54	35.39	4.01	146.63	-31.68	-3.15
1	-18.88	-15.15	-0.37	34.72	4.18	148.07	-31.76	-3.56
2	-16.02	-15.31	-0.38	33.72	4.18	145.90	-32.03	-3.18
3	-17.80	-15.75	-0.39	36.39	4.08	145.63	-31.54	-3.47
4	-15.90	-15.69	-0.30	34.77	4.13	149.16	-31.20	-3.47
5	-18.98	-15.68	-0.58	33.97	4.12	147.36	-31.71	-3.33
6	-15.90	-15.37	-0.49	34.31	4.29	146.96	-30.78	-3.23
7	-14.81	-15.73	-0.31	36.06	4.14	150.82	-32.18	-3.67
8	-15.48	-15.58	-0.58	36.41	4.13	146.32	-31.10	-3.39
9	-16.55	-15.89	-0.37	32.95	3.99	141.73	-30.59	-3.45
10	-18.58	-15.48	-0.48	33.49	4.40	148.23	-30.74	-3.45
11	-15.63	-15.39	-0.38	35.93	4.06	144.15	-30.84	-3.40
12	-17.34	-15.78	-0.58	35.76	4.10	140.54	-32.14	-3.58
13	-16.01	-15.44	-0.51	35.03	4.19	142.21	-31.70	-3.44
14	-19.00	-15.39	-0.63	35.03	4.06	146.35	-31.59	-3.67
15	-16.05	-15.61	-0.51	35.06	4.04	144.24	-31.00	-3.59
16	-1.73	-11.69	0.83	17.68	4.06	115.00	-27.19	-5.03
17	-4.29	-12.41	1.18	16.91	3.80	117.37	-26.45	-5.33
18	-2.73	-12.25	1.21	17.51	3.83	119.84	-28.87	-5.24
19	-1.43	-12.78	1.09	16.03	3.87	118.57	-27.48	-5.28
20	-1.08	-12.82	1.18	16.30	3.94	118.99	-26.64	-4.93
21	-3.12	-12.39	0.98	17.09	3.77	116.73	-27.35	-5.06
22	-2.46	-12.29	1.12	15.90	3.83	119.51	-27.39	-4.75
23	-1.88	-12.16	1.06	16.70	3.81	120.08	-27.74	-5.30
24	-3.35	-12.24	1.09	14.93	3.90	117.65	-27.21	-5.13
25	-1.85	-11.74	1.20	18.98	3.86	118.21	-27.50	-5.19
26	-2.53	-12.60	1.34	18.04	3.83	118.75	-26.64	-5.50
27	-4.14	-11.95	1.01	18.05	3.80	120.99	-27.10	-4.89
28	-2.49	-12.69	1.15	18.06	3.69	117.75	-26.73	-5.11
29	-2.52	-12.69	1.30	15.46	4.07	116.11	-27.32	-5.01
30	-1.51	-12.10	1.15	17.59	3.63	123.35	-27.39	-5.32
31	-4.63	-12.59	1.03	17.39	3.88	119.73	-27.34	-5.17
32	-5.22	1.64	2.40	15.00	2.34	28.20	-5.04	-5.62
33	-5.96	1.34	2.52	14.98	2.36	28.53	-4.99	-5.55
34	-5.47	1.64	2.42	14.89	2.39	29.51	-4.77	-5.28
35	-4.58	1.75	2.52	14.33	2.30	31.32	-5.18	-5.29
36	-6.27	1.70	2.46	14.36	2.42	31.69	-5.05	-5.67
37	-5.85	1.50	2.37	14.83	2.33	29.47	-5.19	-5.62
38	-4.86	1.37	2.45	14.20	2.32	30.31	-5.13	-5.42
39	-6.24	1.66	2.42	14.65	2.30	31.65	-5.26	-5.40

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
0	-10.74	-128.40	-12.06	-27.68	-47.57	129.47	-144.26	5.82
1	-10.79	-126.25	-6.60	-27.37	30.05	129.30	-151.57	0.18
2	-10.74	-126.91	22.40	-24.92	8.00	127.04	-148.75	4.11
3	-10.77	-125.83	-0.47	-27.97	-22.77	127.09	-144.81	-5.16
4	-10.78	-123.99	31.65	-28.65	24.12	128.49	-154.50	13.78
5	-10.58	-125.15	-1.09	-27.15	-24.40	127.57	-149.54	5.15
6	-10.85	-124.37	-9.75	-27.32	-4.55	126.41	-145.43	-4.92
7	-10.65	-127.00	0.46	-27.02	-13.23	129.75	-144.09	-13.95
8	-11.02	-126.29	-13.90	-26.51	-7.75	127.78	-152.92	8.36
9	-10.68	-124.07	-7.82	-27.67	-23.63	127.87	-150.02	8.26
10	-10.83	-124.06	47.68	-28.79	7.39	124.57	-149.34	-3.62

11	-10.80	-128.19	-2.30	-25.63	-6.57	129.60	-150.58	-9.25
12	-10.87	-124.73	-54.90	-24.31	41.58	134.82	-151.79	-4.73
13	-10.82	-127.07	-0.98	-28.09	7.94	130.29	-148.24	-10.15
14	-11.01	-125.00	-2.61	-26.15	-9.59	129.00	-148.98	-19.14
15	-10.69	-125.55	6.16	-27.16	5.81	130.83	-152.90	-9.64
16	-11.43	-133.31	27.27	-18.30	-26.33	148.91	-121.02	7.67
17	-11.74	-134.46	-52.34	-16.89	-23.99	151.23	-120.04	-19.73
18	-11.45	-130.80	18.52	-16.85	-48.77	150.98	-122.92	10.85
19	-11.27	-133.08	-29.76	-17.58	3.69	149.84	-121.37	-4.93
20	-11.27	-134.65	8.68	-16.64	34.24	149.00	-118.06	-9.55
21	-11.71	-137.87	30.52	-14.52	0.44	149.28	-119.23	7.09
22	-11.44	-133.43	27.53	-14.99	8.31	149.32	-122.39	-19.43
23	-11.34	-135.85	-13.46	-14.85	0.51	149.46	-118.80	12.65
24	-11.46	-134.89	-32.30	-18.64	3.14	145.98	-119.97	5.91
25	-10.92	-135.14	-16.97	-16.50	9.98	148.08	-126.66	0.91
26	-11.38	-132.17	38.54	-18.23	23.72	150.56	-114.45	1.06
27	-11.44	-134.82	-43.20	-14.82	-21.37	150.11	-123.74	-7.24
28	-11.64	-136.08	-8.20	-17.05	0.87	153.36	-125.50	18.53
29	-11.25	-132.85	-28.20	-17.77	-16.83	146.70	-118.61	-2.81
30	-11.21	-133.55	-22.12	-16.42	-9.17	150.89	-123.39	7.29
31	-11.50	-135.59	16.35	-17.71	-39.44	149.92	-117.21	-17.87
32	-1.75	-17.34	-4.56	-15.38	-4.53	27.52	-4.76	-15.17
33	-1.66	-18.40	-1.44	-15.69	-11.54	27.41	-2.83	-8.66
34	-1.72	-17.15	3.10	-14.93	-13.85	27.41	0.29	18.03
35	-1.60	-18.29	10.53	-16.01	10.69	27.36	-0.67	-6.10
36	-1.70	-17.23	-4.85	-14.95	7.00	25.49	-2.44	-2.75
37	-1.65	-17.09	-18.55	-15.00	3.78	27.40	1.75	3.17
38	-1.68	-17.34	10.25	-15.93	-2.93	27.08	-9.32	12.10
39	-1.66	-16.95	6.80	-16.81	-3.07	27.33	-0.14	-2.79

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
0	37.14	79.32	-51.17	-1.76	107.54	-136.45	-0.70	49.45
1	39.73	78.51	-53.02	-3.67	107.24	-137.40	-0.92	48.53
2	38.32	79.55	-53.33	6.10	109.08	-137.60	-0.44	47.48
3	43.04	78.76	-49.78	-5.20	109.63	-137.79	-0.41	49.72
4	41.97	78.51	-51.25	-3.82	108.30	-134.67	-0.72	47.28
5	43.91	79.20	-53.25	5.02	104.46	-134.65	-0.72	50.25
6	41.71	79.71	-51.24	9.31	106.65	-135.02	-0.97	48.74
7	44.52	80.29	-48.51	-2.75	107.04	-140.33	0.13	49.50
8	39.20	80.04	-52.02	-1.21	109.14	-135.03	-0.87	48.70
9	42.96	79.10	-48.95	0.10	107.27	-135.84	-1.15	51.11
10	42.98	80.11	-51.37	2.86	107.63	-136.54	-1.02	49.77
11	41.67	78.52	-54.87	9.25	107.78	-137.56	-0.36	49.93
12	41.54	78.66	-49.72	12.58	108.48	-138.05	-1.51	50.82
13	45.53	79.40	-53.00	-0.90	106.29	-138.06	-1.22	48.46
14	41.05	78.05	-50.05	5.44	108.17	-134.77	-1.00	49.65
15	40.71	80.07	-56.73	-3.43	106.52	-136.43	-0.06	50.09
16	37.90	83.28	-64.08	-0.55	90.49	-120.94	4.79	58.73
17	39.35	84.15	-61.34	14.09	91.22	-115.20	4.42	58.38
18	39.25	83.87	-64.28	-2.73	90.15	-119.81	5.17	59.14
19	44.51	83.85	-64.77	-5.83	91.64	-120.52	4.96	59.80
20	39.23	84.11	-64.08	-0.47	90.61	-119.79	4.41	58.04
21	33.64	84.97	-66.24	2.00	90.84	-119.05	4.36	55.65

22	40.09	84.70	-68.59	8.97	89.54	-122.30	4.21	58.89
23	39.94	84.58	-64.95	9.90	90.45	-118.55	4.20	58.47
24	41.44	83.21	-61.49	21.96	86.40	-121.01	5.88	59.02
25	43.13	84.06	-71.09	5.92	91.68	-121.74	5.43	58.39
26	40.08	84.02	-62.57	-16.45	90.01	-124.94	5.37	59.22
27	42.77	83.17	-60.93	-3.80	90.21	-119.34	4.16	59.00
28	38.19	83.14	-68.75	0.93	89.45	-118.83	4.65	58.62
29	40.28	84.08	-66.06	2.04	90.85	-121.14	4.95	58.58
30	38.60	83.57	-63.85	4.02	90.34	-121.73	5.02	59.52
31	38.72	83.98	-62.23	-8.14	90.87	-118.15	4.47	57.21
32	11.57	21.55	-10.03	-6.33	68.62	-1.53	-1.26	5.29
33	12.52	21.60	-9.90	5.14	67.28	-2.03	-1.17	6.06
34	10.52	21.27	-9.92	7.48	62.26	-1.26	-1.10	6.49
35	9.72	21.00	-9.55	4.65	67.09	-2.90	-1.42	5.32
36	10.18	21.30	-10.46	0.53	68.50	-1.87	-1.28	5.86
37	11.41	21.45	-10.21	-5.58	69.19	-2.94	-0.93	6.38
38	10.20	21.55	-10.48	-0.02	72.00	-2.62	-1.45	5.82
39	10.76	21.90	-9.25	2.27	70.02	-3.28	-1.69	5.49

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
0	-12.86	129.34	-0.04	14.92	138.37	-1.55	89.62	30.73
1	-5.85	128.69	0.01	15.19	137.24	-1.35	-3.02	-80.51
2	-4.35	128.38	0.04	15.14	134.51	-1.28	41.45	-64.49
3	-12.07	129.28	0.00	15.99	138.44	-1.79	43.17	-32.41
4	-29.61	130.64	0.05	15.92	138.34	-1.07	-61.32	-48.06
5	-8.28	128.31	0.04	16.59	134.68	-1.42	22.30	2.66
6	19.35	130.30	0.02	13.87	136.49	-1.67	-84.92	0.79
7	24.36	127.83	0.04	15.94	138.86	-1.90	0.65	-3.65
8	-14.01	128.40	0.03	15.30	137.88	-1.44	-4.38	-31.18
9	0.91	129.12	0.02	14.12	135.10	-1.55	69.96	4.18
10	23.68	129.32	0.07	16.18	140.35	-1.81	-95.92	-40.96
11	-17.89	127.87	-0.02	14.63	137.22	-1.64	-53.97	-21.93
12	-16.16	127.54	0.03	15.68	139.52	-1.33	-2.36	-18.42
13	-9.14	128.00	0.04	15.45	137.55	-1.47	-154.65	-37.01
14	6.32	128.51	0.01	15.55	135.27	-1.61	-75.40	17.78
15	-56.04	128.95	-0.01	15.81	140.72	-1.50	15.30	-0.58
16	11.07	103.86	0.32	12.40	106.49	-3.15	-17.66	-6.34
17	1.20	103.57	0.30	11.80	105.25	-3.35	158.30	-11.96
18	15.81	104.02	0.31	12.73	105.76	-3.14	27.25	38.46
19	-38.30	105.03	0.26	13.24	105.45	-3.08	55.65	-53.71
20	-11.59	103.10	0.28	12.71	104.03	-3.53	-163.96	49.02
21	-12.52	102.37	0.36	13.19	102.12	-2.92	-173.39	-5.65
22	-9.62	103.25	0.33	12.39	108.93	-2.99	52.22	-161.59
23	25.92	103.48	0.29	11.74	106.99	-2.55	105.11	-85.85
24	-5.50	103.73	0.35	12.51	104.40	-3.33	69.39	-5.04
25	-10.31	105.43	0.32	12.82	105.99	-3.71	-26.84	-118.93
26	18.71	106.00	0.30	10.96	106.89	-3.08	-32.36	-84.52
27	20.64	104.18	0.31	10.98	105.86	-2.84	-60.72	-27.19
28	-15.68	105.50	0.30	12.70	106.94	-2.97	88.69	23.52
29	-13.79	104.57	0.35	11.66	106.15	-3.25	26.48	-34.70
30	0.58	103.49	0.33	12.68	105.58	-2.79	-41.33	6.57
31	-41.18	104.89	0.34	12.31	105.97	-3.75	29.53	-6.35
32	1.27	44.57	0.12	13.53	42.01	2.37	17.83	3.10

33	6.12	45.81	0.10	13.35	42.85	2.35	-40.89	-6.28
34	-2.92	44.37	0.10	13.49	42.40	2.50	118.60	-8.15
35	-5.72	45.65	0.09	13.59	42.95	2.35	-24.50	42.41
36	8.16	46.19	0.13	13.25	42.19	2.58	-23.66	-14.21
37	-2.52	45.21	0.10	14.30	42.49	2.21	10.46	8.05
38	1.03	44.35	0.08	13.11	42.30	2.30	-19.72	-34.53
39	7.84	44.14	0.07	12.77	41.49	2.14	-87.32	-9.49

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
40	19.90	-2.13	3.40	-23.58	14.75	9.48	3.86	-5.28
41	18.83	-2.16	2.85	-22.88	14.32	8.82	4.18	-4.88
42	20.70	-1.65	3.27	-20.35	15.25	9.24	3.58	-4.35
43	20.25	-1.77	3.20	-22.22	15.27	9.16	3.48	-3.80
44	24.14	-2.06	3.32	-21.09	15.40	8.40	4.18	-4.42
45	18.27	-1.69	3.27	-23.09	14.59	8.75	3.83	-4.73
46	25.79	-2.16	2.67	-23.50	13.93	9.50	3.72	-4.28
47	20.38	-2.29	3.08	-22.68	15.06	8.65	2.52	-5.19
48	50.37	-1.77	3.59	-24.60	18.28	10.50	5.85	-7.96
49	49.85	-2.31	3.47	-28.05	18.13	10.17	7.16	-7.75
50	41.67	-2.30	3.79	-26.10	18.67	10.55	6.25	-8.25
51	42.09	-2.00	3.53	-25.53	18.63	10.87	5.39	-5.86
52	50.69	-2.14	3.14	-26.33	19.35	9.35	6.52	-7.00
53	42.49	-1.43	3.63	-27.22	19.41	9.91	6.84	-7.48
54	53.71	-2.44	3.50	-31.61	18.14	11.45	6.06	-9.48
55	46.99	-2.19	3.48	-29.49	18.31	10.81	6.44	-6.38
56	51.56	-2.37	2.97	-27.93	19.01	11.39	5.33	-6.22
57	46.94	-2.41	3.00	-30.06	17.38	9.09	6.34	-7.10
58	41.31	-2.75	3.65	-25.85	19.08	9.83	6.07	-6.96
59	44.49	-2.22	3.61	-32.55	18.31	10.01	6.66	-7.11
60	45.40	-2.42	3.22	-27.27	18.00	11.33	6.80	-7.85
61	48.69	-2.05	3.04	-27.02	19.11	10.47	6.21	-8.31
62	51.91	-2.10	3.23	-24.02	18.61	10.14	6.24	-8.18
63	49.12	-2.03	3.79	-26.05	19.92	9.39	6.41	-7.48
64	14.82	-1.26	2.54	-14.92	10.38	5.70	4.36	-2.13
65	18.85	-1.68	2.35	-15.32	10.60	5.67	4.25	-2.41
66	12.30	-1.35	2.98	-13.25	10.11	6.36	5.11	-2.88
67	10.43	-0.99	2.47	-13.28	10.38	6.20	5.13	-2.15
68	10.66	-1.07	2.81	-16.65	10.22	5.83	3.98	-2.43
69	14.27	-1.20	2.24	-13.79	9.68	4.91	3.57	-2.61
70	14.63	-0.95	2.44	-13.66	10.38	4.72	4.73	-2.55
71	13.43	-1.33	2.57	-16.19	9.82	5.72	4.41	-2.84
72	12.29	-1.25	2.24	-13.42	9.93	5.64	4.34	-2.66
73	13.65	-1.66	2.41	-16.32	10.52	5.40	4.92	-2.50
74	13.37	-1.27	2.47	-15.49	9.98	5.94	3.65	-1.62
75	14.10	-1.64	2.34	-14.76	10.05	5.40	4.82	-2.77
76	16.53	-1.24	1.83	-18.07	9.51	6.44	4.40	-2.85
77	15.73	-1.15	2.18	-13.81	9.36	5.62	4.61	-2.36
78	17.20	-1.32	2.68	-13.10	9.52	5.70	4.78	-2.81
79	18.10	-1.03	2.61	-14.46	10.11	5.73	4.83	-2.49

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
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40	-0.48	-14.13	-0.26	-4.85	-32.91	4.45	1.62	14.53
41	-2.55	-14.69	-0.26	-2.59	-32.83	4.48	1.64	10.87
42	-2.32	-13.82	-0.25	-3.21	-32.11	4.71	1.92	12.41
43	0.25	-13.86	-0.32	-0.74	-32.76	4.46	2.18	10.08
44	-2.98	-14.88	-0.29	-0.15	-33.19	4.33	1.78	12.83
45	0.26	-14.82	-0.34	-3.73	-35.01	4.39	1.93	11.13
46	-0.55	-13.52	-0.27	-2.21	-32.26	4.23	1.58	13.57
47	-2.58	-14.57	-0.27	-1.27	-32.66	4.42	1.55	11.73
48	-9.49	-17.26	-0.40	-7.16	-36.29	5.87	2.34	25.11
49	-15.26	-17.26	-0.39	-9.71	-37.64	6.04	2.21	24.19
50	-17.74	-16.05	-0.38	-8.14	-37.86	5.44	2.03	27.35
51	-7.51	-17.69	-0.44	-7.78	-36.26	6.10	2.27	26.16
52	-16.09	-16.80	-0.45	-7.39	-35.79	6.14	2.29	22.75
53	-21.20	-18.29	-0.46	-6.47	-37.56	6.01	2.73	22.73
54	-14.84	-18.42	-0.34	-6.94	-35.71	6.25	2.47	23.41
55	-16.75	-17.24	-0.39	-7.01	-38.93	6.20	2.67	26.48
56	-8.39	-22.20	-0.43	-6.67	-40.21	5.83	2.38	22.94
57	-14.66	-19.71	-0.36	-8.63	-40.12	6.05	2.57	23.51
58	-16.77	-15.89	-0.40	-8.32	-38.01	6.03	2.53	23.62
59	-16.13	-20.60	-0.48	-7.39	-36.88	6.09	2.39	23.28
60	-14.21	-16.48	-0.41	-8.28	-39.00	5.52	2.26	23.17
61	-14.30	-22.10	-0.43	-8.26	-37.30	6.59	2.03	22.66
62	-17.72	-18.12	-0.41	-7.89	-37.81	6.13	2.04	25.89
63	-17.68	-18.26	-0.39	-6.30	-36.11	5.40	2.04	21.67
64	2.73	-5.49	-0.33	1.00	-13.27	1.08	2.61	11.70
65	-4.20	-5.79	-0.38	-1.97	-13.62	1.34	2.28	11.01
66	-5.39	-6.55	-0.36	0.72	-12.79	0.98	2.48	13.80
67	-4.50	-5.76	-0.33	-1.87	-13.93	0.96	2.58	11.78
68	-4.01	-5.60	-0.37	-1.22	-14.53	1.28	2.37	12.48
69	-6.29	-5.07	-0.33	0.39	-14.47	1.50	2.31	13.61
70	-6.11	-5.86	-0.40	-0.83	-14.26	1.23	2.63	10.62
71	-8.03	-5.00	-0.32	-0.78	-12.45	1.14	2.62	12.27
72	-6.81	-6.73	-0.38	0.34	-14.42	1.52	2.49	12.46
73	-7.12	-6.95	-0.33	0.12	-13.53	1.04	2.45	10.32
74	-6.57	-3.28	-0.31	0.22	-14.17	1.09	2.41	13.68
75	-6.00	-3.61	-0.33	-0.67	-13.40	1.23	2.67	12.09
76	-5.23	-7.88	-0.38	-0.19	-14.11	1.19	2.52	13.16
77	-7.27	-3.96	-0.32	1.76	-14.15	1.23	2.79	11.77
78	-3.39	-7.77	-0.34	-0.85	-13.00	1.32	2.61	10.48
79	-4.11	-6.10	-0.36	-1.01	-13.69	1.30	2.55	11.87

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
40	-1.34	7.31	20.08	-43.84	7.06	8.83	3.34	1.08
41	-1.10	7.79	19.71	8.41	7.32	8.92	15.86	1.04
42	1.31	8.00	19.19	-20.28	6.83	8.82	24.92	1.08
43	1.45	7.98	19.87	46.25	6.78	8.45	-2.96	1.00
44	0.33	7.34	19.54	-38.90	6.92	8.67	-12.94	1.11
45	1.75	7.10	20.14	-21.37	6.82	8.73	-27.46	1.13
46	0.61	8.00	20.03	-28.34	7.09	8.93	31.83	1.12
47	-0.64	7.87	19.53	-46.57	7.31	8.89	-30.39	1.07
48	-0.15	11.06	25.54	-2.80	12.05	11.55	-13.03	1.27
49	-0.12	11.11	25.52	8.12	12.20	12.65	-6.52	1.32
50	-1.32	10.83	24.11	37.99	12.63	11.38	32.37	1.35

51	0.13	10.49	25.74	8.31	12.13	11.86	-5.37	1.28
52	-1.83	11.17	26.01	33.77	12.42	11.00	34.57	1.29
53	-1.54	11.10	25.30	42.41	11.53	11.16	50.78	1.36
54	0.86	10.54	23.84	-10.95	12.27	11.29	7.27	1.27
55	0.11	11.02	25.47	-32.01	11.40	11.13	0.00	1.32
56	1.25	10.62	26.27	11.39	11.63	11.39	0.65	1.32
57	-1.16	10.55	25.32	76.10	12.37	11.06	17.89	1.41
58	-0.85	11.14	24.88	25.65	11.81	11.47	-34.03	1.31
59	-1.57	10.72	25.21	-19.84	11.67	12.12	-12.65	1.17
60	0.05	10.51	24.90	-78.84	12.63	12.01	-6.74	1.37
61	-1.47	10.82	25.26	46.37	12.04	11.15	-18.96	1.39
62	1.03	10.49	25.79	18.87	12.11	11.18	-7.43	1.35
63	0.25	10.96	26.25	59.58	11.55	11.69	-2.09	1.42
64	-0.42	5.86	11.16	66.99	8.70	6.34	31.58	1.06
65	1.35	5.34	11.18	14.04	8.27	6.43	59.34	1.01
66	0.05	5.54	10.70	-52.21	8.29	6.36	6.84	1.05
67	0.45	5.43	11.31	18.29	7.77	6.55	8.15	1.01
68	-0.61	6.04	9.77	28.80	7.99	6.69	3.61	1.10
69	-0.84	5.57	10.58	42.30	8.11	6.44	-41.16	1.11
70	0.34	5.48	10.22	22.47	8.03	6.57	-21.22	1.11
71	-0.40	6.13	10.13	-19.36	8.25	6.63	-9.41	1.03
72	-0.74	5.96	10.71	15.47	7.99	6.60	-11.79	1.10
73	0.30	5.77	10.78	-35.45	8.02	7.23	-7.79	1.12
74	0.31	6.04	9.85	-2.67	7.43	6.23	-16.70	0.99
75	0.07	5.67	11.17	-30.70	7.82	6.03	-41.32	1.03
76	0.91	6.31	10.37	65.63	7.65	6.38	-19.29	1.11
77	-0.55	5.90	10.02	41.19	7.65	6.85	26.33	1.08
78	0.38	5.93	10.95	-33.73	8.60	6.71	15.83	1.07
79	0.08	5.82	10.18	55.66	8.03	6.07	15.49	1.06

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
40	-1.53	1.02	52.10	-0.48	16.66	0.51	-0.48	3.17
41	1.12	0.85	50.58	-0.66	16.94	-0.20	-1.60	1.47
42	-0.14	0.54	49.36	-0.28	16.62	-0.54	-0.84	2.34
43	1.89	-0.17	50.32	-0.77	16.71	0.11	-0.84	2.40
44	2.72	0.93	51.66	0.03	16.64	-0.52	-0.93	3.31
45	0.90	0.89	50.79	-0.49	16.85	0.29	-0.30	2.02
46	0.92	1.06	49.05	-0.55	15.95	-0.36	-0.75	2.36
47	-1.10	0.50	50.20	-0.34	16.34	-0.29	-0.96	2.86
48	1.43	2.74	66.38	-2.26	21.67	0.95	-5.11	7.11
49	0.16	3.22	63.74	-1.00	21.20	1.23	-4.04	5.34
50	1.56	2.91	65.05	-1.22	20.65	1.57	-3.43	4.36
51	-0.08	2.84	63.90	-2.64	21.54	-0.16	-4.06	5.01
52	-1.22	3.14	64.73	-2.15	20.73	1.26	-3.17	6.76
53	2.12	3.35	63.77	-2.31	20.64	1.90	-2.79	4.15
54	-0.09	3.01	65.03	-2.19	21.60	0.00	-3.75	3.41
55	2.71	3.52	62.85	-2.18	21.41	1.02	-3.95	4.22
56	-1.83	3.25	64.90	-2.23	20.87	1.17	-5.24	6.48
57	-1.32	3.01	65.44	-1.67	20.99	0.86	-4.19	4.77
58	-1.32	3.42	67.01	-1.82	20.89	0.54	-2.96	6.01
59	-1.20	3.13	67.59	-2.06	21.62	0.29	-4.53	7.54
60	3.64	2.66	65.10	-1.51	20.53	2.64	-2.72	4.43
61	-2.11	2.97	65.31	-2.06	21.06	1.06	-5.37	5.61

62	1.09	2.90	65.18	-2.28	21.55	1.93	-3.25	4.31
63	3.76	3.61	62.63	-3.22	19.95	0.38	-4.80	6.77
64	2.01	3.49	35.08	-1.73	7.78	-0.78	0.64	-7.62
65	0.65	3.75	37.57	-1.59	8.16	0.25	1.16	-6.40
66	-0.59	3.12	35.67	-1.59	8.70	-0.65	0.44	-7.70
67	0.84	3.74	36.42	-1.44	8.46	0.59	0.32	-7.58
68	1.82	3.81	36.20	-0.96	8.04	0.50	0.78	-6.98
69	-3.57	3.82	34.70	-1.26	8.63	0.05	0.45	-6.48
70	-0.26	3.34	35.18	-1.92	8.23	-0.19	0.54	-6.45
71	0.42	3.34	36.22	-1.78	7.93	0.71	1.11	-6.50
72	-2.08	3.40	36.98	-1.86	8.30	-0.10	-0.07	-6.00
73	0.88	3.39	36.43	-1.80	8.82	1.05	2.09	-6.28
74	0.13	3.49	37.23	-1.68	7.89	0.03	0.84	-6.37
75	0.72	3.63	34.17	-1.43	8.40	-0.15	1.45	-6.18
76	1.16	3.70	37.28	-1.47	8.24	0.62	0.36	-6.41
77	-1.66	3.54	36.23	-1.32	8.74	-0.46	1.47	-6.34
78	0.57	3.54	37.16	-1.52	8.39	0.23	0.68	-5.44
79	1.19	3.07	36.90	-2.02	8.87	0.19	0.45	-6.90

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
40	-5.34	1.56	2.37	14.21	2.38	29.25	-5.24	-4.95
41	-5.33	1.96	2.51	14.89	2.30	29.91	-4.95	-5.64
42	-4.23	1.47	2.50	14.89	2.39	30.13	-5.49	-5.67
43	-5.61	1.39	2.40	14.13	2.39	30.04	-5.16	-5.43
44	-5.26	1.42	2.47	14.70	2.47	29.51	-5.46	-5.65
45	-4.85	1.38	2.49	14.53	2.35	31.15	-5.49	-5.35
46	-5.17	1.42	2.66	14.21	2.40	31.82	-5.15	-5.79
47	-5.68	1.57	2.53	15.07	2.40	29.24	-5.09	-5.72
48	-6.00	2.58	2.50	14.63	3.20	38.48	-3.62	-5.61
49	-6.67	2.35	2.72	14.89	3.23	37.63	-3.69	-6.09
50	-5.81	2.30	2.52	15.66	3.28	38.95	-4.26	-6.18
51	-5.13	2.44	2.67	15.54	3.18	36.41	-4.09	-5.84
52	-4.42	1.99	2.54	15.71	3.13	39.60	-4.01	-6.18
53	-5.04	2.06	2.66	17.16	3.12	36.76	-4.33	-6.06
54	-6.10	2.53	2.50	16.41	3.27	37.65	-4.36	-6.06
55	-5.57	2.26	2.60	16.96	3.16	40.02	-3.85	-6.00
56	-4.10	1.92	2.64	14.69	3.17	39.48	-4.48	-6.07
57	-4.38	2.29	2.61	15.62	3.22	38.06	-3.97	-6.03
58	-5.50	2.09	2.58	16.36	3.27	40.47	-3.28	-5.79
59	-4.45	2.46	2.78	16.60	3.15	39.11	-3.78	-6.16
60	-3.34	2.51	2.52	15.15	3.23	38.56	-4.02	-5.90
61	-5.04	2.20	2.58	16.64	3.16	38.05	-4.02	-5.69
62	-4.07	2.51	2.61	16.88	3.27	38.29	-3.92	-6.25
63	-5.47	2.46	2.56	17.19	3.20	36.88	-3.81	-5.98
64	-3.26	1.93	2.58	6.64	1.79	15.55	-0.68	-6.62
65	-1.96	1.80	2.75	6.53	1.80	14.47	-1.32	-6.52
66	-2.93	1.95	2.58	6.37	1.80	13.31	-1.55	-6.16
67	-2.63	1.77	2.78	6.77	1.79	14.86	-0.93	-6.19
68	-3.09	2.07	2.62	6.63	1.85	13.31	-0.92	-6.44
69	-2.86	1.95	2.66	6.79	1.74	13.74	-1.00	-6.61
70	-1.71	1.92	2.68	5.20	1.78	15.31	-1.48	-6.46
71	-3.02	1.93	2.72	6.27	1.83	13.72	-1.27	-6.62
72	-2.50	2.08	2.62	6.55	1.75	12.96	-1.31	-6.45

73	-2.41	2.11	2.70	6.32	1.82	15.33	-1.06	-6.63
74	-2.92	2.10	2.73	6.78	1.77	15.18	-1.17	-6.35
75	-2.50	1.93	2.62	6.95	1.85	15.54	-1.22	-6.55
76	-2.98	2.03	2.66	6.13	1.77	14.15	-1.40	-6.31
77	-2.17	1.67	2.73	6.52	1.86	12.99	-0.97	-6.39
78	-3.26	1.85	2.79	5.93	1.78	15.29	-1.24	-6.51
79	-2.63	2.13	2.65	6.26	1.82	14.10	-1.09	-6.49

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
40	-1.63	-17.33	-15.14	-16.17	-29.42	28.83	-1.30	-12.03
41	-1.62	-16.73	15.47	-15.91	-10.45	27.24	-4.18	-15.77
42	-1.76	-17.69	-4.99	-16.03	12.60	28.26	-1.18	19.16
43	-1.73	-16.97	12.09	-15.62	-11.85	26.94	-0.37	0.46
44	-1.61	-17.12	-5.19	-15.90	2.02	26.26	-0.46	-5.74
45	-1.70	-17.11	2.87	-15.47	-1.41	27.27	2.06	2.50
46	-1.71	-18.19	-5.68	-15.33	-1.19	27.62	1.22	2.53
47	-1.69	-17.28	-11.69	-15.44	-22.75	27.70	-1.62	1.81
48	-2.35	-23.98	-5.23	-19.65	22.91	35.28	-12.08	-13.87
49	-2.43	-27.18	5.93	-18.98	-27.08	38.11	-20.01	-4.01
50	-2.31	-25.96	-1.21	-17.47	-3.80	36.94	-16.15	0.74
51	-2.43	-25.54	-5.39	-18.91	2.69	37.51	-17.32	-10.40
52	-2.34	-25.63	-0.54	-16.74	-13.52	38.11	-12.87	1.63
53	-2.28	-26.61	18.39	-19.47	38.54	36.67	-15.14	-8.79
54	-2.24	-25.23	0.96	-18.66	-4.01	37.97	-12.54	3.16
55	-2.46	-26.20	3.85	-20.41	-20.02	38.86	-9.05	-6.67
56	-2.20	-26.94	-5.01	-19.81	20.93	36.49	-12.64	1.44
57	-2.24	-26.94	-1.68	-18.68	-1.71	36.78	-9.21	13.52
58	-2.27	-24.52	0.95	-18.90	-12.42	39.65	-21.43	-15.86
59	-2.33	-25.67	-10.70	-19.74	1.12	39.45	-18.24	7.89
60	-2.30	-26.36	11.17	-18.81	-2.69	39.28	-17.71	-4.19
61	-2.29	-24.66	3.78	-19.38	-19.97	38.21	-17.79	-5.47
62	-2.42	-24.80	11.78	-19.14	3.75	38.29	-17.86	-0.73
63	-2.44	-24.51	-21.78	-18.50	23.88	36.49	-15.04	-0.89
64	-1.83	-13.40	-11.14	-13.47	6.68	20.28	-11.60	8.74
65	-1.62	-15.70	1.68	-12.90	6.44	19.59	-9.48	-7.91
66	-1.80	-13.18	-15.05	-13.18	1.99	18.97	-9.93	-3.74
67	-1.63	-13.75	-7.33	-13.37	-23.16	18.82	-6.33	-1.89
68	-1.76	-13.83	8.46	-12.67	5.10	20.29	-9.76	5.80
69	-1.74	-13.90	1.52	-12.82	-22.21	18.64	-11.50	-0.64
70	-1.78	-14.92	1.24	-12.89	4.97	19.29	-7.71	-4.83
71	-1.90	-13.09	-9.94	-13.40	-18.68	19.39	-5.61	6.31
72	-1.81	-14.30	-9.16	-13.84	13.20	18.53	-6.02	-1.96
73	-1.76	-14.66	-2.27	-12.31	-4.24	18.90	-6.41	14.15
74	-1.79	-13.88	-6.26	-12.21	-0.23	19.15	-10.59	-0.15
75	-1.64	-13.15	2.92	-13.45	-9.89	19.10	-10.12	-5.30
76	-1.77	-13.07	4.78	-13.52	-1.98	19.49	-8.17	-0.11
77	-1.67	-13.32	0.73	-13.84	5.22	20.52	-11.56	-5.77
78	-1.77	-13.99	-4.65	-12.55	-8.03	18.97	-7.84	0.31
79	-1.70	-13.62	3.32	-13.01	5.30	20.11	-7.49	7.78

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
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40	10.47	21.27	-10.21	0.10	68.25	-3.70	-1.40	6.15
41	10.05	21.29	-10.73	-5.80	66.81	-3.54	-1.12	5.55
42	9.65	21.44	-10.41	3.16	64.95	-2.16	-1.26	5.51
43	10.09	20.98	-9.71	-1.86	66.84	-1.27	-1.51	5.68
44	10.76	21.46	-10.07	-3.57	67.32	-1.94	-1.25	6.11
45	12.36	21.09	-11.19	-6.00	66.15	-1.00	-1.45	5.81
46	9.57	21.06	-10.42	-7.61	64.58	-2.32	-1.27	4.69
47	10.48	21.27	-8.46	-5.23	67.39	-2.78	-1.50	5.91
48	19.85	26.07	-17.78	5.77	76.70	-5.97	-1.33	7.34
49	17.76	26.10	-18.03	-11.44	73.40	-5.48	-1.67	5.96
50	19.37	25.92	-17.34	-2.21	75.97	-4.95	-1.61	6.77
51	19.60	26.19	-15.93	1.24	77.29	-6.90	-0.94	6.74
52	20.65	25.47	-17.51	-0.40	74.81	-5.15	-1.53	6.92
53	20.51	24.90	-17.03	6.78	78.03	-4.57	-1.38	7.08
54	19.00	25.89	-16.67	-8.22	78.44	-7.80	-0.90	5.93
55	19.61	26.07	-16.10	-4.31	75.29	-5.28	-0.84	7.33
56	17.19	25.63	-17.16	-4.83	76.12	-6.28	-1.30	7.15
57	16.12	26.62	-18.85	-10.03	76.11	-7.80	-1.48	6.26
58	22.83	25.80	-18.76	0.78	79.84	-6.48	-1.62	6.51
59	18.77	25.66	-19.39	-3.45	75.39	-5.64	-1.89	5.89
60	18.65	26.05	-17.83	9.61	75.07	-4.95	-1.50	7.25
61	20.10	25.50	-17.72	5.46	79.24	-5.64	-0.75	6.04
62	18.06	26.18	-16.77	-0.60	75.60	-5.70	-1.29	5.48
63	20.15	26.14	-18.30	0.52	77.40	-5.65	-1.63	6.15
64	4.17	11.78	-11.14	-6.19	47.76	4.84	-0.10	1.22
65	4.95	11.91	-11.29	-7.40	49.57	5.36	-0.13	0.55
66	5.23	11.69	-11.24	2.25	47.65	3.93	-0.44	1.66
67	4.34	11.68	-12.65	-2.83	44.20	6.00	-0.18	1.83
68	4.77	11.73	-10.81	-7.30	48.16	4.80	0.07	1.50
69	5.63	12.02	-11.26	-5.67	46.34	4.37	-0.32	1.27
70	6.80	11.44	-12.79	4.52	46.72	4.09	-0.01	1.03
71	4.40	11.61	-10.90	3.28	46.91	5.31	-0.15	1.20
72	5.63	11.86	-9.84	-2.76	47.55	4.03	-0.09	1.07
73	5.57	11.65	-11.73	4.00	48.09	5.29	-0.14	1.55
74	6.56	12.02	-11.78	5.11	47.54	4.43	-0.34	1.50
75	4.76	11.61	-12.34	-1.54	46.82	4.11	-0.51	1.09
76	3.83	11.41	-11.62	5.85	45.35	5.55	-0.03	1.42
77	5.23	11.93	-10.66	-2.98	46.92	4.34	-0.39	1.13
78	6.95	12.23	-11.66	2.36	44.43	5.72	-0.37	1.19
79	5.55	11.92	-10.93	2.02	45.36	5.01	-0.28	1.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
40	-13.21	46.06	0.12	12.76	40.97	2.21	-5.81	30.19
41	5.60	43.87	0.12	12.55	40.42	2.20	-5.66	10.59
42	1.30	43.78	0.08	13.33	42.36	2.34	104.95	11.44
43	-7.10	45.74	0.12	12.70	41.80	2.17	53.45	1.42
44	-15.01	43.41	0.08	13.65	40.16	2.34	7.38	7.04
45	-4.50	44.35	0.10	13.38	42.10	2.24	-60.08	29.55
46	-6.08	43.93	0.11	12.88	41.53	2.30	43.67	-37.28
47	-1.42	43.77	0.10	12.83	42.02	2.03	-33.22	-1.93
48	7.40	55.67	0.34	15.31	49.80	3.58	-32.69	-7.42
49	0.03	55.98	0.35	15.00	48.94	3.07	-12.52	-12.64
50	13.40	56.33	0.38	16.14	50.79	2.89	79.93	-57.05

51	-14.93	53.37	0.35	15.22	51.09	2.85	70.33	27.06
52	2.81	54.09	0.35	15.14	49.42	3.15	3.52	-55.33
53	-6.01	56.32	0.35	14.90	50.48	2.85	-83.24	-53.13
54	-22.63	56.03	0.32	15.27	49.90	2.96	131.84	-18.47
55	-12.38	57.33	0.34	14.90	48.67	2.74	61.03	8.29
56	15.92	55.65	0.35	16.30	50.40	2.95	-128.30	-29.08
57	5.86	54.69	0.34	14.83	50.68	3.04	60.23	14.95
58	6.32	56.35	0.37	16.09	50.08	3.27	-61.65	-31.02
59	-4.18	54.81	0.39	15.12	49.89	2.74	48.95	-25.32
60	-1.82	55.12	0.32	16.16	49.81	2.53	23.54	38.64
61	-6.60	56.15	0.31	14.82	50.61	3.48	-15.81	-62.06
62	13.52	55.60	0.36	14.65	50.81	2.75	-55.36	2.84
63	6.26	55.11	0.35	16.22	50.09	2.78	-38.96	-9.46
64	-6.41	21.29	0.06	13.13	25.53	2.19	3.75	28.78
65	5.40	20.54	0.03	12.43	26.21	2.26	-53.85	-56.62
66	-1.36	21.26	0.08	12.95	24.86	1.82	19.76	-12.86
67	2.78	19.53	0.06	13.77	25.62	1.69	-28.92	2.03
68	-2.41	19.89	0.07	12.87	24.40	2.26	4.00	8.27
69	-0.42	20.76	0.08	13.75	25.35	2.40	-7.41	-16.89
70	2.68	21.11	0.06	12.80	26.02	1.63	-34.09	-1.22
71	-2.07	20.29	0.06	13.23	24.64	2.14	-0.73	27.99
72	13.77	19.43	0.03	12.95	25.40	2.05	-41.90	41.49
73	6.38	21.39	0.07	12.67	25.45	2.27	64.46	10.51
74	14.28	20.99	0.05	12.33	26.63	1.89	72.91	18.93
75	11.50	20.47	0.07	13.27	25.94	1.95	95.35	4.58
76	-2.82	20.91	0.06	12.71	25.71	2.04	58.62	12.36
77	-11.93	21.90	0.02	13.81	25.86	2.21	-10.84	12.70
78	-11.93	20.49	0.07	13.10	25.82	2.10	20.80	15.62
79	-4.90	21.84	0.05	13.09	25.78	1.86	-24.02	21.73

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
80	237.62	-19.42	-15.00	22.16	76.44	7.36	55.13	20.11
81	245.81	-19.54	-15.13	18.05	77.32	7.05	55.20	19.51
82	239.03	-19.32	-14.10	14.18	76.56	7.65	54.42	21.17
83	239.85	-19.51	-14.78	16.79	75.94	7.74	53.67	19.67
84	236.54	-19.33	-14.34	17.57	76.25	5.88	54.22	20.28
85	233.79	-19.10	-14.73	18.82	77.02	7.74	55.42	19.18
86	244.29	-19.38	-14.24	21.06	76.84	7.05	53.39	21.57
87	245.46	-19.23	-14.63	19.52	76.89	6.20	55.06	19.29
88	237.27	-19.52	-15.00	11.51	75.62	6.84	54.76	18.98
89	246.30	-19.38	-14.77	19.82	76.40	7.87	53.51	19.31
90	239.99	-19.66	-14.51	23.01	78.38	7.62	54.41	20.58
91	246.68	-19.32	-14.48	19.30	76.38	6.57	54.02	19.59
92	244.90	-18.74	-14.76	19.33	75.50	8.02	53.86	19.84
93	240.25	-19.46	-14.08	15.51	77.25	6.68	54.41	20.96
94	236.25	-19.15	-14.29	15.40	77.92	8.29	54.72	20.92
95	234.53	-19.51	-14.79	15.95	76.96	7.71	53.95	19.60
96	225.80	-14.96	-12.69	32.62	79.10	-8.15	47.15	29.30
97	224.72	-15.06	-12.31	38.11	78.67	-8.34	47.15	28.12
98	219.05	-15.08	-12.67	33.06	78.99	-8.10	46.83	29.35
99	225.28	-15.07	-12.13	34.86	78.98	-7.64	46.64	29.02
100	219.18	-15.13	-12.51	32.24	79.59	-8.61	46.55	28.06
101	223.01	-14.98	-12.59	32.56	78.79	-7.85	47.48	29.74

102	226.92	-15.06	-12.82	35.50	78.13	-8.46	47.53	29.39
103	218.47	-15.15	-12.70	31.58	77.22	-8.28	47.09	28.69
104	217.40	-15.08	-12.47	31.83	77.80	-8.14	47.10	29.56
105	211.59	-15.13	-12.39	34.55	78.57	-8.92	46.36	27.44
106	218.37	-14.92	-13.15	31.41	78.23	-8.59	47.33	28.58
107	213.20	-15.28	-12.69	32.20	78.63	-8.15	47.25	28.56
108	211.17	-15.16	-12.35	33.61	78.13	-8.64	46.45	29.12
109	225.61	-15.22	-12.14	30.36	76.95	-7.72	47.83	29.32
110	225.64	-15.32	-12.32	30.59	79.54	-8.44	47.85	29.54
111	224.91	-15.05	-12.61	33.73	78.88	-8.69	47.23	29.07
112	-89.51	-0.54	3.33	-33.39	1.42	11.15	2.84	-3.79
113	-86.30	-0.64	3.48	-32.33	1.63	11.26	2.88	-3.91
114	-80.17	-0.41	3.36	-33.65	1.40	11.45	2.87	-2.39
115	-86.94	-0.54	3.66	-33.07	1.89	10.91	3.66	-2.75
116	-80.07	-0.67	3.73	-32.76	1.05	11.32	2.95	-2.93
117	-87.45	-0.91	3.55	-31.94	1.72	11.40	3.03	-3.38
118	-80.27	-0.56	3.49	-31.59	1.38	11.19	3.53	-3.49
119	-90.36	-0.70	3.58	-32.58	1.11	10.70	3.54	-3.50

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
80	-186.45	-104.53	-2.49	-80.09	-134.45	10.24	-14.72	95.88
81	-192.97	-107.51	-2.51	-80.49	-133.61	10.69	-14.16	94.57
82	-187.71	-108.95	-2.46	-79.35	-133.97	10.30	-14.36	97.20
83	-196.02	-105.38	-2.44	-78.86	-135.96	9.68	-14.04	92.95
84	-194.61	-105.27	-2.45	-79.94	-133.74	10.20	-14.61	95.05
85	-198.68	-106.99	-2.42	-81.39	-132.72	10.63	-14.47	94.83
86	-194.46	-107.19	-2.42	-82.50	-134.72	10.43	-14.56	95.38
87	-200.13	-106.43	-2.49	-78.22	-134.53	10.86	-14.80	93.82
88	-193.98	-106.70	-2.47	-81.14	-133.25	10.47	-14.52	93.65
89	-196.03	-107.53	-2.45	-80.79	-133.78	9.95	-14.23	95.79
90	-197.87	-106.99	-2.47	-79.13	-135.12	10.49	-14.67	92.70
91	-196.90	-110.92	-2.43	-82.52	-133.55	10.59	-14.29	95.36
92	-197.47	-106.85	-2.47	-82.24	-133.42	10.28	-14.49	95.94
93	-195.18	-104.82	-2.46	-82.04	-134.55	10.47	-14.71	94.09
94	-195.83	-106.77	-2.42	-79.90	-131.30	10.06	-14.41	92.75
95	-195.81	-104.92	-2.43	-82.97	-133.36	10.49	-14.20	92.97
96	-181.47	-115.06	-2.02	-70.47	-131.47	11.38	-17.41	86.99
97	-188.01	-114.36	-2.00	-72.63	-131.68	10.99	-17.24	89.31
98	-180.37	-115.61	-1.98	-70.70	-131.42	10.72	-17.68	87.15
99	-180.37	-114.48	-2.02	-67.98	-131.22	10.86	-17.79	89.21
100	-182.21	-111.90	-2.00	-70.02	-132.86	10.41	-17.62	92.41
101	-178.91	-117.06	-2.03	-71.92	-129.11	11.31	-17.42	90.89
102	-181.48	-113.66	-2.01	-69.24	-130.05	11.09	-17.57	94.75
103	-180.28	-117.23	-2.03	-71.54	-128.29	10.24	-17.24	92.37
104	-185.23	-115.73	-2.01	-69.82	-129.67	11.10	-17.39	87.03
105	-182.95	-111.77	-2.03	-71.93	-130.92	11.05	-17.50	90.66
106	-180.42	-115.91	-2.02	-71.93	-132.11	10.58	-17.60	89.10
107	-184.49	-115.82	-2.04	-70.93	-133.27	10.63	-17.39	92.69
108	-182.04	-114.89	-2.02	-73.68	-130.74	10.27	-17.30	88.48
109	-187.39	-116.94	-2.00	-70.45	-133.71	10.58	-17.37	88.44
110	-183.45	-115.03	-2.00	-72.48	-131.80	10.55	-17.51	87.44
111	-178.97	-116.76	-2.03	-70.39	-131.76	11.18	-17.83	90.02
112	53.55	15.02	-0.01	26.67	-5.88	-9.21	0.67	-27.37

113	55.15	13.52	-0.06	23.88	-7.50	-9.22	0.32	-26.84
114	55.34	13.76	-0.05	25.76	-6.52	-9.00	0.45	-23.41
115	56.13	12.27	-0.04	26.72	-7.43	-8.65	0.23	-25.40
116	58.20	12.54	0.00	27.33	-8.21	-9.11	0.39	-25.47
117	53.40	11.55	-0.04	24.89	-6.52	-9.32	0.47	-26.03
118	57.18	15.39	-0.04	26.82	-6.54	-9.12	0.17	-26.77
119	53.75	13.14	-0.04	25.75	-5.81	-9.26	0.44	-26.64

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
80	1.25	31.48	36.45	5.03	-4.19	-13.70	34.65	3.63
81	1.58	31.20	35.32	3.71	-4.06	-13.75	-25.86	3.60
82	0.39	32.65	36.13	-66.31	-4.02	-13.44	2.89	3.60
83	1.13	32.53	35.72	9.35	-4.30	-13.69	-12.59	3.59
84	1.32	32.05	35.43	77.97	-4.23	-13.75	-11.80	3.64
85	0.40	32.67	35.83	10.64	-4.27	-13.54	-13.35	3.59
86	1.24	31.25	36.79	-0.20	-4.24	-13.69	-7.93	3.61
87	0.63	32.68	36.03	-9.71	-3.66	-13.65	28.19	3.65
88	1.29	31.34	35.88	45.80	-4.03	-13.27	-1.63	3.60
89	-1.15	32.21	35.54	38.92	-4.00	-13.62	-23.70	3.68
90	-0.43	31.99	35.51	-10.68	-4.34	-13.83	-29.95	3.70
91	0.06	31.39	37.08	16.14	-3.99	-13.78	-12.77	3.64
92	-2.56	32.15	36.26	-18.55	-3.82	-13.57	-2.25	3.67
93	-1.69	31.85	35.62	-46.41	-3.96	-13.64	-21.34	3.64
94	0.62	32.69	35.98	-64.99	-4.18	-13.56	-7.76	3.68
95	2.32	32.37	36.68	2.72	-3.94	-13.79	-10.48	3.62
96	1.91	27.17	35.46	31.42	-6.83	-14.35	-4.68	3.04
97	-0.84	28.04	36.09	66.63	-6.64	-14.40	-26.19	3.11
98	-0.22	28.28	35.61	-27.73	-6.88	-14.78	-1.46	3.08
99	0.40	28.54	36.22	-57.05	-6.45	-14.68	10.02	3.12
100	-1.97	28.51	35.25	-1.24	-6.95	-14.54	-11.66	3.11
101	-0.27	28.64	35.64	-32.40	-6.88	-14.72	-4.98	3.09
102	-0.47	27.32	35.97	27.27	-6.83	-14.57	10.59	3.08
103	0.44	28.14	35.71	-30.29	-7.15	-14.77	-9.52	3.05
104	1.54	28.04	35.70	17.22	-7.02	-14.61	-8.51	3.08
105	1.01	27.15	36.02	15.69	-6.67	-14.79	-14.83	3.10
106	-0.91	27.98	35.25	12.21	-6.68	-14.71	-3.81	3.04
107	0.73	26.90	35.44	-45.54	-7.21	-14.30	-8.87	3.12
108	0.21	28.51	35.12	-4.38	-6.82	-14.75	-32.65	3.15
109	1.95	28.09	35.19	-12.43	-7.11	-14.39	17.18	3.04
110	-0.93	28.33	36.14	-55.64	-6.91	-14.73	9.19	3.13
111	1.20	28.73	35.63	-23.48	-6.74	-14.51	0.11	3.05
112	-0.02	-4.26	4.39	-21.84	-1.95	-1.90	4.25	0.50
113	-0.19	-5.10	4.80	-4.81	-1.96	-1.52	0.62	0.54
114	-0.55	-5.07	4.62	20.67	-2.08	-1.80	1.84	0.56
115	-0.12	-4.51	4.51	41.62	-1.85	-1.93	2.57	0.56
116	-0.88	-3.89	4.43	-8.71	-1.84	-1.74	-0.13	0.55
117	-0.21	-4.95	4.50	-8.92	-1.94	-1.54	-3.25	0.54
118	-0.11	-4.62	4.46	15.90	-1.87	-1.91	1.65	0.54
119	-0.29	-4.89	4.58	-6.43	-1.68	-1.77	-5.02	0.52

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
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80	-0.72	13.41	156.23	-22.05	54.69	-21.12	-36.70	-16.35
81	1.23	12.76	154.12	-21.44	53.34	-21.60	-37.59	-15.05
82	1.64	13.48	158.87	-21.61	53.88	-20.64	-36.38	-19.94
83	-0.66	13.19	153.77	-22.15	54.43	-20.25	-34.46	-17.11
84	0.56	13.61	156.37	-21.57	54.48	-19.82	-35.80	-16.25
85	-0.14	13.52	158.07	-21.30	54.41	-20.91	-34.52	-12.68
86	0.41	13.54	156.75	-21.79	53.12	-20.37	-36.33	-13.44
87	-1.00	12.92	156.99	-21.66	53.95	-19.99	-35.01	-12.89
88	-0.16	14.20	155.55	-21.40	54.41	-20.89	-34.07	-15.57
89	-1.85	12.45	156.90	-21.26	53.80	-21.84	-36.75	-17.41
90	0.51	13.56	157.68	-21.81	54.28	-20.43	-34.55	-16.46
91	2.46	13.10	155.86	-22.04	54.75	-20.86	-34.88	-13.51
92	-1.96	13.68	155.14	-22.09	53.98	-20.80	-37.23	-20.04
93	4.18	13.40	158.36	-21.82	54.19	-20.46	-36.36	-16.43
94	3.18	13.43	156.65	-22.12	53.60	-20.05	-35.90	-16.26
95	-1.71	13.55	155.07	-22.08	53.90	-20.33	-36.81	-16.48
96	-1.87	11.55	132.79	-19.54	51.83	-3.17	-27.20	2.49
97	1.90	11.74	132.36	-19.78	51.36	-2.55	-26.79	-1.40
98	2.83	11.98	133.42	-19.47	52.29	-2.64	-26.69	2.37
99	-2.49	11.82	130.84	-19.85	52.62	-2.99	-27.51	0.39
100	-2.51	12.23	132.00	-19.92	52.38	-2.79	-27.31	1.30
101	-0.33	11.25	135.64	-19.17	52.55	-3.21	-28.44	0.79
102	-0.66	11.59	132.50	-19.63	51.89	-2.65	-28.62	0.35
103	4.76	11.46	133.48	-19.67	51.71	-3.21	-25.00	0.87
104	3.28	11.84	132.09	-19.71	52.27	-3.36	-27.45	2.13
105	0.21	11.47	129.46	-19.72	51.54	-2.97	-27.22	-2.11
106	1.65	12.36	134.76	-19.93	52.29	-2.91	-28.74	3.66
107	5.53	11.96	132.76	-19.63	52.03	-3.21	-29.41	-1.15
108	-3.77	11.16	131.28	-19.55	51.37	-2.87	-27.01	1.87
109	-3.12	11.83	134.62	-19.65	52.18	-2.48	-26.46	0.83
110	2.85	12.08	133.21	-19.44	52.47	-3.09	-27.62	-1.18
111	0.07	11.77	131.28	-19.90	52.21	-2.69	-27.81	2.83
112	2.49	-2.66	30.15	-1.56	4.27	-0.80	-2.30	-49.05
113	-0.21	-3.10	31.04	-1.60	4.31	-0.43	-2.86	-49.25
114	1.94	-2.82	31.03	-1.86	4.12	-1.00	-2.97	-50.39
115	1.08	-2.63	31.21	-1.55	3.77	-0.67	-3.06	-50.02
116	-0.40	-2.46	31.05	-1.70	4.43	-0.82	-2.46	-50.67
117	-0.06	-2.74	31.50	-1.52	4.41	-0.77	-2.77	-50.13
118	-0.80	-2.66	31.91	-1.62	3.94	-0.64	-2.97	-49.29
119	-0.25	-2.75	32.83	-1.54	3.88	-0.49	-2.99	-49.55

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
80	-95.69	0.82	-1.10	46.86	-0.54	130.43	-0.31	-6.35
81	-95.93	0.92	-0.98	46.06	-0.58	127.11	0.13	-6.43
82	-96.16	0.86	-1.01	45.27	-0.54	127.80	-0.38	-6.37
83	-95.92	0.73	-0.94	45.00	-0.54	128.49	0.19	-6.34
84	-95.27	0.81	-1.04	44.91	-0.64	126.65	-0.16	-6.39
85	-92.81	0.66	-0.86	47.10	-0.53	130.24	0.06	-6.42
86	-96.29	0.72	-0.91	45.81	-0.46	128.86	-0.40	-6.39
87	-96.00	1.01	-0.93	46.00	-0.58	128.57	0.25	-6.31
88	-94.99	0.62	-0.94	45.13	-0.55	128.39	0.16	-6.34
89	-96.60	0.78	-0.89	45.47	-0.52	126.84	-0.11	-6.40
90	-94.71	1.23	-1.19	45.58	-0.62	128.54	-0.77	-6.34

91	-94.84	0.65	-1.13	45.76	-0.67	128.77	-0.04	-6.34
92	-98.94	0.90	-1.10	46.93	-0.58	126.04	0.20	-6.35
93	-95.55	0.79	-1.08	45.13	-0.51	126.68	0.11	-6.38
94	-95.94	0.89	-0.91	46.23	-0.61	127.94	-0.21	-6.32
95	-95.34	1.31	-1.05	44.16	-0.61	129.03	-0.25	-6.34
96	-107.90	0.76	0.00	43.98	-0.72	132.70	4.65	-5.94
97	-107.59	0.99	-0.20	44.15	-0.85	132.48	4.73	-5.90
98	-107.87	0.85	-0.02	44.18	-0.77	134.56	4.47	-5.93
99	-107.13	0.84	-0.11	44.50	-0.88	132.73	5.06	-5.89
100	-107.14	0.81	-0.29	43.22	-0.80	132.40	4.81	-5.89
101	-107.45	1.18	-0.09	44.08	-0.83	129.23	4.52	-5.89
102	-107.72	0.94	-0.04	43.68	-0.76	128.11	4.76	-5.93
103	-108.87	0.85	-0.20	44.01	-0.84	131.58	4.83	-5.90
104	-106.70	0.96	-0.25	42.66	-0.82	129.83	5.01	-5.99
105	-106.40	0.98	-0.08	43.16	-0.85	130.13	4.95	-5.86
106	-106.63	1.09	-0.06	44.91	-0.82	129.33	4.92	-5.93
107	-107.55	1.00	-0.13	43.78	-0.79	131.18	4.41	-6.02
108	-105.69	1.12	-0.27	44.38	-0.87	131.80	5.34	-5.86
109	-105.87	1.07	0.08	43.11	-0.79	133.55	4.85	-5.86
110	-107.71	0.90	-0.20	43.99	-0.87	133.70	4.92	-6.02
111	-106.95	0.78	-0.10	44.61	-0.76	130.77	4.70	-5.93
112	-4.44	1.70	1.70	0.99	-0.98	-0.86	2.56	-2.05
113	-4.14	1.83	1.66	1.66	-0.97	-1.69	2.43	-2.09
114	-4.10	1.81	1.72	0.63	-0.94	-2.98	2.44	-1.99
115	-4.14	1.83	1.69	0.96	-0.99	-3.20	2.68	-2.10
116	-4.28	1.80	1.62	0.80	-0.98	-3.96	2.41	-2.02
117	-3.84	1.93	1.78	0.41	-1.00	-4.36	2.44	-2.03
118	-4.50	1.70	1.76	0.90	-1.00	-2.13	2.54	-2.05
119	-4.48	1.88	1.66	0.14	-0.95	-2.51	2.49	-2.04

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
80	3.40	19.84	3.52	-78.11	-37.18	-14.13	-71.01	6.01
81	3.35	20.61	-0.60	-78.24	-9.91	-13.29	-75.61	-1.14
82	3.41	20.64	0.82	-76.73	19.98	-13.42	-70.88	-6.79
83	3.46	20.20	-0.67	-78.09	-14.48	-13.16	-78.82	0.37
84	3.44	21.50	0.15	-78.53	-27.77	-12.11	-74.33	-4.33
85	3.42	20.07	-1.55	-77.34	-15.72	-13.50	-71.87	-5.76
86	3.47	19.94	-3.51	-80.03	9.79	-13.72	-76.54	-2.25
87	3.49	20.55	6.58	-77.18	14.38	-13.78	-73.69	-8.28
88	3.47	20.61	10.20	-77.82	-7.27	-12.83	-76.26	-5.58
89	3.41	20.55	2.07	-76.65	5.56	-12.96	-75.26	-0.05
90	3.43	20.12	-1.73	-76.33	-3.90	-13.35	-72.26	5.49
91	3.44	20.60	0.77	-77.71	-11.89	-13.52	-68.06	-2.05
92	3.46	20.20	3.82	-77.63	-0.28	-12.97	-79.25	-10.64
93	3.43	20.64	1.29	-77.80	-18.28	-12.80	-72.32	0.38
94	3.44	20.07	0.60	-76.68	15.97	-13.62	-76.04	6.19
95	3.41	20.74	-1.12	-77.81	2.76	-13.11	-72.34	-2.04
96	3.65	25.16	2.76	-72.70	-2.57	-15.00	-62.60	-2.69
97	3.65	25.10	3.51	-72.06	-1.31	-16.34	-66.97	5.31
98	3.66	25.35	0.27	-73.24	8.88	-15.21	-61.74	2.10
99	3.62	25.43	-5.10	-73.18	-11.54	-15.52	-62.32	1.04
100	3.61	25.28	0.40	-72.53	-1.10	-15.47	-70.69	-4.99
101	3.62	24.49	-3.88	-73.96	-0.97	-15.72	-68.40	2.03

102	3.58	25.52	2.92	-73.28	4.49	-15.85	-61.30	-5.42
103	3.58	25.33	1.26	-73.93	-13.86	-15.93	-64.63	-4.88
104	3.65	25.41	3.75	-73.70	4.01	-15.94	-63.18	5.70
105	3.65	25.65	-0.10	-71.82	-4.99	-15.83	-71.39	-2.33
106	3.63	24.95	1.65	-72.28	-7.68	-15.71	-61.19	0.91
107	3.64	25.07	0.78	-73.83	-7.90	-15.93	-61.00	-4.62
108	3.65	24.83	-0.31	-74.01	5.44	-15.62	-62.16	0.71
109	3.62	24.79	1.58	-72.87	-16.26	-15.41	-54.83	-3.32
110	3.57	24.99	4.36	-73.64	3.01	-16.03	-57.92	2.51
111	3.63	24.88	2.53	-72.63	-2.20	-15.87	-69.05	2.72
112	0.31	4.79	-0.41	-1.00	4.16	-7.16	62.77	-1.29
113	0.36	4.94	-1.85	-0.77	-0.50	-6.81	58.70	2.31
114	0.36	5.23	1.31	-1.09	-3.95	-6.47	60.91	-2.13
115	0.37	5.08	-0.23	-1.37	-5.48	-6.74	63.68	2.69
116	0.37	4.81	-0.94	-1.59	-1.18	-7.42	64.90	2.51
117	0.32	5.26	-0.40	-1.36	7.83	-6.88	69.96	1.83
118	0.34	4.88	-1.21	-1.55	-8.41	-6.50	60.24	2.64
119	0.30	5.04	-0.69	-1.16	-8.28	-6.90	62.01	-0.35

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
80	35.18	-18.92	-15.82	-4.04	123.00	-19.42	-1.04	-51.71
81	34.09	-18.91	-16.07	6.61	118.59	-17.87	-1.15	-52.53
82	36.45	-18.80	-17.25	2.41	120.53	-15.35	-0.15	-51.71
83	35.43	-19.22	-15.75	-0.26	121.11	-14.13	-1.33	-51.60
84	35.61	-19.31	-15.80	3.78	122.22	-13.50	-0.46	-51.12
85	35.84	-19.29	-15.60	-3.07	113.52	-19.69	-1.38	-51.49
86	34.84	-19.11	-15.75	6.88	119.39	-19.36	-1.24	-51.76
87	31.00	-18.93	-16.37	2.25	118.69	-14.80	-0.99	-52.10
88	34.90	-19.56	-16.97	-5.46	121.17	-18.96	-1.06	-52.00
89	34.10	-19.04	-15.91	1.89	118.31	-15.26	-0.99	-51.50
90	34.94	-18.97	-16.30	5.87	119.97	-19.67	-1.19	-52.26
91	29.87	-19.33	-16.96	4.69	119.99	-15.82	-1.76	-51.64
92	34.26	-18.90	-16.35	-0.85	119.09	-13.64	-1.06	-51.17
93	35.63	-18.87	-16.09	-2.07	123.06	-16.23	-0.80	-51.82
94	37.07	-19.29	-16.14	11.96	121.62	-15.06	-1.21	-51.56
95	36.78	-19.43	-16.06	-2.11	119.46	-17.76	-1.23	-51.70
96	21.23	-22.94	-15.56	-2.98	85.63	-20.85	2.54	-46.27
97	24.91	-22.30	-14.76	2.60	81.48	-21.58	2.96	-44.93
98	21.70	-22.33	-14.50	-3.51	88.57	-21.05	1.94	-46.62
99	22.56	-21.99	-14.26	-10.16	85.86	-20.91	2.38	-45.87
100	22.17	-22.94	-14.66	8.34	86.53	-21.30	2.37	-46.47
101	21.08	-23.03	-14.09	-0.27	87.85	-22.21	2.51	-46.73
102	19.67	-22.16	-14.80	3.10	85.87	-21.61	2.82	-47.10
103	22.49	-22.60	-14.67	3.54	87.94	-21.15	2.48	-46.57
104	22.32	-22.83	-15.55	-3.03	89.65	-19.85	2.31	-46.25
105	22.19	-22.25	-14.09	5.15	84.82	-18.96	2.62	-47.25
106	21.33	-22.63	-15.21	-6.48	86.04	-20.04	2.40	-47.84
107	21.61	-22.56	-13.73	2.23	87.92	-19.20	2.10	-46.74
108	19.32	-22.17	-15.69	-1.33	86.07	-18.90	2.44	-46.14
109	21.27	-22.66	-13.80	0.11	86.80	-20.63	2.40	-46.11
110	21.06	-22.97	-14.68	-2.90	86.30	-22.66	2.28	-46.75
111	22.19	-22.73	-15.73	-5.40	87.87	-19.19	2.68	-46.05
112	2.40	-6.68	6.28	-2.72	70.14	32.73	-4.02	-3.00

113	1.66	-6.52	6.68	-1.88	71.24	32.05	-4.00	-1.76
114	2.74	-6.41	6.80	-2.46	72.66	32.14	-4.04	-3.13
115	2.97	-6.78	6.95	-1.73	71.59	31.97	-3.94	-3.26
116	1.87	-6.57	6.62	-2.56	71.55	32.38	-3.96	-3.04
117	2.38	-6.59	7.29	0.80	73.77	32.78	-4.12	-2.99
118	2.23	-6.52	7.40	4.06	73.25	32.26	-4.00	-2.40
119	2.04	-6.41	6.41	-1.61	69.82	31.86	-3.80	-3.61

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
80	7.22	128.52	-1.88	43.51	39.31	15.41	-14.23	23.79
81	16.34	128.59	-1.91	43.36	39.03	15.48	-28.20	5.78
82	-6.02	129.83	-1.88	43.69	37.71	15.66	-12.79	-13.12
83	0.26	129.91	-1.89	43.46	38.12	15.62	39.21	11.40
84	-13.98	128.38	-1.87	43.41	37.06	15.59	13.00	-4.79
85	-20.46	130.06	-1.87	43.02	38.18	15.67	-23.98	-72.21
86	-2.59	127.98	-1.90	43.07	37.21	15.58	-30.51	7.52
87	11.87	128.01	-1.88	43.97	37.46	15.56	-27.53	39.33
88	-8.42	128.67	-1.91	43.50	40.10	15.60	-39.33	-13.06
89	-19.59	130.39	-1.86	43.48	38.13	15.66	-15.58	38.94
90	4.00	129.94	-1.85	43.36	37.51	15.62	-16.66	20.91
91	-15.39	128.60	-1.89	43.80	38.56	15.60	-8.93	13.06
92	13.67	129.15	-1.87	42.93	37.77	15.50	-15.23	5.92
93	3.75	129.33	-1.92	43.30	37.36	15.57	3.32	-17.65
94	-15.76	130.12	-1.87	43.29	37.20	15.39	-11.95	-15.64
95	-20.12	129.29	-1.85	43.60	40.08	15.90	-60.06	-10.83
96	-17.10	114.33	-2.02	39.62	9.75	12.73	52.83	-29.46
97	0.00	113.68	-2.03	39.64	9.45	12.67	-31.55	15.89
98	-5.74	115.79	-2.01	39.91	10.43	13.02	45.48	-21.84
99	1.37	115.20	-2.04	40.55	7.81	12.62	7.80	-103.58
100	4.74	115.96	-2.03	40.13	11.11	12.59	40.56	-60.33
101	9.00	115.12	-2.03	40.05	9.51	12.81	8.19	18.79
102	15.33	113.84	-2.02	39.69	9.57	12.73	29.45	79.46
103	9.99	114.81	-2.04	40.46	9.24	12.55	-24.76	-12.46
104	2.29	114.70	-2.04	40.01	10.31	12.96	8.96	24.34
105	-3.84	114.69	-2.03	39.86	9.44	12.77	-20.44	-33.68
106	-16.27	115.65	-2.04	40.05	9.97	12.69	31.92	-4.97
107	-8.85	114.18	-2.02	40.49	9.01	12.48	6.07	-18.59
108	13.66	113.88	-2.07	39.23	11.45	12.60	38.66	-56.12
109	-8.74	114.98	-2.03	39.73	10.32	12.72	-4.35	-32.95
110	7.85	116.66	-2.01	40.22	9.71	12.66	-0.79	-36.94
111	3.91	115.17	-1.99	39.78	9.93	12.55	9.97	15.23
112	1.14	13.18	-0.14	2.79	24.72	0.07	-16.74	8.49
113	4.25	13.58	-0.14	2.10	24.18	0.17	17.27	13.28
114	-3.59	13.49	-0.16	2.96	24.03	0.26	6.52	41.70
115	5.11	14.28	-0.14	2.68	24.04	0.36	10.37	38.29
116	6.99	13.92	-0.14	2.68	24.65	0.11	10.23	-3.54
117	5.35	14.04	-0.16	2.83	24.69	0.32	-1.11	7.77
118	-14.75	13.75	-0.12	2.85	24.55	0.22	-16.30	0.78
119	7.24	13.73	-0.13	2.73	24.81	0.13	-1.81	10.69

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
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120	-86.29	-0.63	3.88	-32.02	1.84	11.04	2.67	-3.79
121	-85.57	-0.73	3.64	-32.07	1.65	11.44	2.57	-2.83
122	-80.08	-0.26	3.98	-32.67	1.94	10.85	2.31	-3.57
123	-89.15	-0.56	3.69	-32.93	1.34	11.29	2.70	-3.13
124	-93.09	-0.87	3.69	-33.69	1.66	11.05	2.62	-2.62
125	-79.10	-0.35	3.46	-34.24	1.35	11.14	3.71	-2.73
126	-79.96	-0.48	3.30	-34.24	1.28	11.20	2.45	-3.40
127	-88.22	-0.65	3.53	-36.03	1.47	11.34	3.62	-3.03
128	-64.37	-4.04	1.18	-12.23	6.38	4.37	5.12	-0.76
129	-69.64	-4.14	1.10	-13.64	6.02	4.69	5.44	-0.43
130	-63.61	-4.12	0.95	-10.49	6.13	4.01	4.92	-0.32
131	-66.53	-4.06	0.94	-13.49	5.92	4.79	5.35	-0.99
132	-70.96	-4.00	1.10	-13.01	6.39	3.96	5.16	-0.30
133	-67.75	-4.00	1.23	-15.14	6.04	3.95	5.44	-0.43
134	-64.48	-4.20	0.82	-13.91	5.82	4.06	5.23	-0.18
135	-61.97	-4.07	0.68	-13.86	5.92	4.81	5.69	-0.76
136	-68.31	-4.09	0.82	-13.67	5.52	4.21	5.11	-0.62
137	-57.85	-4.15	1.00	-14.37	5.78	4.21	5.63	-1.43
138	-68.14	-4.06	0.87	-13.57	5.80	4.44	5.36	-0.51
139	-71.48	-4.31	1.21	-16.35	6.10	4.03	5.38	-0.61
140	-65.76	-4.29	0.82	-14.07	6.50	4.40	5.54	-0.17
141	-61.81	-4.12	1.10	-12.99	5.91	4.76	4.79	-0.75
142	-66.55	-4.48	1.08	-12.19	5.67	4.64	5.21	-0.94
143	-63.27	-4.03	1.02	-14.10	5.86	4.08	6.23	-1.19
144	-38.06	-4.80	-0.40	-7.99	5.14	1.07	4.09	-2.19
145	-34.73	-5.10	-0.37	-9.37	4.70	0.55	3.89	-1.54
146	-36.22	-4.93	-0.66	-8.77	4.53	0.87	3.70	-1.89
147	-38.97	-4.91	-0.26	-8.19	4.97	0.85	4.47	-1.42
148	-39.77	-4.98	-0.42	-9.31	5.15	0.76	4.33	-2.09
149	-40.83	-4.93	-0.29	-7.86	4.61	0.53	4.00	-1.78
150	-42.55	-4.97	-0.37	-8.84	4.88	1.17	4.23	-1.88
151	-40.43	-4.83	-0.39	-10.02	4.55	0.64	3.83	-2.11
152	-35.57	-4.98	-0.28	-9.08	4.79	1.04	4.22	-1.39
153	-36.15	-4.86	-0.47	-10.73	4.58	1.24	3.96	-1.78
154	-40.29	-4.90	-0.50	-9.89	4.86	1.22	3.86	-2.27
155	-36.78	-5.15	-0.39	-9.03	5.19	0.90	4.12	-1.66
156	-39.53	-5.04	-0.58	-9.80	5.07	0.85	4.06	-1.30
157	-42.15	-4.89	-0.60	-9.48	4.90	0.85	3.62	-2.15
158	-38.22	-5.10	-0.56	-7.54	4.48	0.71	4.17	-1.92
159	-40.67	-5.02	-0.56	-9.52	4.98	0.99	3.72	-1.58

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
120	53.48	10.31	-0.06	26.05	-6.81	-9.45	0.35	-27.38
121	58.44	14.49	-0.03	26.18	-6.15	-9.19	0.45	-24.02
122	55.08	11.57	-0.02	23.35	-9.94	-9.11	0.39	-25.86
123	56.58	12.31	-0.04	26.82	-7.44	-9.01	0.26	-25.55
124	54.35	10.39	-0.05	24.98	-8.46	-9.02	0.44	-25.96
125	53.81	15.02	-0.04	27.99	-7.80	-9.58	0.44	-25.95
126	53.27	14.24	-0.04	27.91	-7.71	-9.26	0.47	-25.63
127	58.03	12.68	-0.01	25.01	-6.40	-8.74	0.40	-28.05
128	39.98	-1.05	-0.33	21.96	-22.04	-4.80	-0.41	-15.66
129	41.22	-2.03	-0.36	22.19	-21.99	-5.06	-0.56	-15.90
130	40.95	-2.24	-0.34	21.97	-22.07	-5.14	-0.57	-19.52

131	40.26	-0.35	-0.34	20.92	-20.71	-5.18	-0.39	-18.20
132	41.86	-2.16	-0.37	22.34	-21.90	-4.85	-0.56	-16.02
133	43.88	-2.36	-0.34	21.20	-21.58	-5.34	-0.51	-16.43
134	36.87	-2.70	-0.33	20.90	-21.32	-5.10	-0.61	-19.40
135	39.83	-2.87	-0.36	18.90	-21.75	-4.79	-0.38	-17.56
136	45.23	-0.60	-0.35	22.43	-21.35	-5.04	-0.36	-17.73
137	41.16	-0.89	-0.36	23.49	-20.47	-5.32	-0.67	-14.10
138	38.19	-1.59	-0.35	21.84	-23.38	-5.16	-0.48	-17.71
139	41.58	-2.39	-0.39	21.34	-24.13	-4.83	-0.53	-17.91
140	42.51	-2.42	-0.38	23.72	-21.10	-5.44	-0.54	-15.10
141	40.30	0.10	-0.36	22.77	-22.23	-5.06	-0.67	-17.92
142	41.11	0.05	-0.35	22.22	-20.58	-4.96	-0.59	-18.54
143	42.21	-2.62	-0.33	21.58	-22.83	-5.20	-0.53	-18.43
144	23.17	3.07	-0.39	11.72	-4.83	-2.20	0.03	-8.73
145	20.93	1.49	-0.37	13.64	-5.71	-2.24	-0.03	-7.18
146	23.18	3.13	-0.37	12.07	-6.08	-2.29	-0.04	-9.87
147	24.10	2.17	-0.37	12.71	-6.07	-1.98	0.18	-8.85
148	21.38	3.16	-0.34	12.79	-5.25	-2.21	-0.02	-7.81
149	23.10	3.56	-0.37	13.95	-6.65	-2.46	-0.06	-9.00
150	19.33	3.00	-0.38	12.67	-5.73	-2.00	-0.07	-7.75
151	25.90	2.23	-0.36	11.71	-6.53	-2.38	0.09	-8.28
152	24.52	2.63	-0.37	14.16	-6.56	-2.29	0.02	-8.56
153	24.20	3.68	-0.34	12.92	-5.60	-2.00	-0.12	-8.80
154	20.10	2.21	-0.36	13.40	-5.76	-2.46	0.13	-8.69
155	23.10	1.82	-0.36	12.73	-5.94	-2.06	0.12	-8.08
156	24.71	2.42	-0.37	12.42	-5.34	-2.50	-0.12	-9.17
157	23.11	3.73	-0.39	12.30	-6.50	-2.19	-0.03	-9.57
158	24.10	3.93	-0.35	14.40	-6.18	-2.17	0.00	-8.30
159	22.82	4.16	-0.34	13.30	-6.05	-2.27	0.10	-7.50

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
120	0.31	-5.03	4.62	-32.56	-1.82	-1.70	4.33	0.53
121	0.72	-4.01	4.85	1.24	-1.88	-1.74	-1.33	0.54
122	0.45	-4.10	4.84	-2.36	-1.67	-1.69	-2.20	0.55
123	0.11	-4.61	4.79	14.98	-1.97	-1.76	-1.86	0.53
124	-0.04	-4.58	4.63	0.19	-1.97	-1.92	1.82	0.55
125	0.56	-4.69	4.83	-20.06	-1.93	-1.77	3.49	0.53
126	-0.20	-4.71	4.49	4.85	-1.85	-1.65	-2.92	0.54
127	-0.06	-4.54	4.62	-14.70	-1.93	-1.83	-4.25	0.55
128	-0.59	-1.46	4.32	-19.70	1.52	0.78	1.79	0.51
129	-0.28	-1.64	4.09	-34.57	1.15	0.54	5.08	0.51
130	0.18	-1.25	4.46	22.73	1.33	0.70	7.53	0.48
131	1.03	-1.55	4.43	8.79	1.40	0.59	5.14	0.47
132	-0.07	-1.65	4.37	-34.22	1.36	0.55	-0.52	0.48
133	0.09	-1.60	4.25	6.16	1.45	0.59	7.45	0.50
134	0.59	-1.67	4.41	-25.72	1.17	0.73	6.02	0.48
135	-1.51	-1.42	4.83	-0.65	1.22	0.56	7.84	0.49
136	-0.64	-1.43	4.64	-25.29	1.16	0.77	2.70	0.47
137	0.38	-1.91	4.26	29.50	1.25	0.50	3.59	0.47
138	-0.51	-1.60	4.68	-16.14	1.52	0.61	-4.79	0.49
139	-0.07	-1.17	4.53	42.81	1.29	0.66	6.75	0.50
140	-0.70	-1.19	4.06	6.12	1.09	0.81	-3.05	0.55
141	-0.06	-1.58	4.40	6.81	1.20	0.47	3.59	0.52

142	-0.30	-1.72	3.98	-8.17	1.33	0.76	4.02	0.50
143	-0.08	-1.32	4.27	-7.54	1.36	0.36	-7.33	0.50
144	0.45	0.82	3.10	11.79	3.12	2.48	-2.30	0.22
145	-0.01	0.44	2.79	-4.24	3.08	2.36	-0.35	0.25
146	0.37	0.43	2.61	-14.16	3.16	2.41	0.10	0.20
147	0.50	0.65	2.30	16.23	3.18	2.47	13.38	0.22
148	-0.08	0.27	2.63	15.55	3.20	2.50	6.03	0.22
149	0.77	0.39	2.89	11.11	3.10	2.45	1.68	0.23
150	-0.07	0.70	2.73	-7.13	3.16	2.45	6.97	0.23
151	-0.14	0.72	2.70	16.50	3.19	2.47	7.78	0.23
152	-0.05	0.58	3.11	5.90	3.31	2.60	5.82	0.21
153	-0.41	0.77	2.48	-3.59	3.23	2.64	1.59	0.21
154	-0.34	0.61	2.63	13.77	3.13	2.70	5.05	0.23
155	-0.58	0.21	2.43	-4.03	3.19	2.42	0.11	0.21
156	-0.25	0.71	2.85	-12.11	3.07	2.63	6.59	0.24
157	0.54	0.53	2.69	2.39	3.20	2.50	-3.10	0.21
158	0.28	0.84	2.99	-14.40	3.08	2.52	-6.42	0.19
159	0.26	0.25	2.33	8.40	3.20	2.36	5.99	0.25

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
120	-3.15	-2.74	32.68	-1.28	4.17	-0.54	-2.97	-49.03
121	0.37	-2.71	32.30	-1.60	4.71	-0.70	-2.92	-49.08
122	-1.57	-2.50	33.88	-1.53	4.70	-0.59	-2.22	-49.68
123	-2.35	-2.72	30.56	-1.41	4.83	-0.77	-2.27	-49.43
124	-1.34	-2.41	29.44	-1.53	4.20	-0.64	-3.11	-49.42
125	-0.38	-2.90	32.38	-1.44	3.98	-0.92	-3.34	-49.54
126	-0.34	-2.87	30.15	-1.49	4.13	-0.79	-3.50	-50.50
127	-1.26	-2.59	31.97	-1.33	3.99	-0.36	-2.67	-50.02
128	1.14	-0.88	32.09	-0.37	9.19	-2.91	-3.97	-36.02
129	0.01	-0.90	29.34	-1.01	8.94	-2.71	-3.49	-35.29
130	-0.35	-0.83	32.11	-0.90	9.10	-3.05	-4.29	-34.94
131	-1.21	-1.12	32.67	-0.71	9.25	-3.00	-3.36	-36.80
132	-0.17	-1.05	30.63	-0.91	8.77	-2.89	-4.80	-36.50
133	1.87	-0.79	31.14	-0.52	9.94	-2.94	-3.83	-36.11
134	-0.47	-1.01	30.51	-0.75	9.13	-3.33	-3.84	-35.71
135	0.30	-0.80	30.54	-1.22	9.54	-2.83	-3.54	-35.94
136	-1.37	-0.96	32.28	-0.65	9.08	-2.91	-3.37	-36.61
137	2.06	-1.05	29.71	-0.50	9.28	-2.99	-3.87	-35.73
138	-0.62	-0.65	28.44	-0.93	9.22	-2.73	-3.47	-36.62
139	-1.69	-0.79	32.28	-0.62	9.13	-2.77	-3.37	-36.46
140	-1.42	-0.48	31.05	-0.86	9.25	-3.21	-3.49	-35.39
141	2.17	-0.73	31.65	-0.97	9.05	-3.07	-4.55	-35.42
142	-1.91	-0.93	29.79	-0.57	8.76	-2.92	-3.90	-35.86
143	0.66	-0.81	33.30	-0.84	9.12	-3.15	-4.24	-37.51
144	1.11	0.42	18.05	0.99	2.98	-0.40	1.34	-22.51
145	0.18	0.19	18.46	0.91	2.43	-0.95	1.18	-21.18
146	-0.04	0.32	17.97	0.82	2.36	-0.36	0.97	-22.37
147	-0.54	0.58	18.34	1.07	2.36	-0.56	0.77	-22.28
148	0.51	0.19	18.43	0.91	2.45	-0.37	0.93	-22.38
149	0.27	0.47	18.05	0.97	2.15	-0.67	1.19	-22.81
150	0.84	0.36	17.25	1.26	2.25	-0.60	1.62	-22.12
151	1.21	0.45	18.18	1.26	2.77	-0.41	1.26	-22.47
152	-1.01	0.39	18.00	1.08	2.27	-0.42	1.60	-23.23

153	-0.96	0.54	17.76	1.28	2.68	-0.65	1.19	-22.35
154	-1.06	0.43	18.31	0.95	1.97	-0.58	1.72	-22.32
155	-1.09	0.56	16.71	0.46	2.04	-0.37	1.65	-21.98
156	-1.60	0.60	18.48	0.96	2.26	-0.62	1.03	-22.74
157	1.04	0.44	16.36	0.86	2.27	-0.58	1.05	-23.32
158	-0.46	0.44	16.89	0.97	2.06	-0.66	1.13	-21.35
159	0.16	0.47	18.03	1.13	2.51	-0.59	1.34	-22.34

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
120	-4.42	1.84	1.76	1.01	-0.98	-2.13	2.45	-2.01
121	-4.37	2.00	1.74	0.71	-0.97	-1.82	2.59	-2.07
122	-3.96	1.87	1.66	0.88	-0.98	-1.29	2.49	-2.04
123	-3.97	1.99	1.80	0.12	-0.92	-1.94	2.43	-2.10
124	-4.29	1.75	1.75	0.79	-0.97	-3.66	2.39	-2.04
125	-4.25	1.90	1.82	1.10	-0.98	-2.15	2.39	-2.09
126	-4.33	1.92	1.73	1.23	-0.96	-3.61	2.43	-2.04
127	-4.28	1.86	1.88	1.40	-1.02	-3.55	2.51	-1.98
128	-7.95	0.11	1.68	8.51	-0.87	10.98	-2.80	-2.96
129	-8.25	0.12	1.63	9.09	-0.88	8.94	-2.75	-3.10
130	-8.36	0.16	1.71	8.72	-0.86	9.22	-2.60	-3.06
131	-8.59	0.00	1.59	7.62	-0.86	11.42	-2.57	-2.99
132	-8.02	0.21	1.68	8.66	-0.87	9.90	-2.69	-3.05
133	-8.27	0.18	1.74	8.45	-0.92	9.04	-2.93	-3.06
134	-8.28	-0.01	1.69	8.82	-0.93	9.85	-2.73	-3.04
135	-8.91	0.14	1.71	8.75	-0.83	9.35	-2.74	-3.04
136	-8.32	0.14	1.68	8.33	-0.91	9.73	-2.51	-3.02
137	-8.23	0.27	1.76	8.53	-0.86	10.43	-2.72	-2.95
138	-8.15	0.11	1.60	8.97	-0.93	10.95	-2.78	-2.98
139	-8.33	0.06	1.70	8.68	-0.89	10.59	-3.01	-3.00
140	-7.64	0.16	1.68	8.39	-0.82	9.51	-2.84	-3.02
141	-9.02	0.15	1.58	8.08	-0.92	10.09	-2.75	-3.09
142	-8.73	0.14	1.68	8.98	-0.89	10.21	-2.67	-3.04
143	-7.95	0.22	1.63	9.15	-0.98	9.25	-3.00	-3.00
144	-2.45	-1.17	0.94	4.61	-0.40	1.78	-3.48	-2.51
145	-2.39	-1.16	1.04	4.32	-0.39	2.64	-3.46	-2.56
146	-1.88	-1.16	0.98	4.39	-0.41	0.79	-3.46	-2.50
147	-2.14	-1.10	1.02	4.42	-0.42	2.73	-3.50	-2.53
148	-2.58	-1.01	0.97	4.80	-0.36	2.52	-3.48	-2.52
149	-2.46	-1.06	0.97	4.32	-0.38	2.31	-3.48	-2.54
150	-2.92	-1.09	1.00	4.66	-0.42	2.15	-3.46	-2.52
151	-2.89	-1.10	0.90	4.61	-0.37	2.57	-3.54	-2.54
152	-2.62	-1.10	0.98	4.88	-0.37	2.24	-3.62	-2.56
153	-3.08	-1.12	1.01	4.37	-0.35	3.16	-3.55	-2.56
154	-2.80	-1.10	0.94	4.54	-0.41	2.04	-3.56	-2.52
155	-2.13	-1.11	0.95	4.54	-0.41	2.28	-3.54	-2.48
156	-2.77	-1.04	0.95	4.39	-0.36	1.86	-3.63	-2.55
157	-3.33	-1.13	1.01	3.85	-0.38	3.13	-3.50	-2.49
158	-2.72	-0.98	0.94	4.57	-0.36	2.95	-3.54	-2.53
159	-2.54	-1.18	1.01	4.96	-0.35	1.73	-3.52	-2.48

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
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120	0.33	4.92	-0.31	-1.59	0.68	-6.93	56.35	1.36
121	0.30	4.77	0.00	-1.50	4.89	-6.96	59.88	-3.65
122	0.34	4.82	4.31	-1.16	-7.78	-6.96	67.78	-1.90
123	0.31	4.96	-0.12	-2.05	-5.38	-7.09	66.01	0.16
124	0.36	4.79	3.53	-0.49	6.42	-6.72	66.01	2.43
125	0.35	5.13	-0.44	-1.41	-1.40	-6.90	63.79	3.73
126	0.33	4.75	-1.99	-1.23	-9.02	-6.86	58.33	3.46
127	0.36	4.73	0.36	-1.53	4.56	-7.29	61.10	3.11
128	-0.07	-1.38	2.12	-8.54	-9.27	-1.78	54.45	-0.97
129	-0.13	-1.34	2.14	-8.46	0.16	-2.53	47.17	-2.51
130	-0.06	-1.74	-2.76	-8.60	2.08	-2.47	57.05	-0.39
131	-0.15	-1.22	0.09	-8.08	5.34	-2.74	64.38	-0.47
132	-0.10	-2.01	-3.16	-8.12	-3.11	-2.73	54.72	3.38
133	-0.13	-1.45	2.02	-8.04	2.46	-2.24	68.54	1.40
134	-0.11	-1.36	-2.75	-7.68	0.87	-2.20	62.77	0.46
135	-0.10	-1.66	0.80	-7.99	-6.50	-2.85	58.11	-3.03
136	-0.10	-1.39	-0.66	-8.05	-0.64	-2.58	60.67	-1.20
137	-0.08	-2.04	-0.70	-8.62	5.92	-2.03	63.08	2.07
138	-0.11	-1.44	-2.66	-8.23	-3.00	-2.64	52.05	-12.58
139	-0.09	-1.34	1.73	-7.92	1.25	-2.09	54.21	-4.23
140	-0.13	-1.72	-1.14	-8.65	3.68	-2.46	60.90	-3.56
141	-0.14	-1.29	-1.05	-8.42	-4.40	-1.80	55.03	-1.73
142	-0.10	-1.33	-4.62	-9.38	-5.19	-2.68	56.49	-0.74
143	-0.09	-1.35	-1.42	-9.07	3.70	-2.50	61.63	0.05
144	-0.44	-6.62	-0.08	-8.43	3.03	2.65	34.50	2.06
145	-0.47	-6.28	1.57	-9.29	-6.12	2.33	29.93	-1.97
146	-0.44	-6.15	-0.35	-8.90	-6.12	2.76	31.77	-0.39
147	-0.47	-6.03	1.02	-8.50	-3.98	2.55	27.37	6.41
148	-0.47	-6.47	0.58	-9.16	-4.25	2.53	29.28	-2.16
149	-0.47	-6.41	0.58	-8.59	-7.69	2.66	32.78	-4.26
150	-0.44	-6.63	0.33	-8.96	5.06	2.56	29.31	0.25
151	-0.45	-6.32	4.26	-8.83	-3.55	2.68	35.36	-1.39
152	-0.44	-6.65	0.05	-8.71	-11.51	2.71	30.30	3.43
153	-0.44	-6.47	-0.37	-8.79	7.18	2.80	30.82	-6.53
154	-0.44	-6.12	0.06	-8.76	3.86	2.95	28.19	-2.98
155	-0.44	-6.12	1.80	-9.11	-9.21	2.68	30.34	6.30
156	-0.46	-6.68	-0.77	-8.69	1.26	2.63	30.94	-0.36
157	-0.47	-6.39	-0.08	-8.15	7.13	2.80	28.23	1.12
158	-0.45	-6.39	-0.75	-8.70	8.69	2.90	31.24	1.50
159	-0.46	-6.57	-1.82	-8.71	-3.29	2.75	28.59	-2.87

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
120	2.78	-6.70	6.71	-4.07	75.91	32.50	-4.02	-2.35
121	1.75	-6.51	6.99	1.30	76.01	32.64	-3.86	-1.98
122	2.33	-6.45	7.21	-2.74	73.44	32.65	-3.94	-3.11
123	2.80	-6.68	7.26	-2.89	72.83	33.33	-3.87	-2.74
124	3.47	-6.47	6.87	1.60	66.53	32.72	-3.73	-1.95
125	1.90	-6.45	7.49	0.13	71.55	33.59	-3.90	-2.40
126	2.54	-6.66	7.75	-4.69	70.82	32.24	-3.83	-3.09
127	1.80	-6.47	6.30	-0.37	75.11	32.34	-3.76	-2.51
128	5.08	-2.11	4.54	-3.75	56.90	24.33	-1.11	-3.41
129	6.93	-2.21	4.69	-1.04	53.20	22.82	-0.86	-3.26
130	5.71	-2.02	4.77	-2.04	59.19	22.72	-1.09	-3.87

131	6.18	-2.16	5.19	-1.55	56.71	22.38	-0.87	-3.38
132	4.92	-2.36	4.70	-3.26	53.43	21.56	-0.71	-3.77
133	5.37	-2.04	5.37	4.33	55.12	23.02	-1.00	-3.41
134	4.82	-2.12	4.95	2.78	57.33	21.93	-1.02	-4.04
135	5.17	-2.10	4.70	2.48	52.38	21.10	-0.78	-3.61
136	5.59	-2.43	5.28	-2.07	55.91	21.57	-1.07	-3.74
137	6.13	-2.32	4.42	-3.71	55.69	22.77	-0.81	-3.72
138	4.85	-2.26	4.81	6.46	55.44	22.67	-0.71	-3.47
139	4.82	-2.22	5.09	2.24	54.98	22.90	-0.79	-3.88
140	6.04	-2.27	4.38	-0.44	59.54	22.51	-0.85	-3.18
141	6.36	-2.39	5.49	-0.20	54.86	21.78	-0.90	-3.71
142	5.02	-2.24	5.11	2.02	57.42	21.87	-0.84	-3.51
143	5.52	-2.22	4.91	1.56	58.08	23.02	-1.00	-3.42
144	-0.21	1.06	1.30	-0.44	38.34	16.26	0.17	-2.55
145	-0.70	1.07	0.82	0.28	39.73	16.77	0.10	-2.79
146	-0.67	0.88	0.94	0.56	41.10	17.62	0.23	-2.75
147	-0.68	1.10	1.41	0.32	40.17	16.63	0.19	-2.42
148	-0.41	1.17	1.07	0.73	37.75	15.98	0.12	-2.83
149	-0.55	1.15	1.39	3.93	39.29	16.75	0.30	-3.12
150	-1.68	1.29	1.09	2.30	41.93	16.56	0.33	-2.48
151	-0.46	1.00	1.37	-0.76	41.86	16.89	0.23	-3.06
152	-1.16	0.71	0.72	-3.59	36.96	15.92	0.09	-2.85
153	-0.73	1.16	1.43	-2.06	37.10	16.71	0.20	-2.38
154	-0.77	1.19	1.48	0.25	39.82	16.00	0.06	-1.96
155	-0.23	1.11	0.91	0.10	39.59	16.68	0.02	-2.58
156	-0.49	1.09	1.59	1.28	37.71	16.57	0.07	-2.28
157	-0.44	1.01	0.76	0.78	41.09	16.19	0.21	-2.43
158	-0.30	1.03	1.11	-2.55	38.76	15.61	0.29	-2.24
159	-1.59	1.10	1.12	-0.03	39.03	16.28	0.16	-2.11

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
120	-3.49	13.40	-0.17	2.77	24.03	0.22	-1.92	8.36
121	-1.56	13.53	-0.15	2.91	24.00	0.19	-0.13	9.30
122	-0.33	13.08	-0.14	2.69	23.29	-0.11	-10.57	21.22
123	4.41	13.72	-0.15	2.98	24.76	0.23	-14.55	8.52
124	-5.43	13.28	-0.15	2.53	23.69	0.36	3.17	30.92
125	-2.68	13.25	-0.13	2.61	24.53	0.23	43.45	1.59
126	7.50	15.48	-0.16	2.78	24.04	0.28	-3.80	25.95
127	4.32	14.32	-0.14	2.77	23.76	0.04	2.42	-25.97
128	-13.42	27.45	-0.24	7.34	17.51	2.47	0.94	9.49
129	10.58	26.06	-0.24	7.23	17.53	2.31	14.43	46.44
130	4.21	24.93	-0.25	7.19	17.44	2.41	-21.84	-14.67
131	1.14	27.25	-0.25	7.89	17.62	2.36	-6.28	-1.76
132	-1.71	25.57	-0.28	7.46	17.35	2.48	0.91	32.36
133	12.79	24.80	-0.28	7.83	17.59	2.47	-0.89	-14.20
134	3.23	24.22	-0.24	7.25	17.33	2.29	7.74	36.11
135	5.29	24.94	-0.25	7.41	17.48	2.41	0.58	13.47
136	-3.92	26.67	-0.25	7.53	17.72	2.42	-8.16	22.44
137	-3.58	26.44	-0.26	7.61	16.62	2.49	-1.47	-16.57
138	4.11	25.58	-0.24	7.62	17.52	2.36	-16.44	-25.70
139	-1.61	25.23	-0.24	7.50	17.71	2.41	-13.21	25.78
140	-2.54	27.55	-0.25	7.87	18.15	2.40	1.36	18.40
141	-12.06	26.71	-0.24	7.31	17.41	2.28	-12.71	9.79

142	1.45	25.83	-0.25	7.40	17.14	2.38	-6.48	-4.14
143	3.09	26.96	-0.25	7.34	17.09	2.50	-5.15	17.91
144	-4.85	8.77	-0.03	7.05	9.88	2.53	14.66	9.71
145	-3.68	8.85	-0.04	7.36	9.83	2.65	13.40	23.05
146	-2.12	8.74	-0.03	7.62	10.26	2.57	-4.56	-13.00
147	0.77	9.60	-0.04	7.20	10.29	2.55	-1.43	5.03
148	1.14	9.42	-0.03	7.40	10.21	2.61	-10.34	1.62
149	1.93	8.85	-0.04	7.68	9.93	2.31	-10.78	11.24
150	-1.34	10.06	-0.04	7.44	10.26	2.50	2.13	6.41
151	3.92	9.28	-0.03	7.41	10.52	2.47	-23.89	21.08
152	1.39	9.20	-0.02	7.35	10.53	2.59	16.81	5.07
153	-8.77	8.14	-0.02	7.68	10.60	2.67	2.72	-9.55
154	-0.43	9.26	-0.04	7.60	10.54	2.54	-2.97	14.19
155	0.07	9.38	-0.03	7.27	10.46	2.65	2.90	14.79
156	-0.81	9.37	-0.05	7.54	10.21	2.54	10.72	0.67
157	-1.46	9.47	-0.05	7.26	10.23	2.60	-7.04	-39.70
158	-5.91	8.46	-0.04	7.27	9.78	2.38	0.29	9.16
159	-3.15	8.99	-0.05	7.41	10.58	2.70	-10.17	2.93

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
160	-183.25	2.14	5.62	-25.74	3.13	10.32	3.56	11.78
161	-200.77	1.99	5.47	-27.03	3.73	10.43	3.14	12.40
162	-189.77	1.88	5.81	-25.08	3.29	10.28	2.50	12.23
163	-180.47	1.92	5.11	-24.66	3.56	9.96	3.94	11.75
164	-190.80	2.23	5.54	-24.51	1.97	10.54	4.33	11.66
165	-202.82	2.32	5.41	-25.31	2.83	10.24	3.98	11.72
166	-191.96	2.42	5.77	-26.22	4.19	10.76	2.92	12.35
167	-202.76	2.17	6.14	-25.94	2.82	10.63	3.41	11.44
168	-200.75	1.98	4.80	-24.85	2.05	10.13	4.60	11.72
169	-203.33	2.07	4.43	-25.07	3.21	10.43	4.22	11.70
170	-210.89	2.59	5.25	-26.95	2.38	10.55	4.53	11.48
171	-182.98	1.78	5.53	-26.40	2.18	10.46	3.14	12.03
172	-198.69	1.99	5.10	-24.36	2.57	10.33	3.05	11.84
173	-188.40	1.90	5.59	-24.79	3.64	10.55	3.01	11.43
174	-214.04	2.14	5.54	-24.92	2.11	10.22	2.96	11.82
175	-184.63	2.21	5.32	-25.17	2.94	10.15	4.37	12.25
176	-200.57	2.15	5.04	-22.81	1.93	9.60	4.05	11.62
177	-192.89	1.82	5.99	-23.73	1.29	9.49	3.28	11.57
178	-203.63	1.64	5.83	-20.94	-0.29	9.36	3.67	12.18
179	-188.18	2.06	5.25	-23.68	1.11	9.95	5.50	11.80
180	-209.99	1.94	5.75	-22.03	1.03	9.56	4.07	11.49
181	-186.37	1.67	5.88	-21.16	0.82	9.68	4.08	11.39
182	-179.41	2.00	4.79	-22.47	1.65	9.46	3.93	11.91
183	-205.64	1.85	6.10	-21.08	0.69	9.89	2.51	12.14
184	-191.97	1.98	5.27	-22.05	0.35	9.36	3.29	11.41
185	-189.61	1.88	5.68	-22.09	1.18	9.34	3.93	11.81
186	-194.95	2.16	5.47	-23.46	2.22	9.99	4.03	11.41
187	-199.69	1.98	5.83	-21.48	0.05	9.70	3.21	11.41
188	-186.98	1.60	5.00	-23.36	1.17	9.80	2.65	11.23
189	-194.99	1.63	5.53	-23.27	1.00	9.60	4.56	12.00
190	-195.44	1.23	5.91	-21.99	1.57	9.26	3.87	12.39
191	-197.60	1.64	5.88	-21.36	0.88	9.76	2.72	10.91
192	-80.99	1.13	5.78	-41.96	-7.61	14.12	4.56	-2.34

193	-89.68	0.98	5.35	-40.67	-7.33	14.32	4.86	-3.09
194	-87.61	1.04	5.68	-40.81	-7.47	14.54	5.72	-3.11
195	-75.78	0.82	5.89	-41.56	-7.69	14.38	4.84	-3.28
196	-81.87	1.14	5.62	-39.90	-7.61	14.36	5.82	-3.01
197	-86.38	1.03	5.52	-40.84	-7.57	14.22	4.74	-2.69
198	-90.67	1.07	5.62	-40.01	-7.85	14.09	4.38	-2.72
199	-85.90	0.90	5.82	-39.91	-7.87	14.58	4.61	-3.11

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
160	93.69	50.35	-0.06	43.76	-7.96	-11.45	-1.65	-39.83
161	94.19	51.22	0.05	41.97	-8.07	-11.77	-1.83	-35.89
162	96.08	50.24	-0.01	44.07	-8.94	-11.94	-1.74	-33.20
163	91.01	50.82	-0.02	41.62	-9.97	-11.61	-1.79	-35.37
164	94.91	51.64	-0.03	44.52	-10.47	-11.14	-1.66	-38.97
165	99.80	49.04	-0.02	49.32	-10.67	-11.51	-1.78	-35.32
166	100.32	49.55	0.01	48.96	-8.32	-11.16	-1.87	-35.89
167	91.74	50.26	-0.03	47.65	-8.62	-11.54	-1.66	-34.51
168	99.23	48.64	-0.05	42.04	-8.21	-11.68	-1.67	-34.09
169	91.16	51.29	-0.01	48.34	-11.59	-11.22	-1.69	-38.04
170	84.86	50.02	0.02	47.47	-8.47	-12.16	-1.65	-39.32
171	104.41	49.99	-0.04	46.22	-7.03	-11.70	-1.61	-36.51
172	95.70	51.92	0.01	43.17	-11.26	-11.33	-1.77	-42.20
173	99.92	49.58	0.00	46.70	-8.98	-12.03	-1.72	-36.09
174	92.96	50.20	-0.05	47.19	-10.20	-11.95	-1.65	-39.79
175	87.99	50.84	-0.06	47.82	-10.64	-11.26	-1.78	-40.49
176	92.57	45.83	-0.03	46.46	-3.54	-10.29	-1.66	-40.48
177	103.07	45.08	-0.08	46.56	-6.48	-10.76	-1.59	-34.80
178	89.56	45.20	-0.04	45.63	-3.64	-10.46	-1.56	-43.17
179	100.68	43.71	-0.03	49.09	-2.93	-10.14	-1.61	-40.29
180	105.61	44.15	-0.06	48.19	-3.47	-9.72	-1.61	-40.95
181	91.34	45.93	-0.02	50.35	-3.56	-10.55	-1.62	-40.77
182	87.76	44.55	0.00	46.54	-7.49	-10.17	-1.51	-43.01
183	97.67	44.55	-0.04	49.44	-4.48	-10.41	-1.59	-36.79
184	93.30	44.01	-0.02	47.30	-2.87	-10.32	-1.78	-37.03
185	101.12	47.15	0.00	45.58	-6.35	-9.75	-1.68	-42.95
186	92.30	43.85	-0.03	49.68	-5.53	-10.36	-1.73	-39.98
187	90.84	43.94	-0.02	44.30	-1.86	-10.72	-1.55	-37.11
188	101.37	42.91	-0.04	48.77	-4.67	-9.64	-1.62	-45.48
189	95.02	44.73	-0.04	45.22	-2.35	-10.64	-1.65	-39.73
190	93.66	43.89	-0.01	45.99	-6.29	-9.88	-1.58	-39.34
191	103.16	44.83	-0.01	47.51	-5.19	-10.19	-1.53	-36.73
192	59.69	57.64	-0.04	16.47	24.54	-9.22	2.64	-37.59
193	69.07	58.35	-0.09	18.14	26.55	-9.27	2.42	-38.58
194	64.30	59.42	-0.09	19.73	26.26	-9.01	2.47	-37.31
195	64.31	57.75	-0.08	18.64	26.38	-9.27	2.49	-39.18
196	66.35	57.50	-0.09	17.61	26.14	-9.63	2.47	-34.66
197	70.49	57.92	-0.09	19.08	25.79	-9.25	2.47	-36.50
198	67.45	59.20	-0.07	19.74	26.43	-9.27	2.38	-38.96
199	69.13	57.68	-0.11	19.25	26.73	-9.21	2.41	-38.46

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
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160	0.05	-6.82	-24.67	-24.74	-2.61	-2.86	-2.71	-1.60
161	0.00	-6.42	-24.61	-59.26	-2.55	-2.53	6.00	-1.60
162	-0.30	-6.70	-24.98	8.36	-2.47	-2.78	20.37	-1.59
163	-0.11	-5.77	-24.59	50.59	-2.62	-2.67	6.97	-1.61
164	0.19	-6.04	-24.78	-12.92	-2.65	-2.69	-11.54	-1.60
165	-0.02	-5.73	-24.78	-25.13	-2.64	-2.84	8.52	-1.63
166	-0.06	-7.15	-24.72	10.36	-2.90	-2.60	11.33	-1.60
167	0.05	-5.02	-24.62	-18.12	-2.65	-2.70	-9.13	-1.60
168	-0.18	-5.74	-24.65	-43.31	-2.61	-2.74	2.60	-1.63
169	0.05	-3.92	-24.69	-0.11	-2.62	-2.81	-0.57	-1.59
170	-0.22	-6.42	-24.74	31.59	-2.54	-2.42	4.93	-1.60
171	0.38	-6.29	-24.79	17.32	-2.47	-2.80	3.16	-1.58
172	-0.20	-4.47	-24.71	-76.32	-2.72	-2.81	14.78	-1.58
173	-0.02	-3.41	-24.55	12.62	-2.59	-2.77	13.40	-1.61
174	-0.19	-6.36	-24.80	-30.09	-2.58	-2.56	11.87	-1.60
175	0.01	-5.58	-24.80	-0.33	-2.57	-2.98	8.06	-1.59
176	-0.21	-7.35	-22.87	-2.52	-2.79	-3.04	-3.04	-1.43
177	0.42	-6.96	-22.77	-14.25	-2.87	-2.93	-3.93	-1.45
178	-0.02	-6.32	-22.67	6.85	-2.92	-3.19	-7.99	-1.45
179	-0.09	-6.74	-22.78	-7.19	-2.85	-2.83	8.51	-1.45
180	0.17	-6.39	-22.66	56.84	-2.89	-2.75	1.92	-1.46
181	0.19	-5.99	-22.80	8.81	-2.85	-2.82	-0.75	-1.45
182	0.14	-5.99	-22.76	24.31	-2.86	-2.90	-4.31	-1.46
183	-0.08	-7.56	-22.53	13.39	-2.79	-2.74	16.60	-1.45
184	-0.11	-6.77	-22.95	19.90	-3.09	-2.74	1.85	-1.47
185	0.08	-6.63	-22.79	28.47	-2.86	-3.04	-8.57	-1.44
186	-0.04	-6.29	-22.94	53.97	-2.89	-3.03	-0.93	-1.44
187	-0.14	-6.33	-22.59	99.70	-3.14	-2.75	-4.33	-1.43
188	0.00	-7.06	-22.71	-0.85	-2.92	-2.89	0.54	-1.45
189	-0.07	-8.13	-22.59	-18.23	-2.91	-3.01	-8.11	-1.44
190	-0.17	-7.08	-22.39	23.48	-2.85	-2.87	-3.17	-1.46
191	-0.06	-7.27	-22.67	41.89	-2.92	-2.76	-3.93	-1.46
192	-0.08	-4.09	-16.51	-9.54	-3.59	-2.93	-0.63	-1.16
193	-0.14	-3.11	-16.34	18.48	-3.60	-3.00	1.15	-1.14
194	0.04	-3.51	-16.64	17.91	-3.51	-2.91	1.10	-1.13
195	-0.17	-3.94	-16.52	15.91	-3.51	-2.89	-1.44	-1.13
196	0.08	-3.88	-16.58	-20.06	-3.47	-2.90	5.56	-1.13
197	0.09	-3.67	-16.49	-16.57	-3.57	-2.88	3.05	-1.13
198	0.05	-3.94	-16.42	-16.93	-3.42	-2.97	1.59	-1.11
199	0.02	-4.02	-16.51	-25.71	-3.60	-2.90	2.24	-1.14

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
160	-1.72	-4.57	-3.31	-5.64	1.91	1.82	6.79	-74.96
161	2.72	-4.07	0.55	-5.64	1.76	1.89	6.87	-73.97
162	-4.32	-4.19	-2.77	-5.46	2.01	1.92	7.05	-73.97
163	-1.39	-4.01	-0.95	-5.48	2.46	1.69	6.74	-74.71
164	-0.61	-4.33	-1.97	-5.78	2.93	1.84	6.85	-75.67
165	2.85	-4.36	-3.46	-5.41	2.81	1.98	6.35	-75.37
166	-1.58	-3.98	-1.06	-5.35	1.94	1.90	6.59	-74.94
167	-0.38	-5.08	-3.13	-5.36	2.77	1.81	6.39	-75.00
168	2.47	-3.87	-0.88	-5.81	2.49	1.71	7.18	-75.15
169	-2.51	-4.45	-6.66	-4.91	1.83	2.03	6.90	-76.07
170	6.57	-3.95	-4.60	-5.75	1.82	2.01	6.96	-73.68

171	2.24	-3.92	-1.15	-5.47	2.71	1.61	6.63	-73.51
172	3.40	-4.54	-2.52	-5.54	2.40	2.00	6.80	-73.48
173	-0.75	-4.31	2.15	-5.78	2.35	1.56	6.88	-74.65
174	3.78	-4.76	-3.69	-5.35	1.90	1.99	7.05	-75.07
175	2.70	-4.13	-3.26	-5.61	1.83	1.74	6.62	-75.22
176	-3.40	-4.70	-1.81	-4.85	1.49	1.10	5.28	-69.91
177	3.89	-4.52	-2.20	-5.42	1.47	1.05	5.44	-68.29
178	-0.89	-4.57	-0.54	-5.06	1.72	1.23	5.32	-68.69
179	0.34	-4.88	-0.94	-5.10	1.60	1.22	5.19	-67.29
180	2.91	-4.43	-2.20	-5.38	1.18	1.28	5.03	-68.04
181	0.20	-4.51	0.35	-5.67	1.11	1.13	5.20	-69.65
182	-5.24	-4.31	-1.03	-5.08	1.14	1.21	4.73	-67.72
183	2.46	-4.05	0.58	-5.67	0.91	1.03	5.01	-68.14
184	-1.50	-4.78	-1.48	-5.34	1.04	1.23	5.58	-68.25
185	0.24	-5.14	0.73	-4.87	1.21	1.05	5.17	-68.14
186	-0.78	-4.30	-4.96	-5.25	1.03	1.33	5.06	-68.46
187	1.72	-5.38	0.77	-4.80	1.35	0.99	5.47	-67.65
188	-0.43	-5.19	-0.03	-5.43	1.03	1.19	5.13	-71.14
189	4.51	-4.64	-3.26	-4.93	0.78	0.95	5.52	-68.34
190	0.38	-4.65	0.48	-4.86	1.19	1.08	5.19	-69.56
191	-3.32	-4.37	4.83	-5.59	1.22	1.00	5.24	-71.35
192	-1.10	-6.19	10.44	-5.27	-10.64	1.78	8.50	-46.49
193	-0.22	-5.99	10.70	-5.10	-10.32	1.84	8.95	-46.44
194	1.19	-6.00	10.41	-5.23	-9.90	1.72	8.65	-45.75
195	-0.99	-6.04	11.34	-5.36	-10.16	1.78	8.59	-44.99
196	-1.78	-5.87	9.88	-5.37	-10.11	1.82	8.86	-46.13
197	0.12	-6.17	11.67	-5.32	-10.56	1.82	8.33	-46.32
198	0.29	-6.03	10.23	-5.05	-10.16	1.82	8.42	-44.92
199	-2.02	-5.95	9.82	-5.22	-9.75	1.57	8.69	-45.94

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
160	-7.10	3.84	2.29	7.25	-1.41	-42.89	-2.47	-2.31
161	-7.16	4.00	2.40	6.79	-1.29	-44.10	-2.43	-2.37
162	-6.88	3.82	2.18	6.67	-1.35	-44.13	-2.49	-2.35
163	-7.18	3.48	2.31	6.99	-1.38	-42.27	-2.53	-2.32
164	-6.67	3.67	2.34	6.50	-1.34	-42.33	-2.46	-2.34
165	-8.06	3.80	2.15	6.43	-1.41	-42.56	-2.48	-2.36
166	-6.97	3.84	2.07	6.30	-1.37	-41.59	-2.35	-2.30
167	-7.50	3.63	2.27	8.27	-1.29	-43.63	-2.40	-2.33
168	-7.23	3.84	2.14	7.07	-1.36	-42.42	-2.44	-2.32
169	-6.82	3.72	2.20	6.12	-1.31	-43.38	-2.53	-2.33
170	-6.81	3.71	2.21	7.09	-1.38	-41.57	-2.45	-2.37
171	-6.97	3.74	2.16	7.13	-1.35	-42.14	-2.49	-2.32
172	-6.01	3.77	2.13	7.48	-1.41	-42.50	-2.42	-2.30
173	-6.58	3.62	2.20	6.64	-1.30	-41.40	-2.46	-2.35
174	-6.09	3.80	2.19	6.27	-1.46	-42.84	-2.53	-2.33
175	-7.31	3.70	2.41	6.68	-1.30	-41.21	-2.56	-2.35
176	-6.83	3.47	2.53	4.38	-1.11	-39.26	-2.01	-2.17
177	-7.14	3.45	2.16	4.46	-1.08	-38.15	-2.01	-2.17
178	-6.84	3.45	2.44	4.82	-1.20	-37.87	-1.96	-2.14
179	-7.17	3.28	2.32	4.76	-1.15	-38.48	-2.04	-2.17
180	-6.73	3.38	2.39	3.85	-1.07	-36.51	-1.99	-2.13
181	-6.45	3.47	2.29	5.05	-0.93	-37.26	-1.92	-2.14

182	-6.41	3.52	2.16	4.22	-1.01	-40.58	-2.04	-2.16
183	-6.37	3.34	2.26	4.47	-1.11	-38.40	-1.98	-2.14
184	-6.07	3.42	2.22	5.10	-1.07	-38.98	-1.95	-2.18
185	-6.69	3.57	2.23	4.02	-1.03	-37.01	-2.01	-2.19
186	-6.10	3.26	2.21	4.04	-1.15	-37.83	-1.92	-2.15
187	-6.41	3.36	2.25	4.21	-1.12	-41.20	-2.00	-2.14
188	-6.48	3.30	2.49	4.27	-1.03	-39.02	-1.98	-2.14
189	-6.73	3.32	2.34	4.37	-1.12	-37.91	-1.96	-2.18
190	-6.61	3.66	2.16	4.43	-1.14	-37.43	-1.86	-2.15
191	-7.03	3.45	2.56	4.75	-1.13	-38.65	-1.94	-2.17
192	14.34	3.23	0.52	-6.89	-0.44	-54.93	-0.40	-0.38
193	15.28	3.27	0.54	-6.69	-0.47	-54.20	-0.34	-0.37
194	15.25	3.25	0.65	-6.55	-0.50	-55.28	-0.39	-0.36
195	15.22	3.25	0.61	-6.99	-0.47	-54.89	-0.36	-0.38
196	15.23	3.32	0.62	-6.76	-0.44	-55.82	-0.36	-0.31
197	14.98	3.30	0.73	-7.01	-0.45	-54.51	-0.36	-0.36
198	14.34	3.25	0.68	-7.27	-0.47	-54.67	-0.36	-0.37
199	14.44	3.17	0.51	-6.82	-0.48	-54.34	-0.37	-0.36

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
160	0.37	6.16	-4.89	-3.95	-0.18	-5.80	216.68	-4.24
161	0.44	6.30	-5.02	-1.35	9.89	-6.94	215.70	0.94
162	0.47	5.80	-2.58	-2.61	8.27	-5.37	202.79	-3.53
163	0.41	6.05	2.06	-4.03	5.60	-6.26	205.63	-4.57
164	0.40	5.86	4.56	-3.05	-1.66	-6.13	201.34	-2.44
165	0.43	6.14	-4.27	-1.20	-3.83	-6.40	219.57	-2.51
166	0.35	6.13	-2.92	-2.09	8.70	-6.09	207.33	1.18
167	0.39	5.87	-2.87	-4.76	8.99	-7.35	200.76	5.54
168	0.38	6.11	2.58	-1.78	-3.50	-5.84	212.19	-5.32
169	0.41	5.65	-0.03	-1.87	9.47	-6.14	212.81	2.41
170	0.44	5.74	-2.44	-1.68	6.05	-6.48	202.50	3.86
171	0.45	6.48	-1.42	-1.59	4.23	-6.51	218.85	-1.59
172	0.38	6.75	2.44	-0.51	1.69	-6.58	204.05	-0.83
173	0.46	5.72	-0.38	-0.25	5.15	-5.50	191.42	0.98
174	0.45	6.33	3.98	-2.61	-5.38	-5.59	205.05	-3.88
175	0.40	5.97	6.85	-2.28	7.55	-6.99	200.65	-0.70
176	0.45	7.06	-0.46	-1.72	1.09	-6.70	193.26	3.23
177	0.44	6.59	0.91	-1.08	-0.45	-7.42	197.52	2.04
178	0.46	7.05	-2.54	-0.60	1.25	-6.31	187.34	0.15
179	0.47	6.80	-3.35	-0.78	2.85	-7.77	196.66	2.32
180	0.45	6.54	-0.56	-2.66	5.28	-7.34	204.22	-2.50
181	0.49	6.72	2.50	-0.06	5.80	-6.91	198.68	-4.05
182	0.51	7.05	-4.15	-2.54	-1.36	-7.48	201.35	0.50
183	0.45	7.01	-1.19	-0.48	3.29	-8.19	205.06	-2.27
184	0.50	6.51	1.42	-1.52	2.04	-7.31	184.98	5.86
185	0.46	6.88	0.90	-0.33	1.45	-7.19	200.08	-1.33
186	0.41	6.38	-0.92	0.12	-2.64	-8.02	193.03	4.59
187	0.48	7.11	-1.69	-1.28	-2.29	-7.25	195.79	-4.40
188	0.44	6.54	3.58	-0.63	5.67	-6.94	198.26	-5.30
189	0.42	6.39	-4.58	-1.29	7.96	-7.29	198.65	-3.19
190	0.46	6.64	-4.68	-1.11	-4.70	-7.46	196.34	3.04
191	0.45	6.81	2.51	-2.38	0.31	-7.21	195.56	1.44
192	0.51	7.88	-0.86	5.86	-1.23	-9.89	31.77	-1.15

193	0.49	7.70	0.24	5.63	-4.70	-9.82	32.51	-0.43
194	0.54	7.71	-0.61	5.51	-5.12	-9.75	32.83	1.99
195	0.49	7.83	0.62	5.83	2.58	-9.67	33.03	0.79
196	0.52	7.78	-1.62	5.24	-1.38	-9.49	37.38	1.85
197	0.54	7.94	0.26	4.88	0.93	-9.66	28.25	-0.89
198	0.48	7.63	-0.16	6.08	1.97	-9.63	26.78	-1.78
199	0.50	7.86	-1.26	5.43	1.97	-9.74	29.21	-1.97

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
160	-15.09	1.43	8.38	2.09	67.10	50.59	-4.05	-2.61
161	-15.99	1.74	8.50	-0.09	74.46	49.89	-3.80	-2.33
162	-15.23	1.34	8.48	-1.80	66.68	50.94	-3.90	-2.70
163	-14.71	1.41	6.35	-3.76	65.56	52.11	-3.89	-3.17
164	-15.64	1.35	8.94	-11.87	69.07	50.02	-3.94	-2.36
165	-15.55	1.51	9.37	4.37	68.67	53.08	-3.92	-2.80
166	-15.40	1.37	8.38	2.83	69.57	52.07	-3.76	-1.98
167	-15.25	1.65	7.20	-1.24	69.80	49.58	-4.03	-3.15
168	-15.21	1.29	7.64	-5.76	65.67	51.80	-3.99	-4.23
169	-15.65	1.44	7.37	0.61	67.24	51.04	-3.94	-3.61
170	-15.53	1.54	7.44	-3.07	72.23	52.70	-4.07	-2.45
171	-14.94	1.54	9.01	-0.48	66.04	51.64	-4.11	-2.67
172	-15.34	1.52	8.43	2.06	71.91	50.70	-4.02	-2.48
173	-14.78	1.57	7.45	-2.85	62.87	50.65	-3.87	-1.81
174	-15.51	1.51	9.59	0.14	65.71	49.51	-4.02	-1.82
175	-15.48	1.26	6.97	7.15	69.96	53.52	-3.95	-3.26
176	-12.35	1.49	9.88	-9.25	57.89	46.03	-3.59	-1.89
177	-12.31	1.61	9.42	-3.73	67.80	45.21	-3.64	-2.45
178	-12.14	1.49	10.23	-0.63	58.49	45.33	-3.73	-2.72
179	-12.13	1.61	10.38	0.24	55.05	46.72	-3.63	-2.00
180	-11.85	1.51	9.89	-9.35	58.31	44.55	-3.63	-3.61
181	-11.39	1.68	10.30	-3.55	55.82	44.55	-3.69	-0.74
182	-11.71	1.58	9.81	-3.65	60.66	45.97	-3.65	-2.52
183	-11.84	1.62	7.87	-3.91	62.48	45.42	-3.52	-2.08
184	-12.30	1.57	9.70	8.09	55.78	46.36	-3.52	-2.01
185	-11.78	1.63	8.33	-0.93	55.84	45.39	-3.58	-2.29
186	-11.46	1.71	9.94	-4.37	60.19	44.13	-3.66	-3.12
187	-12.17	1.61	8.29	-5.53	58.65	44.83	-3.78	-2.80
188	-11.84	1.68	7.48	-2.86	65.02	45.54	-3.61	-2.11
189	-11.77	1.64	9.14	2.22	55.82	44.24	-3.61	-2.37
190	-12.38	1.71	7.76	-4.37	63.61	45.04	-3.67	-2.30
191	-11.86	1.57	8.30	0.81	62.45	45.01	-3.51	-2.05
192	-14.53	1.99	13.18	1.33	106.76	38.07	-5.34	-5.15
193	-13.82	2.08	13.41	1.91	104.39	37.64	-5.16	-4.44
194	-14.37	2.15	13.46	-0.20	110.70	38.01	-5.34	-5.04
195	-13.91	2.12	13.46	0.42	102.96	38.66	-5.33	-4.93
196	-14.24	2.07	14.12	2.08	97.99	38.20	-5.46	-5.31
197	-14.21	2.13	13.13	-1.64	100.89	37.80	-5.44	-4.59
198	-14.39	2.09	13.96	-2.45	103.11	38.34	-5.33	-5.41
199	-13.74	2.08	13.50	-2.52	105.29	38.24	-5.41	-4.78

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
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160	4.80	-3.26	-0.21	3.57	2.42	-0.68	-12.67	-24.28
161	-3.84	-4.72	-0.18	2.94	2.29	-0.37	1.69	35.79
162	2.38	-4.79	-0.18	3.21	2.09	-0.33	4.88	-13.73
163	-0.25	-2.02	-0.17	2.96	2.59	-0.43	27.32	49.32
164	3.73	-5.32	-0.15	3.77	2.77	-0.50	-9.12	57.33
165	0.28	-2.82	-0.17	3.31	2.63	-0.50	22.97	31.74
166	-4.59	-3.20	-0.18	3.25	3.45	-0.72	-15.61	33.03
167	-1.01	-4.60	-0.17	4.16	2.50	-0.33	-27.43	-11.64
168	-4.66	-1.98	-0.19	4.12	2.54	-0.22	-1.92	-10.90
169	3.96	-7.01	-0.20	2.86	2.56	-0.62	17.28	-12.73
170	2.95	-4.19	-0.21	3.43	2.63	-0.19	-14.50	-1.16
171	-7.27	-3.21	-0.20	3.49	2.62	-0.49	1.07	-12.08
172	7.69	-6.61	-0.16	2.74	2.46	-0.33	12.29	-21.91
173	0.38	-3.77	-0.20	3.10	2.67	-0.35	-8.43	25.41
174	1.04	-2.66	-0.18	3.00	2.98	-0.48	-6.63	-62.55
175	1.60	-1.83	-0.18	3.19	2.77	-0.62	-10.72	-93.17
176	-0.52	-7.31	-0.07	2.13	2.59	-0.40	-6.00	14.91
177	-0.77	-6.06	-0.12	1.95	3.14	-0.16	-9.91	-11.31
178	-1.38	-6.04	-0.11	1.95	2.90	-0.10	-14.84	-5.31
179	3.62	-5.16	-0.12	2.88	2.42	-0.29	-2.65	24.60
180	-0.82	-5.22	-0.08	2.50	2.59	-0.01	-9.33	68.11
181	1.07	-4.69	-0.09	1.89	2.66	-0.23	5.36	72.62
182	1.54	-7.98	-0.10	2.35	2.91	-0.19	-19.95	-12.65
183	0.38	-6.18	-0.10	2.44	2.23	-0.51	30.27	7.42
184	-1.62	-7.10	-0.09	1.94	2.99	-0.47	-3.11	-34.92
185	2.42	-5.25	-0.08	2.10	2.45	-0.43	3.54	3.01
186	-5.80	-6.76	-0.10	1.44	2.25	-0.49	-21.05	17.21
187	7.44	-5.48	-0.12	2.46	2.44	-0.27	23.77	30.52
188	-5.44	-6.90	-0.11	2.41	2.25	-0.51	10.61	-0.47
189	0.88	-6.94	-0.09	2.24	2.12	-0.02	20.18	-19.77
190	8.28	-6.66	-0.10	2.54	2.34	-0.33	21.36	57.54
191	-3.19	-5.28	-0.09	2.05	2.63	-0.25	7.47	31.17
192	5.63	-31.18	0.32	-1.71	12.41	0.20	-0.23	7.04
193	2.70	-30.14	0.28	-2.06	12.10	0.16	-29.14	20.81
194	-1.45	-29.85	0.28	-1.81	12.41	0.08	17.53	16.41
195	-2.76	-29.39	0.31	-1.97	12.28	0.17	-0.96	15.02
196	-3.77	-29.74	0.29	-2.19	12.65	0.09	-9.79	31.38
197	-3.05	-29.86	0.30	-1.86	12.48	0.13	-18.07	9.58
198	-2.17	-30.42	0.30	-1.83	12.21	0.18	-5.19	14.38
199	-2.93	-30.44	0.29	-1.97	12.62	0.04	16.43	-0.26

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
200	-84.84	1.13	5.88	-40.53	-7.81	13.76	5.16	-2.61
201	-86.46	1.17	5.92	-41.73	-7.49	14.37	4.96	-2.65
202	-85.65	0.92	5.61	-41.84	-7.73	14.39	4.88	-3.04
203	-83.58	0.92	5.57	-41.07	-7.79	14.00	5.78	-2.87
204	-84.03	0.90	5.59	-40.04	-8.07	15.05	5.36	-2.80
205	-88.01	1.00	5.82	-40.27	-7.37	14.65	5.64	-2.95
206	-91.14	1.09	5.67	-38.94	-7.49	13.90	5.00	-3.05
207	-87.84	1.13	5.79	-39.92	-7.62	14.06	4.61	-3.12
208	-139.32	2.88	8.38	-29.54	-7.87	11.87	0.06	1.37
209	-136.35	2.65	8.20	-29.63	-7.90	11.88	0.14	1.88
210	-136.39	2.88	8.19	-29.62	-7.91	11.54	0.09	1.74

211	-140.86	2.90	8.40	-28.91	-8.02	11.83	1.40	1.81
212	-141.28	2.64	8.02	-30.90	-8.11	11.87	0.91	1.38
213	-147.44	3.03	8.55	-29.96	-8.18	12.28	0.27	0.89
214	-137.06	2.83	8.21	-28.00	-7.95	11.69	-0.13	1.43
215	-136.69	2.81	8.31	-31.18	-8.32	12.24	-0.51	1.76
216	-144.43	3.02	8.72	-30.56	-7.94	11.59	0.02	1.81
217	-137.63	2.89	8.13	-30.66	-7.76	11.71	0.40	1.39
218	-132.91	2.90	8.74	-30.41	-8.04	11.61	0.24	2.17
219	-146.30	2.70	8.31	-32.57	-8.00	12.00	0.33	1.31
220	-141.57	2.69	8.20	-30.05	-8.02	11.79	0.65	1.73
221	-143.00	2.79	8.25	-29.94	-7.79	11.74	-0.08	1.70
222	-138.77	2.83	8.47	-28.20	-7.64	11.82	0.14	1.48
223	-140.99	2.63	8.27	-30.91	-7.93	11.74	0.18	1.55
224	-54.29	0.60	4.08	-17.30	-4.02	6.24	2.53	-0.47
225	-49.96	0.52	4.11	-17.63	-3.57	6.25	2.57	-0.65
226	-53.86	0.63	4.07	-16.88	-3.88	6.15	2.39	-0.17
227	-53.38	0.55	4.12	-16.78	-3.97	6.27	2.64	-0.44
228	-52.04	0.58	4.25	-17.54	-3.97	6.26	2.50	-0.44
229	-49.03	0.59	4.08	-16.65	-3.89	6.26	2.41	-0.68
230	-51.57	0.47	3.92	-17.80	-4.05	6.13	2.83	-0.45
231	-51.91	0.73	4.07	-16.88	-3.72	6.31	2.68	-0.76
232	-53.06	0.54	4.08	-17.42	-3.88	6.13	2.80	-0.66
233	-55.52	0.57	4.00	-17.43	-3.79	6.01	2.67	-0.74
234	-52.68	0.62	3.76	-16.64	-3.85	6.32	2.48	-0.51
235	-57.68	0.48	4.06	-16.80	-3.86	5.94	2.56	-0.56
236	-53.28	0.60	4.17	-17.07	-3.56	6.08	2.93	-0.65
237	-50.64	0.50	4.15	-16.55	-3.98	5.93	2.86	-0.63
238	-54.90	0.57	3.85	-17.16	-3.98	6.12	2.89	-0.43
239	-54.82	0.69	3.85	-16.27	-3.91	6.14	2.75	-0.44

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
200	66.06	57.96	-0.06	20.15	25.87	-9.12	2.49	-40.56
201	68.69	59.17	-0.05	18.98	26.20	-9.42	2.57	-38.06
202	65.93	58.25	-0.06	17.31	26.06	-9.06	2.45	-37.70
203	66.02	57.57	-0.06	17.54	24.57	-9.56	2.48	-37.48
204	62.62	60.56	-0.05	18.59	26.34	-9.47	2.50	-37.12
205	69.42	59.25	-0.07	18.44	25.76	-9.14	2.51	-38.09
206	69.43	58.91	-0.08	17.90	25.60	-9.19	2.56	-38.37
207	66.76	59.12	-0.06	17.58	23.78	-9.08	2.57	-36.53
208	105.25	45.16	0.14	30.65	17.19	-11.84	1.41	-56.05
209	100.97	49.11	0.17	33.24	19.20	-11.96	1.36	-54.82
210	110.23	47.17	0.14	32.19	18.96	-11.95	1.33	-57.81
211	99.46	47.41	0.17	34.11	19.23	-11.51	1.40	-58.91
212	100.89	47.64	0.20	32.97	18.91	-11.64	1.29	-55.22
213	105.89	47.15	0.17	34.74	19.38	-11.71	1.35	-54.68
214	100.66	48.39	0.12	33.05	18.56	-11.70	1.47	-55.39
215	105.40	48.05	0.14	35.47	19.87	-11.73	1.47	-56.98
216	107.76	46.39	0.15	32.78	18.55	-11.55	1.40	-56.60
217	102.67	46.58	0.16	34.43	19.08	-11.72	1.35	-56.71
218	106.31	45.50	0.13	35.30	18.10	-11.87	1.42	-56.77
219	103.64	46.85	0.14	34.19	18.24	-11.49	1.25	-56.12
220	105.64	46.90	0.17	34.57	17.92	-11.69	1.39	-57.90
221	104.76	45.29	0.14	35.97	21.36	-11.91	1.36	-56.05

222	103.79	46.27	0.14	33.43	19.41	-11.53	1.41	-56.73
223	101.88	45.23	0.17	34.42	19.64	-12.02	1.45	-53.76
224	44.53	25.82	-0.06	11.68	10.35	-5.06	0.95	-25.85
225	45.43	25.95	-0.04	11.59	10.33	-4.95	1.01	-27.25
226	42.76	25.29	-0.04	11.55	10.68	-4.99	1.00	-26.64
227	44.83	25.97	-0.05	12.62	10.83	-5.07	0.98	-26.88
228	43.61	25.36	-0.03	12.76	10.30	-4.92	1.10	-27.08
229	46.60	25.62	-0.05	11.49	10.21	-5.09	1.00	-26.17
230	43.98	25.50	-0.06	11.29	10.13	-5.14	0.85	-26.86
231	43.86	25.36	-0.04	11.30	10.41	-4.94	1.01	-25.41
232	46.44	25.68	-0.05	10.66	10.53	-5.00	0.95	-27.32
233	47.20	25.26	-0.02	10.75	11.18	-5.01	0.97	-26.35
234	45.19	26.43	-0.02	11.44	10.41	-5.05	0.94	-25.67
235	46.18	25.03	-0.03	11.83	10.34	-4.91	1.00	-26.54
236	45.28	25.60	-0.04	11.79	10.74	-5.00	0.99	-27.22
237	45.71	26.50	-0.06	11.87	10.86	-5.00	0.99	-28.24
238	44.83	25.30	-0.04	11.05	10.70	-5.05	0.95	-26.50
239	45.99	25.08	-0.06	11.38	10.68	-4.93	0.88	-26.89

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
200	-0.14	-3.90	-16.42	18.94	-3.52	-2.90	2.97	-1.12
201	-0.01	-3.17	-16.60	20.24	-3.57	-2.91	4.30	-1.14
202	0.14	-3.90	-16.67	6.48	-3.59	-2.84	-0.10	-1.11
203	-0.24	-3.89	-16.69	4.00	-3.46	-3.00	-2.50	-1.12
204	-0.10	-3.70	-16.54	0.10	-3.60	-2.94	-1.46	-1.13
205	-0.10	-3.49	-16.60	1.65	-3.53	-2.91	1.37	-1.12
206	0.11	-3.34	-16.64	5.16	-3.55	-2.70	-1.01	-1.15
207	0.02	-4.27	-16.57	-0.97	-3.50	-2.94	9.24	-1.12
208	0.27	-7.69	-16.18	9.78	-3.44	-1.72	2.71	-1.02
209	-0.04	-7.39	-15.94	-9.32	-3.51	-1.63	1.56	-1.04
210	-0.07	-7.15	-16.15	14.88	-3.47	-1.77	-0.45	-1.06
211	0.08	-7.65	-16.49	5.04	-3.52	-1.64	-0.63	-1.05
212	-0.03	-7.56	-16.13	-45.52	-3.43	-1.85	-3.97	-1.04
213	0.07	-6.72	-16.27	-34.23	-3.53	-1.73	-1.11	-1.03
214	0.17	-7.68	-16.15	2.33	-3.41	-1.95	2.65	-1.04
215	0.23	-7.81	-16.12	8.35	-3.41	-1.63	-1.63	-1.05
216	-0.06	-7.14	-16.07	-6.69	-3.47	-1.52	1.36	-1.04
217	-0.18	-7.44	-16.05	-10.69	-3.52	-1.66	4.16	-1.04
218	-0.16	-7.64	-16.02	4.38	-3.55	-1.58	-0.43	-1.03
219	-0.09	-7.83	-16.27	51.83	-3.50	-1.72	2.93	-1.05
220	-0.08	-7.89	-16.36	-26.86	-3.50	-1.95	1.99	-1.04
221	-0.24	-7.62	-16.20	-13.42	-3.47	-1.61	-0.73	-1.07
222	0.22	-7.37	-16.13	-6.98	-3.48	-1.67	5.77	-1.05
223	-0.05	-8.48	-16.17	7.62	-3.47	-1.88	1.22	-1.02
224	0.08	-2.72	-7.93	-0.84	-2.21	-1.51	1.68	-0.53
225	0.06	-2.53	-7.92	0.54	-2.21	-1.54	1.26	-0.55
226	-0.03	-2.53	-8.02	3.16	-2.19	-1.67	1.70	-0.53
227	-0.03	-2.64	-7.78	3.03	-2.22	-1.61	-2.87	-0.53
228	-0.07	-2.56	-7.82	-6.44	-2.22	-1.52	-0.46	-0.53
229	0.06	-2.67	-7.95	7.79	-2.17	-1.65	-0.30	-0.54
230	0.03	-2.67	-7.73	-7.24	-2.21	-1.63	-0.68	-0.53
231	0.08	-2.32	-7.83	-8.27	-2.18	-1.51	0.09	-0.52
232	-0.01	-2.60	-7.85	-3.83	-2.19	-1.56	-0.25	-0.54

233	0.14	-2.88	-7.84	2.24	-2.19	-1.51	0.55	-0.54
234	-0.05	-2.42	-7.90	-4.40	-2.19	-1.60	0.13	-0.54
235	-0.02	-2.55	-7.79	-11.70	-2.22	-1.55	-0.60	-0.53
236	0.19	-2.48	-7.84	12.76	-2.20	-1.54	3.63	-0.53
237	-0.02	-2.83	-7.95	2.25	-2.23	-1.51	-0.31	-0.53
238	-0.12	-2.61	-7.96	13.30	-2.23	-1.60	-0.31	-0.51
239	0.10	-2.52	-7.69	18.92	-2.22	-1.58	0.44	-0.54

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
200	3.99	-5.95	11.60	-5.28	-9.88	1.58	8.77	-45.86
201	0.55	-5.96	10.24	-5.13	-9.75	1.70	8.73	-46.07
202	-1.77	-6.09	11.31	-4.91	-9.69	1.63	8.59	-45.79
203	1.56	-5.98	10.38	-5.02	-10.51	1.79	8.69	-45.10
204	1.56	-6.08	10.34	-5.40	-10.12	1.67	8.71	-45.35
205	2.13	-6.31	10.69	-5.05	-10.29	1.70	8.93	-45.65
206	2.79	-6.28	10.42	-5.44	-10.01	1.75	8.49	-46.18
207	-0.59	-6.19	11.15	-5.23	-10.73	1.85	8.51	-46.18
208	-0.86	-7.96	-3.95	-3.60	-9.52	0.68	11.12	-49.79
209	0.81	-8.49	-5.59	-3.29	-8.62	0.65	11.07	-50.16
210	-1.82	-8.57	-4.86	-3.74	-8.55	0.75	11.01	-50.88
211	-0.18	-8.60	-5.31	-3.43	-8.14	0.70	11.15	-50.58
212	2.61	-8.70	-6.29	-3.63	-8.86	0.61	10.85	-50.54
213	-3.08	-8.69	-3.95	-3.63	-9.03	0.68	11.10	-50.64
214	-1.21	-8.67	-3.89	-3.44	-8.30	0.57	10.92	-50.48
215	-0.63	-8.21	-5.17	-3.56	-8.01	0.58	10.75	-51.30
216	-2.33	-8.45	-6.57	-3.49	-8.58	0.62	10.57	-49.36
217	-1.75	-8.63	-4.28	-3.72	-9.07	0.52	11.17	-51.38
218	-1.16	-8.33	-3.96	-3.48	-8.78	0.65	11.05	-52.15
219	-1.30	-8.25	-5.48	-3.26	-8.99	0.67	10.97	-50.97
220	-2.13	-8.82	-5.83	-3.54	-8.76	0.61	10.94	-50.33
221	0.26	-8.28	-5.24	-4.00	-8.47	0.67	10.96	-50.23
222	1.12	-8.32	-6.96	-3.18	-9.03	0.48	11.17	-52.19
223	2.79	-8.82	-5.29	-3.80	-8.83	0.67	10.90	-50.60
224	0.34	-4.36	2.59	-2.85	-4.74	0.74	5.72	-22.17
225	-0.31	-4.18	2.99	-3.02	-4.17	0.66	5.46	-22.03
226	-0.12	-4.41	2.85	-2.84	-4.73	0.61	5.85	-21.93
227	-0.33	-4.36	3.11	-2.91	-4.70	0.71	5.44	-21.77
228	0.44	-4.44	3.35	-2.94	-4.84	0.57	5.52	-22.25
229	0.33	-4.39	3.32	-2.77	-4.87	0.60	5.59	-22.24
230	-0.06	-4.13	2.39	-2.85	-4.78	0.58	5.64	-21.99
231	0.24	-4.30	4.62	-2.89	-4.63	0.62	5.66	-22.01
232	0.96	-4.38	4.07	-3.03	-4.86	0.60	5.48	-22.08
233	-0.65	-4.41	3.56	-2.99	-4.53	0.68	5.63	-21.70
234	0.36	-4.36	3.33	-2.77	-4.45	0.63	5.65	-21.98
235	-0.71	-4.32	3.30	-2.85	-4.91	0.59	5.67	-22.12
236	-0.29	-4.26	3.75	-2.72	-4.75	0.66	5.69	-22.65
237	0.22	-4.19	2.49	-2.91	-4.69	0.71	5.68	-22.50
238	-1.44	-4.36	3.56	-2.73	-5.08	0.70	5.66	-22.30
239	-0.57	-4.42	2.22	-2.74	-4.73	0.58	5.58	-22.22

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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200	14.32	3.22	0.58	-6.43	-0.50	-53.19	-0.36	-0.36
201	14.99	3.34	0.54	-6.74	-0.49	-54.35	-0.39	-0.36
202	14.82	3.28	0.68	-7.03	-0.52	-54.80	-0.41	-0.36
203	14.71	3.23	0.63	-6.98	-0.46	-54.42	-0.37	-0.36
204	14.96	3.30	0.60	-7.13	-0.47	-55.09	-0.35	-0.36
205	14.23	3.28	0.66	-6.96	-0.49	-55.95	-0.36	-0.36
206	15.18	3.28	0.69	-7.34	-0.50	-55.62	-0.37	-0.35
207	14.69	3.23	0.61	-6.88	-0.52	-53.88	-0.36	-0.35
208	9.47	3.48	1.57	-4.65	-0.96	-45.31	-1.00	-0.74
209	9.10	3.48	1.40	-3.70	-0.97	-43.24	-1.04	-0.68
210	9.37	3.51	1.54	-3.31	-0.90	-42.54	-0.99	-0.72
211	9.83	3.51	1.58	-3.74	-0.92	-42.03	-0.94	-0.72
212	9.76	3.52	1.42	-3.73	-0.93	-43.64	-0.96	-0.71
213	9.51	3.50	1.54	-4.16	-0.90	-44.19	-1.03	-0.72
214	9.23	3.38	1.59	-3.62	-0.89	-43.99	-1.02	-0.71
215	9.61	3.50	1.54	-3.80	-0.93	-44.32	-1.01	-0.74
216	8.98	3.46	1.50	-3.79	-0.98	-44.82	-0.98	-0.72
217	9.21	3.52	1.53	-3.92	-0.92	-44.28	-0.94	-0.73
218	9.94	3.55	1.52	-4.54	-0.91	-41.51	-1.02	-0.72
219	9.63	3.53	1.40	-3.79	-0.93	-43.59	-0.99	-0.73
220	9.23	3.46	1.49	-3.60	-0.89	-45.16	-0.99	-0.73
221	8.94	3.48	1.46	-3.51	-0.91	-44.60	-0.97	-0.72
222	9.57	3.47	1.57	-3.09	-0.87	-43.51	-1.02	-0.73
223	9.36	3.49	1.43	-3.84	-0.88	-43.84	-0.98	-0.72
224	6.36	1.72	0.50	-2.51	-0.38	-24.29	-0.56	-0.24
225	6.20	1.80	0.47	-1.97	-0.35	-25.26	-0.57	-0.23
226	6.18	1.80	0.46	-2.30	-0.37	-24.28	-0.51	-0.24
227	6.37	1.78	0.42	-2.20	-0.34	-23.88	-0.53	-0.23
228	6.45	1.78	0.51	-1.83	-0.37	-24.04	-0.54	-0.23
229	7.00	1.71	0.46	-1.80	-0.35	-24.35	-0.53	-0.23
230	6.17	1.71	0.53	-2.11	-0.33	-25.11	-0.55	-0.21
231	6.09	1.76	0.48	-2.36	-0.36	-23.07	-0.53	-0.23
232	5.82	1.77	0.43	-2.38	-0.37	-24.22	-0.56	-0.22
233	5.85	1.74	0.46	-2.27	-0.32	-24.46	-0.55	-0.25
234	6.48	1.79	0.52	-2.15	-0.37	-23.83	-0.56	-0.25
235	6.59	1.79	0.45	-2.00	-0.36	-24.26	-0.56	-0.23
236	6.53	1.80	0.54	-2.13	-0.35	-24.30	-0.53	-0.25
237	6.32	1.77	0.46	-2.42	-0.32	-24.11	-0.54	-0.23
238	6.42	1.76	0.48	-2.25	-0.34	-24.32	-0.51	-0.22
239	6.80	1.81	0.45	-2.12	-0.34	-25.08	-0.53	-0.24

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
200	0.46	7.85	-0.13	6.00	-0.66	-9.63	27.73	-0.71
201	0.49	7.88	1.56	5.01	5.66	-9.86	37.21	-0.38
202	0.50	7.82	-0.21	5.62	3.39	-9.78	36.01	-3.02
203	0.47	7.87	1.17	5.57	6.66	-9.76	28.22	-1.35
204	0.47	7.73	-1.49	4.73	-0.17	-9.85	34.81	-0.69
205	0.51	7.78	0.66	4.81	0.90	-9.79	27.69	-1.68
206	0.48	7.68	0.46	5.55	0.83	-9.68	33.16	0.42
207	0.49	7.68	1.72	5.43	-3.05	-9.64	30.82	2.49
208	0.15	6.36	-2.14	8.22	-2.43	-9.98	99.11	-0.05
209	0.19	6.61	0.32	8.42	3.47	-9.67	83.51	-0.45
210	0.14	6.61	-1.32	7.86	6.02	-10.02	86.68	1.50

211	0.16	6.66	2.90	7.31	0.41	-9.81	87.51	1.78
212	0.22	6.31	-1.30	7.70	-0.04	-9.87	87.42	-0.67
213	0.15	6.49	-1.79	7.85	5.42	-9.54	85.96	-0.59
214	0.22	6.46	0.30	8.60	2.92	-9.67	90.41	0.31
215	0.20	6.38	-0.86	8.71	4.58	-9.76	83.46	-0.82
216	0.19	6.15	0.25	7.76	-2.22	-9.68	91.93	-0.68
217	0.20	6.57	-1.20	8.05	-1.02	-9.91	86.11	1.29
218	0.17	6.27	-1.14	7.68	2.32	-10.04	91.11	-1.23
219	0.15	6.39	-0.80	7.44	5.12	-9.89	88.36	0.45
220	0.15	6.49	-3.00	8.07	4.47	-9.78	95.29	-0.28
221	0.19	6.40	0.75	7.85	6.16	-9.63	90.57	-0.67
222	0.18	6.41	1.30	8.33	0.11	-9.71	81.50	3.06
223	0.22	6.24	-1.01	8.04	5.86	-9.95	86.36	-0.02
224	0.27	4.44	-0.04	3.08	2.50	-6.38	20.25	-1.03
225	0.24	4.52	-0.34	3.01	-1.29	-6.25	16.20	-0.21
226	0.25	4.55	-0.39	3.51	-0.60	-6.37	18.03	-1.33
227	0.26	4.52	0.03	3.32	-1.50	-6.28	24.04	-0.27
228	0.26	4.70	-0.29	3.18	-0.44	-6.27	17.92	-1.90
229	0.25	4.57	0.92	3.34	1.93	-6.40	22.77	-0.80
230	0.28	4.73	-0.66	2.98	4.13	-6.28	15.11	-0.47
231	0.24	4.58	0.43	3.54	-0.58	-6.26	25.20	0.44
232	0.27	4.64	0.84	3.11	0.14	-6.38	22.13	-0.19
233	0.25	4.51	0.22	3.69	-2.72	-6.35	21.97	0.80
234	0.27	4.51	-0.11	3.55	1.86	-6.42	25.28	-0.06
235	0.26	4.67	1.15	3.35	0.05	-6.42	23.04	-1.52
236	0.27	4.61	-0.75	3.40	-1.85	-6.07	16.17	1.21
237	0.26	4.65	-2.07	3.25	1.95	-6.27	18.98	-0.86
238	0.22	4.73	-0.73	3.27	-0.72	-6.28	20.99	0.14
239	0.24	4.67	1.08	3.59	-0.16	-6.19	18.79	-0.29

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
200	-14.58	2.15	13.52	1.39	98.94	38.13	-5.24	-4.83
201	-14.85	2.12	13.25	-1.46	105.91	38.10	-5.40	-4.96
202	-14.77	2.04	13.42	-0.83	108.07	37.94	-5.43	-4.34
203	-14.29	2.04	13.65	1.22	104.62	38.41	-5.42	-4.76
204	-14.52	2.02	13.38	-2.48	103.69	38.64	-5.38	-5.42
205	-13.83	2.00	13.13	-1.90	100.76	38.11	-5.22	-5.16
206	-14.72	2.09	13.54	1.63	99.44	36.76	-5.42	-5.11
207	-14.24	2.13	13.38	-2.68	106.04	38.62	-5.44	-5.01
208	-18.04	2.04	17.33	-4.13	80.49	45.11	-4.33	-0.42
209	-18.35	1.98	18.32	0.33	75.08	44.55	-4.26	-0.29
210	-18.34	1.91	17.45	-5.83	78.06	45.51	-4.19	0.29
211	-17.86	2.06	18.77	-1.57	76.22	44.80	-4.37	-0.99
212	-18.15	2.07	17.42	3.49	68.79	44.77	-4.29	-0.51
213	-17.90	1.96	18.27	2.48	78.99	45.19	-4.26	-1.08
214	-18.00	2.17	16.74	1.35	81.10	45.50	-4.36	0.34
215	-18.07	2.03	18.09	-0.23	80.85	45.97	-4.42	-0.63
216	-17.82	2.01	17.98	-1.08	76.05	44.86	-4.26	0.54
217	-18.26	1.97	17.85	2.39	76.52	45.77	-4.35	-0.24
218	-18.22	1.93	17.46	-2.42	78.00	44.64	-4.34	-0.75
219	-18.12	2.02	17.20	-2.55	79.55	45.31	-4.45	-0.57
220	-18.32	2.05	18.01	-0.15	73.51	44.83	-4.23	-1.39
221	-18.65	1.85	17.38	1.55	75.38	45.01	-4.24	0.04

222	-18.24	2.01	17.33	3.21	79.54	44.83	-4.33	-0.20
223	-17.86	2.09	17.70	-2.31	78.34	45.15	-4.28	0.51
224	-9.08	1.04	9.54	-0.38	45.23	20.55	-2.26	-2.70
225	-8.99	1.11	9.50	-2.62	47.46	20.68	-2.31	-2.31
226	-8.93	1.09	9.73	-1.20	48.83	20.40	-2.28	-2.68
227	-8.58	1.14	9.31	-0.97	45.76	20.14	-2.25	-2.35
228	-8.73	1.14	9.72	1.59	49.19	20.34	-2.30	-2.72
229	-9.14	1.21	9.50	0.67	47.88	20.41	-2.25	-2.50
230	-8.91	1.09	9.32	0.39	49.39	20.28	-2.25	-2.31
231	-9.06	1.07	9.52	1.36	48.38	20.82	-2.31	-2.84
232	-8.97	1.08	9.76	0.66	47.42	20.49	-2.30	-2.82
233	-8.95	1.16	9.71	-0.24	48.65	20.33	-2.32	-2.82
234	-8.79	1.05	9.83	0.41	50.36	20.50	-2.29	-2.90
235	-8.84	1.09	9.42	0.66	45.98	20.80	-2.30	-2.95
236	-9.18	1.08	9.80	-2.83	45.60	19.75	-2.31	-2.90
237	-9.04	1.19	9.56	1.14	49.97	20.36	-2.42	-2.38
238	-9.17	1.08	9.84	2.18	49.18	19.80	-2.27	-2.61
239	-8.69	1.11	9.76	2.64	47.25	20.56	-2.31	-2.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
200	-1.36	-29.64	0.29	-1.93	12.05	0.24	6.43	22.33
201	3.10	-29.13	0.30	-1.91	12.36	0.07	-7.30	-0.91
202	-5.52	-29.28	0.29	-2.07	12.24	0.09	-9.28	8.15
203	-4.47	-29.11	0.30	-1.96	12.32	0.14	-6.20	8.77
204	-2.98	-29.56	0.32	-1.95	12.15	0.15	7.99	31.91
205	-1.35	-29.64	0.30	-1.68	12.64	0.24	-18.47	-13.38
206	1.26	-30.90	0.30	-1.91	12.30	0.25	6.12	4.06
207	2.52	-29.07	0.29	-1.96	12.47	0.22	3.27	-13.81
208	-0.08	-22.82	0.23	-3.30	8.90	-1.48	9.45	-22.79
209	-0.46	-23.73	0.20	-3.31	9.41	-1.13	3.74	-10.82
210	3.20	-22.42	0.22	-3.21	9.23	-1.46	-3.21	9.83
211	-1.51	-22.56	0.20	-3.15	8.81	-1.43	-20.84	-1.75
212	3.07	-23.26	0.20	-3.21	9.20	-1.39	0.80	-30.30
213	-2.33	-23.08	0.23	-2.69	8.98	-1.30	-19.53	8.55
214	-6.54	-23.51	0.22	-3.41	8.76	-1.14	27.04	6.01
215	-3.79	-21.32	0.21	-3.00	8.93	-1.33	-17.66	-2.79
216	-0.24	-23.24	0.22	-3.03	9.09	-1.24	6.07	24.80
217	3.44	-24.06	0.20	-3.67	8.63	-1.29	-8.15	19.18
218	-1.92	-22.05	0.20	-3.29	9.00	-1.18	6.54	10.41
219	0.30	-22.68	0.21	-3.36	8.67	-1.31	-10.55	1.54
220	-2.80	-23.71	0.20	-3.40	8.54	-1.20	16.87	-20.57
221	-4.02	-23.59	0.23	-3.23	8.89	-1.33	-29.66	-8.53
222	3.71	-22.32	0.22	-3.65	9.42	-1.27	-7.45	20.01
223	4.01	-22.74	0.20	-3.23	9.05	-1.11	-0.92	8.73
224	-0.07	-13.13	0.10	-1.31	4.94	0.10	-2.81	9.80
225	3.00	-13.34	0.13	-1.17	5.09	0.13	9.49	1.20
226	1.83	-13.14	0.12	-1.12	4.86	0.09	-6.43	7.09
227	2.44	-12.89	0.11	-1.27	4.92	0.14	11.11	3.43
228	-1.82	-12.71	0.11	-1.30	5.04	0.23	5.82	2.18
229	2.60	-13.68	0.12	-1.18	4.89	0.13	-13.54	-2.96
230	-0.58	-12.59	0.13	-1.32	5.16	0.14	5.68	2.50
231	-1.66	-13.08	0.12	-1.37	5.20	0.05	0.12	4.41
232	4.51	-13.15	0.12	-1.50	4.62	0.10	-3.69	-7.06

233	1.30	-12.99	0.10	-1.29	5.15	0.15	-8.75	-0.68
234	3.05	-13.52	0.12	-1.26	5.30	0.06	-5.26	-12.29
235	1.63	-13.65	0.11	-1.20	5.12	0.00	-9.46	4.53
236	4.38	-13.38	0.12	-1.23	5.11	0.11	-0.87	-0.28
237	-2.09	-14.06	0.12	-1.23	4.84	0.01	-7.24	-4.51
238	2.60	-13.88	0.11	-1.47	4.69	0.15	-10.41	-5.46
239	1.90	-12.57	0.11	-1.22	5.22	0.13	12.39	8.11

For all Gaussian functions g_{lk} for all images f_k , the exponential factor $\alpha = 1/\epsilon$ is given by $\epsilon = 2.0615528128088303$.

- 5 Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not describe the material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . The hypothesis testing may use as alternative hypothesis that the filter does describe the material classifier. The alternative hypothesis may be given by $H_1: \exists m, m': \mu_m \neq \mu_{m'}$. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each material, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for material for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within: $mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J)$. The hypothesis testing may comprise determining a Mean Sum of Squares between, $mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1)$. The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

- 25 Herein, I is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For exam-

ple, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.05$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

5

Figure 2 shows experimental results of using the spot shape filter to calculate the material feature for different material groups. Specifically, the material feature φ_{2m} is shown for different materials, namely from left to right for a wooden board denoted board, a fabric, a dark skin, a pale skin, and a highly translucent skin denoted trans, and illustrates the separation of skin materials (right) from other materials (left). The spot shape filter passes the hypothesis testing. The p-value computed from the F-test is 0.0076. Thus, the Null-hypothesis H_0 can be rejected with respect to a significance level of 0.01. Further, the alternative hypothesis H_1 can be accepted with respect to a significance level of 0.01. Thus, the image filter is distinguishes at least two material classifiers. In addition, Figure 2 shows the mean value of the feature per material, $\bar{\varphi}_m$, as the line within the respective box. Vertical bars mark the Q1-1.5 interquartile range (IQR) and Q3+1.5 IQR, wherein $IQR = Q3 - Q1$, with Q1 and Q3 being the first and third quartiles. The box marks the Q1 and Q3 quartiles. Outliers are plotted as points. Figure 2 shows, that this material feature separates all skin samples from the non-skin samples. In other words, the expectation values of the Skin Features for skin and non-skin materials differ.

20

Figure 1 moreover shows, in a highly schematic illustration, an exemplary embodiment of the detector 110, wherein the detector 110 specifically may be embodied as a camera 136 and/or may be part of a camera 136. The camera 136 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 1 further shows an embodiment of a detector system 138, which, besides the at least one detector 110, comprises one or more beacon devices 140, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 1 further shows an exemplary embodiment of a human-machine interface 142, which comprises the at least one detector system 138 and, further, an entertainment device 144, which comprises the human-machine interface 142. The figure further shows an embodiment of a tracking system 146 for tracking a position of the object 112, which comprises the detector system 138. The components of the devices and systems shall be explained in further detail below.

35

Fig. 1 further shows an exemplary embodiment of a scanning system 148 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 148 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well

as, optionally, at least one further illumination source, not depicted here. The illumination source 124, generally, is configured for emitting at least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 140 and/or on a surface of the object 112. The scanning system 148 may be designed to
5 generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 148, specifically the detector 110, by using the at least one detector 110.

10 The detector 110, besides the optical sensors 120, comprises the at least one evaluation device 132, having e.g. at least one image analysis device 150 and/or the at least one position evaluation device 152, as symbolically depicted in Fig. 1. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially
15 combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 154 and/or by one or more interfaces, as symbolically depicted in Fig. 1. Further, the one or more connectors 154 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connect-
20 or 154, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 156 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be
25 designed as an article of sports equipment and/or may form a control element or a control device 158, the position of which may be manipulated by a user 160. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 160 himself or herself may be considered as the object 112, the position of which shall be detected.

30 As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 156. Further, the detector 110 may comprise the at least one transfer device 129 such as one or more optical systems, preferably comprising one or more lenses. An opening 162 inside the housing 156, which, preferably, is located concentrically with
35 regard to the optical axis 128 of the detector 110, preferably defines a direction of view 164 of the detector 110. A coordinate system 166 may be defined, in which a direction parallel or anti-parallel to the optical axis 128 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 128 may be defined as transversal directions. In the coordinate system 166, symbolically depicted in Fig. 1, a longitudinal direction is denoted by z, and trans-
40 versal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 142, in order to provide at least one item of information to a machine 168. In the embodiments schematically depicted in Fig. 1, the machine 168 may be a computer and/or may comprise a computer. Other
5 embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 168, such as into the computer.

As outlined above, Fig. 1 also depicts an example of a tracking system 146, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system
10 146 comprises the detector 110 and at least one track controller 170. The track controller 170 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 170 may be an independent device and/or may be fully or partially integrated into the machine 168, specifically the computer, as indicated in Fig. 1 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 142 may form part of an entertainment device 144. The machine 168, specifically the computer, may also form part of the entertainment device 144. Thus, by means of the user 160 functioning as the object 112 and/or by means of the user 160 handling a control device functioning as the object 112, the user 160
20 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Fig. 1 also depicts an example of an inertial measurement unit 172 for use in an electronic device. The inertial measurement unit 172 is adapted to receive data determined by the detector
25 110. The inertial measurement unit 172 further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is
30 adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

The inertial measurement unit 172 may comprise the detector 110 and/or may be connected to the detector 110, via at least one data connection. The evaluation device 132 and/or at least one processing device of the inertial measurement unit 172 may be configured for determining at least one combined distance information, in particular using at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters such as further sensor data from the further
40 sensor of the inertial measurement unit 172.

List of reference numbers

110	detector
112	object
116	sensor element
118	matrix
120	optical sensor
122	light-sensitive area
124	illumination source
126	illumination light beam
128	optical axis
129	transfer device
130	reflection light beam
132	evaluation device
134	divider
136	camera
138	detector system
140	beacon device
142	human-machine-interface
144	entertainment device
146	tracking system
148	scanning system
150	image analysis device
152	position evaluation device
154	connector
156	housing
158	control device
160	user
162	opening
164	direction of view
166	coordinate system
168	machine
170	track controller
172	inertial measurement unit

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Claims

1. A detector (110) for identifying at least one material property m comprising
 - at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein the sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112);
 - at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device (132) is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device (132) is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .
2. The detector (110) according to the preceding claim, , wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifi-

ers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

- 5 3. The detector (110) according to the preceding claim, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.
4. The detector (110) according to any one of the preceding claims, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.
- 10 5. The detector (110) according to any one of the preceding claims, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.
- 15 6. The detector (110) according to any one of the preceding claims, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.
- 20 7. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .
- 25 8. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.
9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.
- 30 10. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.
- 35 11. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.
- 40

12. The detector (110) according to any one of the preceding claims, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element (116), wherein the evaluation device (132) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .
- 10 13. The detector (110) according to any one of the preceding claims, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.
- 15 14. The detector (110) according to any one of the preceding claims, wherein the sensor element (116) comprises at least one CMOS sensor.
- 20 15. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.
- 30 16. The detector (110) according to the preceding claim, wherein the illumination source (124) comprises at least one laser source and at least one diffractive optical element.
- 35 17. A detector system (138), the detector system (138) comprising at least one detector (110) according to any one of the preceding claims, the detector system (138) further comprising at least one beacon device (140) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (140) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
- 40 18. A human-machine interface (142) for exchanging at least one item of information between a user (160) and a machine (168), wherein the human-machine interface (142) comprises at least one detector system (138) according to the preceding claim, wherein the at least one beacon device (140) is adapted to be at least one of directly or indirectly attached to

the user (160) and held by the user (160), wherein the human-machine interface (142) is designed to determine at least one position of the user (160) by means of the detector system (138), wherein the human-machine interface (142) is designed to assign to the position at least one item of information.

- 5
19. An entertainment device (144) for carrying out at least one entertainment function, wherein the entertainment device (144) comprises at least one human-machine interface (142) according to the preceding claim, wherein the entertainment device (144) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (142), wherein the entertainment device (144) is designed to vary the entertainment function in accordance with the information.
- 10
20. A tracking system (146) for tracking a position of at least one movable object, the tracking system (146) comprising at least one detector system (138) according to any one of the preceding claims referring to a detector system, the tracking system (146) further comprising at least one track controller (170), wherein the track controller (170) is adapted to track a series of positions of the object (112) at specific points in time.
- 15
21. A scanning system (148) for determining a depth profile of a scenery, the scanning system (148) comprising at least one detector (110) according to any of the preceding claims referring to a detector, the scanning system (148) further comprising at least one illumination source adapted to scan the scenery with at least one light beam.
- 20
22. A camera (136) for imaging at least one object (112), the camera (136) comprising at least one detector (110) according to any one of the preceding claims referring to a detector.
- 25
23. An inertial measurement unit (172) for use in an electronic device, wherein the inertial measurement unit (172) is adapted to receive data determined by at least one detector (110) according to any one of the preceding claims referring to a detector, wherein the inertial measurement unit (172) further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.
- 30
- 35
- 40 24. A method for determining at least one material property of at least one object (112) by using at least one detector (110) according to any one of the preceding claims referring to a detector, the method comprising the following steps:

a) determining at least one reflection image of the object (112) by using at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122);

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device (132), wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

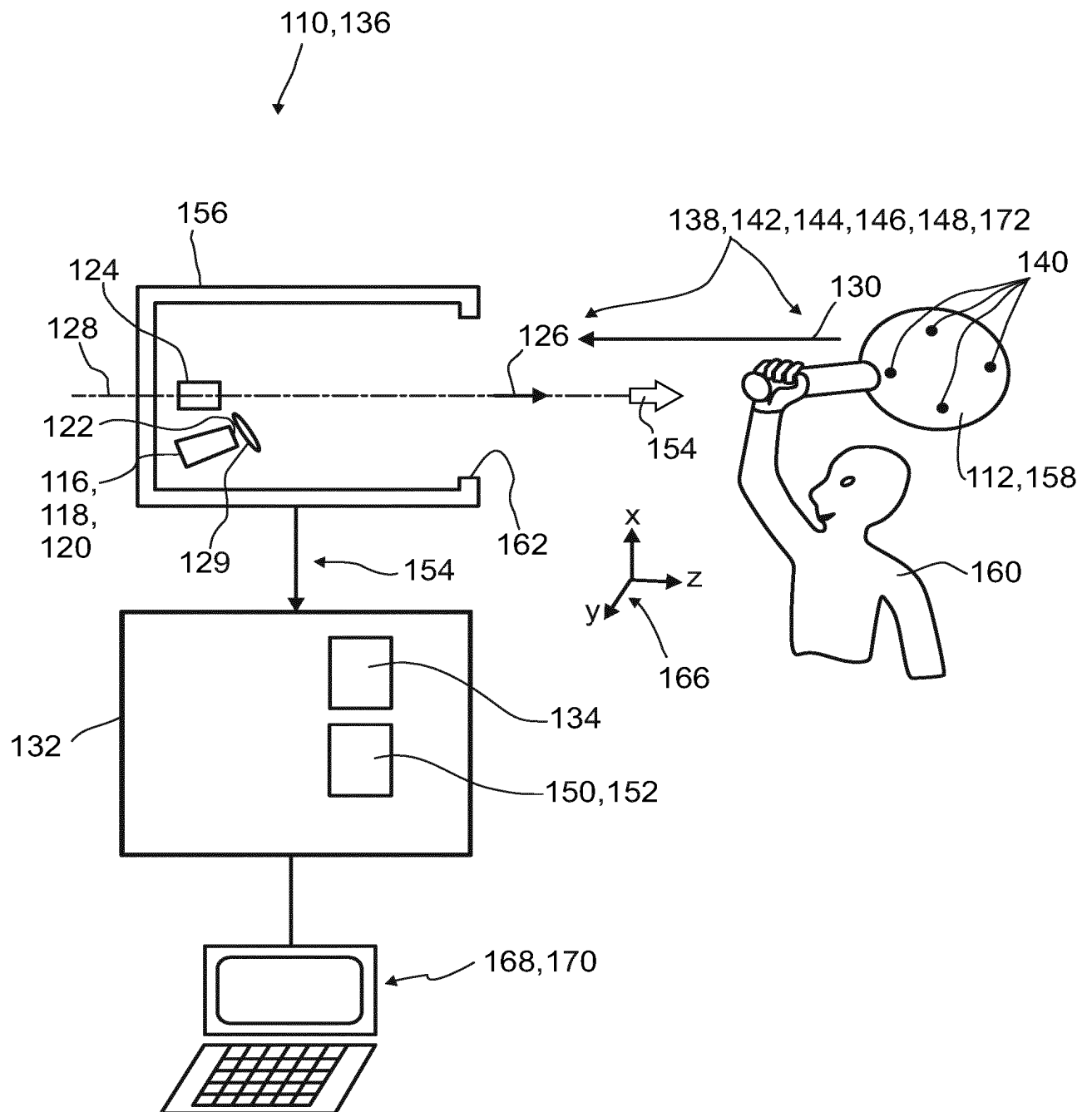
$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

25. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

FIG.1



2/2

FIG.2

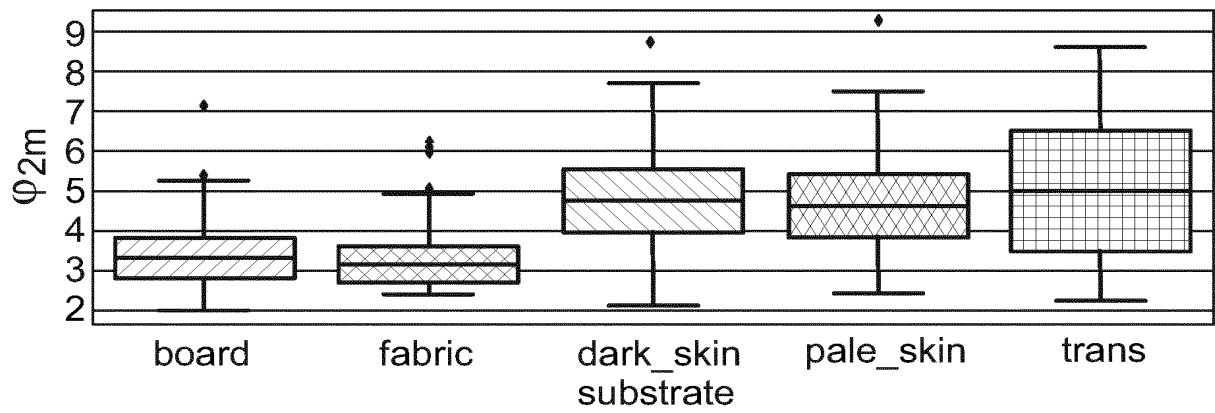
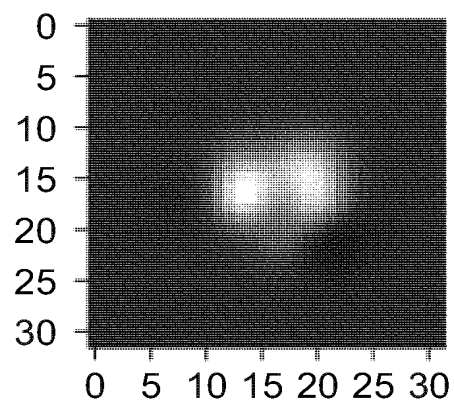


FIG.3



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2020/056759

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S7/48 G01N21/47 G01S17/46 G06K9/00 G06T7/00
G06T7/521

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S G01N G06T G06K G01B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data, COMPENDEX, INSPEC

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	DE 198 46 619 A1 (BYK GARDNER GMBH [DE]) 27 April 2000 (2000-04-27)	1-7, 9-16, 23-25 17-20
Y	column 13, line 45 - column 14, line 66; claims 1,29,7 column 2, lines 31-38 -----	
X	CN 108 363 482 A (JIANGSU SIDIANLING ROBOT CO LTD) 3 August 2018 (2018-08-03)	1-7, 9-14,16, 21,23-25 17-20
Y	claim 1 -----	
X	US 2018/033146 A1 (BLEYER MICHAEL [US] ET AL) 1 February 2018 (2018-02-01)	1-14,16, 22-25 17-20
Y	paragraphs [0013], [0036], [0037]; claims 1,4,6 ----- -/-	



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

13 May 2020

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INTERNATIONAL SEARCH REPORT

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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	WO 2019/042956 A1 (TRINAMIX GMBH [DE]) 7 March 2019 (2019-03-07) claims 19-22,26 -----	17-20

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2020/056759

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		JP 2000131226 A	12-05-2000

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		WO 2018022386 A1	01-02-2018

WO 2019042956 A1	07-03-2019	KR 20200040780 A	20-04-2020
		WO 2019042956 A1	07-03-2019



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17161334.2	16 March 2017 (16.03.2017)	EP
17195427.4	09 October 2017 (09.10.2017)	EP

17195398.7

09 October 2017 (09.10.2017)

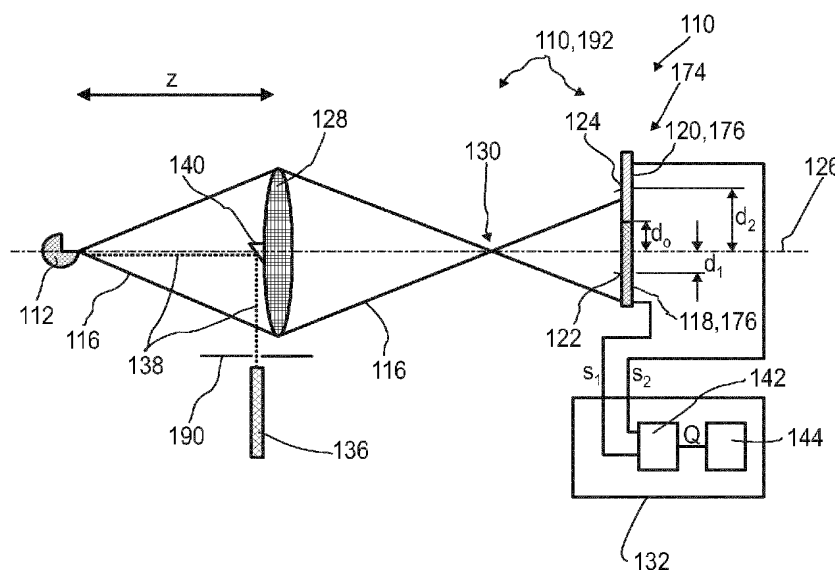
EP

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(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.2



(57) **Abstract:** A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises: -at least two optical sensors (118, 120, 176), each optical sensor (118, 120, 176) having a light-sensitive area (122, 124), wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110) by different spatial offsets, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light beam (116) propagating from the object (112) to the detector (110); and -at least one evaluation device (132) being configured for determining at least one longitudinal coordinate z of the object (112) by combining the at least two sensor signals.

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Published:

- with international search report (Art. 21(3))
- with information concerning incorporation by reference of missing parts and/or elements (Rule 20.6)

Detector for optically detecting at least one object

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system, a readout device for optical storage media and various uses of the detector device. The devices, systems, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

20 Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultraviolet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and for reasons of large-area processing, sensors comprising at least one organic sensor material are being used, as described for example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO 2009/013282 A1.

A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for opti-

cally detecting objects are triangulation systems, by means of which distance measurements can likewise be carried out. Kurt Konolige et al., A Low-Cost Laser Distance Sensor, 2008 IEEE International Conference on Robotics and Automation, Pasadena, CA, USA, May 19-23, 2008, discuss competing technologies using triangulation for planar laser distance sensor (LDS).

5 Structured line devices use a light stripe laser and offset camera to determine range to a set of points. Because the laser energy is spread over a line, it is difficult to achieve accurate range, especially in the presence of ambient light, or with darker objects. Point scan devices for 3D scanning of small objects typically use a scanning mirror to direct a point laser beam and redirect the laser return to an optical receiver. Such devices cannot be miniaturized, and their cost and mechanical fragility will remain high. Centroid point modules typically use position-sensitive devices (PSD). These devices measure the centroid of all light impinging on their surface. Although modulation techniques can be used to offset some of the effects of ambient light, PSDs do not perform well unless the laser spot has a very strong reflection, limiting their use to ranges of a meter or less. Pixel-based point modules search the pixel with maximum signal intensity to determine the position of the light spot on the sensor. Typically CMOS line arrays are used for detection. Konolige et al. introduce a low-cost version of a Pixel-based point module.

US 5 235 377 A describes a measuring apparatus for a camera, wherein light is projected to an object from a projection section. Light reflected on the object is received at a first light receiving section and a second light receiving section arranged to be adjacent to each other. In the first light receiving section, a first and second signals relating to a light receiving position and a quantity of received light are output, and in the second light receiving section, a third signal relating to a quantity of light of the reflected light is output. In a discriminating and calculating section, if it is discriminated that the reflected light is positioned at a place, which is nearer than a predetermined distance, a measuring distance can be calculated from an output ratio of the first signal to the second signal by a measuring calculation circuit. On the other hand, if it is discriminated that the reflected light is positioned at a place, which is farther than the predetermined distance, a measuring distance can be calculated from the output ratio of the sum of the first and second signals to the third signal by the measuring calculation circuit.

US 5 512 997 A describes a distance measuring device having an infrared light-emitting diode by which an infrared light is emitted towards a subject, and first and second linear line sensors on which an image corresponding to light reflected from the subject is formed. The first and second linear line sensors are disposed adjacently and in parallel to each other, and offset by an amount corresponding to half of the predetermined A/D converting step. Analog distance data outputted from the linear line sensors are converted to digital distance data using the predetermined A/D converting step, and then, the digital distance data are added together to obtain a subject distance with a high accuracy.

DE 42 11 875 A1 describes an optical rangefinder with electronic correction for spot eccentricity. The optical rangefinder has an evaluation circuit, which determines if an output distance measurement value is valid or erroneous due to e.g. inclination of object surface. The distance signal is determined by evaluation a position a light spot on a first detector. A second control

signal, which characterizes inclination of the object surface, is determined by evaluating eccentricity of a light spot on a second detector, arranged in a different plane.

WO 2016/051323 A1 describes a detector and a method for optically determining a position of at least one object. The detector comprises at least one optical sensor for determining a position of at least one light beam and at least one evaluation device for generating at least one item of information on a transversal position of the object and at least one item of information on a longitudinal position of the object. The sensor has at least a first electrode and a second electrode. At least one photovoltaic material is embedded in between the first electrode and the second electrode. The first electrode or the second electrode is a split electrode having at least three partial electrodes. The detector and the method can determine three-dimensional coordinates of an object in a fast and efficient way.

In WO 2012/110924 A1, the content of which is herewith included by reference, a detector for optically detecting at least one object is proposed. The detector comprises at least one optical sensor. The optical sensor has at least one sensor region. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area. The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

WO 2014/097181 A1, the full content of which is herewith included by reference, discloses a method and a detector for determining a position of at least one object, by using at least one transversal optical sensor and at least one optical sensor. Specifically, the use of sensor stacks is disclosed, in order to determine a longitudinal position of the object with a high degree of accuracy and without ambiguity.

WO 2015/024871 A1, the full content of which is herewith included by reference, discloses an optical detector, comprising:

- at least one spatial light modulator being adapted to modify at least one property of a light beam in a spatially resolved fashion, having a matrix of pixels, each pixel being controllable to individually modify the at least one optical property of a portion of the light beam passing the pixel;
- at least one optical sensor adapted to detect the light beam after passing the matrix of pixels of the spatial light modulator and to generate at least one sensor signal;
- at least one modulator device adapted for periodically controlling at least two of the pixels with different modulation frequencies; and
- at least one evaluation device adapted for performing a frequency analysis in order to determine signal components of the sensor signal for the modulation frequencies.

US 4,767,211 discloses an apparatus for and a method of measuring a boundary surface of a sample in which a ratio of the light quantity of a part of reflected light from a sample which travels in the vicinity of the optical axis of the reflected light, to the light quantity of another part of the reflected light which is directed to a position deviating from the optical axis by a predetermined distance is used to accurately measure a boundary surface of a sample. Since the accuracy of measurement is increased by using the above ratio, light capable of passing through the sample can be used as incident light. Thus, a deep hole in the surface of the sample and a void such as an air bubble in a living being sample, which cannot be measured by the prior art, can be measured very accurately.

WO 2014/198629 A1, the full content of which is herewith included by reference, discloses a detector for determining a position of at least one object, comprising:

- at least one optical sensor, the optical sensor being adapted to detect a light beam propagating from the object towards the detector, the optical sensor having at least one matrix of pixels; and
- at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

Further, generally, for various other detector concepts, reference may be made to WO 2014/198626 A1, WO 2014/198629 A1 and WO 2014/198625 A1, the full content of which is herewith included by reference. Further, referring to potential materials and optical sensors which may also be employed in the context of the present invention, reference may be made to European patent applications EP 15 153 215.7, filed on January 30, 2015, EP 15 157 363.1, filed on March 3, 2015, EP 15 164 653.6, filed on April 22, 2015, EP 15177275.3, filed on July 17, 2015, EP 15180354.1 and EP 15180353.3, both filed on August 10, 2015, and EP 15 185 005.4, filed on September 14, 2015, EP 15 196 238.8 and EP 15 196 239.6, both filed on November 25, 2015, EP 15 197 744.4, filed on December 3, 2015, the full content of all of which is herewith also included by reference.

Further, reference may be made to detector concepts comparing signals of at least two different sources for determining a position of an object. Thus, as an example, reference may be made to EP 16155834.1, EP 16155835.8 or EP 16155845.7, all filed on February 16, 2016, the full disclosure of which is herewith included by reference.

US 5,235,377 A describes a measuring apparatus for a camera. Light is projected to an object from a projection section, and light reflected on the object is received at a first light receiving section and a second light receiving section arranged to be adjacent to each other. In the first light receiving section, a first and second signals relating to a light receiving position and a quantity of received light are output, and in the second light receiving section, a third signal relating to a quantity of light of the reflected light is output. In a discriminating and calculating sec-

tion, if it is discriminated that the reflected light is positioned at a place, which is nearer than a predetermined distance, a measuring distance can be calculated from an output ratio of the first signal to the second signal by a measuring calculation circuit. On the other hand, if it is discriminated that the reflected light is positioned at a place, which is farther than the predetermined distance, a measuring distance can be calculated from the output ratio of the sum of the first and second signals to the third signal by the measuring calculation circuit.

US 5,512,997 A describes a distance measuring device having an infrared light-emitting diode by which an infrared light is emitted towards a subject, and first and second linear line sensors on which an image corresponding to light reflected from the subject is formed. The first and second linear line sensors are disposed adjacently and in parallel to each other, and offset by an amount corresponding to half of the predetermined A/D converting step. Analog distance data outputted from the linear line sensors are converted to digital distance data using the predetermined A/D converting step, and then, the digital distance data are added together.

DE 42 11 875 A1 describes that a beam from e.g. a collimated laser diode passes through crossed cylinder lenses which focus the rays in both sectional planes and adapt the diameter of the beam to the numerical aperture of an objective lens. A mirror section cuts off half of the incident beam and deflects light scattered from the object to a differential photodiode detector. A beam splitter reflects part of the deflected beam on to a four-quadrant photodetector in an orthogonal plane. A first electronic circuit produces a signal corresponding to the position of the light spot on the detector. A second circuit produces a monitoring or correction signal derived from the eccentricity of the spot position.

US 4,767,211 A describes an apparatus for and a method of measuring a boundary surface of a sample are disclosed in which a ratio of the light quantity of a part of reflected light from a sample which travels in the vicinity of the optical axis of the reflected light, to the light quantity of another part of the reflected light which is directed to a position deviating from the optical axis by a predetermined distance is used to measure a boundary surface of a sample.

Despite the advantages implied by the above-mentioned devices and detectors, several technical challenges remain. Thus, generally, a need exists for detectors for detecting a position of an object in space which is both reliable and may be manufactured at low cost. Specifically, a need exists for 3D-sensing concepts. Various known concepts are at least partially based on using so-called FiP sensors, such as several of the above-mentioned concepts. Therein, as an example, large area sensors may be used, in which the individual sensor pixels are significantly larger than the light spot and which are fixed to a specific size. Still, large area sensors in many cases are inherently limited in the use of the FiP measurement principle, specifically in case more than one light spot is to be investigated simultaneously.

A further challenge using FiP detectors is detector area or active area. Typically, for distance measurements, a large active area of the detector is used or even is required. This area, however, may cause noise problems, specifically when the tetralateral conductivity concept is em-

ployed to build a PSD. This often results in poor signal-to-noise-ratios and slow detector response times due to the large capacitance in conjunction with the series resistance of the detector. A further challenge using FiP detectors is cost of manufacturing. Thus, in many cases, typical FiP sensors are expensive, as compared to e.g. conventional Si-based photodiodes. Further, the evaluation of the measurement results of measurements using FiP-sensors remains an issue, specifically in case the total power of a light beam is unknown. In case a plurality of FiP-sensors is used, located at different positions along an axis of propagation of the light beam, a range of measurement typically is limited to the range in between the two positions of the sensors. Further, many FiP-detectors show a luminance dependency which renders the evaluation of the measurement result more difficult, and, additionally, FiP-measurements in many cases are dependent on a target spot size.

Referring to concepts comparing sensor signals generated by at least two different sources such as at least two different optical sensors, one technical challenge remains a reduction of cost of the overall system and, specifically, of the optical sensors. Thus, in many concepts, special optical sensors have to be used, having a dedicated design and placement, which typically requires setting up an expensive optoelectronic semiconductor manufacturing process and a complex assembly scheme.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide cost-efficient devices and methods which reliably may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and

- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term “at least two optical sensors, each having at least one light sensitive area” refers to configurations with two single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having at least two light sensitive areas. Thus, the term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one bi-cell and/or quadrant diode device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor, may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. Other embodiments, however, are feasible. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photosensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the setup.

As further used herein, a “sensor signal” generally refers to a single signal or a plurality of sensor signals such as a series of sensor signals or a continuous sensor signal, generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the

light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The light beam may propagate from the object towards the detector. As will be outlined in further detail below, the light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source, as an example, may be or may comprise one or more of an external illumination source, an illumination source integrated into the detector or an illumination source integrated into a beacon device being one or more of attached to the object, integrated into the object or held by the object.

As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam specifically may be a Gaussian light beam, as will be outlined in further detail below. Other embodiments are feasible, however.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The light-sensitive areas of the optical sensors may overlap, as visible from the object, or may not overlap, i.e. may be placed next to each other without overlap. The light-sensitive areas may be spaced apart from each other or may directly be adjacent.

The detector has an optical axis. As used herein, the term “optical axis” generally refers to a line of symmetry of the optical setup of the detector. In case the detector has a lens or lens system, such as a lens or lens system being part of a transfer system which will be explained in further detail below, the optical axis, as an example, may be an axis of mirror symmetry or rotational symmetry of the lens or lens system. Thus, as will be outlined in further detail below, the detector may comprise at least one transfer system, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with

the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis, and the condition noted above generally may refer to each beam path independently. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths. In case the optical sensors are distributed over different partial beam paths, the above-mentioned condition may be described such that at least one first optical sensor is located in at least one first partial beam path, being offset from the optical axis of the first partial beam path by a first spatial offset, and at least one second optical sensor is located in at least one second partial beam path, being offset from the optical axis of the second partial beam path by at least one second spatial offset, wherein the first spatial offset and the second spatial offset are different.

The detector may comprise more than two optical sensors. In any case, i.e. in the case of the detector comprising precisely two optical sensors and in the case of the detector comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor being spaced apart from the optical axis by a first spatial offset and at least one second optical sensor being spaced apart from the optical axis by a second spatial offset, wherein the first spatial offset and the second spatial offset differ. In case further optical sensors are provided, besides the first and second optical sensors, these additional optical sensors may also fulfill the condition or, alternatively, may be spaced apart from the optical axis by the first spatial offset, by the second spatial offset or by a different spatial offset. The first and second spatial offsets, as an example, may differ by at least a factor of 1.2, more preferably by at least a factor of 1.5, more preferably by at least a factor of two. The spatial offsets may also be zero or may assume negative values, as long as the above-mentioned conditions are fulfilled.

As outlined above, each light-sensitive area has a geometrical center. As used herein, the term "geometrical center" of an area generally may refer to a center of gravity of the area. As an example, if an arbitrary point inside or outside the area is chosen, and if an integral is formed over the vectors interconnecting this arbitrary point with each and every point of the area, the integral is a function of the position of the arbitrary point. When the arbitrary point is located in the geometrical center of the area, the integral of the absolute value of the integral is minimized. Thus, in other words, the geometrical center may be a point inside or outside the area with a minimum overall or sum distance from all points of the area.

Each geometrical center of each light-sensitive area is spaced apart from the optical axis of the detector, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located. The distance between the geometrical center and the optical axis is denoted by the term "spatial offset".

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As outlined above, the optical sensors specifically may be located in one and the same plane, which, preferably, is a plane perpendicular to the optical axis. Other configurations, however, are possible. Thus, two or more of the optical sensors may also be spaced apart in a direction parallel to the optical axis.

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In a preferred embodiment, which is rather simple and cost-efficiently realizable, the optical sensors are partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector. As used herein, the term "partial diode" may comprise several diodes that are connected in series or in parallel. This example is rather simple and cost-efficiently realizable. Thus, as an example, bi-cell diodes or quadrant diodes are widely commercially available at low cost, and driving schemes for these quadrant diodes are generally known. As used herein, the term "bi-cell diode" generally refers to a diode having two partial diodes in one packaging. Bi-cell and quadrant diodes may have two or four separate light sensitive areas, in particular two or four active areas. As an example, the bi-cell diodes may each form independent diodes having the full functionality of a diode. As an example, each of the bi-cell diodes may have a square or rectangular shape, and the two diodes may be placed in one plane such that the two partial diodes, in total, form a 1 x 2 or 2 x 1 matrix having a rectangular shape. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes or quadrant diode is proposed, as will be outlined in further detail below. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes and the quadrant diode is proposed, as will be outlined in further detail below. Generally, however, the optical sensors specifically may be partial diodes of a quadrant diode, with a center of the quadrant diode being off-centered from the optical axis of the detector. As used herein, the term "quadrant diode" generally refers to a diode having four partial diodes in one packaging. As an example, the four partial diodes may each form independent diodes having the full functionality of a diode. As an example, the four partial diodes may each have a square or rectangular shape, and the four partial diodes may be placed in one plane such that the four partial diodes, in total, form a 2 x 2 matrix having a rectangular or square shape. In a further example, the four partial diodes, in total, may form a 2 x 2 matrix having a circular or elliptical shape. The partial diodes, as an example, may be adjacent, with a minimum separation from one another. However, the separation between adjacent diodes may also be varied to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent diodes may be increased by using an opaque mask. This may reduce the light on one of the diodes relative and/or absolute as compared to the adjacent diode which may increase a quotient of the signal of the two diodes. The rows and columns of the matrix specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the partial diodes are arranged in concentric circles or ellipses about a center

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point, hexagonal arrangements, triangular arrangements, polygonal arrangements or combinations thereof are also feasible, wherein the partial diodes are arranged as regular or irregular polygons such as in hexagons or the like.

- 5 Further, arrangements may be feasible, wherein at least one of the partial diodes of a polygonal pattern may be arranged as star shaped element in a triangular pattern and/or striped element in a rectangular pattern, or the like. Further, arrangements may be feasible, wherein the partial diodes of a polygonal pattern are summarized to a larger, unseparated partial diode, such as star shaped partial diodes in a triangular pattern, striped partial diodes in a rectangular pattern, 10 or the like. Further, separated partial diodes may be summarized or connected to a single partial diode such as the corner partial diodes forming a first partial diode and the non-corner partial diodes forming a second partial diode. Further, for example, the matrix may be a single row of partial diodes. Further, arrangements, where the optical sensors and/or at least one of partial diodes may have a star shape and/or circular shape, and/or cross shape may be feasible. For 15 example, at least two of the partial diodes may be arranged in star shaped manner or cross shaped manner, whereas the other partial diodes may be arranged as such that they complete a square or rectangular shape of the optical sensor, so that a square shaped or rectangular shaped substrate can be employed. Other arrangements are feasible.
- 20 In case a quadrant diode is used, having a 2×2 matrix of partial diodes, the center of the quadrant diode specifically may be off-centered or offset from the optical axis. Thus, as an example, the center of the quadrant diodes, which may be an intersection of the geometrical centers of the optical sensors of the quadrant diode, may be off-centered from the optical axis by at least 0.2 mm, more preferably by at least 0.5 mm, more preferably by at least 1.0 mm or even 2.0 25 mm. Similarly, when using other types of optical sensors setups having a plurality of optical sensors, an overall center of the optical sensors may be offset from the optical axis by the same distance.

- As outlined above, bi-cell and quadrant diodes may have two or four separate light sensitive 30 areas, in particular two or four active areas. The light sensitive areas may be separated by a dividing line. The dividing line, for example, of two Si-diodes, can be arranged parallel to the baseline of the detector, in particular parallel to a spot movement on the optical sensors. However, other arrangements are feasible. For example, the quadrant diode may comprise two dividing lines. The dividing lines may be arranged orthogonal to each other. The orthogonal ar- 35 rangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other. In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device may be adapted to determine a second quotient using at least three or all four sensor signals of the quadrant diode. The two quotients can be formed such that two distinct distance ranges are 40 covered. The two quotient signals for the near field and far field may have an overlap region in which both quotient signals allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient signal may be determined by dividing the sensor signals of two top quadrants, also called top segment, by the sensor signals of two bottom quadrants, also

called bottom segment. Using the quotient signal of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between the top and bottom segment is parallel to the baseline, the quotient signal determined from top segment divided by bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient signal by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals of the quadrants such that distance measurement is possible over a wide range with a large resolution.

The detector may comprise at least one quadrant diode comprising four quadrants. For example, an upper left quadrant of the quadrant diode may be denoted "ol", a lower left quadrant "ul", an upper right quadrant may be denoted "or" and a lower right quadrant may be denoted "ur".

The light spot may move from the left in a small distance range to the left in a far distance range. The detector may comprise at least one neutral density filter. The quadrants of the small distance range may be covered by the neutral density filter. In the small distance range the quotient ol/ul may be used to determine the longitudinal coordinate z. In the far distance range a quotient or/ur and/or a quotient from a combination of all quadrants may be used. The neutral density filter allows that objects in the small distance range are adapted to illuminate the optical sensors brighter by constant amplification without saturating the amplifier. At the same time it is possible to obtain in the far distance range more of the signal. Thus, the neutral density filter allows to enhance dynamic of the detector.

Generally, the light-sensitive areas of the optical sensors may have an arbitrary surface area or size. Preferably, however, specifically in view of a simplified evaluation of the sensor signals, the light-sensitive areas of the optical sensors are substantially equal, such as within a tolerance of less than 10%, preferably less than 5% or even less than 1%. This, specifically, is the case in typical commercially available quadrant diodes.

As further used herein, the term evaluation device generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between sensor signals and/or any secondary signal derived thereof and the longitudinal coordinate. Thus, the evaluation device may be configured for determining at least one combined sensor signal out of the at least two sensor signals, i.e. of the at least one sensor signal of at least one first optical sensor and out of the at least one sensor signal of at least one second optical sensor. As generally used herein, the term “combine” generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least one comparison signal or comparison result. As will be outlined in further detail below, the combined sensor signal or secondary signal may be or may comprise at least one quotient signal.

The evaluation device specifically may be configured for deriving at least one quotient signal Q by one or more of: dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. The evaluation device may further be configured for determining the longitudinal coordinate by evaluating the quotient signal Q . In case more than two sensor signals are provided, as an example, more than one quotient signal may be generated, such as by forming quotient signals of more than one pair of sensor signals.

As an example, Q may simply be determined as

$$Q = s_1/s_2$$

or

$$Q = s_2/s_1,$$

with s_1 denoting a first one of the sensor signals and s_2 denoting a second one of the sensor signals. Additionally or alternatively, Q may be determined as

$$Q = a \cdot s_1 / b \cdot s_2$$

or

$$Q = b \cdot s_2 / a \cdot s_1,$$

with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as

$$Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2),$$

with a, b, c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as

$$Q = s_1 / (s_1 + s_2),$$

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or, as a further example, Q may be determined as

$$Q = (s_1 - s_2) / (s_1 + s_2).$$

10 Other quotient signals are feasible. Thus, as an example, in case more than two optical sensors are provided, the above-mentioned quotient formation may take place between two of the sensor signals generated by these optical sensors or may take place between more than two of the sensor signals. Thus, instead of using the first one of the sensor signals and the second one of the sensor signals in the formulae given above, combined sensor signals may be used for quotient formation.

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The quotient signal Q generally is an example for an asymmetry parameter denoting an asymmetry or an asymmetric distribution of the light spot generated by the light beam on the light-sensitive areas. The quotient of the two or more optical sensors, such as the two or more photodiodes, may provide a combined signal which typically is monotonously dependent on the distance between the detector and the object from which the light beam travels towards the detector, as will be shown by experimental data below. In addition or as an alternative to the quotient signal, other types of combined functions implementing the sensor signals of two or more sensors in the setup of the present invention may be used, which also may show a dependency on the distance between the object and the detector. The asymmetry or the asymmetry parameter of the light spot, as an example, may be an indication of the width of a light beam. If this asymmetry parameter depends on the distance only, the measurement can be used to determine the distance.

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30 The evaluation device may be configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate. For example, the evaluation device is configured for deriving the quotient signal Q by

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$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. As used herein, the term “beam profile” relates to a spa-

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tial distribution, in particular in at least one plane perpendicular to the propagation of the light beam, of an intensity of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_0)$ and beam shape $S(x,y;z_0)$,
 $E(x,y;z_0) = L \cdot S$.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the sensor position used for determining the quotient signal Q . The light-sensitive regions may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may com-

prise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

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The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

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Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

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The evaluation device may be configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

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In one embodiment, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of the optical sensors are arranged as a sensor element having at least two columns, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern. For example, the detector may comprise at least two rows of optical sensors, wherein the rows are arranged on top of each other. The line pattern may be imaged parallel to the rows such that the center of the line pattern is fixed or moves only little relative to a dividing line of the rows. The A1 may correspond to an area above the dividing line and A2 may be an area below the dividing line.

The light beam may be a monochromatic light beam. The light beam may comprise a narrow band of wavelengths, preferably the light beam may comprise a single wavelength. The at least one light source may be adapted to generate at least one monochromatic light beam and/or the detector may comprise at least one filter element adapted to filter a narrow band of wavelength, such as a monochromator.

The illumination source may illuminate the object with a convergent and/or divergent and/or collimated light beam.

The illumination source and the sensor element may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the optical sensors may be arranged in a twisted arrangement.

In typical setups, commercially available quadrant diodes such as quadrant photodiodes are used for positioning, i.e. for adjusting and/or measuring a transversal coordinate of a light spot in the plane of the quadrant photodiode. Thus, as an example, laser beam positioning by using quadrant photodiodes is well known. According to a typical prejudice, however, quadrant photodiodes are used for xy-positioning, only. According to this assumption, quadrant photodiodes are not suitable for measuring distances. The above-mentioned findings, however, using an off-

centered quadrant photodiode with regard to an optical axis of the detector, show otherwise, as will be shown in further measurements below. Thus, as indicated above, in quadrant photodiodes, the asymmetry of the spot can be measured by shifting the quadrant diode slightly off-axis, such as by the above-mentioned offset. Thereby, a monotonously z-dependent function may be generated, such as by forming the quotient signal Q of two or more of the sensor signals of two or more partial photodiodes, i.e. quadrants, of the quadrant photodiode. Therein, in principle, only two photodiodes are necessary for the measurement. The other two diodes may be used for noise cancellation or to obtain a more precise measurement.

In addition or as an alternative to using a quadrant diode or quadrant photodiode, other types of optical sensors may be used. Thus, as will be shown in further detail below, staggered optical sensors may be used.

The use of quadrant diodes provides a large number of advantages over known optical detectors. Thus, quadrant diodes are used in a large number of applications in combination with LEDs or active targets and are widely commercially available at very low price, with various optical properties such as spectral sensitivities and in various sizes. No specific manufacturing process has to be established, since commercially available products may be implemented into the detector according to the present invention.

The detector according to the present invention specifically may be used in multilayer optical storage discs, such as disclosed by international patent application number PCT/IB2015/052233, filed on March 26, 2015. Measurements performed by using the detector according to the present invention specifically may be used in order to optimize the focus position in optical storage discs.

As will be outlined in further detail below, the distance measurement by using the detector according to the present invention may be enhanced by implementing one or more additional distance measurement means into the detector and/or by combining the detector with other types of distance measurement means. Thus, as an example, the detector may comprise or may be combined with at least one triangulation distance measurement device. Thus, the distance measurement can be enhanced by making use of a combination of the measurement principle discussed above and a triangulation type distance measurement. Further, means for measuring one or more other coordinates, such as x- and/or y-coordinates, may be provided.

In case a quadrant diode is used, the quadrant diode may also be used for additional purposes. Thus, the quadrant diode may also be used for conventional x-y-measurements of a light spot, as generally known in the art of optoelectronics and laser physics. Thus, as an example, the lens or detector position can be adjusted using the conventional xy-position information of the quadrant diode to optimize the position of the spot for the distance measurement. As a practical example, the light spot, initially, may be located right in the center of the quadrant diode, which typically does not allow for the above-mentioned distance measurement using the quotient function Q. Thus, firstly, conventional quadrant photodiode techniques may be used for off-centering

a position of the light spot on the quadrant photodiode, such that, e.g., the spot position on the quadrant diode is optimal for the measurement. Thus, as an example, the different off-centering of the optical sensors of the detector may simply be a starting point for movement of the optical sensors relative to the optical axis such that the light spot is off-centered with respect to the optical axis and with respect to a geometrical center of the array of the optical sensors.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned quadrant diode. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like. The case $m=2$, $n=2$ is the case of the quadrant diode or quadrant optical sensor, which, for practical reasons, is one of the preferred cases, since quadrant photodiodes are widely available.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned quadrant diode. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like. The case $m=2$, $n=2$ is the case of the quadrant diode or quadrant optical sensor, which, for practical reasons, is one of the preferred cases, since quadrant photodiodes are widely available.

As a starting point, a geometrical center of the optical sensors within the array may be off-centered from the optical axis, such as by the above-mentioned offset. The sensor array specifically may be movable relative to the optical axis, for example along a gradient, preferably automatically, such as by moving the sensor array, e.g. in a plane perpendicular to the optical axis, and/or by moving the optical axis itself, e.g. shifting the optical axis in a parallel shift and/or tilting the optical axis. Thus, the sensor array may be shifted in order to adjust a position of a light spot generated by the light beam in the plane of the sensor array. Additionally or alternatively, the optical axis may be shifted and/or tilted by using appropriate elements, such as by using one or more deflection elements and/or one or more lenses. The movement, as an example, may take place by using one or more appropriate actuators, such as one or more piezo actuators and/or one or more electromagnetic actuators and/or one or more pneumatic or me-

chanical actuators, which, e.g., move and/or shift the array and/or move and/or shift and/or tilt one or more optical elements in the beam path in order to move the optical axis, such as parallel shifting the optical axis and/or tilting the optical axis. The evaluation device specifically may be adjusted to control a relative position of the sensor array to the optical axis, e.g. in the plane perpendicular to the optical axis. An adjustment procedure may take place in that the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly, moving the array relative to the optical axis, such as by moving the array and/or the optical axis, e.g. by moving the array in the plane to the optical axis until the light spot is off-centered and/or by tilting a lens until the light spot is off-centered. As used therein, a transversal position may be a position in a plane perpendicular to the optical axis, which may also be referred to as the x-y-plane. For the measurement of the transversal coordinate, as an example, the sensor signals of the optical sensors may be compared. As an example, in case the sensor signals are found to be equal and, thus, in case it is determined that the light spot is located symmetrically with respect to the optical sensors, such as in the center of the quadrant diodes, a shifting of the array and/or a tilting of a lens may take place, in order to off-center the light spot in the array. Thus, as outlined above, the off-centering of the array from the optical axis, such as by off-centering the center of the quadrant photodiode from the optical axis, may simply be a starting point in order to avoid the situation which is typical, in which the light spot is located on the optical axis and, thus, is centered. By off-centering the array relative to the optical axis, thus, the light spot should be off-centered. In case this is found not to be true, such that the light spot, incidentally, is located in the center of the array and equally illuminates all optical sensors, the above-mentioned shifting of the array relative to the optical axis may take place, preferably automatically, in order to off-center the light spot on the array. Thereby, a reliable distance measurement may take place.

Further, in a scanning system with a movable light source, the position of the light spot on the quadrant diode may not be fixed. This is still possible, but may necessitate that different calibrations are used, dependent on the xy-position of the spot in the diode.

Further, the use of the above-mentioned quotient signal Q is a very reliable method for distance measurements. Typically, Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, the sensor signals, such as the above-mentioned first sensor signal s_1 and the above-mentioned second sensor signal s_2 , decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, due to the off-centering, the one of the sensor signals decreases more rapidly than the other, since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. By spreading the light spot, however, the portion of the light illuminating the one or more optical sensors outside the center of the light

spot increases, as compared to a situation of a very small light spot. Thus, the quotient of the sensor signals continuously changes, i.e. increases or decreases, with increasing diameter of the light beam or diameter of the light spot. The quotient, further, may further be rendered mainly independent from the total power of the light beam, since the total power of the light beam forms a factor in all sensor signals. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the sensor signals and the size or diameter of the light beam.

Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the quotient signal Q. The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The quotient signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

In view of the technical challenges involved in the prior art documents discussed above, specifically in view of the technical effort which is required for generating the FiP effect, it has to be noted that the present invention specifically may be realized by using non-FiP optical sensors. In fact, since optical sensors having the FiP characteristic typically exhibit a strong peak in the respective sensor signals at a focal point, the range of measurement of a detector according to the present invention using FiP sensors as optical sensors may be limited to a range in between the two positions and which the optical sensors are in focus of the light beam. When using linear optical sensors, however, i.e. optical sensors not exhibiting the FiP effect, this problem, with the setup of the present invention, generally may be avoided. Consequently, the optical sensors

may each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals may be dependent on the total power of illumination of the respective optical sensor and may be independent from a diameter of a light spot of the illumination. It shall be noted, however, that other embodiments are feasible, too.

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The optical sensors each specifically may be semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes. Thus, as opposed to complex and expensive FiP sensors, the present invention simply may be realized by using commercially available inorganic photodiodes, i.e. one small photodiode and
10 one large area photodiode. Thus, the setup of the present invention may be realized in a cheap and inexpensive fashion.

In one embodiment, the detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the
15 bi-cell or quadrant diode may be adapted to generate a so called FiP signal. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. According to the FiP-effect, the longitudinal sensor signal, given the same total power, may exhibit at least one pronounced maximum for one or a plurality of focusings
20 and/or for one or a plurality of specific sizes of the light spot on the sensor region or within the sensor region. For purposes of comparison, an observation of a maximum of the longitudinal sensor signal in a condition in which the corresponding material is impinged by a light beam with the smallest possible cross-section, such as when the material may be located at or near a focal point as affected by an optical lens, may be denominated as a "positive FiP-effect". The
25 negative FiP effect describes an observation of a minimum of the longitudinal sensor signal under a condition in which the corresponding material is impinged by a light beam with the smallest available beam cross-section, in particular, when the material may be located at or near a focal point as effected by an optical lens. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc.
30 may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

At least one of the optical sensors may be adapted to generate at least one sensor signal dependent on a time-of-flight (TOF) the illumination light beam has traveled from the illumination
35 source to the object and the reflection light beam has traveled from the object to the light sensitive area of the optical sensor. The evaluation device may be configured for determining at least one TOF-longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. The optical sensor adapted to generate the TOF sensor signal may be designed as time-of-flight
40 detector. The time-of-flight detector may be selected from the group consisting of: at least one pulsed time-of-flight detector; at least one phase modulated time-of-flight detector; at least one direct time-of-flight detector; at least one indirect time-of-flight detector. For example, the pulsed time-of-flight detector may be at least one range gated imager and/or at least one direct time-of-

flight imager. For example the phase modulated time-of-flight detector may be at least one RF-modulated light source with at least one phase detector. The optical sensor may be adapted to determine a time delay between emission of the illumination light beam by the illumination source and receipt of the reflection light beam.

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For example, the optical sensor adapted to generate the TOF sensor signal may be designed as pulsed time-of-flight detector. The detector may comprise the at least one interruption device, such as at least one shutter element, adapted to generate a pulsed light beam. The optical sensor may be adapted to store the TOF sensor signal dependent on the receiving time of the reflection light beam in a plurality of time windows, in particular subsequent time windows. The optical sensor may be adapted to store dependent on receiving time of the reflection light beam the generated TOF sensor signal in at least one first time window and/or in at least one second time window. The first and second time windows may be correlated with the opening and closure of the interruption device. Duration of first and second time windows may be pre-defined.

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For example, the TOF sensor signal may be stored in the first time window during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may be stored in the second time window. Other durations of time windows are thinkable. The first and the second time window may comprise information about background, signal height and signal shift.

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For example, the optical sensor adapted to generate the TOF sensor signal may be designed as direct time-of-flight imager. The direct time-of-flight imager may comprise the at least one illumination source adapted to generate at least one single laser pulse. The single laser pulse may be reflected back from the object onto the optical sensor. The optical sensor may comprise at least one photodiode, for example at least one Avalanche Photo Diode (APD), such as at least one Si APD, or such as at least one InGaAs APD, or at least one PIN photo detector array, or at least one Single-Photon Avalanche Photodiode (SPAD), adapted to image the reflection light beam. The direct time-of flight imager may be adapted to image at least one image comprising spatial and temporal data.

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For example, the optical sensor adapted to generate the TOF sensor signal may be designed as phase modulated time-of-flight modulator. The phase modulated time-of-flight modulator may be adapted to measure a difference in phase, in particular a phase shift, by determining a correlated signal, for example by multiplying a received signal, i.e. of the reflection light beam, with the emitted signal, i.e. the illumination light beam. A DC component of the correlated signal may comprise an information about the difference in phase. The evaluation device may be adapted to determine the second longitudinal coordinate of the object from the phase difference. For example, the illumination source and the optical sensor adapted to generate the TOF sensor signal may be designed as RF-modulated light source with at least one phase detector. The illumination source may comprise, for example, at least one LED and/or at least one laser. The illumination source may comprise at least one modulation device adapted to modulate the light beam having a pre-defined phase shift. For example, the modulation device may comprise at least one radio frequency module. The radio frequency module may be adapted to modulate the

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illumination beam with an RF carrier. The optical sensor may be adapted to determine a phase shift of the reflective light beam impinging on the optical sensor.

The optical sensor may be designed as and/or may comprise at least one time-of-flight pixel.

5 Preferably, the detector may comprise at least two optical sensors, wherein each optical sensor is designed as and/or comprises at least one TOF pixel. For example, the detector, in particular the optical sensor may comprise a quadrant diode adapted to generate the TOF sensor signal. For example, the detector, in particular the optical sensor, may comprise at least one pixelated TOF-imager.

10 The evaluation device may be configured for determining the at least one TOF longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. As used herein, the term "TOF longitudinal coordinate" refers to a longitudinal coordinate derived from the TOF sensor signal. As described above, the illumination light source may be adapted to periodically generate at least one light pulse. The detector may be adapted to generate the first longitudinal sensor signal for each period. The evaluation device may be adapted to determine in which pulse period the TOF sensor signal was generated using the second longitudinal coordinate. The detector may be adapted to uniquely assign to which period the ToF-signal refers to by using the combined sensor signal. Both, the TOF sensor signal and the combined sensor signal, may be non-monotonous functions of the longitudinal coordinate z_{real} . Thus, the longitudinal coordinate may not be uniquely determined from one of the TOF sensor signal or the combined sensor signal alone and a measurement range may not be restricted to a longitudinal range in which the signals are unique functions of z_{real} . The term "measurement range" generally refers to a range from the object to the detector in which determination of the longitudinal coordinate is possible.

25 The term "longitudinal range" generally refers to a range from the object to the detector in which unique determination of the longitudinal coordinate is possible. Below and/or above certain distances from the object to the detector a determination of the longitudinal coordinate may not be possible. For example, time-of-flight measurements are not possible below a certain distance from the object to the detector, possibly due to the minimum measurement time of the internal clock. Furthermore, in time-of-flight measurements the sensor signal may be unique within a longitudinal period, but the sensor signal may be the same in case integer multiples of the longitudinal period are added such that the determined longitudinal coordinate may be non-unique. Thus, the same TOF sensor signal will be obtained for a distance z_1 and a distance $z_1 + n \cdot z_{1p}$, where n is an integer denoting the longitudinal period and z_{1p} is the longitudinal period of the TOF sensor signal, wherein the distances z_1 and $z_1 + n \cdot z_{1p}$ are within the measurement range. As used herein, the term "longitudinal period" refers to partitions of a period, in particular a distance range, in which the longitudinal coordinate can be unambiguously determined from the TOF sensor signal. Non-unique longitudinal coordinates may be denoted as relative longitudinal coordinates and the unique longitudinal coordinates may be denoted as absolute longitudinal coordinates.

If both the TOF sensor signal $F1$ and the combined sensor signals $F2$ are available, it may be possible to uniquely determine the longitudinal coordinate and to extend the longitudinal range,

as long as each signal pair (F1,F2) corresponds to a unique distance and vice versa. In particular, if a unique signal pair (F1,F2) exists for each longitudinal coordinate and vice versa, the evaluation device may be adapted to determine the unique combined longitudinal coordinate by

(1) selecting at least one first selected signal such as the TOF sensor signal and/or the combined sensor signal and determining non-unique first longitudinal coordinates;

(2) selecting a second selected signal such as the combined signal Q and/or the TOF sensor signal, which was not selected in step (1), and determining non-unique second longitudinal coordinates;

(3) determining whether one of the non-unique first longitudinal coordinates and the non-unique second longitudinal coordinates match up to a predetermined tolerance threshold;

(4) setting a combined unique longitudinal coordinate to be a matching longitudinal coordinate.

In step (1) and step (2), signals may be selected in the given order or may be performed in a different order. For example, in step (1) the TOF sensor signal may be selected and in step (2) the combined signal Q may be selected. In another example, in step (1) the combined sensor signal may be selected, and the non-unique first longitudinal sensor signal may be determined therefrom. In step (2), the TOF sensor signal may be selected.

Additionally or alternatively to step (4), the evaluation device may be adapted to output an error signal in case no matching coordinates are found and/or to output an error signal in case more than one matching coordinates are found. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be stored in a look-up table. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be approximated or described by an analytical function which is evaluated to find the longitudinal coordinate corresponding to a given signal pair.

The evaluation device may comprise at least two memory elements. As used herein, the term "memory element" refers to a device adapted to store information. The evaluation device may be adapted to receive and to store information provided by the optical sensors, for example the at least one first sensor signal. Such information may comprise raw sensor data and/or processed sensor data. For example, the memory element may be adapted to store information for further evaluation by the evaluation device. The memory element may be a volatile or non-volatile memory element.

As described above, the optical sensor may be designed as and/or may comprise at least one ToF pixel. The detector may comprise at least two switches. Each of the switches may be connected to the optical sensor adapted to generate the first sensor signal, for example by at least one connector. In particular, each of the switches may be connected to the ToF pixel. The switches are adapted to provide the TOF sensor signal to one of the memory elements. In particular, the switches may be adapted to let, dependent on receiving time of the reflection light beam, the generated TOF sensor signal pass through one of the switches. For example, the TOF sensor signal may pass one of the switches during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may pass the other

switch. Each of the switches may be controlled by a control signal having a pulse length identical to a pulse length of a light pulse generated by the illumination source. The control signal of one of the switches may be delayed. For example, the delay may correspond to the pulse length of the light pulse. The evaluation device may be adapted to sample and/or store depending on the delay a first part, or fraction, of the TOF sensor signal through a first switch in a first memory element and the other, second part, or fraction, of the TOF sensor signal through a second switch in a second memory element. The evaluation device may be adapted to determine the first longitudinal coordinate by evaluating the first part and second part of the TOF sensor signal. The evaluation device may be adapted to determine the first longitudinal coordinate z_1 by

$$z_1 = \frac{1}{2} \cdot c \cdot t_0 \cdot \frac{S_{12}}{S_{11} + S_{12}} + z_0;$$

wherein c is the speed of light, t_0 is the pulse length of the illumination light beam, z_0 is a distance offset often determined by the resolution of the time measurement, and S_{11} and S_{12} are the first part and second part of the TOF sensor signal, respectively.

As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

As outlined above, specifically, quadrant photodiodes may be used. As an example, commercially available quadrant photodiodes may be integrated in order to provide four optical sensors, such as one or more quadrant photodiodes available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany, such as quadrant Si PIN photodiodes of the type S4349, which are sensitive in the UV spectral range to the near IR spectral range. In case an array of optical sensors is used, the array may be a naked chip or may be an encapsulated array, such as encapsulated in a TO-5 metal package. Additionally or alternatively, a surface mounted device may be used, such as TT Electronics OPR5911 available from TT Electronics plc, Fourth Floor, St Andrews House, West Street Woking Surrey, GU21 6EB, England. It shall be noted that other optical sensors may also be used.

Further, it shall be noted that, besides the option of using precisely one quadrant photodiode, two or more quadrant photodiodes may also be used. Thus, as an example, a first quadrant photodiode may be used for the distance measurement, as described above, providing the two or more optical sensors. Another quadrant photodiode may be used, e.g. in a second partial beam path split off from the beam path of the first quadrant photodiode, for a transversal position measurement, such as for using at least one transversal coordinate x and/or y . The second quadrant photodiode, as an example, may be located on-axis with respect to the optical axis.

Further, it shall be noted that, besides the option of using one or more quadrant photodiodes, one or more quadrant photodiodes or further photodiode arrays may also be replaced or mimicked by separated photodiodes that are arranged or assembled close to each other, preferably in a symmetric shape such as a rectangular matrix, such as a 2×2 matrix. However further arrangements are feasible. In such an arrangement or assembly, the photodiodes may be ar-

5 ranged or assembled in a housing or mount, such as all photodiodes in a single housing or mount or groups of photodiodes in one housing or mount, or each of the photodiodes in a separate housing or mount. Further, the photodiodes may also be assembled directly on a circuit board. In such arrangements or assemblies, photodiodes may be arranged as such that the separation between the active area of the photodiodes, has a distinct value less than one centimeter, preferably less than one millimeter, more preferably as small as possible. Further, to avoid optical reflexes, distortions, or the like that may deteriorate the measurement, the space between the active areas may be either empty or filled with a material, preferably with a light absorbing material such as a black polymer, such as black silicon, black polyoxymethylene, or the like, more preferably optically absorbing and electrically insulating material, such as black ceramics or insulating black polymers such as black silicon, or the like. Further, the distinct value of the photodiode separation may also be realized by adding a distinct building block between the photodiodes such as a plastic separator. Further embodiments are feasible. The replacement of quadrant photodiodes by single diodes arranged in a similar setup such as in a 2 x 2 rectangular matrix with minimal distance between the active areas may further minimize the costs for the optical detector. Further, two or more diodes from a quadrant diode may be connected in parallel or in series to form a single light sensitive area.

20 Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Ga photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer. The optical sensors may comprise differing materials or photovoltaic concepts. As an example, at least one first optical sensor may be a Ge photodiode, whereas at least one second optical sensor may be an InGaAs photodiode. As a further example, the first and second optical sensor may comprise the same material with a differing doping such as a differing doping concentration or doping material.

35 The optical sensors each, independently, may be opaque, transparent or semitransparent. For the sake of simplicity, however, opaque sensors which are not transparent for the light beam, may be used, since these opaque sensors generally are widely commercially available.

The optical sensors each specifically may be uniform sensors having a single light-sensitive area each. Thus, the optical sensors specifically may be non-pixelated optical sensors.

40 The detector may further comprise an illumination source for illuminating the object. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam

propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. As an example, the illuminating light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illuminating light beam may be on the optical axis or off the optical axis. As an example, the illuminating light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

The detector may comprise the at least one illumination source. The illumination source may be adapted to illuminate the object with at least one illumination light beam. The illumination light beam may be fully or partially reflected by the object and may travel back towards the detector. The illumination source may be arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis. As used herein, the term "essentially parallel to the optical axis" refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. The illumination source may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis do not intersect in space. In particular, the illumination may be on-axis. In order to provide such an on-axis illumination, as an example, one or more reflective elements may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms. The reflected light beam may be essentially parallel to the optical axis. Thus, according to the present invention, the object distance is not determined from an angle of the light beam, i.e. a position of the light spot on a sensor, as done in triangulation methods. Instead, the object distance may be determined using light beams essential parallel to the optical axis. Determination of the object distance by a non-triangulation method but by combining the at least two sensor signals, allows arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis.

The illumination source and the optical axis may be separated by a small baseline. As used herein, the term "baseline", also denoted as basis line, refers to a distance, for example in a xy-plane, between the optical axis and the illumination source, in particular a distance between the

optical axis and a z-component of the illumination light beam. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and transfer device. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device. Triangulation systems require a sufficient baseline, however due to the baseline in the near field no detection may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. For example, the detector may be a compact device without further optical elements, wherein the illumination source may be placed as close as possible to an edge of the transfer device. Thus, the baseline may be close to half a diameter of the transfer device, in particular the lens diameter and housings of lens and light source. For example, the detector may be an even more compact device, wherein a mirror, in particular a small mirror, may be positioned in front of the transfer device, in particular in a center, for example a geometrical center, or close to the center of the transfer device, in order to couple the illumination beam into the beam path. Thus, the baseline may be less than half the diameter of the transfer device. The illumination source may be arranged such that the baseline is as small as possible. By arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis and that the illumination source and the optical axis are separated by the small baseline, very compact devices are possible. For example, a distance from the center of the transfer device to the illumination source, in particular along a connecting line from the center of the transfer device to the illumination source, may be preferably less than 2.5 times a distance from the center of the transfer device to an edge of the transfer device, more preferably less than 1.5 times the distance center to edge of the transfer device, and most preferably less 1 times the distance center to edge of the transfer device. The transfer device may have an arbitrary shape, in particular a non-circular shapes are possible. At small distances an aperture of the illumination source may be small and the baseline may be small. At large distances an aperture of the illumination source may be large and the baseline may be small. This is contrarily as in triangulation methods, wherein at large distances a large baseline is necessary. Further, triangulation based systems have a minimum detection range significantly greater than zero, for example such as 20 cm from the detector system, due to necessary spatial extend of the baseline. Such a large baseline may result in that the illuminated light scattered from the object may not reach the light sensitive area of the optical sensor behind the transfer device. In addition, in triangulation based systems, using a small baseline would reduce the minimum detection range, however, at the same time would reduce a maximum detection range. Further, triangulation based systems require a plurality of light sensitive areas and sensor signals, for example sensor signals of at least one detector row.

According to the invention, determination of the longitudinal coordinate z is possible with a reduced number of sensor signals, in particular with less than 20, preferably less than 10 and more preferably less than 5 sensor signals.

5 The illumination source may be adapted to generate at least one illumination pattern for illuminating the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the
10 detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term “pattern” refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term “illumination pattern” refers to a pattern which illuminates the object. The illumination pattern may be generated by ambient light,
20 such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

25 The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light
30 source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which
35 may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate at least one point, e.g. a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source.

40 The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be attached

to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source may be adapted to generate pulsed illumination. The illumination source may be adapted to generate at least one light pulse. As used herein, the term "light pulse" or "pulsed illumination" refers to a light beam limited in time. The light pulse may have a pre-defined length or time duration, for example in the nanoseconds range. For example, the illumination source may be adapted to generate pulses with a pulse length of less than a nanosecond, such as a tenth of a nanosecond, up to a tenth of a second. The illumination source may be adapted to periodically generate the light pulse. For example, the illumination source may be adapted to generate the light pulse with a frequency of 10 Hz to 10 GHz.

The illumination source may be adapted to generate a pulsed light beam. For example, the illumination source may be adapted to generate a continuous illumination light beam and the detector may comprise at least one interruption device adapted to interrupt the illumination, in particular periodically. The interruption device may comprise at least one shutter and/or a beam chopper or some other type of mechanical or electronical periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates at constant speed and which can thus periodically interrupt the illumination. By way of example, the at least one interruption device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

The detector may further comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one diffractive optical element; at least one Fresnel lens; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

The optical sensors may be positioned off focus. As used herein, the term "focus" generally refers to one or both of a minimum extend of a circle of confusion of the light beam, in particular of at least one light beam emitted from one point of the object, caused by the transfer device or a focal length of the transfer device. As used herein, the term "circle of confusion" refers to a light spot caused by a cone of light rays of the light beam focused by the transfer device. The circle of confusion may depend on a focal length f of the transfer device, a longitudinal distance from the object to the transfer device, a diameter of an exit pupil of the transfer device, a longitudinal distance from the transfer device to the light sensitive area, a distance from the transfer device to an image of the object. For example, for Gaussian beams, a diameter of the circle of confusion may be a width of the Gaussian beam. In particular, for a point like object situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from the object into a focus point at the focal length of the transfer device. For non-point like objects situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from at least one point of the object into a focus plane at the

focal length of the transfer device. For point like objects not situated or placed at infinite distance from the detector, the circle of confusion may have a minimum extend at least at one longitudinal coordinate. For non-point like objects not situated or placed at infinite distance from the detector, the circle of confusion of the light beam from at least one point of the object may have a minimum extend at least at one longitudinal coordinate. As used herein, the term “positioned off focus” generally refers to a position other than the minimum extend of a circle of confusion of the light beam caused by the transfer device or a focal length of the transfer device. In particular, the focal point or minimum extend of the circle of confusion may be at a longitudinal coordinate l_{focus} , whereas the position of each of the optical sensors may have a longitudinal coordinate l_{sensor} different from l_{focus} . For example, the longitudinal coordinate l_{sensor} may be, in a longitudinal direction, arranged closer to the position of the transfer device than the longitudinal coordinate l_{focus} or may be arranged further away from the position of the transfer device than the longitudinal coordinate l_{focus} . Thus, the longitudinal coordinate l_{sensor} and the longitudinal coordinate l_{focus} may be situated at different distances from the transfer device. For example, the optical sensors may be spaced apart from the minimum extend of the circle of confusion in longitudinal direction by $\pm 2\%$ focal length, preferably by $\pm 10\%$ focal length, most preferably $\pm 20\%$ focal length. For example, at a focal length of the transfer device may be 20mm and the longitudinal coordinate l_{sensor} may be 19,5 mm, i.e. the sensors may be positioned at 97,5% focal length, such that l_{sensor} is spaced apart from the focus by 2,5% of focal length. The extent of the circle of confusion may be larger than the extent of the optical sensor. For example, the optical sensor may be positioned as such that the circle of confusion extends beyond the optical sensor. The optical sensor may thus only partially evaluate the beam profile of the light beam.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Field Programmable Gate Arrays (FPGAs), or digital signal processors (DSPs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As outlined above, by evaluating the sensor signals, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more additional transversal sensors may be used for determining at least one transversal coordinate of the object. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. These devices may generally also be implemented into the detector according to the present invention.

As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing. The detector specifically may comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensor. The transfer device may comprise one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. Further, if applicable, Fresnel lenses or diffractive optical elements may be used. Therein, generally, the term "transfer device" may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of, preferably at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one bea-

con device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, the at least one beacon device may be adapted to reflect one
5 or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties
10 of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine inter-
15 face comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface
20 is designed to assign to the position at least one item of information.

In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or
25 more of the embodiments disclosed in further detail below. The entertainment device is configured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at spe-
30 cific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

In a further aspect of the present invention, a readout device for optical storage media is proposed. The readout device comprises at least one detector according to any one of the preceding embodiments referring to a detector. As used therein, a readout device for optical storage
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media generally refers to a device which is capable of optically retrieving information stored in optical storage media such as optical storage discs, e.g. CCD, DVD or Blu-ray discs. Thus, the above-described measurement principle of the detector according to the present invention may be used for detecting data modules within an optical storage medium such as in optical storage
5 discs. As an example, in case a reflective data module is present and reflects the illuminating light beam, the detector will not only detect the reflected light beam according to the above-mentioned measurement principle but will also detect a distance between the detector and the reflective data module, i.e. a depth of the reflective data module within the optical storage medium. Thus, as an example, the detector may be used for detecting different layers of information
10 modules or data modules within the optical storage medium. Thereby, as an example, two layer discs or three layer discs or even discs having more than three layers may be generated and read out.

In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light
20 beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of
30 information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers

to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination
5 source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

10 For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source
15 may have a wavelength of 300-500 nm. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may thus, preferably be a
20 constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least
25 one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic
30 material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object.
35 By using at least one of the detectors according to the present invention, at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector.
40 However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as

a front edge or a back edge of the housing, and the single dot may, alternatively, be determined.

Alternatively, the illumination source of the scanning system may emit two individual laser
5 beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions,
10 such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to
15 the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in
20 particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one
25 or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array
30 of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an im-
35 age by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning device. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning
40 devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics

applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with a several corners and surfaces. One or more flexibly movable laser sources may be attached to the scanning system. The one or more laser sources may be moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system may be measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information may be stored in a mobile phone application. The laser sources may be moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and/or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning device may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning device is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning device.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The optional transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The optional transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor.

5 The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least
10 one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination
15 source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively
20 or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic
25 radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence
30 and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be
35 regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the
40 visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. Most preferably, the at least one illumination source is adapted to emit light in the visible spectral

range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm.

5 The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of
10 example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

15 In a further aspect, the present invention discloses an inertial measurement unit for use in an electronic device. The electronic device may be a mobile electronic device. The electronic device may be a camera. The electronic device may be a mobile phone. The inertial measurement unit is adapted to receive data determined by at least one inertial measurement unit comprising at least one detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. As used herein, the term "data determined by the at least one detector" refers to at least
20 one information on the at least one longitudinal coordinate z. The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. As used herein, the term "data determined by the at least one further sensor" refers to at least one information selected from the group consisting of: angle information; speed information; information on turn rate; information on inclination. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position
30 in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed. The inertial measurement unit may comprise at least one processor. The processor may be adapted to evaluate data recorded by the further sensor. In particular, the processor may be adapted to determine and/or calculate one or more of spatial position, spatial orientation, movement and velocity. The inertial measurement unit may comprise a plurality of further sensors. The inertial measurement unit may be adapted to fuse information determined from at least two of the further sensors. The inertial measurement unit may be adapted to fuse information of at least two further sensors by using at least one Kalman filter. As outlined above, the detector may be adapted to provide an absolute measurement of the longitudinal
40 coordinate z. The processor, e.g. the evaluation device as described above, may be adapted to fuse the information of the at least two further sensors considering the longitudinal coordinate z. The various sensor signals may be used within a Kalman filter or a linear quadratic estimated to take into account that each sensor signal is subject to measurement errors and inaccuracies.

Fusion of these sensor signals within a Kalman filter may yield an improved estimate such as for the measurement of the longitudinal coordinate.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used.

The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps are as follows:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light;
- illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

Specifically, evaluating the first and second sensor signal may comprise deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q . Further, the determining of the longitudinal coordinate may comprise evaluating the quotient signal Q .

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine

interface application; a tracking application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for household applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as con-

ventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus, specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head mounted displays.

Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries, obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an

object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of

lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general
5 need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

10 Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic
15 alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems,
20 or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present
25 invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located
30 as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.

35 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors.
40 In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal

sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions.

Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endoscopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric sur-

gery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment.

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Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

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Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe.

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In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

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Further the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

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Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support

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the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

5 Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

10 The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

15 The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

20 Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and/or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to
25 avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the
30 devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the
35 present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such
40 as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-

dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection, such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections, inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft when landing or starting. Therein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or accelerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, ap-

plications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racket, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the present invention may be used in drones or multicopters or the like to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used to recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization

and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concerted movements of several drones or recharging or refueling in the air or the like.

5 Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet-related appliances, child-related appliances, child surveillance, appliance related surveillance, support or service to elderly or ill persons, home
10 security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic
15 robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as cat litter robots, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, window washing robots, toy robots, telepresence robots, social robots providing company to less mobile people, or robots translating and speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots
30 or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate insecure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

35 Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.

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- Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition device, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.
- Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.
- The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or greenhouse or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.
- Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific groups of animals such as bees.
- Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or viruses or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.
- One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present inven-

tion may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further, the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location. Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location-specific information, such as the current position on a map, or information on goods sold, or the like.

Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.

Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices according to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.

As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the

present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or
5 may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an
10 optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction,
15 and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar
20 coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle
25 may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device adapted for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device.
35 Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a
40 stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside the camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based algorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predeter-

mined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

As outlined above, the human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will

be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device.

Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club,

a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object.

The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least

one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

5 The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.
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The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.
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The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of:
20 a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
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As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.
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The proposed devices and methods provide a large number of advantages over known detectors of this kind. Thus, the detector generally may avoid the shortcomings of the known prior art systems disclosed above. Specifically, the detector may avoid the use of FiP sensors, thereby allowing for e.g. using simple and cheap and commercially available semiconductor sensors such as silicon photodiodes. These photodiodes generally do not show a luminance dependen-
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cy, and the method disclosed above is generally independent from the brightness of the scenery and/or the brightness of the light spot on the light beam. Consequently, a range of measurement in terms of luminance or total power of the light beam entering the detector is generally larger in the present invention as compared to many of the devices disclosed above. Further, the measurement by using the detector according to the present invention is generally independent from the target spot size, i.e. from either the size of the object, the size of a light spot projected onto the object or the size of a beacon device being one or more of attached to the object, integrated into the object or held by the object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional option of measuring transversal coordinates, thereby integrating the functionality of a PSD.

The result of the measurement, i.e. the position determined by the detector such as the at least one longitudinal coordinate, may be rendered widely independent from the brightness of the scenery and/or the brightness of the object, the brightness of the at least one Beacon device, the brightness of the at least one illumination source or the total power of the light beam propagating from the object to the detector. Further, due to this independence and wide range of measurement in terms of brightness, reflective objects or non-reflective objects may be used.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention, such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the quotient signal, as mentioned above, contrarily, provides a wide range of a continuously and monotonously decreasing or increasing function which may be used to determine the longitudinal coordinate from the quotient signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the quotient signal from the total power of the light beam, at least as long as no saturation of one or both of the optical sensors is reached, also a very wide range of measurement in terms of brightness may be provided, i.e. in terms of total power of the light beam.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector having

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal

in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and

- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

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Embodiment 2: The detector according to the preceding embodiment, wherein the optical sensors are located in the same plane.

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Embodiment 3: The detector according to any one of the preceding embodiments, wherein the optical sensors are partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector.

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Embodiment 4: The detector according to any one of the preceding embodiments, wherein the optical sensors are part of a sensor array, such as an array comprising 2-4 optical sensors such as a quadrant photodiode.

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Embodiment 5: The detector according to the preceding embodiment, wherein a geometrical center of the sensor array is offset from the optical axis.

Embodiment 6: The detector according to any one of the two preceding embodiments, wherein the sensor array is movable, preferably automatically, relative to the optical axis.

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Embodiment 7: The detector according to the preceding embodiment, wherein the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly, moving the sensor array relative to the optical axis until the light spot is off-centered.

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Embodiment 8: The detector according to any one of the preceding embodiments, wherein the optical sensors are partial diodes of a quadrant diode, with a geometrical center of the quadrant diode being off-centered from the optical axis of the detector.

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the light-sensitive areas of the optical sensors are equal.

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Embodiment 10: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one lens, wherein the optical axis of the detector is an axis of symmetry of the lens.

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Embodiment 11: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are areas of the beam profile at the sensor position, and $E(x,y,z_o)$ denotes the beam profile given at the object distance z_o .

Embodiment 12: The detector according to the preceding embodiment, wherein the light-

5 sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

10 Embodiment 13: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a
15 number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

Embodiment 14: The detector according to any one of the three preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sen-
20 sors with at least one line pattern, wherein $A1$ corresponds to an area with a full line width of the line pattern on the optical sensors, wherein $A2$ is a central area of the line pattern on the optical sensors.

Embodiment 15: The detector according to any one of the three preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the sensor ele-
25 ment with at least one point pattern, wherein $A1$ corresponds to an area with a full radius of a point of the point pattern on the optical sensors, wherein $A2$ is a central area of the point on the point pattern on the optical sensors.

30 Embodiment 16: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the evaluation device is further configured for determining the longitudinal coordinate by evaluating the quotient signal Q .

35 Embodiment 17: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.

40 Embodiment 18: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

Embodiment 19: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

- 5 Embodiment 20: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

- 10 Embodiment 21: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 22: The detector according to any one of the preceding embodiments, wherein the optical sensors are arranged in one and the same beam path of the detector.

- 15 Embodiment 23: The detector according to any one of the preceding embodiments, wherein the optical sensors each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

- 20 Embodiment 24: The detector according to any one of the preceding embodiments, wherein the optical sensors each are uniform sensors having a single light-sensitive area each.

- 25 Embodiment 25: The detector according to any one of the preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

- 30 Embodiment 26: The detector according to any one of the preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object, e.g., an illumination source comprising at least one laser, such as an illumination source providing at least one illuminating light beam being on the optical axis or being not on the optical axis.

- 35 Embodiment 27: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

- 40 Embodiment 28: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

- 45 Embodiment 29: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern; at least one pattern comprising at least one pre-known feature; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 30: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

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Embodiment 31: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one diffractive optical element; at least one Fresnel lens; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

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Embodiment 32: The detector according to any one of the preceding embodiments, wherein the detector comprises the at least one illumination source, wherein the illumination source is adapted to illuminate the object with at least one illumination light beam, wherein the illumination source is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis.

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Embodiment 33: The detector according to any one of the preceding embodiments, wherein the detector comprises the at least one illumination source, wherein the illumination source and the optical axis are separated by a small baseline.

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Embodiment 34: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

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Embodiment 35: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

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Embodiment 36: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

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Embodiment 37: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding

embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

5 Embodiment 38: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

10 Embodiment 39: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 40: A readout device for optical storage media, the readout device comprising at least one detector according to any one of the preceding embodiments referring to a detector.

15 Embodiment 41: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam;
- 20 - illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- 25 - evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

30 Embodiment 42: The method according to the preceding embodiment, wherein evaluating of the sensor signals comprises deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q .

35 Embodiment 43: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use for calibration of a distance measurement; a use in combination with optical data storage and readout.

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Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

- | | | |
|----|---------------------|---|
| 15 | Figures 1 and 2 | show different embodiments of a detector according to the present invention; |
| 20 | Figure 3 | shows an exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera; |
| | Figure 4 | shows an off-centered position of the optical axis of the detector and of a light spot on a quadrant diode; |
| 25 | Figure 5 | shows an alternative embodiment of a detector with staggered optical sensors; |
| 30 | Figures 6 to 8 | show a quotient signal Q of two sensor signals of two photodiodes of a quadrant photodiode as a function of a longitudinal coordinate z of an object under various measurement conditions; |
| | Figures 9A and 9B | show a modification of the embodiment of Figure 2, with an off-axis illumination light beam; |
| 35 | Figures 9C and 9D | show a comparison of two experimental setups using a detector setup according to Figure 9A with a Bi-cell as optical sensors and experimental results; |
| 40 | Figures 10A and 10B | show a further exemplary embodiment of a detector according to the present invention with a small baseline; |
| | Figure 11 A to 11 O | show further exemplary configurations of optical sensors according to the present invention; |

Figures 12A and B show embodiments of determining a longitudinal coordinate z using a line pattern; and

Figures 13A and B show further embodiments of the detector according to the present invention comprising at least one bi-cell.

Detailed description of the embodiments

In Fig. 1, a schematic view of a first embodiment of a detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises a beacon device 114, from which a light beam 116 propagates towards a first optical sensor 118 and a second optical sensor 120. The first optical sensor 118 comprises a first light-sensitive area 122, and the second optical sensor 120 comprises a second light-sensitive area 124. The optical sensors 118, 120, as shown e.g. in Fig. 4, may be part of an array 174 of optical sensors 176, such as the first optical sensor 118 being the optical sensor 176 in the upper left corner of the array 174 and the second optical sensor 120 being the optical sensor 176 in the lower right corner of the array 174. Other choices are feasible. The array 174, as an example, may be a quadrant photodiode 178, and the optical sensors 176 may be partial diodes of the quadrant photodiode 178.

The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110, further, may comprise at least one transfer device 128, such as at least one lens or a lens system, specifically for beam shaping. Consequently, the light beam 116 is focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of the above-mentioned prior art documents, such as to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Fig. 4, the setup of the detector 110 is off-centered in various ways. Thus, a geometrical center 180 of the array 174 may be off-centered from the optical axis 126 by offset d_0 . Further, a geometrical center 182 of the first optical sensor 118 is off-centered from the optical axis 126 by offset d_1 , and a geometrical center 184 of the second optical sensor 120 is off-centered from the optical axis 126 by offset d_2 , wherein $d_1 \neq d_2$. In other words, a light spot 186 is formed, which is unequally distributed over the light-sensitive areas 122, 124.

As will be shown in further detail below, the detector 110 may be configured for automatically establishing the off-centered position of the light spot 186 on the array 174. For this purpose, firstly, the detector 110 may be configured for determining whether the sensor signals generated by the optical sensors 176 of the array 174 are equal. If this should be the case, the detector 110 may be configured to determine that the light spot 186 is centered in the array 174 and, consequently, may shift the light spot 186 out of the geometrical center 180 of the array 174, such as by shifting the whole array 174 in a plane perpendicular to the optical axis 126. For this

purpose, as will be shown in further detail below with respect to Fig. 3, one or more actuators may be provided in the detector 110.

Turning back to the setup of Fig. 1, the first optical sensor 118, in response to the illumination by the light beam 116, generates a first sensor signal s_1 , whereas the second optical sensor 120 generates a second sensor signal s_2 . Preferably, the optical sensors 118, 120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 116 or of the portion of the light beam 116 illuminating their respective light-sensitive areas 122, 124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination. In other words, preferably, the optical sensors 118, 120 do not exhibit the above-described FiP effect.

The sensor signals s_1 and s_2 are provided to an evaluation device 132 of the detector 110. The evaluation device 110, as symbolically shown in Fig. 1, may specifically be embodied to derive a quotient signal Q , as explained above. The quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 112 and/or the beacon device 114, from which the light beam 116 propagates towards the detector 110, as will be explained in further detail with reference to the corrosion signals shown in Fig. 6 to 8 below.

The detector 110, in combination with the at least one beacon device 114, may be referred to as a detector system 134, as will be explained in further detail below with reference to Fig. 3.

In Fig. 2, a modification of the embodiment of Fig. 1 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Fig. 1. Instead of using an active light source, i.e. a beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 comprises at least one illumination source 136. The illumination source 136, as an example, may comprise a laser, whereas, in Fig. 1, as an example, the beacon device 114 may comprise a light-emitting diode (LED). The illumination source 136 may be configured for generating at least one illumination light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The illumination source 136, as an example, may comprise one or more diaphragms 190, such as an adjustable diaphragm 190, e.g. an adjustable iris diaphragm and/or a pin hole.

The setup shown in Fig. 2, as an example, may also be used in or as a readout device 192 for optical storage media. Thus, as an example, the object 112 may be an optical storage medium such as in optical storage discs, e.g. a CCD, DVD or Blu-ray disc. By measuring the presence or non-presence of data storage modules and the depth of the same within the object 112, by using the above-mentioned measurement principle, a data readout may take place.

The light beam 116, specifically, may travel along the optical axis 126 of the detector 110. As shown in Fig. 2, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too, as will be shown in the context of Fig. 9A and 9B below. In order to provide an on-axis illumination, as shown in Fig. 2, as an example, one or more reflective elements 140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Fig. 2 corresponds to the setup in Fig. 1. Thus, again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q, and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the at least one quotient signal Q. It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

It shall further be noted that the embodiments shown in Fig. 1 and 2 simply provide embodiments for determining the longitudinal coordinate z of the object 112. It is also feasible, however, to modify the setups of Fig. 1 and 2 to provide additional information on a transversal coordinate of the object 112 and/or of parts thereof. As an example, e.g. in between the transfer device 128 and the optical sensors 118, 120, one or more parts of the light beam 116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or additional quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 112 and/or of parts thereof. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

Fig. 3 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments shown in Fig. 1 or 2. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 3 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 3 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 148. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises the detector system 134. The components of the devices and systems shall be explained in further detail below.

Fig. 3 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136. The illumination source 136, generally, is configured to emit at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 3 is shown in Fig. 1 and 2 or will be shown, as an alternative embodiment, in Fig. 5 below. Thus, the detector 110, besides the optical sensors 118, 120, comprises at least one evaluation device 132, having e.g. the at least one divider 142 and/or the at least one position evaluation device 144, as symbolically depicted in Fig. 3. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 118, 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Fig. 5. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 118, 120 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In Fig. 3, as an example, one or more reflective elements 140 may be used, for example partially transparent, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

The detector 110 as symbolically shown in the exemplary embodiment of Fig. 3 may also comprise at least one actuator 188 for moving the array 174 of the optical sensors 176 relative to the optical axis 126. As outlined above, for providing this movement, the optical axis 126 may be moved in relation to the array 174 by moving the optical axis 126, by moving the array 174 or both. Thus, as an example, the optical axis may be moved by using one or more of deflecting elements and/or by using the transfer device 128. As a simple example, a lens of the transfer device 128 may be tilted, such as by using one or more actuators 188 (not depicted). Additionally or alternatively, the array 174 may be shifted by the one or more actuators 188, preferably in a plane perpendicular to the optical axis 126. As an example, one or more electromechanical actuators may be used, such as one electromechanical actuator for an x-direction and another

electromechanical actuator for a y-direction. Other embodiments are feasible. Thereby, the above-mentioned off-centering procedure may be implemented for establishing an off-centered situation as shown e.g. in Fig. 4.

5 In the exemplary embodiment shown in Fig. 3, further, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the optical sensors 176, including at least the first optical sensor 118 and the second optical sensor 120. The optical sensors 176 may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 may be comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. A coordinate system 168 may be defined, in which a direction parallel or anti-parallel to the optical axis 126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 126 may be defined as transversal directions. In the coordinate system 128, symbolically depicted in Fig. 5, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the optical sensors 118, 120 as well as, optionally, further optical sensors. The optical sensors 118, 120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 118 covers a portion of the second optical sensor 120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 112 and/or of parts thereof.

35 One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as explained above in the context of Fig. 1, 2 and 4, the evaluation device 132 is configured to evaluate sensor signals provided by the optical sensors 118, 120. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 118, 120 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least

one of these beacon devices 114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 136 is used, the beacon devices 114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the light-sensitive areas 122, 124 of the optical sensors 118, 120. For details of the evaluation, reference may be made to Fig. 1, 2 and 4 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Fig. 3, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Fig. 3 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may be fully or partially integrated into the machine 170, specifically the computer, as indicated in Fig. 3 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In the setup of the detectors 110 as shown in Fig. 1, 2 and 3, the optical sensors 176 are part of an array 174, and all optical sensors 176 may be located in one and the same plane oriented essentially perpendicular to the optical axis 126. As noted in this context, when mentioning “perpendicular” or “essentially perpendicular”, preferably, a 90° orientation is given. However, tolerances may be present, such as angular tolerances of no more than 20°, preferably of no more than 10° or more preferably of no more than 5°. The optical sensors 176, however, not necessarily have to be located in an array 174 and not necessarily have to be located in one and the same plane, as is shown in an alternative setup of the detector 110 shown in Fig. 5. In this figure, only the optical components are shown. For other components, reference may be made to Fig. 1, 2 and 3 above.

As can be seen, in this alternative setup, two or more optical sensors 176 are present, comprising at least one first optical sensor 118 and at least one second optical sensor 120 located in different planes which are offset in a direction of the optical axis 126, also referred to as the z-direction. Further, as can also be seen, the optical sensors 118, 120 may overlap, whereas in the previous embodiments, preferably, no overlap between the optical sensors 176 is given. Apart from these modifications, the functionality and the evaluation of the sensor signals generally corresponds to the embodiment of Fig. 1, 2 and 3 above.

As discussed above, for evaluating the at least two sensor signals of the at least two optical sensors 176 and for deriving an information on the longitudinal position of the object 112 thereof, such as a distance between the detector 110 and the object 112 and/or a z-coordinate of the object 112, preferably, at least one combined sensor signal is generated by the evaluation device 132. The combined sensor signal, as long as this combined sensor signal provides, at least over a measurement range, a unique function of the distance, may be used for deriving the longitudinal coordinate. As an example, the combined sensor signal may be or may comprise at least one quotient signal Q . In Fig. 6 to 8, quotient signals Q of two sensor signals of two optical sensors 176 are shown under various measurement conditions. In each case, the quotient signal Q is denoted on the vertical axis, as a function of the longitudinal coordinate z of the object 112 on the horizontal axis, the latter given in centimeters.

In all experiments, a setup as shown in Fig. 2 was used. As an illumination source 136, in the experiments of Fig. 6 and 7, a 980 nm Picotronic laser source was used, in conjunction with a lens having a focal length of 100 mm. In the experiment of Fig. 8, a Laser Components laser light source having a wavelength of 850 nm was used, in conjunction with a lens having a focal length of 79 mm. In all experiments, the laser beam was aligned on the optical axis 126 via a small prism in front of the lens 128, forming a reflective element 140. A diaphragm 190 in front of the laser source was used to vary the spot size. The quadrant diode 178 was used to measure the reflection of the laser source on different materials. In all experiments, the distance dependency is given by the quotient Q of two adjacent quadrant currents.

In Fig. 6, the laser power was varied during the experiment, from 8 nA laser current, denoted by the dotted line, to 106 nA, denoted by the solid line. Therein, since the laser current typically does not provide a measure for the laser intensity, the laser current indicated therein is a current of a silicon photodetector in a measurement setup in which the laser illuminates a white sheet of paper at a distance of 330 mm from the lens. As is clearly visible, the curves are nearly identical and, at least within this range of variation of the laser power, do not significantly depend on the laser power. This experiment shows that the quotient signal provides a reliable and monotonous function of the longitudinal coordinate, independent from the influence of the brightness of the illumination source.

In Fig. 7, a spot size of the illumination source 136 was varied, by varying the open diameter of the diaphragm 190 in front of the laser. The spot size was varied from 1.5 mm, denoted by the dotted line, to 3.5 mm, denoted by the solid line, in steps of 0.5 mm. As can be seen, up to a

distance of approximately 200 cm, the quotient signal Q does not depend on the spot size and, thus, again, is not negatively affected by this variation.

In Fig. 8, a material of the object 112 illuminated by the laser beam was varied. Therein, the dotted line denotes white paper, the dashed line with the smallest dashes denotes black paper, the dashed line with the medium dashes denotes wood, and the dashed line with the largest dashes denotes an aluminum plate. As can be seen, at least up to a measurement range of approximately 250 cm, the experiment does not strongly depend on the type of material used for the object 112.

The experiments shown in Fig. 6 to 8, thus, clearly demonstrate that the quotient signal Q provides a reliable function of the distance. At least within a range of measurement, the function monotonously rises with the distance. The function is not strongly influenced by the most significant variations which may occur in real life measurements, such as the brightness of the illumination source, the spot size of the illumination source or the material of the object 112. Thus, by evaluating the quotient signal Q of two or more optical sensors 176, reliable distance information may be generated. Thus, as an example, the curves shown in Fig. 6 to 8 directly may be used as calibration curves for the purpose of the evaluation device 132. Other evaluation methods, however, are feasible.

In Fig. 9A and 9B, an alternative embodiment of the detector 110 is shown which is a modification of the setup shown in Fig. 2. Thus, for most elements and optional details as well as further elements not shown in the schematic Figures 9A and 9B, reference may be made to the description of Fig. 2 above.

In Figure 2, the illumination light beam 138, as discussed above, preferably travels along the optical axis 126, i.e. parallel to the optical axis 126 or even on the optical axis 126. In the setup, the position of the center of the light spot 186 typically does not depend on the z-coordinate of the object 112, such as on a distance between the object 112 and the detector 110. In other words, the diameter or equivalent diameter of the light spot 186 changes with the distance between the object 112 and the detector 110 whereas, typically, the position of the light spot 186 on the array 174 does not.

Contrarily, in Fig. 9A and 9B, a setup of the detector 110 is shown in which an illumination light beam 138 travels off-axis, i.e. one or both of at an angle other than 0° with the optical axis 126 or parallel to the optical axis 126 but shifted from the optical axis 126. This embodiment, as will be discussed in further detail below, demonstrates that the method according to the present invention can be further enhanced by increasing the z-dependency of a combined sensor signal. Thus, in Fig. 9A, a side view is shown with two different positions of the object 112, i.e. a first position at z_1 , drawn in solid lines, and a second position at z_2 , drawn in dashed lines. As can be seen, the illumination light beam 138 which, as an example, propagates at an angle of 5° to 30° , e.g. 10° to 20° , with the optical axis 126, hits the object 112 in both cases at different positions. From these points of the object 112 illuminated by the illumination light beam 138,

light beams 116 propagate towards the detector 110, wherein, again, the light beam 116 for the object 112 being located at position z_1 is drawn in solid lines, wherein the light beam 116 for the object 112 being located at position z_2 is drawn in dashed lines.

In Fig. 9B, the array 174, e.g. a quadrant photodiode, is shown in an enlarged fashion. As can be seen in this setup, the position of the light spot 186 moves with the longitudinal position z of the object 112. Thus, not only is the size of the light spot 186 affected by the longitudinal position z but also is the position on the array 174 of the light spot 186 changed. In Fig. 9B, this movement of the light spot 186 is denoted by arrow z .

Consequently, by this movement of the light spot 186, the z -dependency of a combined sensor signal taking into account at least two sensor signals of the optical sensors 176 may be increased. As an example, the four diodes of the array 174, in Fig. 9B, are denoted by D1-D4. The quotient signal Q , as an example, may be formed as $Q = i(D1)/i(D4)$, with $i(D1)$ being the sensor signal of photodiode D1, and $i(D4)$ being the sensor signal of photodiode D4.

As shown in Figure 9B, the quadrant diode may comprise two dividing lines. The dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other. In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device 132 may be adapted to determine a second quotient using at least three or all four sensor signals of the quadrant diode. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotients allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient may be determined by $Q = i(D1+D2)/i(D3+D4)$, wherein the sensor signals of the two top quadrants, also called top segment, are divided by the sensor signals of the two bottom quadrants, also called bottom segment. Using the quotient of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between top and bottom segment is parallel to the baseline, the quotient signal determined from the top segment divided by the bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments, i.e. $Q = i(D1+D3)/i(D2+D4)$, may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals $i(D1)$, $i(D2)$, $i(D3)$ and $i(D4)$ of the quadrants such that distance measurement is possible over a wide range with a large resolution.

In the situation shown in Fig. 2, the position of the light spot 186 does not depend on z . With a change in z , depending on the optical situation, the spot will become larger or smaller, such as by becoming more diffuse or more focused. In case the spot size increases and the spot be-
comes more diffuse, $i(D4)$ will increase more rapidly than $i(D1)$, such that the quotient signal Q decreases.

Contrarily, in the situation of Fig. 9A, both the size and the position of the light spot 186 are dependent on the z -coordinate. Thus, the tendency of the z -dependency of the combined sensor signal such as the quotient signal Q will be increased. In the situation of Figure 2, depending on the z -coordinate, the sensor signal of at least one sensor will increase and simultaneously the sensor signal of at least one different sensor will decrease, resulting in the z -dependent quotient signal Q . In the situation of Fig. 9A, the position dependency of the light spot 186 can result in three different situations depending on the relative position of light source, optical axis, and sensor: Firstly, the position dependency of the light spot may result in a further decrease of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a further increase of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Secondly, the position dependency of the light spot may result in a reduced decrease or even increase of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a reduced increase or even decrease of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Thirdly, the position dependency of the light spot may be as such that the z -dependence of the sensor signals is largely unchanged compared to the situation in Figure 2. However, according to the present invention, object distance is not determined from the position of the light spot on a sensor as done in triangulation methods. Instead, movement of the light spot 186 on the array 174 may be used to enhance dynamic of the sensor signals and or the resulting quotient signal Q which may result in an enhanced dynamic of the z -dependency.

Additionally, as known from the prior art, the sensor signals $i(D1)$, $i(D2)$, $i(D3)$, $i(D4)$ may also be used for determining a transversal position x , y of the object 112. Further, the sensor signals may also be used for verifying the z -coordinate determined by the present invention.

Figure 9C shows a comparison of two experimental setups using a detector setup according to Figure 9A with a Bi-cell as optical sensors 176 with two light sensitive areas. In a first experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 186 may move in parallel to the linear boundary of the two optical sensors 176 of the Bi-cell along a direction of movement 194 in dependence of the object distance. Since the direction of movement 194 of the light spot 186 is in parallel to the linear boundary of the two light sensitive areas in dependence of the object distance, the resulting sensor signals are identical to a situation with no movement of the light spot 186 depending on object distance as shown in Figure 2. In a second experimental setup, depending on the relative

position of the illumination light source, the optical axis and the sensor, the light spot 186 may move as such that the distance of the center of the light spot 186 to the boundary of the two optical sensors 176 of the Bi-cell changes in dependence of the object distance such as a movement orthogonal to the boundary of the two optical sensors 176 such as a movement
5 along a direction of movement 196 in dependence of the object distance. The detector setup allowing movement of the light spot 186 may be a modification of the setup shown in Figure 9A. Thus, for most elements and optional details as well as further elements, reference may be made to the description of Fig. 9A above. In Figure 9C, the optical sensors 176 may be a bi-cell diode.

10 Figure 9D shows experimental results of the comparison of the two experimental setups using a detector setup according to Figure 9A, allowing movement of the light spot 186 according to Figure 9C with movement of the light spot depending on the object distance along directions of movement 194 and 196. Curve 198 shows the dependency of quotient Q on the longitudinal
15 coordinate z for the detector setup allowing movement of the light spot 186 along a direction of movement 194 as shown in Figure 9C which is in parallel to the boundary of the optical sensors of the Bi-Cell and, which is a situation equivalent to Figure 2 without a movement of the light spot depending on the object distance. Curve 200 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup according to Figure 9A and using a detector set-
20 up allowing movement of the light spot 186 with movement of the light spot 186 according to Figure 9C with movement of the light spot depending on the object distance along a direction of movement 196. The experimental setup was as follows: The optical sensors 176 may be a bi-cell diode, in particular a Si-Bi-Cell. The illumination source 136 may be a 950 nm laser with a spot size of 4 mm. The transfer device 128 may have a focal length of 20 mm, e.g. a lens avail-
25 able as Thorlabs Asphere, f= 20 mm. The distance of the object 112 was varied from 0 to 3000 mm. Determination of the longitudinal coordinate z may be possible without allowing movement of the light spot 186. In particular, according to the present invention, movement of the light spot may not be essential for determination of the longitudinal coordinate z. With the detector setup allowing movement of the light spot 186 along a direction 194 or without any movement deter-
30 mination of object distance is possible at very small distance, whereas with movement along a direction 196 determination of object distance is possible for object distance such as distances greater than 500mm.

Figures 10A and 10B show an exemplary embodiment of a detector 110 with a small baseline
35 202. Figure 10A shows side view of the detector 110. The detector 110 may comprise the at least one illumination source 136. The illumination source 136 may be adapted to illuminate the object 112 with at least one illumination light beam 138. The illumination light beam 138 may be fully or partially reflected by the object and may travel back towards the optical sensor 176. The illumination source 136 may be arranged such that a direction of propagation of the illumination
40 light beam is essentially parallel to the optical axis 126. The illumination source 136 may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis 126 do not intersect in space. The illumination source 136 and the optical axis 126 may be separated by a small baseline 202. The illumination source 136 may be

spaced apart from the optical axis 126 by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and transfer device. Figure 10B shows a front view on the illumination light beam 138 and the transfer device 128. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the detector 110 may be a compact device without further optical elements, wherein the illumination source 136 may be placed as close as possible to an edge of the transfer device 128. Thus, the baseline 202 may be close to half a diameter of the transfer device 128, in particular the lens diameter and housings of lens and light source. The illumination source 136 may be arranged such that the baseline 202 is as small as possible. For example, a distance from the center of the transfer device 128 to the illumination source 136, in particular along a connecting line from the center of the transfer device 128 to the illumination source 136, may be preferably less than 2.5 times a distance from the center of the transfer device 128 to an edge if the transfer device 128, more preferably less than 1.5 times the distance center to edge of the transfer device 128, and most preferably less than 1 times the distance center to edge of the transfer device 128.

Figure 11 A to O show further exemplary configurations of optical sensors according to the present invention, in particular top view in direction of propagation of the light beam 116. In Figure 11A, a top view of two rectangular optical sensors is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120. The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset, in particular in a transversal direction y, from the optical axis 126. In Figures 11B and 11C, top view of a large rectangular optical sensor 120 is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120 having a triangle shaped (Figure 11B) or star-shaped (Figure 11C) light-sensitive area. Alternatively, the first optical sensor 118 and the second optical sensor 120 may be located at the same longitudinal coordinate, such as in the same plane, such as as separated light sensitive elements.

In Figures 11 M to O, a top view of two rectangular optical sensors is shown, wherein the first optical sensor 118 and the second optical sensor 120 are rectangular sensors with the same size. In Figures 11 M to O a mask 119 is arranged in front of the first and second optical sensors 118, 120. The mask 119 may be arranged with a different offset from the optical axis 126. The mask 119 may have an arbitrary size and shape, for example, the mask may be rectangular shaped (Figure 11M), triangle shaped (Figure 11N) or star-shaped (Figure 11O). However, other sizes and shapes are feasible. Mask 119 may be adapted to prevent light impinging on the light sensitive areas of the first and second optical sensors 118, 120.

Alternatively to using a mask 119, Figures 11 M to O may be understood as a first optical sensor 118a having rectangular shape (Figure 11M), triangle shape (Figure 11N) or cross-shape or star-shape (Figure 11O) and yielding a first sensor signal and a second optical sensor 120a yielding a second sensor signal, wherein the second optical sensor 120a may be separated into several light sensitive areas, for example four light sensitive areas 121a.

The geometrical centers of the first optical sensor 118 and the second optical sensor 120 are spaced apart from the optical axis 126 of the detector 110. Figure 11K shows two circular shaped optical sensors, wherein the first optical sensor 118 is a small optical sensor in front of the larger second optical sensor 120. Alternatively, the first optical sensor 118 and the second optical sensor 120 may be located at the same longitudinal coordinate, such as in the same plane, such as as separated light sensitive elements. In Figures 11D, the light sensitive area of the first optical sensor 118 is square-shaped, and the light sensitive area of the second optical sensor 120 is rectangular, such that the surface areas in x and y differ. In addition, a center of the first optical sensor 118 and a center of second optical sensor 120 may have different x coordinates such that the optical sensors 118, 120 may have different spatial offset in one or more of x and y direction from the optical axis. In Figure 11H, both the first optical sensor 118 and the second optical sensor 120 may be rectangular. The first optical sensor 118 and the second optical sensor 120 may be arranged such that the center of the first optical sensor 118 and the center of second optical sensor 120 may have different x coordinates and that the surface areas in x and y differ. In Figure 11L, the first optical sensor 118 may have a deviating shape from the shape of the second optical sensor 120 such as a circular or semicircular shape. Figures 11 E, F, G, I, J show a matrix 115 comprising a plurality of partial diodes. In Figures 11 E, F, G the matrix 115 has a rectangular shape, whereas in Figures 11 I and J the matrix 115 has a circular shape. Rows and columns may be arranged equidistant or non-equidistant. In case of equidistant rows and/or columns the matrix 115 may be arranged with a spatial offset to the optical axis 126. The sensor signals of the light sensitive areas of the matrix 115 in Figure 11 E, F, G, I, J may be summarized such as added to form a first sensor signal of a first optical sensor and a second sensor signal of a second optical sensor. As an example, the sensor signals of the light sensitive areas in the corners of the sensor element 115 in Figure 11 G may be added to form a first sensor signal, whereas the remaining light sensitive areas of the sensor element 115 may be added to form a second sensor signal.

Referring to Figures 12A and B, the detector 110 such as the detector as described with respect to Figures 1 to 11 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 118, 120. For example, the detector 110 may comprise a plurality of optical sensors. The detector 110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by the at least two optical sensors 118, 120. In one embodiment, the light beam 116 may illuminate the at least two optical sensors 118, 120 with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 132 may be configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors 118, 120. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 41 A, the light beam 116 propagating from the object 112 to the detector 110 may illuminate the at least two optical sensors 118, 120 with at least one line pattern 2186. The line pattern 204 may be generated by the object 112, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising at least one illumination line pattern. A1 may correspond to an area with a full line width of the line pattern 204. The line pattern 204 on the at least two optical sensors 118, 120 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a line width on the at least two optical sensors 118, 120 is increased. A2 may be a central area of the line pattern 204 on the at least two optical sensors 118, 120. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. Figure 12B show an embodiment, wherein the detector 110 may comprise at least two rows of optical sensors 118, 120, wherein the rows are arranged on top of each other. The line pattern 204 may be imaged parallel to the rows such that the center of the line pattern is fixed or moves only little relative to a dividing line 206 of the rows. The A1 may correspond to an area above the dividing line 206 and A2 may be an area below the dividing line.

Figures 13A and 13B show further embodiments of the detector 110 according to the present invention comprising at least one bi-cell. The illumination source 136, such as a laser source, may generate the illumination light beam 138 illuminating object 112. The reflected light beam 116 may propagate from the object 112 to the transfer device 128 and may impinge on the bi-cell of optical sensors 118, 120. In Figure 13A a side view is shown and in Figure 13B a front view is shown. The detector 110 may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the bi-cell in Figures 45 A and B may be adapted to generate a so called FiP signal. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. The negative FiP effect may be used to tune small image effects at high

distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

List of reference numbers

110	Detector
112	Object
114	beacon device
115	matrix
116	light beam
118	first optical sensor
118a	first optical sensor
119	mask
120	second optical sensor
120a	second optical sensor
121a	light sensitive area
122	first light-sensitive area
124	second light-sensitive area
126	optical axis
128	transfer device
130	focal point
132	evaluation device
134	detector system
136	illumination source
138	illumination light beam
140	reflective element
142	divider
144	position evaluation device
146	camera
148	human-machine interface
150	entertainment device
152	tracking system
154	scanning system
156	connector
158	housing
160	control device
162	User
164	opening
166	direction of view
168	coordinate system
170	machine
172	track controller
174	Array
176	optical sensor
178	quadrant photodiode
180	geometrical center of every

182	geometrical center of first optical sensor
184	geometrical center of second optical sensor
186	light spot
188	actuator
190	diaphragm
192	readout device for optical storage media
194	direction of movement
196	direction of movement
198	curve
200	curve
202	baseline
204	Line pattern
206	Dividing line

Patent claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) having
 - 5 - at least two optical sensors (118, 120, 176), each optical sensor (118, 120, 176) having a light-sensitive area (122, 124), wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110) by different spatial offsets, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light beam (116) propagating from the object (112) to the detector (110); and
 - at least one evaluation device (132) being configured for determining at least one longitudinal coordinate z of the object (112) by combining the at least two sensor signals.
- 15 2. The detector (110) according to the preceding claim, wherein the optical sensors (118, 120, 176) are part of a sensor array (174), wherein a geometrical center (180) of the sensor array (174) is offset from the optical axis (126).
- 20 3. The detector (110) according to the preceding claim, wherein the sensor array (174) is movable relative to the optical axis (126).
4. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for, firstly, determining the transversal position of a light spot (186) generated by the light beam (116) on the sensor array (174) by using the sensor signals and for, secondly, moving the sensor array (174) relative to the optical axis (126) until the light spot (186) is off-centered.
- 25 5. The detector (110) according to any one of the preceding claims, wherein the optical sensors (118, 120, 176) are partial diodes of a quadrant diode (178), with a geometrical center (180) of the quadrant diode (178) being off-centered from the optical axis (126) of the detector (110).
- 30 6. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for deriving at least one quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the evaluation device (132) is further configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
- 35 7. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the at least one quotient signal Q and the longitudinal coordinate.
- 40

8. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (136), wherein the illumination source (136) is adapted to illuminate the object (112) with at least one illumination light beam, wherein the illumination source (136) is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis (126).
9. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises the at least one illumination source (136), wherein the illumination source (136) and the optical axis (126) are separated by a small baseline (202).
10. A detector system (134) for determining a position of at least one object (112), the detector system (134) comprising at least one detector (110) according to any one of the preceding claims, the detector system (134) further comprising at least one beacon device (114) adapted to direct at least one light beam (116) towards the detector (110), wherein the beacon device (114) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
11. A human-machine interface (148) for exchanging at least one item of information between a user (162) and a machine (170), wherein the human-machine interface (148) comprises at least one detector system (134) according to the preceding claim, wherein the at least one beacon device (114) is adapted to be at least one of directly or indirectly attached to the user (162) and held by the user (162), wherein the human-machine interface (148) is designed to determine at least one position of the user (162) by means of the detector system (134), wherein the human-machine interface (148) is designed to assign to the position at least one item of information.
12. An entertainment device (150) for carrying out at least one entertainment function, wherein the entertainment device (150) comprises at least one human-machine interface (148) according to the preceding claim, wherein the entertainment device (150) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (148), wherein the entertainment device (150) is designed to vary the entertainment function in accordance with the information.
13. A tracking system (152) for tracking a position of at least one movable object (112), the tracking system (152) comprising at least one detector system (134) according to any one of the preceding claims referring to a detector system (134), the tracking system (152) further comprising at least one track controller (172), wherein the track controller (172) is adapted to track a series of positions of the object (112) at specific points in time.
14. A scanning system (154) for determining a depth profile of a scenery, the scanning system (154) comprising at least one detector (110) according to any of the preceding claims referring to a detector (110), the scanning system (154) further comprising at least one illumination source (136) adapted to scan the scenery with at least one light beam (116).

15. A camera (146) for imaging at least one object (112), the camera (146) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
16. A readout device (192) for optical storage media, the readout device (192) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
17. A method for determining a position of at least one object (112) by using at least one detector (110), the method comprising the following steps:
- providing at least two optical sensors (118, 120, 176), each optical sensor (118, 120, 176) having a light-sensitive area (122, 124), wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110) by different separations, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light;
 - illuminating the at least two light-sensitive areas (122, 124) of the at least two optical sensors (118, 120, 176) of the detector (110) with a light beam (116) propagating from the object (112) to the detector (110) and, thereby, generating at least two sensor signals; and
 - evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object (112).
18. The method according to the preceding claim, wherein the evaluating the sensor signals comprises deriving at least one quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining the longitudinal coordinate comprises evaluating the quotient signal Q .
19. A use of the detector (110) according to any one of the preceding claims relating to a detector (110), for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface (148) application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use for calibration of a distance measurement; a use in combination with optical data storage and readout.

FIG.1

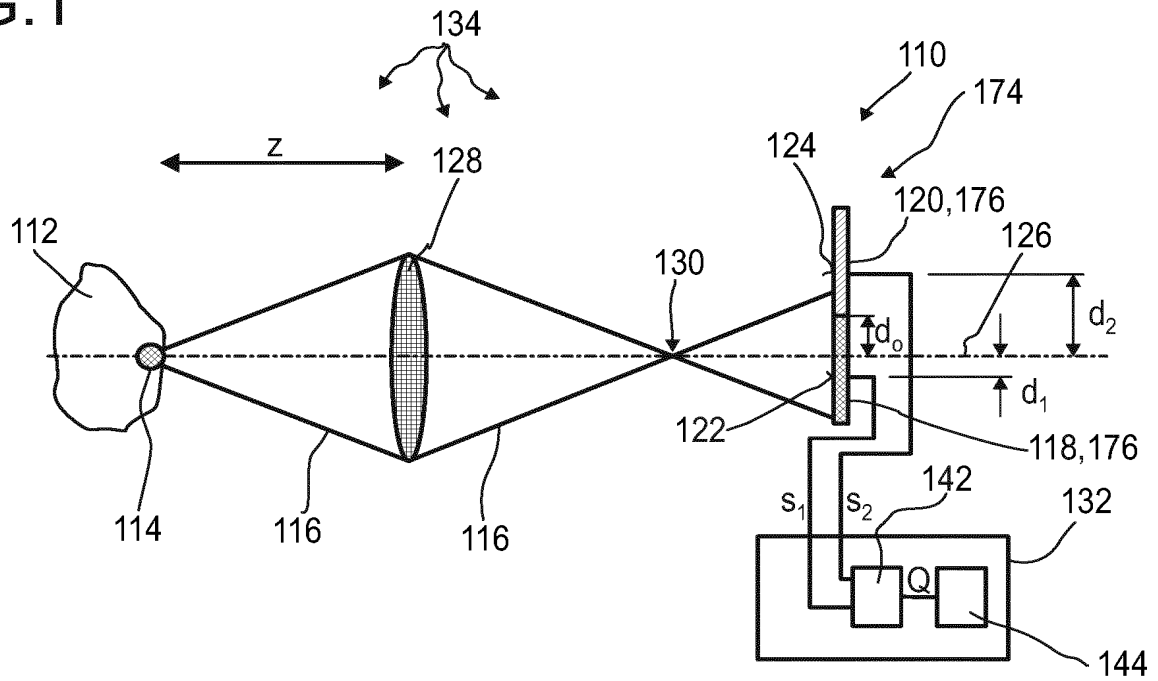


FIG.2

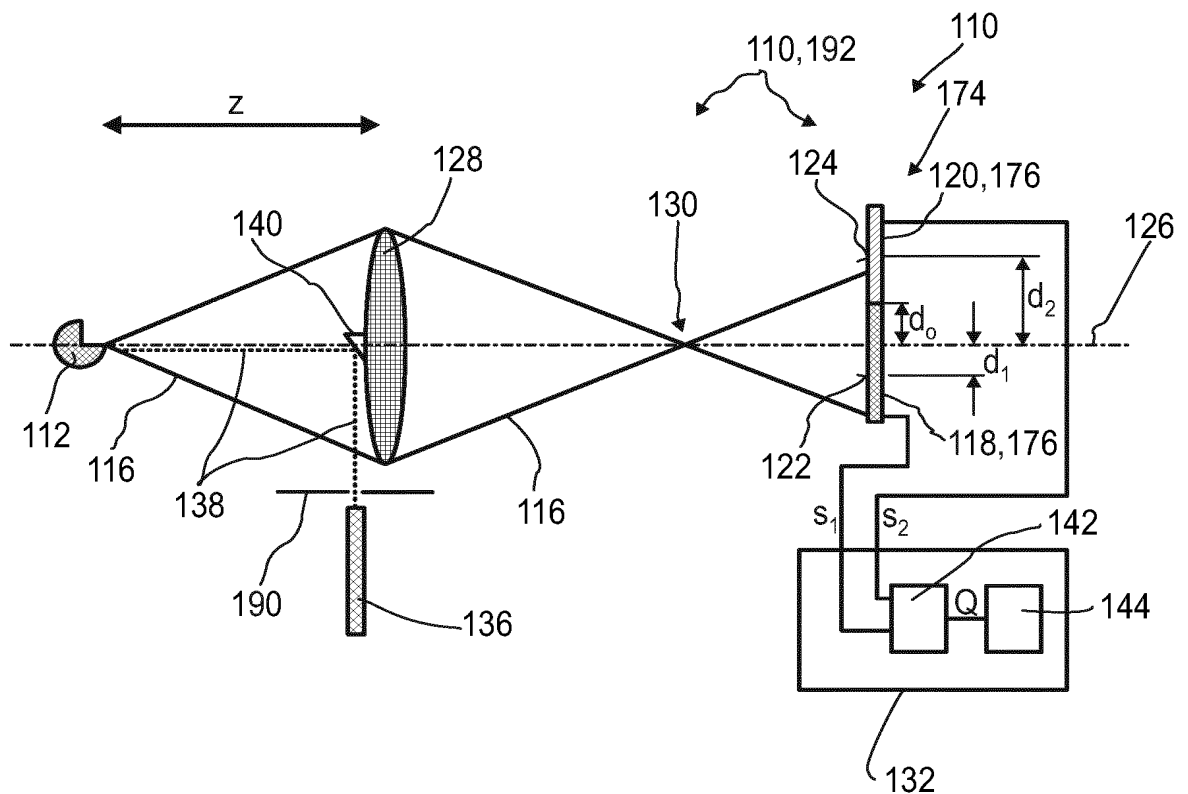


FIG.3

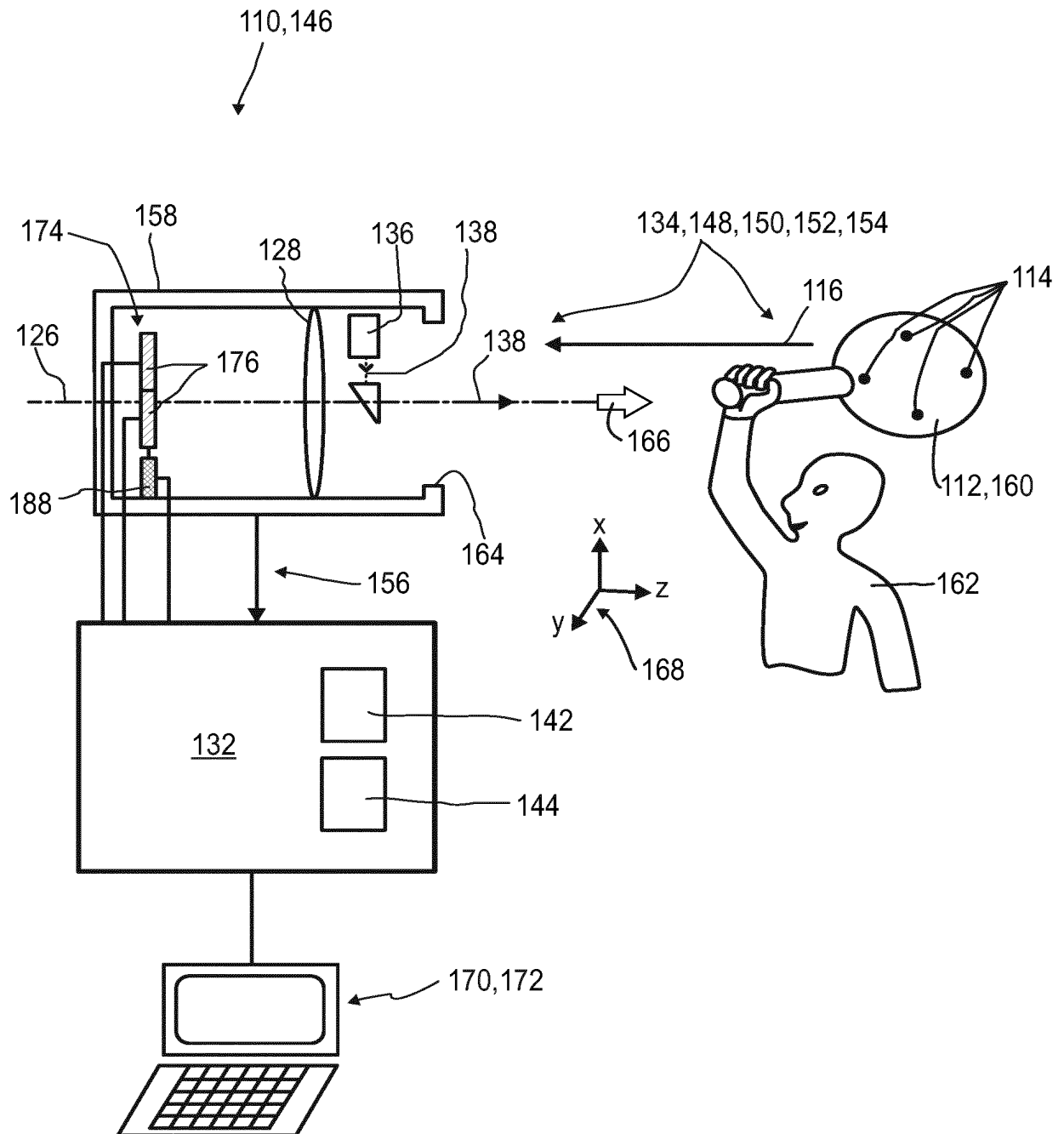


FIG.6

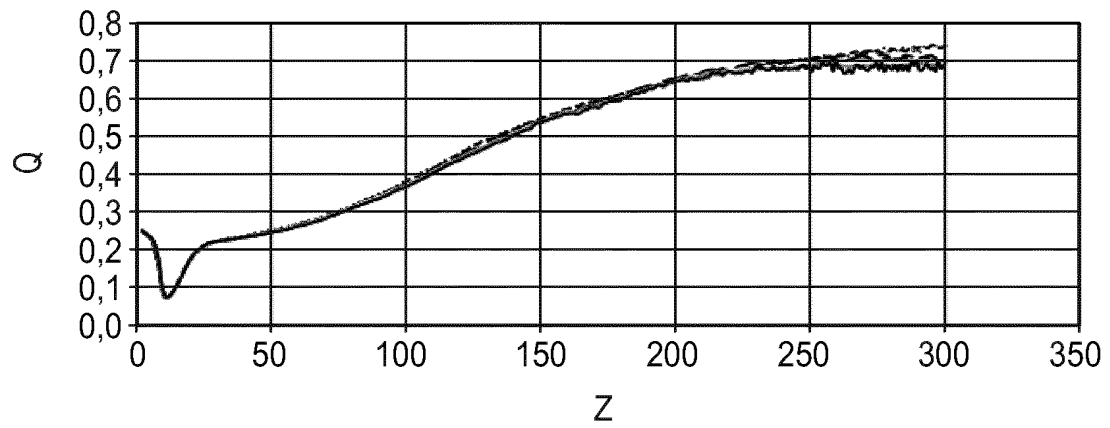


FIG.7

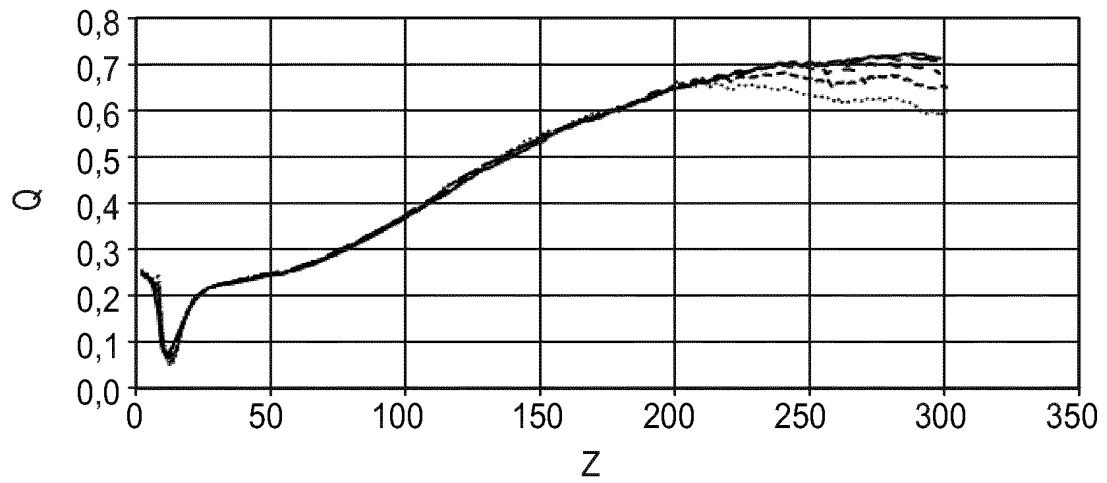


FIG.8

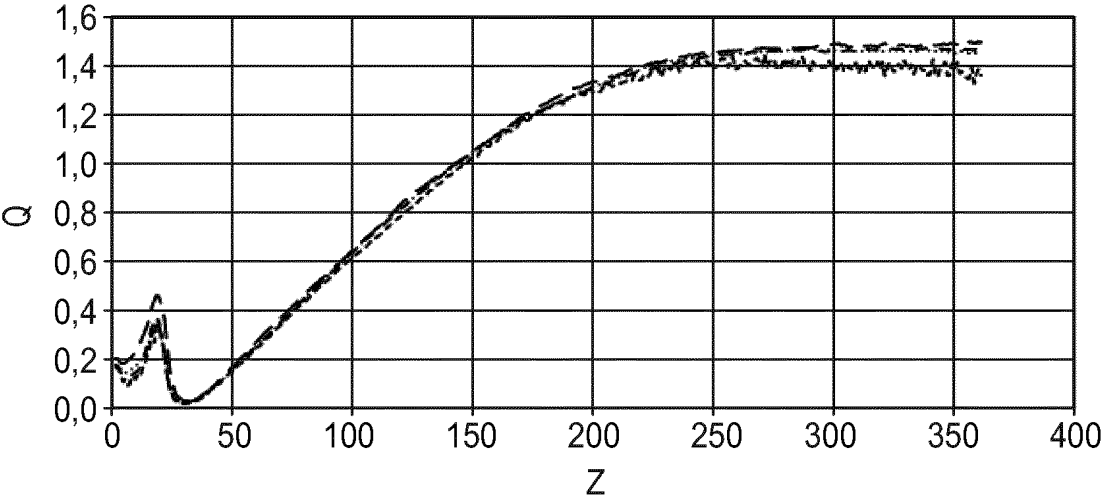


FIG.9A

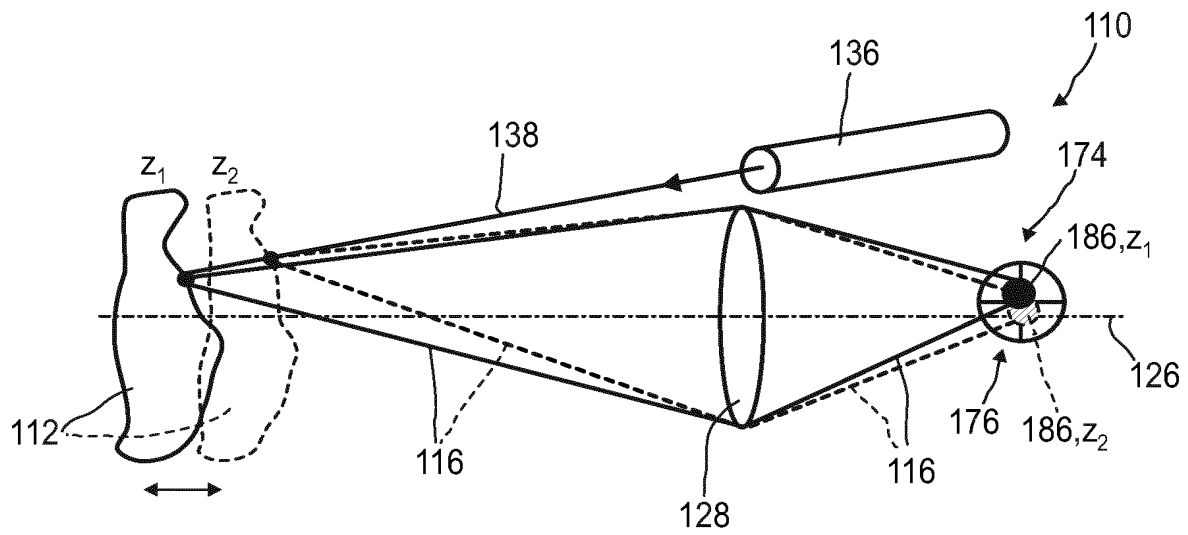


FIG.9B

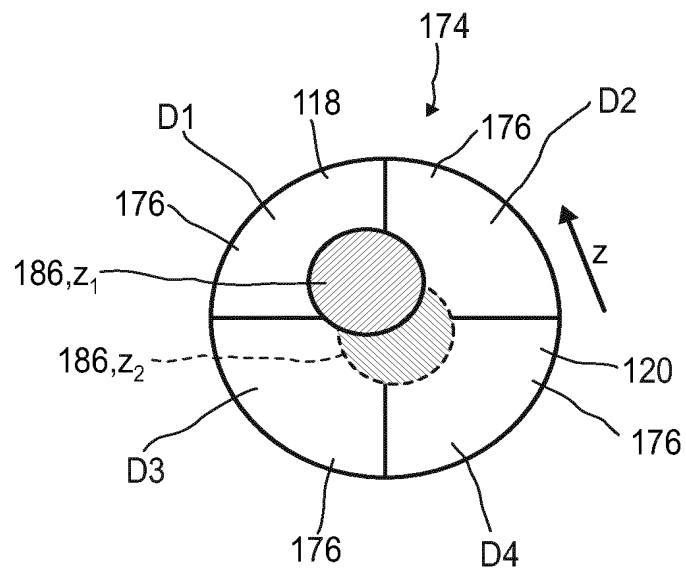


FIG.9C

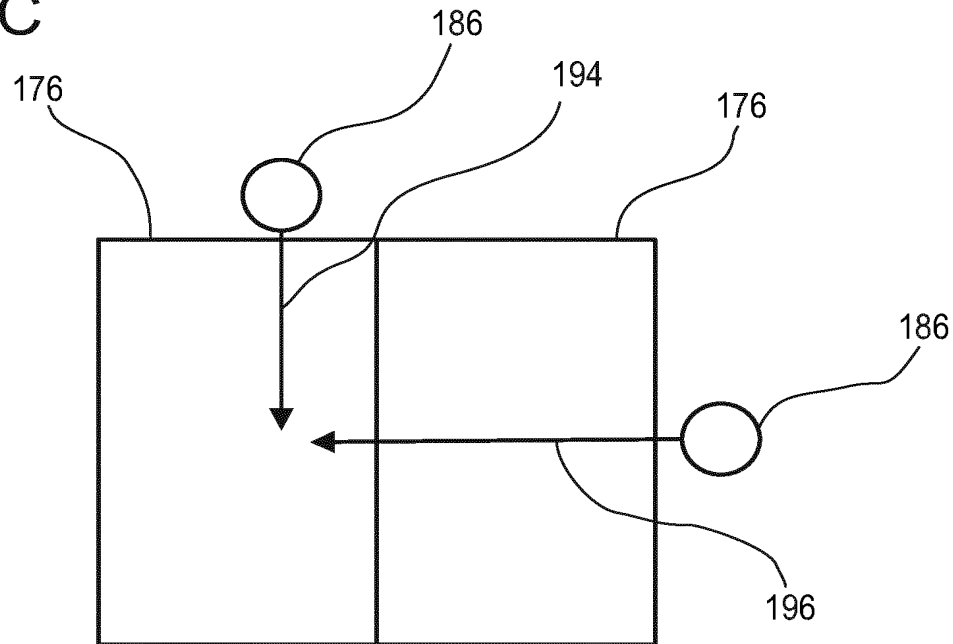


FIG.9D

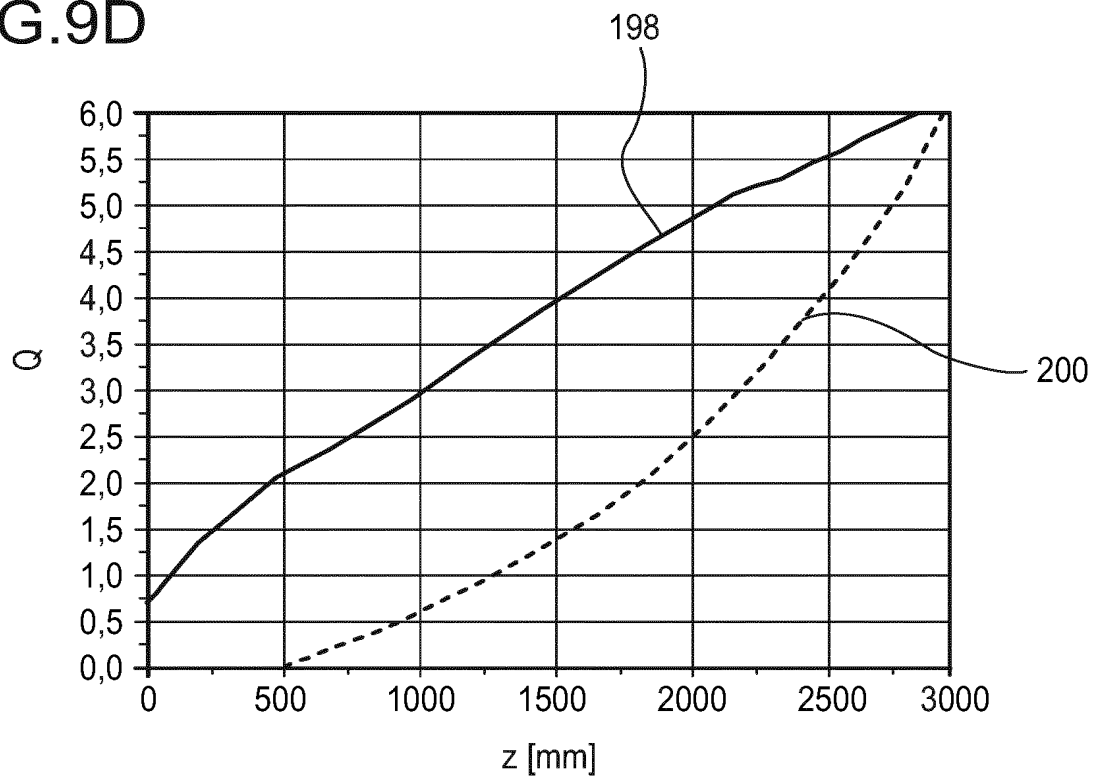


FIG.10A

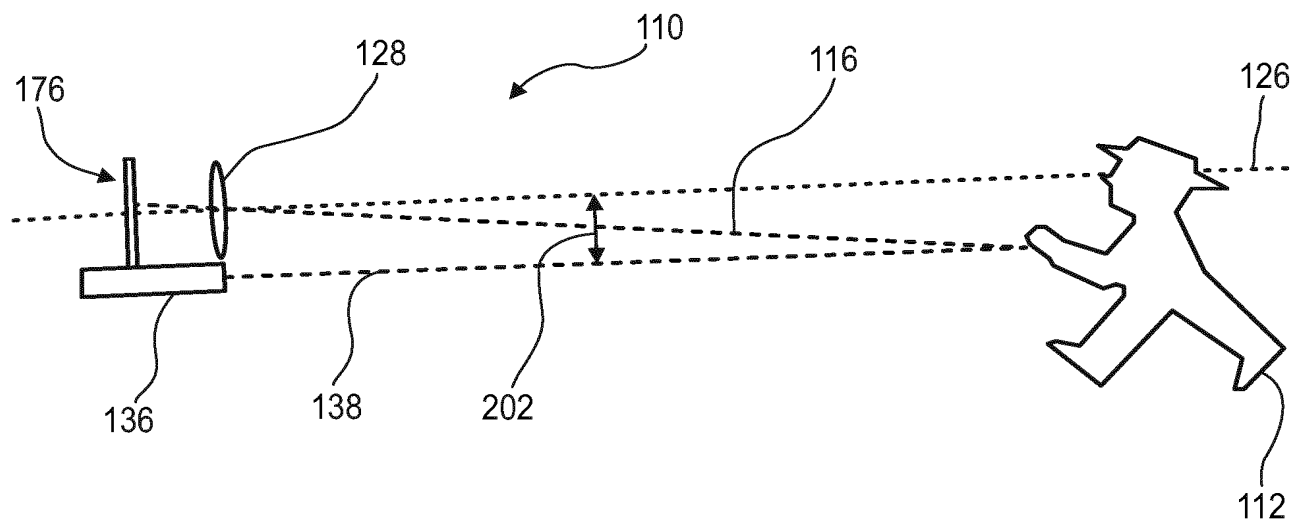


FIG.10B

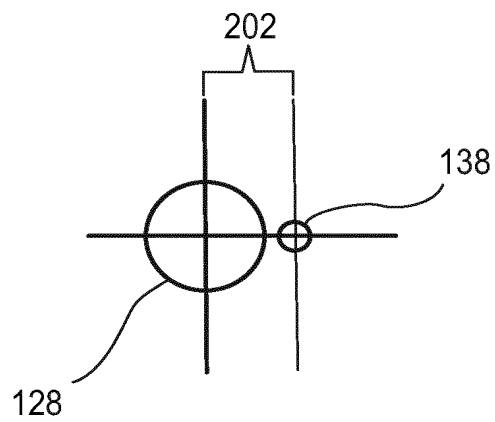


FIG.11

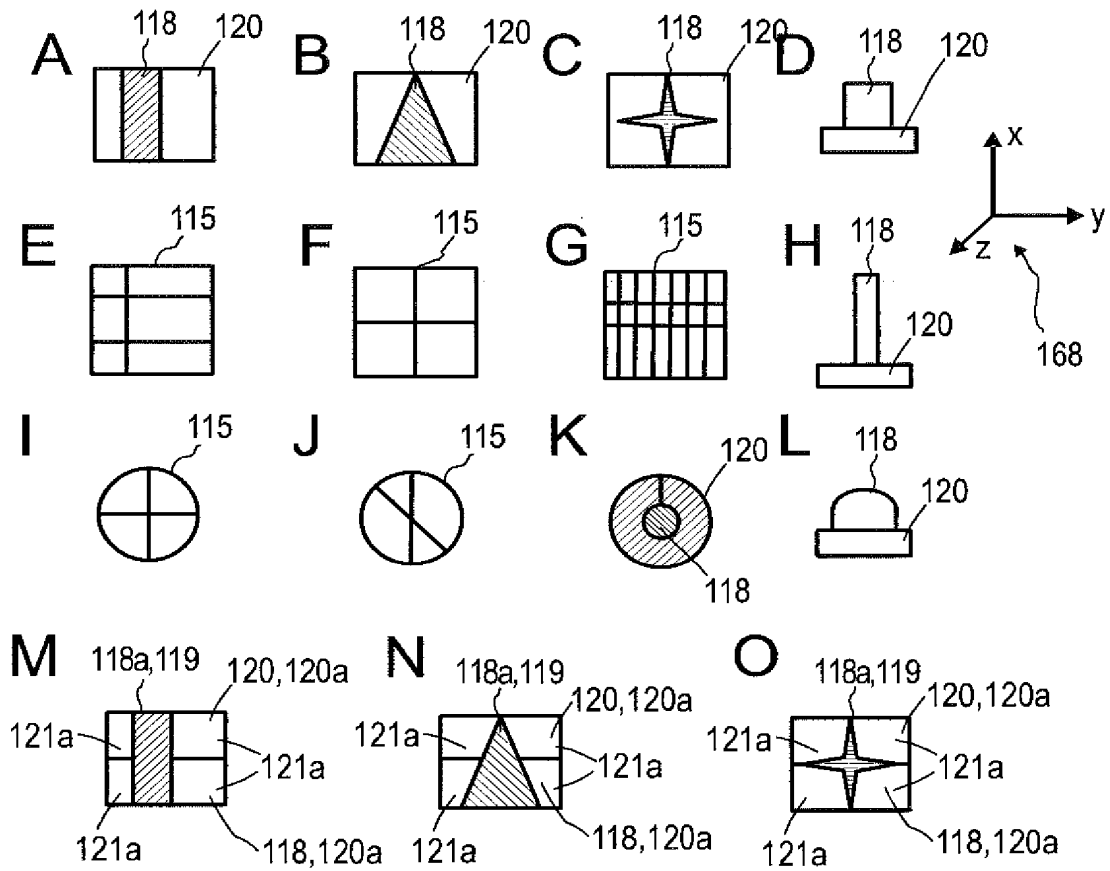


FIG.12A

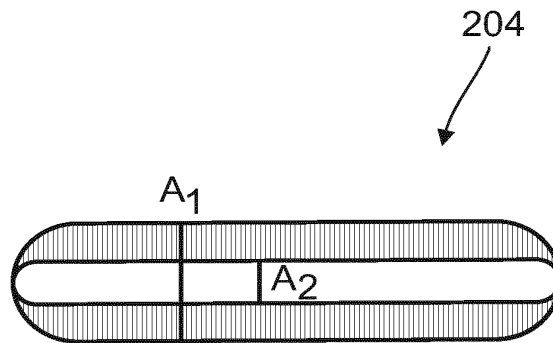


FIG.12B

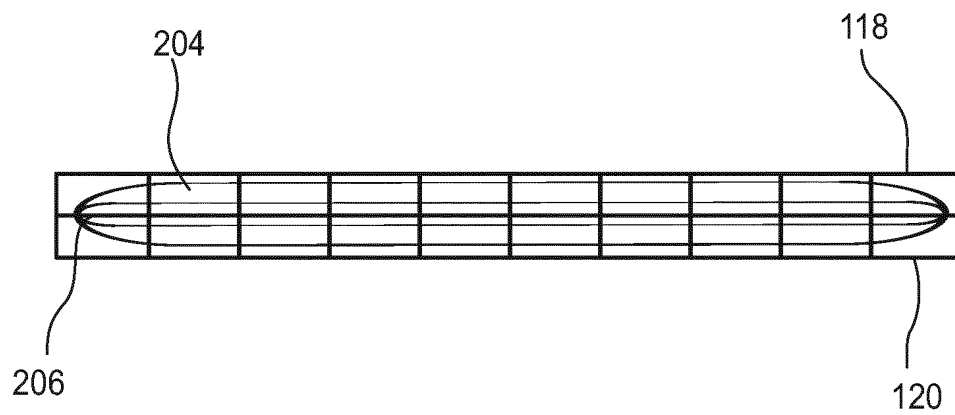


FIG.13A

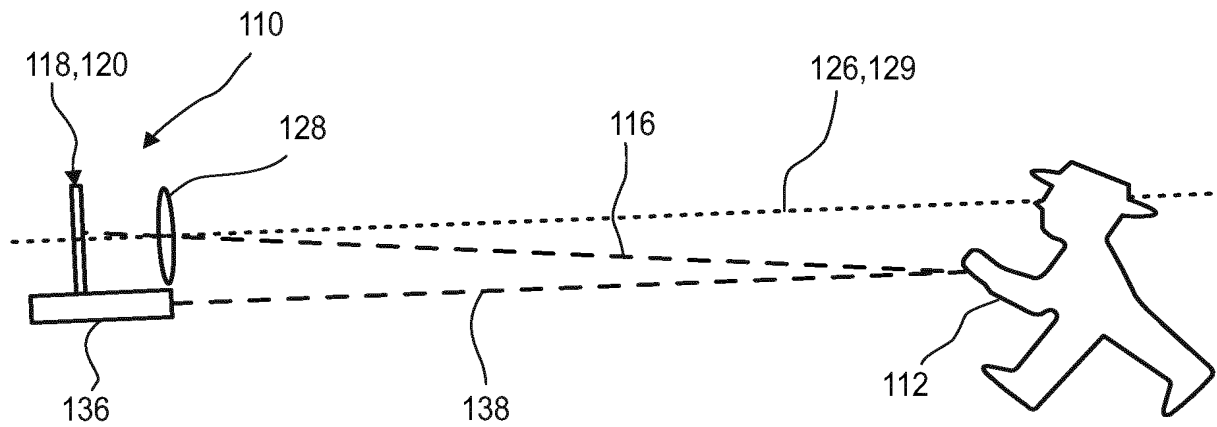
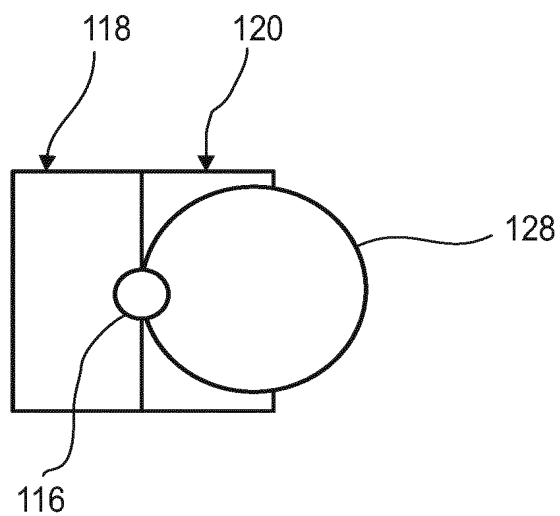


FIG.13B



INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2017/079558

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S7/481 G01S7/48 G01S5/16 G01S11/12
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data, INSPEC

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	EP 2 040 097 A1 (SICK AG [DE]) 25 March 2009 (2009-03-25) paragraphs [0001] - [0003] paragraph [0008] paragraphs [0019] - [0030] figures 1, 2	1-19
A	----- WO 2016/092454 A1 (BASF SE [DE]; BASF CHINA CO LTD [CN]) 16 June 2016 (2016-06-16) page 3, lines 1-13 page 54, line 31 - page 55, line 3 page 58, line 32 - page 61, line 25 page 66, line 41 - page 67, line 20 page 71, lines 24-32 -----	10-16,19



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

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"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

7 February 2018

Date of mailing of the international search report

15/02/2018

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2
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Authorized officer

Hirsch, Stefanie

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2017/079558

Patent document cited in search report		Publication date	Patent family member(s)	Publication date
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			DE 202007013315 U1	22-11-2007
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			KR 20170094197 A	17-08-2017
			US 2017363465 A1	21-12-2017
			WO 2016092454 A1	16-06-2016



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APPLICATION #
18/264,326

RECEIPT DATE / TIME
10/02/2023 04:29:26 PM ET

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27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE Utility - U.S. National Stage under 35 USC 371

PATENT # -

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CUSTOMER # 76656

FIRST NAMED INVENTOR Peter SCHILLEN

INTL. APPLICATION # -

INTL. FILING DATE -

CORRESPONDENCE ADDRESS

AUTHORIZED BY Christopher Goff

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-10-02 Request for Early Publication 27843-2262.pdf	1	Request for Early Publication	96 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2023-10-02 Request for Early Publication 27843-2262.pdf	80AA71DDB70A5FE7A53E8069D05EA412C8CAA5B8E7DBF91D3013DD93A8CEE1DA17703BDD0290D44B225257BE1830965F0293A36D955761976BA2C17DAE8F80C

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Inventor:	Peter Schillen	Art Unit:	TBD
Serial No.:	18/264,326	Confirmation No.:	2486
Date Filed:	August 8, 2023	Examiner:	TBD

Title: OPTICAL SKIN DETECTION FOR FACE UNLOCK

October 2, 2023

REQUEST FOR EARLY PUBLICATION

Commissioner for Patents
Alexandria, VA 22313

Commissioner:

Applicant hereby requests early publication of the above-reference patent application, pursuant to 37 CFR 1.215 and 37 CFR 1.219.

Applicant authorizes the Commissioner to charge any fees due in connection with this response to Deposit Account No. 01-2384 in the name of Armstrong Teasdale LLP. Please reference docket number 27843-2262.

Respectfully submitted,

/Christopher M. Goff/

Christopher M. Goff, Reg. No. 41,785
Armstrong Teasdale LLP
7700 Forsyth Blvd., Suite 1800
St. Louis, Missouri 63105
(314) 621-5070

VIA EFS

MULTIPLE DEPENDENT CLAIM FEE CALCULATION SHEET Substitute for Form PTO-1360 (For use with Form PTO/SB/06)							Application Number 18/264,326		Filing Date				
							Applicant(s) Peter SCHILLEN						
							* May be used for additional claims or amendments						
CLAIMS	AS FILED		AFTER FIRST AMENDMENT		AFTER SECOND AMENDMENT			*		*		*	
	Indep	Depend	Indep	Depend	Indep	Depend		Indep	Depend	Indep	Depend	Indep	Depend

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Total Indep	1		1			
Total Depend	16		15			
Total Claims	17		16		0	

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
18/264,326

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	540
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(i))	16 minus 20 =	*			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1660

APPLICATION AS AMENDED - PART II

	(Column 1)	(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY		
		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)		ADDITIONAL FEE(\$)	RATE(\$)	ADDITIONAL FEE(\$)
AMENDMENT A	Total (37 CFR 1.16(i))	*	Minus **	=	x =		OR	x =	
	Independent (37 CFR 1.16(h))	*	Minus ***	=	x =		OR	x =	
	Application Size Fee (37 CFR 1.16(s))						OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE		
AMENDMENT B	Total (37 CFR 1.16(i))	*	Minus **	=	x =		OR	x =	
	Independent (37 CFR 1.16(h))	*	Minus ***	=	x =		OR	x =	
	Application Size Fee (37 CFR 1.16(s))						OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE		

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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United States Patent and Trademark Office
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/264,326	08/04/2023		1660	27843-2262 202595US01	16	1

CONFIRMATION NO. 2486

FILING RECEIPT



0000000063708846

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Date Mailed: 10/27/2023

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

Peter SCHILLEN, Ludwigshafen am Rhein, GERMANY;
Benjamin GUTHIER, Ludwigshafen am Rhein, GERMANY;
Friedrich SCHICK, Ludwigshafen am Rhein, GERMANY;
Manuel GUENTHER, Zuerich, SWITZERLAND;
Lars DIESSELBERG, Karlsruhe, GERMANY;
Christian LENNARTZ, Ludwigshafen am Rhein, GERMANY;

Applicant(s)

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

Power of Attorney: The patent practitioners associated with Customer Number 76656

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/EP2022/053941 02/17/2022

Foreign Applications (You may be eligible to benefit from the Patent Prosecution Highway program at the USPTO. Please see <http://www.uspto.gov> for more information.)

EUROPEAN PATENT OFFICE (EPO) 21157800.0 02/18/2021 No Access Code Provided

Permission to Access Application via Priority Document Exchange: Yes

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/264,326**

Projected Publication Date: 02/01/2024

Non-Publication Request: No

Early Publication Request: No
Title

OPTICAL SKIN DETECTION FOR FACE UNLOCK
OPTICAL SKIN DETECTION FOR FACE
UNLOCK

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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U.S. APPLICATION NO.	FIRST NAMED INVENTOR	ATTY. DOCKET NO.
18/264,326	Peter SCHILLEN	27843-2262 202595US01
76656 Patent Docket Department Armstrong Teasdale LLP 7700 Forsyth Boulevard Suite 1800 St. Louis, MO 63105		INTERNATIONAL APPLICATION NO. PCT/EP2022/053941
		L.A. FILING DATE 02/17/2022
		PRIORITY DATE 02/18/2021

CONFIRMATION NO. 2486
371 ACCEPTANCE LETTER



OC00000063708851

Date Mailed: 10/27/2023

NOTICE OF ACCEPTANCE OF APPLICATION UNDER 35 U.S.C 371 AND 37 CFR 1.495

The applicant is hereby advised that the United States Patent and Trademark Office, in its capacity as a Designated / Elected Office (37 CFR 1.495), has ACCEPTED the above identified international application for national patentability examination in the United States Patent and Trademark Office.

The United States Application Number assigned to the application is shown above. A Filing Receipt will be issued for the present application in due course. **THE DATE APPEARING ON THE FILING RECEIPT AS THE "FILING DATE or 371(c) DATE" IS THE DATE ON WHICH THE LAST OF THE 35 U.S.C. 371 (c)(1) and (c)(2) REQUIREMENTS HAS BEEN RECEIVED IN THE OFFICE. THIS DATE IS SHOWN BELOW.** The filing date of the above identified application is the international filing date of the international application (Article 11(3) and 35 U.S.C. 363)

08/04/2023

DATE OF RECEIPT OF 35 U.S.C.
371(c)(1) and (c)(2) REQUIREMENTS

The following items have been received:

- Request for Immediate Examination filed on 08/04/2023
- Information Disclosure Statements filed on 08/04/2023
- Copy of the International Search Report filed on 08/04/2023
- Copy of the International Application filed on 08/04/2023
- Inventor's Oath or Declaration filed on 08/04/2023
- Preliminary Amendments filed on 08/04/2023
- Power of Attorney filed on 08/04/2023
- Drawings filed on 08/04/2023
- Application Data Sheet (37 CFR 1.76) filed on 08/04/2023
- Authorization to Permit Access filed on 08/04/2023
- Authorize Access to Search Results filed on 08/04/2023
- Priority Documents filed on 08/04/2023
- U.S. Basic National Fees filed on 08/04/2023

Applicant is reminded that any communications to the United States Patent and Trademark Office must be mailed to the address given in the heading and include the U.S. application no. shown above (37 CFR 1.5)

LAMONT M HUNTER

Telephone: (571) 272-0308



United States Patent and Trademark Office

Under Secretary of Commerce for Intellectual Property and
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A handwritten signature in black ink, reading "Kathi Vidal".

Kathi Vidal

Under Secretary of Commerce for Intellectual Property and
Director of the United States Patent and Trademark Office

A handwritten signature in black ink, reading "Derrick Brent".

Derrick Brent

Deputy Under Secretary of Commerce for Intellectual Property and
Deputy Director of the United States Patent and Trademark Office



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/264,326

RECEIPT DATE / TIME
10/27/2023 12:47:39 PM ET

ATTORNEY DOCKET #
27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCKOPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE Utility - U.S. National Stage under 35 USC 371

PATENT # -

CONFIRMATION # 2486

FILED BY Barbara McGregor

PATENT CENTER # 63098371

FILING DATE 08/04/2023

CUSTOMER # 76656

FIRST NAMED INVENTOR Peter SCHILLEN

INTL. APPLICATION # -

INTL. FILING DATE -

CORRESPONDENCE ADDRESS -

AUTHORIZED BY Christopher Goff

Documents

TOTAL DOCUMENTS: 2

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-10-27 Request for Corrected OFR 27843-2262.pdf	1	Request for Corrected Filing Receipt	102 KB
2023-10-27 Filing Receipt - Marked 27843-2262.pdf	4	Request for Corrected Filing Receipt	299 KB

Digest

DOCUMENT

MESSAGE DIGEST(SHA-512)

2023-10-27 Request for Corrected OFR 27843-2262.pdf	60031894A2C0E90D7F008F0D7A73812E7BCB4C6F3874E8A3E 4830FA7AB4ECF8E271FB00C6DCD0968386E0C81A8C6B1F1D 4AEC83FFCC3E4828DEE9ADC8E535387
2023-10-27 Filing Receipt - Marked 27843-2262.pdf	9660D1C697D720D9BF3DE3CB87BA0E1CF8E5C25FA906BD7F 9835013391914B3255C3D3F056E4C2606C0ADC074CDD311E7 7427E417F21918C5EE05A4DB7796376

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

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National Stage of an International Application under 35 U.S.C. 371

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New International Application Filed with the USPTO as a Receiving Office

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/264,326	08/04/2023		1660	27843-2262 202595US01	16	1

CONFIRMATION NO. 2486

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

FILING RECEIPT



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Date Mailed: 10/27/2023

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Peter SCHILLEN, Ludwigshafen am Rhein, GERMANY;
Benjamin GUTHIER, Ludwigshafen am Rhein, GERMANY;
Friedrich SCHICK, Ludwigshafen am Rhein, GERMANY;
Manuel GUENTHER, Zuerich, SWITZERLAND;
Lars DIESSELBERG, Karlsruhe, GERMANY;
Christian LENNARTZ, Ludwigshafen am Rhein, GERMANY;

Applicant(s)

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

Power of Attorney: The patent practitioners associated with Customer Number 76656

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/EP2022/053941 02/17/2022

Foreign Applications (You may be eligible to benefit from the Patent Prosecution Highway program at the USPTO. Please see <http://www.uspto.gov> for more information.)

EUROPEAN PATENT OFFICE (EPO) 21157800.0 02/18/2021 No Access Code Provided

Permission to Access Application via Priority Document Exchange: Yes

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/264,326**

Projected Publication Date: 02/01/2024

Non-Publication Request: No

Early Publication Request: No
Title

OPTICAL SKIN DETECTION FOR FACE UNLOCK
UNLOCK

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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In the United States Patent & Trademark Office

First Inventor:	Peter SCHILLEN	Art Unit:	
Serial No.:	18/264,326	Confirmation No.:	2486
Date Filed:	August 4, 2023	Examiner:	
Title:	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

October 27, 2023

REQUEST FOR CORRECTED FILING RECEIPT

TO THE COMMISSIONER FOR PATENTS:

Applicant requests that a new Filing Receipt be issued in the above referenced application. Upon review of the Filing Receipt dated October 27, 2022, Applicant noted that the title of the application is incorrect. Applicant noted the mistake after filing the application and filed a Corrected Application Data Sheet on August 9, 2023 correcting the application title. A notated copy of the Filing Receipt is attached hereto.

Although it is believed that there is no fee due for this action, the Commissioner is hereby authorized to charge Deposit Account No. 01-2384 for any outstanding fees incurred in connection with this request. Please reference docket number 27843-2262.

Respectfully submitted,

/Christopher M. Goff/

Christopher M. Goff
USPTO Registration 41,785
ARMSTRONG TEASDALE LLP
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CMG/EMF/ZSW
VIA EFS



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/264,326	08/04/2023		1660	27843-2262 202595US01	16	1

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

CONFIRMATION NO. 2486
REPLACEMENT FILING RECEIPT



0000000063890203

Date Mailed: 11/01/2023

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Peter SCHILLEN, Ludwigshafen am Rhein, GERMANY;
Benjamin GUTHIER, Ludwigshafen am Rhein, GERMANY;
Friedrich SCHICK, Ludwigshafen am Rhein, GERMANY;
Manuel GUENTHER, Zuerich, SWITZERLAND;
Lars DIESSELBERG, Karlsruhe, GERMANY;
Christian LENNARTZ, Ludwigshafen am Rhein, GERMANY;

Applicant(s)

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

Power of Attorney: The patent practitioners associated with Customer Number 76656

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/EP2022/053941 02/17/2022

Foreign Applications (You may be eligible to benefit from the Patent Prosecution Highway program at the USPTO. Please see <http://www.uspto.gov> for more information.)

EUROPEAN PATENT OFFICE (EPO) 21157800.0 02/18/2021 No Access Code Provided

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 10/24/2023

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/264,326**

Projected Publication Date: 02/01/2024

Non-Publication Request: No

Early Publication Request: No

Title

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/264,326	08/04/2023	Peter SCHILLEN	27843-2262 202595US01

CONFIRMATION NO. 2486

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

PUBLICATION NOTICE



0000006778172

Date Mailed: 02/02/2024

Title:OPTICAL SKIN DETECTION FOR FACE UNLOCK

Publication No.US-2024-0037201-A1

Publication Date:02/01/2024

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

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Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Notice of References Cited	Application/Control No. 18/264,326		Applicant(s)/Patent Under Reexamination SCHILLEN et al.	
	Examiner PHUOC TRAN		Art Unit 2668	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-20210049391-A1	2021-02-18	Zou; Xuan	G06T7/62	1/1
*	B	US-20220222464-A1	2022-07-14	Diedrich; Jonathan	G06V40/166	1/1
	C					
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FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
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NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 18/264,326	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

CPC - Searched*		
Symbol	Date	Examiner
G06V40/40, G01S17/894, G06F21/32, G06T7/50, G06V10/141, G06V10/145, G06V10/25, G06V10/54, G06V10/60, G06V10/764, G06V10/82, G06V40/165, G06V40/168, G06V40/172, G06T2207/30201	09/07/2024	PT

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor name search	09/07/2024	PT
See search history printout(s)	09/07/2024	PT

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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Doc code: IDS

PTO/SB/08a (01-10)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 07/31/2012. OMB 0651-0031

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		18264326	
	Filing Date		2023-08-04	
	First Named Inventor	Peter Schillen		
	Art Unit			
	Examiner Name			
	Attorney Docket Number		27843-2262 202595US01	

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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	
	1	20190213309	A1	2019-07-11	MORESTIN et al.		

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	1	2020187719	WO	A1	2020-09-24	TRINAMIX GMBH		☐
	2	2018091649	WO	A1	2018-05-24	TRINAMIX GMBH		☐
	3	2018091638	WO	A1	2018-05-24	TRINAMIX GMBH		☐

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

(Not for submission under 37 CFR 1.99)

Application Number	18264326
Filing Date	2023-08-04
First Named Inventor	Peter Schillen
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2262 202595US01

4	2018091640	WO	A2	2018-05-24	TRINAMIX GMBH		☐
5	2021170791	WO	A1	2021-09-02	TRINAMIX GMBH		☐
6	2021152070	WO	A1	2021-08-05	TRINAMIX GMBH		☐
7	2021214123	WO	A1	2021-10-28	TRINAMIX GMBH		☐

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	1	MASI, et al. "Deep Face Recognition: A Survey" 2018 31st SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI), IEEE, pp. 471-478 (2018)	☐
	2	"Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979, pp. 171-266 (1991)	☐
	3	YU et al, "Face Anti-Spoofing with Human Material Perception", arXiv.org, Cornell University Library, 201 Olin Library Cornell University, Ithaca, NY 14853, 20 pages, (2020)	☐

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EXAMINER SIGNATURE

Examiner Signature	/PHUOC TRAN/	Date Considered	09/20/2024
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		18264326
Filing Date		2023-08-04
First Named Inventor	Peter Schillen	
Art Unit		
Examiner Name		
Attorney Docket Number	27843-2262 202595US01	

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	18264326
Filing Date	2023-08-04
First Named Inventor	Peter Schillen
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2262 202595US01

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2023-09-25
Name/Print	Christopher M. Goff	Registration Number	41785

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Bibliographic Data

Application No: 18/264,326

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged:

/PHUOC TRAN/

Examiner's Signature

Initials

Title:

OPTICAL SKIN DETECTION FOR FACE UNLOCK

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
08/04/2023	382	2668	27843-2262 202595US01
RULE			

APPLICANTS

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY

INVENTORS

Peter SCHILLEN, Ludwigshafen am Rhein, GERMANY

Benjamin GUTHIER, Ludwigshafen am Rhein, GERMANY

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Manuel GUENTHER, Zuerich, SWITZERLAND

Lars DIESSELBERG, Karlsruhe, GERMANY

Christian LENNARTZ, Ludwigshafen am Rhein, GERMANY

CONTINUING DATA

This application is a 371 of PCT/EP2022/053941 02/17/2022

FOREIGN APPLICATIONS

EPO EP21157800.0 02/18/2021

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10/24/2023

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GERMANY

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Patent Docket Department

Armstrong Teasdale LLP

7700 Forsyth Boulevard

Suite 1800

St. Louis, MO 63105

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\$1,660

Doc code: IDS

PTO/SB/08a (01-22)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 05/31/2024. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		18264326	
	Filing Date		2023-08-04	
	First Named Inventor	Peter SCHILLEN		
	Art Unit			
	Examiner Name			
	Attorney Docket Number		27843-2262 202595US01	

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵	

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Examiner Name		
Attorney Docket Number		27843-2262 202595US01

	1	YU et al., "Face Anti-Spoofing with Human Material Perception", Arxiv.org, Cornell University Library, 20 pages (2020)	☐
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Examiner Signature	/PHUOC TRAN/	Date Considered	09/20/2024
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¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	18264326
Filing Date	2023-08-04
First Named Inventor	Peter SCHILLEN
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2262 202595US01

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached certification statement.

☐ The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2023-09-13
Name/Print	Christopher M. Goff	Registration Number	41785

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	282	((("SCHILLEN") near3 ("Peter")) OR ("GUTHIER") near3 ("Benjamin")) OR ("SCHICK") near3 ("Friedrich")) OR ("GUENTHER") near3 ("Manuel")) OR ("DIESELBERG") near3 ("Lars")) OR ("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/09/07 03:49 PM
L2	537832	(G06V40/40 or G01S17/894 or G06F21/32 or G06T7/50 or G06V10/141 or G06V10/145 or G06V10/25 or G06V10/54 or G06V10/60 or G06V10/764 or G06V10/82 or G06V40/165 or G06V40/168 or G06V40/172 or G06T2207/30201).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/09/07 03:50 PM
L3	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authentikat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:03 PM
L4	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authentikat\$4 OR identif\$8 OR	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:04 PM

L5	2	verifi\$8)).clm. 2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:05 PM
L6	0	2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:08 PM
L7	0	(image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:09 PM
L10	329	(image SAME (face OR facial OR skin) SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:14 PM

L11	120	OR skin)) 2 AND (image SAME (face OR facial OR skin) SAME (three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial OR skin))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:15 PM
L12	14	11 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:18 PM
L13	24	10 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:19 PM
L14	10	13 NOT 12	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:26 PM
L15	106	11 NOT 13	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:29 PM
L16	2	("20190213309" OR "20200082155").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2024/09/07 05:27 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Doc code: IDS

PTO/SB/08a (01-22)

Doc description: Information Disclosure Statement (IDS) Filed

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	First Named Inventor	Peter SCHILLEN		
	Art Unit			
	Examiner Name	TBD		
	Attorney Docket Number	27843-2262 202595US01		

U.S.PATENTS							Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	
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	1	20190213309	A1	2019-07-11	MORESTIN et al.		
	2	20200082155	A1	2020-03-12	KALSCHEUR et al.		

If you wish to add additional U.S. Published Application citation information please click the Add button.

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	111164610	CN	A	2020-05-15	SHENZHEN GOODIX TECH CO LTD	Appended with English machine translation	☒

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Art Unit		
Examiner Name	TBD	
Attorney Docket Number	27843-2262 202595US01	

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report and Written Opinion for corresponding PCT/EP2022/053941, mailed June 9, 2022, 12 pages	☐

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EXAMINER SIGNATURE

Examiner Signature	/PHUOC TRAN/	Date Considered	09/20/2024
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Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2023-08-04
Name/Print	Christopher M. Goff	Registration Number	41785

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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18/264,326

08/04/2023

Peter SCHILLEN

27843-2262 202595US01

2486

76656

7590

09/23/2024

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

EXAMINER

TRAN, PHUOC

ART UNIT

PAPER NUMBER

2668

NOTIFICATION DATE

DELIVERY MODE

09/23/2024

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

USpatents@atllp.com

Office Action Summary

Application No.

18/264,326

Applicant(s)

SCHILLEN et al.

Examiner

PHUOC TRAN

Art Unit

2668

AIA (FITF) Status

Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 04 August 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-16 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1-16 is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 04 August 2023 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☒ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Interpretation

2. A “non-transient computer-readable medium” in claim 14 is interpreted to be the same as a “non-transitory computer-readable medium” because the specification does not clearly define the “non-transient computer-readable medium”.
3. Claims 12-14 are considered as independent claims because they simply refer to other claims as a matter of short-hand drafting technique.

4. The following is a quotation of 35 U.S.C. 112(f):

(f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

5. The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

- (A) the claim limitation uses the term “means” or “step” or a term used as a substitute for “means” that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;
- (B) the term “means” or “step” or the generic placeholder is modified by functional language, typically, but not always linked by the transition word “for” (e.g., “means for”) or another linking word or phrase, such as “configured to” or “so that”; and
- (C) the term “means” or “step” or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word “means” (or “step”) in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word “means” (or “step”) in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word “means” (or “step”) are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word “means” (or “step”) are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

6. This application includes one or more claim limitations that do not use the word “means,” but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not preceded by a structural modifier. Such claim limitation(s) is/are: Each “unit” in claims 1-16.

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 101

7. 35 U.S.C. 101 reads as follows:

Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

8. Claim(s) 12, 13 is/are rejected under 35 U.S.C. 101 because the claimed invention is directed to non-statutory subject matter.

The claim 12 does not fall within at least one of the four categories of patent eligible subject matter because claim 12 is directed to a “computer program” without being stored on a non-transitory computer-readable medium. The software computer program per se does not fall within a statutory category.

The claim 13 does not fall within at least one of the four categories of patent eligible subject matter because claim 13 is directed to a "computer-readable storage medium". However, according to paragraph [0145] of the specification, the broadest reasonable interpretation of the "computer-readable storage medium" can encompass a transitory propagating signal which is non-statutory subject matter. See *In re Nuijten*, 500 F.3d 1346, 84 USPQ2d 1495 (Fed. Cir. 2007).

The examiner suggests amending the claim(s) to recite a "non-transitory computer-readable medium" storing a computer program or equivalent. Any amendment to the claims should be commensurate with its corresponding disclosure.

Claim Rejections - 35 USC § 103

9. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

10. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

11. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner

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to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

12. Claim(s) 1-3, 7, 12-16 is/are rejected under 35 U.S.C. 103 as being unpatentable over MORESTIN (US 2019/0213309) in view of Zou (US 2021/0049391).

As to claim 1, MORESTIN discloses a method for face authentication comprising the following steps:

a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one predefined or pre-determined geometrical feature characteristic for faces by using at least one processing unit (para. 0085);

b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin (para. 0036, 0039, 0050, 0053);

c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects (para. 0036, 0050); and

d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object (para. 0086).

MORESTIN is silent on using information of at least four of the reflection features to determine depth levels of 3d object.

Zou teaches using information of at least four of the reflection features to determine depth levels of 3d object (para. 0047, 0104).

It would have been obvious to one of ordinary skill in the art to incorporate Zou's teachings into MORESTIN in order to accurately determine 3D face of the object.

As to claim 2, the combination of MORESTIN and Zou discloses the method according to claim 1, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated (MORESTIN, para. 0081, 0087).

As to claim 3, the combination of MORESTIN and Zou discloses the method according claim 2, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database,

wherein the method comprises identifying the user by comparing the 2D image of the user with the first image (MORESTIN, para. 0087).

As to claim 7, the combination of MORESTIN and Zou discloses the method according to claim 1, wherein the illumination pattern comprises a periodic grid of laser spots (MORESTIN, para. 0051; Zou, para. 0047).

As to claims 12-16, these claims recite features similar to those discussed above. Therefore, they are rejected for reasons similar to those discussed above.

12. Claim(s) 4-6, 8, 9-11 is/are rejected under 35 U.S.C. 103 as being unpatentable over MORESTIN (US 2019/0213309) in view of Zou (US 2021/0049391) and further in view of Diedrich (US 2022/0222464).

As to claim 4, the combination of MORESTIN and Zou is silent on wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.

Diedrich teaches using machine learning to parametrize skin classification model, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input (para. 0056, 0057).

It would have been obvious to one of ordinary skill in the art to incorporate Diedrich's teachings into the combination of MORESTIN and Zou in order to quickly and accurately determine skin of the face.

As to claim 5, the combination of MORESTIN, Zou and Diedrich discloses the method according to claim 4, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model (Diedrich, para. 0056, 0057).

As to claim 6, the combination of MORESTIN, Zou and Diedrich discloses the method according to claim 4, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step at least one region specific parametrized skin classification model is used (Diedrich, para. 0055-0057, Zou, para. 0058, 0104).

As to claim 8, the combination of MORESTIN and Zou is silent wherein the illumination features have wavelengths in a near infrared (NIR) regime.

Diedrich teaches wherein the illumination features have wavelengths in a near infrared (NIR) regime having wavelengths of 940 nm (para. 0031).

It would have been obvious to one of ordinary skill in the art to incorporate Diedrich's teachings into the combination of MORESTIN and Zou in order to accurately determine skin of the face.

As to claim 9, the combination of MORESTIN, Zou and Diedrich discloses the method according to claim 8, wherein the illumination features have wavelengths of 940 nm (Diedrich, para. 0031).

As to claim 10, the combination of MORESTIN, Zou and Diedrich discloses the method according to claim 1, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) and/or for 3D detection in step c) (MORESTIN, para. 0036, 0039, 0050, 0053; Diedrich, para. 0056).

As to claim 11, the combination of MORESTIN, Zou and Diedrich discloses the method according to claim 1, wherein the camera comprises at least one near infrared camera (Diedrich, para. 0031).

Conclusion

13. Any inquiry concerning this communication or earlier communications from the examiner should be directed to PHUOC TRAN whose telephone number is (571)272-7399. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Vu Le can be reached on 571-272-7332. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/PHUOCTRAN/
Primary Examiner, Art Unit 2668

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 18/264,326	Filing Date 08/04/2023	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)	
AMENDMENT	11/13/2024	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA			
	Total (37 CFR 1.16(i))	* 16	Minus	** 20	= 0	x \$100 =	0	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$480 =	0	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE	0	
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =		
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =		
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE		
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE		
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/TRINA RIDDICK/		
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".								
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.								

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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PATENT AND TRADEMARK OFFICE

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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/264,326	11/13/2024 11:36:38 AM Z ET	27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	2486	FILED BY	Barbara McGregor
PATENT CENTER #	67945454	FILING DATE	08/04/2023
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents

TOTAL DOCUMENTS: 3

DOCUMENT		PAGES	DESCRIPTION	SIZE (KB)
2024-11-13 Response to Non-Final Office Action 27843-2262.pdf		8	-	193 KB
2024-11-13 Response to Non-Final Office Action 27843-2262-A....pdf	(1-1)	1	Amendment/Request for Reconsideration-After Non-Final Rejection	105 KB
2024-11-13 Response to Non-Final Office Action 27843-2262-CLM.pdf	(2-5)	4	Claims	142 KB
2024-11-13 Response to Non-Final Office Action 27843-2262-REM.pdf	(6-8)	3	Applicant Arguments/Remarks Made in an Amendment	149 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2024-11-13 Response to Non-Final Office Action 27843-2262.pdf	EA996396201719DAE61A8AF561DE7096022926583C92D97DE03EF0DA67926073449118A69FDFB8D21B20C5B12157097A1E29DA8EB5A03892C68229959F83EFB1
2024-11-13 Response to Non-Final Office Action 27843-2262-A....pdf	3E5BC9A6E8FF19F87B788018F3357B54880309C535F0C13DF365F430D239CEB96287B9E84D7AB4A6862945AC9FFC1158D62ED16497ABA0065A8058912341B6ED
2024-11-13 Response to Non-Final Office Action 27843-2262-CLM.pdf	2507D46D51FC36721532F66597DD91493AA51A79E627A48E3BFC6BB3EF88D1B8858F7F0C4858C5B9DE90012E88FE4358B9DBDA34FAC41FAAF0D656E1EBBB015E
2024-11-13 Response to Non-Final Office Action 27843-2262-REM.pdf	245D0532B00E36920ADA6A945CB9BCC90EE6EB33DCF1739A49DD3807D8A07ECEFO93E646819CF9C1612F0F94A29CD0D8BE7B3BDB2DA6FBC7B093D69C1880EA52

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT & TRADEMARK OFFICE

First Inventor:	Peter Schillen	Art Unit:	2668
Serial No.:	18/264,326	Confirmation No.:	2486
Date Filed:	August 4, 2023	Examiner:	Phuoc Tran
Title:	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

November 13, 2024

RESPONSE TO NON-FINAL OFFICE ACTION

Commissioner for Patents
Alexandria, VA 22313

Commissioner:

In response to the Non-Final Office Action dated September 23, 2024, please consider the following:

Amendments to the Claims begin on page 2.

Remarks begin on page 6.

Conclusion is on page 8.

AMENDMENTS TO THE CLAIMS

1. (Previously Presented) A method for face authentication comprising the following steps:
 - a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
 - b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
 - c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects; and

- d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.
2. (Previously Presented) The method according to claim 1, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.
 3. (Previously Presented) The method according claim 2, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.
 4. (Previously Presented) The method according to claim 1, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.
 5. (Previously Presented) The method according to claim 4, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.
 6. (Previously Presented) The method according to claim 4, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step at least one region specific parametrized skin classification model is used.
 7. (Previously Presented) The method according to claim 1, wherein the illumination pattern comprises a periodic grid of laser spots.

8. (Previously Presented) The method according to claim 1, wherein the illumination features have wavelengths in a near infrared (NIR) regime.
9. (Previously Presented) The method according to claim 8, wherein the illumination features have wavelengths of 940 nm.
10. (Previously Presented) The method according to claim 1, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) and/or for 3D detection in step c).
11. (Previously Presented) The method according to claim 1, wherein the camera comprises at least one near infrared camera.
12. (Currently Amended) A **non-transitory computer-readable medium storing a** computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to claim 1, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d).
13. (Currently Amended) A **non-transitory** computer-readable **storage** medium **storing a computer program** comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to claim 1.
14. (Previously Presented) A non-transient computer-readable medium comprising instructions that, when executed by one or more processors, cause the one or more processors to perform at least steps a) to d) of the method according to claim 1.
15. (Previously Presented) A mobile device comprising at least one camera, at least one illumination unit, and at least one processing unit, the mobile device being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to claim 1.

16. (Previously Presented) A method of using the method for face authentication-according to claim 1, wherein the method comprising using the method for face authentication for biometric presentation attack detection.

REMARKS

I. Status of the Claims

Claims 12 and 13 are amended for clarity. Support can be found at least in Paragraph [0145] of the published application (US 2024/0037201).

Claims 1-16 are pending and under examination. No new matter is added herein.

II. Rejections under 35 U.S.C. § 101

Claims 12 and 13 stand rejected under 35 U.S.C. § 101 as allegedly being directed to non-statutory subject matter.

Without commenting on the appropriateness of the rejections, claims 12-13 are amended consistent with the comments of the Office. It is believed that these amendments overcome the rejections. Accordingly, Applicant respectfully requests withdrawal of the rejection.

III. Rejections under 35 U.S.C. § 103

i. Claims 1-3, 7, 12-16 stand rejected under 35 U.S.C. § 103 as allegedly being unpatentable over Morestin (US 2019/0213309) in view of Zou (US 2021/0049391).

ii. Claims 4-6, 8, 9-11 stand rejected under 35 U.S.C. § 103 as allegedly being unpatentable over Morestin (US 2019/0213309) in view of Zou (US 2021/0049391) and further in view of Diedrich (US 2022/0222464).

The Office acknowledged that “Morestin is silent on using information of at least four of the reflection features to determine depth levels of 3d object” (see Page 7 of the Office Action dated 09/23/24).

However, the Office cited Morestin for allegedly disclosing elements b) and c) of the present claims, based on Paragraphs [0036], [0039], [0050], and [0053] of Morestin (see Pages 6-7 of the Office Action dated 09/23/24).

Applicant respectfully disagrees. Steps b) and c) as presently claimed use a **beam profile analysis** technique. As described in the present application, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials (see Paragraph [0064] of the published application). Beam profile analysis uses a **projected illumination pattern** (see Paragraph [0065] of the published application).

In contrast, Paragraph [0050] of Morestin, as cited by the Office, refers to a **time-of-flight (ToF)** measurement technique within the reflection features which is not related to the beam profile analysis technique as presently claimed. As described in the present application, a well-known working principle of a ToF sensor is sending out light and measuring the time span until receiving the reflected light. But using a projected pattern is **not possible**. Using illumination pattern may be advantageous e.g. in view of covering and therefore allowing to take into account different locations on the face. This may enhance reliability and security of the authentication of the user's face (see Paragraph [0065] of the published application).

Accordingly, in contrast to Morestin, the present claims have the technical effect of allowing for enhancing reliability and security of the authentication of the user's face. Morestin simply proposes a different technique that does not provide this technical effect. Thus, a person having ordinary skill in the art would not have been motivated to modify Morestin by using an entirely different technique, and would have been unable to modify Morestin in this way. Morestin is deficient at least for this reason.

Zou and Diedrich do not cure the deficiencies of Morestin. The Office did not cite either of Zou or Diedrich for disclosing a beam profile analysis technique using a projected illumination pattern as presently claimed. Thus, Zou and Diedrich are deficient at least for this reason.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

CONCLUSION

Applicant requests reconsideration and allowance of all pending claims. The Commissioner is hereby authorized to charge Deposit Account No. 01-2384 for any outstanding fees incurred during the pendency of this application. Please reference docket number 27843-2262.

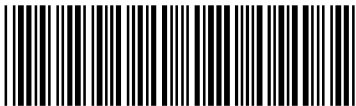
Respectfully submitted,

/Christopher M. Goff/

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USPTO Registration 41,785
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CMG/EMF/ZSW/LMS

VIA EFS

Issue Classification 	Application/Control No. 18/264,326	Applicant(s)/Patent Under Reexamination SCHILLEN et al.	
	Examiner PHUOC TRAN	Art Unit 2668	

CPC						
Symbol					Type	Version
G06V	/	40	/	40	F	2022-01-01
G06T	/	7	/	50	I	2017-01-01
G06V	/	10	/	25	I	2022-01-01
G06V	/	10	/	54	I	2022-01-01
G06V	/	40	/	165	I	2022-01-01
G06V	/	40	/	172	I	2022-01-01
G06V	/	10	/	60	I	2022-01-01
G06V	/	10	/	82	I	2022-01-01
G06V	/	10	/	145	I	2022-01-01
G06V	/	10	/	764	I	2022-01-01
G01S	/	17	/	894	I	2020-01-01
G06V	/	10	/	141	I	2022-01-01
G06V	/	40	/	168	I	2022-01-01
G06F	/	21	/	32	I	2013-01-01
G06T	/	2207	/	30201	A	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	16	
/PHUOC TRAN/ Primary Examiner, Art Unit 2668	07 March 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	2

<i>Issue Classification</i> 	Application/Control No. 18/264,326	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06V	/	40	/ 40
NON-CLAIMED			
/	/		

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	16	
/PHUOC TRAN/ Primary Examiner, Art Unit 2668	07 March 2025	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	2

Issue Classification 	Application/Control No. 18/264,326	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1		10												
	2		11												
	3		12												
	4		13												
	5		14												
	6		15												
	7		16												
	8														
	9														

NONE	Total Claims Allowed:	
(Assistant Examiner)	(Date)	16
/PHUOC TRAN/ Primary Examiner, Art Unit 2668	07 March 2025	O.G. Print Claim(s)
(Primary Examiner)	(Date)	1
		O.G. Print Figure
		2

<i>Search Notes</i> 	Application/Control No. 18/264,326	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

CPC - Searched*		
Symbol	Date	Examiner
G06V40/40, G01S17/894, G06F21/32, G06T7/50, G06V10/141, G06V10/145, G06V10/25, G06V10/54, G06V10/60, G06V10/764, G06V10/82, G06V40/165, G06V40/168, G06V40/172, G06T2207/30201	09/07/2024	PT

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor name search	09/07/2024	PT
See search history printout(s)	09/07/2024	PT
See search history printout(s)	03/07/2025	PT

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
	See search history printout(s)	03/07/2025	PT

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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	282	((("SCHILLEN") near3 ("Peter")) OR ("GUTHIER") near3 ("Benjamin")) OR ("SCHICK") near3 ("Friedrich")) OR ("GUENTHER") near3 ("Manuel")) OR ("DIESELBERG") near3 ("Lars")) OR ("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/09/07 03:49 PM
L2	537832	(G06V40/40 or G01S17/894 or G06F21/32 or G06T7/50 or G06V10/141 or G06V10/145 or G06V10/25 or G06V10/54 or G06V10/60 or G06V10/764 or G06V10/82 or G06V40/165 or G06V40/168 or G06V40/172 or G06T2207/30201).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/09/07 03:50 PM
L3	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:03 PM
L4	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:04 PM

L5	2	verifi\$8)).clm. 2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:05 PM
L6	0	2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:08 PM
L7	0	(image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:09 PM
L10	329	(image SAME (face OR facial OR skin) SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:14 PM

L11	120	OR skin)) 2 AND (image SAME (face OR facial OR skin) SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial OR skin))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:15 PM
L12	14	11 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:18 PM
L13	24	10 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:19 PM
L14	10	13 NOT 12	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:26 PM
L15	106	11 NOT 13	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:29 PM
L16	2	("20190213309" OR "2020082155").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2024/09/07 05:27 PM
L19	7543	2 AND @pd>"20240907"	(US-PGPUB; USPAT)	OR	ON	ON	2025/03/07 08:46 PM
L22	27	19 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME pattern SAME (face OR facial) SAME skin) AND ((beam OR profile) SAME (illuminat\$4 OR light) SAME reflect\$4 SAME feature SAME skin) AND (image SAME (face OR facial OR skin) SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d)) AND	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/03/07 09:01 PM

		(authenticat\$4 OR identif\$8 OR verifi\$8)					
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PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	1	((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME pattern SAME (face OR facial) SAME skin SAME (beam OR profile) SAME reflect\$4 SAME feature SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT)	NEAR	ON	ON	2025/03/07 09:06 PM

Bibliographic Data

Application No: 18/264,326

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged: /PHUOC TRAN/

Examiner's Signature

Initials

Title: OPTICAL SKIN DETECTION FOR FACE UNLOCK

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
08/04/2023	382	2668	
RULE			

APPLICANTS

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CONTINUING DATA

This application is a 371 of PCT/EP2022/053941 02/17/2022

FOREIGN APPLICATIONS

EPO EP21157800.0 02/18/2021

IF REQUIRED, FOREIGN LICENSE GRANTED**

10/24/2023

STATE OR COUNTRY

GERMANY

ADDRESS

FILING FEE RECEIVED

\$1,660



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United States Patent and Trademark Office
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www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

76656 7590 03/17/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

EXAMINER

TRAN, PHUOC

ART UNIT

PAPER NUMBER

2668

DATE MAILED: 03/17/2025

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	08/04/2023	Peter SCHILLEN	27843-2262 202595US01	2486

TITLE OF INVENTION: OPTICAL SKIN DETECTION FOR FACE UNLOCK

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	06/17/2025

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

76656 7590 03/17/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	08/04/2023	Peter SCHILLEN	27843-2262 202595US01	2486

TITLE OF INVENTION: OPTICAL SKIN DETECTION FOR FACE UNLOCK

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	06/17/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
TRAN, PHUOC	2668	382-118000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	08/04/2023	Peter SCHILLEN	27843-2262 202595US01	2486
76656	7590	03/17/2025	EXAMINER	
Patent Docket Department Armstrong Teasdale LLP 7700 Forsyth Boulevard Suite 1800 St. Louis, MO 63105			TRAN, PHUOC	
			ART UNIT	PAPER NUMBER
			2668	
DATE MAILED: 03/17/2025				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 18/264,326	Applicant(s) SCHILLEN et al.	
	Examiner PHUOC TRAN	Art Unit 2668	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to communication(s) filed 11/13/2024.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-16. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
---	--

/PHUOC TRAN/
Primary Examiner, Art Unit 2668

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Reasons for Allowance

2. The following is an examiner's statement of reasons for allowance: Applicant's arguments, on page 7 of the remarks, that the claimed invention is distinguishable from the prior art are persuasive.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to PHUOC TRAN whose telephone number is (571)272-7399. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Vu Le can be reached on 571-272-7332. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional

Art Unit: 2668

questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/PHUOC TRAN/

Primary Examiner, Art Unit 2668



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/264,326	05/20/2025 10:10:28 AM Z ET	27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	2486	FILED BY	Barbara McGregor
PATENT CENTER #	70500993	FILING DATE	08/04/2023
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2025-05-20 Issue Fee Transmittal 27843-2262.pdf	1	Issue Fee Payment (PTO-85B)	107 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-05-20 Issue Fee Transmittal 27843-2262.pdf	34EF7A4B1CF0FA4BEFA51A97023E39CD128DA6D28FE54C90 536ECFB45A931E0915874E7A3FD4BE5018AD164C7EF381858 B97B6745B1DE5F5F09864D2EDD33C8B

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as

described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

76656 7590 03/17/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	08/04/2023	Peter SCHILLEN	27843-2262 202595US01	2486

TITLE OF INVENTION: OPTICAL SKIN DETECTION FOR FACE UNLOCK

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	06/17/2025

EXAMINER	ART UNIT	CLASS-SUBCLASS
TRAN, PHUOC	2668	382-118000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Armstrong Teasdale LLP

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE
TRINAMIX GMBH

(B) RESIDENCE: (CITY and STATE OR COUNTRY)
LUDWIGSHAFEN AM RHEIN, GERMANY

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 012384

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Christopher M. Goff/

Date May 20, 2025

Typed or printed name Christopher M. Goff

Registration No. 41,785



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/264,326	05/20/2025 10:10:28 AM Z ET	27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	2486	FILED BY	Barbara McGregor
PATENT CENTER #	70500993	AUTHORIZED BY	Christopher Goff
CUSTOMER #	76656	FILING DATE	08/04/2023
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Peter SCHILLEN

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
DA / 012384	E20255JA10576923	Barbara McGregor

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1290.00	1	1290.00
TOTAL AMOUNT:				\$1,290.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	06/24/2025	12340630	27843-2262 202595US01	2486

76656 7590 06/11/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 4 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Peter SCHILLEN, Ludwigshafen am Rhein, GERMANY;
Benjamin GUTHIER, Ludwigshafen am Rhein, GERMANY;
Friedrich SCHICK, Ludwigshafen am Rhein, GERMANY;
Manuel GUENTHER, Zuerich, SWITZERLAND;
Lars DIESELBERG, Karlsruhe, GERMANY;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit [SelectUSA.gov](https://selectusa.gov).



(12)发明专利申请

(10)申请公布号 CN 108573203 A

(43)申请公布日 2018.09.25

(21)申请号 201710218218.8

G06K 9/32(2006.01)

(22)申请日 2017.04.05

G06K 9/34(2006.01)

H04L 29/06(2006.01)

(66)本国优先权数据

201710162245.8 2017.03.17 CN

201710161685.1 2017.03.17 CN

(71)申请人 北京旷视科技有限公司

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A座313

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所(普通合伙) 11481

代理人 徐丁峰 戴亚南

(51)Int.Cl.

G06K 9/00(2006.01)

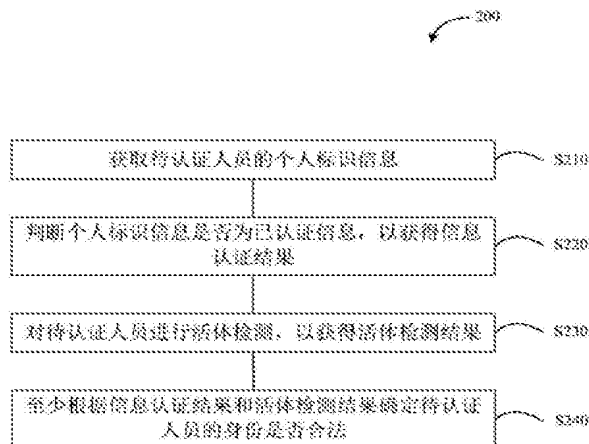
权利要求书8页 说明书58页 附图9页

(54)发明名称

身份认证方法和装置及存储介质

(57)摘要

本发明的实施例提供了一种身份认证方法和装置及存储介质。该方法包括：获取待认证人员的个人标识信息；判断个人标识信息是否为已认证信息，以获得信息认证结果；对待认证人员进行活体检测，以获得活体检测结果；以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。上述方法和装置结合已认证信息判断和活体检测来确定待认证人员的身份是否合法，因此与常规的单纯基于密码或证件进行身份认证的方式相比，根据本发明实施例的身份认证方法和装置的认证结果更准确，可以提高用户鉴权的安全性，从而可以有效保障用户的权益。



1. 一种身份认证方法,包括:
获取待认证人员的个人标识信息;
判断所述个人标识信息是否为已认证信息,以获得信息认证结果;
对所述待认证人员进行活体检测,以获得活体检测结果;以及
至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

2. 如权利要求1所述的身份认证方法,其中,所述个人标识信息为证件号码、姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种。

3. 如权利要求1所述的身份认证方法,其中,在所述获取待认证人员的个人标识信息之前,所述身份认证方法还包括:

输出用于指示所述待认证人员提供预定类型的人员信息的指示信息;
其中,所述个人标识信息为由所述待认证人员提供的人员信息或基于所述人员信息获得。

4. 如权利要求1所述的身份认证方法,其中,所述个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种,

所述获取待认证人员的个人标识信息包括:

获取所述待认证人员的初始信息,其中,所述初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及

基于预定算法对所述初始信息进行变换,以获得所述初始信息的变换值作为所述个人标识信息。

5. 如权利要求1所述的身份认证方法,其中,

所述获取待认证人员的个人标识信息包括:

获取所述待认证人员的证件图像,其中,所述信息认证结果为证件认证结果;

所述对所述待认证人员进行活体检测,以获得活体检测结果包括:

获取所述待认证人员的人脸图像;以及

利用所述人脸图像进行活体检测,以获得所述活体检测结果。

6. 如权利要求5所述的身份认证方法,其中,在所述至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法之前,所述身份认证方法还包括:

利用所述证件图像和/或所述人脸图像执行附加判断操作,以获得附加判断结果;

所述至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法包括:

根据所述证件认证结果、所述活体检测结果和所述附加判断结果确定所述待认证人员的身份是否合法。

7. 如权利要求6所述的身份认证方法,其中,所述附加判断操作包括证件真伪判断操作和/或人脸一致性判断操作,并且所述附加判断结果包括证件真伪判断结果和/或人脸一致性判断结果,

所述证件真伪判断操作包括:判断所述证件图像中的证件是否为真实证件,以获得所述证件真伪判断结果;

所述人脸一致性判断操作包括：根据所述证件图像获取所述待认证人员的证件人脸；以及将所述待认证人员的证件人脸与所述人脸图像中的人脸进行对比，以获得所述人脸一致性判断结果。

8. 如权利要求7所述的身份认证方法，其中，所述根据所述证件图像获取所述待认证人员的证件人脸包括：

从所述证件图像中检测人脸，以获得所述待认证人员的证件人脸。

9. 如权利要求7所述的身份认证方法，其中，所述根据所述证件图像获取所述待认证人员的证件人脸包括：

对所述证件图像进行文字识别，以获得所述证件图像中的文字信息；

基于所述证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息；以及

确定搜索到的、所述匹配的证件信息中的证件人脸为所述待认证人员的证件人脸。

10. 如权利要求7所述的身份认证方法，其中，所述判断所述证件图像中的证件是否为真实证件，以获得所述证件真伪判断结果包括：

提取所述证件图像的图像特征；以及

将所述图像特征输入训练好的证件分类器，以获得所述证件真伪判断结果；

其中，所述证件真伪判断结果为所述证件图像中的证件是真实证件的置信度。

11. 如权利要求7所述的身份认证方法，其中，所述判断所述证件图像中的证件是否为真实证件，以获得所述证件真伪判断结果包括：

从所述证件图像中识别包含证件标识信息的图像块；以及

对所述包含证件标识信息的图像块中的证件标识信息进行识别，以获得所述证件真伪判断结果；

其中，所述证件真伪判断结果为所述证件图像中的证件是真实证件的置信度。

12. 如权利要求6所述的身份认证方法，其中，在所述根据所述证件认证结果、所述活体检测结果和所述附加判断结果确定所述待认证人员的身份是否合法的过程中，所述证件认证结果、所述活体检测结果和所述附加判断结果中的每个结果具有各自的权重系数。

13. 如权利要求5所述的身份认证方法，其中，所述获取所述待认证人员的证件图像包括：

获取在当前拍摄条件下针对所述待认证人员的证件实时采集的预拍摄图像；

实时评估所述预拍摄图像的图像属性；

当所述预拍摄图像的图像属性的评估值小于预设的评估值阈值时，根据所述预拍摄图像的图像属性生成预拍摄提示信息，用于提示所述待认证人员调整其证件的拍摄条件；以及

当所述预拍摄图像的图像属性的评估值等于或大于所述预设的评估值阈值时，保存所述预拍摄图像以将其作为所述证件图像。

14. 如权利要求5所述的身份认证方法，其中，所述判断所述个人标识信息是否为已认证信息，以获得信息认证结果包括：

对所述证件图像进行文字识别，以获得所述证件图像中的文字信息；以及

基于所述证件图像中的文字信息在已认证证件信息数据库中进行搜索，以获得所述证

件认证结果；

其中，所述证件认证结果为所述证件图像中的证件信息是已认证证件信息的置信度。

15. 如权利要求9或14所述的身份认证方法，其中，所述对所述证件图像进行文字识别，以获得所述证件图像中的文字信息包括：

对所述证件图像中的文字进行定位，以获得包含文字的图像块；以及

对所述包含文字的图像块中的文字进行识别，以获得所述证件图像中的文字信息。

16. 如权利要求15所述的身份认证方法，其中，在所述对所述包含文字的图像块中的文字进行识别之前，所述身份认证方法还包括：

将所述包含文字的图像块校正为水平状态。

17. 如权利要求15所述的身份认证方法，其中，在所述对所述包含文字的图像块中的文字进行识别，以获得所述证件图像中的文字信息之后，所述对所述证件图像进行文字识别，以获得所述证件图像中的文字信息还包括：

输出所述证件图像中的文字信息，以供用户查看；

接收所述用户输入的文字修正信息；

将所述文字修正信息所指示的待修正文字与所述证件图像中的文字信息中的对应文字进行对比；以及

如果所述文字修正信息所指示的待修正文字与所述证件图像中的文字信息中的对应文字之间的差异小于预设的差异阈值，则利用所述文字修正信息更新所述证件图像中的文字信息。

18. 如权利要求1所述的身份认证方法，其中，所述对所述待认证人员进行活体检测，以获得活体检测结果包括：

生成活体动作指令，所述活体动作指令用于指示所述待认证人员执行对应的活体动作；

获取实时采集的所述待认证人员的人脸图像；

检测所述人脸图像中的人脸；以及

判断所述人脸图像中的人脸是否执行所述活体动作指令所指示的活体动作，以获得所述活体检测结果。

19. 如权利要求1所述的身份认证方法，其中，所述对所述待认证人员进行活体检测，以获得活体检测结果包括：

生成活体动作指令，所述活体动作指令用于指示所述待认证人员执行对应的活体动作；

获取实时采集的所述待认证人员的人脸图像；

从所采集的所述人脸图像中捕捉所述待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像；以及

将所述待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器，以获得所述活体检测结果。

20. 如权利要求19所述的身份认证方法，其中，所述身份认证方法包括：

获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像，以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像；

从所述真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像；

从所述虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像；以及

利用所述正样本图像和所述负样本图像训练分类器模型，以获得所述皮肤弹性分类器。

21. 如权利要求19所述的身份认证方法，其中，所述从所采集的所述人脸图像中捕捉所述待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像包括：

从所采集的所述人脸图像中选择所述待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像；

利用人脸检测模型对在所述待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行定位；

利用人脸关键点定位模型对在所述待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位；以及

根据定位获得的人脸位置和关键点位置对在所述待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分，以获得所述待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

22. 如权利要求21所述的身份认证方法，其中，所述身份认证方法还包括：

获取样本人脸图像，所述样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的；以及

利用所述样本人脸图像进行神经网络训练，以获得所述人脸检测模型和所述人脸关键点定位模型。

23. 如权利要求1所述的身份认证方法，其中，所述对所述待认证人员进行活体检测，以获得活体检测结果包括：

获取由双目摄像头针对在结构光照射下的所述待认证人员的人脸所采集的人脸图像；以及

根据所述人脸图像确定所述待认证人员的人脸是否属于活体，以获得所述活体检测结果。

24. 如权利要求1所述的身份认证方法，其中，所述对所述待认证人员进行活体检测，以获得活体检测结果包括：

步骤S110：获取针对在检测光的照射下的所述待认证人员的人脸所采集的一个或多个光照图像；

步骤S120：基于所述待认证人员的人脸在所述一个或多个光照图像中所表现的光反射特性确定所述待认证人员的人脸是否属于活体，以获得光照活体验证结果；以及

步骤S130：至少基于所述光照活体验证结果确定所述认证人员是否通过活体验证，以获得所述活体检测结果。

25. 如权利要求24所述的身份认证方法，其中，在照射所述待认证人员的人脸的过程中，所述检测光的模式至少改变一次。

26. 如权利要求25所述的身份认证方法,其中,在照射所述待认证人员的人脸的过程中,所述检测光的模式在每两个连续的时刻之间发生改变。

27. 如权利要求25所述的身份认证方法,其中,在照射所述待认证人员的人脸的过程中,所述检测光的模式是随机改变的或预先设定好的。

28. 如权利要求24所述的身份认证方法,其中,

所述对所述待认证人员进行活体检测,以获得活体检测结果包括:

步骤S140:输出动作指令,其中,所述动作指令用于指示所述认证人员执行相应的动作;

步骤S150:获取针对所述待认证人员的人脸所采集的多个动作图像;

步骤S160:基于所述多个动作图像检测所述待认证人员所执行的动作;以及

步骤S170:根据动作检测结果和所述动作指令确定所述待认证人员的人脸是否属于活体,以获得动作活体验证结果;

所述步骤S130包括:

基于所述光照活体验证结果和所述动作活体验证结果确定所述待认证人员是否通过活体验证,以获得所述活体检测结果。

29. 如权利要求28所述的身份认证方法,其中,所述步骤S170包括:

如果基于在不大于第一预设时间的时段内所采集的所述多个动作图像检测到由所述待认证人员执行的、与所述动作指令所指示的动作一致的动作,则确定所述待认证人员的人脸属于活体,如果基于在所述第一预设时间内所采集的所述多个动作图像未检测到由所述待认证人员执行的、与所述动作指令所指示的动作一致的动作,则确定所述待认证人员的人脸不属于活体。

30. 如权利要求28所述的身份认证方法,其中,

所述身份认证方法还包括:

在每次执行所述步骤S140至所述步骤S170的过程中计数一次,以获得动作验证次数;

在所述步骤S170之后,所述身份认证方法还包括:

如果所述动作活体验证结果表示所述待认证人员的人脸不属于活体,则输出第一错误信息,并判断所述动作验证次数是否达到第一次数阈值,如果所述动作验证次数达到所述第一次数阈值,则转至所述步骤S130,如果所述动作验证次数未达到所述第一次数阈值,则返回所述步骤S140或者在所述步骤S110在所述步骤S140之前执行的情况下返回所述步骤S110,其中,所述第一错误信息用于提示针对所述待认证人员的活体验证失败。

31. 如权利要求24所述的身份认证方法,其中,在所述步骤S110之前,所述身份认证方法还包括:

步骤S108:判断所述待认证人员的图像采集条件是否满足预设要求,如果所述图像采集条件满足所述预设要求,则转至所述步骤S110,其中,所述图像采集条件包括所述待认证人员在图像采集装置的图像采集区域中所处的位置和/或所述待认证人员与所述图像采集装置的相对角度。

32. 如权利要求31所述的身份认证方法,其中,在所述步骤S108之前或与所述步骤S108同时地,所述身份认证方法还包括:

步骤S106:输出第一提示信息,其中,所述第一提示信息用于提示所述待认证人员将脸

部正对所述图像采集装置并靠近所述图像采集装置。

33. 如权利要求32所述的身份认证方法,其中,所述步骤S106包括:通过语音形式、图像形式和文字形式中的一种或多种输出所述第一提示信息。

34. 如权利要求31所述的身份认证方法,其中,所述步骤S108包括:

获取针对所述待认证人员的人脸采集的实时图像;

实时输出用于校准所述图像采集条件的预设区域以及所述实时图像中的人脸区域用于显示;以及

根据所述人脸区域判断所述图像采集条件是否满足所述预设要求,如果所述人脸区域位于预设区域内并且所述人脸区域在所述实时图像中所占比例大于第一预设比例,则确定所述图像采集条件满足所述预设要求,如果所述人脸区域没有位于所述预设区域内或者所述人脸区域在所述实时图像中所占比例不大于所述第一预设比例,则确定所述图像采集条件不满足所述预设要求。

35. 如权利要求34所述的身份认证方法,其中,所述身份认证方法还包括:

如果所述人脸区域在所述实时图像中所占比例不大于所述第一预设比例,则实时输出第一采集提示信息,以提示所述待认证人员与所述图像采集装置靠近。

36. 如权利要求31所述的身份认证方法,其中,所述步骤S108包括:

获取针对所述待认证人员的人脸采集的实时图像;

实时输出用于校准所述图像采集条件的预设区域以及所述实时图像中的人脸区域用于显示;以及

根据在所述人脸区域判断所述图像采集条件是否满足所述预设要求,

如果所述人脸区域位于预设区域内并且所述人脸区域在所述预设区域中所占比例大于第二预设比例,则确定所述图像采集条件满足所述预设要求,如果所述人脸区域没有位于所述预设区域内或者所述人脸区域在所述预设区域中所占比例不大于所述第二预设比例,则确定所述图像采集条件不满足所述预设要求。

37. 如权利要求36所述的身份认证方法,其中,所述身份认证方法还包括:

如果所述人脸区域在所述预设区域中所占比例不大于所述第二预设比例,则实时输出第二采集提示信息,以提示所述待认证人员与所述图像采集装置靠近。

38. 如权利要求34或36所述的身份认证方法,其中,所述身份认证方法还包括:

实时判断所述人脸区域与所述预设区域之间的相对位置关系;以及

基于所述人脸区域与所述预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变所述待认证人员与所述图像采集装置之间的相对位置关系以使得所述人脸区域与所述预设区域靠近。

39. 如权利要求31所述的身份认证方法,其中,所述步骤S108包括:

获取所述图像采集装置的姿态信息;以及

根据所述姿态信息判断所述图像采集装置是否处于竖直放置状态,如果是,则确定所述图像采集条件满足所述预设要求,否则确定所述图像采集条件不满足所述预设要求。

40. 如权利要求31所述的身份认证方法,其中,

所述身份认证方法还包括:

在每次执行所述步骤S110至所述步骤S120的过程中计数一次,以获得光照验证次数;

在所述步骤S120之后,所述身份认证方法还包括:

如果所述光照活体验证结果表示所述待认证人员的人脸不属于活体,则输出第二错误信息,并判断所述光照验证次数是否达到第二次数阈值,如果所述光照验证次数达到所述第二次数阈值,则转至所述步骤S130,如果所述光照验证次数未达到所述第二次数阈值,则返回所述步骤S108或返回所述步骤S110,其中,所述第二错误信息用于提示针对所述待认证人员的活体验证失败。

41.如权利要求24所述的身份认证方法,其中,在所述步骤S110之前或在执行所述步骤S110和所述步骤S120的过程中,所述身份认证方法还包括:

输出第二提示信息,其中,所述第二提示信息用于提示所述待认证人员在第二预设时间内保持不动。

42.如权利要求41所述的身份认证方法,其中,所述第二提示信息为与所述第二预设时间相对应的倒计时信息。

43.如权利要求24所述的身份认证方法,其中,所述检测光通过动态改变向所述待认证人员的人脸发射的光的颜色和/或位置来获得。

44.如权利要求24所述的身份认证方法,其中,所述检测光由显示屏幕发射,所述检测光通过以下方式获得:通过改变所述显示屏幕上显示的内容来动态改变所述显示屏幕发射的光的模式,以向所述待认证人员的人脸发射所述检测光。

45.一种身份认证装置,包括处理器和存储器,其中,所述存储器中存储有计算机程序指令,所述计算机程序指令被所述处理器运行时用于执行以下步骤:

获取待认证人员的个人标识信息;

判断所述个人标识信息是否为己认证信息,以获得信息认证结果;

对所述待认证人员进行活体检测,以获得活体检测结果;以及

至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

46.一种存储介质,在所述存储介质上存储了程序指令,所述程序指令在运行时用于执行以下步骤:

获取待认证人员的个人标识信息;

判断所述个人标识信息是否为己认证信息,以获得信息认证结果;

对所述待认证人员进行活体检测,以获得活体检测结果;以及

至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

47.一种身份认证装置,包括信息采集装置、处理器和存储器,其中,

所述信息采集装置用于采集待认证人员的初始信息;

所述存储器中存储有计算机程序指令,所述计算机程序指令被所述处理器运行时用于执行以下步骤:

获取所述待认证人员的个人标识信息,其中,所述个人标识信息为所述初始信息或基于所述初始信息获得;

判断所述个人标识信息是否为己认证信息以获得信息认证结果;

对所述待认证人员进行活体检测以获得活体检测结果;以及

至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

48. 如权利要求47所述的身份认证装置,其中,所述个人信息采集装置包括图像采集装置和/或输入装置。

49. 一种身份认证装置,包括:

信息采集装置,用于采集待认证人员的初始信息;以及

传输装置,用于将所述初始信息发送至服务器,并从所述服务器接收所述服务器通过以下方式获得的关于所述待认证人员的身份是否合法的认证信息:获取所述待认证人员的个人标识信息,其中,所述个人标识信息为所述初始信息或基于所述初始信息获得;判断所述个人标识信息是否为己认证信息以获得信息认证结果;对所述待认证人员进行活体检测以获得活体检测结果;以及至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

50. 如权利要求49所述的身份认证装置,其中,所述信息采集装置包括图像采集装置和/或输入装置。

51. 一种身份认证装置,包括传输装置、处理器和存储器,其中,

所述传输装置用于从客户端接收待认证人员的初始信息,并将关于所述待认证人员的身份是否合法的认证信息发送至所述客户端;

所述存储器中存储有计算机程序指令,所述计算机程序指令被所述处理器运行时用于执行以下步骤:

获取所述待认证人员的个人标识信息,其中,所述个人标识信息为所述初始信息或基于所述初始信息获得;

判断所述个人标识信息是否为己认证信息以获得信息认证结果;

对所述待认证人员进行活体检测以获得活体检测结果;以及

至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

52. 一种身份认证装置,包括:

信息获取模块,用于获取待认证人员的个人标识信息;

己认证信息判断模块,用于判断个人标识信息是否为己认证信息,以获得信息认证结果;

活体检测模块,用于对待认证人员进行活体检测,以获得活体检测结果;以及

身份确定模块,用于至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

身份认证方法和装置及存储介质

技术领域

[0001] 本发明涉及图像处理领域,更具体地涉及一种身份认证方法和装置及存储介质。

背景技术

[0002] 科技产品使用的社会化成为现代化社会生活的一道靓丽风景线,人们的衣食住行无不与科技密切相关,科技产品已经逐渐应用到社会生活的方方面面,成为现代人类日常生活不可缺少的重要组成部分。然而,人们在享受科学产品带来的福利的同时,也感受到了它所带来的负面问题,例如,信息安全问题。

[0003] 目前许多领域均涉及信息安全问题,尤其在电子商务、移动支付、银行开户等技术领域,信息安全问题尤为突出。具体来说,目前,在上述领域中多数采用密码方式进行用户交互鉴权(也可以称为身份认证),也有采用刷证件的方式进行用户交互鉴权的。上述两种方式都有一定的弊端,前者需要用户牢记密码,而且每次输密码较为繁琐,一旦密码被非法分子盗取,更将给用户造成隐私或财产的损失,而对于后者来说,证件容易被伪造或冒用,其安全性较低。因此,有必要提出一种方便安全的身份认证方法或系统,以应用于电子商务、移动支付、银行开户等技术领域。

发明内容

[0004] 考虑到上述问题而提出了本发明。本发明提供了一种身份认证方法和装置及存储介质。

[0005] 根据本发明一方面,提供了一种身份认证方法。该方法包括:获取待认证人员的个人标识信息;判断个人标识信息是否为己认证信息,以获得信息认证结果;对待认证人员进行活体检测,以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0006] 示例性地,个人标识信息为证件号码、姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种。

[0007] 示例性地,在获取待认证人员的个人标识信息之前,身份认证方法还包括:输出用于指示待认证人员提供预定类型的人员信息的指示信息;其中,个人标识信息为由待认证人员提供的人员信息或基于人员信息获得。

[0008] 示例性地,个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种,获取待认证人员的个人标识信息包括:获取待认证人员的初始信息,其中,初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及基于预定算法对初始信息进行变换,以获得初始信息的变换值作为个人标识信息。

[0009] 示例性地,获取待认证人员的个人标识信息包括:获取待认证人员的证件图像,其中,信息认证结果为证件认证结果;对待认证人员进行活体检测,以获得活体检测结果包括:获取待认证人员的人脸图像;以及利用人脸图像进行活体检测,以获得活体检测结果。

[0010] 示例性地,在至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法之前,身份认证方法还包括:利用证件图像和/或人脸图像执行附加判断操作,以获得附加判断结果;至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法包括:根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法。

[0011] 示例性地,附加判断操作包括证件真伪判断操作和/或人脸一致性判断操作,并且附加判断结果包括证件真伪判断结果和/或人脸一致性判断结果,证件真伪判断操作包括:判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果;人脸一致性判断操作包括:根据证件图像获取待认证人员的证件人脸;以及将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。

[0012] 示例性地,根据证件图像获取待认证人员的证件人脸包括:从证件图像中检测人脸,以获得待认证人员的证件人脸。

[0013] 示例性地,根据证件图像获取待认证人员的证件人脸包括:对证件图像进行文字识别,以获得证件图像中的文字信息;基于证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息;以及确定搜索到的、匹配的证件信息中的证件人脸为待认证人员的证件人脸。

[0014] 示例性地,判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果包括:提取证件图像的图像特征;以及将图像特征输入训练好的证件分类器,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0015] 示例性地,判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果包括:从证件图像中识别包含证件标识信息的图像块;以及对包含证件标识信息的图像块中的证件标识信息进行识别,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0016] 示例性地,在根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法的过程中,证件认证结果、活体检测结果和附加判断结果中的每个结果具有各自的权重系数。

[0017] 示例性地,获取待认证人员的证件图像包括:获取在当前拍摄条件下针对所述待认证人员的证件实时采集的预拍摄图像;实时评估预拍摄图像的图像属性;当预拍摄图像的图像属性的评估值小于预设的评估值阈值时,根据预拍摄图像的图像属性生成预拍摄提示信息,用于提示待认证人员调整其证件的拍摄条件;以及当预拍摄图像的图像属性的评估值等于或大于预设的评估值阈值时,保存预拍摄图像以将其作为证件图像。

[0018] 示例性地,判断个人标识信息是否为已认证信息,以获得信息认证结果包括:对证件图像进行文字识别,以获得证件图像中的文字信息;以及基于证件图像中的文字信息在已认证证件信息数据库中进行搜索,以获得证件认证结果;其中,证件认证结果为证件图像中的证件信息是已认证证件信息的置信度。

[0019] 示例性地,对证件图像进行文字识别,以获得证件图像中的文字信息包括:对证件图像中的文字进行定位,以获得包含文字的图像块;以及对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息。

[0020] 示例性地,在对包含文字的图像块中的文字进行识别之前,身份认证方法还包括:将包含文字的图像块校正为水平状态。

[0021] 示例性地,在对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息之后,对证件图像进行文字识别,以获得证件图像中的文字信息还包括:输出证件图像中的文字信息,以供用户查看;接收用户输入的文字修正信息;将文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字进行对比;以及如果文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字之间的差异小于预设的差异阈值,则利用文字修正信息更新证件图像中的文字信息。

[0022] 示例性地,对待认证人员进行活体检测,以获得活体检测结果包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;检测人脸图像中的人脸;以及判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0023] 示例性地,对待认证人员进行活体检测,以获得活体检测结果包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像;以及将待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器,以获得活体检测结果。

[0024] 示例性地,身份认证方法包括:获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像,以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像;从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像;以及利用正样本图像和负样本图像训练分类器模型,以获得皮肤弹性分类器。

[0025] 示例性地,从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像包括:从所采集的人脸图像中选择待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;利用人脸检测模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行定位;利用人脸关键点定位模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位;以及根据定位获得的人脸位置和关键点位置对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分,以获得待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

[0026] 示例性地,身份认证方法还包括:获取样本人脸图像,样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的;以及利用样本人脸图像进行神经网络训练,以获得人脸检测模型和人脸关键点定位模型。

[0027] 示例性地,对待认证人员进行活体检测,以获得活体检测结果包括:获取由双目摄像头针对在结构光照射下的待认证人员的人脸所采集的人脸图像;以及根据人脸图像确定待认证人员的人脸是否属于活体,以获得活体检测结果。

[0028] 示例性地,对待认证人员进行活体检测,以获得活体检测结果包括:步骤S110:获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像;步骤S120:基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的

人脸是否属于活体,以获得光照活体验证结果;以及步骤S130:至少基于光照活体验证结果确定认证人员是否通过活体验证,以获得活体检测结果。

[0029] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式至少改变一次。

[0030] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式在每两个连续的时刻之间发生改变。

[0031] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式是随机改变的或预先设定好的。

[0032] 示例性地,对待认证人员进行活体检测,以获得活体检测结果包括:步骤S140:输出动作指令,其中,动作指令用于指示认证人员执行相应的动作;步骤S150:获取针对待认证人员的人脸所采集的多个动作图像;步骤S160:基于多个动作图像检测待认证人员所执行的动作;以及步骤S170:根据动作检测结果和动作指令确定待认证人员的人脸是否属于活体,以获得动作活体验证结果;步骤S130包括:基于光照活体验证结果和动作活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0033] 示例性地,步骤S170包括:如果基于在不大于第一预设时间的时段内所采集的多个动作图像检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体,如果基于在第一预设时间内所采集的多个动作图像未检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸不属于活体。

[0034] 示例性地,身份认证方法还包括:在每次执行步骤S140至步骤S170的过程中计数一次,以获得动作验证次数;在步骤S170之后,身份认证方法还包括:如果动作活体验证结果表示待认证人员的人脸不属于活体,则输出第一错误信息,并判断动作验证次数是否达到第一次数阈值,如果动作验证次数达到第一次数阈值,则转至步骤S130,如果动作验证次数未达到第一次数阈值,则返回步骤S140或者在步骤S110在步骤S140之前执行的情况下返回步骤S110,其中,第一错误信息用于提示针对待认证人员的活体验证失败。

[0035] 示例性地,在步骤S110之前,身份认证方法还包括:步骤S108:判断待认证人员的图像采集条件是否满足预设要求,如果图像采集条件满足预设要求,则转至步骤S110,其中,图像采集条件包括待认证人员在图像采集装置的图像采集区域中所处的位置和/或待认证人员与图像采集装置的相对角度。

[0036] 示例性地,在步骤S108之前或与步骤S108同时地,身份认证方法还包括:步骤S106:输出第一提示信息,其中,第一提示信息用于提示待认证人员将脸部正对图像采集装置并靠近图像采集装置。

[0037] 示例性地,步骤S106包括:通过语音形式、图像形式和文字形式中的一种或多种输出第一提示信息。

[0038] 示例性地,步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在实时图像中所占比例大于第一预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在实时图像中所占比例不大于第一预设比例,则确定图像采集条件不满足预设要求。

[0039] 示例性地,身份认证方法还包括:如果人脸区域在实时图像中所占比例不大于第一预设比例,则实时输出第一采集提示信息,以提示待认证人员与图像采集装置靠近。

[0040] 示例性地,步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在预设区域中所占比例大于第二预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在预设区域中所占比例不大于第二预设比例,则确定图像采集条件不满足预设要求。

[0041] 示例性地,身份认证方法还包括:如果人脸区域在预设区域中所占比例不大于第二预设比例,则实时输出第二采集提示信息,以提示待认证人员与图像采集装置靠近。

[0042] 示例性地,身份认证方法还包括:实时判断人脸区域与预设区域之间的相对位置关系;以及基于人脸区域与预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近。

[0043] 示例性地,步骤S108包括:获取图像采集装置的姿态信息;以及根据姿态信息判断图像采集装置是否处于竖直放置状态,如果是,则确定图像采集条件满足预设要求,否则确定图像采集条件不满足预设要求。

[0044] 示例性地,身份认证方法还包括:在每次执行步骤S110至步骤S120的过程中计数一次,以获得光照验证次数;在步骤S120之后,身份认证方法还包括:如果光照活体验证结果表示待认证人员的人脸不属于活体,则输出第二错误信息,并判断光照验证次数是否达到第二次数阈值,如果光照验证次数达到第二次数阈值,则转至步骤S130,如果光照验证次数未达到第二次数阈值,则返回步骤S108或返回步骤S110,其中,第二错误信息用于提示针对待认证人员的活体验证失败。

[0045] 示例性地,在步骤S110之前或在执行步骤S110和步骤S120的过程中,身份认证方法还包括:输出第二提示信息,其中,第二提示信息用于提示待认证人员在第二预设时间内保持不动。

[0046] 示例性地,第二提示信息为与第二预设时间相对应的倒计时信息。

[0047] 示例性地,检测光通过动态改变向待认证人员的人脸发射的光的颜色和/或位置来获得。

[0048] 示例性地,检测光由显示屏幕发射,检测光通过以下方式获得:通过改变显示屏幕上显示的内容来动态改变显示屏幕发射的光的模式,以向待认证人员的人脸发射检测光。

[0049] 根据本发明另一方面,提供一种身份认证装置。该装置包括:信息获取模块,用于获取待认证人员的个人标识信息;已认证信息判断模块,用于判断个人标识信息是否为已认证信息,以获得信息认证结果;活体检测模块,用于对待认证人员进行活体检测,以获得活体检测结果;以及身份确定模块,用于至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0050] 示例性地,个人标识信息为证件号码、姓名、证件人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种。

[0051] 示例性地,身份认证装置还包括:指示信息输出模块,用于在信息获取模块获取待

认证人员的个人标识信息之前,输出用于指示待认证人员提供预定类型的人员信息的指示信息;其中,个人标识信息为由待认证人员提供的人员信息或基于人员信息获得。

[0052] 示例性地,个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种,信息获取模块包括:初始信息获取子模块,用于获取待认证人员的初始信息,其中,初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及变换子模块,用于基于预定算法对初始信息进行变换,以获得初始信息的变换值作为个人标识信息。

[0053] 示例性地,信息获取模块包括:证件图像获取子模块,用于获取待认证人员的证件图像,其中,信息认证结果为证件认证结果;活体检测模块包括:人脸图像获取子模块,用于获取待认证人员的人脸图像;以及检测子模块,用于利用人脸图像进行活体检测,以获得活体检测结果。

[0054] 示例性地,身份认证装置还包括:附加判断模块,用于在身份确定模块至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法之前,利用证件图像和/或人脸图像执行附加判断操作,以获得附加判断结果;身份确定模块包括:身份确定子模块,用于根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法。

[0055] 示例性地,附加判断模块包括证件真伪判断子模块和/或人脸一致性判断子模块,证件真伪判断子模块用于执行证件真伪判断操作,人脸一致性判断子模块用于执行人脸一致性判断操作,并且附加判断结果包括证件真伪判断结果和/或人脸一致性判断结果,

[0056] 证件真伪判断子模块包括:证件真伪判断单元,用于判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果;人脸一致性判子模块包括:证件人脸获取单元,用于根据证件图像获取待认证人员的证件人脸;以及人脸对比单元,用于将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。

[0057] 示例性地,证件人脸获取单元包括:证件人脸检测子单元,用于从证件图像中检测人脸,以获得待认证人员的证件人脸。

[0058] 示例性地,证件人脸获取单元包括:文字识别子模块,用于对证件图像进行文字识别,以获得证件图像中的文字信息;搜索子单元,用于基于证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息;以及证件人脸确定子单元,用于确定搜索到的、匹配的证件信息中的证件人脸为待认证人员的证件人脸。

[0059] 示例性地,证件真伪判断单元包括:特征提取子单元,用于提取证件图像的图像特征;以及输入子单元,用于将图像特征输入训练好的证件分类器,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0060] 示例性地,证件真伪判断单元包括:图像块识别子单元,用于从证件图像中识别包含证件标识信息的图像块;以及标识信息识别子单元,用于对包含证件标识信息的图像块中的证件标识信息进行识别,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0061] 示例性地,在身份确定子模块根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法的过程中,证件认证结果、活体检测结果和附加判断结果中的每个结果具有各自的权重系数。

[0062] 示例性地,证件图像获取子模块包括:采集单元,用于获取在当前拍摄条件下针对

待认证人员的证件实时采集的预拍摄图像;评估单元,用于实时评估预拍摄图像的图像属性;提示单元,用于当预拍摄图像的图像属性的评估值小于预设的评估值阈值时,根据预拍摄图像的图像属性生成预拍摄提示信息,用于提示待认证人员调整其证件的拍摄条件;以及保存单元,用于当预拍摄图像的图像属性的评估值等于或大于预设的评估值阈值时,保存预拍摄图像以将其作为证件图像。

[0063] 示例性地,已认证信息判断模块包括:文字识别子模块,用于对证件图像进行文字识别,以获得证件图像中的文字信息;以及搜索子模块,用于基于证件图像中的文字信息在已认证证件信息数据库中进行搜索,以获得证件认证结果;其中,证件认证结果为证件图像中的证件信息是已认证证件信息的置信度。

[0064] 示例性地,文字识别子模块包括:文字定位单元,用于对证件图像中的文字进行定位,以获得包含文字的图像块;以及文字识别单元,用于对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息。

[0065] 示例性地,身份认证装置还包括:文字校正模块,用于在文字识别单元对包含文字的图像块中的文字进行识别之前,将包含文字的图像块校正为水平状态。

[0066] 示例性地,文字识别子模块还包括:文字输出单元,用于输出证件图像中的文字信息,以供用户查看;修正信息接收单元,用于接收用户输入的文字修正信息;文字对比单元,用于将文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字进行对比;以及文字更新单元,用于如果文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字之间的差异小于预设的差异阈值,则利用文字修正信息更新证件图像中的文字信息。

[0067] 示例性地,活体检测模块包括:指令生成子模块,用于生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;实时图像获取子模块,获取实时采集的待认证人员的人脸图像;人脸检测子模块,用于检测人脸图像中的人脸;以及活体动作执行判断子模块,用于判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0068] 示例性地,活体检测模块包括:指令生成子模块,用于生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;实时图像获取子模块,用于获取实时采集的待认证人员的人脸图像;皮肤区域捕捉子模块,用于从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像;以及输入子模块,用于将待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器,以获得活体检测结果。

[0069] 示例性地,身份认证装置包括:真假图像获取模块,用于获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像,以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;正样本提取子模块,用于从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像;负样本提取子模块,用于从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像;以及训练子模块,用于利用正样本图像和负样本图像训练分类器模型,以获得皮肤弹性分类器。

[0070] 示例性地,皮肤区域捕捉子模块包括:图像选择单元,用于从所采集的人脸图像中选择待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;人脸定位单元,用于利用人脸检测模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行定位;关键点定位单元,用于利用人脸关键点定位模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位;以及皮肤区域获得单元,用于根据定位获得的人脸位置和关键点位置对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分,以获得待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

[0071] 示例性地,身份认证装置还包括:样本图像获取模块,用于获取样本人脸图像,样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的;以及训练模块,用于利用样本人脸图像进行神经网络训练,以获得人脸检测模型和人脸关键点定位模型。

[0072] 示例性地,活体检测模块包括:结构光图像获取子模块,用于获取由双目摄像头针对在结构光照射下的待认证人员的人脸所采集的人脸图像;以及活体确定子模块,用于根据人脸图像确定待认证人员的人脸是否属于活体,以获得活体检测结果。

[0073] 示例性地,活体检测模块包括:光照图像获取子模块,用于获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像;光照活体验证子模块,用于基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果;以及活体验证通过确定子模块,用于至少基于光照活体验证结果确定认证人员是否通过活体验证,以获得活体检测结果。

[0074] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式至少改变一次。

[0075] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式在每两个连续的时刻之间发生改变。

[0076] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式是随机改变的或预先设定好的。

[0077] 示例性地,活体检测模块包括:指令输出子模块,用于输出动作指令,其中,动作指令用于指示认证人员执行相应的动作;动作图像获取子模块,用于获取针对待认证人员的人脸所采集的多个动作图像;动作检测子模块,用于基于多个动作图像检测待认证人员所执行的动作;以及动作活体验证子模块,用于根据动作检测结果和动作指令确定待认证人员的人脸是否属于活体,以获得动作活体验证结果;活体验证通过确定子模块包括:通过确定单元,用于基于光照活体验证结果和动作活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0078] 示例性地,动作活体验证子模块包括:活体确定单元,用于如果基于在不大于第一预设时间的时段内所采集的多个动作图像检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体,如果基于在第一预设时间内所采集的多个动作图像未检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸不属于活体。

[0079] 示例性地,身份认证装置还包括:第一计数模块,用于在每次运行指令输出子模块至运行动作活体验证子模块的过程中计数一次,以获得动作验证次数;活体验证装置还包括:第一验证错误执行模块,用于如果动作活体验证结果表示待认证人员的人脸不属于活

体,则输出第一错误信息,并判断动作验证次数是否达到第一次数阈值,如果动作验证次数达到第一次数阈值,则启动活体验证通过确定子模块,如果动作验证次数未达到第一次数阈值,则重新启动指令输出子模块或者在光照图像获取子模块在指令输出子模块之前运行的情况下重新启动光照图像获取子模块,其中,第一错误信息用于提示针对待认证人员的活体验证失败。

[0080] 示例性地,身份认证装置还包括:条件判断模块,用于判断待认证人员的图像采集条件是否满足预设要求,如果图像采集条件满足预设要求,则启动光照图像获取子模块,其中,图像采集条件包括待认证人员在图像采集装置的图像采集区域中所处的位置和/或待认证人员与图像采集装置的相对角度。

[0081] 示例性地,身份认证装置还包括:第一提示信息输出模块,用于在条件判断模块判断待认证人员的图像采集条件是否满足预设要求之前或与条件判断模块判断待认证人员的图像采集条件是否满足预设要求同时地,输出第一提示信息,其中,第一提示信息用于提示待认证人员将脸部正对图像采集装置并靠近图像采集装置。

[0082] 示例性地,第一提示信息输出模块包括:第一提示信息输出子模块,用于通过语音形式、图像形式和文字形式中的一种或多种输出第一提示信息。

[0083] 示例性地,条件判断模块包括:第一实时图像获取子模块,用于获取针对待认证人员的人脸采集的实时图像;第一实时输出子模块,用于实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及第一判断子模块,用于根据在实时图像中检测出的人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在实时图像中所占比例大于第一预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在实时图像中所占比例不大于第一预设比例,则确定图像采集条件不满足预设要求。

[0084] 示例性地,身份认证装置还包括:第一采集提示输出模块,用于如果人脸区域在实时图像中所占比例不大于第一预设比例,则实时输出第一采集提示信息,以提示待认证人员与图像采集装置靠近。

[0085] 示例性地,条件判断模块包括:第二实时图像获取子模块,用于获取针对待认证人员的人脸采集的实时图像;第二实时输出子模块,用于实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及第二判断子模块,用于根据在人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在预设区域中所占比例大于第二预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在预设区域中所占比例不大于第二预设比例,则确定图像采集条件不满足预设要求。

[0086] 示例性地,身份认证方法还包括:如果人脸区域在预设区域中所占比例不大于第二预设比例,则实时输出第二采集提示信息,以提示待认证人员与图像采集装置靠近。

[0087] 示例性地,身份认证装置还包括:第二采集提示输出模块,用于实时判断人脸区域与预设区域之间的相对位置关系;以及第三采集提示输出模块,用于基于人脸区域与预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近。

[0088] 示例性地,条件判断模块包括:姿态信息获取模块,用于获取图像采集装置的姿态

信息;以及第三判断子模块,用于根据姿态信息判断图像采集装置是否处于竖直放置状态,如果是,则确定图像采集条件满足预设要求,否则确定图像采集条件不满足预设要求。

[0089] 示例性地,身份认证方法还包括:第二计数模块,用于运行光照图像获取模块至运行光照活体验证模块的过程中计数一次,以获得光照验证次数;身份认证装置还包括:第二验证错误执行模块,用于如果光照活体验证结果表示待认证人员的人脸不属于活体,则输出第二错误信息,并判断光照验证次数是否达到第二次数阈值,如果光照验证次数达到第二次数阈值,则启动活体验证通过确定子模块,如果光照验证次数未达到第二次数阈值,则重新启动条件判断模块或重新启动光照图像获取子模块,其中,第二错误信息用于提示针对待认证人员的活体验证失败。

[0090] 示例性地,身份认证装置还包括:第二提示信息输出模块,用于在光照图像获取子模块获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像之前或在光照图像获取子模块获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像及光照活体验证子模块基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体的过程中,输出第二提示信息,其中,第二提示信息用于提示待认证人员在第二预设时间内保持不动。

[0091] 示例性地,第二提示信息为与第二预设时间相对应的倒计时信息。

[0092] 示例性地,检测光通过动态改变向待认证人员的人脸发射的光的颜色和/或位置来获得。

[0093] 示例性地,检测光由显示屏幕发射,检测光通过以下方式获得:通过改变显示屏幕上显示的内容来动态改变显示屏幕发射的光的模式,以向待认证人员的人脸发射检测光。

[0094] 根据本发明另一方面,提供一种身份认证装置,包括处理器和存储器,其中,存储器中存储有计算机程序指令,计算机程序指令被处理器运行时用于执行以下步骤:获取待认证人员的个人标识信息;判断个人标识信息是否为己认证信息,以获得信息认证结果;对待认证人员进行活体检测,以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0095] 示例性地,个人标识信息为证件号码、姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种。

[0096] 示例性地,在计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息之前,计算机程序指令被处理器运行时还用于执行以下步骤:输出用于指示待认证人员提供预定类型的人员信息的指示信息;其中,个人标识信息为由待认证人员提供的人员信息或基于人员信息获得。

[0097] 示例性地,个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种,计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的初始信息,其中,初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及基于预定算法对初始信息进行变换,以获得初始信息的变换值作为个人标识信息。

[0098] 示例性地,计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的证件图像,其中,信息认证结果为证件认证结果;计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检

测结果的步骤包括:获取待认证人员的人脸图像;以及利用人脸图像进行活体检测,以获得活体检测结果。

[0099] 示例性地,在计算机程序指令被处理器运行时所用于执行的至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法的步骤之前,计算机程序指令被处理器运行时还用于执行以下步骤:利用证件图像和/或人脸图像执行附加判断操作,以获得附加判断结果;计算机程序指令被处理器运行时所用于执行的至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法的步骤包括:根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法。

[0100] 示例性地,附加判断操作包括证件真伪判断操作和/或人脸一致性判断操作,并且附加判断结果包括证件真伪判断结果和/或人脸一致性判断结果,证件真伪判断操作包括:判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果;人脸一致性判断操作包括:根据证件图像获取待认证人员的证件人脸;以及将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。

[0101] 示例性地,计算机程序指令被处理器运行时所还用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:从证件图像中检测人脸,以获得待认证人员的证件人脸。

[0102] 示例性地,计算机程序指令被处理器运行时所用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;基于证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息;以及确定搜索到的、匹配的证件信息中的证件人脸为待认证人员的证件人脸。

[0103] 示例性地,计算机程序指令被处理器运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:提取证件图像的图像特征;以及将图像特征输入训练好的证件分类器,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0104] 示例性地,计算机程序指令被处理器运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:从证件图像中识别包含证件标识信息的图像块;以及对包含证件标识信息的图像块中的证件标识信息进行识别,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0105] 示例性地,在计算机程序指令被处理器运行时所用于执行的根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法的步骤的执行过程中,证件认证结果、活体检测结果和附加判断结果中的每个结果具有各自的权重系数。

[0106] 示例性地,计算机程序指令被处理器运行时所用于执行的获取待认证人员的证件图像的步骤包括:获取在当前拍摄条件下针对待认证人员的证件实时采集的预拍摄图像;实时评估预拍摄图像的图像属性;当预拍摄图像的图像属性的评估值小于预设的评估值阈值时,根据预拍摄图像的图像属性生成预拍摄提示信息,用于提示待认证人员调整其证件的拍摄条件;以及当预拍摄图像的图像属性的评估值等于或大于预设的评估值阈值时,保存预拍摄图像以将其作为证件图像。

[0107] 示例性地,计算机程序指令被处理器运行时所用于执行的判断个人标识信息是否为已认证信息,以获得信息认证结果的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;以及基于证件图像中的文字信息在已认证证件信息数据库中进行搜索,

以获得证件认证结果;其中,证件认证结果为证件图像中的证件信息是已认证证件信息的置信度。

[0108] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;检测人脸图像中的人脸;以及判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0109] 示例性地,在计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤之前计算机程序指令被处理器运行时还用于执行以下步骤:获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像,以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像;从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像;以及利用正样本图像和负样本图像训练分类器模型,以获得皮肤弹性分类器。

[0110] 示例性地,在计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤之后,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤还包括:输出证件图像中的文字信息,以供用户查看;接收用户输入的文字修正信息;将文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字进行对比;以及如果文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字之间的差异小于预设的差异阈值,则利用文字修正信息更新证件图像中的文字信息。

[0111] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;检测人脸图像中的人脸;以及判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0112] 示例性地,计算机程序指令被处理器运行时还用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像;以及将待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器,以获得活体检测结果。

[0113] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像,以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像;从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像;以及利用正样本图像和负样本图像训练分类器模型,以获得皮肤弹性分类器。

[0114] 示例性地,计算机程序指令被处理器运行时所用于执行的从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像的步骤包括:从所采集的人脸图像中选择待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;利用人脸检测模型对在待认证人员执行活体动作前的人脸图像和执行活

体动作后的人脸图像中的人脸进行定位;利用人脸关键点定位模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位;以及根据定位获得的人脸位置和关键点位置对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分,以获得待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

[0115] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:获取样本人脸图像,样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的;以及利用样本人脸图像进行神经网络训练,以获得人脸检测模型和人脸关键点定位模型。

[0116] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:获取由双目摄像头针对在结构光照射下的待认证人员的人脸所采集的人脸图像;以及根据人脸图像确定待认证人员的人脸是否属于活体,以获得活体检测结果。

[0117] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:步骤S110:获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像;步骤S120:基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果;以及步骤S130:至少基于光照活体验证结果确定认证人员是否通过活体验证,以获得活体检测结果。

[0118] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式至少改变一次。

[0119] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式在每两个连续的时刻之间发生改变。

[0120] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式是随机改变的或预先设定好的。

[0121] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:步骤S140:输出动作指令,其中,动作指令用于指示认证人员执行相应的动作;步骤S150:获取针对待认证人员的人脸所采集的多个动作图像;步骤S160:基于多个动作图像检测待认证人员所执行的动作;以及步骤S170:根据动作检测结果和动作指令确定待认证人员的人脸是否属于活体,以获得动作活体验证结果;计算机程序指令被处理器运行时所用于执行的步骤S130包括:基于光照活体验证结果和动作活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0122] 示例性地,计算机程序指令被处理器运行时所用于执行的步骤S170包括:如果基于在不大于第一预设时间的时段内所采集的多个动作图像检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体,如果基于在第一预设时间内所采集的多个动作图像未检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸不属于活体。

[0123] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:在每次执行步骤S140至步骤S170的过程中计数一次,以获得动作验证次数;在步骤S170之后,计算机程序指令被处理器运行时还用于执行以下步骤:如果动作活体验证结果表示待认证人员的人脸不属于活体,则输出第一错误信息,并判断动作验证次数是否达到第一次数阈值,如果动作

验证次数达到第一次数阈值,则转至步骤S130,如果动作验证次数未达到第一次数阈值,则返回步骤S140或者在步骤S110在步骤S140之前执行的情况下返回步骤S110,其中,第一错误信息用于提示针对待认证人员的活体验证失败。

[0124] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S110之前,计算机程序指令被处理器运行时还用于执行以下步骤:步骤S108:判断待认证人员的图像采集条件是否满足预设要求,如果图像采集条件满足预设要求,则转至步骤S110,其中,图像采集条件包括待认证人员在图像采集装置的图像采集区域中所处的位置和/或待认证人员与图像采集装置的相对角度。

[0125] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S108之前或与计算机程序指令被处理器运行时所用于执行的步骤S108同时地,计算机程序指令被处理器运行时还用于执行以下步骤:步骤S106:输出第一提示信息,其中,第一提示信息用于提示待认证人员将脸部正对图像采集装置并靠近图像采集装置。

[0126] 示例性地,计算机程序指令被处理器运行时所用于的步骤S106包括:通过语音形式、图像形式和文字形式中的一种或多种输出第一提示信息。

[0127] 示例性地,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在实时图像中检测出的人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在实时图像中所占比例大于第一预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在实时图像中所占比例不大于第一预设比例,则确定图像采集条件不满足预设要求。

[0128] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:如果人脸区域在实时图像中所占比例不大于第一预设比例,则实时输出第一采集提示信息,以提示待认证人员与图像采集装置靠近。

[0129] 示例性地,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在预设区域中所占比例大于第二预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在预设区域中所占比例不大于第二预设比例,则确定图像采集条件不满足预设要求。

[0130] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:如果人脸区域在预设区域中所占比例不大于第二预设比例,则实时输出第二采集提示信息,以提示待认证人员与图像采集装置靠近。

[0131] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:实时判断人脸区域与预设区域之间的相对位置关系;以及基于人脸区域与预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近。

[0132] 示例性地,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取图像采集装置的姿态信息;以及根据姿态信息判断图像采集装置是否处于竖直放置状态,如

果是,则确定图像采集条件满足预设要求,否则确定图像采集条件不满足预设要求。

[0133] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:在每次执行步骤S110至步骤S120的过程中计数一次,以获得光照验证次数;在计算机程序指令被处理器运行时所用于执行的步骤S120之后,计算机程序指令被处理器运行时还用于执行以下步骤:如果光照活体验证结果表示待认证人员的人脸不属于活体,则输出第二错误信息,并判断光照验证次数是否达到第二次数阈值,如果光照验证次数达到第二次数阈值,则转至步骤S130,如果光照验证次数未达到第二次数阈值,则返回步骤S108或返回步骤S110,其中,第二错误信息用于提示针对待认证人员的活体验证失败。

[0134] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S110之前或在计算机程序指令被处理器运行时所用于执行的步骤S110和步骤S120的过程中,计算机程序指令被处理器运行时还用于执行以下步骤:输出第二提示信息,其中,第二提示信息用于提示待认证人员在第二预设时间内保持不动。

[0135] 示例性地,第二提示信息为与第二预设时间相对应的倒计时信息。

[0136] 示例性地,检测光通过动态改变向待认证人员的人脸发射的光的颜色和/或位置来获得。

[0137] 示例性地,检测光由显示屏幕发射,检测光通过以下方式获得:通过改变显示屏幕上显示的内容来动态改变显示屏幕发射的光的模式,以向待认证人员的人脸发射检测光。

[0138] 根据本发明另一方面,提供一种存储介质,在存储介质上存储了程序指令,程序指令在运行时用于执行以下步骤:获取待认证人员的个人标识信息;判断个人标识信息是否为已认证信息,以获得信息认证结果;对待认证人员进行活体检测,以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0139] 示例性地,个人标识信息为证件号码、姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种。

[0140] 示例性地,在程序指令在运行时所用于执行的获取待认证人员的个人标识信息之前,程序指令在运行时还用于执行以下步骤:输出用于指示待认证人员提供预定类型的人员信息的指示信息;其中,个人标识信息为由待认证人员提供的人员信息或基于人员信息获得。

[0141] 示例性地,个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种,程序指令在运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的初始信息,其中,初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及基于预定算法对初始信息进行变换,以获得初始信息的变换值作为个人标识信息。

[0142] 示例性地,程序指令在运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的证件图像,其中,信息认证结果为证件认证结果;程序指令在运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:获取待认证人员的人脸图像;以及利用人脸图像进行活体检测,以获得活体检测结果。

[0143] 示例性地,在程序指令在运行时所用于执行的至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法的步骤之前,程序指令在运行时还用于执行以下步骤:利用证件图像和/或人脸图像执行附加判断操作,以获得附加判断结果;程序指令在运

行时所用于执行的至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法的步骤包括:根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法。

[0144] 示例性地,附加判断操作包括证件真伪判断操作和/或人脸一致性判断操作,并且附加判断结果包括证件真伪判断结果和/或人脸一致性判断结果,证件真伪判断操作包括:判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果;人脸一致性判断操作包括:根据证件图像获取待认证人员的证件人脸;以及将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。

[0145] 示例性地,程序指令在运行时所用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:从证件图像中检测人脸,以获得待认证人员的证件人脸。

[0146] 示例性地,程序指令在运行时所用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;基于证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息;以及确定搜索到的、匹配的证件信息中的证件人脸为待认证人员的证件人脸。

[0147] 示例性地,程序指令在运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:提取证件图像的图像特征;以及将图像特征输入训练好的证件分类器,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0148] 示例性地,程序指令在运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:从证件图像中识别包含证件标识信息的图像块;以及对包含证件标识信息的图像块中的证件标识信息进行识别,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0149] 示例性地,在程序指令在运行时所用于执行的根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法的步骤的执行过程中,证件认证结果、活体检测结果和附加判断结果中的每个结果具有各自的权重系数。

[0150] 示例性地,程序指令在运行时所用于执行的获取待认证人员的证件图像的步骤包括:获取在当前拍摄条件下针对待认证人员的证件实时采集的预拍摄图像;实时评估预拍摄图像的图像属性;当预拍摄图像的图像属性的评估值小于预设的评估值阈值时,根据预拍摄图像的图像属性生成预拍摄提示信息,用于提示待认证人员调整其证件的拍摄条件;以及当预拍摄图像的图像属性的评估值等于或大于预设的评估值阈值时,保存预拍摄图像以将其作为证件图像。

[0151] 示例性地,程序指令在运行时所用于执行的判断个人标识信息是否为已认证信息,以获得信息认证结果的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;以及基于证件图像中的文字信息在已认证证件信息数据库中进行搜索,以获得证件认证结果;其中,证件认证结果为证件图像中的证件信息是已认证证件信息的置信度。

[0152] 示例性地,程序指令在运行时所用于执行的对证件图像进行文字识别,以获得证件图像中的文字信息的步骤包括:对证件图像中的文字进行定位,以获得包含文字的图像块;以及对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息。

[0153] 示例性地,在程序指令在运行时所用于执行的对包含文字的图像块中的文字进行

识别的步骤之前,程序指令在运行时还用于执行以下步骤:将包含文字的图像块校正为水平状态。

[0154] 示例性地,在程序指令在运行时所用于执行的对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息的步骤之后,程序指令在运行时所用于执行的对证件图像进行文字识别,以获得证件图像中的文字信息的步骤还包括:输出证件图像中的文字信息,以供用户查看;接收用户输入的文字修正信息;将文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字进行对比;以及如果文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字之间的差异小于预设的差异阈值,则利用文字修正信息更新证件图像中的文字信息。

[0155] 示例性地,程序指令在运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;检测人脸图像中的人脸;以及判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0156] 示例性地,程序指令在运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像;以及将待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器,以获得活体检测结果。

[0157] 示例性地,程序指令在运行时还用于执行以下步骤:获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像,以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像;从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像;以及利用正样本图像和负样本图像训练分类器模型,以获得皮肤弹性分类器。

[0158] 示例性地,程序指令在运行时所用于执行的从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像的步骤包括:从所采集的人脸图像中选择待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;利用人脸检测模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行定位;利用人脸关键点定位模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位;以及根据定位获得的人脸位置和关键点位置对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分,以获得待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

[0159] 示例性地,程序指令在运行时还用于执行以下步骤:获取样本人脸图像,样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的;以及利用样本人脸图像进行神经网络训练,以获得人脸检测模型和人脸关键点定位模型。

[0160] 示例性地,程序指令在运行时所用于执行的对待认证人员进行活体检测,以获得

活体检测结果的步骤包括:获取由双目摄像头针对在结构光照射下的待认证人员的人脸所采集的人脸图像;以及根据人脸图像确定待认证人员的人脸是否属于活体,以获得活体检测结果。

[0161] 示例性地,程序指令在运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:步骤S110:获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像;步骤S120:基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果;以及步骤S130:至少基于光照活体验证结果确定认证人员是否通过活体验证,以获得活体检测结果。

[0162] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式至少改变一次。

[0163] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式在每两个连续的时刻之间发生改变。

[0164] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式是随机改变的或预先设定好的。

[0165] 示例性地,程序指令在运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:步骤S140:输出动作指令,其中,动作指令用于指示认证人员执行相应的动作;步骤S150:获取针对待认证人员的人脸所采集的多个动作图像;步骤S160:基于多个动作图像检测待认证人员所执行的动作;以及步骤S170:根据动作检测结果和动作指令确定待认证人员的人脸是否属于活体,以获得动作活体验证结果;程序指令在运行时所用于执行的步骤S130包括:基于光照活体验证结果和动作活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0166] 示例性地,程序指令在运行时所用于执行的步骤S170包括:如果基于在不大于第一预设时间的时段内所采集的多个动作图像检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体,如果基于在第一预设时间内所采集的多个动作图像未检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸不属于活体。

[0167] 示例性地,程序指令在运行时还用于执行以下步骤:在每次执行步骤S140至步骤S170的过程中计数一次,以获得动作验证次数;在程序指令在运行时所用于执行的步骤S170之后,程序指令在运行时还用于执行以下步骤:如果动作活体验证结果表示待认证人员的人脸不属于活体,则输出第一错误信息,并判断动作验证次数是否达到第一次数阈值,如果动作验证次数达到第一次数阈值,则转至步骤S130,如果动作验证次数未达到第一次数阈值,则返回步骤S140或者在步骤S110在步骤S140之前执行的情况下返回步骤S110,其中,第一错误信息用于提示针对待认证人员的活体验证失败。

[0168] 示例性地,在程序指令在运行时所用于执行的步骤S110之前,程序指令在运行时还用于执行以下步骤:步骤S108:判断待认证人员的图像采集条件是否满足预设要求,如果图像采集条件满足预设要求,则转至步骤S110,其中,图像采集条件包括待认证人员在图像采集装置的图像采集区域中所处的位置和/或待认证人员与图像采集装置的相对角度。

[0169] 示例性地,在程序指令在运行时所用于执行的步骤S108之前或与程序指令在运行时所用于执行的步骤S108同时地,程序指令在运行时还用于执行以下步骤:步骤S106:输出

第一提示信息,其中,第一提示信息用于提示待认证人员将脸部正对图像采集装置并靠近图像采集装置。

[0170] 示例性地,程序指令在运行时所用于执行的步骤S106包括:通过语音形式、图像形式和文字形式中的一种或多种输出第一提示信息。

[0171] 示例性地,程序指令在运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在实时图像中检测出的人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在实时图像中所占比例大于第一预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在实时图像中所占比例不大于第一预设比例,则确定图像采集条件不满足预设要求。

[0172] 示例性地,程序指令在运行时还用于执行以下步骤:如果人脸区域在实时图像中所占比例不大于第一预设比例,则实时输出第一采集提示信息,以提示待认证人员与图像采集装置靠近。

[0173] 示例性地,程序指令在运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在预设区域中所占比例大于第二预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在预设区域中所占比例不大于第二预设比例,则确定图像采集条件不满足预设要求。

[0174] 示例性地,程序指令在运行时还用于执行以下步骤:如果人脸区域在预设区域中所占比例不大于第二预设比例,则实时输出第二采集提示信息,以提示待认证人员与图像采集装置靠近。

[0175] 示例性地,程序指令在运行时还用于执行以下步骤:实时判断人脸区域与预设区域之间的相对位置关系;以及基于人脸区域与预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近。

[0176] 示例性地,程序指令在运行时所用于执行的步骤S108包括:获取图像采集装置的姿态信息;以及根据姿态信息判断图像采集装置是否处于竖直放置状态,如果是,则确定图像采集条件满足预设要求,否则确定图像采集条件不满足预设要求。

[0177] 示例性地,程序指令在运行时还用于执行以下步骤:在每次执行步骤S110至步骤S120的过程中计数一次,以获得光照验证次数;在程序指令在运行时所用于执行的步骤S120之后,程序指令在运行时还用于执行以下步骤:如果光照活体验证结果表示待认证人员的人脸不属于活体,则输出第二错误信息,并判断光照验证次数是否达到第二次数阈值,如果光照验证次数达到第二次数阈值,则转至步骤S130,如果光照验证次数未达到第二次数阈值,则返回步骤S108或返回步骤S110,其中,第二错误信息用于提示针对待认证人员的活体验证失败。

[0178] 示例性地,在程序指令在运行时所用于执行的步骤S110之前或在程序指令在运行时所用于执行的步骤S110和步骤S120的过程中,程序指令在运行时还用于执行以下步骤:输出第二提示信息,其中,第二提示信息用于提示待认证人员在第二预设时间内保持不动。

[0179] 示例性地,第二提示信息为与第二预设时间相对应的倒计时信息。

[0180] 示例性地,检测光通过动态改变向待认证人员的人脸发射的光的颜色和/或位置来获得。

[0181] 示例性地,检测光由显示屏幕发射,检测光通过以下方式获得:通过改变显示屏幕上显示的内容来动态改变显示屏幕发射的光的模式,以向待认证人员的人脸发射检测光。

[0182] 根据本发明另一方面,提供一种身份认证装置,包括信息采集装置、处理器和存储器,其中,信息采集装置用于采集待认证人员的初始信息;存储器中存储有计算机程序指令,计算机程序指令被处理器运行时用于执行以下步骤:获取待认证人员的个人标识信息,其中,个人标识信息为初始信息或基于初始信息获得;判断个人标识信息是否为己认证信息以获得信息认证结果;对待认证人员进行活体检测以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0183] 示例性地,信息采集装置包括图像采集装置和/或输入装置。

[0184] 示例性地,个人标识信息为证件号码、姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种。

[0185] 示例性地,在计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息之前,计算机程序指令被处理器运行时还用于执行以下步骤:输出用于指示待认证人员提供预定类型的人员信息的指示信息;其中,个人标识信息为由待认证人员提供的人员信息或基于人员信息获得。

[0186] 示例性地,个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种,计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的初始信息,其中,初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及基于预定算法对初始信息进行变换,以获得初始信息的变换值作为个人标识信息。

[0187] 示例性地,信息采集装置包括图像采集装置,图像采集装置还用于采集待认证人员的证件图像和人脸图像;计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的证件图像,其中,信息认证结果为证件认证结果;计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:获取待认证人员的人脸图像;以及利用人脸图像进行活体检测,以获得活体检测结果。

[0188] 示例性地,身份认证装置还包括图像采集装置,图像采集装置用于采集待认证人员的证件图像和人脸图像;计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的证件图像,其中,信息认证结果为证件认证结果;计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:获取待认证人员的人脸图像;以及利用人脸图像进行活体检测,以获得活体检测结果。

[0189] 示例性地,在计算机程序指令被处理器运行时所用于执行的至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法的步骤之前,计算机程序指令被处理器运行时还用于执行以下步骤:利用证件图像和/或人脸图像执行附加判断操作,以获得附加判断结果;计算机程序指令被处理器运行时所用于执行的至少根据信息认证结果和活体检

测结果确定待认证人员的身份是否合法的步骤包括:根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法。

[0190] 示例性地,附加判断操作包括证件真伪判断操作和/或人脸一致性判断操作,并且附加判断结果包括证件真伪判断结果和/或人脸一致性判断结果,证件真伪判断操作包括:判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果;人脸一致性判断操作包括:根据证件图像获取待认证人员的证件人脸;以及将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。

[0191] 示例性地,计算机程序指令被处理器运行时所还用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:从证件图像中检测人脸,以获得待认证人员的证件人脸。

[0192] 示例性地,计算机程序指令被处理器运行时所用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;基于证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息;以及确定搜索到的、匹配的证件信息中的证件人脸为待认证人员的证件人脸。

[0193] 示例性地,计算机程序指令被处理器运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:提取证件图像的图像特征;以及将图像特征输入训练好的证件分类器,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0194] 示例性地,计算机程序指令被处理器运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:从证件图像中识别包含证件标识信息的图像块;以及对包含证件标识信息的图像块中的证件标识信息进行识别,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0195] 示例性地,在计算机程序指令被处理器运行时所用于执行的根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法的步骤的执行过程中,证件认证结果、活体检测结果和附加判断结果中的每个结果具有各自的权重系数。

[0196] 示例性地,图像采集装置进一步用于针对待认证人员的证件实时采集预拍摄图像;计算机程序指令被处理器运行时所用于执行的获取待认证人员的证件图像的步骤包括:获取在当前拍摄条件下针对所述待认证人员的证件实时采集的预拍摄图像;实时评估预拍摄图像的图像属性;当预拍摄图像的图像属性的评估值小于预设的评估值阈值时,根据预拍摄图像的图像属性生成预拍摄提示信息,用于提示待认证人员调整其证件的拍摄条件;以及当预拍摄图像的图像属性的评估值等于或大于预设的评估值阈值时,保存预拍摄图像以将其作为证件图像。

[0197] 示例性地,计算机程序指令被处理器运行时所用于执行的判断个人标识信息是否为已认证信息,以获得信息认证结果的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;以及基于证件图像中的文字信息在已认证证件信息数据库中进行搜索,以获得证件认证结果;其中,证件认证结果为证件图像中的证件信息是已认证证件信息的置信度。

[0198] 示例性地,计算机程序指令被处理器运行时所用于执行的对证件图像进行文字识别,以获得证件图像中的文字信息的步骤包括:对证件图像中的文字进行定位,以获得包含文字的图像块;以及对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信

息。

[0199] 示例性地,在计算机程序指令被处理器运行时所用于执行的包含文字的图像块中的文字进行识别的步骤之前计算机程序指令被处理器运行时还用于执行以下步骤:将包含文字的图像块校正为水平状态。

[0200] 示例性地,在计算机程序指令被处理器运行时所用于执行的包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息的步骤之后,计算机程序指令被处理器运行时所用于执行的对证件图像进行文字识别,以获得证件图像中的文字信息的步骤还包括:输出证件图像中的文字信息,以供用户查看;接收用户输入的文字修正信息;将文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字进行对比;以及如果文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字之间的差异小于预设的差异阈值,则利用文字修正信息更新证件图像中的文字信息。

[0201] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;检测人脸图像中的人脸;以及判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0202] 示例性地,计算机程序指令被处理器运行时还用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像;以及将待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器,以获得活体检测结果。

[0203] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像,以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像;从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像;以及利用正样本图像和负样本图像训练分类器模型,以获得皮肤弹性分类器。

[0204] 示例性地,计算机程序指令被处理器运行时所用于执行的从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像的步骤包括:从所采集的人脸图像中选择待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;利用人脸检测模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行定位;利用人脸关键点定位模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位;以及根据定位获得的人脸位置和关键点位置对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分,以获得待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

[0205] 示例性地, 计算机程序指令被处理器运行时还用于执行以下步骤: 获取样本人脸图像, 样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的; 以及利用样本人脸图像进行神经网络训练, 以获得人脸检测模型和人脸关键点定位模型。

[0206] 示例性地, 身份认证装置还包括结构光源和双目摄像头, 其中, 结构光源用于向待认证人员的人脸发射结构光; 双目摄像头用于针对在结构光照射下的待认证人员的人脸采集人脸图像; 计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测, 以获得活体检测结果的步骤包括: 获取由双目摄像头针对在结构光照射下的待认证人员的人脸所采集的人脸图像; 以及根据人脸图像确定待认证人员的人脸是否属于活体, 以获得活体检测结果。

[0207] 示例性地, 信息采集装置包括图像采集装置, 身份认证装置还包括光源, 其中, 光源用于向待认证人员的人脸发射检测光; 图像采集装置还用于采集待认证人员的人脸在检测光的照射下的一个或多个光照图像; 计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测, 以获得活体检测结果的步骤包括: 步骤S110: 获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像; 步骤S120: 基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体, 以获得光照活体验证结果; 以及步骤S130: 至少基于光照活体验证结果确定认证人员是否通过活体验证, 以获得活体检测结果。

[0208] 示例性地, 身份认证装置还包括光源和图像采集装置, 其中, 光源用于向待认证人员的人脸发射检测光; 图像采集装置还用于采集待认证人员的人脸在检测光的照射下的一个或多个光照图像; 计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测, 以获得活体检测结果的步骤包括: 步骤S110: 获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像; 步骤S120: 基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体, 以获得光照活体验证结果; 以及步骤S130: 至少基于光照活体验证结果确定认证人员是否通过活体验证, 以获得活体检测结果。

[0209] 示例性地, 在照射待认证人员的人脸的过程中, 检测光的模式至少改变一次。

[0210] 示例性地, 在照射待认证人员的人脸的过程中, 检测光的模式在每两个连续的时刻之间发生改变。

[0211] 示例性地, 在照射待认证人员的人脸的过程中, 检测光的模式是随机改变的或预先设定好的。

[0212] 示例性地, 计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测, 以获得活体检测结果的步骤包括: 步骤S140: 输出动作指令, 其中, 动作指令用于指示认证人员执行相应的动作; 步骤S150: 获取针对待认证人员的人脸所采集的多个动作图像; 步骤S160: 基于多个动作图像检测待认证人员所执行的动作; 以及步骤S170: 根据动作检测结果和动作指令确定待认证人员的人脸是否属于活体, 以获得动作活体验证结果; 计算机程序指令被处理器运行时所用于执行的步骤S130包括: 基于光照活体验证结果和动作活体验证结果确定待认证人员是否通过活体验证, 以获得活体检测结果; 图像采集装置还用于采集所述多个动作图像。

[0213] 示例性地, 计算机程序指令被处理器运行时所用于执行的步骤S170包括: 如果基

于在不大于第一预设时间的时段内所采集的多个动作图像检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体,如果基于在第一预设时间内所采集的多个动作图像未检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸不属于活体。

[0214] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:在每次执行步骤S140至步骤S170的过程中计数一次,以获得动作验证次数;在步骤S170之后,计算机程序指令被处理器运行时还用于执行以下步骤:如果动作活体验证结果表示待认证人员的人脸不属于活体,则输出第一错误信息,并判断动作验证次数是否达到第一次数阈值,如果动作验证次数达到第一次数阈值,则转至步骤S130,如果动作验证次数未达到第一次数阈值,则返回步骤S140或者在步骤S110在步骤S140之前执行的情况下返回步骤S110,其中,第一错误信息用于提示针对待认证人员的活体验证失败。

[0215] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S110之前,计算机程序指令被处理器运行时还用于执行以下步骤:步骤S108:判断待认证人员的图像采集条件是否满足预设要求,如果图像采集条件满足预设要求,则转至步骤S110,其中,图像采集条件包括待认证人员在图像采集装置的图像采集区域中所处的位置和/或待认证人员与图像采集装置的相对角度。

[0216] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S108之前或与计算机程序指令被处理器运行时所用于执行的步骤S108同时地,计算机程序指令被处理器运行时还用于执行以下步骤:步骤S106:输出第一提示信息,其中,第一提示信息用于提示待认证人员将脸部正对图像采集装置并靠近图像采集装置。

[0217] 示例性地,计算机程序指令被处理器运行时所用于的步骤S106包括:通过语音形式、图像形式和文字形式中的一种或多种输出第一提示信息。

[0218] 示例性地,所述身份认证装置进一步包括显示装置,其中,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在实时图像中检测出的人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在实时图像中所占比例大于第一预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在实时图像中所占比例不大于第一预设比例,则确定图像采集条件不满足预设要求;图像采集装置还用于获取所述实时图像;显示装置用于显示预设区域以及人脸区域。

[0219] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:如果人脸区域在实时图像中所占比例不大于第一预设比例,则实时输出第一采集提示信息,以提示待认证人员与图像采集装置靠近。

[0220] 示例性地,所述活体验证装置进一步包括显示装置,其中,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在预设区域中所占比例大于第二预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在预设区域中所占比例不大于第二预设比例,则

确定图像采集条件不满足预设要求;图像采集装置还用于获取所述实时图像;显示装置用于显示预设区域以及人脸区域。

[0221] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:如果人脸区域在预设区域中所占比例不大于第二预设比例,则实时输出第二采集提示信息,以提示待认证人员与图像采集装置靠近。

[0222] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:实时判断人脸区域与预设区域之间的相对位置关系;以及基于人脸区域与预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近。

[0223] 示例性地,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取图像采集装置的姿态信息;以及根据姿态信息判断图像采集装置是否处于竖直放置状态,如果是,则确定图像采集条件满足预设要求,否则确定图像采集条件不满足预设要求。

[0224] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:在每次执行步骤S110至步骤S120的过程中计数一次,以获得光照验证次数;在计算机程序指令被处理器运行时所用于执行的步骤S120之后,计算机程序指令被处理器运行时还用于执行以下步骤:如果光照活体验证结果表示待认证人员的人脸不属于活体,则输出第二错误信息,并判断光照验证次数是否达到第二次数阈值,如果光照验证次数达到第二次数阈值,则转至步骤S130,如果光照验证次数未达到第二次数阈值,则返回步骤S108或返回步骤S110,其中,第二错误信息用于提示针对待认证人员的活体验证失败。

[0225] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S110之前或在计算机程序指令被处理器运行时所用于执行的步骤S110和步骤S120的过程中,计算机程序指令被处理器运行时还用于执行以下步骤:输出第二提示信息,其中,第二提示信息用于提示待认证人员在第二预设时间内保持不动。

[0226] 示例性地,第二提示信息为与第二预设时间相对应的倒计时信息。

[0227] 示例性地,检测光通过动态改变向待认证人员的人脸发射的光的颜色和/或位置来获得。

[0228] 示例性地,检测光由显示屏幕发射,检测光通过以下方式获得:通过改变显示屏幕上显示的内容来动态改变显示屏幕发射的光的模式,以向待认证人员的人脸发射检测光。

[0229] 根据本发明另一方面,提供一种身份认证装置,包括:信息采集装置,用于采集待认证人员的初始信息;以及传输装置,用于将初始信息发送至服务器,并从服务器接收服务器通过以下方式获得的关于待认证人员的身份是否合法的认证信息:获取待认证人员的个人标识信息,其中,个人标识信息为初始信息或基于初始信息获得;判断个人标识信息是否为己认证信息以获得信息认证结果;对待认证人员进行活体检测以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0230] 示例性地,信息采集装置包括图像采集装置和/或输入装置。

[0231] 根据本发明另一方面,提供一种身份认证装置,包括传输装置、处理器和存储器,其中,传输装置用于从客户端接收待认证人员的初始信息,并将关于待认证人员的身份是否合法的认证信息发送至客户端;存储器中存储有计算机程序指令,计算机程序指令被处理器运行时用于执行以下步骤:获取待认证人员的个人标识信息,其中,个人标识信息为初

始信息或基于初始信息获得;判断个人标识信息是否为已认证信息以获得信息认证结果;对待认证人员进行活体检测以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0232] 示例性地,个人标识信息为证件号码、姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种。

[0233] 示例性地,在计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息之前,计算机程序指令被处理器运行时还用于执行以下步骤:输出用于指示待认证人员提供预定类型的人员信息的指示信息;传输装置还用于将指示信息发送至客户端以由客户端输出;其中,个人标识信息为由待认证人员提供的人员信息或基于人员信息获得。

[0234] 示例性地,个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种,计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的初始信息,其中,初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及基于预定算法对初始信息进行变换,以获得初始信息的变换值作为个人标识信息。

[0235] 示例性地,计算机程序指令被处理器运行时所用于执行的获取待认证人员的个人标识信息的步骤包括:获取待认证人员的证件图像,其中,信息认证结果为证件认证结果;计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:获取待认证人员的人脸图像;以及利用人脸图像进行活体检测,以获得活体检测结果。

[0236] 示例性地,在计算机程序指令被处理器运行时所用于执行的至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法的步骤之前,计算机程序指令被处理器运行时还用于执行以下步骤:利用证件图像和/或人脸图像执行附加判断操作,以获得附加判断结果;计算机程序指令被处理器运行时所用于执行的至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法的步骤包括:根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法。

[0237] 示例性地,附加判断操作包括证件真伪判断操作和/或人脸一致性判断操作,并且附加判断结果包括证件真伪判断结果和/或人脸一致性判断结果,证件真伪判断操作包括:判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果;人脸一致性判断操作包括:根据证件图像获取待认证人员的证件人脸;以及将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。

[0238] 示例性地,计算机程序指令被处理器运行时所还用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:从证件图像中检测人脸,以获得待认证人员的证件人脸。

[0239] 示例性地,计算机程序指令被处理器运行时所用于执行的根据证件图像获取待认证人员的证件人脸的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;基于证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息;以及确定搜索到的、匹配的证件信息中的证件人脸为待认证人员的证件人脸。

[0240] 示例性地,计算机程序指令被处理器运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:提取证件图像的图像特征;以及将

图像特征输入训练好的证件分类器,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0241] 示例性地,计算机程序指令被处理器运行时所用于执行的判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果的步骤包括:从证件图像中识别包含证件标识信息的图像块;以及对包含证件标识信息的图像块中的证件标识信息进行识别,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0242] 示例性地,在计算机程序指令被处理器运行时所用于执行的根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法的步骤的执行过程中,证件认证结果、活体检测结果和附加判断结果中的每个结果具有各自的权重系数。

[0243] 示例性地,计算机程序指令被处理器运行时所用于执行的获取待认证人员的证件图像的步骤包括:获取在当前拍摄条件下针对所述待认证人员的证件实时采集的预拍摄图像;实时评估预拍摄图像的图像属性;当预拍摄图像的图像属性的评估值小于预设的评估值阈值时,根据预拍摄图像的图像属性生成预拍摄提示信息,用于提示待认证人员调整其证件的拍摄条件;以及当预拍摄图像的图像属性的评估值等于或大于预设的评估值阈值时,保存预拍摄图像以将其作为证件图像;传输装置还用于接收预拍摄图像。

[0244] 示例性地,计算机程序指令被处理器运行时所用于执行的判断个人标识信息是否为已认证信息,以获得信息认证结果的步骤包括:对证件图像进行文字识别,以获得证件图像中的文字信息;以及基于证件图像中的文字信息在已认证证件信息数据库中进行搜索,以获得证件认证结果;其中,证件认证结果为证件图像中的证件信息是已认证证件信息的置信度。

[0245] 示例性地,计算机程序指令被处理器运行时所用于执行的对证件图像进行文字识别,以获得证件图像中的文字信息的步骤包括:对证件图像中的文字进行定位,以获得包含文字的图像块;以及对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息。

[0246] 示例性地,在计算机程序指令被处理器运行时所用于执行的对包含文字的图像块中的文字进行识别的步骤之前计算机程序指令被处理器运行时还用于执行以下步骤:将包含文字的图像块校正为水平状态。

[0247] 示例性地,在计算机程序指令被处理器运行时所用于执行的对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息的步骤之后,计算机程序指令被处理器运行时所用于执行的对证件图像进行文字识别,以获得证件图像中的文字信息的步骤还包括:输出证件图像中的文字信息,以供用户查看;接收用户输入的文字修正信息;将文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字进行对比;以及如果文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字之间的差异小于预设的差异阈值,则利用文字修正信息更新证件图像中的文字信息。

[0248] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;检测人脸图像中的人脸;以及判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0249] 示例性地, 计算机程序指令被处理器运行时还用于执行的对待认证人员进行活体检测, 以获得活体检测结果的步骤包括: 生成活体动作指令, 活体动作指令用于指示待认证人员执行对应的活体动作; 获取实时采集的待认证人员的人脸图像; 从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像; 以及将待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器, 以获得活体检测结果。

[0250] 示例性地, 计算机程序指令被处理器运行时还用于执行以下步骤: 获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像, 以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像; 从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像; 从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像; 以及利用正样本图像和负样本图像训练分类器模型, 以获得皮肤弹性分类器。

[0251] 示例性地, 计算机程序指令被处理器运行时所用于执行的从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像的步骤包括: 从所采集的人脸图像中选择待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像; 利用人脸检测模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行定位; 利用人脸关键点定位模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位; 以及根据定位获得的人脸位置和关键点位置对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分, 以获得待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

[0252] 示例性地, 计算机程序指令被处理器运行时还用于执行以下步骤: 获取样本人脸图像, 样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的; 以及利用样本人脸图像进行神经网络训练, 以获得人脸检测模型和人脸关键点定位模型。

[0253] 示例性地, 计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测, 以获得活体检测结果的步骤包括: 获取由双目摄像头针对在结构光照射下的待认证人员的人脸所采集的人脸图像; 以及根据人脸图像确定待认证人员的人脸是否属于活体, 以获得活体检测结果。

[0254] 示例性地, 计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测, 以获得活体检测结果的步骤包括: 步骤S110: 获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像; 步骤S120: 基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体, 以获得光照活体验证结果; 以及步骤S130: 至少基于光照活体验证结果确定认证人员是否通过活体验证, 以获得活体检测结果。

[0255] 示例性地, 在照射待认证人员的人脸的过程中, 检测光的模式至少改变一次。

[0256] 示例性地, 在照射待认证人员的人脸的过程中, 检测光的模式在每两个连续的时刻之间发生改变。

[0257] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式是随机改变的或预先设定好的。

[0258] 示例性地,计算机程序指令被处理器运行时所用于执行的对待认证人员进行活体检测,以获得活体检测结果的步骤包括:步骤S140:输出动作指令,其中,动作指令用于指示认证人员执行相应的动作;步骤S150:获取针对待认证人员的人脸所采集的多个动作图像;步骤S160:基于多个动作图像检测待认证人员所执行的动作;以及步骤S170:根据动作检测结果和动作指令确定待认证人员的人脸是否属于活体,以获得动作活体验证结果;计算机程序指令被处理器运行时所用于执行的步骤S130包括:基于光照活体验证结果和动作活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果;传输装置还用于将动作指令发送至客户端并从客户端接收多个动作图像。

[0259] 示例性地,计算机程序指令被处理器运行时所用于执行的步骤S170包括:如果基于在不大于第一预设时间的时段内所采集的多个动作图像检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体,如果基于在第一预设时间内所采集的多个动作图像未检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸不属于活体。

[0260] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:在每次执行步骤S140至步骤S170的过程中计数一次,以获得动作验证次数;在步骤S170之后,计算机程序指令被处理器运行时还用于执行以下步骤:如果动作活体验证结果表示待认证人员的人脸不属于活体,则输出第一错误信息,并判断动作验证次数是否达到第一次数阈值,如果动作验证次数达到第一次数阈值,则转至步骤S130,如果动作验证次数未达到第一次数阈值,则返回步骤S140或者在步骤S110在步骤S140之前执行的情况下返回步骤S110,其中,第一错误信息用于提示针对待认证人员的活体验证失败;传输装置还用于将第一错误信息发送至客户端以由客户端输出。

[0261] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S110之前,计算机程序指令被处理器运行时还用于执行以下步骤:步骤S108:判断待认证人员的图像采集条件是否满足预设要求,如果图像采集条件满足预设要求,则转至步骤S110,其中,图像采集条件包括待认证人员在图像采集装置的图像采集区域中所处的位置和/或待认证人员与图像采集装置的相对角度。

[0262] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S108之前或与计算机程序指令被处理器运行时所用于执行的步骤S108同时地,计算机程序指令被处理器运行时还用于执行以下步骤:步骤S106:输出第一提示信息,其中,第一提示信息用于提示待认证人员将脸部正对图像采集装置并靠近图像采集装置;传输装置还用于将第一提示信息发送至客户端以由客户端输出。

[0263] 示例性地,计算机程序指令被处理器运行时所用于的步骤S106包括:通过语音形式、图像形式和文字形式中的一种或多种输出第一提示信息。

[0264] 示例性地,所述身份认证装置进一步包括显示装置,其中,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在实时图像中检测出的人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预

设区域内并且人脸区域在实时图像中所占比例大于第一预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在实时图像中所占比例不大于第一预设比例,则确定图像采集条件不满足预设要求;传输装置还用于从客户端接收实时图像并将预设区域以及人脸区域发送至客户端以由客户端输出用于显示。

[0265] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:如果人脸区域在实时图像中所占比例不大于第一预设比例,则实时输出第一采集提示信息,以提示待认证人员与图像采集装置靠近;传输装置还用于将第一采集提示信息发送至客户端以由客户端输出。

[0266] 示例性地,所述活体验证装置进一步包括显示装置,其中,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在预设区域中所占比例大于第二预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在预设区域中所占比例不大于第二预设比例,则确定图像采集条件不满足预设要求;传输装置还用于从客户端接收实时图像并将预设区域以及人脸区域发送至客户端以由客户端输出用于显示。

[0267] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:如果人脸区域在预设区域中所占比例不大于第二预设比例,则实时输出第二采集提示信息,以提示待认证人员与图像采集装置靠近;传输装置还用于将第二采集提示信息发送至客户端以由客户端输出。

[0268] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:实时判断人脸区域与预设区域之间的相对位置关系;以及基于人脸区域与预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近;传输装置还用于将第三采集提示信息发送至客户端以由客户端输出。

[0269] 示例性地,计算机程序指令被处理器运行时所用于执行的步骤S108包括:获取图像采集装置的姿态信息;以及根据姿态信息判断图像采集装置是否处于竖直放置状态,如果是,则确定图像采集条件满足预设要求,否则确定图像采集条件不满足预设要求。

[0270] 示例性地,计算机程序指令被处理器运行时还用于执行以下步骤:在每次执行步骤S110至步骤S120的过程中计数一次,以获得光照验证次数;在计算机程序指令被处理器运行时所用于执行的步骤S120之后,计算机程序指令被处理器运行时还用于执行以下步骤:如果光照活体验证结果表示待认证人员的人脸不属于活体,则输出第二错误信息,并判断光照验证次数是否达到第二次数阈值,如果光照验证次数达到第二次数阈值,则转至步骤S130,如果光照验证次数未达到第二次数阈值,则返回步骤S108或返回步骤S110,其中,第二错误信息用于提示针对待认证人员的活体验证失败;传输装置还用于将第二错误信息发送至客户端以由客户端输出。

[0271] 示例性地,在计算机程序指令被处理器运行时所用于执行的步骤S110之前或在计算机程序指令被处理器运行时所用于执行的步骤S110和步骤S120的过程中,计算机程序指令被处理器运行时还用于执行以下步骤:输出第二提示信息,其中,第二提示信息用于提示

待认证人员在第二预设时间内保持不动;传输装置还用于将第二提示信息发送至客户端以由客户端输出。

[0272] 示例性地,第二提示信息为与第二预设时间相对应的倒计时信息。

[0273] 示例性地,检测光通过动态改变向待认证人员的人脸发射的光的颜色和/或位置来获得。

[0274] 示例性地,检测光由显示屏幕发射,检测光通过以下方式获得:通过改变显示屏幕上显示的内容来动态改变显示屏幕发射的光的模式,以向待认证人员的人脸发射检测光。

[0275] 根据本发明实施例的身份认证方法和装置及存储介质,结合已认证信息判断和活体检测来确定待认证人员的身份是否合法,因此,与常规的单纯基于密码或证件进行身份认证的方式相比,根据本发明实施例的身份认证方法的认证结果更准确,可以提高用户鉴权的安全性,从而可以有效保障用户的权益。

附图说明

[0276] 通过结合附图对本发明实施例进行更详细的描述,本发明的上述以及其它目的、特征和优势将变得更加明显。附图用来提供对本发明实施例的进一步理解,并且构成说明书的一部分,与本发明实施例一起用于解释本发明,并不构成对本发明的限制。在附图中,相同的参考标号通常代表相同部件或步骤。

[0277] 图1示出用于实现根据本发明实施例的身份认证方法和装置的示例电子设备的示意性框图;

[0278] 图2示出根据本发明一个实施例的身份认证方法的示意性流程图;

[0279] 图3示出根据本发明另一个实施例的身份认证方法的示意性流程图;

[0280] 图4示出根据本发明另一个实施例的身份认证方法的示意性流程图;

[0281] 图5示出根据本发明一个实施例的皮肤弹性分类器的训练步骤的示意性流程图;

[0282] 图6示出用于实现根据本发明另一个实施例的身份认证方法和装置的示例电子设备的示意性框图;

[0283] 图7示出根据本发明一个实施例的活体检测步骤的示意性流程图;

[0284] 图8示出根据本发明另一个实施例的活体检测步骤的示意性流程图;

[0285] 图9示出根据本发明一个实施例的活体检测步骤的实现流程;

[0286] 图10示出根据本发明一个实施例的身份认证装置的示意性框图;以及

[0287] 图11示出根据本发明一个实施例的身份认证系统的示意性框图。

具体实施方式

[0288] 为了使得本发明的目的、技术方案和优点更为明显,下面将参照附图详细描述根据本发明的示例实施例。显然,所描述的实施例仅仅是本发明的一部分实施例,而不是本发明的全部实施例,应理解,本发明不受这里描述的示例实施例的限制。基于本发明中描述的本发明实施例,本领域技术人员在没有付出创造性劳动的情况下所得到的所有其它实施例都应落入本发明的保护范围之内。

[0289] 为了解决上文所述的问题,本发明实施例提供一种身份认证方法和装置及存储介质。该身份认证方法和装置及存储介质结合个人标识信息识别和人脸识别进行身份认证,

以确定待认证人员的身份是否合法,即确定待认证人员是否具有权限进行随后的消费支付等操作。根据本发明实施例的身份认证方法和装置可以方便安全地鉴定待认证人员的身份,是一种安全的交互鉴权方式,可以很好地应用于电子商务、移动支付、银行开户等技术领域。

[0290] 首先,参照图1来描述用于实现根据本发明实施例的身份认证方法和装置及存储介质的示例电子设备100。

[0291] 如图1所示,电子设备100包括一个或多个处理器102、一个或多个存储装置104、输入装置106、输出装置108和图像采集装置110,这些组件通过总线系统112和/或其它形式的连接机构(未示出)互连。应当注意,图1所示的电子设备100的组件和结构只是示例性的,而非限制性的,根据需要,所述电子设备也可以具有其他组件和结构。

[0292] 所述处理器102可以是中央处理单元(CPU)或者具有数据处理能力和/或指令执行能力的其它形式的处理单元,并且可以控制所述电子设备100中的其它组件以执行期望的功能。

[0293] 所述存储装置104可以包括一个或多个计算机程序产品,所述计算机程序产品可以包括各种形式的计算机可读存储介质,例如易失性存储器和/或非易失性存储器。所述易失性存储器例如可以包括随机存取存储器(RAM)和/或高速缓冲存储器(cache)等。所述非易失性存储器例如可以包括只读存储器(ROM)、硬盘、闪存等。在所述计算机可读存储介质上可以存储一个或多个计算机程序指令,处理器102可以运行所述程序指令,以实现下文所述的本发明实施例中(由处理器实现)的客户端功能以及/或者其它期望的功能。在所述计算机可读存储介质中还可以存储各种应用程序和各种数据,例如所述应用程序使用和/或产生的各种数据等。

[0294] 所述输入装置106可以是用户用来输入指令的装置,并且可以包括键盘、鼠标、麦克风和触摸屏等中的一个或多个。

[0295] 所述输出装置108可以向外部(例如用户)输出各种信息(例如图像和/或声音),并且可以包括显示器、扬声器等中的一个或多个。

[0296] 所述图像采集装置110可以采集证件图像(包括视频帧)和/或人脸图像(包括视频帧),并且将所采集的图像存储在所述存储装置104中以供其它组件使用。图像采集装置110可以是摄像头。应当理解,图像采集装置110仅是示例,电子设备100可以不包括图像采集装置110。在这种情况下,可以利用其他图像采集装置采集证件图像和/或人脸图像,并将采集的图像发送给电子设备100。

[0297] 示例性地,用于实现根据本发明实施例的身份认证方法和装置的示例电子设备可以在诸如个人计算机或远程服务器等的设备上实现。

[0298] 本发明实施例提供一种身份认证方法,包括:获取待认证人员的个人标识信息;判断所述个人标识信息是否为己认证信息,以获得信息认证结果;对所述待认证人员进行活体检测,以获得活体检测结果;以及至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

[0299] 下面,将参考图2描述根据本发明一个实施例的身份认证方法。图2示出根据本发明一个实施例的身份认证方法200的示意性流程图。如图2所示,身份认证方法200包括以下步骤。

[0300] 在步骤S210,获取待认证人员的个人标识信息。

[0301] 下面主要以待认证人员的个人标识信息为证件图像中的证件为例来描述本文所述的身份认证方法,在实际使用场景下,个人标识信息并不限于此,其可以为,例如,证件号码、姓名、证件人脸(即从证件图像中检测出的人脸图像)和现场采集人脸(即采用图像采集装置针对待认证人员的人脸采集的人脸图像)中的一种或多种,和/或证件号码的变换值(例如某种散列算法的输出值)、姓名的变换值(例如某种散列算法的输出值)、证件人脸的变换值(例如某种散列算法的输出值)和现场采集人脸的变换值(例如某种散列算法的输出值)中的一种或多种。个人标识信息的类型可以根据特定的使用场景选择,本发明对个人标识信息的类型选取不作具体限定。

[0302] 步骤S210可以包括:获取待认证人员的证件图像。可以将证件图像中的证件信息视为个人标识信息。

[0303] 本文所述的证件可以包括但不限于身份证、驾驶证、护照、社保卡等。

[0304] 证件图像可以是针对待认证人员的证件所采集的图像。示例性地,证件图像可以是摄像头等图像采集装置采集到的原始图像,也可以是对原始图像进行预处理之后获得的图像。

[0305] 证件图像可以由客户端设备(例如包括摄像头的移动终端、远程视频柜员机(Video Teller Machine,VTM)等)发送到电子设备100以由电子设备100的处理器102进行处理,也可以由电子设备100包括的图像采集装置110(例如摄像头)采集并传送到处理器102进行处理。

[0306] 根据本发明实施例,获取待认证人员的证件图像可以包括:获取在当前拍摄条件下针对待认证人员的证件实时采集的预拍摄图像;实时评估预拍摄图像的图像属性;当预拍摄图像的图像属性的评估值小于预设的评估值阈值时,根据预拍摄图像的图像属性生成预拍摄提示信息,用于提示待认证人员调整其证件的拍摄条件;以及当预拍摄图像的图像属性的评估值等于或大于预设的评估值阈值时,保存预拍摄图像以将其作为证件图像。

[0307] 预拍摄是指启动图像采集装置(例如手机、平板电脑等移动终端的摄像头)的拍摄模式并将待拍摄的证件置于图像采集装置的拍摄范围内进行拍摄取景的过程(尚未实际完成照片拍摄)。

[0308] 可选地,可以对拍摄得到的预拍摄图像进行质量评估。例如,在预拍摄过程中,可以实时计算在当前拍摄条件下进行拍摄所得到的预拍摄图像的图像属性。示例性地,拍摄条件可以包括但不限于下列中的一项或多项:证件的摆放位置、证件的摆放角度、图像采集装置的拍摄位置、图像采集装置的拍摄角度等。示例性地,图像属性可以包括但不限于下列中的一项或多项:证件模糊程度、证件轮廓、证件关键部位、证件遮挡情况、证件尺寸、证件文字清晰度等。当图像属性的评估值小于预设的评估值阈值时,认为拍摄到的预拍摄图像不合格。此时,可以根据图像属性生成相应的预拍摄提示信息并将该信息输出,以提示用户调整证件或者图像采集装置的角度、位置等,直到拍摄到合格的预拍摄图像。合格的预拍摄图像即图像属性的评估值等于或大于预设的评估值阈值的预拍摄图像。当在预拍摄模式中拍摄到合格的预拍摄图像时,可以将该预拍摄图像保存下来作为步骤S210所获取的证件图像,用于随后的已认证证件判断等步骤。可选地,当预拍摄图像的某些图像属性不合格时,可以对预拍摄图像进行调整,以使预拍摄图像合格,进而获得所需的证件图像。例如,当预

拍摄图像中的证件大小不合格时,可以对预拍摄图像进行裁剪、缩放等操作,使预拍摄图像中的证件大小合格。

[0309] 在步骤S220,判断个人标识信息是否为已认证信息,以获得信息认证结果。

[0310] 在个人标识信息为证件图像中的证件信息的情况下,已认证信息可以是已认证证件信息,信息认证结果可以是证件认证结果。也就是说,在个人标识信息为证件图像中的证件信息的情况下,步骤S220可以包括:判断证件图像中的证件信息是否为已认证证件信息,以获得证件认证结果。

[0311] 例如,可以从证件图像中识别出与待认证人员的证件相关的一些证件信息,例如身份证上的身份证号、姓名等信息,随后,可以判断这些证件信息是否为已认证证件信息,也就是,判断待认证人员的证件是否为已认证证件。

[0312] 关于已认证证件的已认证证件信息可以存储在数据库中,本文将该数据库称为已认证证件信息数据库。可以基于从证件图像中识别出的证件信息在已认证证件信息数据库中进行搜索,即将从证件图像中识别出的证件信息与已认证证件信息数据库中的已认证证件信息进行对比,以判断证件图像中的证件是否为已认证证件。

[0313] 在一个示例中,已认证证件信息数据库可以存储在本地,例如存储在用于实现身份认证方法和装置的服务器或客户端等设备的存储装置(如图1所示的存储装置104)中。

[0314] 在另一示例中,已认证证件信息数据库可以存储在某些公共服务系统(例如公安系统)的服务器中。用于实现身份认证方法和装置的服务器或客户端等设备可以通过联网对接方式与公共服务系统的服务器通信,从公共服务系统的服务器中搜索证件信息。例如,在公安网上通常有已认证的合法人员的证件信息(备案信息),可以基于在步骤S210中所获取的证件图像在公安网中进行搜索查找,若能查找到与从该证件图像中识别出的证件信息匹配的证件信息(备案信息),则可以确定证件图像中的证件为已认证证件,即为合法证件。

[0315] 在另一示例中,个人标识信息为身份证号。待认证人员可以将其身份证号输入身份认证装置。身份认证装置可以基于接收到的身份证号在已认证人员信息数据库中进行检索,该已认证人员信息数据库可以存储大量已认证人员的身份证号。如果存在匹配的身份证号,说明待认证人员为已认证人员,其个人标识信息为已认证信息。类似地,上述已认证人员信息库可以可选地存储在本地或公共服务系统的服务器中。

[0316] 本领域技术人员可以理解,在个人标识信息为姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值或现场采集人脸的变换值的实施例,身份认证方法的实现方式与在个人标识信息为身份证号的实施例中身份认证方法的实现方式类似,不再赘述。

[0317] 在步骤S230,对待认证人员进行活体检测,以获得活体检测结果。

[0318] 步骤S230可以包括:获取待认证人员的人脸图像;以及利用人脸图像进行活体检测,以获得活体检测结果。

[0319] 人脸图像可以是针对待认证人员的人脸所采集的图像。示例性地,人脸图像可以是摄像头等图像采集装置采集到的原始图像,也可以是对原始图像进行预处理之后获得的图像。

[0320] 人脸图像可以由客户端设备(例如包括摄像头的移动终端)发送到电子设备100,由电子设备100的处理器102进行处理,也可以由电子设备100包括的图像采集装置110(例

如摄像头)采集并传送到处理器102进行处理。步骤S230可以采用任何现有的或将来可能实现的活体检测方法实现,本发明不对此进行限制。示例性地,当人脸图像中的人脸是真实人脸时,认为待认证人员是活体,活体检测结果可以是1,当人脸图像中的人脸是虚假人脸时,认为待认证人员不是活体,活体检测结果可以是0。

[0321] 在步骤S240,至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0322] 在一个示例中,信息认证结果可以是1和0之一,其中,1表示待认证人员的个人标识信息是已认证信息,0表示待认证人员的个人标识信息不是已认证信息。类似地,活体检测结果可以是1和0之一,其中,1表示待认证人员是活体(即通过活体验证),0表示待认证人员不是活体(即未通过活体验证)。示例性地,如果信息认证结果和活体检测结果中的任一者为0,则可以认为待认证人员的身份不合法,即待认证人员鉴权失败,在这种情况下,可以禁止其进行后续的业务操作,例如禁止其进行线上交易或银行开户等操作。

[0323] 在另一示例中,信息认证结果可以是在[0,1]范围内的任何数值,表示待认证人员的个人标识信息是已认证信息的置信度。在这种情况下,可以对信息认证结果和活体检测结果进行诸如加权平均的运算,并根据运算结果衡量待认证人员是否是合法人员。类似的实施例将在下文详细描述,此处不赘述。

[0324] 应理解,图2所示的身份认证方法200的各步骤的执行顺序仅是示例而非限制,例如,步骤S230可以在步骤S210之前、在步骤S210与步骤S220之间、或与步骤S210或S220同时执行。

[0325] 根据本发明实施例的身份认证方法,结合已认证信息判断和活体检测来确定待认证人员的身份是否合法,因此,与常规的单纯基于密码或证件进行身份认证的方式相比,根据本发明实施例的身份认证方法的认证结果更准确,可以提高用户鉴权的安全性,从而可以有效保障用户的权益。上述方法可以很好地应用于各种涉及身份认证的领域,例如电子商务、移动支付或银行业务等领域。

[0326] 示例性地,根据本发明实施例的身份认证方法可以在具有存储器和处理器的设备、装置或者系统中实现。

[0327] 根据本发明实施例的身份认证方法可以部署在图像采集端处,例如,可以部署在诸如银行管理系统等金融系统的图像采集端处或者部署在诸如智能手机、平板电脑等移动终端处。替代地,根据本发明实施例的身份认证方法还可以分布地部署在服务器端(或云端)和客户端处。例如,可以在客户端收集个人标识信息(例如采集证件图像或接收待认证人员输入的证件号码、姓名等)和/或采集人脸图像,客户端将收集到的个人标识信息和/或采集到的人脸图像传送给服务器端(或云端),由服务器端(或云端)进行身份认证。

[0328] 根据本发明实施例,在步骤S210之前,身份认证方法200还可以包括:输出用于指示待认证人员提供预定类型的人员信息的指示信息;其中,个人标识信息为由待认证人员提供的人员信息或基于人员信息获得。

[0329] 由于用于存储已认证信息的数据库(诸如上述已认证证件信息数据库或已认证人员信息数据库等)所存储的已认证信息的类型是已确定的,所以为了顺利实施身份认证,需要用户(即待认证人员)提供类型与数据库所存储的已认证信息的类型一致的个人标识信息。为此,可以输出指示信息来指示待认证人员输入预定类型的人员信息。示例性地,可以

通过文字、图像或语音等形式输出指示信息。

[0330] 例如,可以在显示屏上显示“姓名:”和“身份证号:”这样的指示信息,指示待认证人员在信息类型(即姓名和身份证号)后面的空白处输入对应的人员信息。待认证人员输入的人员信息可以直接作为个人标识信息使用,也可以通过对其进行变换来获得个人标识信息。

[0331] 又例如,可以通过扬声器发出“请出示身份证”这样的语音提示,待认证人员提供自己的身份证之后,可以针对身份证进行图像采集,获得身份证图像。在本示例中,个人标识信息为身份证图像。

[0332] 输出用于指示待认证人员提供预定类型的人员信息的指示信息,有利于获取有价值的、能够与已认证信息进行对比的个人标识信息,从而有利于顺利实施身份认证。此外,指示信息的输出可以提升用户与身份认证装置之间的交互体验。

[0333] 由上文所述可知,个人标识信息可以是诸如证件号码、姓名、证件人脸和现场采集人脸等的原始信息,也可以是对这些原始信息进行变换之后获得的变换值。根据本发明实施例,在个人标识信息为证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种的情况下,步骤S210可以包括:获取待认证人员的初始信息,其中,初始信息为证件号码、姓名、证件人脸和现场采集人脸中的一种或多种;以及基于预定算法对初始信息进行变换,以获得初始信息的变换值作为个人标识信息。

[0334] 在将已认证人员的人员信息存储在数据库中时,可以使用预定算法(例如某种散列算法)对已认证人员的人员信息进行变换,以获得已认证人员的已认证信息。变换方式是任意的,其可以根据需要设定。变换过程可以理解为是一种编码过程。

[0335] 身份认证装置所获取的待认证人员的初始信息通常是没有经过变换的信息,例如证件号码、姓名、证件人脸、现场采集人脸等。为了方便后续与已认证信息的比较,可以采用与生成已认证信息时的算法一致的算法来对初始信息进行变换,以获得个人标识信息。

[0336] 根据本发明实施例,在个人标识信息为证件图像中的证件信息的情况下,在步骤S240之前,身份认证方法200还可以包括:利用证件图像和/或人脸图像执行附加判断操作,以获得附加判断结果;步骤S240可以包括:根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法。

[0337] 附加判断操作可以包括一种或多种针对证件图像中的证件和/或人脸图像中的人脸进行真实性或一致性判断的操作。例如,附加判断操作可以包括证件真伪判断操作和/或人脸一致性判断操作。在已认证证件判断操作和活体检测操作之外,加入一些其他的针对证件图像中的证件和/或人脸图像中的人脸进行真实性或一致性判断的操作,这有利于进一步提高身份认证结果的可靠性,从而可以提高涉及身份认证的 ứng dụng 的安全性。

[0338] 下面举例描述附加判断操作的实施方式。

[0339] 根据一个实施例,附加判断操作可以包括证件真伪判断操作。证件真伪判断操作可以包括:判断证件图像中的证件是否为真实证件,以获得证件真伪判断结果。根据另一个实施例,附加判断操作可以包括人脸一致性判断操作。人脸一致性判断操作可以包括:根据证件图像获取待认证人员的证件人脸;以及将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。下面分别结合图3和图4描述上述两个实施例。

[0340] 图3示出根据本发明另一个实施例的身份认证方法300的示意性流程图。图3描述

的是待认证人员的个人标识信息为证件图像中的证件信息的实施例。图3所示的身份认证方法300的步骤S310、S330至S350在上文关于图2所示的身份认证方法200的步骤S210至S230的描述中已介绍过,在此不再赘述。根据本实施例,在步骤S360之前,身份认证方法300还可以包括步骤S320。在步骤S320中,执行上述证件真伪判断操作。在步骤S360,根据证件认证结果(即信息认证结果)、活体检测结果和证件真伪判断结果确定待认证人员的身份是否合法。

[0341] 在执行证件真伪判断操作之后,可以获得证件真伪判断结果。示例性地,证件真伪判断结果可以是1和0之一,其中,1表示证件图像中的证件是真实证件,0表示证件图像中的证件是虚假证件。当然,证件真伪判断结果还可以是在 $[0,1]$ 范围内的任何数值,表示证件图像中的证件是真实证件的置信度。虚假证件可以是,例如,针对手机或电脑等设备上的屏幕翻拍获得的证件或采用计算机图形技术伪造获得的证件等。

[0342] 示例性地,如果证件认证结果、活体检测结果和证件真伪判断结果中的任何一者为0,则可以确定待认证人员的身份不合法,否则可以确定待认证人员的身份是合法的。

[0343] 下面描述步骤S320的两种示例性实施方式。

[0344] 在一个示例中,步骤S320可以包括:提取证件图像的图像特征;以及将图像特征输入训练好的证件分类器,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0345] 由于图像采集装置的感光元件与显示器之间的波的干涉效应,通过翻拍电脑或手机屏幕上的证件照片而得到的图像上会出现明显的周期性彩色条纹,称为“摩尔纹”。摩尔纹是区别真实证件和翻拍证件的重要线索。由于摩尔纹呈现周期性,因此在频率域中摩尔纹的特性会尤为明显。此外,摩尔纹的颜色也区别于真实证件的颜色。因此,可以基于摩尔纹鉴别证件图像中的证件是否为翻拍证件。

[0346] 示例性地,图像特征可以包括但不限于频谱特征、纹理特征和颜色特征中的至少一个。

[0347] 证件真伪判断操作中所涉及的证件分类器可以是预先采用大量样本证件图像训练好的。示例性地,本文所述的“分类器”可以是任何现有的或将来可能实现的基于机器学习的分类器,例如支持向量机(Support Vector Machine,SVM)等。

[0348] 以身份证为例,证件分类器的训练过程可以包括:采集和标注包含真实身份证的身份证图像和包含翻拍身份证的身份证图像;分别计算包含真实身份证的身份证图像和包含翻拍身份证的身份证图像的频谱信息,以作为各自的图像特征;以包含真实身份证的身份证图像的图像特征作为正样本,以包含翻拍身份证的身份证图像的图像特征作为负样本训练分类器模型,以获得身份证分类器。随后,在实际的身份认证过程中,对于所获取的身份证图像,可以计算其频谱信息作为其图像特征,然后将提取的图像特征输入训练好的身份证分类器,以判断所获取的身份证图像中的身份证是否为翻拍身份证。

[0349] 证件分类器输出的证件真伪判断结果可以为证件图像中的证件是真实证件的置信度。置信度可以是在 $[0,1]$ 范围内的任何一个数值。

[0350] 在另一个示例中,步骤S320可以包括:从证件图像中识别包含证件标识信息的图像块;以及对包含证件标识信息的图像块中的证件标识信息进行识别,以获得证件真伪判断结果;其中,证件真伪判断结果为证件图像中的证件是真实证件的置信度。

[0351] 证件标识信息可以是任何能够标识真实证件的信息。例如,证件标识信息可以包括身份证或社保卡上的国徽图案、某些特殊的防伪标志等。例如,真实证件通常具有比较隐蔽的防伪标志,可以通过识别防伪标志来确定证件的真伪。

[0352] 上述通过图像特征进行证件真伪判断和通过证件标识信息进行证件真伪判断的两个示例可以同时实现,即可以同时基于证件图像的图像特征和其证件标识信息来判断证件的真伪,本领域技术人员可以通过阅读上文的描述来理解这种判断方式的实施方式,此处不进行赘述。

[0353] 应理解,图3所示的身份认证方法300的各步骤的执行顺序仅是示例而非限制,例如,步骤S320可以在步骤S310与步骤S360之间的任一时刻执行。

[0354] 图4示出根据本发明另一个实施例的身份认证方法400的示意性流程图。图4描述的是待认证人员的个人标识信息为证件图像中的证件信息的实施例。图4所示的身份认证方法400的步骤S410至S440在上文关于图2所示的身份认证方法200的步骤S210至S240的描述中已介绍过,此处不再赘述。根据本实施例,在步骤S470之前,身份认证方法400还可以包括步骤S450和S460。在步骤S450和S460中,执行上述人脸一致性判断操作。在步骤S470,根据证件认证结果、活体检测结果和人脸一致性判断结果确定待认证人员的身份是否合法。

[0355] 在步骤S450,根据证件图像获取待认证人员的证件人脸。

[0356] 在一个示例中,步骤S450可以包括:从证件图像中检测人脸,以获得待认证人员的证件人脸。

[0357] 证件上通常包括人脸(典型为人脸照片),本文称为“证件人脸”,以与人脸图像中的人脸进行区分。在证件包括人脸的情况下,可以从证件图像中检测出人脸。这样,可以直接将从证件图像中检测出的人脸作为待认证人员的证件人脸,用于与人脸图像中的人脸进行对比。此外,由上文所述可知,从证件图像中检测出的人脸也可以作为个人标识信息使用。

[0358] 在另一个示例中,步骤S450可以包括:对证件图像进行文字识别,以获得所述证件图像中的文字信息;基于证件图像中的文字信息从已认证证件信息数据库中搜索匹配的证件信息;以及确定搜索到的、匹配的证件信息中的证件人脸为待认证人员的证件人脸。

[0359] 在某些证件上可能不包括人脸,在这种情况下,可以利用已认证证件信息数据库中的证件信息来找出待认证人员的证件人脸。当然,在证件包括人脸的情况下,同样可以采用这种方式找出待认证人员的证件人脸。

[0360] 例如,证件图像是人员X的身份证图像,从身份证图像中可以进行文字识别,以识别身份证号等文字信息,然后可以基于身份证号等文字信息从公安系统的身份证数据库中搜索匹配的身份证信息。如果身份证数据库中已对人员X的身份证信息进行过备案,则可以搜索到匹配的身份证信息。身份证信息中可以典型地包括人员X的身份证号、姓名、性别、人脸照片等基本信息。该人脸照片即为所需的证件人脸。随后,可以将该证件人脸与之前获取的人脸图像中的人脸进行对比。

[0361] 在步骤S460,将待认证人员的证件人脸与人脸图像中的人脸进行对比,以获得人脸一致性判断结果。

[0362] 将通过图像采集装置采集的人脸(即人脸图像中的人脸)与证件人脸进行比对,如果二者之间的相似度大于预设的相似度阈值,则可以认为采集的人脸和证件人脸属于同一

人,否则,可以认为二者不属于同一人。由此,可以获得人脸一致性判断结果。示例性地,一致性判断结果可以是证件人脸与人脸图像中的人脸之间的相似度,即一致性结果可以是在 $[0,1]$ 范围内的任一数值。示例性地,一致性判断结果还可以是1和0之一,其中,1表示证件人脸与人脸图像中的人脸属于同一人,0表示证件人脸与人脸图像中的人脸不属于同一人。

[0363] 示例性地,如果证件认证结果、活体检测结果和人脸一致性判断结果中的任何一者为0,则可以确定待认证人员的身份不合法,否则可以确定待认证人员的身份是合法的。

[0364] 应理解,与图3类似地,图4所示的身份认证方法400的各步骤的执行顺序仅是示例而非限制,例如,步骤S450可以在步骤S420之前、或在步骤S420之后及步骤S430之前、或在步骤S430之后及步骤S440之前、或与步骤S420或S430同时执行,同时步骤S460可以在步骤S440之前、之后或与其同时执行。当然,步骤S450也可以与步骤S440同时执行。

[0365] 上文分别结合图3和图4描述了证件真伪判断操作和人脸一致性判断操作的实施方式,可以理解的是,附加判断操作可以包括证件真伪判断操作和人脸一致性判断操作。也就是说,在身份认证过程中,可以同时对待证件的真伪及人脸的一致性进行判断,并根据证件认证结果、活体检测结果、证件真伪判断结果和人脸一致性判断结果这四种结果确定待认证人员的身份是否合法。

[0366] 根据本发明实施例,在根据证件认证结果、活体检测结果和附加判断结果确定待认证人员的身份是否合法的过程中,证件认证结果、活体检测结果和附加判断结果中的每个结果具有各自的权重系数。

[0367] 可以预先为证件认证结果、活体检测结果和附加判断结果(包括例如上述证件真伪判断结果和/或人脸一致性判断结果)中的每个结果分配权重系数。每个结果的权重系数的大小可以根据需要而定,本发明不对此进行限制。

[0368] 示例性地,可以基于所述权重系数对证件认证结果、活体检测结果和附加判断结果进行加权平均,获得经平均的结果。然后,可以将经平均的结果与预设的阈值进行对比,若其大于阈值,则可以认为待认证人员的身份是合法的,否则认为待认证人员的身份不合法。

[0369] 在待认证人员的个人标识信息为证件图像中的证件信息的情况下,证件认证结果、证件真伪判断结果和人脸一致性判断结果均可以是在 $[0,1]$ 范围内的任一数值。当然,这三者的取值也可以仅是0和1之一。活体检测结果的取值是0和1之一。因此,可以对这些结果进行加权平均或算术平均,求得经平均的结果。然后,将经平均的结果与阈值进行对比。

[0370] 当然,也可以直接将参与身份认证的各种检测或判断结果进行简单的求和,获得总的结果,将该总的结果与阈值进行对比,判断待认证人员的身份是否合法。

[0371] 每个结果具有各自的权重系数,方便身份认证系统根据身份认证中所涉及的每种判断或检测操作的重要性调整其在身份认证过程中的参与度,进而可以提高身份认证的准确度。

[0372] 根据本发明实施例,步骤S220(S330、S420)可以包括:对证件图像进行文字识别,以获得证件图像中的文字信息;以及基于证件图像中的文字信息在已认证证件信息数据库中进行搜索,以获得证件认证结果;其中,证件认证结果为证件图像中的证件信息是已认证证件信息的置信度。

[0373] 如上文所述,可以从证件图像中识别出证件图像中的文字信息,然后可以基于识

别出的文字信息从已认证证件信息数据库中搜索匹配的证件信息。搜索结果典型地是已认证证件信息数据库中存在匹配的证件信息的概率(也可以称为置信度),在这种情况下,证件认证结果可以为搜索结果。

[0374] 根据本发明实施例,对证件图像进行文字识别,以获得证件图像中的文字信息可以包括:对证件图像中的文字进行定位,以获得包含文字的图像块;以及对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息。

[0375] 对证件图像进行文字识别的步骤可以采用任何合适的文字识别方式实现。下面描述文字识别步骤的一种实施方式。

[0376] 首先,可以对证件图像中的文字进行定位,确定文字的位置。然后可以从证件图像中提取包含文字的图像块。示例性地,可以将证件图像输入训练好的神经网络,以对证件图像中的文字进行定位。

[0377] 例如,可以收集大量的样本证件图像,通过人工或机器标注的方式,指明样本证件图像上文字的所在位置。基于标注过的大量的样本证件图像通过机器学习算法训练用于定位文字的位置的神经网络。将实际身份认证过程中所获取的证件图像输入训练好的神经网络,神经网络可以输出该证件图像中的文字的位置,例如,文字所在区域的顶点坐标。

[0378] 随后,可以对包含文字的图像块(文字区域)中的文字进行识别,得到文字识别结果。对包含文字的图像块中的文字进行识别是指将包含文字的图像块的图像内容转化为字符串的过程。示例性地,可以采用传统光学字符识别(Optical Character Recognition, OCR)的方法进行识别:先利用二值化操作分割出每个字符,然后以模板匹配或模式分类的方式识别出所有的字符。

[0379] 可选地,还可以采用滑动窗(sliding window)的识别方法,从证件图像中定位和识别文字,而无需依赖二值化分割的结果。

[0380] 根据本发明实施例,在对包含文字的图像块中的文字进行识别之前,身份认证方法200(300、400)还可以包括:将包含文字的图像块校正为水平状态。

[0381] 在对证件图像中的文字进行定位与对包含文字的图像块中的文字进行识别的步骤之间还可以包括对文字的位置进行调整(校正)的步骤。在实际应用中,证件图像中的证件可能存在一定的倾斜角度,也就是说,包含文字的图像块可能存在一定的倾斜角度。因此,可以在识别证件图像中的文字之前,对包含文字的图像块(即文字所在的区域)进行校正,将其转换为水平平齐的状态。示例性地,由于在对证件图像中的文字进行定位的步骤中,已经获得了证件图像中文字所在区域的四个顶点的坐标,并且已获得四个顶点所包含的图像区域,即包含文字的图像块,因此,只需根据包含文字的图像块的坐标将其旋转到水平状态即可。

[0382] 将包含文字的图像块校正为水平状态可以方便后续对包含文字的图像块中的文字进行识别。

[0383] 根据本发明实施例,在对包含文字的图像块中的文字进行识别,以获得证件图像中的文字信息之后,对证件图像进行文字识别,以获得证件图像中的文字信息还包括:输出证件图像中的文字信息,以供用户查看;接收用户输入的文字修正信息;将文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字进行对比;以及如果文字修正信息所指示的待修正文字与证件图像中的文字信息中的对应文字之间的差异小于预设的差

异阈值,则利用文字修正信息更新证件图像中的文字信息。

[0384] 在文字识别过程中,还可以增加文字修正的步骤。采用OCR方法识别文字时,对于一些生僻字、形近字、字符笔画多的文字会存在识别错误或识别失败的情况。因此,允许用户修改文字可以灵活方便地解决文字识别存在的问题,提高文字识别的精度。此处所述用户可以是本文所述待认证人员,也可以是除待认证人员以外的其他人员,例如身份认证系统的管理人员等。

[0385] 示例性地,可以以文字显示或语音播放等形式输出从证件图像中识别出的文字信息,以供用户查看。当用户发现文字识别结果存在错误时,可以输入文字修正信息。在接收到用户输入的文字修正信息之后,可以将文字修正信息所指示的待修正文字与识别出的对应文字进行对比。如果二者之间的差异小于预设的差异阈值,则可以利用文字修正信息更新证件图像中的文字信息,否则,可以不更新文字信息。例如,如果证件上的一个文字被识别为“一”,用户输入的文字修正信息指示将该文字应修正为“壹”,由于二者差异较大,因此身份认证系统可以拒绝用户的修正要求,不对该文字进行修正。

[0386] 根据本发明实施例,步骤S230(S340和S350、或S430和S440)可以包括:生成活体动作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;检测人脸图像中的人脸;以及判断人脸图像中的人脸是否执行活体动作指令所指示的活体动作,以获得活体检测结果。

[0387] 在一个示例中,可以由客户端设备(例如包括摄像头的移动终端、VTM等)生成并输出活体动作指令,采集人脸图像(包括静态图像或者视频),并利用人脸图像进行活体检测。在另一示例中,可以由云端服务器生成活体动作指令,由客户端设备(例如包括摄像头的移动终端、VTM等)输出活体动作指令并采集人脸图像,随后客户端设备将采集的人脸图像上传到云端服务器,云端服务器检测人脸图像中的人脸,并对人脸的真实性进行判断。如果判断出人脸是真实人脸,则可以认为待认证人员是活体,否则可以认为待认证人员不是活体。示例性地,云端服务器中可以包括训练好的人脸真伪分类器和假脸种类分类器。人脸真伪分类器可以用于判断人脸的真伪,假脸种类分类器可以用于判断在人脸是虚假人脸的情况下该虚假人脸的种类。

[0388] 下面通过示例描述基于人脸图像中的人脸的动作进行活体检测的实施例的实施方式,以便于对本实施例的理解。

[0389] 活体动作指令可以指示待认证人员按照指令做出对应的活体动作。活体动作指令所指示的活体动作可以是单一的静态动作(相当于摆一个姿势),也可以是变化的动作。示例性地,活体动作指令可以在采集人脸图像之前生成和输出,在采集人脸图像的过程中不发生改变。示例性地,活体动作指令还可以是连续的指令序列,即在采集人脸图像的过程中不断生成和输出不同的指令,指示待认证人员跟随指令改变其所做出的活体动作。

[0390] 上述活体动作可以是,例如,手指按压两腮皮肤,嘴中吞气将两腮鼓起,或朗读一段文字等。当待认证人员执行一个或多个活体动作时,可以采集其人脸图像,判断其所执行的活体动作是否合格,若是,则活体检测成功,否则,活体检测失败。例如,如果活体动作指令所指示的活体动作为朗读一段文字,则可以采集朗读过程中的人脸图像,判断人脸图像中的人脸的唇动是否与对应文字的唇动匹配,若匹配,则活体检测成功。

[0391] 根据本发明实施例,步骤S230(S340和S350、或S430和S440)可以包括:生成活体动

作指令,活体动作指令用于指示待认证人员执行对应的活体动作;获取实时采集的待认证人员的人脸图像;从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像;以及将待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像输入皮肤弹性分类器,以获得活体检测结果。

[0392] 当待认证人员执行一个或多个活体动作时,可以从采集到的人脸图像中分别捕捉该待认证人员执行活体动作前和执行活体动作后的皮肤区域图像。待认证人员是否开始执行活体动作可以根据采集到的人脸图像进行判断。例如,可以监测人脸图像中的人脸,根据人脸的状态变化确定待认证人员开始执行活体动作的时间。在该开始时间之前采集的人脸图像作为待认证人员执行活体动作前的人脸图像。随后,可以持续监测人脸图像中的人脸一段时间,该持续时间为活体动作的估计持续时间。在持续时间结束之后所采集的人脸图像作为待认证人员执行活体动作后的人脸图像。可以分别从待认证人员执行活体动作前和执行活体动作后的人脸图像中提取待认证人员执行活体动作前和执行活体动作后的皮肤区域图像。

[0393] 随后,可以将所获得的皮肤区域图像输入皮肤弹性分类器,该皮肤弹性分类器为一个预先训练好的分类模型。例如,如果采集的人脸皮肤是活体皮肤,则该皮肤弹性分类器可以输出1,否则输出0。

[0394] 根据本发明实施例,从所采集的人脸图像中捕捉待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像包括:从所采集的人脸图像中选择待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像;利用人脸检测模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行定位;利用人脸关键点定位模型对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸的关键点进行定位;以及根据定位获得的人脸位置和关键点位置对在待认证人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中的人脸进行区域划分,以获得待认证人员执行活体动作前的皮肤区域图像和执行活体动作后的皮肤区域图像。

[0395] 对皮肤区域图像的提取可以基于已有的人脸检测和人脸关键点定位算法来实现。例如,可以将待认证人员执行活体动作前的人脸图像分别输入训练好的人脸检测模型和人脸关键点定位模型,分别获得人脸位置(例如人脸轮廓点的坐标)和关键点位置(各关键点的坐标)。关键点可以是人脸上的任意点,例如左眼角、鼻尖、左唇角等。当然,关键点也可以是人脸轮廓点。示例性地,可以根据人脸位置和关键点位置将人脸区域切分成一系列三角片元,并且可以将位于下巴,颧骨,两腮等区域的三角片元图像块作为人脸皮肤区域,获得皮肤区域图像。待认证人员执行活体动作后的皮肤区域图像的提取方式与上述方式类似,不再赘述。对于每个人脸图像来说,其区域划分方式、以及所选出的作为人脸皮肤区域的区域数目及每个区域包含的部位均可以根据需要设定,本发明不对此进行限制。

[0396] 在一个示例中,人脸检测模型和人脸关键点定位模型可以采用深度神经网络实现。深度神经网络是一种能够自主学习的网络,采用深度神经网络可以准确、高效地检测和定位人脸图像中的人脸及关键点。

[0397] 根据本发明实施例,身份认证方法200(300、400)还可以包括:获取样本人脸图像,样本人脸图像中的人脸的位置和人脸的关键点的位置是已标注好的;以及利用样本人脸图像进行神经网络训练,以获得人脸检测模型和人脸关键点定位模型。

[0398] 可以预先收集大量(如10000个左右)样本人脸图像,以人工方式在每个样本人脸图像中标注出人脸的眼角,嘴角,鼻翼,颧骨最高点等一系列关键点的位置以及人脸轮廓点的位置。随后,可以使用机器学习算法(如深度学习、或者基于局部特征的回归算法等)将前述标注好的样本人脸图像作为输入进行神经网络训练,从而获得所需的人脸检测模型和人脸关键点定位模型。

[0399] 根据本发明实施例,身份认证方法200(300、400)还可以包括皮肤弹性分类器的训练步骤。下面结合图5描述根据本发明一个实施例的皮肤弹性分类器的训练步骤S500的示意性流程图。

[0400] 如图5所示,皮肤弹性分类器的训练步骤S500包括如下步骤。

[0401] 在步骤S510,获取真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像,以及虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像。

[0402] 在步骤S520,从真实人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取真实人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为正样本图像。

[0403] 在步骤S530,从虚假人员执行活体动作前的人脸图像和执行活体动作后的人脸图像中提取虚假人员执行活体动作前的人脸图像和执行活体动作后的皮肤区域图像作为负样本图像。

[0404] 在步骤S540,利用正样本图像和负样本图像训练分类器模型,以获得皮肤弹性分类器。

[0405] 示例性地,皮肤弹性分类器的训练可以离线进行。可以预先收集真实人员执行规定的活体动作前及执行活体动作后的人脸图像,同时还可以收集虚假人员执行规定的活体动作前及执行活体动作后的人脸图像。虚假人员可以是,例如包含人脸的照片、包含人脸的视频、纸片面具或人脸三维(3D)模型等。

[0406] 从真实人员或虚假人员的人脸图像中提取皮肤区域图像的方式可以参考上文关于实际身份认证过程中的皮肤区域图像的提取方式的描述,在此不再赘述。示例性地,在获得正样本图像和负样本图像之后,可以使用深度学习或SVM等统计学习方法训练分类器模型,从而获得皮肤弹性分类器。

[0407] 根据本发明实施例,步骤S230(S340和S350、或S430和S440)可以包括:获取由双目摄像头针对在结构光照射下的待认证人员的人脸所采集的人脸图像;根据人脸图像确定待认证人员的人脸是否属于活体,以获得活体检测结果。

[0408] 上文所述的根据人脸所做的动作或利用皮肤弹性分类器进行活体检测的方式主要针对一些安全要求较弱的应用场景。针对一些安全要求较高的应用场景,可以选择基于特殊硬件进行活体检测的方式。例如,可以采用双目摄像头采集结构光照射下的人脸图像,以利用采集到的人脸信息和结构光照射信息进行活体检测。

[0409] 在一个示例中,可以基于所采集的结构光照射下的人脸图像,来确定指示结构光在待认证人员脸部的亚表面散射程度的检测参数,随后,可以基于检测参数和预先确定的参数阈值,确定待认证人员的人脸是否为活体。由于3D面具等虚假人脸与真实人脸的亚表面散射程度不同(亚表面散射越强烈时,图像梯度越小,从而扩散程度越小),例如,由一般的纸或塑料等材质制成的面具的亚表面散射程度远弱于真实人脸,而由一般的硅胶等材质

制成的面具的亚表面散射程度远强于真实人脸,因此可以通过对图像扩散程度的判断来区分虚假人脸与真实人脸,从而可以有效防御面具攻击者。

[0410] 在另一示例中,可以根据人脸图像获得待认证人员的人脸的深度信息。此外,还可以获取待认证人员的人脸在结构光照射下形成的光斑图样。可以根据光斑图样获得待认证人员的人脸的纹理信息。随后,可以结合深度信息和纹理信息确定待认证人员的人脸是否属于活体。

[0411] 不同的材质结构在结构光下可以形成不同的光斑图样。根据光斑图样可以获得人脸的纹理信息,即人脸表面的材质性质。如果发现待认证人员的人脸的纹理信息不符合人皮纹理分布规律,则确定待认证人员的人脸不属于活体,判断为面具攻击等。由于攻击者可以使用仿人皮材质的面具实施攻击,因此,即使待认证人员的人脸的纹理信息符合人皮纹理分布规律,也不一定能够确定待认证人员的人脸属于活体,因此可以结合深度信息判断待人员的人脸是否属于活体。从在两个不同视角下采集的人脸图像中可以获得待认证人员的人脸的深度信息。应当理解,真实的人脸通常是有起伏的,例如,其眼睛和鼻子部位的坐标深度是不一样的,差距较大,而用仿人皮材质做的面具起伏很小,眼睛和鼻子部位的坐标深度差距很小。因此,结合深度信息可以进一步判断待认证人员的人脸是否属于活体。

[0412] 因此,在本发明实施例中,可以将双目摄像头和结构光结合,通过双目摄像头采集有结构光图案的3D人脸,然后根据结构光在3D人脸上的亚表面散射程度或者结合人脸的深度信息与结构光在人脸上形成的光斑图样来进行活体检测。

[0413] 根据本发明实施例,身份认证方法中的活体检测过程还可以具有其他的实现方式。下面描述另一种活体检测方式。在该活体检测方式中,基于光反射特性来区分真是人脸与照片或屏幕上的虚假人脸。

[0414] 下面描述身份认证方法中的活体检测步骤和身份认证装置中的活体检测模块的示例性实现方案。需注意,本文所述的“活体验证”与“活体检测”二者含义相同,可以互换使用。

[0415] 图6示出用于实现根据本发明另一个实施例的身份认证方法和装置的示例电子设备600。电子设备600包括一个或多个处理器602、一个或多个存储装置604、输入装置606、输出装置608、图像采集装置610和光源612,这些组件通过总线系统614和/或其它形式的连接机构(未示出)互连。图6所示的电子设备600中的处理器602、存储装置604、输入装置606、输出装置608、图像采集装置610和总线系统614与图1所示的处理器102、存储装置104、输入装置106、输出装置108、图像采集装置110和总线系统112的结构及工作原理类似,不再赘述。

[0416] 光源612可以是能够发光的装置,其可以包括发光二极管等专用光源,也可以包括显示屏幕等非常规光源。在身份认证方法和装置实现于诸如智能手机的移动终端中的情况下,输入装置606、输出装置608和光源612可以是同一显示屏幕。

[0417] 图7示出根据本发明一个实施例的活体检测步骤S700(对应于上述步骤S220、或S340和S350、或S430和S440)的示意性流程图。如图7所示,活体检测步骤S700包括以下步骤。

[0418] 在步骤S710,获取针对在检测光的照射下的待认证人员的人脸所采集的一个或多个光照图像。

[0419] 示例性地,可以利用光源向待认证人员的人脸发射检测光。光源可以由处理器控

制以发光。示例性地,光源可以共用其它发光装置(例如,显示屏幕的至少部分区域、投影仪中的光源)作为光源。又例如,光源还可以是专用光源(例如,按照一定方式布置的一个或多个发光二极管或激光二极管,例如用于摄像头的闪光灯等),也可以为显示屏幕和其他类型的光源的组合等。

[0420] 检测光的模式可以包括但不限于检测光的颜色、发光区域的位置、检测光的强度、检测光的照射角度、检测光的波长、检测光的频率等。

[0421] 示例性地,在照射待认证人员的人脸的过程中,检测光的模式可以不发生变化,也就是说,光源可以采用单一不变的光来照射待认证人员的人脸。例如,在一个优选实施例中,所采用的光源为移动终端的显示屏幕。在显示屏幕上,可以控制每个像素的颜色和亮度等,使得显示屏幕能够发射呈现特定图案的光,诸如结构光,此时,屏幕在特定像素区域显示的特定颜色或亮度可以是根据大量实验数据优化后进行选取的特定的检测光的模式,在此种检测光的模式下,通过与该检测光的模式对应的特定算法能够快速准确地对待验证对象进行活体验证。这种情况下,可以在不变的检测光的照射下采集一个或多个光照图像,基于光照图像进行活体验证。

[0422] 比较可取的是,在照射待认证人员的人脸的过程中,检测光的模式至少改变一次。在这种情况下,可以协调控制检测光的模式改变频率与图像的采集频率,以使得在每种模式的检测光下可以采集到至少一个光照图像。

[0423] 更加可取的是,检测光的模式在每两个连续的時刻之间发生改变。所述時刻可以是预定时段中的任意特定的时间点。例如,检测光的模式可以每隔1秒改变一次。不停改变检测光的模式,可以获得更加丰富的光反射特性信息,有利于更加准确高效地实施基于光反射特性的活体验证。

[0424] 可选地,在照射待认证人员的人脸的过程中,检测光的模式是随机改变的或预先设定好的。在一个示例中,检测光的模式是完全随机地改变的。例如,在一个优选实施例中,所采用的光源为移动终端的显示屏幕。在显示屏幕上,可以控制每个区域的颜色,对于每个区域,每次在预定的RGB值范围内随机选取某一RGB值作为该区域的颜色值进行显示。区域的划分可以任意设定,例如,每个区域可以包括一个或多个像素,两个不同区域的大小可以相同,也可以不同。

[0425] 在另一示例中,检测光的模式可以预先设定好。例如,可以设定检测光共持续照射10秒,每秒改变一种模式,每次发射的检测光的颜色、位置、强度等等均预先设定好。在活体验证过程中,光源可以按照预先设定好的方式依次发射10种模式的检测光。预先设定好的检测光的模式可以是基于在先经验获得的、对活体验证来说比较有效的模式,这有利于提高活体验证的准确率和效率。

[0426] 例如,可以通过动态的改变检测光的发光颜色来动态改变照射到待认证人员的人脸的检测光的模式。又例如,还可以通过动态的改变检测光的发光区域的位置(即,改变检测光的位置)来动态改变照射到待认证人员的人脸的检测光的模式。再例如,还可以通过动态地同时改变检测光的发光颜色以及检测光的发光区域的位置来动态改变照射到待认证人员的人脸的检测光的模式。

[0427] 例如,可以通过改变光源的位置来动态的改变检测光的发光区域的位置,这可以改变检测光照射到待认证人员的人脸的位置。又例如,还可以通过改变光源的出射光线的

角度来动态的改变检测光照射到待认证人员的人脸的位置。

[0428] 在一个优选实施例中,所采用的光源为移动终端的显示屏幕,图像采集装置为该移动终端的与显示屏幕位于同一侧的摄像头(例如前置摄像头)。相对于采用额外的专用光源的方案,本方案采用手机等已有移动终端即可实现,不受外部条件限制,可以将该活体检测方案更好地应用于通过个人移动终端进行远程开户等应用场景。

[0429] 进一步地,在上述优选实施例的进一步优选方案中,所采用的光的模式为光的颜色与发光区域的位置的组合模式,例如:同一时刻在显示屏幕的不同位置发出不同颜色的光,或者,同一时刻在显示屏幕的不同位置发出相同颜色的光但不同时刻发出的光的颜色不同等。采用光的颜色与发光区域的位置的组合模式的方案,相对于选用改变光强等其他光的模式的方案,不仅活体检测效果更好,而且可以降低光对人眼的刺激,从而提升用户体验。

[0430] 在待认证人员的人脸受到检测光的照射的情况下,可以利用图像采集装置(例如电子设备600的图像采集装置610)采集待认证人员的人脸在检测光的照射下的图像,获得所述光照图像。图像采集装置可以由处理器控制以采集图像。图像采集装置将一个或多个光照图像传送到身份认证系统的处理器用于活体验证。示例性地,在同一模式的光的照射下采集到的光照图像的数目可以是一个或多个,本发明不对此进行限制。本领域技术人员可以理解的是,活体验证主要是基于人脸进行的验证,因此,根据本文实施例,在采集光照图像及随后的动作图像和实时图像时,目的在于采集到包含人脸的图像,用于进行活体验证。

[0431] 示例性地,光照图像可以由客户端设备(例如包括摄像头的移动终端、远程视频柜员机(Video Teller Machine,VTM)等)发送到电子设备600以由电子设备600的处理器602进行处理,也可以由电子设备600包括的图像采集装置610(例如摄像头)采集并传送到处理器602进行处理。

[0432] 在步骤S720,基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果。

[0433] 诸如人脸皮肤的人类皮肤是一种漫反射的材质,而且人脸是三维的;与此相反,诸如液晶显示器(LCD)或有机发光二极管(OLED)显示器的显示屏幕可以被认为自发光物体并且还通常包括部分镜面反射分量,而照片等通常是平面的并且也通常包括部分镜面反射分量,而且无论是显示屏幕还是照片,其整体上的反射特性是均匀的且缺乏人脸的三维特性。人脸的光反射特性与显示屏幕或者照片的光反射特性不同,由此可以通过基于待认证人员的人脸的光反射特性判断待认证人员的人脸是否为属于活体。

[0434] 在步骤S730,至少基于光照活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0435] 在一个示例中,可以直接基于光照活体验证结果确定最终的活体检测结果,即,如果光照活体验证结果表示待认证人员的人脸属于活体,则确定待认证人员通过活体验证,活体检测结果可以示例性地为1;如果光照活体验证结果表示待认证人员的人脸不属于活体,则确定待认证人员未通过活体验证,活体检测结果可以示例性地为0。这种活体检测方式的计算量较小,效率较高。在另一示例中,可以结合其他活体验证方式,将光照活体验证结果与基于其他活体验证方式获得的其他活体验证结果综合在一起考虑,以最终确定待认

证人员是否通过活体验证。这种活体检测方式的准确率较高。

[0436] 如上文所述,由于人脸的光反射特性与显示屏幕或照片等物体的光反射特性不同,因此基于光反射特性可以有效区分真实人脸和屏幕上回放的人脸或照片上的人脸。因此,采用根据本发明实施例的活体检测方法的身份认证方法和装置可以有效防御屏幕或照片攻击者,从而可以提高身份认证系统的安全性与用户体验。

[0437] 示例性地,采用根据本发明实施例的活体检测方法的身份认证方法可以在具有存储器和处理器的设备、装置或者系统中实现。

[0438] 根据本发明实施例的身份认证方法可以部署在图像采集端处,例如,可以部署在诸如银行管理系统等金融系统的图像采集端处或者部署在诸如智能手机、平板电脑等移动终端处。替代地,根据本发明实施例的身份认证方法还可以分布地部署在服务器端(或云端)和客户端处。例如,可以在客户端获取个人标识信息、发射光并采集待认证人员的人脸的图像,客户端将获取的个人标识信息和采集到的图像传送给服务器端(或云端),由服务器端(或云端)进行已认证信息判断和活体检测,以获得身份认证结果,并将身份认证结果返回给客户端。服务器相对于客户端具有更大的数据运算能力,将身份认证交由服务器端进行,能够提升认证速度,提升用户体验,此外,由于服务器具备更大的运算速度因此可采用更为复杂的身份认证算法,通过在服务器端进行身份认证能够提升身份认证的准确率。

[0439] 虽然基于光反射特性进行活体验证可以防御屏幕或照片攻击,但是攻击者的攻击方式可能有多种,某些其他的攻击方式有可能突破基于光反射特性进行的活体验证,例如三维仿真面具攻击。对于面具攻击的情况,基于光反射特性的活体验证方法可能无法很好地防御。因此,为了进一步完善活体验证方法,提高活体验证的安全性,可以在基于光反射特性的活体验证的基础之上,进一步结合其他活体验证方式来进行活体检测。下面描述一种示例性的实现方式。

[0440] 图8示出根据本发明另一个实施例的活体检测步骤S800(对应于上述步骤S220、或S340和S350、或S430和S440)的示意性流程图。图8所示的活体检测步骤S800中的步骤S810至S830与图7所示的活体检测步骤S700的步骤S710至S730相对应,本领域技术人员可以参考图7的相关描述理解图8所示的步骤S810至S830,此处不再赘述。根据本实施例,活体检测步骤S800还可以包括步骤S840至S870,并且,步骤S830可以包括:基于光照活体验证结果和动作活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0441] 在步骤S840,输出动作指令,其中,动作指令用于指示待认证人员执行相应的动作。

[0442] 示例性地,可以随机地或按预定规则输出动作指令。动作指令可以包括单独的指令,或由一系列指令组成的指令序列。例如,动作指令可以指示待认证人员点头、摇头、眨眼、张嘴,等等。

[0443] 在步骤S850,获取针对待认证人员的人脸所采集的多个动作图像。

[0444] 在输出动作指令的同时或在输出动作指令之后的一段时间内可以针对待认证人员的人脸进行图像采集,获得多个动作图像。示例性地,多个动作图像可以是连续的视频帧。动作图像同样可以由上述图像采集装置110采集,或者由其他图像采集装置采集。

[0445] 在步骤S860,基于多个动作图像检测待认证人员所执行的动作。

[0446] 示例性地,可以在每个动作图像中进行人脸检测和关键点识别,并基于多个动作

图像中的人脸轮廓和/或人脸关键点来判断人脸所执行的动作,例如,通过识别人脸轮廓和/或人脸关键点在获取到的连续多张动作图像中的变化趋势判断人脸所执行的动作。随后,可以判断人脸所执行的动作与动作指令所指示的动作是否一致。

[0447] 在步骤S870,根据动作检测结果和动作指令确定待认证人员的人脸是否属于活体,以获得动作活体验证结果。

[0448] 示例性地,如果待认证人员在多个动作图像中所执行的动作与动作指令所指示的动作一致,则确定待认证人员的人脸属于活体,如果待认证人员在多个动作图像中所执行的动作与动作指令所指示的动作不一致或者待认证人员在多个动作图像中未执行任何动作(即未检测到待认证人员的动作),则确定待认证人员的人脸不属于活体。当然,上述方式仅是示例,动作活体验证是否通过可以有其他的确定方式,例如,待认证人员在多个动作图像中执行了多个动作,多个动作中包括与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体。

[0449] 示例性地,在步骤S830,如果光照活体验证结果和动作活体验证结果均表示待认证人员的人脸属于活体,则确定待认证人员通过活体验证,如果光照活体验证结果和动作活体验证结果中的任何一者表示待认证人员的人脸不属于活体,则确定待认证人员未通过活体验证。当然,上述方式仅是示例,活体验证是否通过可以有其他的确定方式。

[0450] 应注意,上述基于动作的活体验证步骤(步骤S840-S870)与基于光反射特性的活体验证步骤(步骤S810-S820)的先后执行顺序可以任意设定,本发明不对此进行限制。

[0451] 包含基于动作的活体验证步骤的活体检测方法可以由图像采集端独立执行,例如,可以由诸如银行管理系统等金融系统的图像采集端或者诸如智能手机、平板电脑等移动终端独立执行。替代地,包含基于动作的活体验证步骤的活体检测方法还可以由服务器端(或云端)和客户端合作执行。例如,可以在服务器端或客户端生成动作指令,由客户端采集待认证人员的动作图像,客户端将采集到的动作图像传送给服务器端(或云端),由服务器端(或云端)进行基于动作的活体验证,并将验证结果返回给客户端。

[0452] 基于动作的活体验证方式可以防御诸如面具攻击等攻击方式,与基于光反射特性的活体验证方式结合在一起,可以有效防御多种类的攻击,从而进一步保证采用该活体检测方法的身份认证系统或类似系统的安全,同时也有利于保护用户的信息安全及其权益,具有极其广泛的应用价值和市场前景。

[0453] 根据本发明实施例,步骤S870可以包括:如果基于在不大于第一预设时间的时段内所采集的多个动作图像检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸属于活体,如果基于在第一预设时间内所采集的多个动作图像未检测到由待认证人员执行的、与动作指令所指示的动作一致的动作,则确定待认证人员的人脸不属于活体。

[0454] 随机输出动作指令(例如“请点头”、“请张嘴”等文字或语音指令),以指示待认证人员执行相应的动作(例如点头、张嘴等),对人脸区域关键点进行检测以判断待认证人员所执行的动作与输出的动作指令是否一致。若在第一预设时间内检测到待认证人员所执行的动作与输出的动作指令一致,则确定待认证人员的人脸属于活体;若在第一预设时间内检测到待认证人员所执行的动作与输出的动作指令不一致,或者在第一预设时间内未检测到待认证人员的动作,则可以确定待认证人员的人脸不属于活体。

[0455] 根据本发明实施例,身份认证方法(200、300或400)还可以包括:在每次执行步骤S840至步骤S870的过程中计数一次,以获得动作验证次数;在步骤S870之后,身份认证方法(200、300或400)还可以包括:如果动作活体验证结果表示待认证人员的人脸不属于活体,则输出第一错误信息,并判断动作验证次数是否达到第一次数阈值,如果动作验证次数达到第一次数阈值,则转至步骤S330,如果动作验证次数未达到第一次数阈值,则返回步骤S340或者在步骤S310在步骤S340之前执行的情况下返回步骤S310,其中,第一错误信息用于提示针对待认证人员的活体验证失败。

[0456] 示例性地,可以设置计数器,用于为基于动作的活体验证步骤(步骤S340-S370)的执行次数进行计数,每执行一遍基于动作的活体验证步骤,计数器可以加一。计数器的输出结果即为动作验证次数。在整个活体检测步骤(活体检测步骤S800)终止之后,可以将计数器清零。

[0457] 如果当前的动作活体验证结果表示待认证人员的人脸不属于活体,可以输出第一错误信息。第一错误信息可以提示针对待认证人员的人脸的活体验证失败并提示其重新进行活体验证。如果此时基于动作的活体验证步骤的执行次数(动作验证次数)尚未达到预设的第一次数阈值,则可以尝试重新执行基于动作的活体验证步骤。可选地,如果基于光反射特性的活体验证步骤(步骤S810-S820)在基于动作的活体验证步骤之前执行,则可以直接返回至步骤S810,即重新执行一遍基于光反射特性的活体验证步骤和基于动作的活体验证步骤,以提升活体验证的准确性。

[0458] 第一次数阈值可以是任意合适的值,其可以根据需要设定,本发明不对此进行限制。

[0459] 由于在实际的活体验证过程中,可能有各种意外情况,例如用户未及时做出指定动作、采集的图像不够清晰、人脸检测结果不够准确等,这些情况可能造成用户被误识别为非活体。因此,为了兼顾用户体验和系统安全性,可以设定次数阈值,允许用户在合理范围内多次尝试活体验证。如果在达到次数阈值时用户仍未被正确识别为活体,则可以确定其确实不属于活体。

[0460] 根据本发明实施例,在步骤S710(或S810)之前,身份认证方法(200、300或400)还可以包括:步骤S708:判断待认证人员的图像采集条件是否满足预设要求,如果图像采集条件满足预设要求,则转至步骤S710(或S810),其中,图像采集条件包括待认证人员在图像采集装置的图像采集区域中所处的位置和/或待认证人员与图像采集装置的相对角度。

[0461] 在执行基于光反射特性的活体验证步骤或其他活体验证步骤之前,可以首先对待认证人员的图像采集条件进行检测,判断其是否满足预设要求。可以在只有待认证人员的图像采集条件满足预设要求的情况下才执行后续的基于光反射特性的活体验证步骤或其他活体验证步骤,这样可以保证用于活体验证的图像(包括光照图像、动作图像等)的质量,有利于正确检测图像中的人脸,从而有利于提高活体验证的准确率。

[0462] 根据本发明实施例,在步骤S708之前或与步骤S708同时地,身份认证方法(200、300或400)还可以包括:步骤S706:输出第一提示信息,其中,第一提示信息用于提示待认证人员将脸部正对图像采集装置并靠近图像采集装置。

[0463] 第一提示信息可以采用任何合适的方式输出。示例性地,步骤S706可以包括:通过语音形式、图像形式和文字形式中的一种或多种输出第一提示信息。例如,可以在移动终端

的显示屏幕上输出“请正对屏幕”(正对屏幕相当于正对图像采集装置)这样的文字,或者可以通过移动终端的扬声器发出“请正对屏幕”这样的提示语。

[0464] 示例性地,身份认证方法可以由安装在诸如移动终端等电子设备上的应用(APP)来实现。当用户打开该应用并在随后进入活体检测阶段时,可以开始输出第一提示信息,以提示用户与移动终端保持合适的相对位置关系,以便于移动终端的摄像头采集理想的人脸图像用于活体验证。在一个示例中,第一提示信息可以在待认证人员的图像采集条件满足预设要求之前持续地或间隔性地输出。

[0465] 输出第一提示信息可以便于指导用户及时调整身份认证装置(主要是其中的图像采集装置)与自身的相对位置关系,同时,这种用户与系统的交互也可以提升用户体验。

[0466] 根据本发明实施例,步骤S708可以包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在实时图像中检测出的人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在实时图像中所占比例大于第一预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在实时图像中所占比例不大于第一预设比例,则确定图像采集条件不满足预设要求。

[0467] 在一个示例中,可以根据针对待认证人员的人脸采集的图像来判断待认证人员的图像采集条件是否满足预设要求。例如,移动终端可以通过摄像头采集实时图像并进行人脸检测。人脸检测可以获得人脸区域,人脸区域是包含人脸的图像块。可以根据人脸区域在实时图像中所处的位置以及人脸区域在实时图像中所占比例来判断待认证人员的图像采集条件是否满足预设要求。例如,可以在实时图像中划定预设区域。通过预设区域可以限定待认证人员的人脸在图像采集装置的图像采集区域内的位置。人脸区域的大小可以反映待认证人员与图像采集装置的距离和相对角度。预设区域和第一预设比例均可以根据需要设定,本发明不对此进行限制。

[0468] 例如,如果待认证人员的人脸区域位于预设区域内,但是人脸区域在实时图像中所占比例小于第一预设比例(例如三分之二),则有可能是待认证人员相对于图像采集装置过于倾斜和/或距离图像采集装置较远,此时可以认为图像采集条件不满足预设要求。

[0469] 示例性地,身份认证方法(200、300或400)还可以包括:如果人脸区域在实时图像中所占比例不大于第一预设比例,则实时输出第一采集提示信息,以提示待认证人员与图像采集装置靠近。

[0470] 可选地,第一采集提示信息可以通过语音形式、图像形式和文字形式中的一种或多种输出。例如,若发现人脸区域在实时图像中所占比例不大于第一预设比例,可以在显示屏幕上显示诸如“请靠近摄像头”(或“请靠近手机”)这类提示信息。

[0471] 根据本发明实施例,步骤S708可以包括:获取针对待认证人员的人脸采集的实时图像;实时输出用于校准图像采集条件的预设区域以及实时图像中的人脸区域用于显示;以及根据在实时图像中检测出的人脸区域判断图像采集条件是否满足预设要求,如果人脸区域位于预设区域内并且人脸区域在预设区域中所占比例大于第二预设比例,则确定图像采集条件满足预设要求,如果人脸区域没有位于预设区域内或者人脸区域在预设区域中所占比例不大于第二预设比例,则确定图像采集条件不满足预设要求。

[0472] 在一个示例中,可以根据针对待认证人员的人脸采集的图像来判断待认证人员的

图像采集条件是否满足预设要求。例如,移动终端可以通过摄像头采集实时图像并进行人脸检测。人脸检测可以获得人脸区域,人脸区域是包含人脸的图像块。可以根据人脸区域在实时图像中所处的位置以及人脸区域在预设区域中所占比例来判断待认证人员的图像采集条件是否满足预设要求。例如,可以在显示屏幕上显示预设区域。通过预设区域可以限定待认证人员与屏幕之间的相对位置。人脸区域的大小可以反映待认证人员与图像采集装置的距离和相对角度,例如当检测到人脸距离屏幕由远及近时,在屏幕上实时显示的人脸区域由小变大,并在人脸足够靠近屏幕时使屏幕上显示的人脸区域大小满足预设条件,当然,屏幕上实时显示的人脸区域也可以仅在人脸足够靠近屏幕的条件下调整大小以满足预设条件,在此不作限定。预设区域和第二预设比例均可以根据需要设定,本发明不对此进行限制。

[0473] 例如,如果待认证人员的人脸区域位于预设区域内,但是人脸区域在预设区域中所占比例小于第二预设比例(例如三分之二),则有可能是待认证人员相对于图像采集装置过于倾斜和/或距离图像采集装置较远,此时可以认为图像采集条件不满足预设要求。

[0474] 示例性地,身份认证方法(200、300或400)还可以包括:如果人脸区域在预设区域中所占比例不大于第二预设比例,则实时输出第二采集提示信息,以提示待认证人员与图像采集装置靠近。

[0475] 可选地,第二采集提示信息可以通过语音形式、图像形式和文字形式中的一种或多种输出。例如,若发现人脸区域在预设区域中所占比例不大于第二预设比例,可以在显示屏幕上显示诸如“请靠近摄像头”(或“请靠近手机”)这类提示信息。

[0476] 根据本发明实施例,身份认证方法(200、300或400)还可以包括:实时判断人脸区域与预设区域之间的相对位置关系;以及基于人脸区域与预设区域之间的相对位置关系实时输出第三采集提示信息,以提示改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近。

[0477] 例如,当在移动终端上实现根据本发明实施例的身份认证方法和装置时,可以在移动终端的显示屏上实时显示人脸区域(即从实时图像中提取的包含人脸的图像块)和用于表示预设区域的图标(即在屏幕上实时显示的预设区域)。实时显示人脸区域和用于表示预设区域的图标可以方便用户获知其当前的图像采集条件处于何种状况,与预设要求的差距有多大,以方便用户调整自身或图像采集装置(或包含图像采集装置的身份认证装置)的姿态,以尽快进入后续的活体验证阶段。因此,实时显示人脸区域和用于表示预设区域的图标可以提高用户体验,并且可以提高活体验证的效率。

[0478] 此外,还可以输出第三采集提示信息,来提示用户改变待认证人员与图像采集装置之间的相对位置关系以使得人脸区域与预设区域靠近。可选地,第三采集提示信息可以通过语音形式、图像形式和文字形式中的一种或多种输出。例如,若发现人脸区域没有位于预设区域内,则可以在显示屏幕上显示诸如“请靠近圆圈中心”(预设区域在显示屏幕上以圆圈图标显示)这类提示信息。此外,还可以在显示屏幕上显示从人脸区域指向预设区域的箭头,方便用户了解如何移动自身或图像采集装置以使得人脸区域尽快进入预设区域。上述“请靠近圆圈中心”这类文字提示信息和箭头这类图像提示信息可以同时显示或择一显示。

[0479] 根据本发明实施例,步骤S708可以包括:获取图像采集装置的姿态信息;以及根据

姿态信息判断图像采集装置是否处于竖直放置状态,如果是,则确定图像采集条件满足预设要求,否则确定图像采集条件不满足预设要求。

[0480] 例如,当根据本发明实施例的身份认证方法应用于移动终端场景时,图像采集装置(即移动终端的摄像头)的姿态信息可以利用移动终端内置的陀螺仪传感器和/或加速度传感器等装置来测量。当移动终端处于竖直放置状态时,图像采集装置也处于竖直状态,这种情况下,可以采集到比较理想的人脸图像。因此,可以基于图像采集装置的姿态来衡量待认证人员的图像采集条件是否满足要求。

[0481] 根据本发明实施例,身份认证方法(200、300或400)还可以包括:在每次执行步骤S710至步骤S720(或步骤S810至步骤S820)的过程中计数一次,以获得光照验证次数;在步骤S720(或S820)之后,身份认证方法(200、300或400)还可以包括:如果光照活体验证结果表示待认证人员的人脸不属于活体,则输出第二错误信息,并判断光照验证次数是否达到第二次数阈值,如果光照验证次数达到第二次数阈值,则转至步骤S230(或S330),如果光照验证次数未达到第二次数阈值,则返回步骤S208或返回步骤S210(或S310),其中,第二错误信息用于提示针对待认证人员的活体验证失败。

[0482] 与基于动作的活体验证步骤类似地,对于基于光反射特性的活体验证步骤(图7所示的步骤S710-S720或图8所示的步骤S810-S820)来说,如果确定待认证人员的人脸不属于活体,同样可以尝试重新执行基于光反射特性的活体验证步骤,其原理和优势与基于动作的活体验证步骤类似,此处不再赘述。

[0483] 示例性地,在身份认证方法包括上述步骤S708的情况下,还可以从步骤S708开始重新执行。

[0484] 根据本发明实施例,在步骤S710(或S810)之前或在执行步骤S710(或S810)和步骤S720(或S820)的过程中,身份认证方法(200、300或400)还可以包括:输出第二提示信息,其中,第二提示信息用于提示待认证人员在第二预设时间内保持不动。

[0485] 示例性地,第二预设时间可以是基于光反射特性的活体验证步骤(图7所示的步骤S710-S720或图8所示的步骤S810-S820)的执行时间。在基于光反射特性的活体验证步骤执行时,即在采用检测光照射待认证人员的人脸时,可以提示待认证人员在该期间内保持不动,以免影响图像采集效果及活体验证结果。示例性地,如果待认证人员在第二预设时间内移动导致待认证人员的图像采集条件不再满足预设要求,则可以返回步骤S706或步骤S708,重新执行以下步骤中的一种或多种步骤:判断图像采集条件、输出第一提示信息、输出各种采集提示信息等步骤。

[0486] 示例性地,第二提示信息可以为与第二预设时间相对应的倒计时信息。可选地,倒计时信息可以采用文字、动态图像和语音中的一种或多种形式实现。倒计时信息可以方便用户了解活体检测进展,并且可以提高用户的交互体验。

[0487] 下面结合附图9描述根据本发明实施例的活体检测步骤的一种实现流程。图9所示的应用场景为移动终端。

[0488] 如图9所示,首先,在移动终端的显示屏幕上显示“请正对屏幕”这样的文字,提示用户将脸部正对屏幕,同时在显示屏幕上显示用于表示预设区域的图标(用圆圈表示)和基于实时图像检测到的人脸区域。当用户改变自身人脸的位置和/或姿态,和/或改变移动终端的位置和/或姿态时,可以持续显示“请正对屏幕”这样的文字及用于表示预设区域的图

标,这两种信息可以保持不变。然而,人脸区域的大小和位置都可能变化,因此,可以实时显示不断变化的人脸区域,方便用户查看。随后,当用户的图像采集条件满足预设要求时,可以进入下一阶段,即基于光反射特性的活体验证步骤。

[0489] 在基于光反射特性的活体验证步骤的执行过程中,可以在显示屏幕上显示“请保持不动”这样的文字(如图9的第2和3幅图像所示),提示用户保持不动,同时还可以在显示屏幕上显示倒计时信息。倒计时信息在图9所示的第3幅图像中采用标示在用于表示预设区域的图标(即圆圈)上的带颜色的进度条来表示。

[0490] 当基于光反射特性的活体验证步骤完成之后,可以开始执行基于动作的活体验证步骤。如图9的第4幅图像所示,在显示屏幕上显示“请点头”这样的文字,指示用户执行相应的动作。

[0491] 最后,在显示屏幕上输出最终的活体检测结果,诸如“活体验证通过”这样的文字。

[0492] 根据本发明另一方面,提供一种身份认证装置。图10示出了根据本发明一个实施例的身份认证装置1000的示意性框图。

[0493] 如图10所示,根据本发明实施例的身份认证装置1000包括信息获取模块1010、已认证信息判断模块1020、活体检测模块1030和身份确定模块1040。所述各个模块可分别执行上文中结合图2-9描述的身份认证方法的各个步骤/功能。以下仅对该身份认证装置1000的各部件的主要功能进行描述,而省略以上已经描述过的细节内容。

[0494] 信息获取模块1010用于获取待认证人员的个人标识信息。信息获取模块1010可以由图1所示的电子设备中的处理器102运行存储装置104中存储的程序指令来实现。

[0495] 已认证信息判断模块1020用于判断个人标识信息是否为已认证信息,以获得信息认证结果。已认证信息判断模块1020可以由图1所示的电子设备中的处理器102运行存储装置104中存储的程序指令来实现。

[0496] 活体检测模块1030用于对待认证人员进行活体检测,以获得活体检测结果。活体检测模块1030可以由图1所示的电子设备中的处理器102运行存储装置104中存储的程序指令来实现。

[0497] 身份确定模块1040用于至少根据证件认证结果和活体检测结果确定待认证人员的身份是否合法。身份确定模块1040可以由图1所示的电子设备中的处理器102运行存储装置104中存储的程序指令来实现。

[0498] 本领域普通技术人员可以意识到,结合本文中所公开的实施例描述的各示例的单元及算法步骤,能够以电子硬件、或者计算机软件和电子硬件的结合来实现。这些功能究竟以硬件还是软件方式来执行,取决于技术方案的特定应用和设计约束条件。专业技术人员可以对每个特定的应用来使用不同方法来实现所描述的功能,但是这种实现不应认为超出本发明的范围。

[0499] 图11示出了根据本发明一个实施例的身份认证系统1100的示意性框图。身份认证系统1100包括图像采集装置1110、存储装置1120、以及处理器1130。

[0500] 图像采集装置1110用于采集待认证人员的图像,例如人脸图像、证件图像等。图像采集装置1110是可选的,身份认证系统1100可以不包括图像采集装置1110。在这种情况下,可以利用其他图像采集装置采集用于身份认证的图像,并将采集的图像发送给身份认证系统1100。

[0501] 所述存储装置1120存储用于实现根据本发明实施例的身份认证方法中的相应步骤的计算机程序指令。

[0502] 所述处理器1130用于运行所述存储装置1120中存储的计算机程序指令,以执行根据本发明实施例的身份认证方法的相应步骤,并且用于实现根据本发明实施例的身份认证装置中的信息获取模块1010、已认证信息判断模块1020、活体检测模块1030和身份确定模块1040。

[0503] 在一个实施例中,所述计算机程序指令被所述处理器1130运行时用于执行以下步骤:获取待认证人员的个人标识信息;判断个人标识信息是否为已认证信息,以获得信息认证结果;对待认证人员进行活体检测,以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0504] 此外,根据本发明实施例,还提供了一种存储介质,在所述存储介质上存储了程序指令,在所述程序指令被计算机或处理器运行时用于执行本发明实施例的身份认证方法的相应步骤,并且用于实现根据本发明实施例的身份认证装置中的相应模块。所述存储介质例如可以包括智能电话的存储卡、平板电脑的存储部件、个人计算机的硬盘、只读存储器(ROM)、可擦除可编程只读存储器(EPROM)、便携式紧致盘只读存储器(CD-ROM)、USB存储器、或者上述存储介质的任意组合。

[0505] 在一个实施例中,所述计算机程序指令在被计算机或处理器运行时可以使得计算机或处理器实现根据本发明实施例的身份认证装置的各个功能模块,并且/或者可以执行根据本发明实施例的身份认证方法。

[0506] 在一个实施例中,所述计算机程序指令在运行时用于执行以下步骤:获取待认证人员的个人标识信息;判断个人标识信息是否为已认证信息,以获得信息认证结果;对待认证人员进行活体检测,以获得活体检测结果;以及至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0507] 根据本发明实施例的身份认证系统中的各模块可以通过根据本发明实施例的实施身份认证的电子设备的处理器运行在存储器中存储的计算机程序指令来实现,或者可以在根据本发明实施例的计算机程序产品的计算机可读存储介质中存储的计算机指令被计算机运行时实现。

[0508] 根据本发明另一方面,提供一种身份认证装置,包括信息采集装置、处理器和存储器,其中,信息采集装置用于接收待认证人员的个人标识信息;所述存储器中存储有计算机程序指令,所述计算机程序指令被所述处理器运行时用于执行以下步骤:获取所述待认证人员的个人标识信息,其中,所述个人标识信息为所述初始信息或基于所述初始信息获得;判断所述个人标识信息是否为已认证信息以获得信息认证结果;对所述待认证人员进行活体检测以获得活体检测结果;以及至少根据所述信息认证结果和所述活体检测结果确定所述待认证人员的身份是否合法。

[0509] 示例性地,所述个人信息采集装置包括图像采集装置和/或输入装置。

[0510] 示例性地,所述身份认证装置还可以包括:图像采集装置,用于采集待认证人员的人脸图像;所述处理装置进一步用于利用人脸图像进行活体检测,以获得活体检测结果。

[0511] 示例性地,所述身份认证装置还可以包括:光源,用于向待认证人员的人脸发射检测光;所述图像采集装置进一步用于采集待认证人员的人脸在检测光的照射下的一个或多

个光照图像作为所述人脸图像;所述处理装置进一步用于基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果,并至少基于光照活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0512] 根据本发明另一方面,提供一种身份认证装置,包括:图像采集装置,用于采集待认证人员的个人标识信息;处理装置,用于判断个人标识信息是否为己认证信息以获得信息认证结果,对待认证人员进行活体检测以获得活体检测结果,并至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0513] 示例性地,所述图像采集装置还用于采集待认证人员的人脸图像;所述处理装置进一步用于利用人脸图像进行活体检测,以获得活体检测结果。

[0514] 示例性地,所述身份认证装置还可以包括:光源,用于向待认证人员的人脸发射检测光;所述图像采集装置进一步用于采集待认证人员的人脸在检测光的照射下的一个或多个光照图像作为所述人脸图像;所述处理装置进一步用于基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果,并至少基于光照活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0515] 示例性地,身份认证方法(200、300或400)可以在单独的身份认证装置(例如移动终端等)上实现。上述输入装置可以包括键盘、鼠标、麦克风和触摸屏等中的一个或多个。在个人标识信息为证件号码、姓名、证件人脸、现场采集人脸、证件号码的变换值、姓名的变换值、证件人脸的变换值和现场采集人脸的变换值中的一种或多种的情况下,可以利用输入装置接收用户输入的信息,以获得所需的个人标识信息。在个人标识信息为证件图像中的证件信息的情况下,可以利用图像采集装置采集待认证人员的证件图像,以获得所需的个人标识信息。当然,从证件图像中也可以提取证件号码、姓名、证件人脸这些个人信息,并且可以通过预定算法获得这些个人信息的变换值,也就是说,利用图像采集装置也可以获取证件号码、姓名、证件人脸、证件号码的变换值、姓名的变换值和证件人脸的变换值这类个人标识信息。当然,身份认证装置可以包括输入装置和图像采集装置两者,结合两种信息获取方式来获取个人标识信息。

[0516] 根据本发明另一方面,提供一种身份认证装置,包括:输入装置,用于接收待认证人员的个人标识信息;传输装置,用于将个人标识信息发送至服务器,并从服务器接收服务器通过以下方式获得的关于待认证人员的身份是否合法的认证信息:判断个人标识信息是否为己认证信息以获得信息认证结果,对待认证人员进行活体检测以获得活体检测结果,并至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0517] 示例性地,所述身份认证装置还包括:图像采集装置,用于采集待认证人员的人脸图像;所述传输装置还用于将所述人脸图像发送至服务器,其中,所述服务器利用人脸图像进行活体检测以获得活体检测结果。

[0518] 示例性地,所述身份认证装置还可以包括:光源,用于向待认证人员的人脸发射检测光;所述图像采集装置进一步用于采集待认证人员的人脸在检测光的照射下的多个光照图像作为所述人脸图像;所述传输装置进一步用于将一个或多个光照图像发送至服务器;其中,服务器通过以下方式进行活体检测:基于待认证人员的人脸在一个或多个光照图像

中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果,并至少基于光照活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0519] 根据本发明另一方面,提供一种身份认证装置,包括:图像采集装置,用于采集待认证人员的个人标识信息;传输装置,用于将个人标识信息发送至服务器,并从服务器接收服务器通过以下方式获得的关于待认证人员的身份是否合法的认证信息:判断个人标识信息是否为己认证信息以获得信息认证结果,对待认证人员进行活体检测以获得活体检测结果,并至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法。

[0520] 示例性地,所述图像采集装置还用于采集待认证人员的人脸图像;所述处理装置进一步用于利用人脸图像进行活体检测,以获得活体检测结果。

[0521] 示例性地,所述身份认证装置还可以包括:光源,用于向待认证人员的人脸发射检测光;所述图像采集装置进一步用于采集待认证人员的人脸在检测光的照射下的一个或多个光照图像作为所述人脸图像;所述传输装置进一步用于将一个或多个光照图像发送至服务器;其中,服务器通过以下方式进行活体检测:基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果,并至少基于光照活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0522] 示例性地,身份认证方法(200、300或400)可以在分立的装置上实现,例如在客户端和服务端上实现。在这种情况下,客户端可以包括输入装置和/或图像采集装置以及传输装置。可选地,客户端可以将获取的个人标识信息和采集到的人脸图像(包括光照图像、动作图像、实时图像等)上传到服务器,由服务器进行身份认证。客户端接收到服务器发送的认证信息后,执行认证通过或未通过操作,例如输出关于身份认证通过或未通过的信息,允许或拒绝用户执行后续的业务操作,等等。

[0523] 示例性地,客户端的传输装置可以通过网络或其它技术将个人标识信息和人脸图像发送至服务器,并通过网络或其它技术从服务器接收认证信息。例如,所述网络可以是因特网、无线局域网、移动通信网络等,所述其它技术例如可以包括蓝牙通信、红外通信等。例如,所述服务器可以为通用型服务器或专用型服务器,可以为虚拟服务器或云服务器等。客户端(或服务器)的传输装置可以包括调制解调器、网络适配器、蓝牙发送接收单元或红外发送接收单元等,例如还可以对发送或接收的信息进行编码、解码等操作。

[0524] 由于身份认证的大部分工作在服务器完成,因此可以节省客户端的处理装置的计算资源,由此可以降低对客户端性能的要求以及身份认证装置的制造成本,并可以提升用户的体验。

[0525] 根据本发明另一方面,提供一种身份认证装置,包括传输装置和处理装置,其中,所述传输装置用于从客户端接收待认证人员的个人标识信息,并将关于待认证人员的身份是否合法的认证信息发送至客户端;所述处理装置用于判断个人标识信息是否为己认证信息以获得信息认证结果,对待认证人员进行活体检测以获得活体检测结果,至少根据信息认证结果和活体检测结果确定待认证人员的身份是否合法以获得所述认证信息。

[0526] 示例性地,所述传输装置还用于接收待认证人员的人脸图像;所述处理装置进一步用于利用人脸图像进行活体检测,以获得活体检测结果。

[0527] 示例性地,所述传输装置进一步用于从客户端接收待认证人员的一个或多个光照

图像,其中,一个或多个光照图像是针对在检测光的照射下的待认证人员的人脸所采集获得的;所述处理装置进一步用于基于待认证人员的人脸在一个或多个光照图像中所表现的光反射特性确定待认证人员的人脸是否属于活体,以获得光照活体验证结果,并至少基于光照活体验证结果确定待认证人员是否通过活体验证,以获得活体检测结果。

[0528] 如上文所述,身份认证方法(200、300或400)可以在分立的装置上实现,例如在客户端和服务端上实现。本实施例描述身份认证方法的至少一部分在服务器上实现的一种实现方案。

[0529] 根据本发明实施例的身份认证方法及装置,结合已认证信息判断和活体检测来确定待认证人员的身份是否合法,因此,与常规的单纯基于密码或证件进行身份认证的方式相比,根据本发明实施例的身份认证方法和装置的认证结果更准确,可以提高用户鉴权的安全性,从而可以有效保障用户的权益。上述方法和装置可以很好地应用于各种涉及身份认证的领域,例如电子商务、移动支付或银行业务等领域。

[0530] 尽管这里已经参考附图描述了示例实施例,应理解上述示例实施例仅仅是示例性的,并且不意图将本发明的范围限制于此。本领域普通技术人员可以在其中进行各种改变和修改,而不偏离本发明的范围和精神。所有这些改变和修改意在包括在所附权利要求所要求的本发明的范围之内。

[0531] 本领域普通技术人员可以意识到,结合本文中所公开的实施例描述的各示例的单元及算法步骤,能够以电子硬件、或者计算机软件和电子硬件的结合来实现。这些功能究竟以硬件还是软件方式来执行,取决于技术方案的特定应用和设计约束条件。专业技术人员可以对每个特定的应用来使用不同方法来实现所描述的功能,但是这种实现不应认为超出本发明的范围。

[0532] 在本申请所提供的几个实施例中,应该理解到,所揭露的设备和方法,可以通过其它的方式实现。例如,以上所描述的设备实施例仅仅是示意性的,例如,所述单元的划分,仅仅为一种逻辑功能划分,实际实现时可以有另外的划分方式,例如多个单元或组件可以结合或者可以集成到另一个设备,或一些特征可以忽略,或不执行。

[0533] 在此处所提供的说明书中,说明了大量具体细节。然而,能够理解,本发明的实施例可以在没有这些具体细节的情况下实践。在一些实例中,并未详细示出公知的方法、结构和技术,以便不模糊对本说明书的理解。

[0534] 类似地,应当理解,为了精简本发明并帮助理解各个发明方面中的一个或多个,在对本发明的示例性实施例的描述中,本发明的各个特征有时被一起分组到单个实施例、图、或者对其的描述中。然而,并不应将该本发明的方法解释成反映如下意图:即所要求保护的本发明要求比在每个权利要求中所明确记载的特征更多的特征。更确切地说,如相应的权利要求书所反映的那样,其发明点在于可以用少于某个公开的单个实施例的所有特征的特征来解决相应的技术问题。因此,遵循具体实施方式的权利要求书由此明确地并入该具体实施方式,其中每个权利要求本身都作为本发明的单独实施例。

[0535] 本领域的技术人员可以理解,除了特征之间相互排斥之外,可以采用任何组合对本说明书(包括伴随的权利要求、摘要和附图)中公开的所有特征以及如此公开的任何方法或者设备的所有过程或单元进行组合。除非另外明确陈述,本说明书(包括伴随的权利要求、摘要和附图)中公开的每个特征可以由提供相同、等同或相似目的的替代特征来代替。

[0536] 此外,本领域的技术人员能够理解,尽管在此所述的一些实施例包括其它实施例中所包括的某些特征而不是其它特征,但是不同实施例的特征的组合意味着处于本发明的范围之内并且形成不同的实施例。例如,在权利要求书中,所要求保护的实施例的任意之一都可以以任意的组合方式来使用。

[0537] 本发明的各个部件实施例可以以硬件实现,或者以在一个或者多个处理器上运行的软件模块实现,或者以它们的组合实现。本领域的技术人员应当理解,可以在实践中使用微处理器或者数字信号处理器(DSP)来实现根据本发明实施例的身份认证装置中的一些模块的一些或者全部功能。本发明还可以实现为用于执行这里所描述的方法的一部分或者全部的装置程序(例如,计算机程序和计算机程序产品)。这样的实现本发明的程序可以存储在计算机可读介质上,或者可以具有一个或者多个信号的形式。这样的信号可以从因特网网站上下载得到,或者在载体信号上提供,或者以任何其他形式提供。

[0538] 应该注意的是上述实施例对本发明进行说明而不是对本发明进行限制,并且本领域技术人员在不脱离所附权利要求的范围的情况下可设计出替换实施例。在权利要求中,不应将位于括号之间的任何参考符号构造成对权利要求的限制。单词“包含”不排除存在未列在权利要求中的元件或步骤。位于元件之前的单词“一”或“一个”不排除存在多个这样的元件。本发明可以借助于包括有若干不同元件的硬件以及借助于适当编程的计算机来实现。在列举了若干装置的单元权利要求中,这些装置中的若干个可以是通过同一个硬件项来具体体现。单词第一、第二、以及第三等的使用不表示任何顺序。可将这些单词解释为名称。

[0539] 以上所述,仅为本发明的具体实施方式或对具体实施方式的说明,本发明的保护范围并不局限于此,任何熟悉本技术领域的技术人员在本发明揭露的技术范围内,可轻易想到变化或替换,都应涵盖在本发明的保护范围之内。本发明的保护范围应以权利要求的保护范围为准。

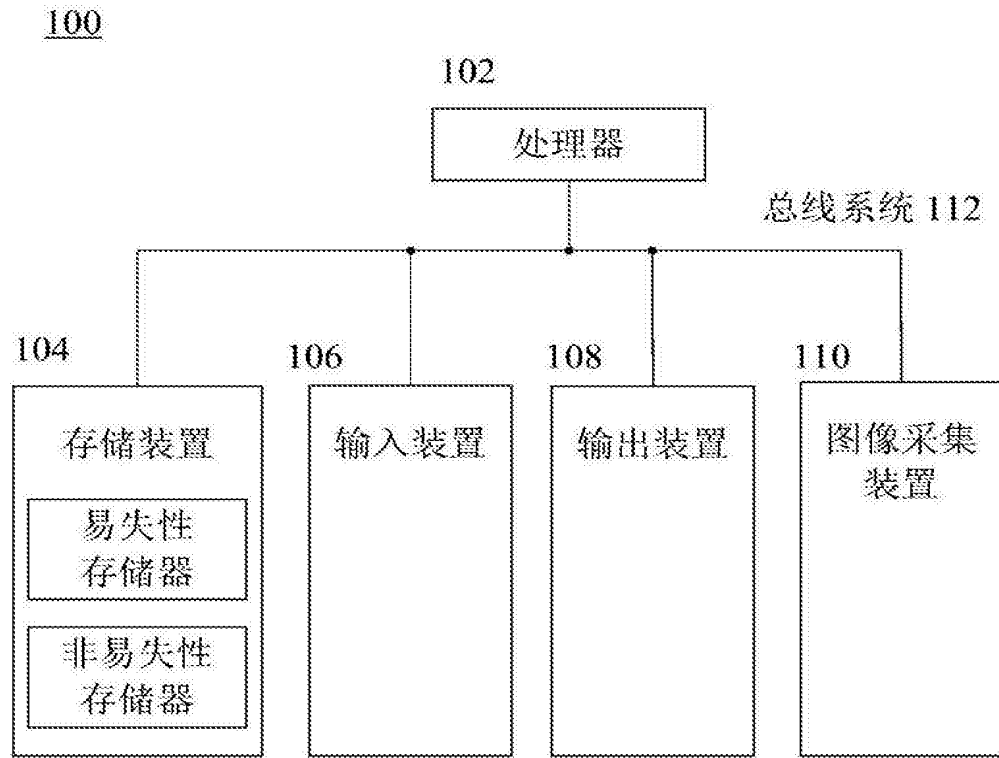


图1

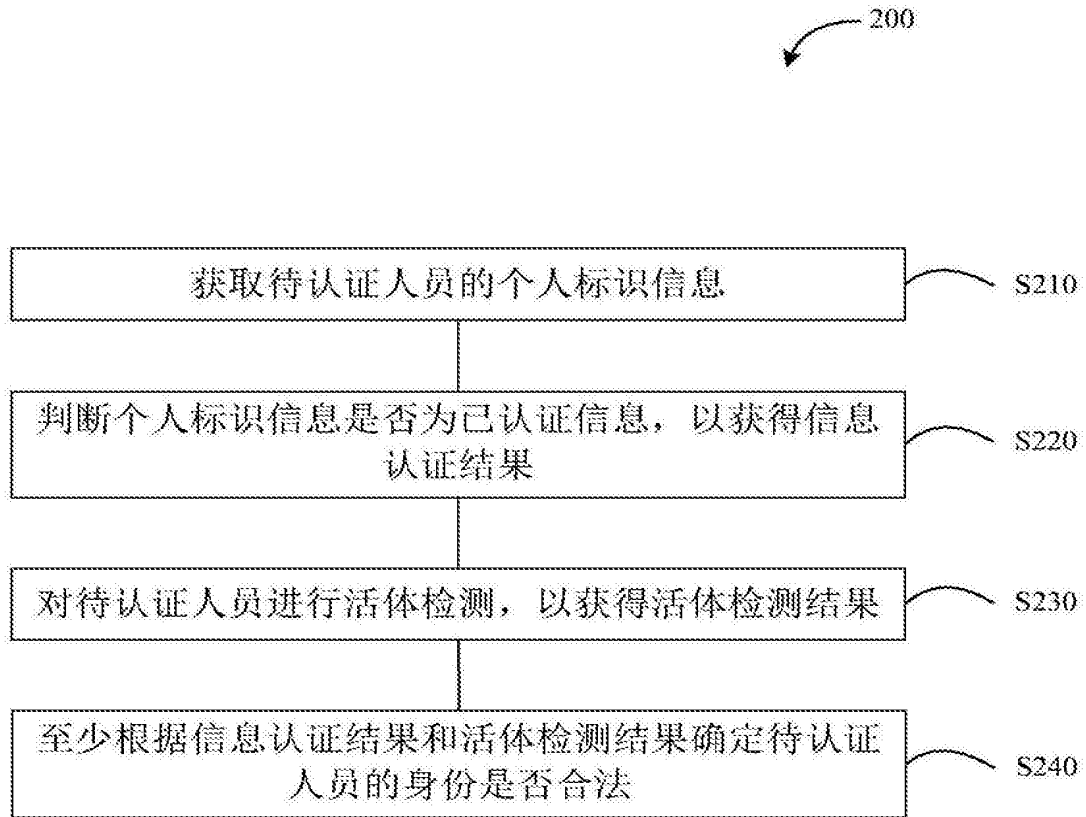


图2

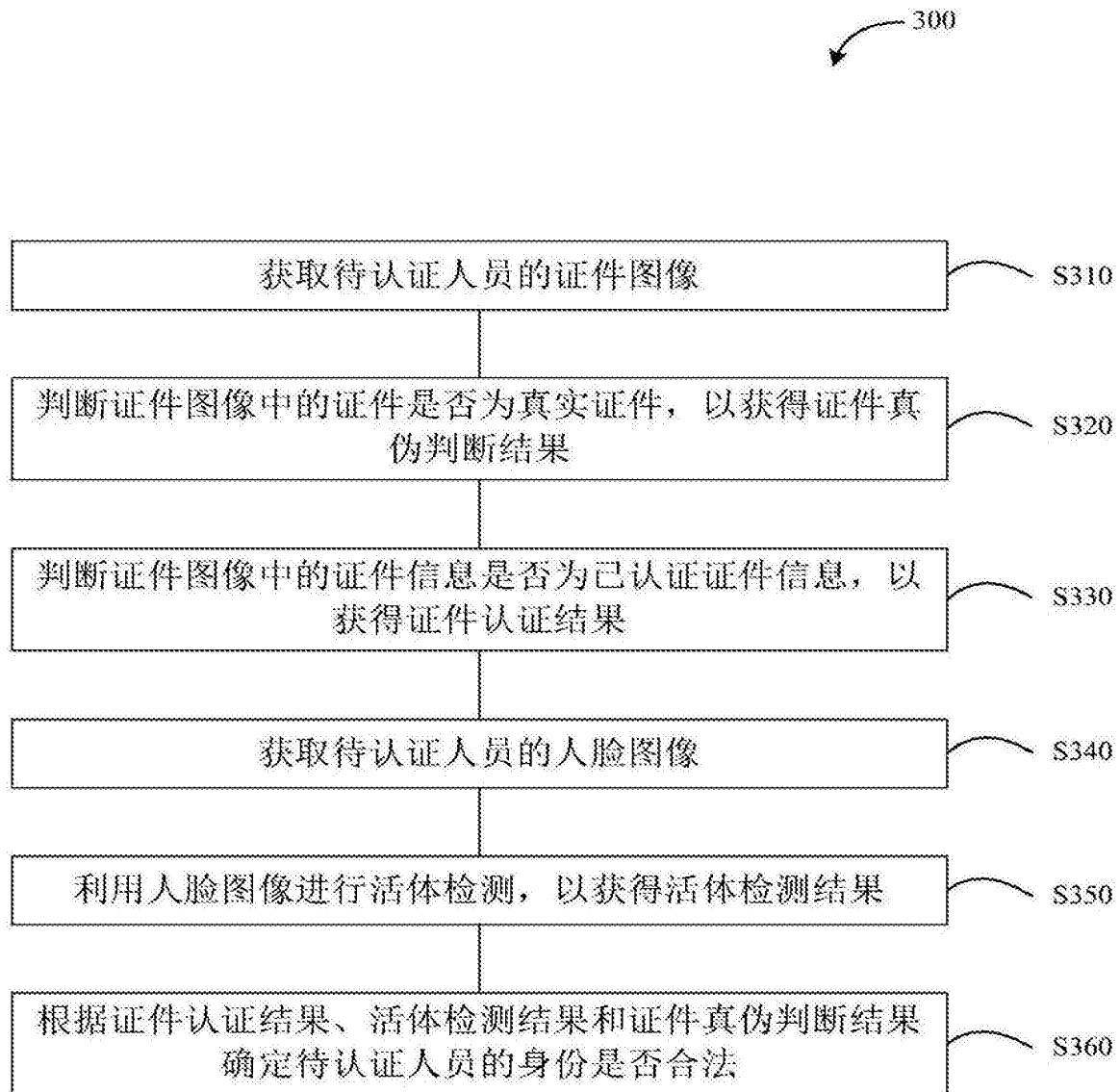


图3

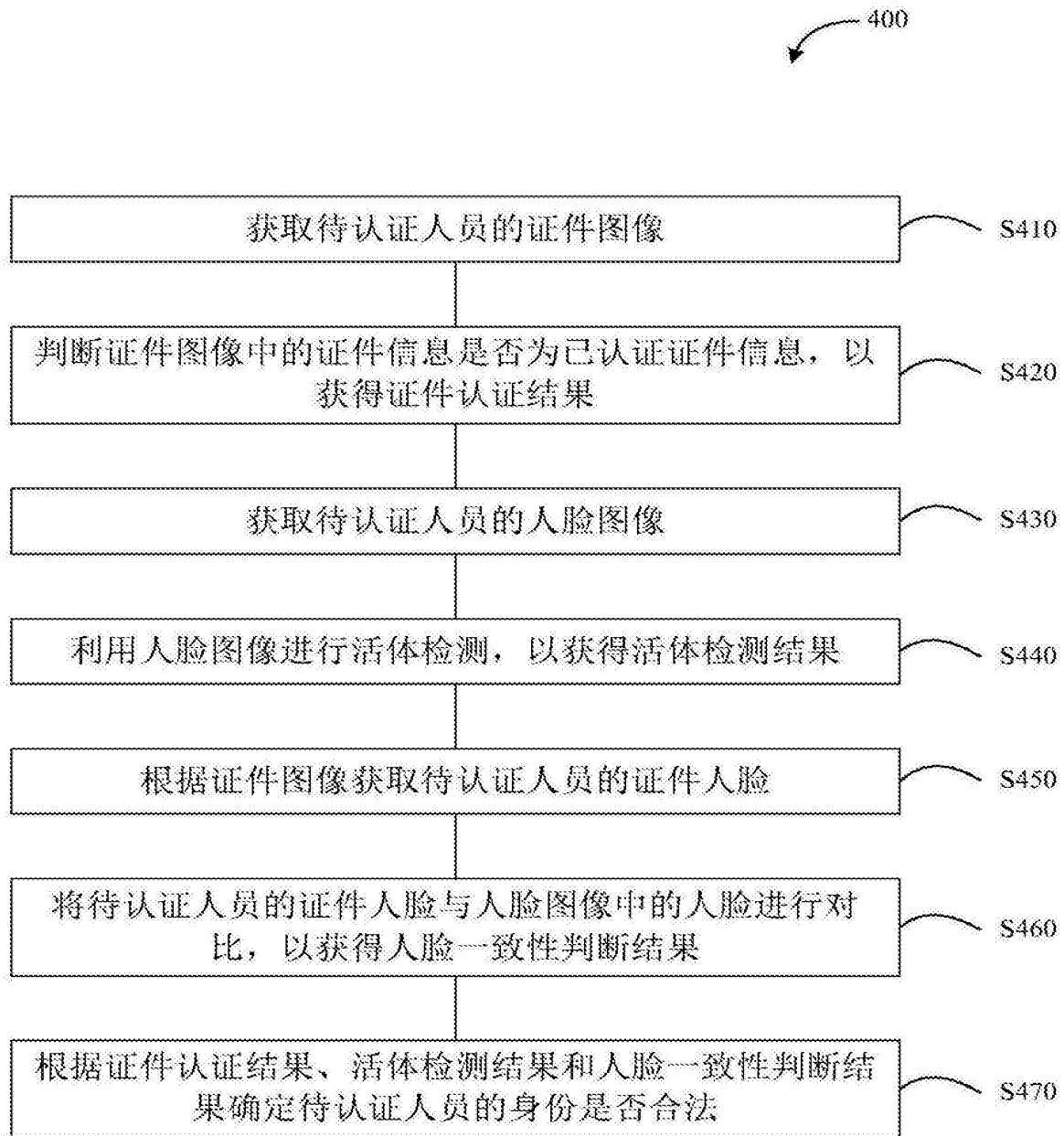


图4

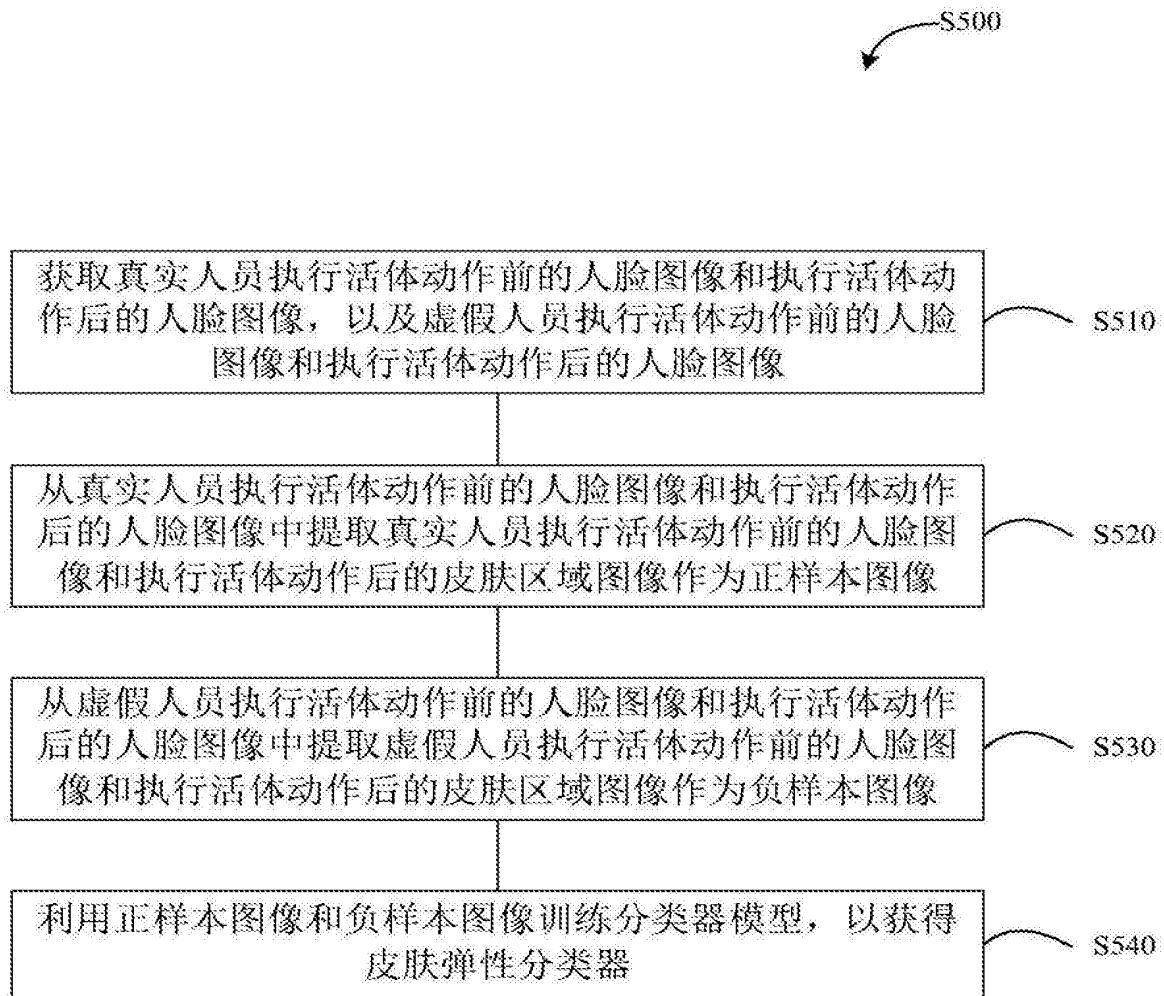


图5

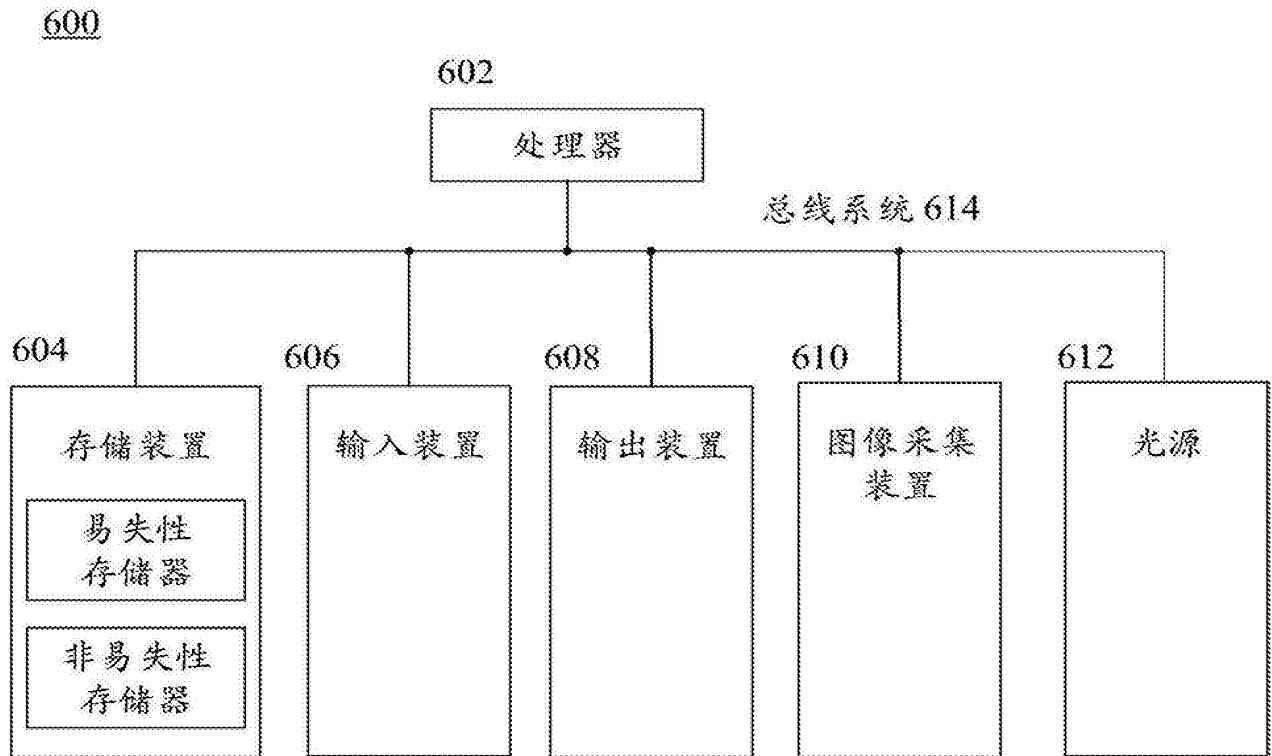


图6

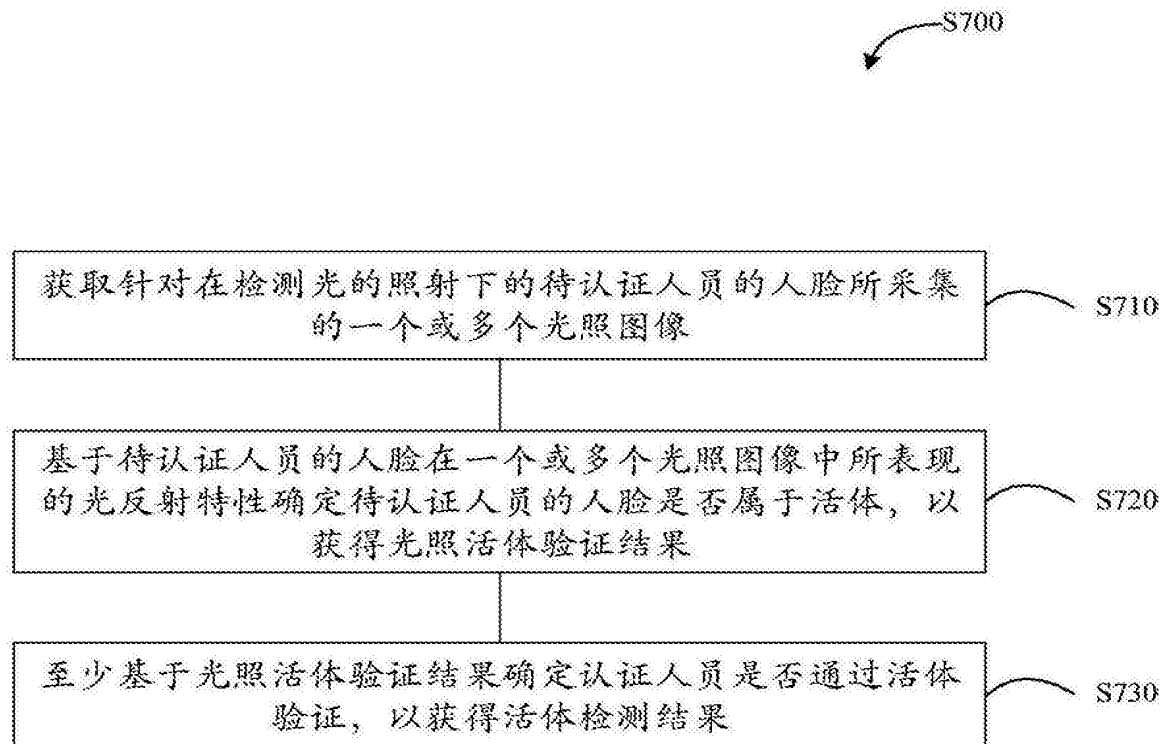


图7

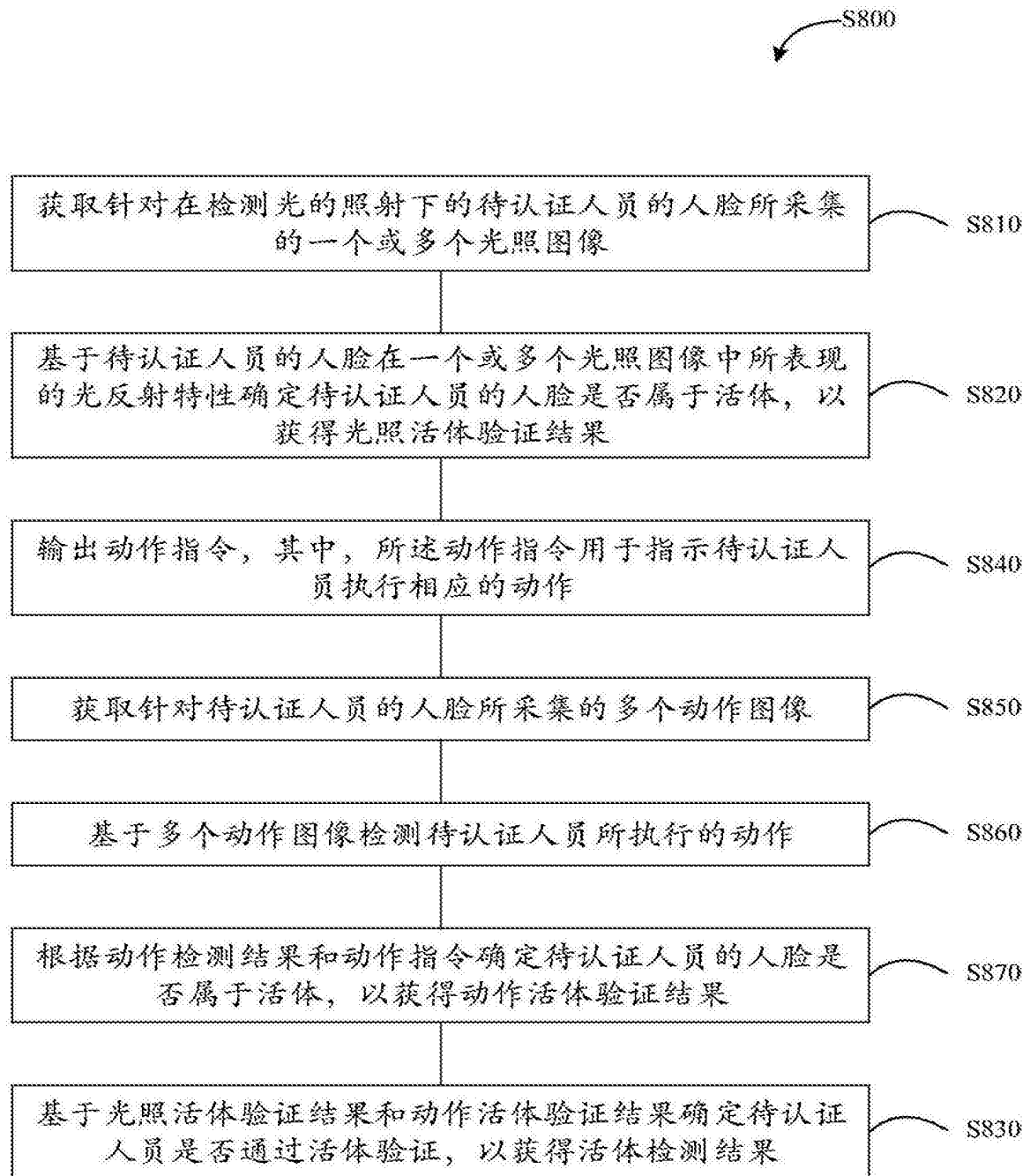


图8

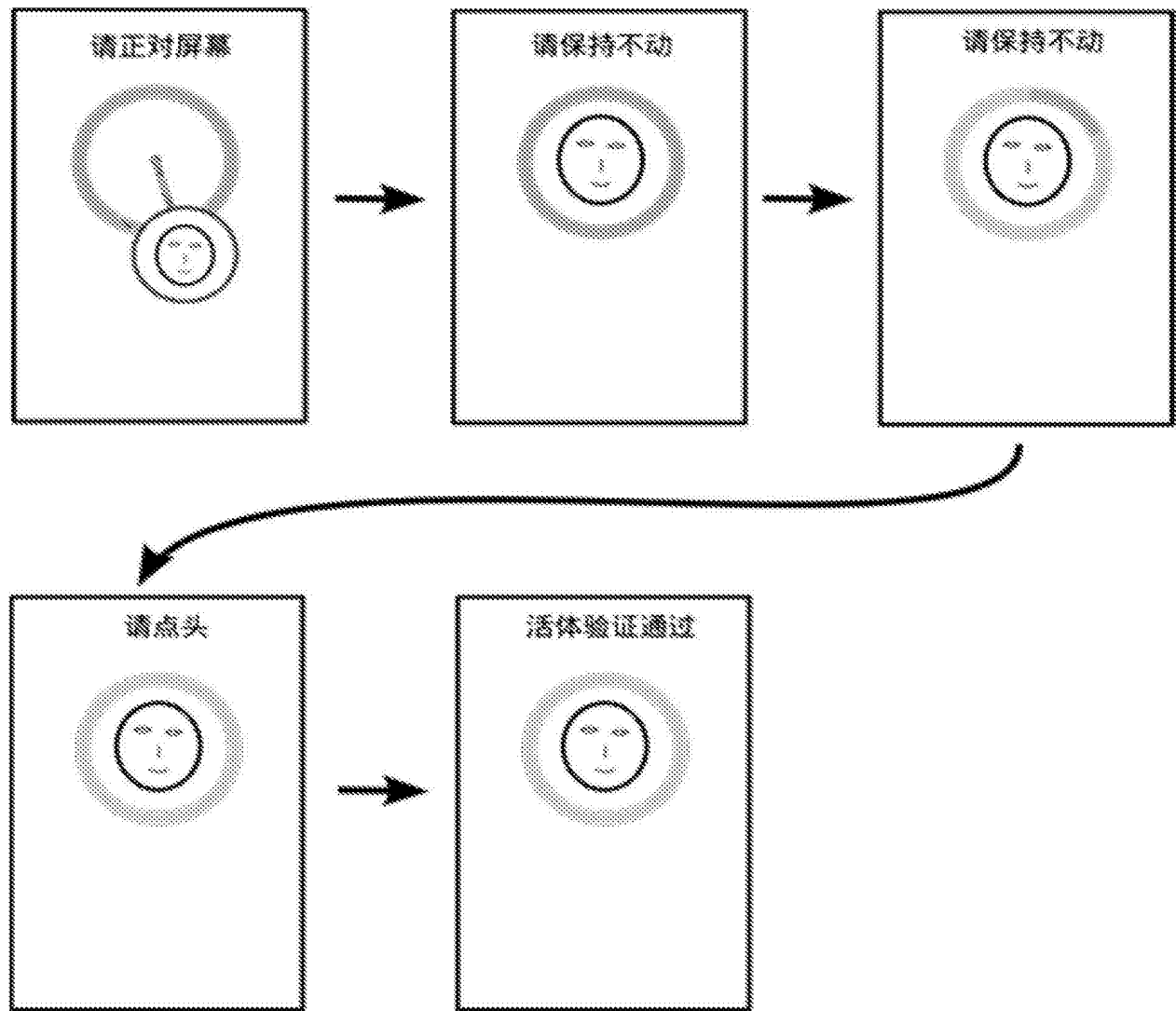


图9

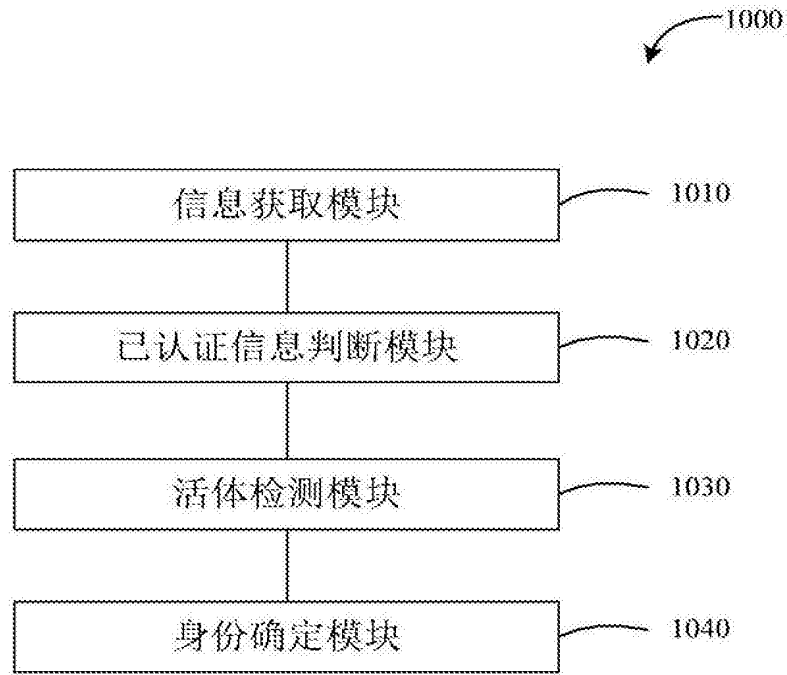


图10

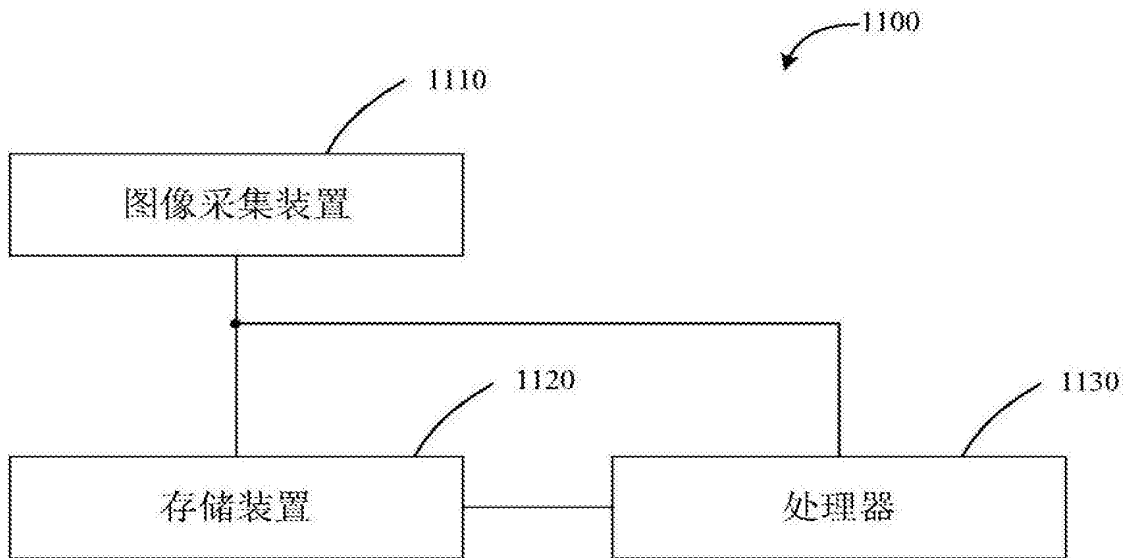


图11

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EPO

SUBJECT OF THE INVENTION

Identity authentication method and device and storage medium

[0001] Technical field**[0002]** The present invention relates to the field of image processing, and more specifically to an identity authentication method and device and storage medium.**[0003]** Background technique**[0004]** The socialization of the use of technology products has become a beautiful landscape of modern social life. People's clothing, food, housing and transportation are all closely related to technology. Technology products have gradually been applied to all aspects of social life and have become an indispensable part of modern human daily life. However, while enjoying the benefits of scientific products, people also feel the negative problems it brings, such as information security issues.**[0005]** At present, many fields involve information security issues, especially in technical fields such as e-commerce, mobile payment, and bank account opening, where information security issues are particularly prominent. Specifically, at present, most of the above-mentioned fields use passwords for user interactive authentication (also referred to as identity authentication), and some use the method of swiping credentials for user interactive authentication. The above two methods have certain drawbacks. The former requires the user to remember the password, and it is more cumbersome to enter the password every time. Once the password is stolen by illegal elements, it will cause the user to lose privacy or property. For the latter, Credentials are easy to be forged or fraudulently used, and their security is low. Therefore, it is necessary to propose a convenient and safe identity authentication method or system to be applied to technical fields such as e-commerce, mobile payment, and bank account opening.**[0006]** Summary of the invention**[0007]** The present invention was made in consideration of the above-mentioned problems. The invention provides an identity authentication method, device and storage medium.

- [0008] According to one aspect of the present invention, an identity authentication method is provided. The method includes: obtaining the personal identification information of the person to be authenticated; judging whether the personal identification information is authenticated information to obtain the information authentication result; performing a living test on the person to be authenticated to obtain the living test result; and at least according to the information authentication result and the living body The test result determines whether the identity of the person to be authenticated is legal.
- [0009] Exemplarily, the personal identification information is one of the ID number, name, ID face, field collected face, ID number conversion value, name conversion value, ID face conversion value, and field collection face conversion value. Kind or more.
- [0010] Exemplarily, before obtaining the personal identification information of the person to be authenticated, the identity authentication method further includes: outputting instruction information for instructing the person to be authenticated to provide a predetermined type of personal information; wherein the personal identification information is provided by the person to be authenticated Personnel information or obtained based on personnel information.
- [0011] Exemplarily, the personal identification information is one or more of the conversion value of the certificate number, the conversion value of the name, the conversion value of the certificate face, and the conversion value of the face collected on site, and obtaining the personal identification information of the person to be authenticated includes : Obtain the initial information of the person to be authenticated, where the initial information is one or more of the credential number, name, credential face, and on-site collected face; and the initial information is transformed based on a predetermined algorithm to obtain the initial information The converted value is used as personal identification information.
- [0012] Exemplarily, obtaining the personal identification information of the person to be authenticated includes: obtaining the credential image of the person to be authenticated, where the information authentication result is the result of the credential authentication; and performing a living test on the person to be authenticated to obtain the living body test result includes: obtaining the person to be authenticated The face image of the person; and the use of the face image for living body detection to obtain the result of the living body detection.
- [0013] Exemplarily, before determining whether the identity of the person to be authenticated is legal at least according to the information authentication result and the living body detection result, the identity authentication method further includes: performing an additional judgment operation using the credential image and/or the face image to obtain the additional judgment result; Determining whether the identity of the person to be authenticated is legal at least based on the information

authentication result and the biological test result includes: determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the biological test result and the additional judgment result.

- [0014] Exemplarily, the additional judgment operation includes a credential authenticity judgment operation and/or a face consistency judgment operation, and the additional judgment result includes a credential authenticity judgment result and/or a face consistency judgment result, and the credential authenticity judgment operation includes: Determine whether the certificate in the certificate image is a real certificate to obtain the result of the authenticity of the certificate; the judgment operation of face consistency includes: obtaining the certificate face of the person to be authenticated according to the certificate image; and comparing the face of the person to be authenticated with the person The faces in the face image are compared to obtain the judgment result of face consistency.

- [0015] Exemplarily, acquiring the credential face of the person to be authenticated according to the credential image includes: detecting the face from the credential image to obtain the credential face of the person to be authenticated.

- [0016] Exemplarily, acquiring the document face of the person to be authenticated according to the document image includes: performing text recognition on the document image to obtain the text information in the document image; searching for matching information from the certified document information database based on the text information in the document image Credential information; and determine that the credential face in the searched and matched credential information is the credential face of the person to be authenticated.

- [0017] Exemplarily, judging whether the document in the document image is a real document to obtain the authenticity judgment result includes: extracting image characteristics of the document image; and inputting the image characteristics into a trained document classifier to obtain the authenticity judgment result of the document ; Among them, the certificate authenticity judgment result is the confidence that the certificate in the certificate image is a real certificate.

- [0018] Exemplarily, determining whether the certificate in the certificate image is a real certificate, to obtain the authenticity judgment result of the certificate includes: identifying the image block containing the certificate identification information from the certificate image; and identifying the certificate identification in the image block containing the certificate identification information The information is identified to obtain the authenticity judgment result of the document; among them, the authenticity judgment result of the document is the confidence that the document in the document image is a real document.

- [0019] Exemplarily, in the process of determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result, and the additional judgment result, each result of the certificate authentication result, the living body detection result and the additional judgment result has its own weight coefficient .
- [0020] Exemplarily, acquiring the credential image of the person to be authenticated includes: acquiring a pre-shooting image that is collected in real time for the credential of the person to be certified under the current shooting conditions; evaluating the image properties of the pre-shooting image in real time; When the evaluation value of is less than the preset evaluation value threshold, the pre-shooting prompt information is generated according to the image attributes of the pre-captured image, which is used to prompt the person to be authenticated to adjust the shooting conditions of his certificate; and when the evaluation value of the image attribute of the pre-captured image is equal to Or when it is greater than the preset evaluation value threshold, save the pre-photographed image to use it as a document image.
- [0021] Exemplarily, determining whether the personal identification information is authenticated information to obtain the information authentication result includes: performing text recognition on the certificate image to obtain the text information in the certificate image; and based on the text information in the certificate image in the authenticated certificate information Search in the database to obtain the certificate authentication result; where the certificate authentication result is the credibility of the certificate information in the certificate image.
- [0022] Exemplarily, performing text recognition on a document image to obtain text information in the credential image includes: positioning text in the credential image to obtain an image block containing text; and recognizing text in the image block containing text To get the text information in the ID image.
- [0023] Exemplarily, before recognizing the text in the image block containing the text, the identity authentication method further includes: correcting the image block containing the text to a horizontal state.
- [0024] Exemplarily, after recognizing the text in the image block containing text to obtain the text information in the credential image, performing text recognition on the credential image to obtain the text information in the credential image also includes: outputting the text in the credential image Text information for users to view; receive text correction information input by the user; compare the text to be corrected indicated by the text correction information with the corresponding text in the text information in the document image; and if the text correction information indicates to be corrected The difference between the text and the corresponding text in the text information in the

credential image is less than the preset difference threshold, and the text correction information is used to update the text information in the credential image.

- [0025]** Exemplarily, performing live body detection on a person to be authenticated to obtain a live body detection result includes: generating a live action instruction, which is used to instruct the person to be authenticated to perform a corresponding live action; and acquiring a face image of the person to be authenticated collected in real time; Detecting the human face in the human face image; and judging whether the human face in the human face image performs the living body motion indicated by the living body motion instruction, so as to obtain the living body detection result.
- [0026]** Exemplarily, performing live body detection on a person to be authenticated to obtain a live body detection result includes: generating a live action instruction, which is used to instruct the person to be authenticated to perform a corresponding live action; and acquiring a face image of the person to be authenticated collected in real time; Capture the skin area image of the person to be authenticated before performing the living action and the image of the skin area after performing the living action from the collected face images; and the skin area image of the person to be authenticated before performing the living action and the skin after performing the living action The regional image is input into the skin elasticity classifier to obtain the result of living body detection.
- [0027]** Exemplarily, the identity authentication method includes: obtaining a face image of a real person before performing a live action and a face image after performing a live action, and a face image of a fake person before performing a live action and a face image after performing a live action. ; From the face image of the real person before the live action and the face image after the live action, the face image of the real person before the live action and the skin area image after the live action are extracted as the positive sample image; from the fake person From the face image before the execution of the living action and the face image after the execution of the living action, the face image of the fake person before the execution of the living action and the skin area image after the execution of the living action are extracted as negative sample images; and the use of positive sample images and negative images The sample image trains the classifier model to obtain the skin elasticity classifier.
- [0028]** Exemplarily, capturing the skin area image of the person to be authenticated before performing the living action and the skin area image after performing the living action from the collected face images includes: selecting the person to be authenticated to perform the living action from the collected face images The previous face image and the face image after performing the live action; use the face detection model to locate the face in the face image before the person to be authenticated performs the live action and the face image after the live action; use The face key point positioning model locates the key points of the face in the face image before the person to be authenticated performs the live action and the face image after the live action; and the face

position and the key point position obtained according to the positioning The face image before the person to be authenticated performs the live action and the face in the face image after the live action is performed are divided into regions to obtain the skin area image before the person to be authenticated performs the live action and the skin after the live action is performed Area image.

- [0029]** Exemplarily, the identity authentication method further includes: obtaining a sample face image, where the position of the face in the sample face image and the position of the key points of the face are marked; and using the sample face image for neural network training , To obtain a face detection model and a face key point positioning model.
- [0030]** Exemplarily, performing live detection of the person to be authenticated to obtain the result of the life detection includes: obtaining the face image collected by the binocular camera on the face of the person to be authenticated under the irradiation of structured light; and determining the person to be authenticated according to the face image. Whether the face of the certified person is a living body, to obtain the result of the living body detection.
- [0031]** Exemplarily, performing a living body detection on a person to be authenticated to obtain a living body detection result includes: Step S110: Obtain one or more illumination images collected for the face of the person to be authenticated under the illumination of the detection light; Step S120: Based on The light reflection characteristics of the face of the person to be authenticated in one or more illumination images determine whether the face of the person to be authenticated is a living body, so as to obtain an illuminated living body verification result; and step S130: the authentication is determined based on at least the illuminated living body verification result Whether the personnel passed the live body verification to obtain the live body test results.
- [0032]** Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed at least once.
- [0033]** Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light changes between every two consecutive moments.
- [0034]** Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is randomly changed or preset.
- [0035]** Exemplarily, performing a living body detection on a person to be authenticated to obtain a living body detection result includes: Step S140: Output an action instruction, where the action instruction is used to instruct the authentication person to perform a corresponding action; Step S150: Obtain a face for the person to be authenticated A plurality of collected motion images; Step S160: Detect the actions performed by the person to be authenticated based on the plurality of

motion images; and Step S170: Determine whether the face of the person to be authenticated belongs to a living body according to the action detection result and the action instruction to obtain the action Living body verification result: Step S130 includes: determining whether the person to be authenticated has passed the living body verification based on the illumination living body verification result and the action living body verification result to obtain the living body detection result.

- [0036] Exemplarily, step S170 includes: determining if an action performed by the person to be authenticated that is consistent with the action indicated by the action instruction is detected based on a plurality of action images collected within a time period not greater than the first preset time. The face of the person to be authenticated belongs to a living body. If no action performed by the person to be authenticated and consistent with the action indicated by the action instruction is detected based on the multiple motion images collected within the first preset time, it is determined to be authenticated. The person's face is not a living body.
- [0037] Exemplarily, the identity authentication method further includes: counting once during each execution of step S140 to step S170 to obtain the number of action verification times; after step S170, the identity authentication method further includes: if the result of the in vivo verification of the action indicates to be authenticated. If the person's face does not belong to a living body, the first error message is output, and it is judged whether the number of action verifications reaches the first number threshold. If the number of action verifications reaches the first number threshold, then go to step S130, if the number of action verifications is not. When the threshold of the first number of times is reached, return to step S140 or return to step S110 if step S110 is executed before step S140, wherein the first error message is used to prompt that the biological verification for the person to be authenticated fails.
- [0038] Exemplarily, before step S110, the identity authentication method further includes: step S108: determine whether the image acquisition conditions of the person to be authenticated meet the preset requirements, and if the image acquisition conditions meet the preset requirements, go to step S110, where the image. The acquisition conditions include the position of the person to be authenticated in the image acquisition area of the image acquisition device and/or the relative angle of the person to be authenticated and the image acquisition device.
- [0039] Exemplarily, before step S108 or simultaneously with step S108, the identity authentication method further includes: step S106: output first prompt information, wherein the first prompt information is used to prompt the person to be authenticated to face the image acquisition device. And close to the image acquisition device.

- [0040] Exemplarily, step S106 includes: outputting the first prompt information in one or more of a voice format, an image format, and a text format.
- [0041] Exemplarily, step S108 includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area for calibrating image acquisition conditions and a face area in the real-time image for display; and according to the face area Determine whether the image acquisition conditions meet the preset requirements. If the face area is located in the preset area and the proportion of the face area in the real-time image is greater than the first preset ratio, it is determined that the image acquisition conditions meet the preset requirements. If the area is not located in the preset area or the proportion of the face area in the real-time image is not greater than the first preset ratio, it is determined that the image acquisition condition does not meet the preset requirement.
- [0042] Exemplarily, the identity authentication method further includes: if the proportion of the face area in the real-time image is not greater than the first preset proportion, outputting first collection prompt information in real time to prompt the person to be authenticated to approach the image collection device.
- [0043] Exemplarily, step S108 includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area for calibrating image acquisition conditions and a face area in the real-time image for display; and The area judges whether the image acquisition conditions meet the preset requirements. If the face area is located in the preset area and the proportion of the face area in the preset area is greater than the second preset ratio, it is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the proportion of the face area in the preset area is not greater than the second preset ratio, it is determined that the image acquisition condition does not meet the preset requirement.
- [0044] Exemplarily, the identity authentication method further includes: if the proportion of the face area in the preset area is not greater than the second preset ratio, outputting second collection prompt information in real time to prompt the person to be authenticated to approach the image collection device.
- [0045] Exemplarily, the identity authentication method further includes: judging the relative position relationship between the face area and the preset area in real time; and outputting the third collection prompt information in real time based on the relative position relationship between the face area and the preset area, so as to It is prompted to change the relative position relationship

between the person to be authenticated and the image acquisition device so that the face area is close to the preset area.

- [0046] Exemplarily, step S108 includes: acquiring the posture information of the image acquisition device; and judging whether the image acquisition device is in a vertically placed state according to the posture information, if it is, it is determined that the image acquisition condition meets the preset requirement, otherwise it is determined that the image acquisition condition is not Meet the preset requirements.
- [0047] Exemplarily, the identity authentication method further includes: counting once during each execution of step S110 to step S120 to obtain the number of light verification times; after step S120, the identity verification method further includes: if the result of the light living body verification indicates that authentication is to be performed If the person's face does not belong to a living body, the second error message is output, and it is judged whether the number of light verification times reaches the second number threshold. If the number of light verification times reaches the second number threshold, then go to step S130, if the number of light verification times does not reach the first If the threshold is two times, return to step S108 or return to step S110, where the second error message is used to prompt that the in-vivo verification for the person to be authenticated has failed.
- [0048] Exemplarily, before step S110 or during the process of step S110 and step S120, the identity authentication method further includes: outputting second prompt information, where the second prompt information is used to prompt the person to be authenticated at the second preset time Keep it still.
- [0049] Exemplarily, the second prompt information is countdown information corresponding to the second preset time.
- [0050] Exemplarily, the detection light is obtained by dynamically changing the color and/or position of the light emitted to the face of the person to be authenticated.
- [0051] Exemplarily, the detection light is emitted by the display screen, and the detection light is obtained by dynamically changing the mode of the light emitted by the display screen by changing the content displayed on the display screen to emit the detection light to the face of the person to be authenticated.
- [0052] According to another aspect of the present invention, an identity authentication device is provided. The device includes: an information acquisition module for acquiring the personal identification information of the person to be authenticated; an authenticated information judging module for

judging whether the personal identification information is authenticated information to obtain information authentication results; a living body detection module for treating The authenticating person performs a living body test to obtain a living body test result; and an identity determination module is used to determine whether the identity of the person to be authenticated is legal at least according to the information authentication result and the living body test result.

- [0053] Exemplarily, the personal identification information is one or more of a certificate number, a name, a certificate face, a conversion value of the certificate number, a conversion value of a name, a conversion value of a certificate face, and a conversion value of a face collected on site.
- [0054] Exemplarily, the identity authentication device further includes: an instruction information output module for outputting instruction information for instructing the person to be authenticated to provide a predetermined type of personal information before the information acquisition module obtains the personal identification information of the person to be authenticated; wherein, The personal identification information is the personnel information provided by the personnel to be certified or obtained based on the personnel information.
- [0055] Exemplarily, the personal identification information is one or more of the conversion value of the certificate number, the conversion value of the name, the conversion value of the certificate face, and the conversion value of the face collected on site, and the information acquisition module includes: an initial information acquisition sub The module is used to obtain the initial information of the person to be authenticated, where the initial information is one or more of the ID number, name, ID face, and on-site collected face; and the transformation sub-module is used to compare the initial information based on a predetermined algorithm The information is transformed to obtain the transformed value of the initial information as the personal identification information.
- [0056] Exemplarily, the information acquisition module includes: a credential image acquisition sub-module for acquiring a credential image of a person to be authenticated, wherein the information authentication result is the credential authentication result; the living body detection module includes: a facial image acquisition sub-module for obtaining The face image of the person to be authenticated; and the detection sub-module, which is used to perform living body detection using the face image to obtain the living body detection result.
- [0057] Exemplarily, the identity authentication device further includes: an additional judgment module, which is used to perform the additional judgment using the ID image and/or the face image before the identity determination module determines whether the identity of the person to be authenticated is legal based on the information authentication result and the living body detection result. Judgment operation to obtain additional judgment results; the identity determination module includes: an identity determination sub-module, which is used to determine whether the identity of

the person to be authenticated is legal based on the certificate authentication result, the living body test result, and the additional judgment result.

- [0058] Exemplarily, the additional judgment module includes a certificate authenticity judgment sub-module and/or a face consistency judgment sub-module. The certificate authenticity judgment sub-module is used to perform the authentication judgment operation of the certificate, and the face consistency judgment sub-module is used to execute The face consistency judgment operation, and the additional judgment result includes the certificate authenticity judgment result and/or the face consistency judgment result,
- [0059] The certificate authenticity judgment sub-module includes: a certificate authenticity judgment unit, which is used to determine whether the certificate in the certificate image is a real certificate to obtain the authenticity judgment result of the certificate; the face consistency judgment sub-module includes: a certificate face acquisition unit, using To obtain the document face of the person to be authenticated according to the document image; and a face comparison unit for comparing the document face of the person to be authenticated with the face in the face image to obtain a face consistency judgment result.
- [0060] Exemplarily, the credential face acquisition unit includes a credential face detection subunit, which is used to detect the face from the credential image to obtain the credential face of the person to be authenticated.
- [0061] Exemplarily, the certificate face acquisition unit includes: a text recognition sub-module for performing text recognition on a certificate image to obtain text information in the certificate image; a search sub-unit for obtaining information from an authenticated person based on the text information in the certificate image Searching for matching certificate information in the certificate information database; and a certificate face determination sub-unit for determining that the certificate face in the searched and matched certificate information is the certificate face of the person to be authenticated.
- [0062] Exemplarily, the credential authenticity judgment unit includes: a feature extraction subunit for extracting image features of a credential image; and an input subunit for inputting image features into a trained credential classifier to obtain a credential authenticity judging result ; Among them, the certificate authenticity judgment result is the confidence that the certificate in the certificate image is a real certificate.
- [0063] Exemplarily, the certificate authenticity judgment unit includes: an image block recognition sub-unit for recognizing the image block containing the certificate identification information from the certificate image; and the identification information recognition sub-unit for identifying the image

block containing the certificate identification information The identification information of the certificate is identified to obtain the authenticity judgment result of the certificate; wherein the authenticity judgment result of the certificate is the confidence that the certificate in the certificate image is a real certificate.

- [0064] Exemplarily, in the process that the identity determination sub-module determines whether the identity of the person to be authenticated is legal based on the certificate authentication result, the living body detection result, and the additional judgment result, each result of the certificate authentication result, the living body detection result, and the additional judgment result has The respective weight coefficient.
- [0065] Exemplarily, the credential image acquisition sub-module includes: an acquisition unit for acquiring pre-shooting images that are collected in real time for the credential of the person to be authenticated under current shooting conditions; an evaluation unit for evaluating image attributes of the pre-shooting images in real time; A unit for generating pre-shooting prompt information according to the image properties of the pre-shooting image when the evaluation value of the image attribute of the pre-shooting image is less than the preset evaluation value threshold, for prompting the person to be authenticated to adjust the shooting conditions of his certificate; and The saving unit is used for saving the pre-shooting image to use it as a credential image when the evaluation value of the image attribute of the pre-shooting image is equal to or greater than the preset evaluation value threshold.
- [0066] Exemplarily, the authenticated information judgment module includes: a text recognition sub-module for recognizing the text in the credential image to obtain text information in the credential image; and a search sub-module for recognizing the text in the credential image based on the text Search in the authentication certificate information database to obtain the certificate authentication result; among them, the certificate authentication result is the credibility of the certificate information in the certificate image.
- [0067] Exemplarily, the text recognition sub-module includes: a text positioning unit, which is used to locate text in a document image to obtain an image block containing text; and a text recognition unit, which is used to perform text recognition on the text in the image block containing text. Recognition to obtain the text information in the image of the certificate.
- [0068] Exemplarily, the identity authentication device further includes: a text correction module for correcting the image block containing the text to a horizontal state before the text recognition unit recognizes the text in the image block containing the text.

- [0069] Exemplarily, the text recognition sub-module further includes: a text output unit for outputting text information in the credential image for the user to view; a correction information receiving unit for receiving text correction information input by the user; a text comparison unit for using To compare the text to be corrected indicated by the text correction information with the corresponding text in the text information in the document image; and a text update unit for if the text to be corrected indicated by the text correction information is used in the text information in the document image If the difference between the corresponding characters is less than the preset difference threshold, the text correction information is used to update the text information in the credential image.
- [0070] Exemplarily, the living body detection module includes: an instruction generation sub-module for generating living body motion instructions, which is used to instruct a person to be authenticated to perform corresponding living body actions; a real-time image acquisition sub-module to obtain real-time collected information of the person to be authenticated Face image; face detection sub-module, used to detect the face in the face image; and living body action execution judging sub-module, used to judge whether the face in the face image performs the living body action indicated by the living body action instruction, To obtain the results of the live test.
- [0071] Exemplarily, the living body detection module includes: an instruction generation sub-module for generating living body action instructions, which are used to instruct the person to be authenticated to perform corresponding living body actions; a real-time image acquisition sub-module for obtaining real-time collected images to be authenticated Person's face image; skin area capture sub-module, used to capture the skin area image of the person to be authenticated before performing the live action and the skin area image after performing the live action from the collected face image; and the input sub-module, using The skin area image before the living body action and the skin area image after the living body action are performed by the person to be authenticated are input into the skin elasticity classifier to obtain the living body detection result.
- [0072] Exemplarily, the identity authentication device includes: a true and false image acquisition module for acquiring a face image of a real person before performing a live action and a face image after performing a live action, as well as a face image and a fake person before performing a live action. The face image after performing the live action; the positive sample extraction sub-module is used to extract the face image and the face image of the real person before performing the live action from the face image of the real person before the live action and the face image after the live action. The skin area image after the live action is taken as the positive sample image; the negative sample extraction sub-module is used to extract the fake person's face image before the live action from the face image before the live action and the face image after the live action The face image and the skin area image after performing the live action are taken as negative sample

images; and the training sub-module is used to train the classifier model using the positive sample image and the negative sample image to obtain the skin elasticity classifier.

- [0073]** Exemplarily, the skin area capturing sub-module includes: an image selection unit for selecting, from the collected face images, the face image of the person to be authenticated before performing the live action and the face image after performing the live action; face positioning; The unit is used to use the face detection model to locate the face in the face image before the person to be authenticated performs the live action and the face image after the live action is performed; the key point positioning unit is used to use the key points of the face The positioning model locates the key points of the face in the face image before the person to be authenticated performs the live action and the face image after the live action; and the skin area obtaining unit is used to obtain the face position and position according to the positioning. The key point position divides the face image before the person to be authenticated performs the live action and the face in the face image after the person performs the live action, so as to obtain the skin area image before the person to be authenticated performs the live action and perform the live action After the skin area image.
- [0074]** Exemplarily, the identity authentication device further includes: a sample image acquisition module for acquiring a sample face image, the position of the face in the sample face image and the position of the key points of the face are marked; and a training module , Used for neural network training using sample face images to obtain a face detection model and a face key point positioning model.
- [0075]** Exemplarily, the living body detection module includes: a structured light image acquisition sub-module for acquiring a face image collected by a binocular camera on the face of a person to be authenticated under the irradiation of structured light; and a living body determination sub-module for To determine whether the face of the person to be authenticated belongs to a living body according to the face image, to obtain a living body detection result.
- [0076]** Exemplarily, the living body detection module includes: an illumination image acquisition sub-module for acquiring one or more illumination images collected for the face of the person to be authenticated under the illumination of the detection light; an illumination living body verification sub-module for Based on the light reflection characteristics of the face of the person to be authenticated in one or more illumination images, it is determined whether the face of the person to be authenticated is a living body, so as to obtain the illumination living body verification result; and the living body verification pass determination sub-module is used to at least Based on the result of the live body verification, it is determined whether the certified person passes the live body verification to obtain the live body test result.

- [0077] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed at least once.
- [0078] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light changes between every two consecutive moments.
- [0079] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is randomly changed or preset.
- [0080] Exemplarily, the living body detection module includes: an instruction output sub-module for outputting action instructions, where the action instruction is used to instruct the authenticator to perform a corresponding action; the action image acquisition sub-module is used to obtain the face of the person to be authenticated The collected multiple action images; the action detection sub-module, which is used to detect the actions performed by the person to be authenticated based on the multiple action images; and the action living verification sub-module, which is used to determine the action of the person to be authenticated based on the action detection results and the action instructions Whether the face belongs to a living body to obtain the result of the live body verification; the live body verification pass determination sub-module includes: a pass determining unit for determining whether the person to be authenticated has passed the live body verification based on the light live body verification result and the action live body verification result to obtain the live body detection result.
- [0081] Exemplarily, the action living body verification sub-module includes: a living body determining unit, which is configured to detect the action executed by the person to be authenticated and the action instruction based on a plurality of action images collected within a time period not greater than the first preset time. If the indicated action is consistent with the action, it is determined that the face of the person to be authenticated belongs to a living body. If based on the multiple action images collected within the first preset time, it is not detected that the person to be authenticated is executed by the person to be authenticated and is instructed by the action instruction If the action is consistent, it is determined that the face of the person to be authenticated does not belong to a living body.
- [0082] Exemplarily, the identity authentication device further includes: a first counting module, configured to count once every time the instruction output sub-module is run to the action biological verification sub-module to obtain the action verification times; the biological verification device further includes: The first verification error execution module is used to output the first error message if the result of the live action verification indicates that the face of the person to be authenticated does not belong to a living body, and determine whether the number of action verifications has reached the first threshold, and if the number of action verifications has reached

The first number of thresholds, the in vivo verification passes the determination sub-module, and if the number of action verifications does not reach the first number of thresholds, the instruction output sub-module is restarted or the illumination image acquisition sub-module runs before the instruction output sub-module Restart the illumination image acquisition sub-module, where the first error message is used to prompt that the in-vivo verification for the person to be authenticated fails.

- [0083]** Exemplarily, the identity authentication device further includes: a condition judgment module for judging whether the image acquisition conditions of the person to be authenticated meet the preset requirements, and if the image acquisition conditions meet the preset requirements, start the illumination image acquisition sub-module, where the image The acquisition conditions include the position of the person to be authenticated in the image acquisition area of the image acquisition device and/or the relative angle of the person to be authenticated and the image acquisition device.
- [0084]** Exemplarily, the identity authentication device further includes: a first prompt information output module, which is used to determine the image acquisition condition of the person to be authenticated before the condition judgment module determines whether the image acquisition condition of the person to be authenticated meets the preset requirements or with the condition judgment module Whether the preset requirements are met or not, the first prompt information is output at the same time, where the first prompt information is used to prompt the person to be authenticated to face the image acquisition device and approach the image acquisition device.
- [0085]** Exemplarily, the first prompt information output module includes: a first prompt information output submodule, configured to output the first prompt information in one or more of a voice form, an image form, and a text form.
- [0086]** Exemplarily, the condition judgment module includes: a first real-time image acquisition sub-module, which is used to acquire real-time images collected for the face of the person to be authenticated; a first real-time output sub-module, which is used to output real-time image acquisition conditions for calibration The preset area and the face area in the real-time image are used for display; and the first judgment sub-module is used to judge whether the image acquisition condition meets the preset requirements according to the face area detected in the real-time image, if the face area is located In the preset area and the proportion of the face area in the real-time image is greater than the first preset proportion, it is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the face area is in the real-time image If the proportion is not greater than the first preset ratio, it is determined that the image acquisition condition does not meet the preset requirements.

- [0087] Exemplarily, the identity authentication device further includes: a first collection prompt output module, configured to output the first collection prompt information in real time to remind the waiting area if the proportion of the face area in the real-time image is not greater than the first preset ratio. The certified person is close to the image capture device.
- [0088] Exemplarily, the condition judgment module includes: a second real-time image acquisition sub-module, which is used to acquire real-time images collected for the face of the person to be authenticated; The preset area and the face area in the real-time image are used for display; and the second judgment sub-module is used to judge whether the image acquisition conditions meet the preset requirements according to the face area, if the face area is located in the preset area and the person If the proportion of the face area in the preset area is greater than the second preset ratio, it is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the proportion of the face area in the preset area is not If the ratio is greater than the second preset ratio, it is determined that the image acquisition condition does not meet the preset requirement.
- [0089] Exemplarily, the identity authentication method further includes: if the proportion of the face area in the preset area is not greater than the second preset ratio, outputting second collection prompt information in real time to prompt the person to be authenticated to approach the image collection device.
- [0090] Exemplarily, the identity authentication device further includes: a second collection prompt output module, used to determine the relative position relationship between the face area and the preset area in real time; and a third collection prompt output module, used based on the face area and the preset area. The relative position relationship between the preset areas outputs third collection prompt information in real time to prompt to change the relative position relationship between the person to be authenticated and the image acquisition device so that the face area is close to the preset area.
- [0091] Exemplarily, the condition judgment module includes: a posture information acquisition module, which is used to obtain posture information of the image acquisition device; and a third judgment sub-module, which is used to judge whether the image acquisition device is in a vertically placed state according to the posture information, and if so, It is determined that the image acquisition conditions meet the preset requirements; otherwise, it is determined that the image acquisition conditions do not meet the preset requirements.
- [0092] Exemplarily, the identity authentication method further includes: a second counting module, configured to count once during the process from running the illumination image acquisition module to the operation of the illumination living body verification module to obtain the number

of illumination verification times; the identity authentication device further includes: a second verification error The execution module is used to output a second error message if the result of the light living body verification indicates that the face of the person to be authenticated does not belong to a living body, and determine whether the number of light verification times reaches the second number threshold, and if the number of light verification times reaches the second number threshold, Then start the living body verification through the confirming sub-module, if the number of light verification times does not reach the second threshold, restart the condition judgment module or restart the light image acquisition sub-module, where the second error message is used to prompt the living body of the person to be authenticated verification failed.

- [0093]** Exemplarily, the identity authentication device further includes: a second prompt information output module, which is used before the illumination image acquisition sub-module acquires one or more illumination images collected for the face of the person to be authenticated under the illumination of the detection light Or the illumination image acquisition submodule acquires one or more illumination images collected for the face of the person to be authenticated under the illumination of the detection light, and the illumination living verification submodule is based on the face of the person to be authenticated in one or more illuminations. In the process of determining whether the face of the person to be authenticated belongs to a living body by the light reflection characteristics shown in the image, the second prompt information is output, where the second prompt information is used to prompt the person to be authenticated to remain motionless for the second preset time .
- [0094]** Exemplarily, the second prompt information is countdown information corresponding to the second preset time.
- [0095]** Exemplarily, the detection light is obtained by dynamically changing the color and/or position of the light emitted to the face of the person to be authenticated.
- [0096]** Exemplarily, the detection light is emitted by the display screen, and the detection light is obtained by dynamically changing the mode of the light emitted by the display screen by changing the content displayed on the display screen to emit the detection light to the face of the person to be authenticated.
- [0097]** According to another aspect of the present invention, there is provided an identity authentication device, including a processor and a memory, wherein computer program instructions are stored in the memory, and the computer program instructions are used to perform the following steps when the processor is run: Identification information; determine whether the personal identification information is authenticated information to obtain the information authentication result; perform a live test on the person to be authenticated to obtain the live test result; and at least

determine whether the identity of the person to be authenticated is legal based on the information authentication result and the live test result .

- [0098] Exemplarily, the personal identification information is one of the ID number, name, ID face, field collected face, ID number conversion value, name conversion value, ID face conversion value, and field collection face conversion value. Kind or more.
- [0099] Exemplarily, before the computer program instructions are executed when the processor is running to obtain the personal identification information of the person to be authenticated, the computer program instructions are also used to perform the following steps when the processor is running: the output is used to instruct the person to be authenticated to provide Indication information of a predetermined type of personnel information; wherein, the personal identification information is personnel information provided by the person to be authenticated or obtained based on personnel information.
- [0100] Exemplarily, the personal identification information is one or more of the conversion value of the certificate number, the conversion value of the name, the conversion value of the certificate face and the conversion value of the face collected on site, and the computer program instructions are used when the processor is running. The step of obtaining the personal identification information of the person to be authenticated includes: obtaining the initial information of the person to be authenticated, where the initial information is one or more of the credential number, name, credential face, and on-site collected face; and The initial information is transformed based on a predetermined algorithm to obtain the transformed value of the initial information as the personal identification information.
- [0101] Exemplarily, the step of obtaining the personal identification information of the person to be authenticated executed when the computer program instructions are executed by the processor includes: obtaining a credential image of the person to be authenticated, wherein the information authentication result is the credential authentication result; the computer program instructions are When the processor is running, the steps of performing live detection on the person to be authenticated to obtain the live detection result include: obtaining a face image of the person to be authenticated; and performing live detection using the face image to obtain the live detection result.
- [0102] Exemplarily, before the computer program instruction is used to execute the step of determining whether the identity of the person to be authenticated is legal based on the information authentication result and the living body test result when the computer program instruction is run by the processor, the computer program instruction is also used to execute the step when the processor is run The following steps: use the certificate image and/or the face image to perform additional judgment operations to obtain additional judgment results; the computer program

instructions used to execute when the processor is running at least determine the identity of the person to be authenticated based on the information authentication result and the living body detection result The legality step includes: determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body test result, and the additional judgment result.

- [0103]** Exemplarily, the additional judgment operation includes a credential authenticity judgment operation and/or a face consistency judgment operation, and the additional judgment result includes a credential authenticity judgment result and/or a face consistency judgment result, and the credential authenticity judgment operation includes: Determine whether the certificate in the certificate image is a real certificate to obtain the result of the authenticity of the certificate; the judgment operation of face consistency includes: obtaining the certificate face of the person to be authenticated according to the certificate image; and comparing the face of the person to be authenticated with the person The faces in the face image are compared to obtain the judgment result of face consistency.
- [0104]** Exemplarily, the step of obtaining the credential face of the person to be authenticated according to the credential image that the computer program instructions are executed when the processor is running includes: detecting the face from the credential image to obtain the credential face of the person to be authenticated .
- [0105]** Exemplarily, the step of obtaining the credential face of the person to be authenticated according to the credential image executed when the computer program instructions are executed by the processor includes: performing text recognition on the credential image to obtain the text information in the credential image; based on the credential image The text information in is searched for matching certificate information from the authenticated certificate information database; and the certificate face in the searched and matched certificate information is determined to be the certificate face of the person to be authenticated.
- [0106]** Exemplarily, when the computer program instructions are executed by the processor to determine whether the document in the document image is a real document, the step of obtaining the authenticity judgment result of the document includes: extracting image characteristics of the document image; and inputting the image characteristics A well-trained document classifier is used to obtain the authenticity judgment result of the document; among them, the authenticity judgment result of the document is the confidence that the document in the document image is a real document.
- [0107]** Exemplarily, when the computer program instructions are executed by the processor to determine whether the certificate in the certificate image is a real certificate, the step of obtaining the

authenticity determination result of the certificate includes: identifying an image block containing the identification information of the certificate from the certificate image ; And the identification of the certificate identification information in the image block containing the certificate identification information to obtain the certificate authenticity judgment result; wherein the certificate authenticity judgment result is the confidence that the certificate in the certificate image is a real certificate.

- [0108] Exemplarily, during the execution of the step of determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result and the additional judgment result, the certificate authentication result and the living body detection are executed when the computer program instructions are executed by the processor. Each of the result and the additional judgment result has its own weight coefficient.
- [0109] Exemplarily, the step of obtaining the credential image of the person to be authenticated executed when the computer program instructions are executed by the processor includes: obtaining a pre-shot image collected in real time for the credential of the person to be authenticated under the current shooting conditions; real-time evaluation of the pre-shot The image attributes of the image; when the evaluation value of the image attribute of the pre-shot image is less than the preset evaluation value threshold, the pre-shot prompt information is generated according to the image attribute of the pre-shot image to prompt the person to be authenticated to adjust the shooting conditions of his certificate; And when the evaluation value of the image attribute of the pre-photographed image is equal to or greater than the preset evaluation value threshold, the pre-photographed image is saved to use it as a credential image.
- [0110] Exemplarily, the step of determining whether the personal identification information is authenticated information executed by the computer program instructions when the processor is running to obtain the information authentication result includes: performing text recognition on the document image to obtain the text information in the document image ; And based on the text information in the credential image to search in the certified credential information database to obtain the credential certification result; wherein the credential credential information in the credential image is the credibility of the certified credential information.
- [0111] Exemplarily, the step of performing text recognition on the credential image to obtain text information in the credential image that the computer program instructions are executed when the processor is running includes: positioning the text in the credential image to obtain an image containing text Block; and to recognize the text in the image block containing the text to obtain the text information in the credential image.
- [0112] Exemplarily, before the computer program instructions are used to perform the step of recognizing text in the image block containing text when the processor is run, the computer program

instructions are also used to perform the following steps when the processor is run: The image block is corrected to a horizontal state.

- [0113]** Exemplarily, after the computer program instructions are executed when the processor is running, the computer program instructions are used when the processor is running, after the step of recognizing the text in the image block containing the text to obtain the text information in the credential image The step of performing text recognition on the credential image to obtain text information in the credential image further includes: outputting text information in the credential image for the user to view; receiving text correction information input by the user; and correcting the text as indicated by the text correction information Compare the text to be corrected with the corresponding text in the text information in the document image; and if the difference between the text to be corrected indicated by the text correction information and the corresponding text in the text information in the document image is less than the preset difference threshold , Use the text correction information to update the text information in the credential image.
- [0114]** Exemplarily, when the computer program instruction is executed when the processor is running, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result includes: generating a living body action instruction, and the living body action instruction is used to instruct the person to be authenticated to execute the corresponding living body Action; obtain the face image of the person to be authenticated collected in real time; detect the face in the face image; and determine whether the face in the face image performs the living body action indicated by the living body action instruction to obtain the living body detection result.
- [0115]** Exemplarily, when the computer program instructions are executed by the processor, they are also used to perform live detection of a person to be authenticated to obtain a live detection result. The steps include: generating a live action instruction, and the live action instruction is used to instruct the person to be authenticated to execute the corresponding Live action; acquire the face image of the person to be authenticated collected in real time; capture the skin area image of the person to be authenticated before performing the live action and the skin area image after performing the live action from the collected face images; and transfer the person to be authenticated The skin area image before the execution of the living body action and the skin area image after the execution of the living body action are input to the skin elasticity classifier to obtain the living body detection result.
- [0116]** Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: obtaining the face image of the real person before performing the live action and the face image after performing the live action, and the face of the fake person before performing the live action Image and face image after performing the live action; extract the face

image of the real person before performing the live action and the skin after performing the live action from the face image of the real person before the live action and the face image after the live action The area image is used as a positive sample image; the face image of the fake person before the execution of the live action and the face image of the fake person before the execution of the live action are extracted from the face image of the fake person before the execution of the live action and the skin area image after the execution of the live action is taken as the negative Sample images; and use positive sample images and negative sample images to train a classifier model to obtain a skin elasticity classifier.

- [0117] Exemplarily, the step of capturing the skin area image of the person to be authenticated before performing the living body action and the skin area image after performing the living body action from the collected face image executed by the computer program instructions when the processor is running includes: From the collected face images, select the face image of the person to be authenticated before performing the live action and the face image after the execution of the live action; use the face detection model to perform the face image and the execution of the live action on the person to be authenticated Position the face in the face image after the action; use the face key point positioning model to perform the key points of the face in the face image before the person to be authenticated performs the live action and the face image after the live action Positioning; and according to the face position and key point position obtained by the positioning, the area of the face image in the face image before the person to be authenticated performs the live action and the face image after the person performs the live action is divided to obtain the execution of the person to be authenticated The skin area image before the living body action and the skin area image after the living body action is performed.
- [0118] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: obtaining a sample face image, where the position of the face and the position of the key points of the face in the sample face image are already marked; and Use sample face images for neural network training to obtain a face detection model and a face key point location model.
- [0119] Exemplarily, when the computer program instructions are executed by the processor to perform the live detection of the person to be authenticated to obtain the result of the live body detection, the steps include: obtaining the face of the person to be authenticated under the structured light by the binocular camera. The collected face image; and determining whether the face of the person to be authenticated belongs to a living body according to the face image, so as to obtain a living body detection result.
- [0120] Exemplarily, the step of performing live body detection of a person to be authenticated by the computer program instructions executed by the processor to obtain a live body detection result

includes: Step S110: Obtain the face of the person to be authenticated under the illumination of the detection light One or more collected illumination images; Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the one or more illumination images, so as to obtain illumination living body verification Result; and step S130: determine whether the certified person passes the in vivo verification based at least on the result of the in vivo illumination verification, so as to obtain the in vivo detection result.

- [0121] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed at least once.
- [0122] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light changes between every two consecutive moments.
- [0123] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is randomly changed or preset.
- [0124] Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result by executing the computer program instructions when the processor is running includes: Step S140: output an action instruction, wherein the action instruction is used to instruct the authentication person to execute the corresponding Step S150: Obtain multiple action images collected for the face of the person to be authenticated; Step S160: Detect actions performed by the person to be authenticated based on the multiple action images; and Step S170: According to the action detection result and the action instruction It is determined whether the face of the person to be authenticated belongs to a living body to obtain the result of the action in vivo verification; the step S130 executed when the computer program instructions are executed by the processor includes: determining whether the person to be authenticated passes or not based on the illumination in vivo verification result and the action in vivo verification result In vivo verification to obtain in vivo test results.
- [0125] Exemplarily, the step S170 executed by the computer program instructions when the processor is running includes: if it is detected based on a plurality of motion images collected within a period of time not greater than the first preset time that the computer program instructions are executed by the person to be authenticated, and If the action indicated by the action instruction is consistent with the action, it is determined that the face of the person to be authenticated belongs to a living body. If based on the multiple action images collected within the first preset time, the action instruction executed by the person to be authenticated is not detected If the indicated actions are

consistent, it is determined that the face of the person to be authenticated does not belong to a living body.

- [0126]** Exemplarily, the computer program instructions are also used to perform the following steps when being executed by the processor: each time step S140 to step S170 are executed to obtain the number of action verifications; after step S170, the computer program instructions are processed When the device is running, it is also used to perform the following steps: if the action verification result indicates that the face of the person to be authenticated is not a living body, the first error message is output, and it is judged whether the number of action verifications reaches the first threshold, if the number of action verifications If the first number threshold is reached, then go to step S130. If the action verification number does not reach the first number threshold, then return to step S140 or return to step S110 if step S110 is executed before step S140, where the first error The information is used to indicate that the in-vivo verification for the person to be certified has failed.
- [0127]** Exemplarily, before the computer program instructions are used to execute step S110 when the processor is running, the computer program instructions are also used to perform the following steps when the processor is running: Step S108: Determine whether the image acquisition conditions of the person to be authenticated meet the predetermined Set requirements, if the image acquisition conditions meet the preset requirements, go to step S110, where the image acquisition conditions include the position of the person to be authenticated in the image acquisition area of the image acquisition device and/or the person to be authenticated and the image acquisition device The relative angle.
- [0128]** Exemplarily, before the computer program instructions are used to execute step S108 when the processor is running or at the same time as the computer program instructions are used to execute step S108 when the processor is running, the computer program instructions are also used when the processor is running The following steps are performed: Step S106: Output first prompt information, where the first prompt information is used to prompt the person to be authenticated to face the image acquisition device and approach the image acquisition device.
- [0129]** Exemplarily, the step S106 used by the computer program instructions when the processor is running includes: outputting the first prompt information in one or more of a voice format, an image format, and a text format.
- [0130]** Exemplarily, the step S108 executed by the computer program instructions when the processor is running includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area used to calibrate the image collection condition and the information in the real-time image The face area is used for display; and according to the

face area detected in the real-time image, it is judged whether the image acquisition conditions meet the preset requirements. If the face area is located in the preset area and the proportion of the face area in the real-time image is greater than The first preset ratio, it is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the proportion of the face area in the real-time image is not greater than the first preset ratio, then the image acquisition conditions are determined Does not meet the preset requirements.

- [0131]** Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: if the proportion of the face area in the real-time image is not greater than the first preset proportion, the first collection prompt information is output in real time to prompt the waiting The certified person is close to the image capture device.
- [0132]** Exemplarily, the step S108 executed by the computer program instructions when the processor is running includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area used to calibrate the image collection condition and the information in the real-time image The face area is used for display; and judging whether the image acquisition conditions meet the preset requirements according to the face area, if the face area is located in the preset area and the proportion of the face area in the preset area is greater than the second preset ratio , It is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the proportion of the face area in the preset area is not greater than the second preset ratio, it is determined that the image acquisition conditions do not meet the preset requirements. Claim.
- [0133]** Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: if the proportion of the face area in the preset area is not greater than the second preset ratio, the second collection prompt information is output in real time to prompt The person to be authenticated is close to the image capture device.
- [0134]** Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: real-time judgment of the relative position relationship between the face area and the preset area; and based on the relative position relationship between the face area and the preset area The third collection prompt information is output in real time to prompt to change the relative position relationship between the person to be authenticated and the image collection device so that the face area is close to the preset area.
- [0135]** Exemplarily, the step S108 executed by the computer program instructions when the processor is running includes: acquiring the posture information of the image acquisition device; and judging whether the image acquisition device is in the upright position according to the posture

information, and if so, determining the image acquisition. The conditions meet the preset requirements, otherwise it is determined that the image acquisition conditions do not meet the preset requirements.

- [0136]** Exemplarily, when the computer program instructions are executed by the processor, they are also used to perform the following steps: count once during each execution of step S110 to step S120 to obtain the number of illumination verification times; and are used when the computer program instructions are executed by the processor. After step S120 is performed, the computer program instructions are also used to perform the following steps when the processor is running: if the result of the light living body verification indicates that the face of the person to be authenticated does not belong to a living body, output a second error message, and determine the number of light verification times. Whether it reaches the second threshold, if the number of light verification reaches the second threshold, go to step S130, if the number of light verification does not reach the second threshold, return to step S108 or return to step S110, where the second error message is It prompts that the in-vivo verification for the person to be certified has failed.
- [0137]** Exemplarily, the computer program instructions are executed by the processor before the steps S110 and S120 are executed when the computer program instructions are executed by the processor or during the steps S110 and S120 executed when the computer program instructions are executed by the processor. The time is also used to perform the following steps: output second prompt information, where the second prompt information is used to prompt the person to be authenticated to remain still within the second preset time.
- [0138]** Exemplarily, the second prompt information is countdown information corresponding to the second preset time.
- [0139]** Exemplarily, the detection light is obtained by dynamically changing the color and/or position of the light emitted to the face of the person to be authenticated.
- [0140]** Exemplarily, the detection light is emitted by the display screen, and the detection light is obtained by dynamically changing the mode of the light emitted by the display screen by changing the content displayed on the display screen to emit the detection light to the face of the person to be authenticated.
- [0141]** According to another aspect of the present invention, a storage medium is provided, and program instructions are stored on the storage medium. The program instructions are used to perform the following steps during operation: obtain personal identification information of a person to be authenticated; determine whether the personal identification information is authenticated. Information to obtain the information authentication result; the person to be authenticated

undergoes a live test to obtain the live test result; and at least the identity of the person to be authenticated is determined based on the information authentication result and the live test result.

- [0142] Exemplarily, the personal identification information is one of the ID number, name, ID face, field collected face, ID number conversion value, name conversion value, ID face conversion value, and field collection face conversion value. Kind or more.
- [0143] Exemplarily, before the program instruction is used to obtain the personal identification information of the person to be authenticated during runtime, the program instruction is also used to perform the following steps at runtime: output information for instructing the person to be authenticated to provide a predetermined type of personnel information Indication information; among them, the personal identification information is the personnel information provided by the person to be authenticated or obtained based on the personnel information.
- [0144] Exemplarily, the personal identification information is one or more of the conversion value of the certificate number, the conversion value of the name, the conversion value of the certificate face, and the conversion value of the face collected on-site. The program instructions are used for execution at runtime. The step of obtaining the personal identification information of the person to be authenticated includes: obtaining the initial information of the person to be authenticated, where the initial information is one or more of the ID number, name, ID face, and on-site collected face; and based on a predetermined algorithm The initial information is transformed to obtain the transformed value of the initial information as the personal identification information.
- [0145] Exemplarily, the step of obtaining the personal identification information of the person to be authenticated executed by the program instruction during runtime includes: obtaining the credential image of the person to be authenticated, wherein the information authentication result is the credential authentication result; The steps performed to perform the living body detection of the person to be authenticated to obtain the living body detection result include: obtaining the face image of the person to be authenticated; and performing the living body detection using the face image to obtain the living body detection result.
- [0146] Exemplarily, before the program instruction is used to execute the step of determining whether the identity of the person to be authenticated is legal based on the information authentication result and the biological test result at least when it is run, the program instruction is also used to perform the following steps during the run: using the credential image And/or the face image performs additional judgment operations to obtain additional judgment results; the step of determining whether the identity of the person to be authenticated is legal at least according to the information authentication result and the living body test result executed by the program instructions during operation includes: authenticating according to the certificate The result, the result of the living

body test, and the additional judgment result determine whether the identity of the person to be authenticated is legal.

- [0147] Exemplarily, the additional judgment operation includes a credential authenticity judgment operation and/or a face consistency judgment operation, and the additional judgment result includes a credential authenticity judgment result and/or a face consistency judgment result, and the credential authenticity judgment operation includes: Determine whether the certificate in the certificate image is a real certificate to obtain the result of the authenticity of the certificate; the judgment operation of face consistency includes: obtaining the certificate face of the person to be authenticated according to the certificate image; and comparing the face of the person to be authenticated with the person The faces in the face image are compared to obtain the judgment result of face consistency.
- [0148] Exemplarily, the step of obtaining the credential face of the person to be authenticated according to the credential image executed by the program instruction during execution includes: detecting the face from the credential image to obtain the credential face of the person to be authenticated.
- [0149] Exemplarily, the step of obtaining the credential face of the person to be authenticated based on the credential image executed by the program instruction during execution includes: performing text recognition on the credential image to obtain text information in the credential image; and based on the text in the credential image The information searches for matching certificate information from the authenticated certificate information database; and determines that the certificate face in the searched and matched certificate information is the certificate face of the person to be authenticated.
- [0150] Exemplarily, the program instruction is used to determine whether the certificate in the certificate image is a real certificate during execution, so as to obtain the authenticity judgment result of the certificate. The steps include: extracting the image characteristics of the certificate image; and inputting the image characteristics into the trained image The certificate classifier is used to obtain the authenticity judgment result of the certificate; wherein the authenticity judgment result of the certificate is the confidence that the certificate in the certificate image is a real certificate.
- [0151] Exemplarily, the step of determining whether the document in the document image is a real document when the program instruction is executed to obtain the authenticity judgment result of the document includes: identifying the image block containing the document identification information from the document image; and The certificate identification information in the image block containing the certificate identification information is identified to obtain the certificate

authenticity judgment result; wherein the certificate authenticity judgment result is the confidence that the certificate in the certificate image is a real certificate.

- [0152] Exemplarily, during the execution of the step of determining whether the identity of the person to be authenticated is legal based on the certificate authentication result, the living body test result, and the additional judgment result that the program instructions are used to execute when the program instruction is running, the certificate authentication result, the living body test result, and the additional Each of the judgment results has its own weight coefficient.
- [0153] Exemplarily, the step of acquiring the credential image of the person to be authenticated executed by the program instruction during execution includes: acquiring a pre-shooting image that is collected in real time for the credential of the person to be authenticated under the current shooting conditions; and evaluating the image of the pre-shooting image in real time Attribute; when the evaluation value of the image attribute of the pre-shot image is less than the preset evaluation value threshold, the pre-shot prompt message is generated according to the image attribute of the pre-shot image to prompt the person to be authenticated to adjust the shooting conditions of the certificate; and When the evaluation value of the image attribute of the captured image is equal to or greater than the preset evaluation value threshold, the pre-captured image is saved to use it as a document image.
- [0154] Exemplarily, the step of determining whether the personal identification information is authenticated information to obtain the information authentication result executed by the program instructions during execution includes: performing text recognition on the document image to obtain the text information in the document image; and The text information in the credential image is searched in the certified credential information database to obtain the credential certification result; where the credential certification result is the credibility of the credential information in the credential image.
- [0155] Exemplarily, the step of performing text recognition on the credential image to obtain text information in the credential image executed by the program instruction during execution includes: positioning the text in the credential image to obtain an image block containing the text; and Recognize the text in the image block containing the text to obtain the text information in the document image.
- [0156] Exemplarily, before the program instructions are used to perform the step of recognizing text in the image block containing text at runtime, the program instructions are also used to perform the following step at runtime: correct the image block containing text to horizontal status.

- [0157]** Exemplarily, after the step of recognizing the text in the image block containing the text to obtain the text information in the credential image, the program instruction is used to execute the credential image The step of performing text recognition to obtain text information in the credential image also includes: outputting text information in the credential image for the user to view; receiving text correction information input by the user; and correcting the text to be corrected indicated by the text and the credential The corresponding text in the text information in the image is compared; and if the difference between the text to be corrected indicated by the text correction information and the corresponding text in the text information in the document image is less than the preset difference threshold, the text correction information is used Update the text information in the ID image.
- [0158]** Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result executed by the program instruction during runtime includes: generating a living body action instruction, the living body action instruction being used to instruct the person to be authenticated to perform the corresponding living body action; obtaining Real-time collection of the face image of the person to be authenticated; detecting the face in the face image; and judging whether the face in the face image performs the living body action indicated by the living body action instruction to obtain the living body detection result.
- [0159]** Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result executed by the program instruction during runtime includes: generating a living body action instruction, the living body action instruction being used to instruct the person to be authenticated to perform the corresponding living body action; obtaining The face image of the person to be authenticated collected in real time; the skin area image of the person to be authenticated before performing the live action and the skin area image after the person performing the live action are captured from the collected face images; and the person to be authenticated before performing the live action The image of the skin area and the image of the skin area after performing the living body action are input into the skin elasticity classifier to obtain the living body detection result.
- [0160]** Exemplarily, the program instructions are also used to perform the following steps during execution: obtaining the face image of the real person before performing the live action and the face image after performing the live action, and the face image and execution of the fake person before performing the live action. Face image after live action; extract the face image of the real person before the live action and the skin area image after the live action from the face image of the real person before the live action and the face image after the live action. Positive sample image; extract the face image of the fake person before performing the live action and the skin area image after performing the live action from the face image of the fake person before performing the live action and the face image after performing the live action as the negative

sample image; And use the positive sample image and the negative sample image to train the classifier model to obtain the skin elasticity classifier.

- [0161] Exemplarily, the step of capturing the image of the skin area of the person to be authenticated before performing the live action and the image of the skin area after performing the live action from the collected face image executed by the program instructions during runtime includes: In the face image, select the face image of the person to be authenticated before performing the live action and the face image after performing the live action; use the face detection model to compare the face image before the person to be authenticated performs the live action and the face image after performing the live action Position the face in the face image; use the face key point positioning model to locate the key points of the face in the face image before the person to be authenticated performs the live action and the face image after the live action; and According to the position of the face and the key point position obtained by the positioning, the face image before the person to be authenticated performs the live action and the face in the face image after the person performs the live action are divided into regions, so as to obtain the person to be authenticated before performing the live action The skin area image and the skin area image after performing the living action.
- [0162] Exemplarily, the program instructions are also used to perform the following steps at runtime: obtain a sample face image, the position of the face in the sample face image and the position of the key points of the face are already marked; and use the sample person Neural network training is performed on the face image to obtain a face detection model and a face key point positioning model.
- [0163] Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result executed by the program instruction at runtime includes: obtaining the binocular camera for the face of the person to be authenticated under the irradiation of structured light. Face image; and determining whether the face of the person to be authenticated belongs to a living body according to the face image, so as to obtain a living body detection result.
- [0164] Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result executed by the program instruction at runtime includes: Step S110: Obtain the collected data for the face of the person to be authenticated under the illumination of the detection light One or more illumination images; Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in one or more illumination images, so as to obtain the result of the illumination living body verification; and Step S130: Determine

whether the authenticator passes the in vivo verification based at least on the result of the in vivo illumination verification, so as to obtain the in vivo detection result.

- [0165] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed at least once.
- [0166] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light changes between every two consecutive moments.
- [0167] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is randomly changed or preset.
- [0168] Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result executed by the program instruction during runtime includes: Step S140: output an action instruction, wherein the action instruction is used to instruct the authentication person to perform a corresponding action; Step S150: Acquire multiple motion images collected for the face of the person to be authenticated; Step S160: Detect actions performed by the person to be authenticated based on the multiple motion images; and Step S170: Determine the person to be authenticated based on the action detection result and the action instruction Whether the person's face belongs to a living body, to obtain the action living body verification result; the step S130 executed by the program instruction during operation includes: determining whether the person to be authenticated passes the living body verification based on the light living body verification result and the action living body verification result to obtain the living body Test results.
- [0169] Exemplarily, the step S170 used for execution of the program instruction during runtime includes: if it is detected based on a plurality of motion images collected within a period of time not greater than the first preset time that the program instruction is executed by the person to be authenticated and is related to the operation instruction. If the indicated action is consistent with the action, it is determined that the face of the person to be authenticated belongs to a living body. If based on the multiple action images collected within the first preset time, it is not detected that the person to be authenticated is executed by the person to be authenticated and is instructed by the action instruction If the action is consistent, it is determined that the face of the person to be authenticated does not belong to a living body.
- [0170] Exemplarily, the program instructions are also used to execute the following steps during execution: counting once during each execution of step S140 to step S170 to obtain the number of action verification times; after step S170 that the program instructions are used to execute during execution , The program instructions are also used to perform the following steps during operation:

If the result of the action in vivo verification indicates that the face of the person to be authenticated does not belong to a living body, output the first error message, and determine whether the number of action verification times reaches the first threshold value, if If the number of action verification times reaches the first number threshold, then go to step S130. If the number of action verification times does not reach the first number threshold, return to step S140 or return to step S110 if step S110 is executed before step S140, where, The first error message is used to indicate that the in-vivo verification for the person to be authenticated has failed.

- [0171] Exemplarily, before the program instructions are used to execute step S110 during runtime, the program instructions are also used to execute the following steps during runtime: Step S108: Determine whether the image acquisition conditions of the person to be authenticated meet the preset requirements, if the image acquisition If the conditions meet the preset requirements, go to step S110, where the image acquisition conditions include the position of the person to be authenticated in the image acquisition area of the image acquisition device and/or the relative angle between the person to be authenticated and the image acquisition device.
- [0172] Exemplarily, before or at the same time as step S108 that the program instructions are used to execute during runtime, the program instructions are also used to execute the following steps during runtime: Step S106: output the first step A prompt message, wherein the first prompt message is used to prompt the person to be authenticated to face the image acquisition device and approach the image acquisition device.
- [0173] Exemplarily, the step S106 executed by the program instruction during runtime includes: outputting the first prompt information in one or more of a voice form, an image form, and a text form.
- [0174] Exemplarily, the step S108 executed by the program instruction during runtime includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area for calibrating image acquisition conditions and a face area in the real-time image Used for display; and judging whether the image acquisition conditions meet the preset requirements according to the face area detected in the real-time image, if the face area is located in the preset area and the proportion of the face area in the real-time image is greater than the first preset If the ratio is set, it is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the proportion of the face area in the real-time image is not greater than the first preset ratio, it is determined that the image acquisition conditions do not meet the preset requirements. Set requirements.
- [0175] Exemplarily, the program instructions are also used to perform the following steps when running: if the proportion of the face area in the real-time image is not greater than the first preset proportion,

the first collection prompt information is output in real time to remind the person to be authenticated and The image capture device is close.

- [0176]** Exemplarily, the step S108 executed by the program instruction during runtime includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area for calibrating image acquisition conditions and a face area in the real-time image Used for display; and judging whether the image acquisition conditions meet the preset requirements according to the face area, if the face area is located in the preset area and the proportion of the face area in the preset area is greater than the second preset ratio, it is determined The image acquisition condition meets the preset requirement. If the face area is not located in the preset area or the proportion of the face area in the preset area is not greater than the second preset ratio, it is determined that the image acquisition condition does not meet the preset requirement.
- [0177]** Exemplarily, the program instructions are also used to perform the following steps when running: if the proportion of the face area in the preset area is not greater than the second preset ratio, the second collection prompt information is output in real time to remind the person to be authenticated Close to the image capture device.
- [0178]** Exemplarily, the program instructions are also used to perform the following steps when running: judging the relative position relationship between the face area and the preset area in real time; 3. Collect prompt information to prompt to change the relative position relationship between the person to be authenticated and the image acquisition device so that the face area is close to the preset area.
- [0179]** Exemplarily, the step S108 executed by the program instruction during runtime includes: acquiring the posture information of the image acquisition device; and judging whether the image acquisition device is in the upright position according to the posture information, and if so, determining that the image acquisition condition satisfies the preset state. Set the requirements, otherwise it is determined that the image acquisition conditions do not meet the preset requirements.
- [0180]** Exemplarily, the program instructions are also used to perform the following steps at runtime: counting once during each execution of step S110 to step S120 to obtain the number of illumination verification times; after the program instructions are used to execute step S120 at runtime , The program instructions are also used to perform the following steps during operation: if the result of the illumination in vivo verification indicates that the face of the person to be authenticated is not a living body, the second error message is output, and it is judged whether the number of illumination verification times reaches the second threshold value, and if the illumination verification result indicates that the face of the person to be authenticated does not belong to a

living body, If the number of verification times reaches the second threshold, then go to step S130. If the number of light verifications does not reach the second threshold, then return to step S108 or return to step S110, where the second error message is used to prompt the in vivo verification for the person to be authenticated failure.

- [0181]** Exemplarily, before the step S110 that the program instructions are used to execute during runtime or during the steps S110 and S120 that the program instructions are used to execute during runtime, the program instructions are also used to execute the following steps during runtime: output The second prompt information, wherein the second prompt information is used to prompt the person to be authenticated to remain still within the second preset time.
- [0182]** Exemplarily, the second prompt information is countdown information corresponding to the second preset time.
- [0183]** Exemplarily, the detection light is obtained by dynamically changing the color and/or position of the light emitted to the face of the person to be authenticated.
- [0184]** Exemplarily, the detection light is emitted by the display screen, and the detection light is obtained by dynamically changing the mode of the light emitted by the display screen by changing the content displayed on the display screen to emit the detection light to the face of the person to be authenticated.
- [0185]** According to another aspect of the present invention, there is provided an identity authentication device, including an information collection device, a processor, and a memory. The information collection device is used to collect initial information of a person to be authenticated; the memory stores computer program instructions and computer program instructions. When the processor is running, it is used to perform the following steps: obtain the personal identification information of the person to be authenticated, where the personal identification information is initial information or obtained based on the initial information; determine whether the personal identification information is authenticated information to obtain the information authentication result; The authenticator performs a live test to obtain a live test result; and at least determines whether the identity of the person to be authenticated is legal based on the information authentication result and the live test result.
- [0186]** Exemplarily, the information collection device includes an image collection device and/or an input device.

- [0187] Exemplarily, the personal identification information is one of the ID number, name, ID face, field collected face, ID number conversion value, name conversion value, ID face conversion value, and field collection face conversion value. Kind or more.
- [0188] Exemplarily, before the computer program instructions are executed when the processor is running to obtain the personal identification information of the person to be authenticated, the computer program instructions are also used to perform the following steps when the processor is running: the output is used to instruct the person to be authenticated to provide Indication information of a predetermined type of personnel information; wherein, the personal identification information is personnel information provided by the person to be authenticated or obtained based on personnel information.
- [0189] Exemplarily, the personal identification information is one or more of the conversion value of the certificate number, the conversion value of the name, the conversion value of the certificate face and the conversion value of the face collected on site, and the computer program instructions are used when the processor is running. The step of obtaining the personal identification information of the person to be authenticated includes: obtaining the initial information of the person to be authenticated, where the initial information is one or more of the credential number, name, credential face, and on-site collected face; and The initial information is transformed based on a predetermined algorithm to obtain the transformed value of the initial information as the personal identification information.
- [0190] Exemplarily, the information collection device includes an image collection device, and the image collection device is also used to collect the credential image and face image of the person to be authenticated; the computer program instructions are executed when the processor is running to obtain the personal identification information of the person to be authenticated The steps include: obtaining the credential image of the person to be authenticated, where the information authentication result is the credential authentication result; the computer program instructions are executed when the processor is running to perform a living body test on the person to be authenticated to obtain the living body test result. The steps include: Obtain the face image of the person to be authenticated; and use the face image to perform live body detection to obtain the live body detection result.
- [0191] Exemplarily, the identity authentication device further includes an image acquisition device, the image acquisition device is used to collect the credential image and the face image of the person to be authenticated; the computer program instructions are executed when the processor is running to obtain the personal identification information of the person to be authenticated The steps include: obtaining the credential image of the person to be authenticated, where the information authentication result is the credential authentication result; the computer program

instructions are executed when the processor is running to perform a living body test on the person to be authenticated to obtain the living body test result. The steps include: Obtain the face image of the person to be authenticated; and use the face image to perform live body detection to obtain the live body detection result.

- [0192]** Exemplarily, before the computer program instruction is used to execute the step of determining whether the identity of the person to be authenticated is legal based on the information authentication result and the living body test result when the computer program instruction is run by the processor, the computer program instruction is also used to execute the step when the processor is run The following steps: use the certificate image and/or the face image to perform additional judgment operations to obtain additional judgment results; the computer program instructions used to execute when the processor is running at least determine the identity of the person to be authenticated based on the information authentication result and the living body detection result The legality step includes: determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body test result, and the additional judgment result.
- [0193]** Exemplarily, the additional judgment operation includes a credential authenticity judgment operation and/or a face consistency judgment operation, and the additional judgment result includes a credential authenticity judgment result and/or a face consistency judgment result, and the credential authenticity judgment operation includes: Determine whether the certificate in the certificate image is a real certificate to obtain the result of the authenticity of the certificate; the judgment operation of face consistency includes: obtaining the certificate face of the person to be authenticated according to the certificate image; and comparing the face of the person to be authenticated with the person The faces in the face image are compared to obtain the judgment result of face consistency.
- [0194]** Exemplarily, the step of obtaining the credential face of the person to be authenticated according to the credential image that the computer program instructions are executed when the processor is running includes: detecting the face from the credential image to obtain the credential face of the person to be authenticated .
- [0195]** Exemplarily, the step of obtaining the credential face of the person to be authenticated according to the credential image executed when the computer program instructions are executed by the processor includes: performing text recognition on the credential image to obtain the text information in the credential image; based on the credential image The text information in is searched for matching certificate information from the authenticated certificate information

database; and the certificate face in the searched and matched certificate information is determined to be the certificate face of the person to be authenticated.

- [0196] Exemplarily, when the computer program instructions are executed by the processor to determine whether the document in the document image is a real document, the step of obtaining the authenticity judgment result of the document includes: extracting image characteristics of the document image; and inputting the image characteristics A well-trained document classifier is used to obtain the authenticity judgment result of the document; among them, the authenticity judgment result of the document is the confidence that the document in the document image is a real document.
- [0197] Exemplarily, when the computer program instructions are executed by the processor to determine whether the certificate in the certificate image is a real certificate, the step of obtaining the authenticity determination result of the certificate includes: identifying an image block containing the identification information of the certificate from the certificate image ; And the identification of the certificate identification information in the image block containing the certificate identification information to obtain the certificate authenticity judgment result; wherein the certificate authenticity judgment result is the confidence that the certificate in the certificate image is a real certificate.
- [0198] Exemplarily, during the execution of the step of determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result and the additional judgment result, the certificate authentication result and the living body detection are executed when the computer program instructions are executed by the processor. Each of the result and the additional judgment result has its own weight coefficient.
- [0199] Exemplarily, the image acquisition device is further used for real-time acquisition of pre-photographed images for the credential of the person to be authenticated; the step of acquiring the credential image of the person to be authenticated executed when the computer program instructions are executed by the processor includes: obtaining the current shooting conditions The following is a pre-shooting image collected in real time for the credentials of the person to be authenticated; real-time evaluation of the image attributes of the pre-shooting image; when the evaluation value of the image attribute of the pre-shooting image is less than the preset evaluation value threshold, according to the image of the pre-shooting image The attribute generates pre-shooting prompt information, which is used to prompt the person to be authenticated to adjust the shooting conditions of his certificate; and when the evaluation value of the image attribute of the pre-shooting image is equal to or greater than the preset evaluation value threshold, the pre-shooting image is saved as ID image.

- [0200] Exemplarily, the step of determining whether the personal identification information is authenticated information executed by the computer program instructions when the processor is running to obtain the information authentication result includes: performing text recognition on the document image to obtain the text information in the document image ; And based on the text information in the credential image to search in the certified credential information database to obtain the credential certification result; wherein the credential credential information in the credential image is the credibility of the certified credential information.
- [0201] Exemplarily, the step of performing text recognition on the credential image to obtain text information in the credential image that the computer program instructions are executed when the processor is running includes: positioning the text in the credential image to obtain an image containing text Block; and to recognize the text in the image block containing the text to obtain the text information in the credential image.
- [0202] Exemplarily, before the computer program instructions are used to perform the step of recognizing text in the image block containing text when the processor is run, the computer program instructions are also used to perform the following steps when the processor is run: The image block is corrected to a horizontal state.
- [0203] Exemplarily, after the computer program instructions are executed when the processor is running, the computer program instructions are used when the processor is running, after the step of recognizing the text in the image block containing the text to obtain the text information in the credential image The step of performing text recognition on the credential image to obtain text information in the credential image further includes: outputting text information in the credential image for the user to view; receiving text correction information input by the user; and correcting the text as indicated by the text correction information Compare the text to be corrected with the corresponding text in the text information in the document image; and if the difference between the text to be corrected indicated by the text correction information and the corresponding text in the text information in the document image is less than the preset difference threshold , Use the text correction information to update the text information in the credential image.
- [0204] Exemplarily, when the computer program instruction is executed when the processor is running, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result includes: generating a living body action instruction, and the living body action instruction is used to instruct the person to be authenticated to execute the corresponding living body Action; obtain the face image of the person to be authenticated collected in real time; detect the face in the face image; and determine whether the face in the face image performs the

living body action indicated by the living body action instruction to obtain the living body detection result.

- [0205] Exemplarily, when the computer program instructions are executed by the processor, they are also used to perform live detection of a person to be authenticated to obtain a live detection result. The steps include: generating a live action instruction, and the live action instruction is used to instruct the person to be authenticated to execute the corresponding Live action; acquire the face image of the person to be authenticated collected in real time; capture the skin area image of the person to be authenticated before performing the live action and the skin area image after performing the live action from the collected face images; and transfer the person to be authenticated The skin area image before the execution of the living body action and the skin area image after the execution of the living body action are input to the skin elasticity classifier to obtain the living body detection result.
- [0206] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: obtaining the face image of the real person before performing the live action and the face image after performing the live action, and the face of the fake person before performing the live action Image and face image after performing the live action; extract the face image of the real person before performing the live action and the skin after performing the live action from the face image of the real person before the live action and the face image after the live action The area image is used as a positive sample image; the face image of the fake person before the execution of the live action and the face image of the fake person before the execution of the live action are extracted from the face image of the fake person before the execution of the live action and the skin area image after the execution of the live action is taken as the negative Sample images; and use positive sample images and negative sample images to train a classifier model to obtain a skin elasticity classifier.
- [0207] Exemplarily, the step of capturing the skin area image of the person to be authenticated before performing the living body action and the skin area image after performing the living body action from the collected face image executed by the computer program instructions when the processor is running includes: From the collected face images, select the face image of the person to be authenticated before performing the live action and the face image after the execution of the live action; use the face detection model to perform the face image and the execution of the live action on the person to be authenticated Position the face in the face image after the action; use the face key point positioning model to perform the key points of the face in the face image before the person to be authenticated performs the live action and the face image after the live action Positioning; and according to the face position and key point position obtained by the positioning, the area of the face image in the face image before the person to be authenticated performs the live action and the face image after the person performs the live action is divided to obtain the

execution of the person to be authenticated The skin area image before the living body action and the skin area image after the living body action is performed.

- [0208] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: obtaining a sample face image, where the position of the face and the position of the key points of the face in the sample face image are already marked; and Use sample face images for neural network training to obtain a face detection model and a face key point location model.
- [0209] Exemplarily, the identity authentication device further includes a structured light source and a binocular camera, wherein the structured light source is used to emit structured light to the face of the person to be authenticated; the binocular camera is used to target the person to be authenticated under the illumination of the structured light The face collects the face image; the computer program instructions are executed when the processor is running to perform the live detection of the person to be authenticated, so as to obtain the live detection result. The steps include: obtaining the binocular camera for the person to be authenticated under the structured light The face image collected by the face; and, according to the face image, it is determined whether the face of the person to be authenticated is a living body, so as to obtain a living body detection result.
- [0210] Exemplarily, the information acquisition device includes an image acquisition device, and the identity authentication device further includes a light source, wherein the light source is used to emit detection light to the face of the person to be authenticated; the image acquisition device is also used to collect the face of the person to be authenticated. One or more illumination images under the irradiation of light; the computer program instructions are used by the processor to execute the living body detection of the person to be authenticated when the processor is running to obtain the living body detection result. The steps include: Step S110: Obtain the illumination for the detection light One or more illumination images collected by the face of the person to be authenticated under the following; Step S120: Determine whether the face of the person to be authenticated is based on the light reflection characteristics of the face of the person to be authenticated in the one or more illumination images Belonging to a living body to obtain an illuminated living body verification result; and step S130: Determine whether the authenticator has passed the living body verification based at least on the illuminated living body verification result to obtain the living body detection result.
- [0211] Exemplarily, the identity authentication device further includes a light source and an image acquisition device, wherein the light source is used to emit detection light to the face of the person to be authenticated; the image acquisition device is also used to collect the face of the person to be authenticated under the illumination of the detection light. One or more illumination

images of the person to be authenticated; the computer program instructions are executed by the processor when the processor is running to perform the living body detection of the person to be authenticated to obtain the living body detection result. The steps include: Step S110: Obtain the object to be authenticated under the illumination of the detection light One or more illumination images collected by the person's face; Step S120: Determine whether the person's face to be authenticated belongs to a living body based on the light reflection characteristics of the person's face shown in the one or more illumination images. Obtain the result of living body verification by illumination; and step S130: Determine whether the authenticator passes the verification by living body at least based on the result of living body verification by illumination, so as to obtain the result of living body detection.

- [0212] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed at least once.
- [0213] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light changes between every two consecutive moments.
- [0214] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is randomly changed or preset.
- [0215] Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result by executing the computer program instructions when the processor is running includes: Step S140: output an action instruction, wherein the action instruction is used to instruct the authentication person to execute the corresponding Step S150: Obtain multiple action images collected for the face of the person to be authenticated; Step S160: Detect actions performed by the person to be authenticated based on the multiple action images; and Step S170: According to the action detection result and the action instruction It is determined whether the face of the person to be authenticated belongs to a living body to obtain the result of the action in vivo verification; the step S130 executed when the computer program instructions are executed by the processor includes: determining whether the person to be authenticated passes or not based on the illumination in vivo verification result and the action in vivo verification result Live body verification to obtain live body detection results; the image acquisition device is also used to collect the multiple motion images.
- [0216] Exemplarily, the step S170 executed by the computer program instructions when the processor is running includes: if it is detected based on a plurality of motion images collected within a period of time not greater than the first preset time that the computer program instructions are executed by the person to be authenticated, and If the action indicated by the action instruction is consistent with the action, it is determined that the face of the person to be authenticated belongs to a living

body. If based on the multiple action images collected within the first preset time, the action instruction executed by the person to be authenticated is not detected. If the indicated actions are consistent, it is determined that the face of the person to be authenticated does not belong to a living body.

- [0217]** Exemplarily, the computer program instructions are also used to perform the following steps when being executed by the processor: each time step S140 to step S170 are executed to obtain the number of action verifications; after step S170, the computer program instructions are processed. When the device is running, it is also used to perform the following steps: if the action verification result indicates that the face of the person to be authenticated is not a living body, the first error message is output, and it is judged whether the number of action verifications reaches the first threshold, if the number of action verifications reaches the first number threshold, then go to step S130. If the action verification number does not reach the first number threshold, then return to step S140 or return to step S110 if step S110 is executed before step S140, where the first error message is output. The information is used to indicate that the in-vivo verification for the person to be certified has failed.
- [0218]** Exemplarily, before the computer program instructions are used to execute step S110 when the processor is running, the computer program instructions are also used to perform the following steps when the processor is running: Step S108: Determine whether the image acquisition conditions of the person to be authenticated meet the predetermined Set requirements, if the image acquisition conditions meet the preset requirements, go to step S110, where the image acquisition conditions include the position of the person to be authenticated in the image acquisition area of the image acquisition device and/or the person to be authenticated and the image acquisition device. The relative angle.
- [0219]** Exemplarily, before the computer program instructions are used to execute step S108 when the processor is running or at the same time as the computer program instructions are used to execute step S108 when the processor is running, the computer program instructions are also used when the processor is running. The following steps are performed: Step S106: Output first prompt information, where the first prompt information is used to prompt the person to be authenticated to face the image acquisition device and approach the image acquisition device.
- [0220]** Exemplarily, the step S106 used by the computer program instructions when the processor is running includes: outputting the first prompt information in one or more of a voice format, an image format, and a text format.
- [0221]** Exemplarily, the identity authentication device further includes a display device, wherein the step S108 performed when the computer program instructions are executed by the processor includes:

acquiring real-time images collected for the face of the person to be authenticated; real-time output for calibration The preset area of the image acquisition condition and the face area in the real-time image are used for display; and according to the face area detected in the real-time image, it is judged whether the image acquisition condition meets the preset requirements, if the face area is within the preset area And the proportion of the face area in the real-time image is greater than the first preset proportion, it is determined that the image acquisition conditions meet the preset requirements, if the face area is not located in the preset area or the proportion of the face area in the real-time image is not If the ratio is greater than the first preset ratio, it is determined that the image acquisition condition does not meet the preset requirements; the image acquisition device is also used to acquire the real-time image; the display device is used to display the preset area and the face area.

[0222] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: if the proportion of the face area in the real-time image is not greater than the first preset proportion, the first collection prompt information is output in real time to prompt the waiting The certified person is close to the image capture device.

[0223] Exemplarily, the living body verification device further includes a display device, wherein the step S108 executed by the computer program instructions when the processor is running includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time output for calibration The preset area of the image acquisition condition and the face area in the real-time image are used for display; and according to the face area to determine whether the image acquisition condition meets the preset requirements, if the face area is located in the preset area and the face area is in the preset area If the proportion of the area is greater than the second preset ratio, it is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the proportion of the face area in the preset area is not greater than the second preset ratio If the ratio is set, it is determined that the image acquisition condition does not meet the preset requirements; the image acquisition device is also used to acquire the real-time image; the display device is used to display the preset area and the face area.

[0224] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: if the proportion of the face area in the preset area is not greater than the second preset ratio, the second collection prompt information is output in real time to prompt The person to be authenticated is close to the image capture device.

[0225] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: real-time judgment of the relative position relationship between the face area and the preset area; and based on the relative position relationship between the face area

and the preset area The third collection prompt information is output in real time to prompt to change the relative position relationship between the person to be authenticated and the image collection device so that the face area is close to the preset area.

- [0226] Exemplarily, the step S108 executed by the computer program instructions when the processor is running includes: acquiring the posture information of the image acquisition device; and judging whether the image acquisition device is in the upright position according to the posture information, and if so, determining the image acquisition The conditions meet the preset requirements, otherwise it is determined that the image acquisition conditions do not meet the preset requirements.
- [0227] Exemplarily, when the computer program instructions are executed by the processor, they are also used to perform the following steps: count once during each execution of step S110 to step S120 to obtain the number of illumination verification times; and are used when the computer program instructions are executed by the processor. After step S120 is performed, the computer program instructions are also used to perform the following steps when the processor is running: if the result of the light living body verification indicates that the face of the person to be authenticated does not belong to a living body, output a second error message, and determine the number of light verification times Whether it reaches the second threshold, if the number of light verification reaches the second threshold, go to step S130, if the number of light verification does not reach the second threshold, return to step S108 or return to step S110, where the second error message is It prompts that the in-vivo verification for the person to be certified has failed.
- [0228] Exemplarily, the computer program instructions are executed by the processor before the steps S110 and S120 are executed when the computer program instructions are executed by the processor or during the steps S110 and S120 executed when the computer program instructions are executed by the processor. The time is also used to perform the following steps: output second prompt information, where the second prompt information is used to prompt the person to be authenticated to remain still within the second preset time.
- [0229] Exemplarily, the second prompt information is countdown information corresponding to the second preset time.
- [0230] Exemplarily, the detection light is obtained by dynamically changing the color and/or position of the light emitted to the face of the person to be authenticated.
- [0231] Exemplarily, the detection light is emitted by the display screen, and the detection light is obtained by dynamically changing the mode of the light emitted by the display screen by changing the

content displayed on the display screen to emit the detection light to the face of the person to be authenticated.

- [0232]** According to another aspect of the present invention, there is provided an identity authentication device, including: an information collection device for collecting initial information of a person to be authenticated; and a transmission device for sending the initial information to a server, and receiving from the server the server through the following Authentication information on whether the identity of the person to be authenticated is obtained by means of authentication: obtain the personal identification information of the person to be authenticated, where the personal identification information is initial information or obtained based on initial information; determine whether the personal identification information is authenticated information to obtain information Authentication result; the person to be authenticated undergoes a live test to obtain the live test result; and at least the identity of the person to be authenticated is determined based on the information authentication result and the live test result.
- [0233]** Exemplarily, the information collection device includes an image collection device and/or an input device.
- [0234]** According to another aspect of the present invention, an identity authentication device is provided, including a transmission device, a processor, and a memory. The transmission device is used to receive initial information of a person to be authenticated from a client, and to determine whether the identity of the person to be authenticated is legal The authentication information is sent to the client; computer program instructions are stored in the memory, and the computer program instructions are used when the processor is running to perform the following steps: obtain the personal identification information of the person to be authenticated, where the personal identification information is initial information or is based on initial Information acquisition; determine whether the personal identification information is authenticated information to obtain the information authentication result; perform a live test on the person to be authenticated to obtain the live test result; and at least determine whether the identity of the person to be authenticated is legal based on the information authentication result and the live test result.
- [0235]** Exemplarily, the personal identification information is one of the ID number, name, ID face, field collected face, ID number conversion value, name conversion value, ID face conversion value, and field collection face conversion value. Kind or more.
- [0236]** Exemplarily, before the computer program instructions are executed when the processor is running to obtain the personal identification information of the person to be authenticated, the computer program instructions are also used to perform the following steps when the processor is running: the output is used to instruct the person to be authenticated to provide Indication information

of a predetermined type of personnel information; the transmission device is also used to send the indication information to the client for output by the client; wherein the personal identification information is the personnel information provided by the person to be authenticated or is obtained based on the personnel information.

- [0237]** Exemplarily, the personal identification information is one or more of the conversion value of the certificate number, the conversion value of the name, the conversion value of the certificate face and the conversion value of the face collected on site, and the computer program instructions are used when the processor is running. The step of obtaining the personal identification information of the person to be authenticated includes: obtaining the initial information of the person to be authenticated, where the initial information is one or more of the credential number, name, credential face, and on-site collected face; and The initial information is transformed based on a predetermined algorithm to obtain the transformed value of the initial information as the personal identification information.
- [0238]** Exemplarily, the step of obtaining the personal identification information of the person to be authenticated executed when the computer program instructions are executed by the processor includes: obtaining a credential image of the person to be authenticated, wherein the information authentication result is the credential authentication result; the computer program instructions are When the processor is running, the steps of performing live detection on the person to be authenticated to obtain the live detection result include: obtaining a face image of the person to be authenticated; and performing live detection using the face image to obtain the live detection result.
- [0239]** Exemplarily, before the computer program instruction is used to execute the step of determining whether the identity of the person to be authenticated is legal based on the information authentication result and the living body test result when the computer program instruction is run by the processor, the computer program instruction is also used to execute the step when the processor is run The following steps: use the certificate image and/or the face image to perform additional judgment operations to obtain additional judgment results; the computer program instructions used to execute when the processor is running at least determine the identity of the person to be authenticated based on the information authentication result and the living body detection result The legality step includes: determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body test result, and the additional judgment result.
- [0240]** Exemplarily, the additional judgment operation includes a credential authenticity judgment operation and/or a face consistency judgment operation, and the additional judgment result includes a credential authenticity judgment result and/or a face consistency judgment result, and the credential authenticity judgment operation includes: Determine whether the certificate in the

certificate image is a real certificate to obtain the result of the authenticity of the certificate; the judgment operation of face consistency includes: obtaining the certificate face of the person to be authenticated according to the certificate image; and comparing the face of the person to be authenticated with the person The faces in the face image are compared to obtain the judgment result of face consistency.

- [0241] Exemplarily, the step of obtaining the credential face of the person to be authenticated according to the credential image that the computer program instructions are executed when the processor is running includes: detecting the face from the credential image to obtain the credential face of the person to be authenticated .
- [0242] Exemplarily, the step of obtaining the credential face of the person to be authenticated according to the credential image executed when the computer program instructions are executed by the processor includes: performing text recognition on the credential image to obtain the text information in the credential image; based on the credential image The text information in is searched for matching certificate information from the authenticated certificate information database; and the certificate face in the searched and matched certificate information is determined to be the certificate face of the person to be authenticated.
- [0243] Exemplarily, when the computer program instructions are executed by the processor to determine whether the document in the document image is a real document, the step of obtaining the authenticity judgment result of the document includes: extracting image characteristics of the document image; and inputting the image characteristics A well-trained document classifier is used to obtain the authenticity judgment result of the document; among them, the authenticity judgment result of the document is the confidence that the document in the document image is a real document.
- [0244] Exemplarily, when the computer program instructions are executed by the processor to determine whether the certificate in the certificate image is a real certificate, the step of obtaining the authenticity determination result of the certificate includes: identifying an image block containing the identification information of the certificate from the certificate image ; And the identification of the certificate identification information in the image block containing the certificate identification information to obtain the certificate authenticity judgment result; wherein the certificate authenticity judgment result is the confidence that the certificate in the certificate image is a real certificate.
- [0245] Exemplarily, during the execution of the step of determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result and the additional judgment result, the certificate authentication result and the living body

detection are executed when the computer program instructions are executed by the processor. Each of the result and the additional judgment result has its own weight coefficient.

- [0246]** Exemplarily, the step of obtaining the credential image of the person to be authenticated executed when the computer program instructions are executed by the processor includes: obtaining a pre-photographed image that is collected in real time for the credential of the person to be authenticated under the current shooting conditions; real-time evaluation The image attributes of the pre-shot image; when the evaluation value of the image attribute of the pre-shot image is less than the preset evaluation value threshold, the pre-shot prompt information is generated according to the image attribute of the pre-shot image, which is used to prompt the person to be authenticated to adjust the shooting of his certificate Condition; and when the evaluation value of the image attribute of the pre-photographed image is equal to or greater than the preset evaluation value threshold, the pre-photographed image is saved to use it as a credential image; the transmission device is also used to receive the pre-photographed image.
- [0247]** Exemplarily, the step of determining whether the personal identification information is authenticated information executed by the computer program instructions when the processor is running to obtain the information authentication result includes: performing text recognition on the document image to obtain the text information in the document image ; And based on the text information in the credential image to search in the certified credential information database to obtain the credential certification result; wherein the credential credential information in the credential image is the credibility of the certified credential information.
- [0248]** Exemplarily, the step of performing text recognition on the credential image to obtain text information in the credential image that the computer program instructions are executed when the processor is running includes: positioning the text in the credential image to obtain an image containing text Block; and to recognize the text in the image block containing the text to obtain the text information in the credential image.
- [0249]** Exemplarily, before the computer program instructions are used to perform the step of recognizing text in the image block containing text when the processor is run, the computer program instructions are also used to perform the following steps when the processor is run: The image block is corrected to a horizontal state.
- [0250]** Exemplarily, after the computer program instructions are executed when the processor is running, the computer program instructions are used when the processor is running, after the step of recognizing the text in the image block containing the text to obtain the text information in the credential image The step of performing text recognition on the credential image to obtain text information in the credential image further includes: outputting text information in the credential

image for the user to view; receiving text correction information input by the user; and correcting the text as indicated by the text correction information Compare the text to be corrected with the corresponding text in the text information in the document image; and if the difference between the text to be corrected indicated by the text correction information and the corresponding text in the text information in the document image is less than the preset difference threshold , Use the text correction information to update the text information in the credential image.

- [0251]** Exemplarily, when the computer program instruction is executed when the processor is running, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result includes: generating a living body action instruction, and the living body action instruction is used to instruct the person to be authenticated to execute the corresponding living body Action; obtain the face image of the person to be authenticated collected in real time; detect the face in the face image; and determine whether the face in the face image performs the living body action indicated by the living body action instruction to obtain the living body detection result.
- [0252]** Exemplarily, when the computer program instructions are executed by the processor, they are also used to perform live detection of a person to be authenticated to obtain a live detection result. The steps include: generating a live action instruction, and the live action instruction is used to instruct the person to be authenticated to execute the corresponding Live action; acquire the face image of the person to be authenticated collected in real time; capture the skin area image of the person to be authenticated before performing the live action and the skin area image after performing the live action from the collected face images; and transfer the person to be authenticated The skin area image before the execution of the living body action and the skin area image after the execution of the living body action are input to the skin elasticity classifier to obtain the living body detection result.
- [0253]** Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: obtaining the face image of the real person before performing the live action and the face image after performing the live action, and the face of the fake person before performing the live action Image and face image after performing the live action; extract the face image of the real person before performing the live action and the skin after performing the live action from the face image of the real person before the live action and the face image after the live action The area image is used as a positive sample image; the face image of the fake person before the execution of the live action and the face image of the fake person before the execution of the live action are extracted from the face image of the fake person before the execution of the live action and the skin area image after the execution of the live action is taken as the negative

Sample images; and use positive sample images and negative sample images to train a classifier model to obtain a skin elasticity classifier.

- [0254] Exemplarily, the step of capturing the skin area image of the person to be authenticated before performing the living body action and the skin area image after performing the living body action from the collected face image executed by the computer program instructions when the processor is running includes: From the collected face images, select the face image of the person to be authenticated before performing the live action and the face image after the execution of the live action; use the face detection model to perform the face image and the execution of the live action on the person to be authenticated Position the face in the face image after the action; use the face key point positioning model to perform the key points of the face in the face image before the person to be authenticated performs the live action and the face image after the live action Positioning; and according to the face position and key point position obtained by the positioning, the area of the face image in the face image before the person to be authenticated performs the live action and the face image after the person performs the live action is divided to obtain the execution of the person to be authenticated The skin area image before the living body action and the skin area image after the living body action is performed.
- [0255] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: obtaining a sample face image, where the position of the face and the position of the key points of the face in the sample face image are already marked; and Use sample face images for neural network training to obtain a face detection model and a face key point location model.
- [0256] Exemplarily, when the computer program instructions are executed by the processor to perform the live detection of the person to be authenticated to obtain the result of the live body detection, the steps include: obtaining the face of the person to be authenticated under the structured light by the binocular camera. The collected face image; and determining whether the face of the person to be authenticated belongs to a living body according to the face image, so as to obtain a living body detection result.
- [0257] Exemplarily, the step of performing live body detection of a person to be authenticated by the computer program instructions executed by the processor to obtain a live body detection result includes: Step S110: Obtain the face of the person to be authenticated under the illumination of the detection light One or more collected illumination images; Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the one or more illumination images, so as to obtain illumination living body verification Result; and step S130: determine whether the

certified person passes the in vivo verification based at least on the result of the in vivo illumination verification, so as to obtain the in vivo detection result.

- [0258] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed at least once.
- [0259] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light changes between every two consecutive moments.
- [0260] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is randomly changed or preset.
- [0261] Exemplarily, the step of performing the living body detection of the person to be authenticated to obtain the living body detection result by executing the computer program instructions when the processor is running includes: Step S140: output an action instruction, wherein the action instruction is used to instruct the authentication person to execute the corresponding Step S150: Obtain multiple action images collected for the face of the person to be authenticated; Step S160: Detect actions performed by the person to be authenticated based on the multiple action images; and Step S170: According to the action detection result and the action instruction It is determined whether the face of the person to be authenticated belongs to a living body to obtain the result of the action in vivo verification; the step S130 executed when the computer program instructions are executed by the processor includes: determining whether the person to be authenticated passes or not based on the illumination in vivo verification result and the action in vivo verification result Live body verification to obtain live body detection results; the transmission device is also used to send action instructions to the client and receive multiple action images from the client.
- [0262] Exemplarily, the step S170 executed by the computer program instructions when the processor is running includes: if it is detected based on a plurality of motion images collected within a period of time not greater than the first preset time that the computer program instructions are executed by the person to be authenticated, and If the action indicated by the action instruction is consistent with the action, it is determined that the face of the person to be authenticated belongs to a living body. If based on the multiple action images collected within the first preset time, the action instruction executed by the person to be authenticated is not detected If the indicated actions are consistent, it is determined that the face of the person to be authenticated does not belong to a living body.
- [0263] Exemplarily, the computer program instructions are also used to perform the following steps when being executed by the processor: each time step S140 to step S170 are executed to obtain the number of action verifications; after step S170, the computer program instructions are processed

When the device is running, it is also used to perform the following steps: if the action verification result indicates that the face of the person to be authenticated is not a living body, the first error message is output, and it is judged whether the number of action verifications reaches the first threshold, if the number of action verifications If the first number threshold is reached, then go to step S130. If the action verification number does not reach the first number threshold, then return to step S140 or return to step S110 if step S110 is executed before step S140, where the first error The information is used to prompt that the in-vivo verification for the person to be authenticated fails; the transmission device is also used to send the first error information to the client for output by the client.

- [0264]** Exemplarily, before the computer program instructions are used to execute step S110 when the processor is running, the computer program instructions are also used to perform the following steps when the processor is running: Step S108: Determine whether the image acquisition conditions of the person to be authenticated meet the predetermined Set requirements, if the image acquisition conditions meet the preset requirements, go to step S110, where the image acquisition conditions include the position of the person to be authenticated in the image acquisition area of the image acquisition device and/or the person to be authenticated and the image acquisition device The relative angle.
- [0265]** Exemplarily, before the computer program instructions are used to execute step S108 when the processor is running or at the same time as the computer program instructions are used to execute step S108 when the processor is running, the computer program instructions are also used when the processor is running Perform the following steps: Step S106: output first prompt information, where the first prompt information is used to prompt the person to be authenticated to face the image acquisition device and approach the image acquisition device; the transmission device is also used to send the first prompt information To the client to be output by the client.
- [0266]** Exemplarily, the step S106 used by the computer program instructions when the processor is running includes: outputting the first prompt information in one or more of a voice format, an image format, and a text format.
- [0267]** Exemplarily, the identity authentication device further includes a display device, wherein the step S108 performed when the computer program instructions are executed by the processor includes: acquiring real-time images collected for the face of the person to be authenticated; real-time output for calibration The preset area of the image acquisition condition and the face area in the real-time image are used for display; and according to the face area detected in the real-time image, it is judged whether the image acquisition condition meets the preset requirements, if the face area is within the preset area And the proportion of the face area in the real-time image is greater than the first preset proportion, it is determined that the image acquisition conditions meet the preset

requirements, if the face area is not located in the preset area or the proportion of the face area in the real-time image is not greater than the first preset ratio, it is determined that the image acquisition conditions do not meet the preset requirements; the transmission device is also used to receive real-time images from the client and send the preset area and the face area to the client to be output by the client for display .

- [0268] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: if the proportion of the face area in the real-time image is not greater than the first preset proportion, the first collection prompt information is output in real time to prompt the waiting The authentication person is close to the image acquisition device; the transmission device is also used to send the first acquisition prompt information to the client for output by the client.
- [0269] Exemplarily, the living body verification device further includes a display device, wherein the step S108 executed by the computer program instructions when the processor is running includes: acquiring a real-time image collected for the face of the person to be authenticated; real-time output for calibration The preset area of the image acquisition condition and the face area in the real-time image are used for display; and according to the face area to determine whether the image acquisition condition meets the preset requirements, if the face area is located in the preset area and the face area is in the preset area If the proportion of the area is greater than the second preset ratio, it is determined that the image acquisition conditions meet the preset requirements. If the face area is not located in the preset area or the proportion of the face area in the preset area is not greater than the second preset ratio If the ratio is set, it is determined that the image acquisition conditions do not meet the preset requirements; the transmission device is also used to receive real-time images from the client and send the preset area and the face area to the client to be output by the client for display.
- [0270] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: if the proportion of the face area in the preset area is not greater than the second preset ratio, the second collection prompt information is output in real time to prompt The person to be authenticated approaches the image acquisition device; the transmission device is also used to send the second acquisition prompt information to the client for output by the client.
- [0271] Exemplarily, the computer program instructions are also used to perform the following steps when the processor is running: judging the relative position relationship between the face area and the preset area in real time; and based on the relative position relationship between the face area and the preset area The third collection prompt information is output in real time to prompt to change the relative position relationship between the person to be authenticated and the image acquisition

device so that the face area is close to the preset area; the transmission device is also used to send the third collection prompt information to the client To be output by the client.

- [0272] Exemplarily, the step S108 executed by the computer program instructions when the processor is running includes: acquiring the posture information of the image acquisition device; and judging whether the image acquisition device is in the upright position according to the posture information, and if so, determining the image acquisition The conditions meet the preset requirements, otherwise it is determined that the image acquisition conditions do not meet the preset requirements.
- [0273] Exemplarily, when the computer program instructions are executed by the processor, they are also used to perform the following steps: count once during each execution of step S110 to step S120 to obtain the number of illumination verification times; and are used when the computer program instructions are executed by the processor. After step S120 is performed, the computer program instructions are also used to perform the following steps when the processor is running: if the result of the light living body verification indicates that the face of the person to be authenticated does not belong to a living body, output a second error message, and determine the number of light verification times Whether it reaches the second number threshold, if the number of light verification times reaches the second number threshold, go to step S130, if the number of light verification times does not reach the second number threshold, return to step S108 or return to step S110, where the second error message is It prompts that the in-vivo verification for the person to be authenticated fails; the transmission device is also used to send the second error information to the client for output by the client.
- [0274] Exemplarily, the computer program instructions are executed by the processor before the steps S110 and S120 are executed when the computer program instructions are executed by the processor or during the steps S110 and S120 executed when the computer program instructions are executed by the processor. The time is also used to perform the following steps: output second prompt information, where the second prompt information is used to prompt the person to be authenticated to stay still within the second preset time; the transmission device is also used to send the second prompt information to the client The terminal can be output by the client.
- [0275] Exemplarily, the second prompt information is countdown information corresponding to the second preset time.
- [0276] Exemplarily, the detection light is obtained by dynamically changing the color and/or position of the light emitted to the face of the person to be authenticated.

- [0277] Exemplarily, the detection light is emitted by the display screen, and the detection light is obtained by dynamically changing the mode of the light emitted by the display screen by changing the content displayed on the display screen to emit the detection light to the face of the person to be authenticated.
- [0278] According to the identity authentication method and device and storage medium of the embodiments of the present invention, the identity of the person to be authenticated is determined by combining the authentication information judgment and the living body detection to determine whether the identity of the person to be authenticated is legal. Therefore, the authentication result of the identity authentication method according to the embodiment of the present invention is more accurate, and the security of user authentication can be improved, thereby effectively protecting the rights and interests of users.
- [0279] Description of the drawings
- [0280] Through a more detailed description of the embodiments of the present invention in conjunction with the accompanying drawings, the above and other objectives, features and advantages of the present invention will become more apparent. The accompanying drawings are used to provide a further understanding of the embodiments of the present invention, and constitute a part of the specification. Together with the embodiments of the present invention, they are used to explain the present invention, and do not constitute a limitation to the present invention. In the drawings, the same reference numerals generally represent the same components or steps.
- [0281] Fig. 1 shows a schematic block diagram of an exemplary electronic device for implementing an identity authentication method and apparatus according to an embodiment of the present invention;
- [0282] Fig. 2 shows a schematic flowchart of an identity authentication method according to an embodiment of the present invention;
- [0283] Fig. 3 shows a schematic flowchart of an identity authentication method according to another embodiment of the present invention;
- [0284] FIG. 4 shows a schematic flowchart of an identity authentication method according to another embodiment of the present invention;
- [0285] Figure 5 shows a schematic flow chart of the training steps of a skin elasticity classifier according to an embodiment of the present invention;

- [0286] Fig. 6 shows a schematic block diagram of an exemplary electronic device for implementing an identity authentication method and apparatus according to another embodiment of the present invention;
- [0287] Figure 7 shows a schematic flowchart of a living body detection step according to an embodiment of the present invention;
- [0288] FIG. 8 shows a schematic flowchart of a living body detection step according to another embodiment of the present invention;
- [0289] Figure 9 shows an implementation process of a living body detection step according to an embodiment of the present invention;
- [0290] Fig. 10 shows a schematic block diagram of an identity authentication device according to an embodiment of the present invention; and
- [0291] Fig. 11 shows a schematic block diagram of an identity authentication system according to an embodiment of the present invention.
- [0292] detailed description
- [0293] In order to make the objectives, technical solutions, and advantages of the present invention more obvious, the exemplary embodiments according to the present invention will be described in detail below with reference to the accompanying drawings. Obviously, the described embodiments are only a part of the embodiments of the present invention, rather than all the embodiments of the present invention, and it should be understood that the present invention is not limited by the exemplary embodiments described herein. Based on the embodiments of the present invention described in the present invention, all other embodiments obtained by those skilled in the art without creative work should fall within the protection scope of the present invention.
- [0294] In order to solve the above-mentioned problems, embodiments of the present invention provide an identity authentication method and device, and a storage medium. The identity authentication method and device and storage medium combine personal identification information recognition and face recognition to perform identity authentication to determine whether the identity of the person to be authenticated is legal, that is, to determine whether the person to be authenticated has the authority to perform subsequent consumption payment and other operations. The identity authentication method and device according to the embodiments of the present invention can conveniently and securely authenticate the identity of the person to be authenticated, is a secure

interactive authentication method, and can be well applied to technical fields such as e-commerce, mobile payment, bank account opening and the like.

- [0295] First, referring to FIG. 1, an exemplary electronic device 100 for implementing the identity authentication method and apparatus and storage medium according to the embodiments of the present invention will be described.
- [0296] As shown in FIG. 1, the electronic device 100 includes one or more processors 102, one or more storage devices 104, an input device 106, an output device 108, and an image acquisition device 110. These components use a bus system 112 and/or other forms. The connecting mechanism (not shown) is interconnected. It should be noted that the components and structure of the electronic device 100 shown in FIG. 1 are only exemplary and not restrictive, and the electronic device may also have other components and structures as required.
- [0297] The processor 102 may be a central processing unit (CPU) or another form of processing unit with data processing capability and/or instruction execution capability, and may control other components in the electronic device 100 to perform desired functions.
- [0298] The storage device 104 may include one or more computer program products, and the computer program products may include various forms of computer-readable storage media, such as volatile memory and/or non-volatile memory. The volatile memory may include random access memory (RAM) and/or cache memory (cache), for example. The non-volatile memory may include, for example, read-only memory (ROM), hard disk, flash memory, and the like. One or more computer program instructions may be stored on the computer-readable storage medium, and the processor 102 may run the program instructions to implement the client functions in the embodiments of the present invention (implemented by the processor) described below. And/or other desired functions. Various application programs and various data, such as various data used and/or generated by the application program, can also be stored in the computer-readable storage medium.
- [0299] The input device 106 may be a device used by the user to input instructions, and may include one or more of a keyboard, a mouse, a microphone, and a touch screen.
- [0300] The output device 108 may output various information (for example, images and/or sounds) to the outside (for example, a user), and may include one or more of a display, a speaker, and the like.
- [0301] The image capture device 110 may capture credential images (including video frames) and/or face images (including video frames), and store the captured images in the storage device 104 for use by other components. The image capture device 110 may be a camera. It should be understood that the image capture device 110 is only an example, and the electronic device 100 may not

include the image capture device 110. In this case, other image capture devices may be used to capture the credential image and/or the face image, and send the captured image to the electronic device 100.

[0302] Exemplarily, the exemplary electronic device used to implement the identity authentication method and apparatus according to the embodiments of the present invention may be implemented on a device such as a personal computer or a remote server.

[0303] The embodiment of the present invention provides an identity authentication method, which includes: obtaining personal identification information of a person to be authenticated; judging whether the personal identification information is authenticated information to obtain an information authentication result; performing a living test on the person to be authenticated, To obtain a live body detection result; and determine whether the identity of the person to be authenticated is legal according to at least the information authentication result and the live body detection result.

[0304] Hereinafter, an identity authentication method according to an embodiment of the present invention will be described with reference to FIG. 2. FIG. 2 shows a schematic flowchart of an identity authentication method 200 according to an embodiment of the present invention. As shown in FIG. 2, the identity authentication method 200 includes the following steps.

[0305] In step S210, the personal identification information of the person to be authenticated is obtained.

[0306] The following mainly uses the personal identification information of the person to be authenticated as the certificate in the credential image as an example to describe the identity authentication method described in this article. In actual use scenarios, the personal identification information is not limited to this, and it can be, for example, a credential number One or more of, name, ID face (that is, the face image detected from the ID image) and on-site collection of the face (that is, the face image collected by the image acquisition device for the face of the person to be authenticated) , And/or the transformed value of the certificate number (such as the output value of a certain hash algorithm), the transformed value of the name (such as the output value of a certain hash algorithm), and the transformed value of the certificate face (such as a certain hash algorithm) One or more of the output value of) and the transformation value of the face collected on site (for example, the output value of a certain hash algorithm). The type of personal identification information can be selected according to specific usage scenarios, and the present invention does not specifically limit the selection of the type of personal identification information.

- [0307] Step S210 may include: obtaining a credential image of the person to be authenticated. The credential information in the credential image can be regarded as personal identification information.
- [0308] The documents described herein may include, but are not limited to, ID cards, driving licenses, passports, social security cards, etc.
- [0309] The credential image may be an image collected for the credential of the person to be authenticated. Exemplarily, the credential image may be an original image collected by an image collecting device such as a camera, or may be an image obtained after preprocessing the original image.
- [0310] The credential image can be sent to the electronic device 100 by a client device (for example, a mobile terminal including a camera, a remote video teller machine (VTM), etc.) to be processed by the processor 102 of the electronic device 100, or it can be processed by the electronic device 100. The included image capture device 110 (for example, a camera) captures and transmits to the processor 102 for processing.
- [0311] According to an embodiment of the present invention, acquiring a credential image of a person to be authenticated may include: acquiring a pre-shooting image that is collected in real time for the credential of the person to be authenticated under current shooting conditions; evaluating image attributes of the pre-shooting image in real time; When the evaluation value of the attribute is less than the preset evaluation value threshold, the pre-shooting prompt information is generated according to the image attributes of the pre-shooting image, which is used to prompt the person to be authenticated to adjust the shooting conditions of his certificate; and when the evaluation value of the image attribute of the pre-shooting image When it is equal to or greater than the preset evaluation value threshold, save the pre-photographed image to use it as a credential image.
- [0312] Pre-shooting refers to the process of starting the shooting mode of the image acquisition device (such as the camera of a mobile terminal such as a mobile phone, tablet computer) and placing the document to be photographed within the shooting range of the image acquisition device for shooting and framing (photo shooting has not been actually completed) .
- [0313] Optionally, the quality of the pre-shooting image obtained by shooting may be evaluated. For example, in the pre-shooting process, the image attributes of the pre-shooting image obtained by shooting under the current shooting conditions can be calculated in real time. Exemplarily, the shooting conditions may include, but are not limited to, one or more of the following: the placement position of the certificate, the placement angle of the certificate, the shooting position of the image capture device, the shooting angle of the image capture device, and so on. Exemplarily, the image

attributes may include, but are not limited to, one or more of the following: degree of document blur, document outline, key parts of the document, document occlusion, document size, document text clarity, etc. When the evaluation value of the image attribute is less than the preset evaluation value threshold, the captured pre-photographed image is considered to be unqualified. At this time, corresponding pre-shooting prompt information can be generated and output according to the image attributes to prompt the user to adjust the angle, position, etc. of the certificate or the image capture device until a qualified pre-shooting image is captured. Qualified pre-captured images are pre-captured images whose image attribute evaluation value is equal to or greater than a preset evaluation value threshold. When a qualified pre-shooting image is captured in the pre-shooting mode, the pre-shooting image can be saved as the certificate image obtained in step S210, and used in subsequent steps such as the certification of the certificate. Optionally, when some image attributes of the pre-photographed image are unqualified, the pre-photographed image can be adjusted to make the pre-photographed image qualified, and then the required document image can be obtained. For example, when the size of the certificate in the pre-photographed image is unqualified, operations such as cropping, zooming, etc. can be performed on the pre-photographed image, so that the size of the certificate in the pre-photographed image is qualified.

- [0314]** In step S220, it is determined whether the personal identification information is authenticated information, so as to obtain an information authentication result.
- [0315]** In the case where the personal identification information is the certificate information in the certificate image, the authenticated information may be the authenticated certificate information, and the information authentication result may be the certificate authentication result. That is, in the case where the personal identification information is the certificate information in the certificate image, step S220 may include: determining whether the certificate information in the certificate image is certified certificate information to obtain the certificate authentication result.
- [0316]** For example, some document information related to the document of the person to be authenticated can be identified from the document image, such as the ID number, name and other information on the ID card, and then it can be judged whether the document information is certified document information, that is, , To determine whether the certificate of the person to be certified is a certified certificate.
- [0317]** The certified certificate information about the certified certificate can be stored in a database, which is referred to as the certified certificate information database in this article. Search in the certified document information database based on the document information identified from the document image, that is, compare the document information identified from the document image with the

certified document information in the certified document information database to determine the document image Whether the certificate in is a certified certificate.

- [0318]** In an example, the authenticated certificate information database may be stored locally, for example, in a storage device (the storage device 104 shown in FIG. 1) of a device such as a server or a client for implementing the identity authentication method and device.
- [0319]** In another example, the certified certificate information database may be stored in a server of some public service system (for example, a public security system). Devices such as servers or clients used to implement the identity authentication method and device can communicate with the server of the public service system through a network connection, and search for credential information from the server of the public service system. For example, on the public security network, there is usually the certificate information (recording information) of a legal person who has been authenticated. You can search the public security network based on the certificate image obtained in step S210. If it can be found and identified from the certificate image If the certificate information matches the certificate information (recording information), it can be determined that the certificate in the certificate image is a certified certificate, that is, a legal certificate.
- [0320]** In another example, the personal identification information is an ID number. The person to be authenticated can enter his ID number into the identity authentication device. The identity authentication device can retrieve the authenticated person information database based on the received ID card number, and the authenticated person information database can store a large number of authenticated person ID numbers. If there is a matching ID number, it means that the person to be authenticated is an authenticated person, and the personal identification information is already authenticated information. Similarly, the above-mentioned authenticated person information database can optionally be stored in a local or a server of a public service system.
- [0321]** Those skilled in the art can understand that the implementation of personal identification information as name, ID face, face collected on site, ID number conversion value, name conversion value, ID face conversion value, or field collection face conversion value In the example, the implementation of the identity authentication method is similar to the implementation of the identity authentication method in the embodiment where the personal identification information is the ID number, and will not be repeated.
- [0322]** In step S230, the person to be authenticated is subjected to a living body detection to obtain a living body detection result.

- [0323] Step S230 may include: acquiring a face image of the person to be authenticated; and performing a living body detection using the face image to obtain a living body detection result.
- [0324] The face image may be an image collected for the face of the person to be authenticated. Exemplarily, the face image may be an original image collected by an image collecting device such as a camera, or may be an image obtained after preprocessing the original image.
- [0325] The face image can be sent by the client device (for example, a mobile terminal including a camera) to the electronic device 100 for processing by the processor 102 of the electronic device 100, or it can be collected by the image acquisition device 110 (for example, a camera) included in the electronic device 100 And transfer to the processor 102 for processing. Step S230 can be implemented by using any existing or possible living body detection method in the future, which is not limited in the present invention. Exemplarily, when the face in the face image is a real face, the person to be authenticated is considered to be a living body, and the live body detection result may be 1, and when the face in the face image is a fake face, the person to be authenticated is considered It is not a living body, and the result of living body test can be 0.
- [0326] In step S240, it is determined whether the identity of the person to be authenticated is legal according to at least the information authentication result and the biological test result.
- [0327] In an example, the information authentication result can be one of 1 and 0, where 1 indicates that the personal identification information of the person to be authenticated is authenticated information, and 0 indicates that the personal identification information of the person to be authenticated is not authenticated information. Similarly, the living body detection result can be one of 1 and 0, where 1 indicates that the person to be authenticated is a living body (that is, passed the living body verification), and 0 indicates that the person to be authenticated is not alive (that is, the person to be authenticated has not passed the living body verification). Exemplarily, if any one of the information authentication result and the living body test result is 0, it can be considered that the identity of the person to be authenticated is illegal, that is, the authentication of the person to be authenticated has failed. In this case, it can be forbidden to perform follow-up Business operations, such as prohibiting online transactions or bank account opening operations.
- [0328] In another example, the information authentication result can be any value in the range of [0,1], indicating the confidence that the personal identification information of the person to be authenticated is authenticated information. In this case, calculations such as weighted average can be performed on the information authentication result and the living body detection result, and whether the person to be authenticated is a legitimate person can be measured according to the

calculation result. Similar embodiments will be described in detail below, and will not be repeated here.

- [0329] It should be understood that the execution order of the steps of the identity authentication method 200 shown in FIG. 2 is only an example and not a limitation. For example, step S230 may be before step S210, between step S210 and step S220, or be combined with step S210 or S220. Simultaneous execution.
- [0330] According to the identity authentication method of the embodiment of the present invention, it is determined whether the identity of the person to be authenticated is legal in combination with the judgment of the authenticated information and the living body detection. Therefore, compared with the conventional method of identity authentication based solely on passwords or certificates, the implementation according to the present invention The authentication result of the identity authentication method of the example is more accurate, and the security of user authentication can be improved, thereby effectively protecting the rights and interests of users. The above method can be well applied to various fields involving identity authentication, such as e-commerce, mobile payment, or banking.
- [0331] Exemplarily, the identity authentication method according to the embodiment of the present invention may be implemented in a device, apparatus, or system having a memory and a processor.
- [0332] The identity authentication method according to the embodiment of the present invention can be deployed at the image collection terminal, for example, can be deployed at the image collection terminal of a financial system such as a bank management system or at a mobile terminal such as a smart phone or a tablet computer. Alternatively, the identity authentication method according to the embodiment of the present invention can also be deployed in a distributed manner on the server side (or the cloud) and the client side. For example, personal identification information can be collected on the client (for example, a credential image is collected or a credential number, name, etc. entered by the person to be authenticated) and/or a face image can be collected. The client can collect the collected personal identification information and/or collect The face image of is sent to the server (or the cloud), and the server (or the cloud) performs identity authentication.
- [0333] According to the embodiment of the present invention, before step S210, the identity authentication method 200 may further include: outputting instruction information for instructing the person to be authenticated to provide a predetermined type of personnel information; wherein the personal identification information is the person information provided by the person to be authenticated Or based on personnel information.

- [0334] Since the type of the authenticated information stored in the database used to store the authenticated information (such as the above-mentioned authenticated certificate information database or the authenticated person information database, etc.) has been determined, in order to successfully implement identity authentication, users (that is, waiting The authentication personnel) provide personal identification information whose type is consistent with the type of authenticated information stored in the database. To this end, instruction information can be output to instruct the person to be authenticated to input a predetermined type of person information. Exemplarily, the instruction information may be output in the form of text, image, or voice.
- [0335] For example, indication information such as "name:" and "ID number:" can be displayed on the display screen to instruct the person to be authenticated to enter the corresponding personnel information in the blank space behind the information type (name and ID number). Personnel information entered by the person to be authenticated can be used directly as personal identification information, or can be transformed to obtain personal identification information.
- [0336] For another example, a voice prompt such as "please show ID card" can be issued through a loudspeaker. After the authenticating person provides his ID card, the ID card can be imaged to obtain the ID card image. In this example, the personal identification information is an ID card image.
- [0337] Outputting instruction information for instructing the person to be authenticated to provide predetermined types of personal information is conducive to obtaining valuable personal identification information that can be compared with the authenticated information, thereby facilitating the smooth implementation of identity authentication. In addition, the output of the instruction information can improve the interactive experience between the user and the identity authentication device.
- [0338] From the foregoing, it can be seen that the personal identification information can be original information such as the ID number, name, ID face, and on-site collected faces, or it can be a transformed value obtained after transforming these original information. According to the embodiment of the present invention, in the case that the personal identification information is one or more of the conversion value of the certificate number, the conversion value of the name, the conversion value of the certificate face, and the conversion value of the face collected on site, step S210 may be Including: obtaining the initial information of the person to be authenticated, where the initial information is one or more of the credential number, name, credential face, and on-site collected face; and transforming the initial information based on a predetermined algorithm to obtain the initial information The converted value of is used as personal identification information.

- [0339] When storing the person information of the authenticated person in the database, a predetermined algorithm (for example, a certain hash algorithm) can be used to transform the person information of the authenticated person to obtain the authenticated information of the authenticated person. The conversion method is arbitrary, and it can be set as required. The transformation process can be understood as a coding process.
- [0340] The initial information of the person to be authenticated obtained by the identity authentication device is usually information that has not been transformed, such as the ID number, name, ID face, and on-site collection of the face. In order to facilitate subsequent comparison with the authenticated information, an algorithm consistent with the algorithm when the authenticated information is generated can be used to transform the initial information to obtain personal identification information.
- [0341] According to the embodiment of the present invention, when the personal identification information is the credential information in the credential image, before step S240, the identity authentication method 200 may further include: using the credential image and/or the face image to perform additional judgment operations to obtain Additional judgment results; Step S240 may include: determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body test result, and the additional judgment result.
- [0342] The additional judgment operation may include one or more operations for authenticity or consistency judgment on the credential in the credential image and/or the face in the face image. For example, the additional judging operation may include a credential authenticity judging operation and/or a face consistency judging operation. In addition to the authentication operation and the living detection operation, the addition of some other operations for the authenticity or consistency judgment of the certificate in the certificate image and/or the face in the face image is conducive to further improving identity authentication. The reliability of the result can improve the security of applications involving identity authentication.
- [0343] The following examples describe the implementation of the additional judgment operation.
- [0344] According to an embodiment, the additional judging operation may include a credential authenticity judging operation. The authenticity judgment operation of the certificate may include: judging whether the certificate in the certificate image is a real certificate, so as to obtain the authenticity judgment result of the certificate. According to another embodiment, the additional judgment operation may include a face consistency judgment operation. The face consistency judgment operation may include: obtaining the document face of the person to be authenticated according to the document image; and comparing the document face of the person to be authenticated with the

face in the face image to obtain the face consistency judgment result. The above two embodiments are described below with reference to FIG. 3 and FIG. 4 respectively.

[0345] FIG. 3 shows a schematic flowchart of an identity authentication method 300 according to another embodiment of the present invention. Figure 3 depicts an embodiment in which the personal identification information of the person to be authenticated is the credential information in the credential image. Steps S310, S330 to S350 of the identity authentication method 300 shown in FIG. 3 have been introduced in the above description of the steps S210 to S230 of the identity authentication method 200 shown in FIG. 2 and will not be repeated here. According to this embodiment, before step S360, the identity authentication method 300 may further include step S320. In step S320, the above-mentioned certificate authenticity judgment operation is performed. In step S360, it is determined whether the identity of the person to be authenticated is legal according to the certificate authentication result (ie, the information authentication result), the living body test result, and the authenticity judgment result of the certificate.

[0346] After the authentication operation of the certificate is performed, the result of the authenticity judgment of the certificate can be obtained. Exemplarily, the authenticity judgment result of the certificate may be one of 1 and 0, where 1 indicates that the certificate in the certificate image is a real certificate, and 0 indicates that the certificate in the certificate image is a false certificate. Of course, the authenticity judgment result of the document can also be any value in the range of [0,1], which indicates the confidence that the document in the document image is a real document. False documents may be, for example, documents obtained by re-photographing the screens on devices such as mobile phones or computers, or documents obtained by forging using computer graphics technology, etc.

[0347] Exemplarily, if any one of the certificate authentication result, the living body detection result, and the certificate authenticity judgment result is 0, it can be determined that the identity of the person to be authenticated is illegal, otherwise it can be determined that the identity of the person to be authenticated is legal.

[0348] Two exemplary implementations of step S320 are described below.

[0349] In an example, step S320 may include: extracting the image features of the document image; and inputting the image features into the trained document classifier to obtain the document authenticity judgment result; wherein the document authenticity judgment result is the document in the document image It is the confidence level of the real document.

[0350] Due to the interference effect of the wave between the photosensitive element of the image acquisition device and the display, the image obtained by retaking the ID photo on the computer

or mobile phone screen will have obvious periodic color stripes, called "moiré". The moiré pattern is an important clue to distinguish between real documents and remakes. Due to the periodicity of the moiré, the characteristics of the moiré in the frequency domain will be particularly obvious. In addition, the color of the moiré pattern is also different from the color of the real document. Therefore, it is possible to identify whether the document in the document image is a remake document based on the moiré pattern.

- [0351] Exemplarily, the image characteristics may include, but are not limited to, at least one of spectral characteristics, texture characteristics, and color characteristics.
- [0352] The credential classifier involved in the credential authenticity judgment operation may be pre-trained using a large number of sample credential images. Exemplarily, the "classifier" described herein may be any existing or possible future classifier based on machine learning, such as Support Vector Machine (SVM) and the like.
- [0353] Taking the ID card as an example, the training process of the ID classifier can include: collecting and labeling the ID card image containing the real ID card and the ID card image containing the reprinted ID card; respectively calculating the ID card image containing the real ID card and the reprinted identity The frequency spectrum information of the ID card image of the ID card is used as the respective image feature; the image feature of the ID card image containing the real ID card is used as the positive sample, and the image feature of the ID card image containing the remake of the ID card is used as the negative sample to train the classifier Model to obtain the ID classifier. Subsequently, in the actual identity authentication process, for the acquired ID card image, its spectrum information can be calculated as its image feature, and then the extracted image feature is input into the trained ID card classifier to determine the acquired ID card Whether the ID card in the image is a remake ID card.
- [0354] The authenticity judgment result of the certificate output by the certificate classifier may be the confidence that the certificate in the certificate image is a real certificate. The confidence level can be any value in the range of [0,1].
- [0355] In another example, step S320 may include: identifying the image block containing the identification information of the certificate from the image of the certificate; and identifying the identification information of the certificate in the image block containing the identification information of the certificate to obtain the authenticity judgment result of the certificate; wherein , The authenticity of the certificate is the degree of confidence that the certificate in the certificate image is a real certificate.

- [0356] The credential identification information can be any information that can identify a real credential. For example, the identification information of the certificate may include the national emblem pattern on the ID card or social security card, some special anti-counterfeiting signs, and so on. For example, real documents usually have relatively hidden anti-counterfeiting signs, and the authenticity of the documents can be determined by identifying the anti-counterfeiting signs.
- [0357] The above-mentioned two examples of document authenticity judgment based on image characteristics and document authenticity judgment based on document identification information can be implemented at the same time, that is, the authenticity of the document can be judged based on the image characteristics of the document image and its identification information at the same time. The skilled person can understand the implementation of this judgment method by reading the above description, and details are not described here.
- [0358] It should be understood that the execution order of the steps of the identity authentication method 300 shown in FIG. 3 is only an example and not a limitation. For example, step S320 can be executed at any time between step S310 and step S360.
- [0359] FIG. 4 shows a schematic flowchart of an identity authentication method 400 according to another embodiment of the present invention. Figure 4 depicts an embodiment in which the personal identification information of the person to be authenticated is the credential information in the credential image. The steps S410 to S440 of the identity authentication method 400 shown in FIG. 4 have been introduced in the above description of the steps S210 to S240 of the identity authentication method 200 shown in FIG. 2 and will not be repeated here. According to this embodiment, before step S470, the identity authentication method 400 may further include steps S450 and S460. In steps S450 and S460, the above-mentioned face consistency judgment operation is performed. In step S470, it is determined whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result, and the face consistency judgment result.
- [0360] In step S450, the credential face of the person to be authenticated is acquired according to the credential image.
- [0361] In an example, step S450 may include: detecting a face from the credential image to obtain the credential face of the person to be authenticated.
- [0362] The document usually includes a face (typically a face photo), which is referred to herein as a "document face" to distinguish it from the face in the face image. In the case that the document includes a human face, the human face can be detected from the document image. In this way, the face detected from the credential image can be directly used as the credential face of the person

to be authenticated for comparison with the face in the face image. In addition, it can be seen from the above that the face detected from the ID image can also be used as personal identification information.

- [0363] In another example, step S450 may include: performing text recognition on the credential image to obtain text information in the credential image; searching for matching credential information from a certified credential information database based on the text information in the credential image; and It is determined that the document face in the searched and matched document information is the document face of the person to be authenticated.
- [0364] Some documents may not include a human face. In this case, the document information in the certified document information database can be used to find the document face of the person to be authenticated. Of course, in the case that the document includes a face, this method can also be used to find the face of the person to be authenticated.
- [0365] For example, the ID image is the ID image of person X. From the ID image, text recognition can be performed to identify text information such as ID number, and then the ID card database of the public security system can be searched for a match based on the text information such as ID number. ID information. If the ID information of person X has been filed in the ID database, matching ID information can be searched. The ID card information may typically include basic information such as the ID number, name, gender, and face photo of the person X. The face photo is the required ID face. Subsequently, the face of the certificate can be compared with the face in the previously acquired face image.
- [0366] In step S460, the credential face of the person to be authenticated is compared with the face in the face image to obtain a face consistency judgment result.
- [0367] Compare the face (that is, the face in the face image) collected by the image acquisition device with the face of the ID. If the similarity between the two is greater than the preset similarity threshold, the collected face can be considered The face of the document belongs to the same person, otherwise, it can be considered that the two do not belong to the same person. Thus, the result of the consistency judgment of the face can be obtained. Exemplarily, the consistency judgment result may be the similarity between the face of the document and the face in the face image, that is, the consistency result may be any value in the range of [0,1]. Exemplarily, the consistency judgment result can also be one of 1 and 0, where 1 means that the face of the document and the face in the face image belong to the same person, and 0 means that the face of the document is different from the face in the face image. Belong to the same person.

- [0368] Exemplarily, if any one of the certificate authentication result, the living body detection result, and the face consistency judgment result is 0, it can be determined that the identity of the person to be authenticated is illegal; otherwise, it can be determined that the identity of the person to be authenticated is legal.
- [0369] It should be understood that, similar to FIG. 3, the execution order of the steps of the identity authentication method 400 shown in FIG. 4 is only an example and not a limitation. For example, step S450 may be before step S420, or after step S420 and before step S430. , Or after step S430 and before step S440, or simultaneously with step S420 or S430, and step S460 can be executed before, after, or simultaneously with step S440. Of course, step S450 can also be executed simultaneously with step S440.
- [0370] The embodiments of the credential authenticity judging operation and the face consistency judging operation are described above with reference to FIGS. 3 and 4 respectively. It is understandable that the additional judging operation may include the credential authenticity judging operation and the face consistency judging operation. That is to say, in the process of identity authentication, the authenticity of the document and the consistency of the face can be judged at the same time, and based on the four results of the certificate authentication result, the living body detection result, the authenticity judgment result of the document, and the face consistency judgment result. This result determines whether the identity of the person to be authenticated is legal.
- [0371] According to the embodiment of the present invention, in the process of determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result and the additional judgment result, each result of the certificate authentication result, the living body detection result and the additional judgment result has its own Weight coefficient.
- [0372] A weight coefficient may be assigned to each result of the certificate authentication result, the living body detection result, and the additional judgment result (including, for example, the above-mentioned certificate authenticity judgment result and/or face consistency judgment result). The size of the weight coefficient of each result can be determined according to needs, and the present invention does not limit this.
- [0373] Exemplarily, the certificate authentication result, the living body detection result, and the additional judgment result may be weighted and averaged based on the weight coefficient to obtain an averaged result. Then, the averaged result can be compared with a preset threshold. If it is greater than the threshold, the identity of the person to be authenticated can be considered legal, otherwise the identity of the person to be authenticated is considered illegal.

- [0374] When the personal identification information of the person to be authenticated is the credential information in the credential image, the credential authentication result, credential authenticity judgment result, and face consistency judgment result can all be any value in the range of $[0,1]$. Of course, the value of these three can also be only one of 0 and 1. The value of the living body test result is one of 0 and 1. Therefore, these results can be weighted or arithmetic averaged to obtain an averaged result. Then, compare the averaged result with the threshold.
- [0375] Of course, it is also possible to simply sum up various detection or judgment results involved in identity authentication to obtain a total result, and compare the total result with a threshold to determine whether the identity of the person to be authenticated is legal.
- [0376] Each result has its own weight coefficient, which is convenient for the identity authentication system to adjust its participation in the identity authentication process according to the importance of each judgment or detection operation involved in identity authentication, thereby improving the accuracy of identity authentication.
- [0377] According to an embodiment of the present invention, step S220 (S330, S420) may include: performing text recognition on the credential image to obtain text information in the credential image; and searching in a certified credential information database based on the text information in the credential image, To obtain the certificate authentication result; where the certificate authentication result is the credibility that the certificate information in the certificate image is the certified certificate information.
- [0378] As described above, the text information in the credential image can be recognized from the credential image, and then matching credential information can be searched from the certified credential information database based on the recognized text information. The search result is typically the probability (also referred to as the confidence level) that matching certificate information exists in the authenticated certificate information database. In this case, the certificate authentication result may be a search result.
- [0379] According to the embodiment of the present invention, performing text recognition on a credential image to obtain text information in the credential image may include: positioning text in the credential image to obtain an image block containing text; and recognizing text in the image block containing text The text is recognized to obtain the text information in the image of the certificate.
- [0380] The step of performing text recognition on the document image can be implemented by any suitable text recognition method. An embodiment of the character recognition step is described below.

- [0381] First, you can locate the text in the credential image to determine the location of the text. Then the image block containing the text can be extracted from the document image. Exemplarily, the credential image can be input to a trained neural network to locate the text in the credential image.
- [0382] For example, a large number of sample ID images can be collected, and the location of the text on the sample ID image can be indicated by manual or machine labeling. A neural network for locating the position of the text is trained through a machine learning algorithm based on a large number of labeled images of the document. The ID image obtained in the actual identity authentication process is input into the trained neural network, and the neural network can output the position of the text in the ID image, for example, the coordinates of the vertex of the area where the text is located.
- [0383] Subsequently, the text in the image block (text area) containing the text can be recognized to obtain the text recognition result. Recognizing the text in the image block containing text refers to the process of converting the image content of the image block containing text into a character string. Exemplarily, a traditional optical character recognition (OCR) method can be used for recognition: first, each character is segmented by a binarization operation, and then all characters are recognized by means of template matching or pattern classification.
- [0384] Optionally, a sliding window recognition method can also be used to locate and recognize characters from the document image without relying on the result of binarization segmentation.
- [0385] According to the embodiment of the present invention, before recognizing the characters in the image block containing the text, the identity authentication method 200 (300, 400) may further include: correcting the image block containing the text to a horizontal state.
- [0386] Between the steps of positioning the text in the credential image and recognizing the text in the image block containing the text, a step of adjusting (correcting) the position of the text may also be included. In practical applications, the credential in the credential image may have a certain oblique angle, that is, the image block containing text may have a certain oblique angle. Therefore, before recognizing the text in the credential image, the image block containing the text (that is, the area where the text is located) can be corrected to convert it to a horizontal and flush state. Exemplarily, in the step of locating the text in the credential image, the coordinates of the four vertices in the area where the text in the credential image is located have been obtained, and the image area contained by the four vertices has been obtained, that is, the area containing the text. The image block, therefore, only need to rotate it to the horizontal state according to the coordinates of the image block containing the text.

- [0387] Correcting the image block containing the text to a horizontal state can facilitate subsequent recognition of the text in the image block containing the text.
- [0388] According to the embodiment of the present invention, after recognizing the text in the image block containing text to obtain text information in the credential image, performing text recognition on the credential image to obtain text information in the credential image further includes: outputting the credential image The text information in the document for the user to view; receive the text correction information input by the user; compare the text to be corrected indicated by the text correction information with the corresponding text in the text information in the document image; and if the text correction information indicates If the difference between the text to be corrected and the corresponding text in the text information in the certificate image is less than the preset difference threshold, the text correction information is used to update the text information in the certificate image.
- [0389] In the text recognition process, you can also add text correction steps. When the OCR method is used to recognize characters, there will be recognition errors or recognition failures for some rare characters, characters with similar shapes, and characters with many strokes. Therefore, allowing users to modify text can flexibly and conveniently solve the problems of text recognition and improve the accuracy of text recognition. The user mentioned here may be the person to be authenticated as described in this article, or other persons other than the person to be authenticated, such as the administrator of the identity authentication system.
- [0390] Exemplarily, the text information recognized from the credential image may be output in the form of text display or voice playback for the user to view. When the user finds that there is an error in the text recognition result, he can input text correction information. After receiving the text correction information input by the user, the text to be corrected indicated by the text correction information can be compared with the recognized corresponding text. If the difference between the two is less than the preset difference threshold, the text correction information can be used to update the text information in the credential image; otherwise, the text information may not be updated. For example, if a text on the certificate is recognized as "one", the text correction information entered by the user indicates that the text should be revised to "one". Because of the large difference between the two, the identity authentication system can reject the user's correction request. The text will not be corrected.
- [0391] According to an embodiment of the present invention, step S230 (S340 and S350, or S430 and S440) may include: generating a living body motion instruction, which is used to instruct the person to be authenticated to perform a corresponding living body action; and obtaining real-time collected person of the person to be authenticated Face image; detecting the face in the face image; and

determining whether the face in the face image performs the living body action indicated by the living body action instruction, so as to obtain the living body detection result.

- [0392] In an example, a client device (for example, a mobile terminal including a camera, a VTM, etc.) may generate and output a living body motion instruction, collect a face image (including a static image or a video), and use the face image for living body detection. In another example, the cloud server may generate the living body motion instruction, and the client device (for example, a mobile terminal including a camera, VTM, etc.) may output the living body motion instruction and collect a face image, and then the client device will collect the face image Upload to the cloud server, the cloud server detects the face in the face image, and judges the authenticity of the face. If it is determined that the human face is a real human face, the person to be authenticated can be considered a living body, otherwise, the person to be authenticated can be considered not a living body. Exemplarily, the cloud server may include a trained face authenticity classifier and a fake face type classifier. The face authenticity classifier can be used to judge the authenticity of the face, and the fake face type classifier can be used to judge the type of the fake face in the case that the face is a fake face.
- [0393] The following describes the implementation of the embodiment of living body detection based on the movement of the human face in the human face image by way of example, so as to facilitate the understanding of this embodiment.
- [0394] The living body action instruction can instruct the person to be authenticated to make the corresponding living body action according to the instruction. The living body action indicated by the living body action instruction can be a single static action (equivalent to a pose) or a changing action. Exemplarily, the living body action instruction may be generated and output before the face image is collected, and no change occurs during the process of collecting the face image. Exemplarily, the living body action instruction may also be a continuous sequence of instructions, that is, different instructions are continuously generated and output in the process of collecting the face image, and the person to be authenticated is instructed to follow the instruction to change the living body action made by him.
- [0395] The above-mentioned living body action can be, for example, pressing the skin of the two cheeks with a finger, swallowing air in the mouth to bulge the cheeks, or reading a paragraph of text aloud. When the person to be authenticated performs one or more living body actions, the face image can be collected to determine whether the living body action performed by the person is qualified. If it is, the living body detection is successful; otherwise, the living body detection fails. For example, if the living body action indicated by the living body action instruction is to read a paragraph of text aloud, the face image during the reading process can be collected to determine whether the lip

movement of the face in the face image matches the lip movement of the corresponding text. If it matches, The live body test is successful.

- [0396] According to an embodiment of the present invention, step S230 (S340 and S350, or S430 and S440) may include: generating a living body motion instruction, which is used to instruct the person to be authenticated to perform a corresponding living body action; and obtaining real-time collected person of the person to be authenticated Face image; capture the skin area image of the person to be authenticated before performing the living action and the image of the skin area after performing the living action from the collected face image; and the skin area image of the person to be authenticated before performing the living action and the execution of the living action The resulting skin area image is input into the skin elasticity classifier to obtain the result of living body detection.
- [0397] When the person to be authenticated performs one or more living actions, the skin area images of the person to be authenticated before and after the living action can be captured from the collected face images. Whether the person to be authenticated starts to perform a living body movement can be judged based on the collected face image. For example, it is possible to monitor the face in the face image, and determine the time when the person to be authenticated starts to perform a living action according to the state change of the face. The face image collected before the start time is used as the face image before the person to be authenticated performs the living action. Subsequently, the face in the face image can be continuously monitored for a period of time, and the duration is the estimated duration of the living body action. The face image collected after the end of the duration is used as the face image after the person to be authenticated performs a living action. The skin area images of the person to be authenticated before and after the person to be authenticated can be extracted from the face images of the person before and after the person to be authenticated.
- [0398] Subsequently, the obtained skin region image can be input to a skin elasticity classifier, which is a pre-trained classification model. For example, if the collected face skin is a living skin, the skin elasticity classifier can output 1, otherwise it outputs 0.
- [0399] According to the embodiment of the present invention, capturing the skin area image of the person to be authenticated before performing the living action and the skin area image after performing the living action from the collected face images includes: selecting the person to be authenticated from the collected face images for execution The face image before the live action and the face image after the live action; the face detection model is used to locate the face in the face image before the person to be authenticated performs the live action and the face image after the live action ; Use the face key point positioning model to locate the key points of the face in the face image before the person to be authenticated performs the live action and the face image after the

live action; and the face position and key obtained according to the positioning The point position divides the face image of the person to be authenticated before performing the live action and the face in the face image after performing the live action to obtain the skin area image of the person to be authenticated before performing the live action and after performing the live action Image of the skin area.

- [0400]** The extraction of the skin area image can be realized based on the existing face detection and face key point location algorithms. For example, the face image of the person to be authenticated before performing the live action can be input into the trained face detection model and the face key point positioning model respectively to obtain the face position (such as the coordinates of the face contour point) and the key point position respectively (The coordinates of each key point). The key point can be any point on the face, such as the corner of the left eye, the tip of the nose, and the corner of the left lip. Of course, the key point can also be a face contour point. Exemplarily, the face area can be divided into a series of triangular fragments according to the position of the face and the position of key points, and the triangular fragments image blocks located in the chin, cheekbones, and cheeks can be taken as the face skin area. Obtain an image of the skin area. The method for extracting the image of the skin area after the person to be authenticated performs a living action is similar to the above method, and will not be repeated. For each face image, the area division method, the number of selected areas as the face skin area and the parts included in each area can be set as needed, and the present invention does not limit this.
- [0401]** In one example, the face detection model and the face key point location model can be implemented by using a deep neural network. A deep neural network is a network that can learn independently. The deep neural network can accurately and efficiently detect and locate the face and key points in the face image.
- [0402]** According to the embodiment of the present invention, the identity authentication method 200 (300, 400) may further include: obtaining a sample face image, where the position of the face and the position of the key points of the face in the sample face image are already marked; and Use sample face images for neural network training to obtain a face detection model and a face key point location model.
- [0403]** A large number of (such as about 10,000) sample face images can be collected in advance, and a series of key points such as the corners of the eyes, the corners of the mouth, the wings of the nose, the highest point of the cheekbones, etc., and the positions of the people can be manually marked in each sample face image. The position of the face contour point. Subsequently, machine learning algorithms (such as deep learning, or regression algorithms based on local features, etc.) can be used to perform neural network training with the aforementioned labeled sample face

images as input, so as to obtain the required face detection model and face key Point positioning model.

- [0404] According to the embodiment of the present invention, the identity authentication method 200 (300, 400) may further include a training step of a skin elasticity classifier. The following describes a schematic flowchart of the training step S500 of the skin elasticity classifier according to an embodiment of the present invention with reference to FIG. 5.
- [0405] As shown in FIG. 5, the training step S500 of the skin elasticity classifier includes the following steps.
- [0406] In step S510, the face image of the real person before performing the living action and the face image after the execution of the living action, as well as the face image of the fake person before the execution of the living action and the face image after the execution of the living action are acquired.
- [0407] In step S520, the face image of the real person before the execution of the living action and the skin area image after the execution of the living action are extracted from the face image of the real person before the execution of the living action and the face image after the execution of the living action are taken as positive sample images.
- [0408] In step S530, the face image of the fake person before the execution of the living action and the skin area image after the execution of the living action are extracted from the face image of the fake person before the execution of the living action and the face image after the execution of the living action as negative sample images.
- [0409] In step S540, the classifier model is trained using the positive sample image and the negative sample image to obtain a skin elasticity classifier.
- [0410] Exemplarily, the training of the skin elasticity classifier can be performed offline. It is possible to collect in advance the face images of real persons before and after performing the specified living actions, and at the same time, it can also collect the face images of the fake persons before and after performing the specified living actions. The fake person may be, for example, a photo containing a human face, a video containing a human face, a paper mask, or a three-dimensional (3D) model of a human face.
- [0411] The method of extracting the skin area image from the face image of the real person or the fake person can refer to the above description of the method of extracting the skin area image in the actual identity authentication process, which will not be repeated here. Exemplarily, after the

positive sample image and the negative sample image are obtained, a statistical learning method such as deep learning or SVM can be used to train the classifier model to obtain a skin elasticity classifier.

- [0412] According to the embodiment of the present invention, step S230 (S340 and S350, or S430 and S440) may include: acquiring a face image collected by a binocular camera on the face of the person to be authenticated under structured light irradiation; according to the face image Determine whether the face of the person to be authenticated is a living body to obtain the result of the living body detection.
- [0413] The above-mentioned methods of performing live body detection based on the actions of the human face or using the skin elasticity classifier are mainly aimed at some application scenarios with weaker safety requirements. For some application scenarios with higher security requirements, you can choose the way of live detection based on special hardware. For example, a binocular camera may be used to collect face images illuminated by structured light, so as to use the collected face information and structured light illumination information for living body detection.
- [0414] In one example, the detection parameter indicating the degree of subsurface scattering of the structured light on the face of the person to be authenticated can be determined based on the collected face image illuminated by the structured light, and then can be based on the detection parameter and the predetermined parameter Threshold, to determine whether the face of the person to be authenticated is a living body. Since 3D masks and other fake faces have different degrees of subsurface scattering from real faces (the stronger the subsurface scattering, the smaller the image gradient and the smaller the degree of diffusion), for example, those made of general paper or plastic The degree of subsurface scattering of the mask is much weaker than that of the real face, while the degree of subsurface scattering of the mask made of general silica gel is much stronger than that of the real face. Therefore, the false face can be distinguished by judging the degree of image diffusion. With real human face, it can effectively defend against mask attackers.
- [0415] In another example, the depth information of the face of the person to be authenticated can be obtained according to the face image. In addition, the spot pattern formed by the face of the person to be authenticated under the irradiation of structured light can also be obtained. The texture information of the face of the person to be authenticated can be obtained according to the light spot pattern. Subsequently, the depth information and texture information can be combined to determine whether the face of the person to be authenticated belongs to a living body.
- [0416] Different material structures can form different spot patterns under structured light. According to the light spot pattern, the texture information of the human face can be obtained, that is, the

material properties of the human face surface. If it is found that the texture information of the face of the person to be authenticated does not conform to the distribution law of human skin texture, it is determined that the face of the person to be authenticated does not belong to a living body, and it is judged as a mask attack. Since the attacker can use a mask made of imitation human skin to carry out the attack, even if the texture information of the person to be authenticated conforms to the distribution law of human skin texture, it may not be possible to determine that the face of the person to be authenticated is a living body, so it can be combined The depth information judges whether the person's face is a living body. The depth information of the face of the person to be authenticated can be obtained from the face images collected in two different perspectives. It should be understood that a real human face usually has ups and downs. For example, the coordinate depths of the eyes and nose are different, and the difference is large. However, a mask made of imitation human leather has very little ups and downs. The coordinate depth gap is very small. Therefore, combining the depth information can further determine whether the face of the person to be authenticated belongs to a living body.

- [0417] Therefore, in the embodiment of the present invention, a binocular camera and structured light can be combined, and a 3D face with a structured light pattern can be collected by the binocular camera, and then according to the degree of subsurface scattering of the structured light on the 3D face, or combined with the human The depth information of the face and the light spot pattern formed by the structured light on the face are used for living body detection.
- [0418] According to the embodiment of the present invention, the living body detection process in the identity authentication method may also have other implementation manners. Another way of living body detection is described below. In this living body detection method, a real human face is distinguished from a fake human face on a photo or screen based on the characteristics of light reflection.
- [0419] The following describes an exemplary implementation scheme of the living body detection step in the identity authentication method and the living body detection module in the identity authentication device. It should be noted that "in vivo verification" and "in vivo detection" mentioned in this article have the same meaning and can be used interchangeably.
- [0420] FIG. 6 shows an example electronic device 600 for implementing the identity authentication method and apparatus according to another embodiment of the present invention. The electronic device 600 includes one or more processors 602, one or more storage devices 604, an input device 606, an output device 608, an image acquisition device 610, and a light source 612. These components are connected through a bus system 614 and/or other forms of connection mechanisms. (Not shown) interconnected. The processor 602, storage device 604, input device 606, output device 608, image acquisition device 110, and bus system 614 in the electronic

device 600 shown in FIG. 6 are the same as the processor 102, storage device 104, and input device shown in FIG. 106. The structure and working principle of the output device 108, the image acquisition device 110 and the bus system 112 are similar, and will not be repeated.

- [0421] The light source 612 may be a device capable of emitting light, which may include a dedicated light source such as a light emitting diode, or may include an unconventional light source such as a display screen. In the case that the identity authentication method and device are implemented in a mobile terminal such as a smart phone, the input device 606, the output device 608, and the light source 612 may be the same display screen.
- [0422] Fig. 7 shows a schematic flowchart of the living body detection step S700 (corresponding to the above-mentioned steps S220, or S340 and S350, or S430 and S440) according to an embodiment of the present invention. As shown in FIG. 7, the living body detection step S700 includes the following steps.
- [0423] In step S710, one or more illumination images collected for the face of the person to be authenticated under the illumination of the detection light are acquired.
- [0424] Exemplarily, a light source can be used to emit detection light to the face of the person to be authenticated. The light source can be controlled by the processor to emit light. Exemplarily, the light source may share other light-emitting devices (for example, at least a part of the display screen, the light source in the projector) as the light source. For another example, the light source may also be a dedicated light source (for example, one or more light-emitting diodes or laser diodes arranged in a certain manner, such as a flashlight for a camera, etc.), or a combination of a display screen and other types of light sources Wait.
- [0425] The mode of the detection light may include, but is not limited to, the color of the detection light, the position of the light-emitting area, the intensity of the detection light, the irradiation angle of the detection light, the wavelength of the detection light, the frequency of the detection light, and the like.
- [0426] Exemplarily, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light may not change, that is, the light source may use a single constant light to illuminate the face of the person to be authenticated. For example, in a preferred embodiment, the light source used is a display screen of a mobile terminal. On the display screen, the color and brightness of each pixel can be controlled, so that the display screen can emit light that presents a specific pattern, such as structured light. At this time, the specific color or brightness of the screen in a specific pixel area can be based on a large number of experiments. After data optimization, a specific detection light mode is selected. In this detection light mode, a specific algorithm

corresponding to the detection light mode can quickly and accurately perform in vivo verification of the object to be verified. In this case, one or more illumination images can be collected under the illumination of the constant detection light, and the living body verification can be performed based on the illumination image.

- [0427] It is preferable that in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed at least once. In this case, the mode change frequency of the detection light and the acquisition frequency of the image can be coordinated and controlled, so that at least one illumination image can be collected under the detection light of each mode.
- [0428] It is more preferable that the mode of detecting light changes between every two consecutive moments. The time can be any specific time point in the predetermined time period. For example, the mode of detecting light can be changed every 1 second. Constantly changing the mode of detecting light can obtain more abundant light reflection characteristic information, which is conducive to more accurate and efficient implementation of living body verification based on light reflection characteristics.
- [0429] Optionally, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is randomly changed or preset. In one example, the pattern of detecting light is changed completely randomly. For example, in a preferred embodiment, the light source used is a display screen of a mobile terminal. On the display screen, the color of each area can be controlled. For each area, a certain RGB value is randomly selected as the color value of the area for display within a predetermined RGB value range each time. The division of regions can be set arbitrarily. For example, each region can include one or more pixels, and the sizes of two different regions can be the same or different.
- [0430] In another example, the mode of detecting light may be preset. For example, the detection light can be set to continuously illuminate for 10 seconds, and one mode is changed every second, and the color, position, and intensity of the detection light emitted each time are preset. In the living body verification process, the light source can sequentially emit 10 patterns of detection light in a preset manner. The preset detection light mode can be a mode that is more effective for in vivo verification based on previous experience, which is beneficial to improve the accuracy and efficiency of in vivo verification.
- [0431] For example, the mode of the detection light irradiated to the face of the person to be authenticated can be dynamically changed by dynamically changing the luminous color of the detection light. For another example, the mode of the detection light irradiated to the face of the person to be authenticated can also be dynamically changed by dynamically changing the position of the light-emitting area of the detection light (that is, changing the position of the detection light). For another

example, it is also possible to dynamically change the mode of the detection light irradiated to the face of the person to be authenticated by dynamically simultaneously changing the light-emitting color of the detection light and the position of the light-emitting area of the detection light.

- [0432]** For example, the position of the light-emitting area of the detection light can be dynamically changed by changing the position of the light source, which can change the position where the detection light irradiates the face of the person to be authenticated. For another example, it is also possible to dynamically change the position where the detection light irradiates the face of the person to be authenticated by changing the angle of the light emitted from the light source.
- [0433]** In a preferred embodiment, the light source used is a display screen of a mobile terminal, and the image acquisition device is a camera of the mobile terminal on the same side of the display screen (for example, a front camera). Compared with the solution that uses an additional dedicated light source, this solution can be implemented using existing mobile terminals such as mobile phones, and is not restricted by external conditions. The living body detection solution can be better applied to application scenarios such as remote account opening through personal mobile terminals .
- [0434]** Further, in a further preferred solution of the above-mentioned preferred embodiment, the light mode used is a combination mode of the color of light and the position of the light-emitting area, for example: different colors of light are emitted at different positions of the display screen at the same time, or , The same color of light is emitted at different positions of the display screen at the same time, but the color of the light emitted at different times is different, etc. The solution that uses the combination mode of the color of light and the position of the light-emitting area, compared with the solution of changing the light intensity and other light modes, not only has a better live detection effect, but also reduces the stimulation of light to the human eye, thereby improving the user experience .
- [0435]** In the case where the face of the person to be authenticated is illuminated by the detection light, an image acquisition device (for example, the image acquisition device 610 of the electronic device 600) can be used to collect the image of the face of the person to be authenticated under the detection light to obtain all述illumination image. The image capture device can be controlled by the processor to capture images. The image acquisition device transmits one or more illuminated images to the processor of the identity authentication system for in vivo verification. Exemplarily, the number of illumination images collected under the illumination of the same mode of light may be one or more, which is not limited in the present invention. Those skilled in the art can understand that the in vivo verification is mainly based on the verification of the human face. Therefore, according to the embodiment of this article, the purpose is to collect the image

containing the human face when the illumination image and the subsequent action image and real-time image are collected. , Used for in vivo verification.

- [0436] Exemplarily, the illuminated image can be sent to the electronic device 600 by the client device (such as a mobile terminal including a camera, a remote video teller machine (VTM), etc.) to be processed by the processor 602 of the electronic device 600, or The image capture device 610 (for example, a camera) included in the electronic device 600 captures and transmits it to the processor 602 for processing.
- [0437] In step S720, it is determined whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in one or more illumination images, so as to obtain the result of the illumination living body verification.
- [0438] Human skin, such as human face skin, is a diffusely reflective material, and human faces are three-dimensional; in contrast, display screens such as liquid crystal displays (LCD) or organic light emitting diode (OLED) displays can be considered self-luminous Objects also usually include partial specular reflection components, while photos, etc. are usually flat and usually include partial specular reflection components, and whether it is a display screen or a photo, its overall reflection characteristics are uniform and lack the three-dimensional characteristics of a human face. . The light reflection characteristics of the human face are different from the light reflection characteristics of the display screen or the photo. Therefore, it is possible to determine whether the face of the person to be authenticated is a living body based on the light reflection characteristics of the face of the person to be authenticated.
- [0439] In step S730, it is determined whether the person to be authenticated has passed the in vivo verification based on at least the result of the in vivo illumination verification, so as to obtain the in vivo detection result.
- [0440] In one example, the final living body detection result can be determined directly based on the result of the live body illumination verification, that is, if the result of the living body illumination verification indicates that the face of the person to be authenticated belongs to a living body, it is determined that the person to be authenticated has passed the living body verification, and the live body detection result can be exemplified Sexually is 1; if the result of the light living body verification indicates that the face of the person to be authenticated does not belong to a living body, it is determined that the person to be authenticated has not passed the living body verification, and the live body detection result can be exemplarily 0. This method of living body detection requires less calculation and higher efficiency. In another example, other in vivo verification methods can be combined to consider the result of the illumination in vivo verification

and other in vivo verification results obtained based on other in vivo verification methods to finally determine whether the person to be authenticated has passed the in vivo verification. This method of living body detection has a higher accuracy rate.

- [0441] As mentioned above, since the light reflection characteristics of a human face are different from the light reflection characteristics of objects such as display screens or photos, the real human face can be effectively distinguished from the human face played back on the screen or the human face on the photo based on the light reflection characteristic. Therefore, the identity authentication method and device using the living body detection method according to the embodiments of the present invention can effectively defend against screen or photo attackers, thereby improving the security and user experience of the identity authentication system.
- [0442] Exemplarily, the identity authentication method using the living body detection method according to the embodiment of the present invention may be implemented in a device, device, or system having a memory and a processor.
- [0443] The identity authentication method according to the embodiment of the present invention can be deployed at the image collection terminal, for example, can be deployed at the image collection terminal of a financial system such as a bank management system or at a mobile terminal such as a smart phone or a tablet computer. Alternatively, the identity authentication method according to the embodiment of the present invention can also be deployed in a distributed manner on the server side (or the cloud) and the client side. For example, the client can obtain personal identification information, emit light, and collect an image of the face of the person to be authenticated. The client transmits the obtained personal identification information and the collected image to the server (or cloud), and the server (Or cloud) to perform authentication information judgment and living body detection to obtain the identity authentication result, and return the identity authentication result to the client. Compared with the client, the server has greater data computing capabilities. By handing the identity authentication to the server, it can increase the authentication speed and improve the user experience. In addition, because the server has greater computing speed, more complex identity authentication can be used Algorithm, by performing identity authentication on the server side, the accuracy of identity authentication can be improved.
- [0444] Although live verification based on light reflection characteristics can defend against screen or photo attacks, there may be multiple attack methods for attackers. Some other attack methods may break through live verification based on light reflection characteristics, such as a three-dimensional simulation mask attack. In the case of a mask attack, the living body verification method based on light reflection characteristics may not be a good defense. Therefore, in order to further improve the living body verification method and improve the safety of the living body

verification, the living body verification can be further combined with other living body verification methods on the basis of the living body verification based on the light reflection characteristic. An exemplary implementation is described below.

- [0445] Fig. 8 shows a schematic flowchart of the living body detection step S800 (corresponding to the above-mentioned steps S220, or S340 and S350, or S430 and S440) according to another embodiment of the present invention. Steps S810 to S830 in the living body detection step S800 shown in FIG. 8 correspond to the steps S710 to S730 of the living body detection step S700 shown in FIG. Steps S810 to S830 will not be repeated here. According to this embodiment, the living body detection step S800 may further include steps S840 to S870, and step S830 may include determining whether the person to be authenticated has passed the living body verification based on the illumination living body verification result and the action living body verification result to obtain the living body detection result.
- [0446] In step S840, an action instruction is output, where the action instruction is used to instruct the person to be authenticated to perform a corresponding action.
- [0447] Exemplarily, the action command may be output randomly or according to a predetermined rule. Action instructions can include individual instructions or a sequence of instructions composed of a series of instructions. For example, the action instruction may instruct the person to be authenticated to nod, shake, blink, open his mouth, and so on.
- [0448] In step S850, multiple motion images collected for the face of the person to be authenticated are acquired.
- [0449] While outputting the action instruction or within a period of time after outputting the action instruction, image collection can be performed on the face of the person to be authenticated to obtain multiple action images. Exemplarily, the multiple motion images may be consecutive video frames. The motion image can also be collected by the above-mentioned image collection device 110, or collected by other image collection devices.
- [0450] In step S860, the actions performed by the person to be authenticated are detected based on a plurality of action images.
- [0451] Exemplarily, face detection and key point recognition can be performed in each action image, and the actions performed by the face can be judged based on the face contours and/or face key points in multiple action images, for example, by Recognize the change trend of the face contour and/or the key points of the face in the acquired continuous action images to determine the action

performed by the face. Subsequently, it can be determined whether the action performed by the face is consistent with the action indicated by the action instruction.

- [0452]** In step S870, it is determined whether the face of the person to be authenticated belongs to a living body according to the action detection result and the action instruction, so as to obtain the action living body verification result.
- [0453]** Exemplarily, if the actions performed by the person to be authenticated in multiple action images are consistent with the actions indicated by the action instructions, it is determined that the face of the person to be authenticated belongs to a living body. If the person to be authenticated performs actions in multiple action images If the action of is inconsistent with the action indicated by the action instruction, or the person to be authenticated does not perform any action in multiple action images (that is, the action of the person to be authenticated is not detected), it is determined that the face of the person to be authenticated does not belong to a living body. Of course, the above method is only an example. There can be other ways to determine whether the action in vivo verification is passed. For example, the person to be authenticated performs multiple actions in multiple action images, and the multiple actions include the actions that are consistent with the actions indicated by the action instructions. Action, it is determined that the face of the person to be authenticated belongs to a living body.
- [0454]** Exemplarily, in step S830, if both the light living body verification result and the action living body verification result indicate that the face of the person to be authenticated belongs to a living body, it is determined that the person to be authenticated passes the living body verification. If the result of the light living body verification result and the action living body verification result are If any one indicates that the face of the person to be authenticated does not belong to a living body, it is determined that the person to be authenticated has not passed the living body verification. Of course, the above method is only an example, and there may be other ways to determine whether the in vivo verification is passed.
- [0455]** It should be noted that the sequence of execution of the above-mentioned action-based living body verification steps (steps S840-S870) and light reflection characteristics-based living body verification steps (steps S810-S820) can be set arbitrarily, and the present invention does not limit this.
- [0456]** The living body detection method including the action-based living body verification step can be independently executed by the image acquisition terminal, for example, it can be executed independently by the image acquisition terminal of a financial system such as a bank management system or a mobile terminal such as a smart phone or a tablet computer. Alternatively, the living body detection method including the action-based living body verification step can also be

executed by the server (or the cloud) and the client in cooperation. For example, action instructions can be generated on the server or client, the client collects the action images of the person to be authenticated, and the client transmits the collected action images to the server (or cloud), and the server (or cloud) performs the Live verification of the action and return the verification result to the client.

- [0457]** Action-based in vivo verification method can defend against attacks such as mask attacks. Combined with the in vivo verification method based on light reflection characteristics, it can effectively defend against various types of attacks, thereby further ensuring that the identity authentication system or similar adopting the in vivo detection method The security of the system is also conducive to protecting the user's information security and rights and interests, and has extremely wide application value and market prospects.
- [0458]** According to an embodiment of the present invention, step S870 may include: if an action performed by the person to be authenticated that is consistent with the action indicated by the action instruction is detected based on a plurality of action images collected within a time period not greater than the first preset time , It is determined that the face of the person to be authenticated belongs to a living body. If no action performed by the person to be authenticated and consistent with the action indicated by the action instruction is detected based on the multiple motion images collected within the first preset time, then Make sure that the face of the person to be authenticated is not a living body.
- [0459]** Randomly output action instructions (such as "please nod", "please open your mouth" and other text or voice instructions) to instruct the person to be authenticated to perform corresponding actions (such as nodding, opening mouth, etc.), and detect key points in the face area to determine the waiting Whether the actions performed by the authentication personnel are consistent with the output action instructions. If it is detected within the first preset time that the action performed by the person to be authenticated is consistent with the output action instruction, it is determined that the face of the person to be authenticated belongs to a living body; if the person to be authenticated is detected within the first preset time If the action of is inconsistent with the output action instruction, or the action of the person to be authenticated is not detected within the first preset time, it can be determined that the face of the person to be authenticated does not belong to a living body.
- [0460]** According to the embodiment of the present invention, the identity authentication method (200, 300, or 400) may further include: counting once during each execution of step S840 to step S870 to obtain the number of action verification times; after step S870, the identity authentication method (200, 300, or 400) may also include: if the result of the action verification indicates that the face of the person to be authenticated does not belong to a living body, outputting the first error message,

and judging whether the number of action verifications reaches the first threshold, if the number of action verifications reaches the first number of times threshold, go to step S330. If the number of action verifications does not reach the first number of times threshold, return to step S340 or return to step S310 if step S310 is executed before step S340, where the first error information is used to indicate that the in-vivo verification for the person to be certified has failed.

- [0461]** Exemplarily, a counter can be set to count the number of executions of the action-based living body verification steps (steps S340-S370), and the counter can be incremented by one each time the action-based living body verification step is executed. The output result of the counter is the number of operation verifications. After the entire living body detection step (the living body detection step S800) is terminated, the counter can be cleared.
- [0462]** If the current live action verification result indicates that the face of the person to be authenticated does not belong to a live body, the first error message may be output. The first error message may prompt that the in vivo verification of the face of the person to be authenticated fails and prompts him to perform the in vivo verification again. If the number of executions of the action-based living body verification step (the number of action verification times) has not reached the preset first number threshold at this time, it may try to re-execute the action-based living body verification step. Optionally, if the living body verification steps based on light reflection characteristics (steps S810-S820) are performed before the action-based living body verification steps, you can directly return to step S810, that is, re-execute the living body verification steps based on light reflection characteristics and Action-based live verification steps to improve the accuracy of live verification.
- [0463]** The first frequency threshold can be any suitable value, which can be set as required, and the present invention does not limit this.
- [0464]** In the actual living body verification process, there may be various unexpected situations, such as the user's failure to make specified actions in time, the captured image is not clear enough, and the face detection result is not accurate enough. These situations may cause the user to be misidentified as a non-living body. . Therefore, in order to take into account both user experience and system security, a threshold for the number of times can be set, allowing users to try in vivo verification multiple times within a reasonable range. If the user is still not correctly identified as a living body when the number threshold is reached, it can be determined that the user does not belong to a living body.
- [0465]** According to the embodiment of the present invention, before step S710 (or S810), the identity authentication method (200, 300, or 400) may further include: Step S708: Determine whether the image acquisition conditions of the person to be authenticated meet the preset requirements, if the image acquisition conditions meet the preset requirements, go to step S710 (or S810), where

the image acquisition conditions include the position of the person to be authenticated in the image acquisition area of the image acquisition device and/or the relative angle between the person to be authenticated and the image acquisition device .

- [0466] Before performing the living body verification step based on the light reflection characteristic or other living body verification steps, the image acquisition conditions of the person to be authenticated can be tested first to determine whether it meets the preset requirements. It is possible to perform subsequent in vivo verification steps based on light reflection characteristics or other in vivo verification steps only when the image acquisition conditions of the person to be authenticated meet the preset requirements, so that the images used for in vivo verification (including light images, motions, etc.) can be guaranteed. The quality of the image, etc.) is conducive to the correct detection of the face in the image, thereby helping to improve the accuracy of the living body verification.
- [0467] According to the embodiment of the present invention, before step S708 or simultaneously with step S708, the identity authentication method (200, 300, or 400) may further include: step S706: output first prompt information, where the first prompt information is used to prompt The authenticating person faces the image acquisition device and approaches the image acquisition device.
- [0468] The first prompt information can be output in any suitable manner. Exemplarily, step S706 may include: outputting the first prompt information in one or more of a voice format, an image format, and a text format. For example, text such as "please face the screen" can be output on the display screen of the mobile terminal (facing the screen is equivalent to facing the image capture device), or a prompt such as "please face the screen" can be issued through the speaker of the mobile terminal language.
- [0469] Exemplarily, the identity authentication method may be implemented by an application (APP) installed on an electronic device such as a mobile terminal. When the user opens the application and then enters the living body detection stage, the first prompt message can be output to prompt the user to maintain a proper relative position relationship with the mobile terminal, so that the camera of the mobile terminal can collect ideal facial images for the living body verification. In an example, the first prompt information may be output continuously or at intervals before the image acquisition condition of the person to be authenticated meets the preset requirement.
- [0470] Outputting the first prompt information can facilitate guiding the user to adjust the relative position relationship between the identity authentication device (mainly the image acquisition device) and

itself in time, and at the same time, this kind of interaction between the user and the system can also improve the user experience.

- [0471]** According to an embodiment of the present invention, step S708 may include: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area for calibrating image acquisition conditions and a face area in the real-time image for display; and The face area detected in the real-time image determines whether the image acquisition conditions meet the preset requirements. If the face area is located in the preset area and the proportion of the face area in the real-time image is greater than the first preset ratio, the image is determined The acquisition condition meets the preset requirement. If the face area is not located in the preset area or the proportion of the face area in the real-time image is not greater than the first preset ratio, it is determined that the image acquisition condition does not meet the preset requirement.
- [0472]** In an example, it can be determined whether the image collection condition of the person to be authenticated meets the preset requirement based on the image collected for the face of the person to be authenticated. For example, the mobile terminal can collect real-time images through a camera and perform face detection. Face detection can obtain a face area, which is an image block containing a face. It can be determined whether the image acquisition conditions of the person to be authenticated meet the preset requirements according to the position of the face area in the real-time image and the proportion of the face area in the real-time image. For example, a preset area can be delineated in the real-time image. The preset area can limit the position of the face of the person to be authenticated in the image capture area of the image capture device. The size of the face area can reflect the distance and relative angle between the person to be authenticated and the image acquisition device. Both the preset area and the first preset ratio can be set as required, and the present invention does not limit this.
- [0473]** For example, if the face area of the person to be authenticated is located in the preset area, but the proportion of the face area in the real-time image is less than the first preset ratio (for example, two thirds), it is possible that the person to be authenticated is relative to the If the image acquisition device is too tilted and/or is far away from the image acquisition device, it can be considered that the image acquisition conditions do not meet the preset requirements.
- [0474]** Exemplarily, the identity authentication method (200, 300, or 400) may further include: if the proportion of the face area in the real-time image is not greater than the first preset proportion, outputting the first collection prompt information in real time to prompt to be authenticated The person is close to the image capture device.
- [0475]** Optionally, the first collection prompt information may be output in one or more of a voice form, an image form, and a text form. For example, if it is found that the proportion of the face area in the

real-time image is not greater than the first preset proportion, a prompt message such as "please approach the camera" (or "please approach the phone") may be displayed on the display screen.

- [0476]** According to an embodiment of the present invention, step S708 may include: acquiring a real-time image collected for the face of the person to be authenticated; real-time outputting a preset area for calibrating image acquisition conditions and a face area in the real-time image for display; and The face area detected in the real-time image determines whether the image acquisition conditions meet the preset requirements. If the face area is located in the preset area and the proportion of the face area in the preset area is greater than the second preset ratio, then confirm The image acquisition condition meets the preset requirement. If the face area is not located in the preset area or the proportion of the face area in the preset area is not greater than the second preset ratio, it is determined that the image acquisition condition does not meet the preset requirement.
- [0477]** In an example, it can be determined whether the image collection condition of the person to be authenticated meets the preset requirement based on the image collected for the face of the person to be authenticated. For example, the mobile terminal can collect real-time images through a camera and perform face detection. Face detection can obtain a face area, which is an image block containing a face. It can be determined whether the image acquisition conditions of the person to be authenticated meet the preset requirements according to the position of the face area in the real-time image and the proportion of the face area in the preset area. For example, a preset area can be displayed on the display screen. The relative position between the person to be authenticated and the screen can be defined by the preset area. The size of the face area can reflect the distance and relative angle between the person to be authenticated and the image acquisition device. For example, when the face is detected from far to nearer, the face area displayed on the screen in real time changes from small to large, and When the face is close enough to the screen, the size of the face area displayed on the screen meets the preset conditions. Of course, the face area displayed on the screen in real time can also be resized to meet the preset conditions only when the face is close enough to the screen. It is not limited here. Both the preset area and the second preset ratio can be set according to needs, and the present invention does not limit this.
- [0478]** For example, if the face area of the person to be authenticated is located in the preset area, but the proportion of the face area in the preset area is less than the second preset ratio (for example, two thirds), it is possible that the person to be authenticated is relatively When the image acquisition device is too tilted and/or is far away from the image acquisition device, it can be considered that the image acquisition conditions do not meet the preset requirements at this time.
- [0479]** Exemplarily, the identity authentication method (200, 300, or 400) may further include: if the proportion of the face area in the preset area is not greater than the second preset ratio, outputting

the second collection prompt information in real time to prompt the waiting The certified person is close to the image capture device.

- [0480] Optionally, the second collection prompt information may be output in one or more of a voice form, an image form, and a text form. For example, if it is found that the proportion of the face area in the preset area is not greater than the second preset ratio, a prompt message such as "please approach the camera" (or "please approach the phone") may be displayed on the display screen.
- [0481] According to the embodiment of the present invention, the identity authentication method (200, 300 or 400) may further include: real-time determination of the relative position relationship between the face area and the preset area; and based on the relative position between the face area and the preset area The relationship outputs the third collection prompt information in real time to prompt to change the relative position relationship between the person to be authenticated and the image acquisition device so that the face area is close to the preset area.
- [0482] For example, when the identity authentication method and device according to the embodiments of the present invention are implemented on a mobile terminal, the face area (that is, the image block containing the face extracted from the real-time image) and the face area can be displayed on the display screen of the mobile terminal in real time. The icon used to represent the preset area (that is, the preset area displayed on the screen in real time). Real-time display of the face area and the icon used to represent the preset area can help users know what their current image acquisition conditions are and how far they are from the preset requirements, so that users can adjust themselves or the image acquisition device (or include The posture of the identity authentication device of the image acquisition device) in order to enter the subsequent living body verification stage as soon as possible. Therefore, real-time display of the face area and the icon used to represent the preset area can improve the user experience, and can improve the efficiency of living body verification.
- [0483] In addition, third collection prompt information can be output to prompt the user to change the relative position relationship between the person to be authenticated and the image collection device so that the face area is close to the preset area. Optionally, the third collection prompt information may be output in one or more of a voice form, an image form, and a text form. For example, if it is found that the face area is not located in the preset area, a prompt message such as "please approach the center of the circle" (the preset area is displayed as a circle icon on the display screen) may be displayed on the display screen. In addition, an arrow pointing from the face area to the preset area can also be displayed on the display screen to facilitate the user to understand how to move himself or the image capture device so that the face area enters the preset area as soon as possible. The text prompt information such as "please approach the center

of the circle" and the image prompt information such as arrows can be displayed at the same time or alternatively.

- [0484] According to the embodiment of the present invention, step S708 may include: acquiring the posture information of the image acquisition device; and judging whether the image acquisition device is in the upright position according to the posture information, if so, determining that the image acquisition condition meets the preset requirements, otherwise determining the image The collection conditions do not meet the preset requirements.
- [0485] For example, when the identity authentication method according to the embodiment of the present invention is applied to a mobile terminal scene, the posture information of the image acquisition device (ie, the camera of the mobile terminal) can be measured by the gyroscope sensor and/or acceleration sensor built in the mobile terminal. . When the mobile terminal is placed in a vertical state, the image acquisition device is also in a vertical state. In this case, an ideal face image can be collected. Therefore, it is possible to measure whether the image acquisition conditions of the person to be authenticated meet the requirements based on the posture of the image acquisition device.
- [0486] According to the embodiment of the present invention, the identity authentication method (200, 300, or 400) may further include: counting once during each execution of step S710 to step S720 (or step S810 to step S820) to obtain the number of light verification times; After step S720 (or S820), the identity authentication method (200, 300, or 400) may further include: if the result of the light living body verification indicates that the face of the person to be authenticated is not a living body, outputting a second error message, and judging the number of light verification times Whether it reaches the second number threshold, if the number of light verification times reaches the second number threshold, go to step S230 (or S330), if the number of light verification times does not reach the second number threshold, return to step S208 or return to step S210 (or S310) , Where the second error message is used to indicate that the in-vivo verification for the person to be authenticated has failed.
- [0487] Similar to the action-based living body verification step, for the living body verification step based on light reflection characteristics (steps S710-S720 shown in FIG. 7 or steps S810-S820 shown in FIG. 8), if the person to be authenticated is determined If the face is not a living body, you can also try to re-execute the living body verification step based on the light reflection characteristics. The principle and advantages are similar to the action-based living body verification step, and will not be repeated here.

- [0488] Exemplarily, in the case where the identity authentication method includes the above-mentioned step S708, it can also be re-executed from step S708.
- [0489] According to the embodiment of the present invention, before step S710 (or S810) or during the execution of step S710 (or S810) and step S720 (or S820), the identity authentication method (200, 300, or 400) may further include: outputting the first 2. Prompt information, where the second prompt information is used to prompt the person to be authenticated to stay still within the second preset time.
- [0490] Exemplarily, the second preset time may be the execution time of the living body verification step (steps S710-S720 shown in FIG. 7 or steps S810-S820 shown in FIG. 8) based on the light reflection characteristics. When the living body verification step based on light reflection characteristics is performed, that is, when the face of the person to be authenticated is illuminated with detection light, the person to be authenticated can be prompted to remain still during the period, so as not to affect the image collection effect and the result of living body verification. Exemplarily, if the person to be authenticated moves within the second preset time and the image acquisition condition of the person to be authenticated no longer meets the preset requirements, then return to step S706 or step S708, and re-execute one or more of the following steps. Steps: judging the image acquisition conditions, outputting the first prompt information, outputting various collection prompt information, and other steps.
- [0491] Exemplarily, the second prompt information may be countdown information corresponding to the second preset time. Optionally, the countdown information can be implemented in one or more forms of text, dynamic images, and voice. The countdown information can facilitate the user to understand the progress of the living body detection, and can improve the user's interactive experience.
- [0492] The following describes an implementation process of the living body detection step according to the embodiment of the present invention with reference to FIG. 9. The application scenario shown in Figure 9 is a mobile terminal.
- [0493] As shown in Figure 9, first, the text "Please face the screen" is displayed on the display screen of the mobile terminal, prompting the user to face the screen directly, and at the same time, an icon ((Represented by a circle) and the face area detected based on real-time images. When the user changes the position and/or posture of his face, and/or changes the position and/or posture of the mobile terminal, the text "Please face the screen" and the icon used to indicate the preset area can be continuously displayed. Both types of information can remain unchanged. However, the size and position of the face area may change. Therefore, the constantly changing face area can be displayed in real time for the user to view. Subsequently, when the user's image acquisition

conditions meet the preset requirements, the next stage can be entered, that is, the living body verification step based on light reflection characteristics.

- [0494] During the execution of the living body verification step based on the light reflection characteristics, the words "please keep still" can be displayed on the display screen (as shown in the second and third images in Figure 9), prompting the user to keep still, At the same time, countdown information can be displayed on the display screen. The countdown information is represented in the third image shown in FIG. 9 by a colored progress bar marked on the icon (ie, circle) for representing the preset area.
- [0495] After the living body verification step based on the light reflection characteristics is completed, the action-based living body verification step can be started. As shown in the fourth image in Figure 9, the text "please nod" is displayed on the display screen, instructing the user to perform the corresponding action.
- [0496] Finally, the final live test results are output on the display screen, such as texts such as "Vitality Verification Passed".
- [0497] According to another aspect of the present invention, an identity authentication device is provided. FIG. 10 shows a schematic block diagram of an identity authentication apparatus 1000 according to an embodiment of the present invention.
- [0498] As shown in FIG. 10, the identity authentication device 1000 according to the embodiment of the present invention includes an information acquisition module 1010, an authenticated information judgment module 1020, a living body detection module 1030, and an identity determination module 1040. The modules can respectively execute the steps/functions of the identity authentication method described above in conjunction with FIGS. 2-9. In the following, only the main functions of the components of the identity authentication device 1000 are described, and the details that have been described above are omitted.
- [0499] The information acquisition module 1010 is used to acquire the personal identification information of the person to be authenticated. The information acquisition module 1010 may be implemented by the processor 102 in the electronic device shown in FIG. 1 running the program instructions stored in the storage device 104.
- [0500] The authenticated information judgment module 1020 is used to judge whether the personal identification information is authenticated information, so as to obtain the information authentication result. The authenticated information judgment module 1020 can be implemented by the processor

102 in the electronic device shown in FIG. 1 running the program instructions stored in the storage device 104.

- [0501] The living body detection module 1030 is used to perform a living body detection on a person to be authenticated to obtain a living body detection result. The living body detection module 1030 may be implemented by the processor 102 in the electronic device shown in FIG. 1 running the program instructions stored in the storage device 104.
- [0502] The identity determination module 1040 is used to determine whether the identity of the person to be authenticated is legal at least according to the certificate authentication result and the living body test result. The identity determination module 1040 may be implemented by the processor 102 in the electronic device shown in FIG. 1 running the program instructions stored in the storage device 104.
- [0503] A person of ordinary skill in the art may be aware that the units and algorithm steps of the examples described in combination with the embodiments disclosed herein can be implemented by electronic hardware or a combination of computer software and electronic hardware. Whether these functions are executed by hardware or software depends on the specific application and design constraint conditions of the technical solution. Professionals and technicians can use different methods for each specific application to implement the described functions, but such implementation should not be considered as going beyond the scope of the present invention.
- [0504] FIG. 11 shows a schematic block diagram of an identity authentication system 1100 according to an embodiment of the present invention. The identity authentication system 1100 includes an image acquisition device 1110, a storage device 1120, and a processor 1130.
- [0505] The image capture device 1110 is used to capture images of the person to be authenticated, such as a face image, a certificate image, and the like. The image acquisition device 1110 is optional, and the identity authentication system 1100 may not include the image acquisition device 1110. In this case, other image acquisition devices may be used to collect images for identity authentication, and the collected images are sent to the identity authentication system 1100.
- [0506] The storage device 1120 stores computer program instructions for implementing the corresponding steps in the identity authentication method according to the embodiment of the present invention.
- [0507] The processor 1130 is used to run the computer program instructions stored in the storage device 1120 to execute the corresponding steps of the identity authentication method according to the embodiment of the present invention, and to implement the steps in the identity authentication device according to the embodiment of the present invention. An information acquisition module

1010, an authenticated information judgment module 1020, a living body detection module 1030, and an identity determination module 1040.

- [0508] In one embodiment, the computer program instructions are used to perform the following steps when the processor 1130 is running: obtain the personal identification information of the person to be authenticated; determine whether the personal identification information is authenticated information to obtain the information authentication result; Perform a live test on the person to be authenticated to obtain a live test result; and at least determine whether the identity of the person to be authenticated is legal based on the information authentication result and the live test result.
- [0509] In addition, according to an embodiment of the present invention, there is also provided a storage medium on which program instructions are stored, and the program instructions are used to execute the identity authentication method according to the embodiment of the present invention when the program instructions are run by a computer or a processor. The corresponding steps are used to implement the corresponding modules in the identity authentication device according to the embodiment of the present invention. The storage medium may include, for example, a memory card of a smart phone, a storage component of a tablet computer, a hard disk of a personal computer, a read-only memory (ROM), an erasable programmable read-only memory (EPROM), a portable compact disk read-only memory (CD-ROM), USB memory, or any combination of the above storage media.
- [0510] In an embodiment, when the computer program instructions are executed by a computer or a processor, the computer or the processor can realize each functional module of the identity authentication device according to the embodiment of the present invention, and/or can execute the embodiment according to the present invention. The identity authentication method.
- [0511] In one embodiment, the computer program instructions are used to perform the following steps at runtime: obtain the personal identification information of the person to be authenticated; determine whether the personal identification information is authenticated information to obtain the information authentication result; Detection to obtain live detection results; and at least to determine whether the identity of the person to be authenticated is legal based on the information authentication result and the live detection result.
- [0512] The modules in the identity authentication system according to the embodiment of the present invention may be implemented by the processor of the electronic device implementing identity authentication according to the embodiment of the present invention running computer program instructions stored in the memory, or may be implemented in accordance with the embodiment of

the present invention. The computer instructions stored in the computer-readable storage medium of the computer program product are implemented when the computer is running.

- [0513] According to another aspect of the present invention, an identity authentication device is provided, including an information collection device, a processor, and a memory, wherein the information collection device is used to receive personal identification information of a person to be authenticated; computer program instructions are stored in the memory, When the computer program instructions are executed by the processor, they are used to perform the following steps: obtain personal identification information of the person to be authenticated, where the personal identification information is the initial information or is obtained based on the initial information; Whether the personal identification information is authenticated information to obtain an information authentication result; perform a live test on the person to be authenticated to obtain a live test result; and determine the to-be-authenticated information at least based on the information authentication result and the live test result Whether the identity of the personnel is legal.
- [0514] Exemplarily, the personal information collection device includes an image collection device and/or an input device.
- [0515] Exemplarily, the identity authentication device may further include: an image collection device for collecting a face image of a person to be authenticated; the processing device is further used for performing a living body detection using the face image to obtain a living body detection result.
- [0516] Exemplarily, the identity authentication device may further include: a light source for emitting detection light to the face of the person to be authenticated; the image acquisition device is further used to collect the detection light on the face of the person to be authenticated One or more illumination images are used as the face image; the processing device is further configured to determine whether the face of the person to be authenticated belongs to the light reflection characteristics of the face of the person to be authenticated in the one or more illumination images. Living body, to obtain a live body verification result under illumination, and determine whether the person to be authenticated passes the live body verification based at least on the result of the live body verification, so as to obtain the live body detection result.
- [0517] According to another aspect of the present invention, an identity authentication device is provided, including: an image acquisition device for collecting personal identification information of a person to be authenticated; a processing device for judging whether the personal identification information is authenticated information to obtain an information authentication result , The person to be authenticated performs a live test to obtain the result of the live test, and at least

determines whether the identity of the person to be authenticated is legal based on the information authentication result and the result of the live test.

- [0518] Exemplarily, the image acquisition device is also used to acquire a face image of a person to be authenticated; the processing device is further used to perform a living body detection using the face image to obtain a living body detection result.
- [0519] Exemplarily, the identity authentication device may further include: a light source for emitting detection light to the face of the person to be authenticated; the image acquisition device is further used to collect the detection light on the face of the person to be authenticated. One or more illumination images are used as the face image; the processing device is further configured to determine whether the face of the person to be authenticated belongs to the light reflection characteristics of the face of the person to be authenticated in the one or more illumination images. Living body, to obtain a live body verification result under illumination, and determine whether the person to be authenticated passes the live body verification based at least on the result of the live body verification, so as to obtain the live body detection result.
- [0520] Exemplarily, the identity authentication method (200, 300, or 400) can be implemented on a separate identity authentication device (such as a mobile terminal, etc.). The aforementioned input device may include one or more of a keyboard, a mouse, a microphone, and a touch screen. The personal identification information is one or more of the ID number, name, ID face, field collected face, ID number conversion value, name conversion value, ID face conversion value, and field collection face conversion value. In this case, the input device can be used to receive the information input by the user to obtain the required personal identification information. In the case that the personal identification information is the credential information in the credential image, the image acquisition device can be used to collect the credential image of the person to be authenticated to obtain the required personal identification information. Of course, personal information such as the certificate number, name, and face of the certificate can also be extracted from the certificate image, and the conversion value of these personal information can be obtained through a predetermined algorithm. In other words, the certificate number, name, and face can also be obtained by using the image acquisition device. Personal identification information such as the certificate face, the conversion value of the certificate number, the conversion value of the name, and the conversion value of the certificate face. Of course, the identity authentication device may include both an input device and an image acquisition device, combining two information acquisition methods to acquire personal identification information.
- [0521] According to another aspect of the present invention, an identity authentication device is provided, including: an input device for receiving personal identification information of a person to be authenticated; a transmission device for sending personal identification information to a server,

and receiving from the server the server through the following Authentication information about whether the identity of the person to be authenticated is legal or not obtained by the method: determine whether the personal identification information is authenticated information to obtain the information authentication result, and the person to be authenticated shall undergo a live test to obtain the live test result, and at least according to the information authentication result and the live body The test result determines whether the identity of the person to be authenticated is legal.

- [0522]** Exemplarily, the identity authentication device further includes: an image acquisition device for collecting a face image of the person to be authenticated; the transmission device is also used for sending the face image to a server, wherein the server uses The face image is subjected to live body detection to obtain the live body detection result.
- [0523]** Exemplarily, the identity authentication device may further include: a light source for emitting detection light to the face of the person to be authenticated; the image acquisition device is further used to collect the detection light on the face of the person to be authenticated A plurality of illumination images are used as the face images; the transmission device is further used to send one or more illumination images to the server; wherein, the server performs the living body detection in the following manner: based on the fact that the face of the person to be authenticated is in one or more The light reflection characteristics shown in each illumination image determine whether the face of the person to be authenticated belongs to a living body to obtain the illumination living body verification result, and at least based on the illumination living body verification result, it is determined whether the person to be authenticated passes the living body verification to obtain the living body detection result.
- [0524]** According to another aspect of the present invention, there is provided an identity authentication device, which includes: an image acquisition device for collecting personal identification information of a person to be authenticated; a transmission device for sending personal identification information to a server, and receiving from the server through the server The authentication information about whether the identity of the person to be authenticated is obtained in the following ways: Determine whether the personal identification information is authenticated information to obtain the information authentication result, and the person to be authenticated shall perform a live test to obtain the live test result, and at least according to the information authentication result and The result of the living body test determines whether the identity of the person to be authenticated is legal.
- [0525]** Exemplarily, the image acquisition device is also used to acquire a face image of a person to be authenticated; the processing device is further used to perform a living body detection using the face image to obtain a living body detection result.

- [0526] Exemplarily, the identity authentication device may further include: a light source for emitting detection light to the face of the person to be authenticated; the image acquisition device is further used to collect the detection light on the face of the person to be authenticated One or more illumination images are used as the face image; the transmission device is further configured to send one or more illumination images to the server; wherein, the server performs living body detection in the following manner: based on the presence of the face of the person to be authenticated The light reflection characteristics shown in or multiple illumination images determine whether the face of the person to be authenticated is a living body, so as to obtain the illumination living body verification result, and at least based on the illumination living body verification result to determine whether the person to be authenticated has passed the living body verification to obtain the living body detection result.
- [0527] Exemplarily, the identity authentication method (200, 300 or 400) can be implemented on separate devices, such as on the client and the server. In this case, the client may include an input device and/or an image acquisition device and a transmission device. Optionally, the client can upload the acquired personal identification information and the collected face images (including lighting images, motion images, real-time images, etc.) to the server, and the server performs identity authentication. After the client receives the authentication information sent by the server, it performs authentication passed or failed operations, such as outputting information about identity authentication passed or failed, allowing or denying the user to perform subsequent business operations, and so on.
- [0528] Exemplarily, the transmission device of the client can send the personal identification information and the face image to the server through the network or other technology, and receive the authentication information from the server through the network or other technology. For example, the network may be the Internet, a wireless local area network, a mobile communication network, etc., and the other technologies may include, for example, Bluetooth communication, infrared communication, and the like. For example, the server may be a general-purpose server or a dedicated server, and may be a virtual server or a cloud server. The transmission device of the client (or server) may include a modem, a network adapter, a Bluetooth transmitting and receiving unit, or an infrared transmitting and receiving unit, etc., for example, it may also perform operations such as encoding and decoding on sent or received information.
- [0529] Since most of the identity authentication work is done on the server, the computing resources of the processing device of the client can be saved, thereby reducing the performance requirements of the client and the manufacturing cost of the identity authentication device, and improving the user experience.

- [0530] According to another aspect of the present invention, there is provided an identity authentication device, including a transmission device and a processing device, wherein the transmission device is used to receive personal identification information of a person to be authenticated from a client, and to determine whether the identity of the person to be authenticated is legal authentication information is sent to the client; the processing device is used to determine whether the personal identification information is authenticated information to obtain the information authentication result, and the person to be authenticated performs a living test to obtain the living test result, at least according to the information authentication result and the living test result, it is determined whether the identity of the person to be authenticated is legal to obtain the authentication information.
- [0531] Exemplarily, the transmission device is also used to receive a face image of the person to be authenticated; the processing device is further used to perform a living body detection using the face image to obtain a living body detection result.
- [0532] Exemplarily, the transmission device is further configured to receive one or more illumination images of the person to be authenticated from the client, where the one or more illumination images are for the face of the person to be authenticated under the illumination of the detection light. The processing device is further used to determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in one or more illumination images, so as to obtain the result of the illumination living body verification, And at least based on the result of the live body verification, it is determined whether the person to be authenticated has passed the live body verification, so as to obtain the live body test result.
- [0533] As described above, the identity authentication method (200, 300, or 400) can be implemented on separate devices, such as on the client and the server. This embodiment describes an implementation scheme in which at least a part of the identity authentication method is implemented on the server.
- [0534] According to the identity authentication method and device of the embodiment of the present invention, the identity of the person to be authenticated is determined by combining the authentication information judgment and living body detection. Therefore, compared with the conventional method of identity authentication based solely on passwords or certificates, according to this The authentication result of the identity authentication method and device of the embodiment of the invention is more accurate, can improve the security of user authentication, and can effectively protect the rights of users. The above-mentioned method and device can be well

applied to various fields involving identity authentication, such as e-commerce, mobile payment, or banking.

- [0535] Although the exemplary embodiments have been described herein with reference to the accompanying drawings, it should be understood that the above-described exemplary embodiments are merely exemplary, and are not intended to limit the scope of the present invention thereto. Those of ordinary skill in the art can make various changes and modifications therein without departing from the scope and spirit of the present invention. All these changes and modifications are intended to be included in the scope of the present invention as claimed in the appended claims.
- [0536] A person of ordinary skill in the art may be aware that the units and algorithm steps of the examples described in combination with the embodiments disclosed herein can be implemented by electronic hardware or a combination of computer software and electronic hardware. Whether these functions are executed by hardware or software depends on the specific application and design constraint conditions of the technical solution. Professionals and technicians can use different methods for each specific application to implement the described functions, but such implementation should not be considered as going beyond the scope of the present invention.
- [0537] In the several embodiments provided in this application, it should be understood that the disclosed device and method may be implemented in other ways. For example, the device embodiments described above are only illustrative. For example, the division of the units is only a logical function division, and there may be other divisions in actual implementation, for example, multiple units or components can be combined or It can be integrated into another device, or some features can be ignored or not implemented.
- [0538] In the instructions provided here, a lot of specific details are explained. However, it can be understood that the embodiments of the present invention can be practiced without these specific details. In some instances, well-known methods, structures, and technologies are not shown in detail, so as not to obscure the understanding of this specification.
- [0539] Similarly, it should be understood that in order to simplify the present invention and help understand one or more of the various aspects of the invention, in the description of the exemplary embodiments of the present invention, the various features of the present invention are sometimes grouped together into a single embodiment. , Or in its description. However, the method of the present invention should not be interpreted as reflecting the intention that the claimed present invention requires more features than those explicitly stated in each claim. More precisely, as reflected in the corresponding claims, the point of the invention is that the corresponding technical problems can be solved with features that are less than all the features of a single

disclosed embodiment. Therefore, the claims following the specific embodiment are thus explicitly incorporated into the specific embodiment, wherein each claim itself serves as a separate embodiment of the present invention.

- [0540]** Those skilled in the art can understand that, in addition to mutual exclusion between the features, any combination of all features disclosed in this specification (including the accompanying claims, abstract, and drawings) and any method or device disclosed in this manner can be used. Processes or units are combined. Unless expressly stated otherwise, each feature disclosed in this specification (including the accompanying claims, abstract and drawings) may be replaced by an alternative feature providing the same, equivalent or similar purpose.
- [0541]** In addition, those skilled in the art can understand that although some embodiments described herein include certain features included in other embodiments but not other features, the combination of features of different embodiments means that they are within the scope of the present invention. Within and form different embodiments. For example, in the claims, any one of the claimed embodiments can be used in any combination.
- [0542]** The various component embodiments of the present invention may be implemented by hardware, or by software modules running on one or more processors, or by a combination of them. Those skilled in the art should understand that a microprocessor or a digital signal processor (DSP) may be used in practice to implement some or all of the functions of some modules in the identity authentication device according to the embodiments of the present invention. The present invention can also be implemented as a device program (for example, a computer program and a computer program product) for executing part or all of the methods described herein. Such a program for realizing the present invention may be stored on a computer-readable medium, or may have the form of one or more signals. Such a signal can be downloaded from an Internet website, or provided on a carrier signal, or provided in any other form.
- [0543]** It should be noted that the above-mentioned embodiments illustrate rather than limit the present invention, and those skilled in the art can design alternative embodiments without departing from the scope of the appended claims. In the claims, any reference signs placed between parentheses should not be constructed as a limitation to the claims. The word "comprising" does not exclude the presence of elements or steps not listed in the claims. The word "a" or "an" preceding an element does not exclude the presence of multiple such elements. The invention can be implemented by means of hardware comprising several different elements and by means of a suitably programmed computer. In the unit claims listing several devices, several of these devices may be embodied in the same hardware item. The use of the words first, second, and third, etc. do not indicate any order. These words can be interpreted as names.

[0544] The above are only specific implementations or descriptions of specific implementations of the present invention. The protection scope of the present invention is not limited thereto. Any person skilled in the art can easily fall within the technical scope disclosed by the present invention. Any change or replacement should be included in the protection scope of the present invention. The protection scope of the present invention shall be subject to the protection scope of the claims.

1. An identity authentication method, including:

Obtain the personal identification information of the person to be authenticated;

Judging whether the personal identification information is authenticated information, so as to obtain an information authentication result;

Perform a living body test on the person to be authenticated to obtain a living body test result; and

At least determining whether the identity of the person to be authenticated is legal according to the information authentication result and the living body test result;

Wherein, the performing a living body test on the person to be authenticated to obtain a living body test result includes:

Step S110: Acquire multiple illumination images collected for the face of the person to be authenticated under the illumination of the detection light, where the detection light is emitted by the display screen, and is irradiating the face of the person to be authenticated. In the process, the pattern of the detection light is changed at least once, and the pattern of the detection light is randomly changed or preset; the detection light dynamically changes the light emitted to the face of the person to be authenticated. The color and/or position of the detection light is obtained in the following manner: by changing the content displayed on the display screen to dynamically change the mode of light emitted by the display screen, so as to show the person to be authenticated. The face emits the detection light;

Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the plurality of illumination images, so as to obtain the result of the illumination living body verification, wherein Light reflection characteristics include: uniformity and three-dimensional characteristics of the face; and

Step S130: Determine whether the certified person has passed the in vivo verification based at least on the illumination in vivo verification result, so as to obtain the in vivo detection result.

2.The identity authentication method according to claim 1, wherein the personal identification information is a certificate number, a name, a certificate face, a face collected on site, a conversion value of a certificate number, a conversion value of a name, and a conversion value of a certificate face And one or more of the transformation values of the face collected on-site.

3.5. The identity authentication method according to claim 1, wherein, before said obtaining the personal identification information of the person to be authenticated, the identity authentication method further comprises:

Outputting instruction information for instructing the person to be authenticated to provide a predetermined type of person information;

Wherein, the personal identification information is person information provided by the person to be authenticated or obtained based on the person information.

4.The identity authentication method according to claim 1, wherein the personal identification information is one or more of a conversion value of a certificate number, a conversion value of a name, a conversion value of a certificate face, and a conversion value of a face collected on site Species,

The obtaining the personal identification information of the person to be authenticated includes:

Obtain the initial information of the person to be authenticated, where the initial information is one or more of a credential number, a name, a credential face, and a face collected on site; and

The initial information is transformed based on a predetermined algorithm to obtain a transformed value of the initial information as the personal identification information.

5.The identity authentication method according to claim 1, wherein:

The obtaining the personal identification information of the person to be authenticated includes:

Acquiring a credential image of the person to be authenticated, where the information authentication result is a credential authentication result;

The performing a living body test on the person to be authenticated to obtain a living body test result includes:

Acquiring the face image of the person to be authenticated; and

The human face image is used for living body detection to obtain the living body detection result.

6.5. The identity authentication method according to claim 5, wherein, before determining whether the identity of the person to be authenticated is legal at least according to the information authentication result and the living body detection result, the identity authentication method further comprises:

Performing an additional judgment operation using the credential image and/or the face image to obtain an additional judgment result;

The determining whether the identity of the person to be authenticated is legal at least according to the information authentication result and the living body detection result includes:

Determine whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result, and the additional judgment result.

7. The identity authentication method according to claim 6, wherein the additional judgment operation includes a credential authenticity judgment operation and/or a face consistency judgment operation, and the additional judgment result includes a credential authenticity judgment result and/or a person The result of face consistency judgment,

The authenticity judgment operation of the certificate includes: judging whether the certificate in the certificate image is a real certificate, so as to obtain the authenticity judgment result of the certificate;

The face consistency judgment operation includes: obtaining the credential face of the person to be authenticated according to the credential image; and comparing the credential face of the person to be authenticated with the face in the face image, To obtain the judgment result of the consistency of the face.

8.8. The identity authentication method according to claim 7, wherein said acquiring the credential face of the person to be authenticated according to the credential image comprises:

The face of the person is detected from the document image to obtain the face of the document of the person to be authenticated.

9.8. The identity authentication method according to claim 7, wherein said acquiring the credential face of the person to be authenticated according to the credential image comprises:

Performing text recognition on the credential image to obtain text information in the credential image;

Searching for matching certificate information from a certified certificate information database based on the text information in the certificate image; and

It is determined that the credential face in the matching credential information that is searched out is the credential face of the person to be authenticated.

10.8. The identity authentication method according to claim 7, wherein said determining whether the certificate in the certificate image is a real certificate to obtain the authenticity determination result of the certificate comprises:

Extract the image features of the credential image; and

Input the image features into the trained certificate classifier to obtain the authenticity judgment result of the certificate;

Wherein, the authenticity judgment result of the certificate is the confidence that the certificate in the certificate image is a real certificate.

11.8. The identity authentication method according to claim 7, wherein said determining whether the certificate in the certificate image is a real certificate to obtain the authenticity determination result of the certificate comprises:

Identify the image block containing the identification information of the certificate from the image of the certificate; and

Identifying the certificate identification information in the image block containing the certificate identification information to obtain the authenticity judgment result of the certificate;

Wherein, the authenticity judgment result of the certificate is the confidence that the certificate in the certificate image is a real certificate.

12. The identity authentication method according to claim 6, wherein, in the process of determining whether the identity of the person to be authenticated is legal according to the certificate authentication result, the living body detection result, and the additional judgment result, the Each of the certificate authentication result, the living body detection result, and the additional judgment result has its own weight coefficient.

13.5. The identity authentication method according to claim 5, wherein said obtaining the credential image of the person to be authenticated comprises:

Acquiring a pre-shooting image that is collected in real time for the credentials of the person to be authenticated under the current shooting conditions;

Evaluate the image attributes of the pre-photographed image in real time;

When the evaluation value of the image attribute of the pre-photographed image is less than the preset evaluation value threshold, the pre-photographing prompt information is generated according to the image attribute of the pre-photographed image to prompt the person to be authenticated to adjust the photographing conditions of the certificate ;as well as

When the evaluation value of the image attribute of the pre-photographed image is equal to or greater than the preset evaluation value threshold, the pre-photographed image is saved to be used as the credential image.

14.5. The identity authentication method according to claim 5, wherein said determining whether said personal identification information is authenticated information to obtain an information authentication result comprises:

Perform text recognition on the credential image to obtain text information in the credential image; and

Searching in a certified certificate information database based on the text information in the certificate image to obtain the certificate authentication result;

Wherein, the certificate authentication result is the credibility that the certificate information in the certificate image is the certified certificate information.

15.The identity authentication method according to claim 9 or 14, wherein said performing text recognition on said credential image to obtain text information in said credential image comprises:

Positioning the text in the document image to obtain an image block containing the text; and

Recognizing the text in the image block containing the text to obtain the text information in the credential image.

16.15. The identity authentication method according to claim 15, wherein before said recognizing the characters in the image block containing the characters, the identity authentication method further comprises:

The image block containing the text is corrected to a horizontal state.

17.The identity authentication method according to claim 15, wherein, after recognizing the text in the image block containing the text to obtain the text information in the credential image, the credential image is Character recognition to obtain the text information in the certificate image also includes:

Output the text information in the credential image for the user to view;

Receiving text correction information input by the user;

Comparing the text to be corrected indicated by the text correction information with the corresponding text in the text information in the credential image; and

If the difference between the text to be corrected indicated by the text correction information and the corresponding text in the text information in the credential image is less than a preset difference threshold, the text correction information is used to update the text in the credential image text information.

18.8. The identity authentication method according to claim 1, wherein, in the process of illuminating the face of the person to be authenticated, the pattern of the detection light is changed between every two consecutive moments.

19.The identity authentication method according to claim 1, wherein:

The performing a living body test on the person to be authenticated to obtain a living body test result includes:

Step S140: output an action instruction, where the action instruction is used to instruct the authenticator to perform a corresponding action;

Step S150: Acquire multiple motion images collected for the face of the person to be authenticated;

Step S160: Detect the actions performed by the person to be authenticated based on the multiple action images; and

Step S170: Determine whether the face of the person to be authenticated belongs to a living body according to the action detection result and the action instruction, so as to obtain an action living body verification result;

The step S130 includes:

It is determined whether the person to be authenticated passes the living body verification based on the illumination living body verification result and the action living body verification result, so as to obtain the living body detection result.

20.The identity authentication method according to claim 19, wherein said step S170 comprises:

If an action performed by the person to be authenticated that is consistent with the action indicated by the action instruction is detected based on the plurality of action images collected within a time period not greater than the first preset time, determine the The face of the person to be authenticated belongs to a living body. If the action performed by the person to be authenticated and indicated by the action instruction is not detected based on the multiple motion images collected within the first preset time A consistent action determines that the face of the person to be authenticated does not belong to a living body.

21.The identity authentication method according to claim 19, wherein:

The identity authentication method further includes:

Count once every time the step S140 to the step S170 are executed to obtain the number of action verification times;

After the step S170, the identity authentication method further includes:

If the action verification result indicates that the face of the person to be authenticated does not belong to a living body, output first error information, and determine whether the action verification times reach the first threshold value, and if the action verification times reach all The first number of times threshold, then go to step S130, if the action verification number does not reach the first number of times threshold, return to step S140 or execute step S110 before step S140 In the case of returning to the step S110, the first error message is used to prompt that the in-vivo verification for the person to be authenticated fails.

22.The identity authentication method according to claim 1, wherein, before the step S110, the identity authentication method further comprises:

Step S108: Determine whether the image acquisition conditions of the person to be authenticated meet the preset requirements, and if the image acquisition conditions meet the preset requirements, go to step S110, where the image acquisition conditions include the The position of the person to be authenticated in the image capture area of the image capture device and/or the relative angle between the person to be authenticated and the image capture device.

23.The identity authentication method of claim 22, wherein, before the step S108 or simultaneously with the step S108, the identity authentication method further comprises:

Step S106: Output first prompt information, where the first prompt information is used to prompt the person to be authenticated to face the image acquisition device and approach the image acquisition device.

24.The identity authentication method according to claim 23, wherein the step S106 comprises: outputting the first prompt information in one or more of a voice format, an image format, and a text format.

25.The identity authentication method according to claim 22, wherein said step S108 comprises:

Acquiring a real-time image collected for the face of the person to be authenticated;

Real-time outputting the preset area used to calibrate the image acquisition conditions and the face area in the real-time image for display; and

Determine whether the image acquisition condition meets the preset requirements according to the face area, if the face area is located in the preset area and the proportion of the face area in the real-time image is greater than the first preset If the ratio is set, it is determined that the image acquisition condition satisfies the preset requirement, if the face area is not located in the preset area or the proportion of the face area in the real-time image is not greater than the The first preset ratio determines that the image acquisition condition does not meet the preset requirement.

26.The identity authentication method according to claim 25, wherein the identity authentication method further comprises:

If the proportion of the face area in the real-time image is not greater than the first preset proportion, first collection prompt information is output in real time to prompt the person to be authenticated to approach the image collection device.

27.The identity authentication method according to claim 22, wherein said step S108 comprises:

Acquiring a real-time image collected for the face of the person to be authenticated;

Real-time outputting the preset area used to calibrate the image acquisition conditions and the face area in the real-time image for display; and

Judging whether the image acquisition condition satisfies the preset requirement according to the face area,

If the face area is located in the preset area and the proportion of the face area in the preset area is greater than the second preset ratio, it is determined that the image acquisition condition satisfies the preset requirement, and if all If the face area is not located in the preset area or the proportion of the face area in the preset area is not greater than the second preset ratio, it is determined that the image acquisition condition does not meet the preset ratio Set requirements.

28.The identity authentication method according to claim 27, wherein the identity authentication method further comprises:

If the proportion of the face area in the preset area is not greater than the second preset ratio, second collection prompt information is output in real time to prompt the person to be authenticated to approach the image collection device.

29.The identity authentication method according to claim 25 or 27, wherein the identity authentication method further comprises:

Determine the relative positional relationship between the face area and the preset area in real time; and

Based on the relative position relationship between the face area and the preset area, third collection prompt information is output in real time to prompt to change the relative position relationship between the person to be authenticated and the image acquisition device so that the The face area is close to the preset area.

30.The identity authentication method according to claim 22, wherein said step S108 comprises:

Acquiring the posture information of the image acquisition device; and

According to the posture information, it is determined whether the image acquisition device is in a vertically placed state, and if so, it is determined that the image acquisition condition meets the preset requirement, otherwise it is determined that the image acquisition condition does not meet the preset requirement.

31.The identity authentication method according to claim 22, wherein:

The identity authentication method further includes:

Count once during each execution of the step S110 to the step S120 to obtain the number of light verification times;

After the step S120, the identity authentication method further includes:

If the result of the light verification in vivo indicates that the face of the person to be authenticated does not belong to a living body, a second error message is output, and it is determined whether the number of light verification times reaches the second threshold value, and if the number of light verification times reaches the The second threshold of the number of times, then go to step S130, if the number of illumination verification times does not reach the threshold of the second number of times, return to the step S108 or return to the step S110, wherein the second error information is It prompts that the in-vivo verification for the person to be authenticated has failed.

32.The identity authentication method of claim 1, wherein, before the step S110 or during the execution of the step S110 and the step S120, the identity authentication method further comprises:

The second prompt information is output, where the second prompt information is used to prompt the person to be authenticated to remain motionless within the second preset time.

33.The identity authentication method according to claim 32, wherein the second prompt information is countdown information corresponding to the second preset time.

34.An identity authentication device includes a processor and a memory, wherein computer program instructions are stored in the memory, and the computer program instructions are used to perform the following steps when the processor is running:

Obtain the personal identification information of the person to be authenticated;

Judging whether the personal identification information is authenticated information, so as to obtain an information authentication result;

Perform a living body test on the person to be authenticated to obtain a living body test result; and

At least determining whether the identity of the person to be authenticated is legal according to the information authentication result and the living body test result;

Wherein, the step of performing a living body detection on the person to be authenticated to obtain a living body detection result executed by the computer program instructions when the processor is running includes:

Step S110: Acquire multiple illumination images collected for the face of the person to be authenticated under the illumination of the detection light, where the detection light is emitted by the display screen, and is irradiating the face of the person to be authenticated. In the process, the pattern of the detection light is changed at least once, and the pattern of the detection light is randomly changed or preset; the detection light dynamically changes the light emitted to the face of the person to be authenticated. The color and/or position of the detection light is obtained in the following manner: by changing the content displayed on the display screen to dynamically change the mode of light emitted by the display screen, so as to show the person to be authenticated. The face emits the detection light;

Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the plurality of illumination images, so as to obtain the result of the illumination living body verification, wherein Light reflection characteristics include: uniformity and three-dimensional characteristics of the face; and

Step S130: Determine whether the certified person has passed the in vivo verification based at least on the illumination in vivo verification result, so as to obtain the in vivo detection result.

35.A storage medium on which program instructions are stored, and the program instructions are used to execute the following steps during operation:

Obtain the personal identification information of the person to be authenticated;

Judging whether the personal identification information is authenticated information, so as to obtain an information authentication result;

Perform a living body test on the person to be authenticated to obtain a living body test result; and

At least determining whether the identity of the person to be authenticated is legal according to the information authentication result and the living body test result;

Wherein, the step of performing a living body detection on the person to be authenticated to obtain a living body detection result executed by the program instruction during execution includes:

Step S110: Acquire multiple illumination images collected for the face of the person to be authenticated under the illumination of the detection light, where the detection light is emitted by the display screen, and is irradiating the face of the person to be authenticated. In the process, the pattern of the detection light is changed at least once, and the pattern of the detection light is randomly changed or preset; the detection light dynamically changes the light emitted to the face of the person to be authenticated. The color and/or position of the detection light is obtained in the following manner: by changing the content displayed on the display screen to dynamically change the mode of light emitted by the display screen, so as to show the person to be authenticated. The face emits the detection light;

Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the plurality of illumination images, so as to obtain the result of the illumination living body verification, wherein Light reflection characteristics include: uniformity and three-dimensional characteristics of the face; and

Step S130: Determine whether the certified person has passed the in vivo verification based at least on the illumination in vivo verification result, so as to obtain the in vivo detection result.

36. An identity authentication device includes an information collection device, a processor, and a memory, wherein:

The information collection device is used to collect initial information of the person to be authenticated;

Computer program instructions are stored in the memory, and the computer program instructions are used to perform the following steps when the processor is running:

Obtaining personal identification information of the person to be authenticated, where the personal identification information is the initial information or is obtained based on the initial information;

Judging whether the personal identification information is authenticated information to obtain an information authentication result;

Perform a living body test on the person to be authenticated to obtain a living body test result; and

At least determining whether the identity of the person to be authenticated is legal according to the information authentication result and the living body test result;

Wherein, the information collection device includes an image collection device, and the identity authentication device further includes a light source, wherein:

The light source is used to emit detection light to the face of the person to be authenticated;

The image acquisition device is also used to acquire one or more illumination images of the face of the person to be authenticated under the illumination of the detection light;

The step of performing a living body detection on the person to be authenticated to obtain a living body detection result executed by the computer program instructions when the processor is running includes:

Step S110: Acquire multiple illumination images collected for the face of the person to be authenticated under the illumination of the detection light, where the detection light is emitted by the display screen, and is irradiating the face of the person to be authenticated. In the process, the pattern of the detection light is changed at least once, and the pattern of the detection light is randomly changed or preset; the detection light dynamically changes the light emitted to the face of the person to be authenticated The color and/or position of the detection light is obtained in the following manner: by changing the content displayed on the display screen to dynamically change the mode of light emitted by the display screen, so as to show the person to be authenticated The face emits the detection light;

Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the plurality of illumination images, so as to obtain the result of the illumination living body verification, wherein Light reflection characteristics include: uniformity and three-dimensional characteristics of the face; and

Step S130: Determine whether the certified person has passed the in vivo verification based at least on the illumination in vivo verification result, so as to obtain the in vivo detection result.

37.The identity authentication device according to claim 36, wherein the information collection device comprises an image collection device and/or an input device.

38.An identity authentication device includes:

Information collection device for collecting initial information of the person to be authenticated; and

The transmission device is configured to send the initial information to a server, and receive from the server authentication information on whether the identity of the person to be authenticated is obtained by the server in the following manner: obtain the individual of the person to be authenticated Identification information, wherein the personal identification information is the initial information or is obtained based on the initial information; judging whether the personal identification information is authenticated information to obtain an information authentication result; performing a living test on the person to be authenticated Obtaining a living body detection result; and determining whether the identity of the person to be authenticated is legal at least according to the information authentication result and the living body detection result;

Wherein, the server performs a living body detection on the person to be authenticated to obtain a living body detection result in the following manner:

Step S110: Acquire multiple illumination images collected for the face of the person to be authenticated under the illumination of the detection light, where the detection light is emitted by the display screen, and is irradiating the face of the person to be authenticated. In the process, the pattern of the detection light is changed at least once, and the pattern of the detection light is randomly changed or preset; the detection light dynamically changes the light emitted to the face of the person to be authenticated The color and/or position of the detection light is obtained in the following manner: by changing the content displayed on the display screen to dynamically change the mode of light emitted by the display screen, so as to show the person to be authenticated The face emits the detection light;

Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the plurality of illumination images, so as to obtain the result of the illumination living body verification, wherein Light reflection characteristics include: uniformity and three-dimensional characteristics of the face; and

Step S130: Determine whether the certified person has passed the in vivo verification based at least on the illumination in vivo verification result, so as to obtain the in vivo detection result.

39.The identity authentication device according to claim 38, wherein the information collection device comprises an image collection device and/or an input device.

40.An identity authentication device includes a transmission device, a processor, and a memory, wherein:

The transmission device is configured to receive initial information of the person to be authenticated from the client, and send authentication information about whether the identity of the person to be authenticated is legal to the client;

Computer program instructions are stored in the memory, and the computer program instructions are used to perform the following steps when the processor is running:

Obtaining personal identification information of the person to be authenticated, where the personal identification information is the initial information or is obtained based on the initial information;

Judging whether the personal identification information is authenticated information to obtain an information authentication result;

Perform a living body test on the person to be authenticated to obtain a living body test result; and

At least determining whether the identity of the person to be authenticated is legal according to the information authentication result and the living body test result;

Wherein, the step of performing a living body detection on the person to be authenticated to obtain a living body detection result executed by the computer program instructions when the processor is running includes:

Step S110: Acquire multiple illumination images collected for the face of the person to be authenticated under the illumination of the detection light, where the detection light is emitted by the display screen, and is irradiating the face of the person to be authenticated. In the process, the pattern of the detection light is changed at least once, and the pattern of the detection light is randomly changed or preset; the detection light dynamically changes the light emitted to the face of the person to be authenticated. The color and/or position of the detection light is obtained in the following manner: by changing the content displayed on the display screen to dynamically change the mode of light emitted by the display screen, so as to show the person to be authenticated. The face emits the detection light;

Step S120: Determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the plurality of illumination images, so as to obtain the result of the illumination living body verification, wherein Light reflection characteristics include: uniformity and three-dimensional characteristics of the face; and

Step S130: Determine whether the certified person has passed the in vivo verification based at least on the illumination in vivo verification result, so as to obtain the in vivo detection result.

41. An identity authentication device includes:

The information acquisition module is used to acquire the personal identification information of the person to be authenticated;

The authenticated information judgment module is used to judge whether the personal identification information is authenticated information to obtain the information authentication result;

The living body detection module is used to perform the living body detection of the person to be certified to obtain the result of the living body detection; and

The identity determination module is used to determine whether the identity of the person to be authenticated is legal at least based on the information authentication result and the living body test result;

Wherein, the living body detection module includes:

The illumination image acquisition sub-module is used to acquire a plurality of illumination images collected for the face of the person to be authenticated under the illumination of the detection light, wherein the detection light is emitted by the display screen and is used to illuminate the face of the person to be authenticated. During the process of the person's face, the pattern of the detection light is changed at least once, and the pattern of the detection light is randomly changed or preset; the detection light is dynamically changed to the person to be authenticated. The color and/or position of the light emitted by the human face is obtained; the detection light is obtained by changing the content displayed on the display screen to dynamically change the mode of the light emitted by the display screen to The face of the person to be authenticated emits the detection light;

The illumination living body verification sub-module is used to determine whether the face of the person to be authenticated belongs to a living body based on the light reflection characteristics of the face of the person to be authenticated in the plurality of illumination images, so as to obtain the result of the illumination living body verification , Wherein the light reflection characteristics include: uniformity and three-dimensional characteristics of a human face; and

The living body verification pass determination sub-module is used to determine whether the authenticator passes the living body verification based at least on the lighted living body verification result, so as to obtain the living body detection result.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		18264326
	Filing Date		2023-08-04
	First Named Inventor	Peter SCHILLEN	
	Art Unit	2668	
	Examiner Name	Phuoc Tran	
	Attorney Docket Number	27843-2262 202595US01	

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	1	108573203	CN	A	2018-09-25	BEIJING KUANGSHI TECHNOLOGY CO., LTD.	English-language translation attached	☒

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number	18264326
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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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***EXAMINER:** Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		18264326
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	Examiner Name	Phuoc Tran	
	Attorney Docket Number		27843-2262 202595US01

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached timing statement under 37 CFR 1.97(e).

☐ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

☒ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

☐ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

☒ No IDS size fee is required under 37 CFR 1.17(v) at this time.

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2025-06-12
Name/Print	Christopher M. Goff	Registration Number	41785

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number	18264326
Filing Date	2023-08-04
First Named Inventor	Peter SCHILLEN
Art Unit	2668
Examiner Name	Phuoc Tran
Attorney Docket Number	27843-2262 202595US01

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



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PETITION TO WITHDRAW AN APPLICATION FROM ISSUE AFTER PAYMENT OF THE ISSUE FEE UNDER 37 CFR 1.313(c)(2) with Assigned Patent Number

APPLICATION #
18264326

ATTORNEY DOCKET #
27843-2262 202595US01

FILING DATE
08/04/2023

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

CONFIRMATION # 2486

FILING DATE 08/04/2023

ART UNIT 2668

FIRST NAMED INVENTOR Peter SCHILLEN

EXAMINER PHUOC TRAN

An application may be withdrawn from issue for further action upon petition by the applicant. To request that the Office withdraw an application from issue, applicant must file a petition under this section including the fee set forth in § 1.17(h) and a showing of good and sufficient reasons why withdrawal of the application from issue is necessary.

APPLICANT HEREBY PETITIONS TO WITHDRAW THIS APPLICATION FROM ISSUE UNDER 37 CFR 1.313(c).

A grantable petition requires the following items:

- (1) Petition fee; and
- (2) One of the following reasons: A grantable petition requires the following items: (a) Unpatentability of one or more claims, which must be accompanied by an unequivocal statement that one or more claims are unpatentable, an amendment to such claim or claims, and an explanation as to how the amendment causes such claim or claims to be patentable;
- (b) Consideration of a request for continued examination in compliance with § 1.114 (for a utility or plant application only); or
- (c) Express abandonment of the application. Such express abandonment may be in favor of a continuing application, but not a CPA under 37 CFR 1.53(d).

Reason for withdrawal

Consideration of a request for continued examination (Quick Path Information Disclosure Statement) - RCE request,

submission, and fee are attached

Applicant claims the following entity status
Regular Undiscounted

Signature

I certify, in accordance with 37 CFR 1.4(d)(4) that I am:
An attorney or agent registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Zachary Wiersma/	Zachary Wiersma	77808



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GRANT LETTER

APPLICATION #	DECISION DATE	DECISION ON PETITION #	ATTORNEY DOCKET NUMBER
18/264,326	06/12/2025		27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

In re Application of Peter SCHILLEN

This is an electronic decision on the petition under , filed June 12, 2025 to withdraw the above-identified application from issue after payment of the issue fee.

The request is **GRANTED**.

Petitioner is advised that the issue fee paid in this application cannot be refunded. If, however, this application is again allowed, petitioner may request that it be applied towards the issue fee required by the new Notice of Allowance.

The application is being referred to Technology Center AU for

Telephone inquired concerning this decision should be directed to the Patent Electronic Business Center (EBC) at 866-217-9197.

Office of Petitions

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/264,326	06/12/2025 03:06:55 PM Z ET	27843-2262 202595US01

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	12340630
CONFIRMATION #	2486	FILED BY	Zachary Wiersma
PATENT CENTER #	70867982	AUTHORIZED BY	-
CUSTOMER #	76656	FILING DATE	08/04/2023
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Peter SCHILLEN

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
DA / 012384	E20256BF07056628	Zachary Wiersma

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1464	PETITIONS REQUIRING THE PETITION FEE SET FORTH IN 37 CFR 1.17(H) (GROUP III)	150.00	1	150.00
1801	REQUEST FOR CONTINUED EXAMINATION (RCE) - 1ST REQUEST (SEE 37 CFR 1.114)	1500.00	1	1500.00
TOTAL AMOUNT:				\$1,650.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

CERTIFICATION AND REQUEST FOR CONSIDERATION OF AN INFORMATION DISCLOSURE STATEMENT FILED AFTER PAYMENT OF THE ISSUE FEE UNDER THE QPIDS PROGRAM

 Non-Provisional Application Number: **18/264,326**

 Filing Date: **2023-08-04**

 First Named Inventor: **Peter SCHILLEN**

 Title of Invention: **OPTICAL SKIN DETECTION FOR FACE UNLOCK**

THE UNDERSIGNED HEREBY CERTIFIES AND REQUESTS THE FOLLOWING FOR THE ABOVE- IDENTIFIED APPLICATION.

1. Consideration is requested of the information disclosure statement (IDS) submitted herewith, which is being filed after payment of the issue fee.
2. Check one of the following boxes:
 - ☒ Each item of information contained in the IDS was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the IDS. See 37 CFR 1.97(e)(1).
 - ☐ No item of information contained in the IDS was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the IDS was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the IDS. See 37 CFR 1.97(e)(2).
 - ☒ See attached certification statement in compliance with 37 CFR 1.97(e).
3. The IDS size fee written assertion under 37 CFR 1.98 is submitted herewith as (i) part of the PTO/SB/08 – Patent Center IDS, or equivalent, or (ii) an additional IDS paper, such as the form PTO/SB/08c or equivalent.
4. If an IDS size fee is due, the Director is authorized to charge the IDS size fee under 37 CFR 1.17(v) in the amount of \$ _____ for the IDS submitted herewith to deposit account _____. (Do not complete if this information is being provided on the form PTO/SB/08c, or equivalent).
WARNING: Failure to include the assertion and/or an authorization to charge to a deposit account the applicable IDS size fee will result in automatic entry of the RCE and the IDS will be placed in the file but not considered.
5. The Director is authorized to charge the IDS timing fee under 37 CFR 1.17(p) to deposit account 012384.
6. A Petition to Withdraw from Issue After Payment of the Issue Fee (37 CFR 1.313(c)(2)), including the petition fee set forth in 37 CFR 1.17(h), is submitted herewith as a **Web-based ePetition**.
WARNING: Failure to use the Web-based ePetition interface will result in automatic entry of the RCE.
7. A request for continued examination (RCE) under 37 CFR 1.114 and the RCE fee under 37 CFR 1.17(e) are submitted herewith.
8. The RCE for a compliant QPIDS will be treated as a "conditional" RCE. In the event the examiner determines that any item of information contained in the IDS necessitates the reopening of prosecution in the application, the undersigned understands that (i) the RCE will be processed and treated as an RCE under 37 CFR 1.114 and therefore (ii) the IDS fee under 37 CFR 1.17(p) will be returned in accordance with 37 CFR 1.97(b)(4). In the event that no item of information in the IDS necessitates reopening prosecution, the undersigned understands that the RCE will not be processed and the RCE fee under 37 CFR 1.17(e) will be returned.
9. This certification and request is being filed as a **Web-based ePetition** and is not accompanied by an amendment to the application. Inclusion of an amendment will result in automatic entry of the RCE.

 Signature **/Christopher M. Goff/**

 Date **2025-06-12**

 Name (Print/Typed) **Christopher M. Goff**

 Practitioner
 Registration Number **41,785**
 (If applicable)

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. The party signing this form must be a listed fee payer for the above-identified deposit account in the USPTO's online fee payment management tool. Submit multiple forms if more than one signature is required and complete below.*

☒ *Total of 1 forms are submitted.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013), <https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>.

Routine uses of the information in this record may include disclosure to: 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law; 2) a federal, state, local, or international agency, in response to its request; 3) a contractor of the USPTO having need for the information in order to perform a contract; 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record; 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record; 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations; 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals; 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)); 9) the Office of Personnel Management (OPM) for personnel research purposes; and 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
18/264,326	06/12/2025 03:06:55 PM Z ET	27843-2262 202595US01

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OPTICAL SKIN DETECTION FOR FACE UNLOCK

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CONFIRMATION #	2486	FILED BY	Zachary Wiersma
PATENT CENTER #	70867982	FILING DATE	08/04/2023
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	-

Documents

TOTAL DOCUMENTS: 5

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
petition-request.pdf	2	ePetition Request Form	29 KB
2025-06-12 QPIDS Form 27843-2262.pdf	2	Quick Path Information Disclosure Statement	230 KB
2025-06-12 SuppIDS 27843-2262.pdf	5	Information Disclosure Statement (IDS) Form (SB08)	1285 KB
FOR_CN108573203 w translation.pdf	198	Foreign Reference	10645 KB
2025-06-12 RCE 27843-2262.pdf	3	Request for Continued Examination (RCE)	1631 KB

Digest

DOCUMENT

MESSAGE DIGEST(SHA-512)

petition-request.pdf	E556E0186C22E5EAC6052A82B508993F1254605444FABAB332 E9D9BDD18B5821813332FB45C9AF46BA6A11372D4C9D78BD 947C2272FEF87F3B3A196F6BA16204
2025-06-12 QPIDS Form 27843-2262.pdf	954179F9283F9BF291D619B145CBCB10B184C734400FE0A672 5A51CF1A0F9BF44FA75E643F3E842120125D1E9EE271D27AF C9FEE998E0B322BC5E29E43E58B38
2025-06-12 SuppIDS 27843- 2262.pdf	5A7213A50D04FA4C14E361477AA64B0011FA03C8067B1BCAA 4D8CB3D5249EE2B725CCBEC56F7112703502A94C8523058B5 26AA7BA5E6331C0FF07CA1D0F2C544
FOR_CN108573203 w translation.pdf	36CF0A91A16B335B7B97D920FF5068D08C372A19E510DD279 DA07DC50AF6CFB8E7C314B6704432900F49CDBDFD25C9A83 578E681D8993C5EE71FEE3AE6309999
2025-06-12 RCE 27843- 2262.pdf	A186A8AF7DDD348465E4F6144C648DDF45EC150072E222508 54DFCCAF68B14DBAFFC679F3FC4AB07D41019011D3D3EBE CF770B674943EC74DA62C70A5386305B

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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United States Patent and Trademark Office

Office of the Chief Financial Officer

Document Code:WFEE

User :Terrance Lawrence

Sale Accounting Date:06/13/2025

Sale Item Reference Number

18264326

Effective Date

06/12/2025

Document Number	Fee Code	Fee Code Description	Amount Paid	Payment Method
I20256CH32059412	1806	SUBMISSION- INFORMATION DISCLOSURE STMT	\$280.00	Deposit Account



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UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	08/04/2023	Peter SCHILLEN	27843-2262 202595US01	2486
76656	7590	06/17/2025	EXAMINER	
Patent Docket Department			TRAN, PHUOC	
Armstrong Teasdale LLP			ART UNIT	PAPER NUMBER
7700 Forsyth Boulevard			2668	
Suite 1800			NOTIFICATION DATE	DELIVERY MODE
St. Louis, MO 63105			06/17/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@atllp.com

CORRECTED
Notice of Allowability

Application No.
18/264,326

Examiner
PHUOC TRAN

Applicant(s)
SCHILLEN et al.

Art Unit
2668

**AIA (First
Inventor to File)
Status** Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to IDS filed 6/12/2025.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☐ The allowed claim(s) is/are _____. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|---|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____. | 6. ☐ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material _____. | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/PHUOC TRAN/
Primary Examiner, Art Unit 2668

Doc code: IDS

PTO/SB/08a (01-25)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2027. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		18264326
	Filing Date		2023-08-04
	First Named Inventor	Peter SCHILLEN	
	Art Unit	2668	
	Examiner Name	Phuoc Tran	
	Attorney Docket Number	27843-2262 202595US01	

U.S. PATENTS							Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Inventor of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	
	1						

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵	
	1	108573203	CN	A	2018-09-25	BEIJING KUANGSHI TECHNOLOGY CO., LTD.	English-language translation attached	☒	

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NON-PATENT LITERATURE DOCUMENTS				Remove
Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		18264326
	Filing Date		2023-08-04
	First Named Inventor	Peter SCHILLEN	
	Art Unit		2668
	Examiner Name	Phuoc Tran	
	Attorney Docket Number		27843-2262 202595US01

	1		☐
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If you wish to add additional non-patent literature document citation information, please click the Add button 

EXAMINER SIGNATURE			
Examiner Signature	/PHUOC TRAN/	Date Considered	06/14/2025

***EXAMINER:** Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

Application Number	18264326		
Filing Date	2023-08-04		
First Named Inventor	Peter SCHILLEN		
Art Unit	2668		
Examiner Name	Phuoc Tran		
Attorney Docket Number	27843-2262 202595US01		

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☒ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached timing statement under 37 CFR 1.97(e).

☐ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

☒ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

☐ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

☒ No IDS size fee is required under 37 CFR 1.17(v) at this time.

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2025-06-12
Name/Print	Christopher M. Goff	Registration Number	41785

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

Application Number	18264326
Filing Date	2023-08-04
First Named Inventor	Peter SCHILLEN
Art Unit	2668
Examiner Name	Phuoc Tran
Attorney Docket Number	27843-2262 202595US01

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



United States Patent and Trademark Office

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Document Code:WFEE

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	07/15/2025	12361760	27843-2262 202595US01	2486

76656 7590 07/02/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/264,326	08/04/2023	Peter SCHILLEN	27843-2262 202595US01	2486
76656	7590	07/15/2025	EXAMINER	
Patent Docket Department			TRAN, PHUOC	
Armstrong Teasdale LLP			ART UNIT	PAPER NUMBER
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Suite 1800			NOTIFICATION DATE	DELIVERY MODE
St. Louis, MO 63105			07/15/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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APPLICATION NO.	ISSUE DATE	PATENT NO.
18/264,326	15-Jul-2025	12361760

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
St. Louis, MO 63105

EGRANT NOTIFICATION

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